

Drug info: Ethosuximide (Zarontin®)

Indications

DOC in ABS.

Dose

R Adult: 500–1500 mg/d. Peds: 10–40 mg/kg/d; MDF = q d. **Supplied:** oral only; capsules 250 mg; syrup 250 mg/5-ml.

Pharmacokinetics (► Table 27.10)

Table 27.10 Pharmacokinetics of ethosuximide

t _{1/2} (half-life)	t _{PEAK} (peak levels)	t _{SS} (steady state)	Therapeutic level (mcg/ml)
adult: 40–70 hrs peds: 20–40 hrs	1–4 hrs	adult: up to 14 days peds: up to 7 days	40–100

Side effects

N/V; lethargy; hiccups; H/A; rarely: eosinophilia, leukopenia, erythema multiforme, Stevens-Johnson syndrome, SLE-like syndrome. Toxic levels→ psychotic behavior.

Drug info: Methsuximide (Celontin®)

Indications

Indicated for absence seizures refractory to other drugs.

Dose

R optimum dose must be determined by trial. Start with 300 mg PO q d, increase by 300 mg at weekly intervals PRN to a maximum of 1200 mg/d. **Supplied:** 150 & 300 mg capsules.

Drug info: Felbamate (Felbatol®)

✱ **CAUTION:** Due to an unacceptably high rate of aplastic anemia and hepatic failure, felbamate (FBM) should not be used except in those circumstances where the benefit clearly outweighs the risk; then, hematologic consultation is recommended by the manufacturer. See Side effects below (also for drug-drug interactions).

FBM is efficacious for monotherapy and adjunctive therapy for partial seizures (complex and secondary generalization), and reduces the frequency of atonic and GTC seizures in Lennox-Gastaut syndrome.

Pharmacokinetics (► Table 27.11)

Table 27.11 Pharmacokinetics of felbamate

t _{1/2} (half-life)	t _{PEAK} (peak levels)	t _{SS} (steady state)	Therapeutic level (mcg/ml)
20–23 hrs	1–3 hrs	5–7 days	not established

Dose

Table 27.12 Effect of felbamate on other ASM levels

ASM	Change in level	Recommended dosing change
phenytoin	↑ 30–50%	↓ 20–33%
carbamazepine	↓ 30% total ↑ 50–60% epoxide	↓ 20–33%
valproic acid	↑ 25–100%	↓ 33%

R: CAUTION see above. Felbamate is *not* to be used as a first-line drug. Patient or guardian should sign informed consent release. Start with 1200 mg/d divided BID, TID, or QID, and decrease other ASMs by one-third. Increase felbamate biweekly in 600 mg increments to usual dose of 1600–3600 mg/d (max: 45 mg/kg/d). Slow down increments and/or reduce other ASMs further if side effects become severe. Administer at upper end of range when used as monotherapy. **Supplied:** (oral only) 400 & 600 mg scored tablets; suspension 600 mg/5-ml.

Side effects

Felbamate has been associated with aplastic anemia (usually discovered after 5–30 wks of therapy) in $\approx$ 2–5 cases per million persons per yr, and hepatic failure (some fatal, necessitating baseline and serial LFTs every 1–2 wks). Other side effects: insomnia, anorexia, N/V, H/A. Felbamate is a potent metabolic inhibitor; thus it is necessary to reduce the dose of phenytoin, valproate or carbamazepine when used with felbamate²² (► Table 27.12, general rule: drop dose by one third).

Drug info: Levetiracetam (Keppra®)

No identified drug-drug interactions. Less than 10% protein bound. Linear pharmacokinetics, no level monitoring needed.

Indications

Adjunctive therapy for partial onset Sz with secondary generalization in patients 4 years of age and older. Myoclonic seizures (juvenile myoclonic epilepsy). Generalized tonic-clonic.

Dose

R start with 500 mg PO BID. Increment by 1000 mg/d q 2 weeks PRN to a maximum of 3000 mg/d. Keppra XR: the same dose of levetiracetam can be converted to Keppra XR for q d dosing.

IV: 500–1500 mg diluted in 100 ml of diluent (LR, D5 W, normal saline) infused over 15 minutes BID.

Supplied: 250, 500, 750 & 1000 mg scored film-coated tabs; 100 mg/ml oral solution. Keppra XR (extended release) 500 mg.

IV: 1 vial (5 ml) contains 500 mg.

Side effects

PO or IV: somnolence and fatigue in 15%. Dizziness in 9%. Asthenia 15% and infection 13% (nasopharyngitis and influenza may or may not have been related).

Keppra XR: somnolence 8%, irritability 6%.

Drug info: Clonazepam (Klonopin®)

A benzodiazepine derivative.

Indications

✗ *Not* a recommended drug for seizures (see below).

Used for myoclonic, atonic, and absence seizures (in absence, less effective than valproate or ethosuximide, and tolerance may develop).

NB: clonazepam usually works very well for several months, and then tends to become less effective, leaving only the sedating effects. Also, many cases have been reported of patients having seizures during withdrawal, including status epilepticus (even in patients with no history of status). Thus, may need to taper this drug over 3–6 months.

Dose

R Adult: start at 1.5 mg/d DIV TID, increase by 0.5–1 mg q 3 d, usual dosage range is 1–12 mg/d (max 20 mg/d); MDF = q d. Peds: start at 0.01–0.03 mg/kg/d DIV BID or TID, increase by 0.25–0.5 mg/kg/d q 3 d; usual dosage range is 0.01–0.02 mg/kg/d; MDF = q d. **Supplied:** oral only; scored tabs: 0.5 mg, 1 mg, 2 mg.

Pharmacokinetics (► Table 27.13)**Table 27.13** Pharmacokinetics of clonazepam

$t_{1/2}$ (half-life)	t_{PEAK} (peak levels)	t_{SS} (steady state)	$t_{\text{D/C}}$ (discontinue)	Therapeutic level (mcg/ml)
20–60 hrs	1–3 hrs	up to 14 days	≈ 3–6 months ^a	0.013–0.072

^aCAUTION: withdrawal seizures are common, see text above**Side effects**

Ataxia; drowsiness; behavior changes.

Drug info: Zonisamide (Zonegran®)**Indications**

A broad spectrum ASM, FDA approved as adjunctive therapy for FOS in adults.

Side effects

Cognitive slowing, especially at higher doses (but less severe than tiagabine).

Dose**R** Initiate therapy with 100 mg PO q PM × 2 wks, then increase dose by 100 mg/d q 2 wks up to 400 mg/d. Bioavailability is not affected by food. Steady state is achieved within 14 days of dosage changes. **Supplied** : 25, 50 & 100 mg capsules. 100 mg/5 ml strawberry flavored liquid.**Drug info: Acetazolamide (Diamox®)**

The anti-epileptic effect may be either due to direct inhibition of CNS carbonic anhydrase (which also reduces CSF production rate) or due to the slight CNS acidosis that results.

Indications

Centricephalic epilepsies (absence, nonfocal seizures). Best results are in absence seizures; however benefit has also been observed in GTC, myoclonic jerk.

Side effects**✖** Do not use in first trimester of pregnancy (may be teratogenic, although this is disputed). The diuretic effect causes renal loss of HCO_3^- (bicarbonate), which may lead to an acidotic state with long-term therapy. A sulfonamide, therefore any typical reaction to this class may occur (anaphylaxis, fever, rash, Stevens-Johnson syndrome, toxic epidermal necrolysis...). Paresthesias: medication should be discontinued.**Dose****R** Adult: 8–30 mg/kg/d in divided doses (max 1 gm/d, higher doses do not improve control). When given with another ASM, the suggested starting dose is 250 mg once daily, and this is gradually increased. **Supplied**: tablets 125, 250 mg. Diamox sequels® are sustained release 500 mg capsules. Sterile cryodesiccated powder is also available in 500 mg vials for parenteral (IV) use.**Drug info: Gabapentin (Neurontin®)**

Although developed to be a GABA agonist, it does not interact at any known GABA receptor. Efficacious for primary generalized seizures and FOS (with or without secondary generalization). Ineffective for absence seizures. Very low incidence of known side effects. No known drug interactions (probably because it is renally excreted). Also used for central pain

Dose

R Adult: 300 mg PO $\times$ 1 day 1; 300 mg BID day 2; 300 mg TID day 3; then increase rapidly up to usual doses of $\approx$ 800–1800 mg per day. Doses of 1800–3600 may be needed in intractable patients.

✖ Dosage must be reduced in patients with renal insufficiency or on dialysis, see Eq (7.1) to estimate. **Supplied:** 100, 300, 400, 600, 800 mg capsules. 50 mg/ml suspension.

Pharmacokinetics (► Table 27.14)

Gabapentin is not metabolized, and 93% is excreted unchanged renally with plasma clearance directly proportional to creatinine clearance.²³ Does not affect hepatic microsomal enzymes, and does not affect metabolism of other ASMs. Antacids decrease bioavailability by $\approx$ 20%, therefore give gabapentin $>$ 2 hrs after the antacid.²⁴

Table 27.14 Pharmacokinetics of gabapentin

$t_{1/2}$ (half-life)	t_{PEAK} (peak levels)	t_{SS} (steady state)	Therapeutic level (mcg/ml)
5–7 hrs ^a	2–3 hrs	1–2 days	not established

^awith normal renal function

Side effects

Somnolence, dizziness, ataxia, fatigue, nystagmus; all reduce after 2–3 weeks of drug therapy. Increased appetite. Not known to be teratogenic.

Drug info: Lamotrigine (Lamictal®)

Anticonvulsant effect may be due to presynaptic inhibition of glutamate release.²³

Efficacious as *adjunctive* therapy for partial seizures (with or without secondary generalization) and Lennox-Gastaut syndrome. Preliminary data suggest it may also be useful as an adjunct for refractory generalized seizures, or as monotherapy for newly diagnosed partial or generalized seizures.²⁵ Also FDA approved for bipolar disorder.

Side effects

Somnolence, dizziness, diplopia. **✖** Serious rashes requiring hospitalization and discontinuation of therapy have been reported (rash usually begins 2 weeks after initiating therapy and may be severe and potentially life-threatening, including Stevens-Johnson syndrome (more of a concern with simultaneous use of valproate), and rarely, toxic epidermal necrolysis (TEN)). Incidence of significant epidermal reaction may be decreased by a slow ramping-up of dosage. May increase seizure frequency in some patients with severe myoclonic epilepsy of infancy.²⁶ Metabolism of lamotrigine is affected by concurrent use of other ASMs.

Dose

R Adult: In adults receiving enzyme-inducing ASMs (PHT, CBZ, or phenobarbital), start with 50 mg PO q d $\times$ 2 wks, then 50 mg BID $\times$ 2 wks, then $\uparrow$ by 100 mg/d q week until the usual maintenance dose of 200–700 mg/d (divided into 2 doses) is reached. For patients on valproic acid (VA) alone, the maintenance dose was 100–200 mg/d (divided into 2 doses), and VA levels drop by $\approx$ 25% within a few weeks of starting lamotrigine. For patients on both enzyme-inducing ASMs and VA, the starting dose is 25 mg PO qod $\times$ 2 wks, then 25 mg qd $\times$ 2 wks, then $\uparrow$ by 25–50 mg/d q 1–2 wks up to a maintenance of 100–150 mg/d (divided into 2 doses). Instruct patients that rash, fever or lymphadenopathy may herald a serious reaction and that a physician should be contacted immediately. Peds: not indicated for use in patients $<$ 16 yrs old due to higher incidence of potentially life-threatening rash in the pediatric population.²³ **Supplied:** 25, 100, 150 & 200 mg tablets. 2, 5 & 25 mg chewable dispersible tablets.

Pharmacokinetics²⁵ (► Table 27.15)

Table 27.15 Pharmacokinetics of lamotrigine

$t_{1/2}$ (half-life)	t_{PEAK} (peak levels)	t_{SS} (steady state)	Therapeutic level (mcg/ml)
24 hrs ^a	1.5–5 hrs	4–7 days	controversial ²⁷

^ahalf-life is shortened to $\approx$ 15 hrs by PHT and CBZ, whereas valproic acid increases it to 59 hrs

Drug info: Vigabatrin (Sabril®, Vigadrone®)

Irreversible GABA transaminase inhibitor, causing an increase in synaptic GABA.

Indications

A narrow spectrum drug used for FOS. Also effective against infantile spasms especially as part of tuberous sclerosis.

✖ May worsen absence and myoclonic seizures in GGE.

Side effects

✖ Causes a progressive and permanent concentric visual field constriction in 30–40% of patients, the higher the risk with the higher the dose and the duration of treatment.

Less severe adverse effects include sedation, fatigue, irritability, behavior changes, psychosis, depression and weight gain.

Dose

Because of visual toxicity, visual fields should be monitored every 3 months.

R Adult: 1500–3000 mg/d. Start at 500 mg BID, titrating upwards by 500 mg/d every week up to a maximum of 1.5 gm BID. Discontinue the drug if significant benefit does not occur in the 1st 3 months.

Drug info: Topiramate (Topamax®)

May block voltage-sensitive sodium channels and enhance GABA activity at GABA-A receptors and attenuate some glutamate receptors.²³

Indications²⁸

A broad spectrum ASM. FDA approved for initial monotherapy for FOS and GTC, but is not a drug of first choice due to cognitive side effects.

Dose

NB: titrate slowly to deal with adverse cognitive effects.

R Adult: start with 50 mg/d and increase slowly up to 200–400 mg/d,²⁹ with no significant benefit noted at dosages > 600 mg/d.³⁰ **Supplied:** 25, 100, & 200 mg tabs.

Pharmacokinetics (► Table 27.16)

30% is metabolized in the liver, the rest is excreted unchanged in the urine.

Table 27.16 Pharmacokinetics of topiramate

t _{1/2} (half-life)	t _{ss} (steady state)	Therapeutic level
19–25 hrs	5–7 days	not established

Side effects

May increase phenytoin concentration by up to 25%. Levels of topiramate are reduced by other ASMs (phenytoin, carbamazepine, valproic acid, and possibly others).

Cognitive impairment (word-finding difficulty, problems with concentration, more pronounced than with lamotrigine...), weight loss, dizziness, ataxia, diplopia, paresthesias, nervousness, and confusion have been troublesome. ≈ 1.5% incidence of renal stones which usually pass spontaneously.²³

Oligohidrosis (reduced sweating) and hyperthermia, primarily in children in association with elevated environmental temperatures and/or vigorous physical activity.

Drug info: Tiagabine (Gabitril®)

A GABA uptake inhibitor, with cognitive problems of a similar frequency to that with topiramate.³¹

R Adult: start with 4 mg/d, increase weekly by 4–8 mg to a maximum of 32–56 mg (divided BID to QID). **Supplied:** 4, 12, 16 & 20 mg tablets.

Drug info: Cenobamate (XCOPRI®)

2 mechanisms of action: 1) blocking the sodium channel (predominantly attenuating the persistent sodium current rather than the transient voltage-gated sodium current), and 2) enhancing GABA activity.

Indications

FDA approved for FOS (it showed exceptional efficacy as adjunctive therapy in this regard⁶).

Dose

R Adult: taken once daily, start at 12.5 mg/d for 2 weeks, then 25 mg/d for 2 weeks, then 50 mg/d for 2 weeks, and then 100 mg/d. Thereafter, increase by 50 mg every 2 weeks to a maximum of 400 mg/d. (NB: slow titration appears to be important to reduce the occurrence of DRESS (drug rash with eosinophilia and systemic symptoms).³² **Supplied:** 12.5, 25, 50, 150 & 200 mg tabs.

Drug info: Lacosamide (Vimpat®)

Enhances slow inactivation of voltage gated sodium channels, affecting only neurons that are depolarized or active over a prolonged period (as in a seizure).

Indications

FOS & GTC. Painful diabetic neuropathy.

Dose

R Adult: start at 100 mg/d, either once at bedtime or divided into 2 daily doses. Increase by 100 mg/d every 1-2 weeks to the range of 200-400 mg/d (maximum: 600 mg/d). **Supplied:** 50, 100, 150 & 200 mg tablets. 200 mg/20 ml single use vial for IV use.

27.4 Withdrawal of antiseizure medications

27.4.1 General information

Most seizure recurrences develop during the first 6 months after ASM withdrawal.³³

27.4.2 Indications for ASM withdrawal

There is no agreement on how long a patient should be seizure-free before withdrawal of antiseizure medications, nor is there agreement on the prognostic value of EEGs and on the best time period over which to withdraw ASMs.

The following is based on a study of 92 patients with *idiopathic* epilepsy, who had been free of seizures for two years.³⁴ Generalization, e.g., to posttraumatic seizures, may not be appropriate. Taper was by 1 "unit" q 2 weeks (where a unit is defined as 200 mg for CBZ or valproic acid, or 100 mg for PHT). Follow-up: mean = 26 mos (range: 6-62).

31 patients (34%) relapsed, with the average time to relapse being 8 mos (range: 1-36). Using actuarial methods, the risk for recurrence is 5.9%/month for 3 months, then 2.7%/month for 3 months, then 0.5%/month for 3 months. Factors found to affect the likelihood of relapse include:

1. seizure type: 37% relapse rate for generalized seizures; 16% for complex or simple partial; 54% for complex partial with secondary generalization
2. number of seizures before control attained: those with ≥ 100 seizures before control had statistically significant higher relapse rate than those with < 100
3. the number of drugs that had to be tried before single drug therapy successfully controlled seizures: 29% if 1st drug worked, 40% if a change to a 2nd drug was needed, and 80% if a change to a 3rd drug was required
4. EEG class ($\gg$ Table 27.17): class 4 had worst prognosis for relapse. Epileptiform discharges on EEG serves to discourage ASM withdrawal³⁵

In a larger randomized study,³⁶ the most important factors identified to predict freedom from recurrent seizures were:

1. longer seizure-free period

Table 27.17 EEG class and seizure relapse rate

Class	EEG description		Relapse rate	No. of relapses/ patients at risk
	Before treatment	Before withdrawal		
1	normal	normal	34%	11/31
2	abnormal	normal	11%	4/35
3	abnormal	improved	50%	2/4
4	abnormal	unchanged	74%	14/19

2. use of only one ASM (vs. multiple ASMs)
3. seizures other than tonic-clonic seizures

27.4.3 Withdrawal times

The recommended withdrawal times in ► Table 27.18 should be used only as guidelines.

Table 27.18 Recommended ASM withdrawal times

ASM	Recommended withdrawal period
phenytoin, valproic acid, carbamazepine	2–4 weeks
phenobarbital	6–8 weeks (25% per week)
clonazepam	3–6 months; see CAUTION (p. 496)

27.5 Pregnancy and antiseizure medications

27.5.1 General information

Women of childbearing potential with epilepsy should undergo counseling regarding pregnancy.³⁷

27.5.2 Birth control

ASMs that induce liver microsomal cytochrome P450 enzymes (► Table 27.19) increase the failure rate of oral contraceptives up to fourfold.³⁸ Patients desiring to use BCPs should employ barrier contraceptive measures until ovulation is consistently suppressed, and they should watch for breakthrough bleeding which may indicate a need for a change in the hormone dosage.³³ Non-oral hormonal contraceptives (e.g., levonorgestrel implant [Norplant®]) circumvent first pass liver degradation but should be combined with a barrier method because of declining effectiveness with time.

Table 27.19 Effect of ASMs on liver cytochrome P₄₅₀^a

Inducers	Noninducers
carbamazepine phenobarbital phenytoin felbamate primidone	valproic acid benzodiazepines gabapentin lamotrigine

^areferences^{33,39}

27.5.3 Complications during pregnancy

Women with epilepsy have more complications with pregnancy than mothers without epilepsy, but > 90% of pregnancies have a favorable outcome.³³

There is an increase in the number of gravid seizures in ≈ 17% (reported range: 17–30%) of epileptic women, which may be due to noncompliance or to changes of free drug levels of ASMs during pregnancy (► Table 27.20). Isolated seizures can occasionally be deleterious, but usually cause no problem. Status epilepticus poses serious risk to mother and fetus during pregnancy and should be treated aggressively.

There is also a slightly increased risk of toxemia (HTN of pregnancy) and fetal loss.

Table 27.20 Changes in free ASM levels during pregnancy⁴⁰

Drug	Change
carbamazepine	↓ 11%
phenobarbital	↓ 50%
phenytoin	↓ 31%
valproic acid	↑ 25%

27.5.4 Birth defects

The incidence of fetal malformations in offspring of patients with a known seizure disorder is $\approx$ 4–5%, or approximately double that of the general population.⁴¹ The degree to which this is due to the use of ASMs vs. genetic and environmental factors is unknown. All ASMs have the potential to cause deleterious effects on the infant. Polytherapy is associated with an increased risk over monotherapy in a more than additive manner.

Generally, the risk of seizures (with possible concomitant maternal and fetal hypoxia and acidosis) is felt to outweigh the teratogenic risk of most ASMs, but this must be evaluated on a case-by-case basis. Occasionally patients may be weaned off ASMs.

Specific drugs

Carbamazepine (CBZ) produced an increased incidence of “minor” malformations (but not of “major” malformations) in one study⁴² (this study may have had methodologic problems), and may increase the incidence of neural tube defects (NTD).⁴³ In utero exposure to phenytoin may lead to the fetal hydantoin syndrome^{13,44} and a child with an IQ lower by $\approx$ 10 points.⁴⁵ Phenobarbital produced the highest incidence of major malformations (9.1%) in one prospective study⁴⁶ and was also associated with most of the increase in fetal death or anomalies in another study.⁴⁷ Valproate (VA) causes the highest incidence of NTD (1–2%¹⁷), which can be detected with amniocentesis and allow an abortion if desired. TID dosing may reduce the risk of NTD (p.492). Benzodiazepines given shortly before delivery can produce the “floppy infant syndrome.”⁴⁸ Similar effects may occur with other sedating ASMs such as phenobarbital.

Drug recommendations

A general consensus is that for most women of childbearing potential who require ASMs, that monotherapy with the lowest dose of CBZ that is effective is the method of choice if the seizure disorder is responsive to it.⁴⁹ If ineffective, then monotherapy with valproic acid (with TID dosing) is currently the recommended second choice. Folate supplementation (after confirming normal B12 levels) should be used in all.

References

- [1] Brodie MJ, Dichter MA. Antiepileptic Drugs. New England Journal of Medicine. 1996; 334:168–175
- [2] Park KM, Kim SE, Lee BI. Antiepileptic Drug Therapy in Patients with Drug-Resistant Epilepsy. J Epilepsy Res. 2019; 9:14–26
- [3] Abou-Khalil BW. Update on Antiseizure Medications 2022. Continuum (Minneapolis Minn). 2022; 28:500–535
- [4] Mattson RH, Cramer JA, Collins JF, et al. Comparison of Carbamazepine, Phenobarbital, Phenytoin, and Primidone in Partial and Secondarily Generalized Tonic-Clonic Seizures. N Engl J Med. 1985; 313:145–151
- [5] Mattson RH, Cramer JA, Collins JF, et al. A Comparison of Valproate with Carbamazepine for the Treatment of Complex Partial Seizures and Secondarily Generalized Tonic-Clonic Seizures in Adults. New England Journal of Medicine. 1992; 327:765–771
- [6] Chung SS, French JA, Kowalski J, et al. Randomized phase 2 study of adjunctive clobazepam in patients with uncontrolled focal seizures. Neurology. 2020; 94:e2311–e2322
- [7] Marson AG, Al-Kharusi AM, Alwaidh M, et al. The SANAD study of effectiveness of valproate, lamotrigine, or topiramate for generalised and unclassifiable epilepsy: an unblinded randomised controlled trial. Lancet. 2007; 369:1016–1026
- [8] Drugs for Epilepsy. Medical Letter. 1986; 28:91–93
- [9] Rambeck B, Boenigk HE, Dunlop A, et al. Predicting Phenytoin Dose - A Revised Nomogram. Ther Drug Monit. 1979; 1:325–333
- [10] Saklad JJ, Graves RH, Sharp WP. Interaction of Oral Phenytoin with Enteral Feedings. J Parent Ent Nutr. 1986; 10:322–323
- [11] Worden JP, Wood CA, Workman CH. Phenytoin and Nasogastric Feedings. Neurology. 1984; 34
- [12] DeToledo JC, Ramsay RE. Fosphenytoin and phenytoin in patients with status epilepticus: improved tolerability versus increased costs. Drug Saf. 2000; 22:459–466
- [13] Buehler BA, Delimont D, van Waes M, et al. Prenatal Prediction of Risk of the Fetal Hydantoin Syndrome. N Engl J Med. 1990; 322:1567–1572
- [14] Public Health Service. Fluoxetine-Phenytoin Interaction. FDA Medical Bulletin. 1994; 24:3–4
- [15] Winkler SR, Luer MS. Antiepileptic Drug Review: Part 1. Surgical Neurology. 1998; 49:449–452
- [16] Oles KS, Waqar M, Penry JK. Catastrophic Neurologic Signs due to Drug Interaction: Tegretol and Darvon. Surgical Neurology. 1989; 32:144–151

- [17] Oakeshott P, Hunt GM. Valproate and Spina Bifida. *Br Med J*. 1989; 298:1300-1301
- [18] Wiggs KK, Rickert ME, Suján AC, et al. Anti-seizure medication use during pregnancy and risk of ASD and ADHD in children. *Neurology*. 2020. DOI: 10.1212/WNL.00000000000010993
- [19] Farwell JR, Lee YJ, Hirtz DG, et al. Phenobarbital for Febrile Seizures - Effects on Intelligence and on Seizure Recurrence. *N Engl J Med*. 1990; 322:364-369
- [20] Wroblewski BA, Garvin WH. Once-Daily Administration of Phenobarbital in Adults: Clinical Efficacy and Benefit. *Arch Neurol*. 1985; 42:699-700
- [21] Davis AG, Mutchie KD, Thompson JA, et al. Once-Daily Dosing with Phenobarbital in Children with Seizure Disorders. *Pediatrics*. 1981; 68:824-827
- [22] Felbamate. *Medical Letter*. 1993; 35:107-109
- [23] Winkler SR, Luer MS. Antiepileptic Drug Review: Part 2. *Surgical Neurology*. 1998; 49:566-568
- [24] Gabapentin - A new anticonvulsant. *Medical Letter*. 1994; 36:39-40
- [25] Lamotrigine for Epilepsy. *Medical Letter*. 1995; 37:21-23
- [26] Guerrini R, Dravet C, Genton P, et al. Lamotrigine and Seizure Aggravation in Severe Myoclonic Epilepsy. *Epilepsia*. 1998; 39:508-512
- [27] Sondergaard Khinchii M, Nielsen KA, Dahl M, et al. Lamotrigine therapeutic thresholds. *Seizure*. 2008; 17:391-395
- [28] Topiramate for Epilepsy. *Medical Letter*. 1997; 39:51-52
- [29] Faught E, Wilder BJ, Ramsay RE, et al. Topiramate Placebo-Controlled Dose-Ranging Trial in Refractory Partial Epilepsy using 200-, 400-, and 600-mg Daily Dosages. *Neurology*. 1996; 46:1684-1690
- [30] Privitera M, Fincham R, Penry J, et al. Topiramate Placebo-Controlled Dose-Ranging Trial in Refractory Partial Epilepsy using 600-, 800-, and 1,000-mg Daily Dosages. *Neurology*. 1996; 46:1678-1683
- [31] Tiagabine for Epilepsy. *Medical Letter*. 1998; 40:45-46
- [32] Sperling MR, Klein P, Aboumatar S, et al. Cenobamate (YKP3089) as adjunctive treatment for uncontrolled focal seizures in a large, phase 3, multicenter, open-label safety study. *Epilepsia*. 2020; 61:1099-1108
- [33] Shuster EA. Epilepsy in Women. *Mayo Clinic Proceedings*. 1996; 71:991-999
- [34] Callaghan N, Garrett A, Goggin T. Withdrawal of Anticonvulsant Drugs in Patients Free of Seizures for Two Years. *N Engl J Med*. 1988; 318:942-946
- [35] Anderson T, Braathen G, Persson A, et al. A Comparison Between One and Three Years of Treatment in Uncomplicated Childhood Epilepsy: A Prospective Study. II. The EEG as Predictor of Outcome After Withdrawal of Treatment. *Epilepsia*. 1997; 38:225-232
- [36] Medical Research Council Antiepileptic Drug Withdrawal Study Group. Randomized study of antiepileptic drug withdrawal in patients in remission. *Lancet*. 1991; 337:1175-1180
- [37] Delgado-Escueta A, Janz D. Consensus Guidelines: Preconception Counseling, Management, and Care of the Pregnant Woman with Epilepsy. *Neurology*. 1992; 42:149-160
- [38] Mattson RH, Cramer JA, Darney PD, et al. Use of Oral Contraceptives by Women with Epilepsy. *Journal of the American Medical Association*. 1986; 256:238-240
- [39] Perucca E, Hedges A, Makki KA, et al. A Comparative Study of the Relative Enzyme Inducing Properties of Anticonvulsant Drugs in Epileptic Patients. *Br J Clin Pharmacol*. 1984; 18:401-410
- [40] Yerby MS, Freil PN, McCormick K. Antiepileptic Drug Disposition During Pregnancy. *Neurology*. 1992; 42:12-16
- [41] Dias MS, Sekhar LN. Intracranial Hemorrhage from Aneurysms and Arteriovenous Malformations during Pregnancy and the Puerperium. *Neurosurgery*. 1990; 27:855-866
- [42] Jones KL, Lacro RV, Johnson KA, et al. Patterns of Malformations in the Children of Women Treated With Carbamazepine During Pregnancy. *N Engl J Med*. 1989; 310:1661-1666
- [43] Rosa FW. Spina Bifida in Infants of Women Treated with Carbamazepine During Pregnancy. *New England Journal of Medicine*. 1991; 324:674-677
- [44] Hanson JW, Smith DW. The Fetal Hydantoin Syndrome. *J Pediatr*. 1975; 87:285-290
- [45] Scolnik D, Nulman I, Rovet J, et al. Neurodevelopment of Children Exposed In Utero to Phenytoin and Carbamazepine Monotherapy. *Journal of the American Medical Association*. 1994; 271:767-770
- [46] Nakane Y, Okuma T, Takahashi R, et al. Multi-Institutional Study of the Teratogenicity and Fetal Toxicity of Antiepileptic Drugs: A Report of a Collaborative Study Group in Japan. *Epilepsia*. 1980; 21:663-680
- [47] Waters CH, Belai Y, Gott PS, et al. Outcomes of Pregnancy Associated with Antiepileptic Drugs. *Archives of Neurology*. 1994; 51:250-253
- [48] Kanto JH. Use of Benzodiazepines During Pregnancy, Labor, and Lactation, with Particular Reference to Pharmacokinetic Considerations. *Drugs*. 1982; 23:354-380
- [49] Saunders M. Epilepsy in Women of Childbearing Age: If Anticonvulsants Cannot be Avoided, Use Carbamazepine. *Br Med J*. 1989; 199

28 Special Seizure Considerations

28.1 New onset seizures

28.1.1 General information

The age adjusted *incidence* of new onset seizures in Rochester, Minnesota was 44 per 100,000 person-years.¹

28.1.2 Etiologies

In patients presenting with a first-time seizure, etiologies include (modified²) (the first 3 items would be considered provoked):

1. following neurologic insult*: either acutely (i.e., < 1 week) or remotely (> 1 week, and usually < 3 mos from insult)
 - a) stroke: 4.2% had a seizure within 14 days of a stroke. Risk increased with severity of stroke³
 - b) head trauma: closed head injury, penetrating trauma (p. 505)
 - c) CNS infection: meningitis, cerebral abscess, subdural empyema
 - d) febrile seizures (p. 509)
 - e) birth asphyxia
2. underlying CNS abnormality*
 - a) congenital CNS abnormalities
 - b) degenerative CNS disease
 - c) CNS tumor: metastatic or primary
 - d) hydrocephalus
 - e) AVM
3. acute systemic metabolic disturbance*
 - a) electrolyte disorders: uremia, hyponatremia, hypoglycemia (especially profound hypoglycemia), hypercalcemia
 - b) drug related, including:
 - alcohol-withdrawal (p. 506)
 - cocaine toxicity (p. 216)
 - opioids (narcotics)
 - phenothiazine antiemetics (p. 149)
 - with administration of flumazenil (Romazicon®) to treat benzodiazepine (BDZ) overdose (especially when BDZs are taken with other seizure lowering drugs such as tricyclic antidepressants or cocaine)
 - phencyclidine (PCP): originally used as an animal tranquilizer
 - cyclosporine: can affect Mg⁺⁺ levels
 - c) eclampsia
4. idiopathic

* Items with an asterisk would be considered provoked seizures.

In 166 *pediatric* patients presenting to an emergency department with either a chief complaint of, or a discharge diagnosis of a first-time seizure⁴:

1. 110 were found to actually have either a recurrent seizure or a non-ictal event
2. of the 56 patients actually thought to have had a first-time seizure
 - a) 71% were febrile seizures
 - b) 21% were idiopathic
 - c) 7% were "symptomatic" (hyponatremia, meningitis, drug intoxication...)

In a prospective study of 244 patients with a new-onset unprovoked seizure, only 27% had further seizures during follow-up.^{2,5} Recurrent seizures were more common in patients with a family seizure history, spike-and-waves on EEG, or a history of a CNS insult (stroke, head injury...). No patient seizure-free for 3 years had a recurrence. Following a second seizure, the risk of further seizures was high.

28.1.3 Evaluation

Adults

A new-onset seizure in an *adult* in the absence of obvious cause (e.g., alcohol withdrawal) should prompt a search for an underlying basis (the onset of idiopathic seizures, i.e., epilepsy, is most common before or during adolescence). An MRI (without and with enhancement) should be performed (CT without and with contrast if MRI cannot be done). A systemic work-up should be done to identify the presence of any provoking factors listed above. If all this is negative, a repeat MRI (or CT if MRI cannot be done) should be done in $\approx$ 6 months and at 1 and possibly 2 years to rule out a tumor which might not be evident on the initial study.

Pediatrics

Among pediatric patients with first-time seizures, laboratory and radiologic evaluations were often costly and not helpful.⁴ A detailed history and physical exam were more helpful.

Management

Management of an adult with the new onset of idiopathic seizures (i.e., no abnormality found on CT or MRI, no evidence of a provoking etiology) is controversial.

28

► **Electroencephalogram (EEG).** An EEG is a brief (20 minute) sample of electrical activity of the brain, and is normal in $> 50\%$ of patients later proven to have epilepsy (poor sensitivity). However, it is highly specific when abnormal.

In one study, an EEG was performed, which, if normal, was followed by a sleep deprived EEG with the following observations⁶:

1. there is substantial interobserver variation in interpreting such EEGs
2. if both EEGs were normal, the 2-yr recurrence rate of seizures was 12%
3. if one or both EEGs showed epileptic discharges, the 2-yr recurrence rate was 83%
4. the presence of nonepileptic abnormalities in one or both EEGs had a 41% 2-yr recurrence rate
5. the recurrence rate with focal epileptic discharges (87%) was slightly higher than for generalized epileptic discharges (78%)

Conclusion: EEGs thus obtained have moderate predictive value, and may be factored into the decision of whether or not to treat such seizures with ASMs.

► **Ambulatory EEG.** Permits home recording typically for 1–3 days (similar to a Holter monitor used for cardiac arrhythmias).

► **Video-EEG.** The patient is monitored with continuous EEG and video recording, usually in a specialized video-EEG monitoring unit (some home video-EEGs have been trialed) to correlate symptoms with electrical activity of the brain. The most accurate means of diagnosing seizures (including ruling out non-epileptic seizures (p.507)). However, with only scalp (surface) electrodes, even video-EEG may miss some seizures.

► **Additional evaluation.** Further evaluation may be needed for uncharacterized disabling episodes. See details (p.1890).

28.2 Posttraumatic seizures

28.2.1 General information

Key concepts

- 2 categories: early (≤ 7 days) and late (> 7 days) after head trauma
- antiseizure medications (ASMs) may be used to prevent early posttraumatic seizures (PTS) in patients at high risk for seizures
- prophylactic ASMs do NOT reduce the frequency of late PTS
- discontinue ASMs after 1 week except for cases meeting specific criteria (see text)

Posttraumatic seizures (PTS) are often divided (arbitrarily) into early (occurring within 1 week of injury) and late (thereafter).⁷ There may be justification for a third category: “immediate,” i.e., within minutes to an hour or so.

28.2.2 Early PTS (≤ 7 days after head trauma)

30% incidence in severe head injury (“severe” defined as: LOC > 24 hrs, amnesia > 24 hrs, focal neuro deficit, documented contusion, or intracranial hematoma) and $\approx 1\%$ in mild to moderate injuries. Occurs in 2.6% of children < 15 yrs of age with head injury causing at least brief LOC or amnesia.⁸

Early PTS may precipitate adverse events as a result of elevation of ICP, alterations in BP, changes in oxygenation, and excess neurotransmitter release.⁹

28.2.3 Late onset PTS (< 7 days after head trauma)

Estimated incidence 10–13% within 2 yrs after “significant” head trauma (includes LOC > 2 mins, GCS < 8 on admission, epidural hematoma...) for all age groups.^{10,11} Relative risk: 3.6 times control population. Incidence in severe head injury > moderate > mild.⁸

The incidence of early PTS is higher in children than adults, but late seizures are much less frequent in children (in children who have PTS, 94.5% develop them within 24 hrs of the injury¹²). Most patients who have not had a seizure within 3 yrs of penetrating head injury will not develop seizures.¹³ Risk of late PTS in children does not appear related to the occurrence of early PTS (in adults: only true for mild injuries). Risk of developing late PTS may be higher after repeated head injuries.

28.2.4 Penetrating trauma

The incidence of PTS is higher with penetrating head injuries than with closed head injuries (occurs in 50% of penetrating trauma cases followed 15 yrs¹⁴).

28.2.5 Treatment

General information

Some early retrospective studies suggested that early administration of PHT prevents early PTS, and reduces the risk of late PTS even after discontinuation of the drug. Later prospective studies disputed this but were criticized for not maintaining satisfactory levels and for lacking statistical power.^{7,11} A prospective double blind study of patients at high risk of PTS (excluding penetrating trauma) showed a 73% reduction of risk of *early* PTS by administering 20 mg/kg loading dose of PHT within 24 hrs of injury and maintaining high therapeutic levels; but after 1 week there was no benefit in continuing the drug (based on intention to treat).¹⁵ Carbamazepine (Tegretol®) has also been shown to be effective in reducing the risk of early PTS.

Phenytoin has adverse cognitive effects when given long-term as prophylaxis against PTS.¹⁶

Treatment guidelines

Based on available information (see above) it appears that:

1. no treatment studied effectively impedes epileptogenesis (i.e., neuronal changes that ultimately lead to late PTS)
2. in high-risk patients ($\triangleright$ Table 28.1), ASMS reduce the incidence of *early* PTS
3. however, no study has shown that reducing early PTS improves outcome¹⁷
4. once epilepsy has developed, continued ASMs reduce the recurrence of further seizures

The following are therefore offered as guidelines.

Table 28.1 High risk criteria for PTS

1. acute subdural, epidural, or intracerebral hematoma (SDH, EDH or ICH)
2. open-depressed skull fracture with parenchymal injury
3. seizure within the first 24 hrs after injury
4. Glasgow Coma Scale score < 10
5. penetrating brain injury
6. history of significant alcohol abuse
7. $\pm$ cortical (hemorrhagic) contusion on CT

Initiation of ASMs

ASMs may be considered for short term use, especially if a seizure could be detrimental. Early post-traumatic seizures were effectively reduced when phenytoin was used for 2 weeks following head injury with no significant increased risk of adverse effects.¹⁸

Acutely, seizures may elevate ICP, and may adversely affect blood pressure and oxygen delivery, and may worsen other injuries (e.g., spinal cord injury in the setting of an unstable cervical spine). There may also be negative psychological effects on the family, loss of driving privileges, and possibly deleterious effects of excess neurotransmitters.⁹

Option: begin ASMs (usually levetiracetam, phenytoin or carbamazepine) within 24 hrs of injury in the presence of any of the high risk criteria shown in ► Table 28.1 (modified^{9,12,15,19}). When using PHT, load with 20 mg/kg and maintain high therapeutic levels. Switch to phenobarbital if PHT is not tolerated.

Discontinuation of ASMs

1. taper ASMs after 1 week of therapy except in the following cases:
 - a) penetrating brain injury
 - b) development of late PTS (i.e., a seizure > 7 days following head trauma)
 - c) prior seizure history
 - d) patients undergoing craniotomy²⁰
2. for patients *not* meeting the criteria to discontinue ASMs after 1 week (see above):
 - a) maintain ≈ 6–12 mos of therapeutic ASM levels
 - b) recommend EEG to rule out presence of a seizure focus before discontinuing ASMs (shown to have poor predictive value, but probably advisable for legal purposes) for the following:
 - repeated seizures
 - presence of high risk criteria shown in ► Table 28.1.

28.3 Alcohol withdrawal seizures

28.3.1 General information

Also, see Alcohol withdrawal syndrome (p.213). The withdrawal syndrome may begin hours after the EtOH peak; see also prevention and treatment (p.213). Ethanol withdrawal seizures are classically seen in up to 33% (some say 75%) of habituated drinkers within 7–30 hours of cessation or reduction of ethanol intake. They typically consist of 1–6 tonic-clonic generalized seizures without focality within a 6-hour period.²¹ Seizures usually occur before delirium develops. They may also occur during intoxication (without withdrawal).

The seizure risk persists for 48 hrs (risk of delirium tremens [DTs] continues beyond that); thus a single loading dose of PHT is frequently adequate for prophylaxis. However, since most EtOH withdrawal seizures are single, brief, and self-limited, PHT has *not* been shown to be of benefit in uncomplicated cases and is thus *usually not indicated*. Chlordiazepoxide (Librium®) or other benzodiazepines (p.213) administered during detoxification reduces the risk withdrawal seizures.²²

28.3.2 Evaluation

The following patients should have a CT scan of the brain, and should be admitted for further evaluation as well as for observation for additional seizures or for DTs:

1. those with their first EtOH withdrawal seizure
2. those with focal findings
3. those having more than 6 seizures in 6 hrs
4. those with evidence of trauma

Other causes of seizure should also be considered, e.g., a febrile patient may require an LP to rule out meningitis.

28.3.3 Treatment

A brief single seizure may not warrant treatment, except as outlined below. A seizure that continues beyond 3–4 minutes may be treated with diazepam or lorazepam, with further measures used as in status epilepticus (p.510) if seizures persist. Loading with phenytoin (18 mg/kg = 1200 mg/70 kg) and long-term treatment is indicated for:

1. a history of previous alcohol withdrawal seizures

- recurrent seizures after admission
- history of a prior seizure disorder unrelated to alcohol
- presence of other risk factors for seizure (e.g., subdural hematoma)

28.4 Nonepileptic seizures

28.4.1 General information

AKA pseudoseizures (this term has fallen out of favor since it may connote voluntary feigning of seizures), with the term psychogenic non-epileptic attacks (PNEA) being preferred for nonepileptic seizures (NES) with a psychologic etiology (PNEAs are real events and may not be under voluntary control).²³

One of the hazards of NES is that patients may end up needlessly taking ASMs, which in some cases may worsen NES. Possible etiologies of NES are given in ► Table 28.2. Most NES are psychogenic.

NES comprise 20–30% of medically refractory presentations resembling seizures. The incidence is ≈ 4 per 1000,000. They may mimic any seizure type.

DDx for seizures:

- psychogenic: comprise 20–90% of patients with intractable seizures referred to epilepsy centers. These patients carry the diagnosis of seizures from 5–7 years. Up to 50% of these may have bona fide seizures at some time as well.²⁴
- tic: can be suppressed, is not repetitive (if repetitive, may be hemifacial spasm)
- movement disorder: myoclonus (can be epileptic or non-epileptic)
 - cataplexy: e.g., with narcolepsy often provoked by laughter or other emotional stimulus (can rarely be caught on EEG, and when it is, it shows REM intrusion into wakefulness)
 - parasomnia: a sleep movement disorder (occurs during sleep). Includes: night terrors (occurs in slow wave sleep, vs. nightmare which occurs in REM), sleep walking, REM behavior disorders (usually occurs in older men), and there is a high probability they will go on to have degenerative brain disease (used to be called paroxysmal nocturnal PNT). Head banging is a benign parasomnia
- syncope: 90% of the time people who faint have myoclonic jerks or shaking²⁵
- TIA

Table 28.2 Differential diagnosis of nonepileptic seizures²³

psychologic disorders (PNEAs)	a) somatoform disorders: especially conversion disorder b) anxiety disorders: especially panic attack and posttraumatic stress disorder c) dissociative disorders d) psychotic disorders e) impulse control disorders f) attention-deficit disorders ^a g) factitious disorders: including Munchausen's syndrome
cardiovascular disorders	a) syncope b) cardiac arrhythmias c) transient ischemic attacks (TIAs) d) breath-holding spells ^a
migraine syndromes	a) complicated migraines ^a b) basilar migraines
movement disorders	a) tremors b) dyskinesias c) tics ^a , spasms d) other (including shivering)
parasomnias & sleep-related disorders	a) night terrors ^a , nightmares ^a , somnambulism ^a b) narcolepsy, cataplexy c) rapid eye movement behavior disorder d) nocturnal paroxysmal dystonia
gastrointestinal disorders	a) episodic nausea or colic ^a b) cyclic vomiting syndrome ^a
other	a) malingering b) cognitive disorders with episodic behavioral or speech symptoms c) medication effects or toxicity d) daydreams ^a

^ausually encountered in children

28.4.2 Differentiating NES from epileptic seizures

General information

Distinguishing between epileptic seizures (ES) and NES is a common clinical dilemma. There are unusual seizures that may fool experts.²⁶ Some frontal lobe and temporal lobe complex partial seizures may produce bizarre behaviors that do not correspond to classic ES findings and may not produce discernible abnormalities with scalp-electrode EEG (and therefore may be misdiagnosed even with video-EEG monitoring, although this is more likely with partial seizures than with generalized). A multidisciplinary team approach may be required.

► Table 28.3 contrasts some features of true seizures vs. NES, and ► Table 28.4 lists some features often associated with NES. However, no characteristics are definitively diagnostic of NES since a number of them may also occur with ES.

Table 28.3 Features of ES vs. NES²⁴

Feature	Epileptic seizure	NES
% males	72%	20%
Clonic UE movement		
in-phase	96%	20%
out-of-phase	0	56%
Clonic LE movement		
in-phase	88%	16%
out-of-phase	0	56%
Vocalizations		
none	16%	56%
start of seizure	24%	44%
middle	60% "epileptic cry"	0
types	only sounds of tonic or clonic respiratory muscle contraction	moans, screams, grunts, snorts, gagging, retching, understandable statements, gasps
Head turning		
unilateral	64%	16%
side-to-side	8% (slow, low amplitude)	36% (violent, high amplitude)

Table 28.4 Features often associated with NES²³

- frequent seizures despite therapeutic ASMs
- multiple different-physician visits
- lingering prodrome or gradual ictal onset (over minutes)
- prolonged duration (>5 mins)
- manifestations altered by distraction
- suggestible or inducible seizures
- intermittent arrhythmic and out-of-phase convulsive activity
- fluctuating intensity and severity during Sz
- side-to-side rolling, pelvic thrusting, wild movements
- bilateral motor activity with preserved consciousness
- nonphysiologic spread of neurologic signs
- absence of labored breathing or drooling after generalized convulsion
- expression of relief or indifference
- crying or whimpering
- no postictal confusion or lethargy
- disproportionate postictal mental status changes
- absence of stereotypy

Features common to both true seizures and NES: verbal unresponsiveness, rarity of automatisms and whole-body flaccidity, rarity of urinary incontinence. Reminder: some seizures can be bizarre and can resemble NES (sometimes called pseudo-pseudoseizures). 10% of patients with psychogenic seizures actually have epilepsy.

Features suggestive of non-epileptic seizures:

1. arching of the back: 90% specific for NES

2. asynchronous movement
3. stop & go: seizures usually build and then gradually subside
4. forced eye closing during entire seizure
5. provoked with stimuli that would not cause a seizure (e.g., tuning fork to the head, alcohol pad to the neck, IV saline...)
6. bilateral shaking with preserved awareness. Exception: supplementary motor area seizures (mesial frontal area)—these seizures are usually tonic (not clonic)
7. weeping (whining): highly specific
8. multiple or variable seizure types (ES is usually stereotypical), fluctuating level of consciousness, denial of correlation of Sz with stress

If any two of the following are demonstrated, 96% of time this will be NES:

1. out-of-phase clonic UE movement
2. out-of-phase clonic LE movement
3. no vocalization or vocalization at start of event

Clinical features in favor of seizures:

1. injuries: especially lateral tongue laceration which is very specific for seizures²⁷
2. very stereotyped
3. duration < 2 minutes
4. eyes open
5. incontinence

History

Attempt to document: prodromal symptoms, precipitating factors, time and environment of Sz, mode and duration of progression, ictal and postictal events, frequency and stereotypy of manifestations. Determine if patient has history of psychiatric conditions, and if they are acquainted with individuals who have ES.

Psychological testing

May help. Differences occur in ES and NES on the Minnesota Multiphasic Personality Inventory (MMPI) scales in hypochondriasis, depression hysteria, and schizophrenia.²⁸

28.5 Febrile seizures

28.5.1 Definitions

► **Febrile seizure.** See reference.²⁹

A seizure in infants or children associated with fever with no defined cause and unaccompanied by acute neurologic illness (includes seizures during post- vaccination fevers). Some may be inherited as in Genetic Epilepsy with Febrile Seizures Plus (GEFS+).

► **Complex febrile seizure.** A convulsion that lasts longer than 15 minutes, is focal, or multiple (more than one convulsion per episode of fever).

► **Simple febrile seizure.** Not complex.

► **Recurrent febrile seizure.** More than one episode of fever associated with seizures.

28.5.2 Epidemiology

See reference.²⁹

Febrile convulsions are the most common type of seizure. Excluding children with pre-existing neurologic or developmental abnormalities, the prevalence of febrile seizures is $\approx 2.7\%$ (range: 2–5% in U.S. children aged 6 mos–6 yrs). The risk for developing epilepsy after a simple febrile seizure is $\approx 1\%$, and for a complex febrile seizure is 6% (9% for prolonged seizure, 29% for focal seizure). An underlying neurological or developmental abnormality or a family history of epilepsy increases the risk of developing epilepsy. The notion that the younger the child with a febrile seizure the greater the risk of epilepsy is unproven.

28.5.3 Treatment

In one study, the IQ in the group treated with phenobarbital was 8.4 points lower (95% confidence interval) than the placebo group, and there remained a significant difference several months after discontinuing the drug.³⁰ Furthermore, there was no significant reduction in seizures in the phenobarbital group. And yet, no other drug really appears well suited to treating this entity: carbamazepine and phenytoin appear ineffective, valproate may be effective but has serious risks in the <2 yrs of age group. Given the low incidence (1%) of having afebrile seizures (i.e., epilepsy) after a *simple* febrile seizure and the fact that ASMs probably do not prevent this development, there is little support for prescribing antiseizure medications in these cases. The recurrence rate of febrile seizures in children with a history of one or more febrile seizure can be reduced by administering diazepam 0.33 mg/kg PO q 8 hrs during a febrile episode (temp > 38.1 °C) and continuing until 24 hrs after the fever subsides.³¹

28.6 Status epilepticus

28.6.1 General information

Key concepts

- definition: > 30 mins of either 1) continuous seizure activity or 2) ≥ 2 sequential seizures without full recovery of consciousness between seizures
- morbidity and mortality are high in untreated status epilepticus (SE)
- most common etiology: patient with known seizure disorder with low ASM levels
- de novo SE in acute illness is considered a manifestation of the illness which should be treated at the same time as the SE
- begin treatment as soon as a seizure persists > 5 minutes
- see treatment measures (p. 512)
- treatment goal: rapid termination of all clinical and electrical seizure activity

► **Definitions.** **Brief seizure:** seizure lasting < 5 minutes.

Prolonged seizure: seizure lasting 5-30 minutes

Status epilepticus: more than 30 minutes of either 1) continuous seizure activity or 2) ≥ 2 sequential seizures without full recovery of consciousness between seizures.³² Status epilepticus lasting > 30 minutes may lead to permanent neurologic damage³³

► **Important things to know about management**

- since 61% of seizures that persist > 5 mins will continue > 1 hour,³⁴ and permanent neurologic damage can occur after 30 minutes, treatment should be initiated once a seizure has persisted > 5 minutes (even though technically this is not long enough to be considered SE yet)
- in patients with no prior seizure history, status epilepticus (SE) is usually a manifestation of illness-related cortical irritation or injury³⁵ and treatment of the underlying disorder (in addition to treating the SE) is critical
- a relapse of seizure in a patient with a known seizure disorder and subtherapeutic ASM levels usually responds to a bolus of the maintenance ASMs. However, true SE should be treated by the standard protocol³⁵
- most cases of convulsive status in adults start as partial seizures that generalize secondarily
- treatment goal: rapid termination (not just reduction) of clinical *and* electrical seizure activity
- prognosis is related to seizure etiology, duration of SE, and patient age
- the choice of 1st and 2nd-line ASMs is arbitrary, *dose*³⁵ and starting treatment in < 30 minutes³⁶ are more important determinants of success in aborting SE

28.6.2 Types of status epilepticus

See reference.³⁷

- **generalized status**
 1. convulsive status epilepticus (SE): repeated generalized (tonic-clonic, tonic-clonic-tonic, or clonic) seizures with persistent postictal neurologic dysfunction between seizures. This is the most common type of SE.³⁸ A medical emergency

2. absence (note: in status, this may present in “epileptic twilight” state)
3. secondarily generalized: accounts for $\approx$ 75% of generalized SE
4. myoclonic
5. atonic (drop attack): especially in Lennox-Gastaut syndrome (p.484)
- partial status (usually related to an anatomic abnormality)
 1. simple (AKA epilepsy partialis continua): not associated with altered mental status
 2. complex (note: in status, this may present in “epileptic twilight” state) most often from frontal lobe focus. Urgent treatment is required (several case reports of permanent deficits following this)
 3. secondarily generalized
- nonconvulsive SE
 1. benign variants (typical absence SE, complex partial SE)
 2. electrical SE during sleep
 3. atypical absence SE
 4. tonic SE (associated with learning disability in children), SE in coma

Alternatively, SE can be broken down as follows:

- with prominent motor effects
- without prominent motor effects
- boundary syndromes (syndromes which combine encephalopathy, behavioral disturbances, delirium, or psychosis with SE-like EEG findings)

28.6.3 Epidemiology

Incidence: 50,000 to 150,000 new cases/year in the U.S. in the outpatient setting.³⁵ Most cases occur in young children (among children, 73% were <5 yrs old³⁹); the next most affected group is patients >60 yrs of age. In $>50\%$ of cases, SE is the patient's first seizure.³⁸ One out of six patients presenting with a first time seizure will present in SE.

Mortality in children is estimated at $<3\%$, but is up to 30% in adults.⁴⁰

28.6.4 Etiologies

The most common causes (the short list):

- low level of prescribed ASM in a patient with a seizure disorder (34%)
- remote symptomatic cause (24%)
- stroke (22%)
- metabolic disturbances (15%)
- hypoxia (13%)

The long list:

1. a patient with a known seizure disorder having low ASM levels for any reason (non-compliance, intercurrent infection preventing PO intake of meds, drug-drug interactions $\rightarrow$ lowering effectiveness of ASMs...)
2. febrile seizures: a common precipitator in young patients. 5–6% of patients presenting with SE have a history of prior febrile seizures
3. stroke: the most commonly identified cause in the elderly
4. CNS infection: in children, most are bacterial, the most common organisms were *H. influenza* and *S. pneumoniae*
5. idiopathic: accounts for $\approx$ one-third (in children, usually associated with fever)
6. epilepsy: is present or is subsequently diagnosed in $\approx 50\%$ of patients presenting with SE. About 10% of adults ultimately diagnosed as having epilepsy will present in SE
7. electrolyte imbalance: hyponatremia (most common in children, usually due to water intoxication³⁹), hypoglycemia, hypocalcemia, uremia, hypomagnesemia...
8. illicit drug intoxication: especially cocaine, amphetamines
9. precipitous drug withdrawal: barbiturates, benzodiazepines, alcohol, or narcotics
10. proconvulsant drugs, including: β -lactam antibiotics (penicillins, cephalosporins), certain antidepressants (bupropion), clonazapine, bronchodilators, immunosuppressants
11. traumatic brain injury: acute as well as old
12. hypoxia/ischemia
13. tumor

In children < 1 yr of age, 75% had an acute cause: 28% were secondary to CNS infection, 30% due to electrolyte disorders, 19% associated with fever.³⁹ In adults, a structural lesion is more likely. In an adult, the most common cause of SE is subtherapeutic ASM levels in a patient with a known seizure disorder.

28.6.5 Morbidity and mortality from SE

Outcomes are related to underlying cause and duration of SE. Mean duration of SE in patients without neurologic sequelae is 1.5 hrs (therefore, proceed to pentobarbital anesthesia before ≈ 1 hour of SE). Recent mortality: < 10–12% (only ≈ 2% of deaths are directly attributable to SE or its complications; the rest are due to the underlying process producing the SE). Mortality is lowest among children (≈ 6%³⁹), patients with SE related to subtherapeutic ASMs, and patients with unprovoked SE.⁴¹ The highest mortality occurs in elderly patients and those with SE resulting from anoxia or stroke.⁴¹ 1% of patients die during the episode itself.

Morbidity and mortality is due to⁴²:

1. CNS injury from repetitive electric discharges: irreversible changes begin to appear in neurons after as little as 20 minutes of convulsive activity. Cell death is very common after 60 mins
2. systemic stress from the seizure (cardiac, respiratory, renal, metabolic)
3. CNS damage by the acute insult that provoked the SE

28.6.6 Treatment

Prehospital phase

1. impending SE: may be heralded by a crescendo in seizures. A 1–3 d course of lorazepam may preempt the development of SE
2. SE treatment may be initiated in the home setting with buccal midazolam or rectal diazepam. For children with SE, midazolam (IM or intranasal or buccal) is probably more effective than diazepam (IV or rectal)⁴⁰ (LevelB)

General treatment measures for status epilepticus

Treatment success, like morbidity/mortality, may be time-dependent. First-line ASM therapy aborts SE in 60% of patients if initiated within the first 30 minutes, and efficacy decreases as seizure duration increases.⁴³ As such, treatment should be initiated as soon as possible and should be directed at stabilizing the patient, rapid termination (not just reduction) of clinical and electrical seizure activity, identifying the cause (determining if there is an acute insult to the brain), and, if possible, also treating any underlying process identified. Treatment often must be initiated prior to the availability of test results to confirm the diagnosis and may even be initiated in the pre-hospital setting. The following is modified from the American Epilepsy Society guidelines⁴⁰

0-5 Minutes

Stabilization phase

1. “ABCs”
 - a) Airway: oral airway if feasible. Turn patient on their side to avoid aspiration
 - b) Breathing: O₂ by nasal cannula or bag-valve-mask. Consider intubation if respirations compromised or if seizure persists > 30 min.
NB: if paralytics are used to intubate, use short acting agents and be aware that muscle paralysis alone may stop visible seizure manifestations, but does not stop the electrical seizure activity in the brain, which can lead to permanent neurologic damage if prolonged
 - c) Circulation: CPR if needed. Large bore proximal IV access (2 if possible: 1 for phenytoin (PHT) (Dilantin®), not necessary if fosphenytoin is available): start with NS KVO
2. begin tracking time from onset of seizure
3. baseline vital signs. Frequent blood pressure checks

4. assess oxygenation: O₂ saturation (via pulse oximeter) or ABG. Administer oxygen via nasal cannula or mask as needed. Intubate if hypoxia or respiratory distress
5. monitor EKG
6. blood glucose: if fingerstick glucose can be obtained immediately and it shows hypoglycemia (glucose < 60 mg/dL), or if no fingerstick glucose can be done, give glucose as below. If it won't delay giving glucose, draw blood for definitive serum glucose first (don't wait for results to give glucose)
 - adult: 100 mg thiamine IV then 50 ml of D50 W IV push.
NB: thiamine is given first because giving glucose in thiamine deficiency (e.g., patients with poor nutrition such as alcoholics) can precipitate Wernicke's encephalopathy (p.214)
 - children ≥ 2 years: 2 ml/kg of D25 W IV push
 - children < 2 years: 4 ml/kg of D12.5 W IV push
7. secure IV access: large bore proximal IV access (2 if possible, 1 for possible phenytoin)
 - use this opportunity to obtain blood for labs: electrolytes, CBC, LFTs, Mg⁺⁺, Ca⁺⁺, ASM levels, toxicology, ABG
 - start maintenance IV fluid with NS @ KVO
8. neurologic exam

As acuity permits

Additional initial measures

- EEG monitor if possible (EEG electrodes typically take 45-60 minutes just to apply)
- head CT (usually without contrast)
- correct any electrolyte imbalance (SE due to electrolyte imbalance responds more readily to correction than to ASMs³⁹)
- if CNS infection is a major consideration, perform LP for CSF analysis (especially in febrile children) unless contraindicated (p.1811). WBC pleocytosis up to 80 × 10⁶/L can occur following SE (benign postictal pleocytosis), and these patients should be treated with antibiotics until infection can be ruled out by negative cultures
- **general meds for unknown patient:**
 - a) naloxone (Narcan®) 0.4 mg IVP (in case of narcotics)
 - b) ± bicarbonate to counter acidosis (1–2 amps depending on length of seizure)
 - c) for neonate < 2 years: consider pyridoxine 100 mg IV push (pyridoxine-dependent seizures constitute a rare autosomal recessive condition that generally presents in the early neonatal period³⁴)

If seizures continue:

5-20 minutes

Initial therapy phase

- **A benzodiazepine is the drug therapy of choice** (Level A)
Choose one of the following (they are equivalent in level of recommendation)
 - IM midazolam: (for weight > 40 kg give 10 mg; wt 13–40 kg give 5 mg) X 1 dose, OR
 - IV lorazepam: 0.1 mg/kg/dose up to max 4 mg/dose, may repeat X 1, OR
 - IV diazepam: 0.15–0.2 mg/kg/dose up to max 10 mg/dose, may repeat X 1
- If none of the 3 above options are available or if all are contraindicated, choose one of:
 - IV phenobarbital: 15 mg/kg/dose, single dose (Level A), OR
 - rectal diazepam (Diastat® gel): 0.2–0.5 mg/kg up to max 20 mg/dose X 1 (Level B), OR
 - intranasal midazolam (Level B) or buccal midazolam (Level B)

If seizures continue:

20-40 minutes

Second therapy phase

- **There is no evidence to guide choice of second therapy** (Level U = lack of studies meeting criteria for levels A, B or C).

Choose one of the following (give as single dose)

- IV fosphenytoin: 20 mg PE/ up to max 1500 mg PE/dose X 1, OR
- IV valproic acid: 40 mg/kg/dose up to max 3000 mg/dose X 1, OR
- IV levetiracetam: 60 mg/kg/dose up to max 4500 mg/dose X 1
- If none of the 3 above options are available or if all are contraindicated:
 - IV phenobarbital (if not already given): 15 mg/kg/dose, single dose (Level B)

If seizures continue:

40-60 minutes

Third therapy phase

- **There is no evidence to guide therapy in the 3rd phase** (Level U).

Options include

- repeat 2nd line therapy, OR
- anesthetic doses of one of thiopental, midazolam, pentobarbital or propofol, together with continuous EEG monitoring

28.6.7 Medications for non-convulsive status epilepticus

In non-convulsive status epilepticus, the first and second line ASMs listed in section 28.6.6 should be utilized. However, many practitioners avoid escalating to the anesthetic options, instead opting for trials of additional ASMs first (carbamazepine, oxcarbazepine, topiramate, lamotrigine, etc).

Absence status epilepticus almost always responds to diazepam.

28.6.8 Myoclonic status epilepticus

Treatment: valproic acid (drug of choice). Place NG, give 20 mg/kg per NG loading dose. Maintenance: 40 mg/kg/d divided.

Can add lorazepam (Ativan®) or clonazepam (Klonopin®) to help with acute control.

References

- [1] Hauser WA, Annegers JF, Kurland LT. Incidence of Epilepsy and Unprovoked Seizures in Rochester, Minnesota, 1935-1984. *Epilepsia*. 1993; 34:453-468
- [2] Hauser WA, Anderson VE, Loewenson RB, et al. Seizure Recurrence After a First Unprovoked Seizure. *New Engl J Med*. 1982; 307:522-528
- [3] Reith J, Jorgensen HS, Nakayama H, et al. Seizures in acute stroke: Predictors and prognostic significance. The Copenhagen stroke study. *Stroke*. 1997; 28:1585-1589
- [4] Landfish N, Gieron-Korthals M, Weibley RE, et al. New Onset Childhood Seizures: Emergency Department Experience. *Journal of the Florida Medical Association*. 1992; 79:697-700
- [5] Hauser WA, Rich SS, Jacobs MP, et al. Patterns of Seizure Occurrence and Recurrence Risks in Patients with Newly Diagnosed Epilepsy. *Epilepsia*. 1983; 24:516-517
- [6] van Donselaar C, Schimsheimer R-J, Geerts AT, et al. Value of the Electroencephalogram in Adult Patients With Untreated Idiopathic First Seizures. *Arch Neurol*. 1992; 49:231-237
- [7] Young B, Rapp RP, Norton JA, et al. Failure of Prophylactically Administered Phenytoin to Prevent Late Posttraumatic Seizures. *J Neurosurg*. 1983; 58:236-241
- [8] Annegers JF, Grabow JD, Groover RV, et al. Seizures After Head Trauma: A Population Study. *Neurology*. 1980; 30:683-689
- [9] Bullock R, Chesnut RM, Clifton G, et al. Guidelines for the Management of Severe Head Injury. 1995
- [10] McQueen JK, Blackwood DHR, Harris P, et al. Low Risk of Late Posttraumatic Seizures Following Severe Head Injury. *J Neurol Neurosurg Psychiatry*. 1983; 46:899-904
- [11] Young B, Rapp RP, Norton JA, et al. Failure of Prophylactically Administered Phenytoin to Prevent

- Early Posttraumatic Seizures. *J Neurosurg.* 1983; 58:231-235
- [12] Hahn YS, Fuchs S, Flannery AM, et al. Factors Influencing Posttraumatic Seizures in Children. *Neurosurgery.* 1988; 22:864-867
 - [13] Weiss GH, Salazar AM, Vance SC, et al. Predicting Posttraumatic Epilepsy in Penetrating Head Injury. *Archives of Neurology.* 1986; 43:771-773
 - [14] Temkin NR, Dikmen SS, Winn HR. Posttraumatic Seizures. *Neurosurgery Clinics of North America.* 1991; 2:425-435
 - [15] Temkin NR, Dikmen SS, Wilensky AJ, et al. A Randomized, Double-Blind Study of Phenytoin for the Prevention of Post-Traumatic Seizures. *N Engl J Med.* 1990; 323:497-502
 - [16] Dikmen SS, Temkin NR, Miller B, et al. Neurobehavioral Effects of Phenytoin Prophylaxis of Posttraumatic Seizures. *JAMA.* 1991; 265:1271-1277
 - [17] Brain Trauma Foundation, Povlishock JT, Bullock MR. Antiseizure prophylaxis. *J Neurotrauma.* 2007; 24:S83-S86
 - [18] Haltiner AM, Newell DW, Temkin NR, et al. Side Effects and Mortality Associated with Use of Phenytoin for Early Posttraumatic Seizure Prophylaxis. *Journal of Neurosurgery.* 1999; 91:588-592
 - [19] Yablon SA. Posttraumatic seizures. *Arch Phys Med Rehabil.* 1993; 74:983-1001
 - [20] North JB, Penhall RK, Hanieh A, et al. Phenytoin and Postoperative Epilepsy: A Double-Blind Study. *J Neurosurg.* 1983; 58:672-677
 - [21] Charness ME, Simon RP, Greenberg DA. Ethanol and the Nervous System. *N Engl J Med.* 1989; 321:442-454
 - [22] Lechtenberg R, Worner TM. Seizure Risk With Recurrent Alcohol Detoxification. *Arch Neurol.* 1990; 47:535-538
 - [23] Chabolla DR, Krahn LE, So EL, et al. Psychogenic Nonepileptic Seizures. *Mayo Clinic Proceedings.* 1996; 71:493-500
 - [24] Gates JR, Ramani V, Whalen S, et al. Ictal Characteristics of Pseudoseizures. *Arch Neurol.* 1985; 42:1183-1187
 - [25] Lempert T, Bauer M, Schmidt D. Syncope: a video-metric analysis of 56 episodes of transient cerebral hypoxia. *Ann Neurol.* 1994; 36:233-237
 - [26] King DW, Gallagher BB, Marvin AJ, et al. Pseudoseizures: Diagnostic Evaluation. *Neurology.* 1982; 32:18-23
 - [27] Benbadis SR, Wolgamuth BR, Goren H, et al. Value of tongue biting in the diagnosis of seizures. *Arch Intern Med.* 1995; 155:2346-2349
 - [28] Henrichs TF, Tucker DM, Farha J, et al. MMPI Indices in the Identification of Patients Evidencing Pseudoseizures. *Epilepsia.* 1988; 29:184-187
 - [29] Verity CM, Golding J. Risk of Epilepsy After Febrile Convulsions: A National Cohort Study. *BMJ.* 1991; 303:1373-1376
 - [30] Farwell JR, Lee VJ, Hirtz DG, et al. Phenobarbital for Febrile Seizures - Effects on Intelligence and on Seizure Recurrence. *N Engl J Med.* 1990; 322:364-369
 - [31] Rosman NP, Colton T, Labazzo J, et al. A Controlled Trial of Diazepam Administered During Febrile Illnesses to Prevent Recurrence of Febrile Seizures. *New England Journal of Medicine.* 1993; 329:79-84
 - [32] Glauser TA. Designing practical evidence-based treatment plans for children with prolonged seizures and status epilepticus. *J Child Neurol.* 2007; 22:385-465
 - [33] Treatment of convulsive status epilepticus. Recommendations of the Epilepsy Foundation of America's Working Group on Status Epilepticus. *Journal of the American Medical Association.* 1993; 270:854-859
 - [34] Abend NS, Dlugos DJ. Treatment of refractory status epilepticus: literature review and a proposed protocol. *Pediatric Neurology.* 2008; 38:377-390
 - [35] Costello DJ, Cole AJ. Treatment of acute seizures and status epilepticus. *Journal of Intensive Care Medicine.* 2007; 22:319-347
 - [36] Eriksson K, Metsaranta P, Huhtala H, et al. Treatment delay and the risk of prolonged status epilepticus. *Neurology.* 2005; 65:1316-1318
 - [37] Varela PN, Spanaki MV, Mirski MA. Status epilepticus: an update. *Curr Neurol Neurosci Rep.* 2013; 13. DOI: 10.1007/s11910-013-0357-0
 - [38] Hauser WA. Status Epilepticus: Epidemiologic Considerations. *Neurology.* 1990; 40:9-13
 - [39] Phillips SA, Shanahan RJ. Etiology and Mortality of Status Epilepticus in Children. *Archives of Neurology.* 1989; 46:74-76
 - [40] Glauser T, Shinnar S, Gloss D, et al. Evidence-Based Guideline: Treatment of Convulsive Status Epilepticus in Children and Adults: Report of the Guideline Committee of the American Epilepsy Society. *Epilepsy Curr.* 2016; 16:48-61
 - [41] Delorenzo RJ, Pellock JM, Towne AR, et al. Epidemiology of Status Epilepticus. *J Clin Neurophysiol.* 1995; 12:312-325
 - [42] Fountain NB, Lothman EW. Pathophysiology of Status Epilepticus. *J Clin Neurophysiol.* 1995; 12:326-342
 - [43] Lowenstein DH. The management of refractory status epilepticus: an update. *Epilepsia.* 2006; 47 Suppl 1:35-40

Part IX

Pain

29 Pain

518

IX

29 Pain

29.1 Major types of pain

Major types of pain:

1. nociceptive pain
 - a) somatic: well localized. Described as sharp, stabbing, aching or cramping. Results from tissue injury or inflammation, or from nerve or plexus compression. Responds to treating the underlying pathology or by interrupting the nociceptive pathway
 - b) visceral: poorly localized. Poor response to primary pain medications
2. deafferentation pain: poorly localized. Described as crushing, tearing, tingling or numbness. Also causes burning dysesthesia numbness often with lancinating pain, and hyperpathia. Unaffected by ablative procedures
3. "sympathetically maintained" pain and the likes, e.g., causalgia (p.525)

29.2 Neuropathic pain syndromes

29.2.1 General information

Definition: Neuropathic pain: pain caused by a lesion of the peripheral and/or central nervous system manifesting with sensory symptoms and signs (Backonja¹ modified from the International Association for the Study of Pain²).

Neuropathic pain syndromes (NPS) are typified by painful diabetic neuropathy (PDN) and postherpetic neuralgia (PHN). Common chronic NPSs are shown in ► Table 29.1,³ divided into central or peripheral nervous system origin of the pain. The pain of PDN and PHN is typically burning and aching, is continuous, and is characteristically refractory to medical and surgical treatment.

Table 29.1 Common neuropathic pain syndromes

Peripheral neuropathic pain

acute & chronic inflammatory demyelinating polyneuropathy (CIDP)
 alcoholic polyneuropathy
 chemotherapy-induced polyneuropathy
 complex regional pain syndrome (CRPS)
 entrapment neuropathies
 HIV sensory neuropathy
 iatrogenic neuralgias (e.g., postthoracotomy pain)
 idiopathic sensory neuropathy
 neoplastic nerve compression or infiltration
 nutritional-deficiency neuropathies
 painful diabetic neuropathy (PDN)
 phantom limb pain
 postherpetic neuralgia (PHN)
 postradiation plexopathy
 radiculopathy
 toxic exposure-related neuropathies
 trigeminal neuralgia
 posttraumatic neuralgias

Central neuropathic pain

cervical spondylotic myelopathy
 HIV myelopathy
 multiple sclerosis-related pain
 Parkinson's disease-related pain
 postischemic myelopathy
 postradiation myelopathy
 poststroke pain
 posttraumatic spinal cord injury pain
 syringomyelia

29.2.2 Medical treatment of neuropathic pain

General information

Treatment traditionally includes narcotic analgesics,⁴ and tricyclic antidepressants (see below). For further details and other treatment measures, see PHN (p. 523).

Tricyclic antidepressants

Use is often limited by anticholinergic and central effects and by limited pain relief.^{5,6} Possibly because serotonin potentiates the analgesic effect of endorphins and elevates pain thresholds, serotonin re-uptake blockers are more effective than norepinephrine re-uptake blockers, e.g., trazodone (Desyrel®) blocks only serotonin. Also useful: amitriptyline (Elavil®) 75 mg daily; desipramine (Norpramin®) 10–25 mg/d; doxepin (Sinequan®) 75–150 mg/d. Some benefit may also derive from the fact that many patients with chronic pain are depressed. **Side effects:** anticholinergic effects and orthostatic hypotension, especially in the elderly. ✖ Not recommended for use in patients with ischemic heart disease.

Gabapentin

Effective in postherpetic neuralgia (PHN) (p.524) and painful diabetic neuropathy. Benefit also reported in pain associated with trigeminal neuralgia, cancer,⁷ multiple sclerosis, HIV-related sensory neuropathy, CRPS, spinal cord injury, postoperative state,⁸ migraine⁹ (a number of these studies may have been sponsored by the manufacturer¹⁰). See also side effects, dosing & availability... (p.496)

Lidocaine patch (Lidoderm®)

May be effective.³ **R:** apply patch for up to 12 hrs/day up to a maximum of 3 patches at a time to the intact skin over the most painful area (may trim patch to appropriate size). **Supplied:** 5% lidocaine (p.525).

Tramadol (Ultram®)

A centrally acting analgesic (p. 147).³

29.3 Craniofacial pain syndromes

29.3.1 General information

Possible pathways for *facial* pain include trigeminal nerve (portio major as well as portio minor [motor root]), facial nerve (usually deep facial pain), and eighth nerve.¹¹ Etiologies (adapted^{12(p 2328),13}):

1. cephalic neuralgias
 - a) trigeminal neuralgia (p.1857)
 - vascular compression of Cr. N. V by the SCA at root entry zone: the most common cause
 - MS: plaque within Cr. N. V nucleus
 - b) glossopharyngeal neuralgia (p. 1873): pain usually in base of tongue and adjacent pharynx
 - c) geniculate neuralgia (p. 1873): otalgia and deep prosopalgia
 - d) tic convulsif (p. 1873): geniculate neuralgia with hemifacial spasm
 - e) occipital neuralgia (p.541)
 - f) superior laryngeal neuralgia: a branch of the vagus, results primarily in laryngeal pain and occasionally pain on the auricle
 - g) sphenopalatine neuralgia
 - h) herpes zoster: pain is continuous (not paroxysmal). Characteristic vesicles and crusting usually follow pain, most often in distribution of V1 (isolated V1 TGN is rare). In rare cases without vesicles, diagnosis may be difficult
 - i) postherpetic neuralgia (p. 1873)
 - j) Ramsay-Hunt syndrome: reactivation of herpes zoster in the geniculate ganglion (AKA herpes zoster oticus). Triad: ipsilateral facial paralysis, ear pain and vesicles on the face, ear, or in the ear
 - k) supraorbital neuralgia (SON) (p.521)
 - l) trigeminal neuropathic pain (AKA trigeminal deafferentation pain)¹³: may follow injuries from sinus or dental surgery, head trauma

- m) trigeminal deafferentation pain: follows trigeminal denervation, including therapeutic measures to treat trigeminal neuralgia¹³
 - n) short-lasting unilateral neuralgiform headache with conjunctival injection and tearing (SUNCT)¹⁴: rare. Usually affects males 23–77 years old. Brief (< 2 minutes) pain (burning, stabbing or shock-like) usually near the eye, occurring multiple times per day. Associated autonomic findings (the “hallmark of SUNCT”): ptosis, conjunctival injection, lacrimation, rhinorrhea, hyperemia. May be due to CPA AVM. Microvascular decompression or trigeminal rhizotomy may be effective in some cases refractory to medical treatment with ASMs or corticosteroids. Note: lacrimation (the most common) or other autonomic signs may occur in V1 trigeminal neuralgia but are usually mild, and appear only in the later stages of the condition and with long lasting attacks.¹⁵ Dramatic lacrimation and conjunctival injection from the onset of symptoms with SUNCT are the best characteristics to distinguish this from trigeminal neuralgia.¹⁶ May also occur in cluster headache (p. 183).
2. ophthalmic pain
 - a) Tolosa-Hunt syndrome (p. 600): painful ophthalmoplegia
 - b) (Raeder’s) paratrigeminal neuralgia (p. 601): unilateral Horner syndrome + trigeminal neuralgia
 - c) orbital pseudotumor (p. 600): proptosis, pain, and EOM dysfunction
 - d) diabetic (oculomotor) neuritis
 - e) optic neuritis
 - f) iritis
 - g) glaucoma
 - h) anterior uveitis
 3. otalgia (see below)
 4. masticatory disorders
 - a) dental or periodontal disease
 - b) nerve injury (inferior and/or superior alveolar nerves)
 - c) temporo-mandibular joint (TMJ) dysfunction
 - d) elongated styloid process
 - e) temporal & masseter myositis
 5. vascular pain syndromes
 - a) migraine headaches: see Migraine (p. 183)
 - simple migraine: includes classic migraine, common migraine
 - complicated migraine: includes hemiplegic migraine, ophthalmoplegic migraine
 - b) cluster H/A (p. 183); subtypes: episodic, chronic, chronic paroxysmal hemicrania
 - c) giant cell arteritis (p. 203) (temporal arteritis). Tenderness over STA
 - d) toxic or metabolic vascular H/A (fever, hypercapnia, EtOH, nitrites, hypoxia, hypoglycemia, caffeine withdrawal)
 - e) hypertensive H/A
 - f) aneurysm or AVM (due either to mass effect or hemorrhage)
 - g) carotidynia: e.g., with carotid dissection (p. 1578)
 - h) basilar dolichoectasia with fifth n. compression or indentation of the pons
 6. sinusitis (maxillary, frontal, ethmoidal, sphenoidal)
 7. dental disease
 8. neoplasm: may cause referred pain or fifth nerve compression
 - a) extracranial
 - b) intracranial tumor: primarily posterior fossa lesions, neoplastic compression of trigeminal nerve usually causes sensory deficit (p. 1859)
 9. atypical facial pain (AFP) (prosopalgia): traditionally a “wastebasket” category used for many things. It has been proposed¹³ to reserve this term for a psychogenic disorder
 10. primary (nonvascular) H/A: including
 - a) tension (muscle contraction) H/A
 - b) posttraumatic H/A

29.3.2 Otolgia

General information

Because of redundant innervation of the region of the ear, *primary* otalgia may have its source in the 5th, 7th, 9th, or 10th cranial nerves or the occipital nerves.¹⁷ As a result, sectioning of the 5th, 9th or 10th nerve or a component of the 7th (nervus intermedius, chorda tympani, geniculate ganglion) has been performed with varying results.¹⁸ Also, microvascular decompression (MVD) of the corresponding nerve may also be done.¹⁹

Work-up includes neurotologic evaluation to rule out causes of secondary otalgia (otitis media or externa, temporal bone neoplasms...). CT or MRI should be done in any case where no cause is found.

Primary otalgia

Primary otalgia is unilateral in most ($\approx 80\%$). Trigger mechanisms are identified in slightly more than half, with cold air or water being the most common.¹⁸ About 75% have associated aural symptoms: hearing loss, tinnitus, vertigo. Pain relief upon cocaineization or nerve block of the pharyngeal tonsils suggests glossopharyngeal neuralgia (p.1873); however, the overlap of innervation limits the certainty.

An initial trial with medications used in trigeminal neuralgia (carbamazepine, phenytoin, baclofen...) (p.1860) is the first line of defense. In intractable cases not responding to pharyngeal anesthesia, suboccipital exploration of the 7th (nervus intermedius) and lower cranial nerves may be indicated. If significant vascular compression is found, one may consider MVD alone. If MVD fails, or if no significant vessels are found, Rupa et al recommend sectioning the nervus intermedius, the 9th and upper 2 fibers of 10th nerve, and a geniculate ganglionectomy (or, if glossopharyngeal neuralgia is strongly suspected, just 9th and upper 2 fibers of 10th).¹⁸

29.3.3 Supraorbital and supratrochlear neuralgia

Anatomy

The supraorbital and supratrochlear nerves arise from the frontal nerve and are 2 of the 5 branches of V1 (ophthalmic division of the trigeminal nerve). The supraorbital nerve is the largest branch. It exits the orbit through the supraorbital notch or foramen, usually within the medial third of the orbital roof (mean distance from exit to medial angle of orbit: 20 mm (range: 5–47)²⁰). The supratrochlear nerve exits the orbit without a foramen or notch 3–38 mm medial to the supraorbital nerve (mean: 15.3 mm²⁰); the most medial branch varies from 8–30 mm lateral to the patient's midline.²⁰

Supraorbital neuralgia characteristics

Trigeminal neuralgia (TGN) may present with pain in the distribution of the supraorbital nerve; however, the supraorbital nerve may be involved in supraorbital neuralgia (SON), a distinct syndrome with different clinical characteristics. SON is a rare condition slightly more common in women, with onset typically 40–50 years of age.²¹ Characteristics²²: 1) unilateral pain in the distribution of the supraorbital nerve ($\approx$ Fig. 104.2), 2) tenderness in the region of the supraorbital notch or along the distribution of the nerve, and 3) temporary relief with nerve block.

The pain is usually chronic-continuous or remitting-intermittent.²¹

SON may be:

1. primary (no identifiable etiology): these cases lack any sensory loss
2. secondary (e.g., due to trauma to the area, or resulting from chronic pressure such as with wearing swim goggles): more common than primary SON. Most cases remit within one year²¹ with elimination of the offending pressure

Supratrochlear neuralgia

Cases of pain isolated to the supratrochlear nerve appear to exist. Supratrochlear neuralgia (STN) may be differentiated from SON by restriction of pain in the more medial forehead, and with relief on blockade of the supratrochlear nerve alone.

Differential diagnosis

1. migraine: suggested by nausea, vomiting, and photophobia
2. associated autonomic activity is rare with SON, and should prompt consideration of cluster H/A (p.183) or SUNCT (p.520)
3. TGN: typical TGN features *lacking* in SON include characteristic triggers and pain consisting exclusively of paroxysmal/ultra-brief electric-shock-like pain
4. hemicrania continua: continuous unilateral pain that tends to be located more posteriorly and is absolutely responsive to indomethacin²²
5. trochleitis: inflammation of the trochlea/superior-oblique muscle complex, may mimic supratrochlear neuralgia with pain of the medial upper orbit extending a short distance to the forehead.²³

The pain is typically exacerbated by supraduction of the eye and to palpation of the trochlea, and is relieved with injection of local anesthetic or by the usually definitive treatment of infiltration of corticosteroids close to the trochlea. Diplopia is rare and minimal

6. nummular (coin-like) H/A²⁴: round or oval 2–6 cm diameter area of pressure-like continuous head pain without underlying structural abnormality. In 13 patients, 9 (70%) the area was located at the parieto-occipital junction. 9 (70%) demonstrated hypoesthesia and touch provoked paresthesias in the affected area

Treatment

Gabapentin (800–2400 mg/d) or pregabalin (150 mg/d) is helpful for some.²⁵

Topical capsaicin (p.1861) applied to the symptomatic area may help.

Refractory cases may respond to rhizotomy with alcohol (providing an average of 8.5 months relief²⁶) or with radiofrequency ablation.

Persistent cases may require exploration and decompression of the nerve by lysing bands overlying the supraorbital notch,²⁷ or, ultimately, to neurectomy (p.1862) which provides an average of 33.2 months relief.²⁸

29.4 Postherpetic neuralgia

29.4.1 General information

Herpes zoster (HZ) (Greek: zoster – girdle) (shingles in lay terms): painful vesicular cutaneous eruptions caused by the herpes varicella zoster virus (VZV) (the etiologic agent of chickenpox, a herpes virus that is distinct from herpes simplex virus). It occurs in a dermatomal distribution over one side of the thorax in ≈ 65% of cases (rarely, infections occur without vesicles, called *zoster sine herpette*). In 20% of cases it involves the trigeminal nerve (with a predilection for the ophthalmic division, called *herpes zoster ophthalmicus*). Pain usually resolves after 2–4 weeks. When the pain persists > 1 month after the vesicular eruption has healed, this pain syndrome is known as postherpetic neuralgia (PHN). PHN can follow a herpes varicella infection in any site and is difficult to treat by any means (medical or surgical). It can occasionally be seen in a limb, and follows a dermatomal distribution (*not* a peripheral nerve distribution). PHN may remit spontaneously, but if it hasn't done so by 6 mos this is unlikely.

29.4.2 Epidemiology

Incidence of herpes zoster is ≈ 125/100,000/year in the general population, or about 850,000 cases per year in the U.S.²⁹ Both sexes are equally affected. There is no seasonal variance. HZ is also more common in those with reduced immunity and in those with a coexistent malignancy (especially lympho-proliferative).^{30,31} PHN occurs in ≈ 10% of cases of HZ.²⁹ Both HZ and PHN are more common in older patients (PHN is rare in age < 40 yrs, and usually occurs in age > 60) and in those with diabetes mellitus. PHN is more likely after ophthalmic HZ than after spinal segmental involvement.

29.4.3 Etiology

It is postulated that the VZV lies dormant in the sensory ganglia (dorsal root ganglia of the spine, trigeminal (semilunar) ganglion for facial involvement) until such time that the patient's immune system is weakened and then the virus erupts. Inflammatory changes within the nerve are present early and are later replaced by fibrosis.

29.4.4 Clinical

PHN is usually described as a constant burning and aching. There may be superimposed shocks or jabs. It rarely produces throbbing or cramping pain. Pain may be spontaneous, or may be triggered by light cutaneous stimulation (allodynia) (e.g., by clothing), and may be relieved by constant pressure. The pain is present to some degree at all times with no pain-free intervals. Scars and pigmentary changes from the acute vesicular eruption are usually visible. It is not known if PHN can follow *zoster sine herpette*. The involved area may demonstrate hypesthesia, hypalgesia, paresthesias and dysesthesias.

29.4.5 Medical treatment

For herpes zoster

Varicella vaccination of older individuals can increase immunity to herpes zoster, but it will be several years before it can be determined if this will reduce PHN.²⁹

Treatment for the pain of the *acute* attack of herpes zoster may be accomplished with epidural or paravertebral somatic (intercostal) nerve block.³² (p 4018)

Oral antiherpetic drugs: Also effective (they shorten the duration of pain), and also reduce the incidence of PHN. They may cause thrombotic thrombocytopenic purpura/hemolytic uremic syndrome (TTP/HUS) when used in severely immunocompromised patients at high doses. These drugs include:

Acyclovir (Zovirax®): poorly absorbed from the GI tract (15–30% bioavailability). **R** 800 mg PO q 4 hrs 5 times/d × 7 d.

Valacyclovir (Valtrex®)³³ is a prodrug of acyclovir and is more completely absorbed and should be equally as effective with fewer daily doses. **R** 1,000 mg PO TID starting within 72 hrs of onset of the rash × 7 days.

Famciclovir (Famvir®): **R** 500 mg PO TID × 7 d.

For post-herpetic neuralgia

General information

Most drugs useful for trigeminal neuralgia (p. 1860) are less effective for PHN. Some treatment alternatives for PHN are summarized in ► Table 29.2. Details of some drugs follows. It is suggested to initiate therapy with lidocaine skin patches (p. 525) since this modality has the lowest potential for serious side effects.²⁹

Table 29.2 Medical treatments for PHN^a

Treatment	Efficacy
PHN treatments that appear effective	
tricyclic antidepressants	widely used for PHN (see text)
lidocaine patch (Lidoderm®) ³⁴	effective, few side effects (p. 525)
intrathecal steroids + lidocaine (see text)	appears very effective, larger studies & long-term follow-up needed
Gabapentin	proven efficacy (see text)
oxycodone CR 10 mg PO BID ⁴	proven efficacy
Treatments of questionable efficacy	
SSRIs ^b	may be effective
SNRIs	may be effective
Tramadol	may be effective
topical capsaicin	controversial (see text)
Iontophoresis	insufficient evidence
nonsteroidal creams	questionable
aspirin suspended in acetone, ether or chloroform	questionable
EMLA cream	questionable
Treatments that are not useful	
dextromethorphan, benzodiazepines, acyclovir, acupuncture	no benefit ³⁵
ketamine (NMDA receptor antagonist)	may be beneficial, but hepatotoxic
Preventative treatment	
oral antiherpetic drugs given during HZ infection	shortens length of HZ, may reduce incidence of PHN
varicella vaccination of older patients	trials of this strategy are in progress ²⁹

^amodified with permission from Rubin M, Relief for postherpetic neuralgia, *Neurology Alert*, 6: 33–4, 2001

^babbreviations: oxycodone CR = controlled release (Oxycontin®); HZ = herpes zoster; PHN = postherpetic neuralgia; SNRIs = serotonin-norepinephrine reuptake inhibitor; SSRIs = selective serotonin reuptake inhibitors (e.g., Prozac®).

Antiseizure medications for post-herpetic neuralgia

Drug info: Gabapentin (Neurontin®)

FDA approved only for partial seizures and postherpetic neuralgia (PHN).

Side effects: dizziness and somnolence (usually during titration, often diminish with time). Ataxia, fatigue, peripheral edema, confusion and depression may occur.

R For PHN, start with 300 mg on Day 1, 300 mg BID on Day 2, and 300 mg TID on Day 3. Dose may be titrated up to 1800 mg/d divided TID. To limit daytime drowsiness, patients may need to start with 100 mg at hs and increase slowly over 3–8. Although doses up to 3600 mg/day (the antiseizure dose) were studied³⁶ there was no significant benefit for PHN over 1800 mg/d. Lower doses are required for renal insufficiency. **Supplied:** 100, 300 & 400 mg capsules; 600 & 800 mg scored tabs. 50 mg/ml suspension.

Drug info: Oxcarbazepine (Trileptal®)

R 150 mg PO BID.

Drug info: Zonisamide (Zonegran®)

R Initiate therapy with 100 mg PO q PM × 2 wks, then increase dose by 100 mg/d q 2 wks up to 400 mg/d. Bioavailability is not affected by food. Steady state is achieved within 14 days of dosage changes. **Supplied:** 100 mg capsules.

29

Tricyclic antidepressants (TCA)

Drug info: Amitriptyline (Elavil®)

Helpful in ≈ 66% of patients at a mean dose of 75 mg/d even without antidepressant effect.⁵ **Side effects:** see Amitriptyline, side effects (p. 572), minimized by starting low and slowly incrementing dose.

R Start with 12.5–25 mg PO q hs, and increase by the same amount q 2–5 days to a maximum of 150 mg/d.

Drug info: Nortriptyline (Pamelor®)

Fewer side effects than amitriptyline.

R Start with 10–20 mg PO q hs, and increase gradually.

Topical treatment

Drug info: Capsaicin (Zostrix®)

A vanillyl alkaloid derived from hot peppers, available without prescription for topical treatment of the pain of herpes zoster and diabetic neuropathy. Beneficial in some patients with either of these conditions (response rate at 8 weeks was 90% for PHN, 71% for diabetic neuropathy, vs. 50% with placebo in either group), although the high placebo response rate is disturbing and many authorities are skeptical.³⁷ Expensive. **Side effects:** include burning and erythema at the application site (usually subsides by 2–4 weeks).

R Manufacturer recommends massaging the medication into the affected area of the skin TID-QID (apply a very thin coat). Some authorities recommend q 2 hr application. Avoid contact with eyes or damaged skin. Supplied as Zostrix® (0.25% capsaicin) or Zostrix-HP® (0.75%).

Drug info: Lidocaine patch 5% (Lidoderm®)

Often better tolerated by elderly patients than TCAs (due to pre-existing cognitive impairments, cardiac disease, or systemic illness).

R Apply up to 3 patches of 5% lidocaine (to cover a maximum of 420 cm²) to intact skin q 12 hrs to cover as much of the area of greatest pain as possible.³⁴

Intrathecal steroids

Over 90% of patients receiving intrathecal methylprednisolone (60 mg) + 3% lidocaine (3 ml) given once per week for up to 4 weeks, reported good to excellent pain relief for up to 2 years.³⁸ This technique was not studied for use in PHN involving the trigeminal nerve. Further clinical trials are needed to verify the efficacy and safety²⁹ (potential long-term side effects include adhesive arachnoiditis).

Surgical treatment

There is no operation that is uniformly successful in treating PHN. Numerous operations have been shown to work occasionally. Procedures that have been tried include:

1. nerve blocks: once PHN is established, nerve blocks provide only temporary relief³⁹
2. cordotomy: although percutaneous cordotomy (p. 1878) may work when the level of PHN is at least 3–4 segments below the cordotomy, this procedure is not recommended for pain of benign etiology because of possible complications and the high likelihood of pain recurrence
3. rhizotomy: including retrogasserian for facial involvement
4. neurectomies
5. sympathectomy
6. DREZ (p. 1886)⁴⁰: often offers good early relief, but recurrence rate is high
7. acupuncture⁴¹
8. TENS
9. spinal cord stimulation (p. 1883)
10. undermining the skin
11. motor cortex stimulation: for *facial* PHN

29.5 Complex regional pain syndrome (CRPS)

29.5.1 General information

The terminology is confusing. Formerly also called *causalgia* (reflex sympathetic dystrophy). The term *causalgia* (Greek: *kausis* – burning, *algos* – pain) was introduced by Weir Mitchell in 1864. It was used to describe a rare syndrome that followed a minority of *partial* peripheral nerve injuries in the American civil war. *Triad*: burning pain, autonomic dysfunction, and trophic changes.

CRPS Type II (AKA major causalgia) follows nerve injury (originally described after high velocity missile injuries). CRPS Type I (AKA reflex sympathetic dystrophy or *causalgia minor*) denoted less severe forms, and has been described after non-penetrating trauma.⁴² Shoulder-hand syndrome and Sudeck's atrophy are other variant designations. In 1916, the autonomic nervous system was implicated by René Leriche, and the term reflex sympathetic dystrophy (RSD) later came into use⁴³ (but RSD may be distinct from causalgia⁴⁴).

Post-op CRPS has been described following carpal tunnel surgery as well as surgery on the lumbar⁴⁵ and cervical spine.

A contemporary definition might be: a disproportionate pain syndrome caused by nerve damage and resultant sympathetic dysfunction.⁴⁶ It may arise from direct injury to a nerve (type 1), or indirectly due to damage to surrounding tissue (type 2).⁴⁷

At best, CRPS must be regarded as a symptom complex, and not as a discrete syndrome nor medical entity (for a cogent editorial on the subject, see the essay by Ochoa⁴⁸). Patients exhibiting CRPS phenomenology are not a homogeneous group, and include⁴⁹:

1. actual CRPS (for these, Mailis proposes the term “physiogenic RSD”): a complex set of neuropathic phenomena that may occur with or without nerve injury
2. medical conditions distinct from CRPS but with signs and symptoms that mimic CRPS: vascular, inflammatory, neurologic...
3. the product of mere immobilization: as in severe pain avoidance behavior, or at times psychiatric disorders
4. part of a factitious disorder with either a psychological basis (e.g., Munchausen's syndrome) or for secondary gain (financial, drug seeking...) i.e., malingering

29.5.2 Pathogenesis

Early theories invoked ephaptic transmission between sympathetics and afferent pain fibers. This theory is rarely cited currently. Another more recent postulate involves norepinephrine released at sympathetic terminals together with hypersensitivity secondary to denervation or sprouting. Many modern hypotheses do not even embrace involvement of the autonomic nervous system in all cases.^{43,44,49}

Thus, many of the alterations seen in CRPS may simply be epiphenomena rather than part of the etiopathogenetic mechanism.

29

29.5.3 Clinical

CRPS may be described as a phenomenology, i.e., a variable complex of signs and symptoms due to multiple etiologies included in this nonhomogeneous group.⁴⁹ No diagnostic criteria for the condition have been established, and various investigators select different factors to include or exclude patients from their studies.

29.5.4 Symptoms

Pain: affecting a limb, usually burning, and prominent in the hand or foot. Onset in the majority is within 24 hrs of injury (unless injury causes anesthesia, then hrs or days may intervene); however, CRPS may take days to weeks to develop. Median, ulnar and sciatic nerves are the most commonly cited involved nerves. However, it is not always possible to identify a specific nerve that has been injured. Almost any sensory stimulus worsens the pain (allodynia is pain induced by a nonnoxious stimulus).

29.5.5 Signs

The physical exam is often difficult due to pain.

Vascular changes: either vasodilator (warm and pink) or vasoconstrictor (cold, mottled blue). Trophic changes (may be partly or wholly due to immobility): dry/scaly skin, stiff joints, tapering fingers, ridged uncut nails, either long/coarse hair or loss of hair, sweating alterations (varies from anhidrosis to hyperhidrosis).

29.5.6 Diagnostic aids

In the absence of an agreed upon etiology or pathophysiology, there can be no basis for specific tests, and the lack of a “gold-standard” diagnostic criteria makes it impossible to verify the authenticity of any diagnostic marker. Numerous tests have been presented as aids to the diagnosis of CRPS, and essentially all have eventually been refuted. Candidates have included:

1. thermography: discredited in clinical practice
2. three-phase bone scan: typical CRPS changes also occur after sympathectomy,⁵⁰ which has traditionally been considered curative of CRPS
3. osteoporosis on X-ray,⁵¹ particularly periarticular demineralization: nonspecific
4. response to sympathetic block (once thought to be the sine qua non for causalgia major and minor, the response sought was relief (complete or significant) with sympathetic block of appropriate trunk (stellate for UE, lumbar for LE)): has failed to hold up once stringent placebo-controlled trials were executed

5. various autonomic tests⁵²: resting sweat output, resting skin temperature, quantitative sudomotor axon reflex test

29.5.7 Treatment

In the absence of a delineated pathophysiology, treatment is judged purely by subjective impression of improvement. CRPS treatment studies have had an unusually high placebo response rate.⁵³ Medical therapy is usually ineffective. Proposed treatments include:

1. tricyclic antidepressants
2. 18–25% have satisfactory long-lasting relief after a series of sympathetic blocks, see Stellate ganglion block (p. 1834) and Lumbar sympathetic block (p. 1835), although one report found no long-lasting benefit in any of 30 patients⁵⁴
3. intravenous regional sympathetic block, particularly for UE CRPS: agents used include *guanethedine*⁵⁵ 20 mg, reserpine, bretylium..., injected IV with arterial tourniquet (sphygmomanometer cuff) inflated for 10 min. If no relief, repeat in 3–4 wks. No better than placebo in several trials⁵⁶.
4. surgical sympathectomy (p. 1853): some purport that this relieves pain in >90% of patients (with a few retaining some tenderness or hyperpathia). Others opine that there is no rational reason to consider sympathectomy since sympathetic blocks have been shown to be no more effective than placebo⁴³
5. spinal cord stimulation: some success has been reported

References

- [1] Backonja MM. Defining neuropathic pain. *Anesth Analg*. 2003; 97:785–790
- [2] Merskey H, Bogduk N. Classification of Chronic Pain: Descriptions of Chronic Pain Syndromes and Definitions of Pain Terms. 2nd ed. Seattle, WA: IASP Press; 1994
- [3] Dworkin RH, Backonja M, Rowbotham MC, et al. Advances in neuropathic pain: diagnosis, mechanisms, and treatment recommendations. *Arch Neurol*. 2003; 60:1524–1534
- [4] Watson CPN, Babul N. Efficacy of oxycodone in neuropathic pain: a randomized trial in postherpetic neuralgia. *Neurology*. 1998; 50:1837–1841
- [5] Watson CP, Evans RJ, Reed K, et al. Amitriptyline versus Placebo in Postherpetic Neuralgia. *Neurology*. 1982; 32:671–673
- [6] Max MB, Lynch SA, Muir J, et al. Effects of Desipramine, Amitriptyline, and Fluoxetine on Pain in Diabetic Neuropathy. *N Engl J Med*. 1992; 326:1250–1256
- [7] Bennett MI, Simpson KH. Gabapentin in the treatment of neuropathic pain. *Palliat Med*. 2004; 18:5–11
- [8] Dierking G, Duedahl TH, Rasmussen ML, et al. Effects of gabapentin on postoperative morphine consumption and pain after abdominal hysterectomy: a randomized, double-blind trial. *Acta Anaesthesiol Scand*. 2004; 48:322–327
- [9] Mathew NT, Rapoport A, Saper J, et al. Efficacy of gabapentin in migraine prophylaxis. *Headache*. 2001; 41:119–128
- [10] Gabapentin (Neurontin®) for chronic pain. *Medical Letter*. 2004; 46:29–31
- [11] Keller JT, van Loveren H. Pathophysiology of the Pain of Trigeminal Neuralgia and Atypical Facial Pain: A Neuroanatomical Perspective. *Clin Neurosurg*. 1985; 32:275–293
- [12] Wilkins RH, Rengachary SS. *Neurosurgery*. New York 1985
- [13] Burchiel KJ. A new classification for facial pain. *Neurosurgery*. 2003; 91:1164–1167
- [14] Pareja JA, Sjaastad O. SUNCT syndrome. A clinical review. *Headache*. 1997; 37:195–195
- [15] Sjaastad O, Pareja JA, Zukerman E, et al. Trigeminal neuralgia. Clinical manifestations of first division involvement. *Headache*. 1997; 37:346–357
- [16] Pareja JA, Baron M, Gili P, et al. Objective assessment of autonomic signs during triggered first division trigeminal neuralgia. *Cephalalgia*. 2002; 22:251–255
- [17] Yeh HS, Tew JM. Tic Convulsif, the Combination of Geniculate Neuralgia and Hemifacial Spasm Relieved by Vascular Decompression. *Neurology*. 1984; 34:682–683
- [18] Rupa V, Saunders RL, Weider DJ. Geniculate Neuralgia: The Surgical Management of Primary Otagia. *Journal of Neurosurgery*. 1991; 75:505–511
- [19] Young RF. Geniculate Neuralgia. *Journal of Neurosurgery*. 1992; 76
- [20] Andersen NB, Bovim G, Sjaastad O. The frontotemporal peripheral nerves. Topographic variations of the supraorbital, supratrochlear and auriculotemporal nerves and their possible clinical significance. *Surgical and Radiologic Anatomy*. 2001; 23:97–104
- [21] Pareja JA, Caminero AB. Supraorbital neuralgia. *Curr Pain Headache Rep*. 2006; 10:302–305
- [22] Headache Classification Committee of the International Headache Society. Classification and diagnostic criteria for headache disorders, cranial neuralgias, and facial pain, 2nd edition. *Cephalalgia*. 2004; 24:9–160
- [23] Pareja JA, Pareja J, Yanguela J. Numbular headache, trochleitis, supraorbital neuralgia, and other epicranial headaches and neuralgias: the epicranias. *Journal of Headache and Pain*. 2003; 4:125–131
- [24] Pareja JA, Caminero AB, Serra J, et al. Numular headache: a coin-shaped cephalgia. *Neurology*. 2002; 58:1678–1679
- [25] Caminero AB, Pareja JA. Supraorbital neuralgia: a clinical study. *Cephalalgia*. 2001; 21:216–223
- [26] Stookey B, Ransohoff J. *Trigeminal Neuralgia: Its History and Treatment*. Springfield, IL: Charles C Thomas; 1959
- [27] Sjaastad O, Stolt-Nielsen A, Pareja JA, et al. Supraorbital neuralgia: on the clinical manifestations and a possible therapeutic approach. *Headache*. 1999; 39:204–212
- [28] Grantham EG, Segerberg LH. An evaluation of palliative surgical procedures in trigeminal neuralgia. *Journal of Neurosurgery*. 1952; 9:390–394
- [29] Watson CPN. A new treatment for postherpetic neuralgia. *New England Journal of Medicine*. 2000; 343:1563–1565
- [30] Loeser JD. Herpes Zoster and Postherpetic Neuralgia. *Pain*. 1986; 25:149–164
- [31] Schimpff S, Serpick A, Stoler B, et al. Varicella-Zoster Infection in Patients with Cancer. *Annals of Internal Medicine*. 1972; 76:241–254
- [32] Youmans JR. *Neurological Surgery*. Philadelphia 1990

- [33] Valacyclovir. Medical Letter. 1996; 38:3-4
- [34] Rowbotham MC, Davies PS, Verkempinck C, et al. Lidocaine patch: double-blind controlled trial of a new treatment method for postherpetic neuralgia. *Pain*. 1996; 65:39-44
- [35] Alper BS, Lewis PR. Treatment of postherpetic neuralgia: a systematic review of the literature. *J Fam Pract*. 2002; 51:121-128
- [36] Rowbotham MC, Harden N, Stacey B, et al. Gabapentin for the treatment of postherpetic neuralgia: A randomized controlled trial. *Journal of the American Medical Association*. 1998; 280:1837-1842
- [37] Capsaicin - A Topical Analgesic. Medical Letter. 1992; 34:62-63
- [38] Kotani N, Kushikata T, Hashimoto H, et al. Intrathecal methylprednisolone for intractable postherpetic neuralgia. *New England Journal of Medicine*. 2000; 343:1514-1519
- [39] Dan K, Higa K, Noda B, et al. Nerve block for herpetic pain. In: *Advances in Pain Research and Therapy*. New York: Raven Press; 1985:831-838
- [40] Friedman AH, Nashold BS. Dorsal Root Entry Zone Lesions for the Treatment of Postherpetic Neuralgia. *Neurosurgery*. 1984; 15:969-970
- [41] Lewith GT, Field J, Machin D. Acupuncture Compared with Placebo in Post-Herpetic Pain. *Pain*. 1983; 17:361-368
- [42] Sternschein MJ, Myers SJ, Frewin DB, et al. Causalgia. *Arch Phys Med Rehabil*. 1975; 56:58-63
- [43] Schott GD. An Unsympathetic View of Pain. *Lancet*. 1995; 345:634-636
- [44] Ochoa JL, Verdugo RJ. Reflex Sympathetic Dystrophy: A Common Clinical Avenue for Somatoform Expression. *Neurol Clin*. 1995; 13:351-363
- [45] Sachs BL, Zindrick MR, Beasley RD. Reflex Sympathetic Dystrophy After Operative Procedures on the Lumbar Spine. *Journal of Bone and Joint Surgery*. 1993; 75A:721-725
- [46] Hsu ES. Practical management of complex regional pain syndrome. *Am J Ther*. 2009; 16:147-154
- [47] Groeneweg G, Huygen FJ, Coderre TJ, et al. Regulation of peripheral blood flow in complex regional pain syndrome: clinical implication for symptomatic relief and pain management. *BMC Musculoskeletal Disord*. 2009; 10. DOI: 10.1186/1471-2474-10-116
- [48] Ochoa JL. Reflex? Sympathetic? Dystrophy? Triple Questioned Again. *Mayo Clinic Proceedings*. 1995; 70:1124-1125
- [49] Mailis A. Is Diabetic Autonomic Neuropathy Protective Against Reflex Sympathetic Dystrophy? *Clin J Pain*. 1995; 11:77-81
- [50] Mailis A, Meindok H, Papagapiou M, et al. Alterations of the Three-Phase Bone Scan After Sympathectomy. *Clin J Pain*. 1994; 10:146-155
- [51] Kozin F, Genant HK, Bekerman C, et al. The Reflex Sympathetic Dystrophy Syndrome. *Am J Med*. 1976; 60:332-338
- [52] Chelimsky TC, Low PA, Naessens JM, et al. Value of Autonomic Testing in Reflex Sympathetic Dystrophy. *Mayo Clinic Proceedings*. 1995; 70:1029-1040
- [53] Ochoa JL. Pain Mechanisms in Neuropathy. *Curr Opin Neurol*. 1994; 7:407-414
- [54] Dotson R, Ochoa JL, Cline M, et al. A Reassessment of Sympathetic Blocks as Long Term Therapeutic Modality for "RSD". *Pain*. 1990; 5
- [55] Hannington-Kiff JG. Relief of Sudek's Atrophy by Regional Intravenous Guanethidine. *Lancet*. 1977; 1:1132-1133
- [56] Blanchard J, Ramamurthy W, Walsh N, et al. Intravenous Regional Sympatholysis: A Double-Blind Comparison of Guanethidine, Reserpine, and Normal Saline. *J Pain Symptom Manage*. 1990; 5:357-361
- [57] Jadad AR, Carroll D, Glynn CJ, et al. Intravenous Regional Sympathetic Blockade for Pain Relief in Reflex Sympathetic Dystrophy: A Systematic Review and a Randomized, Double-Blind Crossover Study. *J Pain Symptom Manage*. 1995; 10:13-20

Part X

Peripheral Nerves

30	Peripheral Nerves	530
31	Entrapment Neuropathies	541
32	Non-Entrapment Peripheral Neuropathies	568



30 Peripheral Nerves

30.1 Peripheral nerves – definitions and grading scales

30.1.1 Peripheral nervous system definition

The peripheral nervous system (PNS) consists of those structures (including cranial nerves III–XII, spinal nerves, nerves of the extremities, and the cervical, brachial and lumbosacral plexi) containing nerve fibers or axons that connect the central nervous system (CNS) with motor and sensory, somatic and visceral, end organs.¹ ► Table 30.1 shows classification of motor and sensory nerves.

Table 30.1 Motor & sensory classification of nerves

Sensory	Sensory & motor	Greatest fiber diameter (mcm)	Greatest conduction velocity (m/sec)	Motor/sensory	Comments
Ia	A-alpha	22	120	motor	large alpha motor neurons of lamina IX (extrafusal) sensory-primary afferents (anulospiral of muscle spindles for proprioception)
Ib	A-alpha	22	120	sensory	Golgi tendon organs, touch & pressure receptors
II	A-beta	13	70	sensory	secondary afferents (flower spray) of muscle spindles, crude touch, pressure receptors, Pacinian corpuscles (vibratory) (to posterior columns ^a)
	A-gamma	8	40	motor	small γ motor neurons of lamina IX (intrafusal)
III	A-delta	5	15	sensory	small, lightly myelinated, fine touch, pressure, pain & temperature
	B	3	14	motor	small, lightly myelinated preganglionic autonomic fibers
IV	C	1	2	motor	all post-ganglionic autonomic
				sensory	unmyelinated pain & temp (to spinothalamic tract)

^athese fibers to the posterior columns are more medial in the root entry zone than the C-fibers (important in DREZ lesions where the goal is to lesion C-fibers and spare A-beta)

30.1.2 Grading strength and reflexes

Muscle strength grading most commonly employs the Royal Medical Research Council of Great Britain (MRC) scale,² a common modification of which is shown in ► Table 30.2. Muscle stretch reflexes may be graded as shown in ► Table 30.3.²

Table 30.2 Muscle grading (modified Medical Research Council (MRC) system)

Grade	Strength	
0	no contraction (total paralysis)	
1	flicker or trace contraction (palpable or visible)	
2	active movement with gravity eliminated	
3	active movement through full ROM against gravity	
4	active movement against resistance; subdivisions →	4 – Slight resistance 4 Moderate resistance 4+ Slight resistance
5	normal strength (against full resistance)	
NT	not testable	

Table 30.3 A sample muscle stretch reflex (deep tendon reflex) grading scale. Various scales are in use. Typically go from 0 to 4+ where 4+ is sustained clonus.

Grade	Definition
0	no contraction (total paralysis)
0.5 +	elicitable only with reinforcement ^a
1 +	low normal
2 +	normal
3 +	more brisk than normal (hyperreflexic)
4 +	hyperreflexic with clonus
5 +	sustained clonus

^aIn the LEs, reinforcement consists of having the patient hook the tips of the fingers of the left hand into the tips of the hooked fingers of the right hand and pulling (Jendrassik maneuver). Reinforcement in the UE's consists of having the patient clench their teeth

30.1.3 Upper motor neuron vs. lower motor neuron

Upper motor neurons (UMN) (first-order motor neurons): some soma reside in the primary motor cortex (precentral gyrus) of the brain. Axons project to LMNs.

Lower motor neurons (LMN) (second-order motor neuron): cell bodies (soma) reside in spinal cord (in anterior gray matter) or in brainstem (for cranial nerve motor nuclei). Axons connect directly to neuromuscular junction of muscles.

See ► Table 30.4 for comparison of weakness due to UMN vs. LMN.

Table 30.4 Upper vs. lower motor neuron paralysis

	Upper motor neuron paralysis	Lower motor neuron paralysis
possible etiologies	stroke (motor strip (p.56), internal capsule...), spinal cord injury, cervical spondylotic myelopathy	herniated intervertebral disc, nerve entrapment syndrome, polio, progressive muscular atrophy (PMA)
muscle tone	initially flaccid; later spastic with clasp-knife resistance	flaccid
tendon reflexes	hyperactive; clonus may be present	absent
pathologic reflexes (e.g., Babinski, Hoffman)	present (after days to weeks)	absent
muscle manifestations	spontaneous spasms may occur; some <i>atrophy of disuse</i> may occur	fibrillations, fasciculations. <i>Atrophy</i> after days to weeks due to trophic influence

30.1.4 Fasciculations vs. fibrillations

Fasciculations are coarse muscle contractions that are visible to the naked eye, whereas fibrillations are *not* visible and require EMG to detect; AKA fibrillation potentials (p.255).

Fasciculations represent discharge of a group of muscle fibers (all or part of an entire motor unit), and occur most often in diseases involving anterior horn cells, including:

1. amyotrophic lateral sclerosis (ALS) (p.191)
2. spinal muscular atrophy (p.1702)
3. polio
4. syringomyelia

30.2 Muscle innervation

30.2.1 Muscles, roots, trunks, cords and nerves of the upper extremities

See ► Table 30.5.

Table 30.5 Muscle innervation—shoulder & upper extremity^a

	Muscle	Action to test	Roots ^b	Trunk ^c	Cord ^d	Nerve
	deep neck	flex, ext, rotation of neck	C1–4	–	–	cervical
	trapezius	elevates shoulder, abducts arm >90°	XI, C3, 4			(spinal acc + roots)
	diaphragm	inspiration	C3–5			phrenic
•	serratus anterior	forward shoulder thrust	C5–7	–	–	long thoracic
	levator scapulae	elevate scapula	C3, 4, 5			dorsal scapular
	rhomboids	adduct & elevate scapula	C4, 5			
	supraspinatus	abduct arm (15–30°)	C4, 5, 6	S		suprascapular
•	infraspinatus	exorotation of humerus	C5, 6	S	–	
	latissimus dorsi	adduct arm	C5, 6, 7, 8			thoracodorsal
	teres major, sub-scapularis		C5–7			subscapular
•	deltoid	abduct arm (30–90°)	C5, 6	S	P	axillary
	teres minor	exorotate & adduct humerus	C4,5			
•	biceps brachii	flex forearm (with hand supinated), & supinate forearm	C5, C6	S	L	musculocutaneous
	coracobrachialis	flex humerus at shoulder	C5–7			
	brachialis	flex forearm	C5, 6			
•	flexor carpi ulnaris	ulnar flexion of wrist	C7, 8, T1	M, I	M	ulnar
•	flexor digitorum profundus III & IV (ulnar part)	flex distal phalanx of Dig 4–5	C7, 8, T1	M, I	M	
	adductor pollicis	thumb adduction	C8, T1		M	
	abductor digiti minimi	abduction Dig 5	C8, T1		M	
	opponens digiti minimi	opposition Dig 5	C7, 8, T1		M	
	flexor digiti minimi brevis	flexion Dig 5	C7, 8, T1		M	
•	interossei	flex proximal phalanx, extend 2 distal phalanges, abduct or adduct fingers	C8, T1	I	M	
	lumbricals 3 & 4	flex proximal phalanges & extend 2 distal phalanges of Dig 4–5	C7, 8			
•	pronator teres	forearm pronation	C6, 7	S,M	L	median
•	flexor carpi radialis	radial flexion of wrist	C6, 7	S,M	L	
	palmaris longus	wrist flexion	C7, 8, T1			
•	flexor digitorum superficialis	flexion middle phalanx Dig 2–5, flex wrist	C7, 8, T1	M, I	M	
•	abductor pollicis brevis	abduct thumb metacarpal	C8, T1	I	M	
	flexor pollicis brevis	flex prox phalanx thumb	C8, T1			

(continued)

Table 30.5 continued

Muscle	Action to test	Roots ^b	Trunk ^c	Cord ^d	Nerve
• opponens pollicis	opposes thumb metacarp	C8, T1	I	M	
lumbricals 1 & 2	flex proximal phalanx & extend 2 distal phalanges Dig 2–3	C8, T1			
• flexor digitorum profundus I & II (radial part)	flex distal phalanx of Dig 2–3; flex wrist	C7, 8, T1	M, I	M	anterior interosseous
• flexor pollicis longus	flex distal phalanx thumb	C7, 8, T1			
• triceps brachii	forearm extension	C6, 7, 8	all	P	radial
• brachioradialis	forearm flexion (with thumb pointed up)	C5, 6	S	P	
• extensor carpi radialis	radial wrist extension	C5, 6	S, M	P	
• supinator	forearm supination	C6, 7	S	P	
• extensor digitorum	extension of wrist & phalanges of Dig 2–5	C7, C8	M, I	P	posterior interosseous (PIN)
• extensor carpi ulnaris	ulnar wrist extension	C7, C8			
• abductor pollicis longus	abduction thumb metacarpal & radial wrist extension	C7, C8	M, I	P	
• extensor pollicis brevis & longus	thumb extension & radial wrist extension	C7, C8			
• extensor indicis proprius	extension Dig 2 & wrist extension	C7, C8			
• pectoralis major: clavicular head	push arm forward against resistance	C5, 6			lateral pectoral
• pectoralis major: sternocostal head	adduct arm	C6, 7, 8			lateral & medial pectoral

^aNB: items marked with a bullet (•) are clinically important muscle/nerves.

NB: Dig @ U.S. digit numbering convention: 1=thumb, 2=index finger, 3=middle, 4=ring, 5=little.

^bMajor innervation is indicated in boldface. Differing opinions exist, most shown are based on reference.³

^cTrunk (trunks of brachial plexus): S = superior, M = middle, I = inferior, all = all three.

^dCord (cords of brachial plexus): P = posterior, L = lateral, M = medial.

30.2.2 Thumb innervation/movement

See ► Table 30.6.

Flexion/extension: occurs in the plane of the palm.

Abduction/adduction: occur in a plane at right angles to palm.

Opposition: bringing the thumb across the hand.

Table 30.6 The 3 innervations of the thumb

Action	Nerve	Muscle(s)
abduction, flexion, opposition ^a	median	abductor pollicis brevis, flexor pollicis brevis, opponens pollicis
adduction	ulnar	adductor pollicis
extension	radial ^b	extensor pollicis brevis & longus

^aoccasional anomalous innervation by ulnar nerve

^bvia the posterior interosseous nerve

30.2.3 Muscles, roots, trunks, cords and nerves of the lower extremities

See ► Table 30.7.

Table 30.7 Muscle innervation—hip & lower extremity^a

Muscle	Action	Roots ^b	Plexus ^c	Nerves
• iliopsoas ^d	hip flexion	L1, 2, 3	L	femoral & L1, 2, 3
sartorius	hip flex & thigh evert	L2, 3		femoral
• quadriceps femoris	leg (knee) extension	L2, 3, 4	L	
pectineus	thigh adduction	L2, 3		obturator
• adductor longus		L2, 3, 4	L	
adductor brevis		L2–4		
adductor magnus		L3, 4		
gracilis		L2–4		
obturator externus	thigh adduction & lateral rotation	L3, 4		
• gluteus medius/minimus	thigh abduction & medial rotation	L4, 5, S1	S	superior gluteal
tensor fasciae lata	thigh flexion	L4, 5		
piriformis	lateral thigh rotation	L5, S1		
• gluteus maximus	thigh abduction (patient prone)	L5, S1, 2	S	inferior gluteal
obturator internus	lateral thigh rotation	L5, S1	S	muscular branches
gemelli		L4, 5, S1	S	
quadratus femoris		L4, 5, S1	S	
• biceps femoris ^e	leg flexion (& assist thigh extension)	L5, S1, 2		sciatic (trunk)
• semitendinosus ^e		L5, S1, 2		
• semimembranosus ^e		L5, S1, 2		
• tibialis anterior	foot dorsiflexion & supination	L4, 5 ^f	S	deep peroneal
• extensor digitorum longus	extension toes 2–5 & foot dorsiflexion	L5, S1		
• extensor hallucis longus (EHL) ^g	great toe extension & foot dorsiflexion	L5 ^f , S1	S	
• extensor digitorum brevis	extension great toe & toes 2–5	L5, S1	S	
• peroneus longus & brevis	P-flex pronated foot & eversion	L5, S1	L/S	superficial peroneal
• posterior tibialis	P-flex supinated foot & inversion	L4, 5	S	tibial
flexor digitorum longus	P-flex sup foot, flex terminal phalanx toes 2–5	L5, S1, 2		
flexor hallucis longus	P-flex sup foot, flex terminal phalanx great toe	L5, S1, 2		
flexor digitorum brevis	flex mid phalanx toes 2–5	S2, 3		
flexor hallucis brevis	flex proximal phalanx great toe	L5, S1, 2		
• gastrocnemius	knee flexion, ankle P-flex	S1, 2	S	
plantaris		S1, 2		
• soleus	ankle P-flex	S1, 2	S	

(continued)

Table 30.7 continued

Muscle	Action	Roots ^b	Plexus ^c	Nerves
• abductor hallucis ^h	(cannot test ^h)	S1, 2	S	
perineal & sphincters	voluntary contract pelvic floor	S2–4		pudendal

NB: items marked with a bullet (•) are clinically important muscle/nerves.

^aAbbreviations: P-flex = plantarflexion, D-flex = dorsiflexion.

^bMajor innervation is indicated in boldface type. e.g., when roots are shown as L4, 5, this indicates L5 is the main innervation, but both L4 & L5 contribute.

^cPlexus: L = lumbar, S = sacral

^dIliopsoas is the term for the combined iliacus and psoas major muscles

^e“hamstrings”: familiar term for the grouped: semitendinosus and semimembranosus (together, the medial hamstrings) and the biceps femoris (lateral hamstrings)

^falthough many references, including some venerable ones, cite AT as being primarily L4, many clinicians agree that L5 innervation is probably more significant

^gEHL is the best L5 muscle to test clinically (although S1 radiculopathy can also weaken this muscle)

^habductor hallucis cannot be tested clinically, but is important for EMG

30.3 Peripheral nerve injury/surgery

30.3.1 Nerve action potentials

Stimulating a healthy nerve fiber with an electrical stimulus of an amplitude and duration that exceeds its threshold will produce a conducted impulse, or nerve action potential (NAP).^{4(p 103)} Medium-sized axons (fibers) have a lower threshold than large ones, which have a lower threshold than small or fine axons.^{4(p 103)}

30.3.2 Use of NAP with lesion in continuity

There is some degree of continuity in ≥ 60% of nerve injuries.^{4(p 104)}

For a lesion in continuity (LIC), if surgical repair is needed, it may be too late if one waits until there is failure of anticipated clinical improvement. Presence of a NAP (regardless of amplitude or latency) distal to an LIC in the first few months after an injury usually indicates that operative intervention will not be needed. For recommended timing to obtain NAP recording, see ► Table 30.8.^{4(p 106)}

Table 30.8 Recommended timing to obtain NAP recording

Injury	Timing
relatively focal contusions	2–4 months
stretch injuries (esp. brachial plexus)	4–5 months
partial injuries & entrapments, compressive lesions and tumors	any time
to identify an area of conduction block (regardless if lesion is from neuropraxia, axonotmesis, or neurotmesis)	acutely

30.3.3 Timing of surgical repair

The longer the distance from the injury site to the functional unit to be reinnervated, the earlier surgical intervention should be considered.^{4(p 74)}

24 month rule^{4(p 74)}: after 24 months of denervation, most muscles cannot recover useful function even with reinnervation. Exceptions: facial muscles, large bulky muscles such as biceps, brachialis, gastrocs, and some lesions in continuity with some preserved innervation.

30.3.4 Brachial plexus

General information

Formed by *ventral* rami (the dorsal rami innervate the paraspinal muscles), most commonly of nerve roots C5–1 (schematically depicted in ► Fig. 30.1).

► Table 30.5 shows action, etc. of specific muscles. Also see ► Fig. 30.1. ↓ indicates that the nerve supplies the muscles listed; ↘ denotes a branch of the preceding nerve.

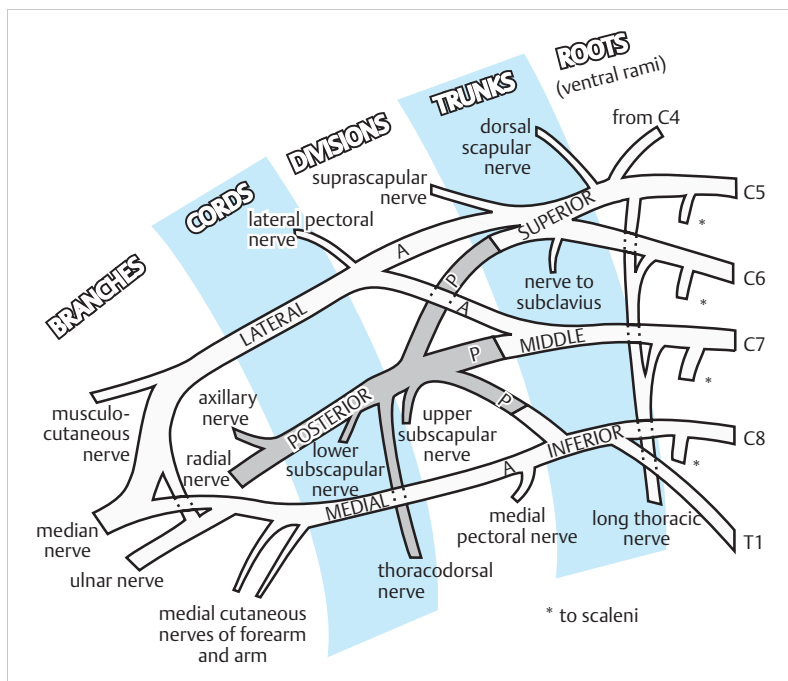


Fig. 30.1 Schematic diagram of the brachial plexus. (By permission: Churchill Livingstone, Edinburgh, 1973, R. Warwick & P. Williams: Gray's Anatomy 35th Edition © Longman Group UK Limited.)

Nerves arising from the brachial plexus

Radial nerve (C5–8)

See ► Fig. 30.2. The radial nerve (and its branches) innervate the extensors of arm and forearm:

- ↓ triceps (all 3 heads)
- ↓ anconeus
- ↓ brachioradialis
- ↓ extensor carpi radialis longus & brevis (latter originates ≈ at terminal branch)
- ↓ supinator (originates near the terminal branch)
- ↓ continues into forearm as **posterior interosseous nerve** (C7, C8)
 - ↓ extensor carpi ulnaris
 - ↓ extensor digitorum
 - ↓ extensor digiti minimi
 - ↓ extensor pollicis brevis & longus
 - ↓ abductor pollicis longus
 - ↓ extensor indicis

Axillary nerve (C5, C6)

See ► Fig. 30.2.

- ↓ teres minor
- ↓ deltoid

Median nerve (C5–1)

See ► Fig. 30.3, also see Martin-Gruber anastomosis (p.540).

- nothing in arm
- all forearm pronators and flexors except the two supplied by ulnar nerve

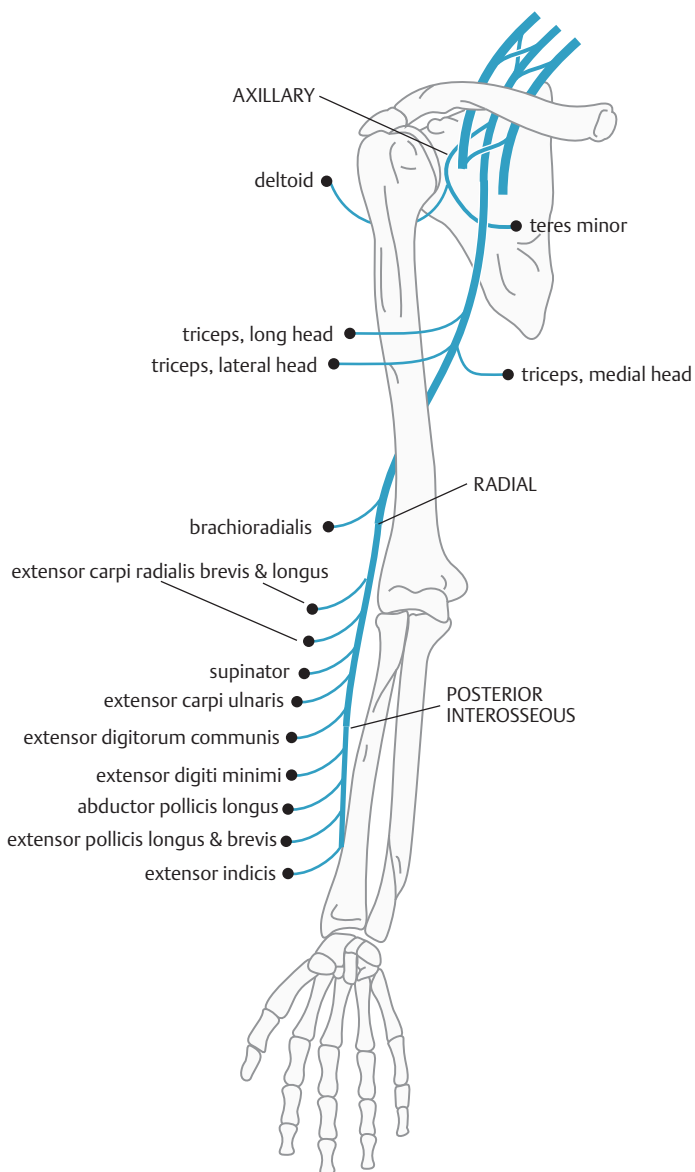


Fig. 30.2 Muscles of the radial and axillary nerves.

- ↓ pronator teres
 - ↓ flexor carpi radialis
 - ↓ palmaris longus
 - ↓ flexor digitorum superficialis
- in the hand ⇒ only the "LOAF muscles"
 - ↓ Lumbricals 1 & 2
 - ↓ Opponens pollicis
 - ↓ Abductor pollicis brevis
 - ↓ Flexor pollicis brevis (C8, T1)
- ↓ branch at or just distal to elbow anterior interosseous nerve (purely *motor*)
 - ↓ flexor digitorum profundus I & II
 - ↓ flexor pollicis longus
 - ↓ pronator quadratus

Ulnar nerve (C8, T1)

See ► Fig. 30.3.

- nothing in arm
- only 2 muscles in forearm:
 - ↓ flexor carpi ulnaris
 - ↓ half of flexor digitorum profundus (parts III & IV)
- all hand muscles excluding "LOAF" muscles (see above), viz.:
 - ↓ adductor pollicis
 - ↓ all interossei (4 dorsal & 3 palmar)
 - ↓ lumbricals 3 & 4
 - ↓ 3 hypothenar muscles: abductor, opponens & flexor digiti minimi
 - ↓ deep part of flexor pollicis brevis (by deep branch of ulnar nerve)
 - ↓ palmaris brevis (by the superficial branch of the ulnar nerve)

Musculocutaneous nerve (C5, C6)

Supplies arm flexors

- ↓ coracobrachialis
- ↓ biceps
- ↓ brachialis
- ↓ lateral cutaneous nerve of the forearm (terminal branch) supplies cutaneous sensation to radial aspect of forearm

Dorsal scapular nerve (C4, C5)

- ↓ rhomboids (major & minor)
- ↓ levator scapulae

Suprascapular nerve (C5, C6)

- ↓ Supraspinatus
- ↓ Infraspinatus

Subscapular nerve (C5–7)

- ↓ teres major
- ↓ subscapularis

Thoracodorsal nerve (C6, C7, C8)

- ↓ latissimus dorsi

Long thoracic nerve (C5–7)

Originates off of proximal nerve roots.

- ↓ serratus anterior (holds scapula to chest wall): lesion → winging of the scapula. To test: patient leans forward against wall with arms outstretched, scapula separates from posterior chest wall if the serratus anterior is not contracting. (note: this describes classic winging of the scapula. A variant of winging can occur with loss of trapezius muscle, e.g., with accessory nerve injury, and typically manifests when the patient pushes forward with the elbow held at the side of the thorax.)

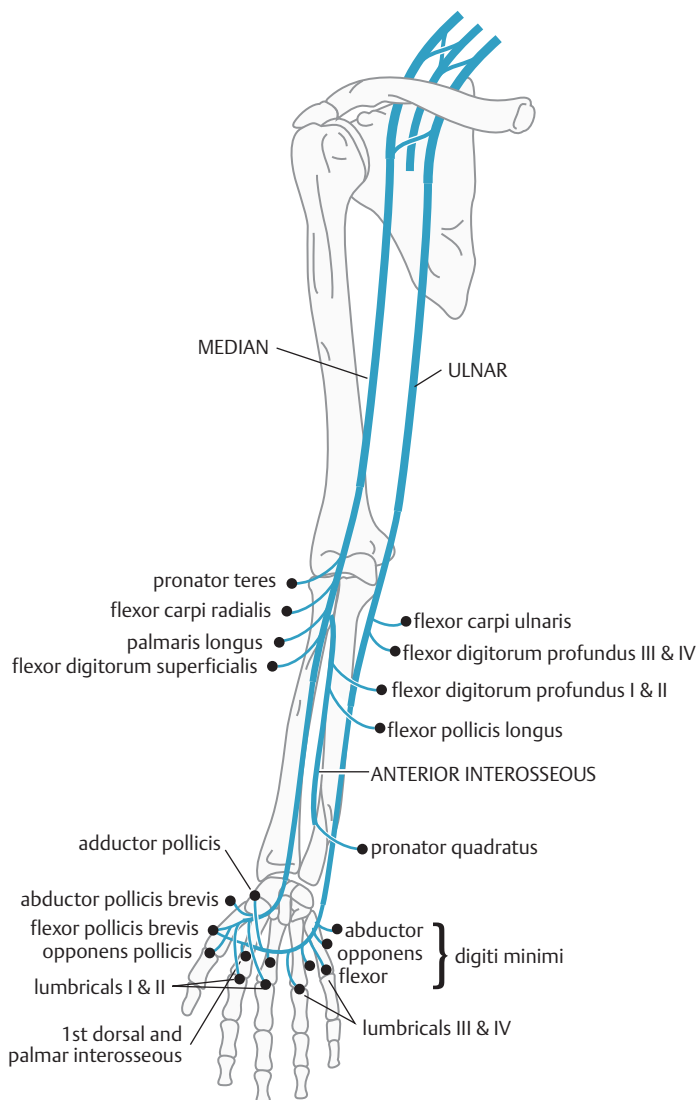


Fig. 30.3 Muscles of the median and ulnar nerves.

Anatomic variants

Martin-Gruber anastomosis

See reference.⁵

Anastomosis between median and ulnar nerves in the forearm found in 16 of 70 (23%) cadavers, bilateral in 3 (19%). Pattern I (90%): 1 anastomotic branch, Pattern II (10%) had 2.

Classification based on the origin from the median nerve: Type a (47.3%) from the branch to the superficial forearm flexor muscles, Type b (10.6%) from the common trunk, and Type c (31.6%) from the anterior interosseous nerve. Pattern II was a duplication of Type c (10.5%). The anastomotic branch was undivided in 15 cases, and divided into two branches in four cases. The anastomosis took an oblique angle or arched course to the ulnar nerve and passed superficial to the ulnar artery in four cases, deep to it in six, and in nine cases it was related to the anterior ulnar recurrent artery.⁵

Richie-Cannieu anastomosis

Motor connections from median to ulnar nerve at the palm. Found in 70% of patients.

References

- [1] Fernandez E, Pallini R, La Marca F, et al. Neurosurgery of the Peripheral Nervous System - Part I: Basic Anatomic Concepts. Surgical Neurology. 1996; 46:47-48
- [2] Dyck PJ, Boes CJ, Mulder D, et al. History of standard scoring, notation, and summation of neuromuscular signs. A current survey and recommendation. Journal of the Peripheral Nervous System. 2005; 10:158-173
- [3] Medical Research Council. Aids to the Examination of the Peripheral Nervous System. London: Her Majesty's Stationery Office; 1976
- [4] Kline DG, Hudson AR. Nerve Injuries: Operative Results for Major Nerve Injuries, Entrapments, and Tumors. Philadelphia: W. B. Saunders; 1995
- [5] Rodriguez-Niedenfuhr M, Vazquez T, Parkin I, et al. Martin-Gruber anastomosis revisited. Clin Anat. 2002; 15:129-134

31 Entrapment Neuropathies

31.1 Entrapment neuropathy – definitions and associations

Entrapment neuropathy is a peripheral nerve injury resulting from compression either by external forces or from nearby anatomic structures. Mechanism can vary from one or two significant compressive insults to many localized, repetitive mild compressions of a nerve. Certain nerves are particularly vulnerable at specific locations by virtue of being superficial, fixed in position, traversing a confined space, or in proximity to a joint. The most common symptom is pain (frequently at rest, more severe at night, often with retrograde radiation causing more proximal lesion to be suspected) with tenderness at the point of entrapment. Referred pain is so common that Frank Mayfield once said that patients with nerve entrapment don't know where the problem is located. Always consider the possibility of a systemic cause. Entrapment neuropathies may be associated with:

1. diabetes mellitus
2. hypothyroidism: due to glycogen deposition in Schwann cells
3. acromegaly
4. amyloidosis: primary or secondary (as in multiple myeloma)
5. carcinomatosis
6. polymyalgia rheumatica (p.206)
7. rheumatoid arthritis: 45% incidence of 1 or more entrapment neuropathies
8. gout

31.2 Mechanism of injury

Brief compression primarily affects myelinated fibers, and classically spares unmyelinated fibers (except in cases of severe acute compression). Acute compression compromises axoplasmic flow which can reduce membrane excitability. Chronic compression affects both myelinated and unmyelinated fibers and can produce segmental demyelination in the former, and if the insult persists, axolysis and wallerian degeneration will occur in both types. The issue of ischemia is more controversial.¹ Some contend that simultaneous venous stasis at the site of compression can produce ischemia, which can lead to edema outside the axonal sheath, which may further exacerbate the ischemia. Eventually, fibrosis, neuroma formation, and progressive neuropathy can occur.

31.3 Occipital nerve entrapment

31.3.1 General information

Greater occipital nerve (nerve of Arnold) is a sensory branch of C2 (► Fig. 112.1 for dermatome). Entrapment presents as occipital neuralgia: pain in occiput usually with a trigger point near the superior nuchal line. Pressure here reproduces pain radiating up along back of head towards vertex. More common in women.

31.3.2 Differential diagnosis

1. headache
 - a) may be mimicked by migraine headache
 - b) may be part of muscle contraction (tension) headache
2. myofascial pain²: the pain may be widely separated from the trigger point
3. vertebrobasilar disease including aneurysm and SAH
4. cervical spondylosis
5. pain from Chiari I malformation (p.295)

31.3.3 Possible causes of entrapment

1. trauma
 - a) direct trauma (including iatrogenic placement of suture through the nerve during surgical procedures, e.g., in closing a posterior fossa craniectomy)

- b) following traumatic cervical extension³ which may crush the C2 root and ganglion between the C1 arch and C2 lamina
- c) fractures of the upper cervical spine
2. atlanto-axial subluxation (AAS) (e.g., in rheumatoid arthritis) or arthrosis
3. entrapment by hypertrophic C1–2 (epistrophic) ligament⁴
4. neuromas
5. arthritis of the C2–3 zygapophyseal joint

31.3.4 Treatment

General information

Σ: Optimal treatment for idiopathic occipital neuralgia

For idiopathic occipital neuralgia: available evidence is from small, retrospective, case series studies and is insufficient to conclude that either local injection or surgery are effective. Nerve blocks with steroids and local anesthetics provide only temporary relief. Surgical procedures such as nerve root decompression or neurectomy may provide effective pain relief for some patients; however, patient-selection criteria for these procedures have not been defined, and recurrence is common.

In idiopathic cases with no neurologic deficit, the condition is usually self limited.

Non surgical treatment

1. greater occipital nerve block with local anesthetic and steroids (see below)
 - a) may provide relief typically lasting $\approx$ 1 month⁵
 - b) is no longer considered diagnostic because it is not sufficiently specific
2. physical therapy: massage and daily stretching exercises
3. TENS unit: provided $\geq$ 50% relief in 13 patients for up to 5 yrs⁶
4. oral antiinflammatory agents
5. centrally acting pain medications: Neurontin, Paxil, Elavil...
6. botulinum toxin injection⁷: although this study had quite a few placebo responders

If these measures do not provide permanent relief in *disabling* cases, surgical treatment may be considered, although caution is advised by many due to poor results.^{2,8} Alcohol neurolysis may be tried. A collar is *not* indicated as it may irritate the condition.

Occipital nerve block

Inject trigger point(s) if one or more can be identified (there is usually a trigger point near the superior nuchal line). The nerve may also be blocked at the point where it emerges from the dorsal neck muscles.

If the pathology is more proximal (e.g., at C2 spinal ganglion), then block of the ganglion may be required. Technique⁹ (done under fluoroscopy): shave hair below the mastoid process; prep with iodine; infiltrate with local; insert a 20 gauge spinal needle midway between C1 and C2, halfway between the midline and the lateral margin of the dorsal neck muscles. Aim rostrally, the final target is the midpoint of the C1–2 joint on AP fluoro, and almost but not touching the inferior articular process of C1. Infiltrate 1–3 ml of anesthetic and check for analgesia in the C2 distribution.

Surgical treatment

1. decompression of C2 nerve root if compressed between C1 and C2⁴
2. in cases of AAS, decompression and atlanto-axial fusion (p. 1778) may work

Surgical treatment options for *idiopathic* occipital neuralgia:

1. peripheral occipital nerve procedures: these may not be effective for proximal compression of the C2 root or ganglion:
 - a) occipital neurectomy (see below)
 - peripheral avulsion of the nerve

- avulsion of the greater occipital nerve as it exits between the transverse process of C2 and the inferior oblique muscle
- b) alcohol injection of greater occipital nerve
- 2. occipital nerve stimulators
- 3. release of the nerve within the trapezius muscle. Immediate results: relief in 46%, improvement in 36%. Only 56% reported improvement at 14.5 mos¹⁰
- 4. intradural division of the C2 dorsal root via a posterior intradural approach
- 5. ganglionectomy

Occipital neurectomy: The occipital nerve usually pierces the cervical muscles $\approx$ 2.5 cm lateral to the midline, just below theinion. Palpation or Doppler localization of the pulse of the accompanying greater occipital artery sometimes helps to locate the nerve. However, relief only occurs in $\approx$ 50%, and recurrence, usually within a year, is common.

31.4 Median nerve entrapment

31.4.1 General information

The two most common sites of entrapment of the median nerve:

1. at the wrist by the transverse carpal ligament: carpal tunnel syndrome (p.546)
2. in upper forearm by pronator teres muscle: pronator teres syndrome (p.548)

31.4.2 Anatomy

Contributing nerve roots: C5 through T1. The median nerve arises from the medial and lateral cords of the brachial plexus ($\blacktriangleright$ Fig. 30.1), and descends the upper arm adjacent to the lateral side of the brachial artery. It crosses to the medial side of the artery at the level of the coracobrachialis. In the cubital fossa, the median nerve passes behind the lacertus fibrosus (bicipital aponeurosis) and enters the upper forearm between the two heads of the pronator teres and supplies this muscle.

Just beyond this point, it branches to form the purely motor anterior interosseous nerve, which supplies all but 2 muscles of finger and wrist flexion. It descends adherent to deep surface of flexor digitorum superficialis (FDS), lying on the flexor digitorum profundus. Near the wrist, it emerges from the lateral edge of FDS becoming more superficial, lying medial to the tendon of flexor carpi radialis, just lateral to and partially under the cover of the palmaris longus tendon. It passes under the transverse carpal ligament (TCL) through the carpal tunnel, which also contains the tendons of the flexor digitorum profundus and superficialis deep to the nerve (9 tendons total, 2 to each finger, 1 to the thumb¹¹). The motor branch arises deep to the TCL, but may anomalously pierce the TCL. It supplies the "LOAF muscles" (Lumbricals 1 & 2, Opponens pollicis, and Abductor and Flexor pollicis brevis).

The TCL attaches medially to pisiform and hook of hamate, laterally to trapezium and tubercles of scaphoid. TCL is continuous proximally with fascia over FDS and antebrachial fascia, distally with the flexor retinaculum of the hand. The TCL extends distally into the palm to $\approx$ 3 cm beyond the distal wrist crease. The palmaris longus tendon, which is absent in 10% of population, partially attaches to the TCL.

Palmar cutaneous branch (PCB) of median nerve: arises from the radial aspect of the median nerve approximately 5.5 cm proximal to styloid process of the radius, underneath the cover of FDS of the middle finger. It crosses the wrist *above* the TCL to provide sensory innervation to the base of the thenar eminence (and is thus spared in carpal tunnel syndrome).

The sensory distribution of the *average* median nerve is shown in $\blacktriangleright$ Fig. 31.1.

31.4.3 Injuries to the main trunk of the median nerve

General information

Above the elbow, the median nerve may rarely be compressed by Struther's ligament (see below). At the elbow and forearm, the median nerve may rarely be trapped at any of three sites: 1) lacertus fibrosus (bicipital aponeurosis),¹² 2) pronator teres, 3) sublimis bridge. Neuropathy may also result from direct or indirect trauma or external pressure ("honeymoon paralysis").¹² Longstanding

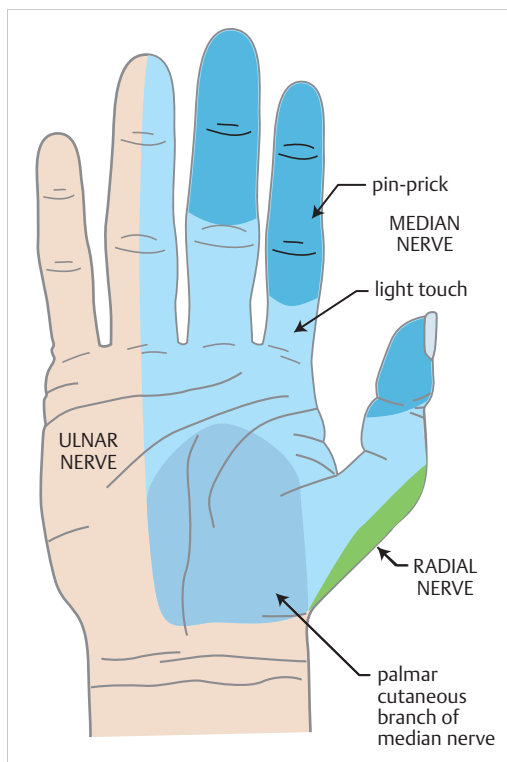


Fig. 31.1 Cutaneous sensory innervation of the hand illustrating the distribution of the median, ulnar & radial nerves in the hand, palmar surface.

Blue areas = median nerve (dark blue = pin-prick, light blue = light touch), green area = radial nerve, remainder = ulnar nerve.

compression of the main trunk of the median nerve produces a “benediction hand” when trying to make a fist (index finger extended, middle finger partially flexed; due to weakness of flexor digitorum profundus I & II).

Struther’s ligament

Distinct from struthers arcade (p.554) which is a normal finding. The supracondylar process (SCP) is an anatomical variant located 5–7 cm above medial epicondyle, present in 0.7–2.7% of population. Struther’s ligament bridges the SCP to the medial epicondyle. The median nerve and brachial artery pass underneath, the ulnar nerve may also. Usually asymptomatic, but occasionally may cause typical median nerve syndrome.

Pronator (teres) syndrome

From direct trauma or repeated pronation with tight hand-grip. Trapped where nerve dives between 2 heads of pronator teres. Causes vague aching and easy fatiguing of forearm muscles with weak grip and poorly localized paresthesias in index finger and thumb. Nocturnal exacerbation is *absent*. Pain in palm distinguishes this from carpal tunnel syndrome (CTS) since the median palmar cutaneous branch (PCB) exits before the TCL and is spared in CTS.

Treat with resting forearm. Surgical decompression indicated for cases that progress while on rest or when continued trauma is unavoidable.

Anterior interosseous neuropathy

General information

Key concepts

- weakness of 3 muscles: FDP I & II, FPL, & pronator quadratus. No sensory loss
- loss of flexion of the distal phalanges of the thumb and index finger (pinch sign)

The anterior interosseous nerve is a purely motor branch of the median nerve that arises in the upper forearm. Anterior interosseous neuropathy (AIN) produces no sensory loss and weakness of the 3 muscles supplied by the nerve:

1. flexor digitorum profundus (FDP) I & II: flexion of distal phalanx of digits 2 & 3
2. flexor pollicis longus (FPL): flexion of distal phalanx of thumb
3. pronator quadratus (in the distal forearm): difficult to isolate clinically

Etiologies of AIN

Include: idiopathic, amyotrophy, ulna/radius fractures, penetrating injuries, forearm lacerations.

Clinical

Symptoms: Patients complain of difficulty grasping small objects between the thumb and the index finger. Idiopathic cases may be preceded with forearm aching.

Physical exam: Sensory: *no* sensory loss.

Strength: digits 1, 2 & 3 are examined individually. The proximal interphalangeal joints are stabilized by the examiner and the patient is asked to flex the DIP. With AIN, there is no significant flexion of the DIP.

Pinch sign: the patient attempts to forcefully pinch the *tips* of the index finger and thumb as in making an “OK” sign (► Fig. 31.2, left); with AIN the terminal phalanges extend and the pulps touch instead of the tips¹³ (► Fig. 31.2, right).

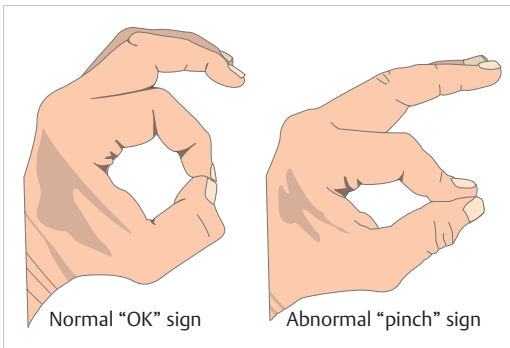


Fig. 31.2 “Pinch sign” seen with anterior interosseous neuropathy (AIN).

Diagnosis

In addition to the physical exam, EMG may be helpful.

EMG: primarily assesses pronator quadratus & flexor pollicis longus (FDP I & II is difficult on EMG because it has dual innervation with the ulnar nerve innervated portion being more superficial than the median nerve innervated portion). Important to evaluate pronator teres (abnormalities suggest involvement more proximal than forearm).

Management

In the absence of an identifiable cause of nerve injury, expectant management is recommended for 8–12 weeks, following which exploration is indicated, which may reveal a constricting band near the origin.

31.4.4 Carpal tunnel syndrome

General information

Key concepts

- the most common compression neuropathy. Involves median nerve in the wrist
- symptoms: tingling in the hand, worse at night and with elevation of hands
- physical exam is not very sensitive:
 - sensory: decreased pinprick in digits 1–3 and the radial half of 4
 - sensitivity: Tinels (tapping on wrist) 60%, Phalens (flexion of wrist) 80%
- electrodiagnostics: sensory latency @ wrist > 3.7 ms is the most sensitive test
- treatment:
 - mild cases: nonsurgical treatment (NSAIDs, neutral position splint...)
 - unresponsive or severe cases (neurologic deficits, duration > 1 year): surgical neurolysis (decompression) of the median nerve at the wrist has 70% satisfaction rate

Carpal tunnel syndrome (CTS) is the most common entrapment neuropathy in the upper extremity.^{14,15,16} Carpal tunnel release (CTR) is one of the most frequently performed hand procedures.^{17,18} The majority of patients have a satisfactory outcome from surgical treatment; see Outcome of surgical treatment (carpal tunnel release) (p.553). The median nerve is compressed within its course through the carpal tunnel just distal to the wrist crease. ► Table 31.1 shows the effect of pressure within the carpal tunnel.

Table 31.1 Pressure within carpal tunnel

Pressure (mm Hg)	Description
<20	normal
20–30	venular flow retarded
30	axonal transport impaired
40	sensory & motor dysfunction
60–80	blood flow ceases

Epidemiology

Usually occurs in middle aged patients. Ratio of female:male = 4:1. It is bilateral in over 50% of cases, but is usually worse in the dominant hand. The prevalence of CTS and UNE is increased in diabetics.

Common etiologies

See reference.¹⁹

In most cases, no specific etiology can be identified. CTS is very common in the geriatric population without any additional risk factors. The following etiologies tend to be more common in younger patients:

1. “classic” CTS: chronic time course, usually over a period of months to years
 - a) trauma: often job-related (may also be associated with avocations)
 - repetitive movements of hand and/or wrist: e.g., carpenter’s
 - repeated forceful grasping or pinching of tools or other objects
 - awkward positions of hand and/or wrist, including wrist extension, ulnar deviation, or especially forced wrist flexion
 - direct pressure over carpal tunnel
 - use of vibrating hand tools
 - b) systemic conditions: in addition to the listed systemic causes of entrapment neuropathies (p.541)—especially rheumatoid arthritis, diabetes—also consider:
 - obesity
 - local trauma
 - pregnancy: 54% remained symptomatic 1 year post-partum, and patients with onset early in pregnancy were less likely to improve²⁰

- mucopolysaccharidosis V
- tuberculous tenosynovitis
- multiple myeloma (p.928) (amyloid deposition in flexor retinaculum)
- the incidence of CTS may be increased in sarcoidosis (especially those with extrapulmonary involvement)²¹

c) local etiologies

- patients with A-V dialysis shunts in the forearm have an increased incidence of CTS, possibly on an ischemic basis (steal and/or venous stasis) or possibly from the underlying renal disorder
- carpal-metacarpal (CMC) osteoarthritis involving the thumb is common after age 40 (and is more prevalent in women), and in addition to the pain (with grasping) & tenderness at the base of the thumb, patients may develop secondary CTS

2. "acute" CTS: an uncommon condition where the symptoms of CTS appear suddenly and severely, usually following some type of exertion or trauma. Etiologies:

- a) median artery thrombosis: < 10% of individuals have a persistent median artery
- b) hemorrhage or hematoma in the transverse carpal ligament

Signs and symptoms

The physical exam for CTS is fairly insensitive. Signs and symptoms may include:

1. dysesthesias:

- a) characteristically patients are awakened at night by a painful numbness in the hand (often described as the "hand falling asleep") that often subjectively feels like a loss of circulation of blood. They often seek relief by: shaking the hand (flick sign²²) or dangling or swinging the hand, opening and closing or rubbing the fingers, running hot or cold water over the hand, slapping the hand on the thigh, or pacing the floor. It may radiate up the forearm, and occasionally as far as the shoulder

- b) daytime activities that characteristically elicit symptoms usually involve prolonged hand elevation: holding a book or newspaper or cell phone to read, driving a car, holding a telephone to the ear, brushing hair, shaving the face

c) distribution of symptoms:

- on palmar side in radial 3.5 fingers (palmar side of thumb, index finger, middle finger, and radial half of ring finger)
- dorsal side of these same fingers distal to the PIP joint
- radial half of palm
- subjective involvement of little finger occurs not infrequently for reasons that are not clear

2. hand weakness, especially grip. Characteristically manifests as difficulty opening jars. May be associated with thenar atrophy (late change, severe atrophy is seldom seen with current awareness of CTS by most physicians). An occasional patient may present with severe atrophy and no history of pain
3. clumsiness of the hand and/or difficulty with fine motor skills: probably due more to numbness than to a motor deficit. Often presents as difficulty buttoning buttons or zipping zippers, putting on earrings, fastening bra straps, handwriting changes...
4. hypesthesia in median nerve sensory distribution: usually best appreciated in finger *tips*, loss of 2-point discrimination may be more sensitive test
5. Phalen's test: 30–60 secs wrist flexion to a 90° angle exaggerates or reproduces pain or tingling. Positive in 80% of cases (80% sensitive)²³
6. Tinel's sign at the wrist: paresthesias or pain in median nerve distribution produced by gently percussing over the carpal tunnel. Positive in 60% of cases. May also be present in other conditions. Reverse Tinel's sign: produces symptoms radiating up the forearm for variable distance
7. ischemic testing: place blood pressure cuff proximal to wrist, inflation x 30–60 seconds may reproduce CTS pain

Differential diagnosis

Differential diagnosis includes (modified²⁴):

1. cervical radiculopathy: coexists in 70% of patients with either median or ulnar neuropathy (C6 radiculopathy may mimic CTS). Usually relieved by rest, and exacerbated by neck movement. Sensory impairment has dermatomal distribution. It has been postulated that cervical nerve root compression may interrupt axoplasmic flow and predispose the nerve to compressive injury distally (the term double-crush syndrome was coined to describe this²⁵), and although this has been challenged,²⁶ it has not been disproven
2. thoracic outlet syndrome (p.581): loss of bulk in hand muscles other than thenar. Sensory impairment in ulnar side of hand and forearm

3. pronator teres syndrome (p.544): more prominent palmar pain than with CTS (median palmar cutaneous branch does not pass through carpal tunnel)
4. de Quervain's syndrome: tenosynovitis of the abductor pollicis longus and extensor pollicis brevis tendons often caused by repetitive hand movements. Results in *pain and tenderness* in the wrist near the thumb. Onset in 25% of cases is during pregnancy, and many in 1st postpartum year. Usually responds to wrist splints and/or steroid injections. NCVs should be normal. Finkelstein's test: the thumb is passively abducted while thumb abductors are palpated, positive if this aggravates the pain²⁷
5. reflex sympathetic dystrophy: may respond to sympathetic block (p.1835)
6. tenosynovitis of any of the flexor ligaments: may occasionally be due to TB or fungus. Usually a long, indolent course. Fluid accumulation may be present

Diagnostic tests

Electrodiagnostics (EDX)

Electromyogram (EMG) and nerve conduction study (NCS) which includes measurement of nerve conduction velocities (NCV): may help confirm the diagnosis of CTS and distinguish it from cervical root abnormalities and from tendonitis.

CTS is predominantly a demyelinating injury, although it can progress to axonal loss.²⁸ Two sensory comparison techniques that clearly agree (either normal or abnormal) are adequate to confirm or refute the diagnosis. For borderline abnormalities, additional sensory comparison testing or the combined sensory index (CSI) can clarify the diagnosis. If sensory responses are absent, the median motor latency in comparison to the ulnar latency can help localize a focal abnormality.²⁹

Practice guideline: Electrodiagnostic criteria for CTS

The practice guideline for CTS recommends diagnostic examination strategies^{30,31,32}:

1. Standard: perform a median sensory nerve conduction study (NCS) across the wrist with conduction distance of 13 to 14 cm. If abnormal, compare to an adjacent sensory nerve in the symptomatic limb
2. Standard: if the initial median sensory NCS across the wrist is normal then additional comparison studies are recommended
3. Guideline: record motor NCS of the median nerve from the thenar muscle and of 1 other nerve in the symptomatic limb
4. Option: supplementary NCS
5. Option: needle electromyography (EMG) of cervical root screen muscles including a thenar muscle

NCV: Electrophysiologic studies support a diagnosis of carpal tunnel syndrome (CTS) using median nerve conduction studies across the transverse carpal ligament. Characteristic abnormalities: prolongation of sensory and motor distal latencies, slowing of the conduction velocity and decreased amplitudes of sensory and motor responses. Guidelines regarding recommended studies are published by the American Association of Neuromuscular and Electrodiagnostic Medicine (AANEM), AAPM&R, and AAN. Adhering to the guidelines, sensitivity is greater than 85% and specificity is greater than 95%.³⁰ Sensory latencies are more sensitive than motor. (Note: although up to 15% of cases may have normal electrodiagnostic studies, great reservation should be exercised in considering operating on CTS with normal sensory NCV and amplitude.)

Normal findings are shown in ► Table 31.2. Abnormal values also listed are a rough guide, but the correlation between severity of EDX findings and symptoms in CTS is not well-established.²⁹ Nevertheless, classification as follows can in part predict the outcome of carpal tunnel release (surgery), with normal and very severe NCS abnormalities having a worse prognosis than patients with moderate NCS abnormalities.^{29,33}

EDX interpretation (reports may also include the degree of slowing in the summary of findings or interpretation^{29,34,35}):

- mild: prolonged (relative or absolute) median nerve sensory latencies with normal motor studies. No evidence of axonal loss.
- moderate: prolonged (relative or absolute) median nerve sensory latencies with prolongation of motor distal latency. No evidence of axonal loss.
- severe: any of the aforementioned NCS abnormalities with evidence of axonal loss on EMG.

Table 31.2 Distal conduction latencies through *carpal tunnel*^a

Degree of involvement ^b	Sensory		Motor	
	latency ^c (mSec)	amplitude (mCV)	latency ^d (mSec)	amplitude (mV)
normal	<3.7	>25	<4.5	>4
mild ^b	3.7–4.0		4.4–6.9	
moderate ^b	4.1–5.0		7.0–9.9	
severe ^b	>5 or unobtainable		>10	

^aassumes normal proximal NCV

^bseverity does not reliably correlate with latency (see text)

^cto index finger. Sensory latency is measured to the peak of the waveform

^dto abductor pollicis brevis

Additional comparison studies for uncertain cases compare median nerve sensory conduction velocity to that of the ulnar nerve (or radial nerve): normal median nerve should be at least 4 m/sec faster than the ulnar; reversal of this pattern suggests median nerve injury. Alternatively, the sensory latencies for the palmar median and ulnar nerves can be compared; the median nerve latency should not be ≥ 0.3 mS longer than the ulnar.

EMG: normal in up to 31% of cases of CTS. In relatively advanced CTS, it may show increased polyphasicity, positive waves, fibrillation potentials, and decreased motor unit numbers on maximal voluntary thenar muscle contraction. EMG may detect cervical radiculopathy if motor involvement is present.

With severe “end stage” CTS, sensory and motor potentials may not be recordable, and EMG is not helpful in localizing (i.e., differentiating CTS from other etiologies).

Laboratory tests

Recommended in cases where an underlying peripheral neuropathy is suspected (e.g., a young individual with no risk factors such as repetitive hand use). This same protocol is a useful initial workup for any case of peripheral neuropathy:

1. thyroid hormone levels (T4 (total or free) & TSH): to R/O myxedema
2. CBC: anemia is common in multiple myeloma
3. electrolytes:
 - a) standard panel (Na, K, Cl, BUN, creatinine, glucose): to R/O chronic renal failure that could cause uremic neuropathy
 - b) HgA1c and blood glucose: R/O diabetes
4. vitamin B12, folate & MMA (methylmalonic acid) levels to R/O clinically significant vitamin B12 deficiency³⁶
5. in cases suspicious for multiple myeloma: (summary, see for details) (p.928)
 - a) 24 hour urine for kappa Bence-Jones protein (p.930)
 - b) bloodwork: SPEP with reflex IFE and FLC (p.930)
 - c) skeletal radiologic survey
6. light-chain assay: in patients with associated kidney failure to R/O light-chain deposition disease (LCDD) which differs from amyloid because light chain immunoglobulins are deposited in the absence of amyloid granules³⁷
7. hTTR: genetic testing for TTR gene mutations in suspected cases (peripheral neuropathy with cardiac and GI symptoms) to rule out hereditary TTR (hTTR) amyloidosis, an autosomal dominant hereditary form of amyloidosis where one of many possible mutations of the TTR (transthyretin) gene causes deposition of abnormal TTR (amyloid) which may produce CTS, often also associated with cardiac and GI symptoms beginning between ages 40–65 years³⁸

Imaging studies

Not routinely done unless a mass lesion is suspected.

Wrist MRI: very sensitive. Findings with CTS include flattening or swelling of the nerve, palmar bowing of the flexor retinaculum. May also demonstrate ganglion cysts, lipomas... Enhancement may occur with hypervascular edema.

Diagnostic ultrasound: faster and less expensive than MRI, and can assess blood flow and changes with different wrist positions. 18 MHz probes may improve images.

Management of CTS

Practice guideline: Management of CTS

American Association of Orthopedic Surgeons (AAOS) Clinical Practice Guideline endorsed by American Association of Neurological Surgeons, Congress of Neurological Surgeons, American Society of Plastic Surgeons, American Academy of PM&R and AANEM³⁹

1. a course of non-operative treatment is an option in patients diagnosed with CTS. Early surgery is an option when there is clinical evidence of median nerve denervation or the patient elects to proceed directly to surgical treatment (Grade C, Level V)
2. another non-operative treatment or surgery is suggested when the current treatment fails to resolve the symptoms within 2–7 weeks (Grade B, Level I and II)
3. there is not sufficient evidence to provide specific treatment recommendations for CTS when found in association with diabetes*, coexisting cervical radiculopathy, hypothyroidism, polyneuropathy, pregnancy, rheumatoid arthritis, and CTS in the workplace (inconclusive)
4. management specifics
 - local steroid injection or splinting is suggested when treating patients with CTS before considering surgery (Grade B, Level I and II)
 - oral steroids or ultrasound are options for treatment of CTS (Grade C, Level II)
 - carpal tunnel release is recommended for treatment of CTS (Grade A, level I)

* Notwithstanding the AAOS recommendations, multiple studies report that the results of carpal tunnel release in diabetics are good even when polyneuropathy is present.^{40,41}

Non-surgical management

Options include:

1. rest
2. medications: non-steroidal anti-inflammatory drugs (NSAIDs), diuretics, and pyridoxine (vitamin B6) have been studied with no evidence of efficacy¹¹
3. treatment of associated conditions (e.g., hypothyroidism or DM) is appropriate, but there is no data as to whether this relieves CTS¹¹
4. **neutral position splints**: alleviates symptoms in > 80% of patients⁴² (usually within a few days) and reduces prolonged sensory latencies.⁴³ Relapse is common (works best when patient does not return to heavy manual labor). A trial of at least 2–4 weeks is recommended
5. **steroid injection**: symptoms improve in > 75% of patients.¹¹ 33% relapse within 15 mos. Repeat injections are possible, but most clinicians limit to 3/year
 - a) use 10–25 mg hydrocortisone. *Avoid local anesthetics* (may mask symptoms of intra-neural injection)
 - b) inject into carpal tunnel (deep to transverse carpal ligament) to *ulnar side* of palmaris longus to avoid median nerve (in patients without palmaris longus, inject in line with fourth digit)
 - c) median nerve injuries have been reported with this technique,⁴⁴ primarily due to intra-neural injection (all steroids are neurotoxic upon intrafascicular injection, and so are some of the carrier agents)
 - d) risk factors for recurrence: severe electrodiagnostic abnormalities, constant numbness, impaired sensation, & weakness or atrophy of thenar muscles¹¹

Surgical treatment

General information

The operation is commonly called a carpal tunnel release (CTR), AKA neurolysis or neuroplasty of the median nerve at the wrist.

Indications

Surgical intervention is recommended for constant numbness, symptoms > 1 year duration, sensory loss, or thenar weakness/atrophy.¹¹ Surgical treatment of cases due to amyloidosis from multiple myeloma is also effective.

With bilateral CTS, in general one operates on the more *painful* hand first. However if the condition is severe in both hands (on EMG) and if it has progressed beyond the painful stage and is only

causing weakness and/or numbness, it may be best to operate on the “better” hand first in order to try and maximize recovery of the median nerve, at least on that side. Simultaneous bilateral procedures may also be done.⁴⁵ In severe cases, nerve recovery may not occur; it may be necessary to wait up to a year to determine extent of recovery.

Applied anatomy

The carpal tunnel is a space in the wrist bounded by the transverse carpal ligament (TCL) as the “roof,” and the “floor” composed of 4 carpal bones: trapezium, trapezoid, capitate and hamate. Through the carpal tunnel passes:

- 4 tendons of the flexor digitorum profundus
- 4 tendons of the flexor digitorum superficialis
- tendon of the flexor pollicis longus
- median nerve

The TCL is contiguous proximally with the deep investing fascia of the forearm, and distally with the flexor retinaculum of the palm of the hand. The fibers of the TCL are oriented transversely, compared to the longitudinal fibers of the flexor retinaculum. 4 points of attachment of the TCL to the carpal bones:

- 2 radial (lateral) points of attachment: scaphoid and trapezium
- 2 ulnar (medial) points of attachment: pisiform and hook of the hamate

Median nerve (MN): positioned slightly to the radial side of midline in the carpal tunnel. 2 branches (KEY: both usually run on the radial side of the MN):

- palmar cutaneous branch (PCB): sensory to skin of base of the thenar eminence (see ► Fig. 31.1) (roots: C6&7). Arises from the radial aspect of the MN proximal to the carpal tunnel and travels along the ulnar aspect of the flexor carpi radialis tendon superficial to the flexor retinaculum/TCL (i.e., passes over, not through, the carpal tunnel), ∴ *spared* in carpal tunnel syndrome
- recurrent motor branch: innervates the muscles of the thenar eminence. AKA thenar motor branch, AKA the “million dollar nerve” because injury to the nerve during carpal tunnel surgery will lead to loss of function of the thumb and a possible malpractice lawsuit. Arises from MN usually distal to the TCL (reports that it arises proximal to the TCL in 20% of cases seem exaggerated)

31

The radial artery and ulnar artery anastomose via a superficial and a deep palmar arch in the palm.

Kaplan's cardinal line: there are multiple variations of the definition of this line. For discussion here, it runs from the base of the thumb web space to the hook of the hamate (see ► Fig. 31.3). The superficial palmar arch, which is vulnerable during carpal tunnel surgery, is distal to this line.

Surgical techniques

A number of techniques are popular, including incision through palm of hand, transverse incision through wrist crease (with or without a retinaculotomy (e.g., Paine retinaculotomy)⁴⁶), and endoscopic techniques (using single or dual incisions). The efficacies of the various approaches have not been compared in an adequately powered randomized study¹¹ and there is no consensus on the superiority of any one technique.^{14,47,48,49} including endoscopic vs. open CTR.

Transpalmar approach (► Fig. 31.3): Magnification (e.g., operating loupes) is helpful.

Incision along an imaginary line extending proximally from the space between digits 3 and 4 (usually stay just to the ulnar side of the interthenar crease to avoid the PCB). The location of the median nerve may also be estimated by the palmaris longus tendon (stay slightly to ulnar side of tendon). Incision starts at distal wrist flexion crease, and the length depends on thickness of hand (it may extend as far distally as Kaplan's cardinal line (p.551)). Optionally: curve ulnarward at proximal wrist flexion crease (to facilitate retraction).

The flexor retinaculum flexor (longitudinally oriented fibers) is incised after the skin, and then the median nerve is carefully approached through the transversely oriented fibers of the TCL with progressively deepening incisions made e.g., with a 15 blade. All approaches to CTS surgery require complete division of the TCL at and distal to the wrist. If tendons of the flexor digitorum superficialis are encountered, you are too deep and need to back out and look more radially (toward the thumb) to find the nerve. In selected cases, the epineurium may be opened; however, internal neurolysis probably does more harm than good and in general should be avoided.

Close with absorbable 4–0 inverted sutures. Approximate skin edges with 4–0 nylon running or interrupted vertical mattress. Pad palm with several fluffs (opened dressing sponge). Cover with Kerlix®.

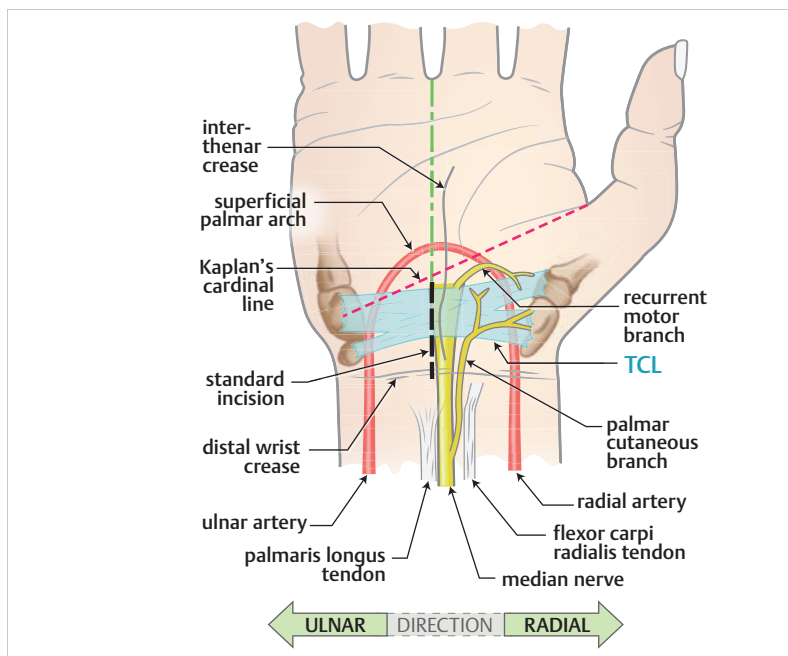


Fig. 31.3 Transpalmar incision for carpal tunnel syndrome surgery (right hand)—broken black line. Broken green line = trajectory of incision (parallel to interspace between digits #3 & 4). Broken red line = Kaplan's cardinal line. TCL = transverse carpal ligament (in blue). NB: the superficial palmar arch (artery) is illustrated, the deep palm arch is generally not at risk and is not shown.

Post-op: wrap hand with thumb exposed. Wrist elevation and rest is recommended for several days. Analgesics for mild-to-moderate pain (e.g., acetaminophen with codeine) for 3–4 days. Sutures are removed at 7–10 days. No heavy work with hand for 2–3 weeks.

Complications of carpal tunnel surgery

See reference.⁵⁰

1. pain due to neuroma formation following transection of **palmar cutaneous branch** (PCB) of the median nerve
 - a) branches of PCB may cross interthenar crease
 - b) avoid by using magnification and making incision slightly to ulnar side of interthenar crease
 - c) treated by ligating this branch where it originates from median nerve in forearm (results in small area of numbness at base of thenar eminence)
2. neuroma of dorsal sensory branch of radial nerve
 - a) caused by extending incision proximally and radially
 - b) may be treated by neurolysis of neuroma
3. injury to recurrent thenar (motor) branch of median nerve
 - a) anomaly may cause nerve to lie above or to pierce TCL
 - b) avoided by: staying to *ulnar side* of midline
4. direct injury to median nerve
5. volar displacement and entrapment of median nerve in healing edges of TCL
6. hypertrophic scar causing compression of median nerve
 - a) usually caused by incision crossing wrist perpendicular to flexion crease
 - b) avoid by not crossing flexion crease, or in cases where necessary (e.g., in releasing Guyon's canal entrapment of ulnar nerve, in tenosynovectomy for rheumatoid arthritis, or in dealing with an anomalous superficialis or palmaris muscle) by crossing wrist obliquely at 45° angle directed toward ulnar side⁵⁰ (see optional extension line in ► Fig. 31.3)

7. failure to improve symptoms
 - a) incorrect diagnosis: if EMG or NCV not done pre-op, they should be done after surgical failure (to R/O e.g., cervical root involvement [look for posterior myotome involvement], or generalized peripheral neuropathy)
 - b) incomplete transection of TCL: the most common cause for failure if diagnosis is correct (also possibility of accessory ligament or fascial band proximal to TCL in cases where division was complete). When this is identified on re-exploration, 75% of patients will be cured or improved after division is completed
8. joint stiffness: caused by excessively long immobilization of wrist and fingers
9. injury to superficial palmar arch (arterial): usually results from “blind” distal division of TCL
10. bowstringing of flexor tendons
11. complex regional pain syndrome AKA reflex sympathetic dystrophy: exact incidence is unknown, reported in 4 of 132 patients in one series (probably too high, most surgeons will see only one or two cases in their career). Treatment with IV phentolamine has been suggested, but most cases are self limited after $\approx$ 2 weeks
12. infection: usually causes exquisite tenderness
13. hematoma: also usually quite painful and tender

Outcome of surgical treatment (carpal tunnel release)

75–90% of patients have symptom resolution or are improved to a satisfactory state following carpal tunnel release.^{17,47,51} Clinical improvement peaks at 6 months post-op,^{52,53,54} although paresthesias may take $\geq$ 9 months to resolve.^{47,55,56,57} 70–90% are free of nocturnal pain.^{58,59}

Diabetic patients: the results of decompression are good in diabetics with CTS even when a generalized peripheral neuropathy is present.⁶⁰ In contrast, ulnar neuropathy at the elbow is often poorly responsive in diabetics (p.557).⁴⁰

Managing surgical treatment failures

Less than satisfactory outcomes following CTR should be categorized as:

1. new symptoms: may include neuropathic pain out of proportion to the surgery, new areas of numbness/paresthesias or marked weakness of the thenar muscles.⁶¹ When present immediately post-op, this suggests iatrogenic injury to branches of the median nerve
2. persistent symptoms (primary failure or failure to improve) defined as symptoms that remain unchanged compared to preop. Etiologies include incorrect initial diagnosis, incomplete release of the transverse carpal ligament, and severe (i.e., irreversible) CTS at initial diagnosis.¹⁴
3. or recurrent symptoms: requires symptom-free interval before return of symptoms (there is no standardization of either the level of recurrent symptoms or length of interval,⁴⁷ although 6 months has been used in some studies⁶¹). Etiologies include circumferential fibrosis around the median nerve, soft tissue adhesions, synovial proliferation, tenosynovitis, ganglions, amyloid deposits, and subtle palmar subluxation.^{14,61}

► **Electrodiagnostic (EDX) studies.** Following CTR, distal motor latency improves after 3 months and 6 months and may continue to improve for up to 2 years.^{53,62,63} Electrophysiologic abnormalities may improve but may not return to normal range after CTR even with clinical improvement.^{47,54} Electrophysiologic studies are most helpful when it is possible to compare to pre-op studies.^{14,17,61,64,65} There are no guidelines or standard recommendations for when to obtain postoperative studies for surgical failures. It is reasonable to obtain repeat studies at 3 to 6 months following carpal tunnel release for persistent symptoms and at onset for new or recurrent symptoms. If the repeat nerve conduction studies are worse or if the EMG needle exam has findings of denervation (fibrillation potentials and positive sharp waves) not previously present, then repeat surgery is indicated.^{17,61} If the preoperative study is not available, repeating the EDX study with comparison at 2 points in time to evaluate for improvement or worsening is advised. Prolonged latencies alone are not an indication for reoperation.⁶⁶

31.5 Ulnar nerve entrapment

31.5.1 General information

Ulnar nerve has components of C7, C8, and T1 nerve roots. Even though this is the second most prevalent entrapment neuropathy after CTS, it is still relatively uncommon. Potential sites of compression:

1. above elbow: possibly by the arcade of Struthers

- at the elbow: retroepicondylar groove ("ulnar groove"): between the medial epicondyle and the olecranon process. Compression by fascia or by dynamic compression or repetitive trauma. This is also the location of the "funny bone" where the nerve can be acutely impacted to produce transient numbness and tingling in the 2 little fingers
- cubital tunnel: just distal to the ulnar groove, under the aponeurosis spanning the heads of the flexor carpi ulnaris (FCU) known as Osborne's ligament⁶⁷ or cubital tunnel retinaculum
- at the point of exit from the FCU
- wrist: Guyon's canal

Etiologies: structural, mechanical or idiopathic.

Motor findings include:

- wasting of the interossei may occur, and is most evident in the first dorsal interosseous (in the thumb web space)
- Wartenberg's sign: one of the earliest findings of ulnar nerve entrapment (abducted little finger due to weakness of the third palmar interosseous muscle—patient may complain that the little finger doesn't make it in when they reach into their pocket)
- Froment's prehensile thumb sign: grasping a sheet of paper between thumb and the extended index finger results in extension of the proximal phalanx of the thumb and flexion of the distal phalanx as a result of substituting flexor pollicis longus (which is spared since it is innervated by anterior interosseous nerve) for the weak adductor pollicis^{68(p 18)}
- claw deformity of the hand (main en griffe): in severe ulnar nerve injuries on attempted finger extension (some have called this "benediction hand," which differs from that with the same name in median nerve injury where the named sign occurs on trying to make a fist. Fingers 4 and 5 and to a lesser extent 3 are hyperextended at the MCP joints (extensor digitorum is unopposed by interossei and "ulnar" lumbricals III & IV) and flexed at the interphalangeal joints (due to pull of long flexor muscles). NB: C8 radiculopathy can also cause benediction sign⁶⁹)

Sensory findings. Disturbance of sensation involving:

- the little finger and ulnar half of the ring finger (► Fig. 31.1)
- sensory loss over the ulnar side of the dorsum of the hand. This will be spared in ulnar nerve entrapment at the wrist (dorsal ulnar cutaneous nerve branches proximal to the wrist)

31.5.2 Injury above elbow

May occur with injury to the medial cord of the brachial plexus.

In the upper arm, the ulnar nerve descends anterior to the medial head of triceps; in 70% of people it passes under arcade of Struthers—distinct from Struther's ligament (p. 544)—a flat, thin, aponeurotic band. This is not normally a point of entrapment, but may cause kinking after ulnar nerve transposition if not adequately divided.^{70(p 1781)}

31.5.3 Ulnar nerve entrapment at elbow (UNE)

General information

Entrapment at or just distal to the elbow produces the cubital tunnel syndrome. (Technically, the cubital tunnel is formed by the fibrous arch between the two heads of the FCU,^{71(p 877)} the proximal entrance to which is just distal to the retrocondylar groove. However, common vernacular usually includes entrapment within the groove itself as being "cubital tunnel syndrome.")

Can also present as so-called tardy ulnar palsy because of delayed onset following bony injury at the elbow, initial case reports occurred ≥ 12 years later, with the majority commencing > 10 years following the original injury. The elbow is the most vulnerable point of the ulnar nerve: here the nerve is superficial, fixed, and crosses a joint. Most cases are idiopathic, although there may be a history of elbow fracture (especially lateral condyle of the humerus, with associated cubitus valgus deformity), dislocation, arthritis, or repeated minor trauma. The aponeurotic arch extending over the ulnar groove and attaching on the medial epicondyle may become thickened and can compress the nerve, especially with elbow flexion.^{71(p 884)} The ulnar nerve may also be injured during anesthesia (p. 575).⁷² In contrast to CTS, which is predominantly demyelinating, UNE has more axonal loss even when chronic.³³

Presentation

Typically presents with discomfort (pain, numbness and/or tingling) in little finger and ulnar half of ring finger, elbow pain, and hand weakness. Early symptoms may be purely motor (see Froment's

sign and claw deformity above), unlike the median nerve where sensory involvement is almost always present. Symptoms may be exacerbated by the cold, and are often somewhat vague and may be described as a loss of finger coordination or clumsiness. Cramping and easy fatiguing of the ulnar innervated muscles of the hand may occur. Pain may *not* be a significant feature, but if present tends to be aching in nature along the ulnar aspect of the elbow or forearm. Atrophy of interossei is common by the time of presentation.

The ulnar nerve is usually tender and may be palpably enlarged in the ulnar groove. Tinel's sign may be positive over the elbow, but this is not very specific.

Grading: the classification system proposed by Stewart⁷³ is shown in ► Table 31.3.

Table 31.3 Stewart classification system for severity of ulnar nerve injury (Stewart⁷³ after Bartels⁷⁴)

Grade	Description
1 (mild)	sensory symptoms ± motor symptoms; ± sensory loss; no muscle atrophy or weakness
2 (moderate)	sensory symptoms with detectable sensory loss. Mild atrophy; 4 or 4+ muscle strength
3 (severe)	usually constant sensory symptoms with detectable sensory loss. Moderate to marked atrophy; 4 - or less muscle strength.

Evaluation

Electrodiagnostic studies

The literature review from AANEM Practice Parameters for EDX studies in ulnar neuropathy at the elbow (UNE) reports sensitivities to range from 37% to 86% with specificities of 95%.^{30,31,32}

EDX interpretation: Reports should include localization and may comment on whether the lesion is predominantly demyelinating or axonal. Reports may use grading classification.⁷⁵

The following suggest a focal lesion involving the ulnar nerve at the elbow. Multiple internally consistent abnormalities are more convincing than isolated abnormalities. These are listed in order of strength of evidence.

Sensory abnormalities of the distal sensory or mixed nerve action potential (NAP), especially loss of amplitude, are not localizing for ulnar neuropathy (in contradistinction to median nerve/carpal tunnel), the motor component of the exam for the ulnar nerve is more useful for localization of the site of entrapment.

31

Practice guideline: Electrodiagnostic criteria for ulnar neuropathy at the elbow

Not all criteria need to be present and an EMG needle exam is not required^{31,76}:

1. absolute motor nerve conduction velocity (NCV) < 50 m/sec from below elbow (BE) to above elbow (AE)
2. drop of NCV > 10 m/sec, comparing BE to wrist segment to the AE to BE segment
3. amplitude of compound motor action potential (CMAP) normally decreases with distance, but a drop > 20% from BE to AE is abnormal (in the absence of anomalous innervation, e.g., Martin-Gruber anastomosis (p.540))
4. if ulnar motor studies are inconclusive (with stimulation at the wrist, above the elbow and below the elbow, while recording from ADQ) the following may be of benefit:
 - NCS recorded from the FDI (first dorsal interosseus muscle, innervated by ulnar nerve)
 - inching study recording latencies and amplitudes at cm increments.
 - with severe UNE and wallerian degeneration, comparison of AE to BE with axilla to AE
 - needle EMG should always include FDI and ulnar innervated forearm muscles (flexor digitorum profundus (FDP) to ring or little finger and/or abductor digiti quinti/minimi [ADQ]). If abnormal then extend to include non-ulnar innervated C8, medial cord, lower trunk muscles and cervical paraspinals to exclude brachial plexopathy/cervical radiculopathy.

The 2 most important parameters that predict a good outcome are preserved compound muscle action potential (CMAP) amplitudes in ulnar hand muscles and conduction block (CB) with slowed conduction velocity across the elbow, which is consistent with demyelination and has a better prognosis.^{28,77} Poor prognosis correlates with small or absent CMAP and no CB consistent with axonal loss.²⁸

Diagnostic ultrasound

Localizing ulnar nerve lesions with electrodiagnostic studies can be difficult. There has been a recent renewed interest in diagnostic ultrasound using high frequency (18 MHz) probes to help with localization, and also for identification of pathology, including nerve swelling, transection,⁷⁸ and neuroma, that exceeds MRI in some aspects and at a lower cost and with faster acquisition time.

Management of ulnar neuropathy at the elbow

There are no Clinical Practice Guidelines for treatment of UNE endorsed by AANEM, AAOS, CNS, AANS, AAPM&R, or Am Society of Plastic Surgeons. A fundamental difficulty in treating UNE are the multiple etiologies and locations so that the natural history and responses to treatment vary widely. The primary management decision is conservative versus surgical treatment.

A Cochrane Database Review concluded the available evidence is not sufficient to identify the best treatment for UNE on the basis of clinical, neurophysiologic and imaging characteristics.^{79,80}

Nonsurgical treatment (see below) may be considered for the patient with intermittent symptoms, no atrophy and mild EDX findings. Surgical intervention has been advised for patients who fail conservative management, although the best nonsurgical management and duration of treatment are not well defined.⁸⁰

Suggested protocol:

- mild or moderate UNE (grade 1 and 2, ► Table 31.3): treat conservatively as studies report improvement or complete recovery in 30–90%.^{33,73} Follow clinically every 2 months to detect deterioration. If worsening occurs, image with CT or MRI. Surgical exploration is indicated regardless of imaging results
- severe (grade 3) UNE: initiate conservative treatment, obtain imaging and f/u in 1 month. If there is worsening or if a structural abnormality is found, or both, then proceed with surgical intervention. If stable or improving and imaging is normal, then follow clinically at monthly intervals. Surgical intervention for worsening.⁷³

There is a probable increased prevalence of UNE in diabetics. The UNE is often more severe with predominant axonal injury, and these patients do not respond well to surgery.⁴⁰

Non-surgical treatment

Patient education on positions to avoid (prolonged elbow bending to $\geq 90^\circ$ flexion). Avoid trauma to the elbow, including resting it on firm surfaces (tables, rigid armrests in motor vehicles...); a soft elbow gel-pad may help. Results are often better when definite traumatic etiology can be identified and eliminated.

Surgical treatment

General approach

Most operations utilize a "lazy omega" skin incision centered over the medial epicondyle, extending at least ≈ 6 cm proximal and distal to the elbow with the central "hump" directed anteriorly. The ulnar nerve is most constant and therefore most easily found immediately at the entrance to the ulnar groove. It may then be followed proximally and distally. Nerve branches that should be preserved include: posterior branches of the medial antebrachial cutaneous nerve (or else numbness or dysesthesias along medial forearm may occur) and branches to the flexor carpi ulnaris (which may branch early). Small articular branches at or proximal to the elbow joint can be preserved with simple decompression but may need to be sacrificed in transposition if they cannot be dissected far enough along the ulnar nerve. Internal neurolysis should be avoided as it may promote intraneural fibrosis.

The choice of one of the options below will determine subsequent steps.

Surgical options primarily consist of:

1. simple nerve decompression without transposition⁸¹ (see below). Includes all of the following:
 - a) at the elbow: division of the cubital tunnel retinaculum
 - b) distal to the elbow: dividing the aponeurosis connecting the two heads of the flexor carpi ulnaris, some advocate resuturing the aponeurosis underneath the nerve
 - c) proximal to the elbow: dividing the medial intermuscular septum (between distal biceps and triceps muscles) and the arcade of Struthers (if present)
 - d) preservation of the branch to the flexor carpi ulnaris and the dorsal cutaneous branch to the hand (arises 5 cm proximal to wrist)

2. nerve decompression and transposition (extent of surgery differs because degree of entrapment varies; all forms of transposition require fashioning a sling to retain the nerve in its new location). Transposition may be to:
 - a) subcutaneous tissue: this leaves the nerve fairly superficial and vulnerable to further trauma
 - b) within the flexor carpi ulnaris muscle (intramuscular transposition): some contend this actually worsens the condition due to intramuscular fibrosis
 - c) a submuscular position: see below
3. medial epicondylectomy. Usually combined with decompression. Normally reserved for patients with a bony deformity
4. sometimes excision of neuroma and possibly jump graft may be required

Submuscular transposition

Placement under pronator teres, within a groove fashioned in the flexor carpi ulnaris (FCU). Usually requires general anesthesia (endotracheal or laryngeal mask airway).

Some key concepts⁸² (p 247,260–5),⁸³:

1. the skin incision must extend at least ≈ 8 cm distal and proximal to the medial epicondyle to mobilize the nerve (spare the medial antecubital cutaneous nerve in the subcutaneous fatty tissue just distal to the elbow)
2. the nerve is mobilized, sparing branches to flexor carpi ulnaris (FCU) and the ulnar flexor profundus branch(es) (usually arise 2–4 cm distal to olecranon)
3. the medial intermuscular septum (between distal biceps and triceps muscles) must be cut in the distal arm to prevent the nerve from being kinked over it
4. the pronator teres muscle must be sectioned completely through just distal to the medial epicondyle
 - a) start by undermining the muscle just distal to the medial epicondyle
 - b) a mosquito hemostat may be passed under the muscle to assist
 - c) the muscle is cut sharply, leaving a cuff to reattach it
5. a trough is cut in the volar aspect of the FCU to accommodate the nerve
6. after the pronator teres is reattached over the nerve, make sure that the nerve can slide back and forth easily under the muscle
7. test the elbow through a range of motion after the transposition to look for snapping of the medial portion of the triceps over the medial epicondyle⁸⁴

31

Transposition vs. decompression

Σ: Ulnar transposition vs. simple decompression

For most cases, simple decompression is recommended over transposition. Possible exceptions include: bony deformity, nerve subluxation

Randomized studies have shown similar success but lower complication rate with simple decompression vs. transposition.^{79,85,86} Advantages of simple decompression include^{74,87} shorter operation that can be done more easily under local anesthesia, avoidance of nerve kinking and muscular fibrosis around the transposed nerve, reduced risk of wound infection⁷⁹ and scar formation,⁷³ and preservation of cutaneous branches, ulnar branches, and nourishing blood vessels (vasa nervorum⁷³) that are sometimes sacrificed with transposition, rendering portions of the nerve ischemic.

Arguments against simple decompression: continued dynamic compression with elbow flexion, possible nerve subluxation (if present pre-op, simple decompression may make this worse; to avoid nerve subluxation and loss of vascular supply with simple decompression, avoid a 360° freeing of the nerve), and incomplete release of pressure points.

Results with surgery

Not as good as with CTS, possibly due in part to the fact that patients tend to present much later. Overall, a good to excellent result is obtained in 60%, a fair result in 25%, and a poor result (no improvement or worsening) in 15%.⁸⁸ (p 2530) These results may be worse in patients with symptoms present > 1 year, with only 30% of these symptomatically improved in one series.⁸¹ Pain and sensory changes respond better than muscle weakness and atrophy.

Patients with comorbidities: lower success rate is seen in older patients and those with certain medical conditions (diabetes, alcoholism...). Ulnar neuropathy at the elbow is often severe in diabetics and predominantly motor with axonal injury and usually does not respond well to surgery.⁴⁰

31.5.4 Entrapment in the forearm

Very rare. Just distal to the elbow, the ulnar nerve exits the retroepicondylar groove to pass under the fascial band (Osborne's ligament) connecting the two heads of the flexor carpi ulnaris (FCU), superficial to the flexor superficialis and pronator teres. Findings with entrapment in the forearm are similar to tardy ulnar nerve palsy (see above).

Surgical treatment consists of steps outlined for nerve distal to the elbow in ulnar nerve decompression (see above). A technique for locating the course of the ulnar nerve distal to the elbow: the surgeon takes the little finger of his/her own hand (using the hand contralateral to the patient's side that is being decompressed) and places the proximal phalanx in the ulnar groove aiming it toward the ulnar side of the wrist.^{82 (p 262)}

31.5.5 Entrapment in the wrist or hand

At the wrist, the terminal ulnar nerve enters Guyon's canal, the roof of which is the palmar fascia and palmaris brevis, and the floor of which is the flexor retinaculum of the palm and the pisohamate ligament.

► **Guyon's canal.** Is *superficial* to the transverse carpal ligament (which overlies the carpal tunnel and compresses the median nerve in carpal tunnel syndrome).

The canal contains no tendons, only the ulnar nerve and artery. At the middle of the canal the nerve divides into a deep and superficial branch. The superficial branch is mostly sensory (except for the branch to palmaris brevis), and supplies hypothenar eminence and ulnar half of ring finger. The deep (muscular) branch innervates hypothenar muscles, lumbricals 3 & 4, and all interossei. Occasionally the abductor digiti minimi branch arises from the main trunk or superficial branch.

Shea and McClain⁸⁹ divided lesions of the ulnar nerve in Guyon's canal into 3 types shown in ► Table 31.4. Injury to the distal motor branch can also occur in the palm and produces findings similar to a Type II injury.

31

Table 31.4 Types of ulnar nerve lesions in Guyon's canal

Type	Location of compression	Weakness	Sensory deficit
Type I	just proximal to or within Guyon's canal	all intrinsic hand muscles innervated by ulnar n.	palmar ulnar distribution ^a
Type II	along deep branch	muscles innervated by deep branch ^b	none
Type III	distal end of Guyon's canal	none	palmar ulnar distribution ^a

^apalmar ulnar distribution: the hypothenar eminence and ulnar half of ring finger, both on the palmar surface only (the dorsum is innervated by the dorsal cutaneous nerve)

^bdepending on the location, may spare hypothenar muscles

Injury is most often due to a ganglion of the wrist,⁹⁰ but also may be due to trauma (use of pneumatic drill, pliers, repetitively slamming a stapler, leaning on palm while riding bicycle). Symptoms are similar to those of ulnar nerve involvement at the elbow, except there will never be sensory loss in the *dorsum* of the hand in the ulnar nerve territory because the dorsal cutaneous branch leaves the nerve in the forearm 5–8 cm proximal to the wrist (sparing of flexor carpi ulnaris and flexor digitorum profundus III & IV is not helpful in localizing because these are so rarely involved even in proximal lesions). Electrodiagnostics are usually helpful in localizing the site of the lesion. Pain, when present, may be exacerbated by tapping over pisiform (Tinel's sign). It may also radiate up the forearm.

Surgical decompression may be indicated in refractory cases. To locate: find the ulnar artery, and the nerve is on the ulnar side of the artery. Controversial whether simple decompression vs. subcutaneous transposition is best; the outcome is similar but there may be more complications in the transposition group,^{85,86} but studies are small.

31.6 Radial nerve injuries

See reference.^{91 (p 1443–45)}

31.6.1 Applied anatomy

The radial nerve arises from the posterior divisions of the 3 trunks of the brachial plexus (► Fig. 30.1). It receives contributions from C5 to C8. The nerve winds laterally along the spiral groove of the humerus where it is vulnerable to compression or injury from fracture.

Distinguish radial nerve injury from injury of posterior cord of brachial plexus by sparing of deltoid (axillary nerve) and latissimus dorsi (thoracodorsal nerve).

31.6.2 Axillary compression

Etiologies: crutch misuse; poor arm position during (drunken) sleep.

31.6.3 Mid-upper arm compression

Etiologies

1. "Saturday night palsy": improper positioning of arm in sleep (especially when drunk and therefore less likely to self-reposition in response to the accompanying discomfort, e.g., due to a bed-mate's head resting on the arm)
2. from positioning under general anesthesia
3. from callus due to old humeral fracture

Clinical

Weakness of wrist extensors (wrist drop) and finger extensors. ★ Key: *triceps* is normal because takeoff of nerve to triceps is proximal to spiral groove. Involvement of distal nerve is variable, may include thumb extensor palsy and paresthesias in radial nerve distribution.

Differential diagnosis

1. isolated wrist and finger extensor weakness can also occur in lead poisoning (usually bilateral, more common in adults)
2. C7 radiculopathy: triceps will be weak

31.6.4 Forearm compression

General information

The radial nerve enters the anterior compartment of the arm just above the elbow. It gives off branches to brachialis, brachioradialis, and extensor carpi radialis (ECR) longus before dividing into the posterior interosseous nerve and the superficial radial nerve. The posterior interosseous nerve dives into the supinator muscle through a fibrous band known as the arcade of Fröhse.

Posterior interosseous neuropathy

Posterior interosseous neuropathy (often referred to by its acronym "PIN") may result from lipomas, ganglia, fibromas, rheumatoid arthritis changes at elbow, entrapment at the arcade of Fröhse (rare), and occasionally from strenuous use of the arm.

Treatment: cases that do not respond to 4–8 weeks of expectant management should be explored, and any constrictions lysed (including arcade of Fröhse).

Radial tunnel syndrome

AKA supinator syndrome. Controversial. The "radial tunnel" extends from just above the elbow to just distal to it, and is composed of different structures (muscles, fibrous bands...) depending on the level.⁹² It contains the radial nerve and its two main branches (posterior interosseous and superficial radial nerves). Repeated forceful supination or pronation or inflammation of supinator muscle attachments (as in tennis elbow) may traumatize the nerve (sometimes by ECR brevis). Characteristic finding: pain in the region of the common extensor origin at the lateral epicondyle on resisted extension of the middle finger which tightens the ECR brevis. May be mistakenly diagnosed as resistant "tennis elbow" (lateral epicondylitis must be excluded). There may also be paresthesias in the distribution of the superficial radial nerve and local tenderness along the radial nerve anterior to the

radial head. Even though the site of entrapment is similar to PIN, unlike PIN, there is usually no muscle weakness. Surgery: rarely required, consists of nerve decompression.⁹²

31.7 Injury in the hand

The distal cutaneous branches of the superficial radial nerve cross the extensor pollicis longus tendon, and can often be palpated at this point with the thumb in extension. Injury to the medial branch of this nerve occurs commonly e.g., with handcuff injuries, and causes a small area of sensory loss in the dorsal web-space of the thumb.

31.8 Axillary nerve injuries

Isolated neuropathy of the axillary nerve may occur in the following situations⁹³:

1. shoulder dislocation: the nerve is tethered to the joint capsule⁹⁴
2. sleeping in the prone position with the arms abducted above the head
3. compression from a thoracic harness
4. injection injury in the high posterior aspect of the shoulder
5. entrapment of the nerve in the quadrilateral space (bounded by the teres major and minor muscles, long head of triceps, and neck of humerus) which contains the axillary nerve and the posterior humeral circumflex artery. Arteriogram may show loss of filling of the artery with the arm abducted and externally rotated

31.9 Suprascapular nerve

31.9.1 General information

The suprascapular nerve is a mixed peripheral nerve arising from the superior trunk of the brachial plexus, with contributions from C5 & C6. There is often a history of shoulder trauma or frozen shoulder. Entrapment results in weakness & atrophy of infra- and supra-spinatus (IS & SS) and deep, poorly localized (referred) shoulder pain (the sensory part of the nerve innervates the posterior joint capsule but has no cutaneous representation).

31.9.2 Etiologies

1. nerve entrapment within the suprascapular notch beneath the transverse scapular (suprascapular) ligament (TSL)⁹⁵
2. repetitive shoulder trauma: may be bilateral when the injury is from activities such as weight-lifting
3. ganglion or tumor⁹⁶ (MRI is the test of choice for imaging these)
4. paralabral cyst from labral tear (the tendon of the long head of the biceps attaches to the superior glenoid labrum; test of choice for labral tears is MR arthrography)

31.9.3 Differential diagnosis

1. pathology in or around shoulder joint⁹⁵
 - a) rotator cuff injuries (distinction may be very difficult)
 - b) adhesive capsulitis
 - c) bicipital tenosynovitis
 - d) arthritis
2. Parsonage-Turner syndrome limited to the suprascapular nerve; see Neuralgic amyotrophy (p.570)
3. the following two etiologies will also produce rhomboid and deltoid weakness and, usually, cutaneous sensory loss:
 - a) cervical radiculopathy (≈ C5)
 - b) upper brachial plexus lesion

31.9.4 Diagnosis

Diagnosis requires temporary relief with nerve block, and EMG abnormalities of SS & IS (in rotator cuff tears, fibrillation potentials will be absent). Transient pain relief with a suprascapular nerve block helps verify the diagnosis.⁹⁷

31.9.5 Treatment

In cases where a mass is not the underlying cause, initial treatment consists of resting the affected UE, PT (including gentle conditioning), NSAIDs, topical capsaicin cream, and sometimes corticosteroid injection.

Surgical treatment is indicated for documented cases that fail to improve with conservative treatment (PT, NSAIDs, steroid/local anesthetic injection...). Position: lateral decubitus. Incision: 2 cm above and parallel to the scapular spine (atrophy of SS facilitates this). Only the trapezius needs to be split along its fibers (caution re spinal accessory nerve). To locate suprascapular notch, follow omohyoid to where it attaches to scapula and palpate just lateral to this. The suprascapular artery and vein pass over the TSL and should be preserved. Elevate the TSL with a dull nerve hook and divide it (exposure of the nerve and/or resection of the bony notch are not necessary).

31.10 Meralgia paresthetica

31.10.1 General information

AKA Originally known as the Bernhardt-Roth syndrome, and sometimes called "swashbuckler's disease," meralgia paresthetica (MP) (Greek: meros – thigh, algos – pain) is a condition often caused by entrapment of the lateral femoral cutaneous nerve (LFCN) of the thigh (a purely sensory branch with contributions from L2 and L3 nerve roots, see ► Fig. 1.16 for distribution), where it enters the thigh through the opening between the inguinal ligament and its attachment to the anterior superior iliac spine (ASIS). Anatomic variation is common, and the nerve may actually pass through the ligament, and as many as four branches may be found. May also be an initial manifestation of diabetes (diabetic neuropathy).

31.10.2 Signs and symptoms

Burning dysesthesias in the lateral aspect of the upper thigh, occasionally just above the knee, usually with increased sensitivity to clothing (hyperpathia). There may be decreased sensation in this distribution. Spontaneous rubbing or massaging the area in order to obtain relief is very characteristic.⁹⁸ MP may be bilateral in up to 20% of cases. Sitting or lying prone usually ameliorates the symptoms.

There may be point tenderness at the site of entrapment (where pressure may reproduce the pain), which is often located where the nerve exits the pelvis medial to the ASIS. Hip extension may also cause pain.

31.10.3 Occurrence

Usually seen in obese patients, may be exacerbated by wearing tight belts or girdles, and by prolonged standing or walking. Recently found in long distance runners. Higher incidence in diabetics. May also occur post-op in slender patients positioned prone, tends to be *bilateral* (p.576).

Possible etiologies are too numerous to list; more common ones include tight clothing or belts, surgical scars post-abdominal surgery, cardiac catheterization (p.577), pregnancy, iliac crest bone graft harvesting, ascites, obesity, metabolic neuropathies, and abdominal or pelvic mass.

31.10.4 Differential diagnosis

1. femoral neuropathy: sensory changes tend to be more anteromedial than MP
2. L2 or L3 radiculopathy: look for motor weakness (thigh flexion or knee extension)
3. nerve compression by abdominal or pelvic tumor (suspected if concomitant GI or GU symptoms)

The condition can usually be diagnosed on clinical grounds. When it is felt to be necessary, confirmatory tests may help (but frequently are disappointing), including:

1. EMG: may be difficult (the electromyographer cannot always find the nerve)
2. MRI or CT/myelography: when disc disease is suspected
3. pelvic imaging (MRI or CT)
4. somatosensory evoked potentials
5. response to local anesthetic injections
6. recent promise of diagnostic ultrasound using high frequency (18 MHz) probes

31.10.5 Treatment

Nonsurgical management

Tends to regress spontaneously, but recurrence is common. Nonsurgical measures achieve relief in $\approx 91\%$ of cases and should be tried prior to considering surgery⁹⁹:

1. remove offending articles (constricting belts, braces, casts, tight garments...)
2. in obese patients: weight loss and exercises to strengthen the abdominal muscles is usually effective, but is rarely achieved by the patient
3. elimination of activities involving hip extension
4. application of ice to the area of presumed constriction $\times 30$ minutes TID
5. NSAID of choice $\times 7$ –10 days
6. capsaicin ointment applied TID (p. 524)
7. lidoderm patches (p. 519) in areas of hyperesthesia may help¹⁰⁰
8. centrally acting pain medications (e.g., gabapentin, carbamazepine...) are rarely effective
9. if the above measures fail, injection of 5–10 ml of local anesthetic (with or without steroids) at the point of tenderness, or medial to the ASIS may provide temporary or sometimes long-lasting relief, and confirms the diagnosis

Surgical treatment

Options include:

1. surgical decompression (neurolysis) of the nerve: higher failure and recurrence rate than neurectomy
2. decompression and transposition
3. selective L2 nerve stimulation
4. division of the nerve (neurectomy) may be more effective, but risks denervation pain, and leaves an anesthetic area (usually a minor nuisance). May be best reserved for treatment failures

31

Technique

See references,^{99,101}

The operation is best performed under general anesthesia. A 4–6 cm oblique incision is centered 2 cm distal to the point of tenderness. Since the course of the nerve is variable, the operation is exploratory in nature, and generous exposure is required. If the nerve can't be located, it is usually because the exposure is too superficial. If the nerve still cannot be found, a small abdominal muscle incision can be made and the nerve may be located in the retroperitoneal area. CAUTION: cases have occurred where the femoral nerve has erroneously been divided.

If neurectomy instead of neurolysis is elected, electrical stimulation should be performed prior to sectioning to rule out a motor component (which would disqualify the nerve as the LFCN). If the nerve is to be divided, it should be placed on stretch and then cut to allow the proximal end to retract back into the pelvis. Any segment of apparent pathology should be resected for microscopic analysis. Neurectomy results in anesthesia in the distribution of the LFCN that is rarely distressing and gradually reduces in size.

A supra-inguinal ligament approach has also been described.¹⁰¹

31.11 Obturator nerve entrapment

Controversial if this exists. The obturator nerve is composed of L2–4 roots. It courses along the pelvic wall to provide sensation to the inner thigh, and motor to the thigh adductors (gracilis and adductors longus, brevis, and magnus). It may be compressed by pelvic tumors, also from the pressure of the fetal head or forceps during parturition.

The result is numbness of the medial thigh and weak thigh adduction.

31.12 Femoral nerve entrapment

Composed of roots L2–4. Entrapment is a rare cause of femoral neuropathy. More commonly due to fracture or surgery. See Femoral neuropathy (p. 573).

31.13 Common peroneal nerve palsy

31.13.1 General information and applied anatomy

The peroneal nerve is the most common nerve to develop acute compression palsy.

Functional anatomy: the sciatic nerve (L4–S3) consists of 2 separate nerves within a common sheath that separate at a variable location in the thigh (the peroneal division of the sciatic nerve is more vulnerable to injury than the tibial division). See ► Fig. 102.2.

1. posterior tibial nerve, or just tibial nerve (AKA medial popliteal nerve), which provides for foot inversion among other motor functions
2. common peroneal nerve (CPN), or just peroneal nerve (AKA lateral popliteal nerve): high injuries may involve the lateral hamstring (short head of the biceps femoris) in addition to the following. The CPN passes behind the fibular head where it is superficial and fixed, making it vulnerable to pressure or trauma (e.g., from crossing the legs at the knee). Just distal to this, the CPN divides into:
 - a) deep peroneal nerve (AKA anterior tibial nerve): primarily motor
 - motor: foot and toe extension (extensor hallucis longus [EHL], anterior tibialis [AT], extensor digitorum longus [EDL])
 - sensory: very small area between great toe and second toe
 - b) superficial peroneal nerve (AKA musculocutaneous nerve)
 - motor: foot eversion (peroneus longus and brevis)
 - sensory: lateral distal leg and dorsum of foot

31.13.2 Causes of common peroneal nerve injury

The most frequent cause of serious peroneal nerve injury is knee injury ± fracture; see also causes of foot drop other than peroneal nerve palsy (p. 1708).

1. entrapment as it crosses the fibular neck or as it penetrates the peroneus longus
2. diabetes mellitus and other metabolic peripheral neuropathies
3. inflammatory neuropathy: including Hansen's disease (leprosy)
4. traumatic: e.g., clipping injury in football players, stretch injury due to dislocating force applied to the knee, fibular fracture, injury during hip or knee replacement surgery
5. penetrating injury
6. masses in the area of the fibular head/proximal lower leg: popliteal fossa cysts (Baker cyst), anterior tibial artery aneurysm¹⁰² (rare)
7. pressure at fibular head: e.g., from crossing the legs at the knee, casts, obstetrical stirrups...
8. traction injuries: severe inversion sprains of the ankle
9. intraneural tumors: neurofibroma, schwannoma, neurogenic sarcoma, ganglion cysts
10. vascular: venous thrombosis
11. weight loss

31.13.3 Findings in peroneal nerve palsy

General information

1. sensory changes (uncommon): involves lateral aspect of lower half of leg
2. muscle involvement: See ► Table 31.5

Table 31.5 Muscle involvement in peroneal nerve palsy

Nerve	Muscle	Action	Involvement
deep peroneal	EHL	great toe dorsiflexion	Most commonly involved ↓ ↓ Least commonly involved (often spared)
	anterior tibialis	ankle dorsiflexion	
	EDL	toe extension	
superficial peroneal	peroneus longus & brevis	foot eversion	

Common peroneal nerve palsy (most common) produces weak ankle dorsiflexion (foot drop) due to anterior tibialis palsy, weak foot eversion, and sensory impairment in areas innervated by deep and superficial peroneal nerve (lateral calf and dorsum of foot). There may be a Tinel's sign with percussion over the nerve near the fibular neck. Occasionally, only the deep peroneal nerve is involved, resulting in foot drop with minimal sensory loss. Must differentiate from other causes of foot drop (p. 1706).

Examination/clinical correlation

See reference.⁸² (p 293)

- buttock level injury: unless the injury is one that permits spontaneous regeneration, prognosis is poor for return of peroneal nerve function even with surgery
- thigh level injury: also difficult to get improvement with surgical repair. Some peroneus function may occur at ≥ 6 mos, early contraction of AT may take ≥ 1 yr
- knee level injury: with successful regeneration, peroneus contraction may begin by 3–5 months. First signs: quivering of muscle lateral to the proximal fibula on attempted foot eversion, or tightening of tendon posterior and behind the lateral malleolus on attempted ankle dorsiflexion

31.13.4 Evaluation

EMG

EMG takes 2–4 weeks from the onset of symptoms to become positive. Stimulate above and below fibular head for prognostic information: if absent in both sites, the prognosis is poor (indicates retrograde degeneration has occurred). Wallerian degeneration takes ≈ 5 days to cause deterioration.

In addition to the expected findings of denervation—PSWs & fibs (p.255)—in the anterior tibialis, evaluate:

1. L5 innervated muscles outside the distribution of the common peroneal nerve:
 - a) posterior tibialis
 - b) flexor digitorum longus
2. L5 muscles whose nerve originates above the knee (these muscles are *spared* in cases of compression of the peroneal nerve at the fibular head due to the fact that the nerve takeoff is proximal to the popliteal fossa):
 - a) biceps femoris (short or long head)
 - b) tensor fascia lata
3. paraspinal muscles: signs of denervation solidifies the location of the lesion as nerve root; not helpful if negative

MRI

May demonstrate causes such as tumor or a ganglion cyst arising from the superior tibiofibular articulation.

31.13.5 Treatment

General information

When treatment can eliminate a reversible cause, the outcome is usually good. Surgical exploration and decompression may be considered when there is no reversible cause or when improvement does not occur.

Nonsurgical management

Bracing: ankle-foot-orthosis (AFO) compensates for loss of ankle dorsiflexion which inserts unobtrusively into a shoe. If this is inadequate, or to stabilize the ankle, a spring-loaded kick-up foot brace built into a shoe may be used. The patient should be instructed in techniques to avoid contracture of the Achilles tendon (heel cord), which would impair ankle dorsiflexion if nerve function returns.

Surgical management

At the level of the popliteal fossa, the skin incision is made just medial to the tendon of the short head of the biceps femoris (lateral hamstring) as the peroneal nerve is best located deep to or slightly medial to this tendon. The incision is carried distally slightly laterally along the surgical neck of the fibula. The biceps femoris is retracted laterally and the nerve is isolated and tagged with a Penrose

drain. The sensory sural nerve branches off the peroneal nerve at variable sites ranging from the sciatic portion of the nerve (proximal to the flexor crease) or distal to this.

In cases of compression, the fascia from the lateral gastrocnemius and soleus overlying the nerve distal to the fibular head is lysed and the nerve is exposed in 360°. As the nerve crosses the fibular neck it divides into superficial and deep branches. The superficial branch travels directly distally to supply the peroneus longus and brevis (foot evertors). The deep branches curve anteriorly to the anterior tibialis, EHL, and toe extensors.

If a graft is needed, the contralateral sural nerve is usually used, which may be supplemented with the ipsilateral sural nerve if needed.

31.14 Tarsal tunnel

31.14.1 General information

Entrapment of (posterior) tibial nerve may occur in the tarsal tunnel, posterior and inferior to *medial malleolus*. The tunnel is covered by the flexor retinaculum (lacinate ligament) which extends downward from the medial malleolus to the tubercle of the calcaneus. There is often (but not necessarily) a history of old ankle dislocation or fracture. The nerve may be trapped at the retinacular ligament. This results in pain and paresthesias in the toes and sole of foot (often sparing the heel because the sensory branches often originate proximal to the tunnel), typically worse at night. May cause clawing of toes secondary to weakness of intrinsic foot muscles. Often caused by fracture or dislocation, also rheumatoid arthritis, rarely tumors.

31.14.2 Exam

Percussion of nerve at medial malleolus produces paresthesias that radiate distally (Tinel's sign). Maximal inversion and eversion of the foot tend to exacerbate. Dorsiflexion-eversion test: examiner maximally everts and dorsiflexes the ankle while dorsiflexing the toes at the MTP joints for 5–10 seconds. Positive test reproduces the pain.

31.14.3 Diagnosis

EMG and NCV studies may help.

31.14.4 Nonsurgical management

External ankle support to improve foot mechanics.

31.14.5 Surgical management

Surgical decompression is indicated for confirmed cases that fail to improve. A curvilinear incision is used, ≈ 1.5 cm posterior and inferior to the medial malleolus. The flexor retinaculum is divided, as are any septa underneath, and the distal branches should be followed until they dive into the muscle.

References

- [1] Neary D, Ochoa JL, Gilliat RW. Subclinical Entrapment Neuropathy in Man. *J Neurol Sci.* 1975; 24:283–298
- [2] Graff-Radford SB, Jaeger B, Reeves JL. Myofascial Pain May Present Clinically as Occipital Neuralgia. *Neurosurgery.* 1986; 19:610–613
- [3] Hunter CR, Mayfield FH. Role of the Upper Cervical Roots in the Production of Pain in the Head. *Am J Surg.* 1949; 48:743–751
- [4] Poletti CE, Sweet WH. Entrapment of the C2 Root and Ganglion by the Atlanto-Epistrophic Ligament: Clinical Syndrome and Surgical Anatomy. *Neurosurgery.* 1990; 27:288–291
- [5] Anthony M. Headache and the greater occipital nerve. *Clin Neurol Neurosurg.* 1992; 94:297–301
- [6] Weiner RL, Reed KL. Peripheral neurostimulation for control of intractable occipital neuralgia. *Neuromodulation.* 1999; 2:217–221
- [7] Freund BJ, Schwartz M. Treatment of chronic cervical-associated headache with botulinum toxin A: a pilot study. *Headache.* 2000; 40:231–236
- [8] Weinberger LM. Cervico-Occipital Pain and Its Surgical Treatment. *Am J Surg.* 1978; 135:243–247
- [9] Bogduk N. Local Anesthetic Blocks of the Second Cervical Ganglion. A Technique with Application in Occipital Headache. *Cephalalgia.* 1981; 1:41–50
- [10] Bovim G, Fredriksen TA, Stolt-Nielsen A, et al. Neurolysis of the greater occipital nerve in cervicogenic headache. A follow up study. *Headache.* 1992; 32:175–179
- [11] Katz JN, Simmons BP. Clinical practice. Carpal tunnel syndrome. *New England Journal of Medicine.* 2002; 346:1807–1812
- [12] Laha RK, Lunsford LD, Dujovny M. Lacertus Fibrosus Compression of the Median Nerve. *Journal of Neurosurgery.* 1978; 48:838–841

- [13] Nakano KK, Lundergan C, Okihira M. Anterior Interosseous Nerve Syndromes: Diagnostic Methods and Alternative Treatments. *Archives of Neurology*. 1977; 34:477-480
- [14] Stewart JD. Median Nerve. In: *Focal Peripheral Neuropathies*. 4th ed. West Vancouver, Canada: JBJ Publishing; 2010:214-239
- [15] Yasargil MG, Antic J, Lაცაგა R, et al. Microsurgical Pterional Approach to Aneurysms of the Basilar Bifurcation. *Surg Neurol*. 1976; 6
- [16] Atroshi I, Gummesson C, Johnsson R, et al. Prevalence of carpal tunnel syndrome in a general population. *JAMA*. 1999; 282:153-158
- [17] Mosier BA, Hughes TB. Recurrent carpal tunnel syndrome. *Hand Clin*. 2013; 29:427-434
- [18] Jain NB, Higgins LD, Losina E, et al. Epidemiology of musculoskeletal upper extremity ambulatory surgery in the United States. *BMC Musculoskelet Disord*. 2014; 15. DOI: 10.1186/1471-2474-15-4
- [19] Feldman RG, Goldman R, Keyserling WM. Classical Syndromes in Occupational Medicine: Peripheral Nerve Entrapment Syndromes and Ergonomic Factors. *Am J Ind Med*. 1983; 4:661-681
- [20] Padua L, Aprile I, Caliendo P, et al. Carpal tunnel syndrome in pregnancy: multiperspective follow-up of untreated cases. *Neurology*. 2002; 59:1643-1646
- [21] Yanardag H, Pamuk ON, Kiziltan M, et al. An increased frequency of carpal tunnel syndrome in sarcoidosis. Results of a study based on nerve conduction study. *Acta Medica (Hradec Kralove)*. 2003; 46:201-204
- [22] Pryse-Phillips W E. Validation of a diagnostic sign in carpal tunnel syndrome. *J Neurol Neurosurg Psychiatry*. 1984; 47:870-872
- [23] Phalen GS. The Carpal Tunnel Syndrome. Clinical Evaluation of 598 Hands. *Clin Ortho Rel Res*. 1972; 83
- [24] Sandzen SC. Carpal Tunnel Syndrome. *Am Fam Physician*. 1981; 24:190-204
- [25] Upton RM, McComas AJ. The Double Crush in Nerve Entrapment Syndromes. *Lancet*. 1973; 11:359-362
- [26] Wilbourn AJ, Gilliat RW. Double-Crush Syndrome: A Critical Analysis. *Neurology*. 1997; 49:21-29
- [27] Rempel DM, Harrison RJ, Barnhart S. Work-Related Cumulative Trauma Disorders of the Upper Extremity. *JAMA*. 1992; 267:838-842
- [28] Robinson LR. How electrodiagnosis predicts clinical outcome of focal peripheral nerve lesions. *Muscle Nerve*. 2015; 52:321-333
- [29] Werner RA, Andary M. Electrodiagnostic evaluation of carpal tunnel syndrome. *Muscle Nerve*. 2011; 44:597-607
- [30] Jablecki CK, Andary MT, Floeter MK, et al. Practice parameter: Electrodiagnostic studies in carpal tunnel syndrome. Report of the American Association of Electrodiagnostic Medicine, American Academy of Neurology, and the American Academy of Physical Medicine and Rehabilitation. *Neurology*. 2002; 58:1589-1592
- [31] Campbell WW. Guidelines in electrodiagnostic medicine. Practice parameter for electrodiagnostic studies in ulnar neuropathy at the elbow. *Muscle Nerve Suppl*. 1999; 8:S171-S205
- [32] American Association of Electrodiagnostic Medicine. Chapter 9: Practice parameter for needle electromyographic evaluation of patients with suspected cervical radiculopathy; Summary statement. *Muscle and Nerve*. 1999; 22:S209-S211
- [33] Bland JD. A neurophysiological grading scale for carpal tunnel syndrome. *Muscle Nerve*. 2000; 23:1280-1283
- [34] Robinson L, Kliot M. Stop using arbitrary grading schemes in carpal tunnel syndrome. *Muscle Nerve*. 2008; 37. DOI: 10.1002/mus.21012
- [35] Bland JD. Stop using arbitrary grading schemes in carpal tunnel syndrome. *Muscle Nerve*. 2008; 38:1527; author reply 1527-1527; author reply 1528
- [36] Leishear K, Boudreau RM, Studenski SA, et al. Relationship between vitamin B12 and sensory and motor peripheral nerve function in older adults. *J Am Geriatr Soc*. 2012; 60:1057-1063
- [37] Ronco PM, Alyanakian MA, Mougnot B, et al. Light chain deposition disease: a model of glomerulosclerosis defined at the molecular level. *J Am Soc Nephrol*. 2001; 12:1558-1565
- [38] Gertz MA, Benson MD, Dyck PJ, et al. Diagnosis, prognosis, and therapy of transthyretin amyloidosis. *J Am Coll Cardiol*. 2015; 66:2451-2466
- [39] Keith MW, Masera V, Chung KC, et al. American Academy of Orthopaedic Surgeons clinical practice guideline on the treatment of carpal tunnel syndrome. *J Bone Joint Surg Am*. 2010; 92:218-219
- [40] Stewart JD. Mononeuropathies in Diabetics. 2012
- [41] Thomsen NO, Cederlund R, Rosen I, et al. Clinical outcomes of surgical release among diabetic patients with carpal tunnel syndrome: prospective follow-up with matched controls. *J Hand Surg Am*. 2009; 34:1177-1187
- [42] Burke DT, Burke MM, Stewart GW, et al. Splinting for carpal tunnel syndrome: in search of the optimal angle. *Archives of Physical Medicine and Rehabilitation*. 1994; 75:1241-1244
- [43] Walker WC, Metzler M, Cifu DX, et al. Neutral wrist splinting in carpal tunnel syndrome: a comparison of night-only versus full-time wear instructions. *Archives of Physical Medicine and Rehabilitation*. 2000; 81:424-429
- [44] Linskey ME, Segal R. Median Nerve Injury from Local Steroid Injection in Carpal Tunnel Syndrome. *Neurosurgery*. 1990; 26:512-515
- [45] Pagnanelli DM, Barrer SJ. Bilateral Carpal Tunnel Release at One Operation: Report of 228 Patients. *Neurosurgery*. 1992; 31:1030-1034
- [46] Pagnanelli DM, Barrer SJ. Carpal Tunnel Syndrome: Surgical Treatment Using the Paine Retinaculotomy. *Journal of Neurosurgery*. 1991; 75:77-81
- [47] Louie D, Earp B, Blazar P. Long-term outcomes of carpal tunnel release: a critical review of the literature. *Hand (N Y)*. 2012; 7:242-246
- [48] Atroshi I, Larsson GU, Ornstein E, et al. Outcomes of endoscopic surgery compared with open surgery for carpal tunnel syndrome among employed patients: randomised controlled trial. *BMJ*. 2006; 332. DOI: 10.1136/bmj.38863.632789.1F
- [49] Atroshi I, Hofer M, Larsson GU, et al. Open compared with 2-portal endoscopic carpal tunnel release: a 5-year follow-up of a randomized controlled trial. *J Hand Surg Am*. 2009; 34:266-272
- [50] Louis DS, Greene TL, Noellert RC. Complications of Carpal Tunnel Surgery. *J Neurosurg*. 1985; 62:352-356
- [51] Louie DL, Earp BE, Collins JE, et al. Outcomes of open carpal tunnel release at a minimum of ten years. *J Bone Joint Surg Am*. 2013; 95:1067-1073
- [52] Guyette TM, Wilgis EF. Timing of improvement after carpal tunnel release. *J Surg Orthop Adv*. 2004; 13:206-209
- [53] Zyluk A, Puchalski P. A comparison of the results of carpal tunnel release in patients in different age groups. *Neurol Neurochir Pol*. 2013; 47:241-246
- [54] Padua L, Lo Monaco M, Padua R, et al. Carpal tunnel syndrome: neurophysiological results of surgery based on preoperative electrodiagnostic testing. *J Hand Surg Br*. 1997; 22:599-601
- [55] Nancollas MP, Peimer CA, Wheeler DR, et al. Long-term results of carpal tunnel release. *J Hand Surg Br*. 1995; 20:470-474
- [56] Katz JN, Fossel KK, Simmons BP, et al. Symptoms, functional status, and neuromuscular impairment following carpal tunnel release. *J Hand Surg Am*. 1995; 20:549-555
- [57] Pensy RA, Burke FD, Bradley MJ, et al. A 6-year outcome of patients who cancelled carpal tunnel surgery. *J Hand Surg Eur Vol*. 2011; 36:642-647
- [58] Katz JN, Keller RB, Simmons BP, et al. Maine Carpal Tunnel Study: outcomes of operative and nonoperative therapy for carpal tunnel syndrome in a

- community-based cohort. *J Hand Surg Am.* 1998; 23:697-710
- [59] Brown RA, Gelberman RH, Seiler JG, 3rd, et al. Carpal tunnel release. A prospective, randomized assessment of open and endoscopic methods. *Journal of Bone and Joint Surgery.* 1993; 75:1265-1275
- [60] Thomsen NO, Rosen I, Dahlin LB. Neurophysiologic recovery after carpal tunnel release in diabetic patients. *Clin Neurophysiol.* 2010; 121:1569-1573
- [61] Jones NF, Ahn HC, Eo S. Revision surgery for persistent and recurrent carpal tunnel syndrome and for failed carpal tunnel release. *Plast Reconstr Surg.* 2012; 129:683-692
- [62] Ginanneschi F, Milani P, Reale F, et al. Short-term electrophysiological conduction change in median nerve fibres after carpal tunnel release. *Clin Neurol Neurosurg.* 2008; 110:1025-1030
- [63] Shurr DG, Blair WF, Bassett G. Electromyographic changes after carpal tunnel release. *J Hand Surg Am.* 1986; 11:876-880
- [64] Schrijver HM, Gerritsen AA, Strijers RL, et al. Correlating nerve conduction studies and clinical outcome measures on carpal tunnel syndrome: lessons from a randomized controlled trial. *J Clin Neurophysiol.* 2005; 22:216-221
- [65] Rotman MB, Enkvetchakul BV, Megerian JT, et al. Time course and predictors of median nerve conduction after carpal tunnel release. *J Hand Surg Am.* 2004; 29:367-372
- [66] Stolp KA. Upper extremity Focal Neuropathies. 2013
- [67] Granger A, Sardi JP, Iwanaga J, et al. Osborne's Ligament: A Review of its History, Anatomy, and Surgical Importance. *Cureus.* 2017; 9. DOI: 10.7759/cureus.1080
- [68] Brazis PW, Masdeu JC, Biller J. Localization in Clinical Neurology. 2nd ed. Boston: Little Brown and Company; 1990
- [69] Harrop JS, Hanna A, Silva MT, et al. Neurological manifestations of cervical spondylosis: an overview of signs, symptoms, and pathophysiology. *Neurosurgery.* 2007; 60:S1-14-20
- [70] Wilkins RH, Rengachary SS. *Neurosurgery.* New York 1985
- [71] Dumitru D. *Electrodiagnostic Medicine.* Philadelphia: Hanley and Belfus; 1995
- [72] Bonney G. Iatrogenic Injuries of Nerves. *Journal of Bone and Joint Surgery.* 1986; 68B:9-13
- [73] Stewart JD. *Ulnar Nerve.* In: *Focal Peripheral Neuropathies.* 4th ed. West Vancouver, Canada: JBJ Publishing; 2010:258-313
- [74] Bartels RHMA, Menovsky T, Van Overbeek JJ, et al. Surgical Management of Ulnar Nerve Compression at the Elbow: An Analysis of the Literature. *Journal of Neurosurgery.* 1988; 89:722-727
- [75] Padua L, Aprile I, Mazza O, et al. Neurophysiological classification of ulnar entrapment across the elbow. *Neurol Sci.* 2001; 22:11-16
- [76] Practice parameter: electrodiagnostic studies in ulnar neuropathy at the elbow. American Association of Electrodiagnostic Medicine, American Academy of Neurology, and American Academy of Physical Medicine and Rehabilitation. *Neurology.* 1999; 52:688-690
- [77] Beekman R, Wokke JH, Schoemaker MC, et al. Ulnar neuropathy at the elbow: follow-up and prognostic factors determining outcome. *Neurology.* 2004; 63:1675-1680
- [78] Cartwright MS, Chloros GD, Walker FO, et al. Diagnostic ultrasound for nerve transection. *Muscle and Nerve.* 2007; 35:796-799
- [79] Caliendo P, La Torre G, Padua R, et al. Treatment for ulnar neuropathy at the elbow. *Cochrane Database Syst Rev.* 2012; 7. DOI: 10.1002/14651858.CD006839.pub3
- [80] Elhassan B, Steinmann SP. Entrapment neuropathy of the ulnar nerve. *J Am Acad Orthop Surg.* 2007; 15:672-681
- [81] Le Roux PD, Ensign TD, Burchiel KJ. Surgical Decompression Without Transposition for Ulnar Neuropathy: Factors Determining Outcome. *Neurosurgery.* 1990; 27:709-714
- [82] Kline DG, Hudson AR. *Nerve Injuries: Operative Results for Major Nerve Injuries, Entrapments, and Tumors.* Philadelphia: W. B. Saunders; 1995
- [83] Janjua RM, Fernandez J, Tender G, et al. Submuscular transposition of the ulnar nerve for the treatment of cubital tunnel syndrome. *Neurosurgery.* 2008; 63:321-4; discussion 324-5
- [84] Spinner RJ, O'Driscoll S W, Jupiter JB, et al. Unrecognized dislocation of the medial portion of the triceps: another cause of failed ulnar nerve transposition. *Journal of Neurosurgery.* 2000; 92:52-57
- [85] Bartels RH, Verhagen WI, van der Wilt GJ, et al. Prospective randomized controlled study comparing simple decompression versus anterior subcutaneous transposition for idiopathic neuropathy of the ulnar nerve at the elbow: Part 1. *Neurosurgery.* 2005; 56:522-30; discussion 522-30
- [86] Biggs M, Curtis JA. Randomized, prospective study comparing ulnar neurolysis in situ with submuscular transposition. *Neurosurgery.* 2006; 58:296-304; discussion 296-304
- [87] Tindall SC. Comment on LeRoux P D, et al.: Surgical Decompression without Transposition for Ulnar Neuropathy: Factors Determining Outcome. *Neurosurgery.* 1990; 27
- [88] Youmans JR. *Neurological Surgery.* Philadelphia 1990
- [89] Shea JD, McClain EJ. Ulnar-Nerve Compression Syndromes at and Below the Wrist. *Journal of Bone and Joint Surgery.* 1969; 51A:1095-1103
- [90] Cavallo M, Poppi M, Martinielli P, et al. Distal Ulnar Neuropathy from Carpal Ganglia: A Clinical and Electrophysiological Study. *Neurosurgery.* 1988; 22:902-905
- [91] Dyck PJ, Thomas PK. *Peripheral Neuropathy.* 2nd ed. Philadelphia: W. B. Saunders; 1984
- [92] Roles NC, Maudsley RH. Radial Tunnel Syndrome: Resistant Tennis Elbow as a Nerve Entrapment. *Journal of Bone and Joint Surgery.* 1972; 54B:499-508
- [93] McKowen HC, Voorhies RM. Axillary Nerve Entrapment in the Quadrilateral Space: Case Report. *Journal of Neurosurgery.* 1987; 66:932-934
- [94] de Laat EAT, Visser CPJ, Coene LNEM, et al. Nerve Lesions in Primary Shoulder Dislocations and Humeral Neck Fractures. *Journal of Bone and Joint Surgery.* 1994; 76B:381-383
- [95] Hadley MN, Sonntag VKH, Pittman HW. Suprascapular Nerve Entrapment: A Summary of Seven Cases. *J Neurosurg.* 1986; 64:843-848
- [96] Fritz RC, Helms CA, Steinbach LS, et al. Suprascapular nerve entrapment: Evaluation with MR imaging. *Radiology.* 1992; 182:437-444
- [97] Callahan JD, Scully TB, Shapiro SA, et al. Suprascapular Nerve Entrapment: A Series of 27 Cases. *Journal of Neurosurgery.* 1991; 74:893-896
- [98] Stevens HI. Meralgia Paresthetica. *Arch Neurol Psychiatry.* 1957; 77:557-574
- [99] Williams PH, Trzil KP. Management of Meralgia Paresthetica. *J Neurosurg.* 1991; 74:76-80
- [100] Devers A, Galer BS. Topical lidocaine patch relieves a variety of neuropathic pain conditions: an open-label study. *Clinical Journal of Pain.* 2000; 16:205-208
- [101] Aldrich EF, Van den Heever C. Suprainguinal Ligament Approach for Surgical Treatment of Meralgia Paresthetica. *Technical Note. J Neurosurg.* 1989; 70:492-494
- [102] Kars HZ, Topaktas S, Dogan K. Aneurysmal Peroneal Nerve Compression. *Neurosurgery.* 1992; 30:930-931

32 Non-Entrapment Peripheral Neuropathies

32.1 General information

Lesions of peripheral nerves which may produce somatic findings (combinations of weakness, sensory disturbance, and/or reflex changes) and/or autonomic disturbances.

- **Mononeuropathy.** A disorder of a single nerve, often due to trauma or entrapment.
- **Mononeuropathy multiplex.** Involvement of 2 or more nerves, usually due to a systemic abnormality (e.g., vasculitis, rheumatoid arthritis, DM...). Treatment is directed at the underlying disorder.

32.2 Etiologies of peripheral neuropathy

A mnemonic for etiologies of peripheral neuropathies is “GRAND THERAPIST” (see ► Table 32.1). Diabetes, alcoholism, and Guillain-Barré (italicized in table) account for 90% of cases. Other etiologies include: arteritis/vasculitis, monoclonal gammopathy (p.574), hepatitis C virus-associated cryoglobulinemia, acute idiopathic polyneuritis, Sjögren’s syndrome (disease), amyloidosis, HIV/AIDS.

Table 32.1 Mnemonic for etiologies of peripheral neuropathy

G-R-A-N-D	T-H-E-R-A-P-I-S-T
Guillain-Barré (p. 193) Renal; uremic neuropathy (p.576) Alcoholism (see below) Nutritional (B12 deficiency...) Diabetes; see below or Drugs (p. 573)	Traumatic Hereditary and HIV Endocrine or Entrapment Radiation Amyloid (p. 576) or AIDS (p. 574) Porphyria (should be under hereditary) or Psychiatric or Paraneoplastic (see below) or Pseudoneuropathy (see below) or PMR (p.206) or Polycythemia vera ¹ Infectious/post-infectious (e.g., Hansen’s disease) Sarcoidosis, see neurosarcoidosis (p. 198), or “Systemic” Toxins, including heavy metals, e.g., lead toxicity (plumbism) (p. 1219)

32.3 Classification

1. inherited neuropathies. Examples include:
 - a) Charcot-Marie-Tooth (CMT) (AKA peroneal muscular atrophy, AKA Hereditary Motor and Sensory Neuropathy [HMSN]): Up to 7 types (the most common form is autosomal dominant, but X-linked recessive forms also exist). CMT Types 1 & 2 together make up the most common inherited disorder of peripheral nerves (up to 40/100,000). The most common forms involve demyelination. Progressive loss of motor (primarily *distal* LE) and, to a lesser degree, sensory function (predominantly proprioception and vibration), with atrophy in UEs & LEs. Earliest findings: pes cavus with hammer toes, foot drop and frequent ankle sprains. Patients are more susceptible to entrapment neuropathies due to underlying compromise of peripheral nerves. Patients with Type 1 usually maintain ability to ambulate, whereas Type 2 usually lose ambulation by their teenage years
 - b) hereditary neuropathy with liability to pressure palsies (HNPP): similar to CMT but due to focal areas of irregular thickening of myelin sheaths (“tomaculous” changes), mild trauma or pressure can produce nerve palsies that may last for months
2. acquired neuropathies: see sections below for details
 - a) acquired pure *sensory* neuropathies (in the absence of autonomic dysfunction) are rare. May be seen with pyridoxine therapy or paraneoplastic syndromes (see below)
 - b) entrapment neuropathies (p. 541)
3. pseudoneuropathy
 - a) definition: psychogenic somatoform disorders or malingering, reproducing the pains, paresthesias, hyperalgesia, weakness, and even objective findings such as changes in color and temperature which may mimic neuropathic symptoms²

32.4 Clinical

32.4.1 Presentation

Peripheral neuropathies can present as loss of sensation, pain, weakness, incoordination and difficulty ambulating as well as with autonomic symptoms.

32.4.2 Evaluation

Initial (screening) work-up for peripheral neuropathies of unknown etiology:

1. bloodwork:
 - a) fasting electrolytes
 - b) Hgb-A1C
 - c) CBC: anemia is common in multiple myeloma
 - d) TSH
 - e) ESR & CRP
 - f) vitamin B₁₂ & methylmalonic acid (p. 1699)
 - g) SPEP with reflex to IFE (serum protein electrophoresis (SPEP), with additional test of serum immunofixation (IFE) if M-spike is present) (see multiple myeloma (p. 930) for details)
 - h) optional (if multifocal motor neuropathy (MMN) (p. 1700) is suspected): Anti GM-1 antibodies
2. electrodiagnostics (EMG/NCV) (p. 254)

32.5 Syndromes of peripheral neuropathy

32.5.1 Length-dependent peripheral neuropathy

The most common clinical pattern related to neuropathy is that of length-dependent peripheral neuropathy. Symptoms are symmetric and begin in the feet (the longest nerves in the body) and gradually progress proximally. Upper limb involvement may occur late or not at all.

32.5.2 Critical illness polyneuropathy (CIP)

AKA neuropathy of critical illness, ICU neuropathy... See DDx under Guillain-Barré syndrome (p. 194).

May occur in up to 70% of septic patients (not all are significantly symptomatic). Affects primarily distal muscles.

Diagnostic criteria:

1. presence of sepsis, multi-organ failure, respiratory failure, or septic inflammatory response syndrome (SIRS)
2. difficulty weaning from ventilator or extremity weakness
3. EMG: ↓ amplitudes of compound muscle action potentials (CMAP) & SNAP
4. widespread muscle denervation potentials
5. normal or only mild increase in serum CPK levels

Recovery occurs in weeks to months (faster than Guillain-Barré).

Treatment is supportive. Complete recovery occurs in 50%.

32.5.3 Paraneoplastic syndromes affecting the nervous system

Occurs in < 1% of cancer patients. Peripheral sensory neuropathy of unknown etiology has been associated with cancer since its earliest description.³ Therefore, in patients with sensory neuropathy of unknown etiology, occult neoplasms should be ruled out. If the work-up is negative, the patient should be followed since up to 35% of patients will be found to have cancer after a mean interval of 28 months after the onset of neuropathy (range: 3–72 months)⁴ (no one particular cancer type predominated, in spite of the fact that historically lung cancer is the most frequent neoplasm associated with sensory neuropathy⁵).

32.5.4 Alcohol neuropathy

Characteristically produces a diffuse sensory neuropathy, with absent Achilles reflexes.

32.5.5 Brachial plexus neuropathy

Evaluation

When the etiology is unclear, check CXR (with apical lordotic view), glucose, ESR, and ANA. If no improvement by ≈ 4 weeks, obtain MRI of the plexus (idiopathic brachial plexitis will usually start to show some improvement by this time; therefore tumor should be ruled out if no improvement).

Differential diagnosis of etiologies of brachial plexopathy

1. Pancoast syndrome or Pancoast tumor AKA superior sulcus tumor. Clinical: various combinations of pain in the shoulder radiating into the upper extremity in the ulnar nerve distribution from involvement of the lower brachial plexus, atrophy of hand muscles, Horner syndrome (p.594), UE edema. Etiologies:
 - a) neoplasms:
 - most common: bronchogenic cancer, usually non-small cell (NSCLC) (squamous cell or adenocarcinoma) arising in the pulmonary apex
 - metastases
 - b) infections
 - c) inflammatory: granulomas, amyloid
2. (idiopathic) brachial plexitis AKA neuralgic amyotrophy: most commonly upper plexus or diffuse (see below)
3. cervical rib
4. viral
5. following radiation treatment: often diffuse (see below)
6. diabetes
7. vasculitis
8. inherited: dominant genetics
9. trauma (p.579)

Neuralgic amyotrophy of the upper extremity

General information

AKA idiopathic brachial plexus neuropathy, AKA (paralytic) brachial neuritis, AKA brachial plexitis, AKA Parsonage-Turner syndrome,⁶ AKA immune-mediated brachial plexus neuropathy, among others. Idiopathic. Not clearly infectious or inflammatory; allergic mechanism possible. Prognosis is generally good. Common patterns: single or multiple mononeuropathy, plexopathy, or some combination. Demographics are shown in ► Table 32.2.

Table 32.2 Neuralgic amyotrophy of the upper extremity

incidence	1.64 per 100,000 population
male:female	2.4:1
age range at onset	3 mos–75 years
prodrome	• ≈ 45% had viral prodrome (URI in 25%) • may follow vaccination
onset	rapid onset of pain or paralysis/paresis
initial symptom	pain in 95%
weakness	• 50% confined to shoulder girdle • 10% confined to a single peripheral nerve
sensory deficit	67%, usually axillary and antebrachial cutaneous
laterality	• 66% unilateral (right side 54%) • 34% bilateral
lab tests	normal

In a review of 99 cases⁷: predominant symptom is acute onset of intense *pain*, with weakness developing simultaneously or after a variable period (70% occur within 2 weeks of pain), usually as the pain lessened.^{6,8} Weakness never preceded pain, onset of weakness was sudden in 80%. Pain was usually constant, and described as “sharp,” “stabbing,” “throbbing,” or “aching.” Arm movement exacerbated the pain, and muscle soreness was noted in 15%. Pain lasted hours to several weeks. Paresthesias occurred in 35%. Pain usually lacked radicular features. When bilateral, weakness is usually asymmetric.

Exam

Weakness or paralysis in 96%, confined to shoulder girdle in 50%. In descending order of involvement: deltoid, spinati, serratus anterior, biceps brachii, and triceps. Winging of the scapula occurred in 20%. Sensory loss occurred in 60% of plexus lesions, of mixed variety (superficial cutaneous and proprioceptive). Sensory loss most common in outer surface of upper arm (circumflex nerve distribution) and radial aspect of forearm. Reflexes were variable.

Overall distribution judged to predominantly involve *upper* plexus in 56%, diffuse plexus in 38%, and lower in 6%.

EMG/NCV

May help localize the portion of the plexus involved, and may detect subclinical involvement of the contralateral extremity. Must wait ≥ 3 weeks from onset for findings. Differentiating from cervical radiculopathy: SNAP (p.255) should be normal in radiculopathy whereas some involvement usually occurs in plexitis. Cervical paraspinals will usually be normal in plexitis (except for very severe cases where there can be some retrograde involvement), and will be abnormal (fibrillations) in radiculopathy (except in cases where there has been enough time that significant recovery has occurred).

Outcome

Functional recovery is better in patients with primarily upper plexus involvement. After 1 year, 60% of upper plexus lesions were functioning normally, whereas none with lower involvement were (later took 1.5–3 years). Rate of recovery estimated to be 36% within 1 year, 75% within 2, and 89% by 3 years. Recurrence was seen in only 5%. No evidence that steroids altered the course of the disease although it is still often prescribed in the acute phase.

Radiation induced brachial plexus neuropathy

Often follows external beam irradiation in the region of the axilla for breast carcinoma. Produces sensory loss with or without weakness. CT or MRI or biopsy may be needed to rule out tumor invasion of the brachial plexus.

32.5.6 Lumbosacral plexus neuropathy**General information**

Analogous to idiopathic brachial plexitis (see above).⁹ It is controversial whether this actually exists in isolation without diabetes. Often starts with LE pain of abrupt onset, followed in days or a few weeks by weakness with or without muscle atrophy. Sensory symptoms are less prominent, and usually involve paresthesias. Objective sensory loss is only occasionally seen. There may be tenderness over the femoral nerve.

Differential diagnosis

May be confused with femoral neuropathy or L4 radiculopathy when quadriceps weakness and wasting occurs. Similarly, L5 radiculopathy or peroneal neuropathy may be erroneously suspected when foot drop is seen. Straight leg raising may occasionally be positive. Conspicuously absent are back pain, exacerbation of pain by Valsalva maneuver or back motion, and significant sensory involvement. See differential diagnosis of foot drop (p.1708) and other causes of sciatica (p.1700).

Etiologies

Other etiologies are similar to that for brachial plexus neuropathy (see above) except that under tumor, a pelvic mass should also be included (check prostate on rectal exam).

Evaluation

Evaluation is as for brachial plexus neuropathy (see above), except that instead of a brachial plexus MRI, a lumbar MRI and pelvic CT should be done to rule out masses.

EMG is key to diagnosis: evidence of patchy denervation (fibrillation potentials, and motor unit potentials that are either decreased in number or increased in amplitude or duration and polyphasic) involving at least 2 segmental levels with *sparing* of the paraspinal muscles is highly diagnostic (once diabetes, etc. have been ruled out).

Outcome

Recovery from pain precedes return of strength. Improvement is generally monophasic, slow (years), and incomplete.

32.5.7 Diabetic neuropathy

General information

≈ 50% of patients with DM develop neuropathic symptoms or show slowing of nerve conduction velocities on electrodiagnostic testing. Neuropathy may sometimes be the initial manifestation of diabetes. Diabetic neuropathy is reduced by tight control of blood glucose.¹⁰

Syndromes

Disagreement exists over the number of distinct clinical syndromes; there is probably a continuum¹¹ and they likely occur in various combinations. Some of the more readily identified syndromes include:

1. primary sensory polyneuropathy: symmetric, affecting feet and legs more than hands. Chronic, slowly-progressive. Often with accelerated loss of distal vibratory sense (normal loss with aging is ≈ 1% per year after age 40). Presents as pain, paresthesias, and dysesthesias. Soles of feet may be tender to pressure. Meralgia paresthetica (p. 561) may be first manifestation
2. autonomic neuropathy: involving bladder, bowel, and circulatory reflexes (resulting in orthostatic hypotension). May produce impotence, impaired micturition, diarrhea, constipation, impaired pupillary light response
3. diabetic plexus neuropathy¹² or proximal neuropathy: possibly secondary to vascular injury to nerves (similar to a diabetic mononeuritis):
 - a) one that occurs in patients > 50 years old with mild diabetes type II that is often confused with femoral neuropathy. Causes severe pain in the hip, anterior thigh, knee, and sometimes medial calf. Weakness of the quadriceps, iliopsoas, and occasionally thigh adductors. Loss of patellar reflex (knee jerk). Possible sensory loss over medial thigh and lower leg. Pain usually improves in weeks, the weakness in months
 - b) ★ diabetic amyotrophy: occurs in similar patient population, often with recently diagnosed DM. Alternative names include¹³: Bruns-Garland syndrome, ischemic mononeuropathy multiplex....¹⁴ Abrupt onset of asymmetric pain (usually deep aching/burning with superimposed lancinating paroxysms, most severe at night) in back, hip, buttocks, thigh, or leg. Progressive weakness in proximal or proximal and distal muscles, often preceded by weight loss. Patellar reflexes are absent or reduced. Sensory loss is minimal. Proximal muscles (especially thigh) may atrophy. EMG findings consistent with demyelination invariably accompanied by axonal degeneration, with involvement of paraspinals and no evidence of myopathy. Symptoms may progress steadily or stepwise for weeks or even up to 18 months, and then gradually resolve. Opposite extremity may become involved during the course or may occur months or years later. Sural nerve biopsy (p. 1832) may suggest demyelination
 - c) diabetic proximal neuropathy (DPN): fairly similar findings to diabetic amyotrophy, except for subacute onset of symmetric LE involvement that usually start with weakness, may be a variant.¹⁵ ► Table 32.3 (adapted¹⁵) compares DPN to diabetic amyotrophy and chronic inflammatory demyelinating polyneuropathy (CIDP)

Treatment

Treatment of Bruns-Garland syndrome is primarily expectant, although immunotherapy (steroids, immune globulin, or plasma exchange) may be considered in severe or progressive cases (efficacy is unproven).¹⁵

For sensory polyneuropathy, good control of blood sugar contributes to reduction of symptoms. Adjunctive agents that have been used include:

1. mexiletine (Mexitil®): start at 150 mg q 8 hrs, and titrate to symptoms to a maximum of 10 mg/kg/d
2. amitriptyline (Elavil®) and fluphenazine (Prolixin®): **R**: start with 25 mg amitriptyline PO q hs and 1 mg fluphenazine PO TID; and work up to 75 mg amitriptyline PO q hs¹⁶ (≈ 100 mg qd amitriptyline alone may also be effective¹⁷). Usefulness has been challenged,¹⁸ but many studies do show benefit.^{17,19} **Side effects**: that may limit use include sedation, confusion, fatigue, malaise, hypomania, rash, urinary retention, and orthostatic hypotension

Table 32.3 Comparison of diabetic amyotrophy, diabetic proximal neuropathy (DPN), & CIDP

Description	Diabetic amyotrophy	DPN	CIDP
Onset	acute	subacute	gradual
Initial symptoms	asymmetric pain→ weakness	symmetric weakness	symmetric weakness
UE weakness	no	uncommon	yes
Sensory loss	minimal	minimal	moderate
Areflexia	LE	LE	generalized
CSF protein	variable	increased	increased
Axonal pathologic changes	common	typical	uncommon
Conduction slowing	patchy	patchy	diffuse
Prognosis	good	good	poor without treatment
Response to immuno-therapy	unknown	possible	yes
Course	monophasic	monophasic	progressive

- desipramine (Norpramin®): more selective blocker of norepinephrine reuptake (which seems more effective for this condition than serotonin reuptake blockers). Effectiveness at mean doses of 110 mg/day ≈ same as amitriptyline and therefore may be useful for patients unable to tolerate amitriptyline.¹⁷ **Side effects:** include insomnia (may be minimized by AM dosing), orthostatic hypotension, rash, bundle branch block, tremor, pyrexia. **Supplied:** 10, 25, 50, 75, 100 & 150 mg tablets
- capsaicin (Zostrix®) (p.524): effective in some
- paroxetine (Paxil®): a selective serotonin reuptake inhibitor (SSRI) antidepressant. **R:** 20 mg PO q AM. If necessary, increase by 10 mg/d q week up to a maximum of 50 mg/day (except in elderly, debilitated, or renal or hepatic failure where maximum is 40 mg/day). **Supplied:** 20 mg (scored) & 30 mg tablets
- gabapentin (Neurontin®) doses of 1800–3600 mg/d produces at least moderate pain relief from painful diabetic neuropathy in 60% of patients²⁰ and was ≈ as efficacious as amitriptyline.²¹ Dosage must be reduced with renal insufficiency. See details (p.496)
- pregabalin (Lyrica®) **R:** start with 50 mg TID and increase up to a maximum of 100 mg PO TID within 1 week in patients with creatinine clearance ≥ 60 ml/min, see Eq (7.1) to estimate. Dosage must be reduced with renal insufficiency. **Supplied:** 25, 50, 75, 100, 150, 200, 225, 300 mg capsules

32.5.8 Drug-induced neuropathy

Many drugs have been implicated as possible causes of peripheral neuropathy. Those that are better established or more notorious include:

- thalidomide; neuropathy may occur with chronic use, and may be irreversible²²
- metronidazole (Flagyl®)
- phenytoin (Dilantin®)
- amitriptyline (Elavil®)
- dapsone: a rare complication reported with use in nonleprosy patients is a reversible peripheral neuropathy that may be due to axonal degeneration, producing a Guillain-Barré-like syndrome (p.193).
- nitrofurantoin (Macrochantin®): may additionally cause optic neuritis
- cholesterol-lowering drugs: e.g., lovastatin (Mevacor®), indapamide (Lozol®), gemfibrozil (Lopid®)
- thallium: may produce tremors, leg pains, paresthesias in the hands and feet, polyneuritis in the LE, psychosis, delirium, seizures, encephalopathy
- arsenic: may produce numbness, burning and tingling of the extremities
- chemotherapy: cisplatin, vincristine...

32.5.9 Femoral neuropathy

Clinical findings

- motor deficits:
 - wasting and weakness of the quadriceps femoris (knee extension)

- b) $\pm$ weakness of iliopsoas (hip flexion): if present, indicates very proximal pathology (lumbar root or plexus lesion) as the branches to the iliopsoas arise just distal to the neural foramina
- 2. diminution of the patellar (knee jerk) reflex
- 3. sensory findings:
 - a) sensory loss over the anterior thigh and medial calf
 - b) pain in same distribution may occur
- 4. mechanical signs: positive femoral stretch test (p. 1252)

Etiologies

- 1. diabetes: the most frequent cause
- 2. femoral nerve entrapment: rare
 - a) may occur secondary to inguinal hernia or may be injured by deep sutures placed during herniorrhaphy
 - b) secondary to prolonged pelvic surgery from retractor compression (usually bilateral)
- 3. intraabdominal tumor
- 4. femoral arterial catheterization: see below
- 5. retroperitoneal hematoma (e.g., in hemophiliac or on anticoagulants)
- 6. during surgery (p. 576)

Differential diagnosis

- 1. L4 radiculopathy: L4 radiculopathy should not cause iliopsoas weakness; see L4 involvement (p. 1702)
- 2. diabetic plexus neuropathy (see above)
- 3. (idiopathic) lumbosacral plexus neuropathy (see above)

32.5.10 AIDS neuropathy

General information

3.3% of patients with AIDS will develop peripheral nerve disorders²³ (whereas none who were just HIV positive developed neuropathy). The most common disorder is distal symmetric polyneuropathy (DSP), usually consisting of vague numbness and tingling, and sometimes painful feet (although it may also be painless). There may be subtle reduction of light touch and vibratory sense. Other neuropathies include mononeuropathies—usually meralgia paresthetica (p. 561)—mononeuropathy multiplex, or lumbar polyradiculopathy. Drugs used to treat HIV can also cause neuropathies (see below).

The DSP in AIDS patients is often associated with CMV infection, *Mycobacterium avium* intracellular infection, or may be due to lymphomatous invasion of the nerve or lymphomatous meningitis. May demonstrate a mixed axonal demyelinating type of neuropathy on electrodiagnostic testing.

Neuropathies associated with drugs used to treat HIV

- 1. nucleoside reverse transcriptase inhibitors
 - a) zidovudine (Retrovir®) (formerly AZT)
 - b) didanosine (ddI; Videx®): (formerly dideoxyinosine) can cause a painful dose-related neuropathy²⁴
 - c) stavudine (d4T; Zerit®): can cause sensory neuropathy which usually improves when d4T is discontinued, and may not recur if restarted at lower dose²⁴
 - d) zalcitabine (ddC; Hivid®): the least potent of the nucleoside analogs, therefore rarely used; dose-related neuropathy can be severe and persistent. More common in patients with DM or didanosine treatment²⁴
- 2. protease inhibitors
 - a) ritonavir (Norvir®): can cause peripheral paresthesias
 - b) amprenavir (Agenerase®): can cause perioral paresthesias

32.5.11 Neuropathy associated with monoclonal gammopathy

General information

Abnormal immunoglobulin protein (paraproteins) are found in the blood.

Monoclonal gammopathies include:

1. (multiple) myeloma (p.928)
2. Waldenstrom's macroglobulinemia
3. non-malignant entities such as monoclonal gammopathy of undetermined significance (MGUS). MGUS patients will develop multiple myeloma (MM) at a rate of 1.5%/year, but the risk of developing a lymphoproliferative disorder before they die is only 11%. Most cases of MM are preceded by MGUS. MGUS can also progress to Waldenstrom's macroglobulinemia, amyloidosis, B-cell lymphoma, or lymphocytic leukemia. Criteria for MGUS:
 - a) monoclonal paraprotein band <30 g/l (which is less than with MM)
 - b) plasma cells <10% on bone marrow biopsy
 - c) no evidence of bone lesions of MM, hypercalcemia, or renal insufficiency related to the paraprotein, and
 - d) no evidence of another β -cell proliferative disorder

Much effort has gone into determining which benign gammopathies are or are not likely to progress, and will not be addressed here.

≈ 10% of patients with neuropathy with no apparent etiology will be determined to have a monoclonal gammopathy (malignant or otherwise).

Etiologies

1. antibodies directed primarily against oligosaccharides of peripheral nerves, e.g., myelin associated glycoprotein (MAG), producing demyelinating neuropathy
2. cryoglobulins may damage vaso-nervorum (small blood vessels nourishing peripheral nerves)
3. in malignant gammopathies, tumor cells can invade the peripheral nerves (lymphomatosis)
4. amyloidosis (p.576): deposition of amyloid in peripheral nerves
5. thalidomide (p.573) used to treat some myelomas, may cause neuropathy

Treatment

1. IgM monoclonal gammopathies: reduce the IgM antibody concentration
2. IgG or IgA monoclonal gammopathies:
 - a) treatment for myeloma-related neuropathy is directed at treating the myeloma
 - b) solitary plasmacytoma: excision or XRT can improve the neuropathy

32.5.12 Perioperative neuropathies

General information

Also, below. Represent ≈ 1/3 of all anesthesia-associated malpractice claims in the U.S.²⁵ Most often involves ulnar nerve or brachial plexus. In many cases, a nerve that is abnormal but asymptomatic may become symptomatic as a result of any of the following factors: stretch or compression of the nerve, generalized ischemia or metabolic derangement. The injury may be permanent or temporary. Occurs almost exclusively in adults.²⁶

Types of perioperative neuropathies

Examples include:

1. ulnar neuropathy: controversial. Often blamed on external nerve compression or stretch as a result of malpositioning. Although this may be true in some cases, in one series this was felt to be a factor in only ≈ 17% of cases.²⁷ Patient-related characteristics associated with these neuropathies are shown in ▶ Table 32.4.²⁸ Many of these patients have abnormal contralateral nerve conduction, suggesting a possible predisposing condition.²⁹ Many patients do not complain of symptoms until > 48 hours post-op^{28,29,30} (if it were due to compression, deficit would be

Table 32.4 Patient-related characteristics in anesthesia-related ulnar neuropathy

male gender
obesity (body mass index ≥ 38)
prolonged post-op bed rest

maximal immediately post-op). Risk may be reduced by padding the arm at, and especially distal to, the elbow, and avoiding flexion of the elbow (especially avoiding $> 110^\circ$ flexion which tightens the cubital tunnel retinaculum), and by reducing the amount of time spent convalescing in the recumbent position with leaning on the elbows³⁰

2. brachial plexus neuropathy: may be mistaken for ulnar neuropathy. May be associated with:
 - a) median sternotomy (most common with internal mammary dissection). Posterior sternal retraction displaces the upper ribs and may stretch or compress the C6 through T1 roots (which are major contributors to the ulnar nerve)
 - b) head-down (Trendelenburg) positions where the patient is stabilized with a shoulder brace. The brace should be placed over the acromioclavicular joint(s), and non-slip mattresses and flexion of the knees may be used as adjuncts²⁶
 - c) prone position (rare): especially with shoulder abduction and elbow flexion with contralateral head rotation²⁶
3. median neuropathy: perioperative median nerve injury may result from stretch of the nerve. Rare. Seems to occur primarily in middle-aged muscular males with reduced extension of the elbows due to muscle mass. This may result in stretching of the nerve after muscle relaxants are given. Padding should be placed under the forearms and hands of these patients to maintain mild elbow flexion²⁶
4. lower extremity neuropathies: most occur in patients undergoing procedures in the lithotomy position.²⁶ Frequency of involvement in a large series of patients undergoing procedures in the lithotomy position³¹: common peroneal 81%, sciatic 15%, and femoral 4%. Risk factors other than position: prolonged duration of procedure, extremely thin body habitus, and cigarette smoking in the preoperative period
 - a) common peroneal neuropathy: susceptible to injury in the posterior popliteal fossa where it wraps around the fibular head. May be compressed by leg holders, which should be padded in this area
 - b) femoral neuropathy: compression of the nerve by self-retaining abdominal wall retractor or rendering the nerve ischemic by occlusion of the external iliac artery.²⁶ Hemorrhage into the iliopsoas muscle may also compress the nerve. Cutaneous branches of the femoral nerve may be injured during labor and/or delivery³² (most are transient)
 - c) sciatic neuropathy: stretch injuries may occur with hyperflexion of the hip and extension of the knee as may occur in some variants of the lithotomy position
 - d) meralgia paresthetica³³: tends to occur bilaterally in young, slender males positioned prone, with operations lasting 6–10+ hours. Onset: 1–8 days post-op. Spontaneous recovery typically occurs over an average of 5.8 months

Management

Once a neuropathy is detected, determine if it is sensory, motor, or both. Pure sensory neuropathies are more often temporary than motor,²⁸ and expectant management for ≈ 5 days is suggested (have the patient avoid postures or activities that may further injure the nerve). Neurologic consultation should be requested for all motor neuropathies and for sensory neuropathies persisting > 5 days²⁶ (EMG evaluation will not usually be helpful earlier than ≈ 3 weeks after onset).

32.5.13 Other neuropathies

Amyloid neuropathy

Amyloid is an insoluble extracellular protein aggregate that can be deposited in peripheral nerves. Amyloidosis occurs in a number of conditions, e.g., in $\approx 15\%$ of patients with **multiple myeloma** (p.928). The neuropathy predominantly produces a progressive autonomic neuropathy and symmetric dissociated sensory loss (reduced pain and temperature, preserved vibratory sense). There is usually less prominent motor involvement. May predispose to pressure injury of nerves—especially carpal tunnel syndrome, see laboratory tests (p.549).

Uremic neuropathy

Occurs in chronic renal failure. Early symptoms include calf cramps (“Charlie horses”), dysesthetic pain in feet (similar to painful diabetic neuropathy) and “restless legs.” Achilles reflexes are lost. A stocking sensory loss is followed later by LE weakness that starts distally and ascends. The offending toxin is not known. Dialysis or renal transplantation relieves the symptoms.

Neuropathy after cardiac catheterization

In a series of $\approx 10,000$ patients followed after femoral artery catheterization³⁴ (e.g., for coronary angiography or angioplasty), neuropathy occurred in 0.2% (with an estimated range in the literature up to $\approx 3\%$). Risk factors identified include patients developing retroperitoneal hematomas or pseudoaneurysms after the procedure, procedures requiring larger introducer sheaths (e.g., angioplasty & stent placement > diagnostic catheterization), excessive anticoagulation (PTT > 90 for at least 12 hours).

Two groups of patients were identified and are shown in ► Table 32.5.

Table 32.5 Neuropathy after cardiac catheterization (N = 9585)³⁴

Catheterization complication	Neurologic complication
Group I (4 patients)	
groin hematoma or pseudoaneurysm	sensory neuropathy in all 4 cases • in distribution of medial & intermediate femoral cutaneous nerves → isolated sensory neuropathy (dysesthesia & sensory loss) of the anterior and medial thigh • no motor deficit
Group II (16 patients)	
large retroperitoneal hematoma	femoral neuropathy • sensory in all 16 cases: dysesthesia of the anterior/medial thigh & medial calf • motor in 13 cases: iliopsoas & quadriceps weakness
	obturator neuropathy in 4 cases • sensory: upper medial thigh • motor: obturator weakness
	lateral femoral cutaneous nerve → meralgia paresthetica

Excruciating pain after the catheterization procedure often preceded the development or recognition of neuropathy.

Treatment

After considering available information, the recommendation is to repair pseudoaneurysms surgically, but to treat the neuropathy conservatively. A case could *not* be made that surgical drainage of hematoma reduced the risk of neuropathy. Weakness from femoral or obturator neuropathy was treated with inpatient rehabilitation.

Outcome

Group I patients all had resolution in <5 mos. In group II, 50% had complete resolution in 2 mos. 6 patients had persistent symptoms, 5 had mild femoral *sensory* neuropathy (1 of whom felt it was at least somewhat disabling), 1 had mild persistent quadriceps weakness and occasionally walks with a cane.

32.6 Peripheral nerve injuries

32.6.1 General information

Anatomy of peripheral nerves

See ► Fig. 32.1. Endoneurium surrounds myelinated and unmyelinated axons. These bundles are gathered into fascicles surrounded by perineurium. The epineurium encases the nerve trunk, containing fascicles separated by interfascicular epineurium or mesoneurium.

Nerve regeneration

Peripheral nerves regenerate ≈ 1 mm/day (about 1 inch/month). Divide this figure into the distance that the nerve has to traverse (from knowledge of anatomy) for guide as to how long to wait before considering failure of therapy (either operative or non-operative). However, this rule may not be

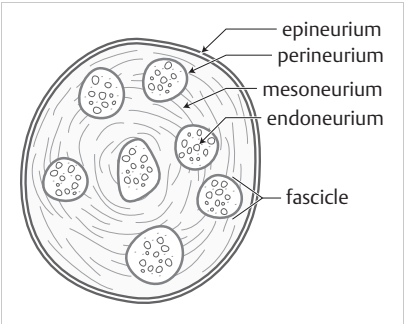


Fig. 32.1 Cross-sectional anatomy of a peripheral nerve.

applicable to long distances ($> \approx 12$ inches), and it may take longer to traverse regions of entrapment, scar or nerve injury. There may also be fibrosis of the muscle beyond salvage.

Peripheral nerve injury classification

See ► Table 32.6.

There are numerous classification systems. The Seddon classification is an older 3-tiered system. The Sunderland system has 5 tiers, essentially dividing axonotmesis into 3 subgroups. Others have added a 6th category as shown in ► Table 32.6.

Table 32.6 Classification of peripheral nerve injury^a

Seddon system	Sunderland system
Neuropraxia	First-degree
Features common to both systems: Physiologic transection (nerve in continuity). Basement membrane intact. Compression or ischemia → local conduction block (impaired axonal transport). ★ No wallerian degeneration ^b . Motor involvement is typically > sensory. Autonomic function is preserved	
Recovers in hours to months; average is 6–8 weeks	Focal demyelination may occur. Recovery is usually complete in 2–3 weeks (not the “1 mm/day rule”)
Axonotmesis	Second-degree
Features common to both systems: Complete interruption of axons and myelin sheaths. Supporting structures (including endoneurium) intact. ★ Wallerian degeneration occurs	Recovers at 1 mm/day as axon follows “tubule.” Sometimes may only be diagnosed retrospectively. Recovery is poor in lesions requiring > 18 months to reach target muscle
	Third-degree
	Endoneurium disrupted, epineurium & perineurium intact. Nerve may not appear seriously damaged on gross inspection. Recovery may range from poor to complete and depends on degree of intrafascicular fibrosis
	Fourth-degree
	Interruption of all neural & supporting elements. Epineurium intact. Grossly: nerve is usually indurated & enlarged
Neurotmesis	Fifth-degree
Nerve completely severed or disorganized by scar tissue. Spontaneous regeneration impossible	Complete transection with loss of continuity
	Sixth-degree ^c
	Mixed lesion. Combination of elements of first through fourth degree. There may be some preserved sensory fascicles (may produce a positive Tinel’s sign)

^acomparing and showing approximate equivalence of Seddon and Sunderland systems

^bwallerian degeneration after British physiologist Augustus Volney Waller (1816–1870), AKA orthograde degeneration, AKA secondary degeneration: degeneration of the axon distal to a focal lesion

^cnot part of original Sunderland system

32.6.2 Brachial plexus injuries

Etiologies

Etiologies include:

1. penetrating trauma
2. traction (stretch injuries): more likely to affect the posterior and lateral cords than the medial cord and median nerve
3. first rib fractures
4. compression by hematoma

Differentiating preganglionic from postganglionic injuries

Initial exam seeks to differentiate preganglionic injuries (proximal to dorsal root ganglion) which cannot be repaired surgically, from postganglionic injuries. Clues to a preganglionic injury include:

1. Horner syndrome: pre-ganglionic injury interrupts white rami communicantes
2. paralysis of serratus anterior (long thoracic nerve): produces winging of scapula
3. paralysis of rhomboids (dorsal scapular nerve)
4. early neuropathic pain suggests nerve root avulsion. MRI or myelogram will show pseudomeningoceles at the avulsed levels
5. EMG: requires ≥ 3 weeks from injury for some findings. Look for:
 - a) denervation potentials in paraspinal muscles due to loss of neural input. The posterior ramus of the spinal nerve originates just distal to the dorsal root ganglion. Due to overlap, cannot localize to a specific segment
 - b) normal sensory nerve action potential (SNAP): preganglionic injuries leave the dorsal ganglion sensory cell body and the distal axon intact, so that normal SNAP can be recorded proximally even in an anesthetic region
6. pseudomeningocele on myelography or MRI: suggests nerve root avulsion (very proximal); however, 15% of pseudomeningoceles are not associated with avulsions, and 20% of avulsions do not have pseudomeningoceles^{35,36}

Types of brachial plexus injuries

(Duchene)-Erb's palsy

Upper brachial plexus injury (C5 & 6, some authors include C7) e.g., from forceful separation of humeral head from shoulder, commonly due to difficult parturition (see below) or motorcycle accident (downward force on shoulder can cause traumatic nerve root avulsion from the spinal cord). Paralysis of deltoid, biceps, rhomboids, brachioradialis, supra- & infraspinatus, and occasionally supinator. C7 involvement produces weak wrist extension.

Motor: arm hangs at side internally rotated & extended at elbow and flexed at the wrist ("Bellhop's tip position"). Hand motion is unaffected.

Klumpke's palsy

Injury to lower brachial plexus (C8 & T1, some authors include C7), from traction of abducted arm e.g., in catching oneself during a fall from a height, or by Pancoast tumor (lung apex tumor—check CXR with apical lordotic view). Characteristic claw deformity (also seen with ulnar nerve injury) with weakness and wasting of small hand muscles. Possible Horner syndrome if T1 involved.

Birth brachial plexus injury (BBPI)

Incidence is 0.3–2.0 per 1000 live births (0.1% in infants with birthweight <4000 gm³⁷). Rarely, a congenital case may be mistaken for BBPI.³⁸ Some contend that the plexus injury may occur when uterine contractions push the shoulder against the mother's pubic bone or with lowering of the shoulder with opposite inclination of the cervical spine.³⁸

Classification of BBPI injuries: Upper plexus injuries are most common, with about half having C5 & C6 injuries, and 25% involving C7 also.³⁹ Combined upper and lower lesions occur in $\approx 20\%$. Pure lower lesions (C7–1) are rare, constituting only $\approx 2\%$ and seen most commonly in breech deliveries. Lesions are bilateral in $\approx 4\%$. A 4-level scale of intensity is shown $\gg$ Table 32.7.⁴⁰

Risk factors:

1. shoulder dystocia
2. high birth weight
3. primiparous mother
4. forceps⁴¹ or vacuum assisted delivery

Table 32.7 Birth brachial plexus injury

Group	Lesion	Manifestation	Spontaneous recovery rate
1	C5 or C6 roots or superior trunk	paralysis of shoulder abduction, elbow flexion & forearm supination. Finger flexion is normal	90%
2	above + involvement of C7 or medial trunk	above + paralysis of finger extensors (but not flexors)	65%
3	above + finger flexors	essentially no hand movement. No Horner syndrome	≈ <50%
4	complete brachial plexus	flail arm + Horner syndrome	0%
	"dominant C7" paralysis variant	selective loss of shoulder abduction & elbow extension	

5. breech presentation⁴²
6. prolonged labor
7. previous birth complicated by BBPI

Management of BBPI: Most surgeons observe all patients until age 3 months. Conservative surgeons may wait up to 9 months. More aggressive surgeons will explore the plexus at age 3 months if not antigravity in deltoid, biceps or triceps. In cases of proven avulsion (pseudomeningocele and EMG indicative of a preganglionic injury), nerve transfers are a valid option at 3 months.⁴³ EMG may show signs of reinnervation, but the recovery may not be robust enough.

Management of brachial plexus injuries

1. most injuries show maximal deficit at onset. Progressive deficit is usually due to vascular injuries (pseudoaneurysm, A-V fistula, or expansile clot); these should be explored immediately
2. clean, sharp, relatively fresh lacerating injuries (usually iatrogenic, scalpel-induced) should be explored acutely and repaired with tension-free end-to-end anastomoses within 24–48 hours (after that, ends will be more edematous and therefore more difficult to suture)
3. penetrating non-missile injuries with severe or complete deficit should be explored as soon as the primary wound heals
4. gunshot wounds (GSW) to the brachial plexus: deficit is usually due to axonotmesis or neurotmesis (see below). Sometimes nerves may be divided. Nerves showing partial function usually recover spontaneously; those with complete dysfunction rarely do so. Surgery is of little benefit for discrete injuries to the lower trunk, medial cord, or C8/T1 roots. Most are managed conservatively for 2–5 months. Indications for surgery are shown in ► Table 32.8

Table 32.8 Indications for neurosurgical intervention in GSW to the brachial plexus⁴⁴

1. complete loss in the distribution of at least one element a) no improvement clinically or on EMG in 2–5 months b) deficit in distribution that is responsive to surgery (e.g., C5, C6, C7, upper or middle trunk, lateral or posterior cords or their outflows) c) injuries with loss only in lower elements are <i>not</i> operated 2. incomplete loss with failure to control pain medically 3. pseudoaneurysm, clot, or fistula involving plexus 4. true causalgia requiring sympathectomy
--

5. traction injuries: incomplete postganglionic injuries tend to improve spontaneously. If recovery is not satisfactory, perform EMG at 4–5 months and explore at 6 months
6. neuromas in continuity: those that do not conduct a SNAP (p.255) have complete internal disruption and require resection and grafting. Methods of repair:
 - a) neurolysis:
 - external neurolysis: most commonly performed in exploration. Value is questionable
 - internal neurolysis: splitting the nerve into fascicles. Not recommended unless a clear neuroma in continuity is found eccentric in the nerve that conducts SNAP
 - b) nerve grafting. Sural nerve is the most commonly used interposition graft following resection of neuroma in continuity

- c) nerve transfers. Donor nerve options:
 - spinal accessory nerve
 - intercostal nerves to musculocutaneous nerve
 - fascicles of the ulnar nerve for the median nerve (Oberlin procedure)
 - anterior interosseus nerve to median nerve

32.7 Missile injuries of peripheral nerves

This section deals primarily with gunshot wounds (GSW). Most injuries from a single bullet are due to shock and cavitation from the missile causing axonotmesis or neurotmesis, and are not from direct nerve transection. Approximately 70% will recover with expectant management.

However, if there is a lack of improvement on serial examinations, including electrodiagnostic studies, intervention should be undertaken by about 5–6 months to avoid further difficulties due to nerve fibrosis and muscle atrophy.

See ► Table 32.8 for indications for surgery for missile injuries of the brachial plexus.

32.8 Thoracic outlet syndrome

32.8.1 General information

The thoracic outlet is a confined area at the apex of the lung bordered by the 1st rib below and the clavicle above through which passes the subclavian artery, vein, and brachial plexus.

Thoracic outlet syndrome (TOS) is a term implying compression of one or more of the enclosed structures producing a heterogeneous group of disorders. TOS tends to be diagnosed more often by general and vascular surgeons than by neurologists and neurosurgeons. Four unrelated conditions with different structures involved:

1. “noncontroversial,” with characteristic symptom complex, reproducible clinical findings, confirmatory laboratory tests. Low incidence⁴⁵
 - arterial vascular: producing arm, hand, and finger pallor and ischemia
 - venous vascular: producing arm swelling and edema
 - true neurologic: compressing the lower trunk or median cord of the brachial plexus (see below)
2. disputed neurologic: includes scalenus anticus syndrome (see below)

32.8.2 Differential diagnosis

1. herniated cervical disc
2. cervical arthrosis
3. lung cancer (pancoast tumor)
4. tardy ulnar nerve palsy
5. carpal tunnel syndrome
6. orthopedic shoulder problems
7. complex regional pain syndrome (reflex sympathetic dystrophy)

32.8.3 True neurologic TOS

General information

A rare condition primarily affecting adult women, usually unilateral.

Neurologic structures involved

1. most common: compression of the C8/T1 roots
2. or proximal lower trunk of the brachial plexus (BP)
3. less common: compression of the median cord of the BP

Etiologies

1. constricting band extending from the first rib to a rudimentary “cervical rib” or to an elongated C7 transverse process
2. scalenus (anticus) syndrome: controversial (see below)
3. compression beneath the pectoralis minor tendon under the coracoid process: may result from repetitive movements of the arms above the head (shoulder elevation and hyperabduction)

Signs and symptoms

1. ★ sensory changes in distribution of median cord (mainly along medial forearm), *sparing* median nerve sensory fibers (pass through upper and middle trunks)
2. hand clumsiness or weakness and wasting, especially abductor pollicis brevis and ulnar hand intrinsic (C8/T1 denervation/atrophy)
3. there may be tenderness over Erb's point (2 to 3 cm above the clavicle in front of the C6 transverse process)
4. may be painless
5. usually unilateral

Confirmatory tests

1. EMG: unreliable (may be negative). Most common abnormality in neurogenic TOS is loss of medial antebrachial cutaneous SNAP (p. 255)
2. MRI does not show bony abnormalities well, but may occasionally demonstrate a kink in the lower BP. Can also rule out conditions that may mimic TOS, such as herniated cervical disc
3. cervical spine X-rays with obliques and apical lordotic CXR may demonstrate bony abnormalities. However, not every cervical rib produces symptoms (some patients with bilateral cervical ribs may have unilateral TOS).

Treatment

Controversial. Conservative treatment (usually including stretching and physical therapy) is about equally as effective as surgery and avoids attendant risks.

Decompression can be achieved by removing the muscles that surround the nerves (scaleneotomy), by transaxillary first rib resection, or both.

32.8.4 Scalenus (anticus) syndrome (disputed neurologic TOS)

Controversial. More commonly diagnosed in the 1940s and 1950s. There is a lack of consensus regarding the pathophysiology (including structures involved), clinical presentation, helpful tests, and optimal treatment. Removal of first thoracic rib is often advocated for treatment, frequently via a transaxillary approach. Unfortunately, injuries, especially to the lower trunk of the brachial plexus, may result from the surgery.

Other variations include an "upper plexus" type for which total anterior scalenectomy is advocated. Again, very controversial.

References

- [1] Poza JJ, Cobo AM, Marti-Masso JF. Peripheral neuropathy associated with polycythemia vera. *Neurologia*. 1996; 11:276–279
- [2] Ochoa JL, Verdugo RJ. Reflex Sympathetic Dystrophy: A Common Clinical Avenue for Somatoform Expression. *Neurol Clin*. 1995; 13:351–363
- [3] Denny-Brown D. Primary Sensory Neuropathy with Muscular Changes Associated with Carcinoma. *Journal of Neurology, Neurosurgery, and Psychiatry*. 1948; 11:73–87
- [4] Camerlingo M, Nemni R, Ferraro B, et al. Malignancy and Sensory Neuropathy of Unexplained Cause: A Prospective Study of 51 Patients. *Archives of Neurology*. 1998; 55:981–984
- [5] McLeod JG, Dyck PJ, Thomas PK. Paraneoplastic Neuropathies. In: *Peripheral Neuropathy*. Philadelphia: W.B. Saunders; 1993:1583–1590
- [6] Turner JW, Parsonage MJ. Neuralgic amyotrophy (paralytic brachial neuritis); with special reference to prognosis. *Lancet*. 1957; 273
- [7] Tsairis P, Dyck PJ, Mulder DW. Natural History of Brachial Plexus Neuropathy: Report on 99 Patients. *Arch Neurol*. 1972; 27:109–117
- [8] Misamore GW, Lehman DE, Parsonage-Turner syndrome (acute brachial neuritis). *Journal of Bone and Joint Surgery*. 1996; 78:1405–1408
- [9] Evans BA, Stevens JC, Dyck PJ. Lumbosacral Plexus Neuropathy. *Neurology*. 1981; 31:1327–1330
- [10] The Diabetes Control and Complications Trial Research Group. The Effect of Intensive Treatment of Diabetes on the Development and Progression of Long-Term Complications in Insulin-Dependent Diabetes Mellitus. *New England Journal of Medicine*. 1993; 329:977–986
- [11] Asbury AK. Proximal Diabetic Neuropathy. *Ann Neurol*. 1977; 2:179–180
- [12] Dyck PJ, Thomas PK. *Peripheral Neuropathy*. 2nd ed. Philadelphia: W. B. Saunders; 1984
- [13] Garland H. Diabetic Amyotrophy. *BMJ*. 1955; 2:1287–1290
- [14] Barohn RJ, Sahenk Z, Warmolts JR, et al. The Bruns-Garland Syndrome (Diabetic Amyotrophy): Revisited 100 Years Later. *Arch Neurol*. 1991; 48:1130–1135
- [15] Pascoe MK, Low PA, Windebank AJ. Subacute Diabetic Proximal Neuropathy. *Mayo Clinic Proceedings*. 1997; 72:1123–1132
- [16] Davis JL, Lewis SB, Gerich JE, et al. Peripheral Diabetic Neuropathy Treated with Amitriptyline and Fluphenazine. *JAMA*. 1977; 21:2291–2292
- [17] Max MB, Lynch SA, Muir J, et al. Effects of Desipramine, Amitriptyline, and Fluoxetine on Pain in Diabetic Neuropathy. *N Engl J Med*. 1992; 326:1250–1256
- [18] Mendel CM, Klein RF, Chappell DA, et al. A Trial of Amitriptyline and Fluphenazine in the Treatment of Painful Diabetic Neuropathy. *JAMA*. 1986; 255:637–639
- [19] Mendel CM, Grunfeld C. Amitriptyline and Fluphenazine for Painful Diabetic Neuropathy. *JAMA*. 1986; 256:712–714

- [20] Backonja M, Beydoun A, Edwards KR, et al. Gabapentin for the symptomatic treatment of painful neuropathy in patients with diabetes mellitus: a randomized controlled trial. *JAMA*. 1998; 280:1831-1836
- [21] Morello CM, Leckband SG, Stoner CP, et al. Randomized double-blind study comparing the efficacy of gabapentin with amitriptyline on diabetic peripheral neuropathy pain. *Arch Intern Med*. 1999; 159:1931-1937
- [22] New Uses of Thalidomide. *Medical Letter*. 1996; 38:15-16
- [23] Fuller GN, Jacobs JM, Guiloff RJ. Nature and Incidence of Peripheral Nerve Syndromes in HIV Infection. *Journal of Neurology, Neurosurgery, and Psychiatry*. 1993; 56:372-381
- [24] Drugs for HIV Infection. *Medical Letter*. 2000; 42:1-6
- [25] Kroll DA, Caplan RA, Posner K, et al. Nerve Injury Associated with Anesthesia. *Anesthesiology*. 1990; 73:202-207
- [26] Warner MA. Perioperative Neuropathies. *Mayo Clinic Proceedings*. 1998; 73:567-574
- [27] Wadsworth TG, Williams JR. Cubital Tunnel External Compression Syndrome. *British Medical Journal*. 1973; 1:662-666
- [28] Warner MA, Marner ME, Martin JT. Ulnar Neuropathy: Incidence, Outcome, and Risk Factors in Sedated or Anesthetized Patients. *Anesthesiology*. 1994; 81:1332-1340
- [29] Alvine FG, Schurrer ME. Postoperative Ulnar-Nerve Palsy: Are There Predisposing Factors? *Journal of Bone and Joint Surgery*. 1987; 69A:255-259
- [30] Stewart JD, Shantz SH. Perioperative ulnar neuropathies: A medicolegal review. *Canadian Journal of Neurological Sciences*. 2003; 30:15-19
- [31] Warner MA, Martin JT, Schroeder DR, et al. Lower-Extremity Motor Neuropathy Associated with Surgery Performed on Patients in a Lithotomy Position. *Anesthesiology*. 1994; 81:6-12
- [32] O'Donnell D, Rottman R, Kotelko D, et al. Incidence of Maternal Postpartum Neurologic Dysfunction. *Anesthesiology*. 1994; 81
- [33] Sanabria EAM, Nagashima T, Yamashita H, et al. Postoperative bilateral meralgia paresthetica after spine surgery: An overlooked entity? *Spinal Surgery*. 2003; 17:195-202
- [34] Kent CK, Moscucci M, Gallagher SG, et al. Neuropathies After Cardiac Catheterization: Incidence, Clinical Patterns, and Long-Term Outcome. *J Vasc Surg*. 1994; 19:1008-1014
- [35] Carvalho GA, Nikkha G, Matthies C, et al. Diagnosis of root avulsions in traumatic brachial plexus injuries: value of computerized tomography myelography and magnetic resonance imaging. *Journal of Neurosurgery*. 1997; 86:69-76
- [36] Hashimoto T, Mitomo M, Hirabuki N, et al. Nerve root avulsion of birth palsy: comparison of myelography with CT myelography and somatosensory evoked potential. *Radiology*. 1991; 178:841-845
- [37] Rouse DJ, Owen J, Goldenberg RL, et al. The effectiveness and costs of elective cesarean delivery for fetal macrosomia diagnosed by ultrasound. *Jama*. 1996; 276:1480-1486
- [38] Gilbert A, Brockman R, Carlouz H. Surgical Treatment of Brachial Plexus Birth Palsy. *Clinical Orthopedics*. 1991; 264:39-47
- [39] Boome RS, Kaye JC. Obstetric Traction Injuries of the Brachial Plexus: Natural History, Indications for Surgical Repair and Results. *Journal of Bone and Joint Surgery*. 1988; 70B:571-576
- [40] van Ouwkerk WJ, van der Sluijs J A, Nollot F, et al. Management of obstetric brachial plexus lesions: state of the art and future developments. *Childs Nerv Syst*. 2000; 16:638-644
- [41] Piatt JH, Hudson AR, Hoffman HJ. Preliminary Experiences with Brachial Plexus Explorations in Children: Birth Injury and Vehicular Trauma. *Neurosurgery*. 1988; 22:715-723
- [42] Hunt D. Surgical Management of Brachial Plexus Birth Injuries. *Developmental Medicine and Child Neurology*. 1988; 30:821-828
- [43] Anand P, Birch R. Restoration of sensory function and lack of long-term chronic pain syndromes after brachial plexus injury in human neonates. *Brain*. 2002; 125:113-122
- [44] Kline DG, Hudson AR. Nerve Injuries: Operative Results for Major Nerve Injuries, Entrapments, and Tumors. Philadelphia: W. B. Saunders; 1995
- [45] Wilbourn AJ. The Thoracic Outlet Syndrome is Overdiagnosed. *Arch Neurol*. 1990; 47:328-330

Part XI

Neurophthalmology and Neurotology

33	Neurophthalmology	586
34	Neurotology	603

33 Neurophthalmology

33.1 Nystagmus

33.1.1 Definition

Involuntary rhythmic oscillation of the eyes, usually conjugate. Most common form is jerk nystagmus, in which the direction of the nystagmus is defined for the direction of the fast (cortical) component (which is *not* the abnormal component). Horizontal or upward gaze-provoked nystagmus may be due to sedatives or ASMs; otherwise vertical nystagmus is indicative of posterior fossa pathology.

33.1.2 Localizing lesion for various forms of nystagmus

1. seesaw nystagmus: intorting eye moves up, extorting eye moves down, pattern then reverses. Lesion in diencephalon. Also reported with chiasmal compression (occasionally accompanied with bitemporal hemianopia in parasellar masses)
2. convergence nystagmus: slow abduction of eyes followed by adducting (converging) jerks, usually associated with features of Parinaud's syndrome. May be associated with nystagmus retractorius (see below) with similar location of lesion
3. nystagmus retractorius: resulting from co-contraction of all EOM's. May accompany convergence nystagmus. Lesion in upper midbrain tegmentum (usually vascular disease or tumor, especially pinealoma)
4. downbeat nystagmus: nystagmus with the fast phase downward while in primary position. Most patients have a structural lesion in the posterior fossa, especially at the *cervicomedullary junction* (foramen magnum (FM)),¹ including *Chiari I malformation*, basilar impression, p-fossa tumors, syringobulbia.² Uncommonly occurs in multiple sclerosis (MS), spinocerebellar degeneration, and in some metabolic conditions (hypomagnesemia, thiamine deficiency, alcohol intoxication or withdrawal, or treatment with phenytoin, carbamazepine or lithium³)
5. upbeat nystagmus: lesion in medulla
6. abducting nystagmus occurs in INO. Lesion in pons (MLF)
7. Brun's nystagmus: lesion in pontomedullary junction (PMJ)
8. vestibular nystagmus: lesion in PMJ
9. ocular myoclonus: lesion in myoclonic triangle
10. periodic alternating nystagmus (PAN): lesion in FM and cerebellum
11. square wave jerks, macro square wave jerks, macro saccadic oscillations. Lesion in cerebellar pathways
12. "nystagmoid" eye movements (not true nystagmus)
 - a) ocular bobbing (p.601): lesion in pontine tegmentum
 - b) ocular dysmetria: overshoot of eye on attempted fixation followed by diminishing oscillations until eye "hones in" on target. Lesion in cerebellum or pathways (may be seen in Friedreich's ataxia)
 - c) ping-pong gaze (p.323)
 - d) "windshield wiper eyes" (p.323)

33.2 Papilledema

33.2.1 General information

AKA choked (optic) disc. The term papilledema is reserved for optic disc swelling caused by increased intracranial pressure, as distinguished from papillitis.

Pathomechanics: elevated ICP is transmitted through the subarachnoid space of the optic nerve sheath to the region of the optic disc → axoplasmic stasis. Elevated ICP will usually obliterate retinal venous pulsation if the pressure is transmitted to the point where the central retinal vein passes through the subarachnoid space (≈ 1 cm posterior to the globe). Papilledema (PPD) may also be dependent on the ratio of retinal arterial to retinal venous pressure, with ratios $<1.5:1$ more commonly associated with papilledema than higher ratios.

Elevated ICP usually causes *bilateral* papilledema (see below for unilateral papilledema).

Papilledema typically takes 24–48 hours to develop following a sustained rise in ICP. It is rarely seen as early as ≈ 6 hours after onset, but not earlier. Papilledema does not cause visual blurring or reduction of visual fields unless very severe and prolonged.

33.2.2 Findings on funduscopy (ophthalmoscopy)

Funduscopy exam is best performed with pharmacologically dilated pupils, usually by an ophthalmologist. Findings include:

- venous engorgement: usually the earliest sign
- loss of venous pulsations
- blurring of optic margins
- elevation of optic disc
- although not part of papilledema, associated findings may include retinal hemorrhages (sub-retinal, pre-retinal, intraretinal), retinal and choroidal folds, Paton's folds, venous tortuosity, retino-choroidal collateral veins, partial macular star

Papilledema is often graded using the Frisén scale (0–5)⁴ (► Table 33.1).

Table 33.1 Modified Frisén grading scale⁵ for papilledema

Frisén grade	Description
0	Normal optic disc • minimal swelling of nasal margin of optic disc • nerve fiber layer (NFL) clear • vessels not obscured • cup, if present, not obscured
1	Minimal papilledema • 230° “C-shaped” swelling of nasal, superior & inferior borders • normal (sharp) temporal margin (temporal gap) • cup, if present, is maintained
2	Low degree of papilledema • elevation of nasal margin • 360° disc swelling (circumferential halo) • no obscuration of major vessels
3	Moderate degree of papilledema • elevation of entire disc • 360° disc swelling • obscuration of ≥ 1 segment of major blood vessel at disc margin • cup may be obscured
4	Marked degree of papilledema • NFL opaque, 360° disc swelling • vessels obscured at disc margin, not completely obscured on disc surface
5	Severe papilledema • NFL opaque, 360° disc swelling • all vessels obscured on disc surface & leaving disc

33.2.3 Imaging findings in papilledema

Papilledema is primarily an ophthalmologic diagnosis. Some imaging correlates that may be associated:

- MRI or CT: flattening of the posterior globe (► Fig. 33.1) or elevation of the optic nerve head are specific findings. An empty sella may be seen with increased intracranial pressure, independent of papilledema
- MRI, CT or ultrasound: dilation of the optic nerve sheath (► Fig. 33.1) (optic nerve sheath hydrops) may be demonstrated (with current improved imaging techniques this is also now being seen more frequently in normal patients), enlargement of Meckel's cave (► Fig. 57.1)

33.2.4 Differential diagnosis

► Mimics of papilledema on funduscopy

1. optic neuritis. Usually associated with more severe visual loss and tenderness to eye pressure over the eye. Etiologies:
 - a) infection: Lyme disease, syphilis, measles, mumps, herpes. Cat-scratch fever (infection with *Bartonella henselae*) can rarely cause optic neuritis (AKA *Bartonella papillitis*)
 - b) inflammatory/granulomatous: sarcoidosis, lupus, Behçet's disease

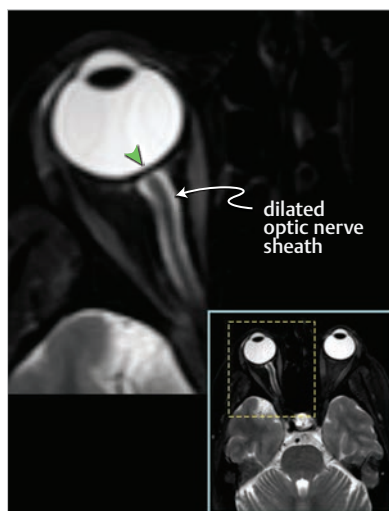


Fig. 33.1 Dilated optic nerve sheath and flattened posterior globe (green arrowhead). Image: T2 axial MRI. Yellow broken line in inset shows from where the detail is taken.

2. pseudopapilledema: anomalous elevation of the optic nerve head. The optic disc may appear swollen, but, unlike true papilledema, the peripapillary vessels are not obscured and retinal venous pulsations are usually present. It may be unilateral or bilateral. Benign conditions that can cause pseudopapilledema include: hyperopia (farsightedness), a small optic cup, buried disc drusen (p. 1659)... Extensive workup is generally not indicated
3. non-arteritic anterior ischemic optic neuropathy (NAAION) can cause optic nerve head swelling
4. optic nerve edema. Can occur e.g., in malignant hypertension

► Etiologies of unilateral papilledema

1. compressive lesions
 - a) orbital tumors: meningiomas...
 - b) tumors of optic nerve sheath: schwannoma
 - c) optic nerve tumors (optic pathway gliomas)
2. local inflammatory disorder
3. Foster Kennedy syndrome (p. 100)
4. demyelinating disease (e.g., multiple sclerosis)
5. elevated ICP in the setting of something that prevents manifestation in the other eye, including:
 - a) blockage that prevents transmission of elevated CSF pressure to that optic disc⁶
 - b) prosthetic eye (artificial eye)

33.3 Visual fields

33.3.1 General information

From a point of fixation, the normal eye can detect stimuli as far as 60° superiorly, 70° inferiorly, 60° nasally, and 100° temporally⁷ (this varies with size, color and brightness of the stimuli). The normal physiological blind spot (caused by absence of light receptors in the optic disc due to penetration of the retina by the optic nerve and vessels) is located on the temporal side of the macular visual area in each eye.

The fibers from the temporal retina pass directly to the ipsilateral lateral geniculate body (LGB), whereas the nasal fibers cross in the optic chiasm (► Fig. 33.2).

33.3.2 Visual field testing

Visual fields may be evaluated by:

1. bedside confrontational testing: detects only gross peripheral field deficits. Using “finger wiggle” with the patient’s gaze fixated on the examiner’s nose while covering one of their eyes is the least

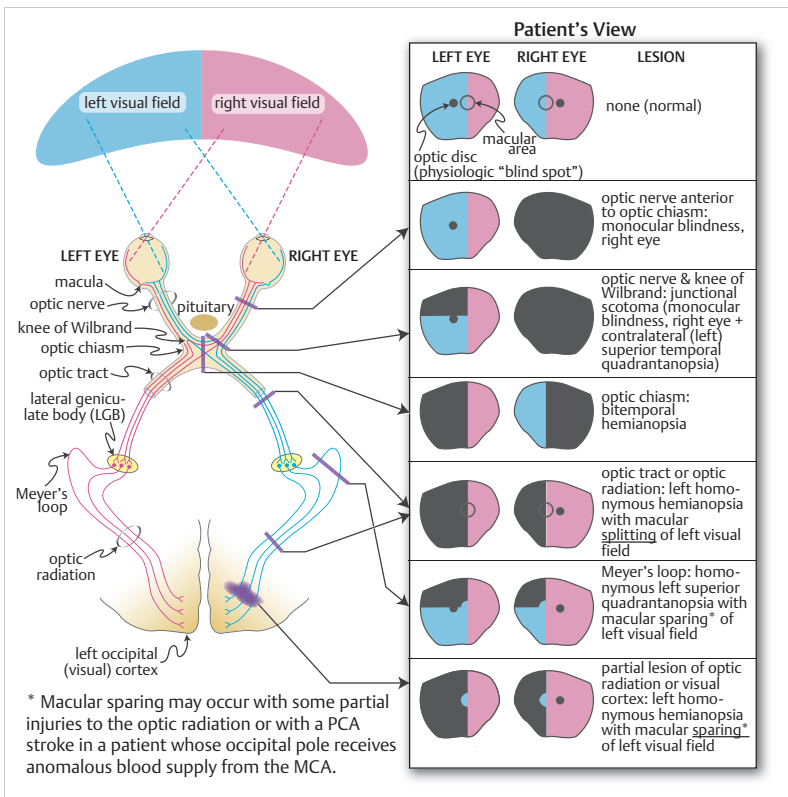


Fig. 33.2 Visual field deficits.

accurate. Presenting simultaneous fingers extended in the temporal fields and asking the patient to add the total number of fingers seen is somewhat better, and assesses homonymous hemianopia

2. formal perimetry

- using a tangent screen (typically using Goldmann perimetry). Stimulus is brought in from non-seeing to seeing field (towards macular vision area) along 8 meridians
- automated perimetry exam: Humphrey visual field (HVF) (Zeiss) or Octopus (Haag-Streit) perimeters. Requires good patient cooperation to be valid

► **Perimetric mean deviation.** With automated perimetry, a “perimetric mean deviation” (PMD) is calculated as an average deviation from age and race matched normal sighted individuals at multiple locations in the visual field. A positive score means the subject sees dimmer stimuli than controls, a negative score means they require brighter stimuli. A PMD of -2 to -7 would be consistent with a mild visual loss.

33.3.3 Visual field deficits

► **Visual field deficits patterns.** Visual field deficit patterns are shown in ► Fig. 33.2.

► **Macular sparing/splitting.** Macular splitting can occur both in lesions anterior or posterior to the lateral geniculate body (LGB). However, macular sparing tends to occur with lesions posterior to the LGB. Homonymous hemianopia with macular sparing usually occurs with lesions of the optic

radiation or infarcts of the primary visual cortex. There is more than 1 way for this to occur: input from the macula is spread over a large portion of the optic radiation and primary visual cortex, and in some cases the occipital pole (primary visual cortex) receives dual blood supply (probably uncommon) or anomalous blood supply from the MCA.

► **Knee of Wilbrand.** Named for Hermann (or Herman) Wilbrand (1851–1935), a German neuro-ophthalmologist, whose surname is often misreported as von Willebrand, Willebrand, or Wildbrand. A 1–2-mm anterior “bend” of the decussating fibers of the optic chiasm extending into the contralateral optic nerve before continuing to the optic tract⁸ (► Fig. 33.2). Initially identified histologically postmortem in subjects who had monocular enucleation. Conventional teaching was that optic nerve injury close to the chiasm produces an anterior chiasmal syndrome, AKA junctional scotoma which consists of ipsilateral central scotoma (from the injury to the optic nerve) and a contralateral superior temporal quadrantanopia (from damage of the decussating “knee” fibers).^{8,9,10} Controversy as to the existence or significance of Wilbrand’s knee initially arose after further cadaveric studies suggested that Wilbrand’s knee is an anatomic artifact resulting from buckling of the decussating fibers into the contralateral optic nerve as the optic nerve and chiasm atrophy following enucleation.⁸ However, advanced optical imaging techniques have demonstrated a forward bend of the anterior inferior decussating fibers in chiasms with no pre-mortem pathology.¹¹ And yet, case series with intraoperative sectioning of the optic nerve at the level of the chiasm have not been shown to develop contralateral visual field deficits.^{12,13}

► **Meyer’s loop.** The inferior portion of the optic radiation loops forward into the posterior temporal lobe before continuing on to the occipital lobe (► Fig. 33.2) and is known as the temporal pathway or Meyer’s loop. Lesions here cause homonymous wedge-shaped defects in the contralateral superior visual field (i.e., bilateral superior quadrantanopsia), so-called “pie in the sky” visual defect.¹⁴

33.4 Pupillary diameter

33.4.1 Pupilodilator (sympathetic)

Pupilodilator muscle fibers are sympathetically innervated and are arranged radially in the iris.

First-order sympathetic nerve fibers arise in the posterolateral hypothalamus, and descend uncrossed in the lateral tegmentum of the midbrain, pons, medulla, and cervical spinal cord to the intermediolateral cell column of the spinal cord from C8–2 (ciliospinal center of Budge-Waller). Here they synapse with lateral horn cells (neurotransmitter: ACh) and give off 2nd order neurons (preganglionics).

Second-order neurons exit the spinal cord at T1 and enter the sympathetic chain and ascend but do not synapse until they reach the superior cervical ganglion, where they give rise to 3rd order neurons.

Third-order neurons (postganglionics) course upward with the common carotid artery. Those that mediate sweat in the face split off with the ECA (exception: fibers to the medial forehead do not branch off¹⁵). The rest travel with the ICA passing over the carotid sinus, entering the skull through the carotid canal, traversing the cavernous sinus where they follow the VI nerve for a short distance before they accompany V1 (ophthalmic division of trigeminal nerve), entering the orbit through the superior orbital fissure with the nasociliary nerve, passing through (without synapsing) the ciliary ganglion. Upon exiting the ciliary ganglion the sympathetic fibers divide, some reach the pupilodilator muscle of the eye as 2 long ciliary nerves (neurotransmitter: NE). Other fibers travel with the ophthalmic artery to innervate the lacrimal gland and Müller’s muscle (the accessory levator muscle of the upper eyelid).

33.4.2 Pupilloconstrictor (parasympathetic)

Pupilloconstrictor muscle fibers are parasympathetically innervated and are arranged as a sphincter (i.e., circular) in the iris.

Parasympathetic preganglionic fibers arise in the Edinger-Westphal nucleus (in high midbrain, superior colliculus level) and are situated peripherally on the intracranial portion of the oculomotor nerve (p. 596).

33.4.3 Pupillary light reflex

Involves cranial nerves II & III, and neurons in 2 brainstem nuclei (Edinger-Westphal and pretectal nuclear complex).

Mediated by rods and cones of the retina which are stimulated by light, and transmit via their axons in the optic nerve. As with the visual path, temporal retinal fibers remain ipsilateral, whereas nasal retinal fibers decussate in the optic chiasm. Fibers subserving the light reflex bypass the lateral geniculate body (LGB) (unlike fibers for vision which enter the LGB) to synapse in the pretectal nuclear complex at the level of the superior colliculus. Intercalating neurons connect to both Edinger-Westphal parasympathetic motor nuclei. The preganglionic fibers travel within the third nerve to the ciliary ganglion as described above under Pupilloconstrictor (parasympathetic).

Monocular light normally stimulates bilaterally symmetric (i.e., equal) pupillary constriction (ipsilateral response is called the direct response, contralateral response is the consensual response).

33.4.4 Pupillary exam

To perform a complete bedside pupillary exam (see following sections for rationale for various aspects of the pupillary exam):

1. measure pupil size in a dimly lit room: anisocoria augmented in the dark indicates the smaller pupil is abnormal and suggests a sympathetic lesion
2. measure pupil size in a lighted room: anisocoria intensified in the light suggests the larger pupil is abnormal and that the defect is in the parasympathetics
3. note the reaction to bright light (direct and consensual)
4. near response (it is necessary to check this only if the light reaction is not good): the pupil normally constricts on convergence, and this response should be greater than the light reflex (accommodation is not necessary, and a visually handicapped patient can be instructed to follow their own finger as it is brought in)
 - a) light-near dissociation: pupillary constriction on convergence but absent light response (Argyll Robertson pupil). Etiologies:
 - classically described in syphilis
 - Parinaud's syndrome (p. 100): dorsal midbrain lesion
 - oculomotor neuropathy (usually causes a tonic pupil as in oculomotor compression, see below): DM, EtOH
 - Adie's pupil: see below
5. swinging flashlight test: alternate the flashlight from one eye to the other with as little delay as possible; watch ≥ 5 seconds for the pupil to redilate (dilation after initial constriction is called pupillary escape and is normal due to retinal adaptation). Normal: direct and consensual light reflexes are equal. Afferent pupillary defect (see below): consensual reflex is stronger than the direct (i.e., pupil is larger on direct illumination than contralateral illumination)

33.4.5 Alterations in pupillary diameter

Anisocoria

General information

► **Definition.** Unequal pupil sizes (usually ≥ 1 mm difference).

► **Note.** An afferent pupillary defect (APD) (even with total blindness in one eye) alone does *not* produce anisocoria (i.e., an APD together with anisocoria indicates two separate lesions).

Evaluation

1. history is critically important. Check for exposure to drugs that affect pupillary size, trauma. Look at old photos (e.g., driver's license) for physiologic anisocoria
2. exam: see Pupillary exam above
3. a non-contrast CT is usually not helpful and can provide a false sense of security

Differential diagnosis

1. physiologic anisocoria: occurs in $\approx 20\%$ of population (more common in people with a light iris). Familial and nonfamilial varieties exist. The difference in pupils is usually < 0.4 mm. The inequality is the same in a light and dark room (or slightly worse in the dark)
2. pharmacologic pupil (see below): the most common cause of *sudden onset* of anisocoria. Drugs include:
 - a) mydriatics (pupillary dilators):
 - sympathomimetics (stimulate the dilator pupillae): usually cause only 1–2 mm of dilation, may react slightly to light. Includes phenylephrine, clonidine, naphazoline (an ingredient in OTC eye drops for allergies), eye contact with cocaine, certain plants (e.g., jimsonweed)

- parasympatholytics (inhibit the sphincter pupillae): cause maximal dilation (up to 8 mm) that does *not* react to light. Includes tropicamide, atropine, scopolamine (including patches for motion sickness), certain plants (e.g., deadly nightshade)
- b) miotics (pupillary constrictors): pilocarpine, organophosphates (pesticides), flea powders containing anticholinesterase
- 3. Horner syndrome: interruption of sympathetics to pupilodilator. The abnormal pupil is the *smaller* (miotic) pupil. If there is ptosis it will be on the side of the *small* pupil. See etiologies, etc. (p. 594)
- 4. third nerve palsy (p. 596). If there is ptosis, it will be on the side of the *large* pupil
 - a) oculomotor neuropathy ("peripheral" neuropathy of the third nerve): usually spares the pupil. Etiologies: DM (usually resolves in $\approx$ 8 weeks), EtOH...
 - b) third nerve compression: tends *not* to spare pupil (i.e., pupil is dilated). Produces loss of parasympathetic tone. Etiologies include:
 - aneurysm:
 - PComA (the most common aneurysm to cause this)
 - basilar bifurcation (occasionally compresses the posterior III nerve)
 - uncal herniation: below
 - tumor
 - cavernous sinus lesions: including cavernous internal carotid aneurysm, carotid-cavernous fistula, cavernous sinus tumors
- 5. Adie's pupil (AKA tonic pupil): see below
- 6. local trauma to the eye: traumatic iridoplegia. Injury to the pupillary sphincter muscle may produce mydriasis or, less often, miosis, shape may be irregular
- 7. pontine lesions
- 8. eye prosthesis (artificial eye) AKA pseudoanisocoria
- 9. occasionally some patients have anisocoria that occurs only during migraine¹⁶
- 10. iritis
- 11. keratitis or corneal abrasion

Marcus Gunn pupil ([relative] afferent pupillary defect [RAPD])

AKA (relative) afferent pupillary defect (APD or RAPD), AKA amaurotic pupil. Finding: consensual pupillary reflex to light is stronger than the direct (normal responses are equal). Contrary to some textbooks, the amaurotic pupil is *not* larger than the other.¹⁷ The presence of the consensual reflex is evidence of a preserved third nerve (with parasympathetics) on the side of the impaired direct reflex. Almost always associated with some visual loss in the affected eye. There are other causes of visual loss that do *not* produce an RAPD, such as a vitreous hemorrhage.

Testing: best detected with the swinging flashlight test (see above).

NB: Pupils should remain equal to each other throughout the test. In a dimly lit room, when you shine a light into a normal eye, both pupils constrict equally. If you move the light from a normal eye to an eye with RAPD, both pupils will dilate slightly because less signal is being transmitted through the pre-chiasmal optic nerve.

Etiologies

Lesion *anterior to the chiasm* ipsilateral to the side of the impaired direct reflex:

1. either in the retina (e.g., retinal detachment, retinal infarct e.g., from embolus)
2. or optic nerve, as may occur in:
 - a) optic or retrobulbar neuritis: commonly seen in MS, but may also occur after vaccinations or viral infections, and usually improves gradually
 - b) trauma to the optic nerve: indirect (p. 1014) or direct
 - c) compression by tumor anterior to the chiasm

Adie's pupil (tonic pupil)

An iris palsy resulting in a dilated pupil, due to impaired postganglionic parasympathetics. Thought to be due to a viral infection of the ciliary ganglion. When associated with loss of all muscle tendon reflexes it is called Holmes-Adie's (is not limited to knee jerks, as some texts indicate). Typically seen in a woman in her twenties.

Slit-lamp exam shows some parts of iris contract and others don't.

These patients exhibit light-near dissociation (see above): in checking near response it is necessary to wait a few seconds.

Denervation supersensitivity: usually occurs after several weeks (not in acute phase). Administer two drops of dilute pilocarpine (0.1–0.125%), a parasympathomimetic, in each eye. Miosis (constriction) will occur in Adie's pupil within 30 minutes (normal pupils will react only to $\approx$ 1% pilocarpine).

Pharmacologic pupil

General information

Follows administration of a mydriatic agent. The mydriatic agent may be “occult” when other care providers have not been alerted that this has been used on a patient (this should always be noted in the chart), or when health care personnel unwittingly inoculate agents, e.g., scopolamine, atropine¹⁸... into a patient's eye or into their own eye. May present with accompanying H/A, and if it is unknown that a mydriatic is involved, this may be misinterpreted e.g., as a warning of an expanding PComA aneurysm.

A pharmacologically dilated pupil is very large (7–8 mm), and is larger than typical mydriasis due to third nerve compression (5–6 mm).

To differentiate pharmacologic pupil from a third nerve lesion: instill 1% pilocarpine (a parasympathomimetic) in both eyes (for comparison). A pharmacologic pupil does *not* constrict, whereas the normal side and a dilated pupil from a third nerve palsy will.

Agents

Drugs intentionally used by physicians to dilate the pupils (e.g., Mydracyl, see below). For other mydriatics, see above.

Management

Option: admit and observe overnight, pupil should normalize.

Using mydriatic agents to produce pupillary dilatation

Indications: to improve the ability to examine the retina. NB: ability to follow bedside examination of pupils will be lost for duration of drug effect. This could mask pupillary dilatation from third nerve compression due to herniation. Always alert other caregivers and place a note in the chart to document that the pupil has been pharmacologically dilated (see above), including the agent(s) used and the time administered.

R: 2 gtt of 0.5% or 1% tropicamide (Mydracyl®) blocks the parasympathetic supply to pupil, and produces a mydriasis that lasts a couple hrs to half a day. This can be augmented with 1 gtt 2.5% phenylephrine ophthalmic (Mydrfrin®, Neofrin®, Phenoptic® and others) which stimulates the sympathetics.

Oculomotor nerve compression

Third nerve compression may manifest initially with a mildly dilated pupil (5–6 mm). Possible etiologies include uncal herniation or expansion of a PComA or basilar bifurcation aneurysm. However, within 24 hours, most of these cases will also develop an oculomotor palsy (with down and out deviation of the eye and ptosis). These pupils respond to mydriatics and to miotic agents (the latter helps differentiate this from a pharmacologic pupil, see above).

Although it is possible for a unilaterally dilated pupil alone to be the initial presentation in uncal herniation, in actuality almost all of these patients will have some other finding, e.g., alteration in mental status (confusion, agitation, etc.) before midbrain compression occurs (i.e., it would be rare for a person undergoing early uncal herniation with a dilating pupil to be awake, talking, appropriate and neurologically intact).

Neuromuscular blocking agents (NMBAs)

Due to the absence of nicotinic receptors on the iris, non-depolarizing muscle blocking agents, such as pancuronium (Pavulon®) normally do not alter pupillary reaction to light¹⁹ except in large doses where some of the first and second order neurons may be blocked.

Paradoxical pupillary reaction

Pupils constrict when light is removed.

1. congenital stationary night blindness
2. Best disease: autosomal dominant hereditary progressive macular dystrophy
3. optic nerve hypoplasia
4. retinitis pigmentosa

33.4.6 Horner syndrome

General information

Horner syndrome (HS) is caused by interruption of sympathetics to the eye and face anywhere along their path; see Pupilodilator (sympathetic) (p.590). Unilateral findings on the involved side in a fully developed Horner syndrome are shown in ► Table 33.2.

Table 33.2 Findings in (complete) Horner syndrome

- miosis (constricted pupil)
- ptosis
- enophthalmos
- hyperemia of eye
- anhidrosis of half of face

Miosis in HS

The miosis (pupillary constriction) in Horner syndrome is only $\approx 2\text{--}3\text{ mm}$. This will be accentuated by darkening the room, which causes the normal pupil to dilate.

Ptosis and enophthalmos

Ptosis is due primarily to paralysis of the superior and inferior tarsal muscles (weakness of the inferior tarsal muscle is technically called “inverse ptosis”). Enophthalmos is due to Müller’s muscle paralysis, which also contributes a maximum of $\approx 2\text{ mm}$ to the ptosis. Ptosis in HS is partial, c.f. complete ptosis, which is due to weakness of levator palpebra superioris, which is not involved in Horner syndrome.

Possible sites of disruption of sympathetics

General information

See also anatomy of 1st, 2nd, and 3rd order sympathetic neurons (p.590).

1st order neuron (central neuron)

Interruption is often accompanied by other brainstem abnormalities. Etiologies of dysfunction: infarction from vascular occlusion (usually PICA), syringobulbia, intraparenchymal neoplasm.

2nd order neuron (preganglionic)

Etiologies of dysfunction: lateral sympathectomies, significant chest trauma, apical pulmonary neoplasms²⁰ (Pancoast tumor), high thoracic or cervical neuroblastoma.

3rd order neuron (postganglionic)

The most common type. Etiologies of dysfunction: neck trauma, carotid vascular disease/studies (e.g., carotid dissections (p.1578)), cervical bony abnormalities, migraine, skull base neoplasms, cavernous sinus lesions (e.g., meningioma). With involvement only of fibers on ICA, anhidrosis does not occur (i.e., sweating is preserved) on ipsilateral face since fibers to facial sweat glands travel with ECA.

Pharmacologic testing in Horner syndrome

Establishing the diagnosis

The following may be used if the diagnosis of Horner syndrome is in doubt. It is not necessary when a pupil lag upon darkening the room can be demonstrated in the affected eye. This test does *not* localize the lesion, e.g., as 1st order, 2nd order or 3rd order, and prevents localizing testing the same day.

Cocaine. **R:** 1 gtt 4% cocaine OU (not the 10% solution that is commonly used in ENT procedures which will also anesthetize the sphincter pupillae, thus preventing miosis), repeat in 10 min. Observe pupils over 30 min. Cocaine blocks the NE re-uptake of postganglionics at the neuroeffector junction. In HS, no NE is released and cocaine will not dilate the pupil (► Fig. 33.3). HS is not present if the pupil dilates normally. Delayed dilatation occurs in partial HS.

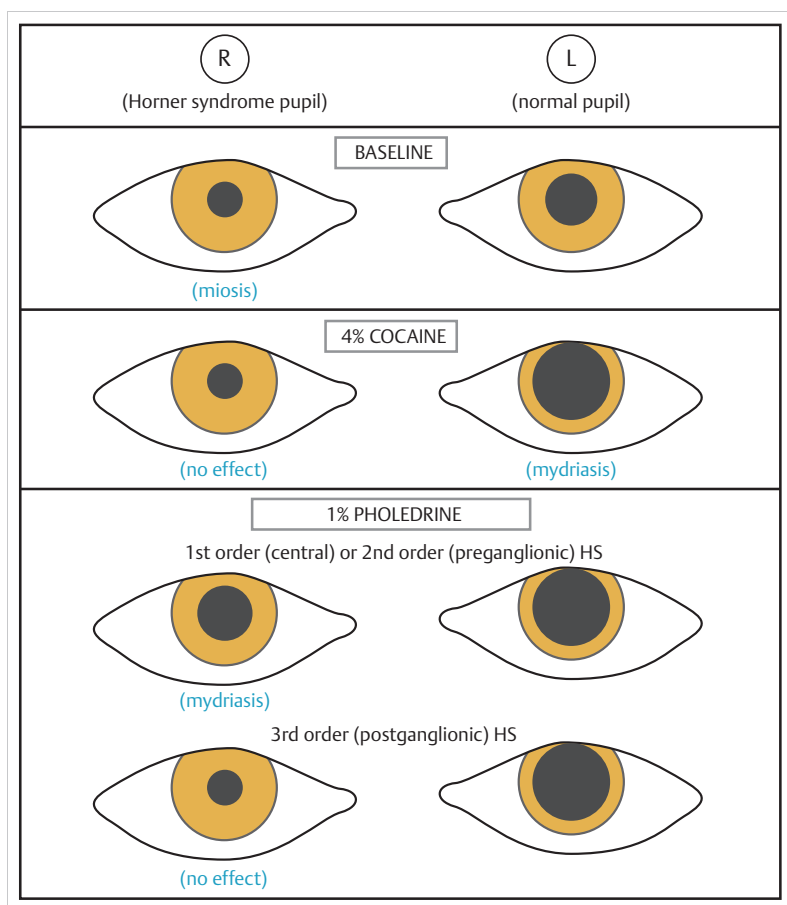


Fig. 33.3 Pharmacologic testing in Horner syndrome (HS).

This figure illustrates the findings in a right-sided HS.

Baseline: the abnormal right pupil will be smaller (miosis). This is accentuated in a darkened room where the unaffected left pupil dilates normally.

Cocaine is used when the diagnosis is uncertain. Only the normal pupil dilates (mydriasis).

Pholedrine is used to distinguish a 3rd order HS (where the affected pupil does not dilate) from a 1st or 2nd order HS (where it dilates some).

See text for details.

1% apraclonidine ophthalmic (Iopidine®): an alpha-agonist, can replace cocaine for establishing the diagnosis. It produces mydriasis in the affected eye due to denervation hypersensitivity in the pupildilator muscle fibers, while the unaffected eye dilates 0–0.5 mm.²¹

Localizing the site of the lesion

First order (central) HS: is generally easy to diagnosis as it is usually accompanied by other hypothalamic, brainstem, or medullary findings.

To differentiate a first or second-order from a third-order HS: 1% *pholedrine* (a substitute for the no longer available hydroxyamphetamine) releases NE from nerve endings at the neuroeffector

junction, causing pupillary dilation except in 3rd order neuron lesions (injured postganglionics do not release NE) (► Fig. 33.3).²² Cannot be used on the same day as the cocaine test.

33.5 Extraocular muscle (EOM) system

33.5.1 Neuroanatomy

Cranial nerves of the EOM system

1. Cr. N. III (oculomotor): innervates the ipsilateral medial rectus (MR), inferior rectus (IR), inferior oblique (IO), and superior rectus (SR)
2. Cr. N. IV (trochlear): innervates the ipsilateral superior oblique (SO), contralateral to the trochlear nucleus (p.598)
3. Cr. N. VI (abducens): innervates the ipsilateral lateral rectus (LR)

Brain anatomy of the EOM system

The frontal eye field (Brodmann's area 8, ► Fig. 1.1) in the frontal lobe, anterior to the primary motor cortex) initiates voluntary (supranuclear) lateral saccadic eye movements ("pre-programmed," rapid, ballistic) to the *opposite* side, involved in suppressing reflexive saccades and generating voluntary, non-visual saccades. It's corticobulbar fibers pass through the genu of the internal capsule to the paramedian pontine reticular formation (PPRF), which controls horizontal gaze, which sends fibers to the ipsilateral abducens/para-abducens (VI) nuclear complex, and via the medial longitudinal fasciculus (MLF) to the contralateral III nucleus to innervate the contralateral MR. Inhibitory fibers go to the ipsilateral third nerve to inhibit the antagonist MR muscle. Thus, the right PPRF controls lateral eye movements to right.

33.5.2 Internuclear ophthalmoplegia

Internuclear ophthalmoplegia (INO) is due to a lesion of the MLF (see above) rostral to the abducens nucleus. Findings in unilateral INO²³ (see ► Fig. 33.4 for illustration):

1. on attempting to look to the side contralateral to the INO:
 - a) the eye ipsilateral to the lesion fails to ADDuct completely
 - b) abduction nystagmus in the contralateral eye (monocular nystagmus) often with some weakness of ABDuction
2. convergence is *not* impaired in isolated MLF lesions (INO is not an EOM palsy)

The most common causes of INO:

1. MS: the most common cause of bilateral INO in young adults
2. brainstem stroke: the most common cause of unilateral INO in the elderly

33.5.3 Oculomotor (Cr. N. III) nerve palsy (OMP)

General information

The oculomotor nerve exits the brainstem ventrally and has two components: motor neurons which originate in the oculomotor nucleus, and more *peripherally situated* parasympathetic fibers which arise from the Edinger-Westphal nucleus. The nerve passes through the cavernous sinus and enters the superior orbital fissure where it divides into a superior division (innervating the superior rectus and the levator palpebrae superioris) and an inferior division (supplying the medial rectus, inferior rectus and inferior oblique). The parasympathetic fibers travel with the inferior division and branch off to the ciliary ganglion where they synapse. Postganglionic fibers enter the posterior globe to innervate the ciliary muscle (relaxes the lens which "thickens" and accommodates for near vision) and the constrictor pupillae muscle.

Oculomotor nerve *motor* palsy causes ptosis with eye deviated "down & out." Nuclear involvement of 3rd nerve is rare. NB: 3rd nerve palsy alone can cause up to 3 mm exophthalmos (proptosis) from relaxation of the rectus muscles.

Also see Painful ophthalmoplegia (p.599) and Painless ophthalmoplegia (p.600). For brainstem syndromes, see Benedikt's syndrome (p.101) and Weber's syndrome (p.101). Also, see Anisocoria (p.591).

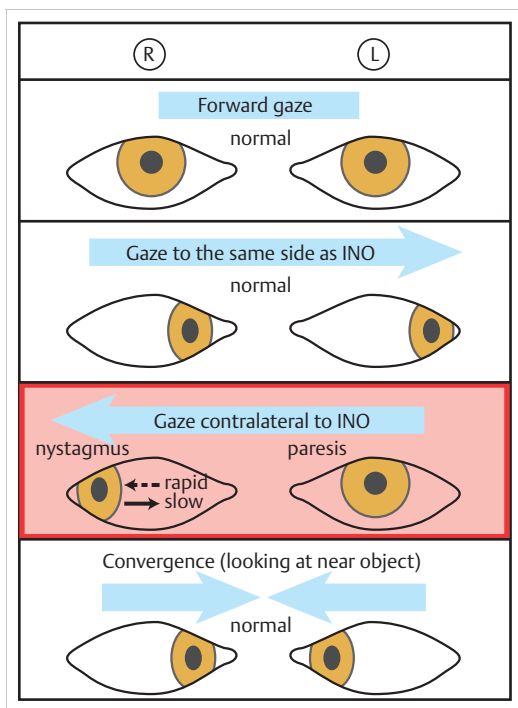


Fig. 33.4 Internuclear ophthalmoplegia (INO). Illustration of gaze findings with a left medial longitudinal fasciculus (MLF) lesion producing a left INO. (Red box illustrates the abnormal findings.)

Non pupil-sparing oculomotor palsy

The rule of the pupil in third nerve palsy

Elucidated in 1958 by Rucker. In effect, the rule states, "Third nerve palsy due to *extrinsic* compression of the nerve will be associated with impaired pupillary constriction." However, it is often overlooked that in 3% the pupil is spared.²⁴

Etiologies

Most cases are due to extrinsic compression of 3rd nerve. Etiologies include:

1. tumor: the most common tumors affecting 3rd nerve:
 - a) chordomas
 - b) clival meningiomas
2. vascular: the most common vascular lesions:
 - a) aneurysms of PComA artery (pupil sparing with aneurysmal oculomotor palsy occurs in < 1%).
 ★ Development of a new 3rd nerve palsy ipsilateral to a PComA aneurysm may be a sign of expansion with the possibility of imminent rupture, and is traditionally considered an indication for urgent treatment
 - b) aneurysms of the distal basilar artery or bifurcation (basilar tip)
 - c) carotid-cavernous fistula (p. 1519): look for pulsatile proptosis and chemosis
3. uncal herniation
4. cavernous sinus lesions: usually cause additional cranial nerve findings (V1, V2, IV, VI); see Cavernous sinus syndrome (p. 1689). Classically the third nerve palsy, e.g., from enlarging cavernous aneurysm, will *not* produce a dilated pupil because the sympathetics which dilate the pupil are also paralyzed^{1 (p 1492)}

Pupil sparing oculomotor palsy (pupil reacts to light)

General information

Usually from intrinsic vascular lesions occluding vaso-nervorum, causing central ischemic infarction. Spares parasympathetic fibers located peripherally in 3rd nerve in 62–83% of cases.²⁴

Etiologies

Etiologies include:

1. diabetic neuropathy
2. atherosclerosis (as seen in chronic HTN)
3. vasculopathies: including giant cell arteritis (p.203) (temporal arteritis)
4. chronic progressive ophthalmoplegia: usually bilateral
5. myasthenia gravis

Rarely, pupil-sparing OMP has been described following an intra-axial lesion, as in a midbrain infarction.²⁵

Other causes of oculomotor palsy

Trauma, uncus herniation, laterally expanding PitNET/adenomas, Lyme disease, cavernous sinus lesions: usually cause additional cranial nerve findings; see Multiple cranial nerve palsies (cranial neuropathies) (p.1687).

Lesions within the orbit tend to affect 3rd nerve branches unequally. Superior division lesion → ptosis and impaired elevation; inferior division lesion → impairment of depression, adduction and pupillary reaction.

33.5.4 Trochlear nerve (IV) palsy

Anatomy: the trochlear nucleus lies ventral to the cerebral aqueduct at the level of the inferior colliculi. Trochlear nerve axons pass *dorsally* around the aqueduct and decussate internally just caudal to the inferior colliculi. The nerve innervates the superior oblique muscle which primarily depresses the adducted eye, but in primary gaze it intorts and secondarily abducts and depresses the globe (i.e., it moves the eye down & out).

Some unique features of the trochlear nerve:

1. the only cranial nerve to decussate internally (i.e., the trochlear nucleus is on the contralateral side to the nerve that exits and to the superior oblique it goes to)
2. the only cranial nerve to exit posteriorly on the brainstem
3. the only cranial nerve passing through the superior orbital fissure that does not pass through the annulus of Zinn (AKA annulus tendineus or annular tendon)

Trochlear palsy results in eye deviation “up and in.” Patients tend to spontaneously tilt the head to the side *opposite* the IV palsy to “intort” the paretic eye and eliminate the diplopia. Diplopia is exacerbated when looking down (e.g., on descending stairs) especially when also looking inward, or when the examiner tilts the head *toward* the paretic side.

Isolated fourth nerve palsy is uncommon. It may occasionally occur with lesions of the cerebral peduncle or injury to the floor of the fourth ventricle near the aqueduct.

33.5.5 Abducens (VI) palsy

Anatomy: the abducens n. (Cr. N VI) leaves the abducens nucleus in the pons just beneath the floor of the 4th ventricle, exiting at the pontomedullary junction. It has a long intracranial course, and passes between the pons and clivus to enter the cavernous sinus via Dorello’s canal.

Abducens palsy produces a lateral rectus palsy (inability to abduct the eye). Clinically produces diplopia that is exaggerated with lateral gaze to the side of the palsy. Etiologies of isolated 6th nerve palsy include²⁶:

1. vasculopathy: including diabetes and giant cell arteritis. Most cases resolve within 3 months (alternative cause should be sought in cases lasting longer)
2. increased intracranial pressure: palsy may occur with increased ICP even in the absence of direct compression of the nerve (a “false localizing” sign in this setting). Postulated to occur due to the fact that the VI nerve has a long intracranial course, which may render it more sensitive to increased pressure. May be bilateral. Etiologies include:

- a) traumatically increased ICP
- b) increased ICP due to hydrocephalus (p.624), e.g., from p-fossa tumor
- c) pseudotumor cerebri syndrome (p.955)
- 3. cavernous sinus lesions: cavernous carotid aneurysm (p.1489), neoplasm (meningioma...), carotid-cavernous fistula (p.1519)
- 4. inflammatory:
 - a) Gradenigo's syndrome (p.601), involvement at Dorello's canal
 - b) sphenoid sinusitis
- 5. intracranial neoplasm: e.g., clivus chordoma, chondrosarcoma
- 6. pseudoabducens palsy: may be due to
 - a) thyroid eye disease: the most common cause of chronic VI palsy. Will have positive forced duction test (eye cannot be moved by examiner)
 - b) myasthenia gravis: responds to edrophonium (Tensilon®) test
 - c) long-standing strabismus
 - d) Duane's syndrome (p.601)
 - e) fracture of the medial wall of the orbit with medial rectus entrapment
- 7. following lumbar puncture (p.1815): almost invariably unilateral
- 8. fracture through clivus (p.1065)
- 9. idiopathic

33.5.6 Multiple extraocular motor nerve involvement

Lesions in cavernous sinus (see below) involve cranial nerves III, IV, VI and V1 & V2 (ophthalmic and maxillary divisions of trigeminal nerve), and spare II and V3.

Superior orbital fissure syndrome: dysfunction of nerves III, IV, VI and V1.

Orbital apex syndrome: involves II, III, IV, VI and partial V1.

4th nerve palsy may result from a contrecoup injury in frontal head trauma.

33.5.7 Painful ophthalmoplegia

Definition

Pain and dysfunction of ocular motility (may be due to involvement of one or more of cranial nerves III, IV, V & VI).

Etiologies

- 1. intraorbital
 - a) inflammatory pseudotumor (idiopathic orbital inflammation): see below
 - b) contiguous sinusitis
 - c) invasive fungal sinus infection producing orbital apex syndrome. Rhinocerebral *mucormycosis* (AKA zygomycosis): sinusitis with painless black palatal or nasal septal ulcer or eschar with hyphal invasion of blood vessels by fungi of the order Mucorales, especially rhizopus.²⁷ Usually seen in diabetic or immunocompromised patients, occasionally in otherwise healthy patients.²⁸ Often involves dural sinuses and may cause cavernous sinus thrombosis
 - d) mets
 - e) lymphoma
- 2. superior orbital fissure/anterior cavernous sinus
 - a) Tolosa-Hunt syndrome: see below
 - b) mets
 - c) nasopharyngeal Ca
 - d) lymphoma
 - e) herpes zoster
 - f) carotid-cavernous fistula
 - g) cavernous sinus thrombosis
 - h) intracavernous aneurysm
- 3. parasellar region
 - a) PitNet/adenoma
 - b) mets
 - c) nasopharyngeal Ca
 - d) sphenoid sinus mucocele
 - e) meningioma/chordoma
 - f) apical petrositis (Gradenigo's syndrome): see below

4. posterior fossa
 - a) PComA aneurysm
 - b) basilar artery aneurysm (rare)
5. miscellaneous
 - a) diabetic ophthalmoplegia
 - b) migrainous ophthalmoplegia
 - c) cranial arteritis
 - d) tuberculous meningitis: may cause ophthalmoplegia, usually incomplete, most often primarily oculomotor nerve

33.5.8 Painless ophthalmoplegia

Differential diagnosis:

1. chronic progressive ophthalmoplegia: pupil sparing, usually bilateral, slowly progressive
2. myasthenia gravis: pupil sparing, responds to edrophonium (Tensilon®) test
3. myositis: usually also produces symptoms in other organ systems (heart, gonads...)

33.6 Neurophthalmologic syndromes

33.6.1 Pseudotumor (of the orbit)

General information

AKA "chronic granuloma" (a misnomer, since true epithelioid granulomas are rarely found). An idiopathic inflammatory disease confined to the orbit that may mimic a true neoplasm. Lymphocytic infiltration of extraocular muscles. Usually unilateral.

Typically presents with rapid onset of proptosis, pain, and EOM dysfunction (painful ophthalmoplegia with diplopia). Often follows URI, may be associated with scleral inflammation. Most commonly involves the superior orbital tissues.

Differential diagnosis

See Orbital lesions (p. 1658) for list.

Key points for Graves' disease (GD): the histologic appearance of GD (hyperthyroidism) may be indistinguishable from pseudotumor. Involvement with GD is usually bilateral.

Treatment

Surgery tends to cause a flare up, and is thus usually best avoided.

Steroids are the treatment of choice. **R:** 50–80 mg prednisone q d. Severe cases may necessitate treatment with 30–40 mg/d for several months.

Radiation treatment with 1000–2000 rads may be needed for cases of reactive lymphocytic hyperplasia.

33.6.2 Tolosa-Hunt syndrome

Nonspecific inflammation in the region of the superior orbital fissure, often with extension into the cavernous sinus, sometimes with granulomatous features. A diagnosis of exclusion. May be a topographical variant of orbital pseudotumor (see above). Clinical diagnostic criteria:

1. painful ophthalmoplegia
2. involvement of any nerve traversing the cavernous sinus. The pupil is usually spared (frequently not the case with aneurysms, specific inflammation, etc.)
3. symptoms last days to weeks
4. spontaneous remission, sometimes with residual deficit
5. recurrent attacks with remissions of months or years
6. no systemic involvement (occasional N/V, due to pain?)
7. dramatic improvement with systemic steroids: 60–80 mg prednisone PO q day (slow taper), relief within about 1 day
8. occasional inflammation of rectus muscle from contiguous inflammation

33.6.3 Raeder's paratrigeminal neuralgia

Two essential components²⁹:

1. unilateral oculsympathetic paresis (AKA partial Horner syndrome [HS]; this usually lacks anhidrosis, and in this syndrome, possibly ptosis also)
2. homolateral trigeminal nerve involvement (usually tic-like pain, but may be analgesia or masseter weakness; pain, if present, must be tic-like and does not include e.g., unilateral head, face or vascular pain)

Localizing value of syndrome: region adjacent to trigeminal nerve in middle fossa. The cause is often not determined, but may rarely be due to aneurysm³⁰ compressing V1 with sympathetics.

33.6.4 Gradenigo's syndrome

AKA apical petrositis. Mastoiditis with involvement of petrous apex (if pneumatized). Usually seen by ENT physicians. Classic triad:

1. abducens palsy: from inflammation of 6th nerve at Dorello's canal, which is where it enters the cavernous sinus just medial to the petrous apex
2. retro-orbital pain: due to inflammation of V1
3. draining ear

33.7 Miscellaneous neurophthalmologic signs

► **Corneal mandibular reflex.** Eliciting the corneal reflex produces a jaw jerk or contralateral jaw movement (ipsilateral pterygoid contraction). A primitive pontine reflex, may be seen in a variety of insults to the brain (trauma, intracerebral hemorrhage...).

► **Duane syndrome.** AKA retraction syndrome: paradoxical innervation causing co-contraction of the lateral and medial rectus muscles on attempted adduction with relaxation on abduction, produces mild enophthalmos with pseudoptosis. May be congenital (e.g., part of one of the following syndromes: acrorenal-ocular syndrome, Okiihiro syndrome...).

► **Hippus.** Rhythmic, irregular pupillary oscillations, changing by ≥ 2 mm. May confuse examination when checking pupillary responses; record the *initial* response. May be normal. No localizing value.

► **Marcus Gunn phenomenon.** Not to be confused with Marcus Gunn pupil (p.592). Opening the mouth causes opening of a ptotic eye (abnormal reflex between proprioception of pterygoid muscles and third nerve). Reverse Marcus Gunn phenomenon: normal eye that closes with opening the mouth. Seen only in patients with peripheral facial nerve injuries, and probably results from aberrant regeneration.

► **Ocular bobbing**³¹. Abrupt, spontaneous, conjugate downward eye deviation with slow return to midposition, 2 to 12 times per min. It is associated with bilateral paralysis of horizontal gaze, including doll's-eyes and calorics. Most commonly seen with destructive lesions of the pontine tegmentum (usually hemorrhage, but also infarction, glioma, trauma), but has also been described with compressive lesions.³² Atypical bobbing is similar except that horizontal gaze is preserved, and can be seen with cerebellar hemorrhage, hydrocephalus, trauma, metabolic encephalopathy...

► **Opsoclonus**³³. (Rare.) Rapid, conjugate, irregular, non-rhythmic (differentiates this from nystagmus) eye movements vertically or horizontally, persist (attenuated) during sleep (opsochoria if dysconjugate). Usually associated with diffuse myoclonus (fingers, chin, lips, eyelid, forehead, trunk and LEs); also, malaise, fatigability, vomiting and some cerebellar findings. Often resolves spontaneously within 4 mos.

► **Oscillopsia.** Visual sensation that stationary objects are swaying side-to-side or vibrating.³⁴ Rarely the sole manifestation of Chiari I malformation³⁵ (often associated with downbeat nystagmus). Other causes include MS, or injury to both vestibular nerves; e.g., aminoglycoside ototoxicity,³⁶ bilateral vestibular neurectomies, see Dandy's syndrome (p.603).

- **Pseudo von Graefe sign.** Lid retraction on downward gaze (true von Graefe sign is lid lag in hyperthyroidism) seen in aberrant nerve regeneration (inferior rectus innervation → activation of levator palpebrae).
- **Optic atrophy.** Chronic, progressive optic atrophy is due to a compressive lesion (aneurysm, meningioma, osteopetrosis...) until proven otherwise.

References

- [1] Wilkins RH, Rengachary SS. Neurosurgery. New York 1985
- [2] Pinel JF, Larmande P, Guegan Y, et al. Down-Beat Nystagmus: Case Report with Magnetic Resonance Imaging and Surgical Treatment. *Neurosurgery*. 1987; 21:736–739
- [3] Williams DP, Troost BT, Rogers J. Lithium-Induced Downbeat Nystagmus. *Archives of Neurology*. 1988; 45:1022–1023
- [4] Frisen L. Swelling of the optic nerve head: a staging scheme. *J Neurol Neurosurg Psychiatry*. 1982; 45:13–18
- [5] Scott CJ, Kardon RH, Lee AG, et al. Diagnosis and grading of papilledema in patients with raised intracranial pressure using optical coherence tomography vs clinical expert assessment using a clinical staging scale. *Arch Ophthalmol*. 2010; 128:705–711
- [6] Sher NA, Wirtschafter J, Shapiro SK, et al. Unilateral Papilledema in 'Benign' Intracranial Hypertension (Pseudotumor Cerebri). *JAMA*. 1983; 250:2346–2347
- [7] Anderson DR, Patella VM. Automated Static Perimetry. 2nd ed. C V Mosby; 1998
- [8] Horton JC. Wilbrand's knee of the primate optic chiasm is an artefact of monocular enucleation. *Trans Am Ophthalmol Soc*. 1997; 95:579–609
- [9] Grzybowski A. Harry Moss Traquair (1875–1954), Scottish ophthalmologist and perimetrist. *Acta Ophthalmol*. 2009; 87:455–459
- [10] Donaldson LC, Eshtiahi A, Sacco S, et al. Junctional Scotoma and Patterns of Visual Field Defects Produced by Lesions Involving the Optic Chiasm. *J Neuroophthalmol*. 2022; 42:e203–e208
- [11] Shin RK, Li TP. Visualization of Wilbrand's knee. *Snowbird*, UT 2013
- [12] Lee JH, Tobias S, Kwon JT, et al. Wilbrand's knee: does it exist? *Surg Neurol*. 2006; 66:11–7; discussion 17
- [13] Zweckberger K, Unterberg AW, Schick U. Prechiasmatic transection of the optic nerve can save contralateral vision in patients with optic nerve sheath meningiomas. *Clin Neurol Neurosurg*. 2013; 115:2426–2431
- [14] Cho J, Liao E, Trobe JD. Visual Field Defect Patterns Associated With Lesions of the Retrochiasmatal Visual Pathway. *J Neuroophthalmol*. 2022; 42:353–359
- [15] Walton KA, Buono LM. Horner syndrome. *Curr Opin Ophthalmol*. 2003; 14:357–363
- [16] Kawasaki A. Physiology, assessment, and disorders of the pupil. *Current Opinion in Ophthalmology*. 1999; 10:394–400
- [17] Walsh FB, Hoyt WF. *Clinical Neuro-Ophthalmology*. Baltimore 1969
- [18] Nakagawa TA, Guerra L, Storgion SA. Aerosolized atropine as an unusual cause of anisocoria in a child with asthma. *Pediatr Emerg Care*. 1993; 9:153–154
- [19] Wijdicks EF. Determining Brain Death in Adults. *Neurology*. 1995; 45:1003–1011
- [20] Lepore FE. Diagnostic Pharmacology of the Pupil. *Clin Neuropharmacol*. 1985; 8:27–37
- [21] Morales J, Brown SM, Abdul-Rahim AS, et al. Ocular effects of apraclonidine in Horner syndrome. *Arch Ophthalmol*. 2000; 118:951–954
- [22] Bates AT, Chamberlain S, Champion M, et al. Pholedrine: a substitute for hydroxyamphetamine as a diagnostic eyedrop test in Horner's syndrome. *J Neurol Neurosurg Psychiatry*. 1995; 58:215–217
- [23] Zee DS. Internuclear ophthalmoplegia: pathophysiology and diagnosis. *Baillieres Clin Neurol*. 1992; 1:455–470
- [24] Trobe JD. Third nerve palsy and the pupil. Footnotes to the rule. *Arch Ophthalmol*. 1988; 106:601–602
- [25] Breen LA, Hopf HC, Farris BK, et al. Pupil-Sparing Oculomotor Nerve Palsy due to Midbrain Infarction. *Arch Neurol*. 1991; 48:105–106
- [26] Galetta SL, Smith JL. Chronic Isolated Sixth Nerve Palsies. *Archives of Neurology*. 1989; 46:79–82
- [27] DeShazo RD, Chapin K, Swain RE. Fungal Sinusitis. *New England Journal of Medicine*. 1997; 337:254–259
- [28] Radner AB, Witt MD, Edwards JE. Acute Invasive Rhinocerebral Zygomycosis in an Otherwise Healthy Patient: Case Report and Review. *Clin Infect Dis*. 1995; 20:163–166
- [29] Mokri B. Raeder's Paratrigeminal Syndrome. *Arch Neurol*. 1982; 39:395–399
- [30] Kashihara K, Ito H, Yamamoto S, et al. Raeder's Syndrome Associated with Intracranial Internal Carotid Artery Aneurysm. *Neurosurgery*. 1987; 20:49–51
- [31] Fisher CM. Ocular Bobbing. *Arch Neurol*. 1964; 11:543–546
- [32] Sherman DG, Salmon JH. Ocular Bobbing with Superior Cerebellar Artery Aneurysm: Case Report. *J Neurosurg*. 1977; 47:596–598
- [33] Smith JL, Walsh FB. Opsoclonus - Ataxic Conjugate Movements of the Eyes. *Arch Ophthalmol*. 1960; 64:244–250
- [34] Brickner R. Oscillopsia: A new symptom commonly occurring in multiple sclerosis. *Arch Neurol Psychiatry*. 1936; 36:586–589
- [35] Gindoff SI, Winfield JA. Oscillopsia and Primary Cerebellar Ectopia: Case Report and Review of the Literature. *Neurosurgery*. 1991; 29:932–936
- [36] Marra TR, Reynolds NC, Stoddard JJ. Subjective Oscillopsia ("Jiggling Vision") Presumably Due to Aminoglycoside Ototoxicity: A Report of Two Cases. *J Clin Neuro Ophthalmol*. 1988; 8:35–38

34 Neurotology

34.1 Dizziness and vertigo

34.1.1 Differential diagnosis of dizziness

► **Near syncope.** Some overlap with syncope; see Syncope and apoplexy (p. 1684)

1. orthostatic hypotension
2. cardiogenic hypotension
 - a) arrhythmia
 - b) valvular disease
3. vasovagal episode
4. hypersensitive carotid sinus; see Syncope and apoplexy (p. 1684)

► **Dysequilibrium**

1. multiple sensory deficits: e.g., peripheral neuropathy, visual impairment
2. cerebellar degeneration

► **Vertigo.** Sensation of movement (usually spinning)

1. inner ear dysfunction
 - a) labyrinthitis
 - b) Meniere disease (see below)
 - c) trauma: endolymphatic leak
 - d) *drugs*: especially aminoglycosides
 - e) **benign (paroxysmal) positional vertigo**¹: AKA *cupulolithiasis*. Attacks of severe vertigo when the head is turned to certain positions (usually in bed). Due to calcium concretions in the semicircular canals. Self limited (most cases do not last > 1 year). No hearing loss
 - f) syphilis
 - g) vertebrobasilar insufficiency (p. 1591)
2. vestibular nerve dysfunction
 - a) vestibular neuronitis: sudden onset of vertigo, gradual improvement
 - b) compression:
 - meningioma
 - vestibular schwannoma: usually slowly progressive ataxia instead of severe vertigo. BAER latencies usually abnormal. CT or MRI usually abnormal
3. **disabling positional vertigo**: as described by Jannetta et al.² *constant* disabling positional vertigo or dysequilibrium, causing ≈ constant nausea, no vestibular dysfunction nor hearing loss (tinnitus may be present). One possible cause is vascular compression of the vestibular nerve which may respond to microvascular decompression
4. brainstem dysfunction
 - a) vascular disease; see Vertebrobasilar insufficiency (p. 1591): less distinct vestibular symptoms, prominent nonvestibular symptoms
 - b) migraine: especially basilar artery migraine
 - c) demyelinating disease: e.g., multiple sclerosis
 - d) *drugs*: antiseizure medications, alcohol, sedatives/hypnotics, salicylates
5. dysfunction of cervical proprioceptors: as in cervical osteoarthritis

► **Poorly defined lightheadedness.** Mostly psychiatric. May also include:

1. hyperventilation
2. hypoglycemia
3. anxiety neurosis
4. hysterical

34.1.2 Vestibular neurectomy

General information

Complete loss of vestibular function from one side is thought to produce transient vertigo due to the mismatch of vestibular input from the two ears. Theoretically, a central compensatory mechanism (the “cerebellar clamp”) results in the amelioration of symptoms. In cases of unilateral *fluctuating* vestibular dysfunction, this compensatory mechanism may be impaired. Unilateral selective vestibular neurectomy

(SVN) may convert the fluctuating or partial loss to a complete cessation of input and facilitate compensation. Bilateral SVN is often complicated by oscillopsia (p.601)—AKA Dandy's syndrome, with difficulty in maintaining balance in the dark due to loss of the vestibulo-ocular reflex—and is to be avoided.

Indications

The two conditions for which SVN is most commonly employed are Meniere disease (see below) and partial vestibular injury (viral or traumatic). SVN may be indicated in disabling cases refractory to medical or non-destructive surgical treatment when vestibular studies demonstrate continued or progressive uncompensated vestibular dysfunction.³

SVN preserves hearing and in Meniere disease is >90% effective in eliminating episodic vertiginous spells ($\approx$ 80% success rate in non-Meniere cases), but is unlikely to improve stability with rapid head movement.

Surgical approaches for SVN

1. retrolabyrinthine, AKA postauricular approach: anterior to sigmoid sinus. Primary choice in patients with Meniere disease who have not had previous endolymphatic sac (ELS) procedures since it permits simultaneous SVN and decompression of the endolymphatic sac. Requires mastoidectomy with skeletonization of the semicircular canals and ELS. The dural opening is bounded anteriorly by the posterior semicircular canal, posteriorly by the sigmoid sinus. Water-tight dural closure is difficult
2. retrosigmoid, AKA posterior fossa, AKA suboccipital approach: posterior to sigmoid sinus. The original approach applied by Dandy in pre-microsurgical era, usually sacrificed hearing, and occasionally facial nerve function. Better results are achieved today with microscopic techniques. Indicated for cases other than Meniere disease where there is no need for identification of the ELS. Also the best approach for positive identification of eighth nerve
3. middle fossa (extradural) approach: the fibers of the vestibular division may be more segregated from the cochlear fibers in the IAC than in the CPA, thus permitting more complete section of the vestibular nerve. May be appropriate for failed response to SVN by the above approaches. Disadvantages: requires temporal lobe retraction, does not allow exposure of ELS, and higher morbidity and risk of damage to facial nerve⁴ than retrolabyrinthine approach

Surgical considerations for selective vestibular neurectomy

(Also, see ► Fig. 1.9)

1. the vestibular nerve is in the superior half of the eighth nerve complex, and is slightly more *gray* in color than the cochlear division (due to less myelin⁵). They may be separated by a small vessel or by an indentation in the bundle
2. facial (VII) nerve:
 - a) whiter than the VIII nerve complex
 - b) lies anterior and superiorly to the VIII nerve
 - c) EMG monitoring of the facial nerve is recommended
 - d) direct stimulation confirms the identification
3. any vessels present on eighth nerve bundle must be preserved to save hearing (primarily, the artery of the auditory canal must be preserved)
4. if no plane of cleavage can be defined between vestibular & cochlear divisions, the superior half of the nerve bundle is divided
5. the endolymphatic sac lies $\approx$ midway between the posterior edge of the internal auditory meatus and the sigmoid sinus

34.2 Meniere disease

34.2.1 General information

Key concepts

- increased endolymphatic pressure
- clinical triad: vertigo, tinnitus & fluctuating hearing loss
- surgical options for failure of medical management include endolymphatic shunt or selective vestibular neurectomy

Probably due to a derangement of endolymphatic fluid regulation (a consistent finding is endolymphatic hydrops: increased endolymphatic volume and pressure with dilatation of endolymphatic spaces), with resultant fistulization into the perilymphatic spaces.

34.2.2 Epidemiology

Incidence: $\approx 8\text{--}46$ per 100,000 population per year.⁶ Most cases have onset between 30 and 60 years of age, rarely in youth or in the elderly.

Prevalence range: 3.5–513 per 100,000 population.⁶ Health claims data found prevalence 190 per 100,000 with a female:male ratio of 1.89 to 1.⁶ May become bilateral in 20%.

34.2.3 Clinical

Clinical triad

1. attacks of violent vertigo (due to vestibular nerve dysfunction): usually the earliest and the most disabling symptom. Nausea, vomiting, and diaphoresis are frequent concomitants. Severe attacks may cause prostration. Vertigo may persist even after complete deafness. Balance is normal between attacks
2. tinnitus: often described as resembling the sound of escaping steam, not a true “ringing”
3. fluctuating low frequency hearing loss: may fluctuate for a periods of weeks to years, and may progress to permanent deafness if untreated (a sensation of fullness in the ear is commonly described⁷; however, this is nonspecific and may occur with hearing loss for any reason)

Other clinical features

Drop attacks (“otolithic crises of Tumarkin”) occasionally occur.

Attack duration: $\approx 5\text{--}30$ minutes (some say 2–6 hours), with a “post-ictal” period of fatigue lasting several hrs.

Frequency: varies from one or two attacks a year to several times per week.

Two subtypes differ from classical form: vestibular Meniere (episodic vertigo with normal hearing) and cochlear Meniere (few vestibular symptoms).

Natural course of syndrome is characterized by periods of remission. Eventually the vertiginous attacks either progress in severity, or “burn out” (being replaced by constant unsteadiness⁷).

Differential diagnosis

Also see Differential diagnosis: Dizziness and vertigo (p.603) for more details.

1. benign (paroxysmal) positional vertigo: AKA *cupulolithiasis*. Self limited (most cases last < 1 year). No hearing loss
2. disabling positional vertigo: *constant* disabling positional vertigo or dysequilibrium, $\approx$ constant nausea, no vestibular dysfunction nor hearing loss (tinnitus may be present)
3. vestibular schwannoma: usually slowly progressive ataxia instead of episodic severe vertigo. BAER latencies usually abnormal. CT or MRI usually positive
4. vestibular neuronitis: sudden onset of vertigo with gradual improvement
5. vertebralbasilar insufficiency (VBI) (p. 1591): less distinct vestibular symptoms, and prominence of nonvestibular symptoms

Diagnostic studies

1. electronystagmography (ENG) with bithermal caloric stimulation usually abnormal, may show blunted thermal responses
2. audiogram: low frequency hearing loss, fairly good preservation of discrimination and loudness recruitment, negative tone decay on impedance testing
3. BAER usually shows normal latencies
4. radiographic imaging (CT, MRI, etc.): no findings in Meniere disease
5. in bilateral cases, a VDRL should be checked to R/O luetic disease

Treatment

Medical treatment

1. reduced intake of salt (strict salt restriction is as effective as any medication) and caffeine

2. diuretics: taken daily until ear fullness abates, then PRN ear pressure (usually once or twice weekly suffices)
 - a) acetazolamide: **R** Diamox® sequels 500 mg PO q d × 1 week, increase to BID if symptoms persist. D/C if paresthesias develop. Do not use during 1st trimester of pregnancy
3. vestibular suppressants
 - a) diazepam (Valium®): probably the most effective
 - b) meclizine HCl (Antivert®): **R** Adult dose for vertigo associated with the vestibular system (during attacks): 25–100 mg/day PO divided. Dose for motion sickness: 25–50 mg PO one hr prior to stimulus. Supplied: 12.5, 25 & 50 mg tabs. **Side effects:** drowsiness
4. vasodilators: postulated to be mediated by increased cochlear blood flow: inhalation of 5–10% CO₂ works well, but relief is short lived

Surgical treatment

Reserved for *incapacitating* cases *refractory* to medical management. When functional hearing exists, procedures that spare hearing are preferred because of high incidence of bilateral involvement. Procedures include:

1. endolymphatic shunting procedures: to mastoid cavity (Arenberg shunt) or to subarachnoid space. Reserved for cases with serviceable hearing. ≈ 65% success rate (see below). If symptoms are relieved ≥ 1 year, then a recurrence would be treated by shunt revision, if < 1 year then vestibular neurectomy
2. direct application of corticosteroids to the inner ear
3. nonselective vestibular ablation (in cases with nonserviceable hearing on the side of involvement)
 - a) surgical labyrinthectomy
 - b) middle ear perfusion with gentamicin
 - c) translabyrinthine section of the 8th nerve
4. selective vestibular neurectomy (p. 603), in cases with serviceable hearing

Outcome

Endolymph shunting procedures

Outcomes from 112 endolymphatic shunting procedures are shown in ► Table 34.1.

Table 34.1 Outcome in 112 endolymphatic-subarachnoid shunts⁷

	Vertigo	Tinnitus	Hearing ^a	Ear pressure
improved	79 (70%) ^b	53 (47%)	19 (17%)	57 (51%)
stable	33 (29%)	49 (43%)	50 (45%)	24 (21%)
worse	(none)	10 (10%)	39 (35%)	31 (28%)

^aimproved hearing considered serviceable (50 dB pure tone, 70% speech discrimination); additional 4 patients had improved but non-serviceable hearing

^b5 patients had recurrence of vertigo after 1–3 years

Neurectomy procedures

Vestibulocochlear nerve section (based on early posterior fossa surgery by Dandy; entire eighth nerve bundle was sectioned in 587 patients; all were deaf post-op): 90% relieved of vertigo, 5% unchanged and 5% worse; 9% incidence of facial paralysis (3% incidence of permanent paralysis).

Selective vestibular nerve section (sparing cochlear portion, 95 patients from Dandy): 10% had improved hearing, 28% unchanged, 48% worse, 14% deaf.

Retrolabyrinthine approach: in 32 patients with Meniere syndrome (25 failed endolymph shunt) responding to survey, 85% had complete relief of vertigo, 6% improved, 9% no relief (one of whom responded to middle fossa neurectomy).⁵

Complications and untoward effects

Patients with little vestibular nerve function pre-op (determined by ENG) usually have little difficulty immediately following vestibular neurectomy; patients with more function may have a transient worsening post-op until they accommodate.

Among 42 patients undergoing retrolabyrinthine approach: none lost hearing as a result of surgery, no facial weakness, one CSF rhinorrhea requiring reoperation, and one meningitis with good outcome.⁵

In post-op failures, check ENG. If any vestibular nerve function is demonstrated on operated side, then the nerve section was incomplete; consider re-operating.

34.3 Facial nerve palsy

34.3.1 Severity grading

Severity of facial palsy is often graded with the House and Brackmann scale (see ► Table 45.3).

34.3.2 Localizing site of lesion

Central facial palsy (AKA supranuclear facial palsy)

The cortical representation for facial movement occurs in the motor strip along the lateral aspect (just above the most inferior opercular portion of the precentral gyrus (► Fig. 1.3)). The keys to differentiating central paralysis (due to *supranuclear* lesions) from peripheral facial palsy are that *central* palsies:

1. are confined primarily to the lower face due to some bilateral cortical representation of upper facial movement
2. may spare emotional facial expression⁸ (e.g., smiling at a joke)

Nuclear facial palsy

The motor nucleus of the seventh nerve is located at the pontomedullary junction. Nuclear VII palsy results in paralysis of all VII nerve motor function. In nuclear facial palsies, other neurologic findings also often occur from involvement of adjacent neural structures by the underlying process (stroke, tumor...); e.g., in Millard-Gubler syndrome (p. 101), there is ipsilateral abducens palsy + contralateral limb weakness. Tumors invading the floor of the 4th ventricle (e.g., medulloblastoma) may also cause nuclear facial palsy (from involvement of facial colliculus in the floor of 4th ventricle).

Facial nerve lesion

Motor fibers ascend within the pons and form a sharp bend (“internal genu”) around the sixth nerve (abducens) nucleus, forming a visible bump in the floor of the 4th ventricle (facial colliculus). The seventh nerve exits from the brainstem at the ponto-medullary junction (► Fig. 2.7) where it may be involved in CPA tumors. It enters the supero-anterior portion of the internal auditory canal (► Fig. 1.9). The geniculate ganglion (“external genu”) is located within the temporal bone. The first branch from the ganglion is the greater superficial petrosal nerve (GSPN), which passes to the pterygopalatine ganglion and innervates the nasal and palatine mucosa and the lacrimal gland of the eye; lesions proximal to this point produce a dry eye. The next branch is the branch to the stapedius muscle; lesions proximal to this point produce hyperacusis. Next, the chorda tympani joins the facial nerve bringing taste sensation from the anterior two-thirds of the tongue. Basal skull fractures may injure the nerve just proximal to this point. Travelling with the chorda tympani are fibers to the submandibular and sublingual glands. The facial nerve exits the skull at the stylomastoid foramen. It then enters the parotid gland, where it splits into the following branches to the facial muscles (cranial to caudal): temporal, zygomatic, buccal, mandibular, and cervical. Lesions within the parotid gland (e.g., parotid tumors) may involve some branches but spare others.

34.3.3 Etiologies

These etiologies produce primarily facial nerve palsy, also see Multiple cranial nerve palsies (cranial neuropathies) (p. 1687). **Note:** 90–95% of all cases of facial palsy are accounted for by the first 3 items: Bell's palsy, herpes zoster oticus, and trauma (basal skull fractures).⁹

1. Bell's palsy (p. 608)
2. herpes zoster oticus (auris) (p. 609)
3. trauma: basal skull fracture
4. birth:
 - a) congenital
 - *bilateral facial palsy (facial diplegia) of Möbius syndrome (p. 1687) : unique in that it affects upper face more than lower face
 - *congenital facial diplegia may be part of facioscapulohumeral or myotonic muscular dystrophy
 - b) traumatic

5. otitis media: with acute otitis media, facial palsy usually improves with antibiotics. With chronic suppurative otitis surgical intervention is required
6. central facial paralysis and nuclear facial paralysis: see Localizing site of lesion above
7. neoplasm: usually causes hearing loss, and (unlike Bell's palsy) *slowly progressive* facial paralysis
 - a) most are either benign schwannomas of the facial or auditory nerve, or malignancies metastatic to the temporal bone. Facial neuromas account for $\approx 5\%$ of peripheral facial nerve palsies¹⁰; the paralysis tends to be slowly progressive
 - b) parotid tumors may involve some branches but spare others
 - c) Masson's vegetant intravascular hemangioendothelioma (p.961)
8. *neurosarcoidosis (p.198): VII is the most commonly affected cranial nerve
9. diabetes: 17% of patients > 40 yrs old with peripheral facial palsy (PFP) have abnormal glucose tolerance tests. Diabetics have 4.5 times the relative risk of developing PFP than nondiabetics¹¹
10. *stage II Lyme disease (p.364)¹²: facial diplegia is a hallmark
11. *Guillain-Barré syndrome: facial diplegia occurs in $\approx 50\%$ of fatal cases
12. occasionally seen in Klippel-Feil syndrome
13. *isolated 4th ventricle (p.436): compression at the facial colliculus

* Items with an asterisk are often associated with facial *diplegia* (i.e., bilateral facial palsy), see also multiple cranial neuropathies (p.1687).

34.3.4 Bell's palsy

General information

Bell's palsy (BP), AKA idiopathic peripheral facial palsy (PFP), is the most common cause of facial paralysis (50–80% of PFPs). Incidence: 150–200/1-million/yr.

Etiology: by definition, PFP is called Bell's palsy when it is not due to known causes of PFP (e.g., infection, tumor, or trauma) and there are no other neurological (e.g., involvement of other cranial nerves) or systemic manifestations (e.g., fever, diabetes, possibly hypertension¹³).¹⁴ Thus, true BP is idiopathic, and is a diagnosis of exclusion. Most cases probably represent a viral inflammatory demyelinating polyneuritis¹⁵ usually due to the herpes simplex virus.¹⁶ Facial palsy due to Lyme disease can usually be recognized on clinical grounds.¹⁷ Severity may be graded on the House & Brackmann grading scale (see ► Table 45.3).

Presentation

A viral prodrome is frequent: URI, myalgia, hypesthesia or dysesthesia of the trigeminal nerve, N/V, diarrhea... Paralysis may be incomplete and remain so (Type I); it is complete at onset in 50% (Type II), the remainder progress to completion in 1 week. Usually exhibits distal to proximal progression: motor branches, then chorda tympani (loss of taste and decreased salivation), then stapedial branch (hyperacusis), then geniculate ganglion (decreased tearing). Associated symptoms are shown in ► Table 34.2, and are usually, but not always, ipsilateral. Herpes zoster vesicles develop in 4% of patients 2–4 days after onset of paralysis; and in 30% of patients 4–8 days after onset. During the recovery phase excessive lacrimation may occur (aberrant nerve regeneration).

Table 34.2 Associated symptoms with Bell's palsy

Symptom	%
facial & retroauricular pain	60
dysgeusia	57
hyperacusis	30
reduced tearing	17

Evaluation

Patients with PFP should be examined at an early stage to optimize outcome.

Electrodiagnostics: EMG may detect re-innervation potentials, aids prognostication. Nerve conduction study: electrical stimulation of the facial nerve near the stylomastoid foramen while recording EMG in facial muscles (a facial nerve may continue to conduct for up to ≈ 1 week even after complete transection).

Management

General measures

Eye protection: protection of the eye is critical. Artificial tears during the day, eye ointment at night, avoid bright light (using dark glasses during the day).

Medical management

Steroids: prednisolone 25 mg p.o. BID $\times$ 10 days, started within 72 hours of onset of symptoms, improves the chances of complete recovery at 3 & 9 months.

Acyclovir: does *not* help (alone or in combination with prednisolone).¹⁸

Surgical management

Surgical decompression: controversial. The definitive study has not been done. Rarely utilized. Indications may include:

1. complete facial nerve degeneration without response to nerve stimulation (although this absence is also used as an argument against surgery⁹)
2. progressively deteriorating response to nerve stimulation
3. no clinical nor objective (nerve testing) improvement after 8 wks (however, in cases where the diagnosis of Bell's palsy is felt to be certain, the active disease will have abated by $\approx$ 14 days after onset⁹)

Prognosis

Essentially all cases show some recovery (if none by 6 mos, other etiologies should be sought). Extent of recovery: 75–80% of cases recover completely, 10% partial, remainder poor. If recovery begins by 10–21 d, tends to be complete; if not until 3–8 wks $\rightarrow$ fair, if not until 2–4 mos $\rightarrow$ poor recovery. If paralysis is complete at onset, 50% will have incomplete recovery. Cases of incomplete paralysis at onset that do not progress to complete paralysis $\rightarrow$ complete recovery; incomplete paralysis at onset that progresses to complete $\rightarrow$ incomplete recovery in 75%. A worse prognosis is associated with: more proximal involvement, hyperacusis, decreased tearing, age $>$ 60 yrs, diabetes, HTN, psycho-neuroses, and aural, facial, or radicular pain.

34.3.5 Herpes zoster oticus facial paralysis

Symptoms are more severe than Bell's palsy, herpetic vesicles are usually present, and antibody titers to varicella-zoster virus rise. These patients have a higher risk of facial nerve degeneration.

34.3.6 Surgical treatment of facial palsy

General information

For cases with focal injury to the facial nerve (e.g., trauma, injury during surgery for CPA tumor...), dynamic reconstruction by nerve anastomoses are usually considered superior to static methods.¹⁹ For nonfocal causes, e.g., Bell's palsy, only "static" methods may be applicable. A functional neural repair is not possible if the facial muscles have atrophied or fibrosed.

Surgical options

Surgical treatment options include:

1. for intracranial injury to facial nerve (e.g., during CPA tumor surgery): intracranial reapproximation (with or without graft) offers the best hope for the most normal facial reanimation
 - a) timing
 - at time of tumor removal (for a divided facial nerve during removal of vestibular schwannoma^{20,21,22}): the best result that can be achieved with this is House-Brackmann Grade III ($\triangleright$ Table 45.3). The operation fails to produce good results in $\approx$ 33% of cases²²
 - in delayed fashion, especially if the nerve was left in anatomic continuity
 - b) techniques
 - direct reanastomosis: difficult due to the frail nature of the VII nerve (especially when it has been stretched by a tumor)
 - cable graft: e.g., using greater auricular nerve²³ or sural nerve

2. extracranial facial nerve anastomosis
 - a) hypoglossal nerve (Cr. N. XII)-facial nerve anastomosis (see below)
 - b) spinal accessory nerve (Cr. N. XI)-facial nerve anastomosis (see below)
 - c) phrenic nerve-facial nerve anastomosis
 - d) glossopharyngeal (Cr. N. IX)-facial nerve anastomosis
 - e) crossface grafting (VII-VII): results have not been very good
3. "mechanical" or "static" means
 - a) facial suspension: e.g., with polypropylene (Marlex®) mesh²⁴
 - b) eye closure techniques (protects the eye from exposure and reduced tearing)
 - tarsorrhaphy: partial or complete
 - gold weights in eyelid
 - stainless-steel spring in eyelid

Timing of surgery

If the facial nerve is known to be interrupted (e.g., transected during removal of vestibular schwannoma) then early surgical treatment is indicated. When the status of the nerve is unknown or if in continuity but not functioning, then several months of observation and electrical testing should be allowed for spontaneous recovery. Very late attempts at anastomosis have less chance for recovery due to facial muscle atrophy.

Hypoglossal nerve-facial nerve (XII-VII) anastomosis

General information

Cannot be used bilaterally in patients with facial diplegia or in those with other lower cranial nerve deficits (or potential for same). In spite of some suggestions to the contrary, sacrificing the XII nerve does create some morbidity (tongue atrophy with difficulty speaking, mastication and swallowing in $\approx 25\%$ of cases, exacerbated when the facial muscles do not function on that side; aspiration may occur if vagus (Cr. N. X) dysfunction coexists with loss of XII).

Not as effective as would theoretically seem possible. The resultant facial reanimation is often less than ideal (may permit mass movement). To avoid severe disappointment, the patient should thoroughly understand the likely side effects and that the facial movement will probably be much less than normal, often with poor voluntary control.

Usually performed in conjunction with anastomosis of the descendens hypoglossi to the distal hypoglossal nerve to try and reduce hemiatrophy of the tongue. Atrophy may also be reduced by using a "jump graft" without completely interrupting XII.²⁵

34

Technique

Position: supine, head turned slightly to the opposite side. **Skin incision:** 6–8 cm incision from just above the mastoid process obliquely downward across the neck to 2 cm below the angle of the jaw. The platysma is opened, and the tip of the mastoid is exposed by incising the insertion of the SCM and using a periosteal elevator. Incise the deep fascia; avoid the parotid gland, which is retracted superiorly. Rongeur the anterior third of the mastoid process (wax any exposed air cells) and identify the facial nerve as it exits the stylomastoid foramen between the mastoid process and the styloid process. Retract the posterior belly of the digastric inferiorly to aid the exposure.

The SCM is retracted laterally until the carotid sheath is identified, revealing the hypoglossal nerve. It loops around the occipital artery at this level (where it gives off the descendens hypoglossi) to pass between the carotid artery and jugular vein. The nerve is freed proximally to the point where it enters the carotid sheath and distally to the submandibular triangle where it is sharply divided.

The facial nerve is divided at the stylomastoid foramen and is approximated to the proximal hypoglossal nerve. The descendens hypoglossi is divided as far distally as possible and is then anastomosed to the distal stump of the hypoglossal nerve.

Variations

1. interposition jump grafts: spares function in the XII nerve (to minimize glottic denervation, the incision of XII should be distal to the descendens hypoglossi²⁵)
 - a) using cutaneous nerve jump graft²⁵
 - b) using muscle interposition jump graft²⁶
2. mobilizing the intratemporal portion of VII out of the fallopian canal (as previously described²⁷) and then anastomosing it using bevelled cuts to a partially incised XII²⁸

Outcome

Results are better if performed early, although good results can occur up to 18 mos after injury. In 22 cases, 64% had good results, 14% fair, 18% poor, and 1 patient had no evidence of reinnervation. In 59% of cases, evidence of reinnervation was seen by 3–6 mos; in the remaining patients with reinnervation improvement was noted by 8 mos.²⁹ Recovery of forehead movement occurs in only ≈ 30%. Return of tone precedes movement by ≈ 3 months.

Spinal accessory nerve-facial nerve (XI-VII) anastomosis

General information

First described in 1895 by Sir Charles Ballance.³⁰ Sacrifices some shoulder movement rather than use of tongue. Initial concerns about significant shoulder disability and pain resulted in the technique of using only the SCM branch of XI³¹; however these problems have not occurred in the majority of patients even with use of the major division.³²

Technique

See reference.³²

Skin incision: curves across the mastoid tip along the anterior margin of the SCM. Strip and remove the anterior third of the mastoid process (wax any exposed air cells), identify the facial nerve and divide it as close to its exit from the stylomastoid foramen as possible. Locate the XI nerve 3–4 cm below the mastoid tip, and divide it distal to the SCM division. Mobilize the free end and anastomose it to the distal stump of VII. Results in loss of trapezius function, which may not cause deficit even if done bilaterally. Alternatively, the SCM branch of XI may be used, sparing the trapezius function; however, the shorter length may be difficult to work with and in some individuals there may only be multiple small branches to the SCM.

34.4 Hearing loss

Two anatomic types: conductive and sensorineural.

34.4.1 Conductive hearing loss

1. patients tend to speak with normal or low volume voice
2. etiologies: anything that interferes with ossicular movement. Included:
 - a) otitis media with middle ear effusion
 - b) otosclerosis
3. clinical findings with unilateral hearing loss (see ► Table 34.3):
 - a) *Weber test* will lateralize to side of hearing loss (Weber test: place a vibrating 256 or 512 Hz tuning fork on the center of the forehead; the sound will lateralize (i.e. sound louder) on the side of conductive hearing loss, or opposite to the side of SNHL)
The “hum test” can provide the same information as the Weber test by having the patient hum, and does not require special equipment and can be done remotely (e.g., over the phone).³³
 - b) *Rinne test* will be abnormal (BC > AC) on the side of hearing loss, called a *negative Rinne* (Rinne test: place a vibrating 256 or 512 Hz tuning fork on the mastoid process; when sound is no longer heard, move the fork to just outside the ear to see if air conduction [AC] is > bone conduction [BC])
4. middle ear impedance measurements are abnormal

Table 34.3 Interpretation of Weber and Rinne test results

Weber	Rinne	Interpretation
nonlateralizing	AC > BC bilat	normal ^a
lateralizes to side A	normal bilaterally (AC > BC)	sensorineural hearing loss (SNHL) side B
lateralizes to side A	abnormal in side A (BC > AC)	conductive hearing loss side A
lateralizes to side A	abnormal in side B (BC > AC)	combined conductive + SNHL side B

^anormal, or symmetric hearing loss

34.4.2 Sensorineural hearing loss (SNHL)

1. patients tend to speak with loud voice
2. clinical findings with unilateral hearing loss (► Table 34.3):
 - a) *Weber test* will lateralize to side of better hearing (Weber test: place a vibrating 256 or 512 Hz tuning fork on the center of the forehead; the sound will lateralize—sound louder—on the side of conductive hearing loss, or opposite to the side of SNHL)
 - b) *Rinne test* will be normal (AC > BC), called a *positive Rinne* (Rinne test: place a vibrating 256 or 512 Hz tuning fork on the mastoid process; when sound is no longer heard, move the fork to just outside the ear to see if air conduction [AC] is > bone conduction [BC])
3. further divided into sensory or neural. Distinguished by otoacoustic emissions (only produced by a cochlea with functioning hair cells) or BSAERS
 - a) sensory: loss of outer hair cells in the cochlea. Etiologies: cochlear damage (usually causes high-frequency hearing loss) from noise exposure, ototoxic drugs (e.g., aminoglycosides), senile cochlear degeneration, viral labyrinthitis. Speech discrimination may be relatively preserved
 - b) neural: due to compression of the 8th cranial nerve. Etiologies: CP angle tumor (e.g., vestibular schwannoma). Typically much greater loss of word discrimination out of proportion to pure tone audiogram abnormalities

Sensory hearing loss may be distinguished from neural hearing loss by

1. otoacoustic emissions which are only produced by a cochlea with functioning hair cells
2. or BSAERS
3. an elevated stapedial reflex threshold out of proportion to PTA abnormalities is also highly diagnostic of a retrocochlear (neural) lesion

References

- [1] Brandt T, Daroff RB. The Multisensory Physiological and Pathological Vertigo Syndromes. *Ann Neurol*. 1980; 7:195–203
- [2] Jannetta PJ, Moller MB, Moller AR. Disabling Positional Vertigo. *N Engl J Med*. 1984; 310:1700–1705
- [3] Arriaga MA, Chen DA. Vestibular Nerve Section in the Treatment of Vertigo. *Contemporary Neurosurgery*. 1997; 19:1–6
- [4] McElveen JT, House JW, Hitselberger WE, et al. Retrolabyrinthine Vestibular Nerve Section: A Viable Alternative to the Middle Fossa Approach. *Otolaryngol Head Neck Surg*. 1984; 92:136–140
- [5] House JW, Hitselberger WE, McElveen J, et al. Retrolabyrinthine Section of the Vestibular Nerve. *Otolaryngol Head Neck Surg*. 1984; 92:212–215
- [6] Alexander TH, Harris JP. Current epidemiology of Meniere's syndrome. *Otolaryngol Clin North Am*. 2010; 43:965–970
- [7] Glasscock ME, Miller GW, Drake FD, et al. Surgical Management of Meniere's Disease with the Endolymphatic Subarachnoid Shunt. *Laryngoscope*. 1977; 87:1668–1675
- [8] Shambaugh GE. Facial Nerve Decompression and Repair. In: *Surgery of the Ear*. Philadelphia: W. B. Saunders; 1959:543–571
- [9] Adour KK. Diagnosis and Management of Facial Paralysis. *N Engl J Med*. 1982; 307:348–351
- [10] Shambaugh GE, Clemis JD, Paparella MM, et al. Facial Nerve Paralysis. In: *Otolaryngology*. Philadelphia: W. B. Saunders; 1973
- [11] Adour KK, Wingerd J, Doty HE. Prevalence of Concurrent Diabetes Mellitus and Idiopathic Facial Paralysis (Bell's Palsy). *Diabetes*. 1975; 24:449–451
- [12] Treatment of Lyme Disease. *Medical Letter*. 1988; 30:65–66
- [13] Abraham-Inpijn L, Devriese PP, Hart AAM. Predisposing Factors in Bell's Palsy: A Clinical Study with Reference to Diabetes Mellitus, Hypertension, Clotting Mechanism and Lipid Disturbance. *Clin Otolaryngol*. 1982; 7:99–105
- [14] Devriese PP, Schumacher T, Scheide A, et al. Incidence, Prognosis and Recovery of Bell's Palsy: A Survey of About 1000 Patients (1974–1983). *Clin Otolaryngol*. 1990; 15:15–27
- [15] Adour KK, Byl FM, Hilsinger RL, et al. The True Nature of Bell's Palsy: Analysis of 1000 Consecutive Patients. *Laryngoscope*. 1978; 88:787–801
- [16] Adour KK, Bell DN, Hilsinger RL. Herpes simplex virus in idiopathic facial paralysis (Bell palsy). *JAMA*. 1975; 233:527–530
- [17] Kuiper H, Devriese PP, de Jongh BM, et al. Absence of Lyme Borreliosis Among Patients With Presumed Bell's Palsy. *Archives of Neurology*. 1992; 49:940–943
- [18] Sullivan FM, Swan IR, Donnan PT, et al. Early treatment with prednisolone or acyclovir in Bell's palsy. *N Engl J Med*. 2007; 357:1598–1607
- [19] Conley J, Baker DC. Hypoglossal-Facial Nerve Anastomosis for Reinnervation of the Paralyzed Face. *Plast Reconstr Surg*. 1979; 63:63–72
- [20] Pluchino F, Fornari M, Luccarelli G. Intracranial Repair of Interrupted Facial Nerve in Course of Operation for Acoustic Neuroma by Microsurgical Technique. *Acta Neurochirurgica*. 1986; 79:87–93
- [21] Stephanian E, Sekhar LN, Janacka IP, et al. Facial Nerve Repair by Interposition Nerve Graft: Results in 22 Patients. *Neurosurgery*. 1992; 31:73–77
- [22] King TT, Sparrow OC, Arias JM, et al. Repair of Facial Nerve After Removal of Cerebellopontine Angle Tumors: A Comparative Study. *Journal of Neurosurgery*. 1993; 78:720–725
- [23] Alberti PWRM. The Greater Auricular Nerve. Donor for Facial Nerve Grafts: A Note on its Topographical Anatomy. *Arch Otolaryngol*. 1962; 76:422–424
- [24] Strelzow VV, Friedman WH, Katsantonis GP. Reconstruction of the Paralyzed Face With the Polypropylene Mesh Template. *Arch Otolaryngol*. 1983; 109:140–144
- [25] May M, Sobol SM, Mester SJ. Hypoglossal-Facial Nerve Interpositional-Jump Graft for Facial Reanimation without Tongue Atrophy. *Otolaryngology - Head and Neck Surgery*. 1991; 104:818–825
- [26] Drew SJ, Fullerton AC, Glasby MA, et al. Reinnervation of Facial Nerve Territory Using a Composite Hypoglossal Nerve-Muscle Autograft-Facial Nerve Bridge. An Experimental Model in Sheep. *Clin Otolaryngol*. 1995; 20:109–117
- [27] Hitselberger WE, House WF, Luetje CM. Hypoglossal-Facial Anastomosis. In: *Acoustic Tumors*:

- Management. Baltimore: University Park Press; 1979:97–103
- [28] Atlas MD, Lowinger DSG. A new technique for hypoglossal-facial nerve repair. *Laryngoscope*. 1997; 107:984–991
- [29] Pitty LF, Tator CH. Hypoglossal-Facial Nerve Anastomosis for Facial Nerve Palsy Following Surgery for Cerebellopontine Angle Tumors. *Journal of Neurosurgery*. 1992; 77:724–731
- [30] Duel AB. Advanced Methods in the Surgical Treatment of Facial Paralysis. *Ann Otol Rhinol Laryngol*. 1934; 43:76–88
- [31] Poe DS, Scher N, Panje WR. Facial Reanimation by XI-VII Anastomosis Without Shoulder Paralysis. *Laryngoscope*. 1989; 99:1040–1047
- [32] Ebersold MJ, Quast LM. Long-Term Results of Spinal Accessory Nerve-Facial Nerve Anastomosis. *J Neurosurg*. 1992; 77:51–54
- [33] Ahmed OH, Gallant SC, Ruiz R, et al. Validity of the Hum Test, a Simple and Reliable Alternative to the Weber Test. *Ann Otol Rhinol Laryngol*. 2018; 127:402–405

Part XII

Tumors of the Nervous and Related Systems

XII

35	Tumor Classification and General Information	616
36	Genetic Tumor Syndromes Involving the CNS	637
37	Adult Diffuse Gliomas	657
38	Pediatric-type Diffuse Tumors	679
39	Circumscribed Astrocytic Gliomas	689
40	Glioneuronal and Neuronal Tumors	706
41	Ependymal Tumors	724
42	Choroid Plexus Tumors	739
43	Embryonal Tumors	744
44	Pineal Tumors and Pineal Region Lesions	758
45	Cranial and Paraspinal Nerve Tumors	768
46	Meningiomas (Intracranial)	803
47	Mesenchymal, Non-meningothelial Tumors	820
48	Melanocytic Tumors and CNS Germ Cell Tumors	829
49	Hematolymphoid Tumors Involving the CNS	840
50	Tumors of the Sellar Region	849
51	PitNET/Adenomas – Clinical Considerations	861
52	Pituitary Tumors – Evaluation	874
53	PitNET/Adenomas – General Management	886
54	PitNET/Adenomas – Surgical Management, Outcome, and Recurrence Management	896

35 Tumor Classification and General Information

35.1 WHO classification of tumors of the nervous system

The fifth edition of the WHO Classification of Tumors of the Central Nervous System (WHO CNS5)¹ published in 2021 is shown in ► Table 35.1 and incorporates recommendations of the cIMPACT-NOW (the Consortium to Inform Molecular and Practical Approaches to CNS Tumor Taxonomy–Not Official WHO).² Some noteworthy general features:

1. tumors are classified by types, some of which are divided into subtypes. Some subtypes are actual variations of the parent tumor, while others are just differing grades of the tumor
2. the suffixes NOS & NEC are used as follows³:
 - NOS (not otherwise specified): indicates that molecular testing was either not done or that the results were uncertain
 - NEC (not elsewhere classified): is used when there is nonconcordance between histologic features and molecular testing, or when results show noncanonical results e.g., as might occur with a new or emerging tumor type
3. molecular alterations are integral to the accurate diagnosis of CNS neoplasms. The most common diagnostic findings are listed in ► Table 35.7
4. in the following sections, descriptions such as histology and molecular biology do not cover the gamut of each tumor—for that the reader is referred to the definitive WHO Classification publication¹—but excerpts are included to show where the tumors fit into the overall scheme and to highlight some features that might be of relevance or interest to the clinician treating patients with these conditions

Table 35.1 WHO classification of tumors of the central nervous system (5th edition)¹ (modified, used with permission from WHO Classification of Tumours Editorial Board. Central Nervous System Tumours. Lyon (France): International Agency for Research on Cancer; 2021. (WHO classification of tumours series, 5th ed.; vol. 6). Available from: <https://tumourclassification.iarc.who.int/>.)

Tumor ^a	WHO grade ^b	ICD-O ^c
Gliomas, glioneuronal tumors, and neuronal tumors (p.657)		
Adult-type diffuse gliomas (p.658)		
astrocytoma, IDH mutant (p.658)		
• astrocytoma, IDH-mutant, grade 2	2	9400/3
• astrocytoma, IDH-mutant, grade 3	3	9401/3
• astrocytoma, IDH-mutant, grade 4	4	9405/3
oligodendroglioma, IDH-mutant & 1p/19q codeleted (p.662)		
• oligodendroglioma, IDH-mutant & 1p/19q codeleted, grade 2	2	9450/3
• oligodendroglioma, IDH-mutant & 1p/19q codeleted, grade 3	3	9451/3
glioblastoma, IDH-wildtype (p.664)	4	9400/3
• giant cell glioblastoma (p.665)	4	
• gliosarcoma (p.665)	4	
• epithelioid glioblastoma (p.666)	4	
Pediatric-type diffuse low-grade gliomas (p.679)		
diffuse astrocytoma, MYB- or MYBL1-altered (p.679)	1	9421/1
angiocentric glioma (p.679)	1	9431/1
polymorphous low-grade neuroepithelial tumor of the young (p.680)	1	9413/0
diffuse low-grade glioma, MAPK pathway-altered (p.682)	1	9421/1
Pediatric-type diffuse high-grade gliomas (p.683)		
diffuse midline glioma, H3 K27-altered (p.683)	4	9385/3
diffuse hemispheric glioma, H3 G34-mutant (p.685)	4	
diffuse pediatric-type high-grade glioma, H3-wildtype & IDH-wildtype (p.686)	4	
infant-type hemispheric glioma (p.687)	N/A	

Table 35.1 continued

Tumor ^a		WHO grade ^b	ICD-O ^c
Circumscribed astrocytic gliomas (p.689)			
	pilocytic astrocytoma (p.689)	1	9421/1
	high-grade astrocytoma with piloid features (p.698)	N/A	9421/3
	pleomorphic xanthoastrocytoma (p.698) (PXA)	2 or 3	9424/3
	subependymal giant cell astrocytoma (p.700) (SEGA)	1	
	chordoid glioma (p.702)	2	9444/1
	astroblastoma, MN1-altered (p.703)	1	9430/3
Glioneuronal and neuronal tumors			
	ganglioglioma (p.706)	1	9505/1
	gangliocytoma (p.707)	1	9492/0
	desmoplastic infantile ganglioglioma (DIG)/ desmoplastic infantile astrocytoma (DIA) (p.708)	1	9412/1
	dysembryoplastic neuroepithelial tumor (p.709) (DNT)	1	9413/0
	diffuse glioneuronal tumor with oligodendroglioma-like features and nuclear clusters (p.711) (<i>provisional entity</i>)	N/A	N/A
	papillary glioneuronal tumor (p.712) (PGNT)	1	9509/1
	rosette-forming glioneuronal tumor (p.713) (RGNT)	1	
	myxoid glioneuronal tumor (p.714)	1	
	diffuse leptomeningeal glioneuronal tumor (p.715) (DLGNT)	2 or 3	9509/3
	multinodular and vacuolating neuronal tumor (p.717) (MVNT)	1	9509/0
	dysplastic cerebellar gangliocytoma (p.716) (DCG) (Lhermitte-Duclos disease)	1	9493/0
	central neurocytoma (p.719)	2	9506/1
	extraventricular neurocytoma (p.720)	2	
	cerebellar liponeurocytoma (p.721)	2	
Ependymal tumors			
	supratentorial ependymoma (p.724)	2 or 3	9391/3
	supratentorial ependymoma, ZFTA fusion-positive (p.725)	2 or 3	9396/3
	supratentorial ependymoma, YAP1 fusion-positive (p.726)	N/A	
	posterior fossa ependymoma (p.727)	2 or 3	9391/3
	posterior fossa group A (PFA) ependymoma (p.730) (PFA)	2 or 3	9396/3
	posterior fossa group B (PFB) ependymoma (p.731) (PFB)	2 or 3	
	spinal ependymoma (p.732)	2 or 3	9391/3
	spinal ependymoma, MYCN-amplified (p.733)	N/A	9396/3
	myxopapillary ependymoma (p.734)	2	9394/1
	subependymoma (p.735)	1	9383/1
Choroid plexus tumors			
	choroid plexus papilloma (p.739) (CPP)	1	9390/0
	atypical choroid plexus papilloma (p.740)	2	9390/1
	choroid plexus carcinoma (p.741)	3	9390/3
Embryonal tumors (p.744)			
Medulloblastomas, molecularly defined (p.750)			
	medulloblastoma, WNT-activated (p.750)	4	9475/3
	medulloblastoma, SHH-activated & <i>TP53-wildtype</i> (p.751)	4	9471/3
	medulloblastoma, SHH-activated & <i>TP53-mutant</i> (p.752)	4	9476/3
	medulloblastoma, non-WNT/non-SHH (p.753)	4	9477/3
Medulloblastomas, histologically defined			
	medulloblastoma, histologically defined (p.748)		9470/3
	• classic medulloblastoma (p.749)	4	N/A

Table 35.1 continued

Tumor ^a		WHO grade ^b	ICD-O ^c
- medulloblastoma, desmoplastic/nodular (p. 749) - medulloblastoma, with extensive nodularity (p. 749) (MBEN) - large cell/anaplastic medulloblastoma (p. 749)	medulloblastoma, desmoplastic/nodular (p. 749)	4	9471/3
	medulloblastoma, with extensive nodularity (p. 749) (MBEN)	4	
	large cell/anaplastic medulloblastoma (p. 749)	4	9474/3
Other CNS embryonal tumors			
	atypical teratoid/rhabdoid tumor (p. 754) (AT/RT)	4	9508/3
	cribriform neuroepithelial tumor (p. 755)	N/A	N/A
	embryonal tumor with multilayered rosettes (p. 756) (ETMR)	4	9478/3
	CNS neuroblastoma, FOXR2-activated	4	9500/3
	CNS tumor with BCOR internal tandem duplication	N/A	
	CNS embryonal tumor NEC/NOS	3 or 4	9473/3
Tumors of the pineal region (p. 758)			
Pineal tumors			
	pineocytoma (p. 762)	1	9361/1
	pineal parenchymal tumor of intermediate differentiation (p. 763) (PPTID)	2 or 3	9362/3
	pineoblastoma (p. 764)	4	9362/3
	papillary tumor of the pineal region (p. 765)	2 or 3	9395/3
Pineal region tumors of uncertain cell of origin			
	desmoplastic myxoid tumor of the pineal region, SMARCB1-mutant	1	N/A
Pineal region tumors of non-pineal cell origin			
	germ cell tumors (see Germ cell tumors section below)		
	other non-pineal tumors (≧ Table 44.1). Includes: ependymomas, epidermoid or dermoid cysts, astrocytomas, paragangliomas...		
Cranial and paraspinal nerve tumors (p. 768)			
	schwannoma (p. 768)		9560/0
	- ancient schwannoma - cellular schwannoma - plexiform schwannoma - epithelioid schwannoma - microcystic/reticular schwannoma	1	
	neurofibroma (p. 769)		9540/0
	- cellular neurofibroma - atypical neurofibroma / atypical neurofibromatous neoplasm of uncertain biological potential (AN/ANNUBP) - plexiform neurofibroma - diffuse neurofibroma - nodular neurofibroma - plexiform neurofibroma - massive soft tissue neurofibroma	1	
	perineurioma (p. 771)	1	9571/0
	hybrid nerve sheath tumor (p. 772)		9563/0
	malignant melanocytic nerve sheath tumor (p. 773)		9540/3
	malignant peripheral nerve sheath tumor (MPNST) (p. 774)	2, 3 or 4	
	cauda equina neuroendocrine tumor (p. 774)		8693/3
Meningiomas			
	meningioma (p. 803)		9530/X
	- meningotheial meningioma - fibrous meningioma - transitional meningioma	1	9531/X
		1	9532/X
		1	9537/X

Table 35.1 continued

Tumor ^a		WHO grade ^b	ICD-O ^c
• psammomatous meningioma • angiomatous meningioma • microcystic meningioma • secretory meningioma • lymphoplasmacyte-rich meningioma • metaplastic meningioma • chordoid meningioma • clear cell meningioma • rhabdoid meningioma • papillary meningioma • atypical meningioma • anaplastic (malignant) meningioma		1	9533/X
		1	9534/X
		1	9531/X
		1	9530/X
		1	9530/X
		1	9530/X
		2	9538/2
		2	9538/2
		3	9538/X
		2 or 3	9538/X
		2	9539/X
		3	9538/X
Mesenchymal, non-meningothelial tumors involving the CNS			
<i>Soft tissue tumors</i>			
Fibroblastic and myofibroblastic tumors			
	solitary fibrous tumor (p.820)	1 - 3	8815/X
Vascular tumors			
	hemangioma (p.821)	1	X
	hemangioblastoma (p.822)	1	9161/1
Skeletal muscle tumors			
	rhabdomyosarcoma		8900/3
Tumors of uncertain differentiation			
	intracranial mesenchymal tumor, FET::CREB fusion-positive		N/A
	CIC-rearranged sarcoma		9367/3
	primary intracranial sarcoma, DICER1-mutant		9480/3
	Ewing sarcoma		9364/3
<i>Chondro-osseous tumors</i>			
Chondrogenic tumors			
	mesenchymal chondrosarcoma		9240/3
	chondrosarcoma		9920/3
<i>Notochordal tumors</i>			
	chordoma (p.825)		9370/3
Melanocytic tumors (p.829)			
Diffuse meningeal melanocytic neoplasms			
	meningeal melanocytosis (p.829)		8728/0
	meningeal melanomatosis (p.829)		8728/3
Circumscribed meningeal melanocytic neoplasms			
	meningeal melanocytoma (p.830)		8728/1
	meningeal melanoma (p.830)		8720/3
Hematolymphoid tumors involving the CNS			
<i>Lymphomas</i>			
CNS lymphomas			
	primary diffuse large B-cell lymphoma of the CNS (p.840)		9680/3
	immunodeficiency-associated CNS lymphomas (p.845)		N/A
	lymphomatoid granulomatosis		9766/1
	• lymphomatoid granulomatosis, grade 1		9766/1
	• lymphomatoid granulomatosis, grade 2		9766/1
	• lymphomatoid granulomatosis, grade 3		9766/3

Table 35.1 continued

Tumor ^a		WHO grade ^b	ICD-O ^c
	intravascular large B-cell lymphoma		9712/3
Miscellaneous rare lymphomas in the CNS			
	MALT lymphoma of the dura		9699/3
	other low-grade B-cell lymphomas of the CNS		
	anaplastic large cell lymphoma (ALK + /ALK-)		9714/3
	T-cell lymphomas		9702/3
	NK/T-cell lymphomas		9719/3
<i>Histiocytic tumors</i>			
	Erdheim-Chester disease		9749/3
	Rosai-Dorfman disease		9749/3
	juvenile xanthogranuloma		9749/1
	Langerhans cell histiocytosis (p. 846)		9751/1
	histiocytic sarcoma		9755/3
Germ cell tumors (p. 831)			
	mature teratoma		9080/3
	immature teratoma		9080/3
	teratoma with somatic-type malignancy		9084/3
	germinoma		9064/3
	embryonal carcinoma		9070/3
	yolk sac tumor		9071/3
	choriocarcinoma		9100/3
	mixed germ cell tumor		9085/3
Tumors of the sellar region (p. 849)⁴			
<i>Tumors of non pituitary origin</i>			
	adamantinomatous craniopharyngioma (p. 849)		9351/1
	papillary craniopharyngioma (p. 850)		9352/1
<i>Tumors of the posterior pituitary gland & infundibulum</i>			
	pituicytoma (p. 853)		9432/1
	granular cell tumor of the sellar region (p. 853)		9582/0
	spindle cell oncocytoma/oncocytic pituicytoma (p. 854)		8290/0
	ependymal pituicytoma		9391/1
<i>Tumors of the anterior pituitary gland (p. 861) (adenohypophyseal tumors) (Pituitary adenoma / pituitary neuroendocrine tumors (PitNET))</i>			
Pituitary neuroendocrine tumors of PIT1-lineage			
	somatotroph PitNET/adenoma (p. 870) (GH & α -subunit secreting)		8272/3
	lactotroph PitNET/adenoma (p. 867) (PRL secreting)		8271/3
	mammomatotroph PitNET/adenoma (p. 870) (GH > PRL & α -subunit secreting)		8272/3
	thyrotroph PitNET/adenoma (p. 871) (TSH- β & α -subunit secreting)		8272/3
	mature plurihormonal PIT1 lineage PitNET/adenoma (GH > PRL, α -subunit, TSH- β secreting)		8272/3
	immature PIT1-lineage PitNET/adenoma (GH, PRL, α -subunit, TSH- β secreting)		8272/3
	acidophil stem cell PitNET/adenoma (PRL > GH secreting)		8272/3
	mixed somatotroph-lactotroph PitNET/adenoma (GH, PRL secreting)		8281/3
Pituitary neuroendocrine tumors of TPIT lineage			
	corticotroph PitNET/adenoma (p. 867) (ACTH & other POMC secreting)		8272/3
Pituitary neuroendocrine tumors of SF1 lineage			
	gonadotroph PitNET/adenoma (p. 870) (α -subunit, FSH- β , LH- β secreting)		8272/3

Table 35.1 continued

Tumor ^a	WHO grade ^b	ICD-O ^c
Pituitary neuroendocrine tumors without distinct lineage differentiation		
null cell PitNET (p.861)/adenoma (non-secretory)		8272/3
Other sellar region tumors		
pituitary blastoma		8273/3
Metastasis to the CNS (p.908)		
metastases to the brain parenchyma (p.867)		
metastases to the spinal cord parenchyma		
metastases to the meninges		

^aTumor type based on the fifth edition of the WHO Classification of Tumors of the Central Nervous System (WHO CNS5)¹

• bulleted tumors are subtypes (in some cases they are differing grades of the parent tumor)

^bgraded according to the 2013 WHO Classification of Tumors of Soft Tissues & Bone.

^cICD-O = morphology code of the International Classification of Diseases for Oncology (ICD-O), third edition, second revision (ICD-O-3.2) International Association of Cancer Registries (IACR) [Internet]. Lyon (France): International Agency for Research on Cancer; updated 2021 Jan 25. Available from (<http://codes.iarc.fr>). The extension after the slash is the "behavior code": /0 = benign; /1 = unspecified, borderline, or uncertain behavior; /2 = carcinoma in situ and grade 3 intrapitithelial neoplasia; /3 = malignant tumors, primary site; /6 = malignant tumors, metastatic site; X = designation and/or grade assigned by characteristics described in that section.

* indicates new codes for 2021 approved by the IARC/WHO Committee for ICD-O in May 2021.

N/A indicates that the WHO has "not assigned" a grade as of the 2021 publication.

35.2 Pediatric brain tumors

35.2.1 General information

Among all childhood cancers, brain tumors are the second only to leukemias in incidence (20%), and are the most common solid pediatric tumor,⁵ comprising 40–50% of all tumors.⁶ Annual incidence of CNS tumors by age is shown in ► Table 35.2.

Table 35.2 Incidence of CNS tumors by age in the U.S. (for the years 2014–2018)⁷

Age (years)	Incidence per 100,000
0–4	6.30
5–9	5.59
10–14	5.94
15–19	7.32

35.2.2 Types of tumors

The common pediatric brain tumors (age <20 years) are gliomas and glioneural tumors (cerebellum, brainstem, and optic nerve) (which account for 45% of primary CNS tumors in this age group), embryonal tumors (12.3%) (67% of which are medulloblastoma), pituitary tumors (6.2%), nerve sheath tumors (4.2%), and craniopharyngiomas (3.8%).⁷

Among patients <20 years of age, the incidence of the following tumors decrease with age: pilocytic astrocytoma, malignant glioma, ependymal tumors, choroid plexus tumors, and embryonal tumors.⁷

35.2.3 Infratentorial vs. supratentorial tumor location

It has traditionally been taught that most pediatric brain tumors (≈ 60%) are infratentorial, and that these are ≈ equally divided among brainstem gliomas, cerebellar astrocytomas, and medulloblastomas. In reality, the ratio of supratentorial to infratentorial tumors is dependent on the specific age group studied, as illustrated in ► Table 35.3.

► Table 35.4 shows the histological breakdown for infratentorial vs. supratentorial tumors.

Astrocytomas are the most common supratentorial tumors in pediatrics as in adulthood.

Table 35.3 Percentage of infratentorial tumors by age

Age	% Infratentorial
0–6 mos	27%
6–12 mos	53%
12–24 mos	74%
2–16 yrs	42%

Table 35.4 Incidence of pediatric brain tumors^a

Tumor type	% of total
infratentorial tumors	54%
cerebellar astrocytomas (p. 693)	15%
medulloblastomas (p. 750)	14%
brainstem gliomas (p. 695)	12%
ependymomas (p. 724) ⁸	9%
supratentorial benign astrocytomas	13%

^adata pooled from 1350 pediatric brain tumors⁹ (p. 368)

35.2.4 Intracranial neoplasms during the first year of life

Brain tumors presenting during the first year of life are a different subset of tumors than those presenting later in childhood. In a busy neurosurgical unit in a children's hospital, they represented ≈ 8% of children admitted with brain tumors, an average of only ≈ 3 admissions per year.¹⁰

90% of brain tumors in *neonates* are of neuroectodermal origin, teratoma being the most common. Some of these tumors may be congenital.¹¹ Other supratentorial tumors include: astrocytoma, choroid plexus tumors, ependymomas, and craniopharyngiomas. Posterior fossa tumors include medulloblastoma and cerebellar astrocytoma.

Many of these tumors escape diagnosis until they are very large in size due to the elasticity of the infant skull, the adaptability of the developing nervous system to compensate for deficits, and the difficulty in examining a patient with limited neurologic repertoire and inability to verbalize or cooperate. The most common presenting manifestations are vomiting, arrest or regression of psychomotor development, macrocrania, poor feeding/failure to thrive. They may also present with seizures.

35.3 Brain tumors—general clinical aspects

35.3.1 Epidemiology

The average annual age-adjusted incidence rate (AAAIR) of all CNS tumors from 2014–2018 in the U.S. was 24.25 per 100,000⁷ The AAAIR was 7.06 for malignant brain tumors, and 17.18 for non-malignant. Approximately 29% of all CNS tumors (including brain tumors) were malignant and 71% were non-malignant.

The 6 histological groups that have a younger median age of diagnosis than other types are: pilocytic astrocytoma (median age = 11 years), choroid plexus tumors (20 years), neuronal and mixed neuronal-glial tumors (26 years), pineal region tumors (32 years), embryonal tumors (8 years), and germ cell tumors (15 years).⁷

35.3.2 Presenting signs and symptoms

For details of presentation, see sections below for supratentorial and infratentorial tumors. Most brain tumors present with:

- progressive neurologic deficit (68%): usually motor weakness (45%). This is in distinction to the more rapid presentation e.g., of stroke or seizure
- headache: a presenting symptom in 54% (see below)
- seizures in 26%. Often focal in onset (due to cortical irritation in the area of the tumor), may generalize secondarily. Uncommon with p-fossa tumors

35.3.3 Focal neurologic deficits associated with brain tumors

In addition to nonfocal signs and symptoms (e.g., seizures, increased ICP...), as with any destructive brain lesion, tumors may produce progressive deficits related to the function of the involved brain. Some characteristic “syndromes”:

1. frontal lobe: abulia, dementia, personality changes. Often nonlateralizing, but apraxia, hemiparesis or dysphasia (with dominant hemisphere involvement) may occur
2. temporal lobe: auditory or olfactory hallucinations, déjà vu, memory impairment. Contralateral superior quadrantanopsia may be detected on visual field testing
3. parietal lobe: contralateral motor or sensory impairment, homonymous hemianopsia. Agnosias (with dominant hemisphere involvement) and apraxias may occur; see Clinical syndromes of parietal lobe disease (p.99)
4. occipital lobe: contralateral visual field deficits, alexia (especially with corpus callosum involvement with infiltrating tumors)
5. posterior fossa: (see above) cranial nerve deficits, ataxia (truncal or appendicular). Seizures generally do not occur with isolated posterior fossa lesions

35.3.4 Headaches with brain tumors

General information

Headache (H/A) may occur with or without elevated ICP. H/A are present equally in patients with primary or metastatic tumor ($\approx 50\%$ of patients¹²). Classically described as being worse in the morning (possibly due to hypoventilation during sleep)—this may actually be uncommon.¹² Often exacerbated by coughing, straining, or (in 30%) bending forward (placing head in dependent position). Associated with nausea and vomiting in 40%, may be temporarily relieved by vomiting (possibly due to hyperventilation during vomiting). These features along with the presence of a focal neurologic deficit or seizure were thought to differentiate tumor H/A from others. However, H/A in 77% of brain tumor patients were similar to tension H/A, and in 9% were migraine-like.¹² Only 8% showed the “classic” brain tumor H/A, two-thirds of these patients had increased ICP.

Etiologies of tumor headache

The brain itself is not pain sensitive. H/A in the presence of a brain tumor may be due to any combination of the following:

1. increased intracranial pressure (ICP): which may be due to
 - a) tumor mass effect
 - b) hydrocephalus (obstructive or communicating)
 - c) mass effect from associated edema
 - d) mass effect from associated hemorrhage
2. invasion or compression of pain sensitive structures:
 - a) dura
 - b) blood vessels
 - c) periosteum
 - d) cranial nerves with sensory function
3. difficulty with vision can cause headache
 - a) diplopia due to dysfunction of nerves controlling extra-ocular muscles
 - direct compression of III, IV, or VI
 - abducens palsy from increased ICP, see diplopia (p.624)
 - internuclear ophthalmoplegia due to brainstem invasion/compression
 - b) difficulty focusing: due to optic nerve dysfunction from invasion/compression
4. extreme hypertension resulting from increased ICP (part of Cushing’s triad)
5. psychogenic: due to stress from loss of functional capacity (e.g., deteriorating job performance)

35.3.5 Supratentorial tumors

Signs and symptoms include¹³:

1. those due to increased ICP (see “increased ICP” under posterior fossa tumors below):
 - a) from mass effect of tumor and/or edema
 - b) from blockage of CSF drainage (hydrocephalus): less common in supratentorial tumors (classically occurs with colloid cyst, may also occur with entrapped lateral ventricle)

2. progressive focal deficits: includes weakness, dysphasia (which occurs in 37–58% of patients with left-sided brain tumors¹⁴): see below
 - a) due to destruction of brain parenchyma by tumor invasion
 - b) due to compression of brain parenchyma by mass and/or peritumoral edema and/or hemorrhage
 - c) due to compression of cranial nerve(s)
3. headache: see above
4. seizures: not infrequently the first symptom of a brain tumor. Tumor should be aggressively sought in an idiopathic first time seizure in a patient > 20 years (if negative, the patient should be followed with repeat studies at later dates). Rare with posterior fossa tumors or pituitary tumors
5. mental status changes: depression, lethargy, apathy, confusion
6. symptoms suggestive of a TIA (dubbed “tumor TIA”) or stroke, may be due to:
 - a) occlusion of a vessel by tumor cells
 - b) hemorrhage into the tumor: any tumor may hemorrhage, see Hemorrhagic brain tumors (p.1612)
 - c) focal seizure
7. in the special case of pituitary tumors (p.861):
 - a) symptoms due to endocrine disturbances
 - b) pituitary apoplexy (p.865)
 - c) CSF leak

Booking the case: Craniotomy for supratentorial tumor

Also see defaults & disclaimers (p. 25). If awake craniotomy is required, see Booking the case: Awake craniotomy (p. 1732).

1. position: (depends on location of tumor)
2. pre-op embolization (by neuroendovascular interventionalist) for some vascular tumors including some meningiomas
3. equipment:
 - a) microscope
 - b) ultrasonic aspirator
 - c) image guidance system
4. blood availability: type and cross 2 U PRBC
5. post-op: ICU
6. consent (in lay terms for the patient—not all-inclusive):
 - a) procedure: surgery through the skull to remove as much of the tumor as is safely possible
 - b) alternatives: nonsurgical management, radiation therapy for some tumors
 - c) complications: usual craniotomy complications (p. 25) plus inability to remove all of the tumor

35.3.6 Infratentorial tumors

Signs and symptoms

Seizures are rare (unlike the situation with supratentorial tumors) since seizures arise from irritation of cerebral cortex.

1. increased intracranial pressure (ICP): most posterior fossa tumors present with signs and symptoms of increased ICP due to hydrocephalus (HCP). These include:
 - a) headache: (see above)
 - b) nausea/vomiting: due either to increased ICP from HCP, or from direct pressure on the vagal nucleus or the area postrema (so-called “vomiting center”)
 - c) papilledema: estimated incidence is $\approx$ 50–90% (more common when the tumor impairs CSF circulation). Chronic increased pressure can cause irreversible blindness from optic nerve atrophy
 - d) gait disturbance/ataxia
 - e) vertigo
 - f) diplopia: may be due to VI nerve (abducens) palsy, which may occur with increased ICP in the absence of direct compression of the nerve

2. S/S indicative of mass effect in various locations within the p-fossa
 - a) lesions in cerebellar *hemisphere* may cause: ataxia of the extremities, dysmetria, intention tremor
 - b) lesions of cerebellar *vermis* may cause: broad based gait, truncal ataxia, titubation
 - c) brainstem involvement usually results in multiple cranial nerve and long tract abnormalities, and should be suspected when nystagmus is present (especially rotatory or vertical)

Evaluation of the patient with a posterior fossa (infratentorial) tumor

See Posterior fossa lesions (p.1089) for differential diagnosis (includes non-neoplastic lesions as well.)

In pediatric patients with a posterior fossa tumor, an MRI of the lumbar spine should be done pre-op to rule out drop mets (post-op there may be artifact from blood).

In adults, most intraparenchymal p-fossa tumors will be metastatic, and work-up for a primary should be undertaken in most cases.

Treatment of associated hydrocephalus

In cases with hydrocephalus at the time of presentation, some authors advocate initial placement of VP shunt or EVD prior to definitive surgery (waiting $\approx$ 2 wks before surgery) because of possibly lower operative mortality.¹⁵ Theoretical risks of using this approach include the following:

1. placing a shunt is generally a lifelong commitment, whereas not all patients with hydrocephalus from a p-fossa tumor will require a shunt
2. possible seeding of the peritoneum with malignant tumor cells e.g., with medulloblastoma. Consider placement of tumor filter (may not be justified given the high rate of filter occlusion and the low rate of "shunt metastases"¹⁶)
3. some shunts may become infected prior to the definitive surgery
4. definitive treatment is delayed, and the total number of hospital days may be increased
5. upward transtentorial herniation (p.325) may occur if there is excessively rapid CSF drainage

Either approach (shunting followed by elective p-fossa surgery, or semi-emergent definitive p-fossa surgery) is accepted. At Children's Hospital of Philadelphia, dexamethasone is started and the surgery is performed on the next elective operating day, unless neurologic deterioration occurs, necessitating emergency surgery.¹⁷

Some surgeons place a ventriculostomy at the time of surgery. CSF is drained only after the dura is opened (to avoid upwards herniation) to help equilibrate the pressures between the infra- and supratentorial compartments. Post-op, the external ventricular drain (EVD) is usually set at a low height ($\approx$ 10 cm above the EAM) for 24 hours, and is progressively raised over the next 48 hrs and should be D/C'd by $\approx$ 72 hrs post-op.

Booking the case: Craniotomy for infratentorial tumor

Also see defaults & disclaimers (p.25) and retromastoid surgery for vestibular schwannomas (p.793).

1. position: (typically either prone or park bench, depending on tumor type/location and surgeon preference)
2. pre-op embolization (by neuroendovascular interventionalist) for some vascular tumors such as hemangioblastoma
3. equipment:
 - a) microscope
 - b) ultrasonic aspirator
 - c) image guidance system (optional)
4. blood availability: type and cross 2 U PRBC
5. post-op: ICU
6. consent (in lay terms for the patient—not all-inclusive):
 - a) procedure: surgery through the skull, remove as much of the tumor as is safely possible
 - b) alternatives: nonsurgical management, radiation therapy for some tumors
 - c) complications: usual craniotomy complications (p.25) plus inability to remove all of the tumor, hydrocephalus, CSF leak

35.4 Management of the patient with a brain tumor

35.4.1 Initial evaluation and management

Approach to the patient with a newly demonstrated or suspected brain tumor

1. imaging:
 - a) obtain an MRI of the brain without and with contrast. A CT without and with contrast is an alternative if the MRI is not available or is contraindicated. In the current era, ensure that a navigation compatible sequence is included for possible surgery
 - b) if the lesion is likely a pituitary tumor, a brain MRI with pituitary protocol without and with contrast should be obtained
 - c) a metastatic workup (p.914) is obtained if there is a possibility that the lesion could represent a metastasis. Indications for this include multiple brain lesions, a solitary round ring-enhancing lesion at the interface between the gray and white matter or in the posterior fossa of an adult, suspicion based on history (patient with known cancer, or symptoms suggestive of cancer such as unexplained weight loss...). If the suspicion of metastases is low, or if the craniotomy needs to be done urgently, the metastatic workup can be deferred until after the craniotomy and then undertaken if the pathology from the surgery is consistent with metastases
 - d) obtain imaging of the entire neuraxis (cervical, thoracic and lumbar spinal MRI without and with contrast in addition to the brain) to rule out drop mets for pediatric patients with a posterior fossa tumor or if the tumor is suspicious for tumors that tend to seed in the CSF (e.g., ependymoma...)
 - e) specialized imaging studies, such as diffusion tensor imaging (DTI) MRI, functional MRI, cerebral angiography... are usually obtained subsequently, based on a number of factors obtained during this initial evaluation phase
 - f) cerebellopontine angle (CPA) tumors: FIESTA MRI and audiology screening should be obtained
 - g) if cerebral abscess is suspected, an infectious workup should be performed
2. medication
 - a) steroids: for lesions with significant or symptomatic edema or mass effect. If not acutely indicated, defer steroids until after biopsy in case the tumor could be a lymphoma (steroids can transiently suppress lymphoma). Typically decadron (see steroid use in brain tumors (p.626))
 - b) antiseizure medications (ASMs): if patient is not on ASMs and has not had a seizure, prophylactic ASMs should not be administered before surgery (see prophylactic antiseizure medications with brain tumors (p.626))
3. labs (in addition to routine labs):
 - a) for tumors that may involve the pituitary, obtain pituitary screening labs (p.874)
 - b) for pineal region tumors: markers are of limited usefulness (p.758). If still desired, one can obtain serum markers for β -hCG & alpha-fetoprotein (AFP), and/or CSF markers (p.832) (usually via LP when not contraindicated by mass effect or hydrocephalus) for β -hCG, AFP, PLAP

Steroid use in brain tumors

The beneficial effect of steroids in metastatic tumors is often much more dramatic than with primary infiltrating gliomas.

Dexamethasone (Decadron®) dose for brain tumors:

- for patients not previously on steroids:
 - adult: 10 mg IVP loading, then 6 mg PO/IVP q 6 hrs.^{18,19} In cases with severe vasogenic edema, doses up to 10 mg q 4 hrs may be used for a short time (<48-72 hours)
 - peds: 0.5-1 mg/kg IVP loading, then 0.25-0.5 mg/kg/d PO/IVP divided q 6 hrs. NB: avoid prolonged treatment because of growth suppressant effect in children
- for patients already on steroids:
 - for acute deterioration, a dose of approximately double the usual dose should be tried
 - see also "stress doses" (p.155)
- patients on steroids should receive prophylaxis against Cushing ulcers (see stress ulcers in neurosurgery (p.134))

Prophylactic antiseizure medications with brain tumors

20-40% of patients with a brain tumor will have had a seizure by the time their tumor is diagnosed.²⁰ Antiseizure medications (ASMs) are indicated in these patients who have had a seizure.

20–45% more will ultimately develop a seizure.²⁰ Prophylactic ASMs do not provide substantial benefit (reduction of risk > 25% for seizure-free survival), and there are significant risks involved. Practice guidelines for ASM use with brain tumors are shown below. Prophylactic ASMs are not indicated for isolated posterior fossa tumors due to the fact that p-fossa tumors by themselves have a low risk of provoking seizures.

Practice guideline: Prophylactic antiseizure medications with brain tumors

Level I²⁰: prophylactic ASMs should not be used routinely in patients with newly diagnosed brain tumors

Level II²⁰: in patients with brain tumors undergoing craniotomy, prophylactic ASMs may be used, and if there has been no seizure, it is appropriate to taper off ASMs starting 1 week post-op

35.4.2 Surgical intervention

Almost all tumors require surgical intervention. Exceptions include situations with an extremely high confidence diagnosis of a slow growing benign lesion based on imaging (e.g., many meningiomas, asymptomatic central neurocytomas...) in which case surveillance mode with serial imaging and monitoring of symptoms is employed (see below).

Goals of surgery are usually cytoreduction (maximal safe resection) with the goal of gross total excision when feasible, but at a minimum for a biopsy for diagnostic, prognostic, and treatment planning purposes.

35.4.3 Surveillance mode

For appropriate tumors with low suspicion of malignancy (e.g., many asymptomatic meningiomas, asymptomatic hormonally inactive pituitary tumors...). Use with caution with suspected low-grade glioma as some of these may be high-grade glioma or encephalitis.²¹

1. an initial follow-up imaging study is usually performed ≈ 3 months after the initial study (so-called "stability scan") to rule-out rapid growth
2. if stability is demonstrated, studies are performed annually thereafter unless a change in symptoms prompts an earlier repeat study

35.5 Chemotherapy agents for brain tumors

35.5.1 General information

General information is presented here. Chemotherapy for some specific tumors is also included in the respective sections devoted to those tumors where appropriate. Some agents used for CNS tumors: see ► Table 35.5.^{22,23}

35.5.2 Alkylating agents

Temozolomide (Temodar®), an oral alkylating agent, is a derivative of Dacarbazine (DTIC®). It is a prodrug which undergoes rapid non-enzymatic conversion at physiologic pH to the active metabolite monomethyl triazenoimidazole carboxamide (MTIC). The mutagenic/cytotoxic effect of MTIC is associated with alkylation (adding an alkyl group, the smallest of which is a methyl group) to DNA at various sites primarily at the O6 and N7 positions on guanine. Cells can repair this damage via O⁶-methylguanine-DNA methyltransferase (p.673) (MGMT), a protein which may be deficient to some degree in various tumors (especially astrocytomas, IDH-mutant (p.658)) which renders them more susceptible to temozolomide.

35.5.3 Nitrosoureas

Excellent BBB penetration (see below). Significant hematopoietic, pulmonary, and renal toxicity.

Table 35.5 Some chemotherapeutic agents used for CNS tumors

Agent	Mechanism
nitrosoureas: BCNU (carmustine), CCNU (lomustine), ACNU (nimustine)	DNA crosslinks, carbamoylation of amino groups
alkylating (methylating) agents: procarbazine, temozolomide (Temodar®) (p.627)	DNA alkylation, interferes with protein synthesis
carboplatin, cisplatin	chelation via intrastrand crosslinks
nitrogen mustards: cyclophosphamide, ifosfamide, cytoxan	DNA alkylation, carbonium ion formation
vinca alkaloids: vincristine, vinblastine, paclitaxel	microtubule function inhibitors
epidophyllotoxins (ETOP-oside, VP16, teniposide, VM26)	topoisomerase II inhibitors
topotecan, irinotecan (CPT-11)	topoisomerase I inhibitors
tamoxifen	protein kinase C inhibitor at high doses
bevacizumab (Avastin®)	anti-VEGF antibody may be useful in vestibular neuromas or recurrent GBM
hydroxyurea bleomycin taxol (paxlitaxol) methotrexate cytosine, arabinoside corticosteroids: dexamethasone, prednisone fluorouracil (FU)	

35.5.4 Combination chemotherapy

PCV: procarbazine, CCNU (lomustine), and vincristine combination therapy is used as either an adjunct (following XRT) or concomitant (simultaneously with XRT) therapy in a number of tumors. Sometimes carmustine (BCNU) is substituted for lomustine.

35.5.5 Blood-brain barrier (BBB) and chemotherapy agents

Traditionally, the BBB has been considered to be a major hindrance to the use of chemotherapy for brain tumors. In theory, the BBB effectively excludes many chemotherapeutic agents from the CNS, thereby creating a “safe haven” for some tumors, e.g., metastases. This concept has been challenged.²⁴ Regardless of the etiology, the response of most brain tumors to systemic chemotherapy is usually very modest, with a notable exception being a favorable response of tumors with deficient MGMT (p.673) activity. Considerations regarding chemotherapeutic agents in relation to the BBB include:

1. some CNS tumors may partially disrupt the BBB, especially malignant gliomas²⁵
2. lipophilic agents (e.g., nitrosoureas) may cross the BBB more readily
3. selective intra-arterial (e.g., intracarotid or intervertebral) injection²⁶: produces higher local concentration of agents which increases penetration of the BBB, with lower associated systemic toxicities than would otherwise occur
4. the BBB may be iatrogenically disrupted (e.g., with mannitol) prior to administration of the agent
5. the BBB may be bypassed by intrathecal administration of agents via LP or ventricular access device, e.g., methotrexate for CNS lymphoma (p.844)
6. biodegradable polymer wafers containing a chemotherapeutic agent may be directly implanted. At present, the only such preparation is Gliadel wafers (p.673) (BCNU and prolifeprosan)

35.5.6 Imaging studies following surgical removal of tumor

At many academic centers, it is common to get a noncontrast CT scan within 6–12 hours of surgery to assess for acute complications (primarily hematoma–intraparenchymal, epidural or subdural, amount of pneumocephalus, hydrocephalus...).

Then to assess the extent of tumor removal, a post-op brain MRI (or CT if MRI cannot be done) without and with contrast should either be obtained within 2–3 days,²⁷ or should be delayed at least ≈ 30 days. The *non*-contrast scan is important to help differentiate blood from enhancement. The contrast images demonstrates areas of enhancement, which may represent residual tumor.

As a result of inflammatory changes due to surgery, enhancement increases in the wall of the resection bed within 24–48 hrs of surgery (so-called “post-operative changes”). After $\approx$ 48 hours, one cannot reliably differentiate contrast enhancement due to postoperative changes from residual. This usually subsides by $\approx$ 30 days,²⁸ but may persist for 6–8 weeks.²⁹ The effect of steroids on contrast enhancement is controversial,^{30,31} and may depend on many factors (including tumor type).

35.6 Intraoperative pathology consultations (“frozen section”)

35.6.1 Accuracy of intraoperative pathology consultations

Accuracy of intraoperative pathologic diagnosis (IPD) can be increased by:

- providing the pathologist with information regarding: patient demographics, clinical history, imaging results, relevant previous pathologic diagnoses, and clinical impression
- larger specimen sizes when possible
- avoiding artifact created by excessive crushing or coagulation

Intraoperative pathology diagnoses (IPD) should be considered preliminary. The final diagnosis differs from the IPD in approximately 3–10% of cases.^{32,33,34} If the IPD does not correlate with the clinical impression, direct discussion with the pathologist may be advisable.

35.6.2 Techniques for intraoperative tissue preparation

► **Touch preparation.** The specimen is gently “touched” with a glass slide which is then rapidly fixed, stained and dehydrated for examination. This technique is particularly useful for tumors with discohesive cells (e.g., lymphoma, PitNET/adenoma).

► **Smear or squash preparation.** A small portion of specimen is smeared or compressed with moderate pressure between two glass slides, rapidly fixed, stained, and dehydrated for examination. This technique can be particularly useful for: multiple sclerosis (identifying histiocytes), visualizing long cell processes in gliomas, and identifying cytoplasmic inclusions or intranuclear pseudoinclusions;³⁵ (p 5–6) The cohesive nature often seen in tumors such as metastases and meningiomas is apparent, as are areas of necrosis.

Σ: Smears are good for cytology, but do not show architecture. Preserve more tissue for permanent pathology than frozen section.

► **Frozen section.** A portion of tissue is rapidly frozen in liquid nitrogen and cut into 4–6 micron sections, mounted on a slide, rapidly fixed, stained, and dehydrated for examination. Unlike touch and smear preparations, this allows more accurate assessment of lesion architecture, cellularity, and interface with adjacent brain tissue. Disadvantages include use of greater tissue with less available for permanent histology (important with small biopsies), as well as artifacts such as freezing ice crystal artifact (which, when present, suggests lesional tissue has been biopsied but limits interpretation of cellularity).³⁵ (p 6) When possible, some tissue should be preserved for processing without freezing to avoid artifacts.

Σ: Frozen section is better for tumor architecture, but creates artifact and uses more tissue.

At the time of frozen section interpretation, consideration for additional studies such as tissue cultures or flow cytometry should be entertained. With minute specimens, a discussion is warranted whether frozen section is required to preserve tissue for permanent studies.

35.6.3 Selected frozen section pitfalls or potential critical diagnoses

- **differentiating low grade gliomas from normal or reactive brain tissue** can be challenging.^{32,33} Increased cellularity (best evaluated at low power), nuclear atypia when present, and increased perineuronal satellitosis can be helpful, though not always readily observed.³⁵ (p 72–3, 174–7) **Pearl:** Secondary structures of Scherer may be useful in identifying gliomas in challenging cases, and include increased perineuronal and perivascular satellitosis (limited perineuronal satellitosis is normal) and accumulation of neoplastic cells in the subpial molecular layer with subpial tumor spread³⁶
- **metastasis vs. glioma:** usually not problematic on frozen section, except occasionally with markedly atypical gliomas with limited sampling.^{32,33,34} In such rare cases immunohistochemistry is helpful

- **astrocytoma vs. oligodendroglioma:** usually not a critical distinction at the time of frozen section diagnosis. However, in part due to frozen section artifact imparted to the nuclei, oligodendrogliomas can be interpreted as astrocytomas on frozen section.³² Smear preparations can at times be helpful due to decreased “freezing” artifact. Also, perineuronal satellitosis may in some cases be more pronounced in oligodendrogliomas. In WHO CNS5, the distinction is dependent on molecular genetics.

Pearl: The “fried egg” appearance in oligodendrogliomas is an artifact of formalin fixation for permanent sections and is NOT present on frozen sections, also hindering intraoperative interpretation

- **glioma grading:** sampling bias, particularly in small biopsy specimens, can lead to undergrading (i.e., underrepresentation of mitotic figures, vascular proliferation or necrosis in the biopsy specimen). Conversely, overgrading of gliomas at the time of intraoperative consultation can occur, including with low-grade childhood gliomas such as pilocytic astrocytomas³²
- **radionecrosis vs. recurrent tumor (higher grade astrocytoma or glioblastoma):** despite the fact that a history of prior radiation therapy is usually known at the time of intraoperative consultation, differentiation of the two entities can at times be difficult.³² Both lesions are often present simultaneously. Identifying obvious tumor cells and palisading necrosis suggests recurrent/residual glioblastoma. Radiation necrosis, which affects primarily white matter, is supported by large geographic areas of necrosis, sclerosis/hyalinization of vessels or fibrinoid necrosis of vessel walls, perivascular lymphocytes, calcifications, and the presence of macrophages
- **ischemic infarct:** may see ischemic red neuron change (if the biopsy includes gray matter), as well as histiocytes (similar to demyelinating lesions). Necrosis may mimic the center of ring enhancing lesions (such as glioblastoma or metastasis), though tumor necrosis will typically involve vessels and lack a macrophage response³⁵ (p 663–6)
- **demyelinating lesions** (e.g., tumefactive MS): primarily affects white matter with discrete borders. Identifying histiocytes at the time of frozen section is critical for the diagnosis. The histiocytes can mimic astrocytes of gliomas on frozen section preparation; smear preparations are particularly helpful in the distinction
- **small cell carcinoma & lymphoma (PCNSL):** accurate frozen section diagnosis may be critical as both conditions typically require only biopsy without debulking unless there is significant mass effect (definitive treatment is chemo-radiation). The two entities may resemble each other on frozen section as well as resembling astrocytomas, oligodendrogliomas, and other types of metastatic carcinomas.^{32,33,34,35} (p 395–7) Touch preparations can be particularly useful in identifying PCNSL
- **meningioma vs. tumors with spindle cells** (e.g., schwannoma): distinguishing between these two can be difficult at times on intraoperative consultation.³² Classic features of meningiomas (whorls, psammoma bodies, intranuclear pseudoinclusions) may be absent, and freeze artifact can create areas that resemble Antoni B fibers.³² In addition, underdiagnosis of malignant meningiomas and sarcomas at the time of frozen section has been noted^{32,33}
- **spinal cord astrocytoma vs. ependymoma:** On occasion these entities can be difficult to distinguish during intraoperative consultation, especially as most spinal cord biopsies for frozen interpretation are minute. Due to the critical surgical implications of the accuracy of the frozen section diagnosis for intramedullary spinal gliomas, a careful discussion should be had between the surgeon and pathologist at the time of intraoperative consultation (refer to section on tumors of the spine and spinal cord)

35.6.4 Tissue preparation for permanent sections

Tissue is processed overnight through a variety of alcohol/xylene steps to remove water. This allows for embedding into paraffin so that thin sections can be cut for slide mounting. The specimens are then rehydrated by essentially reversing the alcohol/xylene steps in preparation for staining, and then dehydrated again for permanent slide coverslipping. This produces better histology with fewer artifacts, allows for processing of larger specimen volumes for evaluation, and allows for application of special stains as needed.

► **Fresh specimens.** Tissue should be sent “fresh” (i.e., without preservatives such as formalin) when the following techniques are needed:

- electron microscopy
- flow cytometry: e.g., when lymphoma is suspected
- muscle
- cultures (typically: aerobic, anaerobic, acid-fast, and fungal) should be sent in situations where infection is a consideration

35.7 Select commonly utilized stains in neuropathology

35.7.1 Organism and special stains

- organism stains:
 - tissue Gram stains (Brown & Brenn, Brown & Hopps)
 - fungi: Periodic acid–Schiff (PAS), Gomori methenamine silver (GMS)
 - acid-fast bacilli (tuberculosis is the most common): Ziehl–Neelsen, Kinyoun, FITE
- special stains:
 - luxol fast blue: stains myelin. Absence of staining highlights demyelinating lesions. Presence within histiocytes demonstrates ingestion of myelin as seen in MS
 - trichrome and reticulin stain: both delineate the sarcomatous component of gliosarcomas. A reticulin stain demonstrates the connective tissue around acini of the normal pituitary gland, a feature that is lost with PitNET/adenomas

35.7.2 Immunohistochemical stains

General information

Staining patterns: An individual tumor may lack a marker that is typically representative of its type. ★ Therefore, a positive stain is typically more significant than a negative stain.³⁷ General staining patterns³⁷ are shown in ► Table 35.6.

Table 35.6 Immunohistochemical staining patterns for nervous system tumor masses of epithelioid cells^a

Neoplasm	Immunohistochemical stain response ^{b,c}					
	GFAP	CAM5.2	EMA	S-100	CgA	Syn
oligodendroglioma	+	–	–		–	0
ependymoma	0		0	+		–
choroid plexus papilloma					0	+
chordoma					–	–
craniopharyngioma		+	+	–	0	0
carcinoma	–				+	+
PitNET		0	–	0		–
paraganglioma			+			–
meningioma				+	–	0
melanoma		–	–	+	–	0
hemangioblastoma	0			0		0

^amodified from McKeever PE. Immunohistochemistry of the Nervous System. In: Dobbs DJ, ed. Diagnostic Immunohistochemistry. Churchill Livingstone, NY, © 2002

^babbreviations: GFAP = glial fibrillary acidic protein, EMA = epithelial membrane antigen, CAM5.2 = cytokeratin CAM5.2, CgA = chromogranin A, syn = synaptophysin

^ca “+” or a “–” sign indicates presence or absence of the stain respectively; “0” entries indicate that the stain is not decisive for that particular tumor

Glial fibrillary acidic protein (GFAP)

Polypeptide, MW = 49,000 Daltons. Stains intermediate filaments classically identified in astrocytes/astrocytic tumors. However, it is also typically expressed in ependymomas, oligodendrogliomas (especially in minigemistocytes and gliofibrillary oligodendrocytes), and some choroid plexus papillomas.³⁸ (p 30–1),³⁹ (p 56,76,83) GFAP is only rarely found outside the CNS in such tissues as Schwann cells, epithelium of the lens, certain liver cells, chondrocytes, etc). ★ It would be unusual for a metastatic lesion to demonstrate staining for GFAP. However, GFAP expression may be limited in certain subsets of GBM (e.g., small cell GBM).³⁹ (p 37)

S-100 protein

A low molecular weight (21,000 Daltons) calcium-binding protein that stains a variety of tissues including glia, neurons, chondrocytes, stellate cells of the adenohypophysis, myoepithelial cells, etc.³⁷ (p 75) Stains a variety of CNS neoplasms such as gliomas (though less specific than GFAP),

ependymomas, chordomas, and craniopharyngiomas.^{37,39} Primary uses in neuropathology include to support the diagnosis of metastatic melanoma, and, in the peripheral nervous system, to confirm the diagnosis of schwannoma or neurofibroma (less intense staining in the latter).^{39 (p 156)}

Clinically has been measured in serum (see below).

Cytokeratins (high and low molecular weight)

A variety of stains (low [e.g., CAM 5.2] and high molecular weight keratin, CK7, CK20, etc.) which stain epithelial cells. Useful to distinguish metastatic carcinoma (positive staining) from primary CNS tumors (note: cytokeratins can be expressed in choroid plexus papillomas, and GBM can express cytokeratins in rare cases).^{39 (p 39–40)} Different keratin staining combinations can be used to suggest possible sites of origin for metastatic tumors.

Epithelial membrane antigen (EMA)

Stains cell membranes in many carcinomas; useful to distinguish hemangioblastoma (negative) from metastatic renal cell carcinoma (positive). Also, meningiomas typically demonstrate positive staining, as do ependymomas.^{39 (p 169)}

MIB-1 (AKA monoclonal mouse anti-human Ki-67 antibody)

The Ki-67 antigen is expressed in all phases of the cell cycle except G0. Available since the early 1990s, it is a valuable marker of cell proliferation but can only be used with fresh-frozen specimens. MIB-1 is a monoclonal antibody developed using recombinant parts of the Ki-67 protein as an immunogen, and can be used on paraffin-embedded sections of fixed tissue. Cells leaving the G0/G1-phase and entering the S-phase (performing DNA synthesis) stain positive with MIB-1 immunohistochemical stain. This stain can be used to compute a semi-quantitative score. A high MIB-1 labeling index denotes high mitotic activity, which often correlates with degree of malignancy. Often used in lymphomas, endocrine tumors, carcinoids, etc. See also use in astrocytomas (p.660) and in meningiomas (p.805).

Neuroendocrine stains

In neuropathology, utilized in central neurocytoma, medulloblastoma, pineocytoma, ganglion cell tumors, paragangliomas, and choroid plexus tumors.³⁷ Metastases that are positive for neuroendocrine stains include small-cell carcinoma of the lung (most common), malignant pheochromocytoma, and Merkel cell tumor.

Includes:

1. chromogranin: stains synaptic vesicles. Perhaps less commonly utilized than synaptophysin in primary CNS tumor
2. synaptophysin: stains synaptic vesicles; has higher sensitivity but lower specificity than chromogranin.^{37 (p 200)} Often positive for central neurocytoma, which typically lacks chromogranin staining.^{39 (p 107)}
3. CD56 (Neural cell adhesion molecule): a family of glycoproteins present in nervous tissue as well as other tissues such as thyroid, liver, etc.^{37 (p 142–3,264)} Frequently used to confirm neuroendocrine differentiation.
4. neuron specific enolase (NSE): sensitive but not specific for neuronal or neuroendocrine differentiation despite its name (often referred to as “neuron non-specific enolase”).^{37 (p 338)} Due to this, it is less often utilized as a neuroendocrine marker.

Cluster of differentiation (CD) markers

A number of immunohistochemical stains that detect antigens on the surface of leukocytes, though many stain other cell types as well. Examples include:

- CD45: General leukocyte marker
- CD3 and CD5: T-cells
- CD20: B-cells
- CD38 and CD138: Plasma cells
- CD68: Histiocytes
- CD56 (Neural cell adhesion molecule): Classically stains natural killer cells, but also a neuroendocrine marker (see above)
- organism-specific immunohistochemical stains are available to detect certain organisms that infect the nervous system including HSV, CMV, and *Toxoplasma gondii*

Metastases that are positive for neuroendocrine stains include: small-cell carcinoma of the lung, malignant pheochromocytoma, Merkel cell tumor. Metastatic small-cell tumors to the brain staining positive for neuroendocrine stains are almost all due to lung primaries (all other primaries are a distant possibility).

35.7.3 Molecular alterations in major CNS tumors

In the WHO CNS5,¹ molecular genetics plays a central role in the classification of CNS tumors. ► Table 35.7 shows key molecular alterations that are involved in the diagnostic classification of major CNS tumors. See reference for more details.

Table 35.7 Key diagnostic genes, mutations, molecules, pathways, and/or combinations in major primary CNS tumors^{1,40} (used with permission from WHO Classification of Tumours Editorial Board. Central Nervous System Tumours. Lyon (France): International Agency for Research on Cancer; 2021. (WHO classification of tumours series, 5th ed.; vol. 6). Available from: <https://tumourclassification.iarc.who.int/>.)

Tumor type		Genes/molecular profiles characteristically altered ^a
Adult-type diffuse gliomas		
	astrocytoma, IDH mutant (p.658)	<i>IDH1, IDH2, ATRX, TP53, CDKN2A/B</i>
	oligodendroglioma, IDH-mutant & 1p/19q codeleted (p.662)	<i>IDH1, IDH2, 1p/19q, TERT promoter, CIC, FUBP1, NOTCH1</i>
	glioblastoma, IDH-wildtype (p.664)	IDH-wildtype, <i>TERT</i> promoter, chromosomes 7/10, <i>EGFR</i>
Pediatric-type diffuse low-grade gliomas		
	diffuse astrocytoma, MYB- or MYBL1-altered (p.679)	<i>MYB, MYBL1</i>
	angiocentric glioma (p.679)	<i>MYB</i>
	polymorphous low-grade neuroepithelial tumor of the young (p.680)	<i>BRAF, FGFR</i> family
	diffuse low-grade glioma, MAPK pathway-altered (p.682)	<i>FGFR1, BRAF</i>
Pediatric-type diffuse high-grade gliomas		
	diffuse midline glioma, H3 K27-altered (p.683)	<i>H3 K27, TP53, ACVR1, PDGFRA, EGFR, EZHIP</i>
	diffuse hemispheric glioma, H3 G34-mutant (p.685)	<i>H3 G34, TP53, ATRX</i>
	diffuse pediatric-type high-grade glioma, H3-wildtype & IDH-wildtype (p.686)	IDH-wildtype, H3-wildtype, <i>PDGFRA, MYCN, EGFR</i> (methylome)
	infant-type hemispheric glioma (p.687)	<i>NTRK</i> family, <i>ALK, ROS, MET</i>
Circumscribed astrocytic gliomas		
	pilocytic astrocytoma (p.689)	<i>KIAA1549-BRAF, BRAF, NF1</i>
	high-grade astrocytoma with piloid features (p.698)	<i>BRAF, NF1, ATRX, CDKN2A/B</i> (methylome)
	pleomorphic xanthoastrocytoma (p.698)	<i>BRAF, CDKN2A/B</i>
	subependymal giant cell astrocytoma (p.700)	<i>TSC1, TSC2</i>
	chordoid glioma (p.702)	<i>PRKCA</i>
	astroblastoma, MN1-altered (p.703)	<i>MN1</i>
Glioneural and neuronal tumors		
	ganglion cell tumors	<i>BRAF</i>
	dysembryoplastic neuroepithelial tumor (p.709)	<i>FGFR1</i>
	diffuse glioneuronal tumor with oligodendrogloma-like features and nuclear clusters (p.711)	chromosome 14, (methylome)
	papillary glioneuronal tumor (p.712)	<i>PRKCA</i>
	rosette-forming glioneuronal tumor (p.713)	<i>FGFR1, PIK3CA, NF1</i>
	myxoid glioneuronal tumor (p.714)	<i>PDGFRA</i>
	diffuse leptomeningeal glioneuronal tumor (p.715)	<i>KIAA1549-BRAF</i> fusion, 1p (methylome)

Table 35.7 continued

Tumor type		Genes/molecular profiles characteristically altered ^a
	multinodular and vacuolating neuronal tumor (p. 717)	MAPK pathway
	dysplastic cerebellar gangliocytoma (Lhermitte-Duclos disease) (p. 716)	<i>PTEN</i>
	extraventricular neurocytoma (p. 720)	<i>FGFR</i> (<i>FGFR1-TACC1</i> fusion), IDH-wildtype
Ependymal tumors		
	supratentorial ependymoma (p. 724)	<i>ZFTA</i> , <i>RELA</i> , <i>YAP1</i> , <i>MAML2</i>
	posterior fossa ependymomas (p. 727)	H3 K27me3, <i>EZH1</i> (methylome)
	spinal ependymomas (p. 732)	<i>NF2</i> , <i>MYCN</i>
Embryonal tumors		
	medulloblastoma, WNT-activated (p. 750)	<i>CTNNB1</i> , <i>APC</i>
	medulloblastoma, SHH-activated (p. 751)	<i>TP53</i> , <i>PTCH1</i> , <i>SUFU</i> , <i>SMO</i> , <i>MYCN</i> , <i>GLI2</i> (methylome)
	medulloblastoma, non-WNT/non-SHH (p. 753)	<i>MYC</i> , <i>MYCN</i> , <i>PRDM6</i> , <i>KDM6A</i> (methylome)
	atypical teratoid/rhabdoid tumor (p. 754)	biallelic inactivation of <i>SMARCB1</i> (or rarely <i>SMARCA4</i>)
	embryonal tumor with multilayered rosettes	<i>C19MC</i> , <i>DICER1</i>
	CNS neuroblastoma, <i>FOXR2</i> -activated	<i>FOXR2</i>
	CNS tumor with <i>BCOR</i> internal tandem duplication	<i>BCOR</i>
desmoplastic myxoid tumor of the pineal region, <i>SMARCB1</i> -mutant		<i>SMARCB1</i>
meningiomas		<i>NF2</i> , <i>AKT1</i> , <i>TRAF7</i> , <i>SMO</i> , <i>PIK3CA</i> ; <i>KLF4</i> , <i>SMARCE1</i> , <i>BAP1</i> in subtypes; H3K27me3; <i>TERT</i> promoter, <i>CDKN2A/B</i> in CNS WHO grade 3
solitary fibrous tumor		<i>NAB2-STAT6</i>
meningeal melanocytic tumors		<i>NRAS</i> (diffuse); <i>GNAQ</i> , <i>GNA11</i> , <i>PLCB4</i> , <i>CYSLTR2</i> (circumscribed)
Tumors of the sellar region		
	adamantinomatous craniopharyngioma	<i>CTNNB1</i>
	papillary craniopharyngioma	<i>BRAF</i>

Abbreviations: CNS, central nervous system; C19MC, chromosome 19 microRNA cluster; IDH, isocitrate dehydrogenase; SHH, sonic hedgehog. Some of these are definitional for specific diagnoses, while others are not definitional but are characteristically altered or not altered. For each tumor type, these distinctions are specified in the WHO CNS5¹ as well as the Essential and Desirable Criteria sections of the individual chapters.

^a In this column, molecules that are definitional (including for those that are wildtype) are listed before others; for those tumor types without specific definitional changes, more commonly altered genes and molecules are listed before others. Most types have characteristic methylome patterns, but "(methylome)" is only listed for those types for which methylome testing offers particular diagnostic guidance, including for designating subtypes (as for Medulloblastoma, SHH-activated; Medulloblastoma, non-WNT/non-SHH; and Diffuse leptomeningeal glioneuronal tumor). H3 is a gene family (eg, H3F3A, HIST1H3B).

35.7.4 Tumor markers used clinically

Human chorionic gonadotropin (hCG)

A glycoprotein, MW = 45,000. Secreted by placental trophoblastic epithelium. Beta chain (β -hCG) is normally present only in the fetus or in gravid or postpartum females; otherwise it indicates disease. Classically associated with choriocarcinoma (uterine or testicular), it is also found in patients with embryonal cell tumors, teratocarcinoma of testis, and others.

CSF β -hCG is 0.5–2% of serum β -hCG in non-CNS tumors. Higher levels are diagnostic of cerebral mets from uterine or testicular choriocarcinoma, or primary choriocarcinoma or embryonal cell carcinoma of pineal (p. 758) or suprasellar region.

Alpha-fetoprotein

Alpha-fetoprotein (AFP) is a normal fetal glycoprotein (MW = 70,000) initially produced by the yolk sac, and later by the fetal liver. It is found in the fetal circulation throughout gestation, and drops rapidly during the first few weeks of life, reaching normal adult levels by age 1 yr. It is detectable only in trace amounts in normal adult males or nonpregnant females. It is present in amniotic fluid in normal pregnancies, and is detectable in maternal serum starting at $\approx$ 12–14 weeks gestation, increasing steadily throughout pregnancy until $\approx$ 32 weeks.⁴¹

Abnormally elevated serum AFP may occur in Ca of ovary, stomach, lung, colon, pancreas, as well as in cirrhosis or hepatitis and in the majority of gravid women carrying a fetus with an open neural tube defect; see Prenatal detection of neural tube defects (p.313). Serum AFP > 500 ng/ml usually means primary hepatic tumor.

CSF-AFP is elevated in some pineal region germ cell tumors (p.759). 16–25% of patients with testicular tumors develop cerebral mets, and elevated CSF AFP levels are reported in some.

Carcinoembryonic antigen (CEA)

A glycoprotein, MW = 200,000. Normally present in fetal endodermal cells. Originally described in the early 1960s in the serum in patients with colorectal adeno-Ca, it is now known to be elevated in many malignant and nonmalignant conditions (including cholecystitis, colitis, diverticulitis, hepatic involvement from any tumor, with 50–90% of terminal patients having elevation).

CSF CEA: levels > 1 ng/ml are reported with leptomeningeal spread of lung Ca (89%), breast Ca (60–67%), malignant melanoma (25–33%), and bladder Ca. May be normal even in CEA-secreting cerebral mets if they don't communicate with the subarachnoid space. Only carcinomatous meningitis from lung or breast Ca consistently elevates CSF CEA in the majority of patients.

S-100 protein

Serum S-100 protein levels rise after head trauma, and possibly after other insults to the brain. Levels may also be elevated in Creutzfeldt-Jakob disease.

References

- [1] Bosman FT, Jaffe ES, Lakhani SR, et al. World Health Organization Classification of Tumors series, vol. 6. In: WHO Classification of Tumors Editorial Board (ed.) World Health Organization classification of tumors of the central nervous system. 5th ed. Lyon, France: International Agency for Research on Cancer; 2021
- [2] Louis DN, Aldape K, Brat DJ, et al. Announcing cIMPACT-NOW: the Consortium to Inform Molecular and Practical Approaches to CNS Tumor Taxonomy. *Acta Neuropathol*. 2017; 133:1–3
- [3] Louis DN, Wesseling P, Paulus W, et al. cIMPACT-NOW update 1: Not Otherwise Specified (NOS) and Not Elsewhere Classified (NEC). *Acta Neuropathol*. 2018; 135:481–484
- [4] Mete O. WHO Classification of Tumors Editorial Board. Pituitary gland. Lyon, France 2022. <https://publications.iarc.fr/Book-And-Report-Series/Who-Classification-Of-Tumours>
- [5] Allen JC. Childhood Brain Tumors: Current Status of Clinical Trials in Newly Diagnosed and Recurrent Disease. *Ped Clin N Am*. 1985; 32:633–651
- [6] Laurent JP, Cheek WR. Brain Tumors in Children. *J Pediatr Neurol*. 1985; 1:15–32
- [7] Ostrom QT, Cioffi G, Waite K, et al. CBTRUS Statistical Report: Primary Brain and Other Central Nervous System Tumors Diagnosed in the United States in 2014–2018. *Neuro Oncol*. 2021; 23:iii1–iii105
- [8] Duffner PK, Cohen ME, Freeman AI. Pediatric Brain Tumors: An Overview. *Ca*. 1985; 35:287–301
- [9] Section of Pediatric Neurosurgery of the American Association of Neurological Surgeons. *Pediatric Neurosurgery*. New York 1982
- [10] Jooma R, Hayward RD, Grant DN. Intracranial Neoplasms During the First Year of Life: Analysis of One Hundred Consecutive Cases. *Neurosurgery*. 1984; 14:31–41
- [11] Wakai S, Arai T, Nagai M. Congenital Brain Tumors. *Surgical Neurology*. 1984; 21:597–609
- [12] Forsyth PA, Posner JB. Headaches in Patients with Brain Tumors: A Study of 111 Patients. *Neurology*. 1993; 43:1678–1683
- [13] Mahaley MS, Mettlin C, Natarajan N, et al. National Survey of Patterns of Care for Brain-Tumor Patients. *J Neurosurg*. 1989; 71:826–836
- [14] Whittle IR, Pringle A-M, Taylor R. Effects of Resective Surgery for Left-Sided Intracranial Tumors on Language Function: A Prospective Study. *Lancet*. 1998; 351:1014–1018
- [15] Albright L, Reigel DH. Management of Hydrocephalus Secondary to Posterior Fossa Tumors. Preliminary Report. *J Neurosurg*. 1977; 46:52–55
- [16] Berger MS, Baumeister B, Geyer JR, et al. The Risks of Metastases from Shunting in Children with Primary Central Nervous System Tumors. *J Neurosurg*. 1991; 74:872–877
- [17] McLaurin RL, Venes JL. *Pediatric Neurosurgery*. Philadelphia 1989
- [18] Galicich JH, French LA. Use of Dexamethasone in the Treatment of Cerebral Edema Resulting from Brain Tumors and Brain Surgery. *Am Pract Dig Treat*. 1961; 12:169–174
- [19] French LA, Galicich JH. The Use of Steroids for Control of Cerebral Edema. *Clinical Neurosurgery*. 1964; 10:212–223
- [20] Glantz MJ, Cole BF, Forsyth PA, et al. Practice Parameter: Anticonvulsant Prophylaxis in Patients with Newly Diagnosed Brain Tumors. Report of the Quality Standards Subcommittee of the American

- Academy of Neurology. *Neurology*. 2000; 54:1886–1893
- [21] Kondziolka D, Lunsford LD, Martinez AJ. Unreliability of contemporary neurodiagnostic imaging in evaluating suspected adult supratentorial (low-grade) astrocytoma. *Journal of Neurosurgery*. 1993; 79:533–536
 - [22] Chicoine MR, Silbergeld DL. *Pharmacology for Neurosurgeons. Part I: Anticonvulsants, Chemotherapy, Antibiotics*. Contemporary Neurosurgery. 1996; 18:1–6
 - [23] Prados MD, Berger MS, Wilson CB. Primary Central Nervous System Tumors: Advances in Knowledge and Treatment. *Ca Cancer J Clin*. 1998; 48:331–360
 - [24] Stewart DJ. A Critique of the Role of the Blood-Brain Barrier in the Chemotherapy of Human Brain Tumors. *Journal of Neuro-Oncology*. 1994; 121–139
 - [25] Salzman M, Broadwell RD, Salzman M. The Blood Brain Barrier. In: *Neurobiology of Brain Tumors*. Baltimore: Williams and Wilkins; 1991:229–250
 - [26] Madajewicz S, Chowhan N, Ttayli A, et al. Therapy for Patients with High Grade Astrocytoma Using Intraarterial Chemotherapy and Radiation Therapy. *Cancer*. 2000; 88:2350–2356
 - [27] Barker FG, Prados MD, Chang SM, et al. Radiation Response and Survival Time in Patients with Glioblastoma Multiforme. *Journal of Neurosurgery*. 1996; 84:442–448
 - [28] Laohaprasit V, Silbergeld DL, Ojemann GA, et al. Postoperative CT Contrast Enhancement Following Lobectomy for Epilepsy. *Journal of Neurosurgery*. 1990; 73:392–395
 - [29] Jeffries BF, Kishore PR, Singh KS, et al. Contrast Enhancement in the Postoperative Brain. *Radiology*. 1981; 139:409–413
 - [30] Gerber AM, Savolaine ER. Modification of Tumor Enhancement and Brain Edema in Computerized Tomography by Corticosteroids: Case Report. *Neurosurgery*. 1980; 6:282–284
 - [31] Hatam A, Bergström M, Yu ZY, et al. Effect of Dexamethasone Treatment in Volume and Contrast Enhancement of Intracranial Neoplasms. *Journal of Computer Assisted Tomography*. 1983; 7:295–300
 - [32] Plesec TP, Prayson RA. Frozen section discrepancy in the evaluation of central nervous system tumors. *Arch Pathol Lab Med*. 2007; 131:1532–1540
 - [33] Shah AB, Muzumdar GA, Chitale AR, et al. Squash preparation and frozen section in intraoperative diagnosis of central nervous system tumors. *Acta Cytol*. 1998; 42:1149–1154
 - [34] Uematsu Y, Owai Y, Okita R, et al. The usefulness and problem of intraoperative rapid diagnosis in surgical neuropathology. *Brain Tumor Pathol*. 2007; 24:47–52
 - [35] Burger PC. *Smears and Frozen Sections in Surgical Neuropathology: A manual*. PB Medical Publishing; 2009
 - [36] Scherer HB. Structural Developments in Gliomas. *American Journal of Cancer*. 1938; 34:333–351
 - [37] McKeever PE, Dabbs DJ. Immunohistochemistry of the Nervous System. In: *Diagnostic Immunohistochemistry*. New York: Churchill Livingstone; 2002:559–624
 - [38] Russell DS, Rubenstein LJ. *Pathology of Tumours of the Nervous System*. 5th ed. Baltimore: Williams and Wilkins; 1989
 - [39] Louis DN, Ohgaki H, Wiestler OD, et al. WHO classification of tumors of the central nervous system. Lyon 2007
 - [40] Louis DN, Perry A, Wesseling P, et al. The 2021 WHO Classification of Tumors of the Central Nervous System: a summary. *Neuro Oncol*. 2021. DOI: 10.1093/neuonc/noab106
 - [41] Burton BK. Alpha-Fetoprotein Screening. *Adv Pediatr*. 1986; 33:181–196

36 Genetic Tumor Syndromes Involving the CNS

36.1 General information

At least 20 genetic syndromes with predisposition for central and peripheral nervous system tumors have been identified (see ► Table 36.1 for some). Several of them used to be grouped under the heading “neurocutaneous disorders” (or phakomatoses), viz., neurofibromatosis types I & II, tuberous sclerosis, and Sturge Weber syndrome (p.652). This section covers some of these syndromes that are more likely to come to the attention of the neurosurgeon.

Table 36.1 Examples of genetic syndromes associated with CNS tumors

Syndrome	Associated CNS tumor(s)
neurofibromatosis type 1 (p. 638)	optic glioma, astrocytoma, neurofibroma
neurofibromatosis type 2 (p. 640)	bilateral vestibular schwannoma, meningioma, ependymoma, astrocytoma
schwannomatosis (p.642)	multiple schwannomas, infrequently meningiomas
tuberous sclerosis (p.644)	subependymal giant cell astrocytoma (SEGA)
von Hippel-Lindau syndrome (p. 646)	hemangioblastoma
Li-Fraumeni syndrome (p. 650)	children: medulloblastoma adult: astrocytoma, PNET
Cowden syndrome (p. 651)	dysplastic cerebellar gangliocytoma
constitutional mismatch repair deficiency syndrome (p. 651)	ultrahypermutated malignant gliomas, CNS embryonal tumors
familial adenomatous polyposis 1 (p. 651) (FAP1) + brain tumors (BTP2)	medulloblastoma WNT-activated
nevroid basal cell carcinoma syndrome (Gorlin syndrome)	desmoplastic/nodular subtype of medulloblastoma
rhabdoid tumor predisposition syndrome (RPTS)	atypical teratoid/rhabdoid tumors (AT/RTs)
Carney complex (p. 652)	malignant melanotic nerve sheath tumor, schwannoma, PitNET/adenoma
DICER1 syndrome	metastatic pleuropulmonary blastoma; pineoblastoma; embryonal tumor with multilayered rosettes; pituitary blastoma; primary intracranial sarcoma, <i>DICER1</i> -mutant
familial paraganglioma syndromes	paraganglioma (including pheochromocytoma)
melanoma-astrocytoma syndrome	astrocytoma, nerve sheath tumors
familial retinoblastoma	pineoblastoma
BAP1 tumor predisposition syndrome	meningioma
Fanconi anemia	medulloblastoma
ELP1-medulloblastoma syndrome	medulloblastoma SHH-activated

36.2 Neurofibromatosis

36.2.1 General information

Neurofibromatosis is a group of neurogenetic disorders with a predisposition to developing multiple nerve sheath tumors. There are as many as 6 distinct types of neurofibromatosis, the two most common of which (NF1 & NF2) are compared in ► Table 36.2. A third, less common type is schwannomatosis (p. 642).

► **Schwannoma vs. neurofibroma.** While similar in many ways, these tumors differ histologically. Schwannomas (formerly: neurilemmomas) arise from schwann cells, which produce myelin. Neurofibromas consist of neurites (axons or dendrites of immature or developing neurons), Schwann's cells, and fibroblasts within a collagenous or myxoid matrix. In contrast to schwannomas which displace axons (centrifugal), neurofibromas are unencapsulated and engulf the nerve of origin

(centripetal). Neurofibromas may occur as solitary lesions, or may be multiple as part of NF1 in the setting of which there is potential for malignant transformation. Both tumors have Antoni A (compact) and Antoni B (loose) fibers, but neurofibromas tend to have more Antoni B fibers and the cells tend to be smaller than those of schwannomas. Markers of mature Schwann cells including S100, SOX10 and collagen IV, are present in both, but to a lesser extent in neurofibromas. A patient ≤ 30 years of age with a vestibular schwannoma is at increased risk of having NF2.

36.2.2 Neurofibromatosis type 1

General information

More common than NF2, representing > 90% of cases of neurofibromatosis.² For a comparison of NF1 and NF2, see ► Table 36.2.

Table 36.2 Comparison of neurofibromatosis 1 & 2¹

current designation →	Neurofibromatosis 1 (NF1) (p. 638)	Neurofibromatosis 2 (NF2) (p. 640)
alternate term (not recommended)	von Recklinghausen's disease	bilateral acoustic NFT AKA MISME syndrome
obsolete term	peripheral NFT	central NFT
U.S. prevalence	100,000 people	≈ 3000 people
incidence	1/3000 births	1/40,000
inheritance	AD	AD
sporadic occurrence	30–50%	> 50%
gene locus	17 (17q11.2)	22 (22q12.2)
gene product	neurofibromin	schwannomin (merlin)
vestibular schwannomas (VS)	uncommon, ≈ never bilateral	bilateral VSs are the hallmark
cutaneous schwannomas	no	70%
Lisch nodules	very common	not associated
cataracts	not associated	60–80%
skeletal anomalies	common (scoliosis...)	not associated
pheochromocytoma	occasional	not associated
MPNST ^a	≈ 2%	not associated
intellectual impairment	in ≥ 50%	not associated
most common associated intramedullary spinal cord tumors	astrocytoma	ependymoma

^amalignant peripheral nerve sheath tumor

Clinical

NF1 should be *suspected* in individuals who have any of the suggestive findings shown in ► Table 36.3 (1988 NIH criteria).

The NIH diagnostic criteria are met in an individual with 2 or more of the features listed in ► Table 36.3 (reference available online at <https://www.ncbi.nlm.nih.gov/books/NBK1109/>).³ In *adults*, the diagnosis is usually made clinically using the NIH criteria which are highly specific and highly sensitive,⁴ molecular genetic testing is rarely needed. In children, more care must be taken in assigning the diagnosis of NF1,³ where molecular analysis may help differentiate NF1 from forms of Noonan syndrome that have multiple neurofibromas.

Associated conditions

1. Schwann-cell tumors on any nerve (but bilateral VSs are virtually nonexistent)
2. spinal and/or peripheral nerve neurofibromas
3. multiple skin neurofibromas
4. aqueductal stenosis (p.273)
5. macrocephaly: secondary to aqueductal stenosis and hydrocephalus, increased cerebral white matter

Table 36.3 Suggestive findings of NF1⁵**Two or more of the following:**

- ≥ 6 café au lait spots^a, each ≥ 5 mm in greatest diameter in prepubertal individuals, or ≥ 15 mm in greatest diameter in postpubertal patients
- ≥ 2 neurofibromas of any type, or one plexiform neurofibroma (neurofibromas are usually not evident until age 10–15 yrs). May be painful
- freckling (hyperpigmentation) in the axillary or intertriginous (inguinal) areas
- optic pathway glioma/pilocytic astrocytoma: see below
- ≥ 2 Lisch nodules: pigmented iris hamartomas that appear as translucent yellow/brown elevations that tend to become more numerous with age
- distinctive osseous abnormality, such as *sphenoid dysplasia* or thinning of long bone cortex with or without pseudarthrosis (e.g., of tibia or radius)
- a first degree relative (parent, sibling, or offspring) with NF1 by above criteria

^acafé au lait spots: hyperpigmented oval light brown skin macules (flat). May be present at birth, increase in number and size during 1st decade. Are present in $>99\%$ of NF1 cases. Rare on face.

6. intracranial tumors: hemispheric astrocytomas are the most common, solitary or multicentric meningiomas (usually in adults). Gliomas associated with NF1 are usually pilocytic astrocytomas. Brainstem astrocytomas include both contrast-enhancing pilocytic lesions and those that are non-enhancing and radiologically diffuse
7. unilateral defect in superior orbit $\rightarrow$ pulsatile exophthalmos
8. neurologic or cognitive impairment: 30–60% have mild learning disabilities
9. kyphoscoliosis (seen in 2–10%, often progressive, which then requires surgical stabilization)
10. visceral manifestations from involvement of autonomic nerves or ganglia within the organ. Up to 10% of patients have abnormal gastrointestinal motility/neuronal intestinal dysplasia related to neuronal hyperplasia within submucosal plexus
11. $\approx 20\%$ develop plexiform neurofibromas: tumors from multiple nerve fascicles that grow along the length of the nerve. Almost pathognomonic for NF1⁶
12. syringomyelia
13. malignant tumors that have increased frequency in NFT: neuroblastoma, ganglioglioma, sarcoma, leukemia, Wilm's tumor, breast cancer⁷
14. pheochromocytoma: is occasionally present
15. "unidentified bright objects" (UBOs) on brain or spinal MRI in 53–79% of patients (bright on T2WI, isointense on T1WI) that may be hamartomas, heterotopias, foci of abnormal myelination or low grade tumors.⁸ Tend to resolve with age

Epidemiology

An autosomal dominant disorder with an incidence of ≈ 1 case per every 3000 births.

Genetics

Simple autosomal dominant inheritance with variable expressivity but almost 100% penetrance after age 5 years. The NF1 gene is on chromosome 17q11.2, which codes for neurofibromin⁹ (neurofibromin is a negative regulator of the RAS oncogene). Loss of neurofibromin results in elevation of growth-promoting signals. The spontaneous mutation rate is high, with 30–50% of cases representing new somatic mutations.¹⁰

Counselling: prenatal diagnosis is possible by linkage analysis only if there are 2 or more affected family members.⁹ 70% of NF1 gene mutations can be detected using protein truncation analysis.

Neurofibromas

Dermal and plexiform neurofibromas are characteristic of NF1. Deep-seated intraneural variants are less common and may produce neurologic symptoms. The lifetime risk of malignant transformation of plexiform neurofibromas to MPNST (p. 774) is $\approx 10\%$.¹¹

Management

1. optic gliomas
 - a) unlike optic gliomas in the absence of NFT, these are rarely chiasmal (usually involving the nerve), are often multiple, and have a better prognosis
 - b) most are non progressive, and should be followed ophthalmologically and with serial imaging (MRI or CT)

- c) surgical intervention probably does not alter visual impairment. Therefore, surgery is reserved for special situations (large disfiguring tumors, pressure on adjacent structures...)
2. other neural tumors in patients with NF1 should be managed in the same manner as in the general population
 - a) focal, resectable, symptomatic lesions should be surgically removed
 - b) MEK inhibitors are approved use in ages 2-18 years of age with for symptomatic, inoperable plexiform neurofibromas,¹² with surgery reserved for cases with increasing ICP
 - c) when malignant degeneration is suspected (rare, but incidence of sarcomas and leukemias is increased³), biopsy with or without internal decompression may be indicated
3. surveillance³
 - a) annual exam by physician familiar with NF1
 - b) annual ophthalmologic exam in children, less frequent in adults
 - c) regular developmental assessment of children
 - d) regular BP monitoring
 - e) MRI for follow-up of clinically suspected intracranial and other internal tumors
 - f) annual mammography in women starting at age 30 years and consider annual breast MRI in women age 30–50 years

36.2.3 Neurofibromatosis type 2 (NF2 AKA bilateral acoustic NFT)

General information

AKA MISME syndrome (acronym for Multiple Inherited Schwannomas, Meningiomas, and Ependymomas).¹³ An autosomal dominant disorder characterized by bilateral vestibular schwannomas (VS), with other neoplastic and dysplastic lesions that primarily affect the nervous system.¹⁴ Multiple meningiomas are the second hallmark, occurring in about 50% of patients. For a comparison of NF1 and NF2, see ► Table 36.2.

There are 2 phenotypic subtypes¹⁵:

1. Wishart phenotype: severe. Early onset, myriad tumors and high tumor burden
2. Gardner phenotype: mild. Later presentation, VS are the only tumors, slower deterioration of hearing

Diagnostic criteria for neurofibromatosis type 2 (NF2)

Diagnostic criteria have been modified from the original NIH guidelines^{16,17} (reference available online at <https://www.ncbi.nlm.nih.gov/books/NBK1201/>) with the revised Manchester criteria¹⁸ shown in ► Fig. 36.1.

NF2 is diagnosed in a proband that meets the consensus diagnostic criteria, or if only 1 criteria is met in the presence of a heterozygous pathogenic variant of the NF2 gene.¹⁹

Clinical

Other clinical features:

1. seizures or other focal deficits
2. skin nodules, dermal neurofibromas, café au lait spots (less common than in NF1)
3. multiple intradural spinal tumors are common (less common in NF1)²⁰; including intramedullary (especially ependymomas) and extramedullary (schwannomas, meningiomas...)
4. retinal hamartomas
5. antigenic nerve growth factor is increased (does not occur with NF1)
6. despite its name, is not associated with neurofibromas

Epidemiology

≈ 1 in 25,000 to 1 in 40,000 are affected. NF2 is considered an adult-onset disease, the average age of onset is 18–24 years.¹⁷ By age 30 years, almost all affected patients will have bilateral VS.

Genetics

Autosomal dominant inheritance. There are a number of NF2 gene mutations that lead to the development of NF2. One is due to a mutation at chromosome 22q12.2, which results in the inactivation of schwannomin (AKA merlin, a semi-acronym for moesin-, ezrin-, and radixin-like proteins), a tumor suppression peptide. NF2 patients with nonsense and frameshift mutations are more likely to have intramedullary tumors (but not any other type of tumor) compared with other mutation types.

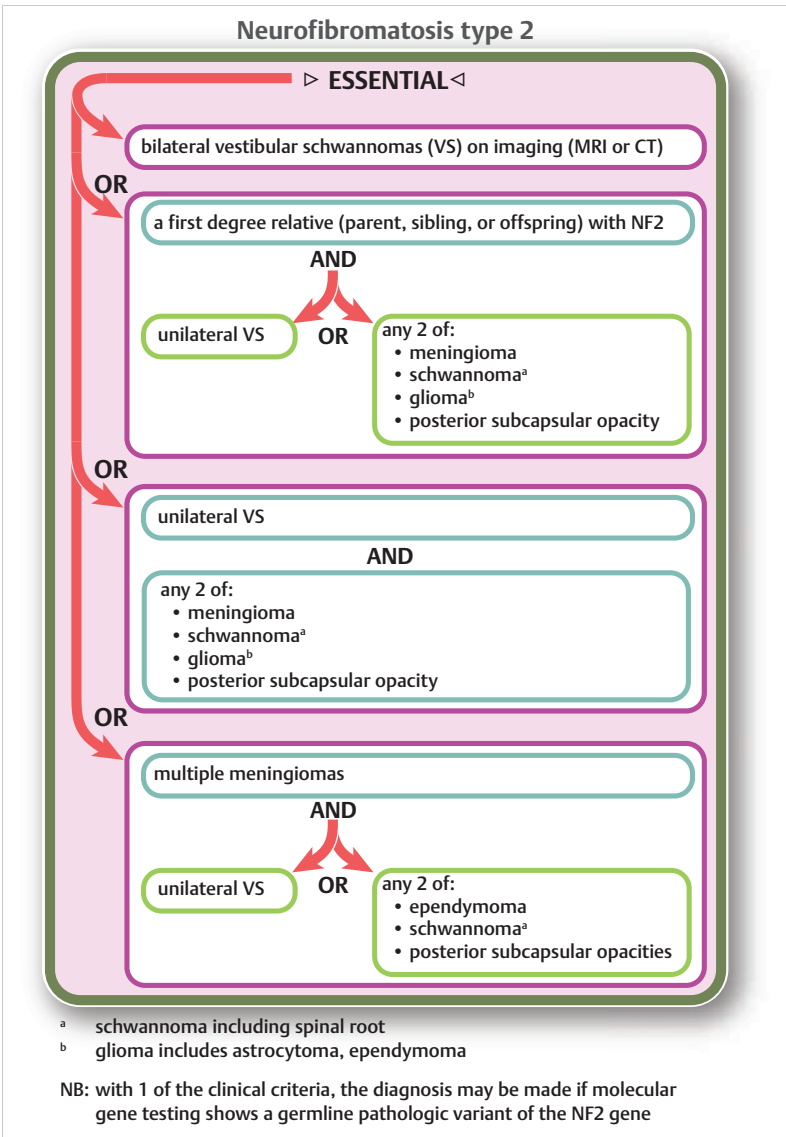


Fig. 36.1 Diagnostic criteria for NF2. (Modified with permission from WHO Classification of Tumors Editorial Board, (ed.) World Health Organization classification of tumors of the central nervous system. 5th ed. Bosman F T, Jaffe E S, Lakhani S R, et al. (series eds.): World Health Organization Classification of Tumors series, vol. 6. International Agency for Research on Cancer, Lyon, France, 2021.)

► **Schwannomas.** Schwannomas associated with NF2 are WHO grade 1 tumors of neoplastic Schwann cells that occur at an earlier age (in the 20s) than sporadic (i.e., non-NF2 associated) schwannomas (in the 50s). Although cutaneous neurofibromas have been reported, in NF2 many of these turn out to be schwannomas on histological analysis.

Management considerations

Management takes into account the fact that most NF2 patients will become deaf at some time during their life.

1. bilateral vestibular schwannomas:
 - a) chance of preserving hearing is best when the VS is small. Thus, one should attempt to remove the smaller tumor. If hearing is serviceable in that ear after surgery, then consider removing the second tumor, otherwise follow the second tumor as long as possible and perform a subtotal removal in an attempt to prevent total deafness
 - b) stereotactic radiosurgery therapy is a treatment alternative
2. prior to surgery, obtain MRI of cervical spine to R/O intraspinal tumors that may cause cord injuries during other operations
3. NB: pregnancy may accelerate the growth of eighth nerve tumors
4. surveillance: for affected or at-risk individuals¹⁷
 - a) annual MRIs beginning age 10-12 years, and continuing until at least the 4th decade of life
 - b) hearing evaluation, including BAER testing
 - c) annual complete eye exam

36.2.4 Schwannomatosis (SWN)

General information

Most cases are associated with alterations in *SMARCB1* or *LZTR1* (both on chromosome 22q11) which in turn is associated with inactivation of the *NF2* gene in the tumors but not in the germline. There are likely other as-yet unidentified mutations that cause the remaining cases of SWN.

SWN typically presents with multiple schwannomas (spinal, cutaneous and cranial), and less frequently with meningiomas (cranial and spinal) with a propensity for spinal and paraspinal nerves, and less frequently the meninges.

There are 2 subtypes:

1. *SMARCB1* schwannomatosis 1: the *SMARCB1* gene is the underlying defect in 50% of familial cases, and < 10% of sporadic cases.
2. *LZTR1* schwannomatosis 2: in those without *SMARCB1* mutation, *LZTR1* mutations were identified in 40% of familial and 25% of sporadic cases

Diagnostic criteria for schwannomatosis

Diagnostic criteria for schwannomatosis are shown in ► Fig. 36.2.²¹

A molecular diagnosis may also be made with the following²²:

- ≥ 2 pathologically proven schwannomas or meningiomas AND genetic studies of at least two tumors with loss of heterozygosity (LOH) for chromosome 22 and two different NF2 mutations; if there is a common *SMARCB1* mutation, this defines *SMARCB1*-associated schwannomatosis
- one pathologically proven schwannoma or meningioma AND germline *SMARCB1* pathogenic mutation

Patients with the following do not qualify for the diagnosis of schwannomatosis²²:

1. germline pathogenic NF2 mutation
2. those fulfilling the diagnostic criteria for NF2 (especially bilateral vestibular schwannomas)
3. those with a first-degree relative with NF2
4. schwannomas only in previous field of radiation therapy

Epidemiology

Annual incidence is 0.58 cases per 1,000,000 persons.²³ There was a family history of SWN in 13% of 87 cases.²⁴

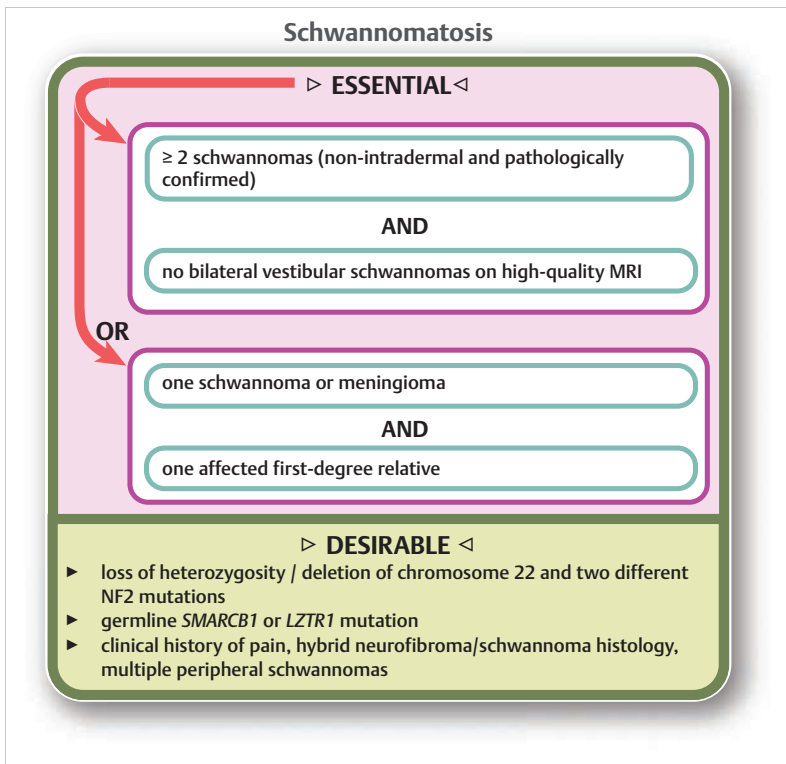


Fig. 36.2 Diagnostic criteria for schwannomatosis. (Modified with permission from WHO Classification of Tumors Editorial Board, (ed.) World Health Organization classification of tumors of the central nervous system. 5th ed. Bosman F T, Jaffe E S, Lakhani S R, et al. (series eds.): World Health Organization Classification of Tumors series, vol. 6. International Agency for Research on Cancer, Lyon, France, 2021.)

Presentation

There is considerable clinical (phenotypical) overlap with NF2, and molecular testing may be necessary to differentiate the two. In distinction to NF2, 68% of SWN patients have chronic pain, and 46% have pain unassociated with a mass.²⁴ It is debatable whether neurologic deficit is common²⁵ or not.²¹ The pain has both neuropathic and nociceptive features and the relationship of the pain to the number, size and location of tumors is unclear.²²

Evaluation

H&P: careful history to assess for 1st degree relatives with SWN or NF2 and physical exam for stigmata of NFT.

Imaging: MRI of the entire neuraxis, including high resolution brain MRI with attention to the internal auditory canals (IACs) with slices ≤ 3 mm thick. This can be accomplished with separate MRIs without and with contrast of the brain (with IACs), cervical, thoracic and lumbar spine, or, when available, with whole-body MRI without contrast using special commercially available software.^{22,26}

Molecular genetic testing: including tumor biopsy for questionable cases (NB: patients with SWN-associated chronic pain should be made aware that risks include possibility of exacerbation of their pain; see below). Appropriate genetic counselling for confirmed cases.

Treatment

Surgery for symptomatic schwannomas may relieve local pain. Surgical risks include nerve injury, and there is anecdotal suggestion that surgery is occasionally associated with worsening of global pain.²²

Prognosis

Life expectancy is near normal, which is higher than with NF2.

36.3 Other genetic tumor syndromes involving the CNS

36.3.1 Tuberous sclerosis complex (TSC)

General information

Key concepts

- most cases are due to spontaneous mutation. Inherited cases are autosomal dominant. Incidence: 1 in 6K–10K live births
- classic clinical triad: seizures, mental retardation, and sebaceous adenomas; the full clinical triad is seen in < 1/3 of cases
- typical CNS finding: subependymal nodules (“tubers”) — which are hamartomas
- common associated neoplasm: subependymal giant cell astrocytoma (SEGA)
- TSC develops if either of these 2 tumor suppressor genes is affected: *TSC1* (on chromosome 9q34) codes for hamartin, and *TSC2* (on chromosome 16p13) encodes tuberin
- CT shows intracerebral calcifications (usually subependymal)

Tuberous sclerosis complex (TSC), AKA Bourneville's disease, is a group of autosomal dominant neurocutaneous disorder characterized by hamartomas and benign neoplastic lesions of many organs including the skin, brain, eyes, and kidneys. In the brain, the hamartomas may manifest as cortical tubers, glial nodules located subependymally or in deep white matter, or subependymal giant cell astrocytomas (SEGA). Associated brain findings include pachygyria or microgyria.

Diagnostic criteria

General information

The diagnosis of TS is based primarily on clinical features, but the variability of phenotype and age at symptom onset can make this challenging. Molecular testing may help when the phenotype is strongly suspicious but a definitive diagnosis cannot be made clinically.

Clinical diagnostic criteria

Clinical diagnostic criteria are shown in ► Table 36.4.

In the infant, the earliest finding is of “ash leaf” macules (hypomelanotic, leaf shaped) that are best seen with a Wood's lamp. Infantile myoclonus may also occur.

In older children or adults, the myoclonus is often replaced by generalized tonic-clonic or partial complex seizures, which occur in 70–80%. Facial adenomas are not present at birth, but appear in > 90% by age 4 yrs (these are not really adenomas of the sebaceous glands, but are small hamartomas of cutaneous nerve elements that are yellowish-brown and glistening and tend to arise in a butterfly malar distribution, usually sparing the upper lip).

Retinal hamartomas occur in ≈ 50% (central calcified hamartoma near the optic disc or a more subtle peripheral flat salmon-colored lesion). A distinctive depigmented iris lesion may also occur.

Genetic diagnostic criteria

Identification of either a *TSC1* or a *TSC2* pathogenic mutation is sufficient to definitively diagnose TS (online resources: for *TSC1*, and for *TSC2*).

Subtypes

1. tuberous sclerosis 1
2. tuberous sclerosis 2

Table 36.4 Clinical diagnostic criteria of tuberous sclerosis complex²⁷

- **Definitive diagnosis** 2 major criteria, or 1 major AND ≥ 2 minor
- **Possible diagnosis** 1 major or ≥ 2 minor

Major criteria

- ≥ 3 hypomelanotic macules ≥ 5 mm diameter
- ≥ 3 angiofibromas or fibrous cephalic plaque
- ≥ 2 ungual fibroma
- shagreen patch
- multiple retinal hamartomas
- cortical dysplasias (including tubers & cerebral white matter radial migration lines)
- subependymal nodules
- subependymal giant cell astrocytoma (SEGA)
- cardiac rhabdomyoma
- lymphangiomyomatosis
- ≥ 2 angiomyolipomas

Minor criteria

- “confetti” skin lesions
- ≥ 4 pits in dental enamel
- ≥ 2 intraoral fibromas
- achromic retinal patch
- multiple renal cysts
- nonrenal hamartomas

Epidemiology/genetics/epigenetics

Incidence: 1 in 6,000–10,000 live births.²⁸ Population prevalence: 1 in 20,000.²⁷

Autosomal dominant inheritance; however, spontaneous mutation accounts for the majority of cases.²⁹ Two distinct tumor suppressor genes have been identified: the *TSC1* gene (located on chromosome 9q34) codes for *TSC1* (AKA hamartin), and the *TSC2* gene (on chromosome 16p13.3) codes for *TSC2* (tuberin). Only 1 gene needs to be affected to develop TSC. These proteins work together to inhibit activation of rapamycin (mTOR).

Genetic counseling for unaffected parents with one affected child: 1–2% chance of recurrence.

Histology

Subependymal nodules (“tubers”) are benign hamartomas that are almost always calcified, and protrude into the ventricles.

► **Subependymal giant cell astrocytoma (SEGA).** A WHO grade 1 tumor—see Subependymal giant cell astrocytoma (SEGA) (WHO grade 1) (p.700). Almost always located at the foramen of Monro. Occurs in 5–15% of patients with TSC.³⁰

Evaluation**Plain skull X-rays**

Not particularly helpful. May show calcified cerebral nodules.

CT scan

See reference.³¹

Intracerebral *calcifications* are the most common (97% of cases) and characteristic finding. Primarily located subependymally along the lateral walls of the lateral ventricles or near the foramina of Monro.

Low density lesions that do not enhance are seen in 61%. Probably represent heterotopic tissue or defective myelination. Most common in occipital lobe.

Hydrocephalus (HCP) may occur even without obstruction. In the absence of tumor, HCP is usually mild. Moderate HCP usually occurs only in the presence of tumor.

Subependymal nodules are usually calcified, and protrude into the ventricle (“candle guttering” described the appearance on pneumoencephalography).

Paraventricular tumors (mostly giant cell astrocytomas; see pathology (p.645)) are essentially the only enhancing lesions in TSC.

MRI

Subependymal tubers are high on T2 and low on T1 and only $\approx 10\%$ enhance.

Low signal in subependymal lesions may represent calcification. SEGA enhance intensely (enhancing subependymal lesions are almost always SEGAs).

Radial bands sign: abnormal signal intensity extending in a radial manner, representing cells of varying degrees of neuronal and astrocytic differentiation as well as difficult-to-classify cells.³²

Treatment

Paraventricular tumors should be followed. Tubers grow minimally, but SEGA progress should be removed if they are symptomatic. A transcallosal approach or ventriculoscopic removal are options.

Infantile myoclonus may respond to steroids. Seizures are treated with ASMs.

Surgery for intractable seizures may be considered when a particular lesion is identified as a seizure focus. Better seizure control, not cure, is the goal in TSC.

Patients ≥ 3 years of age with increasing size of SEGA lesions have had sustained reduction of SEGA volume on everolimus.³³

Prognosis

Lifespan is slightly shortened by TS.³⁴ When death occurs in the second decade of life, it is usually due to brain tumors and status epilepticus, followed by renal abnormalities. After age 40, mortality is usually associated with renal cysts or tumors or lymphangioleiomyomatosis.

36.3.2 Von Hippel-Lindau disease (VHL)**General information****Key concepts**

- disorder with hemangioblastomas (HGB) primarily of cerebellum, retina, brainstem and spinal cord, endolymphatic sac tumors, as well as renal cysts/tumors, pheochromocytomas...
- autosomal dominant, due to germline variants of the VHL tumor suppressor gene on 3p25.3
- expression and age of onset are variable, but $\approx$ always manifests by age 60
- mean age of developing HGBs is at least 10 years younger than sporadic HGBs

An autosomal dominant multisystem neoplastic disorder characterized by a tendency to develop hemangioblastomas (HGB) of the retina, brain and spinal cord (especially nerve roots), renal clear cell carcinoma (RCC), pheochromocytomas, endolymphatic sac tumors, and others^{35,36} (retinal location is 2nd most common after cerebellar, ► Table 36.5). It is caused by germline variants of the VHL tumor suppressor gene on chromosome 3p25.3. The variability of von Hippel-Lindau disease (VHL) has led some to suggest the use of the term hemangioblastomatosis.

Epidemiology

Incidence: 1 in 31,000 to 36,000 live births. $\approx 30\%$ of patients with cerebellar HGB have VHL.²⁸

Genetics

The VHL gene is a tumor suppressor gene on chromosome 3p25.3 that codes for VHL protein, which is part of protein complex VCB-CUL2. Biallelic inactivation (2-hit model) is required for tumor development.²⁸ Most patients inherit the autosomal dominant VHL gene (allele) with the germline mutation from the affected parent and a normal somatic (wild-type) VHL gene from the unaffected parent, with $\approx 95\%$ penetrance by age 60 yrs.^{35,39} However, about 20% of cases result from a spontaneous mutation that occurs in the egg or sperm, or very early in development.⁴⁰

In either case, at some time during their life, the required second mutation occurs in susceptible organs, e.g., brain, kidney, retina... which causes the cell to not make functional VHL protein, which allows the cyst to develop.

4% of patients with the mutant VHL gene are asymptomatic carriers.

Table 36.5 Associations with von Hippel-Lindau disease^a

Common lesions	Frequency in VHL
hemangioblastomas	
• cerebellum (solid or cystic)	80%
• retina	41–59%
• brainstem	10–25%
• spinal cord	10–50%
pancreatic tumors or cysts	22–80%
renal clear cell Ca & cysts	14–60%
polycythemia	9–20% of intracranial HGBs
Rare lesions (pertinent to nervous system)	Frequency in VHL
supratentorial hemangioblastoma	3–6%
cystadenomas of the broad ligament	10% of ♀
papillary cystadenomas of epididymis	25–60% ♂
endolymphatic sac tumors	10–15%
adrenal medullary <i>pheochromocytoma</i> (tends to be bilateral)	7–24%

^asee references^{35,37,38} for more

Subtypes of VHL

See reference.⁴¹**Type I:** any manifestation of VHL (typically HGB and RCC) *except* pheochromocytoma.**Type 2A:** pheochromocytoma is characteristic. RCC is rare.**Type 2B:** high risk of HGB, RCC and pheochromocytoma.**Type 2C:** risk of pheochromocytoma only (without risk of HGB or RCC).

Diagnostic criteria

Formal criteria for diagnosis of VHL have not been published; suggested diagnostic criteria appear here.⁴²

► **Suspected VHL.** VHL should be suspected in an individual (with or without a family history of VHL) who has:

1. retinal angioma, especially at a young age
2. spinal or cerebellar hemangioblastoma
3. adrenal or extra-adrenal pheochromocytoma
4. renal cell carcinoma (RCC), if the individual is < 47 years old, or has a personal or family history of other tumors typical of VHL
5. multiple renal and pancreatic cysts
6. neuroendocrine tumors of the pancreas
7. endolymphatic sac tumors
8. less common: multiple papillary cystadenomas of the epididymis or broad ligament

► **Establishing the diagnosis of VHL.** The diagnosis of VHL is established in a proband with clinical features listed below and/or by identification of a heterozygous germline pathogenic variant in the VHL gene by molecular genetic testing. If the clinical features are absent, a patient with this variant is diagnosed with VHL and should be surveilled as such, even if clinical or radiographic findings are inconclusive.

Clinical diagnostic criteria

- an individual with no known family history of VHL with ≥ 2 characteristic lesions:
 - ≥ 2 HGB of the retina, spine or brain, or a single HGB in association with a visceral manifestation (e.g., multiple renal or pancreatic cysts)
 - renal cell carcinoma
 - adrenal or extra-adrenal pheochromocytoma
 - less common: endolymphatic sac tumors, papillary cystadenomas of the epididymis or broad ligament, or neuroendocrine tumors of the pancreas

- an individual with a family history of VHL with ≥ 1 of the following:
 - retinal angioma
 - spinal or cerebellar hemangioblastoma
 - adrenal or extra-adrenal pheochromocytoma
 - renal cell carcinoma
 - multiple renal and pancreatic cysts

Tumors associated with VHL

1. cerebellar hemangioblastomas (HGB):
 - a) prevalence: 44–72% of VHL patients
 - b) mean age of diagnosis in VHL patients with cerebellar hemangioblastomas is at least 10 years younger than sporadic cerebellar hemangioblastomas
 - c) cysts are commonly associated with cerebellar, brainstem and spinal HGBs
 - d) cysts grow at a faster rate than the HGBs, ∴ symptoms related to mass effect are frequently secondary to the cysts
 - e) cerebellar HGBs were located in the superficial, posterior and superior half of the cerebellar hemispheres⁴³
 - f) 93% of the cerebellar HGBs were located in the cerebellar hemispheres and 7% in the vermis
 - g) the HGBs are also more frequently found in the superficial posterior half of the brainstem and the spinal cord
 - h) the HGBs have multiple sequential growth and quiescent phases
2. spinal cord hemangioblastomas
 - a) occur in 13–44% of VHL patients
 - b) 90% are located cranially within the cervical and thoracic cord. Almost all (96%) of the tumors are located in the posterior half of the spinal cord, 4% are located in the ventral half of the spinal cord. 1–3% are found in the lumbosacral nerve roots
 - c) by way of comparison, 80% of spinal cord HGB are associated with VHL, whereas only 5–31% of cerebellar HGB are associated with VHL
 - d) 95% of symptom-producing spinal HGBs are associated with syringomyelia
3. brainstem hemangioblastomas
 - a) usually located in the posterior medulla oblongata usually around the obex and the region within the area postrema
4. pheochromocytomas (PCC): 20% of PCC are associated with VHL. PCC occur in 7–20% of families with VHL
5. endolymphatic sac tumors (ELST) (the endolymphatic sac is a cul-de sac in the petrous portion of the temporal bone in the middle fossa just deep to the dura that communicates with the sacculus of the inner ear):
 - a) locally invasive benign tumors that occur in 10–15% of VHL patients (30% of these will develop bilateral ELSTs—VHL is the only disease with bilateral ELSTs). Rarely metastasize
 - b) presents with hearing loss in 95% (may be acute (86%) or insidious (14%), tinnitus (90%), vertigo or imbalance (66%), aural fullness (30%), and facial paresthesias (8%))
 - c) mean age of onset of hearing loss: 22 years (range: 12–50)⁴⁴
6. retinal hemangioblastomas⁴⁵
 - a) occur in > 50% of VHL patients. Mean age of presentation: 25 years
 - b) frequently bilateral, multifocal, and recurrent
 - c) often asymptomatic. Visual symptoms occur with progressive growth, edema, retinal detachments, and hard exudates
 - d) typically located in the periphery and near or on the optic disc
 - e) microangiomas measuring a few hundred microns without dilated feeding vessels may be located in the periphery
 - f) retrolbulbar HGB are rare (5.3% in NIH cohort)⁴⁶
 - g) severity of optic disease correlates with CNS and renal involvement
 - h) early diagnosis and treatment with laser photocoagulation, and cryotherapy can prevent visual loss. Low dose external XRT may be an option for refractory cases
7. renal-cell carcinoma (RCC)^{38,47,48,49,50,51,52,53}
 - a) the most common malignant tumor in VHL. Usually a clear cell carcinoma
 - b) lifetime risk for RCC in VHL: $\approx$ 70%.
 - c) the growth rate of RCC is high variable
 - d) RCC is the cause of death in 15–50% of patients

- e) metastases respond poorly to chemotherapy and radiation
- f) bilateral and multiple lesions are common
- g) partial nephrectomy or tumor enucleation is preferred to avoid/delay dialysis and transplantation
- h) nephron- or renal-sparing surgery recommended for tumors less < 3 cm
- i) promising techniques: cryo- and radiofrequency ablation of tumors < 3 cm
- 8. renal cysts^{38,49,52,53,54}
 - a) 50–70% of VHL patients have bilateral and multiple renal cysts
 - b) rarely cause profound renal impairment
 - c) chronic renal failure or renal hypertension not as common as with polycystic kidney disease
- 9. epididymal cystadenomas
 - a) benign lesions that arise from the epididymal duct
 - b) found in 10–60% of male VHL patients
 - c) typically appear in the teenage years
 - d) may cause infertility if bilateral
 - e) may be multiple
- 10. broad ligament cystadenomas
 - a) arise from the embryonic mesonephric duct
 - b) true incidence unknown
 - c) rarely reported and usually not recognized in women with VHL
- 11. pancreatic neuroendocrine tumors and cysts
 - a) 35 to 70% of patients with VHL develop an endocrine tumor or cyst
 - b) pancreatic cysts are generally asymptomatic and often multiple
 - c) pancreatic neuroendocrine tumors are usually non-functional and 8% of them are malignant
 - d) differential diagnosis: pancreatic islet cell tumors, MEN2

Treatment

Medical therapy with belzutifan (Welireg), an hypoxia-inducible factor inhibitor FDA approved for adults with VHL needing therapy for associated renal cell carcinoma (RCC), CNS hemangioblastomas, or pancreatic neuroendocrine tumors not requiring immediate surgery. More effective for the hemangioblastomas and pancreatic neuroendocrine tumors than RCC. Severe anemia and hypoxia can be significant side effects of belzutifan and are treated with transfusions (not erythropoietic agents). As always, follow full prescribing information.

Surgical resection of individual CNS tumors is usually reserved until symptomatic to decrease the number of operations over a lifetime, since the tumors in VHL are usually multiple, tend to recur, and the growth pattern is saltatory. Surgery is the treatment of choice for accessible cystic HGBs. For details, see Treatment, under Hemangioblastoma (p.825).

Stereotactic radiosurgery (SRS)⁵⁵: May provide local control rates of >50% over 5 years. SRS has been recommended for asymptomatic HGB > 5 mm diameter if they are cystic or progressing in size during surveillance.⁵⁶ Cranial treatment plan: using a median dose of 22 Gy (range: 12–40 Gy) prescribed to the median 82% isodose line in 1–4 sessions. In cystic lesions, treatment is confined to the contrast enhancing mural nodule (the cyst wall is not treated). Spinal treatment plan: median dose of 21 Gy (range 20–25 Gy) prescribed to the median 77% isodose line in 1–3 sessions. Radiosurgery is usually contraindicated in hemangioblastomas with a cyst.

Surveillance

Because of the lifetime risk of developing tumors, regular surveillance is needed. Various protocols have been proposed,^{58,59} including those by the NIH³⁸ and the Danish clinical recommendations.⁶⁰ The algorithm recommended by the VHL Family Alliance for patients with VHL and at-risk relatives is shown in ► Table 36.6. (Screening at-risk relatives can be stopped at age 60 years if no abnormalities have been detected.)

Individuals who do not carry the altered gene on DNA testing do not require surveillance.

Prognosis

The lifespan of patients with VHL is decreased (median 49 years). 30–50% die of renal cell Ca (RCC). Metastases from RCC and neurologic complications from cerebellar HGB are the primary causes of death.

Metastases respond poorly to chemotherapy and XRT.

Table 36.6 Health-care provider's surveillance guidelines for patients with or at risk for VHL^a

Age	Surveillance
Any age	DNA testing for VHL marker is available to identify family members at risk
From conception	inform obstetrician of family history of VHL
From birth	check for neurologic disturbance: nystagmus, strabismus, white pupil... & refer to retinologist for abnormal findings. Newborn hearing screening
1–4 years	Annual: retina exam ^b (especially if positive for VHL mutation). Pediatrician to check for neurologic disturbance (nystagmus, strabismus, white pupil, hearing vision...) and abnormal BP
5–15 years	Annual: • PE^c including orthostatic blood pressure measurement, exam for neurologic disturbance (as above) and other retinal problems, hearing disturbances, retina exam^b • test for fractionated metanephrines (p. 940), especially normetanephrine in a "plasma-free metanephrine" blood test or in a 24-hour urine test. If abnormalities detected: abdominal MRI or MIBG scan (p. 940) • abdominal U/S starting at age 8 or earlier if indicated Every 2–3 years: complete audiology exam. Annually if hearing loss, tinnitus or vertigo. If repeated ear infections: thin cut MRI with contrast of the IACs to look for ELST
16+ years	Every 6 months: retina exam ^b Annual: • retina exam^b • PE (including scrotal exam in males), neuro exam • 24^h urine for catecholamines & metanephrines (p. 940). If elevated: abdominal MRI or MIBG scan (p. 940) • quality abdominal U/S (kidneys, pancreas & adrenals) (and at least every other year while not pregnant, abdominal MRI). If abnormal: abdominal MRI or CT (except in pregnancy) Every 2–3 years or if symptoms develop: • MRI without and with contrast on ≥ 1.5 T scanner to look for hemangioblastomas of brain with thin cuts of the posterior fossa with attention to the inner ear/petrous temporal bone to R/O ELST & spine (cervical, thoracic, lumbar). Annually at onset of puberty or before and after pregnancy (only for emergencies during pregnancy) • complete audiology exam. If abnormal, or if tinnitus or vertigo at any time: MRI of IAC to look for ELST
Prior to surgery or childbirth	• blood test or 24^h urine for catecholamines & metanephrines (p. 940) to rule out pheochromocytoma

^aadapted⁵⁷^bdilated eye/retinal exam with indirect ophthalmoscope exam by retinologist familiar with VHL^cabbreviations: PE = physical exam by physician familiar with VHL, ELST = endolymphatic sac tumor

Resources

Genetic screening for VHL can be done at a few centers. Information for patients and families can be found at www.vhl.org.

36 36.3.3 Li-Fraumeni syndrome (LFS)

General information

A rare inherited autosomal dominant disorder of children and young adults with multiple tumors, 80% of which are of the following: soft-tissue sarcomas, osteosarcomas, breast cancer, brain tumors and adrenocortical carcinoma.¹⁴ Usually caused by a germline mutation in the TP53 tumor suppressor gene on chromosome 17p13.1 which occurs in ≈ 1 of every 500 to 5000 live births.

Diagnostic criteria

Essential criterion is pathogenic germline alteration (mutation, rearrangement, or partial/complete deletion) in the TP53 gene.²¹

There are 3 main sets of criteria for recommending TP53 germline testing for diagnosing Li-Fraumeni-like syndrome: 1) classic criteria (shown below), 2) Birch or Eeles criteria for LFS-like syndromes, and 3) the Chompret criteria.²¹

Classic criteria⁶¹:

- sarcoma before age 45 years **AND**
- a first degree relative with any cancer before age 45 years **AND**
- a first or second degree relative with any cancer before age 45 years or a sarcoma at any age

Associated CNS tumors

3.5–14% of tumors are found in the CNS.

CNS tumors show a bimodal age distribution: 1) in children (primarily SHH-activated medulloblastomas, IDH-wildtype high-grade gliomas and choroid plexus carcinomas), and 2) in age 20–40 (primarily IDH-mutant diffuse astrocytic neoplasms).²¹

Prognosis

Medulloblastomas in LFS have a very poor prognosis. Choroid plexus carcinoma in patients with *TP53* mutations fare worse than their sporadic counterparts; likewise for childhood IDH-wildtype astrocytomas occurring in childhood. IDH-mutant astrocytomas in young adults with LFS have similar OS to those without LSF. Close surveillance for tumors is associated with earlier detection and improved outcomes.

36.3.4 Cowden syndrome

General information

AKA *PTEN* hamartoma tumor syndrome. An autosomal dominant syndrome resulting primarily from pathogenic germline mutations of *PTEN*, manifesting as multiple hamartomas derived from all three germ cell layers, accompanied by a high risk of cancers of the breast, thyroid, endometrium, kidneys and colon. Adult onset of dysplastic cerebellar gangliocytoma (p. 716) (Lhermitte-Duclos disease) is pathognomonic for Cowden syndrome. For diagnostic criteria, see reference.²¹

Subtypes

- Bannayan–Riley–Ruvalcaba syndrome
- Proteus syndrome: fewer than 200 confirmed cases worldwide (many undiagnosed cases are likely, especially when the manifestations are mild)

Prognosis

There is inadequate data as to whether tumors in patients with Cowden syndrome have a different prognosis than the same tumors in non-syndromic patients.

36.3.5 Constitutional mismatch repair deficiency syndrome

General information

Originally called brain tumor polyposis syndrome 1 (BTP1) as part of “Turcot syndrome” (both terms now obsolete).

An autosomal recessive syndrome caused by biallelic germline mutations in one of four mismatch repair genes, producing a predisposition to ultrahypermutated malignant gliomas, CNS embryonal tumors, in addition to other systemic cancers of childhood and early adulthood. A different condition (Lynch syndrome 1) is caused by heterozygous carriers.

> 90% present with café-au-lait macules and other dermatologic abnormalities.¹⁴

36.3.6 Familial adenomatous polyposis 1 (FAP1)

General information

FAP1 is an autosomal dominant syndrome resulting from an inactivating germline sequence variant in the tumor suppressor gene *APC* predisposing to gastrointestinal tumors (benign colonic polyposis and colorectal carcinomas). Patients with FAP1 who also develop a primary brain tumor are considered brain tumor polyposis syndrome 2 (BTP2) (see following).

Brain tumor polyposis syndrome 2 (BTP2)

Originally part of the now obsolete “Turcot syndrome.” Extremely rare subset of FAP1 patients. The only primary brain tumor unequivocally associated with BTP2 is medulloblastoma (WNT-activated). Other non-CNS cancers include osteomas, thyroid cancer, and hepatoblastoma. The OS and PFS of patients with BTP2 associated MDB WNT-activated appears similar to sporadic cases, which is excellent.

36.3.7 Carney complex (AKA Carney syndrome)

Obsolete terms include LAMB syndrome.

A rare (> 700 individuals worldwide⁶²) autosomal dominant disorder which in > 70% is associated with a heterozygous inactivating variant in the *PRKAR1A* gene which codes for the type 1 α regulatory (R1 α) subunit of PKA (this variant has $\approx$ 100% penetrance). It is typified by cardiac myxomas, endocrine tumors or overactivity, schwannomas and pigmented skin lesions.⁶² The most common endocrine tumor is primary pigmented nodular adrenocortical disease (PPNAD), which often produces ACTH-independent hypercortisolism (Cushing syndrome).

The primary nervous system association is the potentially lethal malignant melanotic nerve sheath tumor (p.773) (MMNST) which occurs in 8-10% of adults. A somatotroph pituitary adenoma (p.870) is found in 10-18%.²¹

Mean life expectancy is 50 years due to cardiac myxoma, MMNST, surgical complications, and other cancers.

36.3.8 Sturge-Weber syndrome

General information

Key concepts

- cardinal signs: 1) localized cerebral cortical atrophy and calcifications, 2) ipsilateral port-wine facial nevus in the distribution of the ophthalmic branch of the trigeminal nerve (V1)
- contralateral seizures usually present
- caused by somatic mosaicism, not a germline mutation
- skull x-rays classically show "tram-tracking" (double parallel lines) from cortical calcifications

AKA encephalotrigeminal angiomatosis AKA encephalofacial angiomatosis. Sturge-Weber syndrome (SWS) is a rare, congenital neurocutaneous disorder involving the brain, skin, and eye. It is not hereditary and is generally not associated with brain tumors. SWS consists of:

1. cardinal features:
 - a) localized cerebral cortical atrophy and calcifications (especially cortical layers 2 and 3, with a predilection for the occipital and posterior parietal lobes):
 - calcifications appear as curvilinear double parallel lines ("tram-tracking") on plain X-rays
 - cortical atrophy usually causes contralateral hemiparesis, hemiatrophy, and homonymous hemianopia (with occipital lobe involvement)
 - b) ipsilateral port-wine facial nevus (nevus flammeus) (a cutaneous capillary malformation) usually in the distribution of the 1st division of the trigeminal nerve (V1) typically involving the forehead and/or eyelid (rarely bilateral): not always present, alternatively sometimes in V2 or V3 regions⁶³
2. other findings that may be present:
 - a) ipsilateral exophthalmos and/or glaucoma, coloboma of the iris
 - b) oculomeningeal capillary hemangioma
 - c) cerebral venous malformation (leptomeningeal angiomatosis)⁶³
 - d) convulsive seizures: contralateral to the facial nevus and cortical atrophy. Present in most patients starting in infancy
 - e) retinal angiomias
 - f) endocrinopathies: growth hormone deficiency is more common in SWS patients. For suspected or confirmed SWS, screen for this in children $\geq$ age 2 years by measuring serum IGF-1 (p.881)

Diagnostic criteria

Diagnosis is made based on having 2 out of 3 of the following:

- facial port-wine birthmark
- increased intraocular pressure
- leptomeningeal angiomatosis

Patients with only leptomeningeal angiomatosis and no skin or eye involvement are considered to have the intracranial variant of SWS.⁶³

► **Other clinical aspects.** Only 8–20% of patients with facial port-wine birthmarks (with or without ocular involvement) develop neurologic symptoms. Those with port-wine stain only in V2 and V3 have a lower risk of developing symptoms of SWS, and those with bilateral V1 birthmarks have a higher risk ($\approx 35\%$).

Genetics

In contrast to other phakomatoses, SWS is not hereditary. It is caused by somatic mosaicism (two genetically distinct populations of cells brought about by a post-zygotic mutation). A single nucleotide substitution in the *GNAQ* gene on chromosome 9q21 that occurs early after conception in utero is found in affected tissues but not in unaffected tissues.⁶⁴

Epidemiology

Incidence is ≈ 1 in 20,000 to 50,000 live births.⁶⁵

Treatment

Treatment is supportive.

Seizures: frequent and protracted seizures exacerbate the neurologic damage.

- antiseizure medications are the first-line treatment
 1. oxcarbazepine is a common initial drug. Side effects include central hypothyroidism, especially in girls
 2. levetiracetam and topiramate are alternatives
- refractory seizures may require lobectomy or hemispherectomy. Options include: functional hemispherectomy,⁶⁶ anatomic hemispherectomy, and hemispherotomy

Skin lesions: laser treatment (currently, flashlamp-pumped PDL is favored) can lighten the birthmark. It may also reduce hypertrophy of soft and bony tissue.

Endocrinopathies: growth hormone deficiency can be replaced; however, there may be a risk of increasing seizures.⁶³

XRT: complications are common and benefits are lacking.

36.3.9 Neurocutaneous melanosis (NCM)

Background

1. a rare, congenital, nonheritable phakomatosis that usually presents before age 2 years in which large or numerous congenital melanocytic nevi are associated with benign and/or malignant melanocytic tumors of the leptomeninges⁶⁷
2. pathogenesis: neuroectodermal defect during morphogenesis involving melanoblasts of skin and pia mater originating from neural crest cells⁶⁷

Clinical features

1. two-thirds of patients with NCM have giant congenital melanocytic nevi⁶⁷: pigmented nevi that are large, hairy, or both. (The chances that nevi represents NCM is higher when the nevi are located on head, posterior neck, or paravertebral area)
2. one-third have numerous lesions without a single giant lesion⁶⁷
3. virtually all have large cutaneous melanocytic (pigmented) nevi located on the posterior torso⁶⁸
4. neurologic manifestations: usually before age 2 years. Signs of intracranial hypertension (lethargy, vomiting...), focal seizures, motor deficits or aphasia⁶⁷
5. hydrocephalus: in almost 66%. Usually due to obstruction of CSF flow or reduced absorption as a result of thickened leptomeninges⁶⁷

Clinical diagnostic criteria

See reference.⁶⁹

1. large or multiple congenital melanocytic nevi with meningeal melanosis or melanoma
2. absence of cutaneous melanoma, except in patients with benign meningeal lesions (i.e., must rule out meningeal metastases from cutaneous melanoma)
3. no evidence of meningeal melanoma, except in patients with benign cutaneous lesions

Associated conditions

NCM is sometimes associated with

1. neurocutaneous disorders⁶⁷
 - a) Sturge-Weber syndrome (p.652)
 - b) von Recklinghausen's neurofibromatosis (NF1) (p.638)
2. posterior fossa cystic malformations: e.g., Dandy Walker malformation (p.270); occurs in up to 10%. These cases have worse prognosis due to malignant transformation⁶⁷
3. intraspinal lipoma and syringomyelia⁶⁷

Diagnostic testing

1. MRI: T1 and T2 signal shortening produced by melanin. IV gadolinium may demonstrate enhancement of tumor-infiltrated meninges⁶⁷
2. histological exam of CNS lesions shows leptomeningeal melanosis (benign) which develops from the melanocytes of the pia matter. Melanoma (malignant) occurs in 40–62% of cases but distinction has little prognostic significance because of the poor outcome of the symptomatic NCM patient even in the absence of melanoma⁶⁷

Management

The benefit of resecting skin lesions is questionable in the presence of leptomeningeal lesions.⁷⁰ NCM appears refractory to radiation therapy and chemotherapy⁷⁰

Neurosurgical involvement is usually limited to⁶⁹:

1. shunting for hydrocephalus
2. palliative operative decompression if early in the course
3. biopsy for tissue diagnosis in questionable cases

Prognosis

1. when neurological signs are present, prognosis is poor regardless of whether or not malignancy is present
2. > 50% of patients die within 3 years after the first neurologic manifestation⁶⁷

References

- [1] Burger PC, Scheithauer BW. AFIP Atlas of Tumor Pathology. Fourth series. Fascicle 7: Tumors of the Central Nervous System. Washington, D.C.: Armed Forces Institute of Pathology; 2007
- [2] Riccardi VM. von Recklinghausen Neurofibromatosis. N Engl J Med. 1981; 305:1617–1627
- [3] Friedman JM, Adam MP, Ardinger HH, et al. Neurofibromatosis 1. Seattle, WA 1998
- [4] Ferner RE, Gutmann DH. Neurofibromatosis type 1 (NF1): diagnosis and management. Handb Clin Neurol. 2013; 115:939–955
- [5] National Institutes of Health Consensus Development Conference. Neurofibromatosis: Conference Statement. Arch Neurol. 1988; 45:575–578
- [6] Packer RJ, Gutmann DH, Rubenstein A, et al. Plexiform neurofibromas in NF1: toward biologic-based therapy. Neurology. 2002; 58:1461–1470
- [7] Sharif S, Moran A, Huson SM, et al. Women with neurofibromatosis 1 are at a moderately increased risk of developing breast cancer and should be considered for early screening. Journal of Medical Genetics. 2007; 44:481–484
- [8] Sevik RJ, Barkovich AJ, Edwards MS, et al. Evolution of white matter lesions in neurofibromatosis type 1: MR findings. AJR. American Journal of Roentgenology. 1992; 159:171–175
- [9] Karnes PS. Neurofibromatosis: A Common Neurocutaneous Disorder. Mayo Clinic Proceedings. 1998; 73:1071–1076
- [10] Walker L, Thompson D, Easton D, et al. A prospective study of neurofibromatosis type 1 cancer incidence in the UK. British Journal of Cancer. 2006; 95:233–238
- [11] Hirbe AC, Gutmann DH. Neurofibromatosis type 1: a multidisciplinary approach to care. Lancet Neurol. 2014; 13:834–843
- [12] Anderson MK, Johnson M, Thornburg L, et al. A Review of Selumetinib in the Treatment of Neurofibromatosis Type 1-Related Plexiform Neurofibromas. Ann Pharmacother. 2022; 56:716–726
- [13] Martuza RL, Eldridge R. Neurofibromatosis 2: (Bilateral Acoustic Neurofibromatosis). N Engl J Med. 1988; 318:684–688
- [14] Louis DN, Ohgaki H, Wiestler OD, et al. WHO classification of tumors of the central nervous system. Lyon, France 2016
- [15] Parry DM, Eldridge R, Kaiser-Kupfer MI, et al. Neurofibromatosis 2 (NF2): clinical characteristics of 63 affected individuals and clinical evidence for heterogeneity. American Journal of Medical Genetics. 1994; 52:450–461
- [16] Neurofibromatosis. Conference statement. National Institutes of Health Consensus Development Conference. Arch Neurol. 1988; 45:575–578
- [17] Evans DG, Pagon RA, Adam MP, et al. Neurofibromatosis 2. Seattle, WA 1998
- [18] Smith MJ, Bowers NL, Bulman M, et al. Revisiting neurofibromatosis type 2 diagnostic criteria to exclude LZTR1-related schwannomatosis. Neurology. 2017; 88:87–92
- [19] Baser ME, Friedman JM, Joe H, et al. Empirical development of improved diagnostic criteria for neurofibromatosis 2. Genet Med. 2011; 13:576–581
- [20] Egelhoff JC, Bates DJ, Ross JS, et al. Spinal MR Findings in Neurofibromatosis Types 1 and 2. American Journal of Neuroradiology. 1992; 13:1071–1077
- [21] Bosman FT, Jaffe ES, Lakhani SR, et al. World Health Organization Classification of Tumors series, vol. 6. In: WHO Classification of Tumors Editorial Board (ed.) World Health Organization classification of tumors of the central nervous system. 5th ed. Lyon, France: International Agency for Research on Cancer; 2021

- [22] Plotkin SR, Blakeley JO, Evans DG, et al. Update from the 2011 International Schwannomatosis Workshop: From genetics to diagnostic criteria. *Am J Med Genet A*. 2013; 161A:405–416
- [23] Antinheimo J, Sankila R, Carpen O, et al. Population-based analysis of sporadic and type 2 neurofibromatosis-associated meningiomas and schwannomas. *Neurology*. 2000; 54:71–76
- [24] Merker VL, Esparza S, Smith MJ, et al. Clinical features of schwannomatosis: a retrospective analysis of 87 patients. *Oncologist*. 2012; 17:1317–1322
- [25] Li P, Zhao F, Zhang J, et al. Clinical features of spinal schwannomas in 65 patients with schwannomatosis compared with 831 with solitary schwannomas and 102 with neurofibromatosis Type 2: a retrospective study at a single institution. *J Neurosurg Spine*. 2016; 24:145–154
- [26] Plotkin SR, Bredella MA, Cai W, et al. Quantitative assessment of whole-body tumor burden in adult patients with neurofibromatosis. *PLoS One*. 2012; 7. DOI: 10.1371/journal.pone.0035711
- [27] Northrup H, Krueger DA, International Tuberous Sclerosis Complex Consensus Group. Tuberous sclerosis complex diagnostic criteria update: recommendations of the 2012 International Tuberous Sclerosis Complex Consensus Conference. *Pediatr Neurol*. 2013; 49:243–254
- [28] Hottinger AF, Khakoo Y. Neurooncology of familial cancer syndromes. *Journal of Child Neurology*. 2009; 24:1526–1535
- [29] Logue LG, Acker RE, Sienko AE. Best cases from the AFIP: angiomyolipomas in tuberous sclerosis. *Radiographics*. 2003; 23:241–246
- [30] Thiele EA. Managing epilepsy in tuberous sclerosis complex. *Journal of Child Neurology*. 2004; 19:680–686
- [31] McLaurin RL, Towbin RB. Tuberous Sclerosis: Diagnostic and Surgical Considerations. *Pediatr Neurol*. 1985; 12:43–48
- [32] Bernauer TA. The radial bands sign. *Radiology*. 1999; 212:761–762
- [33] Franz DN, Agricola K, Mays M, et al. Everolimus for subependymal giant cell astrocytoma: 5-year final analysis. *Ann Neurol*. 2015. DOI: 10.1002/ana.24523
- [34] Shepherd CW, Gomez MR, Lie JT, et al. Causes of death in patients with tuberous sclerosis. *Mayo Clin Proc*. 1991; 66:792–796
- [35] Catapano D, Muscarella LA, Guarnieri V, et al. Hemangioblastomas of the central nervous system: molecular genetic analysis and clinical management. *Neurosurgery*. 2005; 56:1215–21; discussion 1221
- [36] Glenn GM, Linehan WM, Hosoe S, et al. Screening for von Hippel-Lindau Disease by DNA Polymorphism Analysis. *JAMA*. 1992; 267:1226–1231
- [37] Wanebo JE, Lonser RR, Glenn GM, et al. The natural history of hemangioblastomas of the central nervous system in patients with von Hippel-Lindau disease. *Journal of Neurosurgery*. 2003; 98:82–94
- [38] Butman JA, Linehan WM, Lonser RR. Neurologic manifestations of von Hippel-Lindau disease. *JAMA*. 2008; 300:1334–1342
- [39] Go RCP, Lamiell JM, Hsia YE, et al. Segregation and Linkage Analysis of von Hippel-Lindau Disease Among 220 Descendants from one Kindred. *Am J Human Genet*. 1984; 36:131–142
- [40] Richards FM, Payne SJ, Zbar B, et al. Molecular analysis of de novo germline mutations in the von Hippel-Lindau disease gene. *Hum Mol Genet*. 1995; 4:2139–2143
- [41] Friedrich CA. Genotype-phenotype correlation in von Hippel-Lindau syndrome. *Hum Mol Genet*. 2001; 10:763–767
- [42] van Leeuwen RS, Ahmad S, Links TP, et al. Von Hippel-Lindau Syndrome. Seattle, WA 2000
- [43] Jagannathan J, Lonser RR, Smith R, et al. Surgical management of cerebellar hemangioblastomas in patients with von Hippel-Lindau disease. *Journal of Neurosurgery*. 2008; 108:210–222
- [44] Manksi TJ, Heffner DK, Glenn GM, et al. Endolymphatic sac tumors. A source of morbid hearing loss in von Hippel-Lindau disease. *JAMA*. 1997; 277:1461–1466
- [45] Chew EY. Ocular manifestations of von Hippel-Lindau disease: clinical and genetic investigations. *Transactions of the American Ophthalmological Society*. 2005; 103:495–511
- [46] Meyerle CR, Dahr SS, Wetjen NM, et al. Clinical course of retrolubar hemangioblastomas in von Hippel-Lindau disease. *Ophthalmology*. 2008; 115:1382–1389
- [47] Niemela M, Lemeta S, Summanen P, et al. Long-term prognosis of haemangioblastoma of the CNS: impact of von Hippel-Lindau disease. *Acta Neurochirurgica (Wien)*. 1999; 141:1147–1156
- [48] Choyke PL, Glenn GM, Walther MM, et al. Hereditary renal cancers. *Radiology*. 2003; 226:33–46
- [49] Meister M, Choyke P, Anderson C, et al. Radiological evaluation, management, and surveillance of renal masses in Von Hippel-Lindau disease. *Clinical Radiology*. 2009; 64:589–600
- [50] Maher ER, Kaelin WJ Jr. von Hippel-Lindau disease. *Medicine*. 1997; 76:381–391
- [51] Hes FJ, Feldberg MA. Von Hippel-Lindau disease: strategies in early detection (renal-, adrenal-, pancreatic masses). *European Radiology*. 1999; 9:598–610
- [52] Biscaglia M, Galliani CA, Senger C, et al. Renal cystic diseases: a review. *Advances in Anatomic Pathology*. 2006; 13:26–56
- [53] Truong LD, Choi YJ, Shen SS, et al. Renal cystic neoplasms and renal neoplasms associated with cystic renal diseases: pathogenetic and molecular links. *Advances in Anatomic Pathology*. 2003; 10:135–159
- [54] Bradley S, Dumas N, Ludman M, et al. Hereditary renal cell carcinoma associated with von Hippel-Lindau disease: a description of a Nova Scotia cohort. *Can Urol Assoc J*. 2009; 3:32–36
- [55] Moss JM, Choi CY, Adler JR Jr, et al. Stereotactic radiosurgical treatment of cranial and spinal hemangioblastomas. *Neurosurgery*. 2009; 65:79–85; discussion 85
- [56] Chang SD, Meisel JA, Hancock SL, et al. Treatment of hemangioblastomas in von Hippel-Lindau disease with linear accelerator-based radiosurgery. *Neurosurgery*. 1998; 43:28–34; discussion 34–5
- [57] VHL Alliance. VHLA suggested active surveillance guidelines. 2016. <https://www.vhl.org/wp-content/uploads/2017/07/Active-Surveillance-Guidelines.pdf>
- [58] Constans JP, Meder F, Maiuri F, et al. Posterior fossa hemangioblastomas. *Surg Neurol*. 1986; 25:269–275
- [59] Hes FJ, van der Luit RB. [Von Hippel-Lindau disease: protocols for diagnosis and periodical clinical monitoring. National Von Hippel-Lindau Disease Working Group]. *Ned Tijdschr Geneesk*. 2000; 144:505–509
- [60] Poulsen ML, Budtz-Jorgensen E, Bisgaard ML. Surveillance in von Hippel-Lindau disease (vHL). *Clinical Genetics*. 2009. DOI: 10.1111/j.1399-0004.2009.01281.x
- [61] Li FP, Fraumeni JF Jr, Mulvihill JJ, et al. A cancer family syndrome in twenty-four kindreds. *Cancer Res*. 1988; 48:5358–5362
- [62] Stratakis CA, Raygada M, Adam MP, et al. Carney Complex. Seattle, WA 2003
- [63] Bachur CD, Comi AM. Sturge-Weber syndrome. *Curr Treat Options Neurol*. 2013; 15:607–617
- [64] Shirley MD, Tang H, Gallione CJ, et al. Sturge-Weber syndrome and port-wine stains caused by somatic mutation in GNAQ. *N Engl J Med*. 2013; 368:1971–1979
- [65] Comi AM. Update on Sturge-Weber syndrome: diagnosis, treatment, quantitative measures, and controversies. *Lymphat Res Biol*. 2007; 5:257–264
- [66] Schramm J, Kuczaty S, Sassen R, et al. Pediatric functional hemispherectomy: outcome in 92 patients. *Acta Neurochir (Wien)*. 2012; 154:2017–2028
- [67] Di Rocco F, Sabatino G, Koutzoglou M, et al. Neurocutaneous melanosis. *Child's Nervous System*. 2004; 20:23–28

- [68] DeDavid M, Orlow SJ, Provost N, et al. Neurocutaneous melanosis: clinical features of large congenital melanocytic nevi in patients with manifest central nervous system melanosis. *Journal of the American Academy of Dermatology*. 1996; 35:529-538
- [69] McClelland S, 3rd, Charnas LR, SantaCruz KS, et al. Progressive brainstem compression in an infant with neurocutaneous melanosis and Dandy-Walker complex following ventriculoperitoneal shunt placement for hydrocephalus. Case report. *Journal of Neurosurgery*. 2007; 107:500-503
- [70] Mena-Cedillos CA, Valencia-Herrera A M, Arroyo-Pineda AI, et al. Neurocutaneous melanosis in association with the Dandy-Walker complex, complicated by melanoma: report of a case and literature review. *Pediatric Dermatology*. 2002; 19:237-242

37 Adult Diffuse Gliomas

37.1 Incidence

These are the most common primary intra-axial brain tumors. For a list of tumors in this category, see ► Table 35.1. With the introduction of new diagnostic categories in the WHO CNS5 in 2021, statistics on incidence and prognosis for some of these tumors will have to wait until enough data is accumulated to reflect the fact that prior studies were unknowingly grouping disparate tumors together.

37.2 Risk factors for diffuse gliomas

The best-established causes for brain tumors are syndromic (p.637) (familial diseases...) and radiation therapy used to treat other CNS tumors (most of these are IDH wildtype¹).

► **Cell phones and cancer risk.** The following information is excerpted from the National Cancer Institute (NCI) web site.²

Concern over cell phones and cancer have arisen because cell phones emit radiofrequency electromagnetic radiation (RF-EMR) and the number of cell phones and duration of use has rapidly increased since their introduction. While ongoing studies continue to look into possible ramifications, a summary of findings to date is presented here.

- brain cancer incidence and mortality have remained stable over the past decade
- in May 2011, the WHO issued a warning of possible cancer risk related to the use of devices that emit nonionizing EMR, such as cellular phones³
- the American Cancer Society (ACS) states that the WHO classification means that while RF-EMR could pose some cancer risk, the evidence is not strong enough to be considered causal and further investigation is needed
- the National Institute of Environmental Health (NIEHS) indicates that the weight of current evidence has not conclusively linked cell phone use with any adverse health problems, but that more research is needed
- the US Food and Drug Administration (FDA) commented that the majority of human epidemiologic studies have failed to show a relationship between RF-EMR from cell phones and health problems
- the US Centers for Disease Control (CDC) states that no scientific information definitively answers the question of whether cell phones cause cancer
- the Federal Communications Commission (FCC) concludes that no scientific evidence establishes a causal link between cell phone use and cancer or other illnesses
- the European Commission Scientific Committee on Emerging and Newly Identified Health Risks concluded in 2015 that cell phone RF-EMR exposure did not show an increased risk of brain tumors or other head and neck cancers and that epidemiologic studies do not indicate increased risk for other malignant diseases, including childhood cancer

37.3 General features of gliomas

37.3.1 Neuroradiology

Astrocytomas typically arise in white matter (e.g., centrum semiovale) and traverse through white matter tracts (see Spread (p.658)). See also MR-spectroscopy findings (p.244).

► **CT scan & MRI grading.** Grading gliomas by CT or MRI is imprecise,⁴ but may be used as a preliminary assessment (see ► Table 37.1). Neuroradiologic grading is *not* applicable to pediatric patients with astrocytomas that are not diffuse astrocytomas (e.g., pilocytic astrocytomas).

► **Positron emission tomography (PET) scan.** Astrocytoma, IDH-mutant, grade 2 tumors appear as hypometabolic “cold” spots with fluorodeoxyglucose PET scans. Hypermetabolic “hot” spots suggest grade 3 or 4, and may distinguish the uncommon grade 3 or 4 tumors that do not enhance on MRI from astrocytoma, IDH-mutant, grade 2.

Table 37.1 Grading gliomas by CT or MRI

WHO grade	Typical radiographic findings		Location
astrocytoma, IDH-mutant, grade 2	CT: low density MRI: abnormal signal on T2WI	no enhancement with little or no mass effect	temporal, posterior frontal & anterior parietal lobes
astrocytoma, IDH-mutant, grade 3	complex enhancement ^a		
astrocytoma, IDH-mutant, grade 4	solid enhancement ^a		frontal lobe predilection
glioblastoma, IDH-wildtype	central necrosis with ring enhancement ^b		temporal > parietal > frontal > occipital

^asome may not enhance. Most glioblastomas enhance, but some rare ones do not.^{4,5}
^bthe nonenhancing center may represent necrosis or associated cyst (see Tumor associated cysts (p. 658)). The enhancing ring is cellular tumor; however, tumor cells also extend ≥ 15 mm beyond the ring⁶

37.3.2 Spread

Gliomas may spread by the following mechanisms⁷ (note: <10% of recurrent gliomas recur away from the original site⁸):

- tracking through white matter
 - corpus callosum (CC)
 - through genu or body of CC → bilateral frontal lobe involvement (“butterfly glioma”)
 - through splenium of CC → bilateral parietal or occipital lobes
 - cerebral peduncles → midbrain involvement
 - internal capsule → encroachment of basal ganglion tumors into centrum semiovale
 - uncinate fasciculus → simultaneous frontal and temporal lobe tumors
 - interthalamic adhesion → bilateral thalamic gliomas
- CSF pathways (subarachnoid seeding): 10–25% frequency of meningeal and ventricular seeding by high grade gliomas⁹
- rarely, gliomas may spread systemically

37.3.3 Tumor-associated cysts

Gliomas may have cystic central necrosis, but may also have an associated cyst even without necrosis. When fluid from these cysts is aspirated it can be differentiated from CSF by the fact that it is usually xanthochromic and often clots once removed from the body (unlike e.g., fluid from a chronic subdural). Although they may occur with malignant gliomas, cysts are more commonly associated with pilocytic astrocytomas (p. 693).

37.3.4 Molecular biomarker testing for diffuse gliomas

Recommendations for biomarker testing for diffuse gliomas is summarized in ► Table 37.2¹⁰

37.4 Adult-type diffuse gliomas

37.4.1 Astrocytoma, IDH-mutant (WHO grade 2, 3 or 4) (AIM grades 2–4)

General information

A diffuse astrocytoma with mutation in either the *IDH1* or *IDH2* gene and absence of 1p/19q codeletion. *ATRX* and/or *TP53* mutations are common.

There is a predilection for the supratentorial compartment, usually near or within the frontal lobes (► Fig. 37.3).

Most present with seizures. The slower growth of grade 3 & 4 lesions compared to glioblastoma IDH-wildtype may provide more time for normal brain tissue to accommodate to the mass effect and may account for the longer clinical course (16.8 months).

Table 37.2 Biomarker testing recommendations for diffuse gliomas (DG)¹⁰

Recommendation	Strength of recommendation
• IDH mutational testing must be performed on all DGs	strong
• ATRX status should be assessed in all IDH-mutant DGs unless they show 1p/19q codeletion	strong
• TP53 status should be assessed in all IDH-mutant DGs unless they show 1p/19q codeletion	conditional
• 1p/19q codeletion must be assessed in IDH-mutant DGs unless they show ATRX loss or TP53 mutations	strong
• CDKN2A/B homozygous deletion testing should be performed on IDH-mutant astrocytomas	conditional
• MGMT promoter methylation (p. 673) testing should be performed on all GBM, IDH-WT	strong
• for IDH-mutant DGs, MGMT promoter methylation testing may not be necessary	conditional
• TERT promoter mutation testing may be used to provide further support for the diagnosis of oligodendroglioma and IDH-WT GBM	conditional
• for histologic grade 2-3 DGs that are IDH-WT, testing should be performed for whole chromosome 7 gain/whole chromosome 10 loss, EGFR amplification, and TERT promoter mutation to establish the molecular diagnosis of GBM, IDH-WT, grade 4	strong
• H3 K27M testing must be performed in DGs that involve the midline in the appropriate clinical and pathologic setting	strong
• H3 G34 testing may be performed in pediatric and young adult patients with IDH-WT DGs	conditional
• BRAF mutation testing (V600) may be performed in DGs that are IDH-WT and H3-WT	conditional
• MYB/MYBL1 and FGFR1 testing may be performed in children and young adults with DGs that are histologic grade 2-3 and are IDH-WT and H3-WT	conditional

Abbreviations: ATRX, ATRX chromatin remodeler; BRAF, B-Raf proto-oncogene; CDKN2A, cyclin-dependent kinase inhibitor 2A; CDKN2B, cyclin-dependent kinase inhibitor 2B; DGs, diffuse gliomas; EGFR, epidermal growth factor; FGFR1, fibroblast growth factor receptor 1; GBM, glioblastoma; H3, histone 3; IDH, isocitrate dehydrogenase; MGMT, O-6-methylguanine-DNA methyltransferase; MYB, MYB proto-oncogene; MYBL1, MYB-like; TERT, telomerase reverse transcriptase; TP53, tumor protein p53; WT, wild-type.

Reprinted from Brat DJ, Aldape K, Bridge JA et al "Molecular Biomarker Testing for the Diagnosis of Diffuse Gliomas: Guideline From the College of American Pathologists in Collaboration With the American Association of Neuropathologists, Association for Molecular Pathology, and Society for Neuro-Oncology" (Arch Pathol Lab Med. 2022;146(5):547-574) with permission from Archives of Pathology & Laboratory Medicine. Copyright 2022. College of American Pathologists.

Diagnostic criteria for astrocytoma, IDH-mutant (AIM)

Diagnostic criteria are shown in ► Fig. 37.1.¹¹

Subtypes

See reference.¹¹

- **astrocytoma, IDH-mutant, grade 2 (AIM grade 2)**
 - a) well differentiated
 - b) lacking histologic features of anaplasia, with low or undetectable mitotic activity, and no microvascular proliferation, no necrosis
 - c) no homozygous deletions of CDKN2A* or CDKN2B*
- **astrocytoma, IDH-mutant, grade 3 (AIM grade 3)**
 - a) focal or dispersed anaplasia
 - b) significant mitotic activity
 - c) no microvascular proliferation
 - d) no necrosis
 - e) no homozygous deletions of CDKN2A* or CDKN2B*

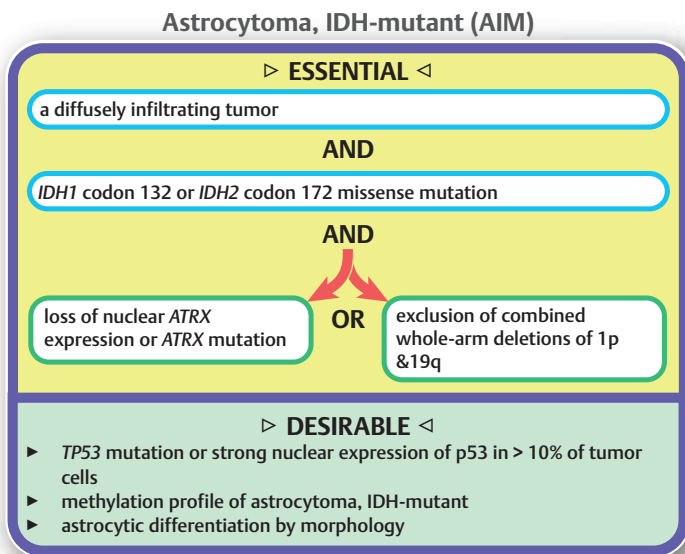


Fig. 37.1 Diagnostic criteria for astrocytoma, IDH-mutant. (Modified with permission from WHO Classification of Tumors Editorial Board, (ed.) World Health Organization classification of tumors of the central nervous system. 5th ed. Bosman F T, Jaffe E S, Lakhani S R, et al. (series eds.): World Health Organization Classification of Tumors series, vol. 6. International Agency for Research on Cancer, Lyon, France, 2021.)

- **astrocytoma, IDH-mutant, grade 4:** (AIM grade 4) any combination of:
 - a) microvascular proliferation
 - b) necrosis
 - c) homozygous deletions of *CDKN2A** and/or *CDKN2B**

* AIM with homozygous deletion of *CDKN2A* and/or *CDKN2B* are more aggressive, with shorter OS and are classified as AIM grade 4 regardless of histology.^{12,13}

Lower grade AIMS can dedifferentiate to a higher grade (p.662).

▶ **Ki-67 (MIB-1) index.** It has been suggested that a MIB-1 index (p.632) ≥ 7–9% is indicative of a high-grade tumor. But this can overlap with grade 2 AIM at the low end or grade 4 AIM at the high end, and can vary within the tumor itself. This, together with the variability between observers and institutions precludes using the Ki-67 index as a sole discriminant between grade 2 & 3 or between grade 3 & 4 AIM.¹⁴

Imaging

AIM grade 2: usually hypodense on CT. Most are hypointense on T1WI MRI, and show high intensity changes on T2WI that extend beyond the tumor volume. See ▶ Fig. 37.2. Most do not enhance on CT or MRI. Calcifications and cysts may be seen.

AIM grades 2 & 3: may demonstrate **T2-FLAIR mismatch** where high signal areas on T2 may be hypointense on FLAIR.¹⁵

With higher grade AIM, there tends to be more edema, enhancement and necrosis (suggested by central hypodensity). Unlike their GBM, IDH-wildtype (▶ Fig. 37.6) counterpart, astrocytoma, IDH-mutant, grade 4 tumors (▶ Fig. 37.3) tend not to have large areas of central necrosis, are often larger at diagnosis, and have less perilesional edema.

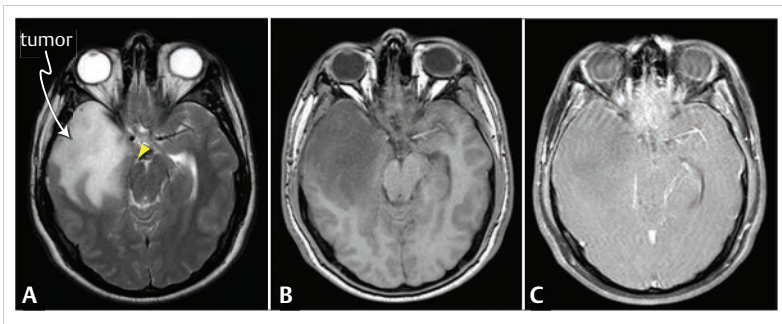


Fig. 37.2 Astrocytoma, IDH-mutant, grade 2 of the right temporal lobe.

Image: axial brain MRI, A: T2, B: T1, C: T1 with contrast.

White matter changes can be seen extending back from the temporal tip on the T2 image (A). Tumor is low intensity on the T1 image (B), and does not enhance with gadolinium contrast (C). The tumor is causing mass effect with compression of the right midbrain (yellow arrowhead), which could progress to uncus herniation.

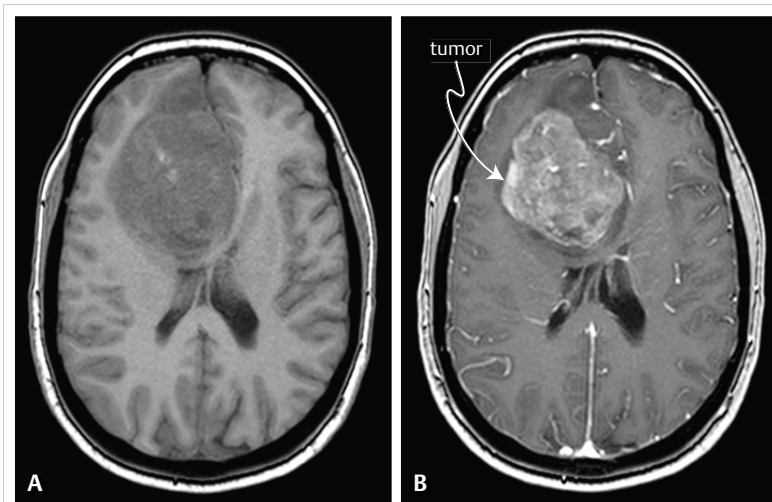


Fig. 37.3 Astrocytoma, IDH-mutant, grade 4 of the right frontal lobe.

Image: axial brain MRI, A: T1 non-contrast, B: T1 with contrast.

Note the location in the frontal lobe, and the lack of central necrosis, both of which are characteristic of astrocytoma, IDH-mutant (compare to GBM, IDH-wildtype, ► Fig. 37.6).

Treatment

See Treatment for adult-type diffuse infiltrating gliomas (p. 668) for grades 2–4.

Prognosis

One cohort demonstrated a median survival of 10.9 years.¹⁶

A recent study¹³ demonstrated that there was no survival benefit of AIM grade 2 over AIM grade 3 as defined by the WHO CNS5 criteria.¹³ They also found that post-operative residual tumor volume and preoperative tumor size were significant prognosticators, and that age was not.

Pre-IDH era data showed worse prognosis with the following¹⁷ (newer prognosticators will need to be identified with known IDH status):

1. age > 40 years (perhaps the most important unfavorable prognosticator)
2. largest tumor diameter $\geq$ 6 cm
3. tumor crossing the midline
4. neurologic deficit prior to surgery
5. subtotal resection

► **Dedifferentiation.** Low-grade AIM are capable of malignant dedifferentiation, and the ultimate behavior of these tumors in adults is usually not benign, as 75% of adult tumors undergo anaplastic progression.

WHO grade 2 astrocytomas tend to undergo malignant transformation more quickly when diagnosed after age 45 years. Once dedifferentiation occurs, median survival is 2–3 years beyond that event (pre-IDH era data¹⁸).

37.4.2 Oligodendroglioma, IDH mutant and 1p/19q-codeleted (WHO grade 2 or 3) (ODG grade 2 or 3)

General information

Key concepts

- slow growing tumor that frequently presents with seizures
- occur primarily in adults, predilection for the frontal lobes > temporal lobe
- by definition: a diffusely infiltrating glioma with codeletion of *BOTH* chromosome arms 1p AND 19q, AND mutation of IDH1 AND/OR IDH2
- 2 grades: WHO 2 and 3
- histology: classic features of “fried egg” cytoplasm (on permanent pathology) and “chicken wire” vasculature are unreliable. Calcifications are common

Location: > 50% occur in the frontal lobes, followed by temporal, parietal, and occipital lobes.

Diagnostic criteria for oligodendroglioma, IDH mutant and 1p/19q-codeleted (ODG)

Diagnostic criteria are shown in ► Fig. 37.4.¹¹

Subtypes

See reference.¹¹

- **oligodendroglioma, IDH mutant and 1p/19q-codeleted, grade 2** (ODG grade 2)
- **oligodendroglioma, IDH mutant and 1p/19q-codeleted, grade 3** (ODG grade 3)

Grading oligodendroglioma

The criteria for distinguishing grade 2 from grade 3 ODG is not well characterized. WHO grade 2 has a more favorable prognosis.

Grade 3 ODGs usually have several of the following features:

- high cellularity
- marked cytological atypia
- brisk mitotic activity
- pathological microvascular proliferation
- necrosis with or without palisading

Other features that may be helpful in borderline cases are proliferation markers such as Ki-67 (MIB1) and clinical or radiologic evidence of rapid symptomatic growth and contrast enhancement.

The presence of several mitotic figures alone is not diagnostic of grade 3

Oligodendroglioma, IDH mutant and 1p/19q-codeleted

▷ ESSENTIAL ◁

a diffusely infiltrating tumor

AND

IDH1 codon 132 or IDH2 codon 172 missense mutation^a

AND

combined whole-arm deletions of 1p and 19q

▷ DESIRABLE ◁

- ▶ DNA methylation profile of oligodendroglioma, IDH-mutant and 1p/19q-codeleted
- ▶ retained nuclear expression of ATRX
- ▶ TERT promoter mutation

^a IDH mutation analysis may not be required when DNA methylome profiling is performed and unequivocally assigns the tumor to the methylation class oligodendroglioma, IDH-mutant and 1p/19q-codeleted.

Fig. 37.4 Diagnostic criteria for oligodendroglioma, IDH mutant and 1p/19q-codeleted. (Modified with permission from WHO Classification of Tumors Editorial Board, (ed.) World Health Organization classification of tumors of the central nervous system. 5th ed. Bosman F T, Jaffe E S, Lakhani S R, et al. (series eds.): World Health Organization Classification of Tumors series, vol. 6. International Agency for Research on Cancer, Lyon, France, 2021.)

Clinical

Presenting symptoms are nonspecific for ODG. Seizures are the presenting symptoms in ≈ two thirds of cases.¹⁹ Other symptoms are nonspecific for ODG, and are more often related to local mass effect and less commonly to ↑ ICP.

Imaging

Calcifications: seen in 28–60% of ODGs on plain radiographs,²⁰ and on 90% of CTs.

CT: hypodense to iso-dense.

MRI: poorly demarcated, hypointense on T1 and hyperintense on T2. Fewer than 20% of grade 2 ODG and over 70% of grade 3 ODG enhance with gadolinium.²¹

Histology

Oligodendroglial tumors cells are isomorphic with round nuclei and are slightly larger than normal oligodendrocytes. Tumors show moderate cellularity, often arranged in sheets.

73% of tumors have microscopic calcifications.²² Isolated tumor cells consistently penetrate largely intact parenchyma; an associated solid tumor component may or may not be present.²³ When a solid portion is present, permanent (paraffin) pathology demonstrates lucent perinuclear halos, giving a “fried egg” or “honeycomb” appearance (which is actually an artifact of *formalin* fixation, and is not present on frozen section which may make diagnosis difficult on frozen). A “chicken-wire” vascular pattern has also been described.²⁴ These features are variable.

Nuclear atypia and an occasional mitotic figure is compatible with an ODG grade 2 tumor.²⁵ But extensive mitotic activity, prominent microvascular proliferation and spontaneous necrosis are indications of anaplasia corresponding to an oligodendroglioma, IDH-mutant 1p/19q codeleted (ODG) grade 3.

16% of hemispheric ODGs are *cystic*²² (cysts generally form from coalescence of microcysts from microhemorrhages, unlike astrocytomas, which actively secrete fluid).

GFAP staining: Since most ODGs contain microtubules instead of glial filaments,²⁶ ODGs usually do not stain for GFAP (p.631), although some do.²⁷

CSF metastases

Metastases reportedly occur in 8–14%, mainly as a microscopic finding,²⁸ but 1% may be a more realistic estimate.²⁰ Symptomatic spinal metastases (drop mets) are even more uncommon.²⁸

Treatment

See Treatment for adult-type diffuse infiltrating gliomas grades 2–4 (p.668).

Prognosis

Median survival with ODG, IDH-mutant & 1p/19q codeletion is 8 years.²⁹

Codeletion of 1p/19q by itself is also associated with longer median survival^{30,31} compared to 6.4 years with astrocytoma, IDH mutant.

ODG grade 2 is an independent favorable prognosticator compared to ODG grade 3.

37.4.3 Glioblastoma, IDH-wildtype (WHO grade 4) (GBM)

General information

The term “glioblastoma *multiforme*” is obsolete,³² although the abbreviation “GBM” persists. This tumor arises de novo, without any identifiable lower grade precursor and was formerly called “primary glioblastoma”. All GBM are now, by definition, IDH-wildtype, and are WHO CNS5 grade 4.¹¹ The putative cell of origin of GBM is a neural precursor cell in the subventricular zone.³³

Glioblastoma accounts for 14% of primary brain tumors, and is the most common adult *malignant* primary brain tumor (comprising 49% of them).³⁴ It is also the most lethal. It primarily affects older adults (mean age at diagnosis ≈ 62 years) with a peak incidence between 55–85 years of age. There is a male predominance of 1.6:1 in the U.S.³⁴

GBM constitutes only 3% of pediatric brain tumors.

Infratentorial GBMs are rare, and often represent subarachnoid dissemination of a supratentorial GBM.³⁵

Diagnostic criteria for glioblastoma, IDH-wildtype

Diagnostic criteria are shown in ► Fig. 37.5.¹¹

Subtypes

See reference.¹¹

- **giant cell glioblastoma** (p.665)
- **gliosarcoma** (p.665)
- **epithelioid glioblastoma** (p.666)

Histopathology

► **Histology.** Glioblastomas are typically infiltrating, highly cellular tumors of poorly differentiated astrocytes with nuclear atypia, cellular pleomorphism, with marked mitotic activity. Characteristic findings include microvascular proliferation and/or necrosis with or without perinecrotic pseudopalisading. As the “multiforme” part of the obsolete term “glioblastoma multiforme” implies, the histopathology of the tumor is highly variable. This feature may make it difficult to establish a reliable diagnosis with small specimens (e.g., needle biopsy).

Cell density is highest in the ring enhancing portion of the tumor, as viable cells may not be found in the necrotic center.

► **Glial fibrillary acidic protein (GFAP).** Most astrocytomas stain positive for GFAP. Astrocytomas that may not stain GFAP positive: some poorly differentiated gliomas, tumors that have a paucity of fibrillary astrocytes (some of these tumors used to be called gemistocytic astrocytomas), since fibrillary astrocytes are required for a positive stain.

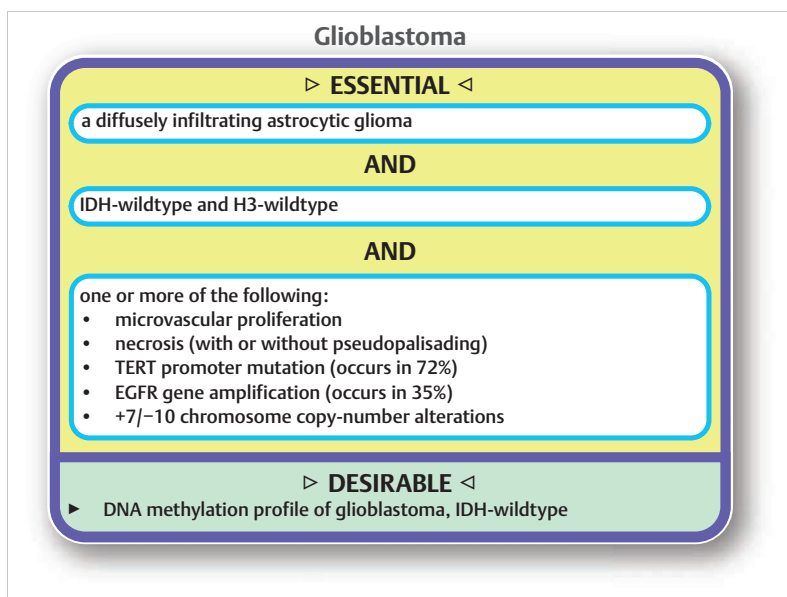


Fig. 37.5 Diagnostic criteria for glioblastoma, IDH-wildtype. (Modified with permission from WHO Classification of Tumors Editorial Board, (ed.) World Health Organization classification of tumors of the central nervous system. 5th ed. Bosman F T, Jaffe E S, Lakhani S R, et al. (series eds.): World Health Organization Classification of Tumors series, vol. 6. International Agency for Research on Cancer, Lyon, France, 2021.)

Tumor location

These tumors tend to favor the deep white matter of the temporal and parietal lobes (► Fig. 37.6) with frequency of locations shown in ► Table 37.3. Spinal cord or cerebellar involvement are rare.

Neuroradiology

IDH-wildtype GBMs tend to have a large area of central necrosis, which produces a ring-enhancement pattern on contrast CT or MRI (► Fig. 37.6) (unlike astrocytoma, IDH-mutant tumors which tend not to have central necrosis (► Fig. 37.3)).

Subtypes of glioblastoma, IDH-wildtype

Giant cell glioblastoma (WHO grade 4)

A rare variant of IDH-wildtype GBM comprising <1% of glioblastomas. Histology features bizarre, multinucleated giant cells and, in some cases, abundant reticulin network.

Tends to develop in younger patients (mean age = 51 years) than classic GBM IDH-wildtype (62 years).

As with GBM IDH-wildtype, these tumors appear to arise de novo with no known precursor.

Mutations of TP53 (in 75–90%) and PTEN (in 33%) are characteristic.

Gliosarcoma (WHO grade 4)

A rare variant of IDH-wildtype GBM, comprising 2–8% of GBMs. Histology features a biphasic tissue pattern consisting of areas of glial differentiation alternating with areas of mesenchymal differentiation.

May arise de novo, or can develop following treatment of a GBM. May also appear in conjunction with ependymoma (ependymosarcoma) and oligodendroglioma (oligosarcoma).

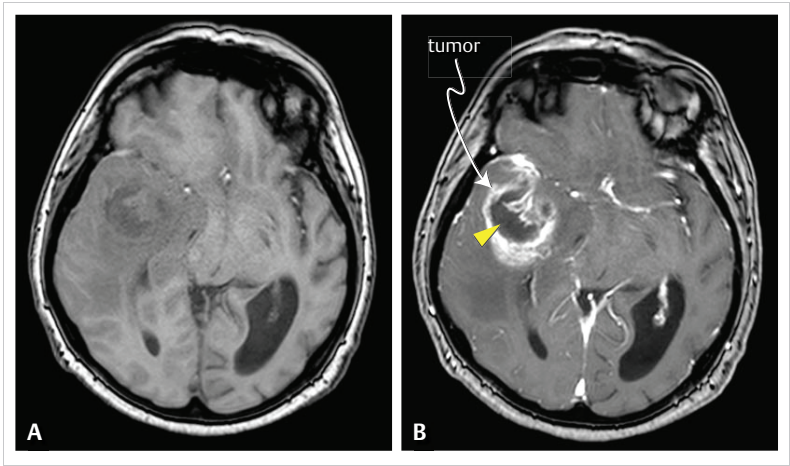


Fig. 37.6 Glioblastoma, IDH-wildtype of the right temporal lobe.
Image: axial brain MRI, A: T1 non-contrast, B: T1 with contrast.
Note the location in the temporal lobe, and the area of central necrosis (dark area indicated by yellow arrowhead) characteristic of GBM, IDH-wildtype.

Table 37.3 Location of glioblastoma IDH-wildtype tumors	
Location	%
temporal	31
parietal	24
frontal	23
occipital	16

Gliosarcomas that are predominantly sarcomatous may enhance homogeneously and can mimic a meningioma.

Epithelioid glioblastoma (WHO grade 4)

A subtype of GBM IDH-wildtype featuring closely spaced epithelioid cells, as well as some rhabdoid cells, in addition to the characteristic GBM findings of mitoses, microvascular proliferation, and necrosis.

- Affects young adults and children, with a predilection for the cerebellum and diencephalon.
- Molecular genetics are notable for a higher incidence (50%) of BRAF V600E mutation.

Treatment

See Treatment for adult-type diffuse infiltrating gliomas (p.668) for grades 2-4.

Prognosis for high grade diffuse gliomas (WHO grades 3 & 4)

Univariate analysis³⁶ shows improved prognosis with the following:

- preoperative performance status: e.g., Karnofsky score (KPS) (p.1640) ≥ 80
- age: younger age is consistently found to be a very significant prognosticator (possibly in part because it is overrepresented with secondary GBM). Patients <45 years in this study fared better
- IDH1 mutation
- methylation of MGMT promoter gene

Survival based on MGMT promotor gene methylation (p.673) (► Table 37.4)³⁷

Table 37.4 Survival with GBM based on MGMT promotor gene status³⁷

	MGMT promoter gene	
	Unmethylated	Methylated
Median OS (months)	12.2	18.2
2-year survival	7.8%	34.1%

Median overall survival among 206 patients: MGMT promoter gene methylation positive patients treated with temozolomide + RT was 21.7 months, RT only 15.3 months; no survival difference between treatments in patients with unmethylated MGMT.

Recursive partitioning analysis (RPA) with grade 4 diffuse glioma: RPA has been used to classify groups with distinct differences in survival. This is useful for refining stratification and phase III study design and can determine which patient subsets will likely benefit from specific treatments (and which may be spared unnecessary treatment).^{38,39} RPA identified 3 classes shown in ► Table 37.5. To read this table, each RPA class can have more than one combination of characteristics to qualify, and each set of characteristics is a row in that class. Caveats for this particular analysis: only IDH1 mutations were assayed (IDH2 mutations are rare in GBM but were not included). There are some implications that appear contradictory (e.g., Class II patients with IDH-wildtype could have GTR or PR/biopsy and end up with the same prognosis).

The European Nomogram GBM Calculator⁴⁰: <http://www.eortc.be/tools/gbmcalculator>. Data from randomized trials by EORTC and NCIC analyzed for the prediction of survival of GBM patients (based on factors such as WHO performance status, mini mental exam score...) which is updated periodically.

Table 37.5 recursive partitioning analysis (RPA) model of survival with AIM grade 4 or GBM^{38,39}

RPA class	n	Median survival (months)	5-year survival	Characteristic				
				MGMT gene	IDH1 gene	Surgery	Age (years)	KPS
I	126	58.5 (range: 40.7–76.3)	13%	methyl	mut			
				methyl	wt	GTR		≥ 90
II	483	21 (range: 18.6–23.3)	3.8%	methyl	wt	GTR		< 90
				methyl	wt	PR/biopsy		
				unmeth		GTR	≥ 50	
				unmeth			< 50	
III	202	14.3 (range: 12.4–16.1)	0	unmeth		PR/biopsy	≥ 50	

Abbreviations: methyl = methylated MGMT promoter gene; unmeth = unmethylated; mut = IDH1 mutated; wt = wildtype; KPS = Karnofsky score (KPS) (p. 1640) at presentation; GTR = gross total resection; PR/biopsy = partial resection or biopsy.

37.4.4 Multiple gliomas

Discussion of multiple gliomatous masses has to acknowledge the concept that a diffuse glioma is an infiltrative disease. Some terms are probably artificial. The term gliomatosis cerebri has been abandoned. Widespread brain invasion involving ≥ 3 lobes, frequently with bilateral involvement and often with posterior fossa extension, is considered a special pattern of spread within several diffuse glioma subtypes.

0.5–35% of GBM are multiple

- multifocal glioblastomas have contiguous pathways of spread by one of the mechanisms previously described (see above (p. 658))
- multiple primary gliomas: some of the following terms are inconsistently used interchangeably: “multicentric,” “multifocal,” and “multiple.” Reported range of occurrence is 2–20% of gliomas^{41,42} (lower end of range ≈ 2–4% is probably more accurate; the higher end of the range is probably accounted for by infiltrative extension⁴³ (p 3117))
 - commonly associated with neurofibromatosis and tuberous sclerosis
 - rarely associated with multiple sclerosis and progressive multifocal leukoencephalopathy
- meningeal gliomatosis: dissemination of glioma throughout the CSF, similar to carcinomatous meningitis (p. 920). Occurs in up to 20% of autopsies on patients with high-grade gliomas. May

present with cranial neuropathies, radiculopathies, myelopathy, dementia, and/or communicating hydrocephalus

► **Treatment considerations for multiple gliomas.** There is little data available. In a nonrandomized study of 25 patients with multifocal glioma,⁴⁴ the 16 patients who underwent debulking did better than the 9 who did not. However, there was significant selection bias in choosing patients suitable for craniotomy.

Biopsy is generally required/recommended to confirm the diagnosis.

Σ: Treatment options with multiple gliomatous masses

Once the diagnosis of multiple gliomatous masses has been ascertained, local therapies (e.g., surgery, interstitial radiation...) are impractical. Whole brain radiation and possibly chemotherapy are indicated. An exception would be to consider debulking a tumor that threatens herniation in a patient deteriorating from mass effect (other conditions such as anticipated length of survival and quality of life would need to be factored in to the decision to operate).

37.5 Treatment for adult-type diffuse gliomas

37.5.1 General information

Tumors included in this section

- astrocytoma, IDH-mutant, grades 2-4 (AIM grades 2-4)
- oligodendroglioma, IDH mutant and 1p/19q-codeleted, grades 2 & 3 (ODG grades 2 & 3)
- glioblastoma, IDH-wildtype (GBM) (all are grade 4)

Initial management

When confronted with an adult with imaging studies suggestive of a diffuse (infiltrating) glioma, see evaluation and initial management (p.626) as outlined for brain tumors in general.

Clinical trials

Eligible patients should be considered for available clinical trials.

37.5.2 Surgical intervention for adult-type diffuse gliomas

General information

Surgical intervention (cytoreduction or biopsy) is indicated for almost all tumors suspected of being diffuse gliomas. Surgery alone is never curative for these infiltrating lesions where there is no margin between brain and tumor. Aside from biopsy, the objective for cytoreduction is maximal safe resection with the goal of gross total resection (see below for definition) when feasible. When maximal safe resection is anticipated to be severely limited or is not desirable at the time, at a minimum a biopsy (open or needle) for diagnostic, prognostic, and treatment planning purposes should be done since clinical and radiographic data are not definitive.⁴⁵ NB: the onset of new postoperative motor or speech deficits reduces OS.⁴⁶

Relative contraindications to surgery

Careful consideration should be exercised in the decision to proceed with surgical debulking (cytoreduction surgery) with the following:

1. extensive dominant lobe high grade glioma
2. lesions with significant bilateral involvement. **Butterfly gliomas** refers to gliomas, typically IDH-wildtype glioblastoma (p.664) (grade 4), that cross into both cerebral hemispheres through the corpus callosum most commonly into frontal (68%) or parietal (8.9%) lobes.⁴⁷ Gliomas can also cross through the commissures. These tumors have a worse prognosis than their unilateral counterparts.⁴⁸ Debulking surgery for the purpose of reducing oncologic burden for these tumors is controversial, and is often not recommended due to the risk of neurologic deficit (in 32% in one

review, including supplementary motor area syndrome (in 5.1%), motor deficit (4.3%) and abulia (2.5%)⁴⁷) with dubious survival benefit (improved survival at 6 months, but no difference at 12 and 18 months⁴⁹). Limited surgery for large bulky portions of these tumors may be a consideration for reducing ICP and risk of herniation (which may occur if edema becomes a major issue during chemoradiation). Yet, there are advocates for maximal safe resection.^{47,48,49} Shunting may be offered for hydrocephalus

3. elderly patients
4. Karnofsky score < 70 (in general, with infiltrating tumors, the neurologic condition on steroids is as good as it is going to get, and surgery rarely improves this)
5. multicentric gliomas
6. patients in poor medical condition

Maximal safe resection – considerations

Maximal safe resection (MSR) is an abstraction. In many cases, maximizing the resection increases the risk of surgery (decreasing the safety), and maximizing safety can mean compromising the extent of resection. It is not possible to know the exact correct stopping point for MSR in every case.

Gross total resection – definition

With diffuse (i.e., infiltrating) tumors, there is no line of demarcation (e.g., a tumor capsule) that separates brain from tumor. Tumor cells can be found beyond the enhancing margins demonstrated on MRI.

Definition of gross total resection (GTR) for diffuse gliomas:

1. lesions that enhance on pre-op imaging (most high-grade (grades 3 & 4) gliomas): complete removal of enhancing portions of the tumor and any nonenhancing portion contained within, as *verified by post-operative contrast imaging* within 48 hours of surgery
2. for nonenhancing lesions (most astrocytoma, IDH-mutant grade 2): removal of high intensity lesion on T2 MRI or very low intensity on T1 MRI, if safe, as *verified on post-operative imaging*⁵⁰

Surgical goals

The objectives of surgery in diffuse astrocytomas are:

1. to obtain adequate tissue for histological and molecular study: can be accomplished with biopsy (open or needle [usually image-guided]) or cytoreductive surgery
2. cytoreduction—maximal safe resection, with a goal of GTR when feasible (see box)
 - a) to reduce tumor burden, which facilitates adjuvant therapy (oncologic goal)
 - b) to reduce mass effect which, when severe, can threaten herniation and can compress and inhibit adjacent brain tissue that is still capable of functioning
 - c) to reduce intracranial pressure (with large lesions)
 - d) may reduce peritumoral edema
 - e) may reduce seizure frequency⁵¹
 - f) for low grade gliomas, to prevent or delay malignant transformation⁵²

NB: Partial resection of a high-grade glioma carries significant risk of postoperative hemorrhage and/or edema (wounded glioma syndrome) with risk of herniation. Furthermore, the benefit of subtotal resection is dubious. Retrospective evidence suggested survival benefits in gross total resection but not with incomplete resection.⁵³ Therefore, surgical excision should be considered carefully when the goal of gross total removal is not feasible.

Greenberg IMHO

A few words about gross total resection

The true effect of extent of resection (EOR) of diffuse gliomas on outcome is not known⁵⁴ and will likely never be known. To make the case for improved outcome with greater EOR, published series compare cases with varying degrees of resection. This begs the question: why would there be any patients with subtotal resections (STR) in a series of surgical cases published by experts in the field? A likely possibility that has not been excluded is that the surgery had to be stopped short of complete resection due to, e.g., concerns about neurologic injury to the patient or due to difficulty identifying tumor—and it may be that these tumors are associated with worse outcome by virtue of their location or possibly even differences in biology, and not solely because of extent of resection. It is

acknowledged that there is a strong *association* between EOR and outcome, but the causality has not been established. And it is unlikely that a trial will ever be conducted in which patients are randomized to intentionally leave portions of tumor behind. We have to be satisfied with the proposition that maximizing the extent of resection *likely* increases OS and PFS. But keep in mind the following:

★ **Caveat:** the onset of a new postoperative motor or speech deficit reduces OS.⁴⁶

Resection – definitions and goals

► **Definitions.** **Extent of resection (EOR)** is reported as the percentage of the original tumor volume that was removed, and should be measured on imaging studies since the estimates made by surgeons based on intra-operative findings are grossly inaccurate. As a metric, EOR fails to communicate the residual tumor burden that requires monitoring and treatment.

A credible case has been made that **residual tumor volume (RTV)** may be a more critical parameter to measure.^{54,55}

► **Resection goals.** The extent of tumor removal and (in an inverse relationship) the residual tumor volume on post-op imaging studies⁵⁴ have a significant correlation with time to tumor progression and median survival.⁵⁶

Whenever feasible, gross total resection (p.669) of tumor with preservation of eloquent and critical structures should be the goal. NB: new onset post-operative deficit (motor or language) reduces OS.⁴⁶

Various thresholds have been quoted for minimal EOR to improve outcome. One study demonstrated that EOR ≥ 97% was *associated* with increased survival time.⁵⁷ Another study showed that the minimum required resection to improve OS and PFS in high grade glioma was 70% EOR, and 5 cm³ RTV.⁵⁵

Technical enhancements at surgery

The extent of resection is limited by the infiltrative nature of diffuse gliomas and their frequent location at or near eloquent brain. The safety of resection may be enhanced using various techniques described below.

Image-guidance (sometimes called stereotactic surgery) is extremely useful especially for deep tumors or in areas bordering on eloquent brain,⁵⁸ and for low-grade diffuse gliomas which may not be readily discernible at the time of surgery.

Intraoperative MRI may also increase the extent of resection, although this is of questionable benefit depending on how it is used.

Intraoperative (diagnostic) ultrasound may help identify tissue with altered echogenicity that differs from the normal brain.

Preoperative mapping functional MRI (fMRI) can localize sensitive areas of brain. Diffusion tensor imaging (DTI) can help localize critical fiber tracts. Mapping can demonstrate when each of these are displaced, that may facilitate preservation. These techniques are even more helpful when merged with image guidance.

Intraoperative mapping to localize critical structures. The motor strip and primary sensory cortex can be mapped with the patient awake or under anesthesia. Mapping the speech area requires awake craniotomy. A meta-analysis of 8091 patients showed that the use of intraoperative stimulation brain mapping achieved more gross total resection with less late severe neurological deficits, and is especially recommended if eloquent areas are involved.⁵⁹ For language function, the extent of resection may be safely increased using speech mapping.⁶⁰ Multicentric gliomas, previously considered unresectable, can also be debulked with the aid of awake intraoperative mapping.⁶¹ Despite this advance, the role of surgery remains limited for widespread gliomas or very deep-seated lesions.

Intraoperative dyes may help distinguish tumor from normal brain and are becoming more widely used primarily for high-grade gliomas. They include indocyanine green (ICG) and 5-aminolevulinic-acid (5-ALA) (Gleolan®).

5-ALA is metabolized into fluorescent porphyrins, which accumulate in malignant glioma cells. This property permits use of ultraviolet illumination during surgery as an adjunct to visualize the tumor. This has been proven with RCT where use of 5-ALA leads to more complete resection (65% vs. 36%, $p < 0.0001$), which translates into a higher 6-month progression free survival (41% vs. 21.1%, $p = 0.0003$) but no effect on OS.⁶²

37.5.3 Treatment of adult diffuse gliomas grade 2 (dLGG)

Tumors included in this section

These patients may be grouped together as “diffuse low grade gliomas” (dLGG)

- astrocytoma, IDH-mutant, grade 2 (AIM grade 2)
- oligodendroglioma, IDH-mutant, 1p/19q codeleted grade 2 (ODG grade 2)

Treatment options

✖ Caution: if any option is being considered that does not include surgical excision, a biopsy is essential to confirm the diagnosis as 43% of tumors that were thought to be dLGG on imaging turned out to have high grade gliomas.⁴⁾

1. surveillance / “watchful waiting” (p.627): no tumor specific treatment. Treat symptomatically (i.e., medical control of seizures and/or perilesional edema), and follow serial neurologic exams and imaging studies, with intervention reserved for progression of symptoms or radiographic progression. This has historically been very controversial and subject to biases. A Norwegian group has convincingly shown significantly lower median overall survival in a center favoring the watchful waiting approach (5.8 years) compared to 14.4 years at a center favoring early resection^{63,64}
2. radiation: XRT is insufficient as monotherapy. As a post-op adjunct, late effects of radiation on cognition need to be considered in long-term survivors with dLGG
3. chemotherapy: the optimal agent and timing of chemotherapy is not well defined. Temozolamide (TMZ) or PCV improves PFS and OS when added to XRT. TMZ is less toxic than PCV, but is inadequate as monotherapy (at least for ODG⁶⁵) and may induce hypermutation (resistant tumor cells)⁶⁶
4. surgery: see “surgical intervention” above
5. combinations of radiation and chemotherapy, with or without surgery

Management recommendations

There is no Class I data. All studies are either retrospective surveys or prospective cohort series where patients were a selected subset of patients with dLGG. With that understanding, early maximal safe surgical resection is currently the recommended initial step for most grade 2 diffuse gliomas in adults^{a, 67,68}

Following maximal safe resection in patients with KPS $\geq$ 60:

- **low-risk patients**—characteristics: age < 40 years and gross total resection.
Recommendation: routine imaging surveillance (e.g., every 3–6 months)
- **high-risk patients**—characteristics: age > 40 or subtotal resection. Tumor size and neurologic deficit are sometimes taken into consideration.
Recommendation: options
 - XRT followed by PCV (p. 628)^{b, 69}
 - XRT + temozolamide (concurrent or after XRT)⁷⁰
 - a period of close monitoring may be considered in select cases

Notes:

^a more aggressive excision is associated with better outcome^{71,72,73} and longer latency to malignant transformation (PFS—progression-free survival).⁷² Even with recurrent diffuse astrocytomas, surgical resection is associated with a survival benefit.

^b RTOG 9802 did not specify how long after XRT the PCV was to be administered.

Surgery for grade 2 diffuse gliomas

See section 37.5.2.

A case for supramaximal resection for dLGG has been made⁷⁴ in which intra-operative mapping is used to excise dLGG plus margins, and in carefully selected patients may result in increased tumor excision. The definition of supramaximal is not precise, and further study is needed to prove superior outcomes.⁷⁵

Adjuvant therapy for grade 2 diffuse gliomas

See discussion of low risk vs. high risk patients following surgery (p.671).

► **Radiation therapy (XRT).** Early post-op radiotherapy has been debated as an adjunct. It has been shown to prolong median progression-free survival from 3.4 to 5.3 years but does not affect overall survival.⁷⁶ Quality of life and cognition were not assessed. Typically given as 54 Gy in fractions of 1.8–2 Gy over 5–6 weeks. In patients with radical tumor resection, early RT did not prolong PFS and is recommended to be deferred until progression. Following incomplete resection, early RT significantly prolongs PFS and disease-specific survival.⁷⁷ Two prospective trials found no difference in OS or PFS between different XRT doses (EORTC trial⁷⁸: 45 Gy in 5 weeks vs. 59.4 Gy in 6.6 weeks; Inter-group study⁷⁹: 50.4 vs. 64.8 Gy). Side effects from whole brain XRT (WBXRT) include leukoencephalopathy and cognitive impairment (see Radiation injury and necrosis (p. 1899)). The frequency of side effects may⁷⁹ or may not⁸⁰ be higher at higher XRT doses.

► **Chemotherapy.** Usually reserved for high-risk patients or low-risk patients with tumor progression. RTOG 9802 showed benefit of PCV (p.628) in PFS and OS in IDH-mutant, grade 2 astrocytomas,⁶⁹ but not for IDH-wildtype.

Temozolomide (Temodar®) initially appeared to be effective in progressive AIM grade 2 tumors (off label use).⁸¹

37.5.4 Treatment of diffuse gliomas, grades 3 & 4

Tumors included in this section

- astrocytoma, IDH mutant grades 3 & 4 (AIM grades 3 & 4)
- oligodendroglioma, IDH-mutant, 1p/19q codeleted grade 3 (ODG grade 3)
- glioblastoma, IDH-wildtype (GBM)

Surgery for newly diagnosed AIM grade 3 & 4 and glioblastoma, IDH-wildtype

See section 37.5.2.

Maximal safe cytoreductive surgery followed by external beam radiation and concurrent temozolomide has become the standard against which other treatments are compared.⁸²

Adjuvant therapy for diffuse glioma grades 3 & 4

Stupp regimen

General information

The standard of care for newly diagnosed grade 3 & 4 gliomas is cytoreductive surgery followed by the “Stupp regimen”⁸² which consists of concomitant and adjuvant radiotherapy and chemotherapy as detailed below. Chemoradiotherapy is started within six weeks after histological diagnosis. Alkylating agents (the current prototypical drug is temozolomide) are more effective in tumors that are deficient in the DNA repair enzyme, MGMT (see below for details).

Radiotherapy (XRT) in the Stupp regimen

XRT in the Stupp regimen consists of fractionated *focal* radiation at a dose of 2 Gy per fraction once daily five days per week over a period of six weeks, for a total dose of 60 Gy, with a 2–3 cm margin of clinical target volume. This is compared to the previous conventional XRT regiment for malignant gliomas of 50–60 Gy (usually 50 Gy to a margin 2–3 cm greater than the enhanced volume on MRI with a boost to the enhancing volume to bring the total to 60 Gy⁸³).

Chemotherapy in the Stupp regimen

Concomitant chemotherapy employs temozolomide (TMZ), an oral alkylating agent that is a prodrug which undergoes rapid non-enzymatic conversion at physiologic pH to the active metabolite monomethyl triazenoimidazole carboxamide (MTIC). The cytotoxic effect of MTIC is associated with alkylation (primarily methylation) of DNA at various sites, including the O⁶ and N⁷ positions on guanine.

Temozolomide dosing

- initial treatment: 75 mg/m²/day, 7 days per week till the end of XRT
- 4 weeks later, start 6 cycles of *adjuvant* chemotherapy with each cycle consisting of 5 days of TMZ repeated every 28 days
 - first cycle dosing: 150 mg/m²/day
 - subsequent cycles dosing: 200 mg/m²/day

Side effects: Temozolomide may cause myelosuppression. It should not be given unless the neutrophil count is $\geq 1.5 \times 10^9$ /L and platelet count $\geq 100 \times 10^9$ /L. For all patients with newly diagnosed

high-grade gliomas for concomitant TMZ and radiotherapy, prophylaxis against *Pneumocystis jirovecii* (formerly *P. carinii*) pneumonia is required for the initial 6-week regimen.

Adjuvant monotherapy with TMZ is also associated with malignant transformation in low grade diffuse gliomas on multivariate analysis.⁸⁴

► **MGMT transcriptional silencing (epigenetics).** Alkylating chemotherapeutic agents (p.627) (e.g., nitrosourea or temozolomide) damage DNA by attaching an alkyl group (the smallest of which is a methyl group) to guanine bases forming O⁶-methylguanine which results in cell death. **MGMT** (O⁶-methylguanine-DNA methyltransferase) is a DNA repair enzyme that specifically removes this cytotoxic alkyl group and restores guanine residues to their native state.^{85,86} Compared to normal cells, ≈ 75% of grade 4 AIGs inherently have reduced MGMT activity (while only 36% of GBM, IDH-wildtype have reduced activity), most likely as a result of epigenetic transcriptional gene silencing through hypermethylation of the CpG islands (promoter areas) of the genome that codes for MGMT, located on chromosome 10q26.^{87,88}

MGMT activity is an independent prognosticator for response to chemotherapy with alkylating agents. Longer survival is observed with higher grade AIGs that have reduced MGMT activity³⁷ (MGMT levels < 30 fmol/mg protein respond better to alkylating agents⁸⁹).

The box below summarizes the implications of gene methylation and MGMT activity.

Notes: testing for promoter gene methylation is done by PCR (polymerase chain reaction) which is may need to be sent out to a lab which typically takes 1–2 weeks for results. Cost is in the range of \$1000 U.S. Measuring MGMT is faster, but does not always produce reliable results, and requires frozen tumor specimens which may not always be available. Although methylation of > 10% of the promoter gene may be reported as “methylated,” it is only when methylation is > 30% that the median OS with Temodar increases significantly to 25.2 months compared to 15.2 months for all others.⁹⁰ Promoter gene methylation may not be as accurate a predictor of outcome as the mRNA expression product.⁸⁹ Analysis of recurrent astrocytomas shows some tumors with methylated MGMT promoter genes may convert to unmethylated following treatment⁹¹ (possibly as a result of Temodar toxicity giving selective advantage to resistant cells).

Σ: Transcriptional silencing of MGMT synthesis

Finding	Comment
Hypermethylation (sometimes reported simply as “methylation”) of the MGMT promoter gene	This naturally occurs on chromosome 10q26 of the DNA in 75% of grade 4 AIGs. It usually leads to reduced levels of MGMT activity (see the next row in this table) which becomes very significant at > 30% methylation
Reduced (or undetectable) MGMT activity	MGMT is an enzyme that repairs the damage to DNA caused by alkylating chemotherapy drugs such as temozolomide (Temodar®). Tumors with reduced MGMT levels are more susceptible to cytotoxic effects of temozolomide than nontumor cells which have normal MGMT levels, and are better able to repair the DNA damage. This results in longer patient survival

Outcomes with the Stupp regimen

The median survival was 14.6 months with the Stupp regimen compared to 12.1 months with radiotherapy alone, with median survival benefit of 2.1 months. The five-year survival rate was 9.8% for the Stupp regimen in contrast to 1.9%.⁹² Regardless of extent of resection and MGMT status, patients receiving the Stupp regimen had longer median survival. Patients with MGMT promoter gene methylation (75% of AIG grade 4 tumors, compare to 36% of GBM IDH-wildtype) had median survival time of 23.4 months compared to 12.6 months in the non-methylated group (see below). In the MGMT unmethylated group, the Stupp regimen only improved median survival from 11.8 months to 12.6 months. Some practitioners extend the adjuvant chemotherapy after the standard six-month regimen until tumor progression is observed; in one study this extended the median survival time from 16.5 months to 24.6 months.⁹³

Other adjuvant treatment options for high-grade AIG and GBM

Gliadel® wafer: carmustine (BCNU) 7.7 mg in a 200 mg prolifeprosan 20 hydrophobic polymer carrier (wafer) that can be applied to resection cavity after tumor excision. The wafers are degraded by

hydrolysis and the drug is released over 2–3 wks. This exposes the tumor to 113 times the concentration of BCNU compared to IV administration. Following tumor removal, up to 8 of the 1.4 cm × 1 mm (dime-sized) wafers are applied to the tumor resection bed at the time of surgery.

It increases median survival to 13.8 months compared with 11.6 months in placebo groups for newly diagnosed malignant gliomas.⁹⁴ It showed no survival benefit for recurrent disease.⁹⁵ Side effects: seizures, cerebral edema, healing complications, intracranial infection.

MRI-guided laser interstitial thermal therapy (LITT): This may be a consideration for patients needing treatment when surgery is not a consideration and when maximal XRT has already been given, oftentimes in the setting of radiation necrosis.⁹⁶

Focused ultrasound (FUS) may become a non-invasive adjuvant therapy option for some deep situated tumors.⁹⁷ Transcranial FUS is not a viable option for tumors near the surface due to the inability to focus the ultrasound beams close to the convexity of the skull.

37.5.5 Response to treatment

Pseudoprogression

Since XRT + temozolomide became part of the standard of care for the treatment of many high grade diffuse gliomas, there has been an increase in progressive contrast-enhancing areas on MRI that mimic tumor progression, typically seen ≤ 3 months after treatment. This phenomenon, called pseudoprogression, occurs in up to 28–60% of patients after XRT + temozolomide treatment. Histologically it resembles radiation necrosis and is believed to be associated with tumor kill by radiation. Increased tumor kill with chemotherapy results in more pseudoprogression; therefore it is more common with temozolomide treatment for tumors with methylated MGMT promoter genes (as in astrocytoma, IDH-mutant) (91%) vs. unmethylated (as in glioblastoma, IDH-wildtype) (41%).⁹⁸

Diagnosis: There is no definitive diagnostic test to distinguish pseudoprogression from true progression or from radiation necrosis. Imaging methods that have been tried: MR perfusion, DWI with higher apparent diffusion co-efficient (ADC), MR spectroscopy, and PET have not achieved high sensitivity and specificity. Monitoring with serial MRIs and clinical exams seems to be an effective strategy. Rule of thumb: MRI changes suggestive of GBM recurrence within the first 6 months after treatment with XRT + temozolomide should be assumed to be pseudoprogression as long as it is within the radiation field.

Management: The MRI findings usually improve without treatment.⁹⁹ If significantly symptomatic, options to ameliorate symptoms include:

1. corticosteroids
2. bevacizumab (Avastin®): often used to treat radiation necrosis¹⁰⁰
3. laser interstitial thermal therapy (LITT): as a salvage technique for *radiation necrosis*, LITT facilitated reduction of steroid dependency with stabilization of KPS, QOL and neurocognitive function¹⁰¹
4. ✖ hyperbaric oxygen: thus far, only case reports and small non-randomized studies support its use

Pseudoresponse

Gadolinium contrast agents used with MRI produce enhancement in areas where the blood-brain barrier (BBB) is disrupted. Just as pseudoprogression may represent increased enhancement without actual tumor regrowth (see above), agents that stabilize the BBB and thereby reduce enhancement may cause underestimation of tumor load or they may mask enhancement-free progression. This is called “pseudoresponse”.¹⁰² It can occur while under treatment with angiogenesis inhibitors (including agents targeting VEGF and VEGF receptors such as bevacizumab). Enhancement may also be reduced by corticosteroids (however, the effect may be variable).

RANO (Response Assessment in Neuro-Oncology) criteria

The MacDonald criteria¹⁰³ were widely used to assess response to treatment in high-grade gliomas. However, it relied on contrast enhancement on CT. Enhancement may be affected by many factors (steroids, inflammation, angiogenesis inhibitors, chemotherapy wafers, gene and viral therapies, brachytherapy...), and can therefore produce pseudoprogression or pseudoresponse (see above). As a result, the MacDonald criteria have been superseded by the RANO (Response Assessment in Neuro-Oncology)¹⁰⁴ criteria (► Table 37.6).

Table 37.6 Response Assessment in Neuro-Oncology (RANO) criteria for high-grade gliomas^a

Complete response

Requires all of the following:

1. complete disappearance of all disease (measurable and nonmeasurable) for ≥ 4 weeks
2. no new lesions
3. stable or improved nonenhancing lesions (on T2/FLAIR)
4. patient off corticosteroids (except for physiologic replacement)
5. clinically stable or improved

Patients with only nonmeasurable disease cannot have complete response (stable disease is best possible response)

Partial response

Requires all of the following:

1. (compared to baseline) $\geq 50\%$ decrease of the sum of products of perpendicular diameters of all measurable enhancing lesions for ≥ 4 weeks
2. no progression of nonmeasurable disease
3. no new lesions
4. stable or improved nonenhancing lesions (on T2/FLAIR) on same or lower doses of corticosteroids compared to doses at baseline scan
5. corticosteroid dose at the time of the scan is no greater than dose at time of baseline scan
6. clinically stable or improved

Patients with only nonmeasurable disease cannot have partial response (stable disease is best possible response)

Stable disease

Requires all of the following:

1. does not qualify for complete response, partial response or progression
2. stable nonenhancing lesions (T2/FLAIR) on same or lower doses of corticosteroids compared to doses at baseline scan
3. if steroid dose was increased for new symptoms and signs without confirming progression on imaging, and subsequent imaging shows that the need for increased steroids was due to disease progression, the last scan considered to show stable disease will be the one obtained when the steroid dose was the same as at the baseline scan

Progression

Defined by any of the following:

1. $\geq 25\%$ increase in sum of products of perpendicular diameters enhancing lesions compared with the smallest lesion measurement at either baseline (if no decrease) or best response, on stable or increasing doses of corticosteroids
2. significant increase in nonenhancing lesions (on T2/FLAIR) on equal or increased doses of steroids compared to dose at baseline or best response after initiation of therapy not caused by comorbid events (e.g., XRT, demyelination, ischemic injury, infection, seizures, post-op changes, or other treatment effects...)
3. any new lesion
4. clear clinical deterioration not attributable to causes other than tumor (e.g., seizures, adverse effects of medication, stroke, infection...) or changes in steroid dose
5. failure to return for evaluation due to death or deterioration
6. or clear progression of nonmeasurable disease

^asee reference¹⁰⁴ for definitions (e.g., measurable and nonmeasurable disease...) and for measurement technique. All lesions (measurable and nonmeasurable) must be assessed using the same technique as baseline imaging.

References

- [1] Lopez GY, Van Ziffle J, Onodera C, et al. The genetic landscape of gliomas arising after therapeutic radiation. *Acta Neuropathol.* 2019; 137:139–150
- [2] National Cancer Institute. Cell Phones and Cancer Risk. 2018. <https://www.cancer.gov/about-cancer/causes-prevention/risk/radiation/cell-phones-fact-sheet>
- [3] World Health Organization. Electromagnetic fields and public health: mobile phones. 2011. <http://www.who.int/mediacentre/factsheets/fs193/en/>
- [4] Kondziolka D, Lunsford LD, Martinez AJ. Unreliability of contemporary neurodiagnostic imaging in evaluating suspected adult supratentorial (low-grade) astrocytoma. *Journal of Neurosurgery.* 1993; 79:533–536
- [5] Chamberlain MC, Murovic J, Levin VA. Absence of Contrast Enhancement on CT Brain Scans of Patients with Supratentorial Malignant Gliomas. *Neurology.* 1988; 38:1371–1373
- [6] Greene GM, Hitchon PW, Schelper RL, et al. Diagnostic Yield in CT-Guided Stereotactic Biopsy of Gliomas. *Journal of Neurosurgery.* 1989; 71:494–497
- [7] Scherer HJ. The Forms of Growth in Gliomas and their Practical Significance. *Brain.* 1940; 63:1–35
- [8] Choucair AK, Levin VA, Gutin PH, et al. Development of Multiple Lesions During Radiation Therapy and Chemotherapy. *J Neurosurg.* 1986; 65:654–658
- [9] Erlich SS, Davis RL. Spinal Subarachnoid Metastasis from Primary Intracranial Glioblastoma Multiforme. *Cancer.* 1978; 42:2854–2864
- [10] Brat DJ, Aldape K, Bridge JA, et al. Molecular Biomarker Testing for the Diagnosis of Diffuse Gliomas. *Arch Pathol Lab Med.* 2022; 146:547–574
- [11] Bosman FT, Jaffe ES, Lakhani SR, et al. World Health Organization Classification of Tumors series, vol. 6. In: WHO Classification of Tumors Editorial Board (ed.) World Health Organization

- classification of tumors of the central nervous system. 5th ed. Lyon, France: International Agency for Research on Cancer; 2021
- [12] Brat DJ, Aldape K, Colman H, et al. cIMPACT-NOW update 5: recommended grading criteria and terminologies for IDH-mutant astrocytomas. *Acta Neuropathol.* 2020; 139:603–608
 - [13] Carstam L, Corell A, Smits A, et al. WHO Grade Loses Its Prognostic Value in Molecularly Defined Diffuse Lower-Grade Gliomas. *Front Oncol.* 2021; 11. DOI: 10.3389/fonc.2021.803975
 - [14] Kleihues P, Louis DN, Scheithauer BW, et al. The WHO classification of tumors of the nervous system. *J Neuropathol Exp Neurol.* 2002; 61:215–25; discussion 226–9
 - [15] Patel SH, Poisson LM, Brat DJ, et al. T2-FLAIR Mismatch, an Imaging Biomarker for IDH and 1p/19q Status in Lower-grade Gliomas: A TCGA/TICIA Project. *Clin Cancer Res.* 2017; 23:6078–6085
 - [16] Reuss DE, Mamatjan Y, Schrimpf D, et al. IDH mutant diffuse and anaplastic astrocytomas have similar age at presentation and little difference in survival: a grading problem for WHO. *Acta Neuropathol.* 2015; 129:867–873
 - [17] Pignatti F, van den Bent M, Curran D, et al. Prognostic factors for survival in adult patients with cerebral low-grade glioma. *Journal of Clinical Oncology.* 2002; 20:2076–2084
 - [18] Shafqat S, Hedley-Whyte ET, Henson JW. Age-Dependent Rate of Anaplastic Transformation in Low-Grade Astrocytoma. *Neurology.* 1999; 52:867–869
 - [19] Zetterling M, Berhane I, Alafuzoff I, et al. Prognostic markers for survival in patients with oligodendroglial tumors: a single-institution review of 214 cases. *PLoS One.* 2017; 12. DOI: 10.1371/journal.pone.0188419
 - [20] Mork SJ, Lindegaard KF, Halvorsen TB, et al. Oligodendroglioma: Incidence and Biological Behavior in a Defined Population. *J Neurosurg.* 1985; 63:881–889
 - [21] Khalid I, Carone M, Dumrongpisutikul N, et al. Imaging characteristics of oligodendroglomas that predict grade. *AJNR Am J Neuroradiol.* 2012; 33:852–857
 - [22] Roberts M, German W. A Long Term Study of Patients with Oligodendrogliomas. *J Neurosurg.* 1966; 24:697–700
 - [23] Daumas-Duport C, Varlet P, Tucker M-L, et al. Oligodendrogliomas: Part I - Patterns of Growth, Histological Diagnosis, Clinical and Imaging Correlations: A Study of 153 Cases. *J Neurooncol.* 1997; 34:37–59
 - [24] Coons SW, Johnson PC, Pearl DK, et al. The Prognostic Significance of Ki-67 Labeling Indices for Oligodendrogliomas. *Neurosurgery.* 1997; 41:878–885
 - [25] Louis DN, Ohgaki H, Wiestler OD, et al. WHO classification of tumors of the central nervous system. Lyon, France 2016
 - [26] Rutka JT, Murakami M, Dirks PB, et al. Role of Glial Filaments in Cells and Tumors of Glial Origin: A Review. *Journal of Neurosurgery.* 1997; 87:420–430
 - [27] Kros JM, Schouten WCD, Janssen PJA, et al. Proliferation of Gemistocytic Cells and Glial Fibrillary Acidic Protein (GFAP)-Positive Oligodendroglial Cells in Gliomas: A MIB-1/GFAP Double Labeling Study. *Acta Neuropathol (Berl).* 1996; 91:99–103
 - [28] Elefante A, Peca C, Del Basso De Caro M L, et al. Symptomatic spinal cord metastasis from cerebral oligodendroglioma. *Neurol Sci.* 2012; 33:609–613
 - [29] Cancer Genome Atlas Research Network, Brat DJ, Verhaak RG, et al. Comprehensive, Integrative Genomic Analysis of Diffuse Lower-Grade Gliomas. *N Engl J Med.* 2015; 372:2481–2498
 - [30] Cairncross JG, Ueki K, Zlatescu MC, et al. Specific Genetic Predictors of Response to Radiotherapy and Survival in Patients with Anaplastic Oligodendrogliomas. *J Natl Cancer Inst.* 1998; 90:1473–1479
 - [31] Smith JS, Perry A, Borell TJ, et al. Alterations of chromosome arms 1p and 19q as predictors of survival in oligodendrogliomas, astrocytomas, and mixed oligoastrocytomas. *J Clin Oncol.* 2000; 18:636–645
 - [32] Louis DN, Ohgaki H, Wiestler OD, et al. WHO classification of tumors of the central nervous system. Lyon 2007
 - [33] Lee JH, Lee JE, Kahng JY, et al. Human glioblastoma arises from subventricular zone cells with low-level driver mutations. *Nature.* 2018; 560:243–247
 - [34] Ostrom QT, Cioffi G, Waite K, et al. CBTRUS Statistical Report: Primary Brain and Other Central Nervous System Tumors Diagnosed in the United States in 2014–2018. *Neuro Oncol.* 2021; 23:iii1–iii105
 - [35] Kopelson G, Linggood R. Infratentorial Glioblastoma: The Role of Neuraix Irradiation. *International Journal of Radiation Oncology and Biological Physics.* 1982; 8:999–1003
 - [36] Yang P, Zhang W, Wang Y, et al. IDH mutation and MGMT promoter methylation in glioblastoma: results of a prospective registry. *Oncotarget.* 2015; 6:40896–40906
 - [37] Hegi ME, Diserens AC, Gorlia T, et al. MGMT gene silencing and benefit from temozolomide in glioblastoma. *N Engl J Med.* 2005; 352:997–1003
 - [38] Wee CW, Kim E, Kim N, et al. Novel recursive partitioning analysis classification for newly diagnosed glioblastoma: A multi-institutional study highlighting the MGMT promoter methylation and IDH1 gene mutation status. *Radiother Oncol.* 2017; 123:106–111
 - [39] Wee CW, Kim IH, Park CK, et al. Validation of a novel molecular RPA classification in glioblastoma (GBM-molRPA) treated with chemoradiation: A multi-institutional collaborative study. *Radiother Oncol.* 2018; 129:347–351
 - [40] Gorlia T, van den Bent MJ, Hegi ME, et al. Nomograms for predicting survival of patients with newly diagnosed glioblastoma: prognostic factor analysis of EORTC and NCIC trial 26981-22981/CE3. *Lancet Oncol.* 2008; 9:29–38
 - [41] Barnard RO, Geddes JF. The Incidence of Multifocal Cerebral Gliomas: A Histological Study of Large Hemispheric Sections. *Cancer.* 1987; 60:1519–1531
 - [42] van Tassel P, Lee Y-Y, Bruner JM. Synchronous and Metachronous Malignant Gliomas: CT Findings. *AJNR.* 1988; 9:725–732
 - [43] Harsh GR, Wilson CB, Youmans JR. Neuroepithelial Tumors of the Adult Brain. In: *Neurological Surgery.* 3rd ed. Philadelphia: W. B. Saunders; 1990:3040–3136
 - [44] Salvati M, Caroli E, Orlando ER, et al. Multicentric glioma: our experience in 25 patients and critical review of the literature. *Neurosurgical Review.* 2003; 26:275–279
 - [45] Berger MS, Apuzzo MLJ. Role of Surgery in Diagnosis and Management. In: *Benign Cerebral Glioma.* Park Ridge, Illinois: American Association of Neurological Surgeons; 1995:293–307
 - [46] McGirt MJ, Mukherjee D, Chaichana KL, et al. Association of surgically acquired motor and language deficits on overall survival after resection of glioblastoma multiforme. *Neurosurgery.* 2009; 65:463–9; discussion 469–70
 - [47] Palmisciano P, Ferini G, Watanabe G, et al. Gliomas Infiltrating the Corpus Callosum: A Systematic Review of the Literature. *Cancers (Basel).* 2022; 14. DOI: 10.3390/cancers14102507
 - [48] Chaichana KL, Jusue-Torres I, Lemos AM, et al. The butterfly effect on glioblastoma: is volumetric extent of resection more effective than biopsy for these tumors? *J Neurooncol.* 2014; 120:625–634
 - [49] Chojak R, Kozba-Gosztyla M, Słycharz K, et al. Impact of surgical resection of butterfly glioblastoma on survival: a meta-analysis based on comparative studies. *Sci Rep.* 2021; 11. DOI: 10.1038/s41598-021-93441-z
 - [50] Kawaguchi T, Sonoda Y, Shibahara I, et al. Impact of gross total resection in patients with WHO

- grade III glioma harboring the IDH 1/2 mutation without the 1p/19q co-deletion. *J Neurooncol.* 2016; 129:505–514
- [51] Chang EF, Potts MB, Keles GE, et al. Seizure characteristics and control following resection in 332 patients with low-grade gliomas. *J Neurosurg.* 2008; 108:227–235
 - [52] van den Bent MJ, Snijders TJ, Bromberg JE. Current treatment of low grade gliomas. *Magazine of European Medical Oncology.* 2012; 5:223–227
 - [53] Kreth FW, Thon N, Simon M, et al. Gross total but not incomplete resection of glioblastoma prolongs survival in the era of radiochemotherapy. *Ann Oncol.* 2013; 24:3117–3123
 - [54] Grabowski MM, Recinos PF, Nowacki AS, et al. Residual tumor volume versus extent of resection: predictors of survival after surgery for glioblastoma. *J Neurosurg.* 2014; 121:1115–1123
 - [55] Chaichana KL, Jusue-Torres I, Navarro-Ramirez R, et al. Establishing percent resection and residual volume thresholds affecting survival and recurrence for patients with newly diagnosed intracranial glioblastoma. *Neuro Oncol.* 2014; 16:113–122
 - [56] Keles GE, Anderson B, Berger MS. The Effect of Extent of Resection on Time to Tumor Progression and Survival in Patients with Glioblastoma Multiforme of the Cerebral Hemisphere. *Surgical Neurology.* 1999; 52:371–379
 - [57] Lacroix M, Abi-Said D, Fourney DR, et al. A multivariate analysis of 416 patients with glioblastoma multiforme: prognosis, extent of resection, and survival. *J Neurosurg.* 2001; 95:190–198
 - [58] Kelly PJ, Apuzzo MJ. Role of Stereotaxis in the Management of Low-Grade Intracranial Gliomas. In: *Benign Cerebral Glioma*. Park Ridge, Illinois: American Association of Neurological Surgeons; 1995:275–292
 - [59] De Witt Hamer PC, Robles SG, Zwinderman AH, et al. Impact of intraoperative stimulation brain mapping on glioma surgery outcome: a meta-analysis. *J Clin Oncol.* 2012; 30:2559–2565
 - [60] De Benedictis A, Moritz-Gasser S, Duffau H. Awake mapping optimizes the extent of resection for low-grade gliomas in eloquent areas. *Neurosurgery.* 2010; 66:1074–84; discussion 1084
 - [61] Terakawa Y, Yordanova YN, Tate MC, et al. Surgical management of multicentric diffuse low-grade gliomas: functional and oncological outcomes: clinical article. *J Neurosurg.* 2013; 118:1169–1175
 - [62] Stummer W, Pichlmeier U, Meinel T, et al. Fluorescence-guided surgery with 5-aminolevulinic acid for resection of malignant glioma: a randomised controlled multicentre phase III trial. *Lancet Oncol.* 2006; 7:392–401
 - [63] Jakola AS, Myrland KS, Kloster R, et al. Comparison of a strategy favoring early surgical resection vs a strategy favoring watchful waiting in low-grade gliomas. *JAMA.* 2012; 308:1881–1888
 - [64] Jakola AS, Skjalsvik AJ, Myrland KS, et al. Surgical resection versus watchful waiting in low-grade gliomas. *Ann Oncol.* 2017; 28:1942–1948
 - [65] Jaekle KA, Ballman KV, van den Bent M, et al. CODEL: phase III study of RT, RT + TMZ, or TMZ for newly diagnosed 1p/19q codeleted oligodendroglioma. Analysis from the initial study design. *Neuro Oncol.* 2021; 23:457–467
 - [66] Yu Y, Villanueva-Meyer J, Grimmer MR, et al. Temozolomide-induced hypermethylation is associated with distant recurrence and reduced survival after high-grade transformation of low-grade IDH-mutant gliomas. *Neuro Oncol.* 2021; 23:1872–1884
 - [67] Tandon A, Schiff D. Therapeutic decision making in patients with newly diagnosed low grade glioma. *Curr Treat Options Oncol.* 2014; 15:529–538
 - [68] Oberheim Bush NA, Chang S. Treatment Strategies for Low-Grade Glioma in Adults. *J Oncol Pract.* 2016; 12:1235–1241
 - [69] Bell EH, Zhang P, Shaw EG, et al. Comprehensive Genomic Analysis in NRG Oncology/RTOG 9802: A Phase III Trial of Radiation Versus Radiation Plus Procarbazine, Lomustine (CCNU), and Vincristine in High-Risk Low-Grade Glioma. *J Clin Oncol.* 2020; 38:3407–3417
 - [70] Baumer BG, Hegi ME, van den Bent MJ, et al. Temozolomide chemotherapy versus radiotherapy in high-risk low-grade glioma (EORTC 22033-26033): a randomised, open-label, phase 3 intergroup study. *Lancet Oncol.* 2016; 17:1521–1532
 - [71] McGirt MJ, Goldstein IM, Chaichana KL, et al. Extent of surgical resection of malignant astrocytomas of the spinal cord: outcome analysis of 35 patients. *Neurosurgery.* 2008; 63:55–60; discussion 60–1
 - [72] Capelle L, Fontaine D, Mandonnet E, et al. Spontaneous and therapeutic prognostic factors in adult hemispheric World Health Organization Grade II gliomas: a series of 1097 cases: clinical article. *J Neurosurg.* 2013; 118:1157–1168
 - [73] Shaw EG, Berkey B, Coons SW, et al. Recurrence following neurosurgeon-determined gross-total resection of adult supratentorial low-grade glioma: results of a prospective clinical trial. *J Neurosurg.* 2008; 109:835–841
 - [74] Duffau H. A new philosophy in surgery for diffuse low-grade glioma (DLGG): oncological and functional outcomes. *Neurochirurgie.* 2013; 59:2–8
 - [75] de Leeuw CN, Vogelbaum MA. Supratotal resection in glioma: a systematic review. *Neuro Oncol.* 2019; 21:179–188
 - [76] van den Bent MJ, Afra D, de Witte O, et al. Long-term efficacy of early versus delayed radiotherapy for low-grade astrocytoma and oligodendroglioma in adults: the EORTC 22845 randomised trial. *Lancet.* 2005; 366:985–990
 - [77] Hanzely Z, Polgar C, Fodor J, et al. Role of early radiotherapy in the treatment of supratentorial WHO Grade II astrocytomas: long-term results of 97 patients. *J Neurooncol.* 2003; 63:305–312
 - [78] Karim ABMF, Maat B, Hatlevoll R, et al. A randomized trial on dose-response in radiation therapy of low-grade cerebral glioma: European Organization for Research and Treatment of Cancer (EORTC) Study 22844. *International Journal of Radiation Oncology and Biological Physics.* 1996; 36:549–556
 - [79] Shaw E, Arusell R, Scheithauer B, et al. Prospective randomized trial of low- versus high-dose radiation therapy in adults with supratentorial low-grade glioma: initial report of a North Central Cancer Treatment Group/Radiation Therapy Oncology Group/Eastern Cooperative Oncology Group study. *Journal of Clinical Oncology.* 2002; 20:2267–2276
 - [80] Laack NN, Brown PD, Ivinik RJ, et al. Cognitive function after radiotherapy for supratentorial low-grade glioma: a North Central Cancer Treatment Group prospective study. *International Journal of Radiation Oncology, Biology, Physics.* 2005; 63:1175–1183
 - [81] Quinn JA, Reardon DA, Friedman AH, et al. Phase II trial of temozolomide in patients with progressive low-grade glioma. *J Clin Oncol.* 2003; 21:646–651
 - [82] Stupp R, Mason WP, van den Bent MJ, et al. Radiotherapy plus concomitant and adjuvant temozolomide for glioblastoma. *New England Journal of Medicine.* 2005; 352:987–996
 - [83] Narayan P, Olson JJ. Management of anaplastic astrocytoma. *Contemporary Neurosurgery.* 2001; 23:1–6
 - [84] Johnson BE, Mazar T, Hong C, et al. Mutational analysis reveals the origin and therapy-driven evolution of recurrent glioma. *Science.* 2014; 343:189–193
 - [85] Margison GP, Kleihues P. Chemical carcinogenesis in the nervous system. Preferential accumulation of O6-methylguanine in rat brain deoxyribonucleic acid during repetitive administration of N-methyl-N-nitrosourea. *Biochem J.* 1975; 148:521–525
 - [86] Goth R, Rajewsky MF. Persistence of O6-ethylguanine in rat-brain DNA: correlation with nervous system-specific carcinogenesis by ethylnitrosourea. *Proc Natl Acad Sci U S A.* 1974; 71:639–643
 - [87] Qian XC, Brent TP. Methylation hot spots in the 5' flanking region denote silencing of the O6-

- methylguanine-DNA methyltransferase gene. *Cancer Res.* 1997; 57:3672–3677
- [88] Esteller M, Hamilton SR, Burger PC, et al. Inactivation of the DNA repair gene O6-methylguanine-DNA methyltransferase by promoter hypermethylation is a common event in primary human neoplasia. *Cancer Res.* 1999; 59:793–797
- [89] Kreth S, Thon N, Eigenbrod S, et al. O-methylguanine-DNA methyltransferase (MGMT) mRNA expression predicts outcome in malignant glioma independent of MGMT promoter methylation. *PLoS One.* 2011; 6. DOI: 10.1371/journal.pone.0017156
- [90] Brigliadori Giovanni, Foca Flavia, Dall'Agata Monia, et al. Defining the cutoff value of MGMT gene promoter methylation and its predictive capacity in glioblastoma. *Journal of Neuro-Oncology.* 2016; 128:333–339
- [91] Wiewrodt D, Nagel G, Dreimüller N, et al. MGMT in primary and recurrent human glioblastomas after radiation and chemotherapy and comparison with p53 status and clinical outcome. *Int J Cancer.* 2008; 122:1391–1399
- [92] Stupp R, Hegi ME, Mason WP, et al. Effects of radiotherapy with concomitant and adjuvant temozolomide versus radiotherapy alone on survival in glioblastoma in a randomised phase III study: 5-year analysis of the EORTC-NCIC trial. *Lancet Oncol.* 2009; 10:459–466
- [93] Roldan Urgoiti G B, Singh AD, Easaw JC. Extended adjuvant temozolomide for treatment of newly diagnosed glioblastoma multiforme. *J Neurooncol.* 2012; 108:173–177
- [94] Westphal M, Ram Z, Riddle V, et al. Gliadel wafer in initial surgery for malignant glioma: long-term follow-up of a multicenter controlled trial. *Acta Neurochir (Wien).* 2006; 148:269–75; discussion 275
- [95] Hart MG, Grant R, Garside R, et al. Chemotherapy wafers for high grade glioma. *Cochrane Database Syst Rev.* 2011. DOI: 10.1002/14651858.CD007294.pub2
- [96] de Franca SA, Tavares WM, Salinet ASM, et al. Laser interstitial thermal therapy as an adjunct therapy in brain tumors: A meta-analysis and comparison with stereotactic radiotherapy. *Surg Neurol Int.* 2020; 11. DOI: 10.25259/SNI_152_2020
- [97] Bunevicius A, McDannold NJ, Golby AJ. Focused Ultrasound Strategies for Brain Tumor Therapy. *Oper Neurosurg (Hagerstown).* 2020; 19:9–18
- [98] Brandes AA, Tosoni A, Spagnoli F, et al. Disease progression or pseudoprogression after concomitant radiochemotherapy treatment: pitfalls in neurooncology. *Neuro Oncol.* 2008; 10:361–367
- [99] Brandsma D, Stalpers L, Taal W, et al. Clinical features, mechanisms, and management of pseudoprogression in malignant gliomas. *Lancet Oncol.* 2008; 9:453–461
- [100] Liao G, Khan M, Zhao Z, et al. Bevacizumab Treatment of Radiation-Induced Brain Necrosis: A Systematic Review. *Front Oncol.* 2021; 11. DOI: 10.3389/fonc.2021.593449
- [101] Ahluwalia M, Barnett GH, Deng D, et al. Laser ablation after stereotactic radiosurgery: a multicenter prospective study in patients with metastatic brain tumors and radiation necrosis. *J Neurosurg.* 2018; 130:804–811
- [102] Clarke JL, Chang S. Pseudoprogression and pseudoresponse: challenges in brain tumor imaging. *Curr Neurol Neurosci Rep.* 2009; 9:241–246
- [103] Macdonald DR, Cascino TL, Schold SC Jr, et al. Response criteria for phase II studies of supratentorial malignant glioma. *J Clin Oncol.* 1990; 8:1277–1280
- [104] Wen PY, Macdonald DR, Reardon DA, et al. Updated response assessment criteria for high-grade gliomas: response assessment in neuro-oncology working group. *J Clin Oncol.* 2010; 28:1963–1972

38 Pediatric-type Diffuse Tumors

38.1 Pediatric-type diffuse low-grade gliomas

38.1.1 Diffuse astrocytoma, MYB- or MYBL1-altered (CNS grade 1)

General information

See reference.¹

A diffusely infiltrating astroglial tumor with *MYB* or *MYBL1* gene alterations.
Comprised of monomorphic cells with bland round, ovoid or spindled nuclei.
Typically presents with seizures that are refractory to medication.

Diagnostic criteria for diffuse astrocytoma, MYB- or MYBL1-altered

Diagnostic criteria are shown in ► Fig. 38.1.¹

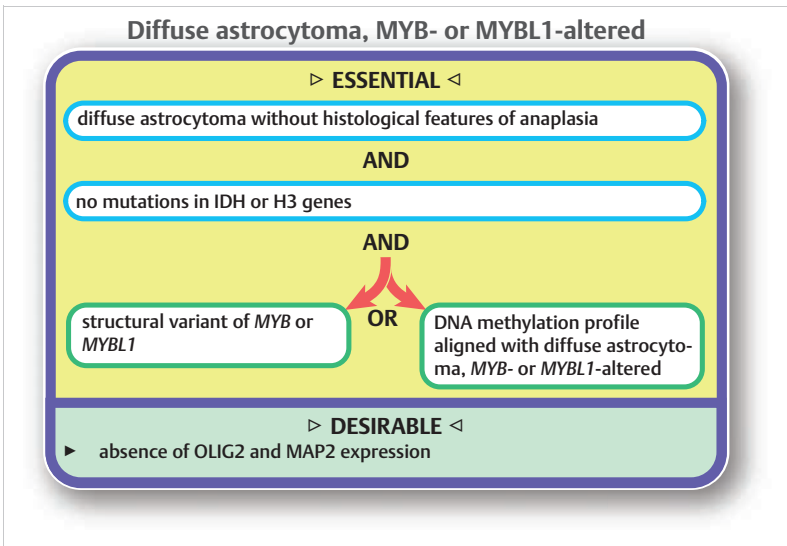


Fig. 38.1 Diagnostic criteria for diffuse astrocytoma, MYB- or MYBL1-altered. (Modified with permission from WHO Classification of Tumors Editorial Board, (ed.) World Health Organization classification of tumors of the central nervous system. 5th ed. Bosman F T, Jaffe E S, Lakhani S R, et al. (series eds.): World Health Organization Classification of Tumors series, vol. 6. International Agency for Research on Cancer, Lyon, France, 2021.)

Subtypes

See reference.¹

- diffuse astrocytoma, MYB-altered
- diffuse astrocytoma, MYBL1-altered

38.1.2 Angiocentric glioma (WHO grade 1)

General information

A static or slow growing WHO grade 1 tumor primarily of children and young adults characterized by angiocentric pattern of growth, monomorphous bipolar cells and features of ependymal differentiation. Nearly all have *MYB::QKI* gene fusion, and the rest usually have another *MYB* alteration.

Behavior is usually indolent, with little change radiographically.

Diagnostic criteria for angiocentric glioma

Diagnostic criteria are shown in ► Fig. 38.2.¹

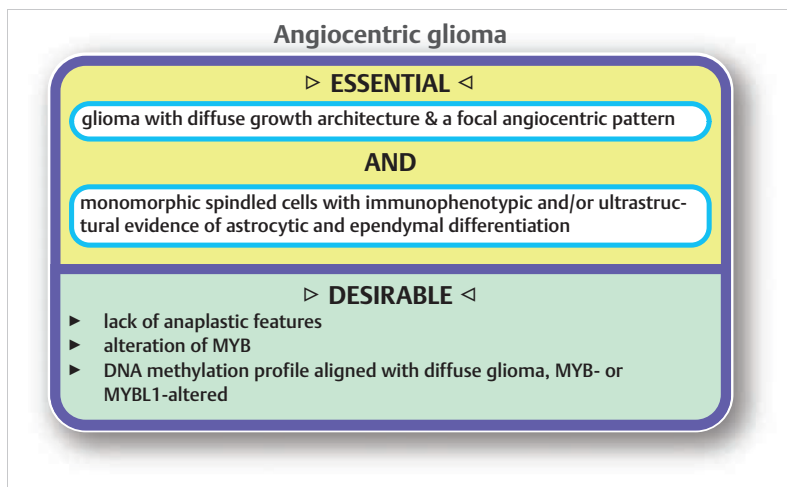


Fig. 38.2 Diagnostic criteria for angiocentric glioma. (Modified with permission from WHO Classification of Tumors Editorial Board, (ed.) World Health Organization classification of tumors of the central nervous system. 5th ed. Bosman F T, Jaffe E S, Lakhani S R, et al. (series eds.): World Health Organization Classification of Tumors series, vol. 6. International Agency for Research on Cancer, Lyon, France, 2021.)

Presentation

Typically present with intractable partial epilepsy.

Imaging

Well circumscribed nonenhancing superficial or cortical based lesion typically in the temporal or frontal lobes, occasionally in the brainstem.

Treatment

Gross total excision can be achieved in most cases and is usually curative.²

38.1.3 Polymorphous low-grade neuroepithelial tumor of the young (PLNTY) (WHO grade 1)

General information

An indolent tumor strongly with diffuse growth patterns, commonly with components resembling oligodendrogliomas, calcifications, CD34 immunoreactivity and MAPK pathway activating genetic abnormalities.

Behavior appears to be consistent with WHO grade 1.

Diagnostic criteria for polymorphous low-grade neuroepithelial tumor of the young

Diagnostic criteria are shown in ► Fig. 38.3.¹

Polymorphous low-grade neuroepithelial tumor of the young

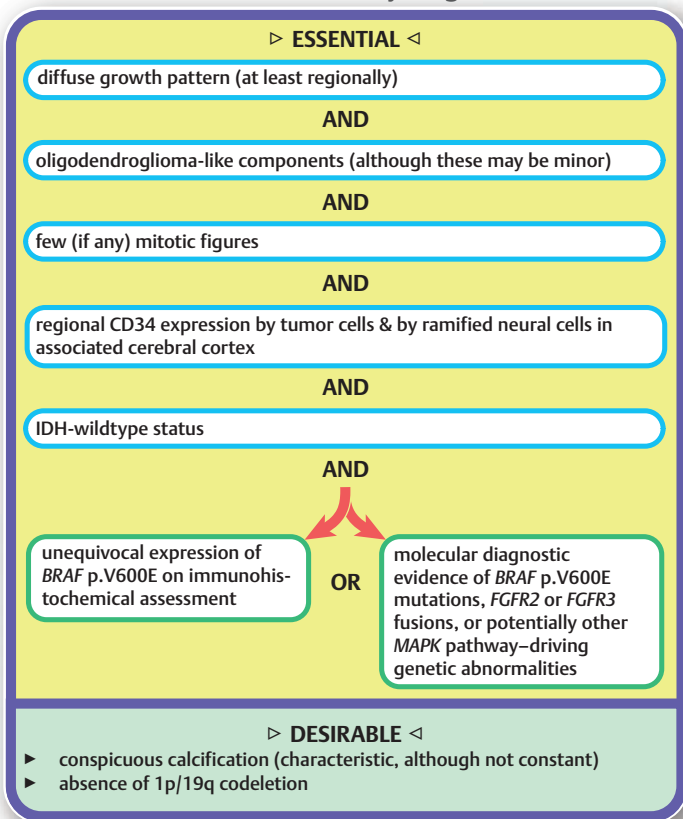


Fig. 38.3 Diagnostic criteria for polymorphous low-grade neuroepithelial tumor of the young. (Modified with permission from WHO Classification of Tumors Editorial Board, (ed.) World Health Organization classification of tumors of the central nervous system. 5th ed. Bosman F T, Jaffe E S, Lakhani S R, et al. (series eds.): World Health Organization Classification of Tumors series, vol. 6. International Agency for Research on Cancer, Lyon, France, 2021.)

Presentation

Typically presents with seizures in young patients.

Imaging

Cortical and subcortical components are typical. 80% involve the temporal lobes, usually on the right side.

Treatment

Tumor excision appears to be effective and improves seizures in most cases.³

38.1.4 Diffuse low-grade glioma, MAPK pathway-altered (WHO grade 1)

General information

A low-grade glioma possessing astrocytic or oligodendroglial morphology and an abnormality in a gene coding for a MAPK pathway protein.

Diagnostic criteria for diffuse low-grade glioma, MAPK pathway-altered

Diagnostic criteria are shown in ► Fig. 38.4.¹

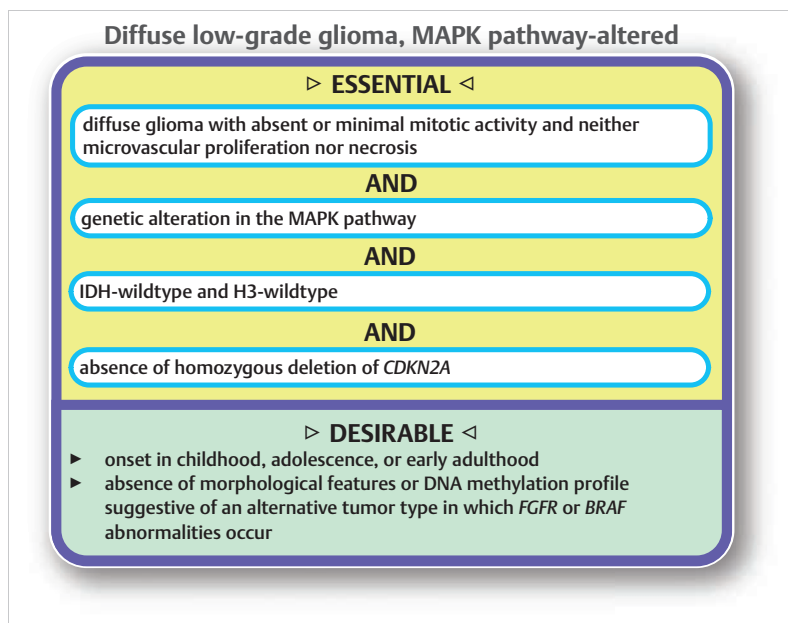


Fig. 38.4 Diagnostic criteria for diffuse low-grade glioma, MAPK pathway-altered. (Modified with permission from WHO Classification of Tumors Editorial Board, (ed.) World Health Organization classification of tumors of the central nervous system. 5th ed. Bosman F T, Jaffe E S, Lakhani S R, et al. (series eds.): World Health Organization Classification of Tumors series, vol. 6. International Agency for Research on Cancer, Lyon, France, 2021.)

Subtypes

See reference.¹

- diffuse low-grade glioma, *FGFR1* tyrosine kinase domain–duplicated
- diffuse low-grade glioma, *FGFR1*-mutant
- diffuse low-grade glioma, *BRAF* p.V600E-mutant

Presentation

Seizures or symptoms related to mass effect, sometimes with increased intracranial pressure.

Imaging

Appearance may resemble that of pilocytic astrocytoma complete with cystic elements, but these tumors tend to be more extensive on T2-FLAIR images.

Treatment

Unresolved issues related to the behavior of this group, and possible subgroups.

38.2 Pediatric-type diffuse high-grade gliomas

38.2.1 Diffuse midline glioma, H3 K27M-altered (WHO grade 4)

General information

An infiltrating midline glioma with loss of H3 p.K28me3 (K27me3) and usually either an H3 c.83A>T p.K28 M (K27M) substitution in one of the histone H3 isoforms, aberrant overexpression of EZHIP, or an *EGFR* mutation.

In children, involvement tends to be in brainstem (formerly called brainstem glioma), pons (formerly called diffuse infiltrating pontine glioma [DIPG]), or bithalamic. In adolescents or adults, they occur unilaterally in the thalamus or in the spinal cord.

Leptomeningeal involvement was found in 40% in autopsy series, and DIPG is the leading cause of brain tumor-related death in children.⁴

Diagnostic criteria for diffuse midline glioma, H3 K27M-altered

Diagnostic criteria are shown in ► Fig. 38.5.¹

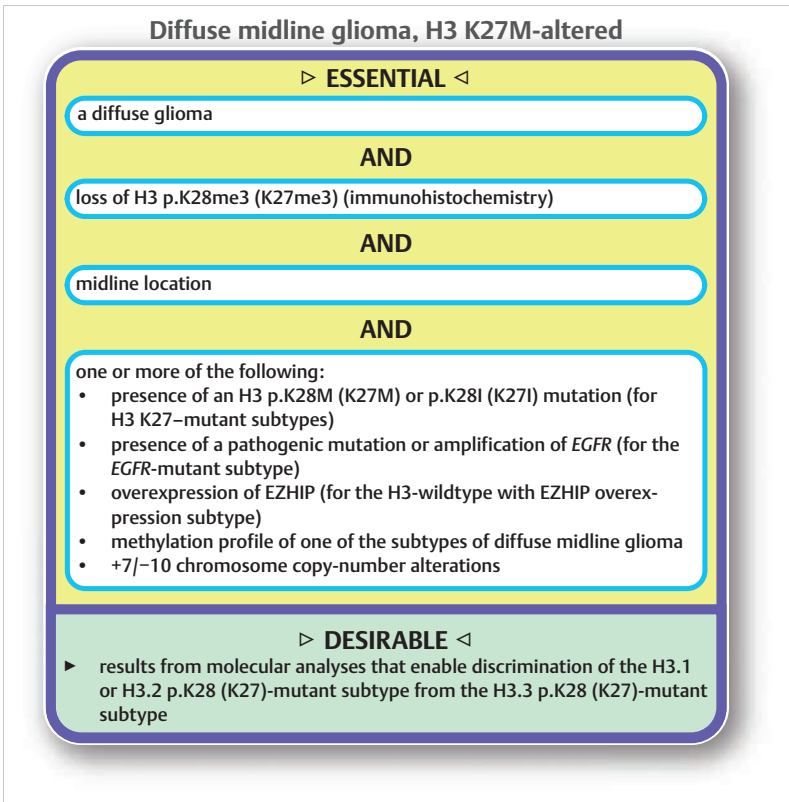


Fig. 38.5 Diagnostic criteria for diffuse midline glioma, H3 K27M-altered. (Modified with permission from WHO Classification of Tumors Editorial Board, (ed.) World Health Organization classification of tumors of the central nervous system. 5th ed. Bosman F T, Jaffe E S, Lakhani S R, et al. (series eds.): World Health Organization Classification of Tumors series, vol. 6. International Agency for Research on Cancer, Lyon, France, 2021.)

Subtypes

See reference.¹

- diffuse midline glioma, H3.3 K27-mutant
- diffuse midline glioma, H3.1 or H3.2 K27-mutant
- diffuse midline glioma, H3-wildtype with EZHIP overexpression
- diffuse midline glioma, *EGFR*-mutant

Presentation

Patients with DMG typically present with a short course with brainstem findings⁵ (triad: multiple cranial nerve palsies, long tract signs and ataxia) or obstructive hydrocephalus. Evidence of thalamic involvement includes signs of increased ICP, motor weakness (e.g., hemiparesis), and gait disturbance.

Evaluation

Given the high rate of leptomeningeal spread, imaging of the entire neuraxis is recommended.

Current trends favor obtaining a biopsy in most cases as early reports of high complications and lack of clinical utility have been debunked.⁶ Biopsy confirms H3 K27M-mutation and can determine if there is MGMT promoter gene methylation.

Imaging

Pontine lesions enlarge the pons (► Fig. 38.6) which is low intensity on T1 and increased intensity on T2. Enhancement is minimal. Compression of the 4th ventricle may produce hydrocephalus. Encasement of the basilar artery is common. There may be an exophytic component.

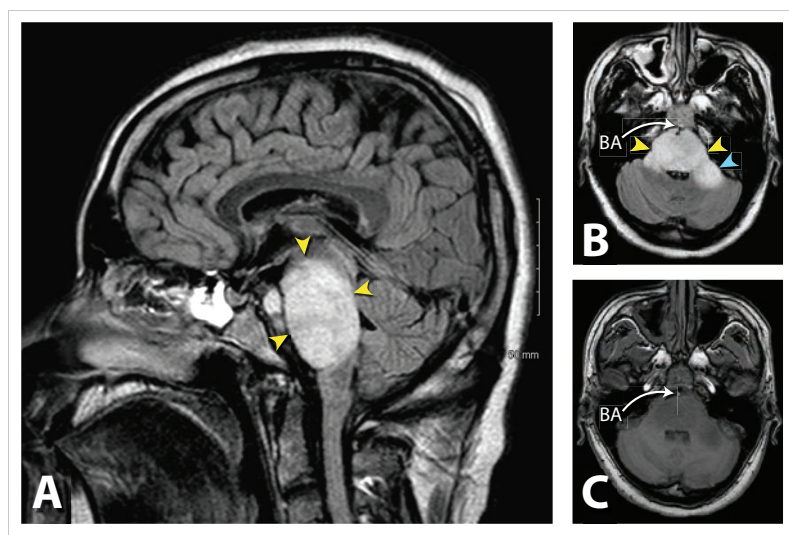


Fig. 38.6 Diffuse midline glioma, H3 K27M-altered.

Tumor is high intensity on FLAIR images and is indicated by the yellow arrowheads. Note the infiltration of the left cerebellar peduncle (blue arrowhead). Abbreviation: BA = the compressed basilar artery.

Image: MRI scans. A: sagittal FLAIR. B: Axial FLAIR. C: Axial T1 with contrast demonstrating lack of tumor enhancement.

Treatment and prognosis

Even with current therapies, 2-year survival is < 10%.⁷

The role for surgical intervention is limited due to the location of these tumors.

Prognosis is better for subtypes involving H3.1 or H3.2 K27-mutation or EZHIP overexpression than the H3.3 K27 mutation.¹

38.2.2 Diffuse hemispheric glioma, H3 G34-mutant (WHO grade 4)

General information

An infiltrative glioma located in the cerebral hemispheres having a missense mutation of the *H3-3A* gene resulting in one of the substitutions of the H3 protein shown in ► Fig. 38.7. The tumor occasionally extends to the midline.

Diagnostic criteria for diffuse hemispheric glioma, H3 G34-mutant

Diagnostic criteria are shown in ► Fig. 38.7.¹

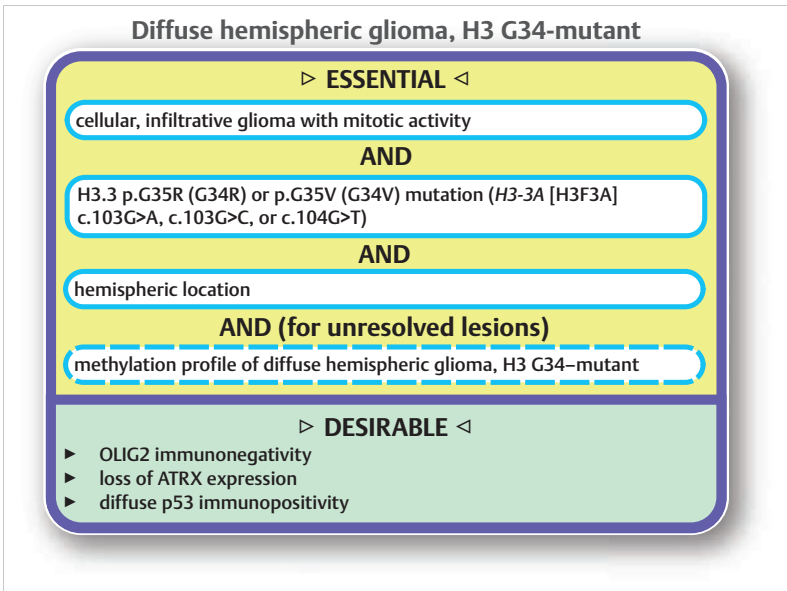


Fig. 38.7 Diagnostic criteria for diffuse hemispheric glioma, H3 G34-mutant. (Modified with permission from WHO Classification of Tumors Editorial Board, (ed.) World Health Organization classification of tumors of the central nervous system. 5th ed. Bosman F T, Jaffe E S, Lakhani S R, et al. (series eds.): World Health Organization Classification of Tumors series, vol. 6. International Agency for Research on Cancer, Lyon, France, 2021.)

Presentation

Non specific (seizures, mass effect from locally involved brain...).

Histology

Appearance similar to GBM (high cellularity, infiltrating margins, extensive mitoses...).

Evaluation

Spine imaging is recommended in addition to brain.

Imaging

Hemispheric lesion that typically contrast enhances. Occasionally multifocal.

Treatment and prognosis

Median PFS is 9 months, with median OS of 18-22 months.⁸

Recurrence after treatment is typically local, although leptomeningeal spread may occur.

MGMT promoter gene methylation (p.673) and absence of oncogene amplification (e.g., *PDGFRA*, *EGFR*, *CDK4*, *MDM2*, *CDK6*, *CCND2*, *MYC*, *MYCN*) may portend a better prognosis for OS.⁸

38.2.3 Diffuse pediatric-type high-grade glioma, H3-wildtype and IDH-wildtype (WHO grade 4)

General information

An infiltrative glioma located in the cerebral hemispheres having a missense mutation of the *H3-3A* gene resulting in one of the substitutions of the H3 protein shown in ► Fig. 38.7. The tumor occasionally extends to the midline.

Diagnostic criteria for diffuse hemispheric glioma, H3 G34-mutant

Diagnostic criteria are shown in ► Fig. 38.8.¹

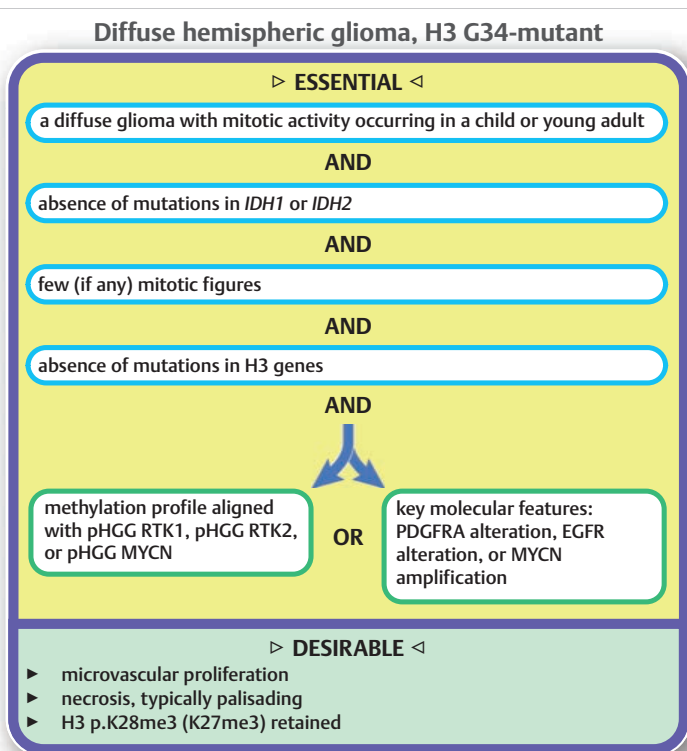


Fig. 38.8 Diagnostic criteria for diffuse pediatric-type high-grade glioma, H3-wildtype and IDH-wildtype. (Modified with permission from WHO Classification of Tumors Editorial Board, (ed.) World Health Organization classification of tumors of the central nervous system. 5th ed. Bosman F T, Jaffe E S, Lakhani S R, et al. (series eds.): World Health Organization Classification of Tumors series, vol. 6. International Agency for Research on Cancer, Lyon, France, 2021.)

Presentation

Nonspecific (seizures, mass effect from locally involved brain...).

Histology

Appearance similar to GBM (high cellularity, infiltrating margins, extensive mitoses...).

Evaluation

Spine imaging is recommended in addition to brain.

Imaging

Hemispheric lesion that typically contrast enhances. Occasionally multifocal.

Treatment and prognosis

Median PFS is 9 months, with median OS of 18-22 months.⁸

Recurrence after treatment is typically local, although leptomeningeal spread may occur.

MGMT promoter gene methylation (p.673) and absence of oncogene amplification (e.g., *PDGFRA*, *EGFR*, *CDK4*, *MDM2*, *CDK6*, *CCND2*, *MYC*, *MYCN*) may portend a better prognosis for OS.⁸

38.2.4 Infant-type hemispheric glioma (WHO grade N/A)

General information

A high-grade hemispheric glioma occurring in early childhood, having receptor tyrosine kinase (RTK) fusions including those in the NTRK family, or in *ROS1*, *ALK*, or *MET*.

Diagnostic criteria for infant-type hemispheric glioma

Diagnostic criteria are shown in ► Fig. 38.9.¹

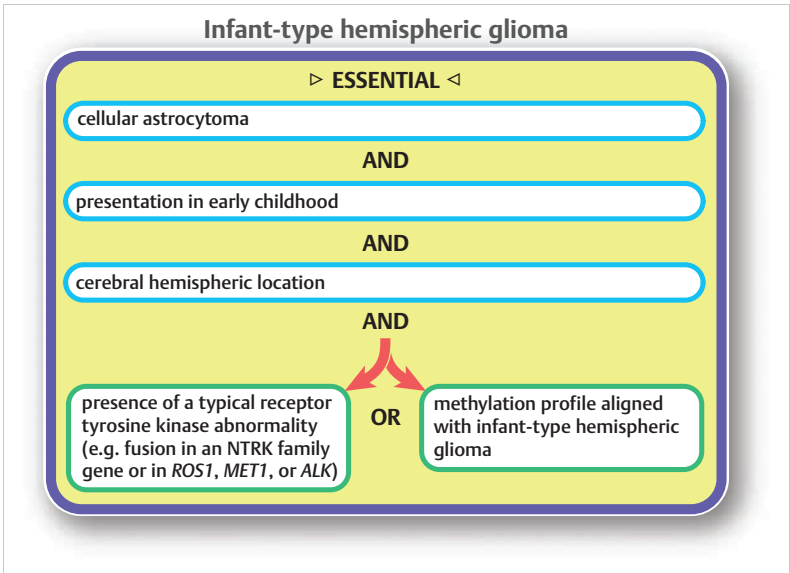


Fig. 38.9 Diagnostic criteria for infant-type hemispheric glioma. (Modified with permission from WHO Classification of Tumors Editorial Board, (ed.) World Health Organization classification of tumors of the central nervous system. 5th ed. Bosman F T, Jaffe E S, Lakhani S R, et al. (series eds.): World Health Organization Classification of Tumors series, vol. 6. International Agency for Research on Cancer, Lyon, France, 2021.)

Subtypes

See reference.¹

- infant-type hemispheric glioma, *NTRK*-altered
- infant-type hemispheric glioma, *ROS1*-altered
- infant-type hemispheric glioma, *ALK*-altered
- infant-type hemispheric glioma, *MET*-altered

Presentation

All known cases occurred in early childhood, usually in the 1st year of life. Typically acute presentation. In infancy, may manifest as agitation or lethargy, possibly with enlarged head circumference.

Histology

Typically well-demarcated, cellular tumors involving brain parenchyma and leptomeninges.

Initial diagnosis was a high-grade glioma or GBM in 84%.¹

Evaluation

Owing to leptomeningeal involvement, spine imaging is recommended in addition to brain.

Treatment and prognosis

There is insufficient data on this recently designated entity. Stereotypically, infants with high-grade gliomas fare better than older children.

References

- [1] Bosman FT, Jaffe ES, Lakhani SR, et al. World Health Organization Classification of Tumors series, vol. 6. In: WHO Classification of Tumors Editorial Board (ed.) World Health Organization classification of tumors of the central nervous system. 5th ed. Lyon, France: International Agency for Research on Cancer; 2021
- [2] Ampie L, Choy W, DiDomenico JD, et al. Clinical attributes and surgical outcomes of angiocentric gliomas. *J Clin Neurosci*. 2016; 28:117–122
- [3] Huse JT, Snuderl M, Jones DT, et al. Polymorphous low-grade neuroepithelial tumor of the young (PLNTY): an epileptogenic neoplasm with oligodendroglioma-like components, aberrant CD34 expression, and genetic alterations involving the MAP kinase pathway. *Acta Neuropathol*. 2017; 133:417–429
- [4] Buczkowicz P, Bartels U, Bouffet E, et al. Histopathological spectrum of paediatric diffuse intrinsic pontine glioma: diagnostic and therapeutic implications. *Acta Neuropathol*. 2014; 128:573–581
- [5] Hoffman LM, Veldhuijzen van Zanten S EM, Colditz N, et al. Clinical, Radiologic, Pathologic, and Molecular Characteristics of Long-Term Survivors of Diffuse Intrinsic Pontine Glioma (DIPG): A Collaborative Report From the International and European Society for Pediatric Oncology DIPG Registries. *J Clin Oncol*. 2018; 36:1963–1972
- [6] Mohanty A. Biopsy of brain stem gliomas: Changing trends? *J Neurosci Rural Pract*. 2014; 5:116–117
- [7] Mackay A, Burford A, Carvalho D, et al. Integrated Molecular Meta-Analysis of 1,000 Pediatric High-Grade and Diffuse Intrinsic Pontine Glioma. *Cancer Cell*. 2017; 32:520–537 e5
- [8] Korshunov A, Capper D, Reuss D, et al. Histologically distinct neuroepithelial tumors with histone 3 G34 mutation are molecularly similar and comprise a single nosologic entity. *Acta Neuropathol*. 2016; 131:137–146

39 Circumscribed Astrocytic Gliomas

39.1 General meaning of “circumscribed”

Circumscribed astrocytic tumors have a more contained growth pattern which differs from the infiltrating gliomas. With diffuse gliomas, there is generally no tumor identifiable margin beyond which there is tumor-free brain.

39.2 Specific tumor types

39.2.1 Pilocytic astrocytomas (PCAs) (WHO grade 1)

General information

Key concepts

- a subgroup of astrocytomas (WHO grade 1) with better prognosis (10-year survival: >95%) than infiltrating fibrillary or diffuse astrocytomas
- the most common astrocytic tumor in children
- average age is lower than for typical astrocytomas (75% of patients are ≤ 20 yrs)
- common locations: cerebellar hemisphere, optic nerve/chiasm, hypothalamus
- CT/MRI appearance: discrete appearing, contrast enhancing lesion. Cerebellar PCAs are classically cystic with mural nodule
- the principal CNS tumor associated with NF1
- histology: biphasic. 1) compacted, 2) loose (myxoid) textured astrocytes with Rosenthal fibers and/or eosinophilic granular bodies
- associated with MAPK pathway gene alterations (typically *KIAA1549::BRAF* gene fusions)
- histology alone may be inadequate for diagnosis; knowledge of patient age & radiographic appearance is critical; there is a risk of overgrading and overtreatment if decisions are made on histology alone
- complete surgical resection, when possible, is usually curative. ★ For cystic PCAs with enhancing mural nodule, only the nodule needs to be resected (the cyst wall is not neoplastic). XRT is used post-op only for nonresectable recurrence or malignant degeneration

Diagnostic criteria for pilocytic astrocytoma and pilomyxoid astrocytoma

Diagnostic criteria for pilocytic astrocytoma are shown in ► Fig. 39.1 and for pilomyxoid astrocytoma in ► Fig. 39.2.¹

Subtypes

See reference.¹

- **pilomyxoid astrocytoma (PMA)**: a tumor of infancy that develops in the hypothalamic/chiasmatic region, and carries a higher rate of recurrence, worse outcome, and tendency for CSF dissemination,² with a case report of extraneural peritoneal mets spread through a VP shunt.³ Because not all PMAs behave differently than a WHO grade 1 PCA, a definite WHO grade was not recommended.⁴ They are comprised of monomorphic piloid cells, have a diffusely myxoid background and increased cellularity compared with classic PCAs and angiocentric cell arrangement. By definition, does not contain Rosenthal fibers or eosinophilic granular bodies.⁴ They may eventually morph into a classic PCA
May also occur in spinal cord
- **pilocytic astrocytoma with histological features of anaplasia**: a proposed term for tumors morphologically similar to PCA but also possess mitotic features ± necrosis. These anaplastic features may be present on initial analysis, or may develop later

Background and terminology

Pilocytic astrocytoma (PCA) is the currently recommended nomenclature for these tumors previously referred to variously as cystic cerebellar astrocytomas and juvenile pilocytic astrocytomas

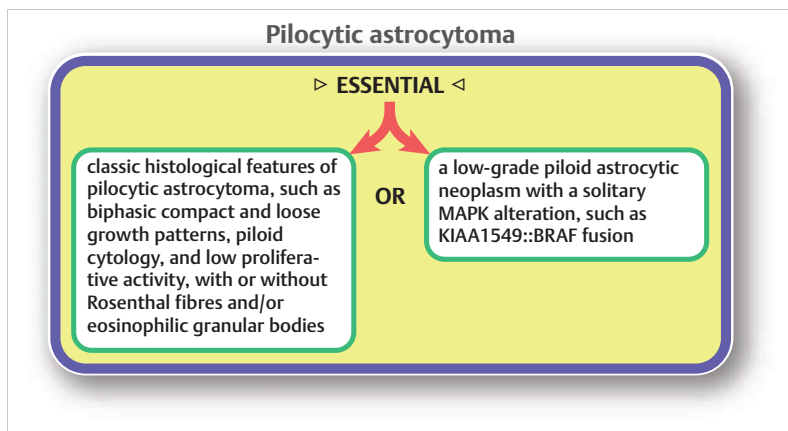


Fig. 39.1 Diagnostic criteria for pilocytic astrocytoma. (Modified with permission from WHO Classification of Tumors Editorial Board, (ed.) World Health Organization classification of tumors of the central nervous system. 5th ed. Bosman F T, Jaffe E S, Lakhani S R, et al. (series eds.): World Health Organization Classification of Tumors series, vol. 6. International Agency for Research on Cancer, Lyon, France, 2021.)

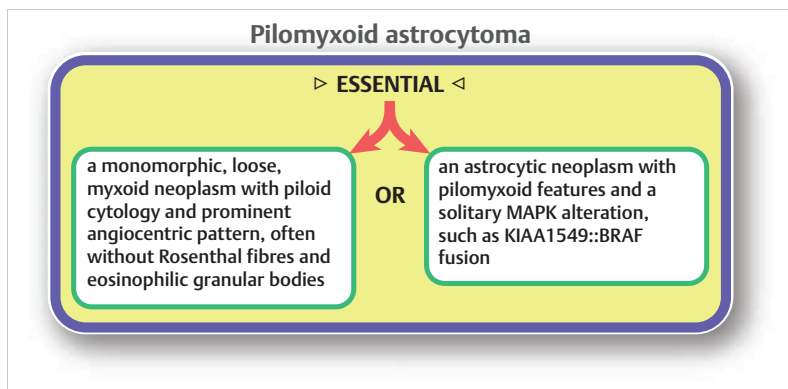


Fig. 39.2 Diagnostic criteria for pilomyxoid astrocytoma. (Modified with permission from WHO Classification of Tumors Editorial Board, (ed.) World Health Organization classification of tumors of the central nervous system. 5th ed. Bosman F T, Jaffe E S, Lakhani S R, et al. (series eds.): World Health Organization Classification of Tumors series, vol. 6. International Agency for Research on Cancer, Lyon, France, 2021.)

(JPA), among others.⁴ Based on location, these tumors may be called optic gliomas, hypothalamic gliomas, and cerebellar PCAs. Treatment decisions vary based on location and neural involvement.

PCAs differ markedly from the infiltrating diffuse astrocytomas in terms of their reduced tendency to invade tissue and their very low propensity for malignant degeneration.

Epidemiology

PCAs comprise 5% of primary brain tumors. They are the most common glioma in pediatrics (age 0–19 years) with an incidence of 0.82/100,000.⁵ The incidence progressively declines after age 15. Slight male predilection.

Usually presents during second decade of life (ages 10–20).⁵ 75% occur in age < 20 years.⁶

Most PCAs are sporadic, but they are the main tumor in neurodevelopmental diseases with germline mutations in the MAPK pathway genes including: NF1 (p.638), Noonan syndrome, and encephalocraniocutaneous lipomatosis.

Location

PCAs arise throughout the neuraxis (2003 study in adults and children⁷):

1. pilocytic astrocytoma of the cerebellum (p.693): 42%. Formerly referred to as cystic cerebellar astrocytoma
2. cerebral hemispheres: 35%. Rare in children, tends to occur more often in older patients (i.e., young adults) than optic nerve/hypothalamic lesions. These PCAs are potentially confused with infiltrating astrocytomas possessing more malignant potential. PCAs are often distinguished by a cystic component with an enhancing mural nodule (would be atypical for a diffuse astrocytoma), and some PCAs have dense calcifications⁸
3. optic gliomas & hypothalamic gliomas: 9%
 - a) PCAs arising in the optic nerve are called optic gliomas (p.694). This is the most common site in neurofibromatosis type 1 (p.638) (NF1) patients (may be bilateral)⁹
 - b) when they occur in the region of the chiasm they cannot always be distinguished clinically or radiographically from so-called hypothalamic gliomas (p.695) or gliomas of the third ventricular region
4. brainstem PCAs: 9%. Usually are fibrillary infiltrating type and only a small proportion are pilocytic. May comprise the majority of the prognostically favorable group described as “dorsally exophytic (p.696)” brainstem gliomas^{9,10}
5. spinal cord: 2%. PCAs may also occur here, but little information is available on these. Again, patients tend to be younger than with infiltrating astrocytomas of the spinal cord

Molecular genetics

Detailed molecular genetics are available.¹¹ For most neurosurgeons, the salient aspects are:

- the most common genetic abnormality in PCAs is in the BRAF gene which activates the MAPK pathway which is involved in nearly all PCAs. This is common in all PCAs, but most prevalent (75%) in cerebellar PCAs
- 15–20% of patients with neurofibromatosis type 1 (NF1) develop PCAs,¹² which are the main CNS tumors associated with NF1. NF1 patients have only 1 NF1 wildtype gene copy. Loss of this copy by a “second hit” (by point mutation, LOH, or DNA hypermethylation¹³) results in overactivation of RAS and MAPK pathway¹⁴

Histology

PCAs are usually histologically WHO grade 1 (however, a rare anaplastic form in adults has been identified,¹⁵ but a WHO grade 3 variant is not officially recognized).

► **Microscopy.** Low to moderate cellularity with two main cell populations (biphasic), best appreciated in cerebellar PCAs:

1. cells with long thin bipolar processes (resembling hairs—hence *pilocytic*)⁹ with Rosenthal fibers⁸ (sausage or corkscrew shaped cytoplasmic eosinophilic inclusion bodies consisting of glial filament aggregates resembling hyaline; stain bright red on Masson trichrome smears)
2. loosely knit tissue comprising stellate astrocytes with microcysts and occasional eosinophilic granular bodies

PCAs easily break through the pia to fill the overlying subarachnoid space. PCAs may also infiltrate into the perivascular spaces. Vascular proliferation is common. Multinucleated giant cells with peripherally located nuclei are common, especially in PCAs of the cerebellum or cerebrum. Mitotic figures may be seen, but are clinically not as ominous as with fibrillary astrocytomas. Areas of necrosis may also be seen. In spite of well-demarcated margins grossly and on MRI, at least 64% of PCAs infiltrate the surrounding parenchyma, especially the white matter¹⁶ (the clinical significance of this is uncertain; one study found no statistically significant decrease in survival¹⁷).

► **Differentiating from a diffuse or infiltrating astrocytoma.** Pathology alone may not be able to differentiate. This may be especially problematic with small specimens obtained e.g., with stereotactic biopsy. A young age may suggest the diagnosis, and knowledge of the radiographic appearance is often critical (see below).

► **Malignant degeneration.** Rare malignant degeneration has been reported, often after many years. Radiation therapy (XRT) had been administered in most cases¹⁸ although this may occur without XRT.¹⁹

Radiographic evaluation

Preoperative MRI of the entire neuraxis (brain, cervical, thoracic, and lumbar spine) without and with contrast is recommended when possible to evaluate for dissemination through the CSF (rare) and in case it proves to be a different pathology.

On CT or MRI, PCAs are usually well circumscribed, 94% enhance with contrast¹⁶ (unlike most low-grade diffuse astrocytomas), frequently have a cystic component with a mural nodule (especially cerebellar PCAs) (see ► Fig. 39.3), and have little or no surrounding edema. The cyst wall may or may not enhance. PCAs involving the optic apparatus are typically fusiform.

Although they may occur anywhere in the CNS, 82% are periventricular.¹⁶ Calcifications are only occasionally present.¹⁶ 4 main imaging patterns of cerebellar or cerebral PCAs are shown in ► Table 39.1.

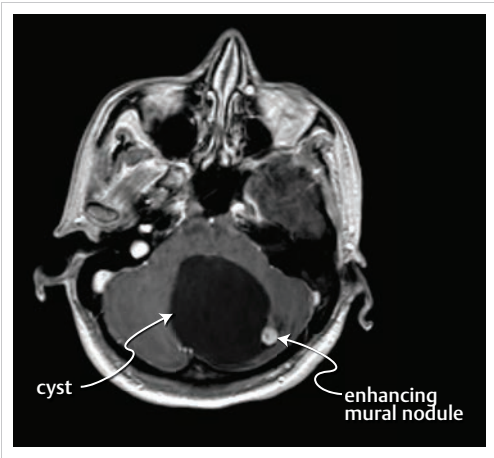


Fig. 39.3 Pilocytic astrocytoma of the posterior fossa. Note characteristic nonenhancing cyst wall with enhancing mural nodule. Image: axial T1 MRI with contrast.

Table 39.1 Common imaging characteristics of cerebellar or cerebral PCAs		
%	Description	
21	nonenhancing cyst with enhancing mural nodule	over 66% are cystic with enhancing mural nodule
46	enhancing cyst with enhancing mural nodule	
16	mass with nonenhancing central area (necrosis)	
17	solid mass with minimal or no cyst	

39 Treatment

Treatment of choice is surgical excision of the maximal amount of the tumor that can be removed without producing deficit. In some, invasion of brainstem or involvement of cranial nerves or blood vessels may limit resection.

► **Cysts.** In tumors composed of a nodule with a true cyst, excision of the nodule is sufficient; the cyst wall need not be removed (the strength of this statement is based on the fact that the wall has been biopsy proven negative for neoplasm in 3 cases²⁰ and that patients do very well after excision of the nodule alone). In tumors with a so-called “false cyst” where the cyst wall is thick and enhances (on CT or MRI), this portion must also be removed.

► **XRT & chemotherapy.** Because of the high 5- and 10-year survival rates together with the high complication rate of radiation therapy over this time interval—see Radiation injury and necrosis (p.1899)—and the fact that many incompletely resected tumors enlarge minimally if at all over periods of 5, 10, or even 20 years, it is recommended to *not* radiate these patients post-op. Rather, they should be followed with serial CT or MRI and be re-operated if there is recurrence.²¹ Radiation therapy is indicated for nonresectable recurrence (i.e., reoperation is preferred if possible) or for recurrence with malignant histology. Chemotherapy is preferable to XRT in younger patients.²²

Prognosis

The natural history of these tumors is slow growth. Survival rates at 5 and 10 years are 95% after surgical resection alone.

Tumor recurrence is relatively common, and although it has been said that they generally occur within ≈ 3 yrs of surgery,²³ this is controversial, and very late recurrences are documented,²¹ violating Collins' law (which says that a tumor may be considered cured if it does not recur within a time period equal to the patient's age at diagnosis + 9 months). Also, some tumors excised partially fail to show further growth, representing a form of cure (long-term progression free survival).

About 20% of cerebellar PCAs develop hydrocephalus, requiring treatment following surgery.²⁴ So-called "drop metastases" occur rarely with PCAs.

Pilocytic astrocytoma (PCA) of the cerebellum

General information

Key concepts

- often cystic, half of these have mural nodule
- usually presents during the second decade of life (ages 10–20 yrs)
- also, see Key concepts for pilocytic astrocytomas in general (p.689)

An informal subtype of pilocytic astrocytoma (PCA) (p.689). Formerly referred to by the nonspecific and confusing term cystic cerebellar astrocytoma. One of the more common pediatric brain tumors ($\approx 15\%$ of primary brain tumors in patients <19 years⁷), comprising 27–40% of pediatric p-fossa tumors.²⁵ (p 367–74),²⁶ (p 3032) They may also occur in adults, where the mean age is lower and the post-operative survival is longer than for diffuse astrocytomas.²⁷

Presentation

Signs and symptoms of cerebellar PCAs are usually those of any p-fossa mass, i.e., those of hydrocephalus (headache, nausea/vomiting...) or cerebellar dysfunction (ataxia, cranial nerve deficits) (see Posterior fossa (infratentorial) tumors (p.624)). Optic pathway gliomas typically present with loss of vision.

Imaging

"Classic" MRI finding: posterior fossa cyst with an enhancing mural nodule (see ► Fig. 39.3). The cyst wall sometimes enhances, usually as a thin rim (biopsy negative for neoplasm, enhancement may be reactive²⁰).

Pathology

The classic "juvenile pilocytic astrocytoma" of the cerebellum is a distinctive entity with its macroscopic cystic architecture and microscopic spongy appearance.⁸ For other microscopic findings, see PCAs in general, above.

These tumors may be solid, but are more often cystic (hence the older term "cystic cerebellar astrocytoma") with a mural nodule. They tend to be large at the time of diagnosis (cystic tumors: 4–5.6 cm diameter; solid tumors: 2–4.8 cm diameter). Cysts contain highly proteinaceous fluid (averaging ≈ 4 Hounsfield units higher density than CSF on CT²⁸).

Treatment guidelines

For cystic cerebellar PCAs, excision of the nodule without removing the cyst wall is sufficient. See treatment of pilocytic astrocytomas in general (p.692).

Also, see Posterior fossa (infratentorial) tumors (p.624) for guidelines regarding hydrocephalus, etc.

Because of the high frequency of BRAF abnormalities, research is ongoing to evaluate the efficacy of BRAF and/or MEK inhibitors.²⁹

Optic pathway glioma (OPG)

General information

Accounts for $\approx$ 2% of gliomas in adults, and 7% in children. The incidence is higher ($\approx$ 25%) in neurofibromatosis type 1 (NF1) (p.637). OPG may occur anywhere along the optic pathway from the optic disc to the occipital cortex.

OPGs may arise in any of the following patterns:

1. one optic nerve (without chiasmal involvement): $>$ 25% of OPGs are confined to the optic disc and nerve³⁰
2. optic chiasm: 40-70% of OPGs.³⁰ Less commonly involved in patients with NF1 than in sporadic cases. Aggressive. 33-60% also invade the hypothalamus or third ventricle
3. multicentric in both optic nerves sparing the chiasm: seen almost exclusively in NF1
4. may occur in conjunction with, or be part of, a hypothalamic glioma (see below)

Pathology

Most OPGs are pilocytic astrocytomas (PCA) and are composed of low-grade (pilocytic) astrocytes. Rarely, a malignant chiasmal glioma occurs.

Presentation

Many OPGs remain asymptomatic.

Painless proptosis is an early sign in lesions involving one optic nerve. Chiasmal lesions produce variable and nonspecific visual defects (usually monocular) without proptosis. Large chiasmal tumors may cause hypothalamic and pituitary dysfunction (see diencephalic syndrome below), and may produce hydrocephalus by obstruction at the foramen of Monro. Gliosis of the optic nerve head may be seen on fundoscopy.

Evaluation

Plain X-rays: not usually helpful. The classic skull X-ray findings are enlargement of the optic canals and a "J"-shaped sella turcica.

CT/MRI: CT scan can readily demonstrate structures within the orbit. In addition to that, MRI also easily demonstrates chiasmal or hypothalamic involvement. On CT or MRI, involvement of the optic nerve produces fusiform enlargement of the nerve usually extending $>$ 1 cm in length. Optic pathway gliomas in patients with NF1 are typically isolated to the optic nerve, whereas those not associated with NF1 usually involve the optic chiasm, may extend outside the optic apparatus, and are frequently cystic. Enhancement is usually only mild to moderate.

Biopsy is not needed in cases with characteristic findings.

Treatment

Tumor involving a single optic nerve, sparing the chiasm, producing proptosis and visual loss should be treated with a transcranial approach with excision of the nerve from the globe all the way back to the chiasm (a transorbital [Kronlein] approach is not appropriate since tumor may be left in the nerve stump). In addition to the anticipated blindness in the involved eye, this may produce a junctional scotoma (p.875).

Chiasmal tumors are generally not treated surgically except for biopsy (especially when it is difficult to distinguish an optic nerve glioma from a hypothalamic glioma), CSF shunting, or to remove the rare ependymic component to try and improve vision.

► **Adjuvant therapy.** Chemotherapy²² (preferred over XRT in younger patients) or XRT. Indications:

1. chiasmal tumors
2. multicentric tumors
3. post-op if tumor is found in the chiasmal stump end of the resected nerve
4. malignant tumor (rare)

Typical XRT treatment planning: 45 Gy in 25 fractions of 1.8 Gy.

Hypothalamic glioma

General information

Pilocytic astrocytomas of the hypothalamus and third ventricular region occur primarily in children. Radiographically, the lesion may have an intraventricular appearance. Many of these tumors have some chiasmal involvement and the distinction from optic nerve glioma cannot be made (see above).

Presentation

Possible presentations include:

- **“diencephalic syndrome”**: a rare syndrome seen in peds, usually caused by an infiltrating glioma of the anterior hypothalamus. Classically:
 - a) cachexia (loss of subcutaneous fat)/failure to thrive, failure to gain weight
 - b) hyperactivity/over-alertness
 - c) an almost euphoric affect (in stark contrast to their gaunt appearance)
- endocrine disturbance
 - a) hypoglycemia
 - b) diabetes insipidus with attendant hypernatremia
 - c) precocious puberty
- symptoms related to hydrocephalus
 - a) macrocephaly
 - b) headache
 - c) nausea/vomiting

Treatment

When complete resection is not possible, adjuvant therapy may be needed as outlined under optic gliomas (above).

Brainstem glioma

General information

Key concepts

- not a homogeneous group. MRI can differentiate malignant from benign lesions
- trend: lower grade tumors tend to occur in the upper brainstem, and higher grade tumors in the lower brainstem/medulla
- usually presents with multiple cranial nerve palsies and long tract findings
- most are malignant, have poor prognosis, and are not surgical candidates
- role of surgery primarily limited to dorsally exophytic lesions and shunting

Brainstem gliomas (BSG) tend to occur during childhood and adolescence (77% are < 20 yrs old, they comprise 1% of adult tumors³¹). BSG are one of the 3 most common brain tumors in pediatrics—see Pediatric brain tumors (p.621)—comprising ≈ 10–20% of pediatric CNS tumors.¹⁰

Presentation

See reference.³²

Upper brainstem tumors tend to present with cerebellar findings and hydrocephalus, whereas lower brainstem tumors tend to present with multiple lower cranial nerve deficits and long tract findings. Due to their invasive nature, signs and symptoms usually do not occur until the tumor is fairly extensive in size.

Signs and symptoms:

1. gait disturbance
2. headache (p.623)
3. nausea/vomiting
4. cranial nerve deficits: diplopia, facial asymmetry
5. distal motor weakness in 30%
6. papilledema in 50%

7. hydrocephalus in 60%, usually due to aqueductal obstruction (often late, except with periaqueductal tumors, e.g., below)
8. failure to thrive (especially in age ≤ 2 yrs)

Pathology

BSG is a heterogeneous group. There may be a tendency towards lower grade tumors in the upper brainstem (76% were low-grade) versus the lower brainstem (100% of the glioblastomas were in the medulla).³³ A cystic component is seen rarely. Calcifications are also rare. 4 growth patterns that can be identified by MRI³⁴ that may correlate with prognosis³⁵:

1. diffuse: all are malignant (most are anaplastic astrocytomas, the rest are glioblastomas). On MRI these tumors extend into the adjacent region in vertical axis (e.g., medullary tumors extend into pons and/or cervical cord) with very little growth towards obex, remaining intra-axial. Diffuse midline glioma, H3 K27M-mutant (p.683) is a specific tumor seen mostly in pediatrics (but also uncommonly in adults)
2. cervicomedullary: most (72%) are low-grade astrocytomas. The rostral extent of these tumors is limited to the spinomedullary junction. Most bulge into the obex of the 4th ventricle (some may have an actual exophytic component)
3. focal: extent limited to medulla (does not extend up into pons nor down into spinal cord). Most (66%) are low-grade astrocytomas
4. dorsally exophytic: may be an extension of "focal" tumors (see above). Many of these may actually be low grade gliomas including:
 - a) pilocytic astrocytomas (p.689)
 - b) gangliogliomas (p.706): very rare. Compared to other BSGs, these patients tend to be slightly older and the medulla is involved more frequently³⁶

Evaluation

MRI

The diagnostic test of choice. MRI evaluates status of ventricles, gives optimal assessment of tumor (CT is poor in the posterior fossa) and detects exophytic component. See ► Fig. 38.6. T1WI: almost all are hypointense, homogeneous (excluding cysts). T2WI: increased signal, homogeneous (excluding cysts). Gadolinium enhancement is highly variable.³⁴

CT

Most do not enhance on CT, except possibly an exophytic component. If there is marked enhancement, consider other diagnoses (e.g., high-grade vermian astrocytoma).

Treatment

Surgery

Biopsy: biopsy for diffuse non-exophytic infiltrating tumors was discouraged³⁷; however, changing trends are leaning towards biopsy even when the MRI shows a diffuse infiltrating brainstem lesion (to look for H3 K27 M mutation (p.683)).³⁸

Treatment is usually non-surgical. Exceptions where surgery may be indicated:

1. tumors with a dorsally exophytic component¹⁰: see below; these may protrude into 4th ventricle or CP angle, tend to enhance with IV contrast, tend to be lower grade
2. some success has been achieved with non-exophytic tumors that are *not* malignant astrocytomas (surgery in malignant astrocytomas is without benefit)³⁵ (detailed follow-up is lacking)
3. shunting for hydrocephalus

Stereotactic biopsy can be transfrontal, or for lesions entering the cerebellar peduncle a transcerebellar approach with the patient positioned on their side may be used.

Dorsally exophytic tumors

These tumors are generally histologically benign (e.g., gangliogliomas) and are amenable to radical subtotal resection. Prolonged survival is possible, with a low incidence of disease progression at short-term follow-up.¹⁰

Surgical goals in exophytic tumors include:

1. enhanced survival by subtotal removal of exophytic component³⁹: broad attachment to the floor of 4th ventricle is typical and usually precludes complete excision (although some "safe entry" zones have been described⁴⁰). An ultrasonic aspirator facilitates debulking
2. establishing diagnosis: radiographic differentiation of exophytic brainstem glioma tumors from other lesions (e.g., medulloblastoma, ependymoma, and dermoids) may be difficult

3. tumors that demonstrate recurrent growth after resection remained histologically benign and were amenable to re-resection¹⁰

Complications of surgery generally consisted of exacerbation of preoperative symptoms (ataxia, cranial nerve palsies...) which usually resolved with time.

Medical

No proven chemotherapeutic regimen. Steroids are usually administered. In pediatrics, there is some indication of response to Temodar® (temozolomide) (p. 627).

Radiation

Traditionally given as 45–55 Gy over a six week period, five days per week. When combined with steroids, symptomatic improvement occurs in 80% of patients.

Possible improved survival with so called “hyperfractionation” where multiple smaller doses per day are used.

Prognosis

Most children with malignant BSG will die within 6–12 months of diagnosis. XRT may not prolong survival in patients with grade 3 or 4 tumors. A subgroup of children have a more slowly growing tumor and may have up to 50% five-year survival. Dorsally exophytic tumors composed of pilocytic astrocytomas may have a better prognosis.

Tectal gliomas

General information

A typically defined diagnosis generally consisting of low-grade astrocytomas. Considered a benign subgroup of brainstem glioma. Because of location, tends to present with hydrocephalus. Was dramatically referred to as “the smallest tumor in the body that can lead to the death of the patient.”⁴¹ Focal neurologic findings are rare—diplopia, visual field deficits, nystagmus, Parinaud’s syndrome (p. 101), ataxia, seizures...—and are often reversible after the hydrocephalus is corrected.

Epidemiology

Comprises ≈ 6% of surgically treated pediatric brain tumors.⁴² Presents primarily in childhood. Median age of patients becoming symptomatic = 6–14 years.⁴²

Pathology

Since many of these are not biopsied, meaningful statistical analysis is not possible. Pathologies identified include: WHO grade 2 diffuse astrocytoma, pilocytic astrocytomas, WHO grade 2 ependymoma, anaplastic astrocytoma, oligodendroglioma, and oligoastrocytoma.

Radiographic evaluation

CT scan detects hydrocephalus, but may miss the tumor in ≈ 50%.⁴³ Calcification on CT has been described in 9–25%.^{43,44}

MRI is the study of choice for diagnosis and follow-up. Typically appears as a mass projecting dorsally from the quadrigeminal plate. Isointense on T1WI, iso- or hyperintense on T2WI.^{42,45} Enhancement with gadolinium occurs in 18% and is of uncertain prognostic significance.

Treatment

General information

Due to the indolent course, open surgery is not recommended. Options include:

1. VP shunt: the standard treatment for years. Long-term results are good with a functioning shunt
2. endoscopic third ventriculostomy: may avoid the need for a shunt. Endoscopic biopsy⁴⁶ may be done at the same time through the same burr hole if it is technically feasible (requires a dilated foramen of Monro, which is often present). Long-term results unknown
3. endoscopic aqueductoplasty (with or without stenting): an option for some. Long-term results unknown

Stereotactic radiosurgery: May be offered for tumor progression (criteria are not defined: radiographic progression may not be associated with clinical deterioration⁴⁵). Dosing should be limited to ≤ 14 Gray at the 50–70% isodose line to avoid radiation-induced side effects.⁴⁷

Prognosis

Tumor progression: described in 15–25%.

Follow-up: no accepted guidelines. Serial neurologic exams and MRIs every 6–12 months has been suggested.⁴²

39.2.2 High-grade astrocytoma with piloid features (HGAP) (WHO grade N/A)

General information

A rare astrocytoma that often shows high-grade piloid and/or glioblastoma-like features, possessing a distinct DNA methylation profile. These tumors may arise de novo, but they can also develop from lower grade astrocytomas, including pilocytic astrocytoma. Prognostic information is limited, with a 5-year OS of 50%.⁴⁸

Diagnostic criteria for high-grade astrocytoma with piloid features

Diagnostic criteria for high-grade astrocytoma with piloid features are shown in ► Fig. 39.4.¹

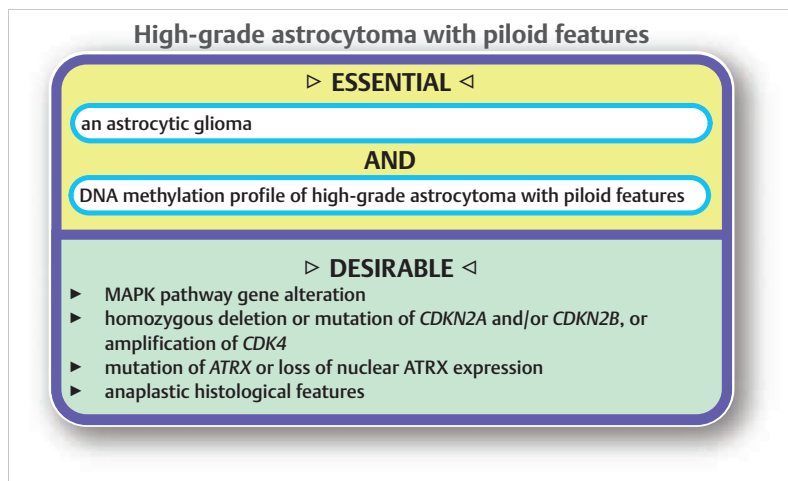


Fig. 39.4 Diagnostic criteria for high-grade astrocytoma with piloid features. (Modified with permission from WHO Classification of Tumors Editorial Board, (ed.) World Health Organization classification of tumors of the central nervous system. 5th ed. Bosman F T, Jaffe E S, Lakhani S R, et al. (series eds.): World Health Organization Classification of Tumors series, vol. 6. International Agency for Research on Cancer, Lyon, France, 2021.)

39.2.3 Pleomorphic xanthoastrocytoma (PXA) (WHO grade 2 or 3)

General information

Key concepts

- a glioma (usually WHO grade 2), possibly from subpial astrocytes → superficial location involving the leptomeninges in > 67%
- 98% supratentorial, most common in children or young adults
- mural nodule with cystic component in 25%
- histology: large pleomorphic cells (xanthomatous [lipid-laden] cells, fibrillary and giant multinucleated astrocytes) often containing eosinophilic granular bodies and reticulin deposition
- characteristically with MAPK pathway gene alterations

- usually circumscribed, occasionally invasive
- WHO grade 2 (if ≥ 5 mitoses per HPF it qualifies as anaplastic, WHO grade 3)
- treatment: maximal safe resection

A low-grade glioma thought to arise from subpial astrocytes, which may explain their superficial location and abundance of reticulin fibers. Over 90% are supratentorial. Predilection for temporal lobes (50%), followed by parietal, occipital & frontal lobes. Most have a cystic component (may be multiloculated, but >90% have a large, single cyst).

Diagnostic criteria for pleomorphic xanthoastrocytoma

Diagnostic criteria for pleomorphic xanthoastrocytoma are shown in ► Fig. 39.5.¹

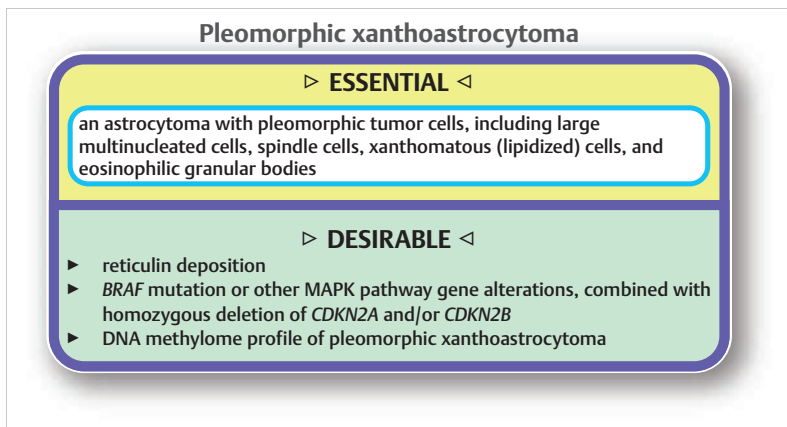


Fig. 39.5 Diagnostic criteria for pleomorphic xanthoastrocytoma. (Modified with permission from WHO Classification of Tumors Editorial Board, (ed.) World Health Organization classification of tumors of the central nervous system. 5th ed. Bosman F T, Jaffe E S, Lakhani S R, et al. (series eds.): World Health Organization Classification of Tumors series, vol. 6. International Agency for Research on Cancer, Lyon, France, 2021.)

Epidemiology

≈ 1% of astrocytomas. Usually occurs in children or young adults (most are < 18 years of age). No gender difference. High prevalence of MAPK pathway alterations in PXA may explain the association with NF1.

Clinical

Usual presentation: seizures. May also produce focal deficit or increased ICP.

Differential diagnosis

1. imaging: meningioma is also superficial with dural tail. PXA may also resemble a low grade diffuse astrocytoma
2. pathology: may be confused with anaplastic astrocytoma

Histology

A compact, superficial tumor, usually circumscribed, occasionally infiltrates cortex. Marked cellular pleomorphism (fibrillary and giant multinucleated astrocytes, large xanthomatous (lipid laden) may cause these tumors to be mistaken for anaplastic astrocytoma. Cells stain for GFAP, bespeaking glial

origin. Abundant eosinophilic granular bodies, reticulin and frequent perivascular chronic inflammatory cells. The reticulin fibers surround two cell types:

1. spindle cells: fusiform cell shape with elongated nuclei
2. pleomorphic cells: round cells with heterochromic, pleomorphic nuclei that may be mononucleated or multinucleated. Variable intracellular lipid content

Grading

Most PXAs are WHO grade 2 with an MIB typically <1%. A high mitotic index (≥ 5 mitoses per 10 HPF) qualifies as WHO grade 3.⁴⁹ Necrosis is common in grade 3 PXAs, but the significance of necrosis when encountered alone is not determined.

Dedifferentiation

PXAs have a tendency to recur and undergo anaplastic change.⁵⁰ There have also been several reported cases of malignant transformation to anaplastic astrocytoma or glioblastoma.⁵¹

Molecular genetics

Almost all PXAs demonstrate abnormal activation of MAPK pathway due to genetic alterations. The most common alteration is in *BRAF* V600E, and when seen in the absence of IDH mutation is strongly supportive of the diagnosis.

Other MAPK pathway gene alterations include homozygous *CDKN2A* and/or *CDKN2B* deletion.

Imaging

The cyst, when present, may partially enhance on CT or MRI. A mural nodule is present in 25%. May have "dural tail" (67% show leptomeningeal involvement, 13% show involvement of all 3 meningeal layers). Peritumoral edema may be mild to moderate, calcifications are rare.⁵²

CT: solid portion of tumor is ill-defined and may be isodense to gray matter.

MRI: T1WI: hypointense cystic component with ill-defined isointense solid component that strongly enhances with gadolinium. T2WI: hyperintense cystic component with ill-defined isointense solid component.

Since PXAs tend to disseminate when they progress, spine MRI is recommended at the time of clinical progression.

Treatment

1. surgery: primary treatment
 - a) gross total resection if it can be accomplished without unacceptable neurologic deficit, otherwise subtotal resection
 - b) extent of resection: most strongly associated with recurrence free survival⁵³
 - c) incomplete resections should be followed since these tumors may grow very slowly over many years before retreatment is necessary, and repeat excision should be considered
2. radiation therapy: controversial
 - a) literature suggests either no difference in overall survival or possibly a trend toward prolonged survival⁵⁴
 - b) considered with: residual disease, high mitotic index, or necrosis
3. chemotherapy: role not defined

Prognosis

PXAs frequently recur and malignant progression is more common than other WHO CNS5 grade 1 or 2 gliomas in young patients.

Grade 3 PXAs fare much worse than grade 2, with 5-year recurrence-free survival of 71% for grade 2 PXAs, and 49% for grade 3 ($P = .0003$).⁵⁵ OS is 54 months for grade 2 vs. 33 for grade 3.⁵⁶

Extent of resection, mitotic index, and necrosis appear to be the best predictors of outcome.^{52,53}

39.2.4 Subependymal giant cell astrocytoma (SEGA) (WHO grade 1)

General information

A circumscribed and often calcified WHO grade 1 tumor having a strong association with tuberous sclerosis (p.644) (it is unclear if SEGAs develop in the absence of this condition).

Diagnostic criteria for subependymal giant cell astrocytoma

Diagnostic criteria for subependymal giant cell astrocytoma are shown in ► Fig. 39.6.¹

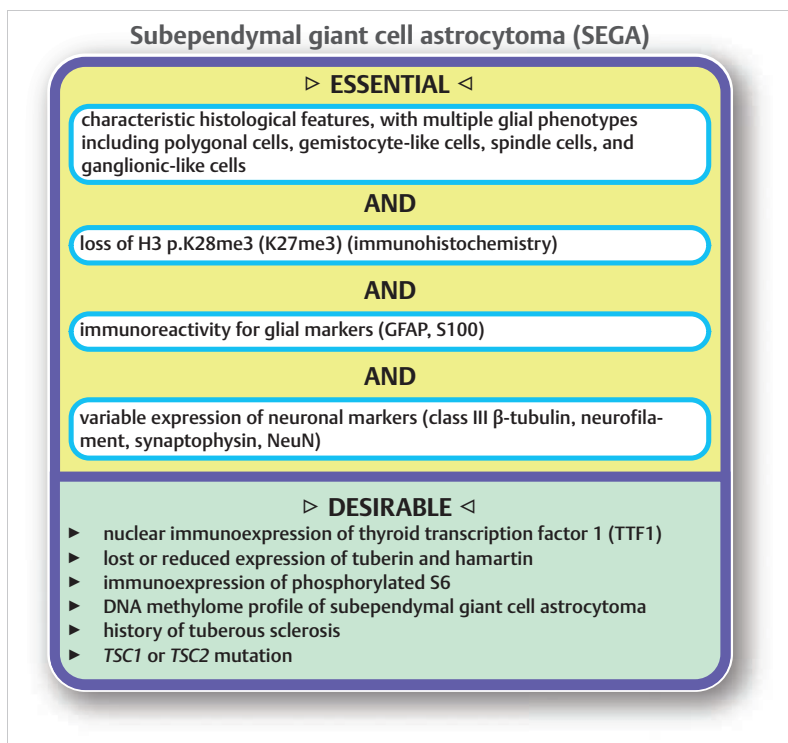


Fig. 39.6 Diagnostic criteria for subependymal giant cell astrocytoma. (Modified with permission from WHO Classification of Tumors Editorial Board, (ed.) World Health Organization classification of tumors of the central nervous system. 5th ed. Bosman F T, Jaffe E S, Lakhani S R, et al. (series eds.): World Health Organization Classification of Tumors series, vol. 6. International Agency for Research on Cancer, Lyon, France, 2021.)

Epidemiology & genetics

Incidence in patients with tuberous sclerosis is 5-15%. SEGAs rarely occur de novo after age $\approx$ 25 years.

SEGAs typically possess biallelic inactivation of *TSC1* (15%) or *TSC2* (56%) with the second hit being deletion or loss of heterozygosity. Deficient tuberin and hamartin expression are observed with these mutations.

Location

Arise from the lateral walls of the lateral ventricles next to the foramen of Monro. Rarely found in the third ventricle.

Histology

A wide variety of phenotypes occur. Large, plump cells resembling gemistocytic astrocytes are common, arrayed in fascicles, sheets and nests. Polygonal cells with ample cytoplasm may be seen. Areas

of necrosis and mitotic figures may be seen, but are not associated with the typical malignant aggressiveness that these features usually denote.⁵⁷ Mean Ki-67 (p.632) is 3%.

Prognosis

Gross total resection is associated with a favorable prognosis. Larger lesions or symptomatic lesions have increased risk of surgical morbidity.⁵⁸

Everolimus may reduce tumor size and progression.⁵⁹

Because of the risk of late recurrence, long-term follow-up is recommended.

39.2.5 Chordoid glioma (WHO grade 2)

General information

A glioma that occurs in the anterior third ventricle, possibly arising from tanycyte ependymal cells of the organum vasculosum of the lamina terminalis.

Diagnostic criteria for chordoid glioma

Diagnostic criteria for chordoid glioma are shown in ► Fig. 39.7.¹

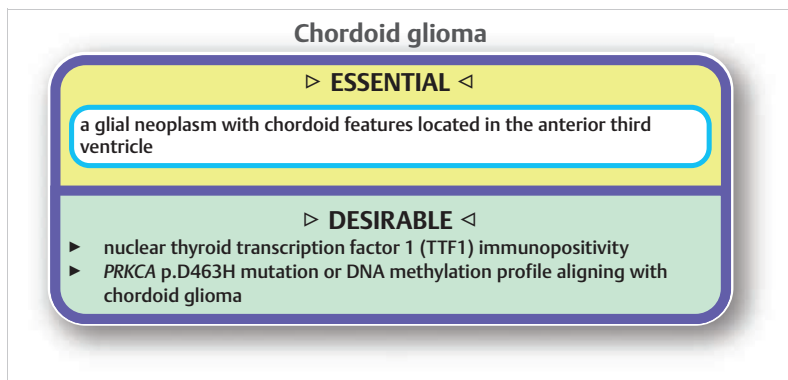


Fig. 39.7 Diagnostic criteria for chordoid glioma. (Modified with permission from WHO Classification of Tumors Editorial Board, (ed.) World Health Organization classification of tumors of the central nervous system. 5th ed. Bosman F T, Jaffe E S, Lakhani S R, et al. (series eds.): World Health Organization Classification of Tumors series, vol. 6. International Agency for Research on Cancer, Lyon, France, 2021.)

Epidemiology & genetics

Comprise <0.1% of primary brain tumors, typically occurring in adults. Female preponderance of 2:1.⁶⁰

Presentation

- obstructive hydrocephalus: with typical symptoms of H/A, N/V, obtundation
- endocrine disturbance (hypothyroidism, amenorrhea, diabetes insipidus) from compression of the hypothalamus visual field deficit from compression of the chiasm
- psychiatric symptoms: personality changes, memory disturbance...

Imaging

MRI shows a well-circumscribed mass isotintense to brain on T1 images with avid enhancement, located in the anterior third ventricle (large tumors may occupy the entire third ventricle).

Histology

Typically comprised of clusters and cords of epithelioid cells embedded in a mucinous stroma. GFAP immunostaining is common. Mitotic activity is absent in most.

Treatment

Maximal safe resection is the primary treatment modality. Attachment to the wall of the 3rd ventricle (hypothalamus) may prevent total removal. Surgical risks include insufficiency of hypothalamic hormones, including diabetes insipidus. Post-op adjuvant therapy with XRT is of uncertain benefit.

39.2.6 Astroblastoma, MN1-altered (WHO grade 1)

General information

A circumscribed glial tumor with MN1 alteration.

Diagnostic criteria for astroblastoma, MN1-altered

Diagnostic criteria for astroblastoma, MN1-altered are shown in ► Fig. 39.8.¹

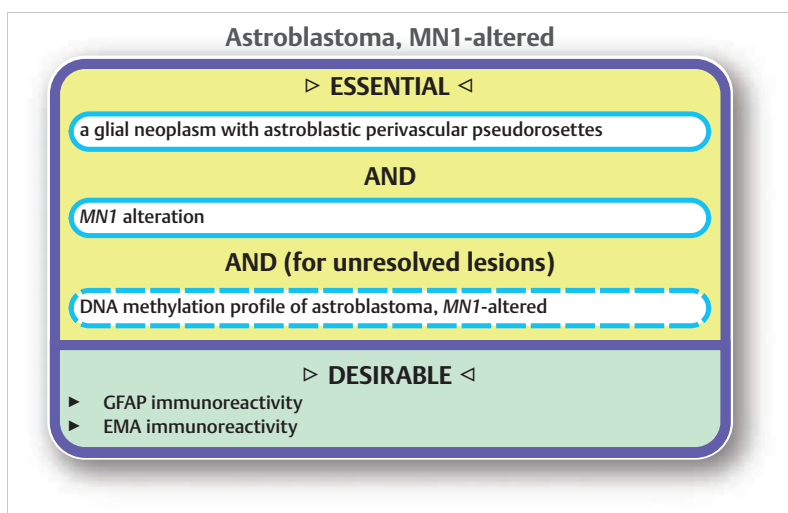


Fig. 39.8 Diagnostic criteria for astroblastoma, MN1-altered. (Modified with permission from WHO Classification of Tumors Editorial Board, (ed.) World Health Organization classification of tumors of the central nervous system. 5th ed. Bosman F T, Jaffe E S, Lakhani S R, et al. (series eds.): World Health Organization Classification of Tumors series, vol. 6. International Agency for Research on Cancer, Lyon, France, 2021.)

Epidemiology & genetics

Occurs almost exclusively in females.⁶¹ Median age of presentation is 15 years (range: 3 months to 40 years).

Location

Primarily in the cerebral hemispheres, typically in the frontal or parietal lobes, but also may occur in the occipital or temporal lobes. Has also been described in the brainstem, intraventricular compartment and spinal cord.

Imaging

Solid or cystic mass that is iso- to hypo-intense to brain on T1 MRI, hyperintense on T2, and heterogeneous enhancement.

Histology

Hallmark is the astroblastic pseudorosette (tumor cells arranged in a radial pattern around blood vessels).

Treatment and prognosis

Maximal safe resection is associated with longer OS.⁶² XRT and chemotherapy are options when surgery is not feasible.

5- and 10-year survival rates are 90% and 50% respectively.

References

- [1] Bosman FT, Jaffe ES, Lakhani SR, et al. World Health Organization Classification of Tumors series, vol. 6. In: WHO Classification of Tumors Editorial Board (ed.) World Health Organization classification of tumors of the central nervous system. 5th ed. Lyon, France: International Agency for Research on Cancer; 2021
- [2] Tihan T, Fisher PG, Kepner JL, et al. Pediatric astrocytomas with monomorphous pilomyxoid features and a less favorable outcome. *Journal of Neuropathology and Experimental Neurology*. 1999; 58:1061–1068
- [3] Arulrajah S, Huisman TA. Pilomyxoid astrocytoma of the spinal cord with cerebrospinal fluid and peritoneal metastasis. *Neuropediatrics*. 2008; 39:243–245
- [4] Louis DN, Ohgaki H, Wiestler OD, et al. WHO classification of tumors of the central nervous system. Lyon, France 2016
- [5] Ostrom QT, Gittleman H, Liao P, et al. CBTRUS statistical report: primary brain and central nervous system tumors diagnosed in the United States in 2007–2011. *Neuro Oncol*. 2014; 16 Suppl 4:iv1–i63
- [6] Wallner KE, Gonzales MF, Edwards MSB, et al. Treatment of juvenile pilocytic astrocytoma. *Journal of Neurosurgery*. 1988; 69:171–176
- [7] Burkhard C, Di Patre PL, Schuler D, et al. A population-based study of the incidence and survival rates in patients with pilocytic astrocytoma. *J Neurosurg*. 2003; 98:1170–1174
- [8] Burger PC, Scheithauer BW. *Atlas of Tumor Pathology, Tumors of the Central Nervous System*. Washington, D.C.: Armed Forces Institute of Pathology; 1994
- [9] Collins VP, Jones DT, Giannini C. Pilocytic astrocytoma: pathology, molecular mechanisms and markers. *Acta Neuropathol*. 2015; 129:775–788
- [10] Pollack IF, Hoffman HJ, Humphreys RP, et al. The Long-Term Outcome After Surgical Treatment of Dorsally Exophytic Brain-Stem Gliomas. *Journal of Neurosurgery*. 1993; 78:859–863
- [11] Park SH, Won J, Kim SI, et al. Molecular Testing of Brain Tumor. *J Pathol Transl Med*. 2017; 51:205–223
- [12] Listerink R, Chawrow J, Greenwald M, et al. Natural history of optic pathway tumors in children with neurofibromatosis type 1: a longitudinal study. *J Pediatr*. 1994; 125:63–66
- [13] Gutmann DH, McLellan MD, Hussain I, et al. Somatic neurofibromatosis type 1 (NF1) inactivation characterizes NF1-associated pilocytic astrocytoma. *Genome Res*. 2013; 23:431–439
- [14] Gutmann DH, McLellan MD, Hussain I, et al. Somatic neurofibromatosis type 1 (NF1) inactivation characterizes NF1-associated pilocytic astrocytoma. *Genome Res*. 2013; 23:431–439
- [15] Fiechter M, Hewer E, Knecht U, et al. Adult anaplastic pilocytic astrocytoma – a diagnostic challenge? A case series and literature review. *Clin Neurol Neurosurg*. 2016; 147:98–104
- [16] Coakley KJ, Huston J, Scheithauer BW, et al. Pilocytic Astrocytomas: Well-Demarcated Magnetic Resonance Appearance Despite Frequent Infiltration Histologically. *Mayo Clinic Proceedings*. 1995; 70:747–751
- [17] Hayostek CJ, Shaw EG, Scheithauer B, et al. Astrocytomas of the Cerebellum: A Comparative Clinicopathologic Study of Pilocytic and Diffuse Astrocytomas. *Cancer*. 1993; 72:856–869
- [18] Schwartz AM, Ghatikar NR. Malignant Transformation of Benign Cerebellar Astrocytoma. *Cancer*. 1990; 65:333–336
- [19] Bernell WR, Kepes JJ, Seitz EP. Late Malignant Recurrence of Childhood Cerebellar Astrocytoma. *Journal of Neurosurgery*. 1972; 37:470–474
- [20] Beni-Adani L, Gomori M, Spektor S, et al. Cyst wall enhancement in pilocytic astrocytoma: neoplastic or reactive phenomena. *Pediatr Neurosurg*. 2000; 32:234–239
- [21] Austin EJ, Alvord EC. Recurrences of Cerebellar Astrocytomas: A Violation of Collins' Law. *Journal of Neurosurgery*. 1988; 68:41–47
- [22] Packer RJ, Lange B, Ater J, et al. Carboplatin and Vincristine for Recurrent and Newly Diagnosed Low-Grade Gliomas of Childhood. *J Clin Oncol*. 1993; 11:850–856
- [23] Bucy PC, Thieman PW. Astrocytomas of the Cerebellum. A Study of Patients Operated Upon Over 28 Years Ago. *Archives of Neurology*. 1968; 18:14–19
- [24] Stein BM, Tenner MS, Fraser RAR. Hydrocephalus Following Removal of Cerebellar Astrocytomas in Children. *Journal of Neurosurgery*. 1972; 36:763–768
- [25] Section of Pediatric Neurosurgery of the American Association of Neurological Surgeons. *Pediatric Neurosurgery*. New York 1982
- [26] Youmans JR. *Neurological Surgery*. Philadelphia 1990
- [27] Ringertz N, Nordenstam H. Cerebellar Astrocytoma. *Journal of Neuropathology and Experimental Neurology*. 1951; 10:343–367
- [28] Zimmerman RA, Bilaniuk CT, Bruno LA, et al. CT of Cerebellar Astrocytoma. *Am J Roentgenol*. 1978; 130:929–933
- [29] Schreck KC, Grossman SA, Pratilas CA. BRAF Mutations and the Utility of RAF and MEK Inhibitors in Primary Brain Tumors. *Cancers (Basel)*. 2019; 11. DOI: 10.3390/cancers11091262
- [30] Binning MJ, Liu JK, Kestle JR, et al. Optic pathway gliomas: a review. *Neurosurg Focus*. 2007; 23. DOI: 10.3171/FOC-07/11/E2
- [31] Packer RJ, Nicholson HS, Vezina LG, et al. Brainstem gliomas. *Neurosurg Clin N Am*. 1992; 3:863–879
- [32] Laurent JP, Cheek WR. Brain Tumors in Children. *J Pediatr Neurosci*. 1985; 1:15–32
- [33] Reigel DH, Scarff TB, Woodford JE. Biopsy of Pediatric Brain Stem Tumors. *Childs Brain*. 1979; 5:329–340
- [34] Epstein FJ, Farmer J-P. Brain-Stem Glioma Growth Patterns. *J Neurosurg*. 1993; 78:408–412

- [35] Epstein F, McCleary EL. Intrinsic Brain-Stem Tumors of Childhood: Surgical Indications. *J Neurosurg.* 1986; 64:11–15
- [36] Garcia CA, McGarry PA, Collada M. Ganglioglioma of the Brain Stem. Case Report. *Journal of Neurosurgery.* 1984; 60:431–434
- [37] Albright AL, Packer RJ, Zimmerman R, et al. Magnetic Resonance Scans Should Replace Biopsies for the Diagnosis of Diffuse Brain Stem Gliomas: A Report from the Children's Cancer Group. *Neurosurgery.* 1993; 33:1026–1030
- [38] Mohanty A. Biopsy of brain stem gliomas: Changing trends? *J Neurosci Rural Pract.* 2014; 5:116–117
- [39] Hoffman HJ, Becker L, Craven MA. A Clinically and Pathologically Distinct Group of Benign Brainstem Gliomas. *Neurosurgery.* 1980; 7:243–248
- [40] Kyoshima K, Kobayashi S, Gibo H, et al. A Study of Safe Entry Zones via the Floor of the Fourth Ventricle for Brain-Stem Lesions. *Journal of Neurosurgery.* 1993; 78:987–993
- [41] Kernohan WJ, Armed Forces Institute of Pathology. Tumors of the central nervous system. In: *Atlas of Tumor Pathology.* Washington, DC 1952:19–42
- [42] Stark AM, Fritsch MJ, Claviez A, et al. Management of tectal glioma in childhood. *Pediatr Neurol.* 2005
- [43] Bogner L, Turjman F, Villanyi E, et al. Tectal plate gliomas. Part II: CT scans and MR imaging of tectal gliomas. *Acta Neurochir (Wien).* 1994; 127:48–54
- [44] Pollack IF, Pang D, Albright AL. The long-term outcome in children with late-onset aqueductal stenosis resulting from benign intrinsic tectal tumors. *J Neurosurg.* 1994; 80:681–688
- [45] Grant GA, Avellino AM, Loeser JD, et al. Management of intrinsic gliomas of the tectal plate in children. A ten-year review. *Pediatr Neurosurg.* 1999; 31:170–176
- [46] Oka K, Kin Y, Go Y, et al. Neuroendoscopic approach to tectal tumors: a consecutive series. *J Neurosurg.* 1999; 91:964–970
- [47] Kihlstrom L, Lindquist C, Lindquist M, et al. Stereotactic radiosurgery for tectal low-grade gliomas. *Acta Neurochir Suppl.* 1994; 62:55–57
- [48] Reinhardt A, Stichel D, Schrimpf D, et al. Anaplastic astrocytoma with piloid features, a novel molecular class of IDH wildtype glioma with recurrent MAPK pathway, CDKN2A/B and ATRX alterations. *Acta Neuropathol.* 2018; 136:273–291
- [49] Fouladi M, Jenkins J, Burger P, et al. Pleomorphic xanthoastrocytoma: favorable outcome after complete surgical resection. *Neuro-oncol.* 2001; 3:184–192
- [50] Weldon-Linne CM, Victor TA, Groothuis DR, et al. Pleomorphic Xanthoastrocytoma: Ultrastructural and Immunohistochemical Study of a Case with a Rapidly Fatal Outcome Following Surgery. *Cancer.* 1983; 52:2055–2063
- [51] Kumar S, Retnam TM, Menon G, et al. Cerebellar hemisphere, an uncommon location for pleomorphic xanthoastrocytoma and lipidized glioblastoma multiformis. *Neurol India.* 2003; 51:246–247
- [52] Pahapill PA, Ramsay DA, Del Maestro RF. Pleomorphic xanthoastrocytoma: case report and analysis of the literature concerning the efficacy of resection and the significance of necrosis. *Neurosurgery.* 1996; 38:822–8; discussion 828–9
- [53] Giannini C, Scheithauer BW, Burger PC, et al. Pleomorphic xanthoastrocytoma: what do we really know about it? *Cancer.* 1999; 85:2033–2045
- [54] Kepes JJ, Rubinstein LJ, Eng LF. Pleomorphic Xanthoastrocytoma: A Distinctive Meningeal Glioma of Young Subjects with Relatively Favorable Prognosis. A Study of 12 Cases. *Cancer.* 1979; 44:1839–1852
- [55] Ida CM, Rodriguez FJ, Burger PC, et al. Pleomorphic Xanthoastrocytoma: Natural History and Long-Term Follow-Up. *Brain Pathol.* 2015; 25:575–586
- [56] Gandham EJ, Goyal-Honavar A, Beno D, et al. Impact of Grade on Survival in Pleomorphic Xanthoastrocytoma and Low Prevalence of BRAF V600E Mutation. *World Neurosurg.* 2022; 164:e922–e928
- [57] Chow CW, Klug GL, Lewis EA. Subependymal Giant-Cell Astrocytoma in Children: An Unusual Discrepancy Between Histological and Clinical Features. *J Neurosurg.* 1988; 68:880–883
- [58] de Ribaupierre S, Dorfmueller G, Bulteau C, et al. Subependymal giant-cell astrocytomas in pediatric tuberous sclerosis disease: when should we operate? *Neurosurgery.* 2007; 60:83–89; discussion 89–90
- [59] Franz DN, Belousova E, Sparagana S, et al. Long-Term Use of Everolimus in Patients with Tuberous Sclerosis Complex: Final Results from the EXIST-1 Study. *PLoS One.* 2016; 11. DOI: 10.1371/journal.pone.0158476
- [60] Ampie L, Choy W, Lamano JB, et al. Prognostic factors for recurrence and complications in the surgical management of primary chordoid gliomas: A systematic review of literature. *Clin Neurol Neurosurg.* 2015; 138:129–136
- [61] Chen W, Soon YY, Pratisneyo PD, et al. Central nervous system neuroepithelial tumors with MN1-alteration: an individual patient data meta-analysis of 73 cases. *Brain Tumor Pathol.* 2020; 37:145–153
- [62] Tauziède-Espariat A, Pages M, Roux A, et al. Pediatric methylation class HGNM1: unresolved issues with terminology and grading. *Acta Neuropathol Commun.* 2019; 7. DOI: 10.1186/s40478-019-0834-z

40 Glioneuronal and Neuronal Tumors

40.1 Glioneuronal tumors

40.1.1 Ganglioglioma (WHO grade 1)

General information

Key concepts

- a well-circumscribed tumor composed of two cell types: ganglion cells (neurons) and glial cells
- extremely rare (<2% of intracranial neoplasms)
- seen primarily in the first 3 decades of life
- characterized by slow growth and a tendency to calcify

A very slow-growing well-differentiated (WHO grade 1) glioneuronal tumor that primarily affects children and young adults. Composed of two types of cells, dysplastic ganglion cells (neurons) in conjunction with neoplastic glial cells (usually astrocytic in any phase of differentiation).¹ The term “ganglioglioma” was introduced in 1930 by Courville.²

An anaplastic ganglioglioma may exist in which the glial component exhibits anaplasia in the form of pronounced mitotic activity with a high Ki-67 proliferation index, necrosis and microvascular proliferation, however, most studies have lacked molecular analysis to exclude other tumor types, and refined data is therefore needed.

Diagnostic criteria for ganglioglioma

Diagnostic criteria for ganglioglioma are shown in ► Fig. 40.1.³

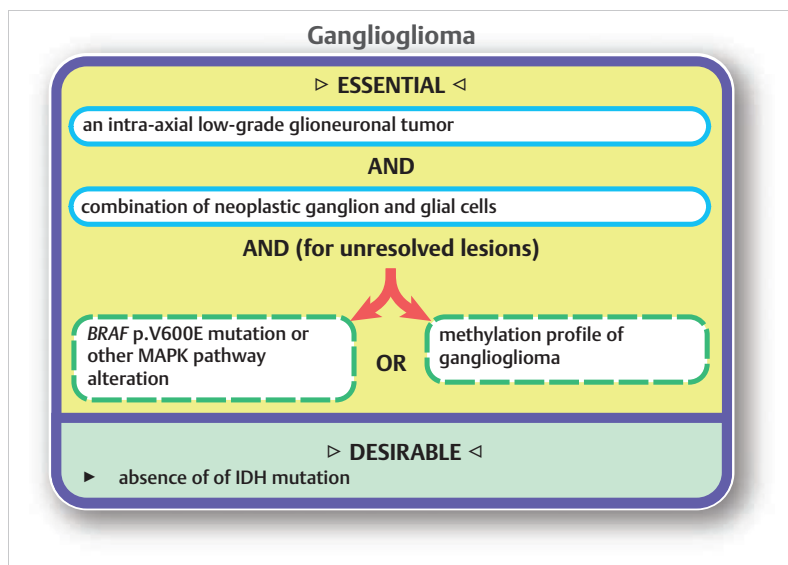


Fig. 40.1 Diagnostic criteria for ganglioglioma. (Modified with permission from WHO Classification of Tumors Editorial Board, (ed.) World Health Organization classification of tumors of the central nervous system. 5th ed. Bosman F T, Jaffe E S, Lakhani S R, et al. (series eds.): World Health Organization Classification of Tumors series, vol. 6. International Agency for Research on Cancer, Lyon, France, 2021.)

Epidemiology

Incidence: Typically quoted⁴ as 0.3–0.6%. One series⁵ found gangliogliomas in 1.3% of all brain tumors (including mets), or 3% of primary brain tumors.

Demographics: Occurs primarily in children and young adults (peak age of occurrence: 11 yrs). Considering only children and young adults, incidence ranges from 1.2–7.6% of brain tumors.⁴

Location

May occur in various parts of the nervous system (cerebral hemispheres, spinal cord, brainstem, cerebellum, pineal region, thalamus, intrasellar, optic nerve, and peripheral nerve have been reported⁴). Most occur above the tentorium, primarily in the temporal lobe (> 70%) or near the 3rd ventricle, in the hypothalamus or in the frontal lobes.⁶ Brainstem gangliogliomas (p.696) occur rarely.

Histology

Mixture of 2 types of neoplastic cells: neuronal (ganglion) and astrocytic (glial).

Grossly: white matter mass, well-circumscribed, firm, with occasional cystic areas and calcified regions. Most dissect easily from brain, but the solid portion may show an infiltrative tendency.⁴

Microscopically: ganglion cells demonstrate nerve cell differentiation, e.g., Nissl substance and axons or dendrites. Pitfall: differentiating neoplastic neurons from neurons entrapped by an invading astrocytoma may be difficult. Also, neoplastic astrocytes may resemble neurons on light microscopy. 2 of 10 patients had areas of oligodendroglioma. Also seen: necrotic areas, minimal calcification, and Rosenthal bodies.⁷

Radiologic evaluation

Neuroradiologic findings are not specific for this tumor.

CT: all of 10 patients had a low density lesion on non-contrast CT.⁵ Tumor calcification are best appreciated on CT and are seen in 30%. Frequently appears cystic on CT, but still may be found to be solid at operation.

MRI: T1WI low to iso-intense with variable enhancement. T2WI hyperintense. Calcifications appear as low signal on both.

CT and MRI: the temporal lobe is favored. Mass effect rare (suggests slow growth). Enhancement varies from none to intense.

Plain skull X-ray: calcification may be noted.

Treatment

Recommendation is maximal safe resection. Close follow-up is recommended, and re-resection should be considered for recurrence. The role of XRT is unknown, and due to the deleterious effects together with the good long-term prognosis, it is not recommended initially but may be considered for recurrence.⁸

Prognosis

These benign tumors have a 97% 7.5-year recurrence-free survival rate.

Prognosis is better with temporal lobe location, gross total excision, and chronic epilepsy.

The correlation between histologic anaplasia and outcome is inconsistent.

The prognosis following subtotal resection of brainstem gangliogliomas is better than for brainstem gliomas as a group.⁶

40.1.2 Gangliocytoma (WHO grade 1)

General information

A rare, slow-growing well-differentiated (WHO grade 1) tumor that primarily affects children, characterized by clusters of neoplastic ganglion cells, frequently with dysplastic features.

Diagnostic criteria for gangliocytoma

Diagnostic criteria for gangliocytoma are shown in ► Fig. 40.2.³

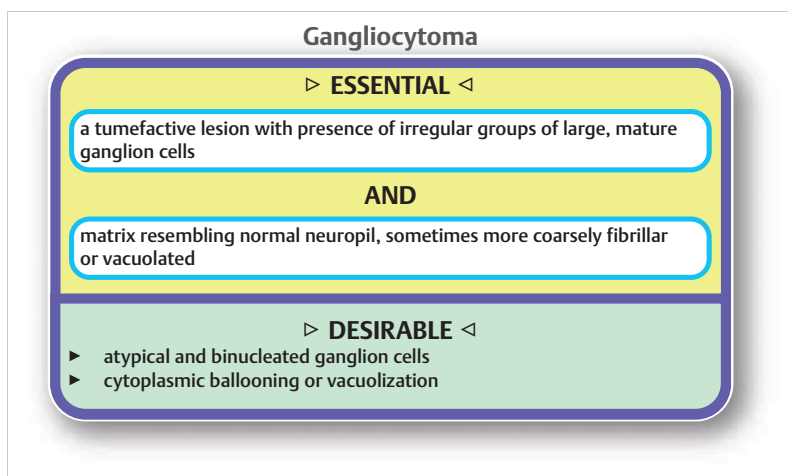


Fig. 40.2 Diagnostic criteria for gangliocytoma. (Modified with permission from WHO Classification of Tumors Editorial Board, (ed.) World Health Organization classification of tumors of the central nervous system. 5th ed. Bosman F T, Jaffe E S, Lakhani S R, et al. (series eds.): World Health Organization Classification of Tumors series, vol. 6. International Agency for Research on Cancer, Lyon, France, 2021.)

Location

As with ganglioglioma, gangliocytomas can occur anywhere in the CNS. Most occur above the tentorium, primarily in the temporal lobe (> 70%) or near the 3rd ventricle, in the hypothalamus or in the frontal lobes.⁶ They have been described in the pituitary.

Prognosis

Gangliocytomas are benign tumors with generally good prognosis. No specific prognostic indicators have been identified.

40.1.3 Desmoplastic infantile ganglioglioma (DIG) & desmoplastic infantile astrocytoma (DIA) (WHO grade 1)

General information

DIA & DIG are benign (WHO grade 1) glioneural tumor with a neuroepithelial population (restricted to either neoplastic astrocytes only (DIA) or to astrocytes with a variably mature neuronal component (DIG)) in a desmoplastic stroma.

DIA & DIG usually occur primarily in the cerebral hemispheres of infants, often as a large cystic lesion, frequently attached to dura. They are driven by MAPK pathway alteration.

These tumors rarely seed through the CSF.⁹

Diagnostic criteria for desmoplastic infantile ganglioglioma & desmoplastic infantile astrocytoma

Diagnostic criteria for desmoplastic infantile ganglioglioma & desmoplastic infantile astrocytoma are shown in ► Fig. 40.3.³

Prognosis

Prognosis is usually favorable with gross total resection, with no relapse in 5-15 years of follow-up. They can rarely recur with anaplastic transformation as long as 8-10 years after initial resection, and the response to re-resection is promising.¹⁰

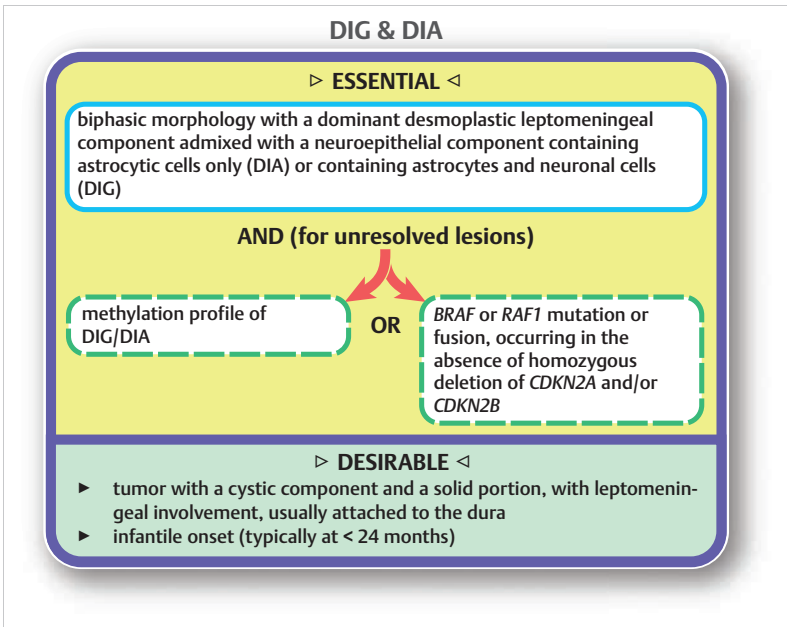


Fig. 40.3 Diagnostic criteria for desmoplastic infantile ganglioglioma & desmoplastic infantile astrocytoma. (Modified with permission from WHO Classification of Tumors Editorial Board, (ed.) World Health Organization classification of tumors of the central nervous system. 5th ed. Bosman F T, Jaffe E S, Lakhani S R, et al. (series eds.): World Health Organization Classification of Tumors series, vol. 6. International Agency for Research on Cancer, Lyon, France, 2021.)

40.1.4 Dysembryoplastic neuroepithelial tumor (DNT) (WHO grade 1)

General information

A WHO grade 1 glioneuronal tumor usually cortically based in the temporal lobe of children and young adults, characterized by a multinodular architecture with neuroglial columns, with a pathognomonic glioneuronal element.

Thought to arise embryologically from the secondary germinal layer (which includes subependymal layer, cerebellar external granular layer, hippocampal dentate fascia, and subpial granular layer).

Most DNTs are sporadic, and result from *FGFR1* alterations present in 40–80% of DNTs which upregulate the MAPK and PI3K pathways. *BRAF* p.V600E mutations are found in up to 50% in some studies.

Diagnostic criteria for dysembryoplastic neuroepithelial tumor

Diagnostic criteria for dysembryoplastic neuroepithelial tumor are shown in ▶ Fig. 40.4.³

Epidemiology

Incidence: not accurately known because the diagnosis may be missed. Estimated range: 0.8–5% of all primary brain tumors. Typically occurs in children and young adults. Slight preponderance in males.

Location

DNTs have a predilection for the temporal lobes (67% of cases) and the frontal lobe (16%), with the remainder occurring in any other location within the cerebral cortex.

Dysembryoplastic neuroepithelial tumor (DNT)

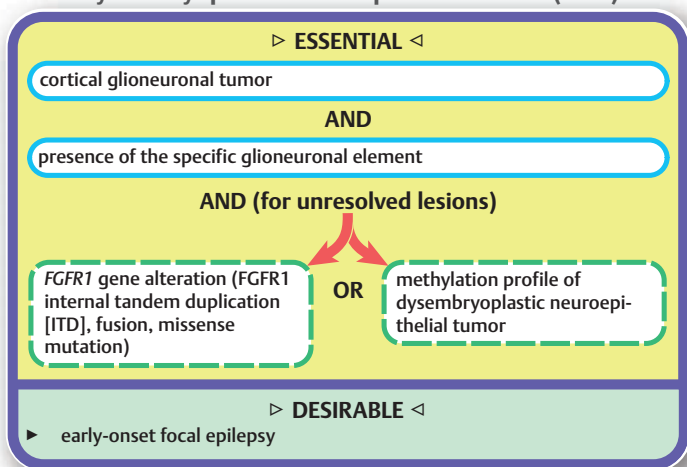


Fig. 40.4 Diagnostic criteria for dysembryoplastic neuroepithelial tumor. (Modified with permission from WHO Classification of Tumors Editorial Board, (ed.) World Health Organization classification of tumors of the central nervous system. 5th ed. Bosman F T, Jaffe E S, Lakhani S R, et al. (series eds.): World Health Organization Classification of Tumors series, vol. 6. International Agency for Research on Cancer, Lyon, France, 2021.)

Histology

Multinodularity at low-power is a key feature with neuroglial columns comprised of bundles of axons aligned perpendicular to the cortical surface. These columns are lined by oligodendroglial like cells incorporated within a mucoid matrix with interspersed neurons.¹¹ Occasionally difficult to differentiate from oligodendroglioma.

Two distinct forms¹² (do not appear to have different prognoses):

1. simple form: glioneural elements consisting of axon bundles perpendicular to the cortical surface, lined with oligodendroglial-like cells that are S-100 positive and GFAP negative. Normal appearing neurons floating in a pale eosinophilic matrix are scattered between these columns (no resemblance to ganglion cells, unlike gangliogliomas)
2. complex form: glioneural elements as described above in the simple form, with glial nodules scattered throughout. The glial component may mimic a low-grade fibrillary astrocytoma. Foci of cortical dysplasia occur

Clinical

Typically associated with longstanding medically intractable seizures, usually complex partial. Symptoms usually begin before age 20.

Imaging

Cortical lesions that traverse the entire cortical mantle with no surrounding edema and no midline mass effect.

CT: hypodense with distinct margins. Deformity of overlying calvaria is common.

MRI: T1WI: hypointense. T2WI: hyperintense, septations may be seen. If there is enhancement, it is usually nodular.

PET scan: hypometabolic with [18F]-fluorodeoxyglucose. Negative [11C]-methionine uptake (unlike all other gliomas).

Prognosis

Seizure control: usually improves after surgery. Degree of control seems to correlate with completeness of removal. Improvement in seizures correlates inversely with the duration of intractable seizures.

Recurrence/continued growth: recurrence after complete removal, or tumor growth after partial resection is rare. Adjuvant treatment (XRT, chemotherapy...) is of no benefit in these benign tumors. Mitoses or endothelial proliferation, seen on occasion, do not affect outcome. Malignant transformation is very rare.

40.1.5 Diffuse glioneuronal tumor with oligodendroglioma-like features and nuclear clusters (DGONC) (WHO grade N/A)

General information

DGONC is a provisional entity for a neuroepithelial tumor with variably differentiated cells, often possessing perinuclear haloes, scattered multinucleated cells and nuclear clusters, having a distinct DNA methylation profile and frequent chromosome 14 monosomy.

Diagnostic criteria for diffuse glioneuronal tumor with oligodendroglioma-like features and nuclear clusters (DGONC)

Diagnostic criteria for DGONC are shown in ► Fig. 40.5.³

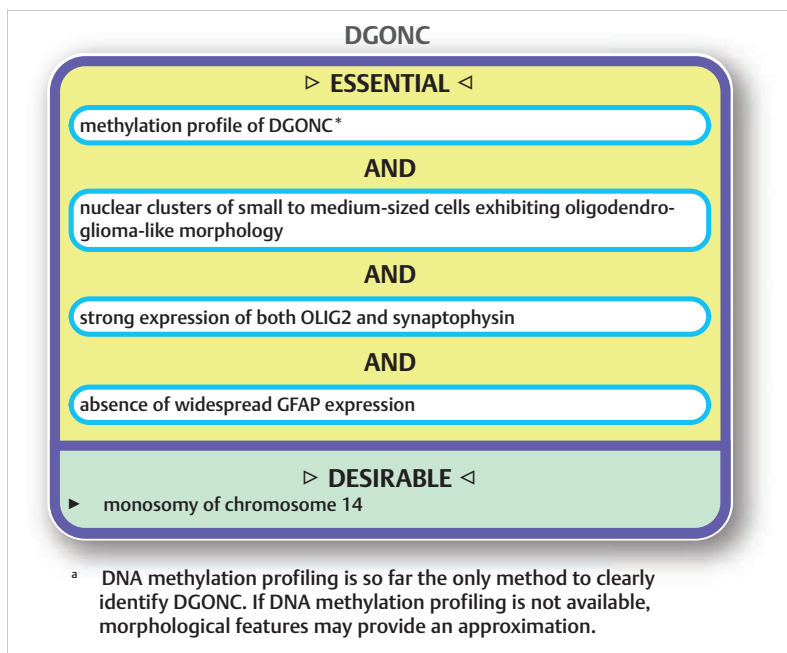


Fig. 40.5 Diagnostic criteria for DGONC. (Modified with permission from WHO Classification of Tumors Editorial Board, (ed.) World Health Organization classification of tumors of the central nervous system. 5th ed. Bosman F T, Jaffe E S, Lakhani S R, et al. (series eds.): World Health Organization Classification of Tumors series, vol. 6. International Agency for Research on Cancer, Lyon, France, 2021.)

Location

In the small number of cases reported, all were supratentorial.^{13,14}

Prognosis

Outcome has been report only for 26 cases which showed 81% 5-year PFS, and 89% 5-year OS.^{13,14} Treatment data are available only for 3 patients.

40.1.6 Papillary glioneuronal tumor (PGNT) (WHO grade 1)

General information

A rare glioneuronal tumor of young adults with a biphasic pattern and variable manifestation of pseudopapillary glial structures and interpapillary neuronal components, and possessing *PRKCA* gene fusion (primarily *SLC44A1::PRKCA* fusion).

Diagnostic criteria for papillary glioneuronal tumor

Diagnostic criteria for papillary glioneuronal tumor are shown in ► Fig. 40.6.³

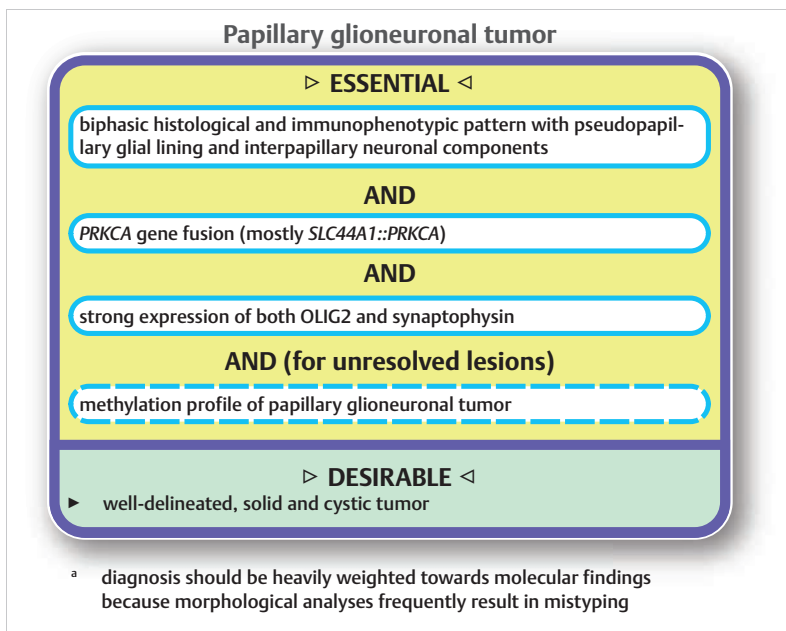


Fig. 40.6 Diagnostic criteria for papillary glioneuronal tumor. (Modified with permission from WHO Classification of Tumors Editorial Board, (ed.) World Health Organization classification of tumors of the central nervous system. 5th ed. Bosman F T, Jaffe E S, Lakhani S R, et al. (series eds.): World Health Organization Classification of Tumors series, vol. 6. International Agency for Research on Cancer, Lyon, France, 2021.)

Location

PGNT are supratentorial, most often in the temporal lobe (28%), frequently in proximity to the lateral ventricle (28%).¹⁵

Imaging

PGNT appear as well demarcated lesions with solid and cystic components and mild mass effect and edema. Portions may enhance with contrast.

Prognosis

The ability to achieve gross total excision is the primary prognostic factor.

40.1.7 Rosette-forming glioneuronal tumor (RGNT) (WHO grade 1)

General information

A rare, slow-growing WHO 1 glioneuronal tumor with two histological components: 1) uniform neurocytes forming rosettes and/or perivascular pseudorosettes, and 2) glial component with piloid and oligodendrogliocyte-like cells resembling a pilocytic astrocytoma.

Although WHO grade 1, they occasionally develop satellite lesions, leptomeningeal involvement or “drop mets” (dissemination of tumor through the CSF to dependent areas, e.g., spinal cord).

Diagnostic criteria for rosette-forming glioneuronal tumor (RGNT)

Diagnostic criteria for rosette-forming glioneuronal tumor are shown in ► Fig. 40.7.³

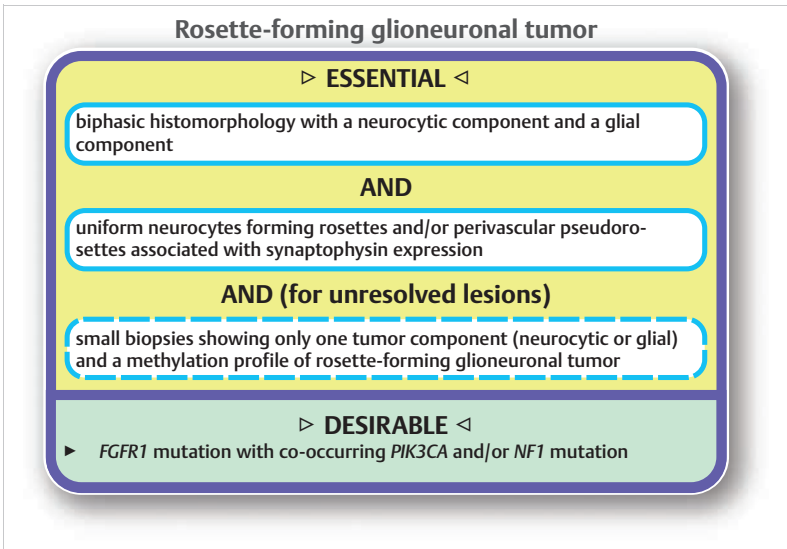


Fig. 40.7 Diagnostic criteria for rosette-forming glioneuronal tumor. (Modified with permission from WHO Classification of Tumors Editorial Board, (ed.) World Health Organization classification of tumors of the central nervous system. 5th ed. Bosman F T, Jaffe E S, Lakhani S R, et al. (series eds.): World Health Organization Classification of Tumors series, vol. 6. International Agency for Research on Cancer, Lyon, France, 2021.)

Epidemiology

The literature on RGNTs is limited to case reports and single center retrospective studies.¹⁶ Slight female preponderance (male:female ratio of 1:1.36).¹⁷ Mean age is 23.57 years (median 22.5, range 4–49).¹⁶ There is no accepted theory for cytogenesis of these tumors.

Location

RGNTs are typically midline, arise from the 4th ventricle in 42%, and extend into adjacent structures.¹⁶ However, they may also occur in the pineal region, third ventricle, optic chiasm, thalamus and temporal lobe. Spinal cord involvement has also been reported.

Histology

RGNTs demonstrate mixed glial and neuronal elements emblematic of tumors in the class of glioneuronal tumors.

Imaging

CT: RGNTs are hypodense.

MRI: appearance is variable. They are relatively well circumscribed. Most have solid and cystic components, but they can be either one alone. They tend to be hyperintense on T2WI, and iso- or hypo-intense on T1WI with variable enhancement.

Treatment

Treatment of choice is of choice is gross total resection (GTR) when possible.

Prognosis

Prognosis for OS is favorable. Recurrence is rare after GTR. However, neurologic deficit following surgery is common. Malignant progression has been reported in 2 cases¹⁶, and may be higher risk with subtotal resection, or with purely solid lesions. RGNTs may disseminate through the CSF.

40.1.8 Myxoid glioneuronal tumor (WHO grade 1)

General information

A rare, low-grade glioneuronal tumor with < 100 cases reported to date, that usually occurs in the septal nuclei, septum pellucidum, corpus callosum, or periventricular white matter. Histologically there is a proliferation of oligodendrocyte-like neoplastic cells contained in a myxoid stroma. There is a recurrent dinucleotide mutation at codon p.K385 in the *PDGFRA* gene.

Diagnostic criteria for myxoid glioneuronal tumor

Diagnostic criteria for myxoid glioneuronal tumor are shown in ► Fig. 40.8.³

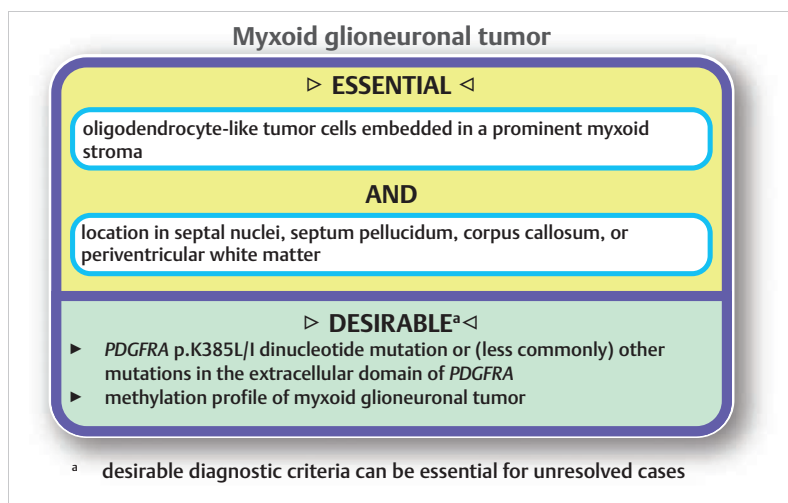


Fig. 40.8 Diagnostic criteria for myxoid glioneuronal tumor. (Modified with permission from WHO Classification of Tumors Editorial Board, (ed.) World Health Organization classification of tumors of the central nervous system. 5th ed. Bosman F T, Jaffe E S, Lakhani S R, et al. (series eds.): World Health Organization Classification of Tumors series, vol. 6. International Agency for Research on Cancer, Lyon, France, 2021.)

Imaging

Well circumscribed tumor that is hypointense on T1 and hyperintense on T2 MRI. On SWI, susceptibility artifact may be seen, suggestive of previous hemorrhages. Septal lesions often produce obstructive hydrocephalus.

Prognosis

These slow growing tumors have a favorable prognosis for OS. A subset may recur locally or disseminate through the CSF, but their behavior remains indolent. Dedifferentiation has not been reported.

40.1.9 Diffuse leptomeningeal glioneuronal tumor (DLGNT) (WHO grade 2 or 3)

General information

A glioneuronal neoplasm comprised of oligodendrocyte-like cells that is characterized by chromosome arm 1p deletion and a MAPK pathway gene alteration, typically *KIAA1549::BRAF* fusion. Leptomeningeal involvement is common.

Diagnostic criteria for diffuse leptomeningeal glioneuronal tumor

Diagnostic criteria for diffuse leptomeningeal glioneuronal tumor are shown in ► Fig. 40.9.³

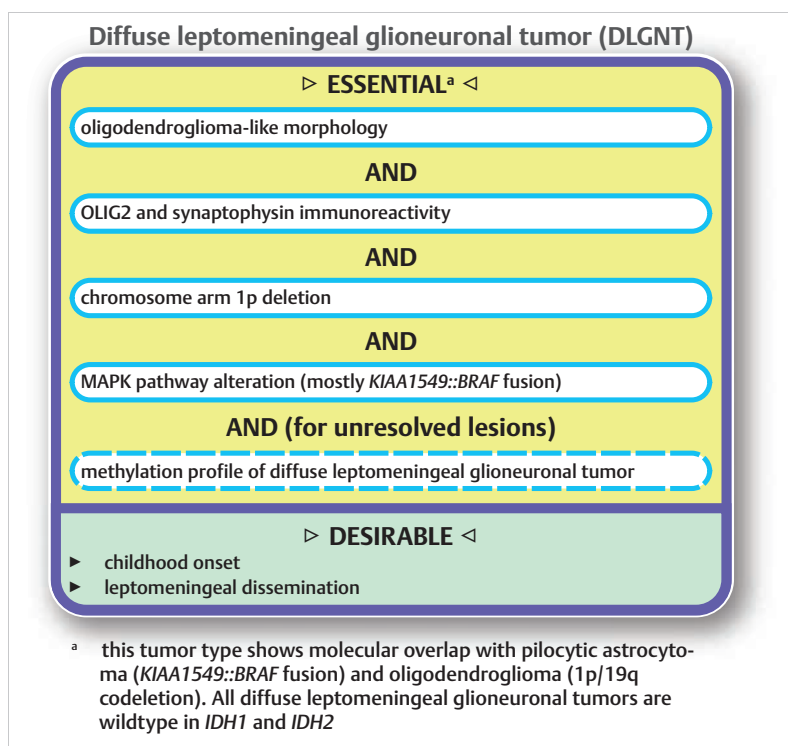


Fig. 40.9 Diagnostic criteria for diffuse leptomeningeal glioneuronal tumor. (Modified with permission from WHO Classification of Tumors Editorial Board, (ed.) World Health Organization classification of tumors of the central nervous system. 5th ed. Bosman F T, Jaffe E S, Lakhani S R, et al. (series eds.): World Health Organization Classification of Tumors series, vol. 6. International Agency for Research on Cancer, Lyon, France, 2021.)

Subtypes

See reference.³

- diffuse leptomeningeal glioneuronal tumor with 1q gain
- diffuse leptomeningeal glioneuronal tumor, methylation class 1 (DLGNT-MC-1)
- diffuse leptomeningeal glioneuronal tumor, methylation class 2 (DLGNT-MC-2)

Grading

Data for grading these tumors is limited. Most DLGNTs are well differentiated low-grade tumors. A subset may show anaplastic features on histology. For cases of conventional DLGNT and DLGNT-MC-1, the behavior has been consistent with WHO grade 2 entities. Whereas tumors with anaplastic features, 1 q gain and/or DLGNT-MC-2 profile have a clinical course more suggestive of a WHO grade 3 lesion.

Location

Predilection for the spinal and intracranial leptomeninges. Intracranially, leptomeningeal growth is typically observed in the posterior fossa, around the brainstem and along the skull base.

Imaging

MRI usually shows diffuse leptomeningeal enhancement and thickening alongside the spinal cord with intracranial extension to the posterior fossa, brainstem and basal cisterns.

Prognosis

Stability or slow growth over many years may still produce significant morbidity.¹⁸ Decreased survival was associated with mitotic activity, a Ki-67 proliferation index of $\geq 4\%$, and microvascular proliferation on initial biopsy.¹⁸

40.1.10 Dysplastic cerebellar gangliocytoma (DCG) (Lhermitte-Duclos disease) (WHO grade 1)

General information

AKA: ganglioneuroma of the cerebellum, purkinjoma, granular cell hypertrophy of the cerebellum, gangliocytoma dysplasticum, hamartoma of the cerebellum.

A cerebellar lesion with features of both a malformation (hamartoma) and a low grade (WHO 1) neoplasm that has a propensity to progress (enlarge) and to recur after surgery. May be focal or diffuse. Cerebellar folia are diffusely enlarged.

Diagnostic criteria for dysplastic cerebellar gangliocytoma

Diagnostic criteria for dysplastic cerebellar gangliocytoma are shown in ► Fig. 40.10.³

Presentation

Typically a middle-aged *adult* with signs and symptoms of a cerebellar mass such as dysmetria, or they may present with hydrocephalus. It can also be an incidental finding. Cranial nerve deficits may also occur.

Epidemiology

Autosomal dominant. Incidence: 1 in 250,000 live births.¹⁹ Most identified cases have been in adults, but patients as young as 3 years have been reported. Strongly associated with Cowden syndrome (p.651) (AKA PTEN hamartoma tumor syndrome) where the incidence is 1 in 200,000.²⁰ Associated with thyroid, breast, and uterine cancer, mucosal neuromas and meningiomas.

Histology

Derangement of normal laminar cellular architecture of the cerebellum with:

1. thickening of the outer molecular cell layer
2. loss of middle Purkinje cell layer
3. infiltration of inner granular cell layer with dysplastic ganglion cells

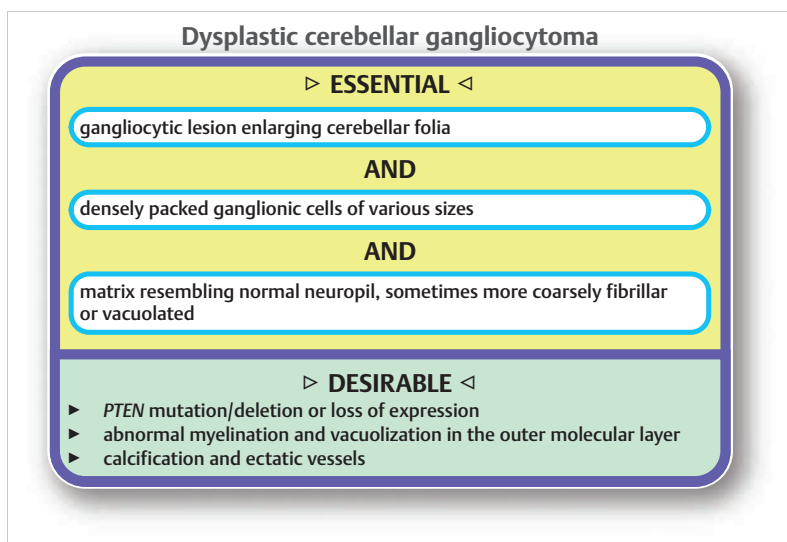


Fig. 40.10 Diagnostic criteria for dysplastic cerebellar gangliocytoma. (Modified with permission from WHO Classification of Tumors Editorial Board, (ed.) World Health Organization classification of tumors of the central nervous system. 5th ed. Bosman F T, Jaffe E S, Lakhani S R, et al. (series eds.): World Health Organization Classification of Tumors series, vol. 6. International Agency for Research on Cancer, Lyon, France, 2021.)

Imaging

CT: hypo- to isodense, nonenhancing lesion with mass effect.

MRI: T1WI: hypo- to isointense. T2WI: hyperintense, heterogeneous. Nonenhancing. Characteristic striated appearance²¹ (tiger stripes) due to widened cerebellar folia. May contain calcifications. DWI: hyperintense. ADC map: hypointense.

NB: in a child with MRI findings of Lhermitte-Duclos disease (LDD) (even if classic), a medulloblastoma is statistically more likely^{22,23} (especially medulloblastoma with extensive nodularity²⁴ (MBEN)).

Treatment

Controversial. A few cases with a benign course have been described.²⁵ Shunting is performed for hydrocephalus. Biopsy is recommended²³ particularly for pediatric cases to rule out medulloblastoma. Surgical excision may be considered when there is significant mass effect.²⁶ Efficacy of XRT is unknown.

Prognosis

Most patients are cured by surgery when indicated. Since DCG may be a harbinger of Cowden syndrome, patients should be monitored for other tumors, e.g., breast and thyroid cancer.

40.2 Neuronal tumors

40.2.1 Multinodular and vacuolating neuronal tumor (MVNT) (WHO grade 1)

General information

A rare tumor (17 case reports as of 2018²⁷) comprised of uniform neuronal elements dispersed in discrete nodules with vacuolar changes. MVNTs favor the temporal lobes and characteristically involve the deep cortical ribbon and superficial white matter.

Diagnostic criteria for multinodular and vacuolating neuronal tumor

Diagnostic criteria for multinodular and vacuolating neuronal tumor are shown in ► Fig. 40.11.³

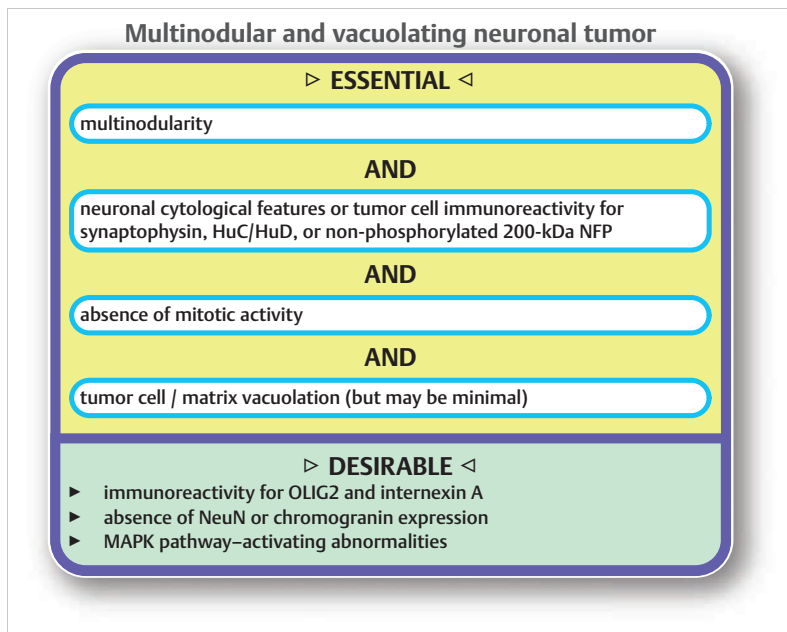


Fig. 40.11 Diagnostic criteria for multinodular and vacuolating neuronal tumor. (Modified with permission from WHO Classification of Tumors Editorial Board, (ed.) World Health Organization classification of tumors of the central nervous system. 5th ed. Bosman F T, Jaffe E S, Lakhani S R, et al. (series eds.): World Health Organization Classification of Tumors series, vol. 6. International Agency for Research on Cancer, Lyon, France, 2021.)

Location

There is a predilection for the temporal lobes (75–80%) and frontal lobes (10–15%).²⁷

Presentation

The most common presentation is seizures (60%) which may be complex partial ± secondary generalization, followed by headache (10–15%).

Prognosis

These are benign neoplasms. Following gross total excision there are no reports of recurrence, and subtotally resected tumors have not progressed.

40.2.2 Central neurocytoma (WHO grade 2)

General information

Key concepts

- rare, WHO grade 2 neuronal tumor primarily seen in young adults
- typically intraventricular, attached to septum pellucidum
- gross total resection can be curative (but some recur even after GTR)
- if MIB-1 labeling index > 2–4%, there is an increased risk of recurrence after subtotal resection
- if MIB-1 labeling is elevated, radiation therapy after subtotal resection can reduce the risk of recurrence

Central neurocytomas are rare WHO grade 2 neuronal tumors. They are generally attached to the septum pellucidum within the lateral ventricles, and less frequently occur within the third ventricle. The most common presentation is with increased intracranial pressure and ventriculomegaly.^{28,29} See also differential diagnosis for intraventricular lesions (p.1667).

Due to an uncertain relation of prognosis to Ki-67 index and anaplastic features, a distinction between a possible grade 1 vs. 2 was not made.³⁰

Diagnostic criteria for central neurocytoma

Diagnostic criteria for central neurocytoma are shown in ► Fig. 40.12.³

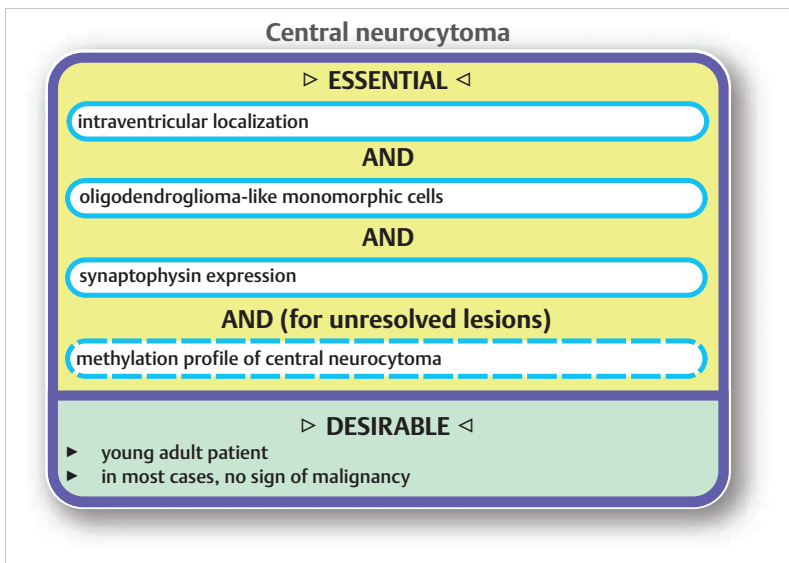


Fig. 40.12 Diagnostic criteria for central neurocytoma. (Modified with permission from WHO Classification of Tumors Editorial Board, (ed.) World Health Organization classification of tumors of the central nervous system. 5th ed. Bosman F T, Jaffe E S, Lakhani S R, et al. (series eds.): World Health Organization Classification of Tumors series, vol. 6. International Agency for Research on Cancer, Lyon, France, 2021.)

Epidemiology

Constitutes 0.1–0.5% of brain tumors. Incidence peaks in the 3rd decade, but they can also present in children and the elderly.³¹ No gender predominates.

Histology

Central neurocytoma cells have small, round nuclei. Cells often have a “fried egg” appearance on H&E staining, which can mimic oligodendroglioma. There are two primary architectures: honeycomb (also mimicking oligodendroglioma) and fibrillary. Rosettes can also be seen. Immunohistochemistry is often positive for neuronal markers synaptophysin and Neu-N.

Electron microscopy: not necessary for diagnosis.³⁰ It has been used by some when the diagnosis is not clear, and demonstrates neuronal features, including: a prominent golgi apparatus, parallel microtubules, and dense core neurosecretory granules.^{28,29}

Molecular genetics: 1p/19q deletion has not been reported in central neurocytoma, but it can be seen in extraventricular neurocytoma.³²

Imaging

CT scan: 25–50% of these tumors show calcifications. Tumors are usually iso- to hyperdense, with hypodense areas that represent cystic degeneration.²⁸

MRI: Tumors appear heterogeneously isointense on T1 and hyperintense on T2. MR spectroscopy often shows a high glycine peak.³³

Moderate to strong contrast enhancement occurs on both CT and MR images.²⁹

Treatment

1. total resection is often curative. After gross total resection, radiation therapy is generally not necessary²⁸
2. subtotal resection can be followed by stereotactic radiosurgery, especially if MIB-1 labeling is $> 2\text{--}4\%$ ^{28,29,33,34}
3. chemotherapy for recurrent and inoperable tumors has been reported with many agents, including: alkylating agents (carmustine, cyclophosphamide, ifosfamide, lomustine), platinum-based agents (carboplatin and cisplatin), etoposide, topotecan, and vincristine^{33,34}
4. after treatment, patients should have long-term follow-up imaging as surveillance for tumor recurrence²⁹

Prognosis

Total resection can be curative.^{28,33} Recurrence risk after subtotal resection varies with the MIB-1 labeling index. In a 2004 study of neurocytomas with MIB-1 labeling $> 2\%$, the 5-year local control rate after subtotal resection alone was 7%, but if subtotal resection was followed with radiation therapy it was 70%.³⁴ In a 2013 study, patients with MIB-1 labeling $< 4\%$ had no recurrence at 4 years after subtotal resection. If MIB-1 labeling was $> 4\%$ at subtotal resection, recurrence was found in 50% of patients at 2 years and 75% at 4 years.³³ Most local recurrences occur within 3–6 years. Recurrence is more common in extraventricular neurocytoma.³⁴

40.2.3 Extraventricular neurocytoma (WHO grade 2)

General information

Extraventricular neurocytoma is a variant of central neurocytoma that is even rarer and that may be located in the cerebral parenchyma, cerebellum, thalamus, brainstem, pineal region, and spinal cord. Histologically, they resemble central neurocytomas but with a wider morphological spectrum. They can be focally infiltrative into surrounding tissue. *FGFR1::TACC1* fusions are frequently identified.

Diagnostic criteria for extraventricular neurocytoma

Diagnostic criteria for extraventricular neurocytoma are shown in ► Fig. 40.13.³

Histology

Histology is similar to central neurocytoma.

Immunohistochemistry is positive for synaptophysin in the cytoplasm and neuropil. Neu-N staining is often positive.

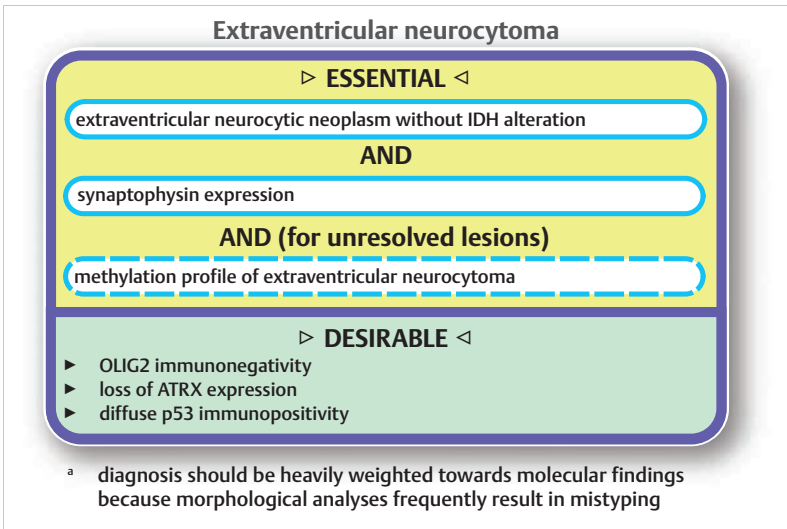


Fig. 40.13 Diagnostic criteria for extraventricular neurocytoma. (Modified with permission from WHO Classification of Tumors Editorial Board, (ed.) World Health Organization classification of tumors of the central nervous system. 5th ed. Bosman F T, Jaffe E S, Lakhani S R, et al. (series eds.): World Health Organization Classification of Tumors series, vol. 6. International Agency for Research on Cancer, Lyon, France, 2021.)

Central neurocytoma can often be identified by its intraventricular location, however for extraventricular neurocytoma the reactivity for synaptophysin without either IDH mutation or 1p/19q codeletion can differentiate this tumor from oligodendroglioma.

Molecular genetics

The most common genetic alterations are *FGFR1::TACC1* fusions which were found in 60%, while other FGFR alterations were present in 13%.³⁵ Neither IDH1/2 mutation nor MGMT promoter gene methylation (p. 673) has been reported.³⁶

Prognosis

The overall prognosis with this low-grade tumor is favorable. The rate of recurrence has been low and seizure control has been good following gross total resection.

40.2.4 Cerebellar liponeurocytoma (WHO grade 2)

General information

Formerly lipomatous medulloblastoma. A rare WHO grade 2 neoplasm with neuronal/neurocytic differentiation and clusters of neoplastic neurocytes with lipidization (resembling adipocytes) that occurs exclusively in cerebellum (typically in the cerebellar hemispheres, and less commonly the midline or CPA) of adults (mean age: 50 years). No gender preference.

Some authors suggest the more inclusive name “Liponeurocytoma” should replace the separate WHO classification of “Cerebellar liponeurocytoma.” Others contend that the supratentorial exceptions are central neurocytomas which contain neurocytes that underwent lipidization, instead of having actual adipose metaplasia.³⁷

Diagnostic criteria for cerebellar liponeurocytoma

Diagnostic criteria for cerebellar liponeurocytoma are shown in ► Fig. 40.14.³

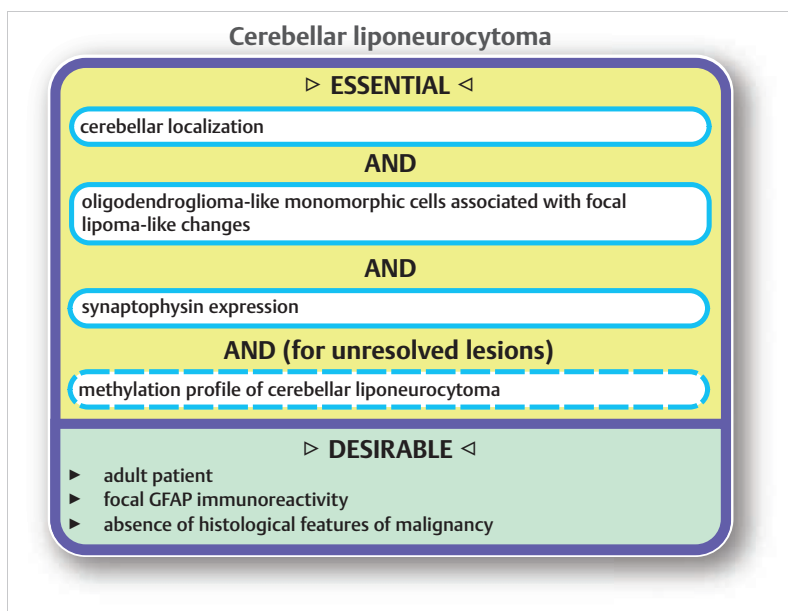


Fig. 40.14 Diagnostic criteria for cerebellar liponeurocytoma. (Modified with permission from WHO Classification of Tumors Editorial Board, (ed.) World Health Organization classification of tumors of the central nervous system, 5th ed. Bosman F T, Jaffe E S, Lakhani S R, et al. (series eds.): World Health Organization Classification of Tumors series, vol. 6. International Agency for Research on Cancer, Lyon, France, 2021.)

Imaging

CT: isodense in areas, with areas of very low density where the fat content is high.

MRI: lipid rich areas are hyperintense on T1 and T2. The suspicion about lipid content is supported when these areas drop out on fat suppression sequences.

Histology

Cells are uniform, neurocytic morphology with regular round to oval nuclei and clear cytoplasm with ill-defined cell membranes arrayed in sheets and lobules. The distinctive feature are lipid containing cells resembling adipocytes.

Synaptophysin (p.632) and MAP-2 immunostaining is consistent and diffuse, focal GFAP staining is common. Usually no mitotic figures. MIB-1 index is usually low (mean of $3.73\% \pm 4.01\%$).³⁸

Prognosis

Cerebellar liponeurocytoma have a favorable prognosis. Following GTR, the 5-year survival rate was 71.3%, PFS rate 61%, mean OS was 16.3 years, and the mean PFS was 10 years.³⁸ Subtotal resection carried a 42% recurrence rate vs. 15% with GTR.³⁸ Post-op radiation improved the PFS from 55% (16/29 patients) to 92% (11 out of 12) with median follow-up of 52 months.³⁸ All recurrences were restricted to the posterior fossa.

References

- [1] Rubinstein LJ. Tumors of the Central Nervous System. Atlas of Tumor Pathology, Second Series, Fascicle 6. Washington, DC: Armed Forces Institute of Pathology; 1972
- [2] Courville CB. Ganglioglioma. Tumor of the Central Nervous System: Review of the Literature and Report of Two Cases. Archives of Neurology and Psychiatry. 1930; 24:439–491
- [3] Bosman FT, Jaffe ES, Lakhani SR, et al. World Health Organization Classification of Tumors series, vol. 6. In: WHO Classification of Tumors Editorial Board (ed.) World Health Organization classification of tumors of the central nervous system. 5th ed. Lyon, France: International Agency for Research on Cancer; 2021
- [4] Demierre B, Stichnoth FA, Hori A, et al. Intracerebral Ganglioglioma. J Neurosurg. 1986; 65:177–182
- [5] Kalyan-Raman UP, Olivero WC. Ganglioglioma: A Correlative Clinicopathological and Radiological Study of Ten Surgically Treated Cases with Follow-Up. Neurosurgery. 1987; 20:428–433
- [6] Garcia CA, McGarry PA, Collada M. Ganglioglioma of the Brain Stem. Case Report. Journal of Neurosurgery. 1984; 60:431–434
- [7] Sutton LN, Packer RJ, Rorke LB, et al. Cerebral Gangliogliomas During Childhood. Neurosurgery. 1983; 13:124–128
- [8] Lang FF, Epstein FJ, Ransohoff J, et al. Central Nervous System Gangliogliomas. Part 2: Clinical Outcome. Journal of Neurosurgery. 1993; 79:867–873
- [9] Darwish B, Arbuckle S, Kellie S, et al. Desmoplastic infantile ganglioglioma/astrocytoma with cerebrospinal metastasis. J Clin Neurosci. 2007; 14:498–501
- [10] Loh JK, Lieu AS, Chai CY, et al. Malignant transformation of a desmoplastic infantile ganglioglioma. Pediatr Neurol. 2011; 45:135–137
- [11] Daumas-Duport C, Scheithauer BW, Chodkiewicz J-P, et al. Dysembryoplastic Neuroepithelial Tumor: A Surgically Curable Tumor of Young Patients with Intractable Seizures. Neurosurgery. 1988; 23:545–556
- [12] Adada B, Sayed K. Dysembryoplastic neuroepithelial tumors. Contemporary Neurosurgery. 2004; 26:1–5
- [13] Deng MY, Sill M, Sturm D, et al. Diffuse glioneuronal tumour with oligodendroglioma-like features and nuclear clusters (DGOnc) - a molecularly defined glioneuronal CNS tumour class displaying recurrent monosomy 14. Neuropathol Appl Neurobiol. 2020; 46:422–430
- [14] Pickles JC, Mankad K, Aizpurua M, et al. A case series of Diffuse Glioneuronal Tumours with Oligodendroglioma-like features and Nuclear Clusters (DGOnc). Neuropathol Appl Neurobiol. 2021; 47:464–467
- [15] Hou Y, Pinheiro J, Sahn F, et al. Papillary glioneuronal tumor (PGNT) exhibits a characteristic methylation profile and fusions involving PRKCA. Acta Neuropathol. 2019; 137:837–846
- [16] Anyanwu CT, Robinson TM, Huang JH. Rosette-forming glioneuronal tumor: an update. Clin Transl Oncol. 2020; 22:623–630
- [17] Schlamann A, von Bueren AO, Hagel C, et al. An individual patient data meta-analysis on characteristics and outcome of patients with papillary glioneuronal tumor, rosette glioneuronal tumor with neuropil-like islands and rosette forming glioneuronal tumor of the fourth ventricle. PLoS One. 2014; 9. DOI: 10.1371/journal.pone.0101211
- [18] Rodriguez FJ, Perry A, Rosenblum MK, et al. Disseminated oligodendroglioma-like leptomeningeal tumor of childhood: a distinctive clinicopathologic entity. Acta Neuropathol. 2012; 124:627–641
- [19] Nelen MR, van Staveren WC, Peeters EA, et al. Germ-line mutations in the PTEN/MMAC1 gene in patients with Cowden disease. Human Molecular Genetics. 1997; 6:1383–1387
- [20] Nelen MR, Kremer H, Konings IB, et al. Novel PTEN mutations in patients with Cowden disease: absence of clear genotype-phenotype correlations. Eur J Hum Genet. 1999; 7:267–273
- [21] Meltzer CC, Smirniotopoulos J G, Jones RV. The striated cerebellum: an MR imaging sign in Lhermitte-Duclos disease (dysplastic gangliocytoma). Radiology. 1995; 194:699–703
- [22] Chen KS, Hung PC, Wang HS, et al. Medulloblastoma or cerebellar dysplastic gangliocytoma (Lhermitte-Duclos disease)? Pediatr Neurol. 2002; 27:404–406
- [23] Someshwar S, Hogg JP, Nield LS. Lhermitte-Duclos disease or neoplasm? Applied Neurology. 2007; 3:37–39
- [24] Suresh TN, Santosh V, Yasha TC, et al. Medulloblastoma with extensive nodularity: a variant occurring in the very young-clinicopathological and immunohistochemical study of four cases. Childs Nerv Syst. 2004; 20:55–60
- [25] Capone Mori A, Hoeltzenbein M, Poetsch M, et al. Lhermitte-Duclos disease in 3 children: a clinical long-term observation. Neuropediatrics. 2003; 34:30–35
- [26] Carlson JJ, Milburn JM, Barre GM. Lhermitte-Duclos disease: case report. J Neuroimaging. 2006; 16:157–162
- [27] Shitara S, Tokime T, Akiyama Y. Multinodular and vacuolating neuronal tumor: A case report and literature review. Surg Neurol Int. 2018; 9. DOI: 10.4103/sni.sni.348.17
- [28] Patel DM, Schmidt RF, Liu JK. Update on the diagnosis, pathogenesis, and treatment strategies for central neurocytoma. J Clin Neurosci. 2013; 20:1193–1199
- [29] Sharma MC, Deb P, Sharma S, et al. Neurocytoma: a comprehensive review. Neurosurg Rev. 2006; 29:270–85; discussion 285
- [30] Louis DN, Ohgaki H, Wiestler OD, et al. WHO classification of tumors of the central nervous system. Lyon, France 2016
- [31] Hassoun J, Soylemezoglu F, Gambarelli D, et al. Central neurocytoma: a synopsis of clinical and histological features. Brain Pathol. 1993; 3:297–306
- [32] Agarwal S, Sharma MC, Sarkar C, et al. Extraventricular neurocytomas: a morphological and histogenetic consideration. A study of six cases. Pathology. 2011; 43:327–334
- [33] Kaur G, Kane AJ, Sughrue ME, et al. MIB-1 labeling index predicts recurrence in intraventricular central neurocytomas. J Clin Neurosci. 2013; 20:89–93
- [34] Rades D, Fehlaue F, Schild SE. Treatment of atypical neurocytomas. Cancer. 2004; 100:814–817
- [35] Sievers P, Stichel D, Schrimpf D, et al. FGFR1:TACC1 fusion is a frequent event in molecularly defined extraventricular neurocytoma. Acta Neuropathol. 2018; 136:293–302
- [36] Myung JK, Cho HJ, Park CK, et al. Clinicopathological and genetic characteristics of extraventricular neurocytomas. Neuropathology. 2013; 33:111–121
- [37] Chakraborti S, Mahadevan A, Govindan A, et al. Supratentorial and cerebellar liponeurocytomas: report of four cases with review of literature. J Neurooncol. 2011; 103:121–127
- [38] Gembruch O, Junker A, Monninghoff C, et al. Liponeurocytoma: Systematic Review of a Rare Entity. World Neurosurg. 2018; 120:214–233

41 Ependymal Tumors

41.1 Introduction to ependymal tumors

The WHO CNS5 classified ependymomas by site as well as by histology and molecular biology features.

41.2 Specific tumor types

41.2.1 Supratentorial ependymoma (WHO grade 2 or 3)

General information

A well-circumscribed supratentorial glioma with small uniform cells having round nuclei that form focal pseudorosettes or ependymal rosettes embedded in a fibrillary matrix. Cells are thought to derive from radial glial cells (bipolar progenitor cells that are thought to be a major source of neurons in the developing nervous system).

The following designations should be used:

- **supratentorial ependymoma, NEC** (not elsewhere classified): when genetic analysis has not identified a pathologic fusion gene involving *ZFTA* (*C11orf95*) or *YAP1*
- **supratentorial ependymoma, NOS** (not otherwise specified): when genetic analysis has not been successful or is not feasible

Diagnostic criteria for supratentorial ependymoma

Diagnostic criteria for supratentorial ependymoma are shown in ► Fig. 41.1.¹

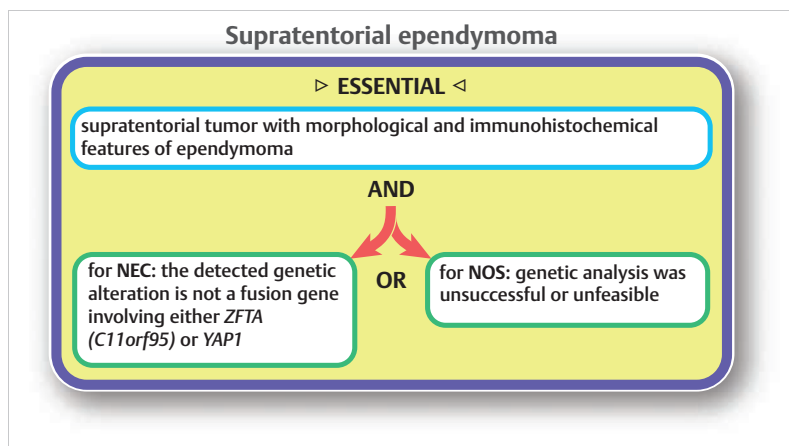


Fig. 41.1 Diagnostic criteria for supratentorial ependymoma. (Modified with permission from WHO Classification of Tumors Editorial Board, (ed.) World Health Organization classification of tumors of the central nervous system. 5th ed. Bosman F T, Jaffe E S, Lakhani S R, et al. (series eds.): World Health Organization Classification of Tumors series, vol. 6. International Agency for Research on Cancer, Lyon, France, 2021.)

Grading

The most important characteristics distinguishing grade 3 supratentorial ependymomas from grade 2 are abundant mitoses and the presence of microvascular proliferation. These are more influential than nuclear pleomorphism or necrosis.² NB: histopathological grading of ependymomas is not highly correlated with overall outcome.

Location

These tumors are found in the cerebral hemispheres. A connection to the ventricular system may or may not be obvious. They favor the frontal or parietal lobes over the temporal or occipital lobes.

Epidemiology

These tumors affect all ages, but the proportion relative to the posterior fossa and spinal ependymomas decreases with age.³ They comprise ≈ one third of intracranial ependymomas.

Prognosis

Most of the data comes from studies performed before molecular testing. The outcome is worse for supratentorial than infratentorial ependymomas in adults.

Gross total removal is the most important prognosticator of OS in adults and children. Post-op XRT lowers the risk of local recurrence.

41.2.2 Supratentorial ependymoma, ZFTA fusion-positive (WHO grade 2 or 3)

General information

A well-circumscribed supratentorial glioma with a *ZFTA* (formerly *C11orf95*) fusion gene (typically *ZFTA* is fused with *RELA*), with small uniform cells having round nuclei containing speckled chromatin and poorly delineated fibrillary cytoplasm. Pseudorosettes are usually not prominent, and true ependymal rosettes embedded are infrequent. There is often a network of branching capillary vasculature. Cells are thought to derive from radial glial cells.

Diagnostic criteria for supratentorial ependymoma, ZFTA fusion-positive

Diagnostic criteria for supratentorial ependymoma, ZFTA fusion-positive are shown in ► Fig. 41.2.¹

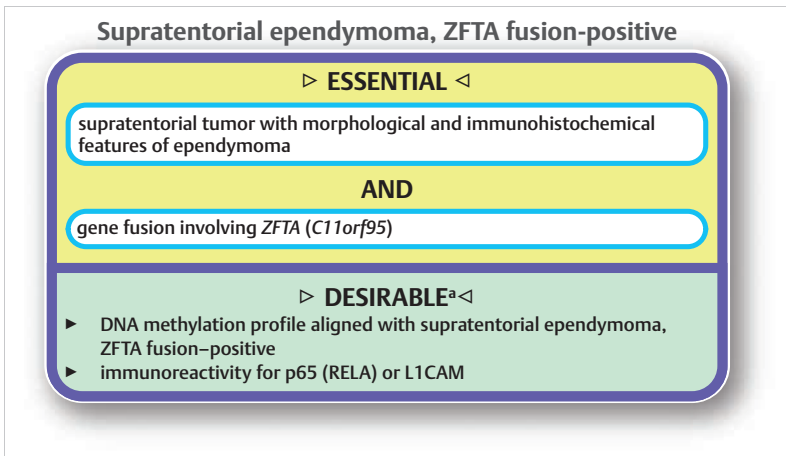


Fig. 41.2 Diagnostic criteria for supratentorial ependymoma, ZFTA fusion-positive. (Modified with permission from WHO Classification of Tumors Editorial Board, (ed.) World Health Organization classification of tumors of the central nervous system. 5th ed. Bosman F T, Jaffe E S, Lakhani S R, et al. (series eds.): World Health Organization Classification of Tumors series, vol. 6. International Agency for Research on Cancer, Lyon, France, 2021.)

Grading

These tumors are considered WHO grade 2 or 3 based on the degree of anaplasia observed.

Location

Most of these tumors are found in the frontal or parietal lobes. They occur infrequently in the thalamus, hypothalamic/third ventricular region, or extra-axially.

Epidemiology

Supratentorial ependymoma, ZFTA fusion-positive tumors comprise the majority of supratentorial ependymomas (20–58% in adults, 66–84% in pediatrics).¹

Prognosis

Preliminary data suggest a worse outcome than with other ependymomas, however, more experience with this recently molecularly defined group will be needed to make any definitive statements. Homozygous deletion of *CDKN2A* and/or *CDKN2B* are independent risk factors for worse OS in a series of supratentorial ependymomas with *ZFTA::RELA* fusions.⁴

41.2.3 Supratentorial ependymoma, YAP1 fusion-positive (WHO grade N/A)

General information

A well circumscribed supratentorial glioma with a *YAP1* fusion gene (typically *YAP1* is fused with *MAMLD1*), with small uniform cells having round nuclei that form focal pseudorosettes or ependymal rosettes embedded in a fibrillary matrix. Cells are thought to derive from PAX6-positive radial glial neural stem cells.

Diagnostic criteria for supratentorial ependymoma, YAP1 fusion-positive

Diagnostic criteria for supratentorial ependymoma, YAP1 fusion-positive are shown in ► Fig. 41.3.¹

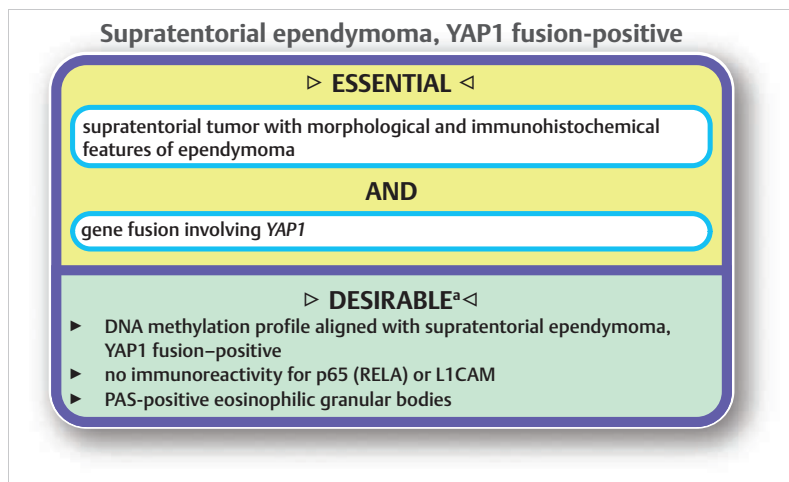


Fig. 41.3 Diagnostic criteria for supratentorial ependymoma, YAP1 fusion-positive. (Modified with permission from WHO Classification of Tumors Editorial Board, (ed.) World Health Organization classification of tumors of the central nervous system. 5th ed. Bosman F T, Jaffe E S, Lakhani S R, et al. (series eds.): World Health Organization Classification of Tumors series, vol. 6. International Agency for Research on Cancer, Lyon, France, 2021.)

Location

These tumors are usually found within or adjacent to the lateral ventricle.

Epidemiology

Supratentorial ependymoma, YAP1 fusion-positive tumors seem to be isolated to young children, and show a M:F ratio of 0.3:1.¹ They account for 6–7.4% of supratentorial ependymomas in pediatric cohorts.¹

Prognosis

Prognosis appears favorable compared to other supratentorial ependymomas (retrospective analysis).¹

41.2.4 Posterior fossa ependymoma (WHO grade 2 or 3)

General information

Key concepts

- usually benign tumors, often fibrillary with epithelial appearance. Perivascular pseudorosettes or ependymal rosettes may be seen
- most often occur in the floor of the 4th ventricle, presenting with hydrocephalus (increased ICP) and cranial nerve VI & VII palsies
- evaluation: includes imaging the entire neuraxis (MRI with and without enhancement: cervical, thoracic, lumbar & brain) because of potential for seeding through CSF
- worse prognosis the younger the patient (especially age < 24 months)
- treatment: the best outcomes are associated with gross total removal (no enhancing tumor on post-op MRI) followed by XRT. XRT may be withheld for age < 3 years due to side effects
- do LP ≈ 2 weeks post-op to send ≈ 10 cc of CSF for cytology for prognostication

A circumscribed posterior fossa glioma of monomorphic round cells with round nuclei that form focal pseudorosettes or ependymal rosettes embedded in a fibrillary matrix. This diagnosis should be employed when molecular analysis cannot assign a molecular group (NEC, not elsewhere classified) or is not feasible or the results were uncertain (NOS, not otherwise specified)

Diagnostic criteria for posterior fossa ependymoma

Diagnostic criteria for posterior fossa ependymoma are shown in ► Fig. 41.4.¹

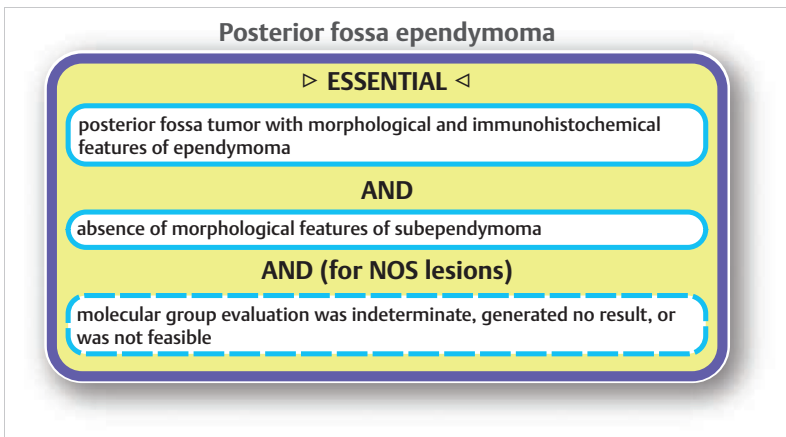


Fig. 41.4 Diagnostic criteria for posterior fossa ependymoma. (Modified with permission from WHO Classification of Tumors Editorial Board, (ed.) World Health Organization classification of tumors of the central nervous system. 5th ed. Bosman F T, Jaffe E S, Lakhani S R, et al. (series eds.); World Health Organization Classification of Tumors series, vol. 6. International Agency for Research on Cancer, Lyon, France, 2021.)

Grading

Should ideally be done in the context of an integrated diagnosis.² The most important characteristics distinguishing grade 3 posterior fossa ependymomas from grade 2 are abundant mitoses and the presence of microvascular proliferation. These are more influential than nuclear pleomorphism or necrosis.² NB: histopathological grading of ependymomas is not highly correlated with overall outcome.

Location

These tumors generally occur in relation to the 4th ventricle (floor, lateral aspect, cerebellar peduncles, or roof).

Spread

Ependymomas have the potential to spread via the CSF through the neuraxis (including to the spinal cord), a process known as “seeding,” resulting in so-called “**drop mets**” in the spinal cord in 11%. The incidence is higher with higher grade.⁵ Systemic spread occurs on rare occasions.

Presentation

Mostly those of posterior fossa mass with increased ICP (from hydrocephalus) and cranial nerve involvement.

► **Symptoms.** Symptoms of increased ICP:

1. headache
2. N/V
3. ataxia or vertigo
4. obtundation/lethargy

Symptoms of cranial nerve involvement: diplopia, facial weakness...

► **Signs.** In infants: increasing head circumference crossing growth curves.

Cranial nerve involvement: invasion of the floor of the 4th ventricle may impact the facial colliculus (a small bump in the floor of the 4th ventricle produced by the internal genu of the VII nerve as it deviates posteriorly around the VI nerve nucleus). This can produce peripheral facial nerve palsy (p.607) (from VII nerve) and/or abducens palsy (p.598) (from VI nucleus).

Imaging

MRI: the imaging study of choice. The entire craniocervical axis should be imaged with and without contrast because of possibility of drop mets. Usually appears as a mass in the floor of fourth ventricle (► Fig. 41.5), or when large it can fill the 4th ventricle and extension through the foramina of Luschka and/or Magendie may be seen. The tumor enhances except for cystic portions or areas of necrosis. Obstructive hydrocephalus is often identified at presentation. May be difficult to distinguish from medulloblastoma radiographically, see differentiating features (p.1647).

CT: not as detailed for evaluation of posterior fossa.

Myelogram: water-soluble contrast myelography with CT is about as sensitive as gadolinium enhanced MRI in detecting “drop mets.” Myelography can also provide CSF for cytology for staging purposes.

Epidemiology

Posterior fossa ependymomas of all of the types can occur at any age, but the median age at presentation is 6 years, with a slight male predominance.

Treatment

Surgical resection

Goal of surgery: maximal possible resection of intracranial portion without causing neurological deficits (since extent of resection is an important prognosticator). Gross total resection may not be possible when invasion of the floor is extensive, or when tumor extends through the foramen of Luschka (bradycardia may prevent GTR).

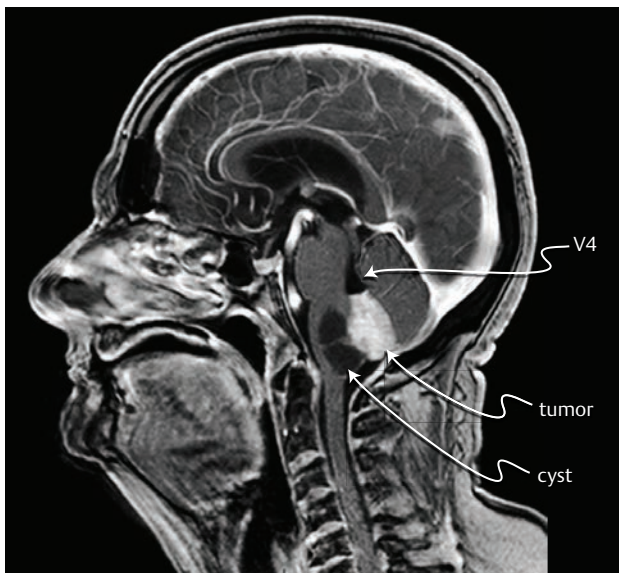


Fig. 41.5 Ependymoma (WHO grade 2) of the posterior fossa in a 71-year-old woman. Here, the tumor arises from the floor of the 4th ventricle and has a cystic component that invades the medulla. It displaces but does not invade the cerebellum. The 4th ventricle (V4) is obstructed and the patient had hydrocephalus.
Image: sagittal T1 contrast-enhanced MRI.

These tumors are usually circumscribed with a covering layer of ependyma, and are only rarely invasive to a significant degree.⁶

Operative morbidity: advise patients/families pre-op of the likelihood of need for post-op gastric feeding tube (G-tube) and tracheostomy (these may be temporary).

2 weeks postoperatively, perform LP to look for "drop mets": 10 cc of CSF is sent for cytology to quantitate (if any) number of malignant cells (may be used to follow treatment). If LP is positive, then by definition there are drop mets. If negative, it is not as helpful (sensitivity is not high). CSF from an EVD is not as sensitive as LP.

Lesions in fourth ventricle region are approached via midline suboccipital craniectomy.

Radiation therapy (XRT)

Ependymomas rank 2nd only to medulloblastomas in radiosensitivity. XRT is administered after surgical excision (survival is improved with post-op XRT^{7,8}: 50% survival time was 2 yrs longer with XRT than without,⁷ and 5-year survival increased from 20–40% without XRT to 40–80% with XRT⁸); however, for patients age < 3 years, see below.

1. cranial XRT
 - a) traditional therapy: 45–48 Gy to tumor bed⁸ (recurrence treated with additional 15–20 Gy)
9(p 2797)
 - b) recent recommendations: 3-D conformal XRT with higher doses (59.4 Gy delivered to tumor bed + 1 cm margins)¹⁰
 - c) intensity modulated proton beam therapy appears equivalent in terms of local control, but may be better at sparing normal tissue¹¹
2. spinal XRT: most radiate only if drop mets or if positive CSF cytology (however, prophylactic spinal is controversial¹²)
 - a) low dose XRT to entire spinal axis (median dose = 30 Gy in one series⁸)
 - b) boost to any regions showing drop mets

3. XRT is undesirable in age < 3 years due to side effects. XRT was avoided in $\approx 30\%$ of patients < 3 years of age with comparable survival when XRT was reserved for treatment failures.^{13,14} This concept of selective XRT may be applicable to older children as well¹⁵

Chemotherapy

Role is very limited.

1. has little impact on newly diagnosed cases. Adjuvant chemo after XRT in patients > 3 years showed no benefit
2. may reduce vascularity of ependymomas, which may facilitate GTR (sometimes in a second stage operation)
3. may be considered for infants < 3 years of age to delay use of XRT (see above)
4. chemo at the time of recurrence may arrest tumor progression for short periods

Prognosis

Extent of resection: the risk of recurrence is highest following subtotal resection. Gross total resection (GTR) (surgical) of primary intracranial tumor followed by craniospinal XRT as outlined above yields 41% 5-year survival.

Treatment failure: tumors tend to recur initially at the site of origin.¹⁶ However, treatment failure in 9–25% of patients is due to drop mets.^{17,18}

Age: peds vs. adults: 5-year survival (5YS) is 20–30% in the pediatric group,^{5,19} compared with up to 80% in adults. Patients 24–35 months old did better (73% 5YS) than those younger than 24 months (26% 5YS) or those older than 36 months (36% 5YS).¹⁷

41.2.5 Posterior fossa group A (PFA) ependymoma (WHO grade 2 or 3)

General information

PFA ependymoma is a circumscribed posterior fossa glioma aligned with the PFA molecular groups of ependymomas (loss of nuclear H3 p.K28me3 (K27me3) expression in tumor cells or by DNA methylation profiling) having small uniform cells with round nuclei that form focal pseudorosettes or ependymal rosettes embedded in a fibrillary matrix. Cells are thought to derive from undifferentiated glial stem or progenitor cell in the developing hindbrain.

Diagnostic criteria for posterior fossa group A (PFA) ependymoma

Diagnostic criteria for posterior fossa group A (PFA) ependymoma are shown in ► Fig. 41.6.¹

Grading

Grading of WHO grade 2 or 3 is based on the degree of anaplasia and reported in an integrated diagnosis.²

Location

These tumors arise more frequently from the roof or lateral aspect of the 4th ventricle.

Presentation

Similar to the presentation of posterior fossa ependymomas (p. 728) in general.

Epidemiology

PFA ependymomas predominantly occur in pre-adolescents with a median age of 3 years at presentation.¹

Prognosis

Prognosis appears less favorable than with PFB ependymomas.²⁰ Extent of resection correlates with outcome.²¹ Tiers of disease risk can be defined using chromosome 1q status and other variables.²¹

Posterior fossa group A (PFA) ependymoma

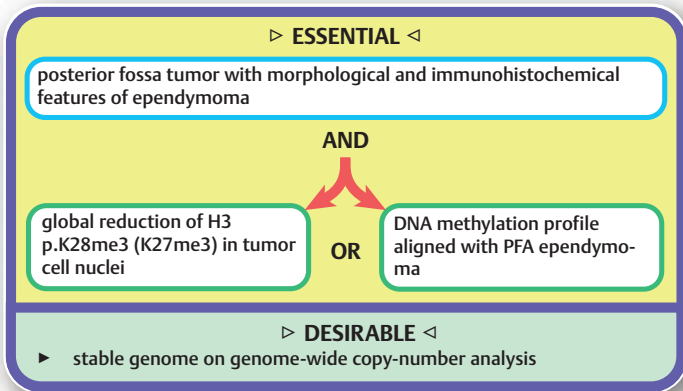


Fig. 41.6 Diagnostic criteria for posterior fossa group A (PFA) ependymoma. (Modified with permission from WHO Classification of Tumors Editorial Board, (ed.) World Health Organization classification of tumors of the central nervous system. 5th ed. Bosman F T, Jaffe E S, Lakhani S R, et al. (series eds.): World Health Organization Classification of Tumors series, vol. 6. International Agency for Research on Cancer, Lyon, France, 2021.)

41.2.6 Posterior fossa group B (PFB) ependymoma (WHO grade 2 or 3)

General information

PFB ependymoma is a circumscribed posterior fossa glioma aligned with the PFB molecular groups of ependymomas (by DNA methylation profiling) having small uniform cells with round nuclei that form focal pseudorosettes or ependymal rosettes embedded in a fibrillary matrix.¹

Diagnostic criteria for posterior fossa group B (PFB) ependymoma

Diagnostic criteria for posterior fossa group B (PFB) ependymoma are shown in ► Fig. 41.7.¹

Grading

Grading of WHO grade 2 or 3 is based on the degree of anaplasia and reported in an integrated diagnosis.² High-grade features, including brisk mitotic activity and microvascular proliferation, were observed in 41% of PFB ependymomas.²⁰

Location

These tumors arise more frequently from the floor of the 4th ventricle.

Presentation

Similar to the presentation of posterior fossa ependymomas (p.728) in general.

Epidemiology

PFB ependymomas are more common in adults and adolescents than in pre-adolescents, with a median age of 30 years (range: 1–72) at presentation.¹

Posterior fossa group B (PFB) ependymoma

▷ ESSENTIAL ◁

posterior fossa tumor with morphological and immunohistochemical features of ependymoma

AND

DNA methylation profile aligned with PFB ependymoma

▷ DESIRABLE ◁

- ▶ chromosomal instability and aneuploidy on genome-wide copy-number analysis
- ▶ retained nuclear expression of H3 p.K28me3 (K27me3)^a

^a retention of nuclear H3 p.K28me3 (K27me3) expression is seen, but is not specific for PFB ependymomas

Fig. 41.7 Diagnostic criteria for posterior fossa group B (PFB) ependymoma. (Modified with permission from WHO Classification of Tumors Editorial Board, (ed.) World Health Organization classification of tumors of the central nervous system. 5th ed. Bosman F T, Jaffe E S, Lakhani S R, et al. (series eds.): World Health Organization Classification of Tumors series, vol. 6. International Agency for Research on Cancer, Lyon, France, 2021.)

Prognosis

Subtotal resection and loss of 13q correlated with a poor prognosis, and gain of 1q did not appear to be correlated to OS.²²

41.2.7 Spinal ependymoma (WHO grade 2 or 3)

General information

Spinal ependymomas are circumscribed intramedullary spinal gliomas having small uniform cells with round nuclei that form focal pseudorosettes or ependymal rosettes embedded in a fibrillary matrix. They typically have low levels of mitoses. Features of myxopapillary ependymoma or subependymoma are absent, by definition. *MYCN* amplification is absent when testing is possible.¹

Diagnostic criteria for spinal ependymoma

Diagnostic criteria for spinal ependymoma are shown in ► Fig. 41.8.¹

Grading

The vast majority of spinal ependymomas are WHO grade 2, with grade 3 tumors being rare. Grade 3 tumors have brisk mitotic activity, high cell density, and tend to invade adjacent spinal cord structures.

Location

These tumors arise anywhere along the spinal canal, but favor cervical or cervicothoracic levels.

Presentation

See presentation of intramedullary spinal cord tumors (p.986).

Epidemiology

Spinal ependymomas comprise 17% of primary spinal tumors in the U.S.²³

Spinal ependymoma

▷ ESSENTIAL ◁

spinal tumor with morphological and immunohistochemical features of ependymoma

AND

absence of morphological features of myxopapillary ependymoma or subependymoma

▷ DESIRABLE ◁

- ▶ DNA methylation profile aligned with spinal ependymoma
- ▶ loss of chromosome 22q
- ▶ no MYCN amplification

Fig. 41.8 Diagnostic criteria for spinal ependymoma. (Modified with permission from WHO Classification of Tumors Editorial Board, (ed.) World Health Organization classification of tumors of the central nervous system, 5th ed. Bosman F T, Jaffe E S, Lakhani S R, et al. (series eds.): World Health Organization Classification of Tumors series, vol. 6. International Agency for Research on Cancer, Lyon, France, 2021.)

Imaging

These are intramedullary tumors that are low intensity on T1 MRI, hyperintense on T2, and enhance readily. They tend to be more centrally located and more readily defined than spinal astrocytomas which are often eccentric and poorly demarcated.²⁴ 22% have a tumor cyst, and 62% are associated with a nontumoral cyst (e.g., syringomyelia).²⁴ Necrosis or calcifications may be seen.

Prognosis

Prognosis for grade 2 spinal ependymomas is favorable with PFS rates of 70% in children and 90% in adults over 5–10 years, and OS of 90% in children and 100% in adults over the same period.²⁵ Unfortunately, late recurrence is not uncommon.²⁶ The prognosis for grade 3 tumors is poor.²⁵

41.2.8 Spinal ependymoma, MYCN-amplified (WHO grade N/A)

General information

MYCN-amplified spinal ependymomas are rare, circumscribed intramedullary spinal gliomas having small uniform cells with round nuclei that form focal pseudorosettes or ependymal rosettes embedded in a fibrillary matrix. Essentially all tumors demonstrate microvascular proliferation, necrosis and brisk mitotic activity. By definition, MYCN amplification is present.¹

Diagnostic criteria for spinal ependymoma, MYCN-amplified

Diagnostic criteria for spinal ependymoma, MYCN-amplified are shown in ► Fig. 41.9.¹

Grading

A WHO grade has not yet been assigned to these tumors.

Location

78% of cases involved the cervical or thoracic spinal cord, and 7% lumbar. They may be intramedullary, intramedullary with exophytic component, or mostly extramedullary. Leptomeningeal is frequent at diagnosis, or eventually occurs during the course.¹

Spinal ependymoma, MYCN-amplified

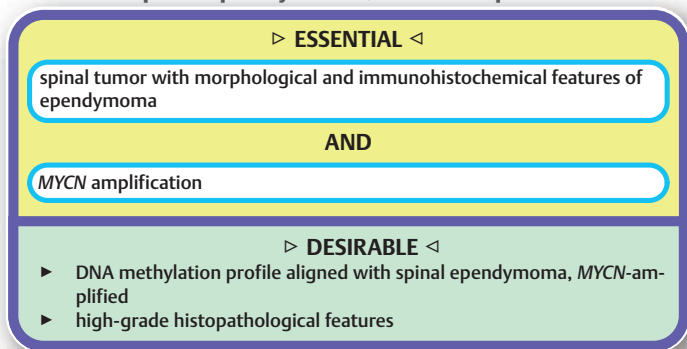


Fig. 41.9 Diagnostic criteria for spinal ependymoma, MYCN-amplified. (Modified with permission from WHO Classification of Tumors Editorial Board, (ed.) World Health Organization classification of tumors of the central nervous system. 5th ed. Bosman F T, Jaffe E S, Lakhani S R, et al. (series eds.): World Health Organization Classification of Tumors series, vol. 6. International Agency for Research on Cancer, Lyon, France, 2021.)

Presentation

See presentation of intramedullary spinal cord tumors (p.986).

Epidemiology

Only 27 cases were reported at the time the WHO CNS5 classification was published. 17 occurred in women, 10 in men. The median age at presentation was 31 years, with a range of 12-56 years.¹

Imaging

These are intramedullary tumors that are low intensity on T1 MRI, hyperintense on T2, and enhance avidly. They tend to be more centrally located and more readily defined than spinal astrocytomas which are often eccentric and poorly demarcated.²⁴ 22% have a tumor cyst, and 62% are associated with a nontumoral cyst (e.g., syringomyelia).²⁴ Necrosis or calcifications may be seen.

Prognosis

Prognosis is poor for PFS and OS. Early spread throughout the neuraxis is frequent. Recurrence occurred in all patients with follow-up.¹

41.2.9 Myxopapillary ependymoma (WHO grade 2)

General information

A distinctive variant of ependymoma. A WHO grade 2 glial neoplasm that essentially only occurs in the conus medullaris (the inferior terminus of the spinal cord) or the filum terminale (connective tissue band that extends inferiorly from the conus medullaris) characterized by tumor cells radially arranged around blood vessels with perivascular myxoid changes and microcyst formation.

Staining for GFAP can help differentiate this tumor from metastatic carcinoma, paragangliomas, schwannomas, chordomas, and myxoid chondrosarcomas.

Diagnostic criteria for myxopapillary ependymoma

Diagnostic criteria for myxopapillary ependymoma are shown in ► Fig. 41.10.¹

Myxopapillary ependymoma

▷ ESSENTIAL ◁

glioma with papillary structures and perivascular myxoid change or at least focal myxoid microcysts

AND

immunoreactivity for GFAP

AND (for unresolved lesions)

DNA methylation profile aligned with myxopapillary ependymoma

▷ DESIRABLE ◁

- ▶ papillary arrangements of tumor cells around vascularized fibromyxoid cores
- ▶ location in the filum terminale or conus medullaris

Fig. 41.10 Diagnostic criteria for myxopapillary ependymoma. (Modified with permission from WHO Classification of Tumors Editorial Board, (ed.) World Health Organization classification of tumors of the central nervous system. 5th ed. Bosman F T, Jaffe E S, Lakhani S R, et al. (series eds.): World Health Organization Classification of Tumors series, vol. 6. International Agency for Research on Cancer, Lyon, France, 2021.)

Treatment

Complete removal is often not possible due to invasion of the spinal cord parenchyma and/or CSF dissemination which is more common in pediatrics.

The high recurrence rate is an argument for early use of post-resection XRT, but proof of benefit in large series is lacking.²⁵

Prognosis

Prognosis is relatively favorable with 10-year survival rates > 90% in adults and children.

41.2.10 Subependymoma (WHO grade 1)

General information

Key concepts

- slow growing, WHO grade 1 tumor
- arise from ependymal lining of, and protrude into, the ventricle. Noninvasive
- one of the only intraventricular tumors (excluding colloid cysts) that does not enhance
- almost always an incidental finding; hydrocephalus when symptomatic
- treatment: most are observed with serial imaging. Gross total removal, when possible, is the goal if intervention is indicated

A slow growing (WHO grade 1) glioma arising from the ependymal lining of the ventricles characterized by clusters of bland to mildly pleomorphic cells in a fibrillary matrix, often with microcysts. Cells are mitotically inactive. 50-60% occur in the 4th ventricle, 30-40% in the lateral ventricles.

The tumors protrude into the ventricle and are noninvasive.

Diagnostic criteria for subependymoma

Diagnostic criteria for subependymoma are shown in ► Fig. 41.11.¹

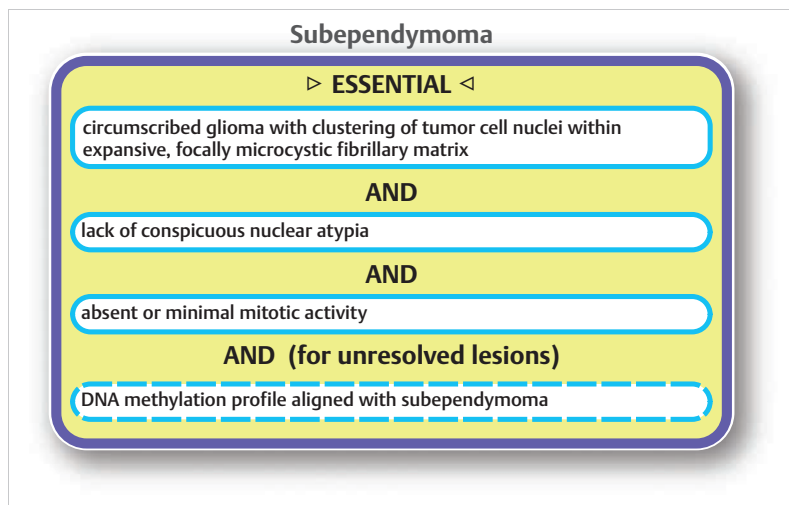


Fig. 41.11 Diagnostic criteria for subependymoma. (Modified with permission from WHO Classification of Tumors Editorial Board, (ed.) World Health Organization classification of tumors of the central nervous system. 5th ed. Bosman F T, Jaffe E S, Lakhani S R, et al. (series eds.): World Health Organization Classification of Tumors series, vol. 6. International Agency for Research on Cancer, Lyon, France, 2021.)

Epidemiology

Reliable data on incidence is difficult to come by since the majority are asymptomatic. A SEER registry database analysis found an incidence of 0.055 cases per 100,000 person years and a M:F ratio of about 2.5:1, and a peak incidence in adults 40–84 years old.²⁷

Subependymomas make up ≈ 8% of ependymal tumors and < 1% of intracranial neoplasms.

Presentation

almost always an incidental finding on imaging performed for some other reason. The unusual exception to this is when a subependymoma causes obstructive hydrocephalus.

Neuroimaging

The typical appearance is a nonenhancing well-demarcated intraventricular mass (► Fig. 41.12). There are differences of opinion of the tendency for subependymomas to enhance; Jelinek et al²⁸ found that they did not. Also see differential diagnosis for intraventricular lesions (p. 1667).

Treatment

Almost always an incidental finding on imaging performed for some other reason. The unusual exception to this is when a subependymoma causes obstructive hydrocephalus.

Tumors meeting the typical imaging characteristics of subependymomas may be monitored with serial imaging. Usually the first follow-up image is obtained 3 months after initial discovery to rule out a mimic with rapid growth. Then, annual imaging should be done for a period of time (e.g., 5 years) and then at a less frequent basis.

Prognosis

Prognosis is excellent. When treatment is indicated, gross total resection (GTR) when possible is almost always curative. When GTR is not possible (e.g., tumors arising from the floor of the 4th ventricle) subtotal resection is usually satisfactory since the tumors are so slow growing.

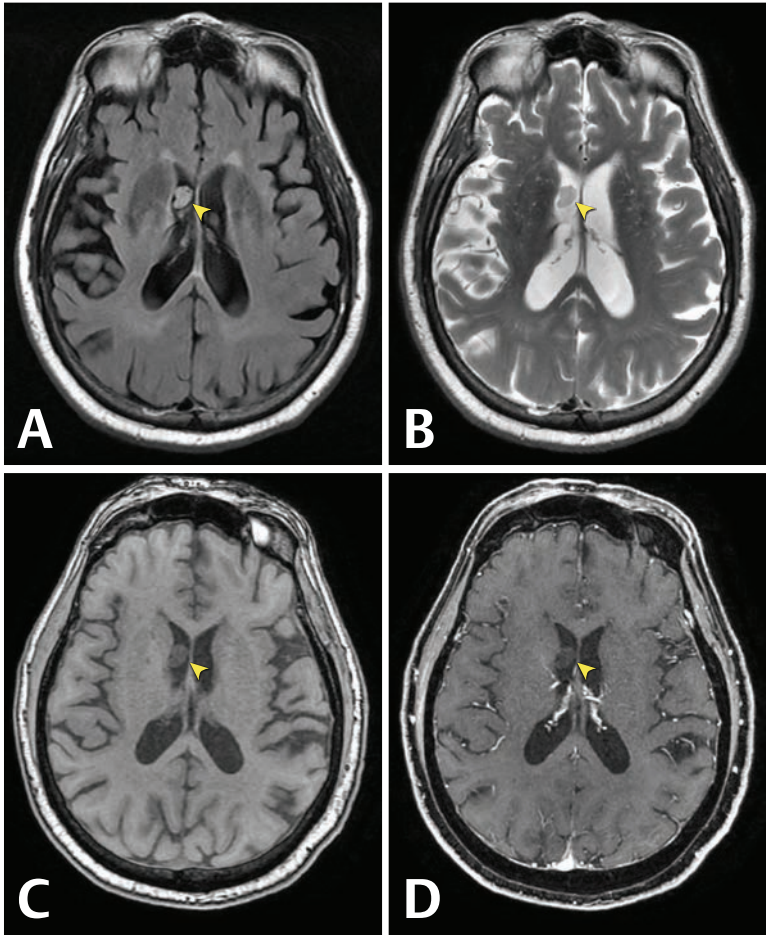


Fig. 41.12 Subependymoma (presumed, not biopsy proven). The lesion (yellow arrowheads) has remained unchanged over 10 years of observation.

Image: axial MRI A: FLAIR, B: T2, C: T1 non-contrast, D: T1 with contrast (note the lack of enhancement).

References

- [1] Bosman FT, Jaffe ES, Lakhani SR, et al. World Health Organization Classification of Tumors series, vol. 6. In: WHO Classification of Tumors Editorial Board (ed.) World Health Organization classification of tumors of the central nervous system. 5th ed. Lyon, France: International Agency for Research on Cancer; 2021
- [2] Louis DN, Perry A, Burger P, et al. International Society Of Neuropathology-Haarlem consensus guidelines for nervous system tumor classification and grading. *Brain Pathol.* 2014; 24:429–435
- [3] Elsamadicy AA, Koo AB, David WB, et al. Comparison of epidemiology, treatments, and outcomes in pediatric versus adult ependymoma. *Neurooncol Adv.* 2020; 2. DOI: 10.1093/noajnl/vdaa019
- [4] Junger ST, Andreiulo F, Mynarek M, et al. CDKN2A deletion in supratentorial ependymoma with RELA alteration indicates a dismal prognosis: a retrospective analysis of the HIT ependymoma trial cohort. *Acta Neuropathol.* 2020; 140:405–407
- [5] Duffner PK, Cohen ME, Freeman AI. *Pediatric Brain Tumors: An Overview.* Ca. 1985; 35:287–301

- [6] Louis DN, Ohgaki H, Wiestler OD, et al. WHO classification of tumors of the central nervous system. Lyon, France 2016
- [7] Mork SJ, Loken AC. Ependymoma: A Follow-Up Study of 101 Cases. *Cancer*. 1977; 40:907–915
- [8] Shaw EG, Evans RG, Scheithauer BW, et al. Postoperative Radiotherapy of Intracranial Ependymoma in Pediatric and Adult Patients. *International Journal of Radiation Oncology and Biological Physics*. 1987; 13:1457–1462
- [9] Youmans JR. *Neurological Surgery*. Philadelphia 1982
- [10] Merchant TE, Mulhern RK, Krasin MJ, et al. Preliminary results from a phase II trial of conformal radiation therapy and evaluation of radiation-related CNS effects for pediatric patients with localized ependymoma. *Journal of Clinical Oncology*. 2004; 22:3156–3162
- [11] MacDonald SM, Safai S, Trofimov A, et al. Proton radiotherapy for childhood ependymoma: initial clinical outcomes and dose comparisons. *International Journal of Radiation Oncology, Biology, Physics*. 2008; 71:979–986
- [12] Vanuytsel L, Brada M. The Role of Prophylactic Apical Irradiation in Localized Intracranial Ependymoma. *International Journal of Radiation Oncology and Biological Physics*. 1991; 21:825–830
- [13] van Veelen-Vincent M L, Pierre-Kahn A, Kalifa C, et al. Ependymoma in childhood: prognostic factors, extent of surgery, and adjuvant therapy. *Journal of Neurosurgery*. 2002; 97:827–835
- [14] Grundy RG, Wilne SA, Weston CL, et al. Primary postoperative chemotherapy without radiotherapy for intracranial ependymoma in children: the UKCCSG/SIOP prospective study. *Lancet Oncol*. 2007; 8:696–705
- [15] Little AS, Sheean T, Manoharan R, et al. The management of completely resected childhood intracranial ependymoma: the argument for observation only. *Child's Nervous System*. 2009; 25:281–284
- [16] Kawabata Y, Takahashi JA, Arakawa Y, et al. Long-term outcome in patients harboring intracranial ependymoma. *Journal of Neurosurgery*. 2005; 103:31–37
- [17] Zacharoulis S, Ji L, Pollack IF, et al. Metastatic ependymoma: a multi-institutional retrospective analysis of prognostic factors. *Pediatric Blood and Cancer*. 2008; 50:231–235
- [18] Foreman NK, Love S, Thorne R. Intracranial ependymomas: analysis of prognostic factors in a population-based series. *Pediatr Neurosurg*. 1996; 24:119–125
- [19] Sutton LN, Goldwein J, Perilongo G, et al. Prognostic Factors in Childhood Ependymomas. *Pediatr Neurosurg*. 1990; 16:57–65
- [20] Pajtler KW, Witt H, Sill M, et al. Molecular Classification of Ependymal Tumors across All CNS Compartments, Histopathological Grades, and Age Groups. *Cancer Cell*. 2015; 27:728–743
- [21] Godfraind C, Kaczmarzka JM, Kocak M, et al. Distinct disease-risk groups in pediatric supratentorial and posterior fossa ependymomas. *Acta Neuropathol*. 2012; 124:247–257
- [22] Cavalli FMG, Hubner JM, Sharma T, et al. Heterogeneity within the PF-EPN-B ependymoma subgroup. *Acta Neuropathol*. 2018; 136:227–237
- [23] Ostrom QT, Cioffi G, Waite K, et al. CBTRUS Statistical Report: Primary Brain and Other Central Nervous System Tumors Diagnosed in the United States in 2014–2018. *Neuro Oncol*. 2021; 23:iii1–iii105
- [24] Koeller KK, Rosenblum RS, Morrison AL. Neoplasms of the spinal cord and filum terminale: radiologic-pathologic correlation. *Radiographics*. 2000; 20:1721–1749
- [25] Benesch M, Frappaz D, Massimino M. Spinal cord ependymomas in children and adolescents. *Childs Nerv Syst*. 2012; 28:2017–2028
- [26] Gomez DR, Missett BT, Wara WM, et al. High failure rate in spinal ependymomas with long-term follow-up. *Neuro Oncol*. 2005; 7:254–259
- [27] Nguyen HS, Doan N, Gelsomino M, et al. Intracranial Subependymoma: A SEER Analysis 2004–2013. *World Neurosurg*. 2017; 101:599–605
- [28] Jelinek J, Smirniotopoulos JG, Parisi JE, et al. Lateral ventricular neoplasms of the brain: Differential diagnosis based on clinical, CT, and MR findings. *AJNR*. 1990; 11:567–574

42 Choroid Plexus Tumors

42.1 General information

Choroid plexus tumors can occur anywhere there is choroid plexus. This is one of several groups of tumors with a younger median age at diagnosis than other histologies.¹

Most are histologically benign.

All may produce drop mets in the CSF, but WHO grade 3 choroid plexus carcinomas do so more commonly.

Malignant degeneration from WHO grade 1 or 2 to grade 3 was seen in 2 out of 124 patients with 59 months mean follow-up.²

42.2 Choroid plexus tumor types

42.2.1 Choroid plexus papilloma (CPP) (WHO grade 1)

General information

A benign (WHO grade 1) intraventricular neoplasm resembling normal choroid plexus, with little or no mitotic activity.

Diagnostic criteria for choroid plexus papilloma

Diagnostic criteria for choroid plexus papilloma are shown in ► Fig. 42.1.³

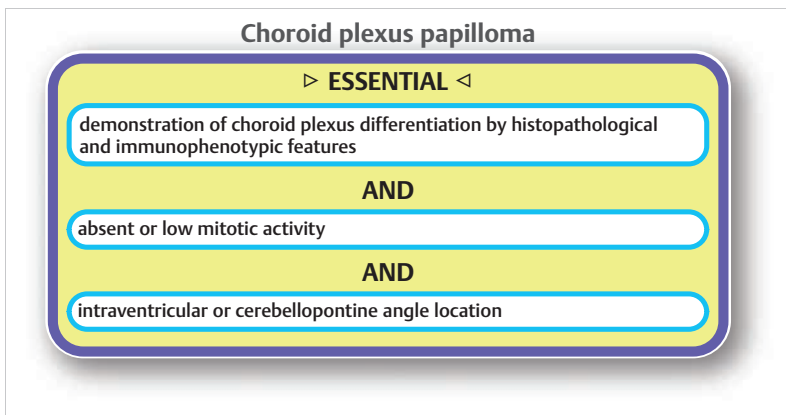


Fig. 42.1 Diagnostic criteria for choroid plexus papilloma. (Modified with permission from WHO Classification of Tumors Editorial Board, (ed.) World Health Organization classification of tumors of the central nervous system. 5th ed. Bosman F T, Jaffe E S, Lakhani S R, et al. (series eds.): World Health Organization Classification of Tumors series, vol. 6. International Agency for Research on Cancer, Lyon, France, 2021.)

Location

CPP are most frequently found in the lateral ventricles (with a predilection for the left side), followed by the fourth ventricles and then third ventricles. Rare ectopic cases have been reported. They may be found in the CPA from extension of choroid plexus through the foramen of Luschka. In adults these tumors are usually infratentorial, whereas in children they tend to occur supratentorially (a reverse from the situation for most other tumors) in the lateral ventricle.⁴ See Intraventricular lesions (p. 1667) for differential diagnosis.

Epidemiology

Although they may occur at any age, 70% of patients are <2 yrs old.⁵ About 80% of lateral ventricle CPPs occur in patients <20 years of age, unlike fourth ventricular CPPs which are evenly distributed among all ages.⁶

CPP comprise 2.1% of all primary and CNS tumors in the U.S.,¹ and 2.1% of brain tumors in individuals <15 years of age, and 10–20% of brain tumors in the first year of life. A portion of these tumors occur in neonates, supporting the hypothesis that some of these are congenital.⁴

Presentation

Most present with symptoms of increased ICP from hydrocephalus (H/A, N/V, craniomegaly), others may present with seizures, subarachnoid hemorrhage (with meningismus), or focal neurologic deficit (hemiparesis, sensory deficits, cerebellar signs, or cranial nerve palsies of III, IV, and VI).

Hydrocephalus, which may result from: overproduction of CSF (although total removal of these tumors does not always cure the hydrocephalus—especially in patients with high CSF protein, hemorrhage from tumor or surgery, or ependymitis), obstruction of CSF outflow, or communicating hydrocephalus from CSF-borne particulates.

Imaging

CT: iso- to hyper-dense.

MRI: T1 isointense, T2 hyperintense.

CT or MRI: intraventricular multilobulated mass, classically with projecting “fronds.” Densely enhancing. Hydrocephalus is common.

Treatment & prognosis

CPP are often cured surgically with total removal.

The operation may be difficult due to fragility of the tumor and bleeding from the choroidal arteries. However, persistence with a second and sometimes even third operation is recommended as 5-year survival rate of 84–97% can be achieved.^{4,7} Postoperative subdural collections after transcortical tumor excision may occur, and may result from a persistent ventriculosubdural fistula, which may require subdural-peritoneal shunting.⁵

Malignant degeneration is rare. There is no role for chemotherapy or radiation for these WHO grade 1 lesions.

42.2.2 Atypical choroid plexus papilloma (atypical CPP) (WHO grade 2)

General information

Atypical CPP is a CPP (p.739) having more mitotic figures than CPP (≥ 2 mitoses per 10 randomly selected HPFs) without frank signs of malignancy seen in choroid plexus carcinoma.⁸ They are more likely to recur than their grade 1 counterpart.

Diagnostic criteria for atypical choroid plexus papilloma

Diagnostic criteria for atypical choroid plexus papilloma are shown in ► Fig. 42.2.³

Location

Unlike CPP which occur equally as often supratentorially as infratentorially, these grade 2 lesions are more common in the lateral ventricles. See Intraventricular lesions (p.1667) for differential diagnosis.

Epidemiology

Median age at diagnosis is younger than CPP (8–10 months vs. 26–35 months). All of the atypical CPP patients were <10–11 years of age.

Atypical choroid plexus papilloma

▷ ESSENTIAL ◁

intraventricular or cerebellopontine angle location

AND

demonstration of choroid plexus differentiation by histopathological and immunophenotypic features

AND

demonstration of ≥ 1 mitosis/mm² in a minimum of 2.3 mm² (equating to ≥ 2 mitoses/10 HPF of 0.23 mm²)

AND

absence of criteria qualifying for the diagnosis of choroid plexus carcinoma

▷ DESIRABLE ◁

- ▶ *in select cases: demonstration of hyperploidy by genome-wide chromosomal copy-number analysis*

Fig. 42.2 Diagnostic criteria for atypical choroid plexus papilloma. (Modified with permission from WHO Classification of Tumors Editorial Board, (ed.) World Health Organization classification of tumors of the central nervous system. 5th ed. Bosman F T, Jaffe E S, Lakhani S R, et al. (series eds.): World Health Organization Classification of Tumors series, vol. 6. International Agency for Research on Cancer, Lyon, France, 2021.)

Imaging

Imaging appears the same as imaging with CPP (p. 740).

Prognosis

Prognosis is intermediate between CPP (grade 1) and choroid plexus carcinoma (grade 3). The definitional requirement of mitotic counts ≥ 2 per 10 HPF increased the 5-year recurrence for atypical CPP by a factor of 5 over CPP.

42.2.3 Choroid plexus carcinoma (CPC) (WHO grade 3)

General information

A malignant (WHO grade 3) tumor of the choroid plexus that invades adjacent brain and is prone to metastasize through the CSF (present in 21% at diagnosis). Occurs primarily in the lateral ventricles of children.

Diagnostic criteria for choroid plexus carcinoma

Diagnostic criteria for choroid plexus carcinoma are shown in ▶ Fig. 42.3.³

Choroid plexus carcinoma

▷ ESSENTIAL ◁

demonstration of choroid plexus differentiation by histopathological and immunophenotypic features

AND

presence of at least four of the following five histological features:

- increased cellular density
- nuclear pleomorphism
- blurring of the papillary pattern with poorly structured sheets of tumor cells
- necrotic areas
- frequent mitoses, usually > 2.5 mitoses/mm² in a minimum of 2.3 mm² (equating to > 5 mitoses/10 HPF of 0.23 mm²)

AND

intraventricular location

▷ DESIRABLE ◁

- ▶ *TP53* mutation analysis
- ▶ methylation profile of choroid plexus carcinoma
- ▶ in select cases: demonstration of hypoploidy by genome-wide chromosomal copy-number analysis

Fig. 42.3 Diagnostic criteria for choroid plexus carcinoma. (Modified with permission from WHO Classification of Tumors Editorial Board, (ed.) World Health Organization classification of tumors of the central nervous system, 5th ed. Bosman F T, Jaffe E S, Lakhani S R, et al. (series eds.): World Health Organization Classification of Tumors series, vol. 6. International Agency for Research on Cancer, Lyon, France, 2021.)

Epidemiology

80% of all CPCs are found in children.

Most arise sporadically, but, since 40% occur as part of Li-Fraumeni syndrome (p.650) with germline *TP53* pathogenic sequence variants, genetic counselling and *TP53* testing is recommended for all patients and their family.³

Imaging

CPCs have heterogeneous signal on T1 & T2 MRI, irregular enhancing margins (suggesting subependymal invasion), edema in the adjacent brain, vascular flow voids in 55%, and often present with hydrocephalus and early CSF dissemination.⁹ Imaging of the entire neuraxis is recommended due to the high incidence of drop mets.

Treatment

Even malignant choroid plexus tumors respond to surgery.

Chemotherapy benefits a subset of patients.¹⁰

XRT was not shown to improve survival in patients undergoing surgery.

Prognosis

3- and 5-year progression free survival rates are 58% and 38% respectively, and overall survival is 83% and 62% for the same time periods.¹¹

Absence of TP53 mutation may be associated with a more favorable outcome.

References

- [1] Ostrom QT, Cioffi G, Waite K, et al. CBTRUS Statistical Report: Primary Brain and Other Central Nervous System Tumors Diagnosed in the United States in 2014-2018. *Neuro Oncol.* 2021; 23:iii1-iii105
- [2] Jeibmann A, Wrede B, Peters O, et al. Malignant progression in choroid plexus papillomas. *Journal of Neurosurgery.* 2007; 107:199-202
- [3] Bosman FT, Jaffe ES, Lakhani SR, et al. World Health Organization Classification of Tumors series, vol. 6. In: WHO Classification of Tumors Editorial Board (ed.) World Health Organization classification of tumors of the central nervous system. 5th ed. Lyon, France: International Agency for Research on Cancer; 2021
- [4] Ellenbogen RG, Winston KR, Kupsky WJ. Tumors of the Choroid Plexus in Children. *Neurosurgery.* 1989; 25:327-335
- [5] Boyd MC, Steinbok P. Choroid Plexus Tumors: Problems in Diagnosis and Management. *Journal of Neurosurgery.* 1987; 66:800-805
- [6] Wolff JE, Sajedi M, Brant R, et al. Choroid plexus tumours. *Br J Cancer.* 2002; 87:1086-1091
- [7] Krishnan S, Brown PD, Scheithauer BW, et al. Choroid plexus papillomas: a single institutional experience. *J Neurooncol.* 2004; 68:49-55
- [8] Jeibmann A, Hasselblatt M, Gerss J, et al. Prognostic implications of atypical histologic features in choroid plexus papilloma. *Journal of Neuropathology and Experimental Neurology.* 2006; 65:1069-1073
- [9] Meyers SP, Khademian ZP, Chuang SH, et al. Choroid plexus carcinomas in children: MRI features and patient outcomes. *Neuroradiology.* 2004; 46:770-780
- [10] Wrede B, Liu P, Wolff JE. Chemotherapy improves the survival of patients with choroid plexus carcinoma: a metaanalysis of individual cases with choroid plexus tumors. *Journal of Neuro-Oncology.* 2007; 85:345-351
- [11] Zaky W, Dhali G, Khatua S, et al. Choroid plexus carcinoma in children: the Head Start experience. *Pediatr Blood Cancer.* 2015; 62:784-789

43 Embryonal Tumors

43.1 General information for embryonal tumors

43

Embryonal tumors are a diverse category of malignancies featuring immature tumor cells that resemble neural progenitors. It is divided into medulloblastomas (relatively common, occurring $\approx$ exclusively in the posterior fossa) and “other” rare non medulloblastoma CNS embryonal tumors (p. 754), many of which exhibit amplification of the C19MC region of chromosome 19.

Many of these tumors were formerly encompassed by the term primitive neuroectodermal tumor (PNET) which has been eliminated from the diagnostic lexicon.¹

43.2 Medulloblastoma (a subset of embryonal tumors), general aspects

43.2.1 General information

Key concepts

- a small-cell embryonal WHO grade 4 neuroepithelial tumor that occurs predominantly in the posterior fossa of children (peak: 1st decade). The most common pediatric brain malignancy
- classified on 4 molecular and 4 histologic criteria
 - 4 molecularly defined types: 1) WNT-activated; 2) SHH-activated (TP53-mutant & -wildtype); 3) non-WNT/non-SHH, group 3; 4) non-WNT/non-SHH, group 4
 - 4 histologic types: 1) classic; 2) desmoplastic/nodular (D/N); 3) extensive nodularity (MBEN); 4) large cell/anaplastic (LC/A)
- brainstem invasion usually limits complete surgical excision
- staging: all patients must be evaluated for “drop mets” (imaging the entire neuraxis with contrast - brain + spine, pre-operatively if possible), and postoperatively with LP for CSF cytology

Medulloblastomas (MDB) are WHO grade 4 embryonal neuroepithelial tumors that typically occur in the posterior fossa (cerebellum or dorsal brainstem) of children. In the cerebellum, they typically arise in the vermis near the apex of the roof of the 4th ventricle (the fastigium [Latin for summit]). They account for 68% of embryonal tumors.²

43.2.2 Seeding and metastases

$\approx$ 10–35% have seeded the craniospinal axis at the time of diagnosis,³ and extraneural mets occur in 5% of patients,⁴ sometimes promoted by shunting⁵ (although this is uncommon⁶).

43.2.3 Clinical

Clinical history is typically brief (6–12 weeks). Predilection for posterior fossa predisposes to early obstructive hydrocephalus. Usual presenting symptoms: H/A, N/V, and ataxia. Infants with hydrocephalus may present with irritability, lethargy, or progressive macrocrania.⁷ Spinal drop mets may produce back pain, urinary retention, or leg weakness.

Common signs: papilledema, truncal and appendicular ataxia, nystagmus, EOM palsies. Macrocrania in infants and young children.

43.2.4 Classification

Classification of medulloblastomas based on a combination of molecular signatures combined with histopathological findings is recommended and provides optimal prognostic information. Molecularly defined MDBs are associated with specific morphological patterns to a high degree.

Molecular genetics: Transcriptome profiling (analysis of all RNA in a cell population), microRNA analysis and methylome profiling allows MDBs to be segregated into these 4 principal types⁸ (which are covered individually in the following sections). NB: on a more granular level, additional subgroups have been identified and will continue to emerge.

1. WNT-activated (p.750)
2. SHH-activated
 - a) SHH-activated, TP53-wildtype (p.751)
 - b) SHH-activated, TP53-mutant (p.752)
3. non-WNT/non-SHH (p.753), group 3
4. non-WNT/non-SHH (p.753), group 4

Histology: 4 histologic types are defined; see detailed description (p.749)

1. classic (p.749)
2. desmoplastic/nodular (p.749) (D/N MDB)
3. extensive nodularity (p.749) (MBEN)
4. large cell/anaplastic (p.749) (LC/A MDB)

The distribution and prognosis of the histologic variants and prognostic information among the molecularly classified MDBs is depicted graphically in ► Fig. 43.1 (this is not the entire gamut, see reference⁹ for more details).

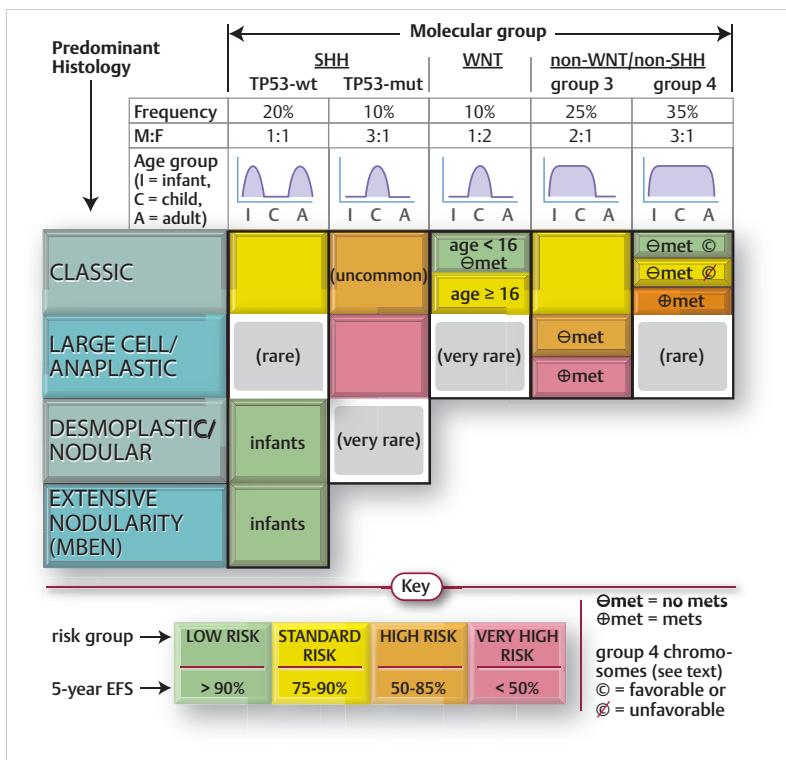


Fig. 43.1 Prognosis and distribution of histologic variants among molecular types of medulloblastoma. Box color indicates prognosis as follows:

5-year EFS rates: **low risk** >90%, **standard risk** 75-90%, **high risk** 50-75%, **very high risk** <50%

(see text for more details).^{8,10,11} No risk is shown for tumors of uncertain clinicopathological significance.

Abbreviations: EFS = event-free survival; MDB = medulloblastoma; SHH = activated sonic hedgehog; TP53 = tumor protein 53; mut = mutant; wt = wildtype; WNT = activated wingless/integrated (signal transduction pathway).

⊙ = favorable chromosomal findings for non-WNT/non-SHH group 4: chromosome 7 gain, chromosome 8 loss, chromosome 11 loss, chromosome 17 gain.

43.2.5 Evaluation

MDBs usually appear as a solid, IV-contrast-enhancing lesion on CT or MRI. Most are located in the midline in the region of the 4th ventricle (laterally situated tumors are more common in adults). Most have hydrocephalus at the time of presentation. Ependymoma (p. 1647) is the main entity from which to differentiate on imaging. MDBs tend to protrude into the 4th ventricle from the posterior aspect (the “roof”) of the ventricle, whereas ependymomas more commonly arise from the floor.

CT: noncontrast → typically hyperdense (due to high cellularity). Contrast → most enhance. 20% have calcifications. CT has poor resolution in the posterior fossa due to bone interference

MRI: Can be variable (► Fig. 43.2). T1WI → hypo- to isointense. T2WI → heterogeneous due to tumor cysts, vessels, and calcifications.¹² Most enhance. MRS: elevated choline and decreased NAA.

Spinal imaging: MRI with IV gadolinium or CT/myelography with water-soluble contrast should be done to rule out “drop mets.” Staging is done either pre-op or within 2–3 weeks of surgery.

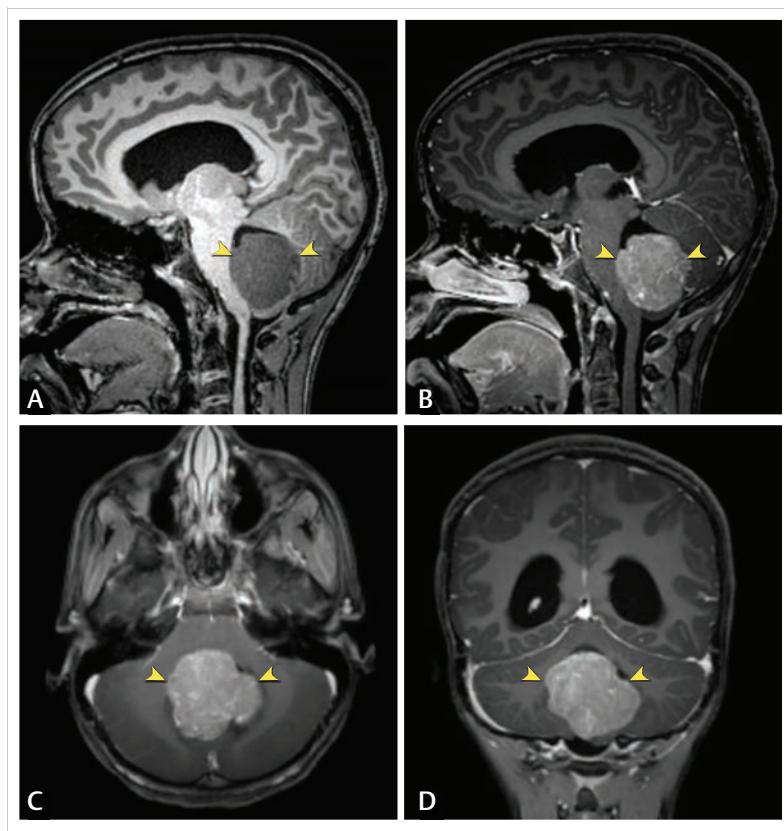


Fig. 43.2 Medulloblastoma. MRI of 10-year-old patient with medulloblastoma (yellow arrowheads) in the 4th ventricle producing obstructive hydrocephalus, presenting with progressive H/A, nausea and acute emesis. Image: A: sagittal T1 FLAIR without contrast, B: sagittal T1 FLAIR with contrast, C: axial T1 with contrast, D: coronal T1 with contrast.

► **Staging.** Staging employs MRI of the entire neuraxis without and with contrast and, postoperatively, CSF cytology obtained by LP. The grading system of Chang et al. (► Table 43.1) is still used.

Table 43.1 Post-operative staging system for medulloblastoma¹³

Stage	Description
M0	no metastasis
M1	microscopic tumor cells in the CSF
M2	gross nodular seeding in the cerebellar/cerebral subarachnoid space or in the third or lateral ventricles
M3	gross nodular seeding in the spinal subarachnoid space
M4	metastasis outside the CNS

43.2.6 Treatment

Surgical intervention

Surgical debulking of as much tumor as possible (without causing neurological injury) to obtain tissue for classification and to improve outcome is considered standard.

Invasion of or attachment to the floor of the fourth ventricle (brainstem in the region of the facial colliculus) often limits excision. It is better to leave a small residual on the brainstem (these patients do fairly well) than it is to chase every last remnant into the brainstem (neurologic deficit is more likely with this).

Surgical exposure of midline cerebellar medulloblastomas requires opening of the foramen magnum, usually removal of the posterior arch of C1, and occasionally the arch of C2. Tumor spread with arachnoidal thickening ("sugar coating") may occur.

Shunts: 30–40% of children require permanent VP shunts following p-fossa resection. The risk of shunt-related seeding has been quoted as high as 10–20%,³ but this is probably overestimated.⁶ In the past, tumor filters were sometimes used. They are less commonly used today because of the high incidence of obstruction.

Post-op adjuvant therapy

Following surgery, stratification of patients into risk groups guides subsequent therapy. One postoperative stratification scheme is shown in ► Table 43.2. Other systems have been published e.g., for pediatrics (age 3–17 years).⁸

In children: chemotherapy can be used to delay the need for XRT in 20–40% of children <3–4 years of age with non-disseminated MBD.¹⁵

► Low-risk and standard-risk patients

- especially in pediatrics, may consider reduced dose XRT: 23.4 Gy with a boost to the primary site of 54–55.8 Gy with adjuvant chemotherapy
- or conventional fractionated XRT (see high-risk below)

► High- and very high-risk patients

- conventional fractionated XRT (once a day) for 54 Gy to the primary tumor and 18 Gy to the craniospinal axis
- chemotherapy¹⁵
 - agents that have been used include: cisplatin, lomustine, vincristine, cyclophosphamide, etoposide, simultaneous high-dose IV methotrexate and/or intrathecal methotrexate of mafosfamide, and/or intraventricular methotrexate
 - in adults: SHH-activated MDBs (which have higher prevalence of PTCH & SMO mutations) may be more responsive to SMO receptor inhibitors and SHH-inhibiting drugs such as vismodegib¹⁵

43.2.7 Prognosis

Prognostic details for specific tumor types will be found in the following sections. Poor prognosticators in general include¹⁶:

- in some cell types younger age (especially if <3 yrs). For other cell types (e.g., desmoplastic/nodular or MBEN, the prognosis for infants is favorable)
- disseminated (metastatic) disease at the time of diagnosis except for MBEN
- inability to perform gross-total removal (especially if residual > 1.5 cm² with localized disease)
- poor Karnofsky performance scale score (p. 1640)

Most common site of recurrence is p-fossa.

Table 43.2 Postoperative risk stratification in medulloblastoma^{14,15,11}**Standard-risk tumors**

- total or near total surgical resection^a
- no CNS metastases on MRI of brain and spine
- no tumor cells on cytospin of lumbar CSF
- no clinical evidence of extra-CNS metastases
- MDB classification
 - SHH-activated, TP53-wildtype with classic histology
 - or non-WNT/non-SHH, group 4, with classic histology and no metastases at diagnosis and no favorable chromosomal findings^b
 - or non-WNT/non-SHH, group 3
 - or WNT with classic histology age ≥ 16 years

Low-risk tumors

- MDB classification
 - WNT-activated group
 - β -catenin mutation (mandatory testing)
 - β -catenin nuclear immunopositivity by immuno-histochemistry & monosomy 6 (optional testing)
 - or SHH-activated, TP53-wildtype in infants with
 - desmoplastic/nodular histology
 - or extensive nodularity histology
- total or near total surgical resection^a

High-risk tumors

- large cell/anaplastic histology non-WNT/non-SHH, group 3 without metastases at diagnosis
- classic histology with
 - SHH-activated, TP53-mutant
 - or non-WNT/non-SHH, group 4 with metastases at diagnosis
- unresectable tumor or residual tumor $> 1.5 \text{ cm}^2$ on axial-plane early post-op MRI
- extra-CNS metastases

Very high-risk tumors

- large cell/anaplastic histology with
 - SHH-activated, TP53-mutant
 - or non-WNT/non-SHH, group 3 with metastases at diagnosis

^a near total resection: $< 1.5 \text{ cm}^2$ on axial-plane early post-op MRI. This criteria is controversial as much of the data is from CT era⁸

^b favorable chromosomal findings for non-WNT/non-SHH group 4: chromosome 7 gain, chromosome 8 loss, chromosome 11 loss, and chromosome 17 gain

Long-term survivors of MDB are at significant risk for permanent endocrinologic, cognitive, and psychological sequelae of treatments. Infants and very young children with MDB remain a difficult therapeutic challenge because they have the most virulent form of the disease and are at highest risk for treatment-related sequelae.

43.3 Medulloblastomas by definition criteria

43.3.1 Medulloblastoma, histologically defined

General information

An embryonal neuroepithelial tumor that occurs in the posterior fossa consisting of small, poorly differentiated cells with a high nuclear to cytoplasmic ratio with profuse mitotic activity and apoptosis. Divided into 4 histologic subtypes described below. On the rare occasion that any subtype shows myogenic and/or melanocytic differentiation, the terms "medulloblastoma" and "melanocytic medulloblastoma" respectively may be used in the description.

Diagnostic criteria for medulloblastoma, histologically defined

Diagnostic criteria for medulloblastoma, histologically defined are shown in ► Fig. 43.3.⁹

Medulloblastoma, histologically defined

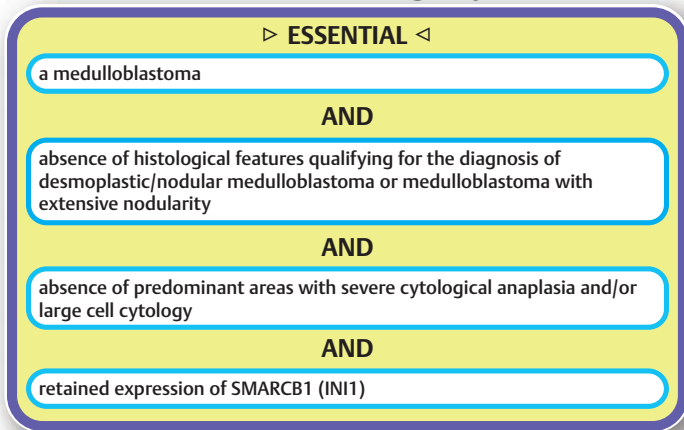


Fig. 43.3 Diagnostic criteria for medulloblastoma, histologically defined. (Modified with permission from WHO Classification of Tumors Editorial Board, (ed.) World Health Organization classification of tumors of the central nervous system. 5th ed. Bosman F T, Jaffe E S, Lakhani S R, et al. (series eds.): World Health Organization Classification of Tumors series, vol. 6. International Agency for Research on Cancer, Lyon, France, 2021.)

Subtypes (histologic/morphologic variants)

- **MDB, classic:** densely packed undifferentiated small round cells with mild-to-moderate nuclear pleomorphism and high mitotic count. Comprise $\approx$ 72% of MDBs. Occur at any age, but predominate in childhood. Found in all 4 molecular MDB types.
- **desmoplastic/nodular MDB:** nodular, reticulin-free zones and intervening densely packed, poorly differentiated cells in an intercellular network of reticulin-positive collagen fibers. Account for $\approx$ 20% of all MDBs. Occurs in cerebellar hemispheres and midline. Bimodal age distribution: 1) young children and adolescents, 2) adults. Associated with Gorlin syndrome in early childhood. The prognosis in early childhood is excellent with surgery + chemotherapy alone, and is likely better than with any other histologic subtype.
- **MDB with extensive nodularity (MBEN):** numerous reticulin-free neurocytic nodules with a neuropil matrix and narrow internodular strands of poorly differentiated tumor cells in a desmoplastic matrix. Occurs primarily in infants. Accounts for $<$ 5% of MDBs overall, but up to 50% of cases in age $<$ 3 years. Outcome is favorable with near 100% 5-year OS in many (but not all) series. $>$ 80% of these occur in the vermis. The presence of metastases at diagnosis in this group does not affect prognosis.
- **large cell/anaplastic MDB:** Undifferentiated cells with marked nuclear pleomorphism, prominent nucleoli, and high mitotic and apoptotic counts. Accounts for 10% of MDBs. Found in all 4 molecular MDB types. The LC/A morphology is an independent prognosticator of poor outcome (5-year PFS is 30–40%) (very high-risk MDB) justifying aggressive adjuvant therapy. Molecular biology of MDB SHH-activated TP53-mut and non-WNT/non-SHH group 3 with MYC amplification can behave more aggressively.

Staging

See postoperative staging (p. 746).

Treatment

For treatment, see treatment for medulloblastomas (p. 747).

43.3.2 Medulloblastomas, molecularly defined

Medulloblastoma, WNT-activated (WHO grade 4)

General information

Typical in older childhood (hardly ever in infancy). Account for $\approx 10\%$ of all MDBs and $\approx 15\%$ of adult MDBs. Male:female = 1:2. Predominant histologic variant: classic (almost all). Favors the cerebellar midline. Putative cell of origin: lower rhombic lip progenitor cells. Frequent genetic mutations: CTNNB1, DDX3X, TP53.

Diagnostic criteria for medulloblastoma, WNT-activated

Diagnostic criteria for medulloblastoma, WNT-activated are shown in ► Fig. 43.4.⁹

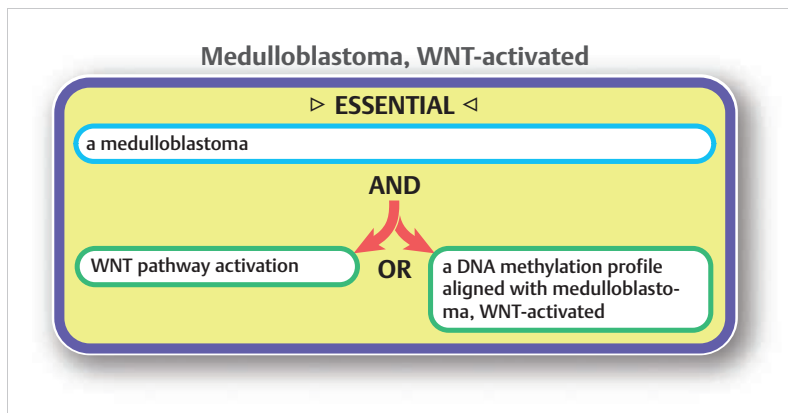


Fig. 43.4 Diagnostic criteria for medulloblastoma, WNT-activated. (Modified with permission from WHO Classification of Tumors Editorial Board, (ed.) World Health Organization classification of tumors of the central nervous system. 5th ed. Bosman F T, Jaffe E S, Lakhani S R, et al. (series eds.): World Health Organization Classification of Tumors series, vol. 6. International Agency for Research on Cancer, Lyon, France, 2021.)

Location

Generally either of the following:

- around the foramen of Luschka, arising from the brainstem or cerebellum
- or in the cerebellar midline, generally contiguous with the brainstem

Extension towards the CPA or cerebellar peduncle occurs in a significant number.

Histology

See section 43.3.1.

- **Classic morphology.** Almost all WNT-activated MDBs have classic morphology.
- **Large cell/anaplastic morphology.** Very rare. Uncertain clinical significance.

Staging

See postoperative staging (p.746).

Treatment

For treatment, see treatment for medulloblastomas (p.747).

Prognosis

In spite of being a grade 4 neoplasm, the prognosis for children is excellent with surgery and adjuvant therapy. In contrast, the prognosis is poor for adults.

Medulloblastoma, SHH-activated, TP53-wildtype (WHO grade 4)

Typical in infancy and adult. Male:female = 1:1. Predominant histologic variant: desmoplastic/nodular. Putative cell of origin: cerebellar granule neuron cell precursors of the external granule cell layer and cochlear nucleus; less likely neural stem cells of the subventricular zone. Frequent genetic mutations: PTCH1, SMO (adults), SUFU (infants), TERT promoter.

Diagnostic criteria for medulloblastoma, SHH-activated, TP53-wildtype

Diagnostic criteria for medulloblastoma, SHH-activated, TP53-wildtype are shown in ► Fig. 43.5.⁹

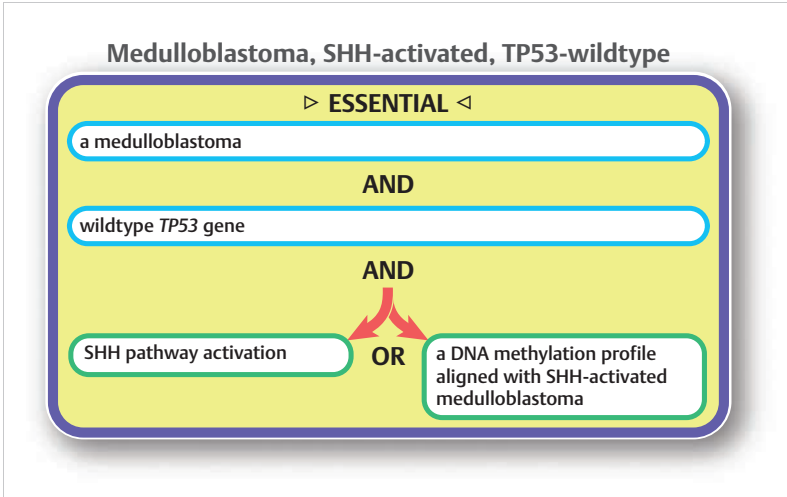


Fig. 43.5 Diagnostic criteria for medulloblastoma, SHH-activated, TP53-wildtype. (Modified with permission from WHO Classification of Tumors Editorial Board, (ed.) World Health Organization classification of tumors of the central nervous system. 5th ed. Bosman F T, Jaffe E S, Lakhani S R, et al. (series eds.): World Health Organization Classification of Tumors series, vol. 6. International Agency for Research on Cancer, Lyon, France, 2021.)

Subtypes

See reference.⁹

Four provisional molecular subgroups are determined by DNA methylation or transcriptome profiling

- SHH-1
- SHH-2
- SHH-3
- SHH-4

Location

Infants: the cerebellar vermis is much more frequently than the cerebellar hemispheres.

Older children and young adults: mainly involves the hemispheres.

Epidemiology

SHH activated medulloblastomas exhibit a bimodal age distribution with one peak in infants and the other in adults. M:F ratio is ≈ 1.5:1.

Histology

See section 43.3.1.

Most MDB, SHH activated TP53-wt tumors are desmoplastic/nodular or MBENS.

► **Classic morphology.** Prognosis: Standard risk tumor.

- ▶ **Large cell/anaplastic morphology.** Rare. Uncertain clinical significance.
- ▶ **Desmoplastic/nodular morphology.** Prognosis: low risk tumor. Prevalent in infants and adults.
- ▶ **Extensive nodularity morphology (MBEN).** Prognosis: low risk tumor. Prevalent in infants.

43

Staging

See postoperative staging (p.746).

Treatment

For treatment, see treatment for medulloblastomas (p.747).

Medulloblastoma, SHH-activated, TP53-mutant (WHO grade 4)**General information**

Typical in childhood. Male:female = 1:1. Predominant histologic variant: large cell/anaplastic. Putative cell of origin: cerebellar granule neuron cell precursors of the external granule cell layer and cochlear nucleus; less likely neural stem cells of the subventricular zone.

Diagnostic criteria for medulloblastoma, SHH-activated, TP53-mutant

Diagnostic criteria for medulloblastoma, SHH-activated, TP53-mutant are shown in ▶ Fig. 43.6.⁹

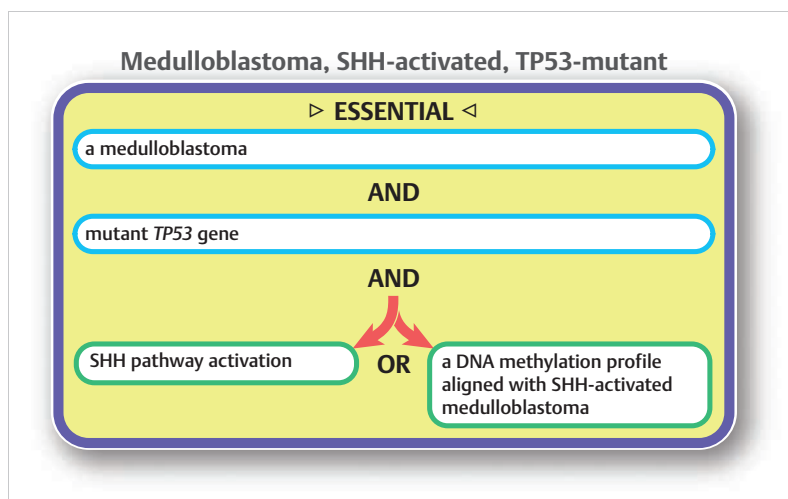


Fig. 43.6 Diagnostic criteria for medulloblastoma, SHH-activated, TP53-mutant. (Modified with permission from WHO Classification of Tumors Editorial Board, (ed.) World Health Organization classification of tumors of the central nervous system. 5th ed. Bosman F T, Jaffe E S, Lakhani S R, et al. (series eds.): World Health Organization Classification of Tumors series, vol. 6. International Agency for Research on Cancer, Lyon, France, 2021.)

Subtypes

The same four subtypes as MDB SHH-activated TP53-wildtype (viz. SHH 1-4).

Histology

See section 43.3.1.

Metastatic disease and MYCN amplification are independent poor prognosticators in children (not infants) and adolescents.

- **Classic morphology.** Prognosis: high-risk tumor. Uncommon.
- **Large cell/anaplastic morphology.** The predominant histologic type (70%). Prognosis: high-risk tumor. Prevalent in ages 7–17 years.
- **Desmoplastic/nodular morphology.** Very rare. Uncertain clinical significance.

Staging

See postoperative staging (p. 746).

Treatment

For treatment, see treatment for medulloblastomas (p. 747).

Medulloblastoma, non-WNT/non-SHH (WHO grade 4)

General information

An embryonal tumor of the cerebellum without a molecular signature. Classified as group 3 or group 4 tumors, with 8 molecular subgroups as determined by DNA methylation profiling. Typical in infancy and childhood. Male:female = 2:1. They occur exclusively in the inferior cerebellum, usually near the midline. Predominant histologic variants: classic; large cell/anaplastic. Putative cell of origin: cerebellar granule neuron cell precursors of the external granule cell layer. Frequent genetic mutations: PVT1-MYC, GF11/GF11B structural variants.

Diagnostic criteria for medulloblastoma, non-WNT/non-SHH

Diagnostic criteria for medulloblastoma, non-WNT/non-SHH are shown in ► Fig. 43.7.⁹

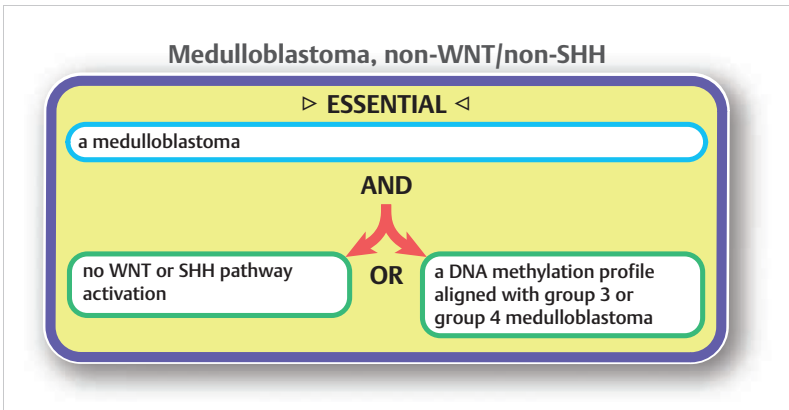


Fig. 43.7 Diagnostic criteria for medulloblastoma, non-WNT/non-SHH. (Modified with permission from WHO Classification of Tumors Editorial Board, (ed.) World Health Organization classification of tumors of the central nervous system. 5th ed. Bosman F T, Jaffe E S, Lakhani S R, et al. (series eds.): World Health Organization Classification of Tumors series, vol. 6. International Agency for Research on Cancer, Lyon, France, 2021.)

Subtypes

There are 8 subgroups of group 3/4 as differentiated by DNA methylation profiling.

Histology

See section 43.3.1.

- **Classic morphology.** The most common histology for this tumor. Prognosis: Standard risk tumor.

► **Large cell/anaplastic morphology.** This histology is more prevalent in group 3 than group 4. Prognosis: high risk tumor.

Staging

See postoperative staging (p. 746).

Treatment

For treatment, see treatment for medulloblastomas (p. 747).

Prognosis

In group 3 MDBs, MYC amplification is associated with poor outcome, but other markers also have prognostic impact.

Among group 4 MDBs, metastatic disease at the time of diagnosis is the strongest indicator of poor outcome. Favorable outcome in group 4 is associated with chromosome 7 gain, chromosome 8 loss, chromosome 11 loss, and chromosome 17 gain.

43.4 CNS embryonal tumors other than medulloblastoma

43.4.1 Atypical teratoid/rhabdoid tumor (AT/RT) (WHO grade 4)

General information

A high-grade malignant WHO grade 4 embryonal tumor of the CNS comprised of poorly differentiated elements and rhabdoid cells which exhibit biallelic inactivation of *SMARCB1* or rarely *SMARCA4* and which can differentiate along neuroepithelial, epithelial and mesenchymal cell lines. Many of these tumors were probably previously misdiagnosed as MDBs.

Diagnostic criteria for atypical teratoid/rhabdoid tumor

Diagnostic criteria for atypical teratoid/rhabdoid tumor are shown in ► Fig. 43.8.⁹

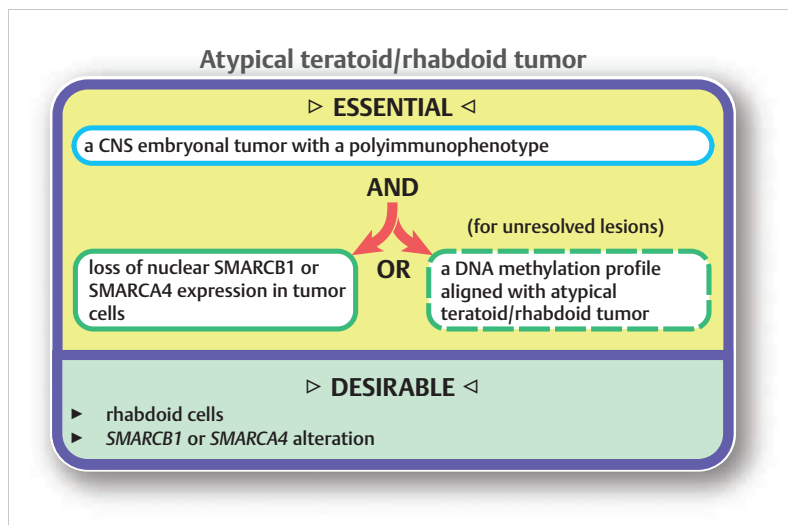


Fig. 43.8 Diagnostic criteria for atypical teratoid/rhabdoid tumor. (Modified with permission from WHO Classification of Tumors Editorial Board, (ed.) World Health Organization classification of tumors of the central nervous system. 5th ed. Bosman F T, Jaffe E S, Lakhani S R, et al. (series eds.): World Health Organization Classification of Tumors series, vol. 6. International Agency for Research on Cancer, Lyon, France, 2021.)

Subtypes

Three molecular subtypes are identifiable on DNA methylation profiling:

- AT/RT-SHH
- AT/RT-TYR
- AT/RT-MYC

Location

Anywhere along the neuraxis. Supratentorial location is more likely with advancing age, and includes the hemispheres, and less often the ventricles, or suprasellar or pineal region. Other locations include cerebellar hemispheres, cerebellopontine angle and brainstem. Spinal cord involvement is rare. 33% have CSF spread at presentation.

Epidemiology

Occurs primarily in infants and children (most are age < 2 years, 33% are ≤ 1 year old, and > 90% are < 5 years). Rare in adults. AT/RT account for 17% of embryonal tumors.²

Familial cases may occur in cohorts with rhabdoid tumor predisposition syndrome.

Staging

See postoperative staging (p. 746).

Prognosis

Although the prognosis is poor, not all AT/RTs have the same behavior. Pending validation studies, prognosis may be predicted based on molecular group, age, location and extent of resection.

43.4.2 Cribriform neuroepithelial tumor (CRINET) (*provisional*)

General information

A *provisional* non-rhabdoid neuroectodermal neoplasm identifiable by cribriform strands and ribbons with loss of SMARCB1 expression.

Diagnostic criteria for cribriform neuroepithelial tumor

Diagnostic criteria for cribriform neuroepithelial tumor are shown in ► Fig. 43.9.⁹

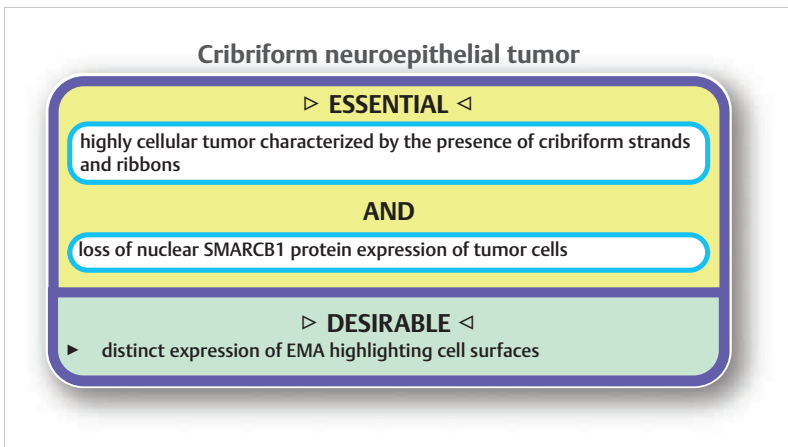


Fig. 43.9 Diagnostic criteria for cribriform neuroepithelial tumor. (Modified with permission from WHO Classification of Tumors Editorial Board, (ed.) World Health Organization classification of tumors of the central nervous system. 5th ed. Bosman F T, Jaffe E S, Lakhani S R, et al. (series eds.): World Health Organization Classification of Tumors series, vol. 6. International Agency for Research on Cancer, Lyon, France, 2021.)

Location

Near any of the ventricles (3rd, 4th, or lateral).

Epidemiology

Median age: 20 months (range: 10-129 months). M:F ratio 1.5:1.¹⁷

Prognosis

In 10 patients with CRINET, estimated mean OS was 125 months.¹⁷

43.4.3 Embryonal tumor with multilayered rosettes (ETMR) (WHO grade 4)

General information

A WHO grade 4 embryonal tumor with multilayered rosettes demonstrating one of three histological patterns (embryonal tumor with abundant neuropil and true rosettes, ependymoblastoma, or medulloepithelioma) typically exhibiting amplification in the C19MC region on chromosome 19 (19q13.42) or, rarely, a *DICER1* mutation. May develop in the cerebrum, brainstem, or cerebellum. Usually affect children < 4 years old (with the vast majority being < 2 years old).

Diagnostic criteria for embryonal tumor with multilayered rosettes

Diagnostic criteria for embryonal tumor with multilayered rosettes are shown in ► Fig. 43.10.⁹

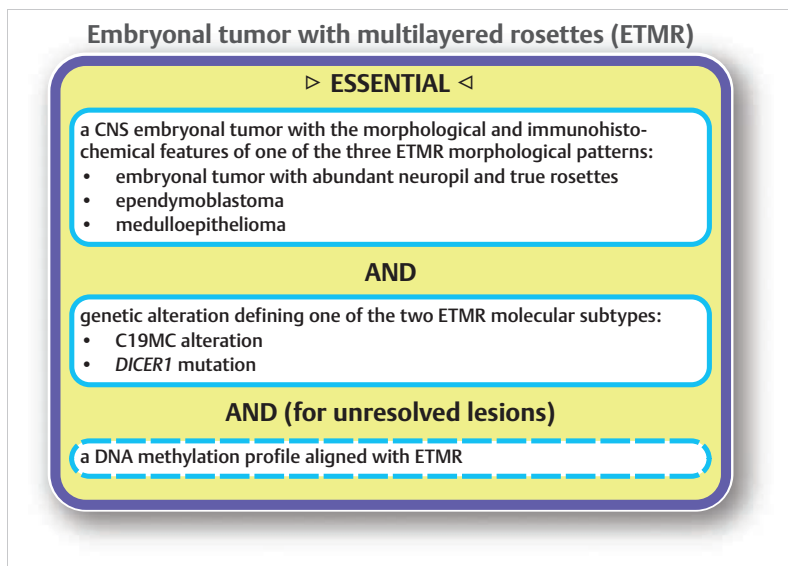


Fig. 43.10 Diagnostic criteria for embryonal tumor with multilayered rosettes. (Modified with permission from WHO Classification of Tumors Editorial Board, (ed.) World Health Organization classification of tumors of the central nervous system. 5th ed. Bosman F T, Jaffe E S, Lakhani S R, et al. (series eds.): World Health Organization Classification of Tumors series, vol. 6. International Agency for Research on Cancer, Lyon, France, 2021.)

Subtypes

- embryonal tumor with multilayered rosettes, C19MC-altered
- embryonal tumor with multilayered rosettes, *DICER1*-mutated

Location

The cerebral hemispheres is the most common location, with 45% located elsewhere including the posterior fossa. Spinal cord involvement is rare.

Epidemiology

Data is sparse due to the rarity of the tumor. Primarily affects children <4 years, with most younger than 2. M:F ratio is close to 1:1.

Staging

See postoperative staging (p. 746).

Prognosis

These aggressive tumors grow rapidly producing precipitous clinical deterioration. Mean OS is 12 months following intensive treatment.⁹ Outcome is worse when metastases are present at diagnosis and with brainstem involvement. Gross total resection, XRT and aggressive chemotherapy are “probably helpful.”

References

- [1] Louis DN, Perry A, Reifenberger G, et al. The 2016 World Health Organization Classification of Tumors of the Central Nervous System: a summary. *Acta Neuropathol.* 2016; 131:803–820
- [2] Ostrom QT, Cioffi G, Waite K, et al. CBRUS Statistical Report: Primary Brain and Other Central Nervous System Tumors Diagnosed in the United States in 2014–2018. *Neuro Oncol.* 2021; 23:iii1–iii105
- [3] Laurent JP, Cheek WR. Brain Tumors in Children. *J Pediatr Neurosci.* 1985; 1:15–32
- [4] Allen JC. Childhood Brain Tumors: Current Status of Clinical Trials in Newly Diagnosed and Recurrent Disease. *Ped Clin N Am.* 1985; 32:633–651
- [5] Kessler LA, Dugan P, Concannon JP. Systemic Metastases of Medulloblastoma Promoted by Shunting. *Surg Neurol.* 1975; 3:147–152
- [6] Berger MS, Baumeister B, Geyer JR, et al. The Risks of Metastases from Shunting in Children with Primary Central Nervous System Tumors. *J Neurosurg.* 1991; 74:872–877
- [7] Park TS, Hoffman HJ, Hendrick EB, et al. Medulloblastoma: Clinical Presentation and Management. *J Neurosurg.* 1983; 58:543–552
- [8] Ramaswamy V, Remke M, Bouffet E, et al. Risk stratification of childhood medulloblastoma in the molecular era: the current consensus. *Acta Neuropathol.* 2016; 131:821–831
- [9] Bosman FT, Jaffe ES, Lakhani SR, et al. World Health Organization Classification of Tumors series, vol. 6. In: WHO Classification of Tumors Editorial Board (ed.) World Health Organization classification of tumors of the central nervous system, 5th ed. Lyon, France: International Agency for Research on Cancer; 2021
- [10] Juraschka K, Taylor MD. Medulloblastoma in the age of molecular subgroups: a review. *J Neurosurg Pediatr.* 2019; 24:353–363
- [11] Goschzik T, Schwalbe EC, Hicks D, et al. Prognostic effect of whole chromosomal aberration signatures in standard-risk, non-WNT/non-SHH medulloblastoma: a retrospective, molecular analysis of the HIT-SIOP PNET 4 trial. *Lancet Oncol.* 2018; 19:1602–1616
- [12] Blaser SI, Harwood-Nash DC. Neuroradiology of pediatric posterior fossa medulloblastoma. *J Neuro-oncol.* 1996; 29:23–34
- [13] Chang CH, Housepian EM, Herbert C Jr. An operative staging system and a megavoltage radiotherapeutic technic for cerebellar medulloblastomas. *Radiology.* 1969; 93:1351–1359
- [14] Louis DN, Ohgaki H, Wiestler OD, et al. WHO classification of tumors of the central nervous system. Lyon, France 2016
- [15] Cambruzzi E. Medulloblastoma, WNT-activated/SHH-activated: clinical impact of molecular analysis and histogenetic evaluation. *Childs Nerv Syst.* 2018; 34:809–815
- [16] Gilbertson RJ. Medulloblastoma: signalling a change in treatment. *Lancet Oncol.* 2004; 5:209–218
- [17] Johann PD, Hovestadt V, Thomas C, et al. Cribriform neuroepithelial tumor: molecular characterization of a SMARCB1-deficient non-rhabdoid tumor with favorable long-term outcome. *Brain Pathol.* 2017; 27:411–418

45 Cranial and Paraspinal Nerve Tumors

45.1 General information

These slow-growing tumors may occur sporadically or as part of a syndromic condition such as neurofibromatosis (type 1 or type 2), schwannomatosis, and Carney complex (p.652). The WHO now considers paragangliomas which involve sympathetic and parasympathetic neuroendocrine cells to be grouped in this category in contrast to the historical classification as neuronal and mixed neuro-glial tumors. Schwannomatosis may be associated with painful schwannomas.

45

45.2 Specific cranial and paraspinal nerve tumors

45.2.1 Schwannoma (WHO grade 1)

General information

A benign nerve sheath tumor exclusively or primarily comprised of differentiated neoplastic Schwann cells. May be sporadic or associated with syndromes. Over 90% are solitary and sporadic, and 50-75% of these show inactivating mutations of the NF2 suppressor gene located at 22q12.2 resulting in loss of expression of the growth inhibiting product, merlin (AKA schwannomin). Multiple schwannomas occur in NF2 (p.640) (bilateral involvement of the vestibular division of Cr N VIII is diagnostic of NF2) and schwannomatosis (p.642). Spinal schwannomas are typical in NF1 (p.638).

Diagnostic criteria for schwannoma

Diagnostic criteria for schwannoma are shown in ► Fig. 45.1.¹

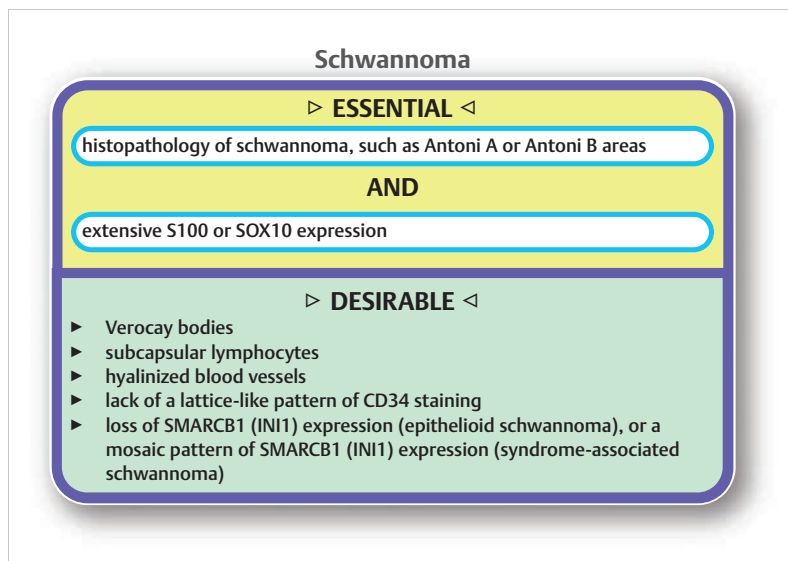


Fig. 45.1 Diagnostic criteria for schwannoma. (Modified with permission from WHO Classification of Tumors Editorial Board, (ed.) World Health Organization classification of tumors of the central nervous system. 5th ed. Bosman F T, Jaffe E S, Lakhani S R, et al. (series eds.): World Health Organization Classification of Tumors series, vol. 6. International Agency for Research on Cancer, Lyon, France, 2021.)

Subtypes

See reference.¹

- **ancient schwannoma**: contains dispersed cells with bizarre nuclei
- **cellular schwannoma**: Antoni A tissue predominates. Verocay bodies are absent. Small areas of necrosis may cause confusion with MPNST (p.774). May contain foci of Ki-67 labeling (the index is usually <20%, but not always)
- **plexiform schwannoma**: often arise in skin or sub-Q tissue. Most are sporadic, but may occur in HF2 or schwannomatosis. Histology may be classic (i.e., biphasic) or cellular subtype
- **epithelioid schwannoma**: demonstrate multilobulated single or nests of epithelioid cells in a myxoid or hyalinized matrix. Most are sporadic
- **microcystic/reticular schwannoma**: the least common subtype. Typically occur in viscera (usually the GI tract) where they are unencapsulated. Named for a rich microcystic network of spindle cells with eosinophilic cytoplasm. Antoni A tissue is common, and they usually lack features of hyalinized blood vessels and Verocay bodies

Location

Commonly arise from peripheral nerves in the skin and subcutaneous tissue of the head and neck, or on flexor surfaces of the limbs. Intracranially, they frequently involve the vestibular division of Cr. N. VIII (vestibular schwannoma (p.777)), but can also arise from Cr. N. V or other cranial nerves. In the spine, they can form “dumbbell”-shaped tumors as they extend out through the neural foramen which constricts them.

Histology

Most are biphasic (composed of 2 cell types):

1. Antoni A tissue: compact. Occasional Verocay bodies (nuclear palisading)
2. Antoni B tissue: loosely arrayed with lipid-laden histiocytes and hyalinized vasculature.

Refer to the comparison of the histology to that of neurofibromas (p.637).

Epidemiology

Peak incidence is the fourth to sixth decades.

Prognosis

The sporadic forms of these benign tumors usually do not recur following gross total resection. However, GTR is often difficult with the cellular and plexiform subtypes. Malignant transformation is exceedingly rare.

45.2.2 Neurofibroma (WHO grade 1)

General information

The most common peripheral nerve sheath tumor. A benign neoplasm composed of neoplastic Schwann cells interspersed with normal cells. They are WHO CNS5 grade 1 except for atypical neurofibromatous neoplasm of uncertain biological potential (ANNUBP), which is not assigned a grade. The finding of multiple or plexiform neurofibromas should prompt consideration of underlying NF1. Complete functional loss of neurofibromin (p.639) is considered a prerequisite for tumor development. Refer to the comparison of the histology to that of schwannomas (p.637).

Diagnostic criteria for neurofibroma

Diagnostic criteria for neurofibroma are shown in ► Fig. 45.2.¹

Subtypes

See reference.¹

- **cellular neurofibroma**
- **atypical neurofibroma / neurofibromatous neoplasm of uncertain biological potential (AN/ANNUBP)**: characterized by 2 or more of the following concerning findings²

Neurofibroma

▷ ESSENTIAL ◁

infiltrative, low-cellularity spindle cell neoplasm associated with a variably myxoid to collagenous stroma and a mixed cell population

▷ DESIRABLE ◁

- ▶ S100 positivity in the Schwann cell population, with a lattice-like CD34 pattern, highlighting the stromal component
- ▶ intraneural localization
- ▶ patient has neurofibromatosis type 1
- ▶ atypical histological features (nuclear enlargement, hypercellularity, architectural loss, mitoses) for atypical neurofibroma / atypical neurofibromatous neoplasm of uncertain biological potential in the setting of neurofibromatosis type 1, often with loss of p16 expression

Fig. 45.2 Diagnostic criteria for neurofibroma. (Modified with permission from WHO Classification of Tumors Editorial Board, (ed.) World Health Organization classification of tumors of the central nervous system. 5th ed. Bosman F T, Jaffe E S, Lakhani S R, et al. (series eds.): World Health Organization Classification of Tumors series, vol. 6. International Agency for Research on Cancer, Lyon, France, 2021.)

- cytological atypia
- hypercellularity
- loss of neurofibroma architecture (on H&E and/or CD34 staining)
- mitotic count > 0.2 mitoses/mm² and < 1.5 mitoses/mm² (equating to > 1 mitosis/50 HPF and < 3 mitoses/10 HPF of 0.51 mm in diameter and 0.2 mm² in area)

AN/ANNUBP are considered premalignant or early malignant lesions that do not meet criteria for MPNST (p. 774) but have an increased risk of progression to MPNST.³ They are not assigned a WHO grade

- **plexiform neurofibroma**
- **diffuse neurofibroma**
- **nodular neurofibroma**
- **massive soft tissue neurofibroma**: consistently benign lesions that may overlie an MPNST

Location

Usually located in the skin, typically in the dermal layer. Less commonly, may involve deeper nerves of medium-size, a nerve plexus or a major nerve trunk. They may also originate from spinal nerve roots.

Epidemiology

Most are sporadic and solitary. Multiple lesions usually occur in the context of NF1 and begin to appear around ages 5–10 years. Plexiform lesions are often congenital.⁴ There is no age or gender bias.

Clinical

Cutaneous lesions are typically soft, mobile and asymptomatic. Deep tumors often produce sensory and/or motor symptoms related to the nerve of origin. The least common manifestation is a plaque-like cutaneous and subcutaneous mass of the head or neck, or as a massive enlargement of the shoulder or pelvis in a patient with NF1.

Prognosis

Isolated cutaneous neurofibromas are uniformly benign. Plexiform neurofibromas, ANNUP and intraneural lesions originating in sizeable nerves may progress to MPNST (p.774). Diffuse neurofibromas of the skin rarely progress.

45.2.3 Perineurioma (WHO grade 1)

General information

A benign tumor comprised exclusively of neoplastic perineural cells.

Diagnostic criteria for intraneural and soft tissue perineurioma

Diagnostic criteria for soft tissue perineurioma are shown in ► Fig. 45.3, and for intraneural perineurioma in ► Fig. 45.4.¹

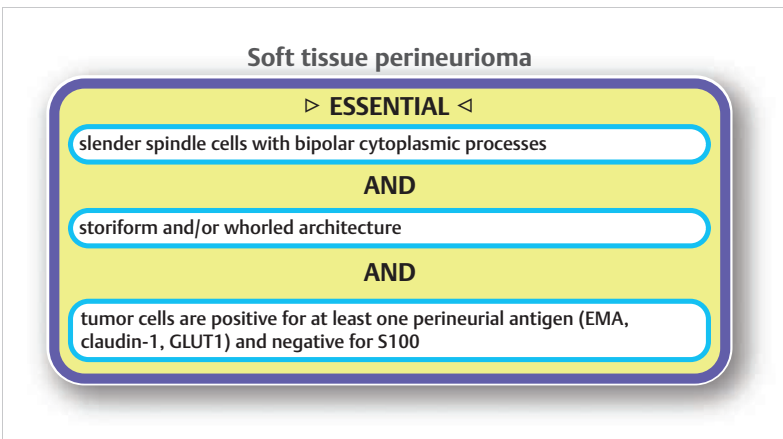


Fig. 45.3 Diagnostic criteria for soft tissue perineurioma. (Modified with permission from WHO Classification of Tumors Editorial Board, (ed.) World Health Organization classification of tumors of the central nervous system. 5th ed. Bosman F T, Jaffe E S, Lakhani S R, et al. (series eds.): World Health Organization Classification of Tumors series, vol. 6. International Agency for Research on Cancer, Lyon, France, 2021.)

Subtypes

See reference.¹

- **soft tissue perineurioma:** uncommon, only $\approx 1\%$ of soft tissue neoplasms. Arise in deep soft tissues and present as a soft tissue mass with nonspecific symptoms. Only rarely can an associated nerve be identified. Almost exclusively benign, but malignant variety does occur. Female:male ratio = 2:1. In males, hands are often affected. Discrete, but not encapsulated, diameter = 1.5–20 cm. Treatment: gross total excision is curative
- **intraneural perineurioma:** rare, only $\approx 1\%$ of nerve sheath tumors. Usually a solitary lesion of adolescence or young adulthood, affecting primarily peripheral nerves of the extremities (cranial nerve involvement is rare). Pseudo-onion bulb formation with cylindrical enlargement of the nerve over 2–10 cm. Mitotic activity is rare, MIB-1 labeling index is low. Chromosome 22 loss is characteristic,⁵ no definite association with NF1 or NF2. No gender preference. Present primarily with weakness and atrophy more often than sensory symptoms of the related nerve. Treatment: conservative sampling of lesion, not resection
- **reticular perineurioma**
- **sclerosing perineurioma**

Intraneural perineurioma

▷ ESSENTIAL ◁

pseudo-onion bulb pattern on cross-section, with axons in the center

AND

tumor cells are positive for at least one perineurial antigen (EMA, claudin-1, GLUT1) and negative for S100

Fig. 45.4 Diagnostic criteria for intraneural perineurioma. (Modified with permission from WHO Classification of Tumors Editorial Board, (ed.) World Health Organization classification of tumors of the central nervous system. 5th ed. Bosman F T, Jaffe E S, Lakhani S R, et al. (series eds.): World Health Organization Classification of Tumors series, vol. 6. International Agency for Research on Cancer, Lyon, France, 2021.)

Prognosis

Intraneural perineuriomas rarely grow lengthwise and they do not extend to adjacent nerves. They do not tend to recur or metastasize.

Soft tissue perineuriomas rarely recur and metastases or malignant progression have not been reported.

Malignant perineurial tumors with high mitotic activity have a poor prognosis and may metastasize.

45.2.4 Hybrid nerve sheath tumors

General information

AKA benign peripheral nerve sheath tumor NOS. Rare, benign peripheral nerve sheath tumors with a mixture of more than one of: schwannoma, neurofibroma, or perineurioma.

Diagnostic criteria for hybrid nerve sheath tumor

Diagnostic criteria for hybrid nerve sheath tumor are shown in ► Fig. 45.5.¹

Hybrid nerve sheath tumor

▷ ESSENTIAL ◁

intermingled features of two types of benign nerve sheath tumors

AND

appropriate immunohistochemical staining for each component

Fig. 45.5 Diagnostic criteria for hybrid nerve sheath tumor. (Modified with permission from WHO Classification of Tumors Editorial Board, (ed.) World Health Organization classification of tumors of the central nervous system. 5th ed. Bosman F T, Jaffe E S, Lakhani S R, et al. (series eds.): World Health Organization Classification of Tumors series, vol. 6. International Agency for Research on Cancer, Lyon, France, 2021.)

Subtypes

See reference.¹

- **schwannoma/perineurioma**: the most common subtype
- **neurofibroma/schwannoma**: the second most common subtype
- **neurofibroma/perineurioma**

Prognosis

Benign neoplasms which rarely recur following excision.

45.2.5 Malignant melanotic nerve sheath tumor (MMNST)

General information

AKA malignant melanotic Schwannian tumor. A peripheral nerve sheath tumor uniformly comprised of neoplastic cells with both Schwann cell and melanocytic features that frequently behaves aggressively. Usually arise from a spinal or autonomic nerve. There is variable association with Carney complex (p.652). Loss of PRKAR1A protein expression with *PRKAR1A* gene mutations are found in the vast majority.

Diagnostic criteria for malignant melanotic nerve sheath tumor

Diagnostic criteria for malignant melanotic nerve sheath tumor are shown in ► Fig. 45.6.¹

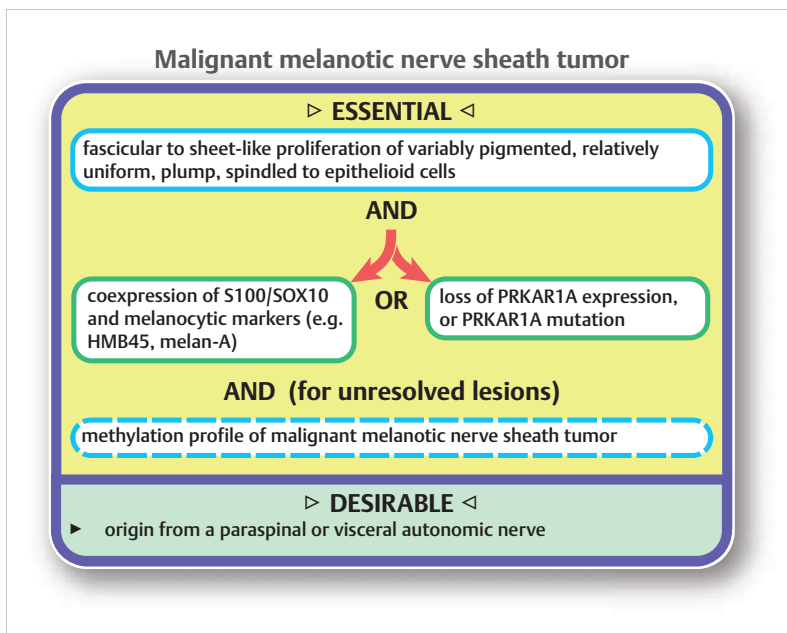


Fig. 45.6 Diagnostic criteria for malignant melanotic nerve sheath tumor. (Modified with permission from WHO Classification of Tumors Editorial Board, (ed.) World Health Organization classification of tumors of the central nervous system. 5th ed. Bosman F T, Jaffe E S, Lakhani S R, et al. (series eds.): World Health Organization Classification of Tumors series, vol. 6. International Agency for Research on Cancer, Lyon, France, 2021.)

Epidemiology

This rare tumor of adulthood tends to occur at a later age with sporadic cases (mean age: 33.2 years) than in those associated with Carney complex (mean age: 22.5 years).⁶ Tumors are multiple in 20%, and these patients have a greater chance of having Carney complex than those with solitary tumors.

Prognosis

Behavior is unpredictable, and 26–44% may recur locally or metastasize even without identifiable malignant features. Recurrence rate is 47% in patients followed for more than 5 years.¹

45

45.2.6 Malignant peripheral nerve sheath tumor (MPNST) (no WHO grade)

General information

A malignant spindle-cell peripheral nerve sheath tumor typically originating from a peripheral nerve, a pre-existing benign nerve sheath tumor, or in a patient with NF1. Limited Schwann cell differentiation is often seen. Characteristic molecular findings are combined genetic inactivation of *NF1*, *CDKN2A* and/or *CDKN2B*, and *SUZ12* or *EED* genes, as well as complex genomic rearrangements.

Approximately 50% are associated with NF1 (p.638) (in NF1 they tend to occur in deep-seated plexiform neurofibromas or large intraneural neurofibromas). The lifetime risk of developing MPNST with NF1 is $\approx$ 9–13%.⁷

MPNSTs may spread along nerves, and 20% metastasize hematogenously, usually to lung.

Diagnostic criteria for malignant peripheral nerve sheath tumor (MPNST)

Diagnostic criteria for MPNST are shown in ► Fig. 45.7.¹

Subtypes

See reference.¹

- **epithelioid malignant peripheral nerve sheath tumor:** very rare (only $\approx$ 5% of MPNSTs)
- **perineurial malignant peripheral nerve sheath tumor:** very rare. Perineurial features of spindle cells and whorls combined with malignant histologic features (frequent mitoses, necrosis...). These are not associated with NF1

Epidemiology

MPNST typically occur in the 20–50 year age range. In children they are usually associated with NF1 where the mean age is about 10 years less than in sporadic cases. They account for 5% of soft tissue sarcomas in patients <20 years old.⁸ Approximately 10% of MPNST are associated with a history of XRT.⁹

Prognosis

GTR, whenever possible, is the preferred treatment. The 5-year OS for sporadic MPNST is 69%, whereas in the setting of NF1 was 35%.¹⁰ Features associated with a more favorable prognosis include^{1,8}:

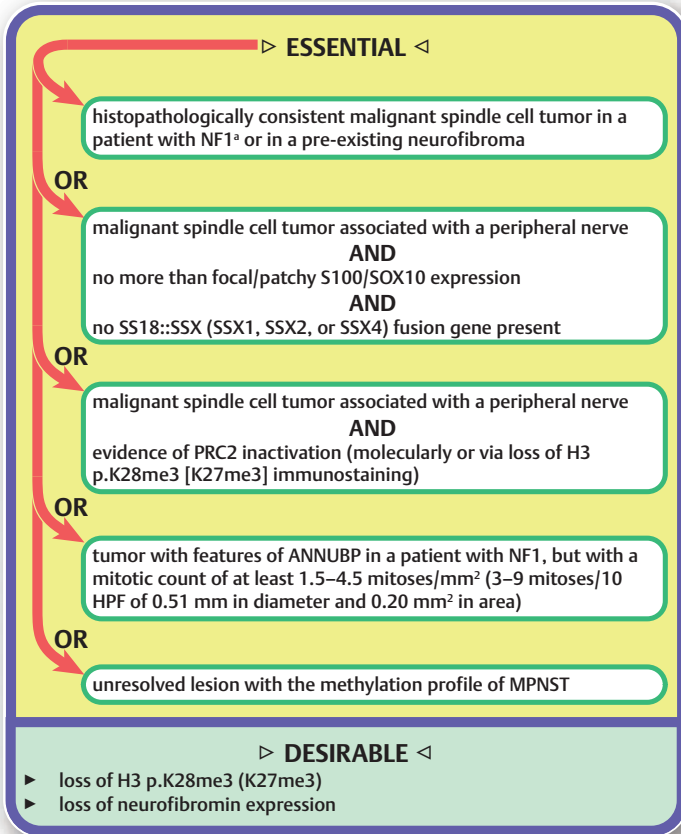
- smaller tumor size
- male gender and non-Hispanic White race
- no metastases at presentation
- GTR
- no association with prior XRT
- epithelioid and perineurial subtypes appear to be less aggressive than conventional MPNST

45.2.7 Cauda equina neuroendocrine tumor (WHO grade 1)

General information

AKA (and formerly) paraganglioma of the cauda equina, or cauda equina paraganglioma.

Malignant peripheral nerve sheath tumor (MPNST)



^a this alteration is highly desirable, and its absence in a tumor otherwise resembling PPTID, should prompt careful consideration as to whether an alternative diagnosis may be more suitable

Fig. 45.7 Diagnostic criteria for MPNST. (Modified with permission from WHO Classification of Tumors Editorial Board, (ed.) World Health Organization classification of tumors of the central nervous system. 5th ed. Bosman F T, Jaffe E S, Lakhani S R, et al. (series eds.): World Health Organization Classification of Tumors series, vol. 6. International Agency for Research on Cancer, Lyon, France, 2021.)

A neuroendocrine tumor originating from specialized neural crest cells in the region of the cauda equina/filum terminale. CSF protein is usually increased.¹¹

Diagnostic criteria for cauda equina neuroendocrine tumor

Diagnostic criteria for cauda equina neuroendocrine tumor are shown in ► Fig. 45.8.¹

Cauda equina neuroendocrine tumor

▷ ESSENTIAL ◁

well-demarcated tumor with Zellballen architecture

AND

synaptophysin or chromogranin immunoreactivity in chief cells

AND

cauda equina location

AND (for unresolved lesions)

methylation profile of cauda equina neuroendocrine tumor*

▷ DESIRABLE ◁

- ▶ S100-positive sustentacular cells
- ▶ cytokeratin-positive chief cells
- ▶ reticulin silver stain showing typical architecture

* the DNA methylation profile and copy-number profiles of cauda equina neuroendocrine tumors are distinct from those of paragangliomas from other locations

Fig. 45.8 Diagnostic criteria for cauda equina neuroendocrine tumor. (Modified with permission from WHO Classification of Tumors Editorial Board, (ed.) World Health Organization classification of tumors of the central nervous system. 5th ed. Bosman F T, Jaffe E S, Lakhani S R, et al. (series eds.): World Health Organization Classification of Tumors series, vol. 6. International Agency for Research on Cancer, Lyon, France, 2021.)

Location

Most spinal paragangliomas occur in the region of the cauda equina, and most of these are intradural and are attached to the filum terminale.

Clinical

Typically present with low back pain and radiculopathy. Numbness, paraparesis and sphincter disturbance are less common. Even less common are cauda equina syndrome or symptoms of increased ICP (including papilledema). Acute paraparesis, spinal subarachnoid hemorrhage,¹¹ or neuroendocrine symptoms due to catecholamine secretion (paroxysmal or sustained hypertension, palpitations, diaphoresis and H/A) are extremely rare.

Imaging

On imaging, these tumors are usually indistinguishable from other common neoplasms of the area (schwannomas, myxopapillary ependymomas). The diagnosis may be suggested preoperatively in the infrequent case demonstrating serpiginous dilated and ectatic blood vessels, sometimes with a “cap sign” (low-intensity rim on T2WI or GRASS or SWI MRI due to hemosiderin from prior hemorrhages).¹¹

Histology

The tumors are well differentiated. Uniform round or polygonal chief (type I) cells congregate in lobules (Zellballen) encompassed by a barely discernible single layer of sustentacular (type II) cells.

Mature ganglion cells and a Schwann cell component (gangliocytic neuroendocrine tumors) are found in $\approx 25\%$. Scattered mitotic figures, areas of focal hemorrhagic necrosis and nuclear pleomorphism may be seen but are not prognostically significant.

Prognosis

The preponderance of these tumors are slow growing. The recurrence rate following GTR is very low.¹² Dissemination of tumor through the CSF may occur.

45.3 Vestibular schwannoma

45.3.1 General information

45

Key concepts

- histologically benign tumor of cranial nerve VIII located in the cerebellopontine angle (CPA)
- usually arises from inferior division (controversial) of the vestibular portion of the VIII nerve
- about 1 in 500 persons will develop this tumor during their lifetime
- 3 most common early symptoms (clinical triad): hearing loss (insidious and progressive), tinnitus (high pitched) and dysequilibrium (true vertigo is uncommon)
- W/U: All patients: ✓ MRI with IAC protocol (p.241), ✓ audiometrics (pure tone audiogram and speech discrimination). In addition for small VSs (≤ 1.5 cm dia): ✓ ENG, ✓ cVEMP, ✓ ABR
- histology: comprised of Antoni A (narrow elongated bipolar cells) and Antoni B fibers (loose reticulated). Compared to most schwannomas, Antoni A fibers predominate
- choice of management option (observation, surgery, XRT or chemotherapy [Avastin®]) depends heavily on tumor size, growth, hearing status, VII function and presence of NF2 (see text)

Owing to the commonality and intricacies, this particular schwannoma receives a section devoted just to this tumor.

Vestibular schwannoma (VS) is a histologically benign Schwann-cell sheath tumor that usually arises from the inferior division of the *vestibular nerve* (not the cochlear portion). VSs arise as a result of the loss of a tumor-suppressor gene on the long arm of chromosome 22 (in sporadic cases this is a somatic mutation; in neurofibromatosis Type 2 (NF2) this is either inherited or represents a new mutation that may then be transmitted to offspring).

Obsolete terms, included for reference, should be avoided^{13,14}: acoustic neuroma, acoustic neurinomas (neurinoma is an obsolete term for schwannoma), neurolemoma or neurilemmoma.

45.3.2 Epidemiology

The most common non-malignant nerve sheath tumor, comprising 8–10% of tumors intracranial tumors.¹⁵ Annual incidence quoted by CBTRUS is 1.51 cases per 100,000 population in the U.S.¹⁶—over the past couple decades the incidence has increased and the typical size at diagnosis has decreased, partially as a result of the proliferation of MRI scans.¹⁷ Other estimates disclose a higher incidence of 3.0–5.2/100,000.^{18,19} The highest incidence occurs in patients aged ≥ 70 years, peaking at 20.6 per 100,000 person-years.¹⁹ There is no gender differential.¹⁹ At least 95% are unilateral.

► **Neurofibromatosis Type 2.** The incidence of vestibular schwannomas (VS) is increased in in neurofibromatosis Type 2 (NF2) (p.640) (central NF2), with bilateral VS being a hallmark. Any patient < 40 yrs old with unilateral VS should be evaluated for NF2. Cytologically, the VSs of NF2 are identical to sporadic cases; however, in NF2 the tumors form grape-like clusters that may infiltrate the nerve fibers (unlike most sporadic VSs which *displace* the eighth nerve).

45.3.3 Pathology

Tumors are composed of Antoni A fibers (narrow elongated bipolar cells) and Antoni B fibers (loose reticulated). Verocay bodies are also seen, and consist of acellular eosinophilic areas surrounded by parallel arrangement of spindle shaped schwann cells (they are not a cell type). Tumor-associated macrophages (TAMs) may be associated with increased risk of recurrence following STR.

45.3.4 Clinical

Symptoms

General information

VSs typically become symptomatic after age 30. Symptoms are shown in ► Table 45.1. The type of symptoms are closely correlated with tumor size. Most initially cause the triad of ipsilateral sensorineural hearing loss, tinnitus and balance difficulties. Larger tumors can cause facial numbness, weakness or twitching, and possibly brainstem symptoms. Rarely, a large tumor may produce hydrocephalus. With current imaging modalities (CT and especially MRI), increasing numbers of smaller lesions are being detected.

Table 45.1 Symptoms in vestibular schwannoma^{15,20,21,22}

Symptom	%
hearing loss	95
tinnitus	55
dizziness/imbalance	61
H/A	32
facial numbness	18
facial weakness	17
diplopia	10
N/V	9
otalgia	9
change of taste	6
vertigo	8
continuous dizziness	3

Acoustic nerve (Cr N VIII) symptoms

Unilateral sensorineural hearing loss, tinnitus and dysequilibrium are the most frequent symptoms. For years it was taken for granted that hearing loss was due to compression and/or stretching of the VIII nerve by the tumor; however, recent evidence suggests possible toxic factors secreted by the tumor cause cochlear damage.^{23,24}

Hearing loss is insidious and progressive in most (c.f. the hearing loss in Meniere disease which fluctuates); however, 10% report sudden hearing loss (see below). 70% have a high frequency loss pattern, and word discrimination is usually affected (especially noticeable in telephone conversation).

The tinnitus is usually high-pitched.

Unsteadiness manifests primarily as difficulty with balance; true vertigo occurs in < 20%.

Sudden hearing loss: The differential diagnosis for sudden hearing loss (SHL) is extensive.²⁵ *Idiopathic* SHL (i.e., no identified etiology: must rule out neoplasm, infection, autoimmune, vascular, and toxic causes) represents 90% of cases²⁶ and occurs in an estimated 5–20 per 100,000 population.²⁷ 1% of patients with SHL will be found to have a VS, and SHL may be the presenting symptom in 1–14% of patients with VS.²⁸ SHL with VS is presumably due to an infarction of the acoustic nerve, or acute occlusion of the cochlear artery. Treatment options for SHL include:

1. steroids: e.g., prednisone 60 mg PO q d × 10 d then tapered.²⁸ Ideally initiated within 2 weeks of onset of SHL²⁶
2. there is no high-quality data supporting addition of antiviral therapy (e.g., famciclovir)^{29,30}
3. ✗ not recommended
 - heparin has been shown *not* to be of help
 - there is also no high quality data to support: rest, restriction of salt, alcohol or tobacco, or use of thrombolytics

Trigeminal (Cr N V) and facial (Cr N VII) nerve symptoms

Otalgia, facial numbness and weakness, and taste changes occur as the tumor enlarges and compresses the fifth and seventh nerves. These symptoms usually do not occur until the tumor is > 2 cm. This highlights an interesting paradox: facial weakness is a rare or late occurrence, even though the 7th nerve is almost always distorted early; whereas facial numbness occurs sooner once trigeminal compression occurs (often in the presence of normal facial movement), despite the fact that the 5th nerve is farther away.³¹ This may be due to the resiliency of motor nerves relative to sensory nerves.

Brainstem and other cranial nerve symptoms

Larger tumors cause brainstem compression (with ataxia, H/A, N/V, diplopia, cerebellar signs, and if unchecked, coma, respiratory depression and death) and lower cranial nerve (IX, X, XII) palsies (hoarseness, dysphagia...). Obstruction of CSF circulation by larger tumors (usually > 4 cm) may produce hydrocephalus with increased ICP.

Rarely, 6th nerve involvement may cause diplopia.

Signs

Hearing loss due to VIII involvement is the earliest cranial nerve finding. 66% of patients have no abnormal physical finding except for hearing loss (for other findings, see ► Table 45.2).

Since hearing loss is sensorineural, **Weber test** (p.611) will lateralize to the uninvolved side, and if there is enough preserved hearing, **Rinne test** (p.611) will be positive (i.e., normal; air conduction > bone conduction) on both sides.

Facial nerve (VII) dysfunction due to VS is uncommon before treatment. Facial nerve function is commonly graded on the House and Brackmann scale (► Table 45.3).

Vestibular involvement causes nystagmus (may be central or peripheral) and abnormal electro-nystagmography (ENG) with caloric stimulation.

Table 45.2 Signs in 131 vestibular schwannomas (excluding hearing loss)¹⁵

Sign	%
abnormal corneal reflex	33
nystagmus	26
facial hypoesthesia	26
facial weakness (palsy)	12
abnormal eye movement	11
papilledema	10
Babinski sign	5

Table 45.3 Clinical grading of facial nerve function (House and Brackmann³²)

Grade	Function	Description
1	normal	normal facial function in all areas
2	mild dysfunction	1. gross: slight weakness noticeable on close inspection; may have very slight synkinesis 2. at rest: normal symmetry and tone 3. motion: a) forehead: slight to moderate movement b) eye: complete closure with effort c) mouth: slight asymmetry
3	moderate dysfunction	1. gross: obvious but not disfiguring asymmetry: noticeable but not severe synkinesis 2. motion: a) forehead: slight to moderate movement b) eye: complete closure with effort c) mouth: slightly weak with maximal effort
4	moderate to severe dysfunction	1. gross: obvious weakness and/or disfiguring asymmetry 2. motion: a) forehead: none b) eye: incomplete closure c) mouth: asymmetry with maximum effort
5	severe dysfunction	1. gross: only barely perceptible motion 2. at rest: asymmetry 3. motion: a) forehead: none b) eye: incomplete closure
6	total paralysis	no movement

45.3.5 Evaluation

Diagnostic studies – overview

1. brain MRI without and with contrast: include navigation and IAC protocol (p. 241) (if available). If MRI is contraindicated, then a CT scan without and with contrast. Tumor size is graded using the Koos system (► Table 45.7)³³
2. temporal bone CT for detailed bony anatomy if surgery is contemplated
3. audiometric evaluation (see text below for explanation & indications):
 - a) pure tone audiogram (see below)
 - b) speech discrimination evaluation (see below)
 - c) patients with small VSs (≤ 1.5 cm dia) also get:
 - ENG: (p. 782) assesses *superior* vestibular nerve
 - VEMP: (p. 782) assesses *inferior* vestibular nerve
 - ABR: (p. 782) prognosticates chance of hearing preservation

45

Audiometric and audiologic studies

General information

Baseline studies are helpful for management treatment decisions and for later comparison and to assess the contralateral ear.

Practice guideline: Otologic & audiologic screening for vestibular schwannomas

- Level III²²: patients with ≥ 10 dB interaural hearing difference at ≥ 2 frequencies or ≥ 15 dB at 1 frequency should have an MRI to look for VS. Selectively doing MRI for ≥ 15 dB interaural difference at 3000 Hz alone may minimize negative MRIs
- Level III²²: MRI is recommended for asymmetric tinnitus, although the yield is low ($< 1\%$ positive for VS)
- Level III²²: MRI is recommended for sudden sensorineural hearing loss; however, the yield is low ($< 3\%$ positive for VS)

Abbreviations: MRI = brain MRI (see imaging guidelines (p. 782)); dB = decibels.

Pure tone audiogram (PTA)

May be useful as first-step screening test. Air conduction assesses the entire system, bone conduction assesses from the cochlea and proximally. PTA assesses the functionality of hearing (to help in treatment decision making) and acts as a baseline for future comparison. The pure tone average (also abbreviated PTA) is a single numerical score that is an *average* of the thresholds for frequencies across the audio spectrum (at 500, 1000 & 2000 Hz). On a standard audiogram, X's denote the left ear (AS) and O's denote the right ear (AD).

Progressive unilateral or asymmetric sensorineural hearing loss of high tones occurs in $> 95\%$ of VSs.³⁴ High-frequency hearing loss also happens to be the most common type of hearing loss with age or with noise-induced sensorineural hearing loss, but is usually symmetrical. Only ≈ 1 in 1000 patients with asymmetric hearing have a VS.¹³ Other causes of asymmetrical sensorineural hearing loss³⁵: other CPA lesions (e.g., meningioma), inner ear lesions, intraaxial lesions (including brainstem infarctions), multiple sclerosis. On hearing screening tests, an unexplained PTA difference from one ear to the other > 10 – 15 dB is suspicious and should be investigated further.

Speech discrimination evaluation

Speech discrimination is maintained in conductive hearing loss, moderately impaired in cochlear hearing loss, and worst with retrocochlear lesions. No longer used for diagnostic purposes; a score of 4% suggests a retrocochlear lesion, as does a score that is worse than would be predicted based on PTA testing (the speech recognition threshold should be similar to PTA thresholds below 4 kHz). Useful in determining serviceability of hearing and prognosticating for hearing preservation surgery.

Table 45.4 Open-set word recognition score

Class	WRS (%)
I	70–100
II	50–69
III	1–49
IV	0

Open-set word recognition score (WRS ► Table 45.4) is a more sensitive measure of communication ability than PTA.

Definition of serviceable hearing

There are many definitions of what constitutes serviceable hearing. Also, even nonserviceable hearing can offer some benefit. If WRS is good ($\geq 70\%$) but PTA is poor, a hearing aid may provide significant benefit.

Two commonly used scoring systems for hearing are shown here:

1. Modified **Gardener-Robertson** system for grading hearing: shown in ► Table 45.5. Class I patients may use a phone on that side, class II patients can localize sounds.
2. The American Academy of Otolaryngology – Head and Neck Surgery Foundation (AAO-HNS) hearing classification system³⁶; shown in ► Table 45.6.

Some definitions of serviceable hearing (see text that follows for details):

1. AAO-HNS class A or B
2. Gardner-Robertson system
 - “50/50 rule”: Gardner-Robertson class I or II (pure tone audiogram threshold ≤ 50 dB and speech discrimination score $\geq 50\%$)
 - some prefer a “70/30 rule” (70% WRS, 30 dB PTA)
3. in a patient with good hearing in the contralateral ear, a speech discrimination score (SDS) of $< 70\%$ in the affected ear is not considered good hearing; whereas if the contralateral ear is totally deaf, an SDS of $\geq 50\%$ can be useful³⁹

Table 45.5 Gardener and Robertson modified hearing classification^a

Class	Pure tone audiogram ^b (dB)	Speech discrimination ^b	Description	Clinical utility
I	0–30	70–100%	good-excellent	serviceable
II	31–50	50–69%	serviceable	
III	51–90	5–49%	non-serviceable	non-serviceable
IV	91–max	1–4%	poor	
V	not testable	0	none	

^amodification³⁷ of the Silverstein and Norrell system³⁸

^bif PTA and speech discrimination score do not qualify in the same class, use the lower class

Boldface grades are generally considered serviceable hearing.

Table 45.6 American Academy of Otolaryngology–Head and Neck Surgery Foundation hearing classification system

Class	Pure tone threshold (dB) ^a	Speech discrimination score ^b (%)	Clinical utility
A	≤ 30	AND ≥ 70	“useful”
B	> 30 AND ≤ 50	AND ≥ 50	
C	> 50	AND ≥ 50	“aidable”
D	any level	< 50	“nonfunctional”

^aaverage of pure tone hearing thresholds by air conduction at 0.5, 1, 2 & 3 kHz

^bspeech discrimination at 40 dB or maximum comfortable loudness

Boldface classes are generally considered serviceable hearing.

Additional audiometric tests that are helpful with small VSs (≤ 1.5 cm diameter)

The ENG (caloric testing) and cVEMP evaluate the superior and inferior division of the vestibular nerve (VN) respectively. The inferior VN is closer to the cochlear nerve than the superior VN (► Fig. 1.9), and small tumors (≤ 4 mm) of the inferior VN tend to be deeper and closer to the cochlear nerve than similarly sized tumors of the superior division which tend to be more superficial and more easily removed.

Electronystagmography (ENG): Only tests the horizontal semicircular canal. assesses the *superior* vestibular nerve which innervates it. Normally, each ear contributes an equal portion of the response. The ENG is considered abnormal if there is $>20\%$ difference between the two sides. Response may be normal with a small tumor arising from the *inferior* division of vestibular nerve. NB: the vestibular nerve may continue to function until almost all of the nerve fibers are affected. Mnemonic for the directionality of the nystagmus (direction is classified based on the fast phase of the nystagmus). COWS (Cold Opposite, Warm Same). NB: this differs from caloric testing for brain death (p. 323) and is a source of confusion.

Vestibular evoked myogenic potential (VEMP): Most commonly cVEMP (“c” for cervical to denote that the muscle recorded in the sternocleidomastoid, oVEMP (ocular) and tVEMP (triceps) are used much less frequently).

cVEMP assesses the *inferior* vestibular nerve by imparting acoustic energy to the saccule.⁴⁰ Independent of hearing (can be done even with deafness from profound sensorineural hearing loss). Electrodes are placed on sternocleidomastoid muscle (SCM).

Auditory brainstem responses (ABR): AKA BAER (p.252). The most common findings are prolonged I-III and I-V interpeak latencies. No longer used for diagnostic purposes (sensitivity is only $\approx 88\text{--}90\%$ (i.e., will miss 10–12% of VSs) and specificity is only 85%). ABR is useful for prognostication—poor wave morphology correlates with lower chance of preserving hearing (even with good hearing).

Radiographic evaluation

Imaging guidelines

Practice guideline: Imaging for vestibular schwannomas

For the full guideline, see reference.⁴¹

Initial pre-op evaluation to detect VS

- Level III⁴¹: high-resolution T2 and contrast-enhanced T1 MRI should be used
- Level III⁴¹: standard T1, T2 FLAIR & DWI MRI sequences in axial, coronal, and sagittal plane may be used

Pre-op surveillance for a known VS if an observation (“watch-and-wait”) approach is being used

- Level III⁴¹: pre-op surveillance for VS growth should be either contrast-enhanced 3-D T1 magnetization prepared rapid acquisition gradient echo (MPRAGE) or high-resolution T2 MRI (including FIESTA or CISS sequences⁹)
- Level III⁴¹: MRIs should be obtained annually for 5 years, with lengthening of the interval after that if stable
- Level III⁴¹: because of more variable growth rates, NF2 patients with VS should be imaged more frequently at first and then annually once the growth rate is determined
- Level III⁴¹: in NF2 patients, consider non-contrast MRIs if high-resolution T2 (including CISS or FIESTA) MRI adequately visualizes the tumor

Pre-op evaluation in preparation for surgery

- Level III⁴¹: T2 MRI may be used to augment visualization of the course of the facial nerve
- Level III⁴¹: counsel adults with cystic VS that their tumors may more often exhibit rapid growth, lower rates of complete resection, and inferior outcome of facial nerve function immediately post-op but similar to noncystic VS over time
- Level III⁴¹: the degree of lateral IAC involvement adversely affects facial nerve and hearing outcomes

Postop evaluation

- Level II⁴¹: post-contrast 3-D T1 MPRAGE should be used for nodular enhancement suspicious for recurrence

- Level III⁴¹: post-op imaging
 - following GTR: an MRI may be considered as late as 1 yr post-op to confirm GTR
 - for < GTR: more frequent MRIs are suggested—annually for 5 years, with adjustments for changes in nodular enhancement. NB: tumor volume decreases by $\approx 35\%$ in the 1st 3 months after STR, and 32% do not enhance in the immediate post-op period⁴²
- Level III⁴¹: after resection of one VS in an NF2 patients with bilateral VS, the growth of the contralateral VS may accelerate and more frequent imaging should be considered until the growth pattern is established

Abbreviations: GTR = gross total resection; STR = subtotal resection; IAC = internal auditory canal.

^a FIESTA (p. 241) or CISS (p. 241) may improve visualization of the tumor and nerves in CSF

Differential diagnosis

See Cerebellopontine angle (CPA) lesions (p. 1647). The major differentials on imaging are:

1. CPA meningioma; see differentiating features (p. 1648)
2. schwannoma of adjacent cranial nerve, including mainly:
 - a) facial nerve (Cr. N. VII) schwannoma: can arise anywhere along the course of Cr. N. VII
 1. imaging: may be dumbbell shaped due to extension from the ICA fundus through the labyrinthine segment and into the geniculate fossa. When isolated to the CPA, they may be indistinguishable from VS⁴³
 2. clinical: typically present with facial weakness (slowly progressive or sudden), often preceded by facial twitching, facial nerve pain, hemifacial spasm and decreased lacrimation.⁴⁴ In patients thought to have Bell's palsy, a facial nerve neuroma was found to be the cause in 5%, and 27% had normal facial nerve function⁴⁵
 - b) trigeminal (Cr. N. V) schwannoma

CT scan

CT with IV contrast is second choice for imaging modality. If MRI is contraindicated and clinical suspicion of VS is strong but the CT is negative, small lesions may be visualized by introducing 3–4 ml of subarachnoid air via lumbar puncture, and scanning the patient with the affected side up (to trap air in region of IAC), non-filling of the IAC is indicative of an intracanalicular mass. Even with air contrast, CT was normal in 6% in Mayo series.¹⁵

Many VSs enlarge the ostium of the IAC (called trumpeting). Normal diameter of the IAC: 5–8 mm. 3–5% of VSs do not enlarge the IAC on CT (percentage is likely higher in small VSs vs. large ones).

Thin-cut temporal bone CT should be obtained for operative planning. Important features to identify:

- for middle fossa approach: bony coverage of geniculate ganglion to identify dehiscence
- for translab approach:
 - extent of pneumatization of the mastoid and position of sigmoid sinus. An anterior sinus with poorly pneumatized mastoids can indicate a tight space for this approach
 - position of the jugular bulb. If high riding, can indicate a tight space in translab approach
- for retrosigmoid transmeatal approach: location and thickness of bone coverage over the posterior semicircular canal and vestibular aqueduct. The extent of peritubular air cells and retro facial air cells needs to be assessed in planning the approach and preventing CSF leaks

MRI

MRI (► Fig. 45.9) is the diagnostic procedure of choice (see Practice guideline (p. 782) for sequence recommendations) with sensitivity close to 98% and almost 0% false positive rate. Characteristic findings: round or oval enhancing tumor centered on IAC. Large VSs (> 3 cm dia) may show cystic appearing areas on CT or MRI; in actuality these areas are usually solid. Adjacent trapped CSF cisterns may also give cystic appearance. In a pilot study, hyperintensity on T2 has been associated with soft and suckable tumors at the time of surgery⁴⁶ and a trend towards better VII function preservation.

Tumor size is graded using the Koos system (► Table 45.7).³³

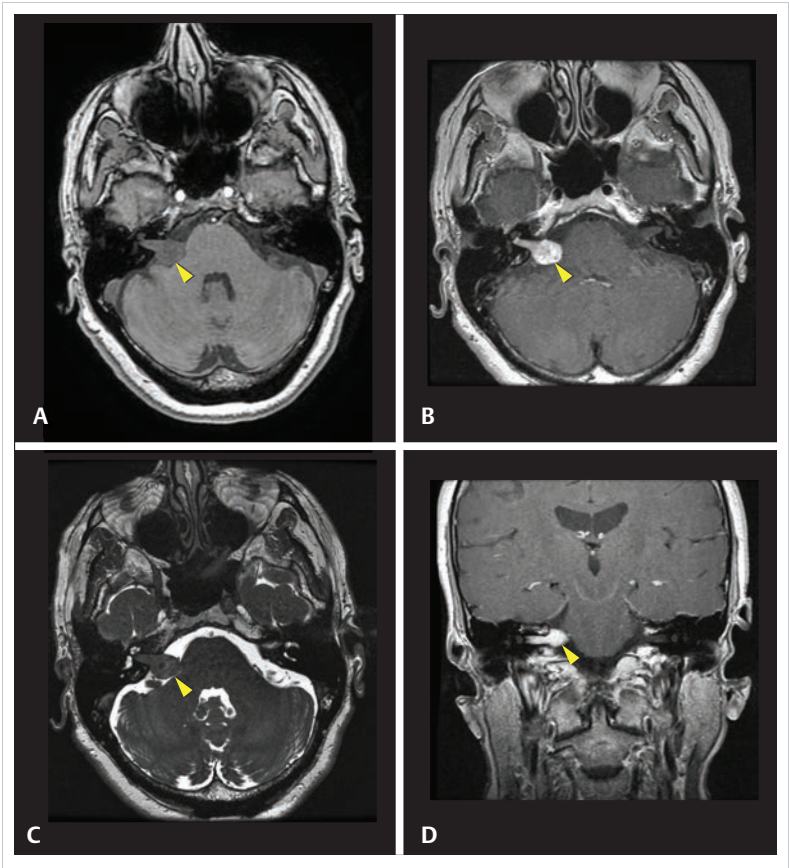


Fig. 45.9 Vestibular schwannoma, right-sided. This Koos grade III tumor is identified by yellow arrowheads. Image: MRI, A: axial T1 sequence, B: axial T1 + contrast, C: axial FIESTA scan, D: coronal T1 + contrast.

Table 45.7 Koos grading system for vestibular schwannomas & approximate corresponding volume ^a		
Grade	Tumor extent ³³	Average tumor volume ^{b 47} (cc)
I	intracanalicular tumor	0.3
II	protrusion into CPA, but not reaching the brainstem	0.6
III	extends to brainstem surface without displacing it	1.8
IV	displaces brainstem and cranial nerves	4.2 ^c

^avolume measurements are not part of the Koos grading system and are included for interest as it may impact stereotactic radiosurgery decisions
^bvolume measurements are from 235 patients undergoing gamma knife treatment and this may be a source of selection bias for volume measurement
^cgrade IV was a very heterogeneous group, and Minderman & Schlegel proposed adding a category V for tumors >6 cc

45.3.6 Management

Management options

Management options include:

1. expectant management ("wait and scan" approach): follow symptoms, hearing (audiometrics) and tumor growth on serial imaging (MRI or CT) (see guidelines (p.782) for imaging recommendations). Intervention is performed for progression. Growth patterns observed:
 - a) little or no growth: applies to most (83%) VSs confined within the IAC (► Table 45.7, Koos grade I) and to 30% extending into CPA (► Table 45.7, Koos grade II) (see natural history of growth below)
 - b) slow growth ≈ 2 mm/yr
 - c) rapid growth: ≥ 10 mm/yr
 - d) a few actually shrink¹⁷
2. radiation therapy (alone, or in conjunction with surgery)
 - a) external beam radiation therapy (EBRT)
 - b) stereotactic radiation
 - stereotactic radiosurgery (SRS) (p.1903): single dose. As there is little difference in radiographic control at different doses, single dose SRS < 13 Gy is recommended for hearing preservation and to minimize new onset or worsening of cranial nerve deficits (Level III⁴⁸)
 - stereotactic radiotherapy (SRT) (p.1903): fractionated
3. surgery: approaches include the following (see below for details)
 - a) retrosigmoid (AKA suboccipital): may be able to spare hearing
 - b) translabrynthine (and its several variations): sacrifices hearing, may be slightly better for sparing VII
 - c) middle fossa approach (extradural subtemporal): only for small lateral VSs
4. chemotherapy: some preliminary promise for progressive NF2-related vestibular schwannomas with bevacizumab (Avastin®), an anti-VEGF (vascular endothelial growth factor) monoclonal antibody (see below). Side effects: hemorrhage occurs in $\approx 7\%$ due to vessel necrosis

Patient/tumor factors influencing management decisions

In addition to the usual factors involved in the decision process with brain tumors (e.g., the patient's general medical condition, age, natural history, etc.) elements unique to VSs include: chances of preserving VII & V nerve function and hearing (in those with serviceable hearing) (all of which are related to tumor size), and the presence of NF2.

Specifics:

1. natural history of growth
 - a) usual quoted range: ≈ 1 – 10 mm/yr. However this can be quite variable
 - b) strictly intracanalicular tumors: only 17% grew outside the meatus (in 552 VSs over 3.6 years mean follow-up (230 were intrameatal at time of diagnosis, 322 had extrameatal extension)¹⁷)
 - c) extrameatal tumors (with extension into CP angle): 30% grew > 2 mm (in 522 VSs over 3.6 years mean follow-up¹⁷)
 - d) VSs that did not grow in the 5 years after diagnosis did not grow after that
 - e) 6% actually decrease in size⁴⁹
2. natural history of hearing function in untreated intracanalicular VSs in AAO-HNS Group A (► Table 45.6) patients. Rule of thumb for patients presenting with serviceable hearing (SH): 75% retain SH at 3 years, 60% at 5 years, and 40% at 10 years.⁵⁰ Details⁵¹:
 - a) 50% deteriorated to a lower class over 4.6 years (loss of ≥ 10 dB PTA or $\geq 10\%$ SDS)
 - b) after 4.6 years of observation, the proportion of patients eligible for hearing preservation treatment (as determined by a word recognition score class I [70–100% SDS]) was reduced to 28% (a 44% reduction) and by AAO-HNS class A to 9% (a 53% reduction)
 - c) the risk of losing hearing was not related to: age, gender, VS size (all tumors were intracanalicular) or tumor sublocalization (fundus, central, porus)
 - d) hearing loss was positively correlated to the absolute volumetric tumor growth rate (tumors that eventually expand out of the IAC have a faster rate and degree of hearing loss compared to tumors remaining in the IAC)
 - e) the risk of losing hearing was significantly lower for patients with 100% word recognition score. Over 4.6 years observation, 89% remained in WRS class I (► Table 45.4) compared to only 43% for patients with only a small (1–10%) loss of WRS at diagnosis
3. size: as tumors exceed 15 mm diameter, treatment complications increase
 - a) significantly lower chance for hearing preservation
 - b) increased incidence of VII injury

4. presence of cysts: cystic tumors may display sudden and dramatic growth¹⁷
5. serviceable hearing: see Definition of serviceable hearing (p.781)
6. hearing in contralateral ear

Surgical factors – hearing preservation following microsurgery (MS) in sporadic VS

The probability of maintaining serviceable hearing following GTR of an *intracanalicular* VS for retrosigmoid vs. middle fossa approach in experienced hands is shown in ► Table 45.8 and is so similar that the choice of approach with regard to hearing preservation should be made on the basis of the expertise of the surgical team.⁵²

If class A hearing (≤ 30 dB PTA, $\geq 70\%$ WRS) is initially preserved following gross total resection of a VS, the probability of maintaining serviceable hearing (≤ 50 dB PTA, $\geq 50\%$ WRS) is 80–100% at 2 years, 80–100% at 5 years, and 70–90% at 10 years.⁵²

Table 45.8 Probability of maintaining SH^a after GTR microsurgery^b ⁵²

Time post op	Probability of preserved SH ^a	
	Retrosigmoid approach	Middle fossa approach
immediate post-op	40-60%	50-70%
2 years	40-60%	40-60%
5 years	30-50%	40-60%
10 years	20-40%	20-40%

^a SH (serviceable hearing): ≤ 50 dB PTA, $\geq 50\%$ WRS

^b following gross total resection (GTR) of a sporadic intracanalicular VS in experienced hands

Practice guidelines for management of sporadic VS

Practice guideline: Hearing preservation outcomes with vestibular schwannomas

For the full guideline, see reference.⁵³

Observation vs. intervention recommendation

- Level III⁴⁸: for VS without tinnitus that are intracanalicular (► Table 45.7, Koos grade I) or < 2 cm, observation^a is recommended since it does not have a negative impact on tumor growth or hearing preservation

Hearing preservation with observation^a in sporadic VS

- Level III⁵³: for VS < 2 cm with serviceable hearing, chances of hearing preservation are > 75 –100% @ 2 years, > 50 –75% @ 5 years, and > 25 –50% @ 10 years
- Level III⁵³: for VS < 2 cm with AAO-HNS class A or GR grade I (► Table 45.5) hearing, chances of hearing preservation are > 75 –100% @ 2 years, and > 50 –75% @ 5 years. Data was insufficient for chances @ 10 years

Hearing preservation following SRS^b in sporadic VS

- Level III⁵³: for serviceable hearing pre-SRS, chances of hearing preservation are > 50 –75% @ 2 & 5 years, and > 25 –50% @ 10 years
- Level III⁵³: for AAO-HNS class A or GR grade I (► Table 45.5) hearing pre-SRS, chances of hearing preservation are > 75 –100% @ 2 years, > 50 –75% @ 5 years, and > 25 –50% @ 10 years

Hearing preservation following middle fossa or retrosigmoid MS in sporadic VS

- Level III⁵³: for sporadic VS < 2 cm with serviceable hearing pre-op, chances of hearing preservation are > 25 –50% immediately post-op, and @ 2, 5 & 10 years post-op
- Level III⁵³: for sporadic VS < 2 cm with AAO-HNS class A or GR grade I (► Table 45.5) hearing pre-SRS, chances of hearing preservation are > 50 –75% immediately post-op and @ 2 & 5 years post op, and > 25 –50% @ 10 years

^a observation consists of serial MRI evaluations as discussed in (p.782)
^b single dose SRS using modern dose planning (generally ≤ 13 Gy to the tumor margin, or ≤ 12 Gy to the tumor margin and cochlear dose ≤ 4 Gy)
 Abbreviations: MS = microsurgery; SRS = stereotactic radiosurgery.

► **Modified Delphi Study Consensus for sporadic VS.** The Modified Delphi Study for sporadic VS published in 2020⁵² issued 103 consensus statements related to this topic, and the reader is highly encouraged to review the results. Some FAQs derived from these statements are presented in ► Table 45.9, along with the strength of the opinion.

Table 45.9 Some FAQs based on the modified Delphi Study on VS ⁵²	
FAQ & Answer	Statement numbers
Q: Is poorer baseline hearing associated with a greater probability of progression to non-serviceable hearing regardless of treatment? A: Yes (strong consensus)	#10, 25, 44
Q: Does larger tumor size increase the probability of progression to non-serviceable hearing? A: Yes (strong consensus)	#7, 24, 40
Q: What is the optimal form of radiation for treating VS? A: There is no clear clinical advantage of fractionated radiation vs. SF-SRS, or for LINAC vs. Gamma radiation. Proton beam and intensity modulated radiation are not established modalities for treating VS (strong consensus)	#4, 74
Q: Is GTR associated with a greater probability of losing SH compared to near-total & subtotal resection? A: Yes (strong consensus)	#31
Q: When should patients undergoing observation be treated? A: Treatment (MS, SF-SRS or fractionated-SRS) should be offered at the first sign of unequivocal tumor growth (strong consensus)	#72
^a regardless of treatment: SF-SRS, MS or observation Abbreviations: GTR = gross total resection; MS= microsurgery; SF-SRS = single fraction stereotactic radiosurgery; SH (serviceable hearing): ≤ 50 dB PTA, $\geq 50\%$ WRS	

Management algorithm

Note: there are no Level I recommendations for most aspects of VS management. Some recommendations here differ from the published guidelines quoted above. The reader is encouraged to review the available information and determine their preferred approach which may differ from either of these strategies.

- small tumors (< 15 mm diameter) with perfect hearing (WRS 100%):
 - observe radiographically (CT or MRI scan) plus serial hearing tests (so-called “wait and scan”):
 - scan: recommend treatment for growth > 2 mm between studies. Scan schedule guide:
 - every 6 months $\times$ 2 years after diagnosis, then (if stable)
 - annually until 5 years after diagnosis, then (if stable)
 - at years 7, 9, and 14 after diagnosis¹⁷
 - annual audiology evaluations
 - hearing deterioration (WRS < 100%) but no growth: see below
 - rationale: in patients with small tumors and normal WRS, comparing the results of hearing preservation following surgery or SRS to the natural history, the conclusion is that *established tumor growth* should be the main determinant for treatment³⁹
- small tumors with serviceable hearing: management is very controversial
 - in general, patients with serviceable hearing but WRS < 100% have a 50% chance of preserving their serviceable hearing (50/50 rule or AAO-HNS class A or B) with either observation, SRS or microsurgical resection. Observation is sstrongly recommended as the initial management⁵²
 - hearing preservation rates better than 50% have been reported for both microsurgery and radiosurgery in very selected patients (smaller tumors, located medially in the IAC, intact pre-op ABR amplitudes and latencies)
 - decisions to try to beat the natural history (50% chance of hearing loss at 10 years) should be very individualized based on tumor factors (size, location) and patient factors (ABR, age, co-morbidities, preferences)

- final management decisions are often dictated by non medical reasons (patient's perception, social situation, financial considerations, support system, etc.)
- medium size tumors (15–25 mm diameter)
 - tumors > 15–20 mm should be treated.^{17,39} This is mostly true for young patients
 - close observation to establish growth is a valid option in older patients or patients with medical co-morbidities
 - complication rate increases and facial outcome worsens with increasing tumor size
 - NF2 patients present a challenge and should be evaluated individually. In general the success rate in the management of their tumors is lower (higher cranial nerve deficit and higher recurrence rate).^{54,55} Early management is considered more favorable for good outcome.⁵⁶ A retrospective study found significant hearing improvement and tumor shrinkage in > 50% of NF2 patients with progressive VS using bevacizumab (Avastin®) (see above)
- large tumors (> 25 mm diameter): treatment is recommended
 - microsurgical resection is favored to reduce mass effect and decompress the brainstem
 - SRS is also useful in larger tumors for older patients or those with significant co-morbidities
 - for tumors > 3 cm maximal p-fossa dimension with brainstem compression: intraoperative monitoring of trigeminal nerve (Cr. N. V), lower cranial nerves (Cr. N. IX, X) and use of SSEPs is not obligatory but should be considered⁵²
 - for tumors > 3 cm diameter with symptomatic hydrocephalus: pre-op placement of EVD is optional. A VP shunt should be placed if the hydrocephalus does not resolve post-op⁵²

Selection options for intervention

Once treatment is elected (see algorithm above), the type of treatment must be selected.

Comparison of microsurgery vs. radiosurgery (SRS)

1. hearing preservation
 - a) for patients with testable preoperative hearing
 - summary: radiosurgery or stereotactic radiation appears to be better at preserving hearing than microsurgery for VS < 2 cm up to 5 years post treatment. The chances of hearing preservation at 10 years is the same with SRS or MS. The difference is minor for tumors < 10 mm and very good preoperative hearing (70% SDS, and 30 dB PTA). The advantage of radiation is more pronounced for larger tumors and greater preoperative hearing loss. Details:
 - SRS: overall, at 2, 5, and 10 years, 50–75%, 50–75%, and 25–50% of the patients maintained their GR hearing class (► Table 45.5). Hearing preservation appears to be related to the radiation dose to the cochlea rather than to the tumor itself⁵⁷
 - microsurgery: hearing preservation is significantly related to the tumor size and to the experience of the surgical team. Hearing preservation in Samii's series of 1000 VS⁵⁵ improved from 24% in the first 200 cases to 49% in later cases. Hearing preservation in microsurgery has improved with the use of direct cochlear nerve monitoring⁵⁸ compared to auditory brainstem responses monitoring only. Hearing preservation in patients with class A, small tumors and direct cochlear nerve monitoring (compound nerve action potential) was 91%.⁵⁹ With microsurgery, the durability of the hearing preservation is also excellent with only 15% of patients with class A hearing after surgery, and 33% of patients with class B hearing after surgery slipping one class at 5 years follow-up⁶⁰
2. facial nerve preservation
 - a) preservation has been excellent with both microsurgery and radiosurgery
 - b) microsurgery: 98.5% overall⁶¹ and 100% in tumors not touching the brainstem. Staged resection has been advocated by some to improve facial nerve preservation in giant VS (> 4–4.5 cm)⁶²
 - c) radiosurgery: 98% of patients.⁶³ The incidence of facial neuropathy has significantly decreased since the SRS dose was decreased to 12–13 Gy. Facial neuropathy in the recent series occurred in patients having received 18–20 Gy
3. trigeminal neuropathy (TGN)
 - a) a complication classically feared in large tumors especially following SRS
 - b) SRS: 7% incidence of TGN (mainly in patients receiving higher doses, i.e., 18 Gy). No patients who received a dose < 13 Gy developed TGN⁶³
 - c) microsurgery: post-op TGN is not reported in most series
4. tumor control (local control rate (LCR)):
 - a) microsurgery: tumor recurrence has been poorly studied. Quoted rates in the literature vary between 0.5% at 6 years⁶¹ and 9.2%⁶⁴
 - b) SRS: long term data are lacking with the currently favored dose of 12–14 Gy to the tumor margin. In 68 patients with sporadic VS with median follow-up of 43.5 months⁶⁵:
 - intracanalicular VS: no growth after SRS (9 patients)

- extracanalicular VS: 97% LCR for VS that grew at a rate < 2.5 mm/year before SRS, vs. 69% LCR for VS that grew ≥ 2.5 mm/year before SRS
- 10–18% of patients show swelling (“pseudogrowth”)^{63,65} following SRS defined as enlargement ≥ 2 mm within 6–12 months that decreased or stabilized in size on follow-up

Vertigo and dizziness

Practice guideline: Balance problems with VS

Data is insufficient to support MS or SRS for balance problems due to VS⁶⁶

45

For patients with episodic vertigo or balance difficulties as the predominant symptom; also, see points under Selection options for intervention (p. 788):

1. remember: patients with VS are susceptible to other causes of vertigo as well, and patients should undergo ENG and functional balance assessment
2. vertigo that is due to the VS is often self-limited, and improves in 6–8 weeks to a reasonably tolerable level with no treatment (patients may do better with so-called “vestibular rehab”)
3. residual dizziness and balance disturbances are common whether stereotactic radiosurgery (SRS) or microsurgery (MS) is used, but are typically less after MS
4. patients’ reported vertigo intensity is improved by any treatment versus observation⁶⁷
5. after SRS: a minimum of 5–6 mos, and sometimes up to 18 months may be required to produce beneficial effects. Symptoms are improved more rapidly after MS than with SRS
6. following MS: The severity of dizziness after MS depends on the preoperative vestibular function on the affected side. If the ipsilateral vestibular function is absent preoperatively, then patients will not suffer from dizziness or nausea post-op. If ipsilateral vestibular function is intact pre-op, then patients may be very dizzy and nauseated in particular for the first 24 hrs
7. conclusion:
 - a) observation may be the best choice for $\approx 20\%$ of patients (“wait and scan”)
 - b) when treatment is desired:
 - surgery is the best choice for most VSs producing vertigo
 - SRS may be the right choice for some, especially: elderly patients (> 70 yrs) with other health problems, for recurrence of VS, for intracanalicular tumors or tumors growing < 2 mm/year, and for individual preference

Hydrocephalus

When hydrocephalus is present, it may require separate treatment with a CSF shunt—see Surgical considerations (p. 791)—and may possibly be done at the same time as surgery for the VS (if surgery for the VS is indicated).

45.3.7 Surgical treatment

Practice guidelines for surgical management of vestibular schwannomas

Practice guideline: Surgical management of vestibular schwannomas

For the full guideline, see reference.⁶⁶

Choice of surgical approach or surgical team

- When serviceable hearing is present: data is insufficient to recommend either middle fossa (MF) or retrosigmoid (RS) approach over the other for complete resection and facial nerve (FN) preservation⁶⁶
- When serviceable hearing is not present: data is insufficient to recommend either the RS or trans-labyrinthine (TL) approach over the other for GTR and FN preservation⁶⁶
- Outcome data is insufficient to recommend a multispecialty team (neurosurgeon and neurotologist) over either specialty working alone⁶⁶

Microsurgery (MS) for small VS

- Data is insufficient to support a firm recommendation that surgery should be the primary treatment for intracanalicular VS < 1.5 cm⁶⁶
- Level III⁶⁶: hearing preservation surgery may be attempted via the MF or RS approach for VS < 1.5 cm with good pre-op hearing

Microsurgery (MS) for VS in patients with neurofibromatosis type 2 (NF2)

- There is insufficient data that MS should be the initial treatment for VS in NF2⁶⁶

Miscellaneous information regarding microsurgery (MS) for VS

- There is insufficient data that subtotal resection followed by SRS provides comparable hearing and FN preservation to patients having GTR⁶⁶
- Level III⁶⁶: trigeminal neuralgia with VS is better relieved with MS than with SRS
- Level III⁶⁶: patients should be counseled that MS after SRS has an increased likelihood of STR and decreased FN function
- Level III⁶⁸: FN monitoring is recommended during MS to improve long-term FN function (strong Delphi consensus⁵²)
- Level III⁶⁸: favorable intraoperative FN response is a reliable predictor of good long-term FN outcome, but a poor response in the presence of an anatomically intact FN does not reliably predict long-term FN function & cannot be used to direct decisions for early reinnervation procedures
- Level III⁶⁸: intraoperative VIII nerve monitoring should be used when hearing preservation surgery is a goal. Update: auditory brainstem reflexes (ABRs) should be utilized, and direct cochlear nerve action potential monitoring should be considered when feasible (strong Delphi consensus⁵²)

Abbreviations: FN = facial nerve; GTR = gross total resection; MF = middle fossa; MS = microsurgery; RS = retrosigmoid; STR = subtotal resection; TL = translabyrinthine approach.

Approaches

General information

Three basic surgical approaches:

1. those with possibility of hearing preservation
 - a) middle fossa (MF): poor access to posterior fossa (see below)
 - b) retrosigmoid (RS) (see below) AKA retrosigmoid-transmeatal approach
2. translabyrinthine (TL): non-hearing preserving (see below)

Excellent results have been reported with each of these approaches. These guidelines assume that the surgical team is comfortable with all three approaches.

Decision algorithm for approach

The choice of approach is dictated by hearing salvageability and tumor size as follows:

1. salvageable hearing (► Table 45.10 for definition and guidelines)
 - a) if tumor is intracanalicular (no extension beyond a few mm into the posterior fossa (CPA); note: differences of opinion exist regarding how much tumor in the CPA can be removed via MF); use the middle fossa approach. Note: some authors exclusively use the retrosigmoid approach for hearing preservation even for these tumors with excellent results
 - b) if tumor extends > few mm into the posterior fossa: use the retrosigmoid approach (it is generally accepted that the cisternal part of the tumor is not well exposed by the middle fossa approach, especially regarding the ability to dissect tumor off the nerves)
2. non salvageable hearing (see ► Table 45.10 for definition and guidelines)
 - a) use translabyrinthine approach or retrosigmoid approach
 - b) either can be used irrespective of tumor size. Surgical team preference is the main deciding factor. Some influencing aspects:
 - a young patient with no cerebellar atrophy might favor a translab approach
 - an anteriorly situated sigmoid sinus and/or a high riding jugular bulb restrict the working space in the translab approach, and might favor a retrosigmoid approach

Table 45.10 Hearing salvageability**Serviceable and unsalvageable hearing****Definition of serviceable hearing**

A generous definition of serviceable hearing: PTA < 50 dB and SDS > 50%^a

Unsalvageable hearing

Serviceable hearing is *unlikely* to be preserved post-op when

1. pre-op SDS < 75%
2. or pre-op PTA loss > 25 dB
3. or pre-op BAER has abnormal wave morphology
4. or tumor > 2–2.5 cm diameter

^asee also other definitions of serviceable hearing (p. 781)

Surgical considerations

General information

The first surgical removal of a vestibular schwannoma was performed over a century ago in 1894.⁶⁹

The facial nerve is pushed forward by the tumor in ≈ 75% of cases (range: 50–80%), but may occasionally be pushed rostrally, less often inferiorly, and rarely posteriorly. It may even continue to function while it is flattened to a mere ribbon on the tumor capsule surface.

Anesthesia with minimal muscle relaxants allows intra-op seventh nerve monitoring. In only ≈ 10% of large tumors is the cochlear nerve a separate band on the tumor capsule, in the remainder it is incorporated into the tumor.

Although total excision of tumor is usually the goal of surgery, facial nerve preservation must take precedence over degree of resection. Near total resection (very small sliver of tumor left on the facial nerve) or subtotal resection are all excellent options if the tumor is tightly adherent to the facial nerve or the brainstem. Both have excellent long term tumor control rate with either observation or post-op radiation.

If hydrocephalus is present, historically standard practice was to place a CSF shunt and wait ≈ 2 weeks before the definitive operation.⁷⁰ While still acceptable, this is less commonly done at present, and shunting or EVD is often performed under the same anesthesia.

Large tumors may be approached by a staged surgical approach to debulk tumor and preserve facial nerve, or a planned subtotal resection followed by radiation. For tumors > 3 cm, such an approach seems to lead to improved results of facial nerve function.⁷¹

The extra anesthesia time involved in translab approach may be detrimental in the elderly.

Middle fossa approach

- indications:
 - a) hearing preservation
 - b) laterally placed tumors
 - c) small tumors (usually < 2.5 cm)
- pros:
 - a) allows drilling and exposure of the IAC all the way to the geniculate ganglion (good for laterally placed tumors)
 - b) basically an extradural subtemporal operation
- cons:
 - a) potential damage to temporal lobe with risk of seizures
 - b) facial nerve is the most superficial nerve in this exposure and therefore the surgeon works “around” the facial nerve (possibility of injury)
- technique summary
 - a) lumbar drain
 - b) usually straight incision, starting in front of the tragus, extending cephalad for 6 cm, held open with a self-retaining retractor
 - c) the temporalis muscle is incised vertically (along the muscle fibers) along the most posterior aspect of the exposure and is reflected anteriorly
 - d) craniotomy: 4 cm × 3 cm
 - e) elevate the middle fossa dura, section the middle meningeal artery. Identify and preserve the greater superficial petrosal nerve (GSPN), arcuate eminence, V3, and true edge of the petrous bone (the false edge is the groove occupied by the superior petrosal sinus)

- f) drill and expose the internal auditory canal all the way to Bill's bar (for tumors extending laterally)
- g) localize the facial nerve with the nerve stimulator
- h) open the IAC dura along the main axis of the IAC, avoiding VII
- i) identify the vestibular, cochlear, and facial nerves
- j) dissect the tumor off the nerves

Translabyrinthine approach

Often preferred by neurotologists.

1. pros & cons: See ► Table 45.11
2. technique summary
 - a) position: supine head rotated to contralateral side, can be done either with pins or doughnut if it is anticipated that no retractors will be used
 - b) prep the abdomen for fat graft (almost always used)
 - c) skin incision should be tailored to the location of the sigmoid sinus (observe location of the sigmoid sinus and pinna of the ear on the pre-op MRI). Usually smaller opening than retrosigmoid approach
 - does not require a craniotomy. For large tumors requiring an "extended translab," 1–2 cm of retrosigmoid dura should be exposed during the mastoidectomy to allow for retraction of the sigmoid sinus
 - dural opening along the IAC after identification of VII with stimulator
 - for large tumor: section the superior petrosal sinus and section the tentorium to gain better intradural exposure
 - closure requires fat graft

Table 45.11 Pros & cons of translabyrinthine approach

Disadvantages	Advantages
• sacrifices hearing (acceptable when hearing is already non-functional or unlikely to be spared by other approach) • may take longer than retrosigmoid approach • possibly higher rate of post-op CSF leak	• early identification of VII may result in higher preservation rate • less risk to cerebellum and lower cranial nerves • patients do not get as "ill" from blood in cisterna magna, etc. (essentially an extracranial approach) • less trauma to musculature ∴ less H/A than retrosigmoid

Booking the case: Translabyrinthine approach for vestibular schwannoma

Also see defaults & disclaimers (p. 25).

1. position: supine with shoulder roll
2. equipment:
 - a) microscope
 - b) high speed drill
 - c) ultrasonic aspirator
3. some surgeons work with neurotologist to assist with the IAC and for follow-up
4. neuromonitoring: facial EMG (does not require EEG tech), SSEPs for tumors involving the brainstem (requires EEG tech)
5. post-op: ICU
6. consent (in lay terms for the patient—not all-inclusive):
 - a) procedure: surgery through an incision behind the ear to remove a tumor growing inside the skull on the nerve to the ear. Possible need for post-op lumbar drain. Fat graft (≈ always used)
 - b) alternatives: nonsurgical management with follow-up MRIs, other surgical approaches, radiation (stereotactic radiosurgery)
 - c) complications: CSF leak with possible meningitis, loss of hearing in ipsilateral ear (if not already lost), paralysis of facial muscles on the side of surgery with possible need for surgical procedures to help correct (correction is often far from perfect), facial numbness, post-op balance difficulties/vertigo, brainstem injury with stroke

Retrosigmoid approach

AKA posterior fossa, AKA suboccipital approach.^{72,73}

- pros:
 - a) familiar to most neurosurgeons ∴ often preferred by neurosurgeons
 - b) quick access to the tumor
 - c) hearing preservation possible
 - d) NOTE: this approach is very versatile. Samii⁵⁵ resected all his acoustic tumors via a retrosigmoid approach; he achieved a significant amount of brain relaxation and improved exposure by using in the sitting position, which is generally not used in the U.S. because of associated complications (p.1737)
- cons:
 - a) cerebellar retraction: not a problem for tumors < 4 cm, provided the craniotomy is sufficiently lateral and the cisterna magna and the CP angle cistern has been opened
 - b) headaches: it has been suggested that headaches are more common following retrosigmoid craniotomy than after translabyrinthine craniotomy. Postulated mechanisms: purely extradural drilling in translab with no bone dust in subarachnoid space. More anterior skin incision in translab and less disruption of the suboccipital musculature and greater occipital nerve

45

Booking the case: Retrosigmoid craniotomy for vestibular schwannoma

Also see defaults & disclaimers (p.25).

1. position: lateral decubitus with tumor side up
2. equipment:
 - a) microscope
 - b) ultrasonic aspirator
 - c) image guided navigation system (if used) (may be helpful for placing skin incision and craniotomy more than for tumor localization)
3. some surgeons work with neurotologist to assist with the IAC and for follow-up
4. neuromonitoring: facial EMG (does not require EEG tech), BAERS, near field monitoring (CNAP: compound nerve action potential)
5. post-op: ICU
6. consent (in lay terms for the patient—not all-inclusive):
 - a) procedure: surgery through an incision behind the ear to remove a tumor growing inside the skull on the nerve to the ear. Possible need for post-op lumbar drain. Possible fat graft (optional)
 - b) alternatives: nonsurgical management with follow-up MRIs, other surgical approaches, radiation (stereotactic radiosurgery)
 - c) complications: CSF leak with possible meningitis, loss of hearing in ipsilateral ear (if not already lost), paralysis of facial muscles on the side of surgery with possible need for surgical procedures to help correct (correction is often far from perfect), facial numbness post-op balance difficulties/vertigo, brainstem injury with stroke. Facial numbness (infrequent)

Technique summary

1. position: lateral decubitus with tumor side up, head in pins rotated (might need shoulder roll), zygomatic arch horizontal. 30° elevation of the head is paramount; see Posterior fossa (suboccipital) craniectomy, Lateral oblique position (p.1738)
2. percutaneous lumbar drain (optional)
3. incision is shaped like the pinna of the ear, 3 finger breaths behind the external auditory canal
4. the craniotomy has to be lateral enough to expose part of the sigmoid and part of the transverse sinuses and to allow a straight line of sight to the lateral end of the IAC
5. to prevent CSF leak, seal all bone edges with bone wax
6. dural opening along the lines of the craniotomy
7. exposure is enhanced by opening the cerebellopontine angle cistern and the cisterna magna under the microscope and draining CSF (20–40 ml of CSF can also be drained via a lumbar subarachnoid catheter)
8. the petrosal vein is often sacrificed at the beginning of the procedure to allow the cerebellum to relax and fall back and to avoid tearing off the transverse sinus. Be careful not to coagulate the SCA that often runs with the petrosal vein

9. using the facial nerve stimulator, the posterior aspect of the tumor is inspected to make sure the facial nerve has not been pushed posteriorly
10. the thin layer of arachnoid that covers most tumors is identified. Vessels within the arachnoid may contribute to cochlear function and may be preserved by keeping them with the arachnoid
11. the plane between tumor and cerebellum may be followed to the brainstem, and occasionally to the VII nerve (this plane is harder to follow once bleeding from tumor debulking occurs)
12. to help locate the origin of the VII nerve at the brainstem, see ► Table 45.12 and CPA anatomy in ► Fig. 1.11
13. the posterolateral tumor capsule is opened, and internal decompression is performed. The tumor is collapsed inward and the capsule is kept intact and is rolled laterally off of VII and is eventually removed. The most difficult area to separate VII from tumor is just proximal to the entrance to the porus acusticus. A general recommendation is to accept a subtotal or near total resection to preserve anatomic continuity of the facial nerve in cases where it is identified by stimulation but because it is so flattened it cannot be seen as a separate structure on the surface of the tumor.
14. after the extracanalicular portion of tumor is removed, the dura over the IAC is incised, and the IAC is drilled open and tumor is removed from this portion. To preserve hearing, the bony labyrinth must not be violated. The posterior semicircular canal (SCC) is the most vulnerable structure (► Fig. 45.10). The vestibule of the SCCs is also at risk but is less likely to be entered. The maximal amount of temporal bone drilling that can be accomplished without entering the posterior SCC can be determined from the pre-op CT. The operculum of the temporal bone is a small step-off palpable with a nerve hook posteriorly from the porus acusticus. It marks the location of the vestibular aqueduct and is a good landmark for the posterior extent of the drilling in the retrosigmoid exposure of the IAC. Measuring the distance from the IAC to the posterior semicircular canal on a pre-op CT and measuring the thickness of the bone overlying the posterior semicircular canal are recommended for safe exposure of the IAC, in particular for hearing preservation. However, opening the labyrinth cannot always be avoided; and any opening should be plugged with bone-wax or muscle.⁷⁴ If the facial nerve is not intact and is not going to be grafted, then the IAC should be plugged, e.g., by bone wax covered with a small piece of hammered muscle (hammering makes the muscle sticky by activating extrinsic clotting factors) and Gelfoam®.

Table 45.12 Aids in localizing VII nerve origin⁷⁶

- VII nerve originates in the pontomedullary sulcus near the lateral end of the sulcus, 1–2 mm anterior to the VIII nerve
- the pontomedullary sulcus ends just medial to the foramen of Luschka (extending from the lateral recess of the IV ventricle, ► Fig. 1.11)
- a tuft of choroid plexus usually extends out of the foramen of Luschka on the posterior surface of IX and X nerve, just inferior to the origin of VII
- the flocculus of the cerebellum projects from the lateral recess into the CPA just posterior to the origin of VII and VIII
- VII origin is 4 mm cephalad and 2 mm anterior to that of the IX nerve

NB: *large tumors*: in some large tumors, the capsule may be adherent to the brainstem and so portions of tumor must be left; recurrence rate among these is ≈ 10–20%.⁷⁵ Large tumors may also involve V superiorly (sometimes VII is pushed up against V), and inferiorly may involve IX, X, and XI. The lower cranial nerves can usually be spared by dissecting them off of the tumor capsule, and protecting them with cottonoids.

Post-op care and care for complications

Practice guideline

Practice guideline: Perioperative nimodipine for VS surgery

Level III⁷⁷: perioperative nimodipine (possibly with the addition of hydroxyethyl starch) should be considered to improve facial nerve functional outcome and may help with hearing outcome following microsurgery

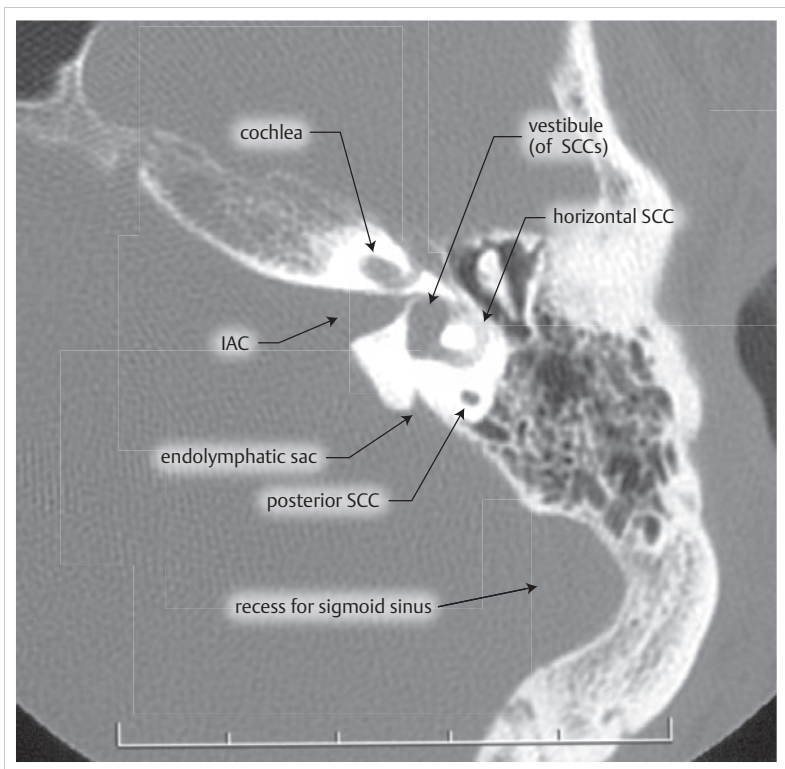


Fig. 45.10 Temporal bone structures. Image: CT scan, axial slice through left petrous bone with bone windows. Provided courtesy of Chris Danner, MD.

Cranial nerve and brainstem dysfunction

Facial nerve (VII)

If eye closure is impaired due to VII dysfunction: **R** natural tears 2 gtts to affected eye q 2 hrs and PRN. Apply Lacrilube® to affected eye and tape it shut q hs. If the eye is also insensate from Cr. N. V impairment, the eye is at risk of exposure and trophic keratitis and immediate consultation with ophthalmology for emergency tarsorrhaphy is indicated. Early tarsorrhaphy should also be considered if there is complete VII palsy with little chance of early recovery.

Facial re-animation (e.g., hypoglossal-facial anastomosis) is performed after 1–2 months if VII was divided, or if no function returns after 1 year with an anatomically intact nerve.

Vestibular nerve (VIII)

Vestibular dysfunction is common post-op, nausea and vomiting due to this (and also intracranial air) is common. Balance difficulties due to this clear rapidly; however, ataxia from brainstem dysfunction may have a permanent component.

Lower cranial nerves

The combination of IX, X, and XII dysfunction creates swallowing difficulties and creates a risk of aspiration.

Brainstem dysfunction

Brainstem dysfunction may occur from dissection of tumor off of the brainstem. This may produce ataxia, contralateral paresthesias in the body. Although there may be improvement, once present, there is often some permanent residual.

CSF fistula

Also, see CSF fistula (cranial) (p.415) for general information. CSF fistula may develop through the skin incision, the ear (CSF otorrhea) through a ruptured tympanic membrane, or via the eustachian tube and then either through the nose (rhinorrhea) or down the back of the throat.

Rhinorrhea may occur through any of the following routes (circled numbers in ► Fig. 45.11):

- ① via the apical cells to the tympanic cavity (TC) or eustachian tube (the most common path)
- entry into the bony labyrinth—reaching the middle ear would require rupture (e.g., of the oval window by overpacking bone wax into the labyrinth)
 - ② through the vestibule of the horizontal semicircular canal (SCC)
 - ③ through the posterior SCC (the posterior SCC is the most common area that is entered by drilling)
- ④ follows the perilyabyrinthine cells and tracts to the mastoid antrum
- ⑤ through the mastoid air cells surgically exposed at the craniotomy site

Most leaks are diagnosed within 1 week of surgery, although one presented 4 years post-op.⁷⁸ They appear to be more common with more lateral unroofing of the IAC.⁷⁸ Meningitis complicates a CSF leak in 5–25% of cases, and usually develops within days of the onset of leak.⁷⁸ Hydrocephalus may promote the development of a CSF fistula.

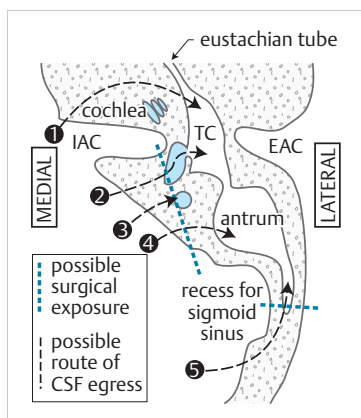


Fig. 45.11 Possible routes for CSF rhinorrhea following vestibular schwannoma surgery (see text). Image: schematic axial slice through right petrous bone. Adapted from Surgical Neurology, Vol. 43, Nutik S L, Korol H W, Cerebrospinal Fluid Leak After Acoustic Neuroma Surgery, 553–7, 1995, with permission from Elsevier Science.

Treatment: 25–35% of leaks stop spontaneously (one series reported 80%).⁷⁸ Treatment options include:

1. surgical treatment: in general, postoperative CSF leak (including rhinorrhea) is best addressed with immediate surgical re-exploration. Non-surgical measures (see following items) may be used to temporize
 - a) in the case of a translabyrinthine approach with absent ipsilateral hearing: to treat rhinorrhea, pack and permanently close the Eustachian tube via a trans tympanic membrane approach. This is very effective and avoids re-opening the surgical incision and removing the previously placed fat graft.
 - b) if hearing is preserved (which excludes translab), every effort should be made to preserve the Eustachian tube function to preserve middle ear function. Re-explore the surgical field, re-wax the aircells and place additional fat graft, fascia, pericranium, or other sealant over the exposed aircells. This aggressive management is the most definitive and rapid treatment, and avoids prolonged bed rest required by placing a lumbar drain and trying to control the leak in a conservative way
2. non-surgical: may be used to temporize, but early surgical treatment is recommended
 - a) evaluate for infection, which is often associated with CSF leak
 - b) elevate HOB

- c) for leaks through the incision, reinforcement with additional sutures may be attempted
 - d) a percutaneous lumbar subarachnoid drain may be tried,^{79,80} although some debate its efficacy,⁷³ and there is a theoretical risk of drawing bacteria into the CNS
 - e) placement of EVD
3. a CSF leak may be an indication of altered CSF hydrodynamics. Most of these patients demonstrate frank ventriculomegaly (hydrocephalus). In some patients the leak may function as a pressure relief valve and thereby ameliorate the ventriculomegaly (i.e., there would be hydrocephalus if there wasn't a leak). Adjunctive CSF shunting is usually also necessary or the repair will be more likely to fail

Outcome and follow-up

Complete surgical removal was reported in 97–99% of cases.⁸¹

Surgical morbidity and mortality

Also see Post-op considerations for p-fossa cranis (p.1744). Estimated frequency of some complications⁸²: the most common complication is CSF leakage in 4–27%⁷⁸ (see above), meningitis in 5.7%, stroke in 0.7%, subsequent requirement for CSF shunt (for hydrocephalus or to treat leak) in 6.5%.

The mortality rate is ~ 1% at specialized centers.^{55,81,83}

Cranial nerve dysfunction

► Table 45.13 shows statistics of VII and VIII cranial nerve preservation following suboccipital removal of VSs in several combined groups of patients. For more details, see below.

Post-radiation cranial neuropathies generally appear 6–18 months following stereotactic radio-surgery (SRS),⁸⁵ and since more than half of these resolve within 3–6 months after the onset, the recommendation is to treat these with a course of corticosteroids.

► **Facial nerve (VII).** See ► Table 45.3 for the House and Brackmann grading scale. Grades 1–3 are associated with acceptable function. Facial nerve preservation is related to tumor size.

Surgery: With the use of modern facial nerve monitoring techniques, anatomical integrity of the facial nerve can be achieved in >90% even for very large tumors and in close to 99% in medium-sized tumors.⁸⁶ In cases where the nerve is so flattened that the tumor has to be left over the nerve to preserve its anatomical integrity, functional outcome of the facial nerve is nevertheless lower, especially with larger tumors. Excellent outcome (HB grade I–III) was only achieved in 75% of the patients with large tumors and 91% in medium-sized tumors.

SRS for tumors ≤ 3 cm diameter: With modern SRS dosimetry (12–13 Gy for patients with serviceable hearing and 13–14 Gy for patients with non-serviceable hearing), the incidence of new facial nerve weakness was 4%.⁸⁷

Table 45.13 Cranial nerve preservation in retrosigmoid removal of VSs ^a		
Size of tumor	Preserved function	
	VII nerve	VIII nerve
< 1 cm	95–100%	57%
1–2 cm	80–92%	33%
> 2 cm	50–76%	6%

^aseries of 135 VSs⁸⁴ (p 729) and other sources⁷⁵ (p 3337),⁸¹

► **Vestibulo-acoustic nerve (VIII).** Patients with unilateral VS and Class I or II hearing (► Table 45.5) comprised ~ 12% of cases in a large series.⁸⁸ Preservation of hearing is critically dependent on tumor size, with little chance of preservation with tumors > 1–1.5 cm diameter. Chances of preserving hearing may possibly be improved by intraoperative brainstem auditory evoked potential monitoring.⁸⁹ In centers treating large numbers of VSs, hearing *preservation* rates of 35–71% can be achieved with tumors < 1.5 cm^{88,90} (although a range of 14–48% may be more realistic⁹¹). Hearing may rarely be improved post-op.⁹²

SRS: for tumors ≤ 3 cm diameter,⁹³ hearing was preserved in 26% of 65 cases with pre-op pure tone threshold <90 dB. Hearing loss has been correlated with increase in tumor size.⁹⁴ NB: there is a high rate of hearing loss at 1 year. SRT: useful hearing was preserved in 93%.⁹⁵

Vestibular nerve function is rarely normal post-op. Attempts at “vestibular” sparing surgery have shown no better results than surgery not specifically addressing this issue. Most patients with

unilateral loss of vestibular nerve function will learn to compensate to a significant degree with input from the contralateral side, if normal. Patients with ataxia as a result of brainstem injury from the tumor or the surgery will have more difficulties post-op. Some patients will seem to do well initially post-op with respect to vestibular nerve function, only to undergo a delayed deterioration several months post-op. These cases likely represent aberrant regeneration of the vestibular nerve fibers and may be extremely difficult to manage. Some experts advocate cutting the vestibular nerve, as for Meniere disease (p.604).

► **Trigeminal nerve (V).** Postoperative trigeminal nerve symptoms occur transiently in 22% and permanently in 11% following microsurgery, similar to the results of SRS.⁹⁶ New facial numbness occurred in 2% with SRT.⁹⁵

► **Lower cranial nerves.** Injuries to IX, X, and XI occur infrequently following surgery on large tumors that distort the nerves and displace them inferiorly against the occipital bone.

Recurrence

Following microsurgery (MS)

Recurrence is highly dependent on extent of removal. However, recurrence can develop in tumors that were apparently totally removed, or when subtotal resection was performed. This can occur many years after treatment. Tumor progression rate following subtotal resection is ≈ 20%.⁹¹ All patients should be followed with imaging (CT or MRI). In older series with up to 15 yrs follow-up, local control rate (LCR) after “total resection” is ≈ 94%. More recent series with MRI follow-up indicate recurrence rates of 7–11% (3–16 yrs follow-up).⁹¹

Use of EBRT

EBRT may improve LCR in incompletely resected tumors as shown in ► Table 45.14 (note: with the long survival expected with benign tumors, post-XRT complications may occur).

Table 45.14 Local control rates of surgery vs. surgery + EBRT for VSs ⁹⁷		
Extent of surgical removal	Local control rate (LCR)	
	Surgery	Surgery + EBRT ^a
gross total	60/62 (97%)	no data
near total (90–99%)	14/15 (93%)	2/2 (100%)
subtotal (<90%)	7/13 (54%)	17/20 (85%) ^a
biopsy only	no data	3/3 (100%)

^awith doses <45 Gy, LCR was 33%; with >45 Gy, LCR was 94%

Microsurgery vs. SRS

In a non-randomized retrospective study⁹⁶ of VSs < 3 cm dia, the *short-term* LCR (median 24 mos follow-up) was 97% for microsurgery vs. 94% for stereotactic radiosurgery (SRS). However, for benign tumors, long-term follow-up is critical (possibly 5–10 years⁸⁵), and this study suggests that the long-term LCR will be better for MS than SRS. SRS studies with long-term follow-up⁹⁸ are not directly comparable because in the cases with longest follow-up, higher radiation doses were used with a resultant higher incidence of radiation complications, and an anticipated better LCR.

Initially there may be temporary enlargement of the tumor accompanied by loss of central contrast enhancement following SRS in ≈ 5% of patients⁹⁹ (with up to 2% of patients showing actual initial tumor growth), and so the need for further treatment after SRS should be postponed until there is evidence of sustained growth.¹⁰⁰ Surgery should therefore be avoided during the interval from 6 to 18 months after SRS because this is the time of maximum damage from the radiation.¹⁰⁰

Although the numbers are small, the rate of VII nerve injury may be higher in patients undergoing MS following SRS failure to achieve LCR compared to cases where microsurgery was the initial procedure^{101,102}; however, this has been disputed.¹⁰⁰ Lastly, there is a potential for malignant transformation of VSs following SRS, including triton tumors^{103,104} (malignant neoplasms with rhabdoid features) or the induction of skull base tumors (which has been reported with external beam radiation¹⁰⁵), as well as the risk of late arterial occlusion (the AICA lies near the surface of VSs), any of which may occur many years later. The observed malignant transformation rate is 0.3% and the annual incidence of malignant transformation is 0.02%.¹⁰⁶

Practice guideline: Malignant transformation of VS after SRS

Level III⁴⁸: patients should be informed that there is a minimal risk of malignant transformation of VS after SRS

45

Treatment for recurrence following microsurgery

Repeat surgery for recurrent VS is an option. One series of 23 patients¹⁰⁷ showed that 6 of 10 patients with moderate or normal VII function maintained at least moderate function after reoperation, 3 patients had increased ataxia, and 1 patient had a cerebellar hematoma. SRS offers a safe and effective long-term management strategy for residual or retained VS after MS.¹⁰⁸ Using SRS for recurrent VSs resulted in progression-free survival rate of 97% at 3 years, 95% at 5 years, and 90% at 10 years. Among patients with any facial dysfunction (► Table 45.3, HB II-VI) at the time of recurrence or progression, 19% had improvement after SRS, and 5.5% with some facial function (HB I-V) developed more facial weakness. Among patients with trigeminal neuropathy, 20% had improvement, 5.8% developed or had worsened trigeminal neuropathy after SRS.

Practice guideline: Retreatment with SRS after SRS

Level III⁴⁸: SRS is safe & effective to re-treat patients with VS progression after initial SRS

Hydrocephalus

May occur following treatment (MS or SRS) for VS, and may even occur years later. The increased CSF pressure may also predispose to development of a CSF fistula.

Quality of life

Being diagnosed with a VS creates anxiety and leads to a temporary reduction in quality of life (QoL). However, once a treatment modality is chosen, observational management for amenable tumors produced the highest QoL.¹⁰⁹

Patients undergoing SRS or observation report a better total Penn Acoustic Neuroma Quality of Life (PANQOL) score and higher PANQOL facial, balance, and pain subdomain scores than those undergoing microsurgical resection. However, the differences in health related QOL outcomes following SRS, observation, and microsurgical resection are small.¹¹⁰

Long term QoL in patients with VS < than 3 cm in diameter was higher in those having gross total resection (GTR) than those with less than GTR despite having similar rates of facial nerve and hearing preservation. This may be due to psychological factors. Notwithstanding, functional preservation should be prioritized.

References

- [1] Bosman FT, Jaffe ES, Lakhani SR, et al. World Health Organization Classification of Tumors series, vol. 6. In: WHO Classification of Tumors Editorial Board (ed.) World Health Organization classification of tumors of the central nervous system. 5th ed. Lyon, France: International Agency for Research on Cancer; 2021
- [2] Miettinen MM, Antonescu CR, Fletcher CDM, et al. Histopathologic evaluation of atypical neurofibromatous tumors and their transformation into malignant peripheral nerve sheath tumor in patients with neurofibromatosis 1—a consensus overview. *Hum Pathol*. 2017; 67:1–10
- [3] Higham CS, Dombi E, Rogiers A, et al. The characteristics of 76 atypical neurofibromas as precursors to neurofibromatosis 1 associated malignant peripheral nerve sheath tumors. *Neuro Oncol*. 2018; 20:818–825
- [4] Pemov A, Li H, Patidar R, et al. The primacy of NF1 loss as the driver of tumorigenesis in neurofibromatosis type 1-associated plexiform neurofibromas. *Oncogene*. 2017; 36:3168–3177
- [5] Emory TS, Scheithauer BW, Hirose T, et al. Intraneural perineuroma. A clonal neoplasm associated with abnormalities of chromosome 22. *American Journal of Clinical Pathology*. 1995; 103:696–704
- [6] Torres-Mora J, Dry S, Li X, et al. Malignant melanotic schwannian tumor: a clinicopathologic, immunohistochemical, and gene expression profiling study of 40 cases, with a proposal for the reclassification of “melanotic schwannoma”. *Am J Surg Pathol*. 2014; 38:94–105
- [7] Evans DG, Huson SM, Birch JM. Malignant peripheral nerve sheath tumors in inherited disease. *Clin Sarcoma Res*. 2012; 2. DOI: 10.1186/2045-3329-2-17
- [8] PDQ Pediatric Treatment Editorial Board. Childhood Soft Tissue Sarcoma Treatment (PDQ®): Health Professional Version. Bethesda, MD 2022

- [9] Foley KM, Woodruff JM, Ellis FT, et al. Radiation-induced malignant and atypical peripheral nerve sheath tumors. *Ann Neurol*. 1980; 7:311–318
- [10] Le Guellec S, Decouvelaere AV, Filleron T, et al. Malignant Peripheral Nerve Sheath Tumor Is a Challenging Diagnosis: A Systematic Pathology Review, Immunohistochemistry, and Molecular Analysis in 160 Patients From the French Sarcoma Group Database. *Am J Surg Pathol*. 2016; 40:896–908
- [11] Nagarjun MN, Savardekar AR, Kishore K, et al. Apoptotic presentation of a cauda equina paraganglioma. *Surg Neurol Int*. 2016; 7. DOI: 10.4103/2152-7806.180093
- [12] Rahimizadeh A, Ahmadi SA, Koshki AM, et al. Paraganglioma of the filum terminal: Case report and review of the literature. *Int J Surg Case Rep*. 2021; 78:103–109
- [13] National Institutes of Health Consensus Development Conference. Acoustic Neuroma: Consensus Statement. Bethesda, MD 1991
- [14] Eldridge R, Parry D. Summary: Vestibular Schwannoma (Acoustic Neuroma) Consensus Development Conference. *Neurosurgery*. 1992; 30:962–964
- [15] Harner SG, Laws ER. Clinical Findings in Patients with Acoustic Neuromas. *Mayo Clin Proc*. 1983; 58:721–728
- [16] Ostrom QT, Cioffi G, Waite K, et al. CEBTRUS Statistical Report: Primary Brain and Other Central Nervous System Tumors Diagnosed in the United States in 2014–2018. *Neuro Oncol*. 2021; 23:iii1–iii105
- [17] Stangerup SE, Caye-Thomasen P, Tos M, et al. The natural history of vestibular schwannoma. *Otol Neurotol*. 2006; 27:547–552
- [18] Marinelli JP, Grossardt BR, Lohse CM, et al. Is Improved Detection of Vestibular Schwannoma Leading to Overtreatment of the Disease? *Otol Neurotol*. 2019; 40:847–850
- [19] Marinelli JP, Beeler CJ, Carlson ML, et al. Global Incidence of Sporadic Vestibular Schwannoma: A Systematic Review. *Otolaryngol Head Neck Surg*. 2022; 167:209–214
- [20] Matthies C, Samii M. Management of 1000 vestibular schwannomas (acoustic neuromas): clinical presentation. *Neurosurgery*. 1997; 40:1–9; discussion 9–10
- [21] Carlson ML, Tveiten OV, Driscoll CL, et al. Long-term dizziness handicap in patients with vestibular schwannoma: a multicenter cross-sectional study. *Otolaryngol Head Neck Surg*. 2014; 151:1028–1037
- [22] Sweeney AD, Carlson ML, Shepard NT, et al. Congress of Neurological Surgeons Systematic Review and Evidence-Based Guidelines on Otologic and Audiologic Screening for Patients With Vestibular Schwannomas. *Neurosurgery*. 2018; 82: E29–E31
- [23] Edvardsson Rasmussen J, Laurell G, Rask-Andersen H, et al. The proteome of perilymph in patients with vestibular schwannoma. A possibility to identify biomarkers for tumor associated hearing loss? *PLoS One*. 2018; 13. DOI: 10.1371/journal.pone.0198442
- [24] Dilwali S, Landegger LD, Soares VY, et al. Secreted Factors from Human Vestibular Schwannomas Can Cause Cochlear Damage. *Sci Rep*. 2015; 5. DOI: 10.1038/srep18599
- [25] Jaffe B. Clinical Studies in Sudden Deafness. *Adv Otorhinolaryngol*. 1973; 20:221–228
- [26] Rauch SD. Clinical practice. Idiopathic sudden sensorineural hearing loss. *N Engl J Med*. 2008; 359:833–840
- [27] Byl FM, Jr. Sudden hearing loss: eight years' experience and suggested prognostic table. *Laryngoscope*. 1984; 94:647–661
- [28] Berenholz LP, Eriksen C, Hirsh FA. Recovery From Repeated Sudden Hearing Loss With Corticosteroid Use in the Presence of an Acoustic Neuroma. *Ann Otol Rhinol Laryngol*. 1992; 101:827–831
- [29] Awad Z, Huins C, Pothier DD. Antivirals for idiopathic sudden sensorineural hearing loss. *Cochrane Database Syst Rev*. 2012. DOI: 10.1002/14651858.CD006987.pub2
- [30] Chandrasekhar SS, Tsai DO, BS, Schwartz SR, et al. Clinical Practice Guideline: Sudden Hearing Loss (Update) Executive Summary. *Otolaryngol Head Neck Surg*. 2019; 161:195–210
- [31] Tarlov EC. Microsurgical Vestibular Nerve Section for Intractable Meniere's Disease. *Clin Neurosurg*. 1985; 33:667–684
- [32] House WF, Brackmann DE. Facial Nerve Grading System. *Otolaryngol Head Neck Surg*. 1985; 93:184–193
- [33] Koos WT, Day JD, Matula C, et al. Neurotopographic considerations in the microsurgical treatment of small acoustic neuromas. *J Neurosurg*. 1998; 88:506–512
- [34] Hardy DG, Macfarlane R, Baguley D, et al. Surgery for Acoustic Neuroma: An Analysis of 100 Translabyrinthine Operations. *J Neurosurg*. 1989; 71:799–804
- [35] Daniels RL, Swallow C, Shelton C, et al. Causes of unilateral sensorineural hearing loss screened by high-resolution fast spin echo magnetic resonance imaging: review of 1,070 consecutive cases. *American Journal of Otology*. 2000; 21:173–180
- [36] Committee on Hearing and Equilibrium of the American Academy of Otolaryngology-Head and Neck Surgery Foundation. Guidelines for the evaluation of hearing preservation in acoustic neuroma (vestibular schwannoma). *Otolaryngol Head Neck Surg*. 1995; 113:179–180
- [37] Gardner G, Robertson JH. Hearing Preservation in Unilateral Acoustic Neuroma Surgery. *Ann Otol Rhinol Laryngol*. 1988; 97:55–66
- [38] Silverstein H, McDaniel A, Norrell H, et al. Hearing Preservation After Acoustic Neuroma Surgery with Intraoperative Direct Eighth Cranial Nerve Monitoring: Part II. A Classification of Results. *Otolaryngol Head Neck Surg*. 1986; 95
- [39] Stangerup SE, Caye-Thomasen P, Tos M, et al. Change in hearing during 'wait and scan' management of patients with vestibular schwannoma. *Journal of Laryngology and Otology*. 2008; 122:673–681
- [40] Murofushi T, Matsuzaki M, Mizuno M. Vestibular evoked myogenic potentials in patients with acoustic neuromas. *Archives of Otolaryngology – Head and Neck Surgery*. 1998; 124:509–512
- [41] Dunn IF, Bi WL, Mukundan S, et al. Congress of Neurological Surgeons Systematic Review and Evidence-Based Guidelines on the Role of Imaging in the Diagnosis and Management of Patients With Vestibular Schwannomas. *Neurosurgery*. 2018; 82: E32–E34
- [42] Heller RS, Joud H, Flores-Milan G, et al. Changing Enhancement Pattern and Tumor Volume of Vestibular Schwannomas After Subtotal Resection. *World Neurosurg*. 2021; 151:e466–e471
- [43] Pathapati D, Barla K, Dayal M, et al. Facial Nerve Schwannoma: The Rare/Giant Mimicker of Vestibular Schwannoma/Neuroma. *Indian J Radiol Imaging*. 2021; 31:510–513
- [44] Latack JT, Gabrielsen TO, Knake JE, et al. Facial nerve neuromas: radiologic evaluation. *Radiology*. 1983; 149:731–739
- [45] Sataloff RT, Frattali MA, Myers DL. Intracranial facial neuromas: total tumor removal with facial nerve preservation: a new surgical technique. *Ear Nose Throat J*. 1995; 74:244–6, 248–56
- [46] Copeland WR, Hoover JM, Morris JM, et al. Use of preoperative MRI to predict vestibular schwannoma intraoperative consistency and facial nerve outcome. *J Neurol Surg B Skull Base*. 2013; 74:347–350
- [47] Minderhann T, Schlegel I. Grading of vestibular schwannomas and corresponding tumor volumes: ramifications for radiosurgery. *Acta Neurochir (Wien)*. 2013; 155:71–4; discussion 74
- [48] Germano IM, Sheehan J, Parish J, et al. Congress of Neurological Surgeons Systematic Review and Evidence-Based Guidelines on the Role of Radiosurgery and Radiation Therapy in the Management of

- Patients With Vestibular Schwannomas. *Neurosurgery*. 2018; 82:E49–E51
- [49] Bederson JB, von Ammon K, Wichmann WW, et al. Conservative Treatment of Patients with Acoustic Tumors. *Neurosurgery*. 1991; 28:646–651
 - [50] Khandalavala KR, Saba ES, Kocharyan A, et al. Hearing Preservation in Observed Sporadic Vestibular Schwannoma: A Systematic Review. *Otol Neurotol*. 2022; 43:604–610
 - [51] Caye-Thomasen P, Dethloff T, Hansen S, et al. Hearing in patients with intracanalicular vestibular schwannomas. *Audiology and Neuro-Otology*. 2007; 12:1–12
 - [52] Carlson ML, Link MJ, Driscoll CLW, et al. Working Toward Consensus on Sporadic Vestibular Schwannoma Care: A Modified Delphi Study. *Otol Neurotol*. 2020; 41:e1360–e1371
 - [53] Carlson ML, Vivas EX, McCracken DJ, et al. Congress of Neurological Surgeons Systematic Review and Evidence-Based Guidelines on Hearing Preservation Outcomes in Patients With Sporadic Vestibular Schwannomas. *Neurosurgery*. 2018; 82:E35–E39
 - [54] Asthagiri AR, Parry DM, Butman JA, et al. Neurofibromatosis type 2. *Lancet*. 2009; 373:1974–1986
 - [55] Samii M, Matthies C. Management of 1000 Vestibular Schwannomas (Acoustic Neuromas): Surgical Management with an Emphasis on Complications and How to Avoid Them. *Neurosurgery*. 1997; 40:11–23
 - [56] Brackmann DE, Fayad JN, Slaterry WH, 3rd, et al. Early proactive management of vestibular schwannomas in neurofibromatosis type 2. *Neurosurgery*. 2001; 49:274–80; discussion 280–3
 - [57] Timmer FC, Hanssens PE, van Haren AE, et al. Gamma knife radiosurgery for vestibular schwannomas: results of hearing preservation in relation to the cochlear radiation dose. *Laryngoscope*. 2009; 119:1076–1081
 - [58] Danner C, Mastrodimos B, Cueva RA. A comparison of direct eighth nerve monitoring and auditory brainstem response in hearing preservation surgery for vestibular schwannoma. *Otol Neurotol*. 2004; 25:826–832
 - [59] Yamakami I, Yoshinori H, Saeki N, et al. Hearing preservation and intraoperative auditory brainstem response and cochlear nerve compound action potential monitoring in the removal of small acoustic neuroma via the retrosigmoid approach. *Journal of Neurology, Neurosurgery, and Psychiatry*. 2009; 80:218–227
 - [60] Wang AC, Chinn SB, Than KD, et al. Durability of hearing preservation after microsurgical treatment of vestibular schwannoma using the middle cranial fossa approach. *J Neurosurg*. 2013; 119:131–138
 - [61] Samii M, Gerganov V, Samii A. Improved preservation of hearing and facial nerve function in vestibular schwannoma surgery via the retrosigmoid approach in a series of 200 patients. *Journal of Neurosurgery*. 2006; 105:527–535
 - [62] Patni AH, Kartush JM. Staged resection of large acoustic neuromas. *Otolaryngology - Head and Neck Surgery*. 2005; 132:11–19
 - [63] Lobato-Polo J, Kondziolka D, Zorzo O, et al. Gamma knife radiosurgery in younger patients with vestibular schwannomas. *Neurosurgery*. 2009; 65:294–300; discussion 300–1
 - [64] Roche PH, Ribeiro T, Khalil M, et al. Recurrence of vestibular schwannomas after surgery. *Prog Neurol Surg*. 2008; 21:89–92
 - [65] Marston AP, Jacob JT, Carlson ML, et al. Pretreatment growth rate as a predictor of tumor control following Gamma Knife radiosurgery for sporadic vestibular schwannoma. *J Neurosurg*. 2017; 127:380–387
 - [66] Hadjipaniayis CG, Carlson ML, Link MJ, et al. Congress of Neurological Surgeons Systematic Review and Evidence-Based Guidelines on Surgical Resection for the Treatment of Patients With Vestibular Schwannomas. *Neurosurgery*. 2018; 82:E40–E43
 - [67] Van Gompel JJ, Patel J, Danner C, et al. Acoustic neuroma observation associated with an increase in symptomatic tinnitus: results of the 2007–2008 Acoustic Neuroma Association survey. *J Neurosurg*. 2013; 119:864–868
 - [68] Vivas EX, Carlson ML, Neff BA, et al. Congress of Neurological Surgeons Systematic Review and Evidence-Based Guidelines on Intraoperative Cranial Nerve Monitoring in Vestibular Schwannoma Surgery. *Neurosurgery*. 2018; 82:E44–E46
 - [69] Pitts LH, Jackler RK. Treatment of Acoustic Neuromas. *New England Journal of Medicine*. 1998; 339:1471–1473
 - [70] Ojemann RG. Microsurgical Suboccipital Approach to Cerebellopontine Angle Tumors. *Clin Neurosurg*. 1978; 25:461–479
 - [71] Porter RG, LaRouere MJ, Kartush JM, et al. Improved facial nerve outcomes using an evolving treatment method for large acoustic neuromas. *Otol Neurotol*. 2013; 34:304–310
 - [72] Rhoton AL Jr. The cerebellopontine angle and posterior fossa cranial nerves by the retrosigmoid approach. *Neurosurgery*. 2000; 47:S93–129
 - [73] Ebersold MJ, Harner SG, Beatty CW, et al. Current Results of the Retrosigmoid Approach to Acoustic Neuroma. *J Neurosurg*. 1992; 76:901–909
 - [74] Tatagiba M, Samii M, Matthies C, et al. The Significance for Postoperative Hearing of Preserving the Labyrinth in Acoustic Neuroma Surgery. *Journal of Neurosurgery*. 1992; 77:677–684
 - [75] Youmans JR. *Neurological Surgery*. Philadelphia 1990
 - [76] Rhoton AL. Microsurgical Anatomy of the Brainstem Surface Facing an Acoustic Neuroma. *Surg Neurol*. 1986; 25:326–339
 - [77] Van Gompel JJ, Agazzi S, Carlson ML, et al. Congress of Neurological Surgeons Systematic Review and Evidence-Based Guidelines on Emerging Therapies for the Treatment of Patients With Vestibular Schwannomas. *Neurosurgery*. 2018; 82:E52–E54
 - [78] Nutik SL, Korol HW. Cerebrospinal Fluid Leak After Acoustic Neuroma Surgery. *Surgical Neurology*. 1995; 43:553–557
 - [79] Symon L, Pell MF. Cerebrospinal Fluid Rhinorrhea Following Acoustic Neuroma Surgery: Technical Note. *J Neurosurg*. 1991; 74:152–153
 - [80] Ojemann RG. Management of Acoustic Neuromas (Vestibular Schwannomas). *Clinical Neurosurgery*. 1993; 40:498–539
 - [81] Sekhar LN, Gormley WB, Wright DC. The Best Treatment for Vestibular Schwannoma (Acoustic Neuroma): Microsurgery or Radiosurgery? *Am J Otol*. 1996; 17:676–689
 - [82] Wiegand DA, Fickel V. Acoustic Neuromas. The Patient's Perspective. Subjective Assessment of Symptoms, Diagnosis, Therapy, and Outcome in 541 Patients. *Laryngoscope*. 1989; 99:179–187
 - [83] Gormley WB, Sekhar LN, Wright DC, et al. Acoustic Neuroma: Results of Current Surgical Management. *Neurosurgery*. 1997; 41:50–60
 - [84] Wilkins RH, Rengachary SS. *Neurosurgery*. New York 1985
 - [85] Flickinger JC, Kondziolka D, Pollock BE, et al. Evolution in Technique for Vestibular Schwannoma Radiosurgery and Effect on Outcome. *International Journal of Radiation Oncology and Biological Physics*. 1996; 36:275–280
 - [86] Samii M, Gerganov VM, Samii A. Functional outcome after complete surgical removal of giant vestibular schwannomas. *J Neurosurg*. 2010; 112:860–867
 - [87] Pollock BE, Driscoll CL, Foote RL, et al. Patient outcomes after vestibular schwannoma management: a prospective comparison of microsurgical resection and stereotactic radiosurgery. *Neurosurgery*. 2006; 59:77–85; discussion 77–85
 - [88] Glasscock ME, Hays JW, Minor LB, et al. Preservation of Hearing in Surgery for Acoustic Neuromas. *Journal of Neurosurgery*. 1993; 78:864–870
 - [89] Ojemann RG, Levine RA, Montgomery WM, et al. Use of Intraoperative Auditory Evoked Potentials to Preserve Hearing in Unilateral Acoustic Neuroma Removal. *J Neurosurg*. 1984; 61:938–948
 - [90] Brackmann DE, House JR III, Hitselberger WE. Technical Modifications to the Middle Cranial Fossa

Approach in Removal of Acoustic Neuromas. Los Angeles, CA 1993

- [91] Pollock BE, Lunsford LD, Flickinger JC, et al. Vestibular Schwannoma Management. Part I. Failed Microsurgery and the Role of Delayed Stereotactic Radiosurgery. *Journal of Neurosurgery*. 1998; 89:944-948
- [92] Shelton C, House WF. Hearing Improvement After Acoustic Tumor Removal. *Otolaryngol Head Neck Surg*. 1990; 103:963-965
- [93] Hirsch A, Norén G. Audiological Findings After Stereotactic Radiosurgery in Acoustic Neuromas. *Acta Otolaryngol (Stockh)*. 1988; 106:244-251
- [94] Flickinger JC, Lunsford LD, Coffey RJ, et al. Radiosurgery of Acoustic Neurinomas. *Cancer*. 1991; 67:345-353
- [95] Selch MT, Pedrosa A, Lee SP, et al. Stereotactic radiotherapy for the treatment of acoustic neuromas. *J Neurosurg*. 2004; 101:362-372
- [96] Pollock BE, Lunsford LD, Kondziolka D, et al. Outcome Analysis of Acoustic Neuroma Management: A Comparison of Microsurgery and Stereotactic Radiosurgery. *Neurosurgery*. 1995; 36:215-229
- [97] Wallner KE, Sheline GE, Pitts LH, et al. Efficacy of Irradiation for Incompletely Excised Acoustic Neuilemmas. *Journal of Neurosurgery*. 1987; 67:858-863
- [98] Noren G, Hirsch A, Mosskin M. Long-Term Efficacy of Gamma Knife Radiosurgery in Vestibular Schwannomas. *Acta Neurochirurgica*. 1993; 122
- [99] Linskey ME, Lunsford LD, Flickinger JC. Neuroimaging of Acoustic Nerve Sheath Tumors After Stereotactic Radiosurgery. *American Journal of Neuroradiology*. 1991; 12:1165-1175
- [100] Pollock BE, Lunsford LD, Kondziolka D, et al. Vestibular Schwannoma Management. Part II. Failed Radiosurgery and the Role of Delayed Microsurgery. *Journal of Neurosurgery*. 1998; 89:949-955
- [101] Slattery WH, Brackmann DE. Results of Surgery Following Stereotactic Irradiation for Acoustic Neuromas. *Am J Otol*. 1995; 16:315-321
- [102] Wiet RJ, Micco AG, Bauer GP. Complications of the Gamma Knife. *Arch Otolaryngol Head Neck Surg*. 1996; 122:414-416
- [103] Yakulis R, Manack L, Murphy AI. Postradiation Malignant Triton Tumor: A Case Report and Review of the Literature. *Arch Pathol Lab Med*. 1996; 120:541-548
- [104] Comey CH, McLaughlin MR, Jho HD, et al. Death From a Malignant Cerebellopontine Angle Triton Tumor Despite Stereotactic Radiosurgery. *Journal of Neurosurgery*. 1998; 89:653-658
- [105] Lustig LR, Jackler RK, Lanser MJ. Radiation-Induced Tumors of the Temporal Bone. *Am J Otol*. 1997; 18:230-235
- [106] Hasegawa T, Kida Y, Kato T, et al. Long-term safety and efficacy of stereotactic radiosurgery for vestibular schwannomas: evaluation of 440 patients more than 10 years after treatment with Gamma Knife surgery. *J Neurosurg*. 2013; 118:557-565
- [107] Beatty CW, Ebersold MJ, Harner SG. Residual and Recurrent Acoustic Neuromas. *Laryngoscope*. 1987; 97:1168-1171
- [108] Huang MJ, Kano H, Mousavi SH, et al. Stereotactic radiosurgery for recurrent vestibular schwannoma after previous resection. *J Neurosurg*. 2017; 126:1506-1513
- [109] Carlson ML, Tombers NM, Kerezoudis P, et al. Quality of Life Within the First 6 Months of Vestibular Schwannoma Diagnosis With Implications for Patient Counseling. *Otol Neurotol*. 2018. DOI: 10.1097/mao.0000000000001999
- [110] Carlson ML, Tveiten OV, Driscoll CL, et al. Long-term quality of life in patients with vestibular schwannoma: an international multicenter cross-sectional study comparing microsurgery, stereotactic radiosurgery, observation, and nontumor controls. *J Neurosurg*. 2015; 122:833-842

46 Meningiomas (Intracranial)

46.1 General information

Meningiomas are a group of tumors that are believed to originate in meningeothelial cells of the arachnoid membrane. They may be intracranial (the most common location), intraorbital, or intraspinal (p.981).

46.2 Meningioma tumor types

46.2.1 Meningioma (WHO grade 1, 2 or 3)

General information

46

Key concepts

- slow growing, extra-axial tumor, usually benign, arise from arachnoid (not dura)
- imaging (MRI or CT): classically broad-based attachment on dura often with dural tail, typically enhance densely, hyperostosis of adjacent bone is common
- MRI: isointense on T1WI, hypointense on T2WI
- 32% of incidentally discovered meningiomas do not grow over 3 years follow-up
- surgical indications: documented growth on serial imaging, symptoms referable to the lesion that are not satisfactorily controlled medically, edema (low density on CT, high signal on T2 MRI) changes in adjacent brain
- most (but not all) are cured if completely removed, which is not always possible
- most commonly located along falx, convexity, or sphenoid bone
- frequently calcified: classic histological finding with this are psammoma bodies

Meningiomas are the most common primary intracranial tumors in adults, representing 36.4% of CNS tumors.¹ They comprise 54.5% of non-malignant tumors.² They are usually slow growing, circumscribed (non-infiltrating), benign lesions. About 10% are histologically malignant and/or rapidly growing. Meningiomas arise from arachnoid cap cells (meningeothelial cells) of the arachnoid layer (not dura). They can be multiple in up to 8% of cases,³ which is more common in neurofibromatosis. Occasionally meningiomas form a diffuse sheet of tumor (**meningioma en plaque**).

They may occur anywhere that arachnoid cells are found (between brain and skull, within ventricles, intraorbital, and along the spinal cord). Ectopic meningiomas may arise within the bone of the skull (primary intraosseous meningiomas)⁴ and others occur in the subcutaneous tissue with no attachment to the skull.

They are characteristically slow growing. Most are asymptomatic (see below).

Psammoma bodies (from the Greek word for sand) are common in some meningioma subtypes, as well as in other tumors, e.g., prolactinomas, schwannomas, papillary thyroid carcinoma... and some non-neoplastic conditions. They are round, microscopic calcified structures having a concentric laminated (onion-like) appearance. The calcification may be dystrophic, or possibly a tumoricidal defense mechanism.⁵ They are eosinophilic, and stain positive for CEA and PAS.

Diagnostic criteria for meningioma

Diagnostic criteria for meningioma are shown in ► Fig. 46.1.⁶

Subtypes

See reference.⁶

There is a wide variety of histological morphology among meningiomas, leading to the 15 subtypes delineated below. The most common ones are meningeothelial, fibrous and transitional. Aggressiveness can vary within each subtype (p.805), but most are benign WHO CNS5 grade 1. However, the chordoid and clear cell subtypes have been assigned grade 2 because of a higher recurrence rate irrespective of the grading criteria discussed in the following section. Most, but not all, rhabdoid meningiomas are considered grade 3.

Meningioma

▷ ESSENTIAL * ◁

classic histopathological features matching at least one of the meningioma subtypes

OR

suggestive histopathological features combined with biallelic inactivation of *NF2* or other classic drivers of conventional meningioma (*TRAF7*, *AKT1*, *KLF4*, *SMO*, *PIK3CA*), clear cell meningioma (*SMARCE1*), or rhabdoid meningioma (*BAP1*)

OR

suggestive histopathological features combined with one of the defined DNA methylation classes of meningioma

▷ DESIRABLE ◁

- ▶ meningeal localization
- ▶ EMA immunoreactivity
- ▶ strong and diffuse SSTR2A immunoreactivity
- ▶ classic copy-number alterations of *NF2*-mutant meningioma, such as monosomy 22/22q in lower-grade meningiomas, with additional losses of 1p, 6, 10q, 14q, and/or 18 in higher-grade meningiomas

* diagnosis frequently requires matching of several criteria

Fig. 46.1 Diagnostic criteria for meningioma. (Modified with permission from WHO Classification of Tumors Editorial Board, (ed.) World Health Organization classification of tumors of the central nervous system. 5th ed. Bosman F T, Jaffe E S, Lakhani S R, et al. (series eds.): World Health Organization Classification of Tumors series, vol. 6. International Agency for Research on Cancer, Lyon, France, 2021.)

- **meningothelial meningioma:** syncytia-like lobules of epithelioid cells with some nuclei appearing to have holes and pseudo-inclusions. May resemble arachnoid cap cells, which may mimic meningotheial hyperplasia which can be seen adjacent to other tumors, e.g., optic gliomas. Whorls and psammoma bodies are less frequent than in fibrous, transitional, and psammomatous subtypes. More common at the skull base than other subtypes
- **fibrous meningioma:** parallel, storiform, or interlacing fascicles of spindle cells in a collagen-rich matrix. Often found over the convexities. Consistency is more rubbery than meningotheial or transitional
- **transitional meningioma:** contains meningotheial and fibrous morphology as well as transitional features. Whorls and psammoma bodies are frequent. Often found over the convexities
- **psammomatous meningioma:** psammoma bodies predominate over meningioma cells (which may be hard to find), and can overlap to form large calcified masses. Often arises in the thoracic spine of middle-aged to elderly women
- **angiomatous meningioma:** small, often hyalinized blood vessels predominate over meningioma cells (which may be hard to find). Angiomatous areas may be scattered in regions consistent with microcystic or metaplastic subtypes, and can manifest nuclear atypia which meets one of the criteria for a higher grade (see next section). Hypervascular specimens may resemble hemangioblastoma. Often produce cerebral edema out of proportion to their size
- **microcystic meningioma:** cells with thin, elongated processes create microcysts with a cobweb-like background. These microcysts may form radiologically identifiable macrocysts. Most are

benign, but degenerative nuclear atypia meets one of the criteria for a higher grade (see next section). Often produce cerebral edema out of proportion to their size

- **secretory meningioma:** foci of cells with gland-like epithelial differentiation and psammoma bodies. Often produce cerebral edema out of proportion to their size
- **lymphoplasmacyte-rich meningioma:** rare. Meningothelial cells are often dominated by chronic inflammatory infiltrates, which may mimic inflammatory conditions having meningothelial hyperplasia
- **metaplastic meningioma:** focal or diffuse mesenchymal features (including one or more of oss-eous, cartilaginous, lipomatous, myxoid and xanthomatous tissue) which are of unknown clinical significance. Features may overlap with angiomatous or microcystic subtypes. Ossification may mimic bone invasion or psammoma bodies of the psammomatous subtype
- **chordoid meningioma** (WHO grade 2): cords or trabeculae of small epithelioid cells in a mucin-rich matrix may resembles a chordoma. Chronic inflammatory infiltrates may be prominent. Typically large supratentorial tumors, with a younger average age of presentation (45 years). A high recurrence rate comparable to atypical meningiomas has caused assignment of WHO grade 2 to these tumors independent of morphology
- **clear cell meningioma** (WHO grade 2): round to polygonal cells with clear, glycogen-rich cytoplasm and prominent PAS-positive staining embedded in perivascular and interstitial collagen arranged in a patternless or sheet architecture. The collagen may coalesce into large acellular structures. Whorls and psammoma bodies are infrequent. Predilection for the CPA and spine (especially the cauda equina) with a young average age (24 years). A high rate of recurrence and CSF seeding has caused assignment of WHO grade 2 to these tumors independent of morphology
- **rhabdoid meningioma:** defined by the identification of rhabdoid cells (plump cells with eccentric nuclei, open chromatin, macronuclei and conspicuous eosinophilic paranuclear inclusions or whorls or waxy spheres). Rhabdoid features alone may be seen in tumors behaving as WHO grade 1 or 2⁷; however, most are highly proliferative and have other malignant features, which together with a high recurrence and mortality rate supports the designation of WHO grade 3 for most, and justifies close follow-up for all. There may be overlap of histologic and genetic features with the papillary subtype
- **papillary meningioma:** defined by a predominant pattern of thin-walled vessels in a perivascular pseudorosette (pseudopapillary) pattern. These tumors have been reported in children and adults, and are commonly associated with peritumoral edema and bony hyperostosis or lysis. Invasion of brain parenchyma and metastases (primarily to lung) are associated with the papillary growth pattern, but in the absence of other higher grade features, focal papillary growth is insufficient for grade 2 or 3 designation
- **atypical meningioma** (WHO grade 2): a meningioma of intermediate grade with features delineated in ► Fig. 46.2, which includes elevated mitotic activity and parenchymal brain invasion (extension along Virchow-Robin spaces does not violate the pia and does not qualify as invasion). Nuclear atypia alone is inadequate to classify as grade 2 as it is often degenerative in nature and does not correlate with outcome. Associated with: male gender, non-skull base location and previous surgery. Recurrence rate is higher following GTR than with grade 1 meningiomas, with further elevation in the presence of bone invasion
- **anaplastic (malignant) meningioma** (WHO grade 3): a meningioma that meets grading criteria shown in ► Fig. 46.3, often exhibiting extensive necrosis and potential for brain parenchymal invasion. Comprise 1-3% of meningiomas. Loss of H3 p.K28me3 occurs in 10-20% of anaplastic meningiomas and correlates with shorter OS

Grading meningiomas

Grading criteria for Grade 2 meningiomas is shown in ► Fig. 46.2, and for Grade 3 meningiomas in ► Fig. 46.3.⁶

These criteria can be applied across all meningioma subtypes, but the CNS WHO grade 2 criteria must be met for a diagnosis of atypical meningioma, and the CNS WHO grade 3 criteria must be met for a diagnosis of anaplastic (malignant) meningioma.

► **Proliferation index.** Due to variation between institutions and observers, it is advised that proliferation indices (e.g., Ki-67 or MIB-1 (p.632)) not be used as the sole discriminant for grading. However, these indices do correlate with prognosis (see ► Table 46.1).

Presentation

Symptoms from mass effect or peritumoral edema depend on the location of the tumor. Some specific locations are associated with well-described symptom complexes (see below).

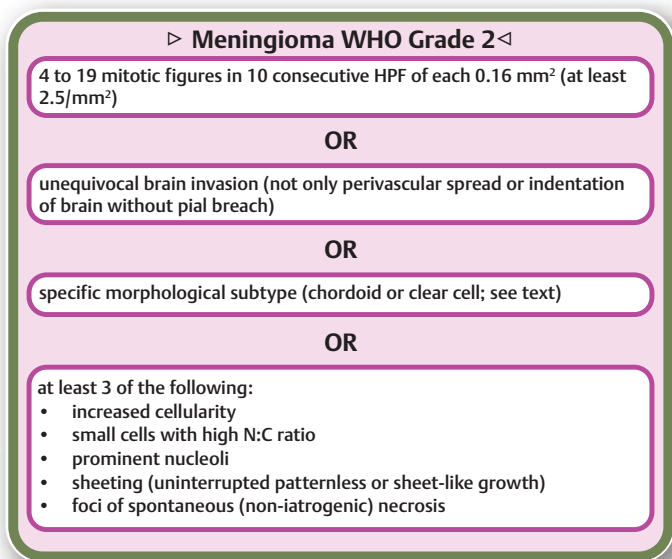


Fig. 46.2 Grading criteria for WHO Grade 2 meningioma.

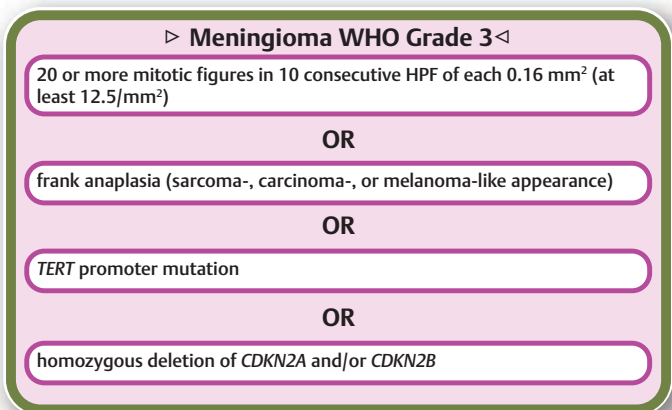


Fig. 46.3 Grading criteria for WHO Grade 3 meningioma.

Table 46.1 Ki-67 proliferation index and recurrence rates in meningiomas^{8a}

Description & WHO grade	Mean Ki-67 index ^a	Recurrence rate
common meningioma (WHO grade 1)	0.7%	9%
atypical meningioma (WHO grade 2)	2.1%	29%
anaplastic meningioma (WHO grade 3)	11%	50%

^anot to be used as sole discriminant for grading (see text)

Seizures may occur with supratentorial meningiomas as a result of irritation of the cerebral cortex. Occasionally patients may feel a mass under their scalp with convexity tumors that invade the bone.

Asymptomatic meningiomas

Meningiomas are the most common primary intracranial tumors, and most remain asymptomatic throughout the patient's life.⁹ The routine use of CT and MRI for numerous indications inevitably results in the discovery of incidental (asymptomatic) meningiomas. In a population-based study (the study population was middle class Caucasians and result, may not be generalizable to other demographic groups),⁹ incidental meningiomas were seen in 0.9% of MRIs. In another series, 32% of primary brain tumors seen on imaging studies were meningiomas, and 39% of these were asymptomatic.¹⁰ Of 63 cases followed for > 1 year with nonsurgical management, 68% showed no increase in size over an average follow-up of 36.6 mos, whereas 32% increased in size over 28 mos average follow-up.¹⁰ Asymptomatic meningiomas with calcification seen on CT and/or hypointensity on T2WI MRI appeared to have a slower growth rate.¹⁰

Common locations

► Table 46.2 lists common locations. Other locations include: CP angle, clivus, planum sphenoidale and foramen magnum. ≈ 60–70% occur along the falx (including parasagittal), along sphenoid bone (including tuberculum sellae), or over the convexity. Childhood meningiomas are rare, 28% are intraventricular, and the posterior fossa is also a common site. Details on some of these appear in the sections that follow.

Table 46.2 Location of adult meningiomas (series of 336 cases¹¹)

Location	%
parasagittal	20.8
convexity	15.2
tuberculum sellae	12.8
sphenoidal ridge	11.9
olfactory groove	9.8
falx	8.0
lateral ventricle (intraventricular)	4.2
tentorial	3.6
middle fossa	3.0
orbital	1.2
spinal	1.2
intrasylvian	0.3
extracalvarial	0.3
multiple	0.9

Sphenoid wing (or ridge) meningiomas

Three basic categories¹²:

1. lateral sphenoid wing (or pterional): behavior and treatment are usually similar to convexity meningioma
2. middle third (or alar)
3. medial (clinoidal): tend to encase the ICA and the MCA as well as cranial nerves in the region of the superior orbital fissure and the optic nerve. May compress brainstem. Total removal is often not possible

Olfactory groove meningiomas

Presentation: usually asymptomatic until they are large. Findings may include:

1. Foster Kennedy syndrome (p. 100): can occur when the tumor is asymmetrically situated to one side. Anosmia (patient is usually unaware of this), ipsilateral optic atrophy, contralateral papilledema (from increased ICP)
2. mental status changes: often with frontal lobe findings (apathy, abulia...) mimicking dementia
3. urinary incontinence
4. posteriorly located lesions may compress the optic apparatus causing visual impairment
5. large lesions may compress the fornix and cause short-term memory loss
6. seizure

The morbidity, mortality, and difficulty in achieving total removal increase significantly for tumors > 3 cm in size.¹³

Pre-op MRA, CTA or angiogram may be helpful to assess location of anterior cerebral arteries relative to the tumor. 70–80% of these get the majority of their blood supply from the anterior ethmoidal artery, which is usually not embolized due to risk to ophthalmic artery (and blindness). If there are substantial middle meningeal feeders, these may be embolized, but the benefit tends to be small.

Parasagittal and falx meningiomas

► Fig. 46.4 shows an example of a parafalcine meningioma (PFM).

PFM are grouped based on their location along the AP direction of the SSS as:

- anterior (ethmoidal plate to coronal suture): 44% (in a series of 75 parasagittal meningiomas¹⁴). Most often present with H/A and mental status changes
- middle (between coronal and lambdoidal sutures): 33%. Most often present as Jacksonian seizure and progressive monoplegia. At the level of the motor strip, a common initial is contralateral foot drop (see ► Fig. 1.3 for location of “ankle” area on motor strip)¹⁵

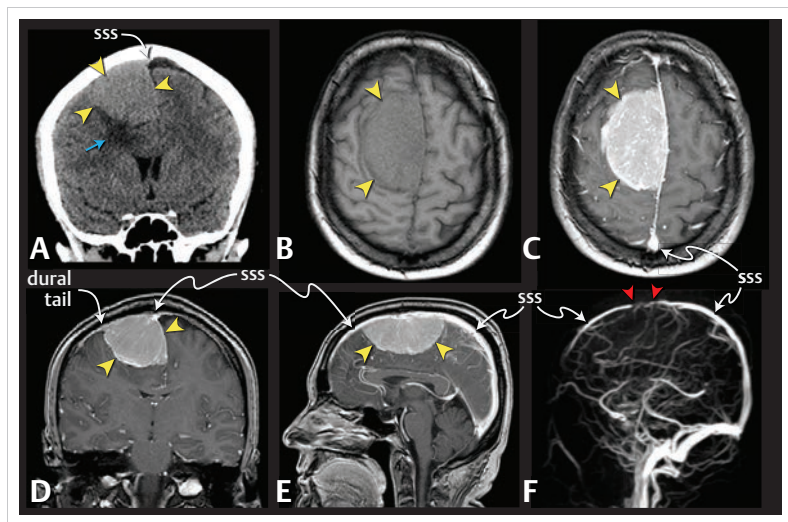


Fig. 46.4 Parafalcine meningioma (yellow arrowheads).

Sample images from a 49-year-old woman with a grade 2 (atypical) meningioma presenting with headache and left-sided weakness.

Image: A: coronal noncontrast brain window CT scan. Blue arrow denotes edema in the adjacent brain tissue.

B: axial T1 MRI without contrast, C: axial T1 MRI with contrast.

D: coronal T1 MRI with contrast. Note the dural tail which is characteristic of meningioma (but not pathognomonic, nor necessary to diagnose).

E: sagittal T1 MRI with contrast.

F: sagittal contrast MRV. Note the lack of opacification of the superior sagittal sinus between the two red arrowheads which is located within the borders of the tumor.

Abbreviations: sss = superior sagittal sinus.

- posterior (lambdoidal suture to torcular Herophili): 23%. Most often present with H/A, visual symptoms, focal seizures, or mental status changes

Up to 50% of PFM invade the superior sagittal sinus (SSS). Classification systems for the extent of SSS invasion includes the one by Sindou et al¹⁶ (► Fig. 46.5).

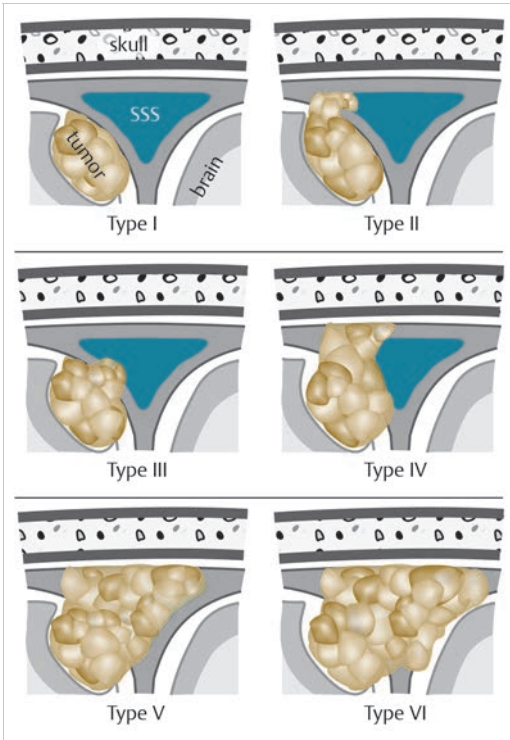


Fig. 46.5 Sindou grading system for meningioma invasion of the superior sagittal sinus.

(Modified from Sindou MP, et al. *J Neurosurg.* 2006;105:514–525.)

Image: schematic coronal sections through the superior sagittal sinus (SSS).

Type I = attachment to lateral wall of sinus.

Type II = invasion of lateral recess.

Type III = invasion of lateral wall.

Type IV = invasion of lateral wall and roof.

Type V = total sinus occlusion, contralateral wall spared.

Type VI = total sinus occlusion, invasion of all walls.

Planum sphenoidale meningiomas

These uncommon meningiomas arise from the flat horizontal portion of the sphenoid bone (► Fig. 46.6) anterior to the chiasmatic sulcus (the depression in the posterior part of the anterior cranial fossa where the optic chiasm perches). An example is shown in ► Fig. 46.7. With large meningiomas arising from the floor of the frontal fossa, it may not be possible to differentiate planum sphenoidale from olfactory groove meningiomas.

Tuberculum sellae meningiomas (TSM)

The site of origin of these tumors is only about 2 cm posterior to that of olfactory groove meningiomas.¹³ The tuberculum sellae is the bony elevation between the chiasmatic sulcus and the sella turcica (see ► Fig. 46.6). By definition, the limbus sphenoidale (which is the anterior margin of the chiasmatic sulcus) is the demarcation between the anterior and middle cranial fossa. Therefore these tumors originate in the middle fossa (unlike planum sphenoidale meningiomas which are in the anterior fossa).

TSMs are notorious for producing visual loss (chiasmal syndrome = primary optic atrophy + bitemporal hemianopsia; see ► Fig. 33.2). When a TSM grows posteriorly into the sella turcica it may be mistaken for a pituitary macroadenoma (see ► Fig. 99.3 for MRI and differentiating features).

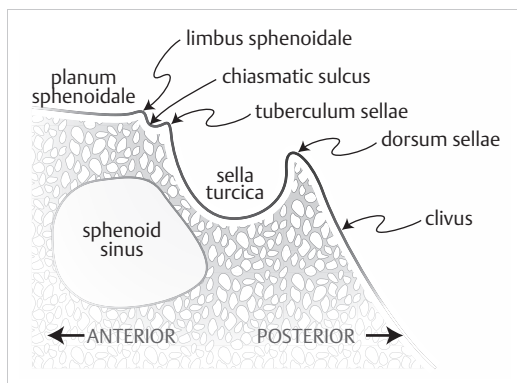


Fig. 46.6 Anatomic locations of planum sphenoidale and tuberculum sellae. The olfactory groove (not shown) is located in the floor of the frontal fossa immediately anterior to the planum sphenoidale. Image: schematic sagittal section through the sella turcica.

Intraventricular meningiomas

Most are located in the atrium of the lateral ventricle (► Fig. 46.8). See also differential diagnosis for intraventricular lesions (p. 1667).

Foramen magnum meningiomas

As with any foramen magnum (FM) lesion (p. 1649); the neurologic symptoms and signs can be very confusing and often do not initially suggest a tumor in this location.

In the French Cooperative Study of 106 FM meningiomas,¹⁷ 31% arose from the anterior lip, 56% were lateral, and 13% arose from the posterior lip of the FM. Most are intradural, but they can be extradural or a combination (the latter 2 have a lateral origin and are often invasive, which makes total removal more difficult).¹⁸ They may be above, below, or on both sides of the vertebral artery.¹⁸

Epidemiology

► Risk factors

- ionizing radiation (typically in doses used for radiation therapy) is the only established environmental risk factor;¹⁹ with higher risk in patients exposed as children (e.g., in treating leukemia). There appears to be genetic susceptibility to developing XRT induced meningiomas²⁰
- the higher incidence in females and the increased risk in postmenopausal women receiving hormonal replacement therapy (HRT)²¹ suggests a hormonal influence
- neurofibromatosis type 1 (NF1) (p. 638) or neurofibromatosis type 2 (NF2) (p. 640)

► **Genetics.** 50-60% of WHO grade 1 meningiomas have mutations of the NF2 tumor suppressor gene on chromosome 22q12.2 (which codes for merlin (p.640)).²²

Genetic mutations tend to be location dependent: convexity meningiomas and most spinal meningiomas harbor a 22q deletion and/or NF2 mutation, skull base lesions carry *AKT1*, *TRAF7*, *SMO* and/or *PIK3CA* mutations.

► **General epidemiology.** As many as 3% of autopsies on patients > 60 yrs of age reveal a meningioma.²³ Meningiomas account for 39% of primary intracranial neoplasms and 54.5% of non-malignant tumors in the U.S.² Incidence peaks at 45 years of age. The median age at diagnosis is 65 years, and the risk increases with age (from an incidence of 1.49/100,000 for ages 20-34 years, to 57.29 for age > 85).² Female:male ratio is 2.3:1 in the U.S.²

Meningiomas comprise 1.7% of primary brain & CNS tumors in patients age 0-14 years, and 4.6% for ages 15-19 years.² 19-24% of adolescent meningiomas occur in patients with neurofibromatosis type 1 (von Recklinghausen's).

Evaluation

See "Imaging" below for specific findings.

Brain MRI, without and with contrast, is the diagnostic study of choice.

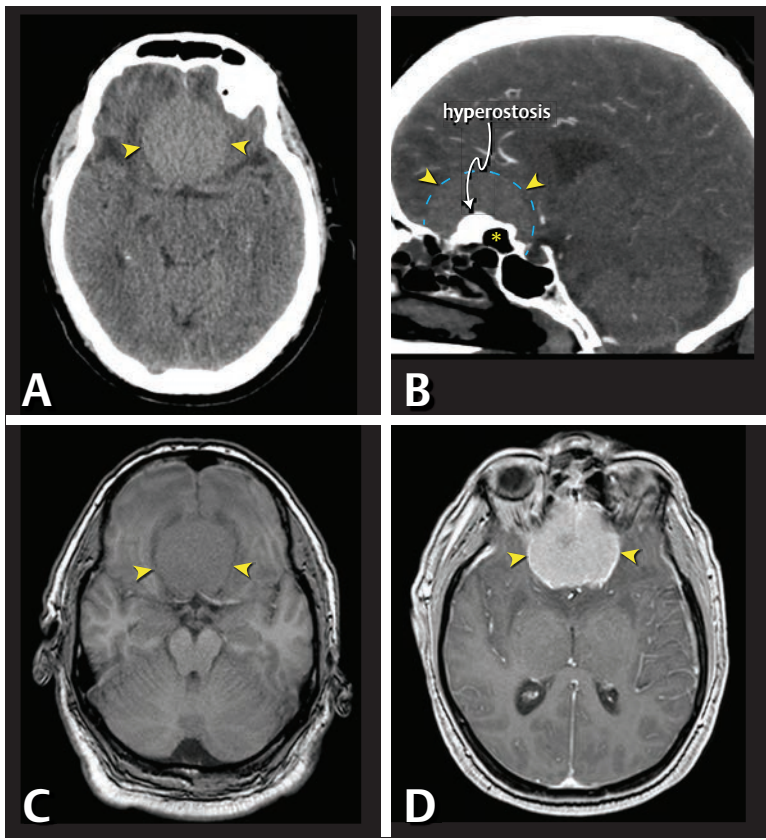


Fig. 46.7 Planum sphenoidal meningioma (yellow arrowheads).

Image: A: axial noncontrast CT scan. The tumor is visible although close to brain in density.

B: sagittal CTA. The tumor is barely visible, and has been outlined by a broken blue line. Note the hyperostosis of bone at the base of the tumor which is common with meningiomas. There is also enlargement of the underlying air sinus (yellow asterisk *), a condition that is termed pneumosinus dilans.

C: axial noncontrast T1 MRI, and D: axial contrast T1 MRI.

CT without and with contrast scan is second choice when MRI cannot be done, and may also be considered when the identification of calcifications or reactive bone changes is considered crucial, or sometimes for surgical planning.

Angiography may be considered when more definitive assessment of dural sinus occlusion is desired, or to determine if pre-op embolization is a viable intervention.

In questionable cases, a metastatic workup (p.914) might be considered to rule out that possibility.

Imaging

CT

(See below for characteristic findings shared between CT and MRI). Without contrast, meningiomas are typically homogeneous, isodense or slightly hyperdense to brain on CT (► Fig. 46.4 panel A).

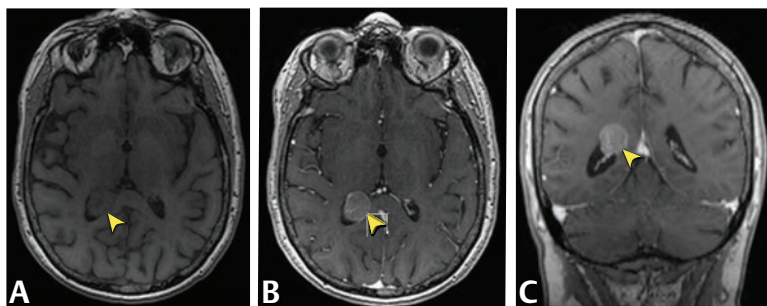


Fig. 46.8 Intraventricular meningioma (yellow arrowheads) in the atrium of the right lateral ventricle. Image: MRI, A: axial T1 without contrast, B: axial T1 with contrast, C: coronal T1 with contrast. The patient presented with entrapped occipital and temporal horns of the right lateral ventricle.

20–30% have areas of calcification (better appreciated on CT than MRI). Non-contrast Hounsfield numbers of 60–70 in a meningioma usually correlate with presence of psammomatous calcifications (p.803). On CT, prostate cancer may mimic meningioma (prostate mets to brain are rare, but prostate frequently goes to bone, and may go to skull and can cause hyperostosis).

MRI

(See below for characteristic findings shared between CT and MRI). Meningiomas may be isointense to brain on T1 or T2 images (making them difficult to visualize) but most enhance with gadolinium. Calcifications appear as signal voids on MRI. MRI gives information regarding patency of dural venous sinuses (accuracy in predicting sinus involvement is $\approx 90\%^{24}$).

Characteristic findings on CT or MRI

1. **broad base of attachment** to the meninges
2. **dural tail**: enhancing extension along the dura where the tumor attaches to the meninges ($\Rightarrow$ Fig. 46.4 panel D) (this is not pathognomonic and can also occur with a lower incidence in many conditions such as pleomorphic xanthoastrocytoma (PXA), Rosai-Dorfman disease, plasmacytoma...). The dural tail may be comprised of tumor itself, reactive fibrovascular tissue, and to a limited extent and in only $\approx$ one third of cases, tumor infiltration of the dura²⁵
3. **hyperostosis** (or sometimes blistering) of adjacent bone is very suggestive of meningioma ($\Rightarrow$ Fig. 46.7 panel B). Usually better appreciated on CT than MRI
4. **edema** of the adjacent brain may be minimal or may be marked especially in certain meningioma subtypes (secretory, angiomatous/microcystic, lymphoplasmacyte-rich, and high-grade meningiomas).²⁶ Significant edema may be considered to be a criteria for surgical excision (p.813). Intraventricular meningiomas: 50% produce extraventricular edema
5. **enhancement**: meningiomas usually enhance densely and uniformly, although there may be cystic areas and/or areas of calcification that do not enhance

Angiography

Classic pattern: appears early in arterial phase, blush persists beyond venous phase (“comes early, stays late”). The distinctive prolonged homogeneous tumor blush can also help confirm the diagnosis.

Meningiomas characteristically have *external* carotid artery feeders. Some notable exceptions that feed from the ICA:

1. low frontal median meningiomas (e.g., olfactory groove): feed from ethmoidal branches of the ophthalmic artery
2. suprasellar meningiomas: may also be fed by large branches of the ophthalmic arteries
3. parasellar meningiomas: tend to feed from the ICA. Secondary vascular supply may be derived from pial branches of the anterior, middle, and posterior cerebral arteries
4. petroclival meningiomas: supplied via the artery of the cerebrium (a branch of the meningohypophyseal trunk) AKA the **artery of Bernasconi & Cassinari** (the so-called “Italian” artery) which is enlarged in lesions involving tentorium (e.g., tentorial meningiomas)

Angiography also gives information about occlusion of dural venous sinuses, especially for parasagittal/falx meningiomas (may be more accurate than MRI for this). Oblique views are often best for evaluating patency of the superior sagittal sinus (SSS). Angiography also provides an opportunity for pre-op embolization (p.815) (see below).

Plain X-rays

May show: calcifications within the tumor (in $\approx 10\%$), hyperostosis or blistering of the skull (including floor of frontal fossa with olfactory groove meningiomas), enlargement of vascular grooves (especially middle meningeal artery). Most of these findings are better appreciated on bone window CT scan.

Differential diagnosis/diagnostic considerations of meningioma

1. pleomorphic xanthoastrocytoma (PXA) (p.698): may mimic meningiomas since they tend to be peripherally located and may have a dural tail
2. solitary fibrous tumor (► Fig. 47.2): (formerly hemangiopericytoma) a rapidly growing tumor
3. gliosarcomas (p.665), especially ones that are predominantly sarcomatous
4. Rosai-Dorfman disease: especially if extracranial lesions are also identified. A connective tissue disorder with sinus histiocytosis and massive painless lymphadenopathy (most have cervical lymphadenopathy). Usually in young adults. Isolated intracranial involvement is rare. MRI: dural-based enhancing mass with signal characteristics similar to meningioma, may have dural tail. Most common intracranial locations: cerebral convexities, parasagittal, suprasellar, cavernous sinus. Pathology: dense fibrocollagenous connective tissue with spindle cells and lymphocytic infiltration, stains for CD68 & S-100. Histiocytic proliferation without malignancy. Foamy histiocytes are characteristic. Surgery and immunosuppressive therapy not effective. Low-dose XRT may be the best option
5. NB: multiple meningiomas suggests the possibility of neurofibromatosis 2 (p.640) (NF2)

Treatment in general

Indications for treatment

Treatment is indicated for meningiomas with any of the following:

1. symptoms that cannot be satisfactorily controlled medically (e.g., refractory seizures)
2. significant growth on serial imaging studies (see ► Table 46.3 for what may constitute "significant growth")
3. more than minimal edema on imaging

"Significant growth": the definition is controversial. Proposed criteria includes any of those shown in ► Table 46.3

Table 46.3 Various proposed criteria for "significant growth" of meningiomas

- 2-5 mm increase in diameter over 1 year^{27*}
- 1 cc increase in volume in 1 year²⁸
- or 15% increase in volume over 1 year²⁹

* the concern with using a linear measurement such as diameter is that the volume of a sphere is proportional to the cube of the radius, so for a small tumor a small change in diameter results in a small increase in volume, but for a large tumor the increase in volume with even a small increase in diameter can be substantial

► **Asymptomatic meningiomas.** Data is lacking to make evidence-based management guidelines. Incidental meningiomas with no brain edema or those presenting only with seizures that are easily controlled medically may be managed expectantly with serial imaging as meningiomas tend to grow slowly, and some may "burn out" and cease growing.

A suggestion is to obtain a follow-up imaging study 3–4 months after the initial study to rule out rapid progression (often indicating a higher grade meningioma or other lesion), and then if growth is not significant (see ► Table 46.3), repeat annually for at least 2–3 years. If stable, follow-up studies could be spaced out further. Treatment would be recommended for significant growth. The development of symptoms should prompt performing a study at the time of the symptoms.

"High-risk meningiomas" (atypical meningiomas, rhabdoid meningiomas, malignant meningiomas, and any meningioma with a high proliferation index) tend to have higher growth rates.

Treatment options

Surgery is the treatment of choice for accessible meningiomas meeting indications for treatment (see above) in patients who are surgical candidates. The perioperative morbidity rate is statistically significantly higher in patients > 70 years old (23%) than in those < 70 (3.5%).¹⁰ See below for details.

Radiation therapy is used most often as adjuvant therapy after surgery for some cases of tumor recurrence or incomplete resection (see post-op radiation (p.814)). Use as a primary therapy might occasionally be considered for patients who are not surgical candidates or for some deep inaccessible tumors.

MRI-guided laser interstitial thermal therapy (LITT): early experience with recurrent meningiomas shows potential, demonstrating early (first few weeks) increase in tumor volume with a subsequent decrease at 3 months and subsequent re-growth in some.³⁰ requiring further study. This may be a consideration for patients needing treatment when surgery is not a consideration and when maximal XRT has already been given.

Focused ultrasound (FUS) may become a non-invasive adjuvant therapy option for some deep situated tumors. Superficially located meningiomas (especially convexity meningiomas) are generally not candidates for transcranial FUS due to the inability to focus the ultrasound beams near the convexity of the skull.

46

Radiation therapy (XRT)

► **As primary treatment.** Generally regarded as ineffective as primary modality of treatment. Also, due to long-term risks of XRT, it is usually avoided for “benign” lesions (WHO grade 1) if there are other effective options. Primary XRT may be considered for symptomatic or progressive tumors if the risks of surgical resection (including risks of patients who are poor surgical candidate) outweigh the benefits.

► **Post-op XRT.** Efficacy of XRT in preventing recurrence is controversial—see box for suggestions.

A retrospective series of 135 non-malignant meningiomas followed 5–15 years post-op at UCSF revealed a recurrence rate of 4% with total resection, 60% for partial resection without XRT, and 32% for partial resection with XRT.³¹ Mean time to recurrence was longer in the XRT group (125 mos) than in the non-XRT group (66 mos). These results suggest that XRT may be beneficial in partially resected meningiomas. Alternatively, one can follow these patients with CT or MRI and use XRT for documented progression if repeat surgery is not desired.

Atypical meningiomas (WHO grade 2): there are no large prospective studies. Retrospective and/or small studies suggest a nonsignificant trend towards longer PFS with post op XRT, but no difference in OS.

Σ: Post-op XRT suggestions (these are not guidelines)

WHO grade 1 meningiomas: post-op XRT is not recommended even with subtotal resection unless there is progressive growth on follow-up imaging and repeat surgery is not viable (e.g., risky tumor location, poor medical condition...) or desired by patient and surgeon.

Post-op XRT is suggested for **high-risk meningiomas** defined as³²:

1. anaplastic (WHO grade 3) regardless of extent of resection
2. new atypical meningiomas (WHO grade 2) with *subtotal* resection (Simpson grade ► Table 46.4 IV - V)
3. recurrent atypical meningiomas regardless of extent of resection

New atypical meningiomas (WHO grade 2) with gross total resection (Simpson grade ► Table 46.4 I - III): patient may be followed and then treated with XRT for recurrence.

Dosing for anaplastic or atypical meningioma: fractionated therapy to a dose of 55–60 Gy.

In addition to the usual side effects of XRT—see Radiation injury and necrosis (p. 1899)—there is a risk of radiation-induced tumors such as astrocytomas.³³

Surgical treatment for meningiomas

General information

Surgical goals: complete resection when possible, with removal of involved meninges and bone when feasible. The Simpson grading system for extent of meningioma resection is shown in ► Table 46.4.

Table 46.4 Simpson grading system for removal of meningiomas⁵⁰

Grade	Degree of removal	Recurrence rate (%)
I	macroscopically complete removal with excision of dural attachment and abnormal bone (including sinus resection when involved)	9
II	macroscopically complete with endothermy coagulation (electrocautery or laser) of dural attachment	19
III	macroscopically complete without resection or coagulation of dural attachment or of its extradural extensions (e.g., hyperostotic bone)	29
IV	partial removal leaving tumor in situ	44
V	simple decompression ($\pm$ biopsy)	N/A

Meningiomas are often very bloody. Preoperative embolization may be helpful for specific tumors (see below). Pre-op autologous blood donation may also be a consideration.

General principles of meningioma surgery³⁴:

1. early interruption of the arterial blood supply to the tumor: unfortunately, the blood supply is often on the deep aspect of the tumor and may be difficult to access early in the resection
2. to avoid excessive retraction on the brain, a common technique consists of
 - a) internal decompression (debulking) of the tumor using ultrasonic aspirator, cautery loops...
 - b) followed by collapsing the tumor capsule into the area of decompression and dissecting the capsule from the brain by cutting and coagulating vascular and arachnoid attachments
3. removal of attached bone and dura when possible

Pre-op tumor embolization: Reduces the vascularity of these often bloody tumors, facilitating surgical removal. Timing of subsequent surgery is controversial. Some advocate waiting 7–10 days to permit tumor necrosis, which simplifies resection.^{35,36} Complications include: hemorrhage (intratumoral and SAH), cranial nerve deficits (usually transient), stroke from embolization through ICA or VA anastomoses, scalp necrosis, retinal embolus, and potentially dangerous tumor swelling. Many meningiomas are not readily amenable to embolization (e.g., olfactory groove meningioma).

Position

As usual, the head should be elevated $\approx 30^\circ$ above the right atrium.

For meningiomas involving the superior sagittal sinus (SSS)³⁷:

- for tumors involving the anterior third of the SSS: supine semi-sitting position
- for tumors of the middle third of the SSS: lateral position with the side of the tumor *down*, the neck tilted 45° toward the upward shoulder
- for tumors of the posterior third of the SSS: prone position

Sinus involvement

Greenberg IMHO

Due to unappreciated venous drainage (see text), attempting to occlude or bypass the middle third of the superior sagittal sinus involved with meningioma is treacherous. Even in expert hands, there is significant risk of venous infarction/sinus occlusion with 8% morbidity and 3% mortality,¹⁶ and complete removal is still not assured.³⁸ Rather than risking venous infarctions, it is almost always preferable to leave residual tumor and follow with serial imaging (especially for WHO grade 1 meningiomas, and with some grade 2 lesions (p. 814)) and consider treating with repeat surgery or radiation therapy as appropriate.

Alternatives for treatment of dural sinus involvement include:

► **Superior sagittal sinus (SSS).** If the tumor occludes the SSS, it has been suggested that the sinus can be resected carefully, preserving veins draining into the patent portions of the sinus. ✖ However, this should be undertaken under advisement since patients still not infrequently develop venous infarcts, probably as a result of loss of venous drainage of even minimal sinus flow, venous channels in the dura adjacent to the sinus, through the skin or bone of the skull, and even the tumor itself may participate. Before ligating the sinus, the lumen should be inspected for a tail of tumor within the sinus.

Partial occlusion of superior sagittal sinus:

1. anterior to the coronal suture, the sinus may usually be divided safely
2. posterior to the coronal suture (or, perhaps more accurately, posterior to the vein of Trolard), it must not be divided or else severe venous infarction will occur
 - a) with superficial involvement (Type I, ▶ Fig. 46.5), tumor may be dissected off the sinus with care to preserve patency
 - b) with extensive involvement:
 - sinus reconstruction: hazardous. Thrombosis rate using venous graft approaches 50%, and is close to 100% with artificial grafts (e.g., Gore-Tex) which should *not* be used
 - it may be best to leave residual tumor, and follow with CT or MRI. If the residual tumor grows, or if the Ki-67 score is high (p.805), SRS may be used; SRS (p.1903) may also be used as initial treatment for tumors that are <2.3–3 cm diameter

▶ **Transverse sinus (TS).** A patent dominant TS must not be suddenly occluded.

46

Convexity meningiomas

NB: the tumor capsule of convexity meningiomas that straddle the Sylvian fissure may be adherent to branched of the middle meningeal artery and must be carefully dissected off of the tumor to avoid damage to them.

The craniotomy should be large enough (ideally 1–2 cm beyond the tumor attachment) to permit excising the invaded dura adjacent to the point of tumor attachment. Dura directly involved with the tumor must be resected, and an autograft (e.g., fascia lata or periosetum from an uninvolved location) or allograft will be required.

▶ **Dural tail.** The dural tail is infrequently comprised of dura invaded by tumor (p.812), and it is not always necessary to remove it all. Current practice is generally to excise as much of the dural tail as is reasonably possible without chasing it for long distances.

▶ **Bone involvement.** Hyperostosis of bone adjacent to a meningioma: occurs in 5–44%.³⁹ Various pathologic mechanisms have been invoked,⁴⁰ and only in some cases is frank tumor invasion present. MRI often shows increased T2 signal extending beyond the limits of hyperostosis. The hyperostotic bone can usually be drilled off the surface of the bone flap which may then be replaced.

Bone invasion: if extensive, the bone flap may have to be discarded and replaced with a cranioplasty (at the time of surgery, e.g., using titanium mesh, or at a later date). Alternatively, the invaded bone may be excised from the center of the flap and the resultant defect is often significantly smaller than the flap and is more easily patched, and the patched flap can then be reimplanted.

A study in which bone biopsies were taken 2 cm outside the hyperostotic margin were negative for tumor in all 12 cases of grade 1 & 2 meningiomas (even where there was increased signal on T2 MRI), whereas in the 2 cases of grade 3 meningioma all biopsies were positive.⁴¹

Sphenoid wing, parasagittal, or falx meningiomas (general principles)

Once tumor is exposed a partial internal debulking is performed. Then the point of attachment (to the falx or sphenoid bone) is peeled away using bipolar cautery to divide feeding vessels. Then the main portion of the tumor may be separated from the brain, with the tumor being avascular once the vascular pedicle has been transected.

Parasagittal and falx meningiomas

The inferior portion of the tumor may adhere to branches of the anterior cerebral artery. Middle or posterior third tumors are exposed using a horseshoe incision based in the direction of the major scalp feeding vessels. Alternatively, the patient may be placed in a lateral position, or the sitting position may be used with Doppler monitoring for air embolism (p.1738). Anterior third tumors are approached using a bicoronal skin incision with the patient supine. For tumors that cross the midline, burr holes are placed to straddle the SSS. For managing superior sagittal sinus involvement, see above.

Since these tumors are often debulked from the inside, removal tends to be bloodier than meningiomas that can be removed in 1 piece. The ability to embolize these tumors pre-op is somewhat limited, but may be an adjunct. For tumors involving the convexity dura, cut through tumor leaving a thin layer on dura. Then remove the now relatively avascular part that impresses the brain. Then make an incision through the dura near the tumor; it tends to be bloody, but once you have control of both sides of the dura you can begin to excise the dura around the tumor (you may need to leave a cuff on SSS if it is involved).

Sphenoid wing meningiomas

A pterional craniotomy (p. 1748) is utilized. The neck is extended to allow gravity to retract the brain off of the floor of the skull.

Lateral sphenoid wing meningiomas: The approach to these tumors is often similar to convexity meningiomas. The height of the skin incision and bone opening should be high enough to encompass the tumor.

Medial sphenoid wing meningiomas: A lumbar drain is used. The head is turned 30° off the vertical. Aggressive extradural removal of sphenoid wing is performed. An FTOZ approach may provide additional exposure. The Sylvian fissure is split widely. The ICA and MCA are often encased by tumor (look for the appearance of "grooves" on the surface of the tumor on MRI, which indicates vessels, e.g., MCA). To locate the ICA, identify MCA branches and follow them proximally into the tumor. The optic nerve is best identified at the optic canal. Avoid excessive retraction of the optic apparatus. The deep portion of the tumor often has numerous small parasitic vessels from the ICA (which makes this part very bloody), and may also invade the lateral wall of the cavernous sinus (which creates risk of cranial nerve deficits with attempted removal). Therefore, the recommendation is to leave some tumor behind and use radiosurgery to deal with it.

Olfactory groove meningiomas

Approached via a bifrontal craniotomy (preserving the periosteum to cover the frontal air sinus and floor of frontal fossa at the end of the case). Small tumors may be approached via unilateral craniotomy on the side with the most tumor.⁴² (p. 3284) For large tumors, a lumbar CSF drain will help with brain relaxation¹³ and the head is rotated 20° to one side to facilitate dissection of the anterior cerebral arteries and optic nerve while preserving visualization of both sides of the tumor involvement.⁴³ The neck is slightly extended. The dura is opened low, and the superior sagittal sinus is ligated and divided at this location. Amputation of the frontal pole should be done if necessary to avoid excessive retraction. Vascular feeding arteries come through the floor of the frontal fossa in the midline. Initially, the anterior tumor capsule is opened and the tumor debulked from within, heading towards the floor of the frontal fossa to interrupt the blood supply. The posterior capsule of the tumor is dissected carefully as this portion of the tumor may encase branches of the anterior cerebral artery, and/or optic nerves and chiasm. A large tumor with suprasellar extension usually displaces the optic nerve and chiasm *inferiorly*.¹³ If necessary, the frontopolar branch and other small branches may be sacrificed without problem.⁴⁴ Periosteum is laid over the floor of the frontal fossa. To hold it in place, one may attempt to suture it to the adjacent dura with a couple of retaining sutures, alternatively a small titanium plate (e.g., "dogbone") can be placed over the flap and screws are placed into the bone of the floor of the frontal fossa (both methods are challenging). Post-op risks include CSF leak through the ethmoid sinuses.

Tuberculum sellae meningiomas

These tumors typically displace both optic nerves posteriorly and laterally.¹³ Occasionally, the nerves are completely engulfed by tumor.

Cerebellopontine angle meningiomas

Usually arise from the meninges covering the petrous bone. May be divided into those that occur anterior to, and those that occur posterior to the IAC.

Foramen magnum meningiomas

Tumors arising from the posterior or posterolateral lip of the foramen magnum (FM) are removed relatively easily. Anterior and lateral FM tumors may be operated by the posterolateral approach, and for anterior tumors,¹⁸ a transcondylar approach may alternatively be used.⁴⁵

With meningiomas below the vertebral artery (VA), the lower cranial nerves are displaced superiorly with the VA. However, when the tumor is above the VA, the position of the lower cranial nerves cannot be predicted.¹⁸

Large tumors may adhere to or encase neurovascular structures, and these should be internally debulked and then dissected free.

Posterior suboccipital approach: Used for meningiomas arising from the posterior lip of the FM or slightly posterolateral.

The patient is positioned prone or three-quarter prone. Neck flexion should be kept to a minimum to avoid brainstem compression by the tumor.⁴⁶ The surgeon must remain vigilant for the PICA and vertebral arteries, which may be encased.

Recurrence after surgery

The most powerful *histopathologic* predictor of recurrence is WHO grade, with grade 1 recurrence rate of 7–25%, grade 2 recur in 29–52%, and grade 3 in 50–94%. Some histologic subtypes are also more prone to recurrence (as noted under subtypes).

The extent of surgical tumor removal is the most important factor in the prevention of post-op recurrence. The Simpson grading system for the extent of meningioma removal is shown in ► Table 46.4. An often overlooked aspect of the Simpson grading system is that it refers exclusively to the removal of *intradural* tumor, and thus leaving the tumor e.g., in the sagittal sinus could still be considered “complete removal.” Recurrence after gross total tumor removal occurred in 11–15% of cases, but was 29% when removal was incomplete (length of follow-up not specified)¹¹; 5-year recurrence rates of 37%⁴⁷–85%⁴⁸ after partial resection are also quoted. The overall recurrence rate at 20 years was 19% in one series,⁴⁹ and 50% in another.⁴⁸ Malignant meningiomas have a higher recurrence rate than benign ones.

Prognosis

5-year survival for patients with meningioma⁵¹: 91.3%. With anaplastic meningiomas, median survival ranges from <2 years to >5 years depending on completeness of removal and use of adjuvant XRT.⁶

References

- [1] Achey RL, Gittleman H, Schroer J, et al. Non-malignant and malignant meningioma incidence and survival in the elderly from 2005–2015 using the Central Brain Tumor Registry of the United States. *Neuro Oncol.* 2018. DOI: 10.1093/neuonc/noy162
- [2] Ostrom QT, Cioffi G, Waite K, et al. CBTUS Statistical Report: Primary Brain and Other Central Nervous System Tumors Diagnosed in the United States in 2014–2018. *Neuro Oncol.* 2021; 23:iii1–iii105
- [3] Sheehy JP, Crookard HA. Multiple Meningiomas: A Long-Term Review. *J Neurosurg.* 1983; 59:1–5
- [4] Kulali A, Ilcayto R, Rahmani O. Primary calvarial ectopic meningiomas. *Neurochirurgia (Stuttg).* 1991; 34:174–177
- [5] Das DK. Psammoma body: a product of dystrophic calcification or of a biologically active process that aims at limiting the growth and spread of tumor? *Diagn Cytopathol.* 2009; 37:534–541
- [6] Bosman FT, Jaffe ES, Lakhani SR, et al. World Health Organization Classification of Tumors series, vol. 6. In: WHO Classification of Tumors Editorial Board (ed.) World Health Organization classification of tumors of the central nervous system. 5th ed. Lyon, France: International Agency for Research on Cancer; 2021
- [7] Vaubel RA, Chen SG, Raleigh DR, et al. Meningiomas With Rhabdoid Features Lacking Other Histologic Features of Malignancy: A Study of 44 Cases and Review of the Literature. *J Neuropathol Exp Neurol.* 2016; 75:44–52
- [8] Kolles H, Niedermayer I, Schmitt C, et al. Triple approach for diagnosis and grading of meningiomas: histology, morphometry of Ki-67/Feulgen stainings, and cytogenetics. *Acta Neurochir (Wien).* 1995; 137:174–181
- [9] Vernooij MW, Ikram A, Tanghe HL, et al. Incidental findings on brain MRI in the general population. *New England Journal of Medicine.* 2007; 357:1821–1828
- [10] Kuratsu J-I, Kochi M, Ushio Y. Incidence and Clinical Features of Asymptomatic Meningiomas. *Journal of Neurosurgery.* 2000; 92:766–770
- [11] Yamashita J, Handa H, Iwaki K, et al. Recurrence of Intracranial Meningiomas, with Special Reference to Radiotherapy. *Surg Neurol.* 1980; 14:33–40
- [12] Cushing H, Eisenhardt L. Meningiomas of the Sphenoidal Ridge. A. Those of the Deep or Clinoidal Third. In: *Meningiomas: Their Classification, Regional Behaviour, Life History, and Surgical End Results.* Springfield, Illinois: Charles C Thomas; 1938:298–319
- [13] Al-Mefty O, Sekhar LN, Janecka IP. Tuberculum Sella and Olfactory Groove Meningiomas. In: *Surgery of Cranial Base Tumors.* New York: Raven Press; 1993:507–519
- [14] Ricci A, Di Vitanantonio H, De Paulis D, et al. Parasagittal meningiomas: Our surgical experience and the reconstruction technique of the superior sagittal sinus. *Surg Neurol Int.* 2017; 8. DOI: 10.4103/2152-7806.198728
- [15] Eskandary H, Hamzel A, Yasamy MT. Foot Drop Following Brain Lesion. *Surgical Neurology.* 1995; 43:89–90
- [16] Sindou MP, Alvernia JE. Results of attempted radical tumor removal and venous repair in 100 consecutive meningiomas involving the major dural sinuses. *J Neurosurg.* 2006; 105:514–525
- [17] George B, Lot G, Velut S. Tumors of the Foramen Magnum. *Neuro-Chirurgie.* 1993; 39:1–89
- [18] George B, Lot G, Boissonnet H. Meningioma of the Foramen Magnum: A Series of 40 Cases. *Surgical Neurology.* 1997; 47:371–379
- [19] Wiemels J, Wrensch M, Claus EB. Epidemiology and etiology of meningioma. *J Neurooncol.* 2010; 99:307–314
- [20] Flint-Richter P, Sadetzki S. Genetic predisposition for the development of radiation-associated meningioma: an epidemiological study. *Lancet Oncol.* 2007; 8:403–410
- [21] Andersen I, Friis S, Hallas J, et al. Hormone replacement therapy increases the risk of cranial meningioma. *Eur J Cancer.* 2013; 49:3303–3310
- [22] Miller RJ Jr, DeCandia ML, Dixon-Mah Y, et al. Molecular targets and treatment of meningioma. *J Neurol Neurosurg.* 2014; 1
- [23] Nakasu S, Hirano A, Shimura T, et al. Incidental Meningiomas in Autopsy Studies. *Surgical Neurology.* 1987; 27:319–322
- [24] Zimmerman RD, Fleming CA, Saint-Louis LA, et al. Magnetic Resonance of Meningiomas. *American Journal of Neuroradiology.* 1985; 6:149–157
- [25] Sotoudeh H, Yazdi HR. A review on dural tail sign. *World J Radiol.* 2010; 2:188–192
- [26] Osawa T, Tosaka M, Nagaiishi M, et al. Factors affecting peritumoral brain edema in meningioma: special histological subtypes with prominently extensive edema. *J Neurooncol.* 2013; 111:49–57
- [27] Yano S, Kuratsu JI, Demonte F, et al. Natural course of untreated meningiomas. In: *Al-Mefty's Meningiomas.* 2nd ed. New York: Thieme Medical Publishers, Inc.; 2011:63–67
- [28] Yoneoka Y, Fujii Y, Tanaka R. Growth of incidental meningiomas. *Acta Neurochir (Wien).* 2000; 142:507–511
- [29] Hashiba T, Hashimoto N, Izumoto S, et al. Serial volumetric assessment of the natural history and growth pattern of incidentally discovered meningiomas. *J Neurosurg.* 2009; 110:675–684

- [30] Rammo R, Scarpace L, Nagaraja T, et al. MR-guided laser interstitial thermal therapy in the treatment of recurrent intracranial meningiomas. *Lasers Surg Med.* 2019; 51:245–250
- [31] Barbaro NM, Gutin PH, Wilson CB, et al. Radiation Therapy in the Treatment of Partially Resected Meningiomas. *Neurosurgery.* 1987; 20:525–528
- [32] Rogers CL, Won M, Vogelbaum MA, et al. High-risk meningioma: Initial outcomes from NRG Oncology/RTOG 0539. *Int J Radiat Oncol Biol Phys.* 2020; 106:790–799
- [33] Zuccarello M, Sawaya R, deCourten-Myers. Glioblastoma Occurring After Radiation Therapy for Meningioma: Case Report and Review of Literature. *Neurosurgery.* 1986; 19:114–119
- [34] Ojemann RG. Management of Cranial and Spinal Meningiomas. *Clinical Neurosurgery.* 1992; 40:321–383
- [35] Chun JY, McDermott MW, Lamborn KR, et al. Delayed surgical resection reduces intraoperative blood loss for embolized meningiomas. *Neurosurgery.* 2002; 50:1231–5; discussion 1235–7
- [36] Kai Y, Hamada J, Morioka M, et al. Appropriate interval between embolization and surgery in patients with meningioma. *AJNR Am J Neuroradiol.* 2002; 23:139–142
- [37] Colli BO, Carlotti CG. Parasagittal meningiomas. *Contemporary Neurosurgery.* 2007; 29:1–8
- [38] Heros RC. Meningiomas involving the sinus. *J Neurosurg.* 2006; 105:511–513
- [39] Dziuk TW, Woo S, Butler EB, et al. Malignant meningioma: an indication for initial aggressive surgery and adjuvant radiotherapy. *J Neurooncol.* 1998; 37:177–188
- [40] Di Cristofori A, Del Bene M, Locatelli M, et al. Meningioma and Bone Hyperostosis: Expression of Bone Stimulating Factors and Review of the Literature. *World Neurosurg.* 2018; 115:e774–e781
- [41] Fathalla H, Tawab MGA, El-Fiki A. Extent of Hyperostotic Bone Resection in Convexity Meningioma to Achieve Pathologically Free Margins. *J Korean Neurosurg Soc.* 2020; 63:821–826
- [42] Youmans JR. *Neurological Surgery.* Philadelphia 1990
- [43] Bogaev CA, Sekhar LN, Sekhar LN, et al. Olfactory groove and planum sphenoidale meningiomas. In: *Atlas of Neurosurgical Techniques.* New York: Thieme Medical Publishers, Inc.; 2006:608–617
- [44] Ojemann RG, Schmidek HH, Sweet WH. Surgical Management of Olfactory Groove Meningiomas. In: *Operative Neurosurgical Techniques.* 3rd ed. Philadelphia: W.B. Saunders; 1995:393–401
- [45] Hakuba A, Tsujimoto T, Sekhar LN, et al. Transcondyle Approach for Foramen Magnum Meningiomas. In: *Surgery of Cranial Base Tumors.* New York: Raven Press; 1993:671–678
- [46] David CA, Spetzler R. Foramen Magnum Meningiomas. *Clinical Neurosurgery.* 1997; 44:467–489
- [47] Mirmanoff RO, Dosoretz DE, Lingood RM, et al. Meningioma: Analysis of Recurrence and Progression Following Neurosurgical Resection. *J Neurosurg.* 1985; 62:18–24
- [48] Adegbite AV, Khan MI, Paine KWE, et al. The Recurrence of Intracranial Meningiomas After Surgical Treatment. *J Neurosurg.* 1983; 58:51–56
- [49] Jaaskelainen J. Seemingly complete removal of histologically benign intracranial meningioma: late recurrence rate and factors predicting recurrence in 657 patients. A multivariate analysis. *Surgical Neurology.* 1986; 26:461–469
- [50] Simpson D. The recurrence of intracranial meningiomas after surgical treatment. *J Neurol Neurosurg Psychiatry.* 1957; 20:22–39
- [51] Mahaley MS, Mettlin C, Natarajan N, et al. National Survey of Patterns of Care for Brain-Tumor Patients. *J Neurosurg.* 1989; 71:826–836

47 Mesenchymal, Non-meningothelial Tumors

47.1 General information

Benign and malignant mesenchymal tumors originating in the CNS are very rare and are organized using terminology and histologic features corresponding to their soft tissue and bone counterparts as outlined in the 2020 WHO Classification of Soft Tissue & Bone Tumors, 5th edition.¹

Primary mesenchymal tumors relating to the brain and spine tend to arise in the meninges rather than in the parenchyma, and occur more often in the supratentorial compartment than in the p-fossa or spine.

47.2 Fibroblastic and myofibroblastic tumors

47.2.1 Solitary fibrous tumor (WHO grade 1, 2 or 3)

General information

A mesenchymal tumor of the fibroblastic type with genomic inversion at the 12q13 locus in most solitary lesions, producing *NAB2* and *STAT6* gene fusion and *STAT6* nuclear expression, with a high chance of recurrence and metastases. Formerly included hemangiopericytomas—an obsolete term. Cells are spindle to ovoid, located amidst dilated thin-walled blood vessels. Calcifications including psammoma bodies have not been observed. There is a case report of a tumor releasing IGF-2 causing hypoglycemia.²

Diagnostic criteria for solitary fibrous tumor

Diagnostic criteria for solitary fibrous tumor are shown in ► Fig. 47.1.³

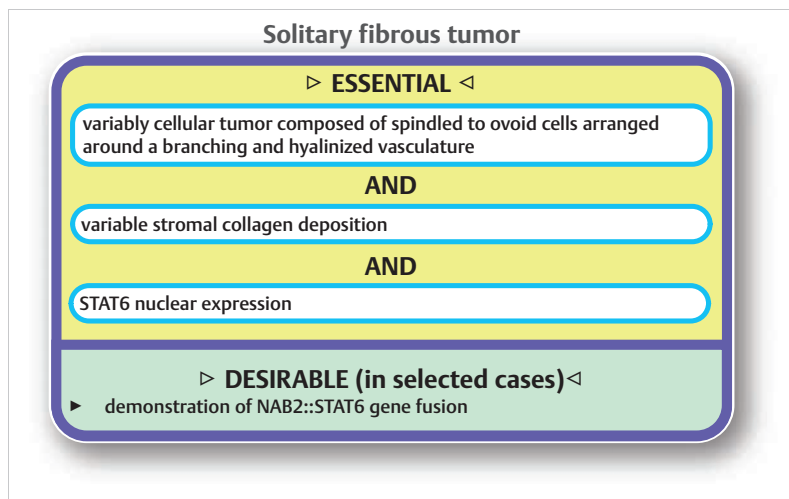


Fig. 47.1 Diagnostic criteria for solitary fibrous tumor. (Modified with permission from WHO Classification of Tumors Editorial Board, (ed.) World Health Organization classification of tumors of the central nervous system, 5th ed. Bosman F T, Jaffe E S, Lakhani S R, et al. (series eds.): World Health Organization Classification of Tumors series, vol. 6. International Agency for Research on Cancer, Lyon, France, 2021.)

Grading

The degree of mitotic activity and the presence of necrosis correlates with prognosis:

- WHO grade 1 (ICD-O 8815/0): hypocellular, collagenized tumor with a classic solitary fibrous tumor phenotype, <2.5 mitoses/mm² (<5 mitoses/10 high powered fields)

- WHO grade 2 (ICD-O 8815/1): more dense. ≥ 2.5 mitoses/mm² (≥ 5 mitoses/10 HPF)
- WHO grade 3 (ICD-O 8815/3): ≥ 2.5 mitoses/mm² (≥ 5 mitoses/10 HPF) with tumoral necrosis

Imaging

Radiographically may be mistaken for a meningioma (► Fig. 47.2).

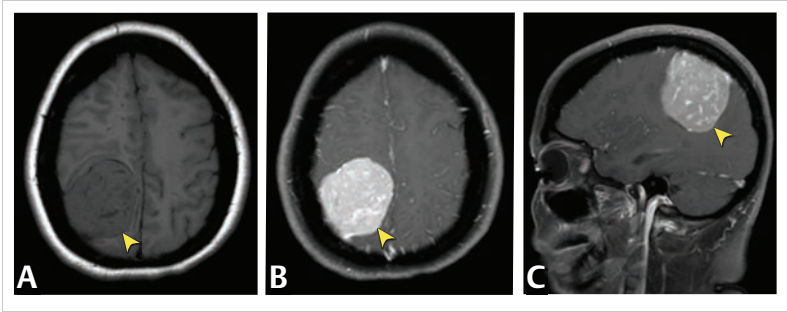


Fig. 47.2 Solitary fibrous tumor (yellow arrowheads), right parietal.

Image: MRI, A: T1 without contrast and B: with contrast; C: sagittal T1 with contrast. Note features that may be mistaken for meningioma: broad base of attachment, iso-intense with brain on T1 & T2 noncontrast sequences, dense enhancement with contrast.

Treatment

Surgery is the primary treatment. Grades 2 & 3 benefit from adjuvant XRT. Chemotherapy is used for metastases or for tumors failing local control measures.

Prognosis

SFTs tumors have a high rate of recurring and/or metastasizing, as late as decades after initial diagnosis. Therefore, long-term follow-up is advised. Unlike SFTs outside the CNS, advanced age does not correlate with worse outcome.

47.3 Vascular tumors

47.3.1 Hemangioma (WHO grade 1)

General information

Hemangiomas are benign neoplastic lesions with multiple closely approximated vessels lined by a single cell layer of bland endothelial cells with very rare mitoses. The vessels range in size from capillary-sized vessels (so-called **capillary hemangioma**, ICD-O 9131/0) to dilated vessels with flattened endothelial lining (so-called **cavernous hemangioma**, ICD-O 9131/0). They form lobules that are each supplied by a separate artery and separated by a fibrous septation. Hemangiomas may be solitary, multiple, or part of a *PIK3CA*-related overgrowth syndrome. Refer to details on vertebral hemangiomas (p.992).

Diagnostic criteria for hemangioma

Diagnostic criteria for hemangioma are shown in ► Fig. 47.3.³

Location

Hemangiomas originate most frequently in the spine (p.992), followed by the skull (p.973), and uncommonly in the brain parenchyma, nerve roots and cauda equina. In the spine, there is a predilection for the thoracic and lumbar regions, and they are often multiple (p.992).

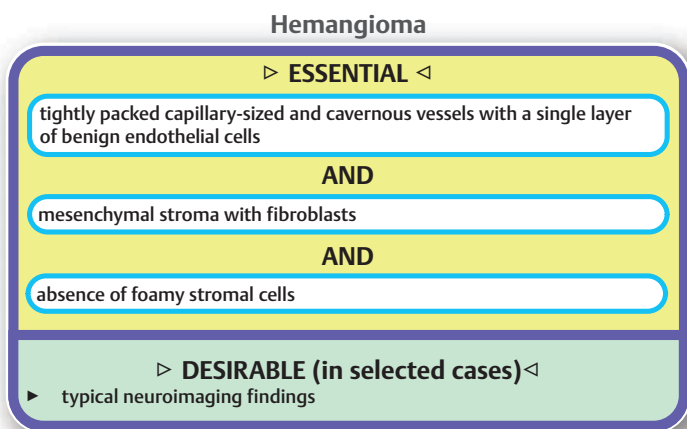


Fig. 47.3 Diagnostic criteria for hemangioma. (Modified with permission from WHO Classification of Tumors Editorial Board, (ed.) World Health Organization classification of tumors of the central nervous system. 5th ed. Bosman F T, Jaffe E S, Lakhani S R, et al. (series eds.): World Health Organization Classification of Tumors series, vol. 6. International Agency for Research on Cancer, Lyon, France, 2021.)

Imaging

In the spine, suggestive findings include polka-dot sign on axial images, or corduroy pattern on longitudinal images (p.992). Calcifications are frequent, and are well demonstrated on CT.

T2-MRI shows mixed signal internally. The surrounding hemosiderin ring is low density on T2 with blooming artifact.

47.3.2 Hemangioblastoma (WHO grade 1)

General information

Key concepts

- highly vascular well-circumscribed benign solid or cystic neoplasm of CNS or retina
- the most common *primary intra-axial* tumor in the adult posterior fossa
- may occur sporadically (70%) or as part of von Hippel-Lindau disease (30%), and any patient with HGB should be screened for VHL
- on imaging, may be solid, or cystic with enhancing mural nodule (for cysts with mural nodule, surgically, the cyst wall does not need to be removed)
- ✓ CBC: may be associated with erythrocytosis (polycythemia)

Hemangioblastomas (HGB) are extremely vascular histologically benign slow growing tumors comprised of neoplastic stromal cells with clear to vacuolated cytoplasm and copious small vessels with characteristic immunohistochemical features (e.g., staining for inhibin) and molecular signatures (e.g., *VHL* alterations). They may occur sporadically or as part of von Hippel-Lindau disease (p.646). HGBs are difficult to distinguish histologically from a renal cell carcinoma (which is common in VHL and adds to the difficulty of this differential).

Diagnostic criteria for hemangioblastoma (HGB)

Diagnostic criteria for hemangioblastoma are shown in ► Fig. 47.4.³

Hemangioblastoma

▷ ESSENTIAL ◁

a tumor composed of large, multivacuolated, and lipidized stromal cells with occasional hyperchromatic nuclei, as well as a rich capillary network

AND

stromal cells with immunohistochemical positivity for markers such as inhibin (at least focally)

OR

loss or inactivation of the VHL gene

OR

in a patient with von Hippel–Lindau syndrome

▷ DESIRABLE ◁

- ▶ in patients with von Hippel–Lindau syndrome, absence of immunohistochemical staining for markers of renal cell carcinoma

Fig. 47.4 Diagnostic criteria for hemangioblastoma. (Modified with permission from WHO Classification of Tumors Editorial Board, (ed.) World Health Organization classification of tumors of the central nervous system. 5th ed. Bosman F T, Jaffe E S, Lakhani S R, et al. (series eds.): World Health Organization Classification of Tumors series, vol. 6. International Agency for Research on Cancer, Lyon, France, 2021.)

Location

Intracranially, they occur almost exclusively in the p-fossa (hemangioblastomas are the most common *primary* intra-axial p-fossa tumor in adults). May occur in the cerebellar hemisphere, vermis, or brainstem. May also occur in the spinal cord (p.986) (1.5–2.5% of spinal cord tumors). Retinal HGB and/or angiomas occur in 6% of patients with cerebellar HGBs.

Epidemiology

HGB represent 1–2.5% of intracranial tumors. They comprise 7–12% of primary p-fossa tumors.⁴

70% of HGBs occur sporadically, but 30% occur as part of von Hippel–Lindau (VHL) disease (p.646).⁵ 5–30% of cases of cerebellar HGB and 80% of spinal HGB are associated with VHL.

Sporadic cases tend to present in the 4th decade, whereas VHL cases present *earlier* (peak in 3rd decade). In sporadic cases, the HGB are solitary and originate in the cerebellum (83–95%), spinal cord (3–13%), medulla oblongata (2%)⁶ or cerebrum (1.5%).⁴ In 65% of patients with VHL the HGB are multiple.

Presentation

S/S of cerebellar HGB are usually those of any p-fossa mass—H/A, N/V, cerebellar findings...; see Posterior fossa (infratentorial) tumors (p.624)—and obstructive hydrocephalus may occur. HGB is rarely documented as a cause of apoplexy due to intracerebral hemorrhage (ICH) (lobar or cerebellar); however, some studies indicate that if cases of ICH are carefully examined, abnormal vessels

consistent with HGB (and occasionally misidentified as AVM) may be found with surprising frequency (in spite of negative CT and/or angiography).⁷

Retinal HGBs tend to be located peripherally, and may hemorrhage and cause retinal detachment. Erythrocytosis may be due to erythropoietin liberated by the tumor.

Histology

Invasion of adjacent neural tissue is rare. Mitotic figures are rare. No true capsule, but usually well circumscribed (narrow zone of infiltration). May be solid, or cystic with a mural nodule (70% of cerebellar lesions are cystic; nodules are very vascular, appear red, are often located near pial surface, and may be as small as 2 mm; cyst fluid is clear yellow with high protein). In cystic lesions, the cyst wall is lined with non-neoplastic compressed cerebellum. The cyst develops because the vessel walls are so thin that they leak water, and proteins don't cross as readily.

Cardinal feature: numerous capillary channels, lined by a single layer of endothelium, surrounded by reticulin fibers (stains positive with reticulin stain). Macrophages stain PAS positive.

Two main component cells:

1. stromal cells: polygonal. Characteristic and distinguishing finding: plentiful lipid laden vacuoles,
2. reactive vascular cells

47

Cyst patterns⁸:

1. no associated cysts: 28%
2. peritumoral cyst alone: 51%
3. intratumoral cyst: 17%
4. peritumoral AND intratumoral cysts: 4%

Evaluation

Patients with a p-fossa HGB (radiologically suspected or histologically proven) should undergo MRI of entire neuraxis because of possibility of spinal HGBs (may be distant from p-fossa lesion; may suggest possibility of VHL).

CT: solid lesions are usually isodense with intense contrast enhancement. Cystic HGBs remain low density with contrast, with the nodule enhancing.

MRI (► Fig. 47.5): preferable to CT due to the predilection of HGBs for the p-fossa. May show serpentine vascular signal voids, especially in the periphery of the lesion. Also, peripheral hemosiderin deposits may occur from previous hemorrhages.⁴

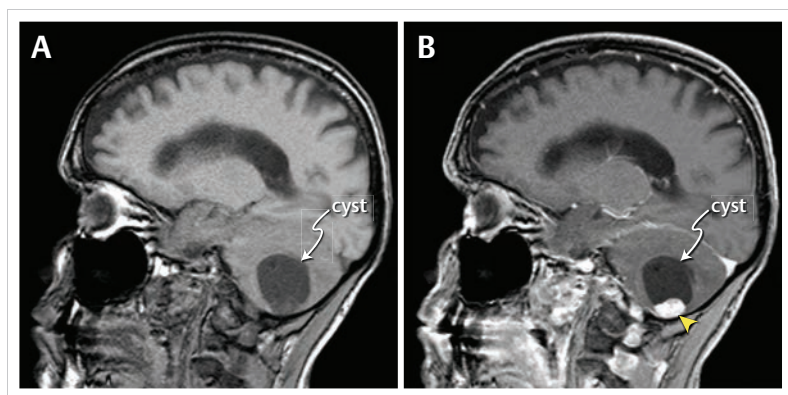


Fig. 47.5 Hemangioblastoma of the posterior fossa. Demonstrates cyst with enhancing mural nodule (yellow arrowhead). Image: Sagittal MRI, A: T1 noncontrast, B: T1 + contrast.

Vertebral angiography: usually demonstrates intense vascularity (most other tumors of the p-fossa are relatively avascular). May be required in HGBs where nodule is too small to be imaged on CT/MRI. 4 patterns: 1) vascular mural nodule on side of avascular cyst, 2) vascular lesion surrounding avascular cyst, 3) solid vascular mass, and 4) multiple, separate vascular nodules.

Labs: often discloses *polycythemia* (no hematopoietic foci within tumor). In cases with suggestive history, labwork to rule out catecholamine production from pheochromocytoma may be indicated; see Endocrine/laboratory studies (p.942).

Treatment

Surgery

Surgical treatment may be curative in cases of sporadic HGB, not in VHL. GTR is considered the optimal treatment.⁹

Preoperative embolization may help reduce the vascularity.

Cystic HGBs require removal of mural nodule (otherwise, cyst will recur). The cyst wall is not removed unless there is evidence of tumor within the cyst wall on MRI (typically thick-walled cysts) or visually at the time of surgery.⁸ 5-ALA fluorescence may aid in visual localization of small heman-gioblastomas within the cyst wall.¹⁰

Solid HGBs tend to be more difficult to remove. They are treated like AVMs (avoid piecemeal removal), working along margin and devascularizing blood supply. A helpful technique is to shrink the tumor by laying a length of bipolar forceps along tumor surface and coagulating. HGBs with attachment to floor of 4th ventricle may be hazardous to remove (cardio-respiratory complications).

Multiple lesions: if ≥ 0.8 –1 cm diameter: may treat as in solitary lesion. Smaller and deeper lesions may be difficult to locate at time of surgery.

Cystic brainstem HGB: the solid nodule of the tumor is removed under the microscope by bipolar-ing and cutting the gliotic adhesions to the parenchyma. ★ Removal of the cyst wall is not necessary. There is often a cleavage between the tumor and the floor of the fourth ventricle which facilitates tumor removal. To reduce bleeding, avoid piecemeal removal. Preserve large draining veins until the arterial feeders to the mural nodule have been isolated and resected.¹¹

Radiation treatment

Effectiveness is dubious. May be useful to reduce tumor size or to retard growth, e.g., in patients who are not surgical candidates, for multiple small deep lesions, or for inoperable brainstem HGB. Does not prevent regrowth following subtotal excision.

Prognosis

Prognosis is excellent following GTR, and is better for sporadic HGB vs. those associated with VHL. No report of malignant change. May spread thru CSF after surgery, but remain benign.

Prognostic factors for *longer* OS in older patients (but not in younger patients) include: GTR vs. STR or biopsy (HR 0.62).

Prognostic factors for *shorter* OS include: age ≥ 40 years (HR 3.9), size, growth pattern, multifocal-ity and brainstem location (vs. cerebellum) (HR 1.9).⁹

47.4 Notochordal tumors

47.4.1 Chordoma

General information

Key concepts

- primary malignant tumor, usually of clivus or sacrum, with high recurrence rate
- histology: characteristic physaliphorous cells (containing intracellular mucin)
- generally slow-growing and radioresistant
- treatment of choice: wide en bloc resection when possible (piecemeal removal carries risk of inducing metastases), proton-beam radiation may help

Rare tumors (incidence of ≈ 0.51 cases/million) of the remnant of the primitive notochord (which normally differentiates into the nucleus pulposus of the intervertebral discs). They represent less than 1% of intracranial tumors and 3% of primary spine tumors.¹²

They are slow growing, locally aggressive and osseodestructive.

Histologically, these tumors are considered low-grade malignancies. However, their behavior is more malignant because of the difficulty of total removal, a high recurrence rate (85% following surgery), and the fact that they metastasize (usually late) in 5–20%,¹³ (often after multiple resections) typically to lung, liver, and bone. Therefore aggressive XRT is usually employed post-op. Malignant transformation into fibrosarcoma or malignant fibrous histiocytoma is rare. Four types have been identified:

1. conventional: median age: 55 years
2. chondroid: extracellular matrix resembles hyaline cartilage. Median age: 45 years
3. dedifferentiated: biphasic tumors comprised of conventional chordoma adjacent to high grade sarcoma. Nuclear and cytokeratin expression are retained in the conventional chordoma areas, but not in the sarcomatous component. Median age: 61 years
4. poorly differentiated: no physaliphorous cells. Median age: 7 years

Diagnostic criteria for chordoma

Diagnostic criteria for chordoma are shown in ► Fig. 47.6.³

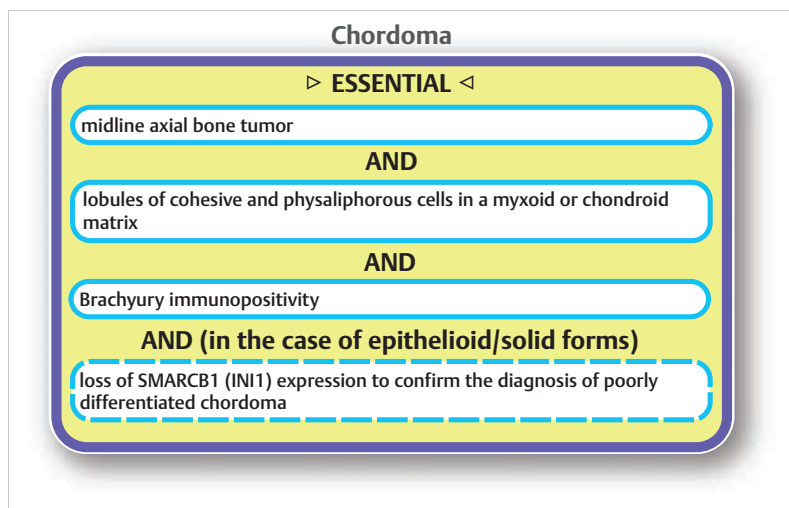


Fig. 47.6 Diagnostic criteria for chordoma. (Modified with permission from WHO Classification of Tumors Editorial Board, (ed.) World Health Organization classification of tumors of the central nervous system. 5th ed. Bosman F T, Jaffe E S, Lakhani S R, et al. (series eds.): World Health Organization Classification of Tumors series, vol. 6. International Agency for Research on Cancer, Lyon, France, 2021.)

Location

Chordomas can arise anywhere along the neuraxis where there is remnant of notochord; however, cases tend to cluster at the two ends of the primitive notochord: 35% cranially¹⁴ in the sphenoid-occipital region (clivus), and 53%¹⁴ in the spine at the sacrococcygeal region.¹⁵ Less commonly, they may occur in the spine above the sacrum.¹⁶

Pathogenesis

The notochord is a flexible cartilage-like embryologic structure that requires the protein brachyury (coded for by the *TBXT* gene) to develop. In most individuals, the notochord regresses in utero and by birth has devolved into the nucleus pulposus (the center of spinal intervertebral discs). In a small number of people, remnants of the notochord persist, usually in the odontoid process or coccyx. The process of neoplastic transformation of these remnants is incompletely understood, but mutations have been identified including duplications of the *TBXT* gene in 27%.¹⁷

Histology

Physaliphorous cells are distinctive, vacuolated cells on histology that probably represent cytoplasmic mucus vacuoles seen ultrastructurally.

Radiographic appearance

Usually lytic, lobular midline lesions with frequent calcifications.¹⁸

MRI: hypointense on T1WI, hyperintense on T2WI.

Enhances on CT or MRI with contrast.¹⁸ Rarely, they may appear as a sclerotic vertebra¹⁹ ("ivory vertebra").

Cranial chordomas

Peak incidence of cranial chordomas is 50–60 years of age. These tumors are rare in patients <30 years of age.²⁰ Male:female distribution is ≈ equal.

Differential diagnosis: Primarily between other cartilaginous tumors of the skull base; see differential diagnosis of other foramen magnum region tumors (p.1649):

1. chondrosarcomas
2. chondromas

Presentation: Usually produces cranial nerve palsies (usually oculomotor or abducens nerve).

Spinal chordomas

General information

Occur primarily in the sacrococcygeal region. Unlike cranial chordomas, sacrococcygeal chordomas show a male predominance,¹⁴ and these patients tend to be older. May also arise in C2. Chordomas constitute over 50% of primary bone tumors of the sacrum. May produce pain, sphincter disturbance, or nerve root symptoms from local nerve root compression. They may occasionally extend cephalad into the lumbar spinal canal. They are usually confined anteriorly by the presacral fascia, and only rarely invades the wall of the rectum.²¹ A firm fixed mass may be palpable between the rectum and the sacrum on a rectal exam.

Evaluation

Characteristic radiographic findings: centrally located destruction of several sacral segments, with an anterior soft-tissue mass that occasionally has small calcifications. CT and MRI show the bony destruction. This is usually difficult to see on plain X-rays. MRI also shows the soft-tissue mass.

Open or CT guided percutaneous *posterior* biopsy can confirm the diagnosis. Transrectal biopsy should be avoided because of the potential of rectal spread of tumor.²²

Chest CT and bone scan: to R/O mets for staging purposes.

Treatment of spinal chordomas

Surgery

Wide en bloc excision with postoperative radiation is usually the best option, although this may also be only temporarily effective. Decompression is best avoided since entering the mass serves to spread tumor (surgically induced metastases) which will then regrow. Chordomas located in C2 are usually not amenable to en bloc resection.²³

Sacral chordomas: The particulars of the surgical procedure are highly dependent on the extent of the lesion. These tumors may spread through the gluteal musculature, and if significant muscular excision is required, then a pedicle-based rectus abdominis flap may be employed. A diverting colostomy may be required if it is necessary to resect the rectum or if a cephalic sacral resection is anticipated.²⁴

For chordomas caudal to the third sacral segment, most agree that a posterior approach is satisfactory. For more rostral lesions, some advocate a combined anterior-posterior approach. However, a posterior approach has also been used for these.²⁴

Adverse effects of sacrectomy: if S2 nerve roots are the most caudal nerve roots spared, there is ≈ 50% chance of normal bladder and bowel control.²⁴ If S1 or more cephalic roots are the most caudal nerve roots spared, most will have impaired bladder control and bowel problems.²⁴

Radiation therapy (XRT)

Best results were obtained with en bloc excision (even if marginal), sometimes combined with high-dose XRT^{16,25} (conventional XRT did not prevent recurrence when incorporated with palliative or

debulking surgery,¹⁶ but it did lengthen the interval to recurrence²⁵). Early radiation was associated with longer survival.²⁶ Higher XRT doses can be used in the sacrococcygeal region (4500–8000 rads) than in the cervical spine (4500–5500 rads) because of concerns of radiation injury to the spinal cord. IMRT and stereotactic radiosurgery have also been used.²³

Proton beam therapy, alone¹³ or combined with high-energy X-ray (photon) therapy,^{27,28} may be more effective than conventional XRT alone. However, proton beam therapy requires travel to one of a limited number of facilities with a cyclotron which may be difficult to arrange for what is typically ≈ 7 weeks of fractionated treatments.

Chemotherapy

Imatinib (Gleevec®) (a tyrosine kinase inhibitor) has some antitumor effect in chordoma.²⁹

Prognosis

The prognosis is worse for dedifferentiated and poorly differentiated with median PFS of 6 months and 4 months respectively, and median OS of 15 months and 13 months. Conventional chordomas have a median PFS of 24 months, and median OS of 48 months.³

Median survival is 6.3 years.²³

References

- [1] WHO Classification of Tumors Editorial Board. Soft Tissue and Bone Tumors. 5th ed. Lyon, France: International Agency for Research on Cancer; 2020; 3. <https://publications.iarc.fr/610>
- [2] Sohda T, Yun K. Insulin-like growth factor II expression in primary meningeal hemangiopericytoma and its metastasis to the liver accompanied by hypoglycemia. *Hum Pathol*. 1996; 27:858–861
- [3] Bosman FT, Jaffe ES, Lakhani SR, et al. World Health Organization Classification of Tumors series, vol. 6. In: WHO Classification of Tumors Editorial Board (ed.) World Health Organization classification of tumors of the central nervous system. 5th ed. Lyon, France: International Agency for Research on Cancer; 2021
- [4] Ho VB, Sminiotopoulos JG, Murphy FM, et al. Radiologic-Pathologic Correlation: Hemangioblastoma. *American Journal of Neuroradiology*. 1992; 13:1343–1352
- [5] Hottinger AF, Khakoo Y. Neurooncology of familial cancer syndromes. *Journal of Child Neurology*. 2009; 24:1526–1535
- [6] Catapano D, Muscarella LA, Guarnieri V, et al. Hemangioblastomas of central nervous system: molecular genetic analysis and clinical management. *Neurosurgery*. 2005; 56:1215–21; discussion 1221
- [7] Wakai S, Inoh S, Ueda Y, et al. Hemangioblastoma Presenting with Intracerebral Hemorrhage. *J Neurosurg*. 1984; 61:956–960
- [8] Jagannathan J, Lonser RR, Smith R, et al. Surgical management of cerebellar hemangioblastomas in patients with von Hippel-Lindau disease. *Journal of Neurosurgery*. 2008; 108:210–222
- [9] Huang Y, Chan L, Bai HX, et al. Assessment of care pattern and outcome in hemangioblastoma. *Sci Rep*. 2018; 8. DOI: 10.1038/s41598-018-29047-9
- [10] Utsuki S, Oka H, Sato K, et al. Fluorescence diagnosis of tumor cells in hemangioblastoma cysts with 5-aminolevulinic acid. *Journal of Neurosurgery*. 2009. DOI: 10.3171/2009.5.JNS08442
- [11] Agrawal A, Kakani A, Vagh SJ, et al. Cystic hemangioblastoma of the brainstem. *J Neurosci Rural Pract*. 2010; 1:20–22
- [12] Wright D. Nasopharyngeal and Cervical Chordoma – Some Aspects of the Development and Treatment. *J Laryngol Otol*. 1967; 81:1335–1337
- [13] Hug EB, Loredi LN, Slater JD, et al. Proton Radiation Therapy for Chordomas and Chondrosarcomas of the Skull Base. *Journal of Neurosurgery*. 1999; 91:432–439
- [14] O'Neill P, Bell BA, Miller JD, et al. Fifty Years of Experience with Chordomas in Southeast Scotland. *Neurosurgery*. 1985; 16:166–170
- [15] Heffelfinger MJ, Dahlin DC, MacCarty CS, et al. Chordomas and Cartilaginous Tumors at the Skull Base. *Cancer*. 1973; 32:410–420
- [16] Boriani S, Chevalley F, Weinstein JN, et al. Chordoma of the Spine Above the Sacrum. Treatment and Outcome in 21 Cases. *Spine*. 1996; 21:1569–1577
- [17] Tarpey PS, Behjati S, Young MD, et al. The driver landscape of sporadic chordoma. *Nat Commun*. 2017; 8. DOI: 10.1038/s41467-017-01026-0
- [18] Meyer JE, Lepke RA, Lindfors KK, et al. Chordomas: Their CT Appearance in the Cervical, Thoracic and Lumbar Spine. *Radiology*. 1984; 153:693–696
- [19] Schwarz SS, Fisher WS, Pulliam MW, et al. Thoracic Chordoma in a Patient with Paraparesis and Ivory Vertebral Body. *Neurosurgery*. 1985; 16:100–102
- [20] Wold LE, Laws ER. Cranial Chordomas in Children and Young Adults. *J Neurosurg*. 1983; 59:1043–1047
- [21] Azzarelli A, Quagliuolo V, Cerasoli S, et al. Chordoma: Natural History and Treatment Results in 33 Cases. *J Surg Oncol*. 1988; 37:185–191
- [22] Mindell ER. Current Concepts Review. Chordoma. *Journal of Bone and Joint Surgery*. 1981; 63A:501–505
- [23] Jiang L, Liu ZJ, Liu XG, et al. Upper cervical spine chordoma of C2–C3. *European Spine Journal*. 2009; 18:293–298; discussion 298–300
- [24] Samson IR, Springfield DS, Suit HD, et al. Operative Treatment of Sacrococcygeal Chordoma. A Review of Twenty-One Cases. *Journal of Bone and Joint Surgery*. 1993; 75:1476–1484
- [25] Klekamp J, Sami M. Spinal Chordomas – Results of Treatment Over a 17-Year Period. *Acta Neurochir (Wien)*. 1996; 138:514–519
- [26] Cheng EY, Özerdemoglu RA, Transfeldt EE, et al. Lumbosacral Chordoma. Prognostic Factors and Treatment. *Spine*. 1999; 24:1639–1645
- [27] Suit HD, Goitein M, Munzenrider J, et al. Definitive Radiation Therapy for Chordoma and Chondrosarcoma of Base of Skull and Cervical Spine. *Journal of Neurosurgery*. 1982; 56:377–385
- [28] Rich TA, Schiller A, Mankin HJ. Clinical and Pathologic Review of 48 Cases of Chordoma. *Cancer*. 1985; 56:182–187
- [29] Magenau JG, Schuetz SM. New targets for therapy of sarcoma. *Current Opinion in Oncology*. 2008; 20:400–406

48 Melanocytic Tumors and CNS Germ Cell Tumors

48.1 Melanocytic tumors

48.1.1 General information

Rare. Diffuse or localized CNS tumors that presumably originate from leptomeningeal melanocytes. Terminology:

1. **circumscribed** melanocytic tumors: rare
 - a) meningeal melanocytoma: may be well differentiated. When increased mitotic activity or parenchymal invasion occurs, these qualify as intermediate grade tumors
 - b) meningeal melanoma: the malignant counterpart to the above
2. **diffuse** melanocytic tumors are defined by involvement of large areas of the subarachnoid space, with or without focal nodularity
 - a) meningeal melanocytosis: benign histology
 - b) meningeal melanomatosis: malignant histology. Aggressive and resistant to XRT

These primary CNS lesions may be distinguished from other pigmented CNS neoplasms (e.g., malignant melanotic nerve sheath tumors) using molecular analysis (► Table 35.7) and methylation profiling.

48.1.2 Diffuse meningeal melanocytic neoplasms

General information

Rare, primary CNS neoplasms that involve the leptomeninges and spread diffusely throughout the subarachnoid space, sometimes with focal or multifocal nodularity, and frequent invasion of the Virchow-Robin (perivascular) spaces. There is a predilection for the temporal lobes, cerebellum, pons, medulla, and spinal cord.

Distinguishing these primary tumors from metastases originating outside the CNS is critical for treatment decisions and prognostication, and often requires molecular testing in addition to microscopic histology.

These tumors frequently occur in the setting of neurocutaneous melanosis (p.653) (a rare neurocutaneous disorder).

Symptoms may be due to hydrocephalus or from dysfunction of invaded brain. Other associations include arachnoid cysts, syringomyelia, other brain tumors (e.g., astrocytoma, ependymoma...), and other structural conditions (e.g., Chiari malformation). Symptoms progress rapidly when malignant transformation occurs.

MRI findings vary with the amount of melanin content, and these tumors may be isodense or hyperintense on T1WI, and hypodense on T2WI.¹

There are case reports of tumor spreading to the peritoneal compartment when ventriculo-peritoneal shunting is used to treat hydrocephalus.²

Diagnostic criteria for diffuse meningeal melanocytic neoplasms

Diagnostic criteria for diffuse meningeal melanocytic neoplasms are shown in ► Fig. 48.1.³

Meningeal melanocytosis

A diffuse or multifocal proliferation of cytologically bland melanocytic cells. See ► Fig. 48.1 for diagnostic criteria. May spread into perivascular spaces without invasion of the brain. Symptoms related to meningeal melanocytosis will develop in 10-15% of patients with large congenital cutaneous melanocytic nevi. Meningeal melanocytosis may be asymptomatic.

Prognosis is poor once symptoms manifest.

Meningeal melanomatosis

Malignant histology distinguishes these tumors from meningeal melanocytosis. CNS invasion is a common finding. See ► Fig. 48.1 for diagnostic criteria. Usually aggressive with poor prognosis.

Diffuse meningeal melanocytic neoplasms

▷ ESSENTIAL ◁

diffuse or multifocal primary meningeal melanocytic neoplasm

AND

- ▶ for **meningeal melanocytosis**: absence of CNS parenchyma invasion, absence of marked cytological atypia, absence of mitotic activity, and absence of necrosis
- ▶ for **meningeal melanomatosis**: invasion of the CNS parenchyma and/or marked cytological atypia and/or mitotic activity and/or necrosis

▷ DESIRABLE ◁

- ▶ in children, meningeal melanocytosis/melanomatosis is often *NRAS*-mutant and rarely *BRAF*-mutant

Fig. 48.1 Diagnostic criteria for diffuse meningeal melanocytic neoplasms. (Modified with permission from WHO Classification of Tumors Editorial Board, (ed.) World Health Organization classification of tumors of the central nervous system. 5th ed. Bosman F T, Jaffe E S, Lakhani S R, et al. (series eds.): World Health Organization Classification of Tumors series, vol. 6. International Agency for Research on Cancer, Lyon, France, 2021.)

48.1.3 Circumscribed meningeal melanocytic neoplasms

General information

These tumors arise from leptomeningeal melanocytes and vary from well-differentiated neoplasms (meningeal melanocytoma) to aggressive, malignancies (meningeal melanoma). An intermediate grade between the two exhibits bland cytology but increased mitotic activity on invasion of parenchyma.

Diagnostic criteria for circumscribed meningeal melanocytic neoplasms

Diagnostic criteria for circumscribed meningeal melanocytic neoplasms are shown in ▶ Fig. 48.2.³

Meningeal melanocytoma

A well developed, solid, noninfiltrating melanocytic tumor that arises from the leptomeninges characterized by epithelioid fusiform, polyhedral or spindled melanocytes with no evidence of anaplasia, necrosis or increased mitoses. Rare reports of malignant transformation.⁴ See ▶ Fig. 48.2 for diagnostic criteria.

Comprises 0.06–0.1% of brain tumors with an estimated incidence of 1 per 10 million.⁵ Most commonly found in the cervical or thoracic spine. Less commonly, they originate in the leptomeninges of the posterior fossa or supratentorial intracranial compartment. A handful of patients with nevus of Ota (AKA oculodermal melanocytosis, which results from incomplete migration of melanocytes from the neural crest to the epidermis) have concurrent meningeal melanocytoma.⁶

Meningeal melanoma

A primary solitary malignant neoplasm that arises from leptomeningeal melanocytes characterized by aggressive growth. Histologically similar to melanomas occurring elsewhere.⁵ See ▶ Fig. 48.2 for diagnostic criteria.

Circumscribed meningeal melanocytic neoplasms

▷ ESSENTIAL ◁

circumscribed/localized primary melanocytic neoplasm in the meninges

AND

- ▶ for **melanocytoma**: limited cytological atypia, (almost) no mitoses, no necrosis, and (in cases of evaluable CNS parenchyma) no CNS invasion
- ▶ for **intermediate-grade melanocytoma**: mitotic count^a of 0.5–1.5 mitoses/mm² and/or CNS invasion, but limited cytological atypia and no necrosis
- ▶ for **melanoma**: mitotic count^a > 1.5 mitoses/mm² and/or necrosis, often accompanied by marked cytological atypia

▷ DESIRABLE ◁

- ▶ demonstration of *GNAQ*, *GNA11*, *PLCB4*, or *CYSLTR2* mutation corroborates the CNS origin of the neoplasm, especially after exclusion of uveal or blue nevus–like melanoma
- ▶ additional molecular markers (*SF3B1*, *EIF1AX*, and *BAP1* mutations; chromosome 3 monosomy; complex copy-number variations) indicating aggressive behavior

^a 1 mm² equates approximately to 5 HPF of 0.5 mm in diameter and 0.2 mm² in area.

48

Fig. 48.2 Diagnostic criteria for circumscribed meningeal melanocytic neoplasms. (Modified with permission from WHO Classification of Tumors Editorial Board, (ed.) World Health Organization classification of tumors of the central nervous system. 5th ed. Bosman F T, Jaffe E S, Lakhani S R, et al. (series eds.): World Health Organization Classification of Tumors series, vol. 6. International Agency for Research on Cancer, Lyon, France, 2021.)

Compared to melanocytoma, these are more pleomorphic, anaplastic, mitotically active and often invade CNS tissue and may demonstrate necrosis. May spread through CSF pathways. Occasional systemic metastases outside the CNS occur.⁷

Incidence of 0.5 per 10 million.⁸ The peak age for this tumor is in the 4th decade (mean age: 43 years) compared to the 7th decade for primary cutaneous melanoma.⁹

They are aggressive and resistant to XRT. However, with GTR, the prognosis is better than with cutaneous melanoma metastatic to the CNS.

48.2 Germ cell tumors of the CNS

48.2.1 General information

Background

A family of morphological and immunophenotypic homologues of germ cell tumors of the gonads and other sites.

80–90% of GCTs occur in the midline, usually in the suprasellar and/or pineal region (simultaneous suprasellar and pineal region lesions is diagnostic of a GCT; so-called synchronous germ cell tumors, which comprise 13% of GCTs). GCTs are highly sensitive to XRT.¹⁰ In the pineal region, these tumors occur predominantly in males. In females, GCTs are more common in the suprasellar region.¹¹ Aside from benign teratomas, all intracranial GCTs are malignant and may metastasize via CSF and systemically.

Epidemiology

GCTs primarily affect children and adolescents. They comprise 2–3% of primary intracranial tumors. Prevalence appears higher in eastern Asia.

Peak incidence: age 10–14 years. Affects males more than females, although in the suprasellar region girls are more prevalent.

CSF tumor markers

GCTs can (but not always) give rise to tumor markers in the CSF (see Tumor markers used clinically (p.634)). ► Table 48.1 summarizes these findings.

- beta-human chorionic gonadotropin (β -hCG): elevated CSF levels are classically associated with choriocarcinomas, but also occur in up to 50% of germinomas (which are more common)
 - alpha-fetoprotein (AFP) is elevated with yolk sac tumors, embryonal carcinoma and occasionally with teratomas
 - placental alkaline phosphatase (PLAP) may be elevated in serum or CSF with intracranial germinomas¹²
- NB: tumor markers alone are no longer considered useful for making a definitive diagnosis or following response to treatment with a pineal region tumor since many of these tumors are mixed cell type and many pineal region tumors do not give rise to markers.

Table 48.1 Occurrence of CSF tumor markers with pineal germ cell tumors

Tumor	β -hCG ^a	AFP	PLAP ^b
choriocarcinoma	≈ 100%	–	–
germinoma	10–50%	–	+
embryonal carcinoma	–	+	–
yolk sac carcinoma	–	+	–
immature teratoma	–	+	–
mature teratoma	–	–	–

^aabbreviations: β -hCG = beta human chorionic gonadotropin, AFP = alpha-fetoprotein, PLAP = placental alkaline phosphatase

^belevated PLAP may also occur in serum

► **General outline of GCTs.** This outline shows the hierarchy of germ cell tumors. Details are provided in following sections.

1. germinoma: malignant tumor of primitive germ cells that occurs in the gonads (called testicular seminomas in males, dysgerminomas in females) or in the CNS. Survival with these is much better than with nongerminomatous tumors
 - *variant*: syncytiotrophoblastic giant cell variant
2. nongerminomatous germ cell tumors (NGGCT) include:
 - a) embryonal carcinoma:
 - b) choriocarcinoma
 - c) yolk sac tumor AKA endodermal sinus tumor (EST): usually malignant
 - d) teratoma. A germ cell tumor composed of at least two of the three germ cell layers: ectoderm, endoderm, and mesoderm. Subclassifications:
 - mature teratoma: adult-type tissue elements. Minimal mitoses. All 3 skin layers
 - immature teratoma: contain immature embryonic or fetal tissue alone or in combination with mature tissues
 - teratoma with somatic-type malignancy: contain a component resulting from malignant transformation of a somatic tissue (usually a carcinoma or sarcoma)
3. mixed germ cell tumor

48.2.2 Mature teratoma

General information

Exclusively comprised of mature somatic tissue, consisting of 2 or more of the 3 skin layers:

1. endoderm: glands lined with respiratory or enteric epithelia (often as dilated cysts) are typical, and rarely GI and bronchial structures, and hepatic or pancreatic tissue

2. mesoderm: smooth and striated muscle, cartilage and bone
3. ectoderm: epidermis, skin appendage organs (sweat glands, hair follicles often with hair), CNS tissue, choroid plexus, salivary gland

Little or no mitotic activity, Ki-67 in the mature tissue may be unexpectedly high.

Diagnostic criteria for mature teratoma

Diagnostic criteria for mature teratoma are shown in ► Fig. 48.3.³

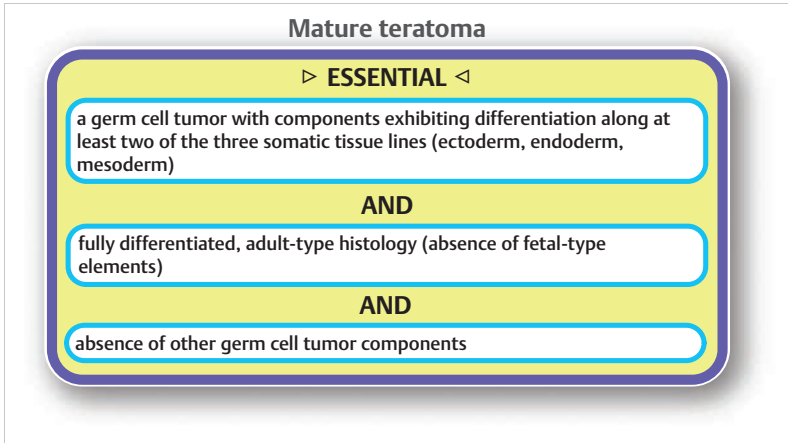


Fig. 48.3 Diagnostic criteria for mature teratoma. (Modified with permission from WHO Classification of Tumors Editorial Board, (ed.) World Health Organization classification of tumors of the central nervous system. 5th ed. Bosman F T, Jaffe E S, Lakhani S R, et al. (series eds.): World Health Organization Classification of Tumors series, vol. 6. International Agency for Research on Cancer, Lyon, France, 2021.)

48.2.3 Immature teratoma

General information

The presence of any incompletely differentiated or fetal-like tissue requires the tumor to be designated as immature teratoma. Hypercellular stromal elements resembling fetal gut and respiratory tissue with increased mitotic activity are common. Other unusual components may include primitive retinal tissue or neural tube elements.

Diagnostic criteria for immature teratoma

Diagnostic criteria for immature teratoma are shown in ► Fig. 48.4.³

48.2.4 Teratoma with somatic-type malignancy

General information

Somatic cancers that may occur include rhabdomyosarcomas and undifferentiated sarcomas, adenocarcinoma resembling GI tract lineage and squamous cell carcinoma. Other rare cancers may also be seen. The diagnosis of somatic-type malignancy should not be made based on cytologic atypia alone.

Diagnostic criteria for teratoma with somatic-type malignancy

Diagnostic criteria for teratoma with somatic-type malignancy are shown in ► Fig. 48.5.³

Immature teratoma

▷ ESSENTIAL ◁

identification of incompletely differentiated elements exhibiting differentiation along at least two of the three somatic tissue lines (ectoderm, endoderm, mesoderm) in a teratoma, or the identification of any such elements within a tumor otherwise qualifying as a mature teratoma

AND

absence of other germ cell tumor components

Fig. 48.4 Diagnostic criteria for immature teratoma. (Modified with permission from WHO Classification of Tumors Editorial Board, (ed.) World Health Organization classification of tumors of the central nervous system. 5th ed. Bosman F T, Jaffe E S, Lakhani S R, et al. (series eds.): World Health Organization Classification of Tumors series, vol. 6. International Agency for Research on Cancer, Lyon, France, 2021.)

Teratoma with somatic-type malignancy

▷ ESSENTIAL ◁

identification of a distinct histological component that has the cytological features, architecture, mitotic activity, and disorderly growth pattern expected of a sarcoma, carcinoma, or other defined type of somatic cancer in a mature or immature teratoma

Fig. 48.5 Diagnostic criteria for teratoma with somatic-type malignancy. (Modified with permission from WHO Classification of Tumors Editorial Board, (ed.) World Health Organization classification of tumors of the central nervous system. 5th ed. Bosman F T, Jaffe E S, Lakhani S R, et al. (series eds.): World Health Organization Classification of Tumors series, vol. 6. International Agency for Research on Cancer, Lyon, France, 2021.)

48.2.5 Germinoma

General information

Large, undifferentiated appearing cells resembling primordial germ cells with ample glycogen-rich clear cytoplasm containing round, vesicular and centrally placed nuclei. The cells are arranged in sheets, lobules or cords with trabeculae. Necrosis is uncommon and mitotic activity is variable.

Some may have syncytiotrophoblastic giant cells seen on squash or smear preparations, but this alone does not qualify them as choriocarcinoma.

Diagnostic criteria for germinoma

Diagnostic criteria for germinoma are shown in ► Fig. 48.6.³

Germinoma

▷ ESSENTIAL ◁

a germ cell tumor containing large tumor cells with typical cytological characteristics

AND

nuclear OCT4 and widespread membranous KIT (or podoplanin [D2-40]) immunoreactivity, or absence of 5-methylcytosine expression

AND

absence of CD30 expression

AND

absence of AFP expression

AND

hCG immunoreactivity in syncytiotrophoblastic giant cells (for the specific diagnosis of germinoma with syncytiotrophoblastic elements)

AND

absence of other germ cell tumor components (except syncytiotrophoblastic giant cells for the specific diagnosis of germinoma with syncytiotrophoblastic giant cells)

Fig. 48.6 Diagnostic criteria for germinoma. (Modified with permission from WHO Classification of Tumors Editorial Board, (ed.) World Health Organization classification of tumors of the central nervous system. 5th ed. Bosman F T, Jaffe E S, Lakhani S R, et al. (series eds.): World Health Organization Classification of Tumors series, vol. 6. International Agency for Research on Cancer, Lyon, France, 2021.)

48.2.6 Embryonal carcinoma

General information

Large epithelioid cells resembling cells of the embryonic germ disc containing vesicular nuclei, macro-nuclei and clear to violet cytoplasm. Mitotic activity is often increased. Necrotic areas are common.

Diagnostic criteria for embryonal carcinoma

Diagnostic criteria for embryonal carcinoma are shown in ► Fig. 48.7.³

48.2.7 Yolk sac tumor

General information

Formerly called endodermal sinus tumor. Primitive appearing epithelial cells that may be associated with loose variably cellular stromal component. Bright eosinophil PAS-positive particles may appear in the cytoplasm of epithelial cells but are not a consistent feature.

Embryonal carcinoma

▷ ESSENTIAL ◁

a germ cell tumor with large epithelioid cells as described in the Histopathology subsection^a

AND

CD30 and OCT4 expression

AND

absent or only focal, non-membranous KIT expression

AND

absence of hCG expression

AND

absence of AFP expression

AND

absence of other germ cell tumor components

▷ DESIRABLE ◁

▶ cytokeratin expression

Fig. 48.7 Diagnostic criteria for embryonal carcinoma.

^a For histopathology details, refer to the WHO blue book³ (Modified with permission from WHO Classification of Tumors Editorial Board, (ed.) World Health Organization classification of tumors of the central nervous system. 5th ed. Bosman F T, Jaffe E S, Lakhani S R, et al. (series eds.): World Health Organization Classification of Tumors series, vol. 6. International Agency for Research on Cancer, Lyon, France, 2021.)

Diagnostic criteria for yolk sac tumor

Diagnostic criteria for yolk sac tumor are shown in ▶ Fig. 48.8.³ AFP (p.635) reactivity in the cytoplasm distinguishes these tumors from other germ cell tumors except for yolk sac tumors and possibly the glandular and hepatocellular components of some teratomas.

48.2.8 Choriocarcinoma

General information

Components of syncytiotrophoblast (the selective barrier between maternal and fetal blood in the placenta) are seen, characterized by giant cells that encompass cytotrophoblastic elements.

Diagnostic criteria for choriocarcinoma

Diagnostic criteria for choriocarcinoma are shown in ▶ Fig. 48.9.³

Yolk sac tumor

▷ ESSENTIAL ◁

a germ cell tumor with epithelioid cells arranged in any of the patterns described in the Histopathology subsection^a, with or without mesenchymal components

AND

absence of other germ cell tumor components

AND

AFP expression

AND

absent or only focal, non-membranous KIT expression

AND

absent or only focal CD30 expression

AND

absence of β -hCG expression

Fig. 48.8 Diagnostic criteria for yolk sac tumor.

^a For histopathology details, refer to the WHO blue book³ (Modified with permission from WHO Classification of Tumors Editorial Board, (ed.) World Health Organization classification of tumors of the central nervous system. 5th ed. Bosman F T, Jaffe E S, Lakhani S R, et al. (series eds.): World Health Organization Classification of Tumors series, vol. 6. International Agency for Research on Cancer, Lyon, France, 2021.)

48.2.9 Mixed germ cell tumor

General information

Combinations of germ cell subtypes which exhibit the same findings as the subtypes in pure form.

Diagnostic criteria for mixed germ cell tumor

Diagnostic criteria for mixed germ cell tumor are shown in ► Fig. 48.10.³

Choriocarcinoma

▷ ESSENTIAL ◁

a germ cell tumor with both syncytiotrophoblastic and cytotrophoblastic elements but no other germ cell tumor components

AND

β-hCG expression

AND

absence of KIT (or podoplanin [D2-40]) expression

AND

absence of AFP expression

AND

absent or only focal CD30 expression

AND

absence of OCT4 expression

Fig. 48.9 Diagnostic criteria for choriocarcinoma. (Modified with permission from WHO Classification of Tumors Editorial Board, (ed.) World Health Organization classification of tumors of the central nervous system. 5th ed. Bosman F T, Jaffe E S, Lakhani S R, et al. (series eds.): World Health Organization Classification of Tumors series, vol. 6. International Agency for Research on Cancer, Lyon, France, 2021.)

Mixed germ cell tumor

▷ ESSENTIAL ◁

a germ cell tumor with at least two distinct germ cell tumor subtypes

Fig. 48.10 Diagnostic criteria for mixed germ cell tumor. (Modified with permission from WHO Classification of Tumors Editorial Board, (ed.) World Health Organization classification of tumors of the central nervous system. 5th ed. Bosman F T, Jaffe E S, Lakhani S R, et al. (series eds.): World Health Organization Classification of Tumors series, vol. 6. International Agency for Research on Cancer, Lyon, France, 2021.)

References

- [1] Smith AB, Rushing EJ, Smirniotopoulos J G. Pigmented lesions of the central nervous system: radiologic-pathologic correlation. *Radiographics*. 2009; 29:1503–1524
- [2] Cajiaba MM, Benjamin D, Halaban R, et al. Metastatic peritoneal neurocutaneous melanocytosis. *Am J Surg Pathol*. 2008; 32:156–161
- [3] Bosman FT, Jaffe ES, Lakhani SR, et al. World Health Organization Classification of Tumors series, vol. 6. In: WHO Classification of Tumors Editorial Board (ed.) World Health Organization classification of tumors of the central nervous system. 5th ed. Lyon, France: International Agency for Research on Cancer; 2021
- [4] Roser F, Nakamura M, Brandis A, et al. Transition from meningeal melanocytoma to primary cerebral melanoma. Case report. *J Neurosurg*. 2004; 101:528–531

- [5] Louis DN, Ohgaki H, Wiestler OD, et al. WHO classification of tumors of the central nervous system. Lyon, France 2016
- [6] Kuo KL, Lin CL, Wu CH, et al. Meningeal Melanocytoma Associated with Nevus of Ota: Analysis of Twelve Reported Cases. *World Neurosurg.* 2019; 127:e311–e320
- [7] Savitz MH, Anderson PJ. Primary Melanoma of the Leptomeninges: A Review. *Mt Sinai J Med.* 1974; 41:774–791
- [8] Helseth A, Helseth E, Unsgaard G. Primary meningeal melanoma. *Acta Oncol.* 1989; 28:103–104
- [9] Gibson JB, Burrows D, Weir WP. Primary Melanoma of the Meninges. *J Pathol Bacteriol.* 1957; 74:419–438
- [10] Sugiyama K, Uozumi T, Kiya K, et al. Intracranial germ-cell tumor with synchronous lesions in the pineal and suprasellar regions: report of six cases and review of the literature. *Surg Neurol.* 1992; 38:114–120
- [11] Hoffman HJ, Ostubo H, Hendrick EB, et al. Intracranial Germ-Cell Tumors in Children. *Journal of Neurosurgery.* 1991; 74:545–551
- [12] Shinoda J, Yamada H, Sakai N, et al. Placental alkaline phosphatase as a tumor marker for primary intracranial germinoma. *J Neurosurg.* 1988; 68:710–720

49 Hematolymphoid Tumors Involving the CNS

49.1 CNS lymphomas

49.1.1 Primary diffuse large B-cell lymphoma of the CNS (CNS-DLBCL)

General information

Key concepts

- may be histologically identical to secondary secondary CNS spread of systemic DLBCL
- suspected with homogeneously enhancing lesion(s) in the central gray matter or corpus callosum (on MRI or CT) especially in AIDS patients
- may present with multiple cranial-nerve palsies
- diagnosis highly likely if tumor seen in conjunction with uveitis
- very responsive initially to steroids → short-lived disappearance ("ghost tumors")
- treatment: usually XRT ± chemotherapy. Role of neurosurgery usually limited to biopsy and/or placement of ventricular access reservoir for chemotherapy, and occasionally to debulking an extremely large tumor causing severe symptoms
- risk factors: immunosuppression (AIDS, transplants), Epstein-Barr virus, collagen vascular diseases

49

AKA primary CNS lymphoma. CNS involvement with lymphoma may arise primarily, or may occur secondarily from a "systemic" lymphoma invading the CNS. CNS-DLBCL is a lymphoma confined to the CNS at presentation that has histological features and many molecular findings that replicate those of systemic lymphomas. 65% are solitary lesions, with the remainder being multifocal.

Diagnostic criteria for primary diffuse large B-cell lymphoma of the CNS (CNS-DLBCL)

Diagnostic criteria for CNS-DLBCL are shown in ► Fig. 49.1.¹

Primary vs. secondary lymphoma

Secondary CNS lymphoma

Non CNS lymphoma is the fifth most common cause of cancer deaths in the U.S.; 63% of new cases are non-Hodgkin's. Secondary CNS involvement usually occurs late in the course. Metastatic spread of systemic lymphoma to the cerebral *parenchyma* occurs in 1–7% of cases at autopsy.²

Primary CNS lymphoma

Older names include: reticulum cell sarcoma and microglioma,³ since they were thought to possibly arise from microglia which were considered part of the reticuloendothelial system.

A rare, malignant primary CNS form of non-Hodgkin lymphoma comprising 0.85–2% of all primary brain tumors and 0.2–2% of malignant lymphomas.⁴ May occasionally metastasize outside the CNS.

Specific types of primary CNS lymphoma

1. diffuse large B-cell lymphoma of the CNS (CNS-DLBCL): excludes any of lymphomas listed below, as well as lymphomas with systemic involvement
2. immunodeficiency-associated CNS lymphoma: immunodeficiency (includes immunosuppression for organ transplantation, IgA deficiency, SLE...) increases the risk for lymphoma. Lymphoma associated with HIV or post-transplant immunosuppression is usually EBV-related
 - a) AIDS-related diffuse large B-cell lymphoma
 - b) EBV + diffuse large B-cell lymphoma, NOS
 - c) lymphomatoid granulomatosis
3. intravascular large B-cell lymphoma
4. miscellaneous rare lymphomas in the CNS
 - a) low-grade B-cell lymphomas

Primary diffuse large B-cell lymphoma of the CNS

▷ ESSENTIAL ◁

biopsy-proven mature large B-cell lymphoma confined to the CNS at presentation

AND

expression of one or more B-cell markers (CD20, CD19, CD22, CD79a, PAX5)

▷ DESIRABLE ◁

- ▶ immunohistochemical phenotype of late germinal-center exit B cells (IRF4 [MUM1]+, BCL6+/-, CD10-); CD10 expression does not exclude the diagnosis, but it is uncommon and indicates possible systemic DLBCL
- ▶ immunohistochemical positivity for BCL2 and MYC
- ▶ absence of EBV-associated markers (in > 97% of cases)
- ▶ molecular detection of a clonal B-cell population in cases in which histology is not definitive, such as corticoid-mitigated CNS-DLBCL

49

Fig. 49.1 Diagnostic criteria for CNS-DLBCL. (Modified with permission from WHO Classification of Tumors Editorial Board, (ed.) World Health Organization classification of tumors of the central nervous system. 5th ed. Bosman F T, Jaffe E S, Lakhani S R, et al. (series eds.): World Health Organization Classification of Tumors series, vol. 6. International Agency for Research on Cancer, Lyon, France, 2021.)

- b) T-cell and NK/T-cell lymphomas
- c) anaplastic large cell lymphoma
5. extranodal marginal zone lymphoma of mucosa associated lymphoid tissue (MALT lymphoma) of the dura

Epidemiology

Incidence of primary CNS lymphoma (PCNSL) in the U.S.: 0.44 per 100,000,⁵ making it the third most common of malignant primary brain tumors (behind GBM & diffuse astrocytoma). In the past 20 years, incidence has increased in patients > 60 years of age.⁶

Male:female ratio = 1.22:1.⁵

Median age at diagnosis: 52 yrs⁷ (younger among immunocompromised patients: ~ 34 yrs).

Most common supratentorial locations: frontal lobes, then deep nuclei; periventricular also common. Infratentorially: cerebellum is the most common location.

Conditions with increased risk of primary CNS lymphomas (PCNSL)

1. collagen vascular disease
 - a) systemic lupus erythematosus
 - b) Sjögren's syndrome: an autoimmune connective tissue disorder
 - c) rheumatoid arthritis
2. immunosuppression
 - a) organ transplant recipients: related to chronic immunosuppression. Falls under category of post transplant lymphoproliferative disease (PTLD) (p.845)⁸
 - b) AIDS^{9,10}: CNS lymphoma occurs in ~ 10% of AIDS patients, and is the first presentation in 0.6%
 - c) severe-congenital immunodeficiency syndrome ("SCIDS")
 - d) possibly increased incidence in the elderly due to reduced competency of immune system

- Epstein-Barr virus¹¹ is associated with a broad spectrum of lymphoproliferative disorders, and is detectable in $\approx 30\text{--}50\%$ of systemic lymphomas; however, it has been associated with almost 100% of PCNSL,¹² especially AIDS-related cases^{13(p 317)}

Presentation

General information

Presentation is similar with primary or secondary CNS lymphoma: the two most common neurologic manifestations are those due to epidural spinal cord compression and those of carcinomatous meningitis (multiple cranial nerve deficits) (p.920). Seizures occur in up to 30% of patients.¹⁴ Most lymphomas in patients first presenting with neurologic symptoms are PCNSL.¹⁵

Neurologic symptoms

- presents with non-focal non-specific symptoms in over 50% of patients; at time of presentation most commonly includes:
 - mental status changes in one-third
 - symptoms of increased ICP (H/A, N/V)
 - generalized seizures in 9%
- focal symptoms in 30–42% of cases:
 - hemimotor or hemisensory symptoms
 - partial seizures
 - multiple cranial-nerve palsies (due to carcinomatous meningitis)
- combination of focal and non-focal symptoms

Signs

- non-focal in 16%:
 - papilledema
 - encephalopathy
 - dementia
- focal findings in 45% of cases:
 - hemimotor or hemisensory deficits
 - aphasia
 - visual field deficits
- combination of focal and non-focal signs

Uncommon but characteristic syndromes

- uveocyclitis, coincident with (in 6% of cases) or preceding the diagnosis of (in 11% of cases) lymphoma
- subacute encephalitis with subependymal infiltration
- MS-like illness with steroid-induced remission

Histology

Characteristic sites: corpus callosum, basal ganglia, periventricular.

The neoplastic cells are identical to those of systemic lymphomas. Most are bulky tumors that are contiguous with the ventricles or meninges.

Histologic distinguishing features: tumor cells form cuffs around blood vessels which demonstrate multiplication of basement membranes (best demonstrated with silver reticulum stain).

Frozen section distorts the cells and may lead to a misdiagnosis of malignant glioma.^{13(p 320)} Immunohistochemical stains differentiate B-cell lymphomas from T-cell lymphomas (B-cell types are more common, especially in PCNSL and in AIDS).

EM shows absence of junctional complexes (desmosomes) that are usually present in epithelial derived tumors.

Intravascular lymphomatosis¹⁶: Formerly: (malignant) angioendotheliomatosis. A rare lymphoma with no solid mass in which malignant lymphoid cells are found in the lumen of small blood vessels in affected organs. CNS involvement is reported in most cases. Presentation is non specific: patients are often febrile, and may present with progressive multifocal cerebrovascular events (including stroke or hemorrhage), spinal cord or nerve root symptoms including cauda equina syndrome (p. 1254), encephalopathy, or peripheral or cranial neuropathies.¹⁷ Initial transient cerebral symptoms may mimic TIAs or seizures. The ESR is often elevated prior to initiation of steroids. Lymphoma cells may be seen in the CSF.

Painful skin nodules or plaques occur in $\approx 10\%$ of cases, generally involving the abdomen or lower extremities, and these cases may be diagnosed with skin biopsy (differential diagnosis here includes angioendotheliomatosis, a benign capillary, and endothelial cell disorder). Otherwise, diagnosis often requires brain biopsy (open or stereotactic), in which involved areas on imaging studies are targeted. Pathology: malignant lymphoid cells distend and occlude small arteries, veins and capillaries with little or no parenchymal extension.^{13(p 324)} Treatment with combination chemotherapy can result in long-term remission in some patients, but early diagnosis before permanent damage occurs is critical (diagnosis is rarely made pre-mortem).

Evaluation

All patients should be assessed (history, physical and laboratory tests as appropriate) for any of the conditions associated with lymphoma (p.841). Since PCNSL is less common than secondary involvement, any patient with CNS lymphoma should have work-up for occult systemic lymphoma which may include:

1. careful physical exam of all lymph nodes (LN)
2. early detection of HIV and reflex CD4+ count nadir are important because HIV increases the risk of PCNSL and infection
3. contrast CT of chest, abdomen & pelvis or whole body PET/CT (evaluation of perihilar and pelvic LN)
4. routine blood (CBC, chemistries) and urine testing
5. LP for CSF (omit if not safe: anticoagulation, thrombocytopenia or intracranial mass effect): for routine tests (cell count, protein/glucose, culture) & cytology, flow cytometry
6. bone marrow biopsy
7. MRI of the entire spine
8. testicular ultrasound in males > 60 years of age (omit if PET scan is done and is negative)
9. ophthalmologic examination (including slit-lamp evaluation of both eyes) in all
 - a) for possible uveitis
 - b) $\approx 28\%$ of patients with PCNSL will also have intraocular lymphoma. Often resistant to methotrexate, but responds to low dose ocular XRT (7–8 Gy)

Diagnostic tests

CT & MRI

► **Findings common to CT/MRI.** On imaging (CT or MRI) 50–60% occur in one or more cerebral lobes (in gray or white matter). 25% occur in deep midline structures (septum pellucidum, basal ganglion, corpus callosum). 25% are infratentorial. 10–30% of patients have multiple lesions at the time of presentation. In contrast, systemic lymphomas that spread to the CNS tend to present with leptomeningeal involvement instead of parenchymal tumors.¹⁸

Almost all PCNSLs enhance (except only 1.1% in immune intact, and 3.2% in immune compromised⁵).

Non-AIDS-related cases tend to enhance homogeneously, whereas AIDS-related cases are more likely to be ring-enhancing (necrotic center) and *multifocal*.^{15,19}

Non-AIDS-related cases: CNS lymphomas should be suspected with homogeneously enhancing lesion(s) in the central gray or corpus callosum. 75% are in contact with ependymal or meningeal surfaces (this together with dense enhancement may produce a “pseudomeningioma pattern”; however, lymphomas lack calcifications and are more likely to be multiple).

► **CT (see also “Findings common to CT/MRI” above).** 60% are hyperdense to brain, only 10% are hypodense. Characteristically, >90% of these tumors enhance; this is densely homogeneous in over 70%. As a result, when rare non-enhancing cases occur it often leads to a delay in diagnosis.²⁰ The appearance of enhanced PCNSL on CT has been likened to “fluffy cotton balls.” There may be surrounding edema²¹ and there is usually mass effect.

There is an almost diagnostic tendency of rapid partial to complete resolution on CT (and even at the time of surgery) following the administration of steroids, earning the nickname of “ghost-cell tumor”^{15,22,23} or disappearing tumor.

► **MRI (see also “Findings common to CT/MRI” above).** No pathognomonic feature. Bright on DWI (restricted diffusion) in most cases, isointense to hypointense on ADC map. May be difficult to discern if tumor is located subependymally (signal characteristics similar to CSF); proton-weighted image may avoid this limitation.

Serologic tests

There are no blood tests that are diagnostic for PCNSL.

Serum lactate dehydrogenase (LDH) elevation indicates rapid cell turnover and is an independent poor prognosticator for non-Hodgkin, follicular and mantle cell lymphomas. In the absence of the diagnosis of lymphoma, this is nonspecific (occurs in liver failure, infection...).¹⁵

CSF studies

LP should only be performed if no mass effect. Some abnormality of routine CSF labs occurs in >80%, but are non-specific. Most common abnormalities are elevated protein (in 45–67%), and increased cell count (in 40%). PCNSL can lower CSF glucose.¹⁵

Cytology is positive for lymphoma cells (preoperatively) in only 2–32%¹⁵ (sensitivity may be higher with leptomeningeal involvement as in non-AIDS patients than with parenchymal involvement commonly seen in AIDS). Sensitivity is higher when ≥ 10.5 ml of CSF is analyzed.¹⁵ Repeating up to 3 LPs may increase yield. The morphology of lymphoma cells can sometimes be difficult to distinguish from inflammatory lymphocytes, but sharp nuclear notches, cytoplasmic irregularity, and increased cell size (>2.5 times the upper limit of normal) are suggestive of lymphoma. Although the diagnosis of lymphoma can be made from CSF, cells obtained in this manner are not adequate for detailed tissue typing which is possible with solid tissue from biopsy, ∴ LP alone rarely suffices.

Flow cytometry provides information about the immunophenotype of lymphocytes.

Immunoglobulin analysis can show a single elevated immunoglobulin heavy chain (IgH) band with lymphoma, compared to a wide band produced by lymphocytes involved in an inflammatory process.¹⁵

Proteomics have shown elevated levels of antithrombin III in the CSF. A cutoff of levels >1.2 mcg/dl is 75% sensitive and 98.7% specific.²⁴

49

Angiography

Rarely helpful. 60% of cases show only an avascular mass. 30–40% show diffuse homogeneous staining or blush.

Treatment

Surgery

Surgical decompression with partial or gross total removal does not alter patient's prognosis. The main indication for surgery:

- biopsy: obtain solid tissue to ascertain that the tumor is a lymphoma, and to determine the type of lymphoma. Stereotactic techniques are often well-suited for these often deep tumors²⁵

Radiation therapy

The standard treatment after tissue biopsy is whole-brain radiation therapy. Doses used tend to be lower than for other primary brain tumors. ≈ 40–50 Gy total are usually given in 1.8–3 Gy daily fractions.

Chemotherapy

General information

In *non-AIDS* cases, chemotherapy combined with XRT prolongs survival compared to XRT alone.²⁶

Methotrexate (MTX)

The addition of intraventricular MTX (rather than just intrathecal via LP) delivered through a ventricular access device (6 doses of 12 mg twice a week, with IV leucovorin rescue) may result in even better survival.²⁷ In the event of an intrathecal MTX overdose (OD), interventions recommended²⁸: ODs of up to 85 mg can be well tolerated with little sequelae; immediate LP with drainage of CSF can remove a substantial portion of the drug (removing 15 ml of CSF can eliminate ≈ 20–30% of the MTX within 2 hrs of OD). This can be followed by ventriculolumbar perfusion over several hours using 240 ml of warmed isotonic preservative-free saline entering through the ventricular reservoir and exiting through a lumbar subarachnoid catheter. For major OD of >500 mg, add intrathecal administration of 2,000 U of carboxypeptidase G2 (an enzyme that inactivates MTX). In cases of MTX OD, systemic toxicity should be prevented by treating with IV dexamethasone and IV (not IT) leucovorin.

Rituximab

Available since 1997 for treatment of refractory systemic B-cell non-Hodgkins lymphoma. Intrathecally, may be more effective for CD33+ lymphomas.

Prognosis

With no treatment, median survival is 1.8–3.3 months following diagnosis.

With radiation therapy,¹⁴ median survival is 10 months, with 47% 1-year median survival, and 16% 2-year median survival. 3-year survival is 8%, and 5-year survival is 3–4%. With intraventricular MTX, median time to recurrence was 41 mos.²⁷ Occasionally, prolonged survival may be seen.²⁹

About 78% of cases recur, usually $\approx$ 15 months after treatment (late recurrences also are seen). Of these recurrences, 93% are confined to the CNS (often at another site if the original site responded well), and 7% are elsewhere.

Although there are individual studies that show trends, there are no prognostic features that consistently correlate with survival.

49.1.2 Immunodeficiency-associated CNS lymphomas (IDA-CNSL)

General information

A family of CNS lymphomas occurring in patients with inherited or acquired immunodeficiency (HIV/AIDS, immunosuppression...). The number of cases related to AIDS has declined due to the effectiveness of newer anti-retroviral therapy. Most immunodeficiency-associated lymphomas are Epstein-Barr virus (EBV) related. EBV-positive DLBCL of the elderly is coupled with immunosenescence and starts to ramp up after age 50 years.³⁰

Imaging demonstrates multiplicity of lesions and necrotic areas more frequently than in immunocompetent patients.

Diagnostic criteria for immunodeficiency-associated CNS lymphomas (IDA-CNSL)

Diagnostic criteria for IDA-CNSL are shown in ► Fig. 49.2.¹

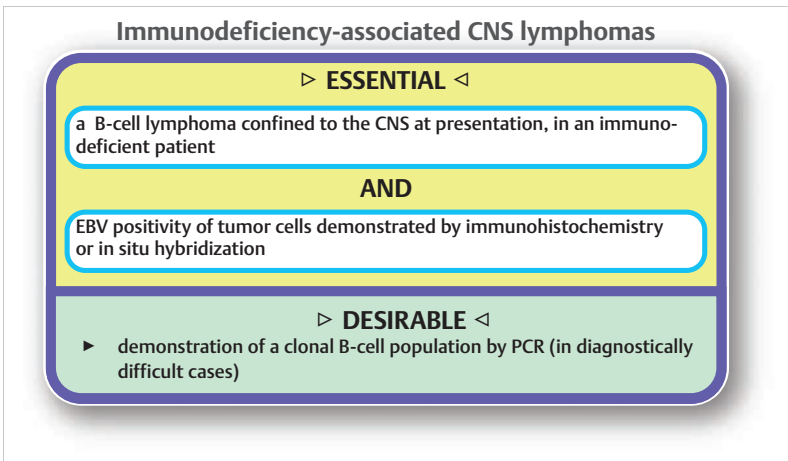


Fig. 49.2 Diagnostic criteria for IDA-CNSL. (Modified with permission from WHO Classification of Tumors Editorial Board, (ed.) World Health Organization classification of tumors of the central nervous system. 5th ed. Bosman F T, Jaffe E S, Lakhani S R, et al. (series eds.): World Health Organization Classification of Tumors series, vol. 6. International Agency for Research on Cancer, Lyon, France, 2021.)

Prognosis

The prognosis is worse than CNS-DLBCL. Although complete remission occurs in 20–50% following XRT, the median survival is only 3–5 months,^{31,32} usually related to AIDS-related opportunistic infection. However, neurologic function and quality of life improve in $\approx$ 75%.³¹

49.2 Histiocytic tumors

49.2.1 General information

Histiocytic disorders may be classified as follows:

1. malignant (true histiocytic lymphoma)
2. reactive (benign histiocytosis)
3. Langerhans cell histiocytosis (LCH)
 - a) **unifocal**: formerly eosinophilic granuloma and/or histiocytosis X. Rare (1/1200 new cases/yr in US). More common in children. Slowly progressing disease. May occur in bone, skin, lungs, or stomach (see below)
 - b) **multifocal unisystem**: seen mostly in children. Fever, bone & skin lesions
 - c) **multifocal multisystem**: formerly Letterer-Siwe disease (a fulminant, malignant lymphoma of infancy).³³ Hand-Schüller-Christian triad: DI (from invasion of pituitary stalk), exophthalmos (from intraorbital tumor), and lytic bone lesions (particularly of cranium).
4. Erdheim-Chester disease
5. Rosai-Dorfman disease
6. juvenile xanthogranuloma
7. histiocytic sarcoma

49.2.2 Langerhans cell histiocytosis of the CNS or meninges

General information

Formerly histiocytosis X (or eosinophilic granuloma, when solitary). A clonal proliferation of Langerhans-type cells involving the CNS, skull or meninges, with or without extra-CNS lesions. 36% of patients develop a leukoencephalopathy pattern, and 8% demonstrate cerebral atrophy.

Diagnostic criteria for Langerhans cell histiocytosis

Diagnostic criteria for Langerhans cell histiocytosis are shown in ► Fig. 49.3.¹

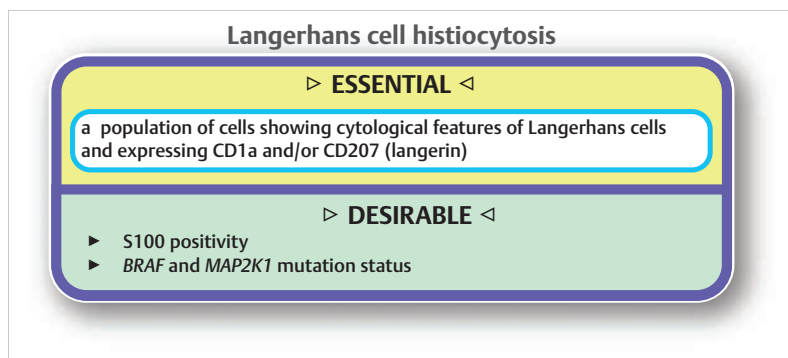


Fig. 49.3 Diagnostic criteria for Langerhans cell histiocytosis. (Modified with permission from WHO Classification of Tumors Editorial Board, (ed.) World Health Organization classification of tumors of the central nervous system. 5th ed. Bosman F T, Jaffe E S, Lakhani S R, et al. (series eds.): World Health Organization Classification of Tumors series, vol. 6. International Agency for Research on Cancer, Lyon, France, 2021.)

Location

There is a predilection for the craniofacial bones and skull base (56%) favoring the parietal and frontal bones.³³ Intracranial involvement may include the hypothalamic-pituitary region (25-50%), meninges (30%) and choroid plexus (6%).¹

Epidemiology

Generally a condition of youth, 70% of patients are <20 yrs of age. In a series of 26 patients,³³ age range was 18 mos–49 yrs (mean: 16 yrs). The incidence among patients <15 years old is 0.5 per 100,000, with a M:F ratio of $\approx 1:2$.³⁴

Clinical

Most common presenting symptom: *tender*, enlarging skull mass (>90%). May be asymptomatic and incidentally discovered on skull X-ray obtained for other reasons. Blood tests were normal in all except 1 who had eosinophilia of 23%.

Evaluation

Skull X-rays

Classic radiographic finding: round or oval non-sclerotic punched out skull lesion with sharply defined margins, involving both inner and outer tables (the disease begins in diploic space), often with bevelled edges. A central bone density is occasionally noted (rare, but diagnostic). No abnormal vascularity of adjacent bone. No periosteal reaction. Differentiate from hemangioma by absence of sunburst appearance.

CT scan

Characteristic appearance of a soft tissue mass within area of bony destruction having a central density.³⁵ Differentiate from epidermoid, which has dense surrounding sclerosis.

Pathology

Gross: pinkish gray to purple lesion extending out of bone and involving pericranium. Dural involvement occurs in only 1 of 26 patients, but with no dural penetration.

Microscopic: numerous histiocytes, eosinophils, and multinucleated cells in a reticulin fiber network. No evidence that this is a result of an infection.

► **Molecular biology.** The most frequent molecular correlate is BRAF p.V600E mutation, seen in $\approx 50\%$.¹

Treatment

There is a tendency toward spontaneous regression; however, most single lesions are treated by curettage. Multiple lesions are usually associated with extracalvarial bony involvement and are often treated with chemotherapy and/or low dose radiation therapy. Very radiosensitive.

Outcome

After a mean 8 years follow-up, 8 patients (31%) developed additional lesions, 5 of these were ≤ 3 yrs of age (all of 5 patients <3 yrs of age)³³ (may suggest a form of multifocal LCH, thus young patients should be followed closely). Recurrences were local in one case, and in others involved other bones (including the skull, femur, lumbar spine) or brain (including the hypothalamus, presenting with diabetes insipidus and growth delay).

References

- [1] Bosman FT, Jaffe ES, Lakhani SR, et al. World Health Organization Classification of Tumors series, vol. 6. In: WHO Classification of Tumors Editorial Board (ed.) World Health Organization classification of tumors of the central nervous system. 5th ed. Lyon, France: International Agency for Research on Cancer; 2021
- [2] Jellinger KA, Radaszkiewicz T. Involvement of the Central Nervous System in Malignant Lymphomas. *Virchows Arch (Pathol Anat)*. 1976; 370:345–362
- [3] Helle TL, Britt RH, Colby TV. Primary Lymphoma of the Central Nervous System. *J Neurosurg*. 1984; 60:94–103
- [4] Alic L, Haid M. Primary Lymphoma of the Brain: A Case Report and Review of the Literature. *J Surg Oncol*. 1984; 26:115–121
- [5] Ostrom QT, Gittleman H, Liao P, et al. CBTRUS Statistical Report: Primary brain and other central nervous system tumors diagnosed in the United States in 2010–2014. *Neuro Oncol*. 2017; 19:v1–v88
- [6] Villano JL, Koshy M, Shaikh H, et al. Age, gender, and racial differences in incidence and survival in primary CNS lymphoma. *Br J Cancer*. 2011; 105:1414–1418
- [7] Murray K, Kun L, Cox J. Primary Malignant Lymphoma of the Central Nervous System: Results of Treatment of 11 Cases and Review of the Literature. *J Neurosurg*. 1986; 65:600–607
- [8] Penn I. Development of Cancer as a Complication of Clinical Transplantation. *Transplant Proc*. 1977; 9:1121–1127

- [9] Levy RM, Bredesen DE, Rosenblum ML. Neurological manifestations of the acquired immunodeficiency syndrome (AIDS): Experience at UCSF and review of the literature. *J Neurosurg.* 1985; 62:475-495
- [10] Jean WC, Hall WA. Management of Cranial and Spinal Infections. *Contemporary Neurosurgery.* 1998; 20:1-10
- [11] Hochberg FH, Miller G, Schooley RT, et al. Central-Nervous-System Lymphoma Related to Epstein-Barr Virus. *N Engl J Med.* 1983; 309:745-748
- [12] MacMahon EME, Glass JD, Hayward SD, et al. Epstein-Barr Virus in AIDS-Related Primary Central Nervous System Lymphoma. *Lancet.* 1991; 338:969-973
- [13] Burger PC, Scheithauer BW, Vogel FS. *Surgical Pathology of the Nervous System and Its Coverings.* 4th ed. New York: Churchill Livingstone; 2002
- [14] O'Neill BP, Illig JJ. Primary Central Nervous System Lymphoma. *Mayo Clinic Proceedings.* 1989; 64:1005-1020
- [15] Scott BrianJ, Douglas VanjaC, Tihan Tarik, et al. A Systematic Approach to the Diagnosis of Suspected Central Nervous System Lymphoma. *JAMA neurology.* 2013; 70:311-319
- [16] Calamia KT, Miller A, Shuster EA, et al. Intravascular Lymphomatosis: A Report of Ten Patients with Central Nervous System Involvement and a Review of the Disease Process. *Advances in Experimental Medicine and Biology.* 1999; 455:249-265
- [17] Glass J, Hochberg FH, Miller DC. Intravascular Lymphomatosis. A Systemic Disease with Neurologic Manifestations. *Cancer.* 1993; 71:3156-3164
- [18] So YT, Beckstead JH, Davis RL. Primary central nervous system lymphoma in acquired immune deficiency syndrome: A clinical and pathological study. *Ann Neurol.* 1986; 20:566-572
- [19] Poon T, Matoso I, Tcherkoff V, et al. CT features of primary cerebral lymphoma in AIDS and non-AIDS patients. *Journal of Computer Assisted Tomography.* 1989; 13:6-9
- [20] DeAngelis LM. Cerebral Lymphoma Presenting as a Nonenhancing Lesion of Computed Tomographic/Magnetic Resonance Scan. *Annals of Neurology.* 1993; 33:308-311
- [21] Enzmann DR, Krikorian J, Norman D, et al. Computed Tomography in Primary Reticulum Cell Sarcoma of the Brain. *Radiology.* 1979; 130:165-170
- [22] Vaquero J, Martinez R, Rossi E, et al. Primary Cerebral Lymphoma: the 'Ghost Tumor'. *J Neurosurg.* 1984; 60:174-176
- [23] Gray RS, Abrahams JJ, Hufnagel TJ, et al. Ghost-cell tumor of the optic chiasm; primary CNS lymphoma. *Journal of Clinical Neuro-Ophthalmology.* 1989; 9:98-104
- [24] Roy S, Josephson SA, Fridlyand J, et al. Protein biomarker identification in the CSF of patients with CNS lymphoma. *J Clin Oncol.* 2008; 26:96-105
- [25] O'Neill BP, Kelly PJ, Earle JD, et al. Computer-Assisted Stereotactic Biopsy for the Diagnosis of Primary Central Nervous System Lymphoma. *Neurology.* 1987; 37:1160-1164
- [26] DeAngelis LM, Yahalom J, Heinemann M-H, et al. Primary Central Nervous System Lymphomas: Combined Treatment with Chemotherapy and Radiotherapy. *Neurology.* 1990; 40:80-86
- [27] DeAngelis LM, Yahalom J, Thaler HT, et al. Combined Modality Therapy for Primary CNS Lymphomas. *J Clin Oncol.* 1992; 10:635-643
- [28] O'Marcaigh AS, Johnson CM, Smithson WA, et al. Successful Treatment of Intrathecal Methotrexate Overdose by Using Ventriculolumbar Perfusion and Intrathecal Instillation of Carboxypeptidase G2. *Mayo Clinic Proceedings.* 1996; 71:161-165
- [29] Hochberg FH, Miller DC. Primary Central Nervous System Lymphoma. *J Neurosurg.* 1988; 68:835-853
- [30] Oyama T, Yamamoto K, Asano N, et al. Age-related EBV-associated B-cell lymphoproliferative disorders constitute a distinct clinicopathologic group: a study of 96 patients. *Clin Cancer Res.* 2007; 13:5124-5132
- [31] Baumgartner JE, Rachlin JR, Beckstead JH, et al. Primary Central Nervous System Lymphomas: Natural History and Response to Radiation Therapy in 55 Patients with Acquired Immunodeficiency Syndrome. *Journal of Neurosurgery.* 1990; 73:206-211
- [32] Formenti SC, Gill PS, Lean E, et al. Primary Central Nervous System Lymphoma in AIDS: Results of Radiation Therapy. *Cancer.* 1989; 63:1101-1107
- [33] Rawlings CE, Wilkins RH. Solitary Eosinophilic Granuloma of the Skull. *Neurosurgery.* 1984; 15:155-161
- [34] Guyot-Goubin A, Donadieu J, Barkaoui M, et al. Descriptive epidemiology of childhood Langerhans cell histiocytosis in France, 2000-2004. *Pediatr Blood Cancer.* 2008; 51:71-75
- [35] Mitnick JS, Pinto RS. CT in the Diagnosis of Eosinophilic Granuloma. *J Comput Assist Tomogr.* 1980; 4:791-793

50 Tumors of the Sellar Region

50.1 General information

The pituitary gland is composed of the adenohypophysis, a neuroendocrine structure with 6 hormone secreting cell types, and the neurohypophysis which is comprised of pituicytes. Tumors can arise from these structures, and other tumors can occur in this region including those from rests of oral ectoderm (craniopharyngiomas) as well as meningiomas, germ cell tumors, chordomas and metastases.

50.2 Sellar region tumors of non pituitary origin

50.2.1 Craniopharyngiomas

Key concepts

- epithelial tumors thought to arise from residual cells of Rathke's pouch (p. 151)
- histologically benign, but the location and propensity to invade critical neural structures makes them malignant in behavior
- initially regarded as 1 tumor with two subtypes, now considered to be two distinct tumors which may be differentiated by gene mutations:
 1. adamantinomatous craniopharyngioma: 95% show *CTNNB1* mutations and aberrant nuclear expression of beta-catenin. Bimodal distribution: 5–15 years & 45–60 years
 2. papillary craniopharyngioma: 1–95% of cases show BRAF V600E mutations. Mean age of 40–55 years.

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Historically, craniopharyngioma was considered a tumor with 2 subtypes, adamantinomatous and papillary. It is now recognized that, although both of these tumors arise in the same region and display squamous lineage, that they are actually distinct tumors with differing epidemiology, radiology, histology, and molecular genetics.¹

They are thought to develop from residual cells of Rathke's pouch (p. 151) (craniopharyngeal duct).

CP do not undergo malignant degeneration, but the fact that they may infiltrate neural tissue can make them difficult to cure (benign in histology, malignant in behavior). CP are distinct from Rathke's cleft cyst (p. 949), but share some similarities.

Epidemiology

0.8% of all brain tumors, with an incidence of 0.19 per 100,000 in the US.² They comprise 5–11% of pediatric brain tumors, making them the most common non-neuroepithelial intracerebral tumor in children.

50.2.2 Adamantinomatous craniopharyngioma (ACP) (WHO grade 1)

General information

A mixed solid and cystic squamous epithelial tumor with stellate reticulum, wet keratin and basal palisades. Up to 95% of cases of this variant shows *CTNNB1* mutations and aberrant nuclear expression of beta-catenin.³ Tend to arise from the anterior superior margin of the pituitary. Fluid in the cysts varies, but usually contains cholesterol crystals.

Diagnostic criteria for adamantinomatous craniopharyngioma

Diagnostic criteria for adamantinomatous craniopharyngioma are shown in ► Fig. 50.1.⁴

Epidemiology

Bimodal age distribution: childhood peak age 5–15 years, adult peak age 45–60 years.³

Adamantinomatous craniopharyngioma

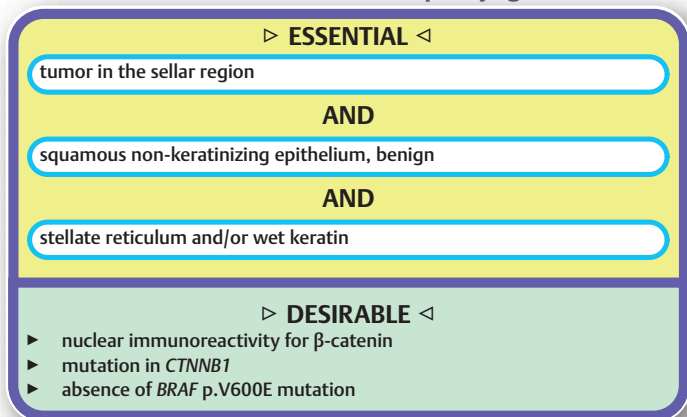


Fig. 50.1 Diagnostic criteria for adamantinomatous craniopharyngioma. (Modified with permission from WHO Classification of Tumors Editorial Board, (ed.) World Health Organization classification of tumors of the central nervous system. 5th ed. Bosman F T, Jaffe E S, Lakhani S R, et al. (series eds.): World Health Organization Classification of Tumors series, vol. 6. International Agency for Research on Cancer, Lyon, France, 2021.)

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Location

Most occur in the region of the sella turcica and pituitary infundibulum. Purely sellar lesions are rare, with $\approx 95\%$ extending into the suprasellar compartment. Pure involvement of the 3rd ventricle is more common with papillary CP.

Imaging

Imaging mnemonic: “90% rule” 90% of adamantinomatous craniopharyngiomas exhibit at least 2 of the following “C” features: cyst formation, prominent calcifications, and contrast enhancement in the cyst walls.⁵

50.2.3 Papillary craniopharyngioma (RCP) (WHO grade 1)

General information

A craniopharyngioma with papillary features that occurs primarily in adults. 81–95% of cases show *BRAF* V600E mutations.³ Usually solid, or partially cystic.

Diagnostic criteria for papillary craniopharyngioma

Diagnostic criteria for papillary craniopharyngioma are shown in ► Fig. 50.2.⁴

Location

Typically arises in the infundibular recess and tuber cinereum of the floor of the 3rd ventricle. Rarely occurs intrasellar.

Epidemiology

Comprise 1.2–4.6% of intracranial tumors, with an incidence of 0.5–1.5 cases per 1 million.⁴ They are less common than adamantinomatous CP, constituting only $\approx 10\%$ of all CP, but 12–33% in adults.⁴

Occurs almost exclusively in adults with a mean age of 40–55 years.

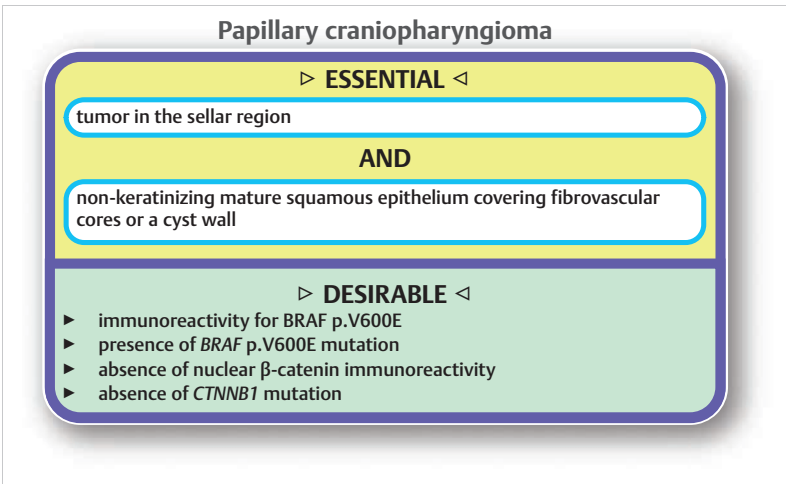


Fig. 50.2 Diagnostic criteria for papillary craniopharyngioma. (Modified with permission from WHO Classification of Tumors Editorial Board, (ed.) World Health Organization classification of tumors of the central nervous system. 5th ed. Bosman F T, Jaffe E S, Lakhani S R, et al. (series eds.): World Health Organization Classification of Tumors series, vol. 6. International Agency for Research on Cancer, Lyon, France, 2021.)

Imaging

Less frequently cystic than adamantinomatous CP. Usually spherical (c.f., lobulated) and infrequently contain calcifications. The pituitary stalk is often thickened.

50.2.4 Vascular anatomy

Arterial supply: usually small feeders from ACA and A-comm, or from ICA and P-comm (do not receive blood from PCA or BA-bifurcation unless blood supply of the floor of the third ventricle is parasitized).

50.2.5 Treatment options

GTR is associated with better PFS than STR alone, but the advantage of GTR over STR followed by adjuvant XRST is not as clear.⁴

BRAF and/or MEK inhibitors show promise in BRAF p.V600E–mutant papillary CP.⁶

50.2.6 Surgical treatment

Pre-op endocrinologic evaluation

As for pituitary tumor (p.876). Hypoadrenalism may be corrected rapidly, but hypothyroidism takes longer; either condition can increase surgical mortality.

Approach

Usually via large right frontotemporal flap as low as possible along base of frontal fossa (lateral sphenoid wing rongeuired/drilled). Approach to tumor is extra-axial, whether subfrontal or frontotemporal. All tumors should be aspirated (even if they appear solid radiographically). Then, with microscope, possible approaches include:

1. subchiasmatic: through space between optic nerves and anterior to chiasm. It was thought that a “prefixed chiasm” (i.e., congenitally short optic nerves with chiasm unusually close to the planum sphenoidale) was more common in patients with CP, making this approach more difficult.

However, in reality the chiasm is probably bowed anteriorly by the tumor within the third ventricle, giving the illusion of a prefixed chiasm in most cases

2. opticocarotid (between right ICA and right optic nerve/tract)
3. lamina terminalis (tumor often needs to be brought down and removed subchiasmatically)^{7,8}
4. lateral to carotid artery
5. transfrontal-transsphenoidal: drill off tuberculum sellae

Alternative approaches to frontotemporal

1. pure transsphenoidal: if dark fluid is aspirated with no CSF evident, it is possible to leave a stent from the tumor cavity to the sphenoid air-sinus to permit continued drainage
2. transcallosal: strictly for tumors limited to the third ventricle
3. a combined subfrontal/pterional approach capitalizes on the advantages of each (head is positioned with slight lateral rotation)
4. extended transnasal transsphenoidal endoscopic resection^{9,10}

Spare the following structures: small arterial feeders to *undersurface* of the chiasm (major supply) and tract; at least a remnant of pituitary stalk (recognized by unique pattern of longitudinal striations which are the long portal veins). If the tumor easily pulls down from above then this is permissible; however, do not pull too hard, or else hypothalamic injury may result.

Post-op

1. steroids: these patients are all considered hypo-adrenal. Give hydrocortisone in physiologic doses (for mineralocorticoid activity) in addition to dexamethasone (glucocorticoid that treats edema) taper. Taper steroids slowly to avoid aseptic (chemical) meningitis
2. diabetes insipidus (DI) (p.902): often shows up early. May be part of a "triphasic response." (p. 125) Best managed initially with fluid replacement. If necessary, use short acting vasopressin (prevents iatrogenic renal shutdown if an SIADH-like phase develops during vasopressin therapy)

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50.2.7 Radiation

Controversial. **Side effects:** include endocrine dysfunction, optic neuritis, dementia. Post-op XRT probably helps prevent regrowth when residual tumor is left behind¹¹; however, in pediatric cases it may be best to postpone XRT (to minimize deleterious effect on IQ), recognizing that reoperation may be necessary for recurrence.

50.2.8 Outcome

Most outcome data are derived from CP as a monolithic entity, and since papillary CP make up only a minority of CPs, their outcome data is sparse. It has been asserted that papillary CP have a better prognosis, but incontrovertible proof of this is lacking.

5–10% mortality in most series, most from hypothalamic injury (unilateral hypothalamic lesions are rarely clinically evident; bilateral injuries may produce hyperthermia and somnolence; damage to anterior osmoreceptors may → loss of thirst sensation).

Five year survival is ≈ 55–85% (range from 30–93% has been reported).

50.2.9 Recurrence

Most recurrences are in <1 year, few >3 yrs (very delayed recurrence usually follow what was thought to be "total" removal). Morbidity/mortality is higher with reoperation.

50.3 Tumors of the neurohypophysis & infundibulum

50.3.1 General information

Description

This group of rare low-grade tumors that originate from pituicytes of the neurohypophysis (posterior pituitary) and infundibulum are probably a spectrum of a single entity, the members of the group being:

1. pituicytoma
2. granular cell tumor of the sellar region (GCT)

3. spindle cell oncocytoma (SCO)

These tumors all show nuclear expression of thyroid transcription factor 1 (TTF1). They are usually misidentified as PitNET/adenomas on imaging, and are rarely considered in the preoperative differential diagnosis.

Epidemiology

The rarity of these tumors has prevented robust epidemiologic analysis. A literature meta-analysis of 270 cases¹² found a mean age of 48 ± 22 years.

Location

These tumors originate in the posterior pituitary or infundibulum, and therefore their location is purely sellar (25%), sellar with suprasellar extension (48%), or suprasellar (23%).¹² Spindle cell oncocytoma can be more invasive (see below).

Presentation

Symptoms from mass effect are similar to other tumors of this location: visual field defects (in 58%) (classically bitemporal hemianopsia), hypopituitarism (34%), and H/A (40%).¹² Diabetes insipidus occurs infrequently (2.4%).

Associated endocrinologic overproduction syndromes include hypercortisolism (4.1%), and acromegaly (2.0%), some of which may be due to synchronous functional PitNET/adenomas.¹³

Imaging

These tumors may appear radiographically identical to adenomas. They are rarely considered in the differential diagnosis pre-op. Isodense on CT and isointense on T1WI MRI, with dense homogeneous enhancement on CT and MRI.

Treatment

GTR frequently cures these tumors, although SCO appears to have a higher recurrence rate. If the diagnosis is suspected pre-op, a transcranial approach is preferred over transsphenoidal because of the vascularity which has prevented total resection in 60–70% of reported cases.¹⁴

XRT may be considered for subtotal resection.¹⁵

50.3.2 Pituicytoma (WHO grade 1)

General information

Rare (mostly case reports). Circumscribed, with bipolar spindle cells arranged in a fascicular or storiform pattern (a cartwheel pattern).^{3,16} Low mitotic activity. Cytoplasmic granules (found in granular cell tumors) are lacking. WHO grade 1. Reported only in adults.

Diagnostic criteria for pituicytoma

Diagnostic criteria for pituicytoma are shown in ► Fig. 50.3.⁴

50.3.3 Granular cell tumor of the sellar region (WHO grade 1)

General information

A circumscribed tumor with nests of large cells having granular, eosinophilic cytoplasm due to copious cytoplasmic lysosomes. Mitotic activity is rare.

While rare, GCTs are the most common primary tumor of the neurohypophysis and pituitary stalk/infundibulum¹⁵ with a predilection for the stalk (causing suprasellar extension). GCTs have been identified in the gastrointestinal tract, genitourinary tract, orbital region as well as in other locations of the central nervous system with no connection to the pituitary gland or hypothalamus (e.g., spinal meninges¹⁷). Female:male ratio $\geq 2:1$. Asymptomatic microscopic clusters of granular cells (tumorettes) are more common, with an incidence up to 17%.¹⁸

They usually follow a slow progression with benign behavior.

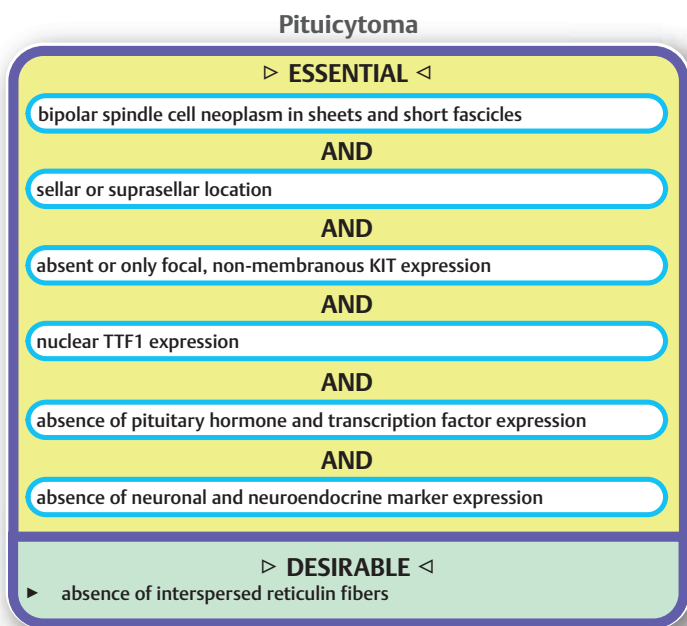


Fig. 50.3 Diagnostic criteria for pituicytoma. (Modified with permission from WHO Classification of Tumors Editorial Board, (ed.) World Health Organization classification of tumors of the central nervous system. 5th ed. Bosman F T, Jaffe E S, Lakhani S R, et al. (series eds.): World Health Organization Classification of Tumors series, vol. 6. International Agency for Research on Cancer, Lyon, France, 2021.)

Diagnostic criteria for granular cell tumor

Diagnostic criteria for granular cell tumor are shown in ▶ Fig. 50.4.⁴

50.3.4 Spindle cell oncocytoma (WHO grade 1)

General information

In addition to intrasellar and suprasellar involvement, there are reports of invasion of the cavernous sinus¹⁹ and floor of the sella.²⁰ Their highly vascular nature makes them susceptible to spontaneous hemorrhage and also to significant blood loss during surgery.

Diagnostic criteria for spindle cell oncocytoma

Diagnostic criteria for spindle cell oncocytoma are shown in ▶ Fig. 50.5.⁴

50.4 Tumors of the adenohypophysis

50.4.1 Pituitary neuroendocrine tumor (PitNET)/pituitary adenoma

General information

A clonal neoplasm of anterior pituitary hormone producing cells.

Granular cell tumor

▷ ESSENTIAL ◁

neoplasm composed of polygonal cells with granular cytoplasm

AND

sellar or suprasellar location

AND

absent or only focal, non-membranous KIT expression

AND

nuclear TTF1 expression

AND

absence of pituitary hormone and transcription factor expression

AND

absence of neuronal and neuroendocrine marker expression

▷ DESIRABLE ◁

- ▶ absence of interspersed reticulin fibers
- ▶ PAS-positive/diastase-resistant
- ▶ CD68 or α1-antitrypsin immunoreactivity

Fig. 50.4 Diagnostic criteria for granular cell tumor. (Modified with permission from WHO Classification of Tumors Editorial Board, (ed.) World Health Organization classification of tumors of the central nervous system. 5th ed. Bosman F T, Jaffe E S, Lakhani S R, et al. (series eds.): World Health Organization Classification of Tumors series, vol. 6. International Agency for Research on Cancer, Lyon, France, 2021.)

Terminology: traditionally, these tumors have been classified as adenomas, but they are actually neuroendocrine tumors since they contain neurosecretory granules. Thus the current nomenclature is pituitary neuroendocrine tumors (PitNETs). To transition the terminology, for the time being they may also be called **PitNET/adenomas** to denote that “pituitary adenoma” is still acceptable, but PitNET is preferred.²¹

Diagnostic criteria for PitNET/adenoma

Diagnostic criteria for PitNET/adenoma are shown in ► Fig. 50.6.⁴

Clinical considerations for PitNET/adenomas

Owing to the intricacies of PitNET/adenomas, a number of sections are devoted to their clinical aspects, starting with section 51.

50.4.2 Invasive pituitary tumors

General information

Tumors of the anterior pituitary can invade surrounding structures. Some of these are PitNETs (≈ 5% of PitNETs, and ≈ 30-40% of *surgically treated* PitNETs are invasive²²). Non-functioning PitNETs have a

Spindle cell oncocytoma

▷ ESSENTIAL ◁

neoplasm composed of polygonal cells with granular cytoplasm

AND

sellar or suprasellar location

AND

absent or only focal, non-membranous KIT expression

AND

nuclear TTF1 expression

AND

absence of pituitary hormone and transcription factor expression

AND

absence of neuronal and neuroendocrine marker expression

▷ DESIRABLE ◁

- ▶ absence of interspersed reticulin fibers
- ▶ antimitochondrial antigen immunoreactivity

Fig. 50.5 Diagnostic criteria for spindle cell oncocytoma. (Modified with permission from WHO Classification of Tumors Editorial Board, (ed.) World Health Organization classification of tumors of the central nervous system. 5th ed. Bosman F T, Jaffe E S, Lakhani S R, et al. (series eds.): World Health Organization Classification of Tumors series, vol. 6. International Agency for Research on Cancer, Lyon, France, 2021.)

higher incidence of invasion.²³ However, more aggressive pituitary tumors can also be locally invasive, and are considered here.

Wilson's anatomic classification for invasive pituitary tumors²⁴ (modified from Hardy^{25,26}) is shown in ► Table 50.1.

At times, a pituitary tumor may displace the medial wall of the cavernous sinus ahead of it without actually perforating this dural structure (i.e., not actually invading the sinus).²⁷ This is difficult to reliably identify on MRI, and the most definitive sign of cavernous sinus invasion is carotid artery encasement.²⁸

The clinical course is variable, with some tumors being more aggressive than others. Occasionally, these tumors grow to sizes >4 cm dia, and these are often very aggressive and follow a malignant course.²⁹

Aggressive pituitary tumors

A subset of invasive pituitary tumors meet the European Society of Endocrinology (ESE) guidelines³⁰ definition of "aggressive pituitary tumors," viz., pituitary tumors that do not respond to standard therapies (surgery, conventional medical treatments and XRT) and present with multiple local recurrences. The WHO does not recognize this category in its current classification⁴ (and the term "atypical adenoma" was dropped in the WHO 2017 classification).

Pituitary carcinoma

The definition of pituitary carcinoma is contentious. Both the ESE and WHO consider the presence of noncontiguous metastases as qualifying as pituitary carcinoma.^{4,21} Only ≈ 0.16% of pituitary tumors

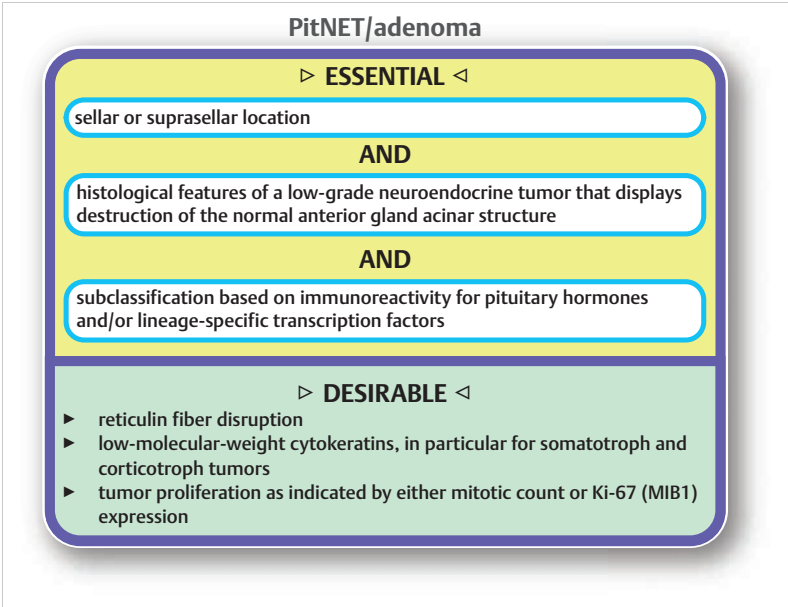


Fig. 50.6 Diagnostic criteria for PitNET/adenoma. (Modified with permission from WHO Classification of Tumors Editorial Board, (ed.) World Health Organization classification of tumors of the central nervous system. 5th ed. Bosman F T, Jaffe E S, Lakhani S R, et al. (series eds.): World Health Organization Classification of Tumors series, vol. 6. International Agency for Research on Cancer, Lyon, France, 2021.)

Table 50.1 Anatomic classification of invasive pituitary tumors (modified Hardy system) ²⁴
Extension
• Suprasellar extension 0: none A: expanding into suprasellar cistern B: anterior recesses of 3rd ventricle obliterated C: floor of 3rd ventricle grossly displaced • Parasellar extension D^a: intracranial (intradural) E: into or beneath cavernous sinus (extradural)
Invasion/Spread
• Floor of sella intact I: sella normal or focally expanded; tumor < 10 mm II: sella enlarged; tumor ≥ 10 mm • Sphenoid extension III: localized perforation of sellar floor IV: diffuse destruction of sellar floor • Distant spread^b V: spread via CSF or blood-borne
^aspecify: 1) anterior, 2) middle, or 3) posterior fossa ^b this qualifies as pituitary carcinoma (see text)

give rise to noncontiguous metastases (spreading systemically or through the CSF).³¹ Although the prognosis with metastases is worse, there are some aggressive tumors with similar behavior even in the absence of metastases.²²

Nomenclature suggestion for pituitary tumor grades

Taking into account the areas of agreement and areas of uncertainty in classification, the suggestion shown in the box has been proposed^{22,32}

Σ: Nomenclature suggestion for pituitary tumor grades^{22,32}

- Noninvasive tumors on MRI (60-65%): **PitNET/adenomas**
- Invasive tumors on MRI (35-40%):
 - Persistent disease that is *controlled* (with or without XRT) after surgery and/or medical treatment: **Invasive PitNet**
 - Persistent disease that is *recurrent* or *progressive* (with or without XRT) after surgery and/or medical treatment³³: **aggressive pituitary tumor** or **invasive PitNET with malignant potential**
 - Persistent disease with noncontiguous metastases³⁴: **Pituitary carcinoma**

^a many of these tumors will exhibit some combination of the following (only 2% exhibit all 3³³): Ki67 ≥ 3%, positive P53 expression and/or mitotic rate > 2/10 HPF

Presentation of invasive pituitary tumors

1. visual system
 - a) most present due to compression of the optic apparatus, usually producing gradual visual deficit (however, sudden blindness is not unheard of)
 - b) extraocular muscle deficits may occur with cavernous sinus invasion, and usually develop after visual loss
 - c) exophthalmos may occur with orbital invasion due to compromise of orbital venous drainage
2. hydrocephalus: suprasellar extension may obstruct one or both foramen of Monro
3. invasion of the skull base may lead to nasal obstruction. CSF rhinorrhea may occasionally be precipitated by tumor shrinkage in response to dopamine agonists (e.g., bromocriptine or cabergoline) as a result of uncovering areas of bone erosion. This carries the risk of ascending meningitis.³⁴
4. invasive pituitary tumors that secrete prolactin (p.863) often present with findings of hyperprolactinemia and with these, the prolactin levels are usually > 1000 ng/ml (caution: giant invasive adenomas with very high prolactin production may have a falsely low PRL due to "hook effect" (p.877))

Management suggestions for invasive pituitary tumors

The European Society of Endocrinology (ESE) treatment guidelines include the following³⁰ (see reference for full details).

1. routine pituitary tumor screening ► Table 52.1 (pituitary hormone assessment and visual field testing)
2. the diagnosis of an aggressive pituitary tumor should be considered with invasive tumors exhibiting an unusually rapid growth rate, or clinically significant growth following optimal standard treatment (surgery, XRT and conventional medical treatment)
3. consider screening for metastatic disease for symptoms outside the sellar region or MRI findings suspicious for local noncontiguous spread
4. initial surgery by a surgeon with extensive experience in pituitary surgery
5. Ki67 proliferative index should be assessed in surgical specimens, and if the Ki67 is ≥ 3%, then p53 immunodetection and mitotic count should be evaluated as potential predictors of tumor aggressiveness
6. consideration of repeat surgery prior to proceeding to other treatment options
7. XRT
 - a) recommended for progression after surgery for non-functioning tumors, or after surgery + standard medical treatment in functioning tumors
 - b) consideration of XRT recommended after surgery for clinically relevant tumor remnant with markers that suggest aggressive behavior (Ki67 ≥ 3% and/or positive P53 and/or mitotic rate > 2/10 HPF)

8. chemotherapy for tumors demonstrating continued growth*
 - a) Stupp regimen (p.672): temozolamide (TMZ) monotherapy using the *adjuvant* dosing regimen (150–200 mg/m² for 5 consecutive days every 28 days) & XRT (if maximal XRT has not yet been given)
 - b) assessment after 3 cycles of TMZ
 1. if there is clinical response, continue TMZ cycles for 5 days every 28 days for 6 months total, and consideration for longer duration if continued benefit is demonstrated
 2. if no response, then a trial with systemic toxic chemotherapy is suggested (no specific agent could be preferentially recommended)
9. for isolated metastases, local regional therapies in addition to systemic treatment as indicated
10. MRIs should be repeated every 3–12 months depending on tumor growth and risk to vital structures

* assessment of MGMT promoter gene methylation (p.673) expression is mentioned as a recommendation, but this may not alter the treatment paradigm.

References

- [1] Muller HL, Merchant TE, Warmuth-Metz M, et al. Craniopharyngioma. *Nat Rev Dis Primers*. 2019; 5. DOI: 10.1038/s41572-019-0125-9
- [2] Ostrom QT, Gittleman H, Liao P, et al. CBTRUS Statistical Report: Primary brain and other central nervous system tumors diagnosed in the United States in 2010–2014. *Neuro Oncol*. 2017; 19:v1–v88
- [3] Louis DN, Ohgaki H, Wiestler OD, et al. WHO classification of tumors of the central nervous system. Lyon, France 2016
- [4] Bosman FT, Jaffe ES, Lakhani SR, et al. World Health Organization Classification of Tumors series, vol. 6. In: WHO Classification of Tumors Editorial Board (ed). World Health Organization classification of tumors of the central nervous system. 5th ed. Lyon, France: International Agency for Research on Cancer; 2021
- [5] Johnson LN, Hepler RS, Yee RD, et al. Magnetic resonance imaging of craniopharyngioma. *Am J Ophthalmol*. 1986; 102:242–244
- [6] Nussbaum PE, Nussbaum LA, Torok CM, et al. Case report and literature review of BRAF-V600 inhibitors for treatment of papillary craniopharyngiomas: A potential treatment paradigm shift. *J Clin Pharm Ther*. 2022; 47:826–831
- [7] Klein HJ, Rath SA. Removal of Tumors of the III Ventricle Using Lamina Terminalis Approach: Three Cases of Isolated Growth of Craniopharyngiomas in the III Ventricle. *Child's Nervous System*. 1989; 5:144–147
- [8] Patterson RH, Denyevich A. Surgical Removal of Craniopharyngiomas by a Transcranial Approach through the Lamina Terminalis and Sphenoid Sinus. *Neurosurgery*. 1980; 7:111–117
- [9] Park HR, Kshettry VR, Farrell CJ, et al. Clinical Outcome After Extended Endoscopic Endonasal Resection of Craniopharyngiomas: Two-Institution Experience. *World Neurosurg*. 2017; 103:465–474
- [10] Cao L, Wu W, Kang J, et al. Feasibility of endoscopic endonasal resection of intrinsic third ventricular craniopharyngioma in adults. *Neurosurg Rev*. 2022; 45:1–13
- [11] Manaka S, Teramoto A, Takakura K. The Efficacy of Radiotherapy for Craniopharyngioma. *Journal of Neurosurgery*. 1985; 62:648–656
- [12] Guerrero-Perez F, Marengo AP, Vidal N, et al. Primary tumors of the posterior pituitary: A systematic review. *Rev Endocr Metab Disord*. 2019; 20:219–238
- [13] Iglesias P, Guerrero-Perez F, Villabona C, et al. Adenohypophyseal hyperfunction syndromes and posterior pituitary tumors: prevalence, clinical characteristics, and pathophysiological mechanisms. *Endocrine*. 2020; 70:15–23
- [14] Gueguen B, Merland JJ, Riche MC, et al. Vascular Malformations of the Spinal Cord: Intrathecal Perimedullary Arteriovenous Fistulas Fed by Medullary Arteries. *Neurology*. 1987; 37:969–979
- [15] Schaller B, Kirsch E, Tolnay M, et al. Symptomatic granular cell tumor of the pituitary gland: case report and review of the literature. *Neurosurgery*. 1998; 42:166–70; discussion 170–1
- [16] Wesseling P, Brat DJ, Fuller GN, et al. Pituitaryoma. In: WHO classification of tumors of the central nervous system. 4th ed. Lyon: International Agency for Research on Cancer; 2007:243–244
- [17] Markesbery WR, Duffy PE, Cowen D. Granular cell tumors of the central nervous system. *J Neuropathol Exp Neurol*. 1973; 32:92–109
- [18] Fuller GN, Wesseling P, Louis DN, et al. Granular cell tumors of the neurohypophysis. In: WHO classification of tumors of the central nervous system. 4th ed. Lyon: International Agency for Research on Cancer; 2007:241–242
- [19] Dahiya S, Sarkar C, Hedley-Whyte ET, et al. Spindle cell oncocytoma of the adenohypophysis: report of two cases. *Acta Neuropathol*. 2005; 110:97–99
- [20] Kloub O, Perry A, Tu PH, et al. Spindle cell oncocytoma of the adenohypophysis: report of two recurrent cases. *Am J Surg Pathol*. 2005; 29:247–253
- [21] Mete O. WHO Classification of Tumors Editorial Board. Pituitary gland. Lyon, France 2022. <https://publications.iarc.fr/Book-And-Report-Series/Who-Classification-Of-Tumors>
- [22] Raverot G, Ilie MD, Lasolle H, et al. Aggressive pituitary tumours and pituitary carcinomas. *Nat Rev Endocrinol*. 2021; 17:671–684
- [23] Lu L, Wan X, Xu Y, et al. Classifying Pituitary Adenoma Invasiveness Based on Radiological, Surgical and Histological Features: A Retrospective Assessment of 903 Cases. *J Clin Med*. 2022; 11. DOI: 10.3390/jcm11092464
- [24] Wilson CB, Tindall GT, Collins WF. Neurosurgical Management of Large and Invasive Pituitary Tumors. In: Clinical Management of Pituitary Disorders. New York: Raven Press; 1979:335–342
- [25] Hardy J, Kohler PO, Ross GT. Transsphenoidal Surgery of Hypersecreting Pituitary Tumors. In: Diagnosis and Treatment of Pituitary Tumors. New York: Excerpta Medica/American Elsevier; 1973:179–194
- [26] Hardy J, Thompson RA, Green R. Transsphenoidal Surgery of Intracranial Neoplasms. In: Advances in Neurology. New York: Raven Press; 1976:261–274
- [27] Laws ER. Comment on Knosp E, et al.: Pituitary Adenomas with Invasion of the Cavernous Sinus Space: A Magnetic Resonance Imaging Classification Compared with the Surgical Findings. *Neurosurgery*. 1993; 33
- [28] Scotti G, Yu CY, Dillon WP, et al. MR Imaging of Cavernous Sinus Involvement by Pituitary Adenomas. *American Journal of Radiology*. 1988; 151:799–806
- [29] Krisht AF. Giant invasive pituitary adenomas. *Contemporary Neurosurgery*. 1999; 21:1–6
- [30] Raverot G, Burman P, McCormack A, et al. European Society of Endocrinology Clinical Practice Guidelines for the management of aggressive pituitary

- tumours and carcinomas. *Eur J Endocrinol.* 2018; 178:G1–G24
- [31] Saeger W, Ludecke DK, Buchfelder M, et al. Pathohistological classification of pituitary tumors: 10 years of experience with the German Pituitary Tumor Registry. *Eur J Endocrinol.* 2007; 156:203–216
- [32] Trouillas J, Burman P, McCormack A, et al. Aggressive pituitary tumours and carcinomas: two sides of the same coin? *Eur J Endocrinol.* 2018; 178:C7–C9
- [33] Raverot G, Dantony E, Beauvy J, et al. Risk of Recurrence in Pituitary Neuroendocrine Tumors: A Prospective Study Using a Five-Tiered Classification. *J Clin Endocrinol Metab.* 2017; 102:3368–3374
- [34] Barlas O, Bayindir C, Hepgul K, et al. Bromocriptine-induced cerebrospinal fluid fistula in patients with macroprolactinomas: report of three cases and a review of the literature. *Surg Neurol.* 1994; 41:486–489

51 PitNET/Adenomas – Clinical Considerations

51.1 General information

Key concepts

- most are benign, however some are locally aggressive and (rarely) metastasize
- presentation (p. 863): most common
 - symptoms from excess hormonal secretion: includes hyperprolactinemia, Cushing's syndrome (↑ cortisol), acromegaly (↑ growth hormone)...
 - mass effect. Most commonly: bitemporal hemianopsia from compression of optic chiasm, or infrequently hypopituitarism from compression of pituitary gland
 - H/A: some H/A are due to increased pressure within the sella
 - incidental finding
 - infrequent presentations: pituitary apoplexy (p. 865), spontaneous CSF rhinorrhea
- work-up for a newly diagnosed intrasellar lesion: endocrine screening, visual field evaluation, imaging to visualize the tumor, the parasellar carotid arteries and the sphenoid sinus (see ► Table 52.1)
- prolactinoma is the only type for which medical therapy (DA agonists) may be the primary treatment (p. 886). For other tumor types, options primarily consist of surgery (usually transsphenoidal, sometimes transcranial), or XRT
- post-op concerns include: diabetes insipidus, adrenal insufficiency, CSF leak, and rarely visual deterioration due to herniation of the optic chiasm into empty sella

See review of pituitary embryology and neuroendocrinology (p.151), and, for a discussion of the change in classification to PitNET/adenomas as well as diagnostic criteria, see section 50.4.1.

Most primary pituitary tumors are benign and arise from the anterior pituitary gland (adenohypophysis) as a monoclonal proliferation. However, some can be locally aggressive and rarely may metastasize.

PitNETs have been classified by a number of schemes, including: by endocrine function (aided by immunostaining), by light microscopy (p.871) with routine histological staining, by electron microscopic appearance, and by molecular biology.

Microadenoma: defined as a pituitary tumor ≤ 1 cm diameter. 50% of pituitary tumors are < 5 mm at time of diagnosis. These may be difficult to find at the time of surgery.

Macroadenomas: pituitary tumors > 1 cm diameter.

51.2 Diagnostic criteria and classification of PitNET/adenoma

Diagnostic criteria for PitNET/pituitary adenoma are shown in ► Fig. 50.6.¹

PitNETs derive from the six adenohypophyseal cell types. Each tumor type may have multiple subtypes which are shown along with their secretion products and clinical syndromes in ► Table 51.1

► **Null cell tumor (adenoma).** A PitNET without cell-type-specific differentiation based on immunohistochemistry, transcription factors, negative PAS stain.³ The term is not synonymous with non-functioning adenomas. No endocrine syndrome is associated with these tumors. They comprise 5–30% of surgically resected adenomas, and < 5% of non-functioning adenomas. They typically occur after age 60 years.

51.3 Epidemiology/pathology

Pituitary tumors represent ≈ 10% of intracranial tumors (incidence is higher in autopsy series). They are most common in the 3rd and 4th decades of life, and affect both sexes equally. Nonfunctioning PitNETs are the most common pituitary tumors.⁴ Cushing's disease and prolactinomas are more common in females, and nonfunctioning, lactotroph and null tumors³ are more common in males.

Table 51.1 Classification of PitNETs^{1,2} (used with permission from WHO Classification of Tumours Editorial Board. Central Nervous System Tumours. Lyon (France): International Agency for Research on Cancer; 2021. (WHO classification of tumours series, 5th ed.; vol. 6). Available from: <https://tumourclassification.iarc.who.int/>.)

Tumor type ^a	Transcription factors	Hormones	Keratin (CAM5.2 or CK18)	Tumor subtypes (if applicable)	Hormone excess syndrome ^b
PIT1-lineage tumors					
Somatotroph tumors	PIT1	GH, α -subunit	perinuclear	densely granulated somatotroph tumor	florid acromegaly
		GH	fibrous bodies (> 70%)	sparsely granulated somatotroph tumor	subtle acromegaly
Lactotroph tumors	PIT1, ER α	PRL (paranuclear)	weak or negative	sparsley granulated lactotroph tumor	hyperprolactinemia ^c
		PRL (diffuse cytoplasmic)	weak or negative	densely granulated lactotroph tumor	hyperprolactinemia ^c
Mammosomatotroph tumor	PIT1, ER α	GH (often predominant), PRL, α -subunit	perinuclear		acromegaly and hyperprolactinemia ^c
Thyrotroph tumor	PIT1, GATA2/3 ^d	α -subunit, TSH- β	weak or negative		hyperthyroidism
Mature PIT1-lineage tumor	PIT1, ER α , GATA2/3 ^d	GH (often predominant), PRL, α -subunit, TSH- β	perinuclear		acromegaly, hyperprolactinemia ^c and hyperthyroidism
Acidophil stem cell tumor	PIT1, ER α	PRL (predominant), GH (focal/variable)	scattered fibrous bodies		hyperprolactinemia ^c and subclinical acromegaly
Immature PIT1-lineage tumor	PIT1, ER α , GATA2/3 ^d	GH, PRL, α -subunit, TSH- β	focal/ variable		acromegaly, hyperprolactinemia ^c and hyperthyroidism
TPIT-lineage tumors					
Corticotroph tumors	TPIT (TBX19), NeuroD1 (β 2)	ACTH and other POMC derivatives	strong	densely granulated corticotroph tumor	florid Cushing, often microtumor
			variable	sparsely granulated corticotroph tumor	subtle Cushing, often macrotumor
			intense ring-like perinuclear	Crooke cell tumor	variable, Cushing
SF1-lineage tumors					
Gonadotroph tumor	SF1, ER α , GATA2/3 ^d	α -subunit, FSH- β , LH- β	variable		hypogonadism (virtually all) or hypergonadism (exceptional)
Tumors without distinct cell lineage					
Unclassified plurihormonal tumors	multiple combinations	multiple combinations	variable		variable
Null cell tumor ^e	none	none	variable		none

^a mixed tumors also occur and can constitute any combination of tumors shown; the most common is mixed somatotroph–lactotroph tumor

^b any tumor type can be clinically non-functioning

^c moderate hyperprolactinaemia can occur with any sellar mass that has suprasellar extension, interrupting hypothalamic tonic dopaminergic inhibition; however, the PRL level rarely exceeds 150 ng/mL; lactotroph tumors usually show a characteristic correlation between tumor size and PRL levels, whereas other PRL-secreting tumors do not

^d GATA2 and GATA3 are paralogues and show cross-reactivity with some available antisera

^e null cell tumor: not synonymous with non-functioning tumor (see text for details)

Most PitNET/adenomas are sporadic,⁵ and arise from a somatic mutation of single progenitor cells. However, PitNETs may also occur as part of syndromes, including:

1. multiple endocrine adenomatosis or neoplasia (MEA or MEN) (especially type I: autosomal dominant inheritance with high penetrance, also involves pancreatic islet cell tumors (which may produce gastrin and hence Zollinger-Ellison syndrome) and parathyroids (hyperparathyroidism), and in which the pituitary tumors are usually nonsecretory)
2. Carney complex (p.652). Autosomal dominant, cardiac and cutaneous myxomas and hyperpigmentation. Associated with PitNET/adenomas that secrete either GH or GH + prolactin
3. familial isolated pituitary adenoma (FIPA). Autosomal dominant. Criteria: ≥ 2 family members with PitNET/adenomas in the absence of MEN or Carney complex. The adenomas may be of the same type (homogeneous FIPA) or of different types (heterogeneous FIPA)
4. McCune-Albright syndrome: mammotroph and somatotroph tumors or hyperplasia
5. tuberous sclerosis: corticotroph tumors

51.4 Differential diagnosis of pituitary tumors

See differential diagnosis (p.1653), which also includes non-neoplastic etiologies.

51.5 Clinical presentation of pituitary tumors

51.5.1 General information

Classically, pituitary tumors are divided into functional (or secreting), and non-functional (AKA endocrine-inactive, which are either nonsecretory or else secrete hormonally inactive products or ones that do not cause endocrinologic symptoms, e.g., gonadotropin).

Possible presentations:

1. endocrine syndromes
 - a) due to hormonal excess (with *functional* tumors), (see below). Primarily, one of the following: prolactin, growth hormone, cortisol, thyroid hormone
 - b) due to hormonal deficiency (p.864) as a result of compression of the normal pituitary
2. mass effect (p.864)
3. incidental finding on imaging done for unrelated reasons (primarily with macroadenomas) in up to 20% of the population⁶
4. pituitary apoplexy (p.865): paroxysmal H/A with nausea and meningismus, endocrinologic and/or neurologic deficit (usually ophthalmoplegia or visual loss)
5. hydrocephalus e.g., when tumors with a large suprasellar component compresses the 3rd ventricle

Unless they are discovered on imaging performed for some other indication ("pituitary incidentaloma"⁷), functional tumors tend to present earlier as a result of symptoms caused by physiologic effects of excess hormones that they secrete⁸ (this applies less to gonadotropins and to prolactinomas in males since the symptoms may be mild or unrecognized). Nonfunctional tumors usually do not present until they are large enough to cause neurologic deficits by mass effect.

51.5.2 Presentation due to hormone oversecretion (secretory tumor)

$\approx 65\%$ of adenomas secrete an active hormone (48% prolactin, 10% GH, 6% ACTH, 1% TSH)⁹:

1. prolactin (p.867): can cause amenorrhea-galactorrhea syndrome in females, impotence in males. Etiologies:
 - a) prolactinoma (p.867): neoplasia of pituitary lactotrophs
 - b) stalk effect (p.877): pressure on the pituitary stalk may reduce the inhibitory control over prolactin secretion causing a modest increase in serum prolactin levels (PRLs)
2. growth hormone (GH): elevated GH is due to a PitNET $> 95\%$ of the time
 - a) in adults: causes acromegaly (p.870)
 - b) in prepubertal children (before epiphyseal plate closure): produces pituitary gigantism (very rare)
3. corticotropin AKA adrenocorticotrophic hormone (ACTH):
 - a) Cushing's disease (endogenous hypercortisolism): see below
 - b) Nelson syndrome (p.869): can develop only in patients who have had an adrenalectomy
4. thyrotropin (TSH) (p.871): secondary (central) hyperthyroidism

- gonadotropins (follicle stimulating hormone (FSH) and less commonly luteinizing hormone (LH)): usually does not produce a clinical syndrome. Rarely produces ovarian hyperstimulation syndrome (p. 867) in women (menstrual disturbance, galactorrhea...) or testicular hypertrophy in men

51.5.3 Presentation due to mass effect

Presentation due to underproduction of pituitary hormones

May be caused by compression of the normal pituitary by large tumors. More common with non-secretory tumors than with secretory tumors. In order of sensitivity to compression (i.e., the order in which pituitary hormones become depressed from mass effect): GH (61–100%), gonadotropins (LH & FSH) (36–96%), TSH (8–81%), ACTH (17–62%),¹⁰ prolactin (mnemonic: Go Look For The Adenoma Please). Chronic deficiency of all pituitary hormones (panhypopituitarism) may produce pituitary cachexia (AKA Simmonds' cachexia).

✖ NB: selective reduction of a single pituitary hormone is very atypical with PitNETs. May occur with autoimmune hypophysitis (p. 1656), which most commonly involves ACTH or ADH (causing diabetes insipidus¹¹—see below)

Deficiency of specific hormones:

- growth hormone deficiency (note: growth hormone stimulation test (p. 882) is more sensitive and specific for GH deficiency than measuring basal GH levels):
 - in children: produces growth delay
 - in adults: produces vague symptoms with metabolic syndrome (decreased lean body mass, centripetal obesity, reduced exercise tolerance, impaired sense of well-being)
 - hypogonadism: amenorrhea (women), loss of libido, infertility
- hypothyroidism: cold intolerance, myxedema, entrapment neuropathies (e.g., carpal tunnel syndrome), weight gain, memory disturbance, integumentary changes (dry skin, coarse hair, brittle nails), constipation, increased sleep demand
- hypoadrenalism: orthostatic hypotension, easy fatigability
- diabetes insipidus: almost never seen preoperatively with pituitary tumors (except possibly with pituitary apoplexy, see below). If DI is present, other etiologies should be sought, including:
 - autoimmune hypophysitis (p. 1656)
 - hypothalamic glioma
 - suprasellar germ cell tumor
- gonadotropin deficiency (hypogonadotropic hypogonadism) with anosmia is part of Kallmann syndrome¹²

Presentation due to mass effect (other than compression of the pituitary gland)

Because they tend to get to a larger size before detection, this is more common with nonfunctioning tumors. Of functional tumors, prolactinoma is the most likely to become large enough to cause mass effect (especially in males or non-menstruating females); ACTH tumor is least likely. Nonspecific symptoms include headaches. ✖ Seizures are rarely attributable to a PitNET and other etiologies should be sought. Mass effect may occur suddenly as a result of expansion with pituitary apoplexy (see below).

Structures are commonly compressed and their manifestations include:

- optic chiasm: from tumor growth superiorly through the diaphragma sella (► Fig. 52.1). Classically produces **bitemporal hemianopsia** (non-congruous) (► Fig. 33.2). May also cause decreasing visual acuity
- involvement of third ventricle may produce obstructive hydrocephalus
- cavernous sinus
 - pressure on cranial nerves contained within (III, IV, V1, V2, VI): ptosis, facial pain, diplopia (see below)
 - occlusion of the cavernous sinus: proptosis, chemosis
 - encasement of the carotid artery by tumor: may cause slight narrowing, but complete occlusion is rare
- invasive adenomas (p. 855) may present as nasal or paranasal masses
- CSF rhinorrhea: usually with invasive adenomas often as a result of erosion of the skull base, sometimes precipitated by shrinkage of these large tumors resulting from medical treatment¹³
- macroadenomas may produce H/A possibly via increased intrasellar pressure

51.5.4 Pituitary apoplexy

General information

Key concepts

- due to expansion of a PitNET from hemorrhage or necrosis
- typical presentation: paroxysmal H/A with nausea and meningismus, endocrinologic and/or neurologic deficit (usually ophthalmoplegia or visual field loss)
- a medical emergency due to potentially lethal adrenal insufficiency and threat of permanent visual loss
- management: immediate administration of fluids and empiric corticosteroids (hydrocortisone) for hemodynamic instability, altered mental status (GCS < 15), visual deficits (acuity or visual fields). Shunt or EVD for hydrocephalus and transsphenoidal decompression within 7 days for visual loss
- not an emergency if asymptomatic

Definition

Neurologic and/or endocrinologic deterioration due to sudden expansion of a mass within the sella turcica.

Etiology

Sudden intrasellar expansion may occur as a result of hemorrhage, necrosis,^{14,15} and/or infarction within a pituitary tumor and adjacent pituitary gland. Occasionally, hemorrhage occurs into a normal pituitary gland or Rathke's cleft cyst.¹⁶

Epidemiology

Pituitary apoplexy (PA) occurs in an estimated 2–12% of patients with pituitary adenomas.¹⁷ In some cases, PA may be the initial presentation of a pituitary tumor.¹⁸ Furthermore, hemorrhage into a pituitary tumor can be asymptomatic.

Clinical features of pituitary apoplexy

Patients often present with abrupt onset of H/A, visual disturbance, and loss of consciousness. The majority of patients present with potentially lethal adrenal insufficiency, and over 50% have visual loss and/or ophthalmoplegia.

Neurologic involvement includes:

1. visual disturbances: one of the most common findings. Includes:
 - a) ophthalmoplegia (unilateral or bilateral): opposite the situation with a pituitary tumor, ophthalmoplegia occurs more often (78%) than visual pathway deficits (52–64%)¹⁹
 - b) one of the typical field cuts (p. 875) seen in pituitary tumors
2. acute secondary adrenal insufficiency (hypocortisolism) occurs in 66% and is the major source of mortality
 - a) fluid and electrolyte disturbances may be caused by vasopressin release from the posterior pituitary
 - b) hemodynamic instability may result from reduced vascular response to catecholamines caused by hypocortisolism
3. reduced mental status (lethargy, stupor, or coma): due to ↑ ICP (from intracranial extension of blood or to hydrocephalus) or hypothalamic involvement
4. cavernous sinus compression can cause venous stasis and/or pressure on any of the structures within the cavernous sinus
 - a) trigeminal nerve symptoms
 - b) proptosis
 - c) ophthalmoplegia (Cr. N. III palsy is more common than VI) in 70% of cases²⁰
 - d) ptosis may be an early symptom^{21,22}

- e) pressure on carotid artery
- f) compression of sympathetics within the cavernous sinus may produce third order (incomplete) Horner syndrome with unilateral ptosis and miosis, but anhidrosis may be absent or limited to the central forehead²³
- g) carotid artery compression may cause stroke or vasospasm
- 5. when hemorrhage breaks through the tumor capsule and the arachnoid membrane into the chiasmatic cistern, signs and symptoms of SAH may be seen
 - a) N/V
 - b) meningismus
 - c) photophobia
- 6. hypothalamic involvement may produce
 - a) hypotension
 - b) thermal dysautoregulation
 - c) cardiac dysrhythmias
 - d) respiratory pattern disturbances
 - e) diabetes insipidus
 - f) altered mental status: lethargy, stupor, or coma
- 7. suprasellar expansion can produce acute hydrocephalus

Evaluation

In cases due to hemorrhage: CT or MRI shows hemorrhagic mass in sella turcica and/or suprasellar region, often distorting the anterior third ventricle.

Cerebral angiography or CTA should be considered in cases where differentiating pituitary apoplexy from aneurysmal SAH is difficult.

Management of pituitary apoplexy

Rapid empiric administration of corticosteroids is indicated for hemodynamic instability, altered mental status (GCS < 15), or visual deficits (acuity or field cuts).²⁰ Typical dose: hydrocortisone 100–200 mg IV bolus followed by 2–4 mg/hr IV infusion.²⁰ ✖ Dexamethasone alone is not recommended as it is deficient in the necessary mineralocorticoid activity. Patients without the aforementioned emergency findings may be evaluated endocrinologically and have corticosteroid replacement if deficient.

In the absence of visual deficits, prolactinomas may be treated with dopamine agonists (bromocriptine, cabergoline...).

Rapid decompression is required for: sudden constriction of visual fields, severe and/or rapid deterioration of visual acuity, or neurologic deterioration due to hydrocephalus. Surgery in ≤ 7 days of pituitary apoplexy resulted in better improvement in ophthalmoplegia (100%), visual acuity (88%), and field cuts (95%) than surgery after 7 days, based on a retrospective study of 37 patients.²⁴ Decompression is usually via a transsphenoidal route (transcranial approach may be advantageous in some cases). Goals of surgery:

1. to decompress the following structures if under pressure: optic apparatus, pituitary gland, cavernous sinus, third ventricle (relieving hydrocephalus)
2. obtain tissue for pathology
3. complete removal of tumor is usually not necessary
4. for hydrocephalus: ventricular drainage is generally required

51.6 Specific types of pituitary tumors

51.6.1 Nonfunctioning PitNET/adenomas (NPPA)

Nonfunctioning PitNET/adenoma (NPPA) is a clinical definition, viz. a pituitary adenoma that does not produce signs or symptoms as a result of hormonal hypersecretion.²⁵ NPPAs are the most common pituitary tumors,⁴ comprising 15–30% of PitNETs.²⁶ They are predominantly derived from gonadotroph cells²⁵ and most actually do secrete gonadotropins or gonadotropin subunits,²⁷ but this is rarely symptomatic (see below). Most are either found incidentally or present due to signs and/or symptoms due to mass effect (p.864). For management see section 53.1.3.

51.6.2 Gonadotropin (FSH, LH) secreting tumors

As indicated above, most of these tumors do not produce clinical symptoms and are therefore considered nonfunctioning adenomas (80–90% of “nonfunctioning adenomas” are gonadotroph adenomas²⁸). Rarely, clinical findings may occur:

- FSH: may cause **ovarian hyperstimulation syndrome** in reproductive-age women causing menstrual irregularities (including amenorrhea) and galactorrhea (as with prolactin) along with ovarian cysts,^{25,29} and rarely testicular hypertrophy and sexual dysfunction in men
- LH: even more rare, reported as causing precocious puberty in 2 boys²⁵

51.6.3 Prolactinomas (lactotroph tumor)

The most common secretory adenoma. Arise from neoplastic transformation of anterior pituitary lactotrophs. See ► Table 52.4 for the differential diagnosis of hyperprolactinemia.

Manifestations of prolonged hyperprolactinemia:

1. females: amenorrhea-galactorrhea syndrome (AKA Forbes-Albright syndrome, AKA Ahumada-del Castillo syndrome). Variants: oligomenorrhea, irregular menstrual cycles. 5% of women with primary amenorrhea will be found to have a prolactin-secreting pituitary tumor.³⁰ Remember: pregnancy is the most common cause of secondary amenorrhea in females of reproductive potential. The galactorrhea may be spontaneous or expressive (only on squeezing the nipples)
2. males: impotence, decreased libido. Galactorrhea is rare (estrogen is also usually required). Gynecomastia is rare. Prepubertal prolactinomas may result in small testicles and feminine body habitus
3. either sex:
 - a) infertility is common
 - b) bone loss (osteoporosis in women, and both cortical and trabecular osteopenia in men) due to a relative estrogen deficiency, not due to the elevated prolactin itself

At the time of diagnosis, 90% of prolactinomas in women are microadenomas, vs. 60% for males (probably due to gender-specific differences in symptoms resulting in earlier presentation in females). Some tumors secrete both prolactin and GH.

51.6.4 Cushing's disease

General information and Cushing's syndrome

Cushing's syndrome (CS) is a constellation of findings caused by hypercortisolism. Cushing's disease (p.867)—endogenous hypercortisolism due to hypersecretion of ACTH by an ACTH-secreting PitNET—is just one cause of CS. The most common cause of CS is iatrogenic (administration of exogenous steroids). Possible etiologies of *endogenous* hypercortisolism are shown in ► Table 51.2. To determine the etiology of CS, see Dexamethasone suppression test (p.879).

Table 51.2 Causes of endogenous hypercortisolism

Site of pathology	Secretion product	Percent of cases	ACTH levels
pituitary corticotroph adenoma: Cushing's disease (p. 867)	ACTH	60–80%	slightly elevated ^a
ectopic ACTH production (p. 868); most are lung tumors, others: pancreas...		1–10%	very elevated
adrenal (adenoma or carcinoma)	cortisol	10–20%	low
hypothalamic or ectopic secretion of corticotropin-releasing hormone (CRH) producing hyperplasia of pituitary corticotrophs; pseudo-Cushing's state (p. 868)	CRH	rare	elevated

^aACTH may be normal or slightly elevated; normal ACTH levels in the presence of hypercortisolism are considered inappropriately elevated

Conversion factors³¹ for ACTH and cortisol between U.S. units and SI units are shown in Eq (51.1) and (51.2).

$$\text{ACTH} : 1 \text{ pg/ml} = 1 \text{ ng/liter} \quad (51.1)$$

$$\text{Cortisol} : 1 \mu\text{g/dl} = 27.59 \text{ nmol/liter} \quad (51.2)$$

Ectopic ACTH secretion

Hypercortisolism may also be due to ectopic secretion of ACTH usually by tumors, most commonly small-cell carcinoma of the lung, thymoma, carcinoid tumors, pheochromocytomas, and medullary thyroid carcinoma. In addition to findings of Cushing's syndrome, patients are typically cachectic due to the malignancy which is usually rapidly fatal.

Prevalence of Cushing's disease

40 cases/million population. ACTH-producing adenomas comprise 10–12% of pituitary adenomas.³² Cushing's disease is 9 times more common in women, whereas *ectopic* ACTH production is 10 times more common in males. Non-iatrogenic CS is 25% as common as acromegaly.

At the time of presentation, over 50% of patients with Cushing's disease have pituitary tumors <5 mm in diameter, which are very difficult to image with CT or MRI. Most are basophilic, some (especially the larger ones) may be chromophobic. Only ≈ 10% are large enough to produce some mass effect, which may cause enlargement of the sella turcica, visual field deficit, cranial nerve involvement, and/or hypopituitarism.

51

Clinical findings in Cushing's disease

Findings are those of Cushing's syndrome (hypercortisolism from any cause) include:

- weight gain
 - generalized in 50% of cases
 - centripetal fat deposition in 50%: trunk, upper thoracic spine ("buffalo hump"), supraclavicular fat pad, neck, "dewlap tumor" (episternal fat), with round plethoric face ("moon facies" AKA "Cushingoid facies") and slender extremities
- hypertension
- ecchymoses and purple striae, especially on flanks, breasts, and lower abdomen
- amenorrhea in women, impotence in men, reduced libido in both
- hyperpigmentation of skin and mucous membranes: due to MSH cross-reactivity of ACTH. Occurs only with elevated ACTH, i.e., Cushing's disease (not Cushing's syndrome) or ectopic ACTH production (also below)
- atrophic, tissue-paper thin skin with easy bruising and poor wound healing
- psychiatric: depression, emotional lability, dementia
- osteoporosis
- generalized muscle wasting with complaints of easy fatigability
- elevation of other adrenal hormones: androgens may produce hirsutism and acne
- sepsis: associated with advanced Cushing's syndrome

Laboratory findings in Cushing's disease

- hyperglycemia: diabetes or glucose intolerance
- hypokalemic alkalosis
- loss of diurnal variation in cortisol levels
- normal or elevated ACTH levels
- failure to suppress cortisol with low-dose (1 mg) dexamethasone test (p.879)
- elevated 24-hour urine-free cortisol

- CRH levels will be low (not commonly measured)

Nelson's syndrome (or Nelson syndrome) (NS)

General information

Key concepts

- a rare condition that follows 10–30% of total bilateral adrenalectomies (TBA) performed for Cushing's disease; see indications for TBA (p. 892)
- classic triad: hyperpigmentation (skin & mucus membranes), abnormal ↑ ACTH, and progression of pituitary tumor (the last criteria is now controversial)
- treatment options: surgery (transsphenoidal or transcranial), XRT, medication

A rare condition that follows 10–30% of total bilateral adrenalectomies (TBA) performed for Cushing's disease; see indications for TBA (p. 891). NS is due to continued growth of corticotroph (ACTH-secreting) adenoma cells. Usually occurs 1–4 years after TBA (range: 2 mos–24 years).³² Theoretical explanation (unproven)³³: following TBA, hypercortisolism resolves, and CRH levels increase back to normal from the (reduced) suppressed state; corticotroph adenomas in patients with NS have an increased & prolonged response to CRH resulting in increased growth. Also, corticotrophs in NS and CD show reduced inhibition by glucocorticoids. It is controversial if some cases may be related to insufficient glucocorticoid replacement after TBA.³²

Manifestations

See reference.³³

- hyperpigmentation (due to melanin stimulating hormone (MSH) cross reactivity of ACTH and actual increased levels of MSH due to increased proopiomelanocortin production). Often the earliest sign that Nelson's syndrome is developing. Look for linea nigra (midline pigmentation from pubis to umbilicus) and hyperpigmentation of scars, gingivae, and areolae. DDx of hyperpigmentation includes: primary adrenal insufficiency (high levels of ACTH), ectopic ACTH secretion, hemochromatosis (more bronze color), jaundice (yellowish)
- tumor growth → increased mass effect (p. 864) or invasion: the most serious consequence. These corticotroph tumors are among the most aggressive of pituitary tumors.³⁴ (p. 545) May produce any of the problems associated with macroadenomas (optic nerve compression, cavernous sinus invasion, pituitary insufficiency, H/A, bony invasion...) as well as necrosis with precipitous intracranial hypertension³⁵; see pituitary apoplexy (p. 865)
- malignant transformation of the corticotroph tumor (very rare)
- hypertrophy of adrenal tissue rests: may be located in the testes → painful testicular enlargement and oligospermia. Rarely the rests can secrete enough cortisol to normalize cortisol levels or even cause a recurrence of Cushing's disease despite the adrenalectomy

Evaluation

- laboratories
 - ACTH > 200 ng/L (usually thousands of ng/L) (normal: usually < 54 ng/L)
 - exaggerated ACTH response to CRH (not required for diagnosis)
 - other pituitary hormones may be affected as with any macroadenoma-causing mass effect (p. 864) and endocrine screening (► Table 52.1) should be done
- formal visual field testing (p. 874): should be done in patients with suprasellar extension or in those being considered for surgery (as a baseline for comparison)

Treatment

See treatment (p. 888).

51.6.5 Acromegaly

General information

Key concepts

- abnormally high levels of growth hormone in an adult. >95% of cases are due to a benign pituitary somatotroph adenoma, >75% are > 10 mm at time of diagnosis
- effects include soft tissue and skeletal changes, cardiomyopathy, colon Ca
- work-up: endocrine tests (p. 874), cardiology consult, colonoscopy
- treatment (p. 889): surgery for most, and then if necessary, medical therapy (p. 889) and/or XRT (p. 893)
- suggested criteria for biochemical cure (p. 903): normal IGF-1, growth hormone level < 5 ng/ml, AND GH nadir of < 1 ng/ml after OGST (p. 882)

Incidence: 3 cases/1-million persons/year. > 95% of cases of excess GH result from a pituitary somatotroph adenoma. Ectopic GH secretion may occur uncommonly with: carcinoid tumor, lymphoma, pancreatic islet-cell tumor. By the time of diagnosis, > 75% of pituitary GH tumors are macroadenomas (> 10 mm dia) with cavernous sinus invasion and/or suprasellar extension.

25% of acromegalics have thyromegaly with normal thyroid studies. 25% of GH adenomas also secrete prolactin. Acromegaly occurs rarely as part of a genetic syndrome, including: multiple endocrine neoplasia type 1 (MEN 1), McCune-Albright syndrome (p. 975), familial acromegaly, and Carney complex (p. 652).³⁶

Clinical

Elevated levels of GH in children before closure of the epiphyseal plates in the long bones (shortly after puberty) produces gigantism. Usually presents in the teen years. Hypertension may also occur.

In adults, elevated GH levels produce acromegaly (age: usually > 50 yrs) with findings that may include^{37,38} (also see ► Table 51.3):

1. skeletal overgrowth deformities
 - a) increasing hand and foot size
 - b) thickened heel pad
 - c) frontal bossing
 - d) prognathism
2. cardiovascular
 - a) cardiac findings (structural and functional): arrhythmias, valvular disease, concentric myocardial hypertrophy
 - b) hypertension (30%)
3. soft tissue swelling (includes macroglossia)
4. glucose intolerance
5. peripheral nerve entrapment syndromes (including carpal tunnel syndrome)
6. debilitating headache
7. excessive perspiration (especially palmar hyperhidrosis)
8. oily skin
9. joint pain
10. sleep apnea
11. fatigue
12. colon cancer: risk is $\approx 2 \times$ risk of general population³⁹

Patients with elevated levels of GH (including partially treated cases) have 2–3 times the expected mortality rate,⁴⁰ primarily due to hypertension, diabetes, pulmonary infections, cancer, and cardiovascular disease (► Table 51.3). Soft-tissue swelling and nerve entrapment may be reversible with normalization of GH levels, but many disfiguring changes and health risks are permanent (see ► Table 51.3 for specifics).

Table 51.3 Risks of long-term exposure to excess growth hormone (GH)⁴⁰

Arthropathy
1. unrelated to age of onset or GH levels 2. usually with longstanding acromegaly 3. reversibility^a: a) rapid symptomatic improvement b) bone & cartilage lesions irreversible
Peripheral neuropathy
1. intermittent anesthetics, paresthesias 2. sensorimotor polyneuropathy 3. impaired sensation 4. reversibility^a: a) symptoms may improve b) onion bulbs (whorls) do not regress
Cardiovascular disease
1. cardiomyopathy a) reduced LV diastolic function b) increased LV mass and arrhythmias c) fibrous hyperplasia of connective tissue 2. HTN: exacerbates cardiomyopathic changes 3. reversibility^a: may progress even with normal GH
Respiratory disease
1. upper airway obstruction: caused by soft tissue overgrowth and decreased pharyngeal muscle tone with sleep apnea in ≈ 50% 2. reversibility^a: generally improves
Neoplasia
1. increased risk of malignancies (especially colon-Ca) & soft-tissue polyps 2. reversibility^a: unknown
Glucose intolerance
1. occurs in 25% of acromegalics (more common with family history of DM) 2. reversibility^a: improves
^a reversibility with normalization of GH levels

51.6.6 Thyrotropin (TSH)-secreting adenomas (thyrotroph adenoma)

General information

Rare: comprise ≈ 0.5–1% of pituitary tumors.^{9,41} Produces central (secondary) hyperthyroidism (note: central hyperthyroidism may also occur with pituitary resistance to thyroid hormones⁴²): elevated circulating T3 and T4 levels, with elevated or inappropriately normal TSH⁴² (TSH should be undetectable in primary hyperthyroidism). Up to 33% of tumors positive for TSH immunostaining are nonsecretory.⁴² Many of these tumors are plurihormonal, but the secondary hormone is usually clinically silent. Most of these tumors are aggressive and invasive and are large enough at presentation to also produce mass effect (especially if prior thyroid ablative procedures have been done, which occurs in up to 60% of cases due to lack of recognition of pituitary abnormality.^{42,43})

Clinical

Symptoms of hyperthyroidism: anxiety, palpitations (due to a-fib), heat intolerance, hyperhidrosis, and weight loss despite normal or increased intake. Signs: hyperactivity, lid lag, tachycardia, irregular rhythm when a-fib is present, hyperreflexia, tremor. Exophthalmos and infiltrative dermatopathy (e.g., pretibial myxedema) are present only in Graves' disease.

51.6.7 Pathological classification of pituitary tumors

Light microscopic appearance of adenomas

Obsolete classification system (presented for completeness). With newer techniques (EM, molecular biology, immunohistochemistry...), many tumors previously considered nonsecretory have been found to have all the components necessary to secrete hormones.

In order of decreasing frequency:

1. chromophobe: most common (ratio of chromophobe to acidophil is 4-20:1). Originally considered "non-secretory," in actuality may produce prolactin, GH, or TSH
2. acidophil (eosinophilic): produce prolactin, TSH, or usually GH
3. basophil → gonadotropins, β -lipotropin, or usually ACTH → Cushing's disease

Classification of adenomas based on secretory products

1. endocrine-active tumors: ~ 70% of pituitary tumors produce 1 or 2 hormones that are measurable in the serum and cause defined clinical syndromes; these are classified based on their secretory product(s)
2. endocrine-inactive (nonfunctional) tumors⁴⁴ (note: a and b constitute the bulk of endocrine-inactive adenomas):
 - a) oncocyoma
 - b) gonadotropin-secreting adenoma
 - c) silent corticotropin-secreting adenoma
 - d) glycoprotein-secreting adenoma

Tumors of the neurohypophysis and infundibulum

The most common tumors encountered in the posterior pituitary are metastases (owing to the rich blood supply and the rarity of posterior pituitary tumors).

See Tumors of the neurohypophysis and infundibulum (p.852) for details of specific tumor types (granular cell tumor, pituicytoma, and spindle cell oncocyoma).

References

- [1] Bosman FT, Jaffe ES, Lakhani SR, et al. World Health Organization Classification of Tumors series, vol. 6. In: WHO Classification of Tumors Editorial Board (ed.) World Health Organization classification of tumors of the central nervous system. 5th ed. Lyon, France: International Agency for Research on Cancer; 2021
- [2] Mete O. Pituitary gland: Chapter 8, Series 8. In: WHO Classification of Tumors Editorial Board (ed.) World Health Organization Classification of Tumors series, Vol. 8. Endocrine and Neuroendocrine tumors, 5th ed. Lyon, France: International Agency for Research on Cancer; 2022. <https://publications.iarc.fr/Book-And-Report-Series/Who-Classification-Of-Tumours>. Date accessed: October 12, 2022
- [3] Moini Jahangir, Badolato C, Ahangari R. Pituitary Tumors. In: Epidemiology of Endocrine Tumors. Elsevier; 2020:151–200
- [4] Cozzi R, Lasio G, Cardia A, et al. Perioperative cortisol can predict hypothalamus-pituitary-adrenal status in clinically non-functioning pituitary adenomas. J Endocrinol Invest. 2009; 32:460–464
- [5] Herman V, Fagin J, Gonsky R, et al. Clonal origin of pituitary adenomas. J Clin Endocrinol Metab. 1990; 71:1427–1433
- [6] Ezzat S, Asa SL, Couldwell WT, et al. The prevalence of pituitary adenomas: a systematic review. Cancer. 2004; 101:613–619
- [7] Aghi MK, Chen CC, Fleseriu M, et al. Congress of Neurological Surgeons Systematic Review and Evidence-Based Guidelines on the Management of Patients With Nonfunctioning Pituitary Adenomas: Executive Summary. Neurosurgery. 2016; 79:521–523
- [8] Ebersold MJ, Quast LM, Laws ER, et al. Long-Term Results in Transphenoidal Removal of Nonfunctioning Pituitary Adenomas. J Neurosurg. 1986; 64:713–719
- [9] Biller BM, Swearingen B, Zervas NT. A decade of the Massachusetts General Hospital Neuroendocrine Clinical Center. Journal of Clinical Endocrinology and Metabolism. 1997; 82:1668–1674
- [10] Fleseriu M, Bodach ME, Tumialan LM, et al. Congress of Neurological Surgeons Systematic Review and Evidence-Based Guideline for Pretreatment Endocrine Evaluation of Patients With Nonfunctioning Pituitary Adenomas. Neurosurgery. 2016; 79:E527–E529
- [11] Abe T, Matsumoto K, Sanno N, et al. Lymphocytic Hypophysitis: Case Report. Neurosurgery. 1995; 36:1016–1019
- [12] Lieblisch JM, Rogol AD, White BJ, et al. Syndrome of anosmia with hypogonadotropic hypogonadism (Kallmann syndrome): clinical and laboratory studies in 23 cases. American Journal of Medicine. 1982; 73:506–519
- [13] Nutkiewicz A, DeFeo DR, Kohout RI, et al. Cerebrospinal Fluid Rhinorrhea as a Presentation of Pituitary Adenoma. Neurosurgery. 1980; 6:195–197
- [14] Reid RL, Quigley ME, Yen SC. Pituitary Apoplexy: A Review. Arch Neurol. 1985; 42:712–719
- [15] Cardoso ER, Peterson EW. Pituitary Apoplexy: A Review. Neurosurgery. 1984; 14:363–373
- [16] Onesti ST, Wisniewski T, Post KD. Pituitary Hemorrhage into a Rathke's Cleft Cyst. Neurosurgery. 1990; 27:644–646
- [17] Shepard Matthew J, Snyder MHarrison, Soldozy Saunson, et al. Radiological and clinical outcomes of pituitary apoplexy: comparison of conservative management versus early surgical intervention. Journal of Neurosurgery. 2021; 135:1310–1318
- [18] Rovit RL, Fein JM. Pituitary Apoplexy, A Review and Reappraisal. J Neurosurg. 1972; 37:280–288
- [19] Liu JK, Couldwell W. Pituitary apoplexy: Diagnosis and management. Contemporary Neurosurgery. 2003; 25:1–5
- [20] Rajasekaran S, Vanderpump M, Baldeweg S, et al. UK guidelines for the management of pituitary apoplexy. Clin Endocrinol (Oxf). 2011; 74:9–20
- [21] Yen MY, Liu JH, Jaw SJ. Ptosis as the early manifestation of pituitary tumour. British Journal of Ophthalmology. 1990; 74:188–191
- [22] Telesca M, Santini F, Mazzucco A. Adenoma related pituitary apoplexy disclosed by ptosis after routine cardiac surgery: occasional reappearance of a dismal complication. Intensive Care Medicine. 2009; 35:185–186
- [23] Walton KA, Buono LM. Horner syndrome. Curr Opin Ophthalmol. 2003; 14:357–363
- [24] Bills DC, Meyer FB, Laws ER Jr, et al. A retrospective analysis of pituitary apoplexy. Neurosurgery. 1993; 33:602–8; discussion 608–9

- [25] Greenman Y, Stern N. Non-functioning pituitary adenomas. *Best Pract Res Clin Endocrinol Metab.* 2009; 23:625–638
- [26] Davis FG, Kupelian V, Freels S, et al. Prevalence estimates for primary brain tumors in the United States by behavior and major histology groups. *Neuro Oncol.* 2001; 3:152–158
- [27] Kwekkeboom DJ, de Jong FH, Lamberts SW. Gonadotropin release by clinically nonfunctioning and gonadotroph pituitary adenomas in vivo and in vitro: relation to sex and effects of thyrotropin-releasing hormone, gonadotropin-releasing hormone, and bromocriptine. *J Clin Endocrinol Metab.* 1989; 68:1128–1135
- [28] Chaidarun SS, Klibanski A. Gonadotropinomas. *Semin Reprod Med.* 2002; 20:339–348
- [29] Kawaguchi T, Ogawa Y, Ito K, et al. Follicle-stimulating hormone-secreting pituitary adenoma manifesting as recurrent ovarian cysts in a young woman—latent risk of unidentified ovarian hyperstimulation: a case report. *BMC Res Notes.* 2013; 6. DOI: 10.1186/1756-0500-6-408
- [30] Amar AP, Couldwell WT, Weiss MH. Prolactinomas: Focus on Indications, Outcomes, and Management of Recurrences. *Contemporary Neurosurgery.* 1989; 21:1–6
- [31] Esposito F, Dusick JR, Cohan P, et al. Early morning cortisol levels as a predictor of remission after transphenoidal surgery for Cushing's disease. *Journal of Clinical Endocrinology and Metabolism.* 2006; 91:7–13
- [32] Banasiak MJ, Malek AR. Nelson syndrome: comprehensive review of pathophysiology, diagnosis, and management. *Neurosurgical Focus.* 2007; 23
- [33] Assie G, Bahurel H, Coste J, et al. Corticotroph tumor progression after adrenalectomy in Cushing's Disease: a reappraisal of Nelson's syndrome. *Journal of Clinical Endocrinology and Metabolism.* 2007; 49:381–386
- [34] Bertagna X, Raux-Demay M-C, Guilhaume B, et al. Cushing's Disease. In: *The Pituitary*. 2nd ed. Malden, MA: Blackwell Scientific; 2002:496–560
- [35] Kasperlik-Zaluska A A, Bonicki W, Jeske W, et al. Nelson's syndrome - 46 years later: clinical experience with 37 patients. *Zentralblatt fur Neurochirurgie.* 2006; 67:14–20
- [36] Cook DM. AACE Medical Guidelines for Clinical Practice for the diagnosis and treatment of acromegaly. *Endocr Pract.* 2004; 10:213–225
- [37] Melmed S. Acromegaly. *N Engl J Med.* 1990; 322:966–977
- [38] Melmed S. Medical progress: Acromegaly. *N Engl J Med.* 2006; 355:2558–2573
- [39] Renehan AG, Shalet SM. Acromegaly and colorectal cancer: risk assessment should be based on population-based studies. *Journal of Clinical Endocrinology and Metabolism.* 2002; 87:1909–1909
- [40] Acromegaly Therapy Consensus Development Panel. Consensus Statement: Benefits Versus Risks of Medical Therapy for Acromegaly. *American Journal of Medicine.* 1994; 97:468–473
- [41] Beck-Peccoz P, Brucker-Davis F, Persani L, et al. Thyrotropin-secreting pituitary tumors. *Endocr Rev.* 1996; 17:610–638
- [42] Clarke MJ, Erickson D, Castro MR, et al. Thyroid-stimulating hormone pituitary adenomas. *J Neurosurg.* 2008; 109:17–22
- [43] Beck-Peccoz P, Persani L. Medical management of thyrotropin-secreting pituitary adenomas. *Pituitary.* 2002; 5:83–88
- [44] Wilson CB. Endocrine-Inactive Pituitary Adenomas. *Clinical Neurosurgery.* 1992; 38:10–31

52 Pituitary Tumors – Evaluation

52.1 History and physical

Directed to identify signs and symptoms of:

1. endocrine hyperfunction, including:
 - a) prolactin: amenorrhea (women), nipple discharge (primarily in women since estrogen is also required), impotence (males)
 - b) thyroid: heat intolerance, proptosis
 - c) growth hormone. Adults: change in ring size or shoe size or coarsening of facial features (acromegaly). Children: gigantism
 - d) cortisol: hyperpigmentation, Cushingoid features (moon facies, buffalo hump...)
2. endocrine deficits due to mass effect (p.864) on pituitary: hypothyroidism, hypocortisolism, growth hormone deficit...
3. visual field deficit: classically bitemporal hemianopsia (► Fig. 33.2) due to compression of the optic chiasm. Use bedside confrontational testing (p.588) as a screening test until formal visual fields can be obtained
4. deficits of cranial nerves within cavernous sinus
 - a) III, IV, VI: disorder of pupil and extraocular muscles
 - b) V1, V2: reduced sensation in forehead, nose, upper lip, and cheek

52.2 Diagnostic tests

52.2.1 Overview

Initial (screening) tests to work-up a patient presenting with a known or suspected pituitary mass are shown in ► Table 52.1. Further testing is indicated for abnormal results or for strong suspicion of specific syndromes.

52.2.2 Vision evaluation

General information

Practice guideline: Visual evaluation with PitNET/adenomas*

Level III²

- evaluation by an ophthalmologist is recommended
- automatic static perimetry is recommended to detect visual field defects
- visual evoked potentials may be useful in certain situations (e.g., when acuity and visual fields cannot be assessed) but the usefulness for routine testing is limited by significant false negative and false positive results
- older patients and those with visual loss > 4 months should be advised that the chances of return of vision post-op are reduced
- optical coherence tomography (OCT): not a practice standard. May be used to document extent of damage to optic disc

* these guidelines were developed for suspected *nonfunctioning* PitNET/adenomas, principles may not be generalizable

At a minimum, formal visual field testing should be done in all patients to detect visual field deficits and to serve as a baseline.

Visual fields

Formal visual field testing:

- by perimetry using a tangent screen (Goldmann perimetry). Use the small red stimulus since desaturation of color is an early sign of chiasmal compression
- or automated perimetry (p.589) (recommended²). E.g., Humphrey perimeter

Table 52.1 Summary of initial (screening) work-up for pituitary tumors (see indicated page for details)

Evaluation	Rationale
Vision to detect visual field deficits and serve as a baseline	
✓ Formal visual fields (most commonly Humphrey automated visual fields (HVF))	• compression of optic chiasm → visual field deficit (classically bitemporal hemianopsia, ► Fig. 33.2)
Endocrine screening	
✓ 8 A.M. cortisol ^a and/or 24-hour urine free cortisol ^a	• cortisol ↓ in hypoadrenalism (primary or secondary) • cortisol ↑ in hypercortisolism (Cushing syndrome) (p.876)
✓ TSH & either free T4 or free T4 index (FTI) ^b (✗ total T4 is not recommended for this purpose ^b)	Hypothyroidism • T4 ↓ & TSH ↑ in primary hypothyroidism (this may cause thyrotroph hyperplasia in the pituitary gland) • T4 ↓ & TSH nl or ↓ in secondary hypothyroidism (as in hypopituitarism) Hyperthyroidism (thyrotoxicosis) • T4 ↑ & TSH ↓ in primary hyperthyroidism • T4 ↑ & TSH ↑ in TSH-secreting PitNETs
✓ prolactin	• ↑ or ↑↑ with prolactinoma • slight ↑ with stalk effect (usually <90 ng/ml)
✓ gonadotropins (FSH, LH) and sex steroids ^c (♀: estradiol, ♂: testosterone)	• ↓ in hypogonadotrophic hypogonadism (from mass effect causing compression of the pituitary gland) • ↑ with gonadotropin-secreting adenoma
✓ insulin-like growth factor-1 (IGF-1) AKA somatomedin-C ^d	• ↑ in acromegaly • ↓ in hypopituitarism (one of the most sensitive markers)
✓ fasting blood glucose	• ↓ in hypoadrenalism (primary or secondary) • ↑ with hyperadrenalism
Radiographic studies (p.882) ✓ either:	
• MRI without & with contrast brain & pituitary protocol (test of choice), include image navigation protocol if navigation is to be used. Some surgeons also get noncontrast brain CT to delineate bony anatomy (especially of sphenoid sinus) • if MRI is contraindicated: CT without & with contrast (with coronal reconstruction) + cerebral angiogram	
^a 8 A.M. cortisol is the best test for hypocortisolism (e.g., to look for pituitary insufficiency); 24-hour urine free-cortisol is the best test for hypercortisolism (p.876) ¹ (e.g., to look for Cushing's syndrome)	
^b free T4 and FTI measure thyroid hormone available to tissues ("ignoring" protein bound hormone)	
^c elevated estrogen or testosterone from a pituitary tumor is extremely rare, and is usually related to exogenous hormone intake or dietary supplements (such as dehydroepiandrosterone (DHEA))	
^d IGF-1 is the primary test for excess growth hormone (GH); direct measurement of GH is unreliable due to pulsatile secretion	

► **Visual field deficit patterns** (► Fig. 33.2). Depends in part on location of chiasm with respect to the sella turcica: the chiasm is located above the sella in most patients, but it is posterior to the sella turcica (postfixed chiasm) in 4% (on imaging); and anterior to the sella (pre-fixed chiasm) in 8%³

1. compression of the optic chiasm:

a) bitemporal hemianopsia that obeys the vertical meridian (macular splitting): classic visual field deficit associated with a pituitary tumor. Due to impingement on crossing nasal fibers in the chiasm

b) other reported patterns that occur rarely: monocular temporal hemianopsia

2. optic nerve compression: more likely in patients with a postfixed chiasm

a) loss of vision in the ipsilateral eye. If carefully sought, there is usually a superior outer (temporal) quadrantanopsia in the contralateral eye⁴ (p.2135).⁵ ("pie in the sky" defect*) from compression of the anterior knee of Wilbrand (p.1478); may also be an early finding even without a post-fixed chiasm (so-called junctional scotoma = monocular visual loss + superior temporal quadrantanopsia in the contralateral eye)

* this "pie in the sky" defect is in the contralateral eye only, unlike the defect with Meyer's loop injuries which is bilateral

b) may produce central scotoma or monocular reduction in visual acuity

3. compression of the optic tract: may occur with a pre-fixed chiasm. Produces homonymous hemianopsia

Ophthalmology consult

A recommended option, see Practice guideline: Visual evaluation with PitNET/adenomas (p.874). In addition to visual field testing, it provides evaluation of the optic disc and other visual deficits and may help provide prognostic information for any visual loss that may be present.

Ancillary visual testing

Visual evoked potentials: usefulness is limited, see Practice guideline: Visual evaluation with pituitary adenomas (p.874).

Optical coherence tomography (OCT): an option, see Practice guideline: Visual evaluation with PitNET/adenomas (p.874) Uses light waves to provide high-resolution information about the thickness of the retina (including the optic disc). Does not utilize ionizing radiation. May be able to provide prognostic information.⁶

52.2.3 Initial endocrinologic evaluation (screening)

Overview

Practice guideline: Endocrine evaluation of PitNET/adenomas*

- Level II⁷:
 - all anterior pituitary axes should be routinely evaluated
 - prolactin levels should be routinely tested
 - cutoff values to initiate thyroid and adrenal replacement may be different in patients with panhypopituitarism (as opposed to patients with isolated deficits)
 - adrenal insufficiency and significant hypothyroidism should be treated pre-op
- Level III⁷: IGF-1 should be routinely tested to rule out subclinical growth hormone hypersecretion

* these guidelines were developed for suspected *nonfunctioning* PitNET/adenomas, principles may not be generalizable

52

See ► Table 52.1. May give indication of tumor type, determines whether any hormones need to be replaced, and serves as a baseline for comparison following treatment. Includes clinical assessment for signs and symptoms, as well as laboratory tests. Screening tests for anterior pituitary hormones should be checked in all patients with pituitary tumors.⁷ Note: selective loss of a *single* pituitary hormone together with thickening of the pituitary stalk is strongly suggestive of autoimmune hypophysitis (p.1656).

1. adrenal axis screening; see tests to assess **cortisol reserve** (p.880)
 - a) cortisol levels generally peak between 7–8 AM. AM cortisol may normally be slightly elevated above the reference range. 8 AM cortisol level: better for detecting hypocortisolism.¹ Normal: 6–18 mcg/100 ml. Interpretation:
 - 8 AM cortisol < 6 mcg/100 ml: suggestive of adrenal insufficiency
 - 8 AM cortisol 6–14 mcg/100 ml: nondiagnostic
 - 8 AM cortisol > 14 mcg/100 ml: adrenal insufficiency is unlikely
 - b) see distinguishing Cushing's disease from ectopic ACTH secretion (p.880) in questionable cases, including differentiating pseudo-Cushing states from Cushing's syndrome
 - c) 24-hour urine free cortisol: more accurate for *hypercortisolism*¹ (almost 100% sensitive and specific, false negative rare except in stress or chronic alcoholism). If not elevated several times above normal, at least 2 additional determinations should be made⁸
2. thyroid axis: the basis for thyroid screening is shown in ► Table 52.2
 - a) screening: T4 level (total or free), thyroid-stimulating hormone (TSH) (AKA thyrotropin). Normal values: free T4 index is 0.8–1.5, TSH 0.4–5.5 mIU/ml, total T4 4–12 mcg/100 ml (NB: be sure to check both T4 *AND* TSH)
 - b) further testing: thyrotropin-releasing hormone (TRH) *stimulation test* (indicated if T4 is low or borderline); check baseline TSH, give 500 mcg TRH IV, check TSH at 30 & 60 mins. Normal response: peak TSH twice baseline value at 30 mins. Impaired response with a low T4 indicates pituitary deficiency. Exaggerated response suggests primary hypothyroidism

3. gonadal axis
 - a) screening:
 - serum gonadotropins: FSH & LH
 - sex steroids: estradiol in women, testosterone in men (measure *total* testosterone)
 - b) further testing: none dependable in differentiating pituitary from hypothalamic disorders
4. prolactin levels (PRLs): for prolactin neurophysiology (p. 151)
 - a) interpretation is shown in ► Table 52.3. See ► Table 52.4 for differential diagnosis of hyperprolactinemia. With prolactinomas, PRL correlates with size of the tumor⁹: if PRL is < 200 ng/ml, ≈ 80% of tumors are microadenomas, and 76% of these will have normal PRL after surgery; if PRL > 200, only ≈ 20% are microadenomas
 - b) blood samples should be obtained midmorning (i.e., not soon after awakening) and not after stress, breast stimulation, or physical examination, which may increase PRLs
 - c) be aware of the following when interpreting PRLs:
 - because of variations in secretion (daily PRL fluctuations can be as high as 30%) and intrinsic inaccuracies of radioimmunoassay, PRLs should be rechecked if there is a reason to question a specific result
 - heterophilic antibodies (seen in individuals routinely exposed to animal serum products) can cause anomalous results
 - **stalk effect**: prolactin is the only pituitary hormone primarily under inhibitory regulation (p. 153). Injury to or compression of the hypothalamus or pituitary stalk from surgery or compression by any type of tumor can cause modest elevation of PRL due to decrease in prolactin inhibitory factors (PIFs). Rule of thumb: the percent chance of an elevated PRL being due to a prolactinoma is ≈ one half the PRL (details (p. 878))
 - lab report stating “prolactin level > X”: if the lab reports the PRL as “> X” (where X is some high value, e.g., 500 ng/ml) instead of specifying an actual number, it usually indicates a PRL so high that it exceeds the upper limits of the assay. Determining the actual number usually requires performing serial dilutions on the specimen until the PRL is in a range that their assay can quantify. Some labs will do this reflexively. If not, contact the lab and ask them to do this (they may be able to be done with the specimen they have, or else the patient will need to have another blood draw). This is important for: treatment decisions: **PRL > 500** usually indicates that surgery alone will not be able to normalize the PRL (see management guidelines (p. 888)); to assess response to treatment: it is essential to know what PRL you are starting with to determine response to medication, surgery, XRT...
 - hook effect: extremely high PRLs may overwhelm the assay (the large numbers of PRL molecules prevent the formation of the necessary PRL-antibody-signal complexes for radioimmunoassay) and produce falsely low results. Therefore, for large adenomas with a normal PRL and a high clinical suspicion of hyperprolactinemia, have the lab perform several dilutions of the serum sample and re-run the PRL
 - macroprolactinemia: occurs when prolactin molecules polymerize and bind to immunoglobulins. Prolactin in this form has reduced biological activity but produces lab values consistent with hyperprolactinemia. Clinical significance is controversial¹⁰; asymptomatic patients usually do not require treatment
5. growth hormone: due to pulsatile secretion of GH:
 - a) IGF-1 (somatomedin-C) level (p. 881) is the recommended initial test (elevated IGF-1 is extremely sensitive for acromegaly)
 - b) checking a single random GH level (p. 881) may not be a reliable indicator and is therefore *not* recommended
6. neurohypophysis (posterior pituitary): deficits are rare with pituitary tumors
 - a) screening: check adequacy of ADH by demonstrating concentration of urine with water deprivation
 - b) further testing: measurement of serum ADH in response to infusion of hypertonic saline

Thyroid screening

Table 52.2 Basis for thyroid screening (T₄ and TSH)

Rationale	T ₄	TSH
Primary hypothyroidism^a (problem with thyroid gland itself) - chronic primary hypothyroidism may produce secondary pituitary hyperplasia (pituitary pseudotumor) indistinguishable from adenoma on CT or MRI. Must be considered in any patient with a pituitary mass^{11,12} - pathophysiology: loss of negative feedback from thyroid hormones causes increased TRH release from the hypothalamus producing secondary hyperplasia of thyrotrophic cells in the adenohypophysis (thyrotroph hyperplasia). The patient may present due to pituitary enlargement (visual symptoms, elevated PRL from stalk effect, enlarged sella turcica on X-rays...) - chronic stimulation from elevated TRH may rarely produce thyrotroph adenomas - labs: T₄ low or normal, TSH elevated (>90–100 in patients presenting with thyrotroph hyperplasia), prolonged and elevated TSH response to TRH stimulation test (see text)	↓	↑
Secondary hypothyroidism^a (insufficient TSH stimulation of thyroid) - pituitary hypothyroidism accounts for only ≈ 2–4% of all hypothyroid cases¹³ - secondary hypothyroidism was found in 8–81% of patients with nonfunctioning pituitary tumors⁷ (pituitary compression reduces TSH) - labs: T₄ low, TSH low or normal, reduced response to TRH stimulation test (see text)	↓	↓ or nl
Primary hyperthyroidism (problem with thyroid gland itself) - etiologies: localized hyperactive thyroid nodule, circulating antibody that stimulates the thyroid, or diffuse thyroid hyperplasia (Graves' disease, AKA ophthalmic hyperthyroidism) - labs: T₄ elevated, TSH subnormal (usually <i>undetectable</i>)	↑	↓
Secondary hyperthyroidism (central hyperthyroidism) - etiologies - TSH-secreting PitNET/adenoma (rare) - pituitary resistance to thyroid hormones (disrupts negative feedback loop) - labs: T₄ elevated, TSH elevated or inappropriately normal	↑	↑ or nl
^a ✖ Caution: replacing thyroid hormone with inadequate cortisol reserves (as may occur in panhypopituitarism) can precipitate adrenal crisis; see management (p.886)		

52

Prolactin screening

Interpretation of prolactin levels (PRL) is shown in ► Table 52.3.

Table 52.3 Significance of prolactin levels^a

PRL (ng/ml)	Interpretation	Situations observed in
3–30 ^b	normal	non-pregnant female
10–400		pregnancy (► Table 52.4)
2–20		postmenopausal female
25 ^b –150	moderate elevation	- prolactinoma "stalk effect" (see text) - other causes^d
>150 ^c	significant elevation	prolactinoma ^d

^anote: ectopic sites of prolactin secretion have rarely been reported (e.g., in a teratoma¹⁴)

^bnormal values vary, use your lab's reference range

^csome authors recommend 200 ng/ml as the cutoff for probable prolactinomas¹⁵

^dfor DDx of hyperprolactinemia, see ► Table 52.4

For differential diagnosis of hyperprolactinemia, see ► Table 52.4.

► **Stalk effect.** Modest elevation of prolactin level (PRL) due to loss of inhibitory effect of PRIF from hypothalamus. There is no Class I evidence to support a threshold to distinguish between stalk effect and a prolactinoma.⁷ Mean PRL in nonfunctioning PitNETs was 39 ng/ml, with a majority of patients with stalk effect having PRL <200 ng/ml.⁷ Rule of thumb: the percent chance of an elevated

Table 52.4 Differential diagnosis of elevated prolactin (PRL) level (hyperprolactinemia)^a

1. pregnancy-related
 - a) during pregnancy^b: 10–400 ng/ml
 - b) postpartum: PRL decreases $\approx$ 50% (to $\approx$ 100 ng/ml) in the first week postpartum, and is usually back to normal in 3 weeks
 - c) in the lactating female: suckling increases prolactin, which is critical for lactogenesis (once initiated, nonpregnant PRLs can maintain lactation). First 2–3 months postpartum: basal PRL = 40–50 ng/ml, suckling $\rightarrow$ increases $\times$ 10–20. 3–6 months postpartum: basal PRLs become normal or slightly elevated, and double with suckling. PRL should normalize by 6 months after weaning
2. PitNET/adenoma
 - a) prolactinoma: larger prolactin microadenomas and macroadenomas usually produce PRL > 100 ng/ml
 - b) stalk effect (p. 877): rule of thumb, the percent chance of an elevated PRL being due to a prolactinoma is equal to one half the PRL
 - c) some tumors secrete both PRL and GH
3. drugs: dopamine receptor antagonists (e.g., phenothiazines, metoclopramide), oral contraceptives (estrogens), tricyclic antidepressants, verapamil, H₂ antagonists (e.g., ranitidine), some SSRIs, in particular paroxetine (Paxil®)¹⁶, carbamazepine rarely produces clinically significant increases¹⁷...
4. primary hypothyroidism: TRH, a prolactin releasing factor (PRF) (p. 153), will be elevated
5. empty sella syndrome (p. 955)
6. transient elevations in human serum prolactin (HSP) levels occur following 80% of generalized motor, 45% of complex partial, and only 15% of simple partial seizures.¹⁸ Peak levels are reached in 15–20 minutes, and gradually return to baseline over the subsequent hour.^{19,20,21}
7. breast or chest-wall trauma/surgery: usually $\leq$ 50 ng/ml
8. excessive exercise: usually $\leq$ 50 ng/ml
9. stress: in some cases the stress of having the blood test is enough to elevate PRL, anorexia nervosa
10. ectopic secretion: reported in renal cell or hepatocellular tumors, uterine fibroids, lymphomas
11. infiltrating hypothalamic tumors
12. renal failure
13. cirrhosis
14. macroprolactinemia: see text

^ahyperprolactinemia from causes other than prolactinomas rarely exceeds 200 ng/ml^balways R/O pregnancy as a cause of amenorrhea and hyperprolactinemia in a female with reproductive potential

PRL being due to a prolactinoma is equal to one half the PRL. Also, a very large tumor with a modest elevation of PRL suggests stalk effect and not a prolactinoma since large prolactinomas usually have a very high PRL. Persistent post-op PRL elevation may occur even with total tumor removal as a result of stalk injury (usually $\leq$ 90 ng/ml; stalk effect is doubtful if PRL > 150). For stalk effect, follow these patients, do not use bromocriptine.

52.2.4 Specialized endocrinologic tests

Cushing's syndrome

Table 52.5 Basis for biochemical tests in Cushing's syndrome (CS)

- normally, low dose DMZ suppress ACTH release through negative feedback on hypothalamic-pituitary axis, reducing urine and serum corticosteroids
- in $\geq$ 98% of cases of Cushing's syndrome, suppression occurs, but at a much higher threshold
- adrenal tumors and most (85–90%) cases of ectopic ACTH production (especially bronchial Ca) will not suppress even with high dose DMZ
- ACTH response to CRH is exaggerated in CS
- DMZ does not interfere with measurement of urinary and plasma cortisol and 17-hydroxycorticosteroids

Tests for hypercortisolism

These tests are used to determine if *hypercortisolism* (Cushing's syndrome, CS) is present or not, regardless of etiology. Usually only needed if the screening 24-hr urine free cortisol (p. 876) is equivocal (the basis of these tests is shown in ► Table 52.5).

1. overnight *low-dose* dexamethasone (DMZ) suppression tests²²:
 - a) overnight low dose test: give DMZ 1 mg PO @ 11 P.M. and draw serum cortisol the next day at 8 A.M. Results:

- cortisol < 1.8 mcg/dl (note: this is the currently accepted normal value; previously it was 5 mcg/dl): Cushing's syndrome is ruled out (except for a few patients with CS who suppress at low DMZ doses, possibly due to low DMZ clearance²³)
 - cortisol 1.8–10 mcg/dl: indeterminate, retesting is necessary
 - cortisol > 10 mcg/dl: CS is probably present. False positives can occur in the so-called pseudo-Cushing's state where ectopic CRH secretion produces hyperplasia of pituitary corticotrophs that is clinically indistinguishable from pituitary ACTH-producing tumors (requires further testing²³). Seen in: 15% of obese patients, in 25% of hospitalized and chronically ill patients, in high estrogen states, in uremia, and in depression. The combined DMZ-CRH test can be used to identify this (see reference²³). False positives also may occur in alcoholics or patients on phenobarbital or phenytoin due to increased metabolism of DMZ caused by induced hepatic microsomal degradation
- b) 2 day low-dose test (used when overnight test is equivocal): give DMZ 0.5 mg PO q 6 hrs for 2 days starting at 6 A.M.; 24 hr urine collections are obtained prior to test and on the 2nd day of DMZ administration. Normal patients suppress urinary 17-hydroxycorticosteroids (OHCS) to less than 4 mg/24 hrs, whereas = 95% of patients with CS have abnormal response (higher amounts in urine)²³
2. 11 PM salivary cortisol: this is the time of the usual cortisol nadir. Test must be run at NIH-approved lab. Accuracy is as good as low-dose DMZ suppression test

Distinguishing Cushing's disease from ectopic ACTH secretion

These tests are used to distinguish primary Cushing's disease (CD) (pituitary ACTH hypersecretion) from ectopic ACTH production and adrenal tumors (may be required since 40% of CD patients have a normal MRI¹)

1. random serum ACTH: if < 5 ng/L indicates ACTH independent CS (e.g., adrenal tumor). Not sensitive or specific due to variability of ACTH levels
2. abdominal CT: usually shows unilateral adrenal mass with adrenal tumors, or normal or bilateral adrenal enlargement in ACTH-dependent cases
3. **high-dose dexamethasone (DMZ) suppression test**: (NB: up to 20% of patients with CD do not suppress with high-dose DMZ. Phenytoin may also interfere with high-dose DMZ suppression²⁴)
 - a) overnight high-dose test: obtain a baseline 8 A.M. plasma cortisol level
 - b) then give DMZ 8 mg PO @ 11 P.M. and measure plasma cortisol level the next morning at 8 A.M.
 - c) in 95% of CD cases plasma cortisol levels are reduced to < 50% of baseline, whereas in ectopic ACTH or adrenal tumors it will usually be unchanged
4. **metyrapone (Metopirone®) test**: performed on an inpatient basis. Give 750 mg metyrapone (suppresses cortisol synthesis) PO q 4 hrs for 6 doses. Most patients with CD will have a rise in 17-OHCS in urine of 70% above baseline, or an increase in serum 11-deoxycortisol 400-fold above baseline
5. corticotropin-releasing hormone (CRH) stimulation test: CD responds to exogenous CRH 0.1 mcg/kg IV bolus with even further increased plasma ACTH and cortisol levels; ectopic ACTH and adrenal tumors do not²⁵
6. inferior petrosal sinus (IPS) sampling (or cavernous sinus sampling is preferred by some): done by interventional neuroradiologist. Uses a microcatheter to measure ACTH levels on each side at baseline, and then at 2, 5, and 10 minutes after stimulation with IV CRH (with simultaneous peripheral ACTH levels at each interval). General information:
 - a) IPS sampling is not needed when the following criteria of CD are met²⁶:
 - ACTH-dependent Cushing's disease
 - suppression with high-dose dexamethasone test (see above)
 - visible PitNET/adenoma on MRI
 - b) may also determine likely side of a microadenoma within the pituitary (thus may be able to avoid bilateral adrenalectomy, which requires lifelong gluco- and mineralo-corticoid replacement and risks Nelson's syndrome (p. 869) in 10–30%). 15–30%¹ of the time this test falsely lateralizes the tumor due to the communication through the circular sinus
 - c) a baseline IPS ACTH to peripheral ACTH ratio > 1.4:1 is consistent with primary Cushing's disease
 - d) a post CRH ratio > 3 is also consistent with primary Cushing's disease
 - e) complication rate: 1–2%, includes puncture of the sinus wall

Assessing cortisol reserve

1. cosyntropin stimulation test²⁷:

- a) draw a baseline cortisol level (fasting is not required; test can be performed at any time of day)
 - b) give cosyntropin (Cortrosyn®) (a potent ACTH analogue) 1 ampoule (250 mcg) IM or IV
 - c) then check cortisol levels at 30 mins (optional) and at 60 mins
 - d) normal response: peak cortisol level > 18 mcg/dl AND an increment > 7 mcg/dl, or a peak > 20 mcg/dl regardless of the increment
 - e) subnormal response: indicates adrenal insufficiency. In primary adrenal insufficiency, pituitary ACTH secretion will be elevated. In secondary adrenal insufficiency, chronically reduced ACTH causes adrenal atrophy and unresponsiveness to acute stimulation with this exogenous ACTH analogue
 - f) normal response: rules out primary and overt secondary adrenal insufficiency, but may be normal in *mild* cases of reduced pituitary ACTH or *early* after pituitary surgery where adrenal atrophy has not occurred. In these cases further testing may be positive: see metyrapone test (p.902) or ITT (see below)
2. insulin tolerance test (ITT): “gold standard” for assessing integrity of the hypothalamic-pituitary-adrenal axis. Cumbersome to do. Abnormal in 80% of CS. Assesses ACTH, cortisol & GH reserve
- a) rationale: an appropriate cortisol increment in response to insulin-induced hypoglycemia suggests patient will also be able to respond to other stresses (acute illness, surgery...)
 - b) contraindications: seizure disorder, ischemic cardiac disease, untreated hypothyroidism
 - c) pre-test preparation: D/C estrogen replacement for 6 weeks prior to test. Have 50 ml of D50 and 100 mg IV hydrocortisone available during test
 - d) protocol: give regular insulin 0.1 U/kg IV push, and draw blood for glucose, cortisol and GH at 0, 10, 20, 30, 45, 60, 90 and 120 mins (monitor blood sugar by fingerstick during test, and give IV glucose if patient becomes symptomatic). If fingerstick blood sugar is not < 50 mg/dl by 30 minutes and patient is asymptomatic, give additional regular insulin 5 U IVP. There must be 2 specimens after adequate hypoglycemia
 - e) results:
 - if adequate hypoglycemia (< 40 mg/dl) was not accomplished: cortisol or GH deficiency cannot be diagnosed
 - normal: cortisol increment > 6 mcg/dl to a peak > 20
 - peak cortisol = 16–20: steroids needed only for stress
 - peak cortisol < 16: glucocorticoid replacement needed
 - Cushing syndrome: increment < 6

Acromegaly

For suspected acromegaly, the most useful test is an IGF-1 level.

1. insulin-like growth factor-1 (IGF-1) (formerly somatomedin-C) level: an excellent integrative marker of average GH secretion. Normal levels depend on age (peaking during puberty), gender, pubertal stage and lab. Typical fasting levels by age are shown in ► Table 52.6. Estrogen may suppress IGF-1 levels

Table 52.6 Normal IGF-1 by age

Age (yrs)	Level (ng/ml)
1–5	49–327
6–8	52–345
9–11	74–551
12–15	143–996
16–20	141–903
21–39	109–358
40–54	87–267
> 54	55–225

2. growth hormone (GH): normal basal fasting level is < 5 ng/ml. In patients with acromegaly, GH is usually > 10 ng/ml but can be normal. Normal basal levels do not reliably distinguish normal patient from GH deficiency.²⁸ Furthermore, due to pulsatile secretion of GH, normal patients may have sporadic peaks up to 50 ng/ml.²⁹ Occasionally acromegaly may be present even with GH

levels as low as 37 pg/ml.³⁰ ∴ random GH levels are not generally useful for diagnosing acromegaly (see above for IGF-1)

3. other tests used uncommonly

- a) oral glucose suppression test (OGST): less precise and more expensive than measuring IGF-1; however, may be more useful than IGF-1 for monitoring initial response to therapy. GH levels are measured at 0, 30, 60, 90, and 120 minutes after a 75 gm oral glucose load. If the GH nadir is not < 1 ng/ml, the patient is acromegalic.^{31,32} GH suppression may also be absent with liver disease, uncontrolled DM & renal failure. ✖ Relatively contraindicated in patients with DM and high glucose levels
 - b) growth-hormone releasing hormone (GHRH) levels: may help diagnose ectopic GH secretion in a patient with proven acromegaly with no evidence of pituitary tumor on imaging. If an extrapituitary source is suspected, chest and abdominal CT and/or MRI should also be obtained³³
 - c) GHRH stimulation test: results may be discordant in up to 50% of patients with acromegaly³¹ and is thus rarely used (as of this writing, pharmaceutical production of GHRF has been discontinued)
4. octreotide scan: SPECT imaging 4 and 24 hours after injection with 6.5 mCi of indium-111 OctreoScan, a somatostatin receptor imaging agent

52.2.5 Radiographic evaluation

General information

Practice guideline: Imaging of PitNET/adenomas*

- Level II³⁴:
 - MRI without and with contrast with dedicated images through the pituitary spaced 1–2 mm apart is the imaging modality of choice
 - thin-cut CT through the sella/sphenoid region and/or CTA may be used to augment the MRI (or may be used in cases where MRI is contraindicated)
- Level III³⁴: 3 T MRI scanners may provide finer anatomic resolution relative to 1.5 T scanners

* these guidelines were developed for suspected *nonfunctioning* PitNET/adenomas, principles may not be generalizable

≈ 50% of pituitary tumors causing Cushing's syndrome are too small to be imaged on CT or MRI at the time of initial presentation (therefore endocrinologic testing is required to prove the pituitary origin). See differential diagnosis of intrasellar lesions (p.1653); some are indistinguishable radiographically.

Normal AP diameter of pituitary gland: female of childbearing age (≈ 13–35 yrs): ≤ 11 mm, for all others normal is ≤ 9 mm. (Note: pituitary glands in adolescent girls may be physiologically enlarged (mean height: 8.2 ± 1.4 mm) as a result of hormonal stimulation of puberty.³⁵)

MRI

MRI is the imaging test of choice for pituitary tumors.

Ordering the MRI:

1. a pituitary protocol (p.241) is especially helpful when searching for a microadenoma
2. for follow-up of macroadenomas: routine coronal and sagittal pituitary MRI without and with contrast with detailed coronal cuts through the pituitary often suffices (i.e., dynamic images are not required unless there is a specific need to visualize the pituitary gland as distinct from the tumor)

Findings: MRI provides information about invasion of cavernous sinus, and about location and/or involvement of parasellar carotids (► Fig. 52.1). MRI may fail to demonstrate microadenomas in 25–45% of cases of Cushing's disease.³⁶ 3 T vs. 1.5 T MRI: based on 5 cases of Cushing's disease, a 3 T MRI showed the adenoma more clearly in 2 cases, in 1 case it showed the tumor on the correct side opposite to where the 1.5 T MRI showed it, and in 2 cases neither 1.5 T nor 3 T MRI could show the microadenoma.³⁷

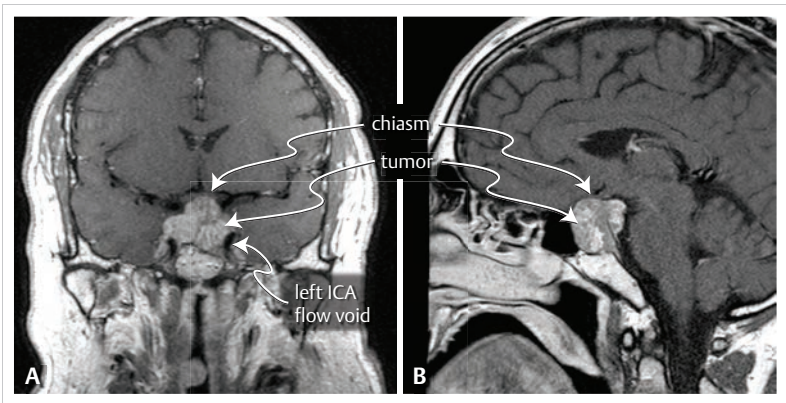


Fig. 52.1 Pituitary macroadenoma.

This 3 cm diameter tumor has grown superiorly through the diaphragma sella and is minimally elevating the optic chiasm. The tumor has invaded both cavernous sinuses, and flow voids within the cavernous segment of the internal carotid arteries (ICAs) are shown.

Image: T1 MRI with contrast. A: Coronal, B: Sagittal.

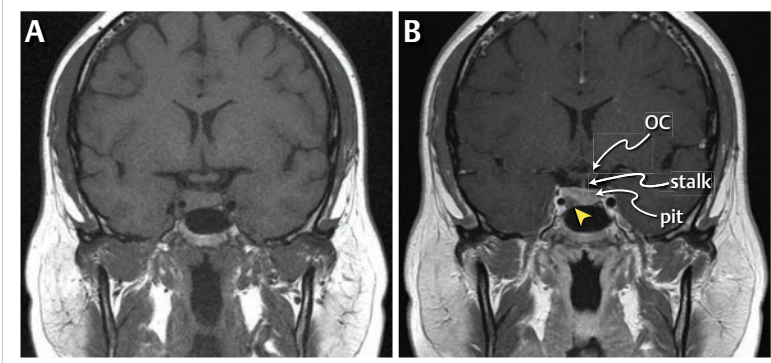


Fig. 52.2 Pituitary microadenoma in a 27-year-old woman with Cushing's disease.

Coronal MRI. A: T1WI noncontrast, B: T1WI with contrast.

The tumor is not visible on the noncontrast image (A), but shows up on the contrast image (B) as a hypointense area (yellow arrowhead) when scanned shortly after the administration of contrast.

Abbreviations: OC = optic chiasm, stalk = pituitary stalk, pit = pituitary gland.

Microadenoma: 75% are low signal on T1WI, and high signal on T2WI (but 25% can behave in any way, including completely opposite to above). Enhancement is very time-dependent. Imaging must be done with 5 minutes of contrast administration to see a discrete microadenoma. Initially, gadolinium enhances the normal pituitary (no blood-brain barrier) but *not* the pituitary tumor (► Fig. 52.2). After ~ 30 minutes, the tumor enhances about the same. Dynamic MRI scans have been used to increase the sensitivity (contrast is injected while the MRI scanner is running).

Neurohypophysis: normally is high signal on T1WI³⁸ (possibly due to phospholipids). Absence of this "bright spot" often correlates with diabetes insipidus as may occur with autoimmune hypophysitis (p. 1656); however, failure to image the bright spot is not uniformly abnormal.

Deviation of the pituitary stalk may also indicate the presence of a microadenoma. Normal thickness of the pituitary stalk is approximately equal to basilar artery diameter. Thickening of stalk is

usually NOT adenoma; differential diagnosis for a thickened stalk: lymphoma, autoimmune hypophysitis (p. 1656), granulomatous disease, hypothalamic glioma.

Cavernous sinus invasion: numerous criteria have been proposed to determine the degree of cavernous sinus invasion which correlates with degree of resectability. Knosp et al.³⁹ utilized 3 lines (MLL, MNL, and LL) as shown in ► Fig. 2.11, and the tumor invasion is graded as shown in ► Table 52.7.

Table 52.7 Knosp et al.³⁹ grading of cavernous sinus invasion as modified by Micko et al.⁴⁰ in 137 tumors grade ≥ 1

Grade	Feature	Percent invasive ^a	GTR	ER
0	tumor does not cross lateral to the MLL	0%		
1	tumor crosses the MLL but not the MNL (intercarotid line)	1.5%	83%	88%
2	tumor crosses MNL but not LL	10%	71%	60%
3A ^b	tumor extends lateral to LL superiorly (above ICA)	26%	85%	67%
3B ^b	tumor extends lateral to LL inferiorly (below ICA)	71%	64%	0%
4	tumor wraps completely around the intracavernous ICA	100%	0%	0%

^apercentage of tumors in the grade that were invasive

^bthe original grade 3 was subdivided into 3A and 3B by Micko et al.⁴⁰

Abbreviations: GTR = gross total resection, ER = endocrinologic remission in functioning tumors, MLL = medial line, MNL = median line (intercarotid line), LL lateral line (see ► Fig. 2.11 for illustration of the lines used)

CT

Generally superseded by MRI. May be appropriate when MRI is contraindicated (e.g., pacemaker). May also be considered as pre-op assessment in cases for transsphenoidal surgery (demonstrates sphenoid septal anatomy with more detail³⁴). If MRI cannot be done, consider also CTA to demonstrate parasellar carotid arteries and in questionable cases to R/O giant aneurysm which may mimic an adenoma (p. 1655).

Calcium in a pituitary tumor usually signifies previous hemorrhage or infarction within the tumor.

Angiography

Sometimes used pre-op for transsphenoidal surgery to localize the parasellar carotids (note: generally supplanted by CTA). NB: MRI also provides this information, and evaluates invasion of cavernous sinuses, usually obviating the need for angiography.

Skull X-ray

A lateral skull X-ray may help define the bony anatomy of the sphenoid sinus in cases where transsphenoidal surgery is contemplated (currently, CT scan is usually preferred).

References

- [1] Chandler WF. Treatment of disorders of the pituitary gland: pearls and pitfalls from 30 years of experience. *Clinical Neurosurgery*. 2008; 56:18–22
- [2] Newman SA, Turbin RE, Bodach ME, et al. Congress of Neurological Surgeons Systematic Review and Evidence-Based Guideline on Pretreatment Ophthalmology Evaluation in Patients With Suspected Nonfunctioning Pituitary Adenomas. *Neurosurgery*. 2016; 79:E530–E532
- [3] Griessenauer CJ, Raborn J, Mortazavi MM, et al. Relationship between the pituitary stalk angle in prefixed, normal, and postfixed optic chiasmata: an anatomic study with microsurgical application. *Acta Neurochir (Wien)*. 2014; 156:147–151
- [4] Walsh FB, Hoyt WF. *Clinical Neuro-Ophthalmology*. Baltimore 1969
- [5] Donaldson LC, Eshtiaqui A, Sacco S, et al. Junctional Scotoma and Patterns of Visual Field Defects Produced by Lesions Involving the Optic Chiasm. *J Neuroophthalmol*. 2022; 42:e203–e208
- [6] Jacob M, Raverot G, Jouanneau E, et al. Predicting visual outcome after treatment of pituitary adenomas with optical coherence tomography. *Am J Ophthalmol*. 2009; 147:64–70 e2
- [7] Fleseriu M, Bodach ME, Tumalian LM, et al. Congress of Neurological Surgeons Systematic Review and Evidence-Based Guideline for Pretreatment Endocrine Evaluation of Patients With Nonfunctioning Pituitary Adenomas. *Neurosurgery*. 2016; 79:E527–E529
- [8] Watts NB. Cushing's Syndrome: An Update. *Contemporary Neurosurgery*. 1995; 17:1–7
- [9] Gillam MP, Molitch ME, Lombardi G, et al. Advances in the treatment of prolactinomas. *Endocrine Reviews*. 2006; 27:485–534
- [10] Olukoga AO. Macroprolactinemia is clinically important. *Journal of Clinical Endocrinology and Metabolism*. 2002; 87:4833–4834
- [11] Bilaniuk LT, Moshang T, Cara J, et al. Pituitary Enlargement Mimicking Pituitary Tumor. *Journal of Neurosurgery*. 1985; 63:39–42

- [12] Atchison JA, Lee PA, Albright L. Reversible Suprasellar Pituitary Mass Secondary to Hypothyroidism. *JAMA*. 1989; 262:3175–3177
- [13] Watanakunakorn C, Hodges RE, Evans TC. Myxedema. A Study of 400 Cases. *Arch Intern Med*. 1965; 116:183–190
- [14] Kallenberg GA, Pesce CM, Norman B, et al. Ectopic Hyperprolactinemia Resulting From an Ovarian Teratoma. *JAMA*. 1990; 263:2472–2474
- [15] Randall RV, Scheithauer BW, Laws ER, et al. Pituitary Adenomas Associated with Hyperprolactinemia. *Mayo Clin Proc*. 1985; 60:753–762
- [16] Cowen PJ, Sargent PA. Changes in plasma prolactin during SSRI treatment: evidence for a delayed increase in 5-HT neurotransmission. *J Psychopharmacol*. 1997; 11:345–348
- [17] Torre DL, Falorni A. Pharmacological causes of hyperprolactinemia. *Ther Clin Risk Manag*. 2007; 3:929–951
- [18] Wyllie E, Luders H, MacMillan JP, et al. Serum Prolactin Levels After Epileptic Seizures. *Neurology*. 1984; 34:1601–1604
- [19] Dana-Haeri J, Trimble MR, Oxley J. Prolactin and Gonadotropin Changes Following Generalized and Partial Seizures. *Journal of Neurology, Neurosurgery, and Psychiatry*. 1983; 46:331–335
- [20] Prichard PB, Wannamaker BB, Sagel J, et al. Serum Prolactin and Cortisol Levels in Evaluation of Pseudoepileptic Seizures. *Ann Neurol*. 1985; 18:87–89
- [21] Laxer KD, Mullooly JP, Howell B. Prolactin Changes After Seizures Classified by EEG Monitoring. *Neurology*. 1985; 35:31–35
- [22] Tyrell JB, Aron DC, Forsham PH, et al. Glucocorticoids and Adrenal Androgens. In: *Basic and Clinical Endocrinology*. 3rd ed. Norwalk: Appleton and Lange; 1991:323–362
- [23] Yanovski JA, Cutler GB, Chrousos GP, et al. Corticotropin-releasing hormone stimulation following low-dose dexamethasone administration: A new test to distinguish Cushing's syndrome from pseudo-Cushing's states. *Journal of the American Medical Association*. 1993; 269:2232–2238
- [24] McCutcheon IE, Oldfield EH, Barrow DL, et al. Cortisol: Regulation, Disorders, and Clinical Evaluation. In: *Neuroendocrinology*. Baltimore: Williams and Wilkins; 1992:117–173
- [25] Chrousos GP, Schulte HM, Oldfield EH, et al. The Corticotropin-Releasing Factor Stimulation Test: An Aid in the Evaluation of Patients with Cushing's Syndrome. *New England Journal of Medicine*. 1984; 310:622–626
- [26] Esposito F, Dusick JR, Cohan P, et al. Early morning cortisol levels as a predictor of remission after transphenoidal surgery for Cushing's disease. *Journal of Clinical Endocrinology and Metabolism*. 2006; 91:7–13
- [27] Watts NB, Tindall GT. Rapid assessment of corticotropin reserve after pituitary surgery. *JAMA*. 1988; 259:708–711
- [28] Abboud CF. Laboratory Diagnosis of Hypopituitarism. *Mayo Clin Proc*. 1986; 61:35–48
- [29] Melmed S. Acromegaly. *N Engl J Med*. 1990; 322:966–977
- [30] Dimaraki EV, Jaffe CA, DeMott-Friberg R, et al. Acromegaly with apparently normal GH secretion: implications for diagnosis and follow-up. *J Clin Endocrinol Metab*. 2002; 87:3537–3542
- [31] Cook DM. AACE Medical Guidelines for Clinical Practice for the diagnosis and treatment of acromegaly. *Endocr Pract*. 2004; 10:213–225
- [32] Melmed S. Medical progress: Acromegaly. *N Engl J Med*. 2006; 355:2558–2573
- [33] Frohman LA. Ectopic hormone production by tumors: growth hormone-releasing factor. *Neuroendocrine Perspect*. 1984; 3:201–224
- [34] Chen CC, Carter BS, Wang R, et al. Congress of Neurological Surgeons Systematic Review and Evidence-Based Guideline on Preoperative Imaging Assessment of Patients With Suspected Nonfunctioning Pituitary Adenomas. *Neurosurgery*. 2016; 79:E524–E526
- [35] Peyster RG, Hoover ED, Viscarello RR, et al. CT Appearance of the Adolescent and Preadolescent Pituitary Gland. *American Journal of Neuroradiology*. 1983; 4:411–414
- [36] Watson JC, Shawker TH, Nieman LK, et al. Localization of Pituitary Adenomas by Using Intraoperative Ultrasound in Patients with Cushing's Disease and No Demonstrable Pituitary Tumor on Magnetic Resonance Imaging. *Journal of Neurosurgery*. 1998; 89:927–932
- [37] Kim LJ, Lekovic GP, White WL, et al. Preliminary Experience with 3-Tesla MRI and Cushing's Disease. *Skull Base*. 2007; 17:273–277
- [38] Kucharczyk W, Davis DO, Kelly WM, et al. Pituitary adenomas: high-resolution MR imaging at 1.5 T. *Radiology*. 1986; 161:761–765
- [39] Knosp E, Steiner E, Kitz K, et al. Pituitary Adenomas with Invasion of the Cavernous Sinus Space: A Magnetic Resonance Imaging Classification Compared with Surgical Findings. *Neurosurgery*. 1993; 33:610–618
- [40] Micko AS, Wohrer A, Wolfsberger S, et al. Invasion of the cavernous sinus space in pituitary adenomas: endoscopic verification and its correlation with an MRI-based classification. *J Neurosurg*. 2015; 122:803–811

53 PitNET/Adenomas – General Management

53.1 Management/treatment recommendations

53.1.1 General information

For pituitary apoplexy (p.865). For large invasive adenomas (p.886).

Note: Prolactinoma is the only pituitary tumor for which medical therapy (dopamine agonist) is the primary treatment modality (in certain cases).

Hormone replacement therapy (HRT)

HRT may be required for patients with documented endocrine deficits pre-op or post-op in cases where pituitary function is not normal. Critical issues:

1. corticosteroids
 - a) indications: inadequate cortisol reserve as demonstrated by failing a cosyntropin stimulation test (viz. failure to achieve a peak cortisol level > 18 mcg/dl in response to cosyntropin) (p.880)
 - b) may start cortisol immediately after bloodwork for cosyntropin test is drawn (do not need to wait for test results)—then, when test results are available, continue or stop therapy based on test results
 - c) **R** physiologic replacement dose: cortisol 20 mg PO q AM and 10 mg PO q 4 PM. Stress doses may be needed (p.155) in some situations
2. thyroid hormone replacement
 - a) **x** thyroid replacement can precipitate adrenal crisis if started before cortisol in a patient with adrenal insufficiency (as may occur in panhypopituitarism)
 - **Δ** do a cosyntropin stimulation test (p.880) and start cortisol
 - thyroid replacement may be initiated after 1 full day of cortisol.
 - R**: start with synthroid 125 mcg/d
 - b) although there are warnings not to do surgery on a hypothyroid patient, the reality is that it takes 3–4 weeks for adequate replacement, and hypothyroid patients frequently undergo surgery before then with no untoward effect
3. testosterone replacement: may increase intratumoral levels of estradiol, which may promote tumor growth. **Δ** wait for stabilization of tumor before starting

53.1.2 Management of large, invasive adenomas

See reference.¹

1. prolactinomas
 - a) dopamine agonists (DA) (p.888) unless there is unstable deficit
 - b) for unstable deficit, or if the tumor does not respond to DAs: debulk the tumor transsphenoidally and then rechallenge with DA therapy
2. tumors secreting growth hormone or ACTH: an aggressive surgical approach is indicated with these tumors since the secretion product is harmful and effective medical adjuvants are lacking
 - a) pre-treat invasive GH-secreting tumors with somatostatin analogue therapy before surgery to reduce surgical risks (general and cardiac)
 - b) elderly patients or tumors > 4 cm diameter: debulk tumor transsphenoidally and/or adjuvant therapy (XRT and/or medications)
 - c) young age and size < 4 cm: radical surgery (may utilize a cranio-orbito-zygomatic skull base approach; may be curative)
3. nonfunctional adenomas
 - a) elderly patient: expectant management is an option, with intervention for signs of progression (radiographic or neurologic)
 - b) central tumor or elderly patient with progression: transsphenoidal tumor debulking and/or XRT (residual tumor in the region of the cavernous sinus may show little or no change over several years, and with these nonfunctional tumors, there is less risk in following them than if there is a harmful secretion product)
 - c) parasellar tumor and/or young age: radical surgery (often not curative)

53.1.3 Nonfunctioning PitNET/adenomas—management

Surgical indications

Practice guideline: Management of nonfunctioning PitNET/adenomas (NFPAs)

Level III²: surgical resection is recommended as the primary treatment modality for *symptomatic** NFPAs

* there was insufficient evidence to make recommendations for asymptomatic NFPAs

Surgical resection is recommended for *symptomatic* NFPAs (based on a large amount of class III data), as opposed to limited class III evidence that showed inconsistent benefits for: observation alone (2 studies), primary XRT (3 studies) or primary medical treatment (8 studies) for improving vision, H/A, hypopituitarism or tumor volume.²

Literature on nonsurgical treatment modalities for NFPAs

- medical management: reported tumor response rates using dopamine agonists (e.g., bromocriptine) in 0–60% of cases, somatostatin analogs (e.g., Octreotide) in 12–40%, and combination therapy in 60%.² These agents are therefore not recommended as primary treatment due to lack of a significant and consistent response²
- XRT: studies have not shown equivalence or superiority of XRT as primary management compared to surgical management. XRT has been demonstrated to be an effective adjunctive therapy for post-op residual tumor or recurrence (p.893)
- natural history: NFPA *microadenomas* have low growth propensity (estimated 3.2–12.5%), whereas ≈ 36% of NFPA *macroadenomas* enlarged (follow-up ranging 32–85 months).^{3,4} Of the enlarging macroadenomas, 45% were asymptomatic. A minority of tumors actually shrink with time. Two studies showed tumor progression in 40–50%, and 21–28% required surgery.^{5,6} Observation only is not recommended for *symptomatic* NFPAs²

Follow-up recommendations for NFPAs managed not meeting surgical criteria

Suggested management for incidental (asymptomatic) NFPAs that do not meet surgical criteria:

1. frequency of follow-up pituitary-MRI imaging⁷
 - a) for microadenoma (≤ 1 cm maximal diameter): years 1, 2, 5 and ± 10 after the initial study, less frequently thereafter
 - b) for macroadenoma (> 1 cm diameter): 6 months after initial study and then, if stable, annually for an additional 3 years, and then less frequently thereafter
2. duration of follow-up MRI: because growth may be very delayed, Dekkers recommended surveillance imaging for a total of 22 years⁵

Surgical indications for nonfunctioning pituitary macroadenomas

1. tumors causing symptoms by mass effect: visual field deficit (classically: bitemporal hemianopia, panhypopituitarism)
2. some surgeons recommend surgery for macroadenomas that elevate the chiasm even in the absence of endocrine abnormalities or visual field deficit because of the possibility of injury to the optic apparatus
3. see invasive pituitary macroadenomas (p.855)
4. acute and rapid visual or other neurologic deterioration. May represent ischemia of the chiasm, or tumor hemorrhage/infarction causing expansion (pituitary apoplexy (p.865)). The immediate concern is irreversible blindness (hypopituitarism, while serious, can be treated with replacement therapy). Visual loss usually requires *emergent* decompression. Some surgeons feel that a transcranial approach is necessary, but transsphenoidal decompression is usually satisfactory^{1,8}
5. to obtain tissue for pathological diagnosis in questionable cases

6. Nelson's syndrome (p. 869)
 - a) surgery (transphenoidal or transcranial): the primary treatment. The aggressiveness of the tumor sometimes requires total hypophysectomy
 - b) XRT (possibly SRS) is used following subtotal excision
 - c) medical therapy is usually ineffective. Agents that could be considered include⁹: dopamine agonists, valproic acid, somatostatin analogues, rosiglitazone, and serotonin agonists

53.1.4 Prolactinomas—management

Management guidelines

Prolactinomas are essentially the only PitNET/adenoma for which medical therapy is sometimes the primary treatment modality.

1. prolactin level (PRL) < 500 ng/ml in tumors that are not extensively invasive (see below for invasive tumors): PRL may be normalized with surgery
2. PRL > 500 ng/ml: the chances of normalizing PRL surgically are very low.¹⁰ Algorithm:
 - a) if no acute *progression* (worsening vision...), an initial attempt at purely medical control should be made as the chances of normalizing PRL surgically with pre-op levels > 500 ng/ml are very low¹⁰ (these tumors may shrink dramatically with bromocriptine)
 - b) response should be evident by 4–6 weeks (significant decrease in PRL, improvement of visual deficits, or shrinkage on MRI)
 - c) if tumor not controlled medically ($\approx 18\%$ will not respond to bromocriptine, an attempt with cabergoline may also be considered): surgery followed by reinstitution of medical therapy may normalize PRL

Medical management with dopamine agonists

Dopamine agonists

Side effects¹¹: (may vary with different preparations) nausea, H/A, fatigue, orthostatic hypotension with dizziness, cold induced peripheral vasodilatation, depression, nightmares and nasal congestion. Side effects are more troublesome during the first few weeks of treatment. Tolerance may be improved by bedtime dosing with food, slow dose escalation, sympathomimetics for nasal congestion, and acetaminophen 1–2 hrs before dosing to reduce H/A. Psychosis and vasospasm are rare side effects that usually necessitate discontinuation of the drug.

Drug info: Bromocriptine (Parlodel®)

A semi-synthetic ergot alkaloid that binds to dopamine receptors (dopamine agonist) on normal and tumor lactotrophs, inhibiting synthesis and secretion of PRL and other cell processes resulting in decreased cell division and growth. Bromocriptine lowers prolactin level, regardless of whether the source is an adenoma or normal pituitary (e.g., as a result of stalk effect), to < 10% of pretreatment values in most patients. It also frequently reduces the tumor size in 6–8 weeks in 75% of patients with macroadenomas, but only as long as therapy is maintained and only for tumors that actually produce prolactin. Only $\approx 1\%$ of prolactinomas continue to grow while the patient is on bromocriptine. Prolactinomas may enlarge rapidly upon discontinuation of the drug. However, permanent normoprolactinemia can occur (see below).

Pregnancy issues: bromocriptine can restore fertility. Continued therapy during pregnancy is associated with a 3.3% incidence of congenital anomalies and 11% spontaneous abortion rate which is the same as for the general population. Estrogen elevation during pregnancy stimulates hyperplasia of lactotrophs and some prolactinomas, but the risk of symptomatic enlargement of microadenomas and totally intrasellar macroadenomas is < 3%, vs. 30% risk for macroadenomas.¹²

Prolonged treatment with bromocriptine may reduce the chances of surgical cure if this should be chosen at a later date. With a microadenoma, one year of bromocriptine may reduce the surgical cure rate by as much as 50%, possibly due to induced fibrosis.¹³ Thus, it is suggested that if surgery is to be done that it be done in the first 6 months of bromocriptine therapy. Shrinkage of large tumors due to bromocriptine may cause CSF rhinorrhea.¹⁴ **Side effects:** see above.

R: start with 1.25 mg (half of a 2.5 mg tablet) PO q hs (nighttime dosing reduces some side effects) (vaginal administration is an alternative). Add additional 2.5 mg per day as necessary (based on prolactin levels (PRLs)), making a dosage change every 2–4 weeks for microadenomas, or every 3–4 days for macroadenomas causing mass effect. Initial recheck of prolactin level after about 4 weeks at a reasonable dose to verify response. ★ To shrink large tumors or for extremely high PRLs, higher

doses are usually needed initially (e.g., 7.5 mg TID for $\approx$ 6 mos), and then lower doses may be able to maintain normal levels (typical maintenance dosage: 5–7.5 mg daily (range: 2.5–15 mg) which may be given as a single dose or divided TID). **Supplied:** 2.5 mg scored tabs; 5 mg capsules.

Drug info: Cabergoline (Dostinex®)

An ergot alkaline derivative that is a selective D2 dopamine agonist (bromocriptine (see above) affects both D2 and D1 receptors).¹⁵ The elimination half-life is 60–100 hrs which usually permits dosing 1–2 times weekly. Control of PRL and resumption of ovulatory cycles may be better than with bromocriptine.¹⁶ **Side effects:** (see above) H/A and GI symptoms are reportedly less problematic than with bromocriptine. **✗** Cardiac valve disease¹⁷ (incidence-rate ratio, 4.9; 95% CI) affecting the mitral, aortic, and tricuspid valves, possibly leading to regurgitation (the drug activates 5-HT2B receptors which induces prolonged mitogenic effects in fibrocytoblasts, which may lead to valvular fibroplasia) which has not been observed at doses used for prolactinomas (it is associated with doses used for Parkinson's disease which are $> 10 \times$ pituitary doses); recommendation: do not discontinue cabergoline for this reason if dose is < 2 mg/wk. **✗** Contraindications: eclampsia or preeclampsia, uncontrolled HTN. Dosage should be reduced with severe hepatic dysfunction.

R: Start with 0.25 mg PO twice weekly, and increase each dose by 0.25 mg every 4 weeks as needed to control PRL (up to a maximum of 3 mg per week). Typical dose is 0.5–1 mg twice weekly. Some combine the total dose and give it once weekly. Initial recheck of prolactin level after about 4 weeks to verify response. **Supplied:** 0.5 mg scored tablets.

Response to medical treatment

Treatment response to DA is assessed with serial prolactin levels as shown in ► Table 53.1. It is uncommon for a prolactinoma to enlarge without an increase in prolactin level.¹⁸

Discontinuation of dopamine agonists: Long-term therapy with DA agonists has some cytotoxic effect on pituitary tissue. In an early report, discontinuation of treatment after 24 months was associated with $> 95\%$ recurrence rate.¹⁹ There is a 20–30% chance of normoprolactinemia off medication in select patients.²⁰

Recommendations²⁰: if response to DA agonist is satisfactory, treat for 1–4 years (microadenomas: check prolactin yearly, macroadenomas are more likely to grow and should be checked more often). Microadenomas or macroadenomas that are no longer visible on MRI are candidates for DA agonist withdrawal. For microadenomas: discontinue the drug; for macroadenomas: slowly taper the drug then discontinue. Recurrence rate is highest during 1st year, $\therefore$ check prolactin levels and clinical symptoms every 3 months during the 1st year. Long-term follow up is required, especially for macroadenomas.

53

Table 53.1 Prolactin level (PRL) with DA agonist treatment

PRL (ng/ml)	Recommendation
< 20	maintain
20–50	reassess dose
> 50	consider surgery

53.1.5 Acromegaly—management

See references.^{21,22,23}

Surgery

Surgery is the primary treatment modality for acromegaly when treatment is indicated.

1. asymptomatic elderly patients do not require treatment since there is little evidence that intervention alters life expectancy in this group
2. if no contraindications, surgery (usually transsphenoidal) is currently the best initial therapy (worse prognosis with macroadenomas), providing more rapid reduction in GH levels and

decompression of neural structures (e.g., optic chiasm) and improving the efficacy of subsequent somatostatin analogues.²³ Surgery is not recommended for elderly patients

3. medical therapy (p.889); reserved for:
 - a) patients not cured by surgery (reoperation doesn't work very often for acromegaly). Note: the definition of "biochemical cure" with acromegaly is not standardized (p.903). Surgery is still helpful for those "not cured" and improves efficacy of other therapies; IGF-1 may take months to normalize after surgery
 - b) or for those who cannot tolerate surgery (e.g., due to cardiomyopathy, severe hypertension, airway obstruction...; these contraindications may improve with medical therapy and then surgery can be reconsidered)
 - c) or for recurrence after surgery or XRT
4. XRT (p.893): for failure of medical therapy. Not recommended as initial treatment. NB: some practitioners use XRT for surgical failure, and employ medical therapy while waiting for XRT to have an effect. GH levels decline very slowly after XRT, see details and side effects (p.888)

Medical therapy

Overview

1. dopamine agonists (DAs): although not mentioned in the AACE guidelines,²⁴ it may be worth trying a DA to see if the tumor responds ($\approx 20\%$ respond). If responsive, DAs are especially well suited for GH tumors that cosecrete PRL
 - a) bromocriptine: (see below) although it benefits only a minority, a first line drug since it is cheaper than pegvisomant or octreotide and is given PO
 - b) cabergoline (see above)
 - c) others: lisuride, depo-bromocriptine (bromocriptine-LAR)
2. somatostatin analogues: indications: as initial medical therapy, or if no response to DAs, some also use this pre-op to improve surgical success rate
 - a) octreotide and octreotide-LAR (see below)
 - b) lanreotide, lanreotide SR and long-acting aqueous gel lanreotide (Autogel)
3. GH antagonists: pegvisomant (see below) considered for failures to above (not a primary therapy)
4. combination therapy: may be more effective than individual drugs. Pegvisomant or octreotide + dopamine agonist if no response to 1 drug alone

Specific agents

Drug info: Bromocriptine (Parlodel®)

Neoplastic somatotrophs may respond fortuitously to dopamine agonists and reduce growth hormone (GH) secretion. Bromocriptine lowers GH levels to <10 ng/ml in 54% of cases, to <5 ng/ml in only $\approx 12\%$. Tumor shrinkage occurs in only $<20\%$. Higher doses are usually required than for prolactinomas. If effective, the drug may be continued but should be periodically withdrawn to assess the GH level; see **Side effects** (p.888). Estimated annual cost: \$3,200 in the U.S.

R For growth hormone tumors that respond to bromocriptine, the usual dosage is 20–60 mg/d in divided doses (higher doses are unwarranted). The maximal daily dose is 100 mg.

Drug info: Octreotide (Sandostatin®)

A somatostatin analogue that is 45 times more potent than somatostatin in suppressing GH secretion but is only twice as potent in suppressing insulin secretion, has a longer half-life (≈ 2 hrs after SQ injection, compared to $\approx$ minutes for somatostatin), and does not result in rebound GH hypersecretion. GH levels are reduced in 71%, IGF-1 levels are reduced in 93%. 50–66% have normal GH levels, 66% achieve normal IGF-1 levels. Tumor volume reduces significantly in about 30% of patients. Many symptoms including H/A usually improve within the first few weeks of treatment. Annual cost to the patient: at least $\approx$ \$7,800 in the U.S. Usually given in combination with bromocriptine.

After 50 mcg SQ injection, GH secretion is suppressed within 1 hr, nadirs at 3 hrs, and remains reduced for 6–8 hrs (occasionally up to 12 hrs). **Side effects:** reduced GI motility and secretion, diarrhea, steatorrhea, flatulence, nausea, abdominal discomfort (all of these usually remit in 10 days), clinically insignificant bradycardia in 15%, cholesterol *cholelithiasis* (in 10–25%) or bile sludge.

Asymptomatic stones require no treatment and routine ultrasonography is not required. Mild hypothyroidism or worsening of glucose intolerance may occur.

R: Start with 50–100 mcg SQ q 8 hrs. Increase up to a maximum of 1500 mcg/d (doses > 750 mcg/d are rarely needed). Average dose required is 100–200 mcg SQ q 8 hrs.

Sandostatin LAR Depot: long acting release (LAR) form given by IM injection. **R:** give a test dose of *short acting* octreotide SQ in the office, and if no reaction (e.g., N/V...) begin LAR injections with 20 mg IM q 4 weeks, increase to 30 mg if GH > 5 mU/L just before 4th dose. Control can be achieved in some patients with dosing q 8–12 weeks.²⁵

Drug info: Pegvisomant (Somavert®)

A genetically engineered competitive GH-receptor antagonist. Treatment for ≥ 12 mos results in normal IGF-1 levels in 97% of patients.²⁶ No change in pituitary tumor size has been observed.²⁷ Indications: failure of somatostatin in patient with GH secreting adenoma (patient is switched to pegvisomant, it is not added to regimen). **Side effects:** significant but reversible liver function abnormalities occur in < 1%. Serum GH typically increases for 5–6 months and then stabilizes, probably as a result of loss of negative feedback on IGF-1 production. Antibodies to GH occurred in 17%, but tachyphylaxis was not observed

R: 5–40 mg/d SQ (dose must be titrated to keep IGF-1 in the normal range, to avoid GH deficiency conditions).

53.1.6 Cushing's disease—management

Management algorithm

1. if pituitary MRI shows a mass: transsphenoidal surgery
2. if pituitary MR is negative (up to 40% of patients with Cushing's disease have negative MRI): perform inferior petrosal sinus (IPS) sampling (p.880)
 - a) if IPS sampling is positive: surgery
 - b) if IPS sampling is negative: look for extra-pituitary source of ACTH (abdominal CT)
3. if pituitary surgery is performed but biochemical cure—see criteria (p.903)—is not obtained with surgery:
 - a) unlike acromegaly, a partial reduction is not helpful to the patient
 - b) consider re-exploration if pituitary source is still suspected
 - c) stereotactic radiosurgery or medical therapy (see below)
 - d) adrenalectomy in appropriate patients (see below)

Transsphenoidal surgery

Transsphenoidal surgery is the treatment of choice for most (medical therapy is inadequate as initial therapy since there is no effective pituitary suppressive medication). Cure rates are ≈ 85% for microadenomas (i.e., tumors ≤ 1 cm dia), but are lower for larger tumors. Even with microadenomas, *hemi-hypophysectomy* on the side of the tumor is usually required for cure (the tumor is difficult to completely extirpate) with attendant increased risk of CSF leak. If this fails, consideration should then be for *total hypophysectomy*. Failure of total hypophysectomy prompts consideration for bilateral adrenalectomy (total hypophysectomy virtually eliminates risk of Nelson's syndrome following adrenalectomy—see below).

Stereotactic radiosurgery

Often normalizes serum cortisol levels. Useful for: recurrence after surgery, inaccessible tumors (e.g., cavernous sinus)²⁸...

Adrenalectomy

Total bilateral adrenalectomy (TBA) corrects hypercortisolism in 96–100%⁹ (unless there is an extra-adrenal remnant), but lifelong gluco- and mineralo-corticoid replacement are required and up to

30% develop Nelson's syndrome (p.869) (incidence reduced by total hypophysectomy or possibly by pituitary XRT).

Indications: continued hypercortisolism with:

1. non-resectable PitNET/adenoma
2. failure of medical therapy to control symptoms after transsphenoidal surgery
3. life-threatening Cushing's disease (CD)
4. CD with no evidence of pituitary tumor; testing should include high-dose DMZ suppression test (p.880) and/or inferior petrosal sinus sampling (p.880)

Follow-up after TBA to rule out Nelson's syndrome: there is no standardized regimen. Suggestion: check serum ACTH levels q 3–6 months $\times$ 1 year, q 6 months $\times$ 2 years, q year thereafter. A pituitary MRI is done if an ACTH level is > 100 ng/L; otherwise, annual MRIs are sufficient²⁹ $\times$ 3 years and then if ACTH levels remain low, get an MRI every other year.

Medical therapy

For patients who fail surgical therapy or for whom surgery cannot be tolerated, medical therapy and/or radiation are utilized. Occasionally may be used for several weeks prior to planned surgery to control significant manifestations of hypercortisolism, e.g., diabetes, HTN, psychiatric disturbances... (p.868).

Ketoconazole (Nizoral®)¹¹: an antifungal agent that blocks adrenal steroid synthesis. The initial drug of choice. Over 75% of patients have normalization of urinary free cortisol and 17-hydroxycorticosteroid levels. **Side effects**: reversible elevations of serum hepatic transaminase (in 15%), GI discomfort, edema, skin rash. Significant hepatotoxicity occurs in 1 of 15,000 patients. See ► Table 8.2 for evidence of adrenal insufficiency.

R Start with 200 mg PO BID. Adjust dosage based on 24-hr urine free cortisol and 17-hydroxycorticosteroid levels. Usual maintenance doses 400–1200 mg daily in divided doses (maximum of 1600 mg daily).

Aminoglutethimide (Cytadren®)¹¹: inhibits the initial enzyme in the synthesis of steroids from cholesterol. Normalizes urinary free cortisol in $\approx$ 50% of cases. **Side effects**: dose-dependent reversible effects include sedation, anorexia, nausea, rash and hypothyroidism (due to interference with thyroid hormone synthesis).

R Start with 125–250 mg PO BID. Effectiveness may diminish after several months and dose escalation may be needed. Generally do not exceed 1000 mg/d.

Metopirone (Metopirone®): inhibits 11- β -hydroxylase (involved in one of the final steps of cortisol synthesis); may be used alone or in combination with other drugs. Normalizes mean daily plasma cortisol in $\approx$ 75%. **Side effects**: lethargy, dizziness, ataxia, N/V, primary adrenal insufficiency, hirsutism, and acne.

R Usual dose range is 750–6000 mg/d usually divided TID with meals. Initial effectiveness may diminish with time.

Mitotane (Lysodren®): related to the insecticide DDT. Inhibits several steps in glucocorticoid synthesis, and is cytotoxic to adrenocortical cells (adrenolytic agent). 75% of patients enter remission after 6–12 months of treatment, and the medication may sometimes be discontinued (however, hypercortisolism may recur). **Side effects**: may be limiting, and include anorexia, lethargy, dizziness, impaired cognition, GI distress, hypercholesterolemia, adrenal insufficiency (which may necessitate supernal doses of glucocorticoids for replacement due to induced glucocorticoid degradation).

R Start with 250–500 mg PO q hs, and escalate dose slowly. Usual dose range is 4–12 g/d usually divided TID–QID. Initial effectiveness may diminish with time.

Cyproheptadine (Periactin®): a serotonin receptor antagonist that corrects the abnormalities of Cushing's disease in a small minority of patients, suggesting that some cases of "pituitary" Cushing's disease are really due to a hypothalamic disorder. Combined therapy with bromocriptine may be more effective in some patients. **Side effects**: sedation & hyperphagia with weight gain usually limit usefulness.

R Usual dosage range: 8–36 mg/d divided TID.

53.1.7 Thyrotropin (TSH)-secreting adenomas—management

General information

1. transsphenoidal surgery has been the traditional first-line treatment.³⁰ These tumors may be fibrous and difficult to remove³¹
2. for incomplete resection: post-op XRT is employed

- if hyperthyroidism persists: medical therapy is added with agents including octreotide, bromocriptine (more effective for tumors that co-secrete PRL), and oral cholecystographic agents (which inhibit conversion of T4 to T3), e.g., iopanoic acid

Medical therapy

Normal and neoplastic anterior hypophyseal thyrotroph cells possess somatostatin receptors and most respond to octreotide (see below). Occasionally, beta-blockers or low-dose antithyroid drugs (e.g., Tapazole® (methimazole) $\approx$ 5 mg PO TID for adults) may additionally be required.

Octreotide (Sandostatin®)

Doses required are usually < than with acromegaly. TSH levels decline by > 50% in 88% of patients, and become normal in $\approx$ 75%. T4 and T3 levels decrease in almost all, with 75% becoming normal. Tumor shrinkage occurs in $\approx$ 33%.

R Start with 50–100 mcg SQ q 8 hrs. Titrate to TSH, T4 and T3 levels.

53.1.8 Gonadotropin secreting adenoma

This pertains to the rare patient who is symptomatic from gonadotropin hypersecretion.

Normal and neoplastic pituitary gonadotrophs have gonadotropin-releasing hormone (GnRH) receptors, and may respond to long-acting GnRH agonists (by down-regulating receptors) or GnRH antagonists, but significant reductions in tumor size does not occur.

53.2 Radiation therapy for PitNET/adenomas

53.2.1 General information

Conventional EBXRT usually consists of 40–50 Gy administered over 4–6 weeks.

Stereotactic radiosurgery (p. 1908) for PitNET/adenomas.

53.2.2 Side effects

Radiation injury to the remaining normal pituitary results in hypocortisolism, hypogonadism, or hypothyroidism in 40–50% of patients after 10 years. It may also injure the optic nerve and chiasm (possibly causing blindness), cause lethargy, memory disturbances, cranial nerve palsies, and tumor necrosis with hemorrhage and apoplexy. Cure rates but also complications are higher after proton beam therapy.

53.2.3 Recommendation

Radiation therapy should *not* be routinely used following surgical removal. Follow patient with yearly MRI. Treat recurrence with repeat operation. Consider radiation if recurrence cannot be removed and tumor continues to grow.

53.2.4 Sellar radiation therapy for specific PitNET/adenoma types

XRT for nonfunctioning pituitary tumors

In one series of 89 *nonfunctioning* pituitary tumors ranging 0.5–5 cm diameter (mean = 2 cm) not totally resected because of involvement of cavernous sinus (or other inaccessible sites), half were treated with radiation therapy (XRT). The recurrence rate was neither lower (and was actually higher) nor later in the XRT group.³² However, another series of 108 pituitary macroadenomas found the recurrence rates shown in ► Table 53.2 which tend to favor radiation therapy.

When used, doses of 40 or 45 Gy in 20 or 25 fractions, respectively, is recommended.³⁴ The oncocytic variant of null cell pituitary tumors appears to be more radioresistant than the nononcocytic undifferentiated cell adenoma.³⁴

Sellar radiation therapy for acromegaly

Not the preferred initial treatment. Works better with lower initial GH levels. In most patients, GH levels begin to fall during the first year after XRT, dropping by $\approx$ 50% after 2 years, and decrease gradually thereafter, reaching $\leq$ 10 ng/ml in 70% of patients after 10 years. It takes up to 20 years for 90%

Table 53.2 Recurrence rate of pituitary tumors removed transsphenoidally^a

Extent of removal	Post-op XRT?	Recurrence rate
subtotal	no	50%
gross total		21%
subtotal	yes	10%
gross total		0
ª108 macroadenomas, 6 mos to 14 years follow-up³³		

of patients to achieve GH levels <5 ng/ml. During this latency period, patients are exposed to unacceptably high levels of GH (octreotide may be used while waiting). Patients are also still at risk for radiation side effects mentioned above. Options include: EBRT, stereotactic radiosurgery (about equally effective). Estimated cost: \$20,000.

Sellar radiation therapy for Cushing's disease

XRT corrects hypercortisolism in 20–40%, and produces some improvement in another 40%. Improvement may not be seen for 1–2 yrs post treatment.

References

- [1] Krisht AF. Giant invasive pituitary adenomas. *Contemporary Neurosurgery*. 1999; 21:1–6
- [2] Lucas JW, Bodach ME, Tumialan LM, et al. Congress of Neurological Surgeons Systematic Review and Evidence-Based Guideline on Primary Management of Patients With Nonfunctioning Pituitary Adenomas. *Neurosurgery*. 2016; 79:E533–E535
- [3] Greenman Y, Stern N. Non-functioning pituitary adenomas. *Best Pract Res Clin Endocrinol Metab*. 2009; 23:625–638
- [4] Huang W, Molitch ME. Management of nonfunctioning pituitary adenomas (NFAs): observation. *Pituitary*. 2018; 21:162–167
- [5] Dekkers OM, Hammer S, de Keizer RJ, et al. The natural course of non-functioning pituitary macroadenomas. *Eur J Endocrinol*. 2007; 156:217–224
- [6] Arita K, Tominaga A, Sugiyama K, et al. Natural course of incidentally found nonfunctioning pituitary adenoma, with special reference to pituitary apoplexy during follow-up examination. *J Neurosurg*. 2006; 104:884–891
- [7] Freda PU, Beckers AM, Katznelson L, et al. Pituitary incidentaloma: an endocrine society clinical practice guideline. *J Clin Endocrinol Metab*. 2011; 96:894–904
- [8] Wilson CB. Endocrine-Inactive Pituitary Adenomas. *Clinical Neurosurgery*. 1992; 38:10–31
- [9] Banasiak MJ, Malek AR. Nelson syndrome: comprehensive review of pathophysiology, diagnosis, and management. *Neurosurgical Focus*. 2007; 23
- [10] Barrow DL, Mizuno J, Tindall GT. Management of Prolactinomas Associated with Very High Serum Prolactin Levels. *Journal of Neurosurgery*. 1988; 68:554–558
- [11] Blevins LS. Medical Management of Pituitary Adenomas. *Contemporary Neurosurgery*. 1997; 19:1–6
- [12] Molitch ME. Pregnancy and the hyperprolactinemic woman. *New England Journal of Medicine*. 1985; 312:1364–1370
- [13] Landolt AM, Osterwalder V. Perivascular Fibrosis in Prolactinomas: Is it Increased by Bromocriptine? *J Clin Endocrinol Metab*. 1984; 58:1179–1183
- [14] Barlas O, Bayindir C, Heggul K, et al. Bromocriptine-induced cerebrospinal fluid fistula in patients with macroprolactinomas: report of three cases and a review of the literature. *Surg Neurol*. 1994; 41:486–489
- [15] Cabergoline for Hyperprolactinemia. *Medical Letter*. 1997; 39:58–59
- [16] Webster J, Piscitelli G, Polli A, et al. A Comparison of Cabergoline and Bromocriptine in the Treatment of Hyperprolactinemic Amenorrhea. *New England Journal of Medicine*. 1994; 331:904–909
- [17] Schade R, Andersohn F, Suissa S, et al. Dopamine agonists and the risk of cardiac-valve regurgitation. *N Engl J Med*. 2007; 356:29–38
- [18] Gillam MP, Molitch ME, Lombardi G, et al. Advances in the treatment of prolactinomas. *Endocrine Reviews*. 2006; 27:485–534
- [19] Johnston DG, Hall K, Kendall-Taylor P, et al. Effect of dopamine agonist withdrawal after long-term therapy in prolactinomas. Studies with high-definition computerised tomography. *Lancet*. 1984; 2:187–192
- [20] Schlechte JA. Long-term management of prolactinomas. *Journal of Clinical Endocrinology and Metabolism*. 2007; 92:2861–2865
- [21] Melmed S. Medical progress: Acromegaly. *N Engl J Med*. 2006; 355:2558–2573
- [22] Acromegaly Therapy Consensus Development Panel. Consensus Statement: Benefits Versus Risks of Medical Therapy for Acromegaly. *American Journal of Medicine*. 1994; 97:468–473
- [23] Colao A, Attanasio R, Pivonello R, et al. Partial surgical removal of growth hormone-secreting pituitary tumors enhances the response to somatostatin analogs in acromegaly. *Journal of Clinical Endocrinology and Metabolism*. 2006; 91:85–92
- [24] Cook DM. AACE Medical Guidelines for Clinical Practice for the diagnosis and treatment of acromegaly. *Endocr Pract*. 2004; 10:213–225
- [25] Turner HE, Thornton-Jones V A, Wass JA. Systematic dose-extension of octreotide LAR: the importance of individual tailoring of treatment in patients with acromegaly. *Clin Endocrinol (Oxf)*. 2004; 61:224–231
- [26] van der Lely AJ, Hutson RK, Trainer PJ, et al. Long-term treatment of acromegaly with pegvisomant, a growth hormone receptor antagonist. *Lancet*. 2001; 358:1754–1759
- [27] Pegvisomant (Somavert) for acromegaly. *Medical Letter*. 2003; 45:55–56
- [28] Sheehan JM, Vance ML, Sheehan JP, et al. Radiosurgery for Cushing's disease after failed transsphenoidal surgery. *J Neurosurg*. 2000; 93:738–742
- [29] Assie G, Bahurel H, Coste J, et al. Corticotroph tumor progression after adrenalectomy in Cushing's Disease: a reappraisal of Nelson's syndrome. *Journal of Clinical Endocrinology and Metabolism*. 2007; 49:381–386
- [30] Clarke MJ, Erickson D, Castro MR, et al. Thyroid-stimulating hormone pituitary adenomas. *J Neurosurg*. 2008; 109:17–22

- [31] Sanno N, Teramoto A, Osamura RY. Long-term surgical outcome in 16 patients with thyrotropin pituitary adenoma. *J Neurosurg.* 2000; 93:194–200
- [32] Ebersold MJ, Quast LM, Laws ER, et al. Long-Term Results in Transsphenoidal Removal of Nonfunctioning Pituitary Adenomas. *J Neurosurg.* 1986; 64:713–719
- [33] Ciric I, Mikhael M, Stafford T, et al. Transsphenoidal Microsurgery of Pituitary Macroadenomas with Long-Term Follow-Up Results. *J Neurosurg.* 1983; 59:395–401
- [34] Breen P, Flickinger JC, Kondziolka D, et al. Radiotherapy for Nonfunctional Pituitary Adenoma: Analysis of Long-Term Tumor Control. *Journal of Neurosurgery.* 1998; 89:933–938

54 PitNET/Adenomas – Surgical Management, Outcome, and Recurrence Management

54.1 Surgical treatment for PitNET/adenomas

54.1.1 Medical preparation for surgery

General principles

1. stress dose steroids: given to all patients during and immediately after surgery
2. hypothyroidism: ideally, hypothyroid patients should have > 4 weeks of replacement to reverse hypothyroidism; however:
 - a) ✗ do not replace thyroid hormone until the adrenal axis is assessed; giving thyroid replacement to a patient with hypoadrenalism can precipitate adrenal crisis. If hypoadrenal, begin cortisol replacement first, may begin thyroid hormone replacement after 24 hours of cortisol
 - b) surgery is done frequently on patients with hypothyroidism and appears to be tolerated well in the vast majority of cases

Pre-op orders

1. for transsphenoidal approach: Polysporin® ointment (PSO) applied in both nostrils the night before surgery
2. antibiotics: e.g., Unasyn® 1.5 gm (1 gm ampicillin + 0.5 gm sulbactam) IVPB at MN & 6 AM
3. steroids, either:
 - a) hydrocortisone sodium succinate (Solu-Cortef®) 50 mg IM at 11 PM & 6 AM. On call to OR: hang 1 L D5LR + 20 mEq KCl/l + 50 mg Solu-Cortef at 75 ml/hr
OR
 - b) hydrocortisone 100 mg PO at MN & IV at 6 AM
4. intra-op: continue 100 mg hydrocortisone IV q 8 hrs

54.1.2 Surgical approaches—overview

Practice guideline: Surgical approaches and technology for nonfunctioning PitNET/adenomas (NFPAs)

Level III¹ recommendations include:

- transsphenoidal surgery (microscopic or endoscopic) is recommended for relief of symptoms
- if microsurgery is used initially, endoscopic visualization afterwards is recommended since it frequently reveals additional resectable tumor
- for invasive NFPAs with significant suprasellar, frontal or temporal involvement: combined transsphenoidal and transcranial surgery is recommended
- intraoperative MRI: may improve overall gross total resection; however, it is not recommended for estimating residual tumor volume because of a variable false-positive rate

✗ There was insufficient evidence to recommend¹:

- neuronavigation
- introduction of intrathecal saline or air to augment delivery of suprasellar tumor during transsphenoidal approach
- to prevent postop CSF leak: perioperative CSF diversion, or a specific dural closure technique

1. transsphenoidal: requires no brain retraction, no external scar (aside from where a fat graft is procured, if used), and is normally an extra-arachnoid approach unless the diaphragma sellae is breached. Usually the procedure of choice. Indicated for microadenomas, macroadenomas without significant extension laterally beyond the confines of the sella turcica, patients with CSF rhinorrhea, and tumors with extension into sphenoid air sinus
 - a) sublabial

- b) trans-nares: for extremely small nares, an alotomy may be used to enlarge the exposure if necessary
2. transethmoidal approach² (p 343–50)
3. transcranial approaches:
 - a) indications: most pituitary tumors are operated by the transsphenoidal technique (see above), even if there is significant suprasellar extension. However, a craniotomy may be indicated for the following³:
 - minimal enlargement of the sella with a large suprasellar mass, especially if the diaphragma sellae is tightly constricting the tumor (producing a “cottage loaf” tumor) and the suprasellar component is causing chiasmal compression⁴(p 124)
 - extrasellar extension into the middle fossa that is larger than the intrasellar component
 - unrelated pathology may complicate a transsphenoidal approach: rare, e.g., a parasellar aneurysm
 - unusually fibrous tumor that could not be completely removed on a previous transsphenoidal approach
 - recurrent tumor following a previous transsphenoidal resection
 - b) choices of approach
 - subfrontal: provides access to both optic nerves. May be more difficult in patients with pre-fixed chiasm
 - frontotemporal (pterional): places optic nerve and sometimes carotid artery in line of vision of tumor. There is also incomplete access to intrasellar contents. Good access for tumors with significant unilateral extrasellar extension
 - ✖ subtemporal: usually not a viable choice. Poor visualization of optic nerve/chiasm and carotid. Does not allow total removal of intrasellar component

54.1.3 Transsphenoidal surgery

Booking the case: Transsphenoidal surgery

Also see defaults and disclaimers (p. 25) and pre-op orders (p. 896).

1. position: supine, horseshoe head rest or (especially if image-guided navigation is used) pin headholder
2. equipment:
 - a) microscope
 - b) C-arm (if used)
 - c) image-guided navigation system (if used)
 - d) endoscopy cart for cases performed endoscopically (surgeon preference)
3. instrumentation: transsphenoidal instrument set (usually includes speculum, curettes, long instruments including bipolars)
4. some surgeons use ENT to perform the approach and closure and for follow-up
5. post-op: ICU
6. consent (in lay terms for the patient—not all-inclusive):
 - a) procedure: removal of pituitary tumor through the nose, possible placement of fat graft from abdomen
 - b) alternatives: surgery through the skull (trans-cranial), radiation
 - c) complications: CSF leak with possible meningitis, problems with pituitary hormones which may sometimes be permanent (which would require lifetime replacement therapy), injury to optic nerve with visual loss, injury to carotid artery with possible bleeding and/or stroke

54

Technique

General information

For pre- and post-op orders, see below.

Details of the surgery are beyond the scope of this text, see references.^{4,5,6,7}

Intraoperative disasters

Usually related to loss of landmarks.⁴ Can be minimized using intraoperative navigation or fluoroscopy to verify location.

- injury of carotid artery:
 - typically injured in lateral aspect of opening. Bone may be dehiscent over the ICA
 - signaled by profuse arterial bleeding
 - can usually be packed off, if fat/fascia graft from thigh or abdomen is available it may be used; otherwise use e.g., woven Surgicel®
 - the operation is halted, and a STAT post-op arteriogram must be done
 - if a pseudoaneurysm or site of injury is identified angiographically, it must be eliminated before a potentially lethal hemorrhage; accomplished either by endovascular techniques or by surgical trapping with clips above and below
- injury of basilar artery: with pulling on tissue adherent to the artery or reaching through sella and rongeur
- opening through the clivus and erroneous biopsy of the pons
- opening through the floor of the frontal fossa with injury to the olfactory nerves and entry into inferior frontal lobes

Overview of the procedure

1. lumbar drain: may be used with some macroadenomas to inject fluid in order to help bring the tumor down (p.896), also may be used for post-op CSF drainage (p.899) following transsphenoidal repair of CSF fistula
2. medications (in addition to pre-op meds (p.896)): intraoperatively 100 mg hydrocortisone IV bolus followed by 100 mg dripped in over 8 hrs
3. positioning
 - a) elevate thorax 10–15°: reduces venous pressure
 - b) head stabilization: if image-guided navigation is to be used, the head is placed in a Mayfield headholder, or a headband with registration array is attached and the head is placed on a horseshoe head rest. If no image guidance, a horseshoe headrest may be used
 - c) position option 1: surgeon standing to right of patient
 - shoulder-roll
 - top of head canted slightly to left
 - neck position: For microscope: extend neck slightly with the head in either a Mayfield head-holder or on a horseshoe headrest. For endoscope: do not extend neck (more comfortable for holding instruments)
 - ET tube positioned down and to patient's *left* (to get it out of the way)
 - microscope: observer's eyepiece on the *left*
 - d) position option 2: surgeon standing above patient's head: patient head pointing straight up towards ceiling, neck slightly extended
 - e) abdomen or right thigh is prepped for possible fat graft if needed
4. C-arm fluoro: image-guided navigation can eliminate the need for fluoro. Orient the C-arm for a true lateral by aligning the mandibular rami and/or by superimposing the floor of the left and right frontal fossae. If this proves difficult, lay a Penfield 4 on the nasion oriented from lateral canthus to lateral canthus, then aim the fluoro to shoot "down the barrel" of the Penfield 4
5. after approach to floor of sella is complete (see below), outline the upper and lower boundaries of the sella using image navigation or fluoro using an instrument (e.g., suction tip)
6. opening the sellar floor:
 - a) starting the opening: open *exactly* in the midline using the nasal septum as a landmark (NB: the septum of the sphenoid sinus is unreliable as a midline indicator, and often curves inferiorly towards one of the carotid arteries)
 - macroadenomas may have thinned the bone to the point that it just flakes off
 - otherwise, use a bayoneted chisel or high-speed diamond burr to start the opening
 - b) use a Kerrison rongeur to expand the opening. ✖ CAUTION: stay away from the extreme lateral sella to avoid entering the cavernous sinus or injuring the carotid artery
7. coagulate the dura centrally in an "X" pattern (*NOT* "+" pattern) with bipolar cautery. Macroadenomas may cause yellowish discoloration of the dura directly over the tumor
8. consider aspirating through dura with a 20 gauge spinal needle to R/O large venous sinus (dura often has bluish discoloration), giant aneurysm (p. 1655), or empty sella
9. incise the dura in the "X" pattern in the midline with a #11 scalpel on a bayoneted handle
10. tumor removal for **macroadenoma**:
 - a) gently bring tumor into the field with ring curettes, and remove with pituitary rongeurs or aspirate with suction. Some tumors are very fibrous and may be difficult to remove
 - b) do not *pull* on the lateral component of the tumor with pituitary rongeur due to risk of injuring carotid artery

- c) if the suprasellar component will not come down, it may be brought down by having the anesthesiologist inject 5 ml aliquots of saline into a previously placed lumbar drain while monitoring blood pressure and pulse^{4(p 135),8}
 - d) once the tumor is debulked internally, try to develop a plane between the tumor capsule and the pituitary. A good place to start looking is inferiorly where the dura can be separated from the tumor capsule and then followed on the surface. Sometimes the tumor capsule cannot be removed due to severe bleeding
 - e) complete tumor removal is often not possible, and the goal of the surgery then is “containment,” with intractable post op re-growth treated with XRT
 - f) endoscopic techniques and image-guided navigation may be employed to assist in removal of macroadenomas
11. tumor removal for microadenoma
- a) if the side of the tumor is known, begin exploration of the gland on that side by making incision with #11 blade and using a dissector to try and locate the tumor (like a “grain of rice in a blueberry”)
 - b) for Cushing’s disease, if no tumor is identified on pre-op MRI⁹:
 - intraoperative ultrasound may help localize tumor in $\approx 70\%$ of cases¹⁰ (a specialized U/S probe is required)
 - if IPS sampling showed a lateralizing ACTH gradient: start with a paramedian incision on the side of the higher ACTH gradient; if no adenoma is encountered, the contralateral paramedian and then midline incisions are used to explore the gland
 - if IPS sampling and MRI do not suggest tumor location: the gland is explored sequentially with 2 paramedian incisions and then a midline incision
 - if the adenoma cannot be found, a hemihypophysectomy is performed on the side of higher ACTH levels if IPS sampling shows a lateralizing gradient, or on the side with more suspicious tissue on frozen section. Total hypophysectomy is not routinely performed⁹
 - c) most adenomas are purplish-gray and easily aspirated; however, some may be more fibrous. The normal pituitary gland is firm and rubbery (the adenohypophysis is orange-pink, the neurohypophysis is a whitish-gray), and normally does not curette very easily
 - d) use image guidance or fluoro to determine approximate location of diaphragma sellae. Do not go cephalad to this to avoid a CSF leak, to avoid entering the circular venous sinus in the dura here, and to avoid trauma to the optic chiasm
12. after removal of macroadenoma, check depth of tumor bed on fluoro or image guidance, and make sure it correlates with approximate tumor volume on pre-op MRI
13. the sella may be packed in a number of ways,⁷ one method:
- a) if CSF leak occurs: place muscle or fat in defect within sella. Some recommend against the use of muscle because it always putrefies.^{4(p 129)} Do not overpack to avoid recreating mass effect with the graft
 - b) recreate the floor of the sella using nasal cartilage placed within the sella. Alternatively, a nonporous Medpor® polyethylene transsphenoidal sellar implant (Porex Surgical Products <http://www.porexurgical.com>) may be used
 - c) if CSF leak occurs pack sphenoid sinus with fat from abdomen (option: fat with fascia on surface)
 - d) fibrin glue may optionally be used to help hold any of these components in place
 - e) elaborate packing for closure is not mandatory if no CSF leak occurs during surgery

Approach to sphenoid sinus for microscopic removal

Often done by ENT co-surgeon. One method:

1. insert temporary speculum into nose. For this discussion, the right nares will be described
2. use endoscope to locate middle concha. Follow this posteriorly to identify os into sphenoid sinus, which is usually located posterior to and slightly above the posterior extent of the middle concha
3. inject local anesthetic with epinephrine to blanch mucosa
4. insert sickle knife into os with sharp side facing the septum (medially) and incise the mucosa as the knife is drawn outward
5. use a Freer to dissect the thusly created mucosal flaps off the medial septum (pull one up, the other down)
6. break through the posterior part of the septum so that both sides of the floor of the sphenoid sinus are exposed. Cartilage or bone from this step is saved to use later in reconstructing the floor of the sella if desired
7. open the floor of the sphenoid and take it all the way to the right os (you will probably not see the left os)

8. place the Hardy speculum or equivalent
9. strip mucosa off the walls of the sphenoid sinus using a Blakely and *slow* pulling motion

Endoscopic tumor removal

Advantages over microscopic removal: visualization is better, especially within tumor bed.

Disadvantages: most neurosurgeons have less facility with endoscopes compared to ENT surgeons. Lack of 3D visualization (may be overcome using 3D endoscopes). Need for single-handed technique (may be overcome by having assistant hold endoscope, or using an endoscope holder, e.g., Mitaka), need for a binasal approach if it is desired to use two hands for the actual surgery.

Recommendations (p.896): Either endoscopic or microscopic approach is acceptable; however, after microscopic approach it is recommended that an endoscope be used at the time of the same surgery to visualize additional resectable tumor.¹

54.1.4 Perioperative complications

1. hormonal imbalance:
 - a) acute post-op concerns:
 - alterations in ADH: transient abnormalities are common—see typical post-op patterns (p.902) – including DI. DI lasting > 3 mos is uncommon
 - cortisol deficiency → hypocortisolism → Addisonian crisis if severe
 - b) long-term: hypopituitarism in ≈ 5% (retrospective series¹¹)
 - TSH deficiency → hypothyroidism → (rarely) myxedema coma if severe
 - adrenal insufficiency
 - deficiency of sex hormones → hypogonadotropic hypogonadism
2. secondary empty sella syndrome (chiasm retracts into evacuated sella → visual impairment). Re-operation with gentle packing of fat into the sella may support the chiasm
3. hydrocephalus with coma¹²; may follow removal of tumors with suprasellar extension (trans-sphenoidally or transcranially). Consider ventriculostomy placement if hydrocephalus is present (even if not symptomatic). Possible etiologies:
 - a) traction on the attached 3rd ventricle
 - b) cerebral edema due to vasopressin release from manipulation of the pituitary and/or stalk
 - c) tumor edema following resection
4. infection
 - a) pituitary abscess^{13,14}
 - b) meningitis
5. CSF rhinorrhea (fistula): 3.5% incidence¹⁵
6. carotid artery rupture: rare. May occur intraoperatively (see above) or in delayed fashion after surgery, often ≈ day 10 post-op (due to breakdown of fibrin around carotid, or possibly due to rupture of a pseudoaneurysm created at surgery)
7. entry into cavernous sinus with possible injury of any structure within
8. nasal septal perforation

54.1.5 Frontotemporal (pterional) approach

A right-sided approach is usually employed (less risk to dominant hemisphere). Exceptions: when the left eye is the side of worse vision; if there is predominant left-sided tumor extension; if there is other pathology on the left (e.g., aneurysm).

Positioning is the same as for an ACoA aneurysm, as seen in ► Fig. 106.1, with the head turned 60° to the side. The frontal lobe is elevated, and the temporal tip is gently retracted posteriorly. Bridging veins to the temporal tip must be coagulated to avoid rupture, as for any pterional approach. The approach is similar to that for an ACoA aneurysm (i.e., more emphasis is placed on frontal lobe elevation than on temporal tip retraction), except that unlike ACoA aneurysm, exposure of the ICA is not needed because proximal control is not necessary.

The tumor capsule can usually be seen between the two optic nerves. The capsule is coagulated with bipolar cautery and is incised. The tumor is then debulked from within. By staying within the capsule, risk of injury to the pituitary stalk and optic chiasm is minimized. Significant amounts of tumor can be removed by aspiration if it is soft and suckable.

✖ Caution: the blood supply to the optic chiasm is from the inferior aspect. Skeletonizing the chiasm or attempting to tease away tumor adherent to it may worsen vision.

54.1.6 Postoperative management

“DI watch” (diabetes insipidus watch)

Following any surgery involving the pituitary gland, patients are at risk of developing DI (inadequate secretion of ADH) which can produce dangerous dehydration and hypernatremia (elevation of serum sodium $[Na^+]$). Therefore, these patients place on so-called “DI watch.” One needs to be familiar with the 3 possible post-op patterns of ADH abnormalities (see typical patterns (p.902)). DI watch consists of:

1. strict intake & output (I's & O's) q 1 hr
2. urine specific gravity (USG) q 4^h and any time urine output (UO) > 250 ml/hr
3. serum electrolytes q 6 hours: watching primarily $[Na^+]$

Diagnostic criteria for DI: UO > 250 ml/hr × 2 hrs, and SG < 1.005.

DI is not present if USG > 1.005. In DI, USG is usually < 1.003.

Post-op orders

1. strict intake & output (I's & O's) q 1 hr
2. urine specific gravity (USG) q 4^h and any time urine output (UO) > 250 ml/hr X 2 hours
3. activity: BR with HOB @ 30°
4. diet: ice chips PRN, advance to clear liquids as tolerated. For transsphenoidal approach: patient is not to drink through a straw (to avoid negative pressure on sphenoid sinus with risk of promoting CSF fistula)
5. for transsphenoidal approach: no incentive spirometry (for same reason as no straw)
6. IVF
 - a) basal IVF: IV D5 1/2 NS + 20 mEq KCl/L at appropriate rate (75–100 ml/hr)
 - b) NB: patients normally receive IV fluids at the beginning of an operation to “fill the tank” to avoid hypotension as a result of muscle relaxation. It is normal for patients to have a post-op diuresis as they eliminate excess fluids. As long as the urine SG is > 1.003 and urine output (UO) is < 250 ml/hr, aggressive fluid administration is usually not warranted
 - c) if UO > 250 ml/hr X 2 hrs and USG < 1.005, the patient probably has diabetes insipidus (DI). If this occurs, in addition to the basal IVF, replace UO that is over the IV basal rate ml for ml with ½ NS and consider starting medication (see “meds” below)
 - d) if UO is high and patient is not clearly in DI, an option is to replace only ≈ 2/3 of UO > basal IV rate with ½ NS
7. meds
 - a) antibiotics: continue antibiotics until nasal packing is removed
 - b) steroids (some surgeons routinely use post-op steroids until the adequacy of endogenous steroids is established, especially with Cushing's disease, see below). If this is EITHER:
 - hydrocortisone 50 mg IM/IV q 6 hrs, on POD #2 change to prednisone 5 mg PO q 6 hrs × 1 day, then 5 mg PO BID, D/C on POD #6
 - OR
 - hydrocortisone 50 mg IM/IV/PO BID, taper 10 mg/dose/day to physiologic dose of 20 mg q AM and 10 mg q PM until adrenal axis assessed
 - c) if DI develops, attempt to keep up with fluid loss with oral and/or IV fluids (see above). If rate is too high to continue IV or PO replacement (> 300 cc/hr × 4 hrs or > 500 cc/hr × 2 hrs), check urine SG and if < 1.005 then give a vasopressin preparation (see below, or see ► Table 5.7). ✖ Caution: danger of overtreating in case of triphasic response (p. 125). Use EITHER:
 - 5 U aqueous vasopressin (Pitressin®) IVP/IM/SQ q 6 hrs PRN
 - OR
 - desmopressin (DDAVP®) injection SQ/IV titrated to UO. Usual adult dose: 0.5–1 ml (2–4 mcg) daily in 2 divided doses
 - ✖ AVOID tannate oil suspension (withdrawn from U.S. market in 1993), because it is a long acting preparation (and may overtreat) and has erratic absorption
 - THEN: (for transsphenoidal approach or when patient is taking po for transcranial approach) when nasal packs out, EITHER

intranasal DDAVP (100 mcg/ml): range 0.1–0.4 ml (10–40 mcg) intranasally BID (typically 0.2 ml BID) PRN
OR

- clofibrate (Atromid S®) 500 mg PO QID (does not always work)
- 8. labs: electrolytes with osmolality q 6 hrs, 8 A.M. serum cortisol on post-op day 1
- 9. imaging: (this is one paradigm, but is not standard of care)
 - a) non-contrast CT scan of the brain within 6–12 hours of surgery to look for acute complications (hematoma, hydrocephalus, pneumocephalus...)
 - b) pituitary-MRI without and with contrast within 72 hours of surgery to assess degree of tumor removal
- 10. nasal packs (if used for transsphenoidal approach): remove on post-op day 3–6

Urinary output: patterns of postoperative diabetes insipidus

Manage diabetes insipidus (DI) as described above in post-op orders. Post-op DI generally follows one of three patterns¹⁶; see Diabetes insipidus for details (p. 124):

- a) transient DI: lasts until $\approx$ 12–36 hrs post-op then normalizes
- b) “prolonged” DI: lasts months, or rarely may be permanent
- c) “triphasic response” (least common). 3 stages:
 1. DI (short duration): due to injury to posterior pituitary
 2. normalization or SIADH-like picture: due to release of ADH from neuron endings from hypothalamus. It is during this phase that there is a risk of severe iatrogenic hyponatremia from overtreatment initiated during the initial DI phase
 3. DI (long-term)

Discontinuation of steroids post-op

Simple management schemes

Options:

- taper and stop hydrocortisone 24–48 hrs post-op. Then, check 6 AM serum cortisol level 24 hrs after discontinuing hydrocortisone and interpret the results as shown in ► Table 54.1.¹⁷
- if there is any question about reserve, the patient can be discharged on hydrocortisone 50 mg PO q AM and 25 mg PO q 4 PM until adrenal reserve can be formally assessed (p.902)

Table 54.1 Interpretation of 6 AM cortisol levels

6 AM cortisol	Interpretation	Management
≥ 9 mcg/dl	normal	no further tests or treatment
3–9 mcg/dl	possible ACTH deficiency	place patient on hydrocortisone ^a (p. 154)
≤ 3 mcg/dl	ACTH deficient	
^a perform cosyntropin stimulation test (p. 880) 1 month post-op; D/C steroids if normal; if subnormal, then permanent replacement required		

Assessment of postoperative ACTH (corticotropin) reserve

Simple assessment protocol for patients who go home on hydrocortisone and were not on it pre-op.

- taper hydrocortisone over 2–3 weeks down to 20 mg PO q AM and 10 mg q 4 PM (a little higher than maintenance to provide for some stress coverage) for several days
- then hold the PM dose and check an 8 AM serum cortisol the next day
- to avoid adrenal insufficiency in patients with incompetent reserve: as soon as the blood is drawn have the patient take their morning cortisol dose and resume regular dosing until the test results are available
- if this 8 AM cortisol shows any significant adrenal function, then taper the patient off hydrocortisone

Metypapone (Metopirone®) test

This test more accurately assesses the hypothalamic-pituitary-adrenal axis and is useful if there is suspicion of reduced reserve of pituitary ACTH production. Metypapone inhibits 11- β -hydroxylation in the adrenal cortex, reducing production of cortisol and corticosterone with concomitant increase of serum 11-deoxycortisol precursors and its 17-OHCS metabolites which appear in the urine. In response, a normal pituitary increases ACTH production.

1. all patients should have a cosyntropin stimulation test (p.880) first to rule out primary adrenal insufficiency.
2. ✖ do not do this test if there is known primary adrenal insufficiency.
3. ✖ do not do this test as an outpatient.
4. test protocol
 - a) give 2–3 grams metyrapone PO at midnight
 - b) check serum 11-deoxycortisol level the next morning
 - c) a normal response is a 11-deoxycortisol level > 7 mcg/dl
 - d) CAUTION: in patients with very little reserve, the reduced cortisol may provoke adrenal insufficiency (this test is safer than the higher doses used for urinary 17-OHCS testing)

Postoperative CT/MRI scan

Immediate post-op imaging is often done to rule-out complications and to assess degree of removal (see post op orders (p.901)). After that, it is recommended to wait 3–4 months post-op for routine follow-up imaging (usually MRI) as it may take this long for the maximal height of the pituitary gland to return to its baseline position (even with total tumor removal)¹⁸

54.2 Outcome following transsphenoidal surgery

54.2.1 General information

In a series of 108 macroadenomas, gross total removal was unusual in tumors with > 2 cm suprasellar extension.¹⁵

54.2.2 Visual deficit

In cases with compression of the optic apparatus, there can be significant improvements in vision following surgery.^{15,19}

54.2.3 Biochemical outcome for hormonally active tumors

Prolactinomas

In a series of 108 macroadenomas, endocrinologic cure was attained in 25% of prolactin-secreting tumors.¹⁵

Acromegaly

Criteria of biochemical cure

The criteria for biochemical cure of acromegaly is not standardized. There may be a discord between IGF-1 levels and mean GH levels.²⁰ Many use a GH cutoff level; range of levels described: < 2.5 – 5 ng/ml. Others feel that an elevated IGF-1 represents lack of cure even if GH < 5 . However, normal IGF-1 levels may not be mandatory.²¹ Still others require a normal IGF-1 AND a normal response to an oral glucose suppression test (OGST) (p.882).

Low GH levels that do not also suppress to < 1 ng/ml after an OGST are considered controlled but not cured (even with normal IGF-1 levels).²² If asymptomatic, expectant management with close follow-up is recommended.²²

Σ: Biochemical cure criteria for acromegaly after treatment

Biochemical cure criteria for acromegaly is not standardized. Recommendation²²:

1. IGF-1 levels within age-matched reference range
2. basal (morning) serum GH level < 5 ng/ml, AND GH nadir < 1 ng/ml in OGST

Outcome with acromegaly

Transsphenoidal surgery results in biochemical cure in 85% of cases with adenomas < 10 mm diameter, no evidence of local invasion, and random GH levels < 40 ng/ml pre-op. Overall, $\approx 50\%$ of all

acromegalics undergoing transsphenoidal surgery had a biochemical cure.²³ Only 30% of macroadenomas and very few with marked suprasellar extension have surgical cure. Patients not cured with surgery require lifelong medical suppression. These tumors may also recur years later after apparent cure. Patients should be monitored every 6–12 months for recurrence.²²

Cushing's disease

There are numerous methodologies for determining biochemical cure for Cushing's disease. One difficulty is that exogenous steroids are often given post-op to avoid potential hypoadrenalism or Addisonian crisis or for nausea, and this can obscure the evaluation. Some options:

1. immediate post-op early morning cortisol levels⁹:
 - a) all steroids are withheld post-op (including dexamethasone as an antiemetic) unless biochemical and/or clinical evidence of hypocortisolism (clinical signs: nausea, anorexia, H/A, arthralgias). ✖ Requires close monitoring and administration of steroids if symptoms develop
 - b) serum ACTH and cortisol levels are drawn between 6 AM and 9 AM on post-op days 1 & 2
 - c) early remission defined as a lowest cortisol level ≤ 140 nmol/L (≤ 5 mcg/dl)
 - 97% (31/32) patients with early remission had sustained remission with mean follow-up of 32 months
 - only 12.5% (1/8) without early remission showed evidence of sustained remission
 - this has been used to select patients for possible early re-exploration
 - early ACTH levels usually drop, but do not consistently become subnormal and are not reliable in predicting sustained remission⁹
2. provocative tests
 - a) overnight low-dose dexamethasone suppression test: an AM cortisol level on post-op day 3 that is ≤ 8 mcg/dl after an overnight 1 mg dexamethasone suppression test is predictive of sustained remission in 97%²⁴
 - b) CRH stimulation test²⁵
3. measurements usually conducted 3 days to 2 weeks post-op following 24 hours of steroid cessation after initial post-op coverage with glucocorticoids
 - a) 24-hour urinary free cortisol
 - b) serum cortisol: the criteria of a cortisol level < 50 nmol/l (< 1.8 mcg/dl)^{26,27,28} is probably too stringent^{9,29,30}
 - c) serum ACTH

The overall remission rate since 1980 is 64–93%, with the highest rates (86–98%) in patients with noninvasive microadenomas identifiable on MRI.⁹

Following effective treatment, all of the following usually improve but may not normalize:

1. HTN and hyperglycemia: within ≈ 1 year
2. osteoporosis related to CD: over ≈ 2 years
3. psychiatric symptoms

Thyrotropin (TSH)-secreting adenomas

Following debulking, small amounts of residual tumor may continue to produce sufficient TSH for hyperthyroidism to persist.³¹ Following surgery + XRT, only $\approx 40\%$ achieve a cure (defined as no residual tumor at surgery or on imaging, and normal free T3 with TSH levels at or below normal).

54.3 Follow-up suggestions for PitNET/adenomas

54.3.1 Nonfunctioning PitNET/adenomas

Post treatment follow-up

Due to the lack of active hormonal products that can be followed with serial lab tests, follow-up relies primarily on serial imaging and monitoring for signs of mass effect from growing tumors (e.g., increasing visual field defects).

Practice guideline: Posttreatment follow-up evaluation for nonfunctioning PitNET/adenomas

Radiologic follow-up

Level III³²

1. T1WI fat sat & T2WI MRI should be included in imaging of NFPAs after surgery or XRT
2. long-term surveillance for tumor regrowth or recurrence is recommended
3. patients with radiologically-proven gross total resection of NFPA require less frequent monitoring than those with subtotal resection
4. the first post-op imaging should be obtained 3–4 months post-op

Level inconclusive recommendations³²

1. there is insufficient evidence to make recommendations for frequency of imaging or length of surveillance after surgery or XRT for NFPAs
2. there is insufficient evidence to make recommendations for the timing of the first post-XRT imaging

Endocrinologic follow-up

Level III³²

1. evaluation for endocrinologic pituitary dysfunction is recommended after surgery and/or XRT for NFPAs
2. post-op adrenal function evaluation is recommended on post-op day 2, week 6, and 12 months
3. perioperative corticosteroid supplementation is recommended for patients with hypocortisolemia pre-op or on post-op day 2
4. post-op endocrinologic follow-up is not recommended beyond 1 year in patients with normal pituitary function
5. endocrinologic follow-up after surgical resection of NFPAs is recommended indefinitely in patients with abnormal pituitary function or those who undergo XRT of NFPAs
6. surveillance of serum sodium for 48 hours after surgery and on post-op days 7–8 is recommended to avoid symptomatic post-op hyponatremia

Level inconclusive recommendations³²

1. there is insufficient information to make recommendations on the detection and treatment of post-op diabetes insipidus (DI)
2. there is insufficient evidence to make recommendations for frequency of endocrinologic evaluation after surgery or XRT for NFPAs

Ophthalmologic follow-up

Level III³²

- ophthalmologic follow-up to assess visual fields and acuity is recommended after surgery and/or XRT for NFPAs

Level inconclusive recommendations³²

- there is insufficient evidence to make recommendations on the frequency or duration of ophthalmologic follow-up

Integrated follow-up

Level inconclusive recommendations³²

- there is insufficient evidence to make recommendations on integration of imaging, endocrinologic and ophthalmologic follow-up after surgery or XRT for NFPAs

54.4 Recurrent PitNET/adenomas

Recurrence incidence: $\approx 12\%$, with most recurring 4–8 years post-op (in the same series¹⁵).

For tumors demonstrating significant regrowth or symptoms following initial resection, consideration for re-resection may be given. Once the tumor is debulked, consideration should be given to XRT, either immediately following the second operation, or, if recurrence after a second operation then almost certainly after a third debulking.

References

- [1] Kuo JS, Barkhoudarian G, Farrell CJ, et al. Congress of Neurological Surgeons Systematic Review and Evidence-Based Guideline on Surgical Techniques and Technologies for the Management of Patients With Nonfunctioning Pituitary Adenomas. *Neurosurgery*. 2016; 79:E536–E538
- [2] Schmidek HH, Sweet WH. *Operative Neurosurgical Techniques*. New York 1982
- [3] Wilson CB. Inactive Pituitary Adenomas. *Clinical Neurosurgery*. 1992; 38:10–31
- [4] Powell M, Lightman SL. *Management of Pituitary Tumours: A Handbook*. New York 1996
- [5] Hardy J. Transsphenoidal Hypophysectomy. *J Neurosurg*. 1971; 34:582–594
- [6] Kern EB, Pearson BW, McDonald TJ, et al. The Transseptal Approach to Lesions of the Pituitary and Parasellar Region. *Laryngoscope*. 1979; 89:1–34
- [7] Spaziante R, de Divitiis E, Cappabianca P. Reconstruction of the Pituitary Fossa in Transsphenoidal Surgery: An Experience of 140 Cases. *Neurosurgery*. 1985; 17:453–458
- [8] Zhang X, Fei Z, Zhang J, et al. Management of Non-functioning Pituitary Adenomas with Suprasellar Extension by Transsphenoidal Microsurgery. *Surgical Neurology*. 1999; 52:380–385
- [9] Esposito F, Dusick JR, Cohan P, et al. Early morning cortisol levels as a predictor of remission after transsphenoidal surgery for Cushing's disease. *Journal of Clinical Endocrinology and Metabolism*. 2006; 91:7–13
- [10] Watson JC, Shawker TH, Nieman LK, et al. Localization of Pituitary Adenomas by Using Intraoperative Ultrasound in Patients with Cushing's Disease and No Demonstrable Pituitary Tumor on Magnetic Resonance Imaging. *Journal of Neurosurgery*. 1998; 89:927–932
- [11] Fatemi N, Dusick JR, Mattozo C, et al. Pituitary hormonal loss and recovery after transsphenoidal adenoma removal. *Neurosurgery*. 2008; 63:709–18; discussion 718–9
- [12] Decker RE, Chalif DJ. Progressive Coma After the Transsphenoidal Decompression of a Pituitary Adenoma with Marked Suprasellar Extension: Report of Two Cases. *Neurosurgery*. 1991; 28:154–158
- [13] Domingue JN, Wilson CB. Pituitary Abscesses. *J Neurosurg*. 1977; 46:601–608
- [14] Robinson B. Intrasellar Abscess After Transsphenoidal Pituitary Adenectomy. *Neurosurgery*. 1983; 12:684–686
- [15] Ciric I, Mikhael M, Stafford T, et al. Transsphenoidal Microsurgery of Pituitary Macroadenomas with Long-Term Follow-Up Results. *J Neurosurg*. 1983; 59:395–401
- [16] Verbalis JG, Robinson AG, Moses AM. Postoperative and Post-Traumatic Diabetes Insipidus. *Front Horm Res*. 1985; 13:247–265
- [17] Watts NB, Tindall GT. Rapid assessment of corticotropin reserve after pituitary surgery. *JAMA*. 1988; 259:708–711
- [18] Teng MM, Huang CI, Chang T. The pituitary mass after transsphenoidal hypophysectomy. *AJNR Am J Neuroradiol*. 1988; 9:23–26
- [19] Cohen AR, Cooper PR, Kupersmith MJ, et al. Visual Recovery After Transsphenoidal Removal of Pituitary Adenoma. *Neurosurgery*. 1985; 17:446–452
- [20] Turner HE, Thornton-Jones VA, Wass JA. Systematic dose-extension of octreotide LAR: the importance of individual tailoring of treatment in patients with acromegaly. *Clin Endocrinol (Oxf)*. 2004; 61:224–231
- [21] Ayuk J, Clayton RN, Holder G, et al. Growth hormone and pituitary radiotherapy, but not serum insulin-like growth factor-I concentrations, predict excess mortality in patients with acromegaly. *Journal of Clinical Endocrinology and Metabolism*. 2004; 89:1613–1617
- [22] Cook DM. AACE Medical Guidelines for Clinical Practice for the diagnosis and treatment of acromegaly. *Endocr Pract*. 2004; 10:213–225
- [23] Davis DH, Laws ER, Ilstrup DM, et al. Results of Surgical Treatment for Growth Hormone-Secreting Pituitary Adenomas. *Journal of Neurosurgery*. 1993; 79:70–75
- [24] Chen JC, Amar AP, Choi S, et al. Transsphenoidal microsurgical treatment of Cushing's disease: post-operative assessment of surgical efficacy by application of an overnight low-dose dexamethasone suppression test. *Journal of Neurosurgery*. 2003; 98:967–973
- [25] Nishizawa S, Oki Y, Ohta S, et al. What can predict postoperative "endocrinological cure" in Cushing's disease? *Neurosurgery*. 1999; 45:239–244
- [26] Trainer PJ, Lawrie HS, Verhelst J, et al. Transsphenoidal resection in Cushing's disease: undetectable serum cortisol as the definition of successful treatment. *Clinical Endocrinology (Oxf)*. 1993; 38:73–78
- [27] Rees DA, Hanna FW, Davies JS, et al. Long-term follow-up results of transsphenoidal surgery for Cushing's disease in a single centre using strict criteria for remission. *Clinical Endocrinology (Oxf)*. 2002; 56:541–551
- [28] Yap LB, Turner HE, Adams CB, et al. Undetectable postoperative cortisol does not always predict long-term remission in Cushing's disease: A single centre audit. *Clinical Endocrinology (Oxf)*. 2002; 56:25–31
- [29] Simmons NE, Alden TD, Thorner MO, et al. Serum cortisol response to transsphenoidal surgery for Cushing disease. *Journal of Neurosurgery*. 2001; 95:1–8
- [30] Rollin GA, Ferreira NP, Junges M, et al. Dynamics of serum cortisol levels after transsphenoidal surgery in a cohort of patients with Cushing's disease. *Journal of Clinical Endocrinology and Metabolism*. 2004; 89:1131–1139
- [31] Sanno N, Teramoto A, Osamura RY. Long-term surgical outcome in 16 patients with thyrotropin pituitary adenoma. *J Neurosurg*. 2000; 93:194–200
- [32] Ziu M, Dunn IF, Hess C, et al. Congress of Neurological Surgeons Systematic Review and Evidence-Based Guideline on Posttreatment Follow-up Evaluation of Patients With Nonfunctioning Pituitary Adenomas. *Neurosurgery*. 2016; 79:E541–E543

Part XIII

Other Tumors and Tumor-like Conditions

55	Metastases to the CNS	908
56	Other Tumors, Cysts, and Tumor-Like Lesions	935
57	Pseudotumor Cerebri Syndrome (PTCS)	955
58	Tumors and Tumor-Like Lesions of the Skull	972
59	Tumors of the Spine and Spinal Cord	979

55 Metastases to the CNS

55.1 Cerebral metastases

55.1.1 General information

Key concepts

- brain metastases are the most common brain tumor seen clinically
- at the time of onset of neurologic symptoms, 70% will be multiple on MRI
- with solitary brain lesions in a patient with a history of cancer, biopsy should almost always be done since 11% of these lesions will not be mets
- although median survival with maximal treatment is only 8 months (similar to GBM), long-term survivors do occur

55.1.2 Metastases to the brain

Cerebral metastases are the most common brain tumor seen clinically, comprising slightly more than half of brain tumors (if one considers only imaging studies, they comprise $\approx 30\%$). In the U.S., the annual incidence of new cases of metastases is up to 170,000,¹ compared to 17,000 for primary brain tumors. 15–30% of patients with cancer (Ca) develop cerebral mets.² In patients with no Ca history, a cerebral met was the presenting symptom in 15%; of these, 43–60% will have an abnormal chest X-ray (CXR)^{3,4} (showing either a bronchogenic primary or other mets to lung).

In 9% of cases, a cerebral met is the only detectable site of spread. Cerebral mets occur in only 6% of pediatric cancers.

The route of metastatic spread to the brain is usually hematogenous, although local extension can also occur.

Solitary mets

- CT (p.914): at the time of *neurologic* diagnosis, 50% are solitary on CT^{5,6}
- MRI: if the same patients have an MRI, <30% will be solitary⁷
- on autopsy: mets are solitary in one-third of patients with brain mets, and 1–3% of solitary mets occur in the brainstem⁸

Increasing incidence of cerebral mets: may be due to a number of factors including:

1. increasing length of survival of cancer patients⁹ as a result of improvements in treatment of systemic cancer
2. enhanced ability to diagnose CNS tumors due to availability of CT and/or MRI
3. many chemotherapeutic agents used systemically do not cross the blood-brain barrier (BBB) well, providing a “haven” for tumor growth there
4. some chemotherapeutic agents may transiently weaken the BBB and allow CNS seeding with tumor

55.1.3 Metastases of primary CNS tumors

Spread via CSF pathways

CNS tumors that more commonly spread via CSF pathways include the following (when these tumors spread to the spinal cord, they are often called “drop mets”):

1. high grade gliomas (p.667) (10–25%)
2. embryonal tumors, primarily medulloblastoma (p.750)
3. ependymoma (p.724) (11%)
4. choroid plexus tumors (p.739)
5. pineal region tumors
 - a) germ cell tumors (p.831)
 - b) pineocytoma (p.762) and pineoblastoma (p.764)
6. rarely:
 - a) oligodendrogliomas (p.662) ($\approx 1\%$)

- b) hemangioblastomas (p.822)
- c) primary CNS melanoma (p.830)

Extraneural spread

Although most CNS tumors do not spread systemically, there is some potential for extraneural spread with the following tumors:

1. medulloblastoma: the most common primary responsible for extraneural spread. May spread to lung, bone marrow, lymph nodes, abdomen
2. meningioma: rarely goes to heart or lungs
3. malignant astrocytomas rarely metastasize systemically
4. ependymomas
5. pineoblastomas
6. meningeal sarcomas
7. choroid plexus tumors
8. tumors that spread through CSF pathways (see above) may spread via a CSF shunt (e.g., to peritoneum with VP shunt or hematogenously with a VA shunt); however, this risk is probably quite small¹⁰

55.1.4 Location of cerebral mets

Intracranial metastases may be either parenchymal ($\approx 75\%$) or may involve the leptomeninges in a carcinomatous meningitis (p.920). 80% of solitary metastases are located in the cerebral hemispheres.

The highest incidence of parenchymal mets is posterior to the Sylvian fissure near the junction of temporal, parietal, and occipital lobes (presumably due to embolic spread to terminal MCA branches).¹¹ Many tend to arise at the gray/white-matter interface.

The cerebellum is a common site of intracranial mets, and is the location in 16% of cases of solitary brain mets. It is the most common p-fossa tumor in adults, thus "a solitary lesion in the posterior fossa of an adult is considered a metastasis until proven otherwise." Spread to the posterior fossa may be via the spinal epidural venous plexus (Batson's plexus) and the vertebral veins.

55.1.5 Primary cancers in patients with cerebral metastases

General information

Accurately ascertaining the source of cerebral metastases in the U.S. is difficult because of lack of detailed coding.¹² In over 2,700 adults with a primary cancer undergoing autopsy at Sloan-Kettering, the sources of cerebral metastases are shown in ► Table 55.1. Sources of brain metastases in pediatrics is shown in ► Table 55.2.

In adults, lung and breast Ca together account for > 50% of cerebral mets.

55

Table 55.1 Sources of cerebral mets in adults (autopsy data)

Primary	%
lung Ca	44
breast	10
kidney (renal cell) ^a	7
GI (colorectal)	6
melanoma ^b	3
undetermined	10

^aan uncommon tumor that metastasizes frequently to brain (in 20–25% of cases)

^b16% in older series¹³

Table 55.2 Sources of cerebral mets in peds

neuroblastoma
rhabdomyosarcoma
Wilm's tumor

In patients with a metastatic brain tumor as the initial presentation (i.e., undiagnosed primary) in 5–10% of cancers, and compared to patients with a known primary, there is about the same number of brain lesions, but there was an increased frequency of extracranial mets.¹⁴ In up to 26% of cases, the primary tumor was never identified.¹⁴

The autopsy incidence of cerebral mets for various types of primary cancers at Sloan-Kettering Cancer Center is shown in ► Table 55.3.

Table 55.3 Autopsy incidence of cerebral mets for given primary cancers

Primary	% with cerebral mets
lung	21
breast	9
melanoma	40
lymphoma	1
• Hodgkin's	0
• non-Hodgkin's	2
GI	3
• colon	5
• gastric	0
• pancreatic	2
GU	11
• kidney (renal)	21
• prostate ^a	0
• testes	46
• cervix	5
• ovary	5
osteosarcoma	10
neuroblastoma	5
head and neck	6

^auncommon, but does occur

Lung cancer

The lungs are the most common source of cerebral mets, and these are usually multiple. The lung primary may be so small as to render it occult.

Necropsy demonstrates cerebral mets in up to 50% of patients with small-cell lung Ca (SCLC) and non-squamous, non-small-cell lung Ca.¹⁵

Cigarette smoking is responsible for 88% of all types of lung cancer combined (odds ratio for developing any type of lung cancer with cigarette smoking is 13.4), 91% of squamous cell (OR = 18.8), 89% of small cell (OR = 14.3), 95% of large cell (OR = 34.3), and 82% of adenocarcinoma (OR = 7.9).¹⁶

Small-cell lung cancer (SCLC)

AKA "oat cell" Ca. A neuroendocrine tumor. 95% arise in proximal airways, usually in mainstem or lobar bronchi. Typically younger (27–66 years) than other lung Ca. Median survival: 6–10 months. Considered a systemic disease. Staged in 1 of 2 categories:

1. limited: confined to an area of the chest that can be encompassed by a single radiation port
2. extensive: metastasis outside the thorax or intrathoracic disease that cannot be contained in a single radiation port

Although SCLC comprises only ≈ 20% of primary lung cancers, it is more likely to produce cerebral mets than other bronchogenic cell types (brain mets are found in 80% of patients who survive 2 yrs after diagnosis of SCLC).⁹

Treatment

Very radiosensitive.

No identified brain mets: prophylactic cranial irradiation (PCI) with WBXRT reduces the incidence of symptomatic brain mets and increases survival (disease-free & overall),^{17,18} Typically 25 Gy in 10 fractions.

Brain mets: surgical resection considered for immediately life-threatening large lesions, XRT is used otherwise. Multiple SCLC brain lesions: XRT (initial treatment 30 Gy in 10 fractions) + chemotherapy.

Treatment of primary: usually not resected. Treated with chemotherapy ± XRT.

Recurrent brain mets after failure of initial treatment: 20 Gy in 10 fractions.

Non-small-cell lung cancer (NSCLC)

Includes: adenocarcinoma (the most common NSCLC), large cell, squamous cell, bronchoalveolar. Retrospective analysis of patients with NSCLC completely resected from the lung found a 6.8% first recurrence rate in the brain.¹⁵ Staged with typical TNM system. Prognosis better than SCLC.

Treatment of lung primary:

1. grades I, II, IIIA: resection
2. higher grades (e.g., distal mets, excluding single brain met): XRT + chemotherapy

Staging studies for known lung primary

1. PET scan: can detect small malignancies. Useful in NSCLC to determine eligibility of resection of primary. Not useful in initial evaluation of SCLC
2. chest CT: usually includes adrenals and liver (thus abdomen and pelvis CT not necessary)
3. bone scan
4. brain: CT or MRI

When metastatic lung cancer is the suspected source of a newly diagnosed brain lesion, the lung lesion should be biopsied (if technically feasible) to rule out SCLC before obtaining tissue from the cerebral mass because XRT is the primary treatment for SCLC brain metastases.

Melanoma

General information

Melanoma: the 5th most common cancer in men, 7th in women. Incidence is increasing. Most common sites of origin of melanoma metastases: skin, retina, brain—primary CNS melanoma (p.830)—nail bed. The primary site cannot be identified in up to ≈ 14% of cases.¹⁹ Extremely difficult to locate primary sites: intraocular, GI mucosa.

Brain mets are found in 10–70% of patients with metastatic melanoma in clinical studies, and in 70–90% on autopsy of patients who died from melanoma. Patients with melanoma who have neurosurgical lesions typically presented 14 months after primary lesion was identified. Once cerebral mets of melanoma are detected, median survival is ≤ 6 months^{20,21,22} and the mets contributed to the death in 94% of cases.²³ A small group with survival > 3 yrs had a single surgically treated met in the absence of other visceral lesions.

Evaluation

Metastatic melanoma to the brain classically causes pia/arachnoid involvement on imaging. Hemorrhagic involvement is common.

CT: lesions may be slightly hyperdense to brain on unenhanced CT due to melanin. Enhancement is less constant than for other mets (e.g., bronchogenic Ca).

MRI: decreased signal on T2WI surrounded by intense halo of edema. Enhancing T1WI lesions in a patient with melanoma is highly suggestive of melanoma metastases.

Systemic work-up: systemic disease determines ultimate survival after treatment of melanoma mets to the brain in 70% of patients. ∴ search for systemic mets should be done, including: CT of chest/abdomen/pelvis & bone scan. PET scan may be more sensitive for detecting metastatic spread than CT when there are clinical signs that the tumor has spread²⁴; *except for the brain*, where brain MRI is more sensitive than CT or PET.

Treatment

Suggested algorithm for patients with metastatic melanoma to the brain (adapted,²⁵ see ► Fig. 55.1).

Patients with **Karnofsky performance scale (KPS) score** (p.1640) < 70 are likely to be poor surgical candidates.

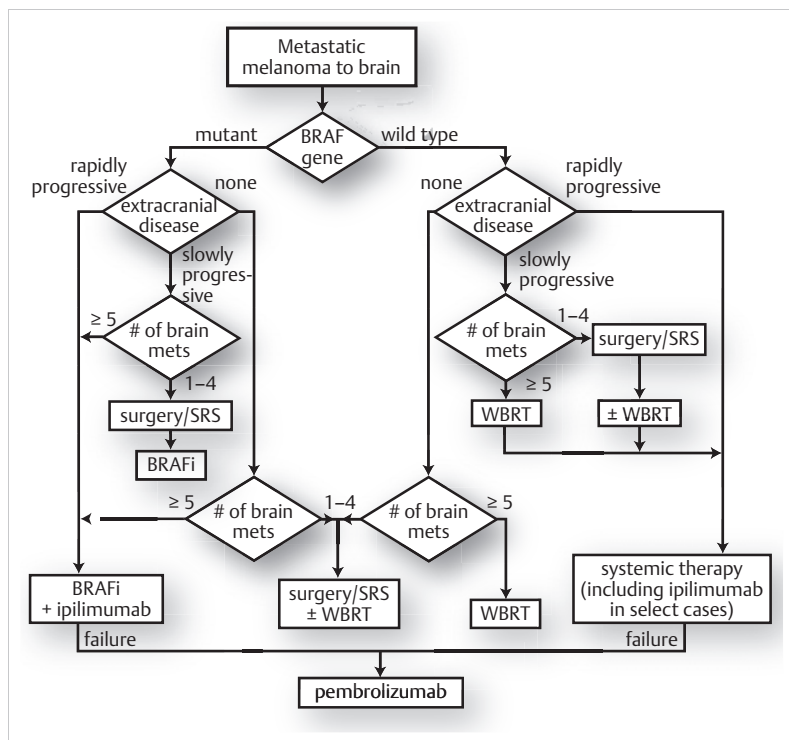


Fig. 55.1 Suggested algorithm for patients with metastatic melanoma to the brain (adapted²⁵).

Abbreviations: BRAFi = BRAF inhibitors, SRS = stereotactic radiosurgery (p.1903), WBRT = whole brain radiation therapy.

Some key points:

1. patients with rapidly progressive systemic disease: treat the systemic disease with chemotherapy first, before dealing with the brain mets
2. patients without systemic disease and 1–4 mets are candidates for surgery if all mets are accessible and can all be removed²⁶ (this is a general principle for brain mets, not just for melanoma). SRS is an alternative

► Surgical indications

1. patients with 1–4 CNS metastases that can be completely resected when systemic disease is absent or slowly progressive: long-term survival is possible
2. patients with intracranial mets that cannot be completely removed or with uncontrolled systemic disease may be surgical candidates for the following:
 - a) for symptomatic relief: e.g., lesion causing painful pressure
 - b) life-threatening lesion: e.g., large p-fossa lesion with 4th ventricle compression
 - c) for hemorrhagic lesion causing symptoms by mass effect from the clot

► **Whole-brain radiation therapy (WBRT).** Melanoma is typically radioresistant. WBRT provides 2–3 month survival benefit and may be considered for palliation in patients with multiple mets that preclude complete excision or SRS.

► **Stereotactic radiosurgery (SRS).** Considered for ≤4 lesions, all ≤3 cm in diameter, that are surgically inaccessible, with limited or quiescent systemic involvement. Relative contraindications: hemorrhagic lesions, lesions with significant mass effect surrounding edema.

► Chemotherapy

1. alkylating agents:
 - a) dacarbazine, formerly the gold-standard treatment for melanoma. About equally as effective as its newer orally administered analog temozolomide (Temodar®). Response rate: 10–20%
 - b) Fotemustine appeared promising in phase II trials but only 6% responded in phase III (vs. 0% for dacarbazine)²⁷
2. immunotherapy:
 - a) ipilimumab (Yervoy®): monoclonal antibody against cytotoxic T lymphocyte antigen-4 (CTLA-4) antigen. More effective in patients who do not require corticosteroids
 - b) interleukin-2 (IL-2): has shown minimal activity in brain mets, and trials have usually excluded patients with untreated or uncontrolled brain mets due to risk of cerebral edema and hemorrhage from capillary leak^{28,29,30}
3. BRAF inhibitors (BRAFi): inhibits BRAF kinase (a protein that participates in regulation of cell division and differentiation), useful in tumors with BRAF oncogene mutation (as opposed to BRAF wildtype) which is common in melanoma
 - a) dabrafenib: phase II trial (NCT01266967)³¹
 - b) vemurafenib: promising results in heavily treated patients. Phase II trial (NCT01378975)³²
4. anti-PD-1 drug (monoclonal antibody to PD-1 programmed cell death receptor): pembrolizumab (Keytruda) approved for advanced or unresectable melanoma not responding to other drugs³³

Outcome

1. in a patient with a single brain met (any type) and good Karnofsky performance score (> 70) and no evidence of extracranial disease, surgery + XRT had a median survival of 40 weeks vs. 15 weeks for XRT alone^{34,35}
2. for melanoma, retrospective studies have shown a benefit of treatment with either surgery or SRS only when all brain lesions are completely treated (selection bias possible in these studies)^{35, 36,37,38}
3. predictors of poor outcome in melanoma:
 - a) > 3 brain mets²¹
 - b) development of brain mets after the diagnosis of extracranial disease²¹
 - c) elevated lactate dehydrogenase > 2 × normal²²
 - d) presence of bone metastases²²
 - e) multiple brain mets and extensive visceral disease³⁹

Renal-cell carcinoma

AKA hypernephroma. Usually associated with spread to lungs, lymph nodes, liver, bone (high affinity for bone), adrenals, and contralateral kidney before invading the CNS (thus, this tumor rarely presents as isolated cerebral metastases). Look for hematuria, abdominal pain, and/or abdominal mass on palpation or CT. Response to XRT is only ≈ 10%.

Esophageal cancer

Median survival is 4.2 months based on a review of 26 cases.⁴⁰ Solitary brain met with good Karnofsky score and surgical treatment may indicate a better prognosis.

55.1.6 Clinical presentation

As with most brain tumors, signs and symptoms are usually slowly progressive compared to those from vascular events (ischemic or hemorrhagic infarcts), which tend to be sudden in onset and slowly resolve, or electrical events (seizures), which tend to be sudden in onset and rapidly resolve. Metastases may also present acutely by hemorrhaging, causing an intraparenchymal hematoma (IPH). There are no findings that would allow differentiation of a metastatic tumor from a primary neoplasm on clinical grounds.

Signs and symptoms include:

1. those due to increased ICP from mass effect and/or blockage of CSF drainage (hydrocephalus):
 - a) headache (H/A): the most common presenting symptom, occurs in ≈ 50%
 - b) nausea/vomiting
2. focal deficits:
 - a) due to compression of brain parenchyma by mass and/or peritumoral edema (e.g., monoparesis without sensory disturbance)
 - b) due to compression of cranial nerve

3. seizures: occur only in $\approx$ 15% of cases
4. mental status changes: depression, lethargy, apathy, confusion
5. hemorrhage (p. 1612): producing IPH, often with H/A and sudden onset of focal deficit, obtundation may occur with large hemorrhages
6. symptoms suggestive of a TIA (dubbed "tumor TIA") or stroke, may be due to:
 - a) occlusion of a vessel by tumor cells
 - b) hemorrhage into the tumor, especially common with metastatic melanoma, choriocarcinoma, and renal-cell carcinoma⁴¹; see Hemorrhagic brain tumors (p. 1612). May also occur due to decreased platelet count

55.1.7 Evaluation

Imaging studies (CT or MRI)

Metastases usually appear as "non-complicated" masses (i.e., round, well circumscribed), often arising at the gray/white junction ($\blacktriangleright$ Fig. 55.2). Characteristically, profound white matter edema ("fingers of edema") reach deep into the brain from the tumor, usually more pronounced than that seen with primary (infiltrating) brain tumors. When multiple lesions are present (on CT or MRI of brains with multiple mets) Chamber's rule applies: "Whoever counts the most mets is right." Mets usually enhance, and must be considered in the differential diagnosis of a ring-enhancing lesion.

MRI is more sensitive than CT, especially in the posterior-fossa (including brain stem). Detects multiple mets in up to $\approx$ 20% of patients who appear to have single mets on CT.²

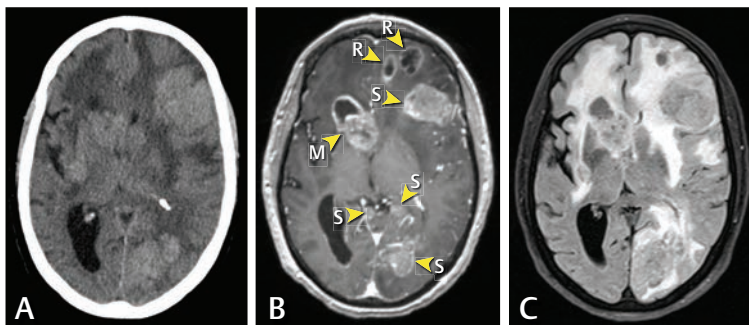


Fig. 55.2 Cerebral metastases.

65-year-old woman with small cell cancer of the lung metastatic to the brain.

Image: A: axial non-contrast head CT obtained at admission. B: axial MRI T1 + contrast, showing multiple metastatic lesions (yellow arrowheads) (which were also present in the cerebellum, not shown), some are solid (S), some are ring enhancing (R), and some are mixed (M). C: axial T2 MRI demonstrating characteristic extensive white-matter "fingers of edema."

Lumbar puncture

Relatively contraindicated when there is a cerebral mass (may be considered once mass lesion has been ruled out). May be most useful in diagnosing carcinomatous meningitis (p. 920), and may be of limited help in diagnosing CNS lymphoma.

Metastatic work-up with suspected brain metastases

When metastatic disease is suspected based on imaging, a search for a primary site and assessment for other metastatic lesions may be considered since it may provide alternative sites for tissue for histologic diagnosis, and it may guide treatment (e.g., widely disseminated metastases may preclude aggressive therapy). It is also undertaken to look for a primary and other metastases when biopsy is consistent with brain metastases.

Metastatic work-up should include:

1. CT of the chest (more sensitive than CXR), abdomen and pelvis. Evaluates for primary and additional mets (to lung, adrenals, liver... CT has superseded CXR)
2. radionuclide bone scan: for patients with bone pain or bone lesions or for tumors that tend to produce osseous metastases (especially: prostate, breast, kidney, thyroid & lung)
3. mammogram in women
4. prostate specific antigen (PSA) in men
5. PET scan: can detect small malignancies

Cancer of unknown primary site (CUP): If the metastatic work-up (see above) is negative, the pathology of a metastatic brain lesion as determined by biopsy may implicate specific primary sites.

Small-cell carcinoma metastatic to the brain is most likely from the lung. These tumors stain positive for neuroendocrine stains (p.632).

Adenocarcinoma: lung is the most common primary. Other sources: GI (mostly colon), breast. The primary site may remain occult even after extensive evaluation in up to 88%.⁴² Immunostaining has been tried to identify the primary site but has not been found to be widely useful.

55.1.8 Management

General information

With optimal treatment, median survival of patients with cerebral mets is still only $\approx$ 26–32 weeks, therefore management is mostly palliative. Also see Outcome (p.919) for comparison of various treatments.

Confirming the diagnosis

NB: 11% of patients with abnormalities on brain CT or MRI with a history of cancer (within past 5 yrs) do not have cerebral metastases.³⁴ Differential diagnoses include: primary brain tumor (glioblastoma, low grade astrocytoma), abscess, and nonspecific inflammatory reaction. If non-surgical treatment (e.g., chemotherapy or XRT) is being contemplated, the diagnosis should first be confirmed by biopsy in almost all cases.

Management decisions

Prognostication

This is critical since many treatment decisions depend on overall prognosis.

RTOG RPA: Radiation Therapy Oncology Group recursive partitioning analysis classification⁴³ (see ► Table 55.4). Conclusion: the specific tumor type, length of time since diagnosis, etc. are not as important prognostically as the Karnofsky Performance Scale (KPS) score (p.1640).

The applicability of RPA to melanoma mets to the brain is controversial (it has been both validated⁴⁴ and disputed³⁸).

RPA Class 3 patients have been shown to be unlikely to benefit from any of numerous treatment modalities studied. Class 1 are more likely to benefit. Most patients are Class 2, for whom benefit is unclear.

Table 55.4 Radiation Therapy Oncology Group recursive partitioning analysis (RPA) classification for patients with brain mets (from 1200 patients with $\geq$ 1 brain mets undergoing XRT)⁴³

RPA class	Description	Median survival (mos) ^a
1	• KPS^b $\geq$ 70 and • age $<$ 65 years and • controlled^c or absent primary tumor with the brain the only site of metastasis	7
2	• all others (i.e., not RPA class 1 or 3)	4
3	• KPS $<$ 70	2

^afor patients undergoing XRT

^bKPS = Karnofsky Performance Scale score (p.1640)

^ccontrolled = stable disease over 3 months observation

Management algorithm

► Table 55.5 shows a summary of management suggestions (details appear in following sections).

Also, surgical excision may be considered for patients with completely resectable brain mets who are candidates for chemotherapy with interleukin-2 (IL-2) for systemic disease (e.g., for renal-cell Ca or melanoma) since there are case reports of significant cerebral edema associated with use of this drug in the presence of cerebral mets.

Table 55.5 Management suggestions for cerebral metastases^a

Clinical situation		Management
unknown primary or unconfirmed diagnosis		stereotactic biopsy for ≈ all patients if surgical excision is not a consideration
uncontrolled widespread systemic cancer & obviously short life expectancy and/or poor performance status, KPS (p. 1640) ≤ 70		(biopsy as indicated above) + WBXRT or no treatment
metastases known to be extremely radiosensitive (lymphoma, small cell lung cancer, germ cell tumor, multiple myeloma)		XRT
Stable systemic disease & KPS > 70		
solitary met	symptomatic or threatening, and safely resectable ^b lesion	surgical excision ^{c, d} + WBXRT ^e or surgical excision ^{c, d} + SRS ^e
	asymptomatic, small, or not safely resectable ^b	WBXRT ± SRS boost
multiple mets	single large lesion that is life threatening or producing mass effect	surgery for the large lesion + WBXRT for the rest
	≤ 3 lesions: symptomatic & can all be removed	surgery + WBXRT or SRS + WBXRT
	≤ 3 lesions: cannot all be removed	WBXRT or SRS + WBXRT
	> 3 lesions: with no mass effect requiring surgery	WBXRT ⁴⁵

^a adapted Pollock et al.⁴⁶ and the 2019 Congress of Neurological Surgeons Systematic Review and Evidence-Based Guidelines^{47,48}

^b safely resectable lesion implies surgery can be performed without undue risk of worsening neurologic condition (including hemiparesis, aphasia...) that would lower KPS and thereby shorten longevity

^c surgery + WBXRT is superior to WBXRT alone (Level 1); surgery + SRS is superior to SRS alone (Level 3)⁴⁷

^d surgical excision: class III data suggests gross total resection is superior to subtotal resection in RPA class I patients, and that with single mets that en bloc resection reduces the risk of leptomeningeal disease compared to piecemeal resection⁴⁷

^e NB: used alone for patients with 1-3 mets, SRS is better for survival and neurocognition but is worse for local and distant control compared to WBXRT

Abbreviations: KPS = Karnofsky Performance Scale score (p. 1640); SRS = stereotactic radiosurgery; WBXRT = whole brain radiation therapy.

Medical management

Initial treatment

- antiseizure medications:** e.g., Keppra® (levetiracetam) starting with 500 mg PO or IV q 12 hours. Generally not indicated for posterior fossa lesions due to low epileptogenicity with these. Guidelines: ASMs showed a nonstatistically significant reduction of early seizures, and had no benefit for seizures at 3 months or for survival.^{49,50} Guideline conclusions: no compelling data for prophylactic ASM pre-op or post-op. Non-guideline opinion: use is more justifiable in patients who actually have a seizure
- corticosteroids:** many symptoms are due to peritumoral edema (which is primarily vasogenic), and respond to steroids within 24–48 hrs. This improvement is not permanent, and prolonged steroid administration may produce side effects; see Possible deleterious side effects of steroids (p. 626). Guidelines: steroids are recommended for temporary relief of symptoms due to increased ICP or edema related to mets (Level 3).⁵¹ There is insufficient evidence to make recommendations for use with asymptomatic lesions without mass effect.⁵¹

Dexamethasone is the drug of choice (Level 3)⁵¹

R for mild-to-moderate symptoms in a patient who is not already on steroids: dexamethasone (Decadron®) 4-8 mg/d (Level 3)⁵¹ usually given in divided doses. Higher doses are not more effective and side effects are more common⁵²

R for severe symptoms consider higher dose of 16 mg/d (Level 3).⁵¹

Regardless of regimen used, taper steroids as rapidly as tolerated to the lowest effective dose for symptom control (Level 3).⁵¹

3. H2 antagonists (e.g., ranitidine 150 mg PO q 12 hrs) or proton pump inhibitor (omeprazole)

Chemotherapy

See discussion of Limitations of chemotherapy in the brain (p.627). If multiple lesions of known small-cell Ca are detected on cerebral imaging, treatment of choice is radiation plus chemotherapy.

Radiation therapy

General information

NB: not all brain lesions in cancer patients are mets (see above).

In patients not considered for surgery, steroids and radiation may be palliative. H/A are usually reduced, and in ~ 50% of cases symptoms improve or completely resolve.⁵³ This does not result in local control for the majority of these patients and they frequently succumb from progressive brain disease.

"Radiosensitivity" of various metastatic tumors to whole brain radiation therapy (WBXRT) are shown in ► Table 55.6.

The usual dose is 30 Gy in 10 fractions given over 2 weeks. With this dose, 11% of 1-yr survivors and 50% of 2-yr survivors develop severe dementia.

Table 55.6 "Radiosensitivity" of brain metastases to WBXRT

Radiosensitivity	Tumor
Radiosensitive ³⁴	• small-cell lung Ca • germ cell tumors • lymphoma • leukemia • multiple myeloma
Moderately sensitive	• breast
Moderately resistant	• colon • non small-cell lung cancer
Highly resistant ^a	• thyroid • renal cell (10% respond) • malignant melanoma • sarcoma • adenocarcinoma

^aSRS may be better than WBXRT for these

Prophylactic cranial irradiation

Prophylactic cranial irradiation after resection of small-cell lung carcinoma (SCLC) reduces relapses in brain, but does not affect survival.⁵⁴

Post-op radiation therapy

WBXRT is usually recommended following craniotomy for metastatic disease,⁵⁵ especially with SCLC where "micro-metastases" are presumed to be present throughout brain. (**Note:** some centers do not routinely administer post-op WBXRT (except for very radiosensitive tumors such as SCLC) but instead follow patients with serial imaging studies and administer XRT only when metastases are documented.)

Optimal dose is controversial. Early reports recommended 30–39 Gy over 2–2.5 weeks (3 Gy fractions) with or without surgery.⁵⁶ This is acceptable in patients not expected to live long enough to get long-term radiation effects. Recent recommendations are for smaller daily fractions of 1.8–2.0 Gy to reduce neurotoxicity.⁵⁷ These low doses are also associated with a higher rate of recurrent brain metastases.⁵⁸ Since 50 Gy are needed to achieve >90% control of micrometastases, some use

45–50 Gy WBXRT, plus a boost to the tumor bed to bring the total treatment up to 55 Gy, all with low fractions of 1.80–2.0 Gy.⁵⁹

Stereotactic radiosurgery

Inconsistent in its ability to reduce tumor size. Some retrospective studies show results comparable to surgery.⁶⁰ Others do not.⁶¹ Does not obtain tissue for histological analysis, and generally cannot be used for lesions > 3 cm. Also, see Stereotactic radiosurgery (p.919).

Surgical management

Solitary lesions

Indications favoring surgical excision of a solitary lesion:

1. primary disease quiescent
2. lesion accessible
3. lesion is symptomatic or life-threatening
4. primary tumor known to be relatively radioresistant (excision is rarely indicated for untreated brain metastases from SCLC because of its radiosensitivity)
5. for recurrent SCLC following XRT
6. diagnosis unknown: alternatively consider biopsy, e.g., stereotactic biopsy

Surgical resection in patients with progressive systemic disease and/or significant neurologic deficit is probably unjustified.⁶² Also, in newly diagnosed cancer patients, craniotomy may delay systemic treatment for weeks and the ramifications of this need to be considered.

Multiple lesions

Patients with multiple metastases generally have much worse survival than those with solitary lesions.⁵⁷ Multiple metastases are usually treated with XRT without surgery. However, if total excision of *all* mets is feasible, then even multiple mets may be removed with survival similar to those having a single met removed²⁶ (also see ► Table 55.5 for summary). If only incomplete excision is possible (i.e., cannot remove all mets, or portions of 1 or more must be left behind) then there is no improvement in survival with surgery, and XRT alone is recommended. The mortality of removing > 1 met at a single sitting is not statistically significantly higher than removing a single met.

Situations where surgery may be indicated for multiple mets⁶³:

1. one particular and accessible lesion is clearly symptomatic and/or life threatening (life-threatening lesions include p-fossa and large temporal lobe lesions). This is palliative treatment to reduce the symptom/threat from that particular lesion
2. multiple lesions that can all be completely removed (see above)
3. no diagnosis (e.g., no identifiable primary): consider stereotactic biopsy

Stereotactic biopsy

Considered for:

1. lesions not appropriate for open surgery. Includes cases with no definite diagnosis and:
 - a) deep lesions
 - b) multiple small lesions
2. patients not candidates for surgical resection
 - a) poor medical condition
 - b) poor neurologic condition
 - c) active or widespread systemic disease
3. to ascertain a diagnosis
 - a) when another diagnosis is possible: e.g., no other sites of metastases, long interval between primary cancer and detection of brain mets...
 - b) especially if nonsurgical treatment modalities are planned (see above)

Intraoperative considerations for surgical removal

Most lesions present themselves on the surface of the brain or through the dura. For lesions not visible on the surface nor palpable immediately beneath the surface, intraoperative ultrasound or stereotactic techniques may be used to localize the lesion.

Metastases usually have a well defined border, thus a plane of separation from normal brain may be exploited, often allowing gross total removal.

55.1.9 Outcome

General information

► Table 55.7 lists factors associated with better survival regardless of treatment. Also, the prognosis gets worse as the number of mets increases.⁴⁶ Median survival even with best treatment in some studies is only ≈ 6 months. To put this into perspective, this is worse than with glioblastoma.

Table 55.7 Factors associated with better prognosis for brain mets (with any treatment)

- Karnofsky score^a (KPS) > 70
- age < 60 yrs
- metastases to brain only (no systemic mets)
- absent or controlled primary disease
- > 1 yr since diagnosis of primary
- the fewer the number of brain mets
- female gender

^athe KPS (p. 1640) is probably the most important predictor; those with a score of 100 had median survival > 150 weeks

Natural history

By the time that neurologic findings develop, median survival among untreated patients is ≈ 1 month.⁶⁴

Steroids

Using steroids alone (to control edema) doubles survival⁶⁵ to 2 mos (**Note:** this is based largely on pre-CT era data, and the tumors were therefore probably larger than in current studies⁶⁶).

Whole brain radiation therapy (WBXRT)

WBXRT + steroids increases survival to 3–6 mos.²⁶ 50% of deaths are due to progression of intracranial disease.

Surgery $\pm$ WBXRT

Recurrence of tumor was significantly less frequent and more delayed with the use of post-op WBXRT.⁵⁵ Length of survival was unchanged with supplemental use of WBXRT. There is also an additional loss of cognitive function in many cases, and patients are rarely independent after WBXRT.

In 33 patients treated with surgical resection of single mets and post-op WBXRT⁶⁷: median survival was 8 months; with 44% 1-yr survival. If no evidence of systemic Ca, 1-yr survival is 81%. If systemic Ca is present (active or inactive), 1-yr survival is 20%. Patients with solitary mets and no evidence of active systemic tumor have the best prognosis.^{33,62} With total removal, no recurrence nor new parenchymal mets occurred within 6 months, and the major cause of death was progression of Ca outside the CNS. A randomized trial verified the improved longevity and quality of survival of patients with solitary mets undergoing surgical excision plus WBXRT vs. WBXRT alone (40 weeks vs. 15 weeks median survival).³⁴ The surgical mortality was 4% ($\approx$ same as 30-day mortality in the XRT-only group). More patients treated with WBXRT alone die of their brain mets than those who underwent surgery. Following total removal and post-op WBXRT, 22% of patients will have recurrent brain tumor at 1 year.⁵⁷ This is better than surgery without XRT (with reported failure rates of 46%⁵⁷ and 85%⁵⁸).

Stereotactic radiosurgery (SRS)

There has not been a randomized study to compare surgery to SRS. *Retrospective* studies suggest that SRS may be comparable to surgery.^{60,68} However, a prospective (non-randomized, retrospectively matched) study⁶¹ found a median survival of 7.5 mos with SRS vs. 16.4 mos with surgery, and a higher mortality from cerebral disease in the SRS group (with the mortality due to the SRS treated lesions and not new lesions). A local control rate of $\approx 88\%$ has been reported, with one study also recommending WBXRT following the SRS for better regional control.⁶⁹

Actuarial control rates at 1 year following SRS + WBXRT were 75–80% and appear to be similar to surgery + WBXRT.⁴⁶ However, SRS was unreliable in reducing tumor size.

Multiple mets

Patients with multiple mets that were totally removed have a survival that is similar to those having single mets surgically removed²⁶ (see above).

Recurrent metastases

Surgery is recommended for intracranial recurrence after initial surgery or SRS (Level 3).⁴⁷

55.1.10 Carcinomatous meningitis

General information

Carcinomatous meningitis (CM) AKA (lepto)meningeal carcinomatosis (LMC). Found in up to 8% of patients autopsied with systemic cancer. CM may be the presenting finding in up to 48% of patients with cancer (before the diagnosis cancer is known). Most common primaries: breast, lung, then melanoma.⁷⁰ (p 610–2) Always include *lymphomatous meningitis* in the differential diagnosis; see CNS lymphoma (p. 840).

Clinical

Simultaneous onset of findings in multiple levels of neuraxis. Multiple cranial nerve findings are frequent (in up to 94%, most common: VII, III, V, and VI), usually progressive. Most frequent symptoms: H/A, mental status changes, lethargy, seizure, ataxia. Non-obstructive hydrocephalus is also common. Painful radiculopathies can occur with “drop mets.”

Diagnosis

Lumbar puncture

Perform only after mass lesion has been ruled out with cranial CT or MRI. Although the initial LP may be normal, CSF is eventually abnormal in > 95%.

CSF should be sent for:

1. cytology to look for malignant cells (requires ≈ 10 ml for adequate evaluation for CM). Repeat if negative (45% positive on first study, 81% eventually positive after up to 6 LPs). May need to pass CSF through a millipore filter
2. bacterial and fungal cultures (including unusual organisms, e.g., cryptococcus)
3. tumor markers: carcinoembryonic antigen, alpha-fetoprotein
4. protein/glucose: elevated protein is the most common abnormality. Glucose may be as low as ≈ 40 mg% in about a third of patients

MRI

Contrast enhanced MRI is more sensitive in showing meningeal enhancement.⁷¹

CT

May show (mild) ventricular dilatation, enhancement of basal cisterns. Sulcal enhancement may also occur with involvement of the convexities.

Myelography

Spinal seeding (“drop mets”) will produce filling defects on myelography.

Survival

Untreated: < 2 months. With radiation therapy + chemotherapy: median survival is 5.8 mos (range 1–29). Chemotherapy may be given intrathecally. About half of patients die of CNS involvement, and half die of systemic disease.

55.2 Spinal epidural metastases

55.2.1 General information

Key concepts

- suspected in a cancer patient with back pain that persists in recumbency
- occurs in $\approx 10\%$ of all cancer patients
- 80% of primary sites: lung, breast, GI, prostate, melanoma, and lymphoma
- many treatments reduce pain. Surgery + XRT in selected cases increases chances of preserving ambulation & produces a modest improvement in survival
- if no neurologic compromise or bony instability, usual treatment: biopsy (CT- or fluoro-guided) followed by XRT (surgical indications ► Table 55.11)
- surgery not helpful for: total paralysis > 8 hrs, loss of ambulation > 24 hrs, and not recommended for prognosis < 3–4 months survival, poor medical condition (poor PFTs...), or radiosensitive tumor

Spinal epidural metastases (SEM) occur in up to 10% of cancer patients at some time,⁷² and are the most common spinal tumor. 5–10% of malignancies present initially with cord compression.⁷³ For other etiologies of spinal cord compression, see items marked with a dagger (†) under Myelopathy (p. 1696).

Routes of metastasis to spine:

1. arterial
2. venous: via spinal epidural veins (Batson's plexus⁷⁴)
3. perinervous (direct spread)

The usual route of spread is hematogenous dissemination to the vertebral body with erosion back through pedicles and subsequent extension into the epidural space (i.e., *anterior* epicenter). Less commonly may initially metastasize to lateral or posterior aspect of canal. Most metastases (mets) are epidural, only 2–4% are intradural, and only 1–2% are intramedullary. Distribution between cervical, thoracic, and lumbar spine is proportional to the length of the segment, thus the thoracic spine is the most common site (50–60%).

55.2.2 Primary tumors that metastasize to the spine

► Table 55.8 shows primary tumor types that give rise to SEM. The majority are common primaries that tend to metastasize to bone (lung, breast, prostate, renal-cell, and thyroid). Rare tumors that

Table 55.8 Sources of spinal epidural metastases causing cord compression

Site of primary	Series A	Series B ^a	Series C ^b
lung	17%	14%	31%
breast	16%	21%	24%
prostate	11%	19%	8%
kidney (renal-cell)	9%		1%
unknown site	9%	5%	2%
sarcoma	8%		2%
lymphoma	6%	12%	6%
GI tract	6%		9%
thyroid	6%		
melanoma	2%		4%
others (including multiple myeloma)	13%	29% ^c	13%

^aseries B: retrospective study of 58 patients undergoing MRI evaluation for SEM⁷²

^bseries C: 75 patients with SEM out of 140 patients evaluated prospectively for back pain⁷⁶

^cin series B, "other" includes GI, GU, skin, ENT, CNS

may go to bone include the myxoid subtype of liposarcoma⁷⁵ (17% of these patients develop bone mets, 5-year median survival is 16%).

55.2.3 Presentation

Pain: the most common initial symptom. Occurs in up to 95% of patients with SEM.^{77,78} Types of pain:

1. local pain: typically aching, experienced at the level of involvement. Increased pain with *recumbency* (especially at night) is characteristic
2. radicular: tends to be sharp or shooting, referred into dermatome of the involved nerve root. Commonly bilateral in thoracic region
3. mechanical: usually exacerbated by movement

Neck-flexion, straight-leg-raising, coughing, sneezing, or straining may also aggravate the pain.

Motor or autonomic dysfunction: the second most common presentation. Up to 85% of patients have weakness at the time of diagnosis. Leg stiffness may be an early symptom. Bladder dysfunction (urinary urgency, hesitancy, or retention) is the most common autonomic manifestation; others include constipation or impotence.

Sensory dysfunction: anesthesia, hypesthesia, or paresthesias usually occur with motor dysfunction. Cervical or thoracic cord involvement may produce a sensory level.

Other presentations: pathologic fracture. Bone metastases can sometimes produce hypercalcemia (a medical emergency).

The greater the neurologic deficit when treatment is initiated, the worse the chances for recovery of lost function. 76% of patients have weakness by the time of diagnosis.⁷² 15% are paraplegic on initial presentation, and <5% of these can ambulate after treatment. Median time from onset of symptoms to diagnosis is 2 months.⁷⁹

Metastases to the upper cervical spine

For differential diagnosis, see Foramen magnum lesions (p.1649), and Axis (C2) vertebra lesions (p.1678).

Metastases to the C1–2 region comprise only $\approx 0.5\%$ of spinal mets.⁸⁰ They typically present initially with suboccipital and posterior cervical pain, and as the lesion progresses patients develop a characteristic pain that makes it difficult to sit up (some will hold their heads in their hands to stabilize it). Possibly as a result of the capacious spinal canal at this level, only $\approx 11\text{--}15\%$ of patients present with neurologic symptoms. 15% develop spinal cord compression,⁸¹ and quadriplegia from atlantoaxial subluxation occurred in $\approx 6\%$.⁸¹

Anterior approaches for stabilization at this location are difficult. Pathologic fractures due to osteoblastic types of tumors (e.g., prostate, some breast) may heal with radiation treatment and immobilization. For others, good pain relief and stabilization may be achieved with radiation followed by posterior fusion.⁸¹

55.2.4 Evaluation and management of epidural spinal metastases

General information

There is no difference in outcome between lesions above or below the conus; thus spinal cord, conus medullaris, or cauda equina mets are considered together here as epidural spinal cord compression (ESCC). Features that help distinguish conus lesions from cauda equina are shown in ► Table 55.9.

Grading function

There is prognostic significance in the presenting neurologic condition. Grading scales such as that of Brice and McKissock (► Table 55.10) have been proposed, but are not widely used. The ASIA grading scale is more commonly applied.

Diagnostic tests

MRI in evaluating SEM

MRI without and with contrast is the diagnostic test of choice in most situations.

Table 55.9 Features distinguishing conus lesions from cauda equina lesions with metastases⁸²

	Conus medullaris lesions	Cauda equina lesions
spontaneous pain	rare; when present, is usually bilateral & symmetric in perineum or thighs	may be most prominent symptom; severe; radicular type; in perineum, thighs & legs, back or bladder
sensory deficit	saddle; bilateral; usually symmetric; sensory dissociation	saddle; no sensory dissociation; may be unilateral & asymmetric
motor loss	<i>symmetric</i> ; not marked; fasciculations may be present	<i>asymmetric</i> ; more marked; atrophy may occur; fasciculations rare
autonomic symptoms (including bladder dysfunction, impotency...)	prominent early	late
reflexes	only ankle jerk absent (preserved knee jerk)	ankle jerk & knee jerk may be absent
onset	sudden and bilateral	gradual and unilateral

Table 55.10 Grading spinal cord function with spinal metastases (Brice & McKissock)⁸³

Group	Grade	Description
1	mild	patient able to walk
2	moderate	able to move legs, but not antigravity
3	severe	slight residual motor and sensory function
4	complete	no motor, sensory, or sphincter function below level of lesion

MRI findings in spinal epidural metastases:

1. vertebral mets are slightly hypointense compared to normal bone marrow on T1WI, and are slightly hyperintense on T2WI
2. axial cuts typically show lesion involving the posterior vertebral body with invasion into one or both pedicles
3. when myelopathy or radiculopathy are present, there is usually tumor extension into the spinal canal (may not occur in lesions presenting only with local pain)
4. DWI images may help differentiate osteoporotic compression fracture from pathologic fracture⁸⁴

Plain X-rays

Most spinal mets are osteolytic, but at least 50% of the bone must be eroded before plain X-rays will be abnormal.⁸⁵ Not very specific. Possible findings: pedicle erosion (defect in “owl’s eyes” AKA “winking owl sign” on LS or thoracic spine AP view) or widening, pathological compression fracture, vertebral body (VB) scalloping, VB *sclerosis*, osteoblastic changes (may occur with prostate Ca, Hodgkin’s disease, occasionally with breast Ca, and rarely with multiple myeloma).

Plain CT in evaluating SEM

Very good for bone detail. Often helpful for surgical planning. By itself, has low sensitivity for spinal cord compression by tumor. Sensitivity is increased with intrathecal contrast (CT-myelogram).

CT-myelogram (CT-myelo)

Indicated when MRI cannot be done (contraindications, unavailability...).

Advantages over MRI:

- Can obtain CSF (when performing LP to inject contrast) for cytological study
- Excellent bony detail
- Can be performed in patients with pacemaker/AICD, claustrophobia...

Disadvantages of myelography over MRI⁷²:

- Invasive
- May require second procedure (C1–2 puncture) if there is a complete block (providers proficient in this technique are becoming fewer)

- Risk of neurologic deterioration from LP in patient with complete block
- Cannot detect lesions that do not cause bony destruction or distortion of the spinal subarachnoid space
- Up to 20% of patients with SEM have at least two sites of cord compression, MRI can evaluate region between two complete blocks, myelography cannot
- Cannot demonstrate paraspinous lesions
- Does not image spinal cord parenchyma

Positron emission tomography (PET) scan

PET scan using [18F]-fluorodeoxyglucose may be used for whole-body work-up for bone mets in patients with known cancer.⁸⁶ Sensitivity is high, but spatial resolution and specificity are low, so often must be used with CT and/or MRI.

Metastatic work-up for patients with suspected spine metastases

- CT of chest, abdomen and pelvis: assess tumor burden, staging, prognostication (which factors into decisions regarding surgery). Has superseded CXR to rule out lung lesion (primary or other mets)
- bone scan: looks for other sites of skeletal involvement
- serum prostate specific antigen (PSA) in males
- mammogram in females
- for multiple myeloma (p.930)
- careful physical exam of lymph nodes

Management algorithm

General information

Management is dependent on the degree and rapidity of neurologic involvement.⁸² Patients may be categorized into one of the three groups that follows, which determines the subsequent steps. In a patient with suspected spine mets, the goals of management are:

- assessment of neurologic involvement and timeline of neurologic changes
- delineation of the degree of spinal involvement
- determination of a histologic diagnosis: this affects management
- preservation or restoration of neurologic function
- preservation or restoration spinal stability
- controlling pain

The tools that are employed in the assessment and stabilization phase are listed under diagnostic tests above. The section that follows discusses the rapidity with which they are implemented.

A metastatic work-up (p.924) is undertaken as time permits (a preliminary work-up, e.g., CXR and physical exam, may be all that can be initially obtained for patients in Group I, whereas more complete work-up can be done in others).

Group I—rapid progression or severe deficit

Group I characteristics

Signs/symptoms of new or progressive (hours to days) cord compression (e.g., urinary urgency, ascending numbness). These patients have a high risk of rapid deterioration and require immediate evaluation.

Management

1. dexamethasone (DMZ) (Decadron®): reduces pain in 85%, may produce transient neurologic improvement. Optimal dose is not known. No difference was found comparing 100 mg IV bolus to 10 mg.⁸⁷ Suggestion: 10 mg IV or PO q 6 hrs × 72 hrs, followed by lower dose of 4–6 mg q 6 hrs. Steroids may temporarily mask lymphoma (on imaging and at surgery); however, in this group the benefit of giving steroids usually outweighs this pitfall
2. radiographic evaluation
 - a) STAT MRI (above)
 - b) plain X-rays of entire spine: 67–85% will be abnormal (see above)

- c) if time permits, plain CT scan through involved levels and at least 2 levels above and below to evaluate bone for surgical planning
 - d) **emergency myelogram:** indicated if MRI cannot be done (include possible C1–2 puncture on the consent). Start with a so-called “blockogram” to R/O complete block: instill small volume of contrast, e.g., iohexol (Omnipaque™) (p. 230) via LP and run the dye all the way up the spinal column; CSF is usually xanthochromic with complete block, see Froin’s syndrome (p. 987)
 - if there is not a complete block: withdraw 10 cc of CSF and send for *cytology*, protein & glucose. One may then inject more contrast to complete the study
 - if complete block: do *not* remove CSF (pressure shifts via LP caused neurologic deterioration in ≈ 14% of patients with complete block,⁸⁸ whereas there was no deterioration after C1–2 puncture). In some cases, contrast can be “squeezed” past a “complete” block by injecting 5–10 ml of room air through a millipore filter.⁸⁹ Alternatively, perform a lateral C1–2 puncture (p. 1819) and instill water soluble contrast to delineate the superior extent of the lesion
 - with myelography, epidural lesions classically produce hourglass deformity with smooth edges if block is incomplete, or paintbrush effect (feathered edges) if block is complete, unlike the sharp margins (capping or meniscus sign) of intradural extramedullary lesion, or fusiform cord widening of intramedullary tumors
 - *bone scan* if time permits. Abnormal in ≈ 66% of patients with spine mets
3. treatment based on results of radiographic evaluation
- a) if no epidural mass: treat primary tumor (e.g., systemic **chemotherapy**). Local radiation therapy (XRT) to bony lesion if present. Analgesics for pain
 - b) if epidural lesion, either surgery *or* start XRT (usually 30–40 Gy in 10 treatments over 7–10 d with ports extending 2 levels above and below lesion). XRT is usually as effective as laminectomy with fewer complications; for further discussion see Treatment for SEM (p. 926). Thus, *surgery* instead of XRT is considered only for the indications shown in ► Table 55.11
 - c) **urgency** of treatment (surgery or XRT) is based on degree of block and rapidity of deterioration:
 - if > 80% block or rapid progression of deficit: emergency treatment ASAP (if treating with XRT instead of surgery, continue DMZ next day at 24 mg IV q 6 hrs x 2 days, then taper during XRT over 2 wks)
 - if < 80% block: treatment on “routine” basis (for XRT, continue DMZ 4 mg IV q 6 hrs, taper during treatment as tolerated)

Table 55.11 Indications for surgery for spinal metastases

Indications
- unknown primary and no tissue diagnosis (CT guided needle biopsy is an option for accessible lesions). NB: lesions such as spinal epidural abscess can be mistaken for metastases⁹³ - spinal instability (see the Spine Instability Neoplastic Score (SINS) ► Table 55.12) - deficit due to spinal deformity or compression by bone rather than by tumor (e.g., due to compression fracture with collapse and retropulsed bone) - radio-resistant tumors (e.g., renal-cell carcinoma, melanoma...) or progression during XRT (usual trial: at least 48 hrs, unless significant or rapid deterioration) - recurrence after maximal XRT - rapid neurologic deterioration
Relative contraindications
- very radiosensitive tumors (multiple myeloma, lymphoma...) not previously radiated - total paralysis (Brice and McKissock group 4) > 8 hours duration, or inability to walk (B&M group > 1) for > 24 hrs duration (after this, there is essentially no chance of recovery and surgery is not indicated) - expected survival: ≤ 3–4 months - multiple lesions at multiple levels - patient unable to tolerate surgery: for patients with lung lesions, check PFTs

Group II—mild and stable signs and symptoms

Group II characteristics

Mild and stable signs/symptoms of cord compression (e.g., isolated Babinski), or either plexopathy or radiculopathy without evidence of cord compression. Admit and evaluate within 24 hrs.

Management

1. for suspected ESCC, manage as in Group I except on less emergent basis. Use low dose dexamethasone (DMZ) unless radiographic evaluation shows > 80% block or if suspicion of lymphoma is high and tissue will be obtained relatively soon
2. for radiculopathy alone (radicular pain, weakness, or reflex changes in one myotome or sensory changes in one dermatome); if plain X-rays show bony lesion then 70–88% will have ESCC on myelography. If the plain film is normal, only 9–25% will have ESCC. Obtain MRI or myelogram and manage as for suspected ESCC
3. for plexopathy (brachial or lumbosacral): pain is the most common early symptom, distribution not limited to single dermatome, commonly referred to elbow or ankle. May mask coexistent radiculopathy, distinguish by EMG (denervation of paraspinal muscles occurs in radiculopathy) or presence of proximal signs and symptoms (Horner syndrome in cervical region, ureteral obstruction in lumbar region). Management:
 - a) MRI is initial diagnostic procedure (CT if MRI unavailable): C4 through T4 for brachial plexopathy, L1 through pelvis for lumbosacral plexopathy
 - b) if CT shows bony lesion or paraspinal mass (with negative CT, plain films and bone scan are rarely helpful; however, if done, and plain X-ray shows malignant appearing bony lesion, or if bone scan shows vertebral abnormality, perform MRI or myelogram within 24 hrs) (give dexamethasone if ESCC suspected or MRI/myelogram delayed). Management as in Group I based on degree of block, XRT ports extended laterally to include any mass shown on CT
 - c) if no bony nor paraspinal lesion on MRI/CT, primary treatment of plexus tumor; analgesics for pain

Group III—pain without neurologic involvement**Group III characteristics**

Back pain without neuro signs/symptoms. Can be evaluated as outpatient over several days (modify based on ability of patient to travel, reliability, etc.).

Treatment for SEM**Treatment goals and outcome**

No treatment for SEM significantly prolongs life, with the possible exception of surgery for renal cell carcinoma with a solitary spine metastasis.⁹⁰ Treatment goals are palliative: pain control, preservation of spinal stability, and maintenance of sphincter control and ability to ambulate.

The most important factor affecting prognosis, regardless of treatment modality, is ability to walk at the time of initiation of therapy. Loss of sphincter control is a poor prognosticator and is usually irreversible.

The main decision is between surgery + post-op XRT, or XRT alone. As yet, no chemotherapy has been found to be useful for SEM (may help with primary). Surgery alone appears least effective for pain control (36%, compared to 67% for surgery + XRT, and 76% for XRT alone).⁹¹ Surgery has the attendant complications of anesthetic risk, post-op pain, wound problems in 11% (further complicated by radiation),⁹¹ and mortality in 5–6% after laminectomy and 10% after anterior approach with stabilization.⁹² Therefore, surgery appears best reserved for situations described in ► Table 55.11.

Medical therapy

Chemotherapy is ineffective for SEM.

Bisphosphonates reduce the risk of vertebral compression fractures (VCF) by $\approx$ 50%, but the effect seems to abate after $\approx$ 2–3 years.

Promising agents undergoing trials include: denosumab (Prolia®), a RANK ligand (RANKL) inhibitor (p. 1212) that may counteract RANKL, which is overexpressed in response to lytic bony metastases.⁹⁴ The efficacy seems better than the bisphosphonates.

Vertebroplasty/kyphoplasty

Vertebroplasty/kyphoplasty (p. 1212) reduces pain associated with pathologic fractures in up to 84%⁹⁵ with an associated increase in functional outcome.⁹⁶ Kyphoplasty appears to offer comparable pain relief to vertebroplasty with lower rates of cement leakage.⁹⁶

Relative contraindication: spinal cord compression. Unless the diagnosis has already been verified, a biopsy should be taken through one of the pedicles prior to injecting PMMA.

Radiation therapy

Radiosensitive tumors: ▶ Table 55.6 lists radiosensitivity of metastatic tumors (to brain or spine). Other radiosensitive tumors that metastasize to the spine include: myxoid liposarcoma.⁹⁷

Treatment⁹⁸: Dose: range = 25–40 Gy. Typical plan: 30 Gy delivered in 3 Gy fractions over 10 days (2 working weeks) to ports extending at least 1 vertebral level above and below the extent of the lesion. Timing: for initial treatment, try to start XRT within 24 hours of diagnosis; for post-op XRT, within about 14 days following surgery.

There is a theoretical risk of radiation induced edema causing or accelerating neurologic deterioration. This has not been borne out by experimental studies with the usual small daily fractions utilized. Deterioration is more likely to be due to tumor progression.⁹⁹ The spinal cord is usually the dose limiting structure in treating SEM.

Increased doses are being made possible with the application of the added precision of stereotactic radiosurgery techniques to spinal metastases.¹⁰⁰

Surgical treatment

See ▶ Table 55.11 for indications for surgery.

The Spine Instability Neoplastic Score (SINS)¹⁰¹ ▶ Table 55.12 was originally devised to help medical oncologists and radiation oncologists determine when surgical consultation should be sought, and to help spine surgeons assess the stability of various types of metastatic spine involvement. The decision to recommend surgery is not solely determined by stability, and must take into account prognosis, tumor type, systemic tumor burden, medical condition of the patient...

The SINS ranges from 0 to 18, and is interpreted as shown in ▶ Table 55.13

Pre-op embolization by interventional radiologist may facilitate resection with less blood loss for highly vascular tumors such as: renal-cell, thyroid, and hepatocellular. Blood supply is through the

Table 55.12 Spine Instability Neoplastic Score (SINS)¹⁰¹ (see ▶ Table 55.13 for interpretation of total score)

Category	Finding	Points	Score
Mobility of involved spinal segment	junctional (occiput-C2, C7-T2, T11-L1, L5-S1)	3	(0 - 3)
	mobile spine (C3-6, L2-4)	2	
	semi-rigid spine (T3-10)	1	
	rigid spine (S2-5)	0	
Pain	pain relieved with recumbency and/or pain with movement/loading of the spine	3	(0 1 3)
	occasional, non-mechanical pain	1	
	no pain	0	
Bone lesion type	lytic	2	(0 - 2)
	mixed lytic/blastic	1	
	blastic	0	
Radiographic spinal alignment	subluxation or translation	4	(0 2 4)
	de novo deformity (kyphosis and/or scoliosis) ^a	2	
	normal	0	
Vertebral body involvement (anterior and middle column)	> 50% collapse	3	(0 - 3)
	< 50% collapse	2	
	no collapse with > 50% VB involved	1	
	none of the above	0	
Posterior column involvement ^b	bilateral	3	(0 1 3)
	unilateral	1	
	none	0	
SINS; range: 0 (stable) to 18 (unstable) → TOTAL			(0 - 18)

^a requires prior imaging or comparison of upright with recumbent imaging

^b posterior column = facets, pedicles or CV joint (see Three column model of the spine, ▶ Fig. 72.1). Involvement = fracture or replacement by tumor

Table 55.13 Stability based on SINS ► Table 55.12

SINS	Stability
0-6	stable
7-12	"indeterminate" or possibly "impending instability"
13-18	unstable

intercostal arteries, and care must be taken to avoid embolizing vessels providing significant blood supply to the spinal cord, especially the artery of Adamkiewicz (p.87).

Approaches

Laminectomy alone is *poor* for spinal metastases when the pathology is *anterior* to the cord because of poor access to the tumor and the destabilizing effect of laminectomy when metastatic involvement of the vertebral body is significant.^{102,103}

Deterioration in one of the 3 major criteria (pain, continence, ambulation) occurred in 26% of patients treated with *laminectomy* alone, 20% of laminectomy + XRT, and 17% of XRT alone (roughly comparable). There is a 9% incidence of spinal instability⁹¹ following laminectomy *without* stabilization.

In a randomized controlled trial by Patchell et al,¹⁰⁴ approaches directed at the location of the tumor (e.g., costotransversectomy, transthoracic approach...) with stabilization where necessary, produced better results than simple laminectomy, and surgery + XRT was superior to XRT alone (see ► Table 55.14). This study found a modest increase in survival, but more significant maintenance or regaining of lost ambulation. However, operative mortality with anterior decompression and stabilization was ≈ double (10%) that of laminectomy with (5%) or without (6%) stabilization in a literature review.⁹²

Table 55.14 Comparing surgery + XRT to XRT alone¹⁰⁴

Result	XRT	Surgery + XRT
Ambulatory after treatment	57%	84%
Days ambulatory after treatment	13	122
Ambulatory after treatment when nonambulatory before treatment	19%	62%
Mean survival (days)	100	126

Solitary spinal metastases with indolent tumors (e.g., renal cell Ca) may be candidates for attempted cure with en bloc resection (total spondylectomy).^{105,106}

Laminectomy is still appropriate with isolated involvement of the posterior elements. For anterior pathology, if the posterior elements are intact, a transthoracic approach with corpectomy and stabilization (e.g., with methylmethacrylate and Steinmann pins,¹⁰⁷ or with cage graft and lateral plate) followed by XRT improves neurologic function in ≈ 75% and pain in ≈ 85%. A posterolateral approach (e.g., costotransversectomy) may be used for anterolateral tumor.¹⁰⁸ Combining a corpectomy and removal of the pedicle and posterior elements destabilizes the spine; therefore, posterior instrumentation prior to performing the corpectomy is required, followed by cage graft.^{109,110,111,112,113,114,115} To access a VB via a costotransversectomy, the rib of the like numbered VB and the one below need to be removed.

55.3 Hematopoietic tumors

55.3.1 Multiple myeloma

General information

Multiple myeloma (MM) (sometimes referred to simply as myeloma) is a neoplasm of a single clone of plasma cells characterized by proliferation of plasma cells in bone marrow, infiltration of adjacent tissues with mature and immature plasma cells, and the production of an immunoglobulin, usually monoclonal IgG or IgA (referred to collectively as M-protein¹¹⁶). Circulating pre-myeloma cells lodge in appropriate microenvironments (e.g., in bone marrow) where they differentiate and expand. Although MM is often referred to in the context of “metastatic lesions” to bone, it is also sometimes considered a primary bone tumor. If only a single lesion is identified, then it is referred to as a plasmacytoma (see below).

Epidemiology

In the U.S., incidence is $\approx 1\text{--}2$ per 100,000 in Caucasians, and is $\approx$ twice that in black people. MM accounts for 1% of malignancies, and 10% of hematologic cancers. The peak age of occurrence is 60–70 yrs of age, with $<2\%$ of patients being <40 yrs old. Slightly more common in males. Monoclonal gammopathy without MM occurs in $\approx 0.15\%$ of the population, and in long-term follow-up, 16% of these develop MM with an annual rate of 0.18%.¹¹⁷

Presentation

General information

MM presents as a result of the following (italicized items are characteristic for MM):

- proliferation of plasma cells: interferes with normal immune system function $\rightarrow$ increased *susceptibility to infection*
- bone involvement
 - bone marrow involvement $\rightarrow$ destruction of hematopoietic capacity $\rightarrow$ normocytic normochromic *anemia*, leukopenia, thrombocytopenia
 - bone resorption
 - $\rightarrow$ weakening of the bone $\rightarrow$ *pathologic fractures* (see below)
 - $\rightarrow$ *hypercalcemia* (present initially in 25% of MM patients, see below)
 - swelling or local tenderness of bone
 - bone pain*: characteristically induced by movement, and absent at rest
 - spinal involvement
 - invasion of spinal canal in $\approx 10\%$ of cases $\rightarrow$ spinal cord compression $\rightarrow$ myelopathy
 - nerve root compression (radiculopathy)
- overproduction of certain proteins by plasma cells. May lead to:
 - hyperviscosity syndrome
 - cryoglobulinemia
 - amyloidosis
 - renal failure*: multifactorial, but monoclonal light chains play a role

Skeletal disease

MM involvement is by definition multiple, and is usually restricted to sites of red marrow: ribs, sternum, spine, clavicles, skull, or proximal extremities. Lesions of the spine and/or skull are the usual reasons for presentation to the neurosurgeon.

Bone resorption in MM is not due simply to mechanical erosion by plasma cells. Increased osteoclastic activity has been observed.

Plasma cell tumors of the skull involving the cranial vault usually do not produce neurologic symptoms. Cranial nerve palsies can arise from skull base involvement. Orbital involvement may produce proptosis (exophthalmos).

Neurologic involvement

Neurologic manifestations can occur as a result of:

- tumor involvement of bone causing compression (see above)
 - tumor in spine with compression of spinal cord or nerve roots
 - tumor in skull with compression of brain or cranial nerves
- deposition of amyloid within the flexor retinaculum of the wrist $\rightarrow$ carpal tunnel syndrome; the median nerve itself does not contain amyloid, and therefore responds well to surgical division of the transverse carpal ligament (p.550)
- diffuse progressive sensorimotor polyneuropathy: occurs in 3–5% of patients with MM
 - about half are due to amyloidosis (p.576)
 - polyneuropathy can also occur without amyloidosis, especially in the rare osteosclerotic variant of MM
- multifocal leukoencephalopathy has been described in MM¹¹⁸
- hypercalcemia: may produce a dramatic encephalopathy with confusion, delirium, or coma. Neurologic symptoms of hypercalcemia associated with MM are more common than in hypercalcemia of other etiologies
- very rare: intraparenchymal metastases¹¹⁹

Evaluation

The diagnostic criteria for MM is shown in ► Table 55.15. Tests that may be used in evaluating patients with MM or suspected MM include:

- 24 hour urine for kappa Bence-Jones protein. A monoclonal immunoglobulin light chain (MW: 22–24 kDa) present in 75–80% of patients with MM (may also occur in other conditions). Typically kappa, occasionally lambda. Monoclonal proteins cannot be detected in the urine or serum of $\approx 1\%$ of MM patients; two or more monoclonal bands are produced in ≈ 0.5 –2.5% of patients with MM.¹²⁰
- bloodwork:
 - SPEP with reflex IFE and FLC
Explanation: serum protein electrophoresis (SPEP) looks for an M-spike (identifies monoclonal immunoglobulin gammopathies), if present (reflex) immunoelectrophoresis (IEP) AKA immunofixation (IFE) looks for IgG kappa band; free light chain (FLC) assay looks for K & λ light chains. These 3 assays together are highly sensitive for MM and related plasma cell disorders
 - CBC: anemia eventually develops in most patients with MM as myeloma cells crowd out normal bone marrow cells. It is usually of moderate severity (Hgb ≈ 7 –10 gm%) with a low reticulocyte count
 - serum creatinine: for prognostication
- skeletal radiologic survey. Characteristic X-ray finding: multiple, round, “punched-out” (sharply demarcated) lytic lesions in the bones typically involved (see above). Osteosclerotic lesions are seen in < 3% of patients with MM. Diffuse osteoporosis may also be seen
- technetium-99m nuclear bone scan is usually *negative* in untreated MM (due to rarity of spontaneous new bone formation) and is less sensitive than conventional radiographs. Therefore it is not usually helpful except perhaps to implicate etiologies other than MM to explain the observed findings. After treatment, bone scan may become positive as osteoblastic activity ensues (“flare” response)
- bone marrow biopsy: virtually all MM patients have “myeloma cells” (mutated plasma cells). Although sensitive, this is not specific and other diagnostic criteria should be sought

Table 55.15 Criteria for diagnosis of MM^a

- | |
|--|
| 1. Cytologic criteria |
| <ol style="list-style-type: none"> marrow morphology: plasma cells and/or myeloma cells $\geq 10\%$ of 1000 or more cells biopsy proven plasmacytoma |
| 2. Clinical and laboratory criteria |
| <ol style="list-style-type: none"> myeloma protein (M-component) in serum (usually > 3 gm/dl) or urine IEP osteolytic lesions on X-ray (generalized osteoporosis qualifies if marrow contains > 30% plasma or myeloma cells) myeloma cells in ≥ 2 peripheral blood smears |

^adiagnosis requires¹²¹: 1A & 1B, or 1A or 1B and 2A, 2B, or 2C

Treatment

Many aspects of treatment fall into the purview of the oncologist (see review¹¹⁷). Some aspects pertinent to neurosurgical care include:

- XRT (p. 917): MM is very radiosensitive. Focal XRT for pain due to readily identifiable bone lesions may allow pathologic fractures to heal and is effective in spinal cord compression due to tumor
- mobilization; immobilization due to pain and fear of pathologic compression fractures leads to further detrimental increases in serum calcium and weakness
- pain control: mild pain often responds well to salicylates (contraindicated in thrombocytopenia). Local XRT is also effective (see below)
- percutaneous kyphoplasty (p. 1213) may be used for some spine lesions (preferred over vertebroplasty because of reduced potential of spreading neoplasm)
- therapy for hypercalcemia: bisphosphonates usually improve symptoms related to this
- bisphosphonates (p. 1364) inhibit bone resorption and rapidly reduces hypercalcemia. Pamidronate is currently preferred over older agents
- bortezomib (Velcade®): the first proteasome inhibitor, indicated for treatment of refractory MM

Prognosis

Untreated MM has a 6-month median survival. Solitary plasmacytoma has a 50% 10-year survival. If there is an solitary site of involvement but M-protein is present (i.e., essentially a plasmacytoma except for the M-protein), elimination of the M-protein following XRT indicates a 50–60% chance of remaining free of MM; if the M-protein doesn't resolve, there is a high chance of developing MM.

55.3.2 Plasmacytoma

General information

A neoplasm of a single clone of plasma cells similar to multiple myeloma (see above) but meeting the following criteria:

1. there must be no other lesions on complete skeletal survey (*not* bone scan)
2. bone marrow aspirate must show no evidence of myeloma
3. and serum and urine electrophoresis should show no M-protein

MM will develop in 55–60% of patients with a solitary plasmacytoma in 5 years, and in 70–80% by 10 yrs.

Treatment

1. local XRT provides good local control rates
2. percutaneous kyphoplasty (p. 1213): preferred over vertebroplasty because of reduced potential of spreading neoplastic cells
3. surgery for instability

References

- [1] Johnson JD, Young B. Demographics of brain metastasis. *Neurosurg Clin N Am*. 1996; 7:337–344
- [2] Mintz AP, Cairncross JG. Treatment of a Single Brain Metastasis. The Role of Radiation Following Surgical Excision. *Journal of the American Medical Association*. 1998; 280:1527–1529
- [3] Voorhies RM, Sundaresan N, Thaler HT. The Single Supratentorial Lesion: An Evaluation of Preoperative Diagnosis. *J Neurosurg*. 1980; 53:364–368
- [4] Patchell RA, Posner JB. Neurologic Complications of Systemic Cancer. *Neurol Clin*. 1985; 3:729–750
- [5] Zimm S, Galen L, Wampler GL, et al. Intracerebral Metastases in Solid-Tumor Patients: Natural History and Results of Treatment. *Cancer*. 1981; 48:384–394
- [6] DeAngelis LM. Management of Brain Metastases. *Cancer Invest*. 1994; 12:156–165
- [7] Davis PC, Hudgins PA, Peterman SB, et al. Diagnosis of Cerebral Metastases: Double-Dose Delayed CT versus Contrast-Enhanced MR Imaging. *American Journal of Neuroradiology*. 1991; 12:293–300
- [8] Weiss HD, Richardson EP. Solitary Brainstem Metastasis. *Neurology*. 1978; 28:562–566
- [9] Nugent JL, Bunn PA, Matthews MJ, et al. CNS Metastases in Small-Cell Bronchogenic Carcinoma: Increasing Frequency and Changing Pattern with Lengthening Survival. *Cancer*. 1979; 44:1885–1893
- [10] Berger MS, Baumeister B, Geyer JR, et al. The Risks of Metastases from Shunting in Children with Primary Central Nervous System Tumors. *J Neurosurg*. 1991; 74:872–877
- [11] Kindt GW. The Pattern of Location of Cerebral Metastatic Tumors. *J Neurosurg*. 1964; 21:54–57
- [12] Gavrilovic IT, Posner JB. Brain metastases: epidemiology and pathophysiology. *J Neurooncol*. 2005; 75:5–14
- [13] Vieth RG, Odum GL. Intracranial Metastases and their Neurosurgical Treatment. *J Neurosurg*. 1965; 23:375–383
- [14] Agazzi S, Pampallona S, Pica A, et al. The origin of brain metastases in patients with an undiagnosed primary tumour. *Acta Neurochir (Wien)*. 2004; 146:153–157
- [15] Figlin RA, Piantadosi S, Feld R, et al. Intracranial Recurrence of Carcinoma After Complete Resection of Stage I, II, and III Non-Small-Cell Lung Cancer. *N Engl J Med*. 1988; 318:1300–1305
- [16] Barbone F, Bovenzi M, Cavallieri F, et al. Cigarette smoking and histologic type of lung cancer in men. *Chest*. 1997; 112:1474–1479
- [17] Auperin A, Arriagada R, Pignon JP, et al. Prophylactic cranial irradiation for patients with small-cell lung cancer in complete remission. Prophylactic Cranial Irradiation Overview Collaborative Group. *N Engl J Med*. 1999; 341:476–484
- [18] Slotman B, Faivre-Finn C, Kramer G, et al. Prophylactic cranial irradiation in extensive small-cell lung cancer. *N Engl J Med*. 2007; 357:664–672
- [19] Solis OJ, Davis KR, Adair LB, et al. Intracerebral Metastatic Melanoma: CT Evaluation. *Comput Tomogr*. 1977; 1:135–143
- [20] Zakrzewski J, Geraghty LN, Rose AE, et al. Clinical variables and primary tumor characteristics predictive of the development of melanoma brain metastases and post-brain metastases survival. *Cancer*. 2011; 117:1711–1720
- [21] Davies MA, Liu P, McIntyre S, et al. Prognostic factors for survival in melanoma patients with brain metastases. *Cancer*. 2011; 117:1687–1696
- [22] Staudt M, Lasithiotakis K, Leiter U, et al. Determinants of survival in patients with brain metastases from cutaneous melanoma. *Br J Cancer*. 2010; 102:1213–1218
- [23] Sampson JH, Carter JH, Friedman AH, et al. Demographics, Prognosis, and Therapy in 702 Patients with Brain Metastases from Malignant Melanoma. *Journal of Neurosurgery*. 1998; 88:11–20
- [24] Swetter SM, Carroll LA, Johnson DL, et al. Positron emission tomography is superior to computed

- tomography for metastatic detection in melanoma patients. *Ann Surg Oncol*. 2002; 9:646–653
- [25] Carlino MS, Fogarty GB, Long GV. Treatment of Melanoma Brain Metastases: A New Paradigm. *The Cancer Journal*. 2012; 18:208–212
 - [26] Bindal RK, Sawaya R, Leavens ME, et al. Surgical treatment of multiple brain metastases. *Journal of Neurosurgery*. 1993; 79:210–216
 - [27] Avril MF, Aamdal S, Grob JJ, et al. Fostemustine compared with dacarbazine in patients with disseminated malignant melanoma: a phase III study. *J Clin Oncol*. 2004; 22:1118–1125
 - [28] Guirguis LM, Yang JC, White DE, et al. Safety and efficacy of high-dose interleukin-2 therapy in patients with brain metastases. *J Immunother*. 2002; 25:82–87
 - [29] Lochead R, McKhann G, Hankinson T, et al. High dose systemic interleukin-2 for metastatic melanoma in patients with treated brain metastases. *Journal of Immunotherapy*. 2004; 27
 - [30] Majer M, Jensen RL, Shrieve DC, et al. Biochemotherapy of metastatic melanoma in patients with or without recently diagnosed brain metastases. *Cancer*. 2007; 110:1329–1337
 - [31] ClinicalTrials.gov identifier: NCT 01266967. A Study of GSK 2118436 in BRAF Mutant Metastatic Melanoma to the Brain (Break MB). 2014. <https://clinicaltrials.gov/ct2/show/results/NCT01266967>
 - [32] ClinicalTrials.gov identifier: NCT 01378975. A Study of Vemurafenib in Metastatic Melanoma Patients With Brain Metastases. 2015. <https://clinicaltrials.gov/ct2/show/NCT01378975?term=NCT01378975&rank=1&view=results>
 - [33] U.S. Food and Drug Administration (FDA). FDA approves Keytruda for advanced melanoma. 2014. <https://wayback.archive-it.org/7993/20170112023823/http://www.fda.gov/NewsEvents/Newsroom/PressAnnouncements/ucm412802.htm>
 - [34] Patchell RA, Tibbs PA, Walsh JW, et al. A Randomized Trial of Surgery in the Treatment of Single Metastases to the Brain. *New England Journal of Medicine*. 1990; 322:494–500
 - [35] Vecht CJ, Haaxma-Reiche H, Noordijk EM, et al. Treatment of single brain metastasis: radiotherapy alone or combined with neurosurgery? *Ann Neurol*. 1993; 33:583–590
 - [36] Sampson JH, Carter JH Jr, Friedman AH, et al. Demographics, prognosis, and therapy in 702 patients with brain metastases from malignant melanoma. *J Neurosurg*. 1998; 88:11–20
 - [37] Fife KM, Colman MH, Stevens GN, et al. Determinants of outcome in melanoma patients with cerebral metastases. *J Clin Oncol*. 2004; 22:1293–1300
 - [38] Eigentler TK, Figl A, Krex D, et al. Number of metastases, serum lactate dehydrogenase level, and type of treatment are prognostic factors in patients with brain metastases of malignant melanoma. *Cancer*. 2011; 117:1697–1703
 - [39] Gupta G, Robertson AG, MacKie RM. Cerebral metastases of cutaneous melanoma. *Br J Cancer*. 1997; 76:256–259
 - [40] Song Z, Lin B, Shao L, et al. Brain metastases from esophageal cancer: clinical review of 26 cases. *World Neurosurg*. 2014; 81:131–135
 - [41] Kondziolka D, Bernstein M, Resch L, et al. Significance of Hemorrhage into Brain Tumors: Clinicopathological Study. *J Neurosurg*. 1987; 67:852–857
 - [42] Shildt RA, Kennedy PS, Chen TT, et al. Management of patients with metastatic adenocarcinoma of unknown origin: a Southwest Oncology Group study. *Cancer Treatment Reports*. 1983; 67:77–79
 - [43] Gaspar L, Scott C, Rotman M, et al. Recursive partitioning analysis (RPA) of prognostic factors in three Radiation Therapy Oncology Group (RTOG) brain metastases trials. *Int J Radiat Oncol Biol Phys*. 1997; 37:745–751
 - [44] Morris SL, Low SH, A'Hern RP, et al. A prognostic index that predicts outcome following palliative whole brain radiotherapy for patients with metastatic malignant melanoma. *Br J Cancer*. 2004; 91:829–833
 - [45] Nieder C, Andratschke N, Grosu AL, et al. Recursive partitioning analysis (RPA) class does not predict survival in patients with four or more brain metastases. *Strahlenther Onkol*. 2003; 179:16–20
 - [46] Pollock BE. Management of Patients with Multiple Brain Metastases. *Contemporary Neurosurgery*. 1999; 21:1–6
 - [47] Nahed BV, Alvarez-Breckenridge C, Brastianos PK, et al. Congress of Neurological Surgeons Systematic Review and Evidence-Based Guidelines on the Role of Surgery in the Management of Adults With Metastatic Brain Tumors. *Neurosurgery*. 2019; 84:E152–E155
 - [48] Ammirati M, Nahed BV, Andrews D, et al. Congress of Neurological Surgeons Systematic Review and Evidence-Based Guidelines on Treatment Options for Adults With Multiple Metastatic Brain Tumors. *Neurosurgery*. 2019; 84:E180–E182
 - [49] Forsyth PA, Weaver S, Fulton D, et al. Prophylactic anticonvulsants in patients with brain tumour. *Can J Neurol Sci*. 2003; 30:106–112
 - [50] Chen CC, Rennett RC, Olson JJ. Congress of Neurological Surgeons Systematic Review and Evidence-Based Guidelines on the Role of Prophylactic Anticonvulsants in the Treatment of Adults With Metastatic Brain Tumors. *Neurosurgery*. 2019; 84:E195–E197
 - [51] Ryken TC, Kuo JS, Prabhu RS, et al. Congress of Neurological Surgeons Systematic Review and Evidence-Based Guidelines on the Role of Steroids in the Treatment of Adults With Metastatic Brain Tumors. *Neurosurgery*. 2019; 84:E189–E191
 - [52] Vecht CJ, Hovestadt A, Verbiest HB, et al. Dose-effect relationship of dexamethasone on Karnofsky performance in metastatic brain tumors: a randomized study of doses of 4, 8, and 16 mg per day. *Neurology*. 1994; 44:675–680
 - [53] Horton J. Treatment of Metastases to the Brain. 1984
 - [54] Jackson DV, Richards F, Cooper MR, et al. Prophylactic Cranial Irradiation in Small Cell Carcinoma of the Lung: A Randomized Study. *JAMA*. 1977; 237:2730–2733
 - [55] Patchell RA, Tibbs PA, Regine WF, et al. Postoperative radiotherapy in the treatment of single metastases to the brain: a randomized trial. *Journal of the American Medical Association*. 1998; 280:1485–1489
 - [56] Kramer S, Hendrickson F, Zelen M, et al. Therapeutic Trials in the Management of Metastatic Brain Tumors by Different Time/Dose Fraction Schemes. *Natl Cancer Inst Monogr*. 1977; 46:213–221
 - [57] DeAngelis LM, Mandell LR, Thaler HT, et al. The Role of Postoperative Radiotherapy After Resection of Single Brain Metastases. *Neurosurgery*. 1989; 24:798–804
 - [58] Smalley SR, Schray MF, Laws ER, et al. Adjuvant Radiation Therapy After Surgical Resection of Solitary Brain Metastasis: Association with Pattern of Failure and Survival. *International Journal of Radiation Oncology and Biological Physics*. 1987; 13:1611–1616
 - [59] Shaw E. Comment on DeAngelis L M, et al.: The Role of Postoperative Radiotherapy After Resection of Single Brain Metastases. *Neurosurgery*. 1989; 24:804–805
 - [60] Sills AK. Current treatment approaches to surgery for brain metastases. *Neurosurgery*. 2005; 57:S24–32; discussion S1–4
 - [61] Bindal AK, Bindal RK, Hess KR, et al. Surgery versus radiosurgery in the Treatment of Brain Metastasis. *Journal of Neurosurgery*. 1996; 84:748–754
 - [62] Smalley SR, Laws ER, O'Fallon JR, et al. Resection for Solitary Brain Metastasis: Role of Adjuvant Radiation and Prognostic Variables in 229 Patients. *Journal of Neurosurgery*. 1992; 77:531–540
 - [63] Tobler WD, Sawaya R, Tew JM. Successful Laser-assisted Excision of a Metastatic Midbrain Tumor. *Neurosurgery*. 1986; 18:795–797
 - [64] Markesbery WR, Brooks WH, Gupta GD, et al. Treatment for Patients with Cerebral Metastases. *Archives of Neurology*. 1978; 35:754–756

- [65] Ruderman NB, Hall TC. Use of Glucocorticoids in the Palliative Treatment of Metastatic Brain Tumors. *Cancer*. 1965; 18:298-306
- [66] Posner JB. Surgery for Metastases to the Brain. *N Engl J Med*. 1990; 322:544-545
- [67] Galicich JH, Sundaresan N, Thaler HT. Surgical Treatment of Single Brain Metastasis: Evaluation of Results by CT Scanning. *Journal of Neurosurgery*. 1980; 53:63-67
- [68] Alexander E, Moriarty TM, Davis RB, et al. Stereotactic Radiosurgery for the Definitive Noninvasive Treatment of Brain Metastases. *Journal of the National Cancer Institute*. 1995; 87:34-40
- [69] Fuller BG, Kaplan ID, Adler J, et al. Stereotactic Radiosurgery for Brain Metastases: The Importance of Adjuvant Whole Brain Irradiation. *International Journal of Radiation Oncology and Biological Physics*. 1992; 23:413-418
- [70] Wilkins RH, Rengachary SS. *Neurosurgery*. New York 1985
- [71] Sze G, Soletsky S, Bronen R, et al. MR Imaging of the Cranial Meninges with Emphasis on Contrast Enhancement and Meningeal Carcinomatosis. *AJNR*. 1989; 10:965-975
- [72] Godersky JC, Smoker WRK, Knutzen R. Use of MRI in the Evaluation of Metastatic Spinal Disease. *Neurosurgery*. 1987; 21:676-680
- [73] Livingston KE, Perrin RG. The neurosurgical management of spinal metastases causing cord and cauda equina compression. *Journal of Neurosurgery*. 1978; 49:839-843
- [74] Batson OV. The Function of the Vertebral Veins and Their Role in the Spread of Metastases. *Ann Surg*. 1940; 112
- [75] Schwab JH, Boland P, Guo T, et al. Skeletal metastases in myxoid liposarcoma: an unusual pattern of distant spread. *Ann Surg Oncol*. 2007; 14:1507-1514
- [76] Rodichok LD, Ruckdeschel JC, Harper GR, et al. Early Detection and Treatment of Spinal Epidural Metastases: The Role of Myelography. *Ann Neurol*. 1986; 20:696-702
- [77] Bach F, Larsen BH, Rhode K, et al. Metastatic spinal cord compression. Occurrence, symptoms, clinical presentations and prognosis in 398 patients with spinal cord compression. *Acta Neurochirurgica (Wien)*. 1990; 107:37-43
- [78] Helwig-Larsen S, Sorensen PS. Symptoms and signs in metastatic spinal cord compression: a study from first symptom until diagnosis in 153 patients. *European Journal of Cancer*. 1994; 30A:396-398
- [79] Levack P, Graham J, Collie D, et al. Don't wait for a sensory level; listen to the symptoms: a prospective audit of the delays in diagnosis of malignant cord compression. *Clinical Oncology (Royal College of Radiologists)*. 2002; 14:472-480
- [80] Sherk HH. Lesions of the Atlas and Axis. *Clinical Orthopedics*. 1975; 109:33-41
- [81] Nakamura M, Toyama Y, Suzuki N, et al. Metastases to the upper cervical spine. *Journal of Spinal Disorders*. 1996; 9:195-201
- [82] Portenoy RK, Lipton RB, Foley KM. Back Pain in the Cancer Patient: An Algorithm for Evaluation and Management. *Neurology*. 1987; 37:134-138
- [83] Brice J, McKissock W. Surgical Treatment of Malignant Extradural Spinal Tumors. *Br Med J*. 1965; 1:1341-1344
- [84] Li KC, Poon PY. Sensitivity and specificity of MRI in detecting spinal cord compression and in distinguishing malignant from benign compression fractures of vertebrae. *Magnetic Resonance Imaging*. 1988; 6:547-556
- [85] Gabriel K, Schi D. Metastatic spinal cord compression by solid tumors. *Seminars in Neurology*. 2004; 24:375-383
- [86] Francken AB, Hong AM, Fulham MJ, et al. Detection of unsuspected spinal cord compression in melanoma patients by 18F-fluorodeoxyglucose-positron emission tomography. *European Journal of Surgical Oncology*. 2005; 31:197-204
- [87] Vecht CJ, Haaxma-Reiche H, van Putten WL, et al. Initial bolus of conventional versus high-dose dexamethasone in metastatic spinal cord compression. *Neurology*. 1989; 39:1255-1257
- [88] Hollis PH, Malis LI, Zappulla RA. Neurological Deterioration After Lumbar Puncture Below Complete Spinal Subarachnoid Block. *J Neurosurg*. 1986; 64:253-256
- [89] Lee Y-Y, Glass JP, Wallace S. Myelography in Cancer Patients: Modified Technique. *AJR*. 1985; 145:791-795
- [90] Jung ST, Ghert MA, Harrelson JM, et al. Treatment of osseous metastases in patients with renal cell carcinoma. *Clin Orthop Relat Res*. 2003; 223-231
- [91] Findlay GFG. Adverse Effects of the Management of Malignant Spinal Cord Compression. *J Neurol Neurosurg Psychiatry*. 1984; 47:761-768
- [92] Witham TF, Khavkin YA, Gallia GL, et al. Surgery insight: current management of epidural spinal cord compression from metastatic spine disease. *National Clinics of Practical Neurology*. 2006; 2:87-94
- [93] Danner RL, Hartman BJ. Update of Spinal Epidural Abscess: 35 Cases and Review of the Literature. *Rev Infect Dis*. 1987; 9:265-274
- [94] Mundy GR. Metastasis to bone: causes, consequences and therapeutic opportunities. *Nat Rev Cancer*. 2002; 2:584-593
- [95] Fournier DR, Schomer DF, Nader R, et al. Percutaneous vertebroplasty and kyphoplasty for painful vertebral body fractures in cancer patients. *Journal of Neurosurgery*. 2003; 98:21-30
- [96] Bouza C, Lopez-Cuadrado T, Cediñ P, et al. Balloon kyphoplasty in malignant spinal fractures: a systematic review and meta-analysis. *BMC Palliat Care*. 2009; 8. DOI: 10.1186/1472-684X-8-12
- [97] Reitan JB, Kaalhus O. Radiotherapy of liposarcomas. *Br J Radiol*. 1980; 53:969-975
- [98] Faul CM, Flickinger JC. The use of radiation in the management of spinal metastases. *Journal of Neuro-Oncology*. 1995; 23:149-161
- [99] Rubin P. Extradural Spinal Cord Compression by Tumor: Part I. Experimental Production and Treatment Trials. *Radiotherapy*. 1969; 93:1243-1248
- [100] Rock JB, Ryu S, Yin FF, et al. The evolving role of stereotactic radiosurgery and stereotactic radiation therapy for patients with spine tumors. *J Neurooncol*. 2004; 69:319-334
- [101] Fisher CG, DiPaola CP, Ryken TC, et al. A novel classification system for spinal instability in neoplastic disease: an evidence-based approach and expert consensus from the Spine Oncology Study Group. *Spine (Phila Pa 1976)*. 2010; 35:E1221-E1229
- [102] Onimus M, Schraub S, Bertin D, et al. Surgical Treatment of Vertebral Metastasis. *Spine*. 1986; 11:883-891
- [103] Cooper PR, Errico TJ, Martin R, et al. A Systematic Approach to Spinal Reconstruction After Anterior Decompression for Neoplastic Disease of the Thoracic and Lumbar Spine. *Neurosurgery*. 1993; 32:1-8
- [104] Patchell RA, Tibbs PA, Regine WF, et al. Direct decompressive surgical resection in the treatment of spinal cord compression caused by metastatic cancer: a randomized trial. *Lancet*. 2005; 366:643-648
- [105] Fournier DR, Abi-Said D, Rhines LD, et al. Simultaneous anterior-posterior approach to the thoracic and lumbar spine for the radical resection of tumors followed by reconstruction and stabilization. *Journal of Neurosurgery*. 2001; 94:232-244
- [106] Sakaura H, Hosono N, Mukai Y, et al. Outcome of total en bloc spondylectomy for solitary metastasis of the thoracolumbar spine. *Journal of Spinal Disorders*. 2004; 17:297-300
- [107] Sundaresan N, Galicich JH, Lane JM, et al. Treatment of Neoplastic Epidural Cord Compression by Vertebral Body Resection and Stabilization. *Journal of Neurosurgery*. 1985; 63:676-684
- [108] Overby MC, Rothman AS. Anterolateral Decompression for Metastatic Epidural Spinal Cord Tumors: Results of a Modified Costotransversectomy Approach. *Journal of Neurosurgery*. 1985; 62:344-348

- [109] Shaw B, Mansfield FL, Borges L. One-Stage Posterolateral Decompression and Stabilization for Primary and Metastatic Vertebral Tumors in the Thoracic and Lumbar Spine. *Journal of Neurosurgery*. 1989; 70:405–410
- [110] Akeyson EW, McCutcheon IE. Single-stage posterior vertebrectomy and replacement combined with posterior instrumentation for spinal metastasis. *Journal of Neurosurgery*. 1996; 85:211–220
- [111] Fournay DR, Abi-Said D, Lang FF, et al. Use of pedicle screw fixation in the management of malignant spinal disease: experience in 100 consecutive cases. *Journal of Neurosurgery*. 2001; 94:25–37
- [112] Wang JC, Boland P, Mitra N, et al. Single-stage posterolateral transpedicular approach for resection of epidural metastatic spine tumors involving the vertebral body with circumferential reconstruction: results in 140 patients. *Journal of Neurosurgery (Spine)*. 2004; 1:287–298
- [113] Hunt T, Shen FH, Arlet V. Expandable cage placement via a posterolateral approach in lumbar spine reconstructions: technical note. *Journal of Neurosurgery (Spine)*. 2006; 5:271–274
- [114] Snell BE, Nasr FF, Wolfla CE. Single-stage thoracolumbar vertebrectomy with circumferential reconstruction and arthrodesis: surgical technique and results in 15 patients. *Neurosurgery (Operative Neurosurgery)*. 2006; 58:263–269
- [115] Sciubba DM, Gallia GL, McGirt MJ, et al. Thoracic kyphotic deformity reduction with a distractible titanium cage via an entirely posterior approach. *Neurosurgery*. 2007; 60:223–231
- [116] Keren DF, Alexanian R, Goeken JA, et al. Guidelines for Clinical and Laboratory Evaluation of Patients with Monoclonal Gammopathies. *Arch Pathol Lab Med*. 1999; 123:106–107
- [117] Bataille R, Harousseau J-L. Multiple Myeloma. *New England Journal of Medicine*. 1997; 336:1657–1664
- [118] McCarthy J, Proctor SJ. Cerebral Involvement in Multiple Myeloma. Case Report. *J Clin Pathol*. 1978; 31:259–264
- [119] Norum J, Wist E, Dahil IM. Cerebral Metastases from Multiple Myeloma. *Acta Oncol*. 1991; 30:868–869
- [120] Foerster J, Lee GR, Bithell TC, et al. Multiple Myeloma. In: Wintrobe's Clinical Hematology. 9th ed. Philadelphia: Lea and Febiger; 1993:2219–2249
- [121] Costa G, Engle RL, Schilling A, et al. Melfalan and Prednisone: An Effective Combination for the Treatment of Multiple Myeloma. *American Journal of Medicine*. 1973; 54:589–599

56 Other Tumors, Cysts, and Tumor-Like Lesions

56.1 Other tumors

56.1.1 Olfactory neuroblastoma (ONB)

General information

Formerly called esthesioneuroblastoma, originally described in 1924, AKA olfactory esthesioneuroblastoma, AKA esthesioneurocytoma, AKA olfactory placode tumor.¹ A rare nasal neoplasm with an incidence 0.4 per 1,000,000 people.² Believed to arise from the olfactory neural crest cells in the upper nares, it is considered to be malignant. These tumors occur over a wide age range (3 to 90 years), with a bimodal peak between the second and third decade and a second peak in the sixth and seventh decades.

Imaging

MRI: isointense with brain on T1-weighted imaging and intermediate to high signal intensity on T2-weighted imaging and enhance heterogeneously with gadolinium. Signal characteristics may mimic meningioma. For higher stage lesions, the cribriform plate may be eroded; better seen on thin cut CT. The most important factor determining resectability is intracranial extension. Magnetic resonance aids in the distinction between extradural tumor, dural invasion, or parenchymal brain invasion. None of these are specific to this tumor.

Differential diagnosis

Includes SNUC, nasal melanoma, nasal squamous cell carcinoma, and meningioma.

Diagnosis

Endoscopic biopsy is typically performed in the otolaryngology office prior to surgery.

A clinical oncology exam should be performed, and if there is suspicion for metastatic disease a PET scan, which is sensitive for metastatic disease, should be ordered.

Clinical classification systems

The modified Kadish system³ (which added category D to the original Kadish system⁴) is shown in ► Table 56.1. This classification appears to correlate with survival.³ Alternative systems by Biller et al⁵ and Dulguerov and Calcaterra⁶ (see ► Table 56.1) attempt to subdivide the Kadish C classification; however, the more popular modified Kadish system is more frequently used.

Table 56.1 Clinical classification systems for olfactory neuroblastoma

Modified Kadish ³	Biller et al ⁵	Dulguerov and Calcaterra ⁶
A: Confined to Nasal Cavity	T1: Nasal/Paranasal Sinuses	T1: Nasal/Paranasal Sinuses
B: Extends to Paranasal Sinus	T2: Periorbital/Anterior Fossa Extension	T2: Erosion of Cribriform Plate
C: Local Extension (orbit or cribriform plate)	T3: Brain Involvement, Resectable Margins	T3: Periorbital/Anterior Fossa Extension
D: Distant Metastasis	T4: Unable to Obtain Negative Margins—Unresectable	T4: Brain Involvement

Pathologic grading

Hyams grading, a system used to define all upper respiratory tract carcinomas, is utilized to assess nuclear pleomorphism, mitotic activity, rosette presence, and necrosis, and summates these to produce Hyams 1–4 classification.⁷ It has been shown in meta-analysis, as well as in large series, that Hyams grade 1 and 2 predict benign disease course, as compared to Hyams 3 and 4, which predict poor disease course. It is recommended that grading be performed in all cases.^{8,9}

Treatment

Primary treatment is controversial. Some institutions believe in upfront combined radiation therapy and chemotherapy prior to craniofacial resection. However, most practice initial surgery, which classically consisted of endoscopic resection with negative margins for Kadish A and B lesions, and craniofacial resection (bifrontal craniotomy with associated lateral rhinotomy) for Kadish C and D lesions. However, with the advent of endoscopic techniques, the lateral rhinotomy is often replaced with a purely endoscopic approach, unless there is inferior lateral orbital or maxillary involvement, in which case the lateral rhinotomy is frequently used. Finally, some institutions are now managing Kadish stages purely endoscopically, unless they are unable to get negative margins at the time of surgery, in which case conversion to an open approach is performed or SRS is performed; however, this is controversial.

Outcome

Median overall survival is typically 7.2 ± 0.7 years.⁸

Mean progression free survival is 4.8 ± 0.7 years. The 5 and 10 year survivals are 63% and 40%.⁸

Population-based analysis of the Surveillance Epidemiology and End Results (SEER) database confirms that Kadish staging, lymph node involvement, and age at diagnosis have significant prognostic value.¹⁰ These findings have been confirmed in a large meta-analysis recently published by Kane et al in 2010.⁹ Furthermore, higher Hyams grading (grades 3 and 4) correlate with a poorer prognosis.^{8,9}

Salvage treatment: For patients with recurrent disease, this typically occurs in 2 patterns: that of intracranial recurrence or those with distant metastasis.^{11,12} Intracranial recurrence is typically treated with repeat transcranial resection; however, stereotactic radiosurgery is a viable option.^{11,12} In patients with distant metastasis, those with cervical lymph node metastasis should undergo modified radical neck dissection to understand the extent of the disease. This typically leads to chemotherapy, of which platinum based therapies remain the standard of therapy at this time.^{11,14,15}

56.1.2 Epidermoid and dermoid tumors

General information

AKA epidermoid or dermoid cysts.

Both are usually developmental, benign tumors that may arise when retained ectodermal implants are trapped by two fusing ectodermal surfaces. The growth rate of these tumors is linear, like skin (rather than exponential, as with neoplastic tumors).

They may occur in the following locations:

1. calvaria: skull involvement (p.973) occurs when ectodermal rests are included in the developing cranium, epidural extension may occur with growth
2. intracranial: the most common sites include
 - a) suprasellar: commonly produce bitemporal hemianopsia and optic atrophy, and only occasionally pituitary (endocrine) symptoms (including DI)
 - b) Sylvian fissure: may present with seizures
 - c) cerebellopontine angle (CPA): may produce trigeminal neuralgia, especially in young patient
 - d) basilar-posterior fossa: may produce lower cranial nerve findings, cerebellar dysfunction, and/or corticospinal tract abnormalities
 - e) within the ventricular system: occur within the 4th ventricle more commonly than any other
3. scalp
4. within the spinal canal:
 - a) most arise in the thoracic or upper lumbar spine
 - b) epidermoids of the lower lumbar spine may occur iatrogenically following LP; see Lumbar puncture (p.1811)
 - c) dermoids of the spinal canal are usually associated with a dermal sinus tract (p.286) and may produce recurrent bouts of spinal meningitis.

Comparison of dermoids and epidermoids

Distinguishing features between the two tumors are shown in ► Table 56.2.

Table 56.2 Comparison of epidermoids and dermoid

Feature	Epidermoid	Dermoid
frequency	0.5–1.5% of brain tumors	0.3% of brain tumors
lining	stratified squamous epithelium	also include dermal appendage organs (hair follicles and sebaceous glands)
contents	keratin, cellular debris, cholesterol, occasional hair	same as epidermoids, plus hair and sebum
location	more common laterally (e.g., CP angle)	more commonly near midline
associated anomalies	tend to be isolated lesions	associated with other congenital anomalies in up to 50% of cases
meningitis	may have recurrent episodes of chemical meningitis (p.342)	may have repeated bouts of bacterial meningitis

Epidermoid cysts

General information

Key concepts

- usually arise from ectoderm trapped within or displaced into the CNS
- predilection for: CP angle, 4th ventricle, suprasellar region, spinal cord
- sometimes AKA cholesteatoma (not to be confused with cholesterol granuloma)
- grow at linear rate (unlike exponential rate of true neoplasms)
- imaging: CSF-like mass (hi-signal on DWI MRI is the best test to differentiate)
- may produce chemical meningitis
- treatment: surgical excision. XRT has no role

AKA cholesteatoma (not cholesterol granuloma [see below]), AKA pearly tumor, AKA ectodermal inclusion cyst (see ► Table 56.2 for comparison to dermoids). Although epidermoids and cholesteatomas are histologically identical (both arise from epithelium entrapped in an abnormal location, epidermoids are intradural, cholesteatomas are extradural), the term cholesteatoma is most often used to describe the lesion in the middle ear where the entrapped epithelium usually arises from chronic middle ear infections which lead to a retraction pocket (rarely, may instead be congenital).

► May arise from any of the following¹⁶:

1. displaced dorsal midline ectodermal cell rests trapped during neural tube closure between gestational weeks 3–5
2. multipotential embryonic cell rests
3. epithelial cell rests carried to the CPA with the developing otic vesicle
4. epidermal cells displaced into CNS, e.g., by LP—see Lumbar puncture (p. 1811)—or repeated percutaneous cranial subdural taps¹⁷

Epidemiology

Epidermoids comprise 1% of intracranial tumors,¹⁸ and ≈ 7% of CPA tumors. Peak age of occurrence: 40 years. No gender difference.

Histology

Epidermoids are lined by stratified squamous epithelium, and contain keratin (from desquamated epithelium), cellular debris, and cholesterol.¹⁹ Growth occurs at a linear rate like normal skin, unlike the exponential growth of true neoplasms.²⁰ The cyst contents may be liquid or may have a flaky consistency. They tend to spread along normal cleavage planes and surround vital structures (cranial nerves, ICA...). Bony destruction occurs in a minority, usually with larger tumors. Rare degeneration to squamous cell cancer²¹ primarily in cases of repeated recurrences after multiple surgeries.

Distinction from cholesterol granuloma

Epidermoid cysts are sometimes mistakenly equated with cholesterol granulomas,²² possibly because of the similarity between the terms cholesteatoma and cholesterol granuloma. However,

these are distinct lesions.²³ Cholesterol granulomas usually occur following chronic inflammation (usually in pneumatized portions of the temporal bone: petrous apex, mastoid air cells, middle ear space). Some differences are delineated in ► Table 56.3.

Table 56.3 Characteristics of epidermoid & cholesteatoma vs. cholesterol granuloma

Feature	Epidermoid	Cholesteatoma	Cholesterol granuloma
origin	ectodermal cells in abnormal location (within CNS, intradural)	(within ear, extradural)	chronic inflammatory cells surrounding cholesterol crystals (? from breakdown of RBC membranes)
precursor	usually congenital, occasionally acquired, e.g., after LP (p. 1815)	usually acquired (following chronic infection ? due to epithelial cells from tympanic membrane), occasionally congenital	chronic middle ear infection or idiopathic hemotympanum
symptoms	vary depending on location	chronic hearing loss, ear drainage, pain, or numbness around ear	usually involve vestibular or cochlear dysfunction
imaging (may not reliably distinguish among these)	CT: low density; no enhancement; bone erosion in only 33% MRI: T1WI: intensity slightly > CSF; T2WI: tumor & CSF similar high intensity		CT: homogeneous & isodense; rim enhancement; extensive destruction of petrous bone MRI: increased signal on both T1WI and T2WI
gross appearance	pearly white		brown (from hemosiderin)
microscopic pathology ²⁴	hyperkeratotic cyst lined with stratified squamous epithelium		fibroblastic proliferation, hemosiderin-laden macrophages, cholesterol clefts, giant cell reaction
ideal treatment	aggressive near-total excision		subtotal resection followed by drainage & restoration of pneumatization ²⁵

Presentation

1. may present as any mass lesion in the same location
2. CPA lesions can produce V, VII, or VIII neuropathies
3. symptoms of hydrocephalus: hydrocephalus may be obstructive or communicating due to chemical meningitis
4. recurrent episodes of chemical meningitis caused by rupture of the cyst contents, which may lead to hydrocephalus
 - a) symptoms include fever and meningeal irritation
 - b) CSF shows pleocytosis, hypoglycorrhachia, elevated protein, and negative cultures. Cholesterol crystals may be seen and can be recognized by their amorphous birefringent appearance

Imaging

MRI (► Fig. 56.1): epidermoids mimic CSF on T1WI (low signal, may be slightly > CSF) and T2WI (high signal). Other tumors are also high signal on T2WI, but most enhance with contrast on T1WI (epidermoids do not enhance). An epidermoid may pass from the posterior fossa through the incisura to the middle fossa.

Diffusion weighted imaging (DWI) is the best test to differentiate epidermoids from CSF (e.g., as in similar appearing arachnoid cyst). Epidermoids show intense signal on DWI as a result of restriction of water movement.

Treatment

Caution when removing epidermoid cysts to minimize spilling contents as they are quite irritating and may cause severe chemical meningitis (see above). Berger¹⁶ advocates intraoperative irrigation with hydrocortisone (100 mg/L of LR) to reduce the risk of post-op communicating hydrocephalus. Perioperative IV steroids and copious saline irrigation during surgery may provide similar results. The tumor is not in the cyst wall, and the surgical plan is generally to remove as much as possible

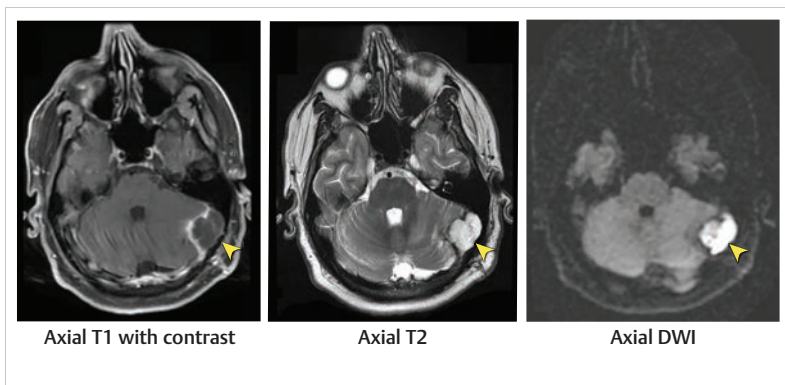


Fig. 56.1 Epidermoid (yellow arrowheads) of the left posterior fossa on MRI. Note: the epidermoid is bright on the DWI, whereas an arachnoid cyst would be dark (as is the CSF).

but to leave capsule adherent to critical structures such as brainstem and blood vessels as the morbidity of removal is high and a small residual does not preclude satisfactory outcome.

In spite of adequate removal, it is not unusual to see persistent brainstem distortion on post-op imaging.²³ Post-op radiation is not indicated as the tumor is benign and XRT does not prevent recurrence.²⁶

Dermoid cysts

Much of the description is covered above (p. 936). Imaging findings are shown in ► Fig. 56.2

56.1.3 Paraganglioma (WHO grade 1)

General information

AKA chemodectoma (obsolete), AKA glomus tumor. A WHO grade 1 neuroendocrine tumor that arises from specialized neural crest cells associated with autonomic ganglia, comprised of uniform cells with neural differentiation conglomerated into compact nests (Zellballen) surrounded by a capillary network.

Paragangliomas in the CNS are rare, and they primarily occur in the cauda equina/filum terminale or jugulotympanic regions. Outside the CNS, paragangliomas are often designated by a name that confers the site of origin, as shown in ► Table 56.4.

These tumors arise from paraganglion cells (not chemoreceptor cells as previously thought; therefore the term chemodectoma is rarely used). Slow growing tumors (< 2 cm in 5 years). Histologically benign (< 10% associated with lymph node involvement or distant spread). Most contain secretory granules on EM (mostly epinephrine and norepinephrine, and these tumors may occasionally secrete these catecholamines with risk of life-threatening HTN and/or cardiac arrhythmias).

Glomus tumors may occur in 2 patterns:

1. familial: non multicentric. Up to 50%
2. nonfamilial: may be multicentric (metachronous) 5%

Pheochromocytoma

General information

Located in the adrenal gland. May be sporadic, or as part of familial syndrome (von Hippel-Lindau disease (p. 646), MEN 2A and 2B, and neurofibromatosis). Consider genetic testing if age at diagnosis is < 50 years for mutations of VHL and other genetic abnormalities (RET, SDHS, SDHB, SDHC²⁷).

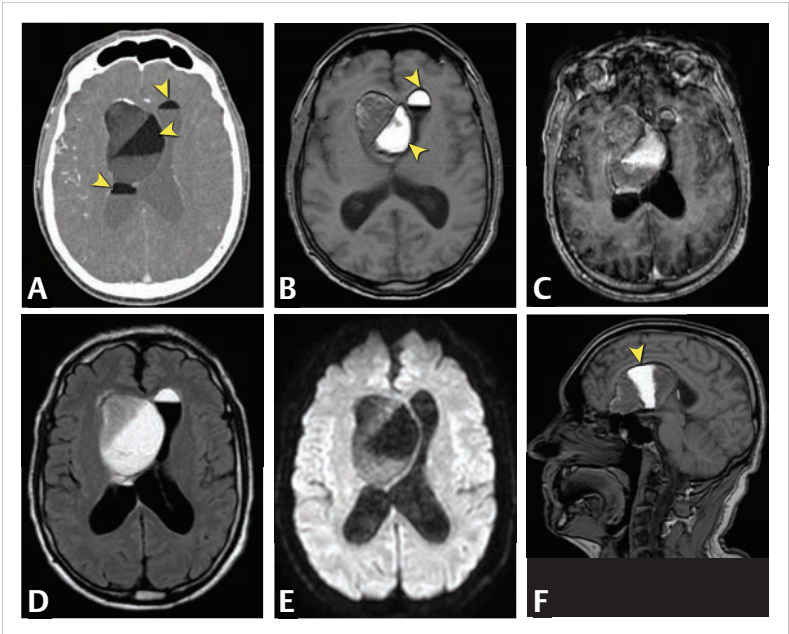


Fig. 56.2 Intraventricular dermoid cyst.
At surgery, the cyst was filled with cholesterol crystals, fat (yellow arrowheads), and hair.
Image: A: Axial CTA. B: Axial T1 noncontrast MRI. C: Axial T1 + contrast MRI (note the lack of significant enhancement with contrast). D: Axial FLAIR MRI. E: Axial DWI MRI. F: Sagittal T1 noncontrast MRI.

Table 56.4 Paraganglioma designation based on site of origin

Site	Designation
carotid bifurcation (most common)	carotid body tumors
auricular branch of vagus (middle ear)	glomus tympanicum
superior vagal ganglion (jugular foramen)	glomus jugulare
inferior vagal (nodose) ganglion (nasopharynx at skull base) (least common)	glomus intravagale (AKA glomus vagale)
adrenal medulla & sympathetic chain	pheochromocytoma

Laboratory studies

1. fractionated plasma metanephrines: 96% sensitivity, 85% specificity.²⁸ More sensitive than serum catecholamines with sporadic elevations. Pheochromocytoma is ruled out if plasma normetanephrine (NMN) < 112 pg/ml and metanephrine (MN) < 61 pg/ml. Highly suspicious if NMN > 400 pg/ml or MN > 236 pg/ml
2. 24 hr urine collection for: total catecholamines (epinephrine and norepinephrine) and metanephrines (88% sensitivity, 99.7% specificity²⁹). **Note:** testing for vanillylmandelic acid (VMA) is no longer done as it does not measure fractionated metanephrines.
3. where elevation is found, a clonidine suppression test can be done. Normal response consists of a fall in plasma catecholamines to ≤50% of baseline and below 500 pg/ml (there will be a reduction in essential hypertension, but no change with pheochromocytoma or other tumor production)

Imaging

Indicated when laboratory tests confirm pheochromocytoma.

MRI with contrast is preferred over CT.

CT may be used when MRI is contraindicated, but is less sensitive, especially for lesions <1 cm diameter.

¹²³I-MIBG (iodine-123-meta-iodobenzylguanidine) scintigraphy detects extra-adrenal pheochromocytomas with 83–100% sensitivity, 95–100% specificity. If not available, ¹³¹I-MIBG may be used with 77–90% sensitivity, 95–100% specificity.

Carotid body tumors

General information

Possibly the most common paraganglioma (pheochromocytoma may possibly be more common). Approximately 5% are bilateral; the incidence of bilaterality increases to 26% in familial cases (these are probably autosomal dominant).

Clinical

Usually present as painless, slow growing mass in upper neck. Large tumors may → cranial nerve involvement (especially vagus and hypoglossal). May also cause stenosis of ICA → TIAs or stroke.

Evaluation

1. carotid angiogram: demonstrates predominant blood supply (usually external carotid, with possible contributions from vertebral and thyrocervical trunk). May also detect bilateral lesions. Characteristic finding: splaying of bifurcation
2. MRI (or CT): evaluates extent, and assesses for intracranial extension

Treatment

Resection reported to carry a high complication rate, including stroke (8–20%) and cranial nerve injury (33–44%). Mortality rate is 5–13%.

Glomus tumors

General information

Glomus tumors may be subdivided into glomus jugulare and glomus tympanicum tumors. Glomus jugulare tumors arise from the jugular bulb (in the jugular foramen at the junction of the sigmoid sinus and jugular vein). Glomus tympanicum tumors are centered higher than glomus jugulare. Glomus tumors are rare (0.6% of all head and neck tumors), yet the glomus tympanicum is the most common neoplasm of the middle ear. Glomus jugulare tumors (GJT) arise from glomus bodies, usually in the area of the jugular bulb, and track along vessels. May have finger-like extension into the jugular vein (which may embolize during resection).³⁰ Most are slow growing, although rapidly growing tumors do occur.

Vascular supply: very vascular. Main feeders of GJT are from the external carotid (especially inferior tympanic branch of ascending pharyngeal artery, and branches of posterior auricular, occipital, and internal maxillary), with additional feeders from petrous portion of the ICA. Glomus tympanicum tumors feed from the auricular artery.

Epidemiology

Female:male ratio is 6:1. Bilateral occurrence is almost nonexistent.

Pathology

General information

Histologically indistinguishable from carotid body tumors. May invade locally, both through temporal bone destruction and especially along pre-existing pathways (along vessels, eustachian tube, jugular vein, carotid artery). Intradural extension is rare. Malignancy, although rare, may occur. These tumors rarely metastasize.

Secretory properties

These tumors usually possess secretory granules (even the functionally inactive tumors) and may actively secrete catecholamines (similar to pheochromocytomas, occur in only 1–4% of GJT³¹). Norepinephrine will be elevated in functionally active tumors since glomus tumors lack the methyltransferase needed to convert this to epinephrine. Alternatively, serotonin and kallikrein may be released,

and may produce a carcinoid-like syndrome (bronchoconstriction, abdominal pain and explosive diarrhea, violent H/A, cutaneous flushing, hypertension, hepatomegaly, and hyperglycemia).³² During surgical manipulation, these tumors may also release histamine and bradykinin, causing hypotension and bronchoconstriction.³³

Clinical

Symptoms

Patients commonly present with hearing loss and pulsatile tinnitus. Dizziness is the third most common symptom. Ear pain may also occur.

Signs

Hearing loss may be conductive (e.g., due to obstruction of the ear canal) or sensorineural, due to invasion of the labyrinth often with accompanying vertigo (the eighth nerve is the most common cranial nerve involved). Various combinations of palsies of cranial nerves IX, X, XI, and XII occur—see Jugular foramen syndromes (p. 102)—with occasional VII palsy (usually from involvement within the temporal bone). Ataxia and/or hydrocephalus can occur with massive lesions that cause brainstem compression. Occasionally patients may present with symptoms due to secretory products (see below).

Otoscopic exam → pulsatile reddish-blue mass behind eardrum (occasionally, lamentably biopsied by ENT physician with possible ensuing massive blood loss).

Differential diagnosis

See Cerebellopontine angle (CPA) lesions (p. 1647). The major differential is neurilemmomas (vestibular schwannomas), both enhance on CT. A cystic component and extrinsic compression of the jugular bulb are characteristic of neurilemmomas. Angiography will differentiate difficult cases.

Evaluation

Neurophysiologic testing

Audiometric and vestibular testing should be performed.

Imaging

1. CT or MRI used to delineate location and extent of tumor; CT is better for assessing bony involvement of the skull base
2. angiography: confirms diagnosis (helping to rule out vestibular schwannoma), and ascertains patency of contralateral jugular vein in event that jugular on side of tumor must be sacrificed; jugular bulb and/or vein are usually partially or completely occluded

Endocrine/laboratory studies

See also details (p. 940).

Classification

A number of classification schemes have been proposed. The modified Jackson classification is shown in ► Table 56.5.

Table 56.5 Modified Jackson classification of glomus tumors³⁴

Type	Description	Intracranial extension
I	small; involves jugular bulb, middle ear & mastoid	none
II	extends under IAC	possible
III	extends into petrous apex	possible
IV	extends beyond petrous apex into clivus or infratemporal fossa	possible

Treatment

Surgical resection is usually simple and effective for small tumors confined to the middle ear. For larger tumors that invade and destroy bone, the relative role of surgery and/or radiation is not fully determined. With large tumors, surgery carries the risk of significant cranial nerve palsies.

Medical management

General information

For tumors that actively secrete catecholamines, medical therapy is useful for palliation or as adjunctive treatment before embolization or surgery. Alpha and beta blockers given before embolization or surgery blocks possibly lethal blood pressure lability and arrhythmias. Adequate blockade takes $\approx$ 2–3 weeks of alpha blocker and at least 24 hours of beta blocker therapy; in emergency, 3 days of treatment may suffice.

Alpha blockers

Reduce BP by preventing peripheral vasoconstriction.

1. phenoxybenzamine (Dibenzylin®): long acting; peak effect 1–2 hrs. Start with 10 mg PO BID and gradually increase to 40–100 mg per day divided BID
2. phentolamine (Regitine®): short acting. Usually used IV for hypertensive crisis during surgery or embolization.

R: 5 mg IV/IM (peds: 1 mg) 1–2 hrs pre-op, repeat PRN before and during surgery

Beta blockers

Reduces catecholamine-induced tachycardia and arrhythmias (may also prevent hypotension that might occur if only alpha blockade is used). These drugs are not always needed, but when used 6 NB: these drugs must *not* be started before starting alpha-blockers (to prevent hypertensive crisis and myocardial ischemia).

1. propranolol (Inderal®): **R:** oral dose is 5–10 mg q 6 hrs. IV dose for use during surgery is 0.5–2 mg slow IVP
2. labetalol (Normodyne®) (p. 131): may have some efficacy in blocking α 1 selective and β non-selective (potency < propranolol)

Serotonin, bradykinin, histamine release blockers

These agents may provoke bronchoconstriction that does not respond to steroids, but may respond to inhaled β -agonists or inhaled anticholinergics. Somatostatin may be used to inhibit release of serotonin, bradykinin, or histamines. Since this drug has a short half-life, it is preferable to give octreotide (p. 890) 100 mcg sub-Q q 8 hrs.

Radiation therapy

XRT may relieve symptoms and stop growth in spite of persistence of tumor mass. 40–45 Gy in fractions of 2 Gy has been recommended.³⁵ Lower doses of $\approx$ 35 Gy in 15 fractions of 2.35 Gy appear as effective and have fewer side effects.³⁶ Generally used as primary treatment only for large tumors or in patients too elderly or infirmed to undergo surgery. Some surgeons pretreat 4–6 mos preoperatively with XRT to decrease vascularity³⁷ (controversial).

Embolization

1. generally reserved for large tumors with favorable blood supply (i.e., vessels that can be selectively embolized with no danger of particles passing through to normal brain)
2. post-embolization tumor swelling may compress brainstem or cerebellum
3. may be used preoperatively to reduce vascularity. Performed 24–48 hours pre-op (not used prior to that, because of post-embolization edema)
4. caution with actively secreting tumors that may release vasoactive substances (e.g., epinephrine) upon infarction from the embolization
5. may also be used as primary treatment ($\pm$ radiation) in patients who are not surgical candidates. In this case, XRT is only palliative, as tumor will develop new blood supply
6. absorbable (Gelfoam®) and non-absorbable (Ivalon®) materials have been used

Surgical treatment

General information

The tumor is primarily extradural, with extremely vascular surrounding dura.

Suboccipital approach may cause dangerous bleeding and usually results in incomplete resection. Team approach by a neurosurgeon in conjunction with a neuro-otologist (and possibly head and neck surgeon) has been advocated.³⁸ This approach utilizes an approach to the skull base through the neck.

ECA feeders are ligated early, followed rapidly by draining veins (to prevent systemic release of catecholamines).

Sacrifice of the jugular vein (JV) is tolerated if the contralateral JV is patent (often, the ipsilateral JV will already be occluded).

Surgical complications and outcome

The most common complications are CSF fistula, facial nerve palsy, and varying degrees of dysphagia (from dysfunction of lower cranial nerves). Dysfunction of any of the cranial nerves VII thru XII can occur, and a tracheostomy should be performed if there is any doubt of lower nerve function, and a gastrostomy feeding tube may be needed temporarily or permanently. Lower cranial nerve dysfunction also predisposes to aspiration, the risk of which is also increased by impaired gastric emptying and ileus that may occur due to reduced cholecystokinin (CCK) levels post-op. Excessive blood loss can also occur.

Even after gross total tumor removal, recurrence rate may be as high as one third.^{37,39}

56.2 Cyst like lesions

56.2.1 Colloid cyst

General information

Key concepts

- slow-growing benign tumor comprising < 1% of intracranial tumors
- classically occurs in the anterior 3rd ventricle, blocking foramina of Monro bilaterally → obstructive hydrocephalus involving only the lateral ventricles (≈ pathognomonic)
- enhances minimally or not at all on CT/MRI
- natural history: risk of *sudden* death has been described, but is controversial. A small number of patients will present with acute obstructive hydrocephalus and progress to death despite appropriately rapid intervention
- treatment, when indicated, is surgical. Main options: microsurgery (transcallosal, transcortical/transventricular (only if hydrocephalus)), ventriculoscopic

AKA neuroepithelial cysts. Most commonly found in the third ventricle in the region of the foramina of Monro, but may be seen elsewhere, e.g., in septum pellucidum.⁴⁰

Epidemiology

Estimated incidence is 3.2 per 1,000,000 per year,⁴¹ with a prevalence of about 1 in 5800.⁴² They comprise 0.5–1.5% of all intracranial tumors.⁴¹ Usual age of diagnosis: 20–50 yrs.

There is evidence that some cases are part of familial clusters which appear to have primarily mother-to-daughter transmission.⁴²

Pathogenesis

Origin: unknown. Implicated structures include: paraphysis (evagination in roof of third ventricle, rudimentary in humans), diencephalic ependyma in the recess of the postvelar arch, ventricular neuroepithelium.

Comprised of a fibrous epithelial-lined wall filled with either mucoid or dense hyaloid substance. A slow growing, benign tumor.

Clinical signs and symptoms

Presenting symptoms or indications neuroimaging in patients found to have a colloid cyst in one series are shown in ► Table 56.6. 46% of their symptomatic patients were found to have hydrocephalus which is responsible for the most common presenting symptoms (H/A, N/V, blurred vision, ataxia, and cognitive decline).⁴³ A small number of patients will present with acute obstructive hydrocephalus (estimates vary from 3–35%) and despite rapid institution of CSF diversion and accompanying critical care will progress to death in 5–38%⁴³ (see “Sudden death” below).

Most cysts < 1 cm diameter do not produce hydrocephalus and are asymptomatic.

► **Sudden death.** A high rate of sudden death has been reported with colloid cysts (20% in pre-CT era⁴⁴), but is probably overestimated. One prevailing obsolete theory was that these tumors are mobile and thus could shift position and acutely block CSF flow with resultant herniation. More

Table 56.6 Presenting symptoms or indication for neuroimaging in 163 patients with colloid cyst⁴³

Feature	Symptomatic (n=65)	Incidental ^a (n=98)
headache	58 (89%)	24 (25%)
nausea/vomiting	15 (23%)	5 (5%)
vision change	14 (22%)	2 (2%)
dizziness/ataxia	13 (20%)	6 (6%)
cognitive decline	9 (14%)	11 (11%)
syncope	6 (9%)	5 (5%)
seizure	4 (6%)	2 (2%)
bradycardia	4 (6%)	0
sensory change	3 (5%)	1 (1%)
memory decline	3 (5%)	0
urinary incontinence	2 (3%)	0
trauma	1 (2%)	27 (28%)
general medical	1 (2%)	12 (12%)
weakness	1 (2%)	3 (3%)
general neurological	0	8 (8%)
stroke	0	8 (8%)
cancer staging	0	6 (6%)
intoxication	0	2 (2%)
other	0	1 (1%)

^aMost colloid cysts in the incidental group were discovered during routine neuroimaging for unrelated conditions (trauma, stroke, nonspecific H/A, ALS...).

Careful analysis reveals that nearly all patients have progressive symptoms preceding what is then seen as a precipitous deterioration. Changes in CSF dynamics resulting from procedures (LP, ventriculography...) may have also contributed in some cases.⁴⁵ Another proposed mechanism is disturbance of hypothalamic-mediated cardiovascular reflex control.⁴⁵

Given that sudden death, while infrequent, may occur with colloid cysts, and that the literature is rife with references to this, it is imperative that the existence of this small possibility should be expounded to the patient (and this discussion should be documented) and treatment decisions should involve patient input.

Evaluation

Imaging (MRI or CT) demonstrates the tumor usually located in the anterior 3rd ventricle (► Fig. 56.3). Here, it often blocks both foramina of Monro causing almost pathognomonic hydrocephalus involving only the lateral ventricles (sparing the 3rd and 4th). Differential diagnosis includes basilar artery aneurysms, hamartomas, primary or secondary neoplasm, and xanthogranulomas.⁴⁶

MRI: usually the optimal imaging technique. However, there are cases where cysts are isointense on MRI and CT is superior⁴⁷ (scrutinize the midline T1WI MRI images). When the lesion is identifiable, MRI clearly demonstrates the location of the cyst and relation to nearby structures, usually obviating an angiogram. MRI appearance: variable. Usually hyperintense on T1WI, hypointense on T2WI. Some data suggests that symptomatic patients are more likely to display T2 hyperintense cysts on MRI, indicating higher water content which may reflect a propensity for continued cyst expansion.⁴⁸ Enhancement: minimal, sometimes involving only capsule.

CT scan: findings are variable. Most are hyperdense (however, iso- and hypodense colloid cysts occur), and about half enhance slightly. Density may correlate with viscosity of contents; hyperdense cysts were harder to drain percutaneously.⁴⁹ CT is usually not quite as good as MRI, especially with isodense cysts. These tumors calcify only rarely.

✗ LP: *contraindicated* prior to placement of shunt due to risk of herniation.

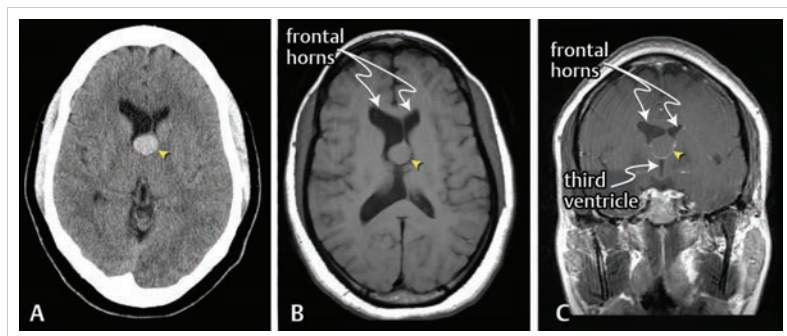


Fig. 56.3 Colloid cyst (yellow arrowheads) of the third ventricle.

Image: A: axial noncontrast CT scan. B: axial noncontrast T1 FLAIR MRI. C: coronal T1 contrast enhanced MRI showing relation to the normal sized 3rd ventricle and the frontal horns of the lateral ventricles (in this patient, the right lateral ventricle is enlarged, the left is not). Note: the colloid cyst does not enhance.

Treatment – general information and management options

General information

Optimal treatment remains controversial. Initially, shunting without treating the cyst was advocated.⁵⁰ The nature of the obstruction (both foramina of Monro) requires *bilateral* ventricular shunts (or, unilateral shunt with fenestration of the septum pellucidum). Presently, one form or another of direct surgical treatment of the tumor is usually recommended for some or all of the following reasons:

1. to prevent shunt dependency
2. to reduce the possibility of tumor progression
3. the mechanism of sudden neurologic deterioration may be due to factors such as cardiovascular instability from hypothalamic compression and not due to hydrocephalus

Management options

A meta-analysis of 1278 patients comparing endoscopic and various microsurgical techniques found that the microsurgical group had a significantly greater extent of resection (96.8% vs. 58.2%), lower rates of recurrence (1.48% vs. 3.91%), and lower rates of reoperation than the endoscopic group (0.38% vs. 3.0%). Both groups had similar rates of mortality (1.4% vs. 0.6%) and shunt dependency (6.2% vs. 3.9%). Overall, the complication rate was lower in the endoscopic group than in the microsurgical group (10.5% vs. 16.3%). Within the microsurgery group, the transcallosal approach had a lower overall morbidity rate (14.4%) than the transcortical approach (24.5%).⁵¹

56

Using natural history for treatment decisions

A review of 58 asymptomatic patients (average age 57 years) with incidentally discovered colloid cysts of the third ventricle with mean follow up of 79 months demonstrated the incidence of symptomatic worsening at 2-, 5-, and 10-year follow-up to be 0%, 0%, and 8%, respectively. Of the 34 patients who obtained follow up imaging, 32 demonstrated no change in cyst size or ventricular caliber. The average age of these patients was significantly higher than that of the patients undergoing surgery for symptomatic lesions (57 vs. 41) and thus may reflect a patient cohort with differing natural histories.⁵²

NB: Many “asymptomatic” patients may have headaches at diagnosis. Careful evaluation of headache etiology (i.e., posttraumatic, migrainous, tension, etc.) should be undertaken to determine whether the headaches are due to the colloid cyst or if the cyst is asymptomatic.

Pollock et al⁴⁸ reviewed 155 patients with newly diagnosed colloid cysts, performing a recursive partitioning analysis, and divided patients into three classes as shown in ► Table 56.7.

Additionally, a significantly greater number of Class III patients had hyperintense cyst contents on T2 weighted imaging (44% vs. 13%), and symptomatic patients were more likely to have increased T2 signal (44% vs. 8%) compared to their asymptomatic counterparts. The authors suggest that asymptomatic patients with low T2 signal-containing cysts may reflect a group with low potential for cyst

expansion and development of cyst-related symptoms (even in the presence of ventriculomegaly); thus they may represent a population that may be safely managed in a nonoperative fashion.⁴⁸ Nevertheless, most surgeons would recommend surgery for a patient with ventriculomegaly and symptoms such as headache even if they may not be definitely related.

Table 56.7 Classes of colloid cyst based on recursive partitioning analysis

Class	Characteristic			Symptomatic patients (% of total)	Treatment options
	Age (years)	Cyst diameter	Ventricles		
I	any	≤ 10 mm	normal	12%	May be monitored clinically and with serial imaging (CT or MRI)
II	> 50	≤ 10 mm	ventriculomegaly	50%	If asymptomatic, may be monitored clinically and with serial imaging (CT or MRI)
III	any	> 10 mm	normal	85%	Surgical removal is recommended
	≤ 50	any	ventriculomegaly		
	> 50	> 10 mm	ventriculomegaly		

Surgical treatment

Surgical options

Also see Approaches to the third ventricle.

1. transcallosal approaches: not dependent on dilated ventricles. Higher incidence of venous infarction or forniceal injury (see below)
2. transcortical approach: higher incidence of post-op seizures ($\approx 5\%$). Not feasible with normal sized ventricles (e.g., in patient with VP shunt)
3. stereotactic drainage: see below
4. ventriculoscopic removal: see below

Transcallosal approach

Access to the 3rd ventricle via either the foramen of Monro or by interforaminal approach. Since colloid cysts tend to occur exactly at the foramen of Monro, it is *rarely* necessary to enlarge the foramen to locate the tumor. See Transcallosal approach to lateral or third ventricle (p. 1758).

Endoscopic (ventriculoscopic) resection

Ventriculoscopic resection has been shown to have a reduced incidence of operative morbidity and overall surgical treatment cost (including reduced length of stay) when compared to microsurgery.⁵³ It is argued that this should be the initial procedure of choice, with craniotomy reserved for treatment failures.⁵⁴ However, endoscopic resection is associated with a significantly increased incidence of residual coagulated cyst-wall remnants compared to microsurgery. It remains to be seen whether this will ultimately result in an increased rate of reoperation, and how that will impact the overall patient outcome and costs.

Stereotactic drainage of colloid cysts

May be useful,⁵⁵ especially in patients with normal ventricles from shunting, but the contents may be too viscous,⁵⁶ and the tough capsule may make blind penetration difficult. Total or even subtotal aspiration may not require further treatment in some patients; however, recurrence rate is higher than with surgical removal.⁵⁷

Early morbidity was relatively high from this procedure (not widely reported in literature), possibly from vascular injury or mechanical trauma; this has improved. May be more feasible with intraoperative ventriculography⁵⁸ or with a

Two features that correlate with *unsuccessful* stereotactic aspiration⁵⁹:

1. high viscosity: correlates with hyperdensity on CT (low viscosity correlated with hypo- or iso-dense CT appearance; no MRI finding correlated with viscosity)
2. deflection of the cyst from tip of aspirating needle due to small size

Stereotactic technique⁶⁰:

1. insertion point of stereotactic needle is just anterior to right coronal suture
2. start with sharp-tipped 1.8 mm probe, and advance to 3–5 mm beyond target site (to accommodate for displacement of cyst wall)
3. use a 10 ml syringe and apply 6–8 ml of negative aspiration pressure
4. if this does not yield any material, repeat with a 2.1 mm probe
5. although complete cyst evacuation is desirable, if this cannot be accomplished an acceptable goal of aspiration is re-establishment of patency of the ventricular pathways (may be verified by injecting 1–2 cc of iohexol)

56.2.2 Pineal cysts (PCs)

General information

Usually an incidental finding (i.e., not symptomatic), seen on $\approx 4\%$ of MRIs⁶¹ or on 25–40% of autopsies⁶² (many are microscopic). The most common ones are intra-pineal glial-lined cysts with diameter < 1 cm. Etiology is obscure; PCs are nonneoplastic, and may be due to ischemic glial degeneration or due to sequestration of the pineal diverticulum. They have been regarded as benign, but the natural history is not known with certainty.⁶³ PCs may contain clear, slightly xanthochromic, or hemorrhagic fluid. In rare cases, they may enlarge, and like other pineal region masses, may become symptomatic by causing hydrocephalus by aqueductal compression,⁶⁴ gaze paresis⁶⁵ including Parinaud's syndrome (p. 101), or hypothalamic symptoms.

Positional H/As have been attributed to PCs; the theory is that the cyst could intermittently compress the vein of Galen and/or Sylvian aqueduct.⁶⁶ This remains unproven since asymptomatic compression of the vein of Galen and the quadrigeminal plate has been demonstrated on MRI.⁶⁷

Imaging

May escape detection on CT because the cyst fluid density is often similar to CSF. MRI T1WI ($\blacktriangleright$ Fig. 56.4) shows round or ovoid abnormality in region of pineal recess, signal varies with protein content (isointense or slightly hyperintense). T2WI occasionally shows increased intensity.⁶³ Gadolinium occasionally enhances the cyst wall with a maximum thickness of 2 mm. Irregularities of the wall with nodular enhancement suggests the lesion is not a benign pineal cyst.

Epidermoid-dermoid cysts may also occur in the pineal region, and are larger and have different signal characteristics on MRI.

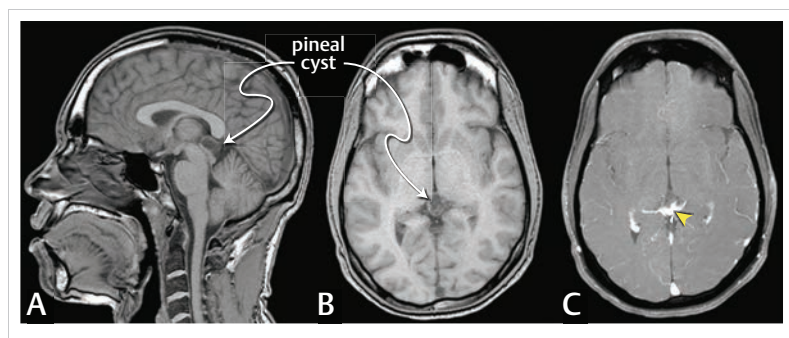


Fig. 56.4 Pineal cyst.

Dimensions: 1.5 cm AP, 1 cm height, 1.2 cm mediolateral. Yellow arrowhead marks contrast enhancement, probably representing the pineal gland.

Image: MRI, A: sagittal T1 noncontrast, B: axial T1 noncontrast, C: axial T1 with contrast.

Management

► Asymptomatic PCs. Asymptomatic PCs < 2 cm diameter with typical appearance (wall: ≤ 2 mm thick and no irregularities or nodular enhancement): are generally considered not to grow, but the natural history is not known with certainty. Risk of growing is $\approx 4\%$ with 6 months median follow up

in adults,⁶⁸ and 11% in children with 10 months median follow-up.⁶⁹ Based on this, it seems a little hard to justify not providing initial follow-up to reassure the patient and the practitioner.

A reasonable approach might consist of a 3-month follow-up MRI after the initial scan to check for rapid growth (as might occur with a neoplasm) and then if stable, annual imaging studies for an arbitrary period (some suggest 1 year⁶⁸; 3 years is probably reasonable) to assure stability.

► **Symptomatic lesions or those that show changes on MRI.** Surgery to relieve symptoms and/or to obtain a diagnosis.

► **Hydrocephalus.** Surgical options include:

1. CSF diversion: recommended only for lesions with appearance of typical PC as it does not obtain tissue for pathology. May not relieve gaze disturbance from direct pressure on tectal plate
 - a) CSF shunt
 - b) endoscopic third ventriculostomy (ETV) (p. 1828). A few cases of regression of PCs after ETV have been reported⁷⁰
2. aspiration only (stereotactic or endoscopic): may not get enough tissue for diagnosis
3. cyst excision (open or endoscopic): relieves symptoms and establishes diagnosis. Low morbidity⁷¹

56.2.3 Rathke's cleft cyst

General information

Rathke's cleft cysts (RCC) AKA pars intermedia cysts, are slow-growing nonneoplastic lesions that are thought to be remnants of Rathke's pouch (p. 151). They are primarily intrasellar, suprasellar extension is common, purely suprasellar presentation is rare.⁷² RCC are found incidentally in 13–23% of necropsies.⁷³ Since the adenohypophysis arises from proliferation of the anterior wall of Rathke's pouch, RCCs and PitNET/adenomas have a similar lineage and rarely they are found together.⁷⁴ RCC are often discussed in contrast to craniopharyngiomas (CP) (see above). Some features are compared in ► Table 56.8. Recurrence after treatment is common.

Table 56.8 Comparison of craniopharyngioma to Rathke's cleft cyst

Feature	Craniopharyngioma	Rathke's cleft cyst
site of origin	anterior superior margin of pituitary	pars intermedia of pituitary
cell lining	stratified squamous epithelium	single layer cuboidal epithelium
cyst contents	cholesterol crystals	may be clear or may resemble motor oil
surgical treatment	total removal is the goal	partial excision and drainage ⁷⁵
cyst wall	thick	thin

Clinical presentation

Some symptom may be a result of mass effect; however, other symptoms may be related to inflammation that is associated with some RCCs.

- most are asymptomatic, discovered incidentally on imaging obtained for unrelated reasons
- headache occurs in 44–81% of symptomatic RCCs,⁷⁶ with sudden onset of severe H/A in up to 16%⁷⁷
- 30–60% develop endocrine disturbance,⁷⁶ often with hypogonadism in males (producing fatigue and decreased libido), menstrual irregularities and galactorrhea in premenopausal females or pan-hypopituitarism in postmenopausal females. Diabetes insipidus is more common with symptomatic RCCs (up to 37%) than with pituitary adenomas (possibly due to associated inflammation)
- visual field deficit is present in 11–67%⁷⁶

Evaluation

General workup

Workup for these lesions is the same as for sellar/suprasellar masses (p. 874) in general.

1. up to 60% have pituitary dysfunction. Endocrine workup (p. 876) is as for any sellar mass
2. visual field evaluation (p. 874)

Imaging

RCCs are typically midline, without stalk deviation.

CT: RCCs usually appear as low-density cystic lesions. One half show capsular enhancement. RCCs infrequently can erode the skull base.

MRI appearance is variable (► Fig. 56.5).⁷⁵ Typically high signal on T2WI. On T1WI they may be hyperintense (correlates with RCCs having proteinaceous mucinous contents) or hypointense (seen with RCCs containing clear low protein fluid). An intracystic nodule that is high signal on T1WI low signal on T2WI is described in > 75% of RCCs.⁷⁸ Rule of thumb: a sellar lesion with a nodule is usually an RCC. There is no internal enhancement.

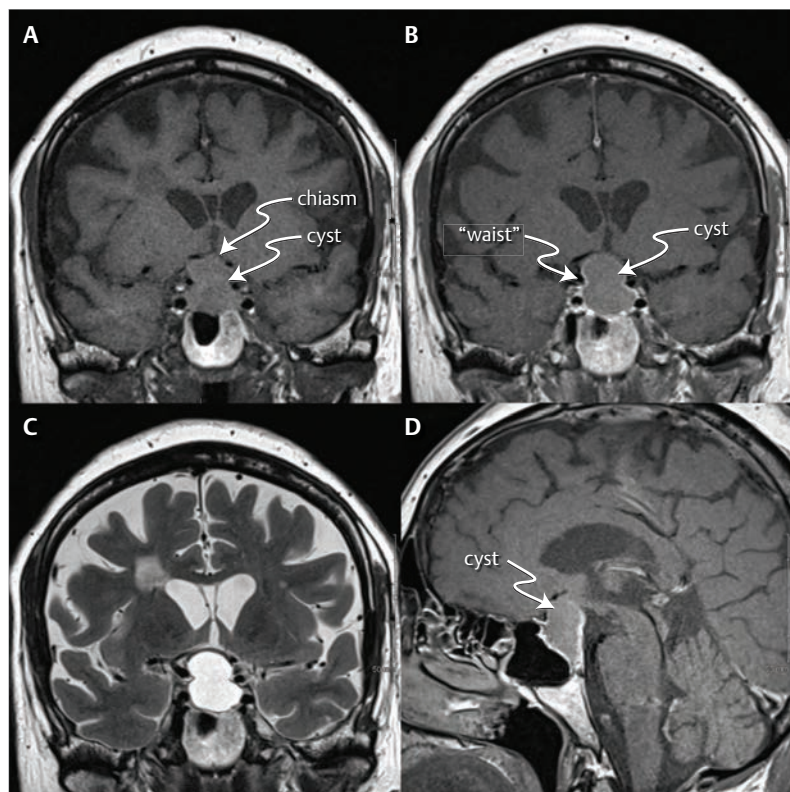


Fig. 56.5 Rathke cleft cyst.

Note: enhancement of the cyst wall but no enhancement within the cyst. The diaphragma sella creates a “waist” in the cyst where it extends into the suprasellar cistern. The optic chiasm drapes over the top of the cyst (not well visualized in this MRI).

Image: MRI. A: noncontrast T1 coronal, B: contrast enhanced T1 coronal, C: noncontrast T2 coronal, D: contrast enhanced T1 sagittal.

Natural history

Among 61 incidentally discovered RCCs, 69% showed no growth over 9 years.⁷⁹ In another series of 94 cases, only 5% increased in size and 16% actually decreased in size with 27 months mean follow-up.⁸⁰

Management

Incidental (i.e., asymptomatic) lesions are followed with serial imaging.

Symptomatic RCCs are usually drained, most transsphenoidally, either microscopically or endoscopically, usually combined with cyst fenestration.

► **Reduction of recurrence rate.** Methods to reduce recurrence include techniques that promote persistence of continued drainage by maintaining patency of the fenestration. There is not sufficient evidence to recommend one method over another. Techniques include:

1. mucosal coupling: the epithelium of the RCC and the mucosa of the sphenoid sinus are connected⁸¹
2. stenting of the fenestration, e.g., with a bioabsorbable steroid eluting stent (e.g., PROPEL® Mini (Intersect ENT, Menlo Park, CA, USA) mometasone furoate (a synthetic antiinflammatory corticosteroid) implant, 370 mcg)
3. reduced recurrence rate has been quoted with complete cyst wall removal; however, the incidence of post-op endocrine dysfunction may be higher with this approach⁷⁶
4. ✖ there is no strong evidence to support the use of hydrogen peroxide or alcohol irrigation in an effort to reduce the recurrence rate.

Surgical outcome

Complete cyst decompression occurs in the vast majority of cases (up to 97%⁷⁹), with improved vision in 83–97%^{79,82} in those presenting with visual disturbance. Headache improves in 71%, and endocrinopathies improve in 33–94%

Morbidity:

1. CSF leak in ≈ 10% (higher rates with extension outside the sella)
2. diabetes insipidus (DI): permanent DI occurs in up to 9% after cyst drainage, and from to 19–69% with cyst wall resection⁷⁶
3. other risks seen with any transsphenoidal operation: internal carotid artery injury

Recurrence of cyst: reported rate is as high as 42%, but 16–18% may be more typical over 2–5 years. Higher rates of recurrence may be associated with: purely suprasellar location, inflammation and reactive metaplasia in the cyst wall, cyst infection, and use of a fat graft within the cyst cavity.

56.3 Empty sella syndrome

56.3.1 General information

Empty sella syndrome (ESS) can be “primary,” “secondary,” or congenital.

MRI is the imaging modality of choice (► Fig. 56.6). The sellar contents will follow imaging characteristics of CSF. The pituitary infundibulum may be seen traversing the sella (**infundibulum sign**),⁸³

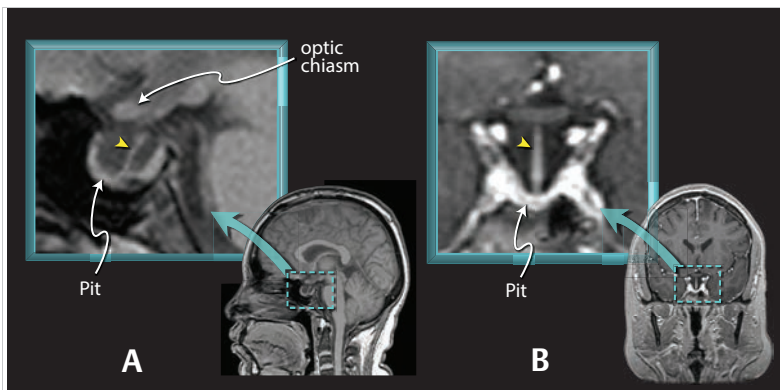


Fig. 56.6 Empty sella.

The infundibulum (AKA pituitary stalk) (yellow arrowheads) traverses the sella (infundibulum sign) to the pituitary gland, which is compressed against the floor of the sella.

Image: unenhanced T1 MRI, A: sagittal. B: coronal. Broken lines in inset shows from where the details are taken.

Abbreviation: Pit = pituitary gland.

which distinguishes empty sella from an intrasellar arachnoid cyst which displaces the infundibulum).

56.3.2 Primary empty sella syndrome

General information

Occurs in the absence of prior treatment of a pituitary tumor (medical, surgical, or XRT). Herniation of the arachnoid membrane into the sella turcica⁸⁴ which can act as a mass, probably as a result of repeated CSF pulsation. The pituitary gland may become compressed against the floor, and in the long term the sella itself can become enlarged (see Sella turcica (p.227) for normal dimensions).

Frequent association: female sex (female:male ratio=5:1), obesity, and HTN. The frequency of intrasellar arachnoid herniation is higher in patients with pituitary tumors and in those with increased intracranial pressure for any reason—including pseudotumor cerebri (p.955)—than in the general population.

These patients usually present with symptoms that do not suggest an intrasellar abnormality including: headache (the most common symptom), dizziness, seizures... Occasionally patients may develop CSF rhinorrhea,⁸⁵ deterioration of vision (acuity or field deficit resulting from kinking of optic chiasm due to herniation into the sella), or amenorrhea-galactorrhea syndrome.

Clinically evident endocrine disturbances are rare with primary ESS; however, up to 30% have abnormal pituitary function tests, most commonly reduced growth hormone secretion following stimulation. Mild elevation of prolactin (PRL) and reduction of ADH may occur, probably from compression of the stalk. These patients show a normal PRL rise with TRH stimulation (whereas patients with prolactinomas do not).

Treatment

Surgical treatment is usually not indicated, except in the case of CSF rhinorrhea. In this setting, it is necessary to determine if there is increased ICP, and if so, if there is an identifiable cause. Simple shunting for hydrocephalus runs the risk of producing tension pneumocephalus from air drawn in through the former leak site. This may necessitate transsphenoidal repair with simultaneous external lumbar drainage, to be converted to a permanent shunt shortly thereafter. Hyperprolactinemia may be treated, e.g., with bromocriptine (p.888) if it interferes with gonadal function.

56.3.3 Secondary empty sella syndrome

Entities associated with secondary empty sella syndrome:

1. following trauma⁸⁶
2. after successful transsphenoidal removal or XRT for a pituitary tumor⁸⁶
3. any cause of increased intracranial pressure, including: pseudotumor cerebri (p.955), Chiari malformation

Often presents with visual deterioration due to herniation of the optic chiasm into the empty sella. There may be hypopituitarism from the underlying cause.

Visual deterioration may be treated with chiasmpexy (propping up the chiasm), usually by transsphenoidal approach and packing the sella with fat, muscle or cartilage. Usually done endoscopically.⁸⁷ Chiasmpexy appears to be better for improving visual field deficits than loss of visual acuity.⁸⁸

References

- [1] Berger L, Luc G, Richard D. L'Esthesioneuroepitheliome Olfactif. Bull Assoc Franc Etude Cancer. 1924; 13:410–421
- [2] Theilgaard SA, Buchwald C, Ingeholm P, et al. Esthesioneuroblastoma: a Danish demographic study of 40 patients registered between 1978 and 2000. Acta Otolaryngol. 2003; 123:433–439
- [3] Chao KS, Kaplan C, Simpson JR, et al. Esthesioneuroblastoma: the impact of treatment modality. Head Neck. 2001; 23:749–757
- [4] Kadish S, Goodman M, Wang CC. Olfactory neuroblastoma. A clinical analysis of 17 cases. Cancer. 1976; 37:1571–1576
- [5] Biller HF, Lawson W, Sachdev VP, et al. Esthesioneuroblastoma: surgical treatment without radiation. Laryngoscope. 1990; 100:1199–1201
- [6] Dulguerov P, Calcaterra T. Esthesioneuroblastoma: the UCLA experience 1970–1990. Laryngoscope. 1992; 102:843–849
- [7] Hyams V. Tumors of the upper respiratory tract and ear. Washington, D.C.: Armed Forces Institute of Pathology; 1988
- [8] Van Gompel JJ, Giannini C, Olsen KD, et al. Long-term outcome of esthesioneuroblastoma: hyams grade predicts patient survival. J Neurol Surg B Skull Base. 2012; 73:331–336
- [9] Kane AJ, Sughrue ME, Rutkowski MJ, et al. Posttreatment prognosis of patients with esthesioneuroblastoma. J Neurosurg. 2010; 113:340–351
- [10] Gardner G, Robertson JH. Hearing Preservation in Unilateral Acoustic Neuroma Surgery. Ann Otol Rhinol Laryngol. 1988; 97:55–66

- [11] Dias FL, Sa GM, Lima RA, et al. Patterns of failure and outcome in esthesioneuroblastoma. *Arch Otolaryngol Head Neck Surg*. 2003; 129:1186–1192
- [12] Gore MR, Zanation AM. Salvage Treatment of Local Recurrence in Esthesioneuroblastoma: A Meta-analysis. *Skull Base*. 2011; 21:1–6
- [13] Van Gompel JJ, Carlson ML, Pollock BE, et al. Stereotactic radiosurgical salvage treatment for locally recurrent esthesioneuroblastoma. *Neurosurgery*. 2013; 72:332–9; discussion 339–40
- [14] Foote RL, Morita A, Ebersold MJ, et al. Esthesioneuroblastoma: the role of adjuvant radiation therapy. *Int J Radiat Oncol Biol Phys*. 1993; 27:835–842
- [15] Kim HJ, Cho HJ, Kim KS, et al. Results of salvage therapy after failure of initial treatment for advanced olfactory neuroblastoma. *J Craniomaxillofac Surg*. 2008; 36:47–52
- [16] Berger MS, Wilson CB. Epidermoid Cysts of the Posterior Fossa. *J Neurosurg*. 1985; 62:214–219
- [17] Gutin PH, Boehm J, Bank WO, et al. Cerebral convexity epidermoid tumor subsequent to multiple percutaneous subdural aspirations. Case report. *J Neurosurg*. 1980; 52:574–577
- [18] Guidetti B, Gagliardi FM. Epidermoids and Dermoid Cysts. *J Neurosurg*. 1977; 47:12–18
- [19] Fleming JFR, Botterrell EH. Cranial Dermoid and Epidermoid Tumors. *Surg Gynecol Obstet*. 1959; 109:57–79
- [20] Alvord EC. Growth Rates of Epidermoid Tumors. *Annals of Neurology*. 1977; 2:367–370
- [21] Link MJ, Cohen PL, Breneman JC, et al. Malignant squamous degeneration of a cerebellopontine angle epidermoid tumor. Case report. *J Neurosurg*. 2002; 97:1237–1243
- [22] Sabin HI, Bardi LT, Symon L. Epidermoid Cysts and Cholesterol Granulomas Centered on the Posterior Fossa: Twenty Years of Diagnosis and Management. *Neurosurgery*. 1987; 21:798–803
- [23] Altschuler EM, Jungreis CA, Sekhar LN, et al. Operative Treatment of Intracranial Epidermoid Cysts and Cholesterol Granulomas: Report of 21 Cases. *Neurosurgery*. 1990; 26:606–614
- [24] Friedman I. Epidermoid Cholesteatoma and Cholesterol Granuloma: Experimental and Human. *Ann Otol Rhinol Laryngol*. 1959; 68:57–79
- [25] Chang P, Fagan PA, Atlas MD, et al. Imaging destructive lesions of the petrous apex. *Laryngoscope*. 1998; 108:599–604
- [26] Keville FJ, Wise BL. Intracranial Epidermoid and Dermoid Tumors. *J Neurosurg*. 1959; 16:564–569
- [27] van Nderveen FH, Gaal J, Favier J, et al. An immunohistochemical procedure to detect patients with paraganglioma and pheochromocytoma with germline SDHB, SDHC, or SDHD gene mutations: a retrospective and prospective analysis. *Lancet Oncol*. 2009; 10:764–771
- [28] Kudva YC, Sawka AM, Young WF Jr. Clinical review 164: The laboratory diagnosis of adrenal pheochromocytoma: the Mayo Clinic experience. *Journal of Clinical Endocrinology and Metabolism*. 2003; 88:4533–4539
- [29] de Jong WH, Eisenhofer G, Post WJ, et al. Dietary influences on plasma and urinary metanephrines: implications for diagnosis of catecholamine-producing tumors. *Journal of Clinical Endocrinology and Metabolism*. 2009; 94:2841–2849
- [30] Chretien PB, Engelman K, Hoyer RC, et al. Surgical Management of Intravascular Glomus Jugulare Tumor. *Am J Surg*. 1971; 122:740–743
- [31] Jackson CG, Harris PF, Glasscock MEI, et al. Diagnosis and Management of Paragangliomas of the Skull Base. *Am J Surg*. 1990; 159:389–393
- [32] Farrior JB, Hyams VJ, Benke RH, et al. Carcinoid Apudoma Arising in a Glomus Jugulare Tumor: Review of Endocrine Activity in Glomus Jugulare Tumors. *Laryngoscope*. 1980; 90:110–119
- [33] Jensen NF. Glomus Tumors of the Head and Neck: Anesthetic Considerations. *Anesth Analg*. 1994; 78:112–119
- [34] Jackson CG, Glasscock ME, Nissen AJ, et al. Glomus Tumor Surgery: The Approach, Results, and Problems. *Otolaryngol Clin North Am*. 1982; 15:897–916
- [35] Kim J-A, Elkon D, Lim M-L, et al. Optimum Dose of Radiotherapy for Chemodectomas of the Middle Ear. *International Journal of Radiation Oncology and Biological Physics*. 1980; 6:815–819
- [36] Cummings BJ, Beale FA, Garrett PG, et al. The Treatment of Glomus Tumors in the Temporal Bone by Megavoltage Radiation. *Cancer*. 1984; 53:2635–2640
- [37] Spector GJ, Fierstein J, Ogura JH. A Comparison of Therapeutic Modalities of Glomus Tumors in the Temporal Bone. *Laryngoscope*. 1976; 86:690–696
- [38] Schmidek HH, Sweet WH. *Operative Neurosurgical Techniques*. New York 1982
- [39] Hatfield PM, James AE, Schulz MD. Chemodectomas of the Glomus Jugulare. *Cancer*. 1972; 30:1164–1168
- [40] Ciric I, Zivin I. Neuroepithelial (Colloid) Cysts of the Septum Pellucidum. *J Neurosurg*. 1975; 43:69–73
- [41] Hernesiemi J, Leivo S. Management outcome in third ventricular colloid cysts in a defined population: a series of 40 patients treated mainly by transcallosal microsurgery. *Surg Neurol*. 1996; 45:2–14
- [42] Giantini-Larsen A M, Garton ALA, Villamater FN, et al. Familial colloid cysts: not a chance occurrence. *J Neurooncol*. 2022; 157:321–332
- [43] Beaumont TL, Limbrick DD Jr, Rich KM, et al. Natural history of colloid cysts of the third ventricle. *J Neurosurg*. 2016; 125:1420–1430
- [44] Guner M, Shaw MDM, Turner JW, et al. Computed Tomography in the Diagnosis of Colloid Cyst. *Surg Neurol*. 1976; 6:345–348
- [45] Ryder JW, Kleinschmidt BK, Keller TS. Sudden Deterioration and Death in Patients with Benign Tumors of the Third Ventricle Area. *J Neurosurg*. 1986; 64:216–223
- [46] Tatter SB, Ogilvy CS, Golden JA, et al. Third ventricular xanthogranulomas clinically and radiologically mimicking colloid cysts. Report of two cases. *J Neurosurg*. 1994; 81:605–609
- [47] Mamourian AC, Cromwell LD, Harbaugh RE. Colloid Cyst of the Third Ventricle: Sometimes More Conspicuous on CT Than MR. *American Journal of Neuroradiology*. 1998; 19:875–878
- [48] Pollock BE, Schreiner SA, Huston J, 3rd. A theory on the natural history of colloid cysts of the third ventricle. *Neurosurgery*. 2000; 46:1077–81; discussion 1081–3
- [49] El Khoury C, Brugieres P, Decq P, et al. Colloid cysts of the third ventricle: are MR imaging patterns predictive of difficulty with percutaneous treatment? *AJNR Am J Neuroradiol*. 2000; 21:489–492
- [50] Torkildsen A. Should Extirpation be Attempted in Cases of Neoplasm in or Near the Third Ventricle of the Brain? Experiences with a Palliative Method. *J Neurosurg*. 1948; 5:249–275
- [51] Sheikh AB, Mendelson ZS, Liu JK. Endoscopic versus microsurgical resection of colloid cysts: a systematic review and meta-analysis of 1,278 patients. *World Neurosurg*. 2014; 82:1187–1197
- [52] Pollock BE, Huston J, 3rd. Natural history of asymptomatic colloid cysts of the third ventricle. *Journal of Neurosurgery*. 1999; 91:364–369
- [53] Beaumont TL, Limbrick DD, Patel B, et al. Surgical management of colloid cysts of the third ventricle: a single-institution comparison of endoscopic and microsurgical resection. *J Neurosurg*. 2022; 1–9
- [54] Apuzzo MJ. Comment on Garrido E, et al.: Cerebral Venous and Sagittal Sinus Thrombosis After Transcallosal Removal of a Colloid Cyst of the Third Ventricle: Case Report. *Neurosurgery*. 1990; 26
- [55] Bosch DA, Rahn T, Backlund EO. Treatment of Colloid Cyst of the Third Ventricle by Stereotactic Aspiration. *Surg Neurol*. 1978; 9:15–18
- [56] Rivas JJ, Lobato RD. CT-Assisted Stereotactic Aspiration of Colloid Cysts of the Third Ventricle. *J Neurosurg*. 1985; 62:238–242
- [57] Mathiesen T, Grane P, Lindquist C, et al. High Recurrence Rate Following Aspiration of Colloid Cysts in the Third Ventricle. *Journal of Neurosurgery*. 1993; 78:748–752
- [58] Musolino A, Fosse S, Munari C, et al. Diagnosis and Treatment of Colloid Cysts of the Third Ventricle by Stereotactic Drainage. Report on Eleven Cases. *Surg Neurol*. 1989; 32:294–299

- [59] Kondziolka D, Lunsford LD. Stereotactic Management of Colloid Cysts: Factors Predicting Success. *J Neurosurg.* 1991; 75:45–51
- [60] Hall WA, Lunsford LD. Changing Concepts in the Treatment of Colloid Cysts. An 11-Year Experience in the CT Era. *J Neurosurg.* 1987; 66:186–191
- [61] Di Costanzo A, Tedeschi G, Di Salle F, et al. Pineal Cysts: An Incidental MRI Finding? *Journal of Neurology, Neurosurgery and Psychiatry.* 1993; 56:207–208
- [62] Hasegawa A, Ohtsubo K, Mori W. Pineal Gland in Old Age: Quantitative and Qualitative Morphological Study of 168 Human Autopsy Cases. *Brain Research.* 1987; 409:343–349
- [63] Torres A, Krisht AF, Akouri S. Current Management of Pineal Cysts. *Contemporary Neurosurgery.* 2005; 27:1–5
- [64] Maurer PK, Ecklund J, Parisi JE, et al. Symptomatic Pineal Cysts: Case Report. *Neurosurgery.* 1990; 27:451–454
- [65] Wisoff JH, Epstein F. Surgical Management of Symptomatic Pineal Cysts. *Journal of Neurosurgery.* 1992; 77:896–900
- [66] Klein P, Rubinstein LJ. Benign Symptomatic Glial Cysts of The Pineal Gland: A Report of Seven Cases and Review of the Literature. *Journal of Neurology, Neurosurgery and Psychiatry.* 1989; 52:991–995
- [67] Mamourian AC, Towfighi J. Pineal Cysts: MR Imaging. *American Journal of Neuroradiology.* 1986; 7:1081–1086
- [68] Nevins EJ, Das K, Bhojak M, et al. Incidental Pineal Cysts: Is Surveillance Necessary? *World Neurosurg.* 2016; 90:96–102
- [69] Jussila MP, Olsen P, Salokorpi N, et al. Follow-up of pineal cysts in children: is it necessary? *Neuroradiology.* 2017; 59:1265–1273
- [70] Di Chirico A, Di Rocco F, Velardi F. Spontaneous regression of a symptomatic pineal cyst after endoscopic third-ventriculostomy. *Childs Nerv Syst.* 2001; 17:42–46
- [71] Berhouma M, Ni H, Delabar V, et al. Update on the management of pineal cysts: Case series and a review of the literature. *Neurochirurgie.* 2015; 61:201–207
- [72] Wenger Markus, Simko Marian, Markwalder Regula, et al. An entirely suprasellar Rathke's cleft cyst: case report and review of the literature. *Journal of Clinical Neuroscience.* 2001; 8:564–567
- [73] Maggio WW, Cail WS, Brookeman JR, et al. Rathke's Cleft Cyst: Computed Tomographic and Magnetic Resonance Imaging Appearances. *Neurosurgery.* 1987; 21:60–62
- [74] Nishio S, Mizuno J, Barrow DL, et al. Pituitary Tumors Composed of Adenohypophyseal Adenoma and Rathke's Cleft Cyst Elements: A Clinicopathological Study. *Neurosurgery.* 1987; 21:371–377
- [75] Voelker JL, Campbell RL, Muller J. Clinical, Radiographic, and Pathological Features of Symptomatic Rathke's Cleft Cysts. *Journal of Neurosurgery.* 1991; 74:535–544
- [76] Han SJ, Rolston JD, Jahangiri A, et al. Rathke's cleft cysts: review of natural history and surgical outcomes. *J Neurooncol.* 2014; 117:197–203
- [77] Benveniste RJ, King WA, Walsh J, et al. Surgery for Rathke cleft cysts: technical considerations and outcomes. *J Neurosurg.* 2004; 101:577–584
- [78] Byun WM, Kim OL, Kim D. MR imaging findings of Rathke's cleft cysts: significance of intracystic nodules. *AJNR Am J Neuroradiol.* 2000; 21:485–488
- [79] Aho CJ, Liu C, Zelman V, et al. Surgical outcomes in 118 patients with Rathke cleft cysts. *J Neurosurg.* 2005; 102:189–193
- [80] Sanno N, Oyama K, Tahara S, et al. A survey of pituitary incidentaloma in Japan. *Eur J Endocrinol.* 2003; 149:123–127
- [81] Kino H, Akutsu H, Tanaka S, et al. Endoscopic endonasal cyst fenestration into the sphenoid sinus using the mucosa coupling method for symptomatic Rathke's cleft cyst: a novel method for maintaining cyst drainage to prevent recurrence. *J Neurosurg.* 2019; 1–11
- [82] Lillehei KO, Widdel L, Astete CA, et al. Transsphenoidal resection of 82 Rathke cleft cysts: limited value of alcohol cauterization in reducing recurrence rates. *J Neurosurg.* 2011; 114:310–317
- [83] Haughton VM, Rosenbaum AE, Williams AL, et al. Recognizing the empty sella by CT: the infundibulum sign. *AJR Am J Roentgenol.* 1981; 136:293–295
- [84] Kaufman B. The "empty" sella turcica - A manifestation of the intrasellar subarachnoid space. *Radiology.* 1968; 90:931–941
- [85] Perani D, Scotti G, Colombo N, et al. Spontaneous CSF rhinorrhea through the lamina cribrosa associated with primary empty sella. *Italian Journal of Neurological Sciences.* 1984; 5:167–172
- [86] Lee WM, Adams JE. The Empty Sella Syndrome. *J Neurosurg.* 1968; 28:351–356
- [87] Alvarez Berastegui G R, Raza SM, Anand VK, et al. Endonasal endoscopic transsphenoidal chiasmectomy using a clival cranial base cranioplasty for visual loss from massive empty sella following macroprolactinoma treatment with bromocriptine: case report. *J Neurosurg.* 2015; 1–7
- [88] Fouad W. Review of empty sella syndrome and its surgical management. *Alexandria Journal of Medicine.* 2011; 47:139–147

57 Pseudotumor Cerebri Syndrome (PTCS)

57.1 General information

Key concepts

- increased ICP in the absence of a mass, hydrocephalus, CNS infection, or malignant hypertension
- primarily a neuro-ophthalmological (due to visual loss) and neurologic (due to H/A) condition
- a preventable cause of (often permanent) blindness from optic nerve atrophy
- may be primary (idiopathic intracranial hypertension) or secondary (to other conditions)
- more common in women of childbearing age who are either obese (BMI ≥ 30 kg/m²) or who have recently gained weight than in the general population
- diagnostic criteria for pseudotumor cerebri syndrome (PTCS):
 - papilledema
 - ICP elevation ≥ 25 cm CSF on a properly performed LP
 - normal CSF analysis (no pleocytosis, elevated protein, hypoglycorrhachia, abnormal cytology, or other indication of infection or malignancy)
 - no focal neurologic deficit. Allowed exception: abducens (CN VI) palsy
 - no intracranial mass, infection, hydrocephalus, or hypertensive encephalopathy
 - normal neuroimaging. Allowed exceptions: slit-like ventricles and/or findings associated with increased intracranial pressure (empty sella (p. 951), transverse sinus stenosis, or intraorbital findings including flattening of the posterior globe (p. 587), dilated optic nerve sheath (p. 587) ...)
- pseudotumor cerebri syndrome without papilledema (p. 958) (PTCSWOP) is a variant with less risk to vision and more refractory H/A. May not be the same condition
- symptoms: headache is present in > 90% of patients. However, class I evidence suggests that H/A does not correlate with opening pressure (OP) on LP
- risk of blindness correlates best with progressive visual field constriction (enlarging blind spot is characteristic) diagnosed on visual field testing, and is not reliably correlated to duration of symptoms, degree of papilledema, H/A, Snellen visual acuity, or number of recurrences
- recommended work-up:
 - ✓ brain MRI (without and with contrast) to include craniocervical junction (to R/O Chiari malformation)
 - ✓ brain MRV to evaluate dural sinuses for patients who are male or not obese, or who have progression of visual deficit despite treatment
 - ✓ LP: measure opening pressure (OP) in lateral decubitus position, and send CSF at a minimum for cell count, protein and glucose, cytology and culture (aerobic, anaerobic, fungal, and TB)
 - ✓ BP (blood pressure) to R/O hypertensive encephalopathy
 - ophthalmologic eval. Priorities: ✓ visual fields and ✓ papilledema
 - Secondary: ✓ acuity, color vision, intraocular pressure, EOM testing for abducens palsy
 - assess for common associated conditions: obstructive sleep apnea, hypervitaminosis A or use of retinoids, iron deficiency anemia...
- long-term treatment depends on weight loss and appropriate H/A treatment
- neurosurgical involvement is generally restricted to
 - acute intervention for threatened vision (in cases of fulminant PTCS (p. 967))
 - treating surgical conditions that are not PTSC (tumors, subdural hematomas, hydrocephalus, cerebral hemorrhages...)
 - in a shunted patient, diagnosing and correcting overshunting if present
 - in limited circumstances: treating refractory H/A accompanied by increased ICP, preferably under advisement of a multidisciplinary panel if available
- nonsurgical treatment:
 - weight loss: the only disease modifying treatment. 6-15% weight loss usually resolves papilledema. Dietary consultation and/or bariatric surgery may help
 - medical management: provides modest improvement. Acetazolamide (Diamox®) is initial drug of choice. Alternative: Topamax
- procedural interventions: not recommended unless vision threatened. Primarily:
 - optic nerve sheath fenestration (ONSF): best for visual loss without H/A
 - CSF shunt: better than ONSF for refractory H/A associated with visual loss

- stenting for bilateral focal transverse sinus stenosis: controversial. Requires antiplatelet drugs X $\approx$ 6 mos afterwards, which precludes other procedures (e.g., ONSF, shunting, LP...) during that time. Possible indications: a gradient > 8 mmHg (some say 5) across the stenosis in the presence of refractory increased ICP

First described in 1897 by Heinrich Quincke,¹ who called it “serous meningitis.”

The nomenclature has varied widely over the years, and terms used include²: otitic hydrocephalus, pseudotumor cerebri (which was abandoned for a time and is now embraced), idiopathic intracranial hypertension (IIH) and the repudiated term which should be avoided “benign intracranial hypertension” (because loss of vision and reduced quality of life from debilitating headaches are not benign).

The following taxonomy has been endorsed in the U.S.³ and abroad⁴:

► **Pseudotumor cerebri syndrome (PTCS).** A group of conditions characterized by increased intracranial pressure with no evidence of intracranial mass, hydrocephalus, infection (e.g., meningitis, especially chronic ones such as fungal meningitis), or hypertensive encephalopathy (see ► Table 57.3 for diagnostic criteria) which encompasses:

1. idiopathic intracranial hypertension (IIH) AKA primary pseudotumor cerebri. Includes: patients with obesity, recent weight gain, polycystic ovarian syndrome and thin children
2. secondary pseudotumor cerebri. Underlying causes include: conditions that increase intracranial venous pressure (dural venous sinus thrombosis, jugular vein thrombosis, superior vena cava syndrome...), certain medications and medical conditions (see below)

Either primary or secondary PTSC patients may have either:

1. papilledema (p.586): this is considered “pseudotumor cerebri syndrome” or
2. “pseudotumor cerebri syndrome without papilledema (p.958)” (PTCSWOP) which may be diagnosed in patients without papilledema (Frisen stage 0 ► Table 33.1) meeting PTCSWOP criteria in ► Table 57.3

57.2 Epidemiology

1. female:male ratio reported ranges from 2:1 to 8:1 (no gender difference in juvenile form)
2. obesity (BMI > 30 kg/m²) is reported in 11–90% of cases and is not as prevalent in men with PTCS⁵
3. incidence among obese women of childbearing years^{6,7}: 19–21/100,000 (compared to incidence in general population²: 1–2/100,000). Also occurs in patients with recent weight gain (5–15% weight gain in the past year)⁸
4. peak incidence in 3rd decade (range: 1–55 years). 37% of cases are in children, 90% of these are age 5–15 years
5. PTCS is rare in age < 3 years or > 60 years: in these age groups other causes of increased ICP should be strongly considered³
6. frequently self limited: recurrence rate: 9–43%
7. severe visual deficits develop in 4–12%, unrelated to duration of symptoms, degree of papilledema, headache, visual obscuration, and number of recurrences.⁹ Perimetry is the best means to detect and follow visual loss

57.3 Natural history

Spontaneous resolution may occur, sometimes within months, but usually after $\approx$ 1 year. Papilledema persists in $\approx$ 15%. *Permanent* visual loss occurs in 2–24% (depending on criteria used and degree to which it is sought). Persistent H/A may occur. PTCS recurs in $\approx$ 10% after initial resolution.¹⁰

57.4 Associated conditions

57.4.1 General information

Some cases of PTCS are idiopathic (IIH), i.e., primary. The remainder are secondary to another condition, e.g., transverse sinus thrombosis, hypervitaminosis A... (see below). Many conditions cited as

being associated with PTCS may be coincidental. Four criteria suggested to establish a cause-effect relationship are shown in ► Table 57.1.¹¹

Table 57.1 Criteria for causality of PTCS by another condition¹¹

1. meets Dandy's criteria (► Table 57.4)
2. the condition should be proven to increase ICP
3. treatment of the condition should improve the PTCS
4. properly controlled studies should show an association between the condition and PTCS

► Table 57.2 shows a scale¹² to rank the likelihood of association between various conditions and PTCS that is somewhat dated.

Table 57.2 Conditions that may be associated with PTCS (modified¹²)

Proven association

Meets 4 criteria from ► Table 57.1

- obesity. Recent weight gain (5-15% weight gain in the year prior to the diagnosis of PTCS was associated with increased risk of PTCS)⁸

Likely association

Meets 3 criteria from ► Table 57.1

- drugs: keprone, lindane
- hypervitaminosis A or use of retinoids

Probable association

Meets 2 criteria from ► Table 57.1

- steroid withdrawal^a
- thyroid replacement in children
- ketoprofen & indomethacin in Bartter syndrome
- hypoparathyroidism
- Addison's disease^a
- uremia
- iron deficiency anemia
- drugs: tetracycline, nalidixic acid, Danazol, lithium, amiodarone, phenytoin, nitrofurantoin, ciprofloxacin, nitroglycerin

Possible association

Meets 1 criteria from ► Table 57.1

- menstrual irregularity
- oral contraceptive use^b
- Cushing's syndrome
- vitamin A deficiency
- minor head trauma
- Behçet syndrome

Unlikely association

Meets none of the criteria in ► Table 57.1

- hyperthyroidism
- steroid use
- immunization

Unsupported association

- pregnancy
- menarche

^amay respond to steroids

^bmay be associated with dural sinus thrombosis, see text

Other conditions not included in this list that meet minimal criteria but are unconfirmed in case-control studies² include:

1. other drugs: isotretinoin (Accutane®), trimethoprim-sulfamethoxazole, cimetidine, tamoxifen, fluoroquinolones¹³
2. systemic lupus erythematosus (SLE)
3. obstructive sleep apnea (OSA): hypercarbia increases ICP

Conditions that may be related by virtue of increased pressure in the dural sinuses (see below):

1. otitis media with petrosal extension (so-called otitic hydrocephalus)
2. radical neck surgery with resection of the jugular vein
3. hypercoagulable states

57.4.2 Venous hypertension and sinovenous abnormalities

Venous hypertension has often been proposed as a unifying underlying cause of PTCS. Abnormalities of the dural sinuses, including thrombosis, stenosis,¹⁴ obstruction, or elevated pressure (reaching levels as high as 40 mm Hg) have been demonstrated in a number of studies. While these findings may underlie a number of cases, in many cases they may actually be epiphenomena (e.g., venous hypertension may be due to compression of the transverse sinuses by elevated intracranial pressure¹⁵), and it is unlikely that such abnormalities will explain all cases.

Bilateral sinovenous stenosis was seen (using 3D gadolinium-enhanced MRI venography) in 27 of 29 patients with PTCS and in only 4 of 59 controls.¹⁴

57.4.3 CSF fistula (leaks)

A subgroup of PTCS patients also have CSF fistula.¹⁶ It is possible that the fistula arises as a result of chronic increased ICP (many years) causing erosion of the skull base and dural defects.¹⁷ CSF fistula may actually protect against papilledema in patients with PTCS,¹⁸ and cases have been reported where repair of the fistula results in increased ICPs to levels similar to PTCS without CSF fistula.¹⁷ Furthermore, fistula repair has a higher rate of failure if the elevated ICP is not also treated.¹⁷

57.5 Diagnostic criteria

Diagnostic criteria are summarized in ► Table 57.3. Details of the criteria components appear in following sections.

Modified²¹ Dandy's criteria²² are still referenced especially in research articles (e.g., the IIHTT²¹), and are shown in ► Table 57.4.

► **Pseudotumor cerebri syndrome without papilledema (PTCSWOP).** May be diagnosed in patients without papilledema (Frisen stage 0 ► Table 33.1) meeting criteria in ► Table 57.3. Diagnosing PTCSWOP can be difficult, and subtle findings should be ruled out by an experienced trained observer.²³ Papilledema may not be discernible in the presence of optic atrophy.

This may not be the same disease entity as PTCS.³ Compared to patients with papilledema, PTCSWOP patients typically have:

1. lower opening pressure on LP: mean: 30.9 cm CSF (which is still abnormally elevated), vs. 37.3 in those with papilledema
2. higher incidence of photopsia (flashes of light)
3. higher incidence of non-physiologic visual field constriction²³ which tend not to progress over time^{23,24}
4. lower incidence of abducens palsy
5. higher incidence of normal visual fields on presentation²³
6. H/A that are more refractory to treatment^{23,25}
7. preserved venous pulsation on funduscopy²³

57.6 Clinical findings

57.6.1 Symptoms

See references.^{11,26,27,28}

1. headache (H/A), see details (p.960)
 - a) the most common symptom, reported in > 90%²⁹ (range: 76–99%)
 - b) results in significant disability and reduced quality of life³⁰
2. tinnitus: 52–61%. The causal relationship with PTCS has been demonstrated by resolution of these symptoms with reduction of CSF pressure. Usually pulse synchronous. Described as rushing noise. May be unilateral (in these, tinnitus may be reduced by ipsilateral jugular vein compression + ipsilateral head rotation)
3. visual loss, see details (p.961)
 - transient visual obscuration (TVO): 68–72%

Table 57.3 Diagnostic criteria for pseudotumor cerebri syndrome (PTCS)³

Criteria	PTCS		PTCS without papilledema	
	Definite	Probable	Definite	Suggested
papilledema ^a	+	+	-	-
normal neurologic exam ^b	+	+	+	+
normal neuroimaging ^c	+	+	+	+
normal CSF composition ^d	+	+	+	+
OP > 25 cm CSF on LP ^e	+	-	+	+
abducens nerve palsy (p. 598) ^f			+	-
≥ 3 neuroimaging criteria ^g			-	+

^a (see text (p. 960) for description) may be asymmetric or (uncommonly) unilateral. NB: papilledema may not be reliably diagnosable in the presence of optic atrophy

^b exception: cranial nerve abnormalities (usually VI or VII). For other deficits (e.g., encephalopathy, focal deficit...) other diagnoses should be sought

^c MRI (or CT if MRI cannot be done) without and with contrast. May show findings of increased ICP (see g below), but must show normal brain parenchyma without hydrocephalus, mass or structural lesion, and

- for typical patient (specifically: obese (BMI ≥ 30 kg/m²) female adult or adolescent¹⁹):
no abnormal meningeal enhancement
- for others (men, non-obese women, prepubescent children, and patients with progressive visual loss despite therapy):
no abnormal meningeal enhancement and normal MRV

NB: meningeal enhancement may occasionally be seen shortly after an LP

^d none of the following: pleocytosis, elevated protein, hypoglycorrhachia, abnormal cytology or other indication of infection or malignancy

^e opening pressure on a properly performed LP (p. 962) (see text):
in adults, or children who are not sedated or obese: ≥ 25 cm CSF,
in children who are sedated or obese: ≥ 28 cm CSF

^f unilateral or bilateral, may be complete or incomplete. Produces esotropia (inward eye deviation)

^g ≥ 3 neuroimaging criteria suggestive of increased ICP:

1. empty sella (► Fig. 56.6 (85% of PTCS had partial or complete empty sella)²⁰)
2. flattening of the posterior globe (► Fig. 33.1)
3. optic nerve sheath distension (► Fig. 33.1) with or without a tortuous optic nerve
4. transverse venous sinus (TVS) stenosis (but not thrombosis) is present in most cases of PTCS³ (see MRV (p. 964) for details)

Table 57.4 Modified modified Dandy's criteria for PTCS

- signs & symptoms of increased ICP
- no localizing signs (other than possible Cr. N VI palsy^a in an otherwise awake and alert patient)
- increased CSF pressure without chemical or cytological CSF abnormalities
- normal neuroimaging (especially no intracranial mass) aside from small ventricles and findings of increased ICP

^a abducens palsy may result from ↑ ICP (p. 598)

- blurry vision: 32%
 - afferent visual pathway injury: often reversible but may be permanent
4. diplopia: 18-30%. More common in adults, usually horizontal due to VI nerve palsy
 5. less common or uncharacteristic symptoms
 - a) neck stiffness (30-50%) or neck pain (42%)
 - b) ataxia: 4-11%
 - c) acral paresthesias: 25%
 - d) back pain: 53%
 - e) nausea: 32% (actual vomiting is less common)
 - f) retrobulbar eye pain on eye movements
 - g) arthralgia: 11-18%
 - h) dizziness: 32-52%
 - i) fatigue
 - j) reduced olfactory acuity

NB: Worsening of any of the above symptoms with postural changes that increase ICP (bending over, Valsalva maneuver...) is characteristic in PTCS.

► **Headache in PTCS.** According to The International Classification of Headache Disorders, 3rd edition (ICHD-3),^{31 (p 99)} H/A attributable to PTCS is a H/A in a patient with a nonvascular intracranial disorder known to cause H/A (here: increased ICP documented by OP ≥ 25 cm on LP) that is not better accounted for by another ICHD-3 diagnosis, and fulfills at least 2 of the following*

1. H/A developed in temporal relation to PTCS or led to its discovery
2. H/A worsens in parallel to worsening of PTCS (i.e., increasing ICP) and/or improves in parallel to improvement of PTCS (e.g., by reducing ICP)
3. H/A are typical for PTCS

* This definition has been challenged, since recent Class I evidence in a study of 165 PTCS patients with papilledema and mild vision loss²⁹ found:

1. there are no typical H/A characteristics for PTCS. H/A phenotype in PTCS changes over time, and is often mixed. 93% of patients with H/A in the IIHTT had features consistent with definite or probable migraine (68%) or tension H/A (25%). Non-migraine H/A phenotypes may be exacerbated by increased ICP
2. H/A in PTCS did not correlate with opening pressure, BMI, papilledema grade or use of acetazolamide. 62% of patients continued to have H/A after successful treatment of increased CSF pressure³²
3. ★ despite the fact that patients and providers perceive that H/As are linked to CSF pressure and that lowering CSF pressure will improve H/A control, evidence demonstrates that H/A and CSF pressure are clinically independent features of PTCS

NB: the findings in patients recruited to the study may not be generalizable e.g., to a patient who presents to the E/R with H/A and an occluded shunt.

Medication overuse headache (MOH): medication overuse (which can lead to medication overuse H/A) was found in 37% of H/A patients in the IIHTT.²⁹ MOH diagnostic criteria^{31 (p 122)}:

1. H/A occurring ≥ 15 days/month in a patient with pre-existing primary H/A, developing as a result of regular overuse of H/A medication ≥ 10 days/month (for opioids, combined preparations, ergotamines or triptans) or ≥ 15 days/month (for NSAIDs, APAP, ASA)
2. regular overuse for > 3 months
3. not better accounted for by another ICHD-3 diagnosis

MOH usually (but not always) resolve after medication overuse stops.

57.6.2 Signs

General information

- signs are generally restricted to the visual system (see below)
- ★ paradoxically *absent*: altered level of consciousness in spite of high ICP
- the presence of a focal neurologic deficit, or any unexplained cranial nerve deficit (except for abducens palsy) should prompt a search for an alternative diagnosis to PTCS
- the infantile form of PTCS may have only enlarging OFC, is frequently self-limited, and usually requires only follow-up without specific treatment

Eye findings in PTCS

Ophthalmologic consultation is generally recommended, especially in questionable cases.

1. papilledema
 - a) the sine qua non of PTCS
 - b) if papilledema is absent, PTCS without papilledema (PTCSWOP) may be diagnosed if the diagnostic criteria for that are met (► Table 57.3^{3,23,33})
 - c) findings: optic disc elevation, blurring of the disc margin, peripillary halo, obscuration by nerve fiber layer of blood vessels crossing the disc margin, venous distension, and overlying of peripapillary hemorrhages, exudates, or cotton-wool spots³
 - d) usually bilateral, may be asymmetric, infrequently unilateral¹⁴
 - e) may be mild (subtle nerve fiber elevation)
 - f) may be absent in cases of optic atrophy or resolved prior optic disc swelling³

- g) other etiologies that must be ruled out include: optic neuritis, anterior ischemic optic neuropathy, neuroretinitis, pseudopapilledema (p.588) (includes optic nerve head drusen)
2. abducens nerve (Cr. N. VI) palsy: 20%; a false localizing sign (p.598). The esotropia ranges from <5 prism diopters dysconjugate angle in primary gaze to >50¹⁰
3. ★ visual field defect: correlates with visual loss more closely than other findings (see below)
4. visual acuity: relatively insensitive assessment of visual function in PTCS (see below)
5. loss of color vision: not sensitive for early papilledema-related optic atrophy

57.6.3 Visual loss in PTCS

General information

Prevalence in PTCS: 48–68% (lower numbers generally come from population-based samples).

Visual loss in PTCS may occur early or late, and is not reliably correlated to duration of symptoms, papilledema, H/A, visual acuity (as tested with a standard Snellen eye chart), or number of recurrences. Visual loss is usually gradual and spares central vision until late in the course³⁵ and thus may not be noticed by the patient until profound. In some cases visual loss can be sudden.

Visual field deficits are the most sensitive means for detecting visual loss in PTCS.

Pathomechanics

Increased ICP is transmitted along the optic nerve sheath → circumferential compression of the retinal ganglion cell axons at the level of the lamina cribrosa.¹⁰

Manifestations

1. transient visual obscurations (TVO): graying or blacking out of vision. Lasts ≈ 1 second. Uni- or bilateral. Typically occur with eye movement, bending over, or Valsalva maneuver. Directly proportional to severity of papilledema. Frequency of TVOs parallels ICP elevation, but doesn't correlate with permanent visual loss
2. ★ visual field deficits: the most sensitive measure of visual impairment in PTCS, found in 96% of 50 patients in one prospective series.³⁶ Visual field defects are diagnosed by perimetry (p.589) (e.g., Goldmann or Humphrey). Automated perimetry can be quantified (to generate a measurement for comparison) as the perimetric mean deviation (PMD) (p.589)
 - a) enlarged blind spots are the classic field defect in PTCS, but are rarely noticed by the patient until there is encroachment on macular vision (fixation)³⁵
 - b) early changes³⁵: constriction of fields usually with nasal quadrant defect (“nasal step”) and arcuate defects (a curved visual defect) neither of which cross the horizontal meridian
 - c) late: central vision is affected. Findings include: concentric constrictions, enlarged blind spot, cecentral scotomas...³⁵
3. less common visual symptoms³⁷: blurry vision (32%), diplopia (18%) (usually horizontal, may be due to abducens palsy)

57.7 Differential diagnosis

1. true mass lesions: tumor, cerebral abscess, subdural hematomas, rarely gliomatosis cerebri may be undetectable on CT and will be misdiagnosed as PTCS
2. hydrocephalus: typically ventricles are enlarged. Patients generally have obtundation, lethargy, vomiting... Some patients with reduced cerebral compliance may develop increased ICP without ventriculomegaly
3. cranial venous outflow impairment (some authors include these in IIH,³⁸ others do not)
 - a) dural sinus thrombosis (p.1594)
 - b) congestive heart failure
 - c) superior vena cava syndrome
 - d) unilateral or bilateral jugular vein or sigmoid sinus³⁹ obstruction
 - e) hyperviscosity syndromes
 - f) Masson's vegetant intravascular hemangioendothelioma⁴⁰: an uncommon, usually benign lesion that may rarely involve the neuraxis (including intracranial occurrence). Not definitely neoplastic. Organizing thrombi develop endothelialized projections into the vessel lumen. Must be distinguished from other conditions such as angiosarcoma
4. Chiari I malformation (CIM) (see Chiari-pseudotumor cerebri syndrome (p.301))
 - a) CIM may produce clinical findings similar to PTCS

- b) 6-21% of PTCS patients have tonsillar descent ≥ 5 mm below the FM on MRI^{38,41,42}
- c) $\approx 5\%$ of patients with CIM have papilledema¹⁰
- d) if severe, LP is contraindicated
- e) in some cases, tonsillar descent may be a result of intracranial hypotension (p. 421) (low CSF pressure) possibly as a result of a spinal CSF leak⁴³
- 5. infection (CSF will be abnormal in most of these): encephalitis, arachnoiditis, meningitis (especially basal meningitis or granulomatous infections, e.g., syphilitic meningitis, chronic cryptococcal meningitis), chronic brucellosis, cat scratch disease (in rare cases may produce neuroretinitis⁴⁴)
- 6. inflammatory conditions: e.g., neurosarcoidosis (p. 198), SLE
- 7. vasculitis: e.g., Behçet's syndrome
- 8. metabolic conditions: e.g., lead poisoning
- 9. **malignant hypertension**: (DBP ≥ 120 mm Hg or SBP ≥ 180 mm Hg⁴⁵). May produce H/A and bilateral optic disc edema, which can be indistinguishable from papilledema. May also produce hypertensive encephalopathy (p. 202). Check BP in all PTCS suspects
- 10. meningeal carcinomatosis
- 11. Guillain-Barré syndrome (p. 193): CSF protein is usually elevated
- 12. glaucoma: visual field defects of PTCS mimic those of glaucoma except that glaucoma does not typically have an enlarged blind spot³⁵
- 13. following head trauma

57.8 Evaluation

57.8.1 Overview

Investigations that may help confirm the diagnosis, or rule out conditions that may mimic PTCS, or identify underlying causes of secondary PTCS:

- 1. history to identify other possible causes of increased ICP (► Table 57.2)
- 2. VS: check BP to R/O malignant HTN (which can cause optic disc edema, see above)
- 3. ophthalmologic evaluation. Includes: dilated funduscopic exam for papilledema (see below) and formal visual field testing (to rule-out visual loss)
- 4. routine serum labs: CBC (to rule-out anemia), electrolytes, PT/PTT
- 5. imaging:
 - a) cerebral MRI without and with contrast for all patients (see details below). CT without and with contrast is performed if MRI is not an option
 - b) contrast MRV (p. 964): obtain in atypical patients (pre-pubertal children, males, non-obese females) or in patients with progressive visual loss despite therapy. In practice, this is usually ordered along with the initial MRI
- 6. LP: measure opening pressure (OP) with unsedated patient in the lateral decubitus position (see below), and obtain CSF for laboratory analysis (no proven benefit of large volume tap)
- 7. evaluation for obstructive sleep apnea (hypercarbia increases ICP)
- 8. W/U for systemic conditions (e.g., sarcoidosis or SLE if other findings are suggestive, e.g., cutaneous nodules, hypercoagulable state...)

57.8.2 Ophthalmologic evaluation

A thorough exam consists of all of the following. The first 2 items are the most critical.

- 1. **dilated funduscopy exam** for papilledema $\pm$ fundus photographs
- 2. **visual field testing** may be done at bedside initially by a proficient examiner, however formal testing using quantitative perimetry with evaluation of size of blind spot and determination of perimetric mean deviation (PMD) (p. 589) should be done as soon as practical
- 3. visual acuity testing
- 4. color vision testing
- 5. intraocular pressure measurement

57.8.3 Lumbar puncture

If contraindications to LP (p. 1811) have been ruled out on neuroimaging, perform an LP to 1) measure opening pressure and, 2) to obtain CSF for analysis. The benefit of removing a large volume of CSF (≥ 20 ml) is uncertain.³

Opening pressure (OP): key requirements for a properly performed LP to diagnose PTCS:

- position: lateral decubitus position (requisite for diagnosis,³) measurement is most accurate in this position without knee flexion. Other positions may increase ICP:
 - prone position: typically used for LP under fluoroscopy, may ↑ intra-thoracic pressure → decrease venous return from the brain → increased ICP
 - sitting position: increases measured pressure due to column of CSF above the needle
 - if the LP needle has to be inserted with the patient prone or sitting, once CSF is coming through the needle, replace the stylet and reposition the patient to the lateral decubitus and then remove the stylet to accurately measure OP
 - knee flexion may increase ICP (controversial in children⁴⁶)
- avoid sedation which may cause hypoventilation → hypercapnia → increased ICP⁴⁷
- keep the base of the manometer level with the right cardiac atrium
- avoid valsalva maneuver⁴⁸ (including patient crying or forced knee-chest position)
- magnitude: the higher the OP, the more likely that it is pathologically elevated.³⁷ Suggested cutoffs:
 - OP < 25 cm CSF: normal
 - ★ OP ≥ 25 cm CSF: required for diagnosis of PTCS by criteria,³ however there is some dissent
 - OP ≥ 30 cm CSF: almost universally accepted as abnormally elevated
 - in prepubertal children: OP ≥ 28 cm CSF (or ≥ 25 cm if the child is not sedated and not obese) meets diagnostic criteria³

NB: OP on LP is a snapshot of CSF pressure. Diurnal variations in CSF pressure may occasionally cause a falsely low (i.e., normal) reading. ∴ If clinical suspicion is high, an LP at a different time of day or continuous ICP monitoring may be required
- CSF analysis to rule out infection (especially chronic, e.g., fungus, TB, or Lyme disease), inflammation (e.g., sarcoidosis, SLE), or neoplasm (e.g., carcinomatous meningitis). Minimum:
 - protein/glucose
 - cell count
 - routine and fungal cultures
 - cytology if suspicion of carcinomatous meningitis

57.8.4 MRI of the brain

Intraparenchymal abnormalities are usually absent or minimal.

Slit ventricles are common (however, the incidence may be no higher in PTCS than in age-matched controls⁴⁹).

Tonsillar ectopia (Chiari I) is more prevalent in PTCS than controls (see above)

Most findings are those of increased ICP (not specific just for PTCS):

- empty sella (p.951) in 30–70%. Includes partially empty sella or decreased pituitary height
- dural sinuses stenosis: (MRV is more sensitive—see below for findings)
- Meckel's cave enlargement⁵⁰ ▶ Fig. 57.1 or compression⁵¹ which may depend on the patency of the opening (porus trigeminus)

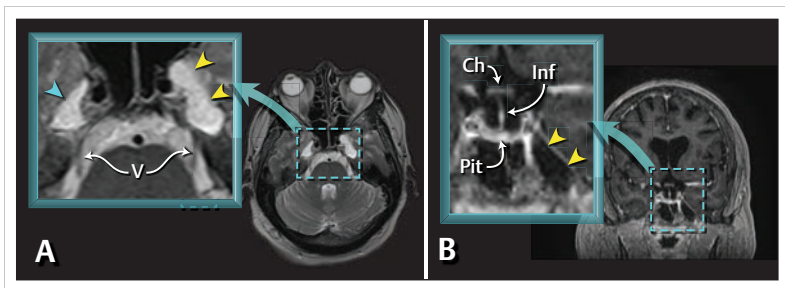


Fig. 57.1 Dilated Meckel's cave.

Image: MRI A: axial T2, B: coronal T1. Meckel's cave is mildly dilated on the right (blue arrowhead), and markedly dilated on the left (yellow arrowheads). The patient also has an empty sella demonstrating the "infundibulum sign" in B (see also ▶ Fig. 56.6).

Abbreviations: Ch = optic chiasm; Inf = infundibulum (pituitary stalk); Pit = pituitary gland; V = trigeminal nerve (Cr.N. V) as it crosses the CPA cistern.

- d) intraorbital findings consistent with increased ICP include¹⁰:
- a) flattening of the posterior globe/sclera: (► Fig. 33.1) occurs in 80%
 - b) enhancement of the prelaminar optic nerve (the prelaminar layer is essentially the middle layer of the retina): in 50%
 - c) distention of the perioptic subarachnoid space (optic nerve sheath): (► Fig. 33.1) in 45%
 - d) vertical tortuosity of the orbital optic nerve: in 40% (the intraorbital optic nerve normally has a slight "S" shape)
 - e) intraocular protrusion of the prelaminar optic nerve head: in 30%

57.8.5 MRV (magnetic resonance venography)

MR venography (MRV) has largely replaced conventional venography to rule out dural sinus or venous thrombosis or AV fistula.

Indications: atypical PTCS patients (i.e., patients who are not both obese and female) or those with progressive visual loss despite therapy. If the first MRV is negative or equivocal and the suspicion of sinus thrombosis is high, a CT venogram, catheter angiogram or second MRV is indicated.

Sequences: maximum intensity projection (MIP) reconstructions from gadolinium-enhanced MRV is recommended, since time-of-flight MRV tends to overestimate stenosis.⁵²

Findings: attenuation/stenosis (but not thrombosis) of dural venous sinuses is present in most cases of PTCS.³ Includes bilateral transverse venous sinus (TVS) stenosis or stenosis of a dominant TVS (normal transverse sinuses are often asymmetric, with the right usually larger/dominant) (► Fig. 57.2). Bilateral TVS stenosis is uncommon in healthy individuals⁵³

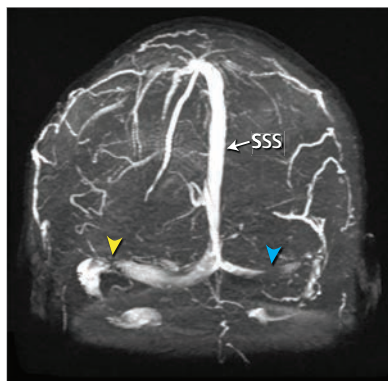


Fig. 57.2 Transverse sinus stenosis.

Image: coronal gadolinium MRV showing focal high-grade stenosis within the distal third of the dominant (larger) right transverse sinus (yellow arrowhead) and a longer area of stenosis in the non-dominant left transverse sinus (blue arrowhead). Abbreviation: SSS = superior sagittal sinus.

CT of the brain

MRI and MRV are preferred. Brain CT scan is less sensitive, but may be used if MRI is not an option. CT without and with IV contrast is usually adequate to R/O intracranial mass as a possible cause of intracranial hypertension. CT venography (CVT) may evaluate dural sinus compression.

57.9 Treatment and management

57.9.1 Treatment goals

- protect vision
- minimize headache morbidity

57.9.2 Interventional options

Overview

Initial management: usually consists of attempt at weight loss and pharmacologic therapy. However some patients will have acute (fulminant (p.967)) or progressive or persistent symptoms and require more aggressive intervention. Treatment options are shown below, management guidelines follow (p.967).

► **Treatment options.** This is only a catalogue of options. For *guidelines* on how to use them, see the list (p.967) that follows.³⁷

1. weight loss
2. medical treatment
 - a) fluid and salt restriction, usually combined with carbonic anhydrase inhibitors
 - b) carbonic anhydrase inhibitors and other diuretics: slow CSF production (see below)
 - c) headaches should be characterized by phenotype and treated accordingly (p.960) by a practitioner experienced in H/A management
 - d) ✖ corticosteroids are not recommended due to the lack of sound clinical evidence.³⁷ Prolonged use may also exacerbate weight gain and increase the risk of venous thrombosis
3. treat obstructive sleep apnea (OSA) if present (hypercarbia increases ICP): e.g., CPAP
4. *surgical therapy* only for cases refractory to above, or where visual loss is progressive or is severe initially or unreliable patient
 - a) CSF diversion (shunts): see below
 - b) optic nerve sheath fenestration (p.966) (ONSF)
 - c) procedures that may be reserved for very specific situations
 - serial LPs: remove up to 30 ml to halve OP, perform qod until OP < 20 cm CSF, then decrease to q wk. Using large gauge needle (e.g., 18 Ga) may help promote a (desirable) post-LP CSF leak into subcutaneous tissues. LPs may be difficult in obese patients. Not recommended routinely.³⁷
 - **Side effects:** include sciatica from nerve root irritation, acquired cerebellar tonsillar herniation (p.456), spinal H/A (from intracranial hypotension), back pain, risk of spinal hematoma (epidural or subdural)
 - decompressive craniectomy, usually subtemporal decompression (p.967)
5. endovascular procedures (p.966): transverse sinus stenting

Weight loss

This is the only disease-modifying therapy for PTCS. Patients with BMI ≥ 30 kg/m² should be counseled regarding weight management in a sensitive manner. The amount of weight loss required to arrest PTCS is not known. 15% weight loss was required in one cohort,⁵⁴ 6% weight loss resulted in 3 Frisen papilledema grade improvement (► Table 33.1) in a smaller study.⁵⁵

NB: resolution of PTCS is too slow for acutely threatened vision. Also, symptoms often recur if the weight is regained.

- dieting: uncontrolled studies⁵⁶ suggest that this is effective, but it is difficult to accomplish and maintain
- bariatric surgery: gastric bypass, laparoscopic banding... may help

Diuretics

1. carbonic anhydrase (CA) inhibitors:
 - a) **acetazolamide** (Diamox®): has been extensively studied.^{21,57,58,59} Conclusion: when combined with a low-sodium weight-reduction diet, acetazolamide provides a modest improvement in papilledema, ICP and QOL²¹ but its use is often limited by side effects.³⁷ Has not been shown to be effective for treating H/A alone.

R start at 125–250 mg PO q 8–12 hrs, or long-acting Diamox Sequels® 500 mg PO BID. Increase by 250 mg/day until symptoms improve, side effects occur. 1 gm/d is usually tolerated,⁶⁰ 3 gm/d is a usual upper limit but doses up to 4 gm/day may be needed⁵⁹ (usually in very obese patients)

Duration: no consensus. Expert opinion: treat for up to 2 years

Side effects: at high doses: acral paresthesias, nausea, metabolic acidosis, altered taste, renal calculi, drowsiness, diarrhea. Rare: Stevens-Johnson syndrome, toxic epidermal necrolysis, agranulocytosis.

✖ contraindications: allergy to sulfa or a history of renal calculi. Use under advisement of high-risk obstetrician during pregnancy due to claims of teratogenicity (which are disputed)
 - b) topiramate (Topamax®): antiseizure medication with secondary inhibition of CA. **R** start at 50 mg/d,⁶¹ with weekly dose escalation by 25–50 mg/d up to 100–150 mg/d.⁶¹ May help weight reduction by suppressing appetite
 - c) methazolamide has been used in place of acetazolamide. May also be teratogenic

2. *furosemide* (Lasix®): used by some as an alternative, role is uncertain
 - a) start: 160 mg per day in adults, adjust per symptoms and eye exam (not to CSF pressure)
 - b) if ineffective, double (320 mg/day)
 - c) monitor K⁺ levels and supplement as needed

Shunts

The literature regarding shunt type is mainly observational and case series based. When considering shunting, complications (shunt infection or obstruction, shunt migration, low pressure H/A (p.462), subdural hematoma (p.464), mortality (low risk but not negligible)...) should be taken into account.

1. ventriculoperitoneal (VP) shunt is recommended due to reported lower revision rate^{25,62}
 - a) navigational techniques may facilitate ventricular catheter placement in the presence of small or slit-like ventricles (which are common in PTCS⁶³)
 - b) a programmable valve with an antisiphon device should be considered to help manage and avoid low pressure H/A
 2. lumbar shunt: usually lumboperitoneal; see insertion technique (p.1830). Often more difficult in obese patients. May need a horizontal-vertical valve (p.458) to prevent H/A from intracranial hypotension. Alternative: lumbopleural shunt. Visual loss has been reported in patients with functioning LP shunts.⁶⁴ Other potential complications include: back pain, radiculopathy, tonsillar herniation (secondary Chiari malformation) and scoliosis (p.456). Contraindicated in the presence of lumbar spinal arachnoiditis
 3. cisterna magna shunt: has been described, not commonly used. May shunt to vascular system
- **Important facts about shunts in PTCS**
- shunts are very effective for *acute* treatment of H/A in PTCS (95% have relief in the 1st month)²⁵
 - some patients will achieve long-term relief, however, at 3 years, ~ 50% of PTCS patients will have recurrence of severe H/A despite a working shunt²⁵
 - patients without papilledema had a higher rate of recurrence of H/A (75% by 3 years)²⁵
 - 80% of shunts require revision by 36 months²⁵

Optic nerve sheath fenestration (ONSF)

See references.^{65,66,67}

A surgical procedure usually performed by an ophthalmologist to create a window (fenestration) or series of slits in the optic nerve sheath immediately posterior to the globe. This may help by allowing egress of CSF thereby reducing pressure on the optic nerve within the sheath, and may also cause scarring that prevents Performed via medial transconjunctival approach (50%), a superomedial lid crease incision (31%) a lateral orbitotomy (10%).⁶⁸

Generally better for protection of vision and reversal of papilledema than for other symptoms (fails to improve H/A in 33–50%⁶⁹). No reported mortality. May reverse or stabilize visual deterioration⁷⁰ and sometimes (but not always) lowers ICP (by continued CSF filtration) and may protect the contralateral eye (if not, contralateral ONSF must be performed). Has succeeded in cases where visual loss progressed after LP shunting,⁶⁴ possibly due to poor communication between orbital and intracranial subarachnoid spaces. **Potential adverse effects:** includes pupillary dysfunction, peripapillary hemorrhage, chemosis, chorioretinal scarring,⁷¹ diplopia (usually self-limited) from medial rectus disruption, worsening of vision (central retinal artery occlusion is a risk). Needs to be repeated in up to 6%.¹⁰

Endovascular procedures

Transverse sinus stenting. Transverse sinus (TVS) stenosis is a common epiphenomenon of increased ICP and in many cases may be a result of and not the cause of the increase. Endovascular placement of a stent across the stenosis to re-expand the lumen has been reported to improve symptoms after other measures fail. Stenting requires antiplatelet therapy afterwards (usually for >6 months), which may preclude other intervention (e.g., shunting) if it fails. Most studies are observational and case series. Current guidelines consider the role of neurovascular stenting as not yet established³⁷; however, since their publication there has been a recent trend to move to unilateral TVS stenting for PTCS patients with persistent increased ICP and papilledema who have failed to respond adequately to medical therapy who have bilateral TVS stenosis with an elevated TVS pressure gradient (a gradient >8 mm Hg across the stenosis has been arbitrarily recommended,^{72,73} some use 5 as the cutoff).^{37,74}

In a series of 32 PTCS patients,⁷⁵ 90% had a pressure gradient >8 mmHg, with bilateral TVS narrowing. Following stenting, papilledema was eliminated in 96% and 81% were able to discontinue acetazolamide. 6% had to have a repeat procedure.

Decompressive craniectomy

Usually subtemporal decompression, or less frequently suboccipital decompression.

Subtemporal decompression: older treatment advocated by Dandy. Fell out of favor (possibly because of risk of post-op seizures and painful bulging at decompression site), but may still have a place in the armamentarium⁷⁶ (e.g., patient with slit ventricles and in whom lumbar-peritoneal shunting is not feasible). Steps:

1. "silver-dollar size" craniectomies (approximately 1 inch = 25.4 mm) under the temporalis muscle to the floor of the middle fossa are made (usually bilaterally) through a linear incision. Kessler et al⁷⁶ used larger 6–8 cm diameter unilateral craniectomies; this larger craniectomy may explain their high incidence (62%) of H/A and tense bulging decompression site necessitating CSF diversion
2. the dura is opened
3. the brain is covered with an absorbable sponge (e.g., Gelfoam®)
4. the fascia and muscle are closed in a watertight fashion
5. a short course of antiseizure medications is used due to risk of post-op seizures

57.9.3 Guidelines for management

Adapted from the UK Guidelines³⁷ (there are currently no U.S. guidelines).

General measures

1. weight loss (p.965): the only known disease-modifying intervention. Should be attempted in all patients with BMI ≥ 30 kg/m²
2. ophthalmologic evaluation: at a minimum, dilated exam for papilledema and visual field assessment. Then, when possible, thorough ophthalmologic evaluation (p.962) should be done
3. stop possible offending drugs: e.g., vitamin A
4. brain imaging within 24 hours:
 - MRI or CT: to rule-out surgical conditions including subdural hematoma, hydrocephalus, Chiari malformation, true mass (tumor, hemorrhage)...
 - venography: MRV or CTV. Primarily for "atypical patients" (patients who are male or not obese) or patients who have progression of visual deficit despite treatment
5. properly performed LP (p.962) (after brain imaging rules out contraindications to LP (p.1811), which include Chiari malformation, obstructive hydrocephalus...) or shunt tap (in patients with shunt)
 - a) indications:
 1. patients who have not yet been diagnosed with PTCS
 2. patients with clinical signs of infection (meningitis, shunt infection...)
 3. patients with refractory H/A due to documented elevated CSF pressure (p.968)
 - b) objectives of LP/shunt tap:
 1. measure CSF pressure
 2. obtain CSF for analysis: routine labs. Also include CSF culture if clinical suspicion of infection
6. follow newly diagnosed patient at least two years with repeat imaging (e.g., MRI) to R/O occult tumor
7. see sections immediately below for guidelines pertaining to specific scenarios

For visual loss with PTCS

By definition of PTCS, these patients have papilledema. In addition to general measures (above):

1. ★ **fulminant PTCS**. Suggested definition: < 4 weeks between symptom onset to severe loss of acuity or visual field, and worsening of vision over a few days.⁷⁷ Severe visual is not always defined, but likely includes PMD (p.589) more negative than about -7. Occurs in $\approx$ 2–3% of PTSC patients.⁷⁷ Necessitates rapid intervention to protect vision:
 - a) immediate
 1. lumbar drain: when definitive treatment (shunting or ONSF) will not occur within 24 hrs, temporize by using a lumbar drain to decrease ICP
 2. start acetazolamide or topamax (see below)
 - b) optic nerve sheath fenestration or CSF diversion (shunting or revision of existing shunt) ASAP. These measures work well in the short term
 - c) start or continue efforts at weight loss to modify the underlying disease

Shunting may relieve both H/A and papilledema simultaneously. ONSF may be more effective in relieving visual problems (the failure rate may be lower than the shunt malfunction rate) but is not as good for the H/A.

2. mild *stable* visual loss (no immediate threat to vision)
 - a) start or continue efforts at weight loss to modify the underlying disease
 - b) consider medical therapy
 - acetazolamide (p.965) plus low-sodium weight-reduction diet. Provides modest visual field improvement compared to diet alone in patients with mild visual loss,⁷⁸ and at 6 months improved QOL measures.⁷⁹ Caution re metabolic acidosis, kidney stones, pregnancy... (see acetazolamide (p.965))
 - or
 - topiramate (p.965): may help by reducing CSF production (carbonic anhydrase effect) and by suppressing appetite which may aid weight loss
 - c) evaluate H/A phenotype (including medication overuse H/A), treat accordingly
 - d) ophthalmological follow-up: based on papilledema grade and visual field status as shown in ► Table 57.5
 - e) shunt or ONSF for significant visual deterioration on follow-up

Table 57.5 Recommended ophthalmologic follow-up schedule^{a, 37}

Papilledema grade	Visual field status			
	Normal	Abnormal & improving	Abnormal & stable	Abnormal & worsening
Atrophic ^b			4-6 mos	≤ 4 wks
Mild	6 mos	3-6 mos	3-4 mos	
Moderate	3-4 mos	1-3 mos	1-3 mos	≤ 2 wks
Severe			≤ 4 weeks	≤ 7 days

^a once papilledema resolves, follow-up is not necessary except for asymptomatic patients who require longer follow-up since they will not be aware if there is recurrence

^b atrophic papilledema: visible optic nerve head pallor due to nonviable optic nerve axons. Precludes diagnosis of papilledema

For H/A with PTCS and no visual loss (or mild/stable visual loss)

"Despite significant headache morbidity in IIH, there are no randomized controlled trials to guide headache management in IIH."³⁷ In addition to general measures (above):

1. rule-out overshunting (p.462) in patients with shunts (low-pressure H/A, thickened meninges on imaging...). Consider revision if present
2. LP or shunt tap only for clinical suspicion of infection
3. a multidisciplinary approach involving a practitioner with expertise in H/A management is recommended to assess H/A phenotype (including medication overuse H/A) and to direct treatment specific for that phenotype
4. medication: judicious use of *non-opioid* medical therapy to treat H/A in the short-term (for the first few weeks) may help. Medication overuse headache (p.960) (MOH) can occur with long-term use and is treated by elimination of overuse⁸⁰
5. greater occipital nerve block: may be considered, but effectiveness is controversial
6. ✖ *not* recommended for H/A alone (without visual loss):
 - medication to ↓ ICP: acetazolamide has not been shown to be effective for H/A alone (see above)
 - CSF shunting or shunt revision*:³⁷ limited evidence of efficacy. 68% continue to have H/A at 6 months, and 79% at 2 years, 28% develop low pressure H/A.⁸¹ Failure of shunting may be attributable to inadequate medical treatment of a migrainous component or medication overuse H/A
 - optic nerve sheath fenestration (ONSF)
 - serial LPs are not recommended for H/A alone³⁷
 - neurovascular stenting is not recommended for H/A³⁷

* Shunting or shunt revision may be indicated for refractory H/A, and then only after elevated ICP is documented (e.g., on ICP monitoring) and preferably under advisement of a multidisciplinary team (neurologist, ophthalmologist, sleep medicine, and neurosurgeon)

For pseudotumor cerebri syndrome without papilledema (PTCSWOP)

For PTCSWOP (p.958),

1. "risk of visual loss has not been identified and does not seem to develop over the disease course."³⁷ Caution: papilledema may not be reliably discerned if optic atrophy is present
2. counselling as appropriate for weight management (p.965) as with PTCS
3. H/A should be managed as with PTCS (see above)
4. ✖ surgical intervention (shunting, shunt revision, or ONSF) to control increased ICP should not be considered unless advised by a multidisciplinary team

Asymptomatic PTCS patients

Treatment of asymptomatic PTCS patients is controversial and challenging as there is no reliable predictor for visual loss. Close follow-up with serial formal visual field evaluation is necessary. Intervention is recommended in unreliable patients, or whenever visual fields deteriorate. It is possible to lose vision without H/A or papilledema.

For PTCS in children and adolescents

1. may be seen with withdrawal of steroids used for asthma
2. search for and correction of underlying etiology (offending drugs listed above, hypercalcemia, cancer...)
3. acetazolamide has been used with success

For PTCS in pregnancy

1. women who first present with PTCS during pregnancy: resolution of PTCS following delivery is common
2. weight management to maintain weight gain that is appropriate for the gestational age of the fetus as recommended by the ACOG⁸²
3. no specific mode of delivery should be recommended based on the diagnosis of PTCS
4. women who become pregnant during therapy:
 - a) 1st trimester: observation, limitation of weight gain, serial LPs.
✖ Acetazolamide should be avoided because of teratogenicity in rodents
 - b) 2nd & 3rd trimester: acetazolamide has been used safely, but involvement of high-risk obstetrician specialist is advised
 - c) topiramate should not be used in pregnancy
5. if acute vision loss is threatened, serial lumbar punctures (p.965) may be used to temporize until CSF diversion or ONSF can be performed

For pseudopapilledema

Pseudopapilledema (p.588) (associated with drusen, etc.) in the absence of intracranial hypertension needs no intervention.¹⁰ Reassurance and H/A management are recommended.

References

- [1] Quincke HL. Meningitis serosa. Sammlung Klinischer Vorträge. 1893; 67
- [2] Radhakrishnan K, Ahlskog JE, Garrity JA, et al. Idiopathic intracranial hypertension. Mayo Clinic Proceedings. 1994; 69:169–180
- [3] Friedman DI, Liu GT, Digre KB. Revised diagnostic criteria for the pseudotumor cerebri syndrome in adults and children. Neurology. 2013; 81:1159–1165
- [4] Johnston I, Owler B, Pickard J. The Pseudotumor Cerebri Syndrome: Pseudotumor Cerebri, Idiopathic Intracranial Hypertension, Benign Intracranial Hypertension and Related Conditions. Cambridge, England: Cambridge University Press; 2007
- [5] Digre KB, Corbett JJ. Pseudotumor Cerebri in Men. Arch Neurol. 1988; 45:866–872
- [6] Durcan FJ, Corbett JJ, Wall M. The Incidence of Pseudotumor Cerebri: Population Studies in Iowa and Louisiana. Arch Neurol. 1988; 45:875–877
- [7] Radhakrishnan K, Ahlskog JE, Cross SA, et al. Idiopathic Intracranial Hypertension (Pseudotumor Cerebri): Descriptive Epidemiology in Rochester, Minn, 1976 to 1990. Archives of Neurology. 1993; 50:78–80
- [8] Daniels AB, Liu GT, Volpe NJ, et al. Profiles of obesity, weight gain, and quality of life in idiopathic intracranial hypertension (pseudotumor cerebri). Am J Ophthalmol. 2007; 143:635–641
- [9] Rush JA. Pseudotumor Cerebri: Clinical Profile and Visual Outcome in 63 Patients. Mayo Clin Proc. 1980; 55:541–546
- [10] Bejjani GK, Cockerham KP, Pless M, et al. Idiopathic intracranial hypertension. Contemporary Neurosurgery. 2002; 24:1–8
- [11] Giuseffi V, Wall M, Siegel PZ, et al. Symptoms and disease associations in idiopathic intracranial hypertension (pseudotumor cerebri): a case-control study. Neurology. 1991; 41:239–244

- [12] Digre KB. Epidemiology of idiopathic intracranial hypertension. 1992
- [13] Sodhi M, Sheldon CA, Carleton B, et al. Oral fluoroquinolones and risk of secondary pseudotumor cerebri syndrome: Nested case-control study. *Neurology*. 2017; 89:792–795
- [14] Farb RI, Vanek I, Scott JN, et al. Idiopathic intracranial hypertension: The prevalence and morphology of sinovenous stenosis. *Neurology*. 2003; 60:1418–1424
- [15] King JO, Mitchell PJ, Thomson KR, et al. Manometry combined with cervical puncture in idiopathic intracranial hypertension. *Neurology*. 2002; 58:26–30
- [16] Schlosser RJ, Woodworth BA, Wilensky EM, et al. Spontaneous cerebrospinal fluid leaks: a variant of benign intracranial hypertension. *Ann Otol Rhinol Laryngol*. 2006; 115:495–500
- [17] Chaaban MR, Illing E, Riley KO, et al. Spontaneous cerebrospinal fluid leak repair: a five-year prospective evaluation. *Laryngoscope*. 2014; 124:70–75
- [18] Aaron G, Doyle J, Vaphiades MS, et al. Increased intracranial pressure in spontaneous CSF leak patients is not associated with papilledema. *Otolaryngol Head Neck Surg*. 2014; 151:1061–1066
- [19] Friedman DI, Jacobson DM. Idiopathic intracranial hypertension. *Journal of Neuro-Ophthalmology*. 2004; 24:138–145
- [20] Yuh WT, Zhu M, Taoka T, et al. MR imaging of pituitary morphology in idiopathic intracranial hypertension. *J Magn Reson Imaging*. 2000; 12:808–813
- [21] The NORDIC Idiopathic Intracranial Hypertension Study Group Writing Committee, Wall M, McDermott MP, et al. Effect of acetazolamide on visual function in patients with idiopathic intracranial hypertension and mild visual loss: the idiopathic intracranial hypertension treatment trial. *JAMA*. 2014; 311:1641–1651
- [22] Smith JL. Whence pseudotumor cerebri? *J Clin Neuroophthalmol*. 1985; 5:55–56
- [23] Digre KB, Nakamoto BK, Warner JE, et al. A comparison of idiopathic intracranial hypertension with and without papilledema. *Headache*. 2009; 49:185–193
- [24] Kathol RG, Cox TA, Corbett JJ, et al. Functional visual loss. Follow-up of 42 cases. *Arch Ophthalmol*. 1983; 101:729–735
- [25] McGirt MJ, Woodworth G, Thomas G, et al. Cerebrospinal fluid shunt placement for pseudotumor cerebri-associated intractable headache: predictors of treatment response and an analysis of long-term outcomes. *J Neurosurg*. 2004; 101:627–632
- [26] Ahlskog JE, O'Neill BP. Pseudotumor Cerebri. *Ann Int Med*. 1982; 97:249–256
- [27] Round R, Keane JR. The minor symptoms of increased intracranial hypertension: 101 patients with benign intracranial hypertension. *Neurology*. 1988; 38:1461–1464
- [28] Markey KA, Mollan SP, Jensen RH, et al. Understanding idiopathic intracranial hypertension: mechanisms, management, and future directions. *Lancet Neurol*. 2016; 15:78–91
- [29] Friedman DI, Quiros PA, Subramanian PS, et al. Headache in Idiopathic Intracranial Hypertension: Findings From the Idiopathic Intracranial Hypertension Treatment Trial. *Headache*. 2017; 57:1195–1205
- [30] Digre KB, Bruce BB, McDermott MP, et al. Quality of life in idiopathic intracranial hypertension at diagnosis: IIH Treatment Trial results. *Neurology*. 2015; 84:2449–2456
- [31] Headache Classification Committee of the International Headache Society (IHS) The International Classification of Headache Disorders, 3rd edition. *Cephalalgia*. 2018; 38:1–211
- [32] Friedman DI, Rausch EA. Headache diagnoses in patients with treated idiopathic intracranial hypertension. *Neurology*. 2002; 58:1551–1553
- [33] Wang SJ, Silberstein SD, Patterson S, et al. Idiopathic intracranial hypertension without papilledema: a case control study in a headache center. *Neurology*. 1998; 51:245–249
- [34] Sher NA, Wirtschafner J, Shapiro SK, et al. Unilateral Papilledema in 'Benign' Intracranial Hypertension (Pseudotumor Cerebri). *JAMA*. 1983; 250:2346–2347
- [35] Rowe FJ, Sarkies NJ. Assessment of visual function in idiopathic intracranial hypertension: a prospective study. *Eye (Lond)*. 1998; 12 (Pt 1):111–118
- [36] Wall M, George D. Idiopathic Intracranial Hypertension: A Prospective Study of 50 Patients. *Brain*. 1991; 114:155–180
- [37] Mollan SP, Davies B, Silver NC, et al. Idiopathic intracranial hypertension: consensus guidelines on management. *J Neurol Neurosurg Psychiatry*. 2018; 89:1088–1100
- [38] Johnston I, Hawke S, Halmagyi M, et al. The Pseudotumor Syndrome: Disorders of Cerebrospinal Fluid Circulation Causing Intracranial Hypertension Without Ventriculomegaly. *Archives of Neurology*. 1991; 48:740–747
- [39] Powers JM, Schnur JA, Baldrée ME. Pseudotumor Cerebri due to Partial Obstruction of the Sigmoid Sinus by a Cholesteatoma. *Arch Neurol*. 1986; 43:519–521
- [40] Wen DY, Hardten DR, Wirtschafner JD, et al. Elevated Intracranial Pressure from Cerebral Venous Obstruction by Masson's Vegetant Intravascular Hemangioendothelioma. *Journal of Neurosurgery*. 1991; 75:787–790
- [41] Banik R, Lin D, Miller NR. Prevalence of Chiari I malformation and cerebellar ectopia in patients with pseudotumor cerebri. *J Neurol Sci*. 2006; 247:71–75
- [42] Aiken AH, Hoots JA, Saindane AM, et al. Incidence of cerebellar tonsillar ectopia in idiopathic intracranial hypertension: a mimic of the Chiari I malformation. *AJNR Am J Neuroradiol*. 2012; 33:1901–1906
- [43] Mukherjee S, Kalra N, Warren D, et al. Chiari I malformation and altered cerebrospinal fluid dynamics—the highs and the lows. *Childs Nerv Syst*. 2019; 35:1711–1717
- [44] Cunningham ET, Koehler JE. Ocular bartonellosis. *Am J Ophthalmol*. 2000; 130:340–349
- [45] Whelton PK, Carey RM, Aronow WS, et al. 2017 ACC/AHA/AAPA/ABC/ACPM/AGS/APHA/ASH/ASPC/NMA/PCNA Guideline for the Prevention, Detection, Evaluation, and Management of High Blood Pressure in Adults: Executive Summary: A Report of the American College of Cardiology/American Heart Association Task Force on Clinical Practice Guidelines. *Hypertension*. 2018; 71:1269–1324
- [46] Avery RA, Mistry RD, Shah SS, et al. Patient position during lumbar puncture has no meaningful effect on cerebrospinal fluid opening pressure in children. *J Child Neurol*. 2010; 25:616–619
- [47] Avery RA, Shah SS, Licht DJ, et al. Reference range for cerebrospinal fluid opening pressure in children. *N Engl J Med*. 2010; 363:891–893
- [48] Neville L, Egan RA. Frequency and amplitude of elevation of cerebrospinal fluid resting pressure by the Valsalva maneuver. *Can J Ophthalmol*. 2005; 40:775–777
- [49] Jacobson DM, Karanjia PN, Olson KA, et al. Computed Tomography Ventricular Size has no Predictive Value in Diagnosing Pseudotumor Cerebri. *Neurology*. 1990; 40:1454–1455
- [50] Aaron GP, Illing E, Lambertsen Z, et al. Enlargement of Meckel's cave in patients with spontaneous cerebrospinal fluid leaks. *Int Forum Allergy Rhinol*. 2017; 7:421–424
- [51] Degnan AJ, Levy LM. Narrowing of Meckel's cave and cavernous sinus and enlargement of the optic nerve sheath in Pseudotumor Cerebri. *J Comput Assist Tomogr*. 2011; 35:308–312
- [52] Farb RI, Vanek I, Scott JN, et al. Idiopathic intracranial hypertension: the prevalence and morphology of sinovenous stenosis. *Neurology*. 2003; 60:1418–1424
- [53] Kelly LP, Saindane AM, Bruce BB, et al. Does bilateral transverse cerebral venous sinus stenosis exist in patients without increased intracranial pressure? *Clin Neurol Neurosurg*. 2013; 115:1215–1219
- [54] Sinclair AJ, Burdon MA, Nightingale PG, et al. Low energy diet and intracranial pressure in women

- with idiopathic intracranial hypertension: prospective cohort study. *BMJ*. 2010; 341. DOI: 10.1136/bmj.c2701
- [55] Johnson LN, Krohel GB, Madsen RW, et al. The role of weight loss and acetazolamide in the treatment of idiopathic intracranial hypertension (pseudotumor cerebri). *Ophthalmology*. 1998; 105:2313–2317
 - [56] Newberg B. Pseudotumor Cerebri Treated by Rice/Reduction Diet. *Arch Intern Med*. 1974; 133:802–807
 - [57] Ball AK, Howman A, Wheatley K, et al. A randomised controlled trial of treatment for idiopathic intracranial hypertension. *J Neurol*. 2011; 258:874–881
 - [58] Piper RJ, Kalyvas AV, Young AM, et al. Interventions for idiopathic intracranial hypertension. *Cochrane Database Syst Rev*. 2015. DOI: 10.1002/14651858.CD003434.pub3
 - [59] Smith SV, Friedman DI. The Idiopathic Intracranial Hypertension Treatment Trial: A Review of the Outcomes. *Headache*. 2017; 57:1303–1310
 - [60] ten Hove MW, Friedman DI, Patel AD, et al. Safety and Tolerability of Acetazolamide in the Idiopathic Intracranial Hypertension Treatment Trial. *J Neuroophthalmol*. 2016; 36:13–19
 - [61] Celebisoy N, Gokcay F, Sirin H, et al. Treatment of idiopathic intracranial hypertension: topiramate vs acetazolamide, an open-label study. *Acta Neurol Scand*. 2007; 116:322–327
 - [62] Kalyvas AV, Hughes M, Koutsarnakis C, et al. Efficacy, complications and cost of surgical interventions for idiopathic intracranial hypertension: a systematic review of the literature. *Acta Neurochir (Wien)*. 2017; 159:33–49
 - [63] Hahn FJ, McWilliams FE. The Small Ventricle in Pseudotumor Cerebri: Demonstration of the Small Ventricle in Benign Intracranial Hypertension. *CT*. 1978; 2:249–253
 - [64] Kelman SE, Sergott RC, Cioffi GA, et al. Modified Optic Nerve Decompression in Patients with Functioning Lumboperitoneal Shunts and Progressive Visual Loss. *Ophthalmology*. 1991; 98:1449–1453
 - [65] Brouman ND, Spoor TC, Ramocki JM. Optic Nerve Sheath Decompression for Pseudotumor Cerebri. *Arch Ophthalmol*. 1988; 106:1384–1390
 - [66] Sergott RC, Savino PJ, Bosley TM. Modified Optic Nerve Sheath Decompression Provides Long-Term Visual Improvement for Pseudotumor Cerebri. *Arch Ophthalmol*. 1988; 106:1384–1390
 - [67] Corbett JJ, Nerad JA, Tse D, et al. Optic Nerve Sheath Fenestration for Pseudotumor Cerebri: The Lateral Orbitotomy Approach. *Arch Ophthalmol*. 1988; 106:1391–1397
 - [68] Sobel RK, Syed NA, Carter KD, et al. Optic Nerve Sheath Fenestration: Current Preferences in Surgical Approach and Biopsy. *Ophthalmic Plast Reconstr Surg*. 2015; 31:310–312
 - [69] Banta JT, Farris BK. Pseudotumor cerebri and optic nerve sheath decompression. *Ophthalmology*. 2000; 107:1907–1912
 - [70] Kelman SE, Heaps R, Wolf A, et al. Optic Nerve Decompression Surgery Improves Visual Function in Patients with Pseudotumor Cerebri. *Neurosurgery*. 1992; 30:391–395
 - [71] Spoor TC, Ramocki JM, Madion MP, et al. Treatment of Pseudotumor Cerebri by Primary and Secondary Optic Nerve Sheath Decompression. *Am J Ophthalmol*. 1991; 112:177–185
 - [72] Ahmed RM, Wilkinson M, Parker GD, et al. Transverse sinus stenting for idiopathic intracranial hypertension: a review of 52 patients and of model predictions. *AJNR Am J Neuroradiol*. 2011; 32:1408–1414
 - [73] Cappuzzo JM, Hess RM, Morrison JF, et al. Transverse venous stenting for the treatment of idiopathic intracranial hypertension, or pseudotumor cerebri. *Neurosurg Focus*. 2018; 45. DOI: 10.3171/2018.5.FOCUS18102
 - [74] Higgins JN, Owler BK, Cousins C, et al. Venous sinus stenting for refractory benign intracranial hypertension. *Lancet*. 2002; 359:228–230
 - [75] Reid Kate, Winters HStephen, Ang Timothy, et al. Transverse Sinus Stenting Reverses Medically Refractory Idiopathic Intracranial Hypertension. *Frontiers in Ophthalmology*. 2022; 2. DOI: 10.3389/fopht.2022.885583
 - [76] Kessler LA, Novelli PM, Reigel DH. Surgical treatment of benign intracranial hypertension—subtemporal decompression revisited. *Surg Neurol*. 1998; 50:73–76
 - [77] Thambisetty M, Lavin PJ, Newman NJ, et al. Fulminant idiopathic intracranial hypertension. *Neurology*. 2007; 68:229–232
 - [78] Wall M, Kupersmith MJ, Kiebertz KD, et al. The idiopathic intracranial hypertension treatment trial: clinical profile at baseline. *JAMA Neurol*. 2014; 71:693–701
 - [79] Bruce BB, Digre KB, McDermott MP, et al. Quality of life at 6 months in the Idiopathic Intracranial Hypertension Treatment Trial. *Neurology*. 2016; 87:1871–1877
 - [80] deSouza RM, Toma A, Watkins L. Medication overuse headache - An under-diagnosed problem in shunted idiopathic intracranial hypertension patients. *Br J Neurosurg*. 2015; 29:30–34
 - [81] Sinclair AJ, Kuruvath S, Sen D, et al. Is cerebrospinal fluid shunting in idiopathic intracranial hypertension worthwhile? A 10-year review. *Cephalalgia*. 2011; 31:1627–1633
 - [82] Committee on Obstetric Practice. . Weight Gain During Pregnancy. 2013. <https://www.acog.org/clinical/clinical-guidance/committee-opinion/articles/2013/01/weight-gain-during-pregnancy>

58 Tumors and Tumor-Like Lesions of the Skull

58.1 Skull tumors

58.1.1 General information

See Skull lesions (p.1661) for differential diagnosis and evaluation (including non-neoplastic lesions). Considering only tumors, the differential diagnosis includes:

1. benign tumors
 - a) osteoma: see below
 - b) hemangioma: see below
 - c) dermoid and epidermoid tumors: see below
 - d) chondroma: occur mainly in conjunction with the basal synchondroses
 - e) meningioma involving the skull
 - f) aneurysmal bone cyst
2. malignant tumors: malignancy is suggested by a single large or multiple (> 6) small osteolytic lesions with margins that are ragged, undermined, and lacking sclerosis¹
 - a) bone metastases to the skull. Common ones include:
 - prostate
 - breast
 - lung
 - kidney
 - thyroid
 - lymphoma
 - multiple myeloma/plasmacytoma (p.928)
 - direct extension of head and neck tumors, e.g., squamous cell carcinoma of the scalp
 - b) chondrosarcoma
 - c) osteogenic sarcoma
 - d) fibrosarcoma

58.1.2 Osteoma

General information

Osteomas are the most common primary bone tumor of the calvaria. They are benign, slow-growing lesions that occur commonly in the cranial vault, mastoid and paranasal air sinuses, and the mandible. Lesions within air sinuses may present as recurrent sinusitis. More common in females, highest incidence is in 6th decade. Triad of Gardner's syndrome: multiple cranial osteomas (of calvaria, sinuses, and mandible), colonic polyposis, and soft-tissue tumors.

See Localized increased density or hyperostosis of the calvaria (p. 1664) for differential diagnosis.

Pathology

Consists of osteoid tissue within osteoblastic tissue, surrounded by reactive bone. Difficult to distinguish from fibrous dysplasia (p.975).

Radiographic evaluation

Skull X-ray: round, sclerotic, well-demarcated, homogeneous dense projection. Usually arise from outer table of skull (inner table less common). May be compact or spongy (spongy osteoma may be radiolucent). Unlike meningiomas, diploë are preserved and vascular channels are not increased.

Osteomas are "hot" on nuclear bone scan.

Treatment

Asymptomatic lesions may simply be followed. Surgery may be considered for cosmetic reasons, or if pressure on adjacent tissues produces discomfort. Lesions involving only the outer table may be removed, leaving the inner table intact.

58.1.3 Hemangioma

General information

Comprise $\approx 7\%$ of skull tumors.¹ These benign tumors commonly occur in the skull (discussed here) and spine (p.992). Two types: cavernous (most common) and capillary (rare).

Radiographic evaluation

Skull X-ray: characteristically shows a circular lucency with honeycomb or trabecular pattern (seen in $\approx 50\%$ of cases) or radial trabeculations producing a sunburst pattern (seen in $\approx 11\%$ of cases).¹ Sclerotic margins are evident in only $\approx 33\%$.

CT: hypodense lesion with sclerotic spaced trabeculations. Nonenhancing.

Bone scan: typically hot.

Treatment

Accessible lesions may be cured by en bloc excision or curettage. The gross appearance is of a hard, blue-domed mass beneath the pericranium. Radiation may be considered for inaccessible tumors.

58.1.4 Epidermoid and dermoid tumors of the skull

General information

See also epidermoids and dermoids in general (p.973).

Dermoids and epidermoids are benign inclusion cysts of ectoderm that may involve skull and underlying dural venous structures or brain. They may become infected. Primary skull involvement is rare and occurs when ectodermal rests are entrapped in the developing skull, which causes these tumors to arise within the diploë and expand both inner and outer tables. Because they are not neoplastic, they grow at a linear rate (instead of exponential). Usually midline.

Epidermoid tumors contain only the outer layer of skin, and are therefore lined with stratified squamous epithelium and the resultant byproduct, keratin.

Dermoid tumors contain all elements of skin including hair follicles (which may produce hair in the tumor) and dermal glands (sebaceous glands (apocrine) and sweat glands (eccrine)).²

Teratomas are true neoplasms and may also contain bone, cartilage, teeth, and nails.

Presentation

These lesions may present as a result of mass effect from continued growth.

They may rupture (more common with dermoids than epidermoids), and can cause chemical meningitis (from the irritating properties of fat and/or keratin), or, if infected, bacterial meningitis.

Radiographic evaluation

1. skull X-ray: these osteolytic lesions have well-defined, sclerotic margins
2. some imaging is required to evaluate possible intracranial involvement
 - a) CT: the lesions are hypodense (keratin contains fats), and non-enhancing
 - b) MRI: like CSF they are low intensity on T1WI and high signal on T2WI, but unlike CSF they are high signal on DWI > MRI (p.938)

Treatment

Treatment is surgical. Radiation and chemotherapy are not indicated.

When possible, the goal is to avoid rupture during removal in order to avoid chemical and/or bacterial meningitis.

Bone margins are curetted. Search must be made for a tract leading to the intracranial cavity which must be followed if found. Preparation for dural sinus repair must be made for lesions overlying the sagittal sinus (including torcular Herophili).

Endoscopic surgery may be an option for some skull base lesions.

58.1.5 Langerhans cell histiocytosis

Formerly called histiocytosis X (or eosinophilic granuloma, when solitary). May occur in the skull as a result of direct invasion of craniofacial bone and skull base, or from meninges (p.846).

58.1.6 Squamous cell carcinoma of the scalp involving the skull

It is not unusual for plastic surgeons to request neurosurgical assistance in managing squamous cell carcinoma of the scalp that involves the calvaria over the convexity. For first time surgical intervention (i.e., non-recurrence) without resectable mass or frank invasion through the inner table of the bone, neurosurgical management usually involves burring down the outer table in the area of involvement until normal appearing bone is encountered, taking care not to penetrate the inner table.

58.2 Non-neoplastic skull lesions

58.2.1 General information

Includes:

1. osteopetrosis (p.1689)
2. Paget's disease of the skull
3. hyperostosis frontalis interna (see below)
4. fibrous dysplasia (p.975)

58.2.2 Hyperostosis frontalis interna

General information

See differential diagnosis (p.1664). Hyperostosis frontalis interna (HFI) is a benign irregular nodular thickening of the inner table of the frontal bone that is almost always bilateral. The midline is spared at the insertion of the falx. Unilateral cases have been reported,³ and in these cases one must R/O other etiologies such as meningioma, calcified epidural hematoma, osteoma, fibrous dysplasia, an epidural fibrous tumor,⁴ or Paget's disease.

Epidemiology

The incidence of HFI in the general population is $\approx 1.4\text{--}5\%$.³ HFI is more common in women (female: male ratio may be as high as 9:1) with an incidence of 15–72% in elderly women. A number of possible associated conditions have been described (most are unproven), the majority of which are metabolic, earning it the alias of metabolic craniopathy. Associated conditions include:

1. Morgagni's syndrome (AKA Morgagni-Stewart-Morel syndrome): headache, obesity, virilism, and neuropsychiatric disorders (including mental retardation)
2. endocrinologic abnormalities
 - a) acromegaly (p.870)⁵ (elevated growth hormone levels)
 - b) hyperprolactinemia⁵
3. metabolic abnormalities
 - a) hyperphosphatemia
 - b) obesity
4. diffuse idiopathic skeletal hyperostosis (DISH) (p.1373)

Clinical

HFI may present without symptoms as an incidental finding on radiographic evaluation for other reasons. Many signs and symptoms have been attributed to HFI, including: hypertension, seizures, headache, cranial nerve deficits, dementia, irritability, depression, hysteria, fatigability, and mental dullness. The incidence of headache may be statistically higher in patients with HFI than in the general population.⁶

Evaluation

Blood tests to R/O some of the above noted conditions may be indicated in appropriate cases: check growth hormone, prolactin, phosphate, alkaline phosphatase (to R/O Paget's disease).

Plain skull X-ray shows thickening of the frontal bone with characteristic sparing of the midline. Spread to parietal and occipital bone occasionally occurs.

CT demonstrates the lesion, which usually causes 5–10 mm of bone thickening, but as much as 4 cm has been reported.

Bone scan: usually shows moderate uptake in HFI (generally not as intense as with bone mets). Also, indium-111 leukocyte scan (commonly used to detect occult infection) will show accumulation in HFI (a false positive).^{7,8}

Treatment

In spite of a large number of published descriptive works in the medical literature, primarily in the early and mid-20th century, little has been written about treatment of cases where symptoms are suspected to be due to HFI. In one report, removal of the thickened bone was accomplished without evidence of dural adhesions, and with improvement in the presenting hysteria.³

Surgical technique

One technique described consists of using the craniotome to excise the thickened portion of the bone (a plain skull X-ray may be used to make a template), and then the thickened bone is thinned down with a high-speed drill, and the bone flap is then replaced. Alternatively, a cranioplasty with methylmethacrylate or custom-made implant fabricated using CT data may be performed.

58.2.3 Fibrous dysplasia

General information

Key concepts

- non-neoplastic condition where bone is replaced by expansile fibrous connective tissue
- malignant transformation to osteosarcoma or other sarcoma occurs in < 1%
- common sites of involvement: ribs, proximal femur, craniofacial bone
- associated endocrinopathies are common: includes precocious puberty or growth hormone excess
- may be monostotic, polyostotic, or part of McCune-Albright syndrome (triad: café au lait spots, endocrinopathy (e.g., precocious puberty) and polyostotic fibrous dysplasia)
- common presentation: may be incidental, cosmetic, hearing loss, rarely visual loss
- treatment:
 - observation is often the best initial management. Prophylactic surgery for optic nerve compression is not justified due to the typical slow progression and the high risk of visual loss with surgery
 - medical therapy may include calcitonin, bisphosphonates, ? RANK ligand inhibitors
 - treat endocrinopathies (reduces morbidity including visual loss with GH excess)
 - ✗ XRT is not recommended

Fibrous dysplasia (FD) is a non-neoplastic benign condition in which normal bone is replaced by fibrous connective tissue and immature woven bone that is weaker than normal bone and tends to expand. Most lesions occur in the ribs, proximal femur or craniofacial bones (especially the maxilla). Frequent femoral neck fractures can lead to a varus deformity (so-called “shepherd’s crook deformity”)

Molecular genetics: FD is not heritable. It results from somatic activating mutations in the α subunit of the stimulatory G protein encoded by the GNAS locus at 20q13.2-q13.3.

Types of involvement

Localization of involvement

3 patterns of involvement

1. monostotic fibrous dysplasia (MFD): most common. The most frequent site of involvement is the zygomatic-maxillary complex
2. polyostotic fibrous dysplasia (PFD): 25% with this form have > 50% of the skeleton involved with associated fractures and skeletal deformities
3. PFD as part of McCune-Albright syndrome (and its variants). Triad:
 - a) café au lait spots which tend to occur on one side of the midline and tend to be more jagged than those seen in neurofibromatosis (p.638) and fewer in number

- b) endocrinopathy: including precocious puberty (primarily in females) and growth hormonesecretion
- c) polyostotic fibrous dysplasia (PFD)

Skull involvement occurs in 27% of MFD. In PFD and MAS, craniofacial involvement occurs in 90% of cases and the anterior skull base is involved in > 95%.⁹

3 forms of fibrous dysplasia lesions

1. cystic (the lesions are not actually cysts in the strict sense): widening of the diploë usually with thinning of the outer table and little involvement of the inner table. Typically occurs high in calvaria
2. sclerotic: usually involves skull base (especially sphenoid bone) and facial bones
3. mixed: appearance is similar to cystic type with patches of increased density within the lucent lesions

Epidemiology

True incidence is unknown since many cases are asymptomatic. FD constitutes ≈ 7% of benign bone lesions.¹⁰ MFD is probably more common than PFD, but the actual ratio depends on the screening methodology. MAS is more common in females.

Clinical

Clinical manifestations of craniofacial fibrous dysplasia lesions include:

1. incidental finding (i.e., asymptomatic)
2. local pain or tenderness: the lesions are not tender, but overlying stretched periosteum may be. Associated aneurysmal bone cysts (ABCs) may be painful
3. local swelling (rarely: marked distortion resembling aneurysmal bone cyst)
4. changes due to facial deformity/asymmetry
 - a) cosmetic: frontal bossing, hypertelorism
 - b) orbital involvement may cause: proptosis, vertical dystopia (asymmetry in vertical alignment of the eyes), visual loss (p.976)
 - c) nasal congestion
 - d) mandibular involvement: asymmetry which may produce malocclusion
5. may predispose to pathologic fractures when FD lesions occur in long bones
6. cranial nerve-related manifestations:
 - a) hearing loss: temporal bone involvement may obliterate the external auditory canal or may restrict movement of middle ear ossicles. Temporal bone involvement occurs in > 70% of cases of craniofacial PFD and MAS and is uncommon in MFD¹¹
 - b) visual loss: an uncommon but recognized sequela of FD as a result of compression of the optic nerve in the optic canal, more common in children
 - c) trigeminal neuralgia¹²
 - d) facial nerve palsy: rare. Compression typically occurs within Fallopiian canal and/or IAC
7. seizures: rare presentation
8. serum alkaline phosphatase is elevated in about 33%, calcium levels are normal
9. darkened hair pigmentation overlying skull lesions
10. spontaneous scalp hemorrhages
11. rarely associated with Cushing's syndrome, acromegaly

Evaluation

A major objective is to determine if the patient has a single lesion (MFD) or multiple lesions as in PFD or MAS.

1. history:
 - onset and nature of symptoms
 - rapidity of progression
 - history of pathologic fractures (clue for other FD lesions)
 - indications of endocrinopathies: age of menarche in females (R/O precocious puberty), growth abnormalities (R/O growth hormone excess)
2. physical exam: look for and ask about skin discoloration (café-au-lait spots)
3. diagnostic studies:
 - skeletal survey (total body X-rays) or bone scan if additional lesions are suspected

- non-contrast head CT with thin cuts (► Fig. 58.1): characteristic ground glass appearance of craniofacial FD on X-rays and CT is due to the thin spicules of woven bone. With age, lesions morph into a mixed radio-dense/radio-lucent appearance. In prepubertal patients with PFD or MAS a homogeneous radio-dense appearance is more common.
- panorex and/or dental films when there is involvement around the teeth

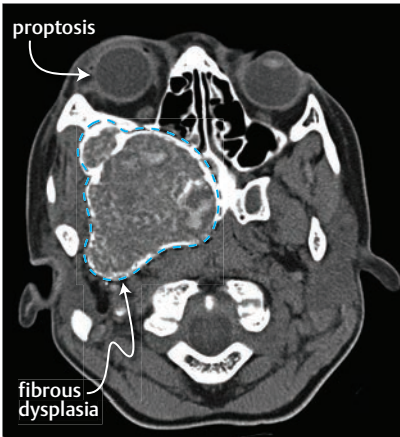


Fig. 58.1 Fibrous dysplasia of the right middle cranial fossa (outlined by blue broken line). Image: axial CT scan, bone window.

Natural history

FD lesions are usually slowly progressive.

Rapid progression may occur:

- infrequently, in PFD or MAS prior to puberty. Progression usually slows when skeletal maturity is reached.
- malignant transformation: occurs in < 1% of cases.¹¹ Typically to osteosarcoma, but other sarcomas are possible (fibrosarcoma, chondrosarcoma...)
- associated expansile lesions:
 - aneurysmal bone cyst (ABC)
 - mucocele: when the ostium to a sinus becomes occluded by fibrous overgrowth
- osteomyelitis: often difficult to cure in the presence of FD

The most disfiguring and symptomatic lesions occur in patients with poorly controlled excess growth hormone (GH).¹³ Therefore excess GH should be managed aggressively

Craniofacial FD may be categorized as¹¹:

1. quiescent: no progression. Typically with smaller lesions
2. non-aggressive: slow growing
3. aggressive: rapid growth, associated pain, pathologic fractures, malignant transformation...

Management

Observation is often the best first management option. Delay surgery until skeletal maturity (around age 10–12 years) if possible.

Cure may be possible with smaller lesions in MFD but is unlikely with PFD or MAS. Local procedures (mostly orthopedic) are used for appendicular deformities or bone pain that is refractory to other treatment. Incompletely resected lesions are subject to regrowth.

Consult an endocrinologist if there is any suspicion of endocrinopathy.

Biopsy of an accessible lesion by the appropriate surgical specialist should be considered when the diagnosis is in doubt.¹¹ Bleeding may be brisk due to vascularity. Biopsy does not promote growth of the lesion. Histology does not predict biological behavior.

Quiescent craniofacial lesions without complaints of facial deformity may be monitored with annual evaluations which should include: patient reported symptoms, neurologic exam including

sensory testing, photographs and facial CT for the first 2 visits and then less frequently based on prior CT results and clinical findings.

Non-aggressive or quiescent lesions producing bothersome facial disfigurement may be treated by a craniofacial surgeon. If practical, it is best to wait until the patient reaches skeletal maturity.

When optic canal involvement is documented, annual ophthalmological exams should be conducted. The diagnosis of optic neuropathy is made when there is a visual defect or if 2 out of 3 exams (contrast sensitivity, color vision and funduscopy exam) are abnormal. Optic neuropathy in FD is more common in patients with growth hormone excess.^{13,14} Evidence suggest aggressive management of GH excess reduces the risk of optic neuropathy.

When there is temporal bone involvement, regular otolaryngology exams (including microscopy for EAC stenosis) should be performed to maintain patency of EAC (from bone, cerumen or the rare cholesteatoma). Annual audiology evaluations are recommended during periods of active lesion growth.

Pain is a common complaint, and is more prevalent in the lower extremities than with craniofacial involvement. It does not correlate with disease burden.

Calcitonin may be used for widespread lesions with bone pain and/or high serum alkaline phosphatase levels.

Bisphosphonates (e.g., alendronate, pamidronate or zoledronic acid): mixed results for craniofacial pain control and growth rate reduction in FD.

RANK ligand inhibitors e.g., denosumab: uncertain role in FD.¹⁵

✱ Radiation therapy is not recommended (risk of XRT induced tumors).

► **Neurosurgical involvement.** May be required for skull lesions producing refractory pain or neurologic symptoms or for rapidly growing associated lesions (e.g., ABC). Calvarial lesions may be treated with curettage and cranioplasty. Once the dysplastic bone is disconnected from the skull, the loss of vascular supply usually renders it inactive. Skull base lesions often require a multidisciplinary approach.

Acute visual deterioration associated with an expansile lesion near the optic canal should be treated with high-dose glucocorticoids and rapid surgical decompression.¹⁴ However, vision is usually preserved despite optic canal narrowing with FD,⁹ and visual loss may follow injudicious surgical intervention due to the intolerance of the optic nerve to surgical trauma. Rule out other possible explanations for visual loss before operating.

Image guidance may be helpful when drilling down exuberant bone to decompress the optic canal. Copious irrigation (e.g., with irrigating drill) is critical to avoid thermal damage. Endoscopic techniques may be useful adjuncts, but the distorted anatomy requires an experienced practitioner.

References

- [1] Thomas JE, Baker HL. Assessment of Roentgenographic Lucencies of the Skull: A Systematic Approach. *Neurology*. 1975; 25:99-106
- [2] Smirniotopoulos J G, Chiechi MV. Teratomas, dermoids, and epidermoids of the head and neck. *Radiographics*. 1995; 15:1437-1455
- [3] Hasegawa T, Ito H, Yamamoto S, et al. Unilateral Hyperostosis Frontalis Interna: Case Report. *Journal of Neurosurgery*. 1983; 59:710-713
- [4] Willison CD, Schochet SS, Voelker JL. Cranial Epidural Fibrous Tumor Associated with Hyperostosis: A Case Report. *Surgical Neurology*. 1993; 40:508-511
- [5] Fulton JD, Shand J, Ritchie D, et al. Hyperostosis frontalis interna, acromegaly and hyperprolactinemia. *Postgrad Med J*. 1990; 66:16-19
- [6] Bavazzano A, Del Bianco PL, Del Bene E, et al. A statistical evaluation of the relationships between headache and internal frontal hyperostosis. *Research & Clinical Studies in Headache*. 1970; 3:191-197
- [7] Floyd JL, Jackson DE, Carretta R. Appearance of Hyperostosis Frontalis Interna on Indium-111 Leukocyte Scans: Potential Diagnostic Pitfall. *J Nucl Med*. 1986; 27:495-497
- [8] Oates E. Spectrum of Appearance of Hyperostosis Frontalis Interna on In-111 Leukocyte Scans. *Clin Nucl Med*. 1988; 13:922-923
- [9] Lee JS, FitzGibbon E, Butman JA, et al. Normal vision despite narrowing of the optic canal in fibrous dysplasia. *N Engl J Med*. 2002; 347:1670-1676
- [10] Parekh SG, Donthini-Rao R, Ricchetti E, et al. Fibrous dysplasia. *J Am Acad Orthop Surg*. 2004; 12:305-313
- [11] Lee JS, FitzGibbon EJ, Chen YR, et al. Clinical guidelines for the management of craniofacial fibrous dysplasia. *Orphanet J Rare Dis*. 2012; 7 Suppl 1:S2-S2
- [12] Finney HL, Roberts TS. Fibrous dysplasia of the skull with progressive cranial nerve involvement. *Surg Neurol*. 1976; 6:341-343
- [13] Cutler CM, Lee JS, Butman JA, et al. Long-term outcome of optic nerve encasement and optic nerve decompression in patients with fibrous dysplasia: risk factors for blindness and safety of observation. *Neurosurgery*. 2006; 59:1011-7; discussion 1017-8
- [14] Boyce AM, Glover M, Kelly MH, et al. Optic neuropathy in McCune-Albright syndrome: effects of early diagnosis and treatment of growth hormone excess. *J Clin Endocrinol Metab*. 2013; 98:E126-E134
- [15] Boyce AM, Chong WH, Yao J, et al. Denosumab treatment for fibrous dysplasia. *J Bone Miner Res*. 2012; 27:1462-1470

59 Tumors of the Spine and Spinal Cord

59.1 Spine tumors – general information

15% of primary CNS tumors are intraspinal (the intracranial:spinal ratio for astrocytomas is 10:1; for ependymomas it is 3–20:1).¹ There is disagreement over the prevalence, prognosis, and optimal treatment. Most primary CNS spinal tumors are benign (unlike the case with intracranial tumors). Most present by compression rather than invasion.²

59.2 Compartmental locations of spinal tumors

May be classified into 3 groups based on the compartment involved. Although metastases may be found in each area, they are most commonly extradural. Frequencies quoted below are from a general hospital, extradural lesions are less common in neurosurgical clinics because many of these tumors are managed by oncologists without requiring neurosurgical involvement.

1. extradural (ED) (55%): arise outside cord in vertebral bodies or epidural tissues
2. intradural extramedullary (ID-EM) (40%): arise in leptomeninges or roots. Primarily meningiomas and neurofibromas (together = 55% of ID-EM tumors)
3. intramedullary spinal cord tumors (IMSCT) (p.984), 5%: arise in SC substance. Invade and destroy tracts and gray matter

► **Spinal lymphoma.** Lymphoma may occur in any of or all 3 compartments.

1. epidural
 - a) metastatic or secondary lymphoma: the most common form of spinal lymphoma. Spinal involvement occurs in 0.1–10% of patients with non-Hodgkin's lymphoma
 - b) primary spinal epidural non-Hodgkin's lymphoma: rare. Completely epidural with no bony involvement. The existence of this entity is controversial, and some investigators feel that it represents extension of undetected retroperitoneal or vertebral body lymphoma. May have a better prognosis than secondary lymphoma³
2. intramedullary
 - a) secondary (p.986) (metastatic)
 - b) primary: very rare (see below)

59.3 Differential diagnosis: spine and spinal cord tumors

59.3.1 General information

See also Myelopathy (p.1696) for a list including not only tumors but also *nonneoplastic* causes of spinal cord dysfunction (e.g., spinal meningeal cyst, epidural hematoma, transverse myelitis...).

59.3.2 Extradural spinal cord tumors (55%)

Arise in vertebral bodies or epidural tissues

1. metastatic: comprise the majority of ED tumors
 - a) most are osteolytic (cause bony destruction): see Spinal epidural metastases (p.921). Common ones include:
 - lymphoma: most cases represent spread of systemic disease (secondary lymphoma); some cases may be primary (see below)
 - lung
 - breast
 - prostate
 - b) metastases that may be osteoblastic:
 - in men: prostate Ca is the most common
 - in women: breast Ca is the most common
2. primary spinal tumors (very rare)
 - a) chordomas (p.825)
 - b) osteoid osteoma (p.990)
 - c) osteoblastoma (p.990)
 - d) **aneurysmal bone cyst (ABC)**: an expansile tumor-like osteolytic lesion consisting of a highly vascular honeycomb of blood-filled cavities separated by connective tissue septa, surrounded by a

thin cortical bone shell which may expand. Comprise 15% of spine tumors.⁴ Etiology is controversial. May arise from preexisting tumor (including: osteoblastoma, giant cell tumor, fibrous dysplasia, chondrosarcoma) or following acute fracture. In the spine, there is a tendency to involve primarily the posterior elements. Peak incidence is in the second decade of life. Treatment usually consists of intralesional curettage. High recurrence rate (25–50%) if not completely excised

- e) chondrosarcoma: a malignant tumor of cartilage. Lobulated tumors with calcified areas
- f) osteochondroma (AKA chondroma AKA osteocartilaginous exostosis): benign tumors of bone that arise from mature hyaline cartilage. Most common during adolescence. An enchondroma is a similar tumor arising within the medullary cavity
- g) vertebral hemangioma (p.992)
- h) **giant cell tumors (GCT)** of bone (p.995) : AKA osteoclastoma
- i) giant cell (reparative) granuloma: AKA solid variant of ABC.⁵ Related to GCT. Occurs primarily in mandible, maxilla, hands and feet, but there are case reports of spine involvement.^{5,6} Not a true neoplasm—more of a reactive process. Treatment: curettage. Recurrence rate: 22–50%, treated with re-excision
- j) brown tumor of hyperparathyroidism
- k) osteogenic sarcoma: rare in spine
- 3. miscellaneous
 - a) plasmacytoma (p.931)
 - b) multiple myeloma (p.928)
 - c) **unifocal Langerhans cell histiocytosis (LHC), formerly eosinophilic granuloma**: osteolytic defect with progressive vertebral collapse; LHC is one cause of **vertebra plana** (p.1679). C-spine is the most commonly affected region. Individual LHCs associated with systemic conditions (Letterer-Siwe or Hand-Schüller-Christian disease) are treated with biopsy and immobilization. Collapse or neurologic deficit from compression may require decompression and/or fusion. Low-dose XRT may also be effective^{7,8}
 - d) Ewing's sarcoma: aggressive malignant tumor with a peak incidence during the second decade of life. Spine mets are more common than primary spine lesions. Treatment is mostly palliative: radical excision followed by XRT (very radiosensitive) and chemotherapy⁹
 - e) chloroma: focal infiltration of leukemic cells
 - f) angioliipoma: ≈ 60 cases reported in literature
 - g) neurofibromas (p.982): most are intradural, but some are extradural, usually dilate neural foramen (dumbbell tumors)
 - h) Masson's vegetant intravascular hemangioendothelioma (p.961) ¹⁰

59.3.3 Intradural extramedullary spinal cord tumors (40%)

1. meningiomas: usually intradural, but may be partly or, in 15%, wholly extradural; see below
2. neurofibromas: usually intradural, but may be partly or wholly extradural
3. many lipomas are extramedullary with intramedullary extension
4. miscellaneous: only ≈ 4% of spinal metastases involve this compartment

59.3.4 Intramedullary spinal cord tumors (5%)

1. astrocytoma (p.985): 30%
2. ependymoma (p.984): 30%, including myxopapillary ependymoma (p.985)
3. miscellaneous: 30%, includes:
 - a) malignant glioblastoma
 - b) dermoid. In addition to the general population, dermoids present in a delayed fashion following ≈ 16% of myelomeningocele (MM) closures.¹¹ An iatrogenic etiology has been debated¹²; however, a case of a congenital dermoid in a newborn with MM¹³ indicates that the origin is not always from incompletely excised dermal elements at the time of MM closure
 - c) epidermoid
 - d) teratoma
 - e) lipoma
 - f) hemangioblastoma (p.986)
 - g) neuroma (very rare intramedullary)
 - h) syringomyelia (not neoplastic)
4. extremely rare tumors
 - a) lymphoma
 - b) oligodendroglioma
 - c) cholesteatoma

- d) intramedullary metastases: comprises only $\approx$ 2% of spinal mets
- e) solitary fibrous tumors of the spinal cord: recognized in 1996. Probable mesenchymal origin. May also occur extramedullary (less common). Treatment is complete surgical excision. Prognosis is unclear¹⁴

59.4 Intradural extramedullary spinal cord tumors

59.4.1 Spinal meningiomas

See reference.¹⁵

Epidemiology

Peak age: 40–70 years. Female:male ratio = 4:1 overall, but the ratio is 1:1 in the lumbar region. 82% thoracic, 15% cervical, 2% lumbar. 90% are completely intradural, 5% are extradural, and 5% are both intra- and extradural. 68% are lateral to the spinal cord, 18% posterior, 15% anterior. Multiple spinal meningiomas occur rarely.

Clinical

Symptoms ($\blacktriangleright$ Table 59.1.)

Signs prior to surgery (only 1 of 174 patients was intact)¹⁵:

1. motor
 - a) pyramidal signs only: 26%
 - b) walks with aid: 41%
 - c) antigravity strength: 17%
 - d) flexion-extension with gravity removed: 6%
 - e) paralysis: 9%
2. sensory
 - a) radicular: 7%
 - b) long tract: 90%
3. sphincter deficit: 51%

Table 59.1 Symptoms of spinal meningiomas

Symptom	At onset	At time of first surgery
local or radicular pain	42%	53%
motor deficits	33%	92%
sensory symptoms	25%	61%
sphincter disturbance		50%

Outcome

Recurrence rate with complete excision is 7% with a minimum of 6 years follow-up (relapses occurred from 4 to 17 years post-op).¹⁵

59.4.2 Spinal schwannomas

General information

Key concepts

- slow growing benign tumors
- most (75%) arise from the dorsal (sensory) rootlets
- early symptoms are often radicular
- recurrence is rare after total excision (except in neurofibromatosis)

Incidence: 0.3–0.4/100,000/yr. Most occur sporadically and are solitary, but they may also be associated with neurofibromatosis (p.637), primarily type 2 (NF2), but can occur with type 1.

Configurations

Most are entirely intradural, but 8–32% may be completely extradural^{16,17}; 1–19% are a combination, 6–23% are dumbbell, and 1% are intramedullary.

► **Dumbbell tumors.** Definition: tumors that develop an “hourglass” shape as a result of an anatomic barrier encountered during growth. Not all dumbbell tumors are schwannomas, e.g., neuroblastoma (p.637). Most have a contiguous intraspinal, foraminal (usually narrower), and extraforaminal components (widening of the neural foramen is a characteristic finding, can be recognized even on plain films, and speaks to the longstanding benign nature of the lesion). The waist may also be due to a dural constriction.

Asazuma et al¹⁸ classification system for dumbbell spinal schwannomas is shown in ► Fig. 59.1.

Type I tumors are intradural and extradural and are restricted to the spinal canal. The constriction occurs at the dura.

Type II are all extradural, and are subclassified as: IIa = do not extend beyond the neural foramen, IIb = inside spinal canal + paravertebral, IIc = foraminal + paravertebral.

Type IIIa are intradural and extradural foraminal, IIIb are intradural and extradural paravertebral.

Type IV are extradural and intravertebral. Type V are extradural and extralaminar with laminar invasion. Type VI show multidirectional bone erosion.

Craniocaudal spread: IF and TF designate the number of intervertebral foramina and transverse foramina involved, respectively (e.g., IF stage 2 = 2 foramina).

Schwannomas involving C1 & C2: May involve vertebral arteries and require additional caution.

Clinical

Patients typically present with local pain.

Neurologic deficits develop late. Tumors may cause radiculopathy (from nerve root compression), myelopathy (from spinal cord compression), radiculomyelopathy (from compression of both), or cauda equina syndrome (for tumors below conus medullaris).

Pathology

Composed of Antoni A (compact, interwoven bundles of long, spindly Schwann cells) and Antoni B tissue (sparse areas of Schwann cells in a loose eosinophilic matrix).

Surgical approaches

See reference.¹⁹

Posterior approaches: Types I, IIa, IIIa, some upper cervical IIIb, and some VI are generally amenable to a posterior approach. IIa and IIIa usually require total facetectomy for complete removal.¹⁸ Reconstruction with instrumentation may be needed if substantial posterior disruption occurs.

Strategy for large tumors: a nerve stimulator can be used to identify a “safe zone” where the capsule can be incised without going through nerve tissue. Then the tumor is debulked internally. It is often not possible to separate tumor from nerve as the tumor passes into the neural foramen. Here it is not unusual to end up with a thin capsule of tumor attached to strands of nerve, and this typically results in satisfactory function.

Anterior and combined anterior/posterior approaches: Asazuma et al¹⁸ recommend a combined approach for Type IIb, IIc, and IIIb lesions where the extraforaminal extension is large (viz. beyond the vertebral arteries). Reconstruction with instrumentation was required for some tumors (= 10% of all patients treated) which were type IV (2 patients), IIIb (1 pt), and VI (1 pt).

Nerve sacrifice

It is usually possible to preserve some fascicles of the nerve root, although sometimes section of the entire nerve root is required. New deficits may not occur, especially with large tumors, since involved fascicles are often nonfunctional, and adjacent roots may compensate. The risk for motor deficit is higher for schwannomas than for neurofibromas, for cervical vs. lumbar tumors, and for cervical tumors with extradural extension.

Outcome

Recurrence is rare following gross total excision, except in the setting of NF2.

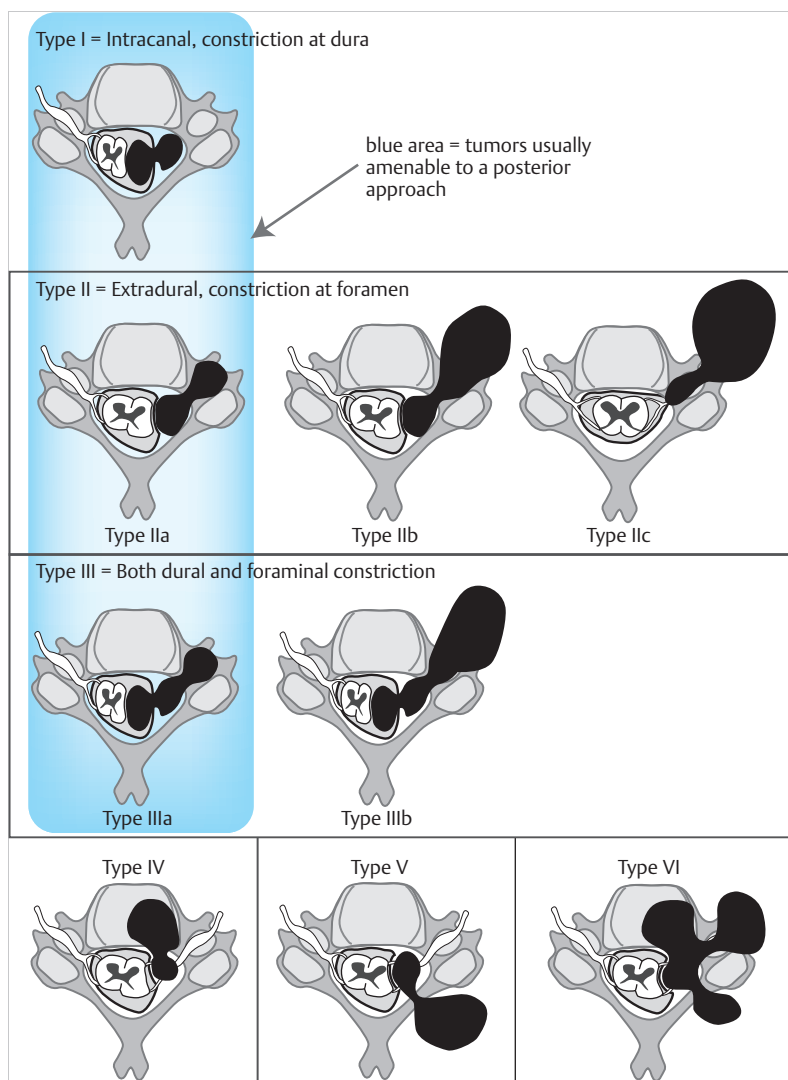


Fig. 59.1 Classification of dumbbell spinal tumors. (Modified with permission from Asazuma T, Yoshiaki T, Hirofumi M, et al. Surgical strategy for cervical dumbbell tumors based on a three-dimensional classification. *Spine* 2003;29(1):E10–4.)

59.5 Intramedullary spinal cord tumors

59.5.1 Types of intramedullary spinal cord tumors

The following list excludes metastases (see below) and lipomas (of questionable neoplastic origin,²⁰ and most are actually extramedullary intradural, see below). **Note:** in pediatrics, astrocytoma and ependymoma constitute 90% of intramedullary spinal cord tumors (IMSCT).

1. astrocytoma (nonmalignant): 30% (the most common IMSCT outside the filum terminale²) tend to be eccentric
2. ependymoma: 30%, tend to be more central, more uniform dense enhancement
3. miscellaneous: 30%, including:
 - a) malignant glioblastoma
 - b) dermoid
 - c) epidermoid (including iatrogenic from LP without stylet)^{21,22}
 - d) teratoma
 - e) hemangioblastoma (see below)
 - f) hemangioma
 - g) neuroma (very rarely intramedullary)
 - h) extremely rare tumors
 - primary lymphoma (only 6 case reports, all non-Hodgkin type²³)
 - oligodendroglioma, only 38 cases in world literature²⁴
 - cholesteatoma
 - paraganglioma
 - primary spinal embryonal tumor ("spinal PNET") (p.744)²⁵
 - pilomyxoid astrocytoma (p.689)
 - metastasis

59.5.2 Differential diagnosis

Also see DDx for Myelopathy (p.1696).

1. neoplasm (tumor): (see above for list). Enhancement: 91% enhance²⁶; of the 9% that do not, most were astrocytomas, 1 was a subependymoma; enhancement did not correlate with grade
2. *nonneoplastic* lesions
 - a) vascular lesions (e.g., AVM): serpiginous linear flow-void. Spinal angiography may be useful²
 - b) demyelinating disease (e.g., multiple sclerosis):
 - usually does not extend > 2 vertebral levels
 - cord lesions in MS are most common in the cervical region
 - c) inflammatory myelitis
 - d) paraneoplastic myelopathy
 - e) diseases causing pain over certain body segments (e.g., cholecystitis, pyelonephritis, intestinal pathology). To differentiate from these, look for dermatomal distribution, increase with Valsalva maneuver, and accompanying sensory and/or motor changes in LEs that suggest cord/radicular lesion. Radiographic studies are frequently required to differentiate
 - f) diseases of vertebral structures, e.g., Paget's disease, giant cell tumors of bone (p.995), etc.

59.5.3 Specific types of intramedullary spinal cord tumors

Ependymoma

General information

Key concepts

- the most common glioma of lower cord, conus, and filum (most ependymomas in conus and filum are myxopapillary ependymomas). More common in adults
- evaluation: includes imaging the entire neuraxis (usually with enhanced MRI: cervical, thoracic, lumbar, and brain) because of potential for seeding through CSF
- associated cysts are common
- treatment: surgical excision (most are encapsulated)

See WHO diagnostic criteria (p.732). The most common glioma of the lower spinal cord, conus and filum (see below). Slow-growing. Benign. Slight male predominance; slight peak in 3rd to 6th decade. Over 50% in filum, next most common location is cervical. Histologically: papillary, cellular, epithelial, or mixed (in filum, myxopapillary ependymoma is most common, see below). Cystic degeneration in 46%. May expand spinal canal in filum.²⁷ Usually encapsulated and minimally vascular (papillary; may be highly vascular; may cause SAH). Symptoms present > 1 yr prior to diagnosis in 82% of cases.²⁸

Myxopapillary ependymoma

Myxopapillary ependymomas are a subtype of ependymoma that tend to occur at the conus medullaris, the filum terminale or cauda equina (► Fig. 59.2). They are WHO grade 1. Usually solitary. Histology: papillary, with microcystic vacuoles, mucosubstance; connective tissue. No anaplasia, but CSF dissemination occurs rarely (can seed intracranially following removal of spinal tumor²⁹). Denovo intracranial lesions also occur rarely. Rare reports of systemic mets.¹ Outside the CNS, may occur in sacrococcygeal subcutaneous tissues from heterotopic rests of ependymal cells.³⁰

Surgical removal of filum tumors consists of coagulating and dividing the filum terminale just above and below the lesion—see Distinguishing features of the filum terminale intraoperatively (p.292)—and excising it in total. The filum is first cut *above* the lesion to prevent retraction upwards.

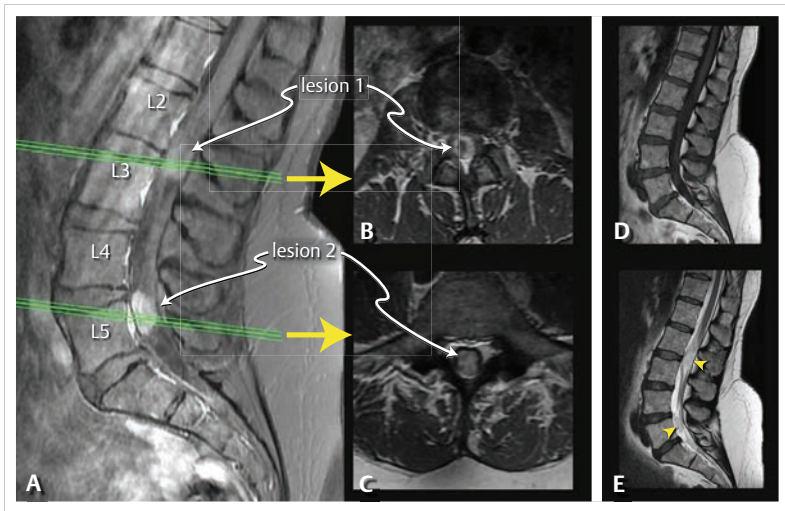


Fig. 59.2 Myxopapillary ependymoma. Two myxopapillary ependymomas of the lumbar spine. Image: contrast T1 MRI. A: midline sagittal, B: axial slice through upper lesion located near the conus, C: axial slice through lower lesion located in the cauda equina. D: unenhanced sagittal T1 image, the lesions are isointense with their surroundings. E: lesions (yellow arrowheads) are fairly easily identified on this noncontrast T2WI sagittal image.

Astrocytoma

Uncommon in first year. Peak: 3rd–5th decade. Male:female = 1.5:1. The ratio of low-grade:high-grade = 3:1 in all ages.²⁷ Occurs at all levels, thoracic most common, then cervical. 38% are cystic; cyst fluid usually has high protein.

Dermoid and epidermoid

Epidermoids are rare before late childhood. Slight female predominance. Cervical and upper thoracic rare; conus common. Usually ID-EM, but conus/cauda equina may have IM component (completely IM lesions rare).

Lipoma

May occur in conjunction with spinal dysraphism, see Lipomyeloschisis (p.284). The following considers lipomas that occur in the absence of spinal dysraphism.

Peak occurrence: 2nd, 3rd, and 5th decade. Technically hamartomas. No sex predominance. Usually ID-EM (a sub-type is truly IM and essentially replaces the cord³¹), cervicothoracic region is the most common location. NB: unlike other IMSTs, most common symptom is ascending mono- or paraparesis (c.f. pain). Sphincter disturbance is common with low lesions. Local subcutaneous masses or dimples are frequent. Malis recommends early subtotal removal at about 1 year of age in asymptomatic patient.³¹ Superficial extrasacral removal is inadequate, as patients then develop dense scarring intraspinally leading to fairly rapid severe neurological damage with poor salvageability even after the definitive procedure.

Hemangioblastoma

(► Fig. 59.3). Usually non-infiltrating, well demarcated, may have cystic caps. 80% of patients with spinal hemangioblastoma will have von Hippel-Lindau disease (p.646). Cannot incise nor core because of vascularity. Requires microsurgical approach similar to AVM, possibly with intraoperative hypotension.

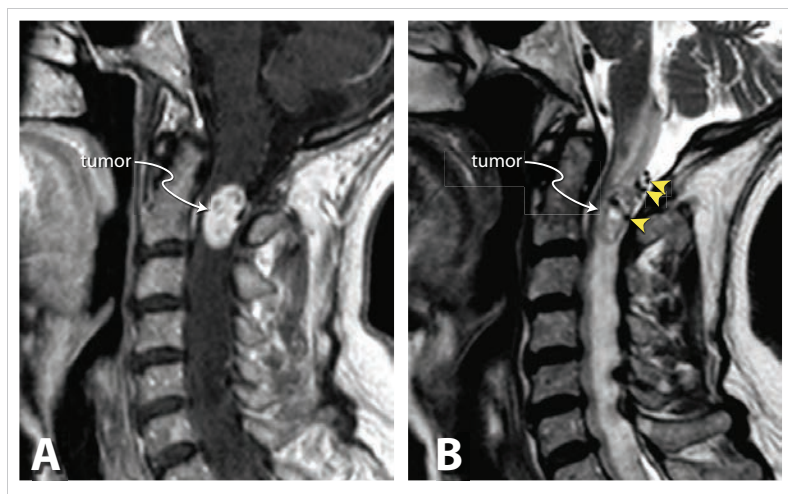


Fig. 59.3 Intramedullary hemangioblastoma of the cervical spinal cord.

Sagittal MRI. A: T1 with contrast. B: T2.

Flow voids appear dark on the T2 image, with several of them denoted by yellow arrowheads.

Metastases

Most spinal mets are extradural. Intramedullary metastases are rare,³² accounting for 3.4% of symptomatic metastatic spinal cord lesions.³³ Primaries include: small-cell lung Ca,³⁴ breast Ca, malignant melanoma, lymphoma and colon Ca.^{33,35} Ca rarely presents first as an intramedullary spinal met.

59.5.4 Presentation

1. pain: the most common complaint. Almost always present in filum tumors (exception: lipomas).²¹ Possible pain patterns:
 - a) radicular: increases with Valsalva maneuver and spine movement. Suspect SCT if dermatome is unusual for disc herniation
 - b) local: stiff neck or back, Valsalva maneuver increases pain.
- ★ Pain during recumbency ("nocturnal pain") is classic for SCT

- c) medullary (as in syrinx): oppressive, burning, dysesthetic, non-radicular, often bilateral, unaffected by Valsalva maneuver
- 2. motor disturbances
 - a) weakness is 2nd or 3rd most common complaint. Usually follows sensory symptoms temporally
 - b) children present most frequently with gait disturbances
 - c) syringomyelic syndrome: suggests IMSCT. Findings: UE segmental weakness, decreased DTR, dissociative anesthesia (see below)
 - d) long-tract involvement → clumsiness and ataxia (distinct from weakness)
 - e) atrophy, muscle twitches, fasciculations
- 3. non-painful sensory disturbances
 - a) dissociated sensory loss: decreased pain and temperature, preserved light touch, as in Brown-Séquard syndrome (p. 1135). There is disagreement whether this is common² or uncommon³⁶ in IMSCT. ± non-radicular dysesthesias (early), with upward extension³⁷
 - b) paresthesias: either radicular or “medullary” distribution
- 4. sphincter disturbances
 - a) usually urogenital (anal less common) → difficulty evacuating, retention, incontinence, and impotence. Early in conus/cauda equina lesions, especially lipomas (pain not prominent)
 - b) sphincter dysfunction common in age < 1 yr due to frequency of lumbosacral lesions (dermoids, epidermoids, etc.)
- 5. miscellaneous symptoms:
 - a) scoliosis or torticollis
 - b) SAH
 - c) visible mass over spine

Time course of symptoms

Onset usually insidious, but abruptness occurs (benign lesions in children occasionally progress in hours). The onset is often erroneously attributed to coincidental injury. Temporal progression has been divided into 4 stages³⁸:

1. pain only (neuralgic)
2. Brown-Séquard syndrome
3. incomplete transectional dysfunction
4. complete transectional dysfunction

Note: 78% (of 23) ependymomas, 74% (of 42) gliomas, all 7 dermoids, and 50% (of 8) lipomas reached the latter 2 stages before diagnosis (not affected by location in cross-sectional nor longitudinal dimension of SC (excludes conus lesions—more frequently diagnosed in 1st stage) (a pre-CT study)).

59.5.5 Diagnosis

It is usually difficult to distinguish IMSCT, ID-EM, and ED on clinical grounds.² Schwannomas often start with radicular symptoms that later progress to cord involvement. Most IMSCTs are located posteriorly in cord, which may cause sensory findings to predominate early.²⁰

Diagnostic studies

MRI: mainstay of diagnosis. Ependymomas enhance intensely and are often associated with hemorrhage and cysts. Cord edema may mimic a cyst.

Plain radiographs: vertebral body destruction, enlarged intervertebral foramina, or increases in interpedicular distances suggests ED SCT.

Lumbar puncture: Elevated protein is the most common abnormality¹ seen in ≈ 95%. The reported range with primary IMSCTs is 50–2,240 mg%. Glucose is normal except with meningeal tumor. SCT can cause complete block, indicated by:

- Froin's syndrome: clotting (due to fibrinogen) and xanthochromia of CSF
- Queckenstedt's test (failure of jugular vein compression to increase CSF pressure, which it normally does in the absence of block)
- barrier to flow of myelographic contrast media

Myelography (p.925): classically shows fusiform cord widening (may be normal early). Distinct from ED tumors, which produce hourglass deformity (with incomplete block) or paintbrush effect

(with complete block), or ID-EM tumors, which produce a capping effect with a sharp cutoff (meniscus sign).

CT: some IMSCTs enhance with IV contrast. Myelo-CT distinguishes IMSCT from ID-EM (poor in differentiating IMSCT subtypes).

Spinal angiography: rarely indicated, except in hemangioblastoma (may be suspected on myelography or MRI by linear serpiginous structures). MRI often obviates this test.

59.5.6 Management

General information

Asymptomatic lesions may be followed since there is significant risk of neurologic deficit with surgery. For symptomatic lesions, surgery should be performed as soon as possible (generally not as an emergency) after diagnosis since surgical results correlate with the preoperative neurologic condition, and it makes no sense to follow the patient as they develop progressive neurologic deficit³⁹ (some of which may be irreversible).

Astrocytomas: For low grade lesions, if a plane can be developed between the tumor and spinal cord (when it can, it usually consists of a thin gliotic layer traversed by small blood vessels and adhesions²⁰), an attempt at total excision is an option.⁴⁰ For high-grade astrocytomas or for low-grade astrocytomas without a plane of separation, biopsy alone or biopsy plus limited excision is recommended.⁴⁰

For high-grade lesions, post-op XRT ($\pm$ chemotherapy) is recommended.⁴⁰ XRT is not supported following radical resection of low grade gliomas.⁴⁰

Ependymomas: Gross total removal should be attempted. XRT is not recommended following gross total removal.⁴⁰

59.5.7 Technical surgical considerations

Key concepts

In the surgical removal of IMSCT

- in almost all cases, IMSCTs should be debulked from within using ultrasonic aspirator or laser (to minimize manipulation of neural tissue), and no attempt should be made initially to develop a plane between tumor and spinal cord (even for ependymoma, which of the 3 most common IMSCTs is the only one that actually has such a plane)
- ★ the key to removing IMSCTs is knowing when to stop. If MEPs or D-waves are being monitored, it is suggested that tumor removal should be discontinued if the amplitudes drop to $\leq 50\%$ of baseline

1. position: usually prone, well padded and also securely taped to avoid undesirable movement if MEP monitoring is to be used. Other options include: lateral oblique, sitting
2. if a cystic component is suspected, partial aspiration with a 25 Ga needle once the spinal cord is exposed will decrease the pressure (do not try to completely aspirate the contents because that can make it more difficult to locate the tumor).⁴¹ If the cyst forms a "cap" at either end of the tumor, the dura does not need to be opened over the cyst, since drainage can be accomplished with removal of tumor
3. adjunctive options include:
 - a) electrophysiologic monitoring (p.989)
 - b) intraoperative ultrasound: also controversial,⁴² favored by some experts. Astrocytomas are usually isoechoic with spinal cord, whereas ependymomas are usually hyperechoic
4. a myelotomy is performed either in the midline or just to one side of the dorsal midline to avoid the posteromedian vein. Alternatively, if the tumor is known to be very superficial off the midline (which may be confirmed by ultrasound), entry may be made there. Tumors may cause distortion and displacement of the midline—look for dorsal root entry zones on both sides to identify the midline as the midpoint between root entry zones
5. 6-0 silk sutures may be placed through the pial edge to gently retract the spinal cord open. Standard sized (i.e., non-micro) bayonet forceps can be used to gently spread tissues
6. copious irrigation is used whenever bipolar cautery is employed on the tumor/spinal cord in order to minimize transference of heat to the spinal cord. Monopolar cautery should not be used⁴¹

7. either laser or ultrasonic aspiration (USA) is used to debulk tumor from within until the glial-tumor interface is reached. Charring from laser may make it more difficult to recognize the glial-tumor interface than USA, and the laser tends to be slower when debulking larger tumors
8. watertight dural closure is critical

► **Electrophysiologic monitoring.** Intraoperative electrophysiologic monitoring (SSEP and motor evoked potentials [MEPs]⁴³ and D-waves) is an aid in determining when to stop.

SSEPs almost always degrade with the initial myelotomy and do not correlate well with motor outcome⁴⁴ (which is critical),^{42,45} and postoperative motor deficit may occur in spite of unaltered intraoperative SSEPs.^{43,44} Conversely, SSEPs may be lost without motor deficit.

MEPs are useful, but since they can only be obtained intermittently they do not give real-time warning of a developing problem, and they often cause violent muscle contractions which can be disruptive to delicate surgery. There is a risk of seizures with transcranial MEPs, however none of 322 monitored patients undergoing *spinal cord* surgery had a seizure.⁴⁶

Direct waves (D-waves): a single transcranial stimulus of 80–100 mA of 0.5–1 mS duration with a frequency 0.5–2 Hz produces a response that can be recorded by an electrode in the epidural or subdural space of the spinal cord.⁴⁷ They may be monitored continuously and since they do not need averaging they can alert the surgeon immediately when deterioration starts to occur.⁴⁸ They can determine the level of a deficit by moving the electrode to different levels, but they cannot determine laterality and cannot be used below T12 (due to too few corticospinal tract fibers) or in children less than ≈ 4 years age (due to incomplete maturation of motor pathways).⁴⁸ Deterioration to <50% of baseline is highly correlated with severe post-op deficit.

59.5.8 Prognosis

No well-designed studies give long term functional results with microsurgery, laser, and radiotherapy. Better results occur with lesser initial deficits.²⁰ Recurrence depends on totality of removal, and on growth pattern of the specific tumor.

Ependymoma: total extirpation improves functional outcome, and myxopapillary ependymomas fare better than the “classic” type.²⁸ Best functional outcome occurs with modest initial deficits, symptoms < 2 years duration,⁴⁹ and total removal. Survival is independent of extent of excision.

Astrocytomas: radical removal is rarely possible (a cleavage plane is unusual even with a microscope). Long term functional results are poorer than with ependymomas. There is 50% recurrence rate in 4–5 yrs.

59.6 Primary bone tumors of the spine

59.6.1 General information

Types of tumors.

1. metastatic: the most common malignancy of spine
 - a) common osteolytic metastatic tumors (p.921) include:
 - lung
 - breast
 - prostate
 - lymphoma: most cases represent spread of systemic disease (secondary lymphoma); however, some may be primary (p.979)
 - plasmacytoma (p.928)
 - multiple myeloma (p.928)
 - Langerhans cell histiocytosis: see differentiating features (p.980)
 - b) metastases that may be osteoblastic:
 - in men: prostate Ca is the most common
 - in women: breast Ca is the most common
 - c) Ewing’s sarcoma (p.980)
 - d) chloroma: focal infiltration of leukemic cells
2. primary spinal tumors (very rare)
 - a) benign
 - vertebral hemangioma (p.980)
 - osteoid osteoma (p.990)
 - osteoblastoma (p.990)
 - aneurysmal bone cyst (p.979): cavity of highly vascular honeycomb surrounded by a thin cortical shell which may expand
 - osteochondroma (chondroma) (p.980)

- giant cell tumors of bone (p.995): AKA osteoclastoma. Almost always benign with pseudo-malignant behavior
- b) malignant
 - chondrosarcoma (p.980)
 - chordomas (p.825)
 - osteogenic sarcoma: rare in spine

59.6.2 Osteoid osteoma and osteoblastoma

General information

Key concepts

- both are benign bone tumors
- histologically identical, differentiation depends on size (≤ 1 cm = osteoid osteoma, > 1 cm = osteoblastoma)
- can occur in the spine and may cause neurologic symptoms (esp. osteoblastoma)
- high cure rate with complete excision

Two types of benign osteoblastic lesions of bone: osteoid osteoma (OO) and benign osteoblastoma (BOB), see ► Table 59.2. They are indistinguishable histologically, and must be differentiated based on size and behavior.

Characteristically cause night pain and pain relieved by aspirin (see Clinical below).

Table 59.2 Comparison of osteoid osteoma and benign osteoblastoma⁵⁰

	Osteoid osteoma	Benign osteoblastoma
percent of primary bone tumors	3.2%	
percent of primary vertebral tumors	1.4%	
percent that occur in spine	10%	35%
size limitations	≤ 1 cm	> 1 cm
growth pattern	confined, self limiting	more extensive, may extend into spinal canal
potential for malignant change?	no	rare
location within spine (83 patients)		
• % in cervical spine	27%	25%
• % in thoracic spine		35%
• % in lumbar region	59%	35%
location within vertebra (81 patients)		
• lamina only	33%	16%
• pedicle only	15%	32%
• articular facet only	19%	0
• vertebral body (VB) only	7%	5%
• transverse process only	6%	8%
• spinous process	5%	5%
• > 1 element of neural arch	6%	19%
• combined posterior elements & VB	0	11%

Osteoblastoma is a rare, benign, locally recurrent tumor with a predilection for spine that may rarely undergo sarcomatous change (to osteosarcoma,⁵¹ only a handful of known cases of this). More vascular than OO.⁵²

Differential diagnosis

Lesions with similar symptoms and increased uptake on radionuclide bone scan:

1. benign osteoblastoma

2. osteoid osteoma: more pronounced sclerosis of adjacent bone than BOB
3. osteogenic sarcoma: rare in spine
4. aneurysmal bone cyst (p.979): typically trabeculae in central, lucent region
5. unilateral pedicle/laminar necrosis

Clinical

See ► Table 59.3 for signs and symptoms. Tenderness confined to vicinity of the lesion occurs in $\approx 60\%$. 28% of patients with BOB presented with myelopathy. OO presented with neurologic deficit in only 22%.

Table 59.3 Signs and symptoms in 82 patients⁵⁰

Finding	Osteoid osteoma	Benign osteoblastoma
pain on presentation	100%	100%
pain increased by motion	49%	74%
pain increased by Valsalva	17%	36%
nocturnal pain	46%	36%
pain relieved by aspirin	40%	25%
radicular pain	50%	44%
scoliosis	66%	36%
neurologic abnormalities	22%	54%
myelopathy	0	28%
weakness	12%	51%
atrophy	9%	15%

Evaluation

Bone scans are a very sensitive means for detecting these lesions. Once localized, CT or MRI may better define the lesion in that region.

Caution regarding needle biopsy: if the lesion turns out to be osteosarcoma, the contaminated needle tract can result in worse prognosis.

Osteoid osteoma

Radiolucent area with or without surrounding density, often isolated to pedicle or facet. May not show up on tomograms.

Osteoblastoma

Most are expansile, destructive lesions, with 17% having moderate sclerosis. 31% have areas of ↑ density, 20% surrounded by calcified shell. Often a contralateral spondylolysis.⁵¹

Treatment

In order to obtain a cure, these lesions must be *completely* excised. The role of radiation therapy is poorly defined in these lesions, but is probably ineffective.⁵¹

Osteoid osteoma

Cortical bone may be hardened and thickened, with granulomatous mass in underlying cavity.

Osteoblastoma

Hemorrhagic, friable, red to purple mass well circumscribed from adjacent bone. Complete excision → complete pain relief in 93%. Curettage only → pain relief, with more likely recurrence. Recurrence rate with total excision is $\approx 10\%$.

59.6.3 Osteosarcoma

The most common primary bone cancer. More common in children, usually occurring near the ends of long bones, but also in the mandible, pelvis, and rarely in the spine.⁵³ Spinal osteosarcoma usually

occurs in the lumbosacral region in males in their 40s, sometimes arising from areas of osteoblastoma or Paget's disease. If a percutaneous biopsy reveals osteosarcoma, the contaminated needle tract can increase the difficulty of subsequent surgery. Poor prognosis, median survival = 10 months.⁵³

59.6.4 Vertebral hemangioma

General information

Key concepts

- the most common primary spine tumor
- benign, but a small number behave aggressively
- rarely symptomatic (< 1.2%), symptoms typically from compression fracture, and, rarely, neural compression from bone expansion or soft tissue mass
- MRI: small lesions are hyperintense on T1WI and T2WI. Larger ones may be hypointense. CT or X-ray: striations (corduroy pattern) or "honeycomb" or polka dot pattern. Bone scan: usually do not have increased uptake
- treatment: incidental lesions require no treatment or routine follow-up. Biopsy when mets or malignancy are a strong consideration. Treatment options (when indicated): XRT, embolization, vertebroplasty (better than kyphoplasty), surgery

Vertebral hemangiomas (VH), AKA spinal hemangioma, cavernous hemangioma (of the vertebral body), or hemangiomatous angioma. Benign lesions of the spine. The most common primary tumor of the spine (10–12% of primary spinal bone tumors). Estimated incidence: 9–12%,^{54,55} with a 10–12% prevalence in the general population.⁵⁴ 70% are solitary, 30% are multiple (up to 5 levels may be involved, often noncontiguous). Lumbar and lower thoracic spine are the most common locations; cervical and sacral lesions are rare. Lesions involve only the vertebral bodies in ≈ 25%, posterior spinal arch in ≈ 25%, and both areas in ≈ 50%. Rarely, aggressive hemangiomas may compromise neural structures (► Fig. 59.7).

Occasional cases of purely extradural lesions have been described.⁵⁶ Intramedullary spinal cord lesions are even less common.⁵⁷ Typically found in post-pubertal females.

Malignant degeneration does not occur. Mature thin-walled blood vessels of varying sizes replace normal marrow, producing hypertrophic sclerotic bony trabeculations oriented in a rostral-caudal direction in one of two forms: cavernous (venous) or capillary (difference in subtype carries no prognostic significance).

Presentation

1. incidental (i.e., asymptomatic) lesions
2. symptomatic: only 0.9–1.2% are symptomatic. There may be a hormonal influence (unproven) that may cause symptoms to increase with pregnancy (could also be due to increased blood volume and/or venous pressure)⁵⁸ or to vary with the menstrual cycle and may explain why symptoms rarely occur before puberty
 - a) pain: occasionally VH may present with pain localized to the level of involvement with no radiculopathy. However, pain is more often due to other pathology (compression fracture, herniated disc, spinal stenosis...) rather than the VH itself
 - b) progressive neurologic deficit: this occurs rarely, and usually takes the form of thoracic myelopathy. Deficit may be caused by the following mechanisms
 - subperiosteal (epidural) growth of tumor into the spinal canal
 - expansion of the bone (cortical "blistering") with widening of the pedicles and lamina producing a "bony" spinal stenosis
 - compression by vessels feeding or draining the lesion
 - compression fracture of the involved vertebra (very rare)⁵⁹
 - spontaneous hemorrhage producing spinal epidural hematoma⁶⁰ (also very rare)
 - spinal cord ischemia due to "steal"

Imaging

Typical imaging features of VH are shown in ► Table 59.4.⁶¹

Table 59.4 Typical features of VH⁶¹

Typical features of VH include:

1. lesion limited primarily to vertebral body
2. vertical striation/honeycomb appearance/polka-dot sign on CT
3. hypointense on T1 and hyperintense on T2 MRI
4. enhancement on contrast MRI
5. restriction to 1 vertebral level, and when soft tissue mass invades the spinal canal, the soft tissue mass is < 1 vertebral body in length

Plain X-rays: classically show coarse vertically oriented striations (corduroy pattern) or a “honeycomb” appearance. At least $\approx$ one-third of the VB must be involved to produce these findings on plain X-ray (see ► Fig. 59.4 for demonstration of this finding on sagittal CT).

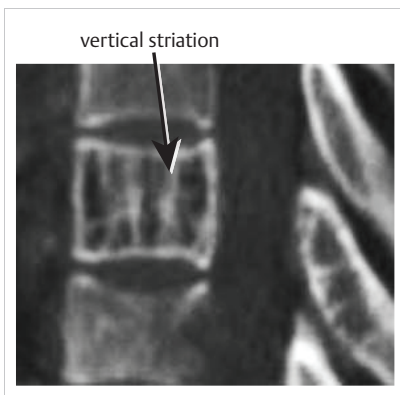


Fig. 59.4 Vertebral hemangioma.

Image: sagittal CT, bone window. Note the vertical striations (“corduroy pattern”).

Bone scan: findings on multiphase⁹⁹Tc-MDP bone scintigraphy and PET scan are highly variable and are therefore not helpful.⁶² If the lesion is “cold” (does not light up), it is likely not a metastasis.

MRI: small hemangiomas are focal, round, and hyperintense on T1WI and T2WI (► Fig. 59.5). More extensive lesions can be hypointense. MRI may help distinguish lesions that tend not to evolve (mottled increased signal on T1WI and T2WI, possibly due to adipose tissue) from those that tend to be symptomatic (isointense on T1WI, hyperintense on T2WI). Axial MRI may show “salt & pepper” appearance. VHs enhance due to high vascularity.

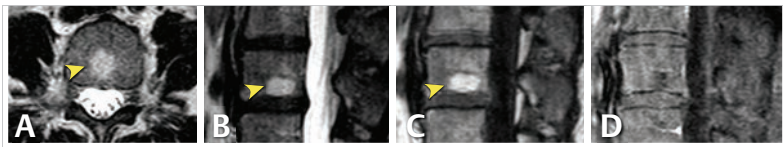


Fig. 59.5 Small round vertebral hemangioma (yellow arrowheads).

Image: MRI scans, A: axial T2, B: sagittal T2, C: sagittal T1 with contrast (note lack of enhancement of the lesion).

CT: diagnostic procedure of choice for the bony involvement. Sagittal images may show vertical striations representing thickened trabeculae (► Fig. 59.4) while axial images show multiple high density dots within the bone (“Polka-dot sign”⁶³) representing cross-sections through the trabeculae (► Fig. 59.6). NB: similar-appearing features may be seen in osteoporosis.

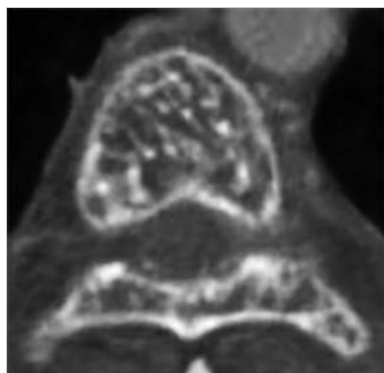


Fig. 59.6 Large vertebral hemangioma.
Image: axial CT bone window demonstrating "polka-dot sign."

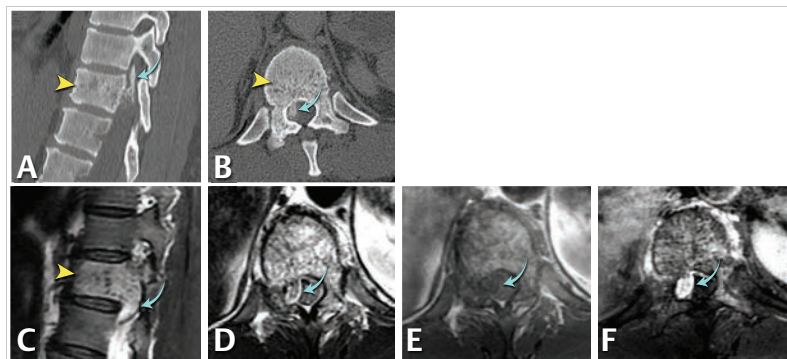


Fig. 59.7 Aggressive vertebral hemangioma.

Image: CT scan, A: sagittal through left side of spinal canal, B: axial.

MRI, C: sagittal T2 through left side of spinal canal, D: axial T2, E: axial T1, F: axial T1 + contrast.

Note the mottling of the VBs (yellow arrowheads), the extension of the lesion into the spinal canal (curved blue arrow), and the intense contrast enhancement.

Management

Management guidelines:

1. asymptomatic VH with typical features (► Table 59.4) without concern for malignancy: no treatment or routine follow-up is required^{54,61}
2. biopsy: may be indicated in cases where malignancy (including metastases) is a concern or when there are atypical features on imaging (see typical features, ► Table 59.4).⁶¹ In spite of their highly vascular nature, the risk of bleeding complications with CT guided biopsy is extremely low. An alternative to biopsy would be serial imaging, but the literature is sparse on this topic
3. those presenting with pain or neurologic deficit
 - a) **radiation therapy**: VH are radiosensitive and undergo sclerotic obliteration. With aggressive VH, there is a risk of vertebral collapse following XRT⁶⁴
 1. as a stand-alone treatment: for lesions causing pain without neuro deficit, XRT provided complete relief in 60-87.5%, and partial relief in nearly all.^{65,66} Improvement in pain may be delayed up to years.

With signs/symptoms of mild myelopathy (e.g. ASIA impairment scale ► Table 68.13 group D), some studies have shown clinical improvement with XRT. One review recommended a median total dose of 34 Gy, with a median single doses of 2.0 Gy.⁶⁶ Radiographic evidence of response may⁶⁶ or may not⁵⁴ occur

2. as an adjunct following incomplete resection to decrease recurrence.⁶⁷ Total dosage should be ≤ 40 Gy to reduce the risk of radiation myelopathy
- b) **embolization**: can provide immediate relief of pain, and may be used alone⁶⁸ or as a pre-op surgical adjunct to reduce operative blood loss. Risks: spinal cord infarction if major radicular artery—e.g., artery of Adamkiewicz (p.87)—is embolized, and it may increase the risk of future fracture (not statistically significant). The decision whether to embolize pre-op is up to the surgeon^{68,69,70,71}
- c) **vertebroplasty** (p.1212): has been used for painful VH or following XRT. May be better than kyphoplasty for VH because kyphoplasty destroys the trabecular bone, but no direct comparisons have been made. May be performed at the same time as CT-guided biopsy.^{72,73} Not known to be effective for VH with neurologic deficit, but may be used as an adjunct to surgery^{74,75}
- d) **surgery**: generally regarded as the primary treatment for VH producing severe or rapidly progressive neurologic deficit, or for painful lesions that fail to respond to above measures. Recurrence rate is low (see below) and resolution of symptoms is high. When needed, surgery can stabilize the spine with fusion. Surgery may be discussed with the patient as an option for treatment of VH presenting only with pain if the comparative risk profile is preferred over the other options above, especially for lesions with high risk of instability

Surgical treatment

For surgical indications, see above. Recommendations for surgical management are shown in ► Table 59.5.

Intralesional resection results in a low recurrence rate (3% with 3.9 years mean follow-up) (i.e., the increased risk with en bloc resection is not warranted),⁷⁶ and resolution of symptoms is high. Pre-operative embolization is a viable option.

Major risks of surgery: blood loss, destabilization of the spine requiring fusion, neurologic deficit (during surgery, or postoperatively usually from epidural hematoma).

With subtotal resection, recurrence rate is 20–30% at 2 years, which was decreased to 7% using post-op XRT.^{54,69} Surgery often requires fusion due to the common involvement of the anterior and middle columns of the thoracic or lumbar spine.^{69,76,77,78}

Table 59.5 Recommendations for surgical management of VH^a 54

VH involvement	Approach	Post-op XRT?
posterior elements only	radical excision via posterior approach	not for total excision
VB involvement with anterior canal compression (with or without ST in canal)	anterior corpectomy with strut graft	
VB involved but no expansion, ST in lateral canal	laminectomy with removal of soft-tissue	follow serial CT, give XRT if VB expansion or ST expansion
extensive involvement of anterior and posterior vertebral elements with circumferential bone expansion, no ST compression	laminectomy	either XRT, or close follow-up with CT and XRT for ST recurrence or progressive VB expansion
extensive anterior and posterior involvement with ST in anterior canal	anterior corpectomy with strut graft	

^aabbreviations: XRT = radiation treatment; ST = soft-tissue component of VH; VB = vertebral body.

59.6.5 Giant cell tumors of bone

AKA osteoclastoma (cells arise from osteoclasts). In the same general category as aneurysmal bone cysts. Typically arise in adolescence. Most common in knees and wrists. Those that come to the attention of the neurosurgeon generally arise in the skull (especially the skull base, and in particular the sphenoid bone), or in the vertebral column ($\approx 4\%$ occur in sacrum).

Pathology

Lytic with bony collapse. Almost always benign with pseudomalignant behavior (recurrence is common, and pulmonary mets can occur).

Evaluation

Soft tissues are best evaluated with MRI. Spine CT is critical to assess degree of bony destruction and for surgical planning purposes.

Work-up includes chest CT because of possibility of pulmonary mets.

Treatment

Intratumoral curettage, possibly aided by pre-op embolization. Recurrence rate with this treatment (even if resection is subtotal) is only $\approx 20\%$. Role of XRT is controversial⁷ because of the possibility of malignant degeneration (therefore use XRT only for non-resectable recurrence).

Osteoclast inhibiting drugs have met with some success following subtotal resection:

1. bisphosphonates: e.g., pamidronate (p. 1364)
2. RANKL inhibitors: e.g., denosumab (Prolia®) (p. 926))

For gross residual disease after resection, re-resection is a consideration.

Cryosurgery with liquid nitrogen has been employed in long bones. Its use is limited in neurosurgical cases because of risk of injury to adjacent neural structures (brain, spinal cord) and cryotherapy induced fractures, although it has been described for use in the sacrum.⁷⁹

Close follow-up is required due to propensity for recurrence. MRI or CT initially q 3 months is suggested.

References

- [1] Kopelson G, Linggood RM, Kleinman GM, et al. Management of Intramedullary Spinal Cord Tumors. *Radiology*. 1980; 135:473-479
- [2] Adams RD, Victor M. Intraspinal Tumors. In: Principles of Neurology. 2nd ed. New York: McGraw-Hill; 1981:638-641
- [3] Lyons MK, O'Neill BP, Kurtin PJ, et al. Diagnosis and Management of Primary Spinal Epidural Non-Hodgkin's Lymphoma. *Mayo Clinic Proceedings*. 1996; 71:453-457
- [4] Liu JK, Brockmeyer DL, Dailey AT, et al. Surgical management of aneurysmal bone cysts of the spine. *Neurosurg Focus*. 2003; 15
- [5] Suzuki M, Satoh T, Nishida J, et al. Solid variant of aneurysmal bone cyst of the cervical spine. *Spine*. 2004; 29:E376-E381
- [6] Neviaser JS, Eisenberg SH. Giant cell reparative granuloma of the cervical spine; case report. *Bulletin of the Hospital for Joint Diseases*. 1954; 15:73-78
- [7] Dunn EJ, Davidson RI, Desai S, et al. Diagnosis and Management of Tumors of the Cervical Spine. In: *The Cervical Spine*. 2nd ed. Philadelphia: JB Lippincott; 1989:693-722
- [8] Menezes AH, Sato Y. Primary Tumors of the Spine in Children - Natural History and Management. *Concepts Pediatr Neurosurg*. 1990; 10:30-53
- [9] Grubb MR, Currier BL, Pritchard DJ, et al. Primary Ewing's Sarcoma of the Spine. *Spine*. 1994; 19:309-313
- [10] Porter DG, Martin AJ, Mallucci CL, et al. Spinal Cord Compression Due To Masson's Vegetant Intravascular Hemangioendothelioma: Case Report. *Journal of Neurosurgery*. 1995; 82:125-127
- [11] Scott RM, Wolpert SM, Bartoshesky LE, et al. Dermoid tumors occurring at the site of previous myelomeningocele repair. *Journal of Neurosurgery*. 1986; 65:779-783
- [12] Storts BB. Are dermoid and epidermoid tumors preventable complications of myelomeningocele repair? *Pediatr Neurosurg*. 1994; 20:160-162
- [13] Ramos E, Marlin AE, Gaskill SJ. Congenital dermoid tumor in a child at initial myelomeningocele closure: an etiological discussion. *J Neurosurg Pediatrics*. 2008; 2:414-415
- [14] Metellus P, Bouvier C, Guyotat J, et al. Solitary fibrous tumors of the central nervous system: clinicopathological and therapeutic considerations of 18 cases. *Neurosurgery*. 2007; 60:715-22; discussion 722
- [15] Solero CL, Fornari M, Giombini S, et al. Spinal meningiomas: review of 174 operated cases. *Neurosurgery*. 1989; 25:153-160
- [16] Seppala MT, Haltia MJ, Sankila RJ, et al. Long-term outcome after removal of spinal schwannoma: a clinicopathological study of 187 cases. *J Neurosurg*. 1995; 83:621-626
- [17] Conti P, Pansini G, Mouchaty H, et al. Spinal neurinomas: retrospective analysis and long-term outcome of 179 consecutively operated cases and review of the literature. *Surg Neurol*. 2004; 61:34-43; discussion 44
- [18] Asazuma T, Toyama Y, Maruiwa H, et al. Surgical strategy for cervical dumbbell tumors based on a three-dimensional classification. *Spine*. 2004; 29:E10-E14
- [19] Gottfried ON, Binning MJ, Schmidt MH. Surgical Approaches to Spinal Schwannomas. *Contemporary Neurosurgery*. 2005; 27:1-8
- [20] Stein B. Intramedullary Spinal Cord Tumors. *Clin Neurosurg*. 1983; 30:717-741
- [21] Stern WE. Localization and Diagnosis of Spinal Cord Tumors. *Clin Neurosurg*. 1977; 25:480-494
- [22] DeSouza AL, Kalsbeck JE, Mealey J, et al. Intraspinal Tumors in Children. A Review of 81 Cases. *J Neurosurg*. 1979; 51:437-445
- [23] Hautzer NW, Ayiesimoju A, Robitaille Y. Primary Spinal Intramedullary Lymphomas: A Review. *Ann Neurol*. 1983; 14:62-66
- [24] Alvisi C, Cerisoli M, Giuloni M. Intramedullary Spinal Gliomas: Long Term Results of Surgical Treatment. *Acta Neurochirurgica*. 1984; 70:169-179
- [25] Kumar R, Reddy SJ, Wani AA, et al. Primary spinal primitive neuroectodermal tumor: case series and review of the literature. *Pediatr Neurosurg*. 2007; 43:1-6
- [26] White JB, Miller GM, Layton KF, et al. Nonenhancing tumors of the spinal cord. *J Neurosurg Spine*. 2007; 7:403-407
- [27] Dorwart RH, LaMasters DL, Watanabe TJ, et al. Tumors. In: *Computed Tomography of the Spine and Spinal Cord*. San Anselmo: Clavadel Press; 1983:115-131
- [28] Mork SJ, Loken AC. Ependymoma: A Follow-Up Study of 101 Cases. *Cancer*. 1977; 40:907-915
- [29] Tzerakis N, Georgakoulas N, Kontogeorgos G, et al. Intraparenchymal myxopapillary ependymoma: case report. *Neurosurgery*. 2004; 55
- [30] Helwig EB, Stern JB. Subcutaneous sacrococcygeal myxopapillary ependymoma. A clinicopathologic study of 32 cases. *Am J Clin Pathol*. 1984; 81:156-161
- [31] Malis LI. Intramedullary Spinal Cord Tumors. *Clin Neurosurg*. 1978; 25:512-539

- [32] Smaltino F, Bernini FP, Santoro S. Computerized Tomography in the Diagnosis of Intramedullary Metastases. *Acta Neurochirurgica*. 1980; 52:299-303
- [33] Edelson RN, Deck MDF, Posner JB. Intramedullary Spinal Cord Metastases. *Neurology*. 1972; 22:1222-1231
- [34] Murphy KC, Feld R, Evans WK, et al. Intramedullary Spinal Cord Metastases from Small Cell Carcinoma of the Lung. *J Clin Onc*. 1983; 1:99-106
- [35] Jellinger K, Kothbauer P, Sunder-Plassmann, et al. Intramedullary Spinal Cord Metastases. *J Neurol*. 1979; 220:31-41
- [36] Stein B. Surgery of Intramedullary Spinal Cord Tumors. *Clin Neurosurg*. 1979; 26:473-479
- [37] Sebastian PR, Fisher M, Smith TW, et al. Intramedullary Spinal Cord Metastasis. *Surg Neurol*. 1981; 16:336-339
- [38] Nittner K, Olivecrona H, Tonniss W. *Handbuch der Neurochirurgie*. New York: Springer-Verlag; 1972:1-606
- [39] Post KD, Stein BM, Schmidek HH, et al. Surgical Management of Spinal Cord Tumors and Arteriovenous Malformations. In: *Operative Neurosurgical Techniques*. 3rd ed. Philadelphia: W.B. Saunders; 1995:2027-2048
- [40] Nadkarni TD, Rekatte HL. Pediatric Intramedullary Spinal Cord Tumors: Critical Review of the Literature. *Child's Nervous System*. 1999; 15:17-28
- [41] Greenwood J. Surgical Removal of Intramedullary Tumors. *J Neurosurg*. 1967; 26:276-282
- [42] Albright AL. Pediatric Intramedullary Spinal Cord Tumors. *Child's Nervous System*. 1999; 15:436-437
- [43] Morota N, Deletis V, Constantini S, et al. The Role of Motor Evoked Potentials During Surgery for Intramedullary Spinal Cord Tumors. *Neurosurgery*. 1997; 41:1327-1336
- [44] Kothbauer P, Deletis V, Epstein FJ. Intraoperative Spinal Cord Monitoring for Intramedullary Surgery: An Essential Adjunct. *Pediatric Neurosurgery*. 1997; 26:247-254
- [45] Albright AL. Intraoperative Spinal Cord Monitoring for Intramedullary Surgery: An Essential Adjunct? *Pediatric Neurosurgery*. 1998; 29
- [46] Ulkatan S, Jaramillo AM, Tellez MJ, et al. Incidence of intraoperative seizures during motor evoked potential monitoring in a large cohort of patients undergoing different surgical procedures. *J Neurosurg*. 2017; 126:1296-1302
- [47] Kothbauer KF. Intraoperative neurophysiologic monitoring for intramedullary spinal-cord tumor surgery. *Neurophysiol Clin*. 2007; 37:407-414
- [48] Park JH, Hyun SJ. Intraoperative neurophysiological monitoring in spinal surgery. *World J Clin Cases*. 2015; 3:765-773
- [49] Guidetti B, Mercuri S, Vagnozzi R. Long-Term Results of the Surgical Treatment of 129 Intramedullary Spinal Gliomas. *J Neurosurg*. 1981; 54:323-330
- [50] Janin Y, Epstein JA, Carras R, et al. Osteoid Osteomas and Osteoblastomas of the Spine. *Neurosurgery*. 1981; 8:31-38
- [51] Amacher AL, Eltomey A. Spinal Osteoblastoma in Children and Adolescents. *Childs Nervous System*. 1985; 1:29-32
- [52] Lichtenstein L, Sawyer WR. Benign Osteoblastoma. *J Bone Joint Surg*. 1964; 46A:755-765
- [53] Shives TC, Dahlin DC, Sim FH, et al. Osteosarcoma of the spine. *J Bone Joint Surg Am*. 1986; 68:660-668
- [54] Fox MW, Onofrio BM. The natural history and management of symptomatic and asymptomatic vertebral hemangiomas. *J Neurosurg*. 1993; 78:36-45
- [55] Healy M, Herz DA, Pearl L. Spinal Hemangiomas. *Neurosurgery*. 1983; 13:689-691
- [56] Richardson RR, Cerullo LJ. Spinal Epidural Cavernous Hemangioma. *Surgical Neurology*. 1979; 12:266-268
- [57] Cosgrove GR, Bertrand G, Fontaine S, et al. Cavernous Angiomas of the Spinal Cord. *Journal of Neurosurgery*. 1988; 68:31-36
- [58] Tekköt IH, Açıköz B, Saglam A, et al. Vertebral Hemangioma Symptomatic During Pregnancy - Report of a Case and Review of the Literature. *Neurosurgery*. 1993; 32:302-306
- [59] Graham JJ, Yang WC. Vertebral Hemangioma with Compression Fracture and Paraparesis Treated with Preoperative Embolization and Vertebral Resection. *Spine*. 1984; 9:97-101
- [60] Kosary IA, Brahm J, Shacked I, et al. Spinal Epidural Hematoma due to Hemangioma of Vertebra. *Surgical Neurology*. 1977; 7:61-62
- [61] Wang B, Zhang L, Yang S, et al. Atypical Radiographic Features of Aggressive Vertebral Hemangiomas. *J Bone Joint Surg Am*. 2019; 101:979-986
- [62] Gaudino S, Martucci M, Colantonio R, et al. A systematic approach to vertebral hemangioma. *Skeletal Radiol*. 2015; 44:25-36
- [63] Persaud T. The polka-dot sign. *Radiology*. 2008; 246:980-981
- [64] Tarantino R, Donnarumma P, Nigro L, et al. Surgery in extensive vertebral hemangioma: case report, literature review and a new algorithm proposal. *Neurosurg Rev*. 2015; 38:585-92; discussion 592
- [65] Athana AK, Tandon SC, Pant GC, et al. Radiation therapy for symptomatic vertebral haemangioma. *Clin Oncol (R Coll Radiol)*. 1990; 2:159-162
- [66] Heyd R, Seegenschmiedt M H, Rades D, et al. Radiotherapy for symptomatic vertebral hemangiomas: results of a multicenter study and literature review. *Int J Radiat Oncol Biol Phys*. 2010; 77:217-225
- [67] Jiang L, Liu XG, Yuan HS, et al. Diagnosis and treatment of vertebral hemangiomas with neurologic deficit: a report of 29 cases and literature review. *Spine J*. 2014; 14:944-954
- [68] Premat K, Clarencon F, Cormier E, et al. Long-term outcome of percutaneous alcohol embolization combined with percutaneous vertebroplasty in aggressive vertebral hemangiomas with epidural extension. *Eur Radiol*. 2017; 27:2860-2867
- [69] Piper K, Zou L, Li D, et al. Surgical Management and Adjuvant Therapy for Patients With Neurological Deficits From Vertebral Hemangiomas: A Meta-Analysis. *Spine (Phila Pa 1976)*. 2020; 45:E99-E110
- [70] Yao KC, Malek AM. Transpedicular N-butyl cyanoacrylate-mediated percutaneous embolization of symptomatic vertebral hemangiomas. *J Neurosurg Spine*. 2013; 18:450-455
- [71] Acosta FL Jr, Sanai N, Chi JH, et al. Comprehensive management of symptomatic and aggressive vertebral hemangiomas. *Neurosurg Clin N Am*. 2008; 19:17-29
- [72] Boschi V, Pogorelec Z, Gulán G, et al. Management of cement vertebroplasty in the treatment of vertebral hemangioma. *Scand J Surg*. 2011; 100:120-124
- [73] Saracen A, Kotwica Z. Vertebroplasty (PVP) is effective in the treatment of painful vertebral hemangiomas. *Acta Orthop Belg*. 2018; 84:105-107
- [74] Hu W, Kan SL, Xu HB, et al. Thoracic aggressive vertebral hemangioma with neurologic deficit: A retrospective cohort study. *Medicine (Baltimore)*. 2018; 97. DOI: 10.1097/MD.0000000000001275
- [75] Blecher R, Smorgick Y, Anekstein Y, et al. Management of symptomatic vertebral hemangioma: follow-up of 6 patients. *J Spinal Disord Tech*. 2011; 24:196-201
- [76] Goldstein CL, Varga PP, Gokaslan ZL, et al. Spinal hemangiomas: results of surgical management for local recurrence and mortality in a multicenter study. *Spine (Phila Pa 1976)*. 2015; 40:656-664
- [77] Vasudeva VS, Chi JH, Groff MW. Surgical treatment of aggressive vertebral hemangiomas. *Neurosurg Focus*. 2016; 41. DOI: 10.3171/2016.5.FOCUS.16169
- [78] Kato S, Kawahara N, Murakami H, et al. Surgical management of aggressive vertebral hemangiomas causing spinal cord compression: long-term clinical follow-up of five cases. *J Orthop Sci*. 2010; 15:350-356
- [79] Marcove RC, Sheth DS, Brien EW, et al. Conservative surgery for giant cell tumors of the sacrum. The role of cryosurgery as a supplement to curettage and partial excision. *Cancer*. 1994; 74:1253-1260

Part XIV

Head Trauma

60	Head Trauma – General Information, Grading, Initial Management	1000
61	Concussion, High-Altitude Cerebral Edema, Cerebrovascular Injuries	1019
62	Neuromonitoring in Head Trauma	1036
63	Skull Fractures	1062
64	Traumatic Hemorrhagic Conditions	1071
65	Gunshot Wounds and Non-Missile Penetrating Brain Injuries	1093
66	Pediatric Head Injury	1098
67	Head Injury – Long-Term Management, Complications, Outcome	1106